

A History of Western Philosophy

Ralph McInerny

Volume I

Foreword / Acknowledgements

- Part I: Presocratic Philosophy
 - **Chapter I: Before Philosophy**
 - A. The Quarrel Between Philosophy and Poetry
 - B. The Theological Poets
 - C. Greek Primitive Religion
 - **Chapter II: The Ionians**
 - A. Thales of Miletus
 - B. Anaximander of Miletus
 - C. Anaximenes of Miletus
 - D. Xenophanes
 - E. Heraclitus of Ephesus
 - **Chapter III: The Italians**
 - A. The Pythagoreans
 - 1) Pythagoras of Samos
 - 2) Pythagorean Doctrines
 - B. Parmenides of Elea
 - C. Zeno of Elea
 - D. Melissus of Samos
 - **Chapter IV: Empedocles of Acragas**
 - A. *On Nature*
 - B. *Purifications*
 - **Chapter V: Anaxagoras of Clazimene**
 - **Chapter VI: Atomism**
 - Diogenes of Apollonia
 - **Chapter VII: The Sophists**
 - A. Protagoras of Abdera
 - B. Gorgias of Leontini
 - C. Prodicus of Ceos
 - D. Some Other Sophists
 - E. Concluding Summary
- Part II: The Classical Period
 - **Chapter I: Socrates**
 - A. His Life
 - B. The Character of Socrates

- C. The Doctrine of Socrates
 - D. The Socratic Schools
- **Chapter II: Plato**
 - A. The Man and His Work
 - B. The Doctrine of Forms
 - C. The Crisis in Plato's Thought
 - D. Plato's Natural Doctrine
 - E. Plato's View of Man
- **Chapter III: Aristotle**
 - A. The Man and his Work
 - B. The Nature and Division of Philosophy
 - C. Aristotle's Logic
 - D. Aristotle's Philosophy of Nature
 - E. Moral and Political Philosophy
 - F. First Philosophy
- **Part III: The Hellenistic Period**
 - **Chapter I: Epicureanism**
 - A. Canonic
 - B. Physics
 - C. Ethics
 - D. The History of the School
 - **Chapter II: The Stoics**
 - A. Logic
 - B. Physics
 - C. Ethics
 - D. The Roman Stoics
 - **Chapter III: Sceptics and the New Academy**
 - A. Pyrrho of Elis
 - B. Timon of Philus
 - C. Arcesilaus
 - D. Carneades of Cyrene
 - E. Some Later Sceptics
 - F. The Subsequent History of the Academy
 - **Chapter IV: Neoplatonism**
 - A. Revival of Pythagoreanism.
 - B. Plotinus
 - C. After Plotinus
- **Bibliography**
 - Readings for Part One
 - A. Sources
 - B. General Studies
 - C. Particular Studies
 - Readings for Part Two
 - A. Sources and Studies: Plato
 - B. Sources and Studies: Aristotle
 - Readings for Part Three
- **Index to Volume I**

Volume II

- Part I: The Age of Augustine
 - **Chapter I: Faith and Philosophy**
 - **Chapter II: Saint Augustine**
 - A. The Man and His Work
 - B. Philosophy and the Arts
 - C. Philosophy and Beatitude
 - D. Criticism of Platonism
 - E. What Is Man?
 - F. God
 - G. Creation
 - H. *The City of God*
 - I. Conclusion
 - **Chapter III: Denis the Areopagite**
 - **Chapter IV: Boethius**
 - A. The Man and His Work
 - B. Faith and Reason
 - C. Division of Philosophy
 - D. The Status of Universals
 - E. Plato or Aristotle?
 - F. Conclusion
 - **Chapter V: Cassiodorus, Isadore, Bede**
- Part II: The Carolingian Renaissance
 - **Chapter I: Alcuin and Rhabanus Maurus**
 - A. Charlemagne and the Schools
 - B. Alcuin (735-804)
 - C. Fredegisus of Tours
 - D. Rhabanus Maurus (784-856)
 - E. The Carolingian Heritage
 - **Chapter II: John Scotus Erigena**
 - A. His Life and Works
 - B. Faith and Philosophy
 - C. The Division of Nature
 - **Chapter III: Other Ninth and Tenth Century Figures**
 - A. Heiric of Auxerre (c.835 - c.887)
 - B. Remigius of Auxerre (c.841 - c.908)
 - C. Gerbert of Aurillac (c.940-1003)
- Part III: The Twelfth Century
 - **Chapter I: Introduction**
 - **Chapter II: Saint Anselm of Canterbury**
 - A. The Man and His Work
 - B. Faith and Reason
 - C. The Proof of God's Existence
 - D. Anselm and Dialectics

- **Chapter III: Peter Abelard**
 - A. The Man and His Work
 - B. Abelard's Logic
 - C. Faith and Reason
 - D. Abelard's Ethics
- **Chapter IV: The School of Chartres**
 - A. From Fulbert to Bernard
 - B. Gilbert of Poitiers (1076-1154)
 - C. William of Conches (c.1080 - c.1184)
 - D. Thierry of Chartres (died before 1155)
 - E. Clarenbald of Arras (died c.1160)
 - F. John of Salisbury (1110-1180)
- **Chapter V: Monastic Thought**
 - A. Hugh of St. Victor (1096-1141)
 - B. Other Victorines
 - C. Bernard of Clairvaux (1090-1153)
 - D. Other Figures
- **Chapter VI: Dominicus Gundissilinus**
 - A. What Is Philosophy?
 - B. The Division of Philosophy
- **Part IV: The Thirteenth Century**
 - **Chapter I: The Background**
 - A. The Universities
 - B. Translations
 - C. Islamic Philosophy
 - **Chapter II: The Beginnings**
 - A. William of Auvergne (c.1180-1249)
 - B. Alexander of Hales (c.1185-1245)
 - C. Robert Grosseteste (1175-1253)
 - **Chapter III: Albert the Great**
 - A. The Man and His Work
 - B. Faith and Reason
 - C. Conclusion
 - **Chapter IV: Roger Bacon**
 - A. His Life and Work
 - B. The Opus majus
 - C. Conclusion
 - **Chapter V: Saint Bonaventure**
 - A. The Man and His Work
 - B. The Nature of Philosophy
 - C. Simultaneity of Knowledge and Belief
 - D. Is Philosophy Autonomous?
 - E. The Division of Philosophy
 - F. The Divine Ideas
 - G. The Nature of Illumination
 - H. Proofs of God's Existence
 - I. Creation and Universal Hylomorphism

- J. Conclusion
- **Chapter VI: Saint Thomas Aquinas**
 - A. The Man and His Work
 - B. Philosophy and Theology
 - C. The Division of Philosophy
 - D. Logic
 - E. Natural Philosophy
 - F. Metaphysics
 - G. Moral Philosophy
 - H. Thomas and His Time
- **Chapter VII: Conclusion**
- **Part V: The Fourteenth Century**
 - **Chapter I: Introduction**
 - **Chapter II: John Duns Scotus**
 - A. The Man and His Work
 - B. Being and God
 - C. Faith and Reason
 - **Chapter III: William of Ockham**
 - A. The Man and His Work
 - B. Knowledge
 - C. Logic
 - D. Metaphysics
- **Index to Volume II**

Foreword

This history of ancient philosophy tries to give a comprehensive but wholly introductory sketch of a difficult and changing historical terrain. We are still learning about the beginnings of philosophy and the scholarly contributions to our knowledge mount almost menacingly, intimidating one who would attempt an over-all simplified presentation. Writing a memo in anticipation of the Libyan battles, Churchill predicted that renown awaited the commander who would restore artillery to its proper place on the battle field: later he seemed as pleased with his phrasing of the claim as of its fulfillment. Perhaps a relieved welcome, if not renown, awaits an introductory history which is not studded with the artillery of footnotes apprising the bewildered neophyte of esoteric studies on the fine points of recent scholarship in the period he is encountering for the first time. It is my feeling that there is little point in cluttering an introductory work with such references: the teacher does not need them and the student is not ready for them. Better unabashedly to popularize the period so as to make it as immediately and painlessly accessible as can honestly be done. The short reading lists at the back of the book will enable the interested reader to begin study in that scholarship on which such books as this are based. Of course, in the narrative, broad divergences of interpretation are mentioned and occasionally even adjudicated, but in every instance the attitude has been irenic and permissive. It is an Aristotelian axiom that we must begin any study with a confused view of the whole and this volume provides only a first step in the study of ancient philosophy.

The present work was not conceived to fill some glaring gap in the works available for classroom use; there is a plethora of good histories of ancient philosophy. This effort differs from some in the manner indicated in the preceding paragraph; it differs from others in being more brief; it differs from all, hopefully, in the style of its approach which may appeal to student and teacher alike. It is difficult to resist the impulse to put what one has learned into his own words even when what he knows is neither a private possession nor a personal discovery. In the course of teaching the history of ancient philosophy at the University of Notre Dame, on campus as well as in Moreau Seminary, I amassed folders of notes, made sketches of chapters, had visions of a volume. When an opportunity came to prepare this book for Henry Regnery Company I was willing if not wholly ready to accept it. The result, being actual, seems almost a betrayal of the shimmering possibility I had cherished. But that is often the way with actualities. I shall now let my imagination play on the possibility

that this book will be of some aid to teacher and student in courses in the history of ancient philosophy. That hope, at once modest and immense, is why it was written.

Acknowledgements

I would like to express my gratitude to the following publishers for their permission to quote copyrighted material: To the Clarendon Press, Oxford, for quotations from *The Works of Aristotle Translated into English*, 12 volumes, edited by W. D. Ross, 1905-1952, and from E. R. Dodds, *Proclus' Elements of Theology*, 1933; To Harvard University Press for quotations from the following volumes in the Loeb Classical Library: *Diogenes Laertius' Lives of the Eminent Philosophers*, 2 volumes, edited by R. D. Hicks, 1950; H. G. Evelyn-White, *Hesiod, The Homeric Hymns and Homeric* (1943); *Sextus Empiricus*, 4 volumes, edited by R. G. Bury, 1933-1949. To the Editors of the *Encyclopedia Britannica* for permission to quote from the MacKenna and Page translation of the *Enneads* of Plotinus which appears in *Great Books of the Western World*. To Cambridge University Press for generous permission to quote from G. S. Kirk and J. E. Raven, *The Presocratic Philosophers*, Cambridge, 1957. To Basil Blackwell for permission to quote from Kathleen Freeman, *Ancilla to the Presocratic Philosophers*, Oxford, 1948. To Appleton-Century-Crofts for permission to quote from *Selections from Hellenistic Philosophy* by Gordon Clark, 1940. To Princeton University Press for permission to quote from Philip Wheelwright, *Heraclitus*, Princeton, 1959. To Columbia University Press for permission to quote from *Anaximander and the Origins of Greek Cosmology* by Charles H. Kahn, 1960.

Unless otherwise indicated, all quotations from the dialogues of Plato are taken from the 19th century translation by Benjamin Jowett. Quotations from the letters of Plato are from the L. A. Post translation by permission of The Clarendon Press, Oxford.

Volume I

Part I: Presocratic Philosophy

Chapter I

Before Philosophy

In presenting the history of philosophy from its beginnings to Plotinus, we are assuming that philosophy did indeed have a beginning and that it is possible to pass a more or less satisfactory judgment as to when this took place. In the records and tradition which have come down to us, Thales of Miletus is said to be the first philosopher; accordingly, if we examine what he is said to have done and taught, we can formulate a notion of what philosophy meant for the Greeks -- even before their word "philosophy" existed. In doing so, however, we are explicitly or implicitly contrasting Thales with his predecessors, by definition non-philosophers. An examination of the prior state of affairs will sharpen our understanding of what philosophy itself is.

The procedure suggested seems wonderfully simple, but it is no easy matter to follow it out to the desired term. An examination of the activities and writings of the predecessors of Thales turns up a good many ways of viewing man and the world not wholly different from those which have come to be called philosophical. In the absence of a sharp line of demarcation in the documents and tradition, we might approach the past armed with our notion of what philosophy is and, when we find something answering to it, say: here is where philosophy begins. Obviously such a method could produce as many opinions on the identity of the first philosopher as there are different contemporary views on the nature of philosophy. The method may be made less arbitrary by accepting the view of some important Greeks that philosophy arose out of myth, religion, or poetry. Yet it is possible -- and indeed frequently done -- to understand this opposition in terms of what we mean by myth, religion, and poetry, and doubt arises as to whether the transition described is the one that historically occurred.

The fact that some ancient Greeks themselves spoke of oppositions between philosophy and other pursuits, for example, myth and poetry, and seem to

imply, when they do so, that non-philosophy and philosophy are related not only absolutely but chronologically as well, suggests the possibility of a defensible statement of what philosophy was for the Greeks, as well as of the state of affairs out of which it arose. By pursuing such oppositions we will not find ourselves provided with so clearcut a distinction that all philosophy can be placed on one side of a line and all non-philosophy on the other, but we will have poles which will enable us to evaluate particular documents. And then we will be able to see why the Greeks thought Thales was the first philosopher. All we shall do here is to briefly document the opposition in question, say a few things about the supposed non-philosophers, and leave it to the sequel to show whether early philosophers are set off from their predecessors in the way claimed.

A. The Quarrel Between Philosophy and Poetry

In the tenth book of the *Republic*, having decided that poetry will have to be banished from the ideal city he is describing, Plato says, "But, lest poetry should convict us of being harsh and unmannerly, let us tell her further that there is a long-standing quarrel between poetry and philosophy." (607B; Cornford) It is not difficult to document this quarrel from the side of philosophy.⁽¹⁾

Xenophanes says: "Homer and Hesiod have attributed to the gods everything that is a shame and reproach among men, stealing and committing adultery and deceiving each other."⁽²⁾ "But mortals consider that the gods are born, and that they have clothes and speech and bodies like their own." (#170) With Xenophanes and Plato, the charge against the poets is reduced to the way the gods are treated; this suggests that philosophers speak more accurately of the gods, that theology and philosophy are somehow ultimately connected. Heraclitus also criticizes the type of religion which is celebrated by Homer and Hesiod.

They vainly purify themselves with blood when they are deified with blood, as though one who had stepped into mud were to wash with mud; he would seem to be mad, if any of men noticed him doing this. Further, they pray to these statues, as if one were to carry on a conversation with houses, not recognizing the true nature of gods or demi-gods. (Kirk and Raven, #224)

Thus very early philosophy entered the arena of public opinion to correct the abuses and practices of religion and to make statements about the gods. Equally it showed a concern with the actions of men, and thus implied that philosophy provides a guide for conduct, if not a way of life.⁽³⁾ Philosophy, then, is not so

much ordered to expunging religion as it is meant to purify it by its rationally defensible statements about the gods and rites which would not demean man in his worship of the gods.

If philosophers are critical of the poets' theological remarks, they take no single attitude towards poetic myths. "I can tell you, Socrates, that, when the prospect of dying is near at hand, a man begins to feel some alarm about things that never troubled him before. He may have laughed at those stories [mythoi] they tell of another world and of punishments there for wrongdoing in this life; but now the soul is tormented by a doubt whether they may not be true." (*Republic*, 330D) There is a juxtaposition of poets and makers of myth (*Ibid.*, 329D), such that one can state the opposition between philosophy and poetry as one between philosophy and myth. And all *mythos* means in these remarks of Plato is a story or narrative. Still because myth is grouped with poetry and poetry with statements about religion, we must inquire into both the poetry in question and the religion it reflects. In Plato the opposition between philosophy and myth is not clear, since his own employment of myth is notorious and self-avowed. The following exchange from the *Protagoras* is a good example. "Shall I, as an elder, speak to you as a younger man in an apologue or myth, or shall I argue out the question? To this several of the company answered that he should choose for himself. Well then, he said, I think that the myth will be more interesting" (320G). Whatever his own practice, however, Plato here and elsewhere (e.g., *ibid.*, 324D; *Gorgias*, 523A; *Timaeus* 23E) distinguishes between *mythos* and *logos*. The latter is characteristically philosophical whereas the former is poetical.

When we turn to Aristotle, the opposition between philosophy and myth sharpens, but there is also present an indication of what they have in common. Note how the following text states the opposition.

The disciples of Hesiod and all the theologians have been satisfied with explanations that seem to them credible, but that make no sense to us. For when they present the principles as gods and say that anything that has not tasted nectar and ambrosia is born mortal, it is clear that they are using words which, though familiar enough to them, are explanations completely above our heads. If the gods take nectar and ambrosia for the sake of pleasure, their doing so does not explain their being; and if the gods do so for the sake of their very being, how could beings who need nourishment be eternal? But why should we examine seriously the spurious wisdom of myths? We must look for information to those who use the language of proof, and we must ask them why it is that if

all things consist of the same elements some are by nature eternal, whereas others perish (*Metaphysics*, 1000a5-23).

Those who fabricate myths do not use the language of proof or demonstration; the opposition is between speaking mythically and apodictically. When one fails to make use of the "language of proof," talk becomes like that of the poet, a lapse of which Aristotle thought Plato had been guilty. (See *Metaphysics* 991a18 ff.) As we shall see later, Aristotle also argues that myth and Philosophy have things in common; his final position seems to suggest a graded scale of argumentation with poetry at one extreme and apodictic proof at the other. Aristotle quotes with approval the line "Bards tell many a lie" (*Metaphysics*, 983a3), but his developed view on that point must be sought in the *Poetics* (Chaps. 24-5).

B. The Theological Poets

Notice that it is Homer and Hesiod who are the object of the critical remarks the philosophers direct at poetry, although popular religion also comes in for criticism. Why do the philosophers consider Homer and Hesiod important enough to be singled out for special attention? The answer to this question sheds light on Greek culture both before and during the golden age of philosophy. Until recently students in America usually knew Homer only through laboriously wrestling with a small portion of the Greek text of the *Iliad*, often reproduced in editions containing one or several books (of the twenty-four) surrounded by learned notes, ingenious word studies, and a general aura of Teutonic scholarship. Sometimes despite the method the student caught glimpses of the poem's beauty and could therefore perhaps appreciate that the *Iliad* and the *Odyssey* were composed for oral delivery and were to be memorized. It was not unusual for the Greek schoolboy to have his Homer by heart, that is at least substantial portions of the two epics. There is nothing comparable in our own times to the influence Homer had on the Greeks. Even Plato, to whose criticism we have already alluded, is forced to acclaim Homer as the most divine of the poets. Another ancient view, that of Herodotus, pays tribute to Homer and Hesiod. "Homer and Hesiod composed a poetical theogony for the Hellenes, gave the gods their significant names, assigned to them their proper honors and arts, and indicated the various kinds of them."

The *Iliad*, concerned with the fall of Troy, opens with a quarrel between Agamemnon and Achilles over two captive girls. Achilles loses both the argument and his girl, and enraged at the Greeks for the injustice, withdraws

from the battle. Things go badly for the Greeks and Agamemnon asks Achilles to rejoin the battle. He refuses, but lends his armor to his friend, Patroclus, who does brilliantly until he is killed by Hector, son of Troy's king, Priam. The death of Patroclus moves Achilles to return to the battle; he slays Hector, and drags his body around the walls of Troy. His grief undimmed, he returns the body of Hector to Priam for burial out of pity for the old man.

The inadequacy of these remarks cannot be conveyed simply by saying that no great poem can be replaced by a paraphrase. Obviously, we have not even begun to suggest the richness of action in the epic nor will we try to do Homer's poetry the poor service of our praise. What we have hinted at may be termed the terrestrial or human plane of the epic; there is another plane, that of the gods, whose actions, rivalries and involvement in the acts of men is an essential part of the story Homer is telling. The names of these Olympian gods are familiar to everyone: Zeus, his wife Hera, Aphrodite, Ares, Athene, etc. The world of Homer fairly swarms with gods and not very exemplary gods at that. They quarrel, they fight, they deceive one another; they are at once involved in human affairs and disclaim responsibility for the evil men do. All this may seem perplexing to a modern man, much more so than it did to Homer's critics in antiquity. For while Zeus has some sort of supremacy over the other Olympians, he is not the oldest of the gods and has surprising limitations on his power. Thus his deception by Hera is an element in the beginning of the Trojan war, and he is confessedly limited by fate or *moira*. (*Il.* XVI, 431 ff.) The parents of the gods are Kronos and Rhea, and their three sons are Zeus, Hades and Poseidon, each of whom has been allotted a portion of the world as his province (*moira*). Thus Poseidon speaks (*Il.*, XV, 185 ff.):

For we are three brothers, born of Kronos and Rhea, Zeus and I and Hades, the lord of the dead. And in three lots were all things divided, and each took his appointed domain. When we cast the lots, to me fell the hoary sea, that I should dwell therein forever; and Hades drew the misty darkness, and Zeus the broad heaven among the bright air and the clouds: the earth and high Olympus are yet common to all.^{4}

Fate or destiny is above the gods and all must bow to it. In Homer, fate is not something which detracts from the freedom and responsibility of human acts; it is rather an expression of the seriousness of our acts, all of which we will be held accountable for. Particularly is this true in the case of pride (*hybris*); when a man transcends the limits of his estate an inevitable retribution follows. We should not be misled, then, by the intervention of the gods in the epics of

Homer. Such intervention is never looked upon as fixing the human action in a set pattern. The evil consequences of a man's actions cannot be blamed on the gods. Zeus says in the *Odyssey* (I.26), "Alack, see how mortals lay blame upon the gods. For they say that evils come from us; but it is they who, from the blindness of their own hearts, have sorrows beyond what is ordained."

Recognition of the divine dimension in the Homeric epics does not mean there is a systematic theology in Homer, nor that all gods are Olympian personalities. Although the genealogy just sketched could lead us to believe that Kronos is the origin of all the gods, yet Homer speaks of Okeanos as the source (*genesis*) of all the gods, and Okeanos as a river which surrounds the world. So too the sky (Ouranos) and earth (Gaia) are gods, sleep is a god, the winds are gods, as is justice, and so on. Is it possible to find the divine in Homer? Hack⁽⁵⁾ suggests that any power or influence on human life is likely to be called divine, and that such powers are immortal and always have cosmic significance; they play a role in the history of the universe. But it is not Homer's purpose to develop a theory about the divine and the interrelationships between the gods. He is telling a story about human conduct seen against the background of a world where injustice does not go unpunished, where the deeds of men have consequences which are inexorable. This motif is clear from the opening lines of the *Iliad*.

Sing, goddess, the destructive wrath of Achilles, son of Peleus, which brought countless sufferings upon the Greeks and hurled many valiant souls of heroes to Hades and made them the prey of dogs and birds and yet the will of Zeus was all the while being done. (I, 1-5)

Achilles' wrath at the loss of Briseis to Agamemnon leads to his refusal to fight, and is destined to have consequences which cannot be avoided. Many will die because of his refusal, among them his friend, Patroclus; and Achilles himself, when he has dragged the body of Hector around Troy and delivered it to Priam, feels compassion for the bereaved father. The will of Zeus mentioned in the passage is not the arbitrary will of Hera's husband but rather that to which Zeus too is subject -- fate or destiny. In the *Odyssey* also, the punishment of the suitors and Odysseus' reunion with Penelope show the triumph of justice. The world of Homer is above all a moral world -- a world of law and justice, both of which transcend the quarrels of the Olympians. Moreover, his later criticisms of the gods he depicts remind us that Homer does not always approve of their activities.

What in Homer is hardly more than the background of human action becomes in Hesiod's *Theogony* the major object of concern. Who are the gods; what are the relationships between them; how did the world and man come into being? The muses, daughters of Zeus and Memory, provide the answers.

And they, uttering their immortal voice, celebrate in song first of all the reverend race of the gods from the beginning, those whom Earth and wide heaven begot, and the gods sprung of these, givers of good things. Then, next, the goddesses sing of Zeus -- the father of gods and men, as they begin and end their strain -- how much he is the most excellent among the gods and supreme in power. And again, they chant the race of men and strong giants, and gladden the heart of Zeus. . . . (*Theogony*, 43-51).¹⁶¹

The poem has a threefold burden. First, the coming into being of the world, the cosmogony which is spoken of in terms of the first race of the gods. Secondly, the sequence of generations of the gods is given, the theogony proper. Thirdly, the story of how Zeus gained supremacy over the other gods. We are faced here with a shifting notion of divinity. Hesiod puts us on guard against confusing the first race of the gods with the anthropomorphic gods most prominent in Homer. Indeed, we find in the prologue to the *Theogony* evidence of a critical attitude towards what the muses sing, since the muses themselves observe, "We know how to speak many false things as though they were true; but we know, when we will, to utter true things." (27-8)

These lines indicate a critical attitude toward the traditional stories concerning the gods, and Hesiod's approach to the Olympian gods is such that the preceding cosmogony is seen as all but identical with the later efforts of the Philosophers.

We have seen that Homer speaks of Okeanos as the source of all the gods. The word he uses (*genesis*) suggests a giving birth, and we might feel that the other gods are sons of Okeanos in the way that later gods are said to be sons of Zeus. Hesiod does not put Okeanos first, but he too speaks in terms of generation.

Verily at the first Chaos came to be, but next wide-bosomed Earth, the ever-sure foundation of Tartarus in the depth of the wide-pathed Earth, and Love, fairest among the deathless gods, who unnerves the limbs and overcomes the mind and wise counsels of all gods and all men within them. From Chaos came forth Erebus and black Night: but of Night were born Aether and Day whom she conceived and bore from union in love with Erebus. And Earth first bore

starry heaven, equal to herself, to cover her on every side, and to be an ever-sure abiding place for the blessed gods. And she brought forth long hills, graceful haunts of the goddess-nymphs who dwell among the glens of the hills. She bore also the fruitless deep with her raging swell, Pontus, without love or marriage. (116-132)

By saying that Chaos first came to be, Hesiod clearly does not mean that first there was chaos in our meaning of that term since chaos is not unqualifiedly first -- it came to be. What is Chaos? The only meaning it has here is gap or opening and Hesiod is probably saying that the beginning of the world as we know it occurred when earth was separated, presumably from sky, though sky is later spoken as if its becoming were distinct from that of earth. Despite this reduplication, we can understand Hesiod as saying that at first earth and sky were one and then were separated -- that is, chaos, or the gap between them, came to be. Elsewhere (*Theogony* 700) he speaks of chaos in this sense, and Aristophanes uses the word to mean that in which, or through which, birds fly. (*Birds*, 192) Thus, "came to be" (*genet'*) seems to mean "to be separated," or the phrase can mean, separation came to be. This is a straightforward kind of remark which leads us to conclude that, although Earth and Sky and the rest are spoken of as gods here, they are nevertheless plain old earth and sky as well. That Hesiod, in this passage, is striving for a non-anthropomorphic explanation is also suggested by the denial accompanying the description of how earth gives birth to the sea without love or marriage. In other words, here is a birth which is not a birth in the human sense, but results in the separation of sea and dry land. Perhaps this passage should be collated with that in which the muses tell of the ambiguity of their tales.

Hesiod is not necessarily spurning, in some sharp and definitive way, the mythical and anthropomorphic approach of Homer. The very passage before us is one of mixed quality since, if the giving birth to sea by earth is said to be without love or marriage, earth bore Aether and Day from a union in love with Erebus; what is more, Love or Eros is also spoken of as coming into being as if things once separated needed a principle of union to beget other things. Nevertheless, the non-anthropomorphic picture of the world which emerges from these lines is one with earth below and sky above. In the gap between, night and day come to be; the sky is starry and the earth hilly with dry land separated from the sea. This picture must also accommodate Eros and the goddess-nymphs in the glens. In other words, while we seem to be reading of sky and earth, hills and sea, night and day, stars and atmosphere, we are also told of nymphs and love, the former personified, the latter almost so. This

ambiguous cosmogony prompts the judgment that Hesiod deserves to be numbered among the philosophers, though his description trails mythical elements. Resistance to this interpretation is sometimes based on a failure to recognize troublesome elements in unquestioned philosophical accounts, while insistence on the mythical in early philosophical statements has led to an important generalization concerning the origins of philosophy, the relation of philosophy to myth.⁽⁷⁾

Immediately following the cosmogonical passage quoted, the theogony proper begins; earth and heaven become parents in the usual sense and their progeny is listed. In this third generation such familiar gods as Okeanos, Rhea, Themis, Memory, and Kronos are born. Sky, or Ouranos, is not unequivocally proud of his offspring and keeps some hidden away in earth (Gaia), an outrage the latter finds difficult to countenance. She urges Kronos to revenge the injustice; he does so by castrating his father as he lies upon the earth, throwing his members into the sea. In their flight, drops of blood fall on earth and giants spring up, and from the drowned members Aphrodite rises from the foam of the sea. The generations of the gods are stars and planets, winds and seasons, the emotions of man and the evils which plague him, and the familiar Olympians. All in all, it is an attempt at a systematic theology which can account for everything and everyone hitherto called divine and which ends with Zeus as the chief god of Mount Olympus. As in Homer, Zeus, together with Hades and Poseidon is a son of Kronos and Rhea. Kronos is depicted as devouring his sons as soon as they are born lest someone replace him as king of the gods. Predictably Rhea looks darkly on this and when she is about to give birth to Zeus; seeks for some way to prevent the usual outcome for her offspring. She consults her parents, earth and sky, who presumably are familiar with such marital difficulties, and they spirit her away to Crete where she gives birth to Zeus. In place of the new-born Zeus, a wrapped stone is rushed to Kronos; he swallows it. Subsequently Rhea induces him to cough up all his sons; and when the stone comes forth first, it is set up by Zeus for the veneration of mortals. All that now remains for Zeus is to vanquish the Titans and Typhaeus, the fire-breathing monster. When he does so, his supremacy is complete.

We have still to consider the origin of man as the Greeks saw it. A somewhat melancholy account is given in another poem of Hesiod, *Works and Days*. A famous passage (ll. 110-201) tells of the five ages of man. First, the gods made a golden race of men who lived when Kronos reigned supreme; their life was without toil or care but eventually they died out. Their spirits still dwell on earth and they are kindly guardians of mortals. Secondly, a silver generation, less

noble than the first, was made. It took a hundred years for a child to reach maturity, and their prime was brief and filled with sinning because of their foolishness. Their spirits dwell in the underworld and are worthy of men's honor. Thirdly came a race of bronze men, a war-like breed. These killed one another off and were sent to Hades. As with the second and third, the fourth generation of men was made by Zeus, and it was a race of heroes, god-like men. These are they who fought at Troy, for example, and some of them live now long the shores of Okeanos, ruled over by Kronos; their life is similar to that attributed to the golden race. Lastly comes the race of iron, those men of whom Hesiod is one, though to his great sorrow. Upon these men the gods lay troubles, though some good is given with evil. This race will degenerate to the point where children will be born old and grey-haired and then die out. In the meantime, they are known for their injustice, their lack of respect, the ease with which they break their oaths. The time will come when the gods will desert this race, and only evil will remain.

This story of the races of men parallels in its way that of the races of the gods, and in *Works and Days* Hesiod urges his brother, a prodigal son, to turn himself to hard work -- farm work, which is not romanticised, though it is preferred in no uncertain terms to the life of the seafaring man. Man's lot is one of toil, of doing what he has to do; let the knowledge of that be its own reward.

The position of Hesiod vis-a-vis the beginning of philosophy is difficult to assess. While the *Theogony* gives a story of the coming into being of the world in terms which reveal the priority of certain deities other than the Olympians, Hesiod is not simply giving an account of how things came to be. For if the latter stages of the *Theogony* embrace the Olympian gods, the cosmogony is a rational account of a religion more primitive than the Olympian; it is the depiction of a myth of creation which antedates the Olympians and has much in common with non-Greek views of the origin of things. Viewed as a defense of a more fundamental myth -- one originally embodied in ritual -- this cosmogony appears to separate itself quite definitively from philosophical accounts. These do not so much constitute interpretations of myth as replacements of it. Nevertheless, it has been argued that it is just the relation of Hesiod's cosmogony to myth which makes it so much like philosophical accounts.

C. Greek Primitive Religion

The myths and rituals of the barbarians (in the Greek sense) contemporary with Homer make a grim and depressing story, and we turn from them to the bright

world of Homer with no little relief. How unlike other ancient peoples the Greeks seem. For all their anthropomorphic defects, the gods of Olympus are out in the open, probably not really believed in, but a conscious and convenient poetic fiction to overlay the mystery and difficulty of human life. It all seems so sun-bathed and reasonable -- unlike the dreary rites of Egypt and Babylon so dark and primitive and inhuman for being all too human.

This attitude towards the Greeks -- the conviction that they had, so to speak, no dark, irrational, and primitive side -- is one that has been dispelled by recent additions to our historical knowledge. The Olympian religion, the state festivals, are seen to cover, but not wholly conceal, a religion which is literally of the earth, earthy. We can see an indication of this in Hesiod's account of the generations of the gods in which Zeus and the other Olympians are represented as replacing an earlier generation of gods, gods whose mother is the earth. The victory, moreover, involves the imprisonment of various Titans, giants, and monsters in the darkest regions below the earth. This suggests a polar opposition between the Olympian and the subterranean or earth gods or, as it is usually put, between the Olympian and Cthonic or earth religions. The point to remember is that the Olympian gods do not so much replace the underground gods as that their cults are grafted on those of the Cthonic deities. It is noteworthy, too, that Hesiod reserves the appellation "givers of good things" to the Olympian deities, for the Cthonic gods are rather looked upon as doers of evil to be placated.

In such works as Jane Harrison's *Prolegomena to the Study of Greek Religion*,^[8] ample evidence is presented for the view that the cults of the Olympian deities were grafted on to an already existing cult which was that of a local deity, and very often a Cthonic or underworld god. The Olympian rite seems to have consisted in an offering to the god, say, of an animal, part of which was burnt in sacrifice and the rest eaten so that the day was turned into a feast and various contests were held. This offering to the god was made to enlist his help. But there is another side to such rites, where the attitude is rather that of urging the god to leave the cultists alone, not to heap evils on them. For example, an examination of the feast of Zeus Meilichios -- Zeus of placations -- reveals that the original ceremony had nothing to do with Olympian Zeus at all, but rather is a cult of a snake, an apt underworld god, and the idea was to get rid of the snake god and the evil he represented. The sacrifice of an animal was not to share the meat with the god, but to burn it entirely; and the whole business was carried out, not with an air of cheerful festivity, but with revulsion. Thus, by adding Meilichios as epithet to Zeus, we see how the Greeks gradually replaced

the of the Cthonic deity with that of the Olympian, of the primitive superstition with that of the above board anthropomorphic god. With their imperfections, these Olympian deities form an important part of what sets the Greeks off.

Further, the layers of gods systematized in Hesiod's *Theogony*, for instance, represent successive invasions of what became Hellas. Zeus and the other Olympians represent the ascendancy of the Hellenes whose gods were then grafted on to the objects of superstitious cult of the conquered peoples. This explains the hyphenated deities which abound in Greek mythology, whereby local deities are identified with Zeus or another of the Olympians. This pre-eminence of the Olympian is visible in Homer, while in Hesiod another step is taken which brings us to the threshold of what came to be called philosophy.

What precisely is the step Hesiod has taken? The movement is from Cthonic or underworld gods -- objects of superstition and placation and aversion -- to the Olympian gods, full-blown anthropomorphic projections, recognizable persons if somehow supermen are more or less conscious personifications of natural forces. In Homer all this is background, part of an interpretation of what is basically the human world, the stage of actions whose consequences have to be accepted. In the *Theogony*, the gods are themselves the objects of concern and they are invoked to explain, not just the realm of action, but the make-up of the world around us. In the cosmogonical passage we analysed, the world of nature is explained; but the gods and what become gods are the principles of explanation. Perhaps this is what Aristotle has in mind when he calls Hesiod a theologian.

The myth and ritual, then, which precede philosophy are, respectively, anthropomorphic and emotional attempts to adjust to the world; the sympathetic magic of the rite, the attribution of the observed world to deities, represent a first attempt at an explanation. Moreover, in Hesiod, there is a preoccupation with putting order into the chaos of existing mythical accounts. If the attempt is unsuccessful, it nevertheless provides a relatively stable jumping-off point for the efforts which came to be called philosophical. As we turn next to the earliest philosophers, notice the imperfect line of demarcation between the Ionian thinkers and their poetic and theological predecessors. Anything like a precise elucidation of the distinction between myth and philosophy must await our consideration of the figures of the classical period and its sequel.

{1} See R. L. Nettleship, *Lectures on the Republic of Plato* (London: Macmillan, 1937), pp. 340ff.

{2} From G. S. Kirk and J. E. Raven, *The Presocratic Philosophers* (Cambridge, England: University Press, 1957), #169.

{3} See C. J. de Vogel, "What Philosophy Meant for the Greeks," *International Philosophical Quarterly*, Vol. 1, No. 1, pp. 35-57.

{4} For these passages from Homer, see F. M. Cornford, *Greek Religious Thought* (London: S. M. Dent & Sons, 1923).

{5} R. K. Hack, *God in Greek Philosophy* (Princeton: Princeton University Press, 1931).

{6} We cite Hesiod from H. G. Evelyn-White, *Hesiod: The Homeric Hymns and Homeric* (Cambridge: Loeb Classical Library, 1943).

{7} See F. M. Cornford, *From Religion to Philosophy* (New York: Harper Torchbook, 1957).

{8} (New York: Meridian Book Edition, 1955); also see Gilbert Murray, *Five Stages of Greek Religion* (Garden City: Doubleday Anchor, 1955), and A. J. Festugiere, *Personal Religion Among the Greeks* (Berkeley: University of California Press, 1954).

Chapter II

The Ionians

First we shall examine three thinkers, all natives of Miletus and each seeking a basic nature or stuff as the ground of the visible universe. Xenophanes of Colophon does not seem to share the interest of Thales, Anaximander and Anaximenes in the physical world, but his attitude towards the official religion and his obvious acquaintance with the efforts of the natural Philosophers is what makes him of interest here. Heracitus is difficult to classify. There is a temptation to see him as an erratic but genuine natural Philosopher and thereby reduce the import of his ethical utterances; on the other hand, it is easy to succumb to the view that he is primarily a moralist and that his cosmological fragments are unimportant. In treating his dark and difficult dicta, we shall try to strike a balance between these extremes.

A. Thales of Miletus

While Thales of Miletus is traditionally hailed as the first philosopher -- a designation we find in Aristotle -- most things concerning him are matters of dispute. We are not certain when he lived or whether he wrote; nor is there anything like general agreement as to the meaning of the doctrines attributed to him by later authors. Our main source for his doctrine is Aristotle. Herodotus has a number of things to say about his life and what he is reputed to have done, but neither Aristotle nor Herodotus seems to have much more than hearsay to go on; both express some doubt as to what is said about Thales. What underpins the doctrine attributed to him is particularly open to conjecture. Diogenes Laertius⁽⁹⁾ account of Thales teems with anecdotes, most of which are nowadays rejected; one safe fact recounted by him, however, is that Thales always finds a place in the changing lists of the seven sages of the ancient world.

Herodotus and others speak of Thales' knowledge of astronomy, and it is the mention of Thales' prediction of an eclipse of the sun during the war between the Medes and the Lydians which enables us to fix the time in which he lived. The eclipse in question is thought to have been that of 585 B.C. In the ancient doxographical tradition, it was a simple matter to move from such an important accomplishment to the date of Thales' birth and death -- he is said to have had a long life -- the date of the eclipse locates Thales early in the sixth century before Christ.

Did Thales' prediction of the eclipse involve knowing what an eclipse really is? This question refers to the tradition that Thales spent time in Egypt where he learned geometry from priests and brought it to Greece.⁽¹⁰⁾ Connected with the possibility of such a sojourn is the view that Thales "actually measured the pyramids by their shadows, having observed the time when our own shadow is equal to our height." (Diogenes Laertius I,27) Moreover, a theory on the flooding of the Nile is ascribed to Thales and recorded by Herodotus (II,20), which makes a visit to Egypt at least probable.

We have mentioned that Thales probably learned geometry from priests in Egypt. The Greeks, from the time of Herodotus (II,4,109), had a tendency to speak glowingly of the wisdom of the East -- of Egypt and Babylon. Aristotle bears witness to this penchant in the beginning of his *Metaphysics*. Connecting the rise of wisdom with leisure, he writes: "Hence it was in Egypt that the mathematical arts were first developed; for there the priestly caste was set apart as a leisure class." (981b23-5) What we know of Babylonian and Egyptian mathematics gives little support to the view that the Greeks could have borrowed geometry from them, certainly nothing on a plane with the geometry of Euclid.⁽¹¹⁾ The renown of Thales as a geometer is based on the tradition that he computed the heights of the pyramids and the distance of ships from shore. But neither of these feats demands a knowledge of geometrical science, though of course the problems involved are later seen as simple applications of known geometry. The mathematics of the Egyptians appears to have been a matter of more or less crude calculation, no more than this need be attributed to Thales. Proclus, in his commentary on the first book of Euclid's *Elements*, reports the view of Eudemus that Thales knew that two triangles are equal then they have one side and the two adjacent angles equal. The calculation of the distance of ships from shore was thought to depend upon the truth of this proposition. Since a simple rule of calculation would suffice to solve the problem, there is no need to think that Thales knew geometry in the rich sense of science or that he had learned it from the Egyptians. What he could have gotten from them, directly or indirectly, is a rule of calculation, such as that recorded in the Rhind papyrus.⁽¹²⁾

The same reservation must be made about Thales' prediction of the eclipse. Such a prediction can be made without knowing the cause of the eclipse; and since this was certainly not known by the immediate successors of Thales at Miletus -- and it does not seem likely that such an important bit of knowledge could have been lost so soon -- it seems safest to hold that Thales himself had no knowledge of the true nature of the eclipse. Priests in Babylonia had

compiled records of eclipses for religious purposes and could have gained a knowledge of a cycle of solstices within which eclipses could be predicted to occur at certain intervals. Since the Greeks traveled a great deal, it is not at all unlikely that Thales gained access to these records and made his prediction on their basis. This leaves unexplained, however, the implication that his prediction was exact although Herodotus seems to suggest only that Thales said that an eclipse would occur in a given year. This relative inexactness would not, of course, detract from the wonder the actual occurrence elicited. The fact that it came about on the day of an important battle, though presumably explainable only in terms of chance, would serve to increase the wonder and make Thales himself the object of a good deal of adulation.

These remarks are not intended to minimize the reputation Thales enjoyed in antiquity nor the role he plays in the history of thought. The esteem in which Thales was held in ancient times has a wider base than we have hitherto indicated. He is pictured as urging the Ionians to unify and name a single capitol (Herodotus, 1,170), and as having averted the streams of a river to make it fordable by King Croesus and his army (Herodotus is somewhat dubious about this incident). He is credited as well with the discovery of the Little Bear as an aid to navigation; indeed, the book ascribed to Thales was called "The Nautical Star Guide." The picture that emerges is one of a legendary sage, statesman, engineer, geometer, astronomer; so great was his reputation that any man of great practical wisdom came to be called "a veritable Thales" (Aristophanes, *Birds*, 1009). In Plato, Thales becomes the type of the absent-minded professor or philosopher.

I will illustrate my meaning, Theodorus, by the jest which the clever witty Thracian handmaid is said to have made about Thales, when he fell into a well as he was looking up at the stars. She said that he was so eager to know what was going on in heaven, that he could not see what lay before his feet. This is a jest which is equally applicable to all philosophers." (*Theaetetus*, 174A).

The incident is not considered to be historical, no more than that recorded by Aristotle which shows the other side of the coin. In this story, Thales, knowing that it was to be a good year for olives, obtained a corner on all the olive presses in the country and hired them out at a handsome profit when the crop came in. "Thus demonstrating that it is easy for philosophers to be rich, if they wish, but that it is not in this that they are interested." (*Politics*, I,II)

But it is his doctrines which have won for Thales the title of the first philosopher -- doctrines for the knowledge of which we are indebted almost exclusively to Aristotle. The three ascribed to Thales may be stated thus: (1) Water is the nature of all things. (2) All things have soul in them. (3) The all is divine. The passages are so brief it is worth letting Aristotle speak himself.

(1) Thales, the founder of this kind of philosophy, says that the principle is water [and therefore declared the earth to be on water] perhaps taking the supposition from the fact that the nutriment of all things is moist and that heat comes to be and is sustained by the moist, that from which they come to be is the principle of things . . . He also noticed that the seeds of everything have a moist nature and that water is the beginning of the growth of moist things . . . Thales at any rate is said to have explained the principles and origins of things in this way. (*Metaphysics*, I, 3, 983b20-984a2)

(2) Thales seems also, from what they say, to have supposed that soul was something moving, if he said that the stone possesses soul because it moves iron. (*De anima*, I, 2, 405a19)

(3) And some say that soul pervades everything, for which reason, perhaps, Thales thought that all things are full of gods. (*De anima*, I, 5, 411a7)

Notice that in each of these passages there is an indication that Aristotle is dependent on reports and not on any written work of Thales himself. If there was a book or books, we would expect a more positive tone; moreover, when Aristotle attempts to give reasons which might underlie what Thales is reported to have said, he has to settle for probability and his conjectures are framed in terms of his own more advanced understanding. Although these passages tend to bolster the view that no written work of Thales was known in Aristotle's time, Galen gives the following as a direct quote from Thales: "Water is the substrate and all things are derived from it; the manner has already been described by me in Book One." But this sounds very much like a later description of what Thales said.

Of the doctrines attributed to Thales, the first includes the view that water is the principle (*arche*) of all things and that the earth floats on water. This last point, mentioned as an aside in the *Metaphysics*, is criticised in Aristotle's *De Caelo*. (II,13,294a28):

Others say the earth rests on water. For this is the most ancient account, which they say was given by Thales the Milesian, that it stays in place by floating like a log or some other such thing . . . as though the same argument did not apply to the water supporting the earth as to the earth itself.

To call water a principle in Aristotle's sense of "principle," namely "that from which a thing comes and which remains within it" (*Metaphysics*, V, 1,1013a4), is most likely to go beyond what Thales meant. Perhaps we can put it in the most general terms by saying that Thales held that water is somehow involved in the origin or becoming of things.

What prompted him to take this stand? The reasons Aristotle gives as possible ones are all biological. Nutriment is moist and seeds are moist. Another supporting factor was his observation that corpses dry out. Burnet thought the idea would have been suggested to Thales by meteorological rather than biological considerations. For example, he would have noticed that water is now liquid, now solid, now a mist, and this would have suggested a cosmological view, since neither air nor fire -- certainly not earth -- appears in this diversity of states. Water is drawn up in evaporation and descends in rain; in ancient times it may even have been thought to turn to earth because of the Nile delta. While Burnet's contention that biological considerations could not have influenced Thales must be ruled out, for reasons given by Kirk and Raven (p. 89) as well as by Freeman,^[13] there is no need to rule out Burnet's own suggestion. In any case, we can see why Thales is considered the founder of science and was so thought of in antiquity. He sought to name what underlies the diverse things around us, that from which all things take their origin. Water seemed to him to fill the bill and plausible reasons can be adduced for his choice. In putting it this way, we do not intend to overlook Aristotle's reminder that the primacy of water had a long history before Thales, particularly in mythology.

Some think that those ancients who, long before the present generation, were the first to theologize, had a similar idea of nature, because they presented Ocean and Tethys as the parents of becoming and water as that by which the gods swore, which these people styled the Styx. (983b27-32)

Aristotle takes the primitiveness and antiquity of the opinion to be questionable, but the mention of theology is noteworthy in view of what we considered earlier. Other doctrines attributed to Thales include the view that soul pervades the universe -- that is, all things are alive, a view said to be suggested by the magnet and by amber. Notice that the general proposition is based on observation of

the magnet; if something as seemingly inanimate as a stone has soul (i.e., a power to move) in it, well, what might not be alive? The view is also said to be based on amber which becomes active only when rubbed. (Diogenes Laertius, I, 24) As Freeman remarks:

It has been thought odd that he should posit 'life' in all inanimate objects on the strength of the magnet, which was a unique manifestation; but if he treated amber and got the same manifestation, it may be that he thought that all objects had the same power if one knew how to invoke it; and that he therefore thought that the whole Cosmos was a living thing, nourished by the life-giving water of which it was composed, and that each particular object in it was likewise alive. (pp. 53-4)

All things are full of gods or daemons. The note of divinity is power as well as immortality; and it seems to be as much the former as the latter which connects this remark (also quoted by Plato, *Laws*, X, 899B, though not there attributed to Thales) to that which says all things have soul in them. There is a force or power -- call it soul -- which pervades all things and from which they take their origin; it is water.

The putting together of these three things -- water, soul, god, or, abstractly, nature, life, divinity -- is something which we cannot ignore in any appraisal of Thales as the first philosopher. The connection or the identification of these three with mythical thought is one which many scholars feel is too easily overlooked when we stress the first doctrine and let the other two fade away or find their explanation solely in the function of water as principle. This, however, is a question wider than the interpretation of Thales.

We can say, in conclusion, that Thales himself is a somewhat mythical figure. Remarkable engineering feats, political wisdom, uncanny calculations, a cosmology -- all these are attributed to Thales, but by way of legend or hearsay. In written accounts, there does not seem to be one sentence that can be pointed to with certainty as the written or spoken words of Thales. Hence, inevitably guesswork attends any assessment of his scientific or philosophical importance. One thing at least is certain. The beginning of philosophy is shrouded in obscurity.

B. Anaximander of Miletus

In 547-6 B.C. Anaximander was sixty-four and he died soon after. Thus, he was not a great deal younger than Thales of whom, according to tradition, he was a kinsman, student and successor at the "school" of Miletus. Tradition tended to describe in terms of later history the relationships between the early philosophers, and we need not take too literally the talk of a school of Miletus and of masters and disciples. The very least we must say is that Anaximander carried on what was considered to have begun with Thales, that he was younger than Thales and a citizen of Miletus. Of course, it is not pure conjecture to say that Anaximander knew and learned from Thales, given the considerable reputation of the latter.

Anaximander was the first one known to the later Greeks to have ventured a written account of Nature. The title was thought to be just that, *On Nature*; but it was common to attribute a book of that title to each of the ancients Aristotle designated as physical philosophers. A number of other specifically titled works were said to have been written by Anaximander, but we can have no certitude that they were actually written by him. What we can be sure of, however, is that he did write; for a sentence of his is preserved by Simplicius in his commentary on Aristotle's *Physics*, and it is thought that Simplicius in his turn is indebted for the information to Theophrastus, Aristotle's disciple. It is with that fragment that we shall begin our consideration of Anaximander. A few remarks on the difficulties of interpretation provide a concrete example of the character of our sources for thinkers prior to Parmenides. More importantly, we shall use the doctrine of the fragment to control our other more indirect information though, of course, not all of the latter should be considered operative in the fragment.

Anaximander . . . said that the principle and element of things is the Boundless, having been the first to introduce this very term 'principle;' he says that "it is neither water nor any other of the so-called elements, but some different, boundless nature, from which all the heavens arise and the world within them; out of those things whence is the generation for existing things, into these again does their destruction come to be, according to necessity; for they make amends and give reparation to one another for their offence, according to the disposition of time,' speaking of them thus in rather poetical terms. It is clear that, having observed the change of the four elements into one another, he did not think fit to make any one of these the material substratum, but something else besides these. (After Kahn)⁽¹⁴⁾

In placing the quotation marks where we have, we are adopting the interpretation of Kahn; the more common interpretation would restrict the direct quote in such a way that it begins "out of these things. . . ." That a direct quote, whatever its length, is involved in this passage from Simplicius seems assured by the comment on the poetical style of Anaximander. Those who feel the quotation is shorter than we have made it point out that, since Theophrastus, like Aristotle himself, inevitably sees early philosophy from a Peripatetic viewpoint we must be on our guard against attributing to the earliest philosophers notions elaborated only much later. In the present instance, "generation" and "corruption" ("destruction" in the given translation) are taken to be technical terms of later philosophy and said not to have been used by the pre-Socratics. Kahn (pp. 168-78) has argued that these terms, in a sense close to that Anaximander requires, are used even in pre-philosophical literature (we have seen that Homer uses *genesis*, Hesiod *genet'*) and that it is not utterly impossible that these very words and, at the least, the thoughts they convey are Anaximandrian. If his arguments are valid, the passage gives us a solid textual base in Anaximander for much of what has been traditionally ascribed to him.

The doctrine of Anaximander is often epitomized by observing that, while Thales gave water as the origin or principle of everything in the universe, his pupil Anaximander said that none of the elements could serve such a function and that consequently it must rather be some boundless or indefinite (*apeiron*) nature. The passage brings this doctrine immediately to the fore and we must ask what Anaximander meant by the boundless and what relation this bore to the elements. We notice that Simplicius speaks of the four elements, which is perhaps a later restriction of their number. What could Anaximander's own view of the elements have been?

At the end of the quotation, Simplicius gives a reason for Anaximander's choice of the boundless as the origin of things, namely that, having seen that the elements change into one another, Anaximander would have concluded that no one of them could be the source of all else. There is a passage in Aristotle which makes the same point and is thought to have been written with Anaximander in mind.

But yet, nor can the infinite body be one and simple, whether it be, as some say, that which is beside the elements, from which they generate the elements, or whether it be expressed simply. For there are some people who make what is beside the elements the infinite substance; for the elements are opposed to each other (for example, air is cold, water moist, and fire hot), and if one of these

were infinite the rest would already have been destroyed. But, as it is, they say that the infinite is different from these, and that they come into being from it. (*Physics*, III, 5, 204b22ff.)

The elements are considered to be opposites which change into one another; the boundless of Anaximander is not one of the elements because then it would seem necessary that sooner or later all things would change into it. Not being an element, the boundless is not opposed to any of the things that are, to any of the elements which are in opposition to one another. There seem to be two notes of the boundless, namely, indeterminateness in quality or nature and boundlessness in extent -- that which cannot be traversed. It is this latter sense which accords best with previous usage of the term *apeiron*, we are told, and indeed it answers best to the later discussion of infinity. Indefiniteness in quality seems to follow from the denial that the boundless is one of the elements.

From the boundless nature are said to arise the heavens and the worlds within them.

For some posit one substance only, and this some posit as water, some as air, some as fire, some as finer than water and thicker than air; which they say surrounds all the heavens, being infinite. (*De Caelo*, 111,5)

The boundless here seems in the present state of things to be a kind of enclosure for the heavens. "And this is the divine; for it is immortal and indestructible, as Anaximander says;" it is said "to be the beginning of the other things and to surround all things and to steer all." (*Physics*, 111,4) It seems that Anaximander taught that things had their beginning when the opposites "separated off" (*Physics*, 1,4) from the boundless nature due to the eternal motion of the latter.

He says that that which is productive from the eternal of hot and cold was separated off at the coming to be of this world, and that a kind of sphere of flame from this was formed round the air surrounding the earth, like the bark round a tree. When this was broken off and shut off in certain circles, the sun and the moon and the stars were formed. (Ps.-Plutarch, *Strom.* 2; Kirk and Raven)

The picture suggests the separation of fire and mist from the boundless with the fire encircling the mist like bark or skin. At the core of the air or mist, the earth condensed and its shape is that of cylinder whose diameter is to its height in a

proportion of three to one. The fire encircling air bursts, forming wheels of fire enclosed by air. The earth is at the center of things, not floating on water as for Thales, but it is where it is from considerations of geometrical symmetry. Men live on one side of the cylinder of earth and the sea is what remains of the original mist. The heavenly bodies are simply the fire, disclosing itself through holes in the wheels formed in the way indicated a moment ago. Eclipses are explained as the temporary closing of these holes in the fire-encircling wheels of mist. Since Anaximander explained eclipses in this way, it is thought to be highly unlikely that Thales had hit upon the true explanation earlier.

With this sketch of Anaximander's picture of the universe, we can turn once more to our basic text. Just as the position of the earth is dictated by the notion of geometrical symmetry -- if it is at the center, why should it go elsewhere? -- so the alterations of the opposites separated off from the boundless are seen in terms of a proportion expressed by a judicial metaphor. "Out of those things from which is generation for existing things, into these again does their destruction take place" -- the plural here is sign enough that the passage does not say that as all things come from the boundless nature so do they return to it, but rather, the elements originally separated off are such that one comes to be from another and ceases to be in the reverse change. If we think of day coming to be from night and then once more giving way to night, Anaximander asks us to see something like injustice in the coming to be, an imbalance which is righted when day is destroyed by night. In some such way, the elements are related and the rhythm from hot or cold and back again is seen as injustice and retribution, according to necessity, according to the disposition of time. The world is thus looked upon as governed by a law likened to human justice; proportion is achieved in time. One wants to see here a connection with the geometric inspiration operative in the view of the place of earth and in the proportion of its dimensions. The interchange of opposites everywhere observable in the world is what arrests Anaximander's attention in the extant fragment, and Simplicius' comment on his style must, in the light of the previous chapter, arrest ours. The "rather poetical terms" of Anaximander refer to the justice metaphor. The opposites Anaximander has in mind are first of all the hot and cold, namely fire and air, and then wet and dry, corresponding to water and earth. We have recognized here what were to become, with Empedocles, the four elements, but there is no cogent reason for saying that the Empedoclean doctrine is already taught by Anaximander. Indeed, Aristotle tells us that Empedocles was the first to speak of four elements. We should add that a striking point of continuity with Thales is found in Anaximander's teaching that living things come from the moist element.

The view that some boundless, unlimited, indefinite thing was the first stage in the coming to be of the world and even now surrounds and steers the universe is something of a giant step beyond Thales. This is true if Anaximander made his choice from a consideration of the consequences of singling out one of the elements as the origin and beginning of all else. Moreover, the sentiment expressed by the fragment is that the ceaseless changes in the world around us are governed by a law likened to that of the courts and attributed to the divine which steers all things. In his cosmological teachings, the heavenly bodies are explained in terms of wheels rotating above the earth, with the sun ring being the farthest from earth; the aperture through which what we call the sun is visible is said to be approximately the diameter of the earth cylinder. The moon ring is closer and then comes the star wheel which, of course, has many openings.

There are far fewer anecdotes connected with the name of Anaximander than with that of Thales. We might mention the story that he set up a gnomon at Sparta, that is, an instrument for measuring time, presumably erected on an inscribed surface on which the hour and the seasons could be read. He is also credited with having made a map of the known world.

C. Anaximenes of Miletus

Citizen of Miletus, pupil of Anaximander, Anaximenes is the last major figure of the Milesian school. That he wrote a book is known from the description of his style ("... he used simple and unextravagant Ionic speech." [Diogenes Laertius, 11,3]) and from a remaining fragment. His continuity with Thales and Anaximander is found in his choice of the material principle. "Anaximenes and Diogenes make air, rather than water, the material principle above the other simple bodies." (*Metaphysics*, 1,3) Air took on the characteristic of Anaximander's primary stuff, namely, infinity, and a new method of origination is hit upon by Anaximenes which is more determinate than the "separating off" of Anaximander.

Anaximenes, son of Eurystratus, of Miletus, a companion of Anaximander, also says that the underlying nature is one and infinite like him, but not undefined as Anaximander said but definite, for he identifies it as air; and it differs in its substantial nature by rarity and density. Being made finer, it becomes fire, being made thicker it becomes wind, then cloud, then (when thickened still more) water, then earth, then stones; and the rest come into being from these. He too

makes motion eternal, and says that change, also, comes about through it.
(Simplicius, *Physics*, 24,26; Kirk and Raven)

We are also told that Anaximenes made "gods and divine things" come from air. A first form of air is such that it is invisible; it becomes perceptible insofar as it is hot or cold or wet -- forms taken on because of the changing density of air. Thus Anaximenes has hit upon a stuff from which the basic elements and consequently all else can be derived. He indicates the method of such derivation, namely, the condensation and rarefaction of the basic material. By making air the boundless, Anaximenes seems to imply that he recognizes the two meanings of the term and intends it only in the quantitative sense -- there is an inexhaustible supply of air -- but not in the sense of qualitative indetermination. If the elements are simply different states of the basic stuff, we might wonder why it is designated as air, since air could be explained as a different state of fire or earth. It may be that Anaximenes is here influenced by Anaximander and the other meaning of "infinite," for of all the elements air seems the least determined. The comparison of air and breath in the extant fragment suggests a more anthropomorphic motive for Anaximenes' choice.

A primary stuff from which the other elements arise by a change of density, and the difference of density seems joined with the notion of temperature, since the hot and cold are caused by rarefaction and condensation. Condensed air is cold; expanded air is hot. Anaximenes is said to have offered proof for this by observing that when we blow on our hand with compressed lips, the stream of air is cold, while when the mouth is open our breath feels warm on the hand. Aristotle was to reject this by pointing out that when the lips are puckered, we are blowing the air in front of our face onto our hand, whereas when the mouth is open, it is the warmth of our breath that we feel. What is of interest here is both the appeal to an easily conducted experiment to ground the point and the resultant scale of elements which differ in density and, accordingly, in temperature. Moreover, unlike the "separating out" process taught by Anaximander, the principle of change among the elements that Anaximenes chose enables the process to go in either direction with equal ease.

Our earth is formed by the condensation of air. In shape it is cylindrical; and Anaximenes spoke of it as riding on air, thereby rejoining Thales who had thought earth needed some support. The flatness of the earth is used to explain its buoyancy; it presses down on the air beneath it and is thereby supported like a cosmic hovercraft or, better, kite. In much the same way, it is their flatness which explains the heavenly bodies; they are borne upon the air and, indeed,

can be blown from their courses by strong winds. In the heavens there are said to be fiery bodies as well as earthy ones. This is difficult to interpret, since Anaximenes is said to have given the earth as the origin of heavenly bodies; the sun is earth and gets its heat from the swiftness of its motion. It has been conjectured that the bits of earth which differ from the heavenly bodies were appealed to for an explanation of eclipses. Anaximenes denied that the heavenly bodies pass under the earth, as was the case with Anaximander's wheels of fire; at night the sun goes out of sight behind mountains in the north and the earth is apparently thought to be raised at its northern end as well. This does not seem to accord well with the doctrine of the flatness of the earth nor with the doctrine -- also attributed to Anaximenes -- that holds the sky is a hemisphere which fits snugly to the edges of the earth somewhat like an overturned cup set on a diminutive saucer. The bodies are said to swing above the earth as a cap spins on the head, an allusion which has called forth much ingenuity from commentators. There are as well fixed stars, studding the surface of the heavens.

The following passage from Aetius is thought to contain a fragment of Anaximenes' writings.

Anaximenes . . . said that air is the principle of existing things; for from it all things come to be and into it they are again dissolved, 'As our soul,' he says, 'being air holds us together and controls us, so does wind and air enclose the whole world.' Air and wind mean the same thing here. (Diels, B2; Kirk and Raven)

We added quotation marks around the words thought to be those of Anaximenes. What is the intent of the simile? Perhaps what it means is something like this. We require air to breathe and are surrounded by an inexhaustible supply of it. Now air is the origin of all things in the world and the world is surrounded by an inexhaustible supply of air which can be drawn in and, by rarefaction and condensation, produce many things. If this is the meaning of the comparison, we might ask if Anaximenes conceived the world as some kind of giant animal, alive and breathing much like ourselves. Although no certain answer is possible, in each of the Milesians there is an identification of the material principle and of the divine; in Thales and Anaximenes, soul and life are also referred to as the primal stuff. It is this which leads to the view that the mythological cosmologies only gradually cease to influence the efforts of the first philosophers.

D. Xenophanes

Xenophanes, first non-Milesian we will consider -- like Thales, Anaximander and Anaximenes -- was an Ionian. He was a native of Colophon and 570 B.C. is the likely year of his birth. Tradition has it that he was expelled from his native city and spent the rest of a very long life wandering throughout Greece, particularly in the western part. He tells us he left Colophon in his twenty-fifth year and was still on the move at the age of ninety-two. "Seven and sixty are now the years that have been tossing my cares up and down the land of Greece; and there were then twenty and five years more from my birth up, if I know how to speak truly about these things." (Diogenes Laertius, IX,2) Several towns in Sicily are mentioned in the tradition as well as Elea, on the Italian peninsula, which has led to the assertion that he was the founder of the Italian or Eleatic school of philosophy. Although Xenophanes is much influenced by the Milesian school, whose doctrines he could have known as a boy, there are significant differences between him and his Ionian predecessors, not the least of which is the fact that he wrote in verse. In his wanderings, Xenophanes declaimed his own poetry; some have thought that he was a Homeric rhapsode, i.e., one who publicly recited the Homeric epics. Inevitably, a work *On Nature* was attributed to Xenophanes, but this seems unlikely since the natural world was not as such a major concern of his. His poetry has been described as satire, doubtless due in part to his attacks on Homer from whom, as he said, all men have learned from the beginning. This attack on earlier poets is aimed principally at their depiction of the gods and it is in his theological *obiter dicta* that we find Xenophanes' importance for the beginnings of philosophy.

We have seen the change in the discussion about the gods which takes place in Hesiod. The *Theogony* attempts to derive the Olympian gods from earlier generations by a method which is either unabashedly that of human reproduction or something modeled on it, with the possibility that Hesiod was attempting to achieve a notion of becoming that escaped the limits of anthropomorphism. Despite this effort at a systematic theology, Hesiod's statements about the gods do not satisfy; and it is this that Xenophanes may be thought of as insisting on first of all. "Homer and Hesiod have attributed to the gods all things that are a shame and blameworthy among men, stealing and committing adultery and deceiving each other." (Fr. 11) "But men consider that the gods are born, and that they have clothes and speech and bodies like their own." (Fr. 14) This complaint of Xenophanes -- that the gods of the epics are allowed to do things for which men would be punished and that these same epics were the chief instrument of instruction of the young -- was destined to

find a responsive echo in later writers until, in the early books of the *Republic*, it received its masterly statement. It is not merely the description of the gods in terms of what is reprehensible in men that bothers Xenophanes, however; the more innocuous anthropomorphism which attributes generation, dress, bodies and speech to the gods also earns his censure, for it is this that leads to an utterly provincial attitude towards the divine. "The Ethiopians say their gods are snub-nosed and black, the Thracians that theirs have blue eyes and red hair." (Fr. 16) What Xenophanes is getting at in his negative way is that the divine should not be localized, so that there is a god or gods of the Greeks, and other gods for the different barbarian peoples. We have already seen that this was a sentiment in some ways shared by Xenophanes' countrymen, since they made great efforts to reduce the numerous gods of local cults to the Olympian deities. Moreover, in the *Iliad*, Homer does not think of the Olympians as the gods of the Greeks alone. Despite this, the Homeric deities are still made in the image of man. When we consider the animal gods of the Egyptians and the snake god of Othonic religion, we might wonder how, with those in mind, Xenophanes would have rephrased the following remark. "But if cattle and horses or lions had hands, or were able to draw with their hands and do the works that men can do, horses would draw the forms of the gods like horses, and cattle like cattle, and they would make their bodies such as they each had themselves." (Fr. 15) We can imagine that Xenophanes would only show greater disgust for men who fashioned gods after the animals. The import of these censures of Xenophanes is that anthropomorphism must be abandoned in talking about the divine. But Xenophanes' influence is not confined to negative statements -- to what we must not say of the gods; he has also more positive remarks.

Thales and the other Milesians applied the note of divinity to the underlying nature and have nothing to say of any god even remotely resembling the Homeric deities. From this silence we can conclude that they had either rejected such gods as anthropomorphic or, at the very least, that they saw no need to accord a cosmological function to such imaginative entities. As the quotations indicate, Xenophanes did not content himself with a switch of interest away from the divine; indeed, he may be said to differ from the Milesians in this above all: that divinity is his major concern.

Nonetheless, we may feel that the Milesians' search for unity had its effect on Xenophanes. "One god, greatest among gods and men, in no way similar to mortals either in body or in thought." (Fr. 23) Let us see what Xenophanes has to say of this greatest of gods.

Always he remains in the same place, moving not at all; nor is it fitting for him to go to different places at different times, but without toil he shakes all things by the thought of his mind. (Fr. 26,25)

No doubt inevitably, Xenophanes' more affirmative remarks about god proceed by way of denying him what he conceives as imperfections. God is immovable and unchanging, primarily in terms of place; the reason is that it would not be fitting for god to go from place to place to accomplish his desires. Rather, he operates without toil, simply by thinking a thought. The model for Xenophanes' statement is the king immobile on his throne, for whom it is not fitting to run his own errands. Still there is no need to see a latent anthropomorphism in Xenophanes' theology; even if it were present, what transcends the world of man is the dictum that god accomplishes his effects by his thoughts. Not that Xenophanes wants us to think of god as somehow parceled out in his being. He has no limbs distinct from one another, certainly; but neither are his faculties multiple. "All of him sees, all thinks, all hears." (Fr. 24)

How seriously can we take this talk of one god? If we take Xenophanes' pronouncements as indicative of an unequivocal monotheism, we run into the difficulty of explaining why he called this god the "greatest among gods and men." Obviously there is either one god or many. There are several ways of handling this problem. One is to take the mention of many gods as a concession on Xenophanes' part to the polytheism of the multitude. In this view, Xenophanes, while holding to his conviction that there is but one god -- supreme and quite unlike man -- nevertheless makes use of the familiar gods to speak of the widespread power of the one god. Thus, the rainbow is the god Iris, and this would mean that this striking phenomenon is only one manifestation of the divine power.

It is possible, on the other hand, to doubt seriously that Xenophanes gives us anything like a clear-cut view on the one and the many as applied to the divine. One finds it all too easy to read the fragments of Xenophanes as if they referred to a transcendent deity like the Judaeo-Christian God. To get a true picture, we must take into account Aristotle's judgment. In the *Metaphysics* (986b21 ff.) he writes, "Xenophanes, however, who first expounded the theory of unity (Parmenides is said to have been his disciple), made no clear statement and seems not to have understood either material or formal explanation; but, gazing at the whole sky, he says: 'Unity is God.'" Later in this passage, Xenophanes is dismissed, together with Melissus, for being too crude. We see in this remark the suggestion of an affinity between Xenophanes and Parmenides, an affinity

bolstered by the conjecture that the former was the teacher of the latter. Quite possibly Aristotle was here influenced by Plato's remark in the *Sophist* (242D) "Our Eleatic tribe, beginning from Xenophanes and even before, explains in myths that what we call all things are actually one." But Plato's references to his predecessors are seldom objective, and the remark in question is not meant to convey any historical fact. Xenophanes' relation to the Eleatic school aside, what are we to make of his one god? It is clear that, as Aristotle understands him, Xenophanes is saying that the one or the all -- this is, the world -- is divine, and god is coextensive with the universe. This shifts the ground entirely, and we are bound to think of the Milesian attribution of divinity to the stuff out of which everything comes -- the stuff which permeates the universe. On this, Xenophanes himself says in the fragments that, "All things come from the earth and in earth all things end." (Fr. 27) "All things are earth and water that come into being and grow." (Fr. 29) These words show we are faced with a view very much like those of Thales, Anaximander and Anaximenes, but with a change of emphasis to the divinity of the stuff from which things come. Burnet aptly comments that Xenophanes would have been quite amused to learn he would gain the reputation of a theologian in later times.

Aristotle, however, does not give us a license for this interpretation -- at least not as an exclusive view. What is perhaps most important in Aristotle's account of Xenophanes is that the wandering poet made nothing clear. In other words, Aristotle can be taken as drawing our attention to the many contradictions in the statements of Xenophanes. God, for Xenophanes, has a body and yet he moves all things by intellect; god is motionless and yet all things move -- can the all be god? By saying that Xenophanes made nothing clear, then, Aristotle appears to recognize that some of the poet's remarks suggest the interpretation of a transcendent deity, others that the world is god, and that the incompatibility of these two lines of thought vitiates the effort of Xenophanes to arrive at a clear position.

Surely, if the notion of a transcendent god, clearly other than the corporeal world, were obvious in the doctrine of Xenophanes, Aristotle would have seized on it as an indication of a truth he himself wished to establish. But Xenophanes does not get the consideration from Aristotle that Anaxagoras does. This shows that, unless Aristotle was here uncharacteristically insensitive to a hint of the truth in his predecessors, the doctrine of a transcendent deity was hopelessly obscured in the writings of Xenophanes.

Difficult though it is to settle on Xenophanes' positive contribution to philosophical theology, he is not thereby bereft of all importance. His eloquent rejection of the naive anthropomorphism of the earlier poets was at least an important adjunct to the efforts of the natural philosophers to lay aside the seductive myth explanation and turn to the things themselves. His critique of the Olympian gods is accompanied by an obviously sincere belief in divinity; he is clearly calling for a purification of belief rather than its rejection. While the Milesian's retention of the notion of divinity in speaking of the ultimate stuff may seem ambiguous (and even indicative of a kind of conscious hypocrisy to hide his atheism), Xenophanes' attitude towards the divine is clearly that of a man convinced. It is for this reason that we can confidently reject the guess that Xenophanes was a public reciter of the Homeric epics. The man who emerges from the fragments is not one who could declaim the very poems he thought conveyed a gross and reprehensible picture of the gods.

From the side of natural philosophy, Xenophanes' importance may lie principally in creating a climate in which the new science was welcomed throughout Greece. The fragments which speak of the derivation of things from water, of living things from water and earth, i.e., mud, are clearly reminiscent of Thales and Anaximander. Nevertheless, Xenophanes, apparently made direct contributions to natural science, by way of observation and interpretation. In one of his fragments, he observes that water oozes from the ceilings of caves, which may have been taken to suggest that water is indeed in everything since it shows up in such unlikely places. More importantly, Xenophanes reports on the finding of fossils of fish imbedded in rock far inland, and of shells and seaweed found in many landlocked places. These are taken as indicative of a time when earth and water were mixed, a time which was followed by a period of separation which will lead finally to a return to water; and so on in cyclic progression.

A final consideration should be drawn from another theme of the fragments of Xenophanes.

"There never was nor will be a man who has certain knowledge about the gods and about all the things I speak of. Even if he should chance to say the complete truth, yet he himself knows not that it is so." (Fr. 34)

We have here a conviction of the limitations of human knowledge which can be looked on once more as a criticism of earlier attempts to give the genealogy of the gods.

"Yet the gods have not revealed all things to men from the beginning; hut by seeking, men find out better in time." (Fr. 18)

Although Xenophanes seems to request that his own remarks be taken only as resembling the truth, not as conveying it whole, more likely than not, he principally intends to censure the presumption of the earlier poets.

It may seem somewhat surprising that Xenophanes, the first we have considered who wrote in verse, is the first to level an explicit criticism at the poets. Their anthropomorphism is the main object of his attack; and by pointing it out with the sharpness he did, he is implicitly calling for another kind of approach to the things that are. Even if he himself makes at best but slight contributions to philosophical knowledge, he had an important role in the movement of thought then gaining momentum, and which he, in the course of his long life, saw moving steadily away from the kind of assessment of reality found in the poems of Homer and Hesiod. His negative role seems easy to describe. When we seek to determine his positive contributions, however, we encounter difficulties which baffled even the Greeks and continue to provide grounds for conflicting interpretations today.

E. Heraclitus of Ephesus

Heraclitus, an Ephesian who lived out his life in his native town, was in his prime between the years 504 and 501 B.C. According to the doxographical tradition, this would place his birth about 540 B.C. and his death around 480 B.C. All we can be sure of is that Heraclitus was active in the year 500 B.C. Of his life we know little. It is said that he refused an hereditary kingship in Ephesus in favor of his younger brother; and, on the basis of the fragments, we get a picture of a proud misanthrope, bitterly critical of the multitude.

That Heraclitus wrote is certain from the wealth of quotations from him found throughout ancient literature. When these are brought together, we have a list of approximately 120 fragments. The question naturally arises whether these were originally in one book or many; or, as has also been suggested, simply individual utterances. The difficulty with this last interpretation is that one of the fragments seems to suggest a connected plan.

Of the *Logos* which is as I describe it men always prove to be uncomprehending, both before they have heard it and when once they have heard it. For although all things happen according to this *Logos* men are like people of no experience,

even when they experience such words and deeds as I explain, when I distinguish each thing according to its constitution and declare how it is; but the rest of men fail to notice what they do after they wake up just as they forget what they do when asleep. (Fr. 1)

This would indicate that the fragments we have formed part of whatever literary plan he contemplated. It is something else again, however, to agree with the tradition recorded by Diogenes Laertius (IX,5) according to which Heraclitus' book was entitled *On Nature* and contained three divisions, the first dealing with the universe, the second with politics, the last with theology. We have already seen that anyone whom Aristotle considered to have contributed to natural philosophy was assigned a book with the generic title *On Nature*.

If it is always hazardous to attempt the construction of a coherent doctrine from a few direct quotes and the comments of ancient writers, the matter becomes a good deal more complicated in the case of Heraclitus. Even in antiquity he had a reputation for opaqueness, and "the obscure" was usually appended to his name. The fragments are largely gnomic, oracular utterances, highly paradoxical, replete with metaphors and puns. Aristotle tells us that Heraclitus is difficult to understand because his sayings are difficult to punctuate. (*Rhetoric* 111,5) The contents of several of the fragments suggest that his was a studied obscurity.

The Sibyl with raving mouth utters solemn, unadorned, unlovely words, and reaches over a thousand years with her voice, thanks to the god in her. (Fr. 92)
The lord whose oracle is at Delphi neither utters nor hides his meaning, but shows it by a sign. (Fr. 93)

Since Heraclitus does not have a high opinion of men's ability to understand what he has to say, it is not unlikely that he deliberately chose his arresting style to sting his readers to think. He does not advocate knowledge of many things -- polymathy -- since this does not make one wise (if it did, he suggests, Hesiod, Xenophanes and others would have been wise [Fr. 40]); Heraclitus would draw our attention to the one thing which will guide us through the maze of particular understandings. "Men who love wisdom should acquaint themselves with many particulars." (Fr. 35) He uses the term *Logos* to convey this central point, and by it he does not mean what he says precisely as what *he* says. "Listening not to me, but to the *Logos*, it is wise to agree that all things are one." (Fr. 50) Heraclitus is not communicating a private vision, but drawing attention to what is public and common. "Therefore it is necessary to follow the

common; but although the *Logos* is common, the many live as though they had a private understanding." (Fr. 2)

What is the common *Logos* which is the burden of the Heraclitean fragments? Heraclitus' remark that the all is one is reminiscent of the Milesian philosophers; now when we consider the role that Heraclitus assigns to fire and that, living but a few miles from Miletus, he would have been acquainted with the teachings of Thales and his followers, it is all too easy to conclude that we have here a different choice for the primal stuff out of which all things come to be and into which they return.

All things are an exchange for fire and fire for all things, as wares for gold and gold for wares. (Fr. 90) The transformations of fire are, first, sea; and half the sea becomes earth, half the lightning flash. (Fr. 31) Such remarks as these have led to the listing of Heraclitus as the fourth in a sequence which exhausts the possibility of choices for the underlying nature, given the list of the five elements -- Thales: water, Anaximander: the boundless; Anaximenes: air; Xenophanes: earth (?); Heraclitus: fire. There is clearly something to be said for this interpretation as the fragments indicate; the difficulty is that it tends to make us overlook what is most characteristic of Heraclitus. For, while it is true that water and air, for example, have rather startling properties attributed to them by the Milesians, something more than this seems to be operative in Heraclitus' remarks about fire. If fire plays a role similar to that of water and air in Milesian cosmologies, it also is a symbol of what the word *Logos* means. The unity in all things that Heraclitus sees is not simply that of an indestructible stuff, the whence and whither of whatever is, but the unity of a law, of proportion, of balance and harmony. It is this what we must see; and the best approach is through the fragments whose paradoxical tone almost seems to defy understanding.

The note of paradox is sounded even in the fragments which speak of the undertaking of the inquiry itself. We have seen Heraclitus say that knowledge of many things does not make a man wise (Fr. 40) and that men who love wisdom should be acquainted with many things. (Fr. 35) So, too, he says, "Nature loves to hide," (Fr. 123) and, "The things of which there can be sight, hearing, learning, these are what I especially prize." (Fr. 55) "Eyes and ears are bad witnesses to men having barbarian souls." (Fr. 101a) Heraclitus seems to be saying that a multiplicity of knowledge without a unifying goal is pointless; that nature is difficult to know, but reveals itself to careful observation, if we are able to read the testimony of the senses. If the senses speak an alien tongue, if, as we

should say, it is all Greek to us (or all barbarian to a Greek), then nature will remain hidden. If we read correctly, we will see that all things are one and our wisdom will be one. "Wisdom is one thing. It is to know the thought by which all things are steered through all things." (Fr. 41) This, Heraclitus finds lacking in the teaching of others. "Of all whose discourses I have heard, there is not one who attains to understanding that wisdom is apart from all." (Fr. 108) What is the one thing?

"This world, which is the same for all, has not been made by any god or man, but it always has been, is, and will be, an ever-living fire, kindling itself by regular measures and going out by regular measures." (Fr. 30) Fire emerges here as of central importance for Heraclitus, but its importance is somewhat overshadowed by the notion of proportionate give and take which is also present in this fragment. There is a balance in the coming and going of fire and indeed of all things and once this is recognized, opposed things seem not so opposed since they are part of a harmony or proportion. "Sea is the most pure and the most polluted water; for fish it is drinkable and healthy; for men it is undrinkable and harmful." (Fr. 61) "Disease makes health pleasant; hunger satiety, weariness rest." (Fr. 111) Much more is involved here than the relativity expressed by Xenophanes: "If god had not made yellow honey, men would consider figs to be sweeter than they do." (Fr. 38) The difference is clear in the following, much quoted remark. "The path up and the path down is one and the same." (Fr. 60) Before looking at the possible cosmological intent of that dark saying, let us consider another statement. "And as the same thing there exists in us living and dead and the waking and the sleeping and young and old: for these having changed around are those, and those changed around are these." (Fr. 88) Heraclitus' concern is with change as taking place between opposites. But since the change binds the opposites together and the change can go in either direction, what originally appear utterly other are seen to be in some way the same. This unity of opposites can be interpreted first of all in a cosmological sense. Things taken together are whole and not whole, something which is being brought together and brought apart, which is in tune and out of tune; out of all things there comes a unity, and out of a unity all things. (Fr. 10) Notice that the emphasis here is not on a common substrate. Anaximander tended to look on change as the encroachment of one element on the territory of another so that what results from change is an instance of injustice calling for retribution, that is, corruption. That Heraclitus has a different view -- one that has sometimes been taken as an implicit criticism of Anaximander -- is clear from the fragments. "It is necessary to know that war is common and right is strife and that all things happen by strife and necessity." (Fr. 80) "War is the

father and king of all .. ." (Fr. 53) Strife and encroachment is not an aberration, not unjust; the warring of things with one another is precisely justice. Heraclitus wants to find unity in the strife itself. "Men do not apprehend how being brought apart, it is brought together with itself: there is a back-stretched connection, as in the bow and lyre." As the two hands of the bowman pull apart from one another, the tips of the bow come together, and we must see in this tension of opposites rectitude and justice.

But what has this metaphor to tell us of the natural world? In what is without a doubt his best known fragment, Heraclitus says, "You cannot step twice into the same river; for fresh waters are continually flowing on." (Fr. 91,12) The sameness of the river depends upon the ceaseless change of its constituent parts. So too the universe is one and the same in the ceaseless warring of its components; the way up and the way down are one and the same. It is possible, of course, to interpret this quite simply as meaning that ascent and descent are accomplished along the same road, that one does one or the other depending on his starting point. There is as well, however, a cosmological interpretation insofar as it refers to the emanation of all things from fire and their subsequent return to this source.

Fragment 31 indicates that from fire, sea comes to be, and that earth and what is called the lightning flash come from the sea. Is this process irreversible? If that is the downward path from fire, is there an upward path at the term of which all things disappear into fire? There are partisans of both viewpoints.

The Stoics, who taught that our world would end in a fiery conflagration, found support for their view in Heraclitus. The Stoic view was that this conflagration was a periodic one, occurring at the end of what was called the Great Year, which was sometimes said to be a period of 18,000 years, sometimes 10, sometimes 800. The last figure was arrived at by taking 30 years as representing a human generation and multiplying it by 360. That is, the Great Year is a year of human generations. In the theory that interests us, the world is destroyed by fire at the end of the Great Year and is replaced by another which has a duration of one Great Year, and so on and on. It was generally held in ancient times that this was the view of Heraclitus. The apparently opposed view expressed in Fragment 30 was explained by saying that Heraclitus is not there talking of a particular world, but the pattern or order (cosmos) involved in any world, and this is indeed unchanging. The cycle is explained in terms of the downward and upward paths. Heraclitus is faced with the fiery heavenly bodies, the dry land, and the sea. Rain comes from above, from the fiery region and the land seems

to come in some way from the sea. Earth returned to sea when islands sank and when new springs and streams welled up from below, washing away the earth. The sea was drawn up again in the process of evaporation. When the upward path is looked upon as total and cataclysmic, we have the *ecpyrosis*, the consuming of the world by fire. Did Heraclitus teach this? Aristotle (*De Caelo*, I,10) seems to say so and after Theophrastus the judgment becomes fairly fixed.

What are the arguments against it? Kirk gives five.⁽¹⁵⁾ (1) *Ecpyrosis* goes contrary to the whole tenor of Heraclitean thought as expressed in the fragments. The unity of opposites, balance, constant strife without ultimate victory -- these seem to underpin the notion of *Logos*. Homer is rebuked by Heraclitus for thinking strife unnatural. "Homer was wrong in saying, 'Would that strife might perish from among gods and men!'" (2) *Ecpyrosis* would entail the abandonment of the balance and measure in the exchange of fire with all things. (3) Fragment 30 quite clearly speaks of *this* world or order. (4) Plato (*Sophist*, 242 D) is clear in saying that unity and multiplicity coexist and do not succeed one another. (5) Even some Stoics doubted this interpretation of Heraclitus.

Wheelwright addresses himself to each of these five points⁽¹⁶⁾ (1) He finds this the strongest argument, but thinks it not unassailable.

If the dominance of fire in an *ecpyrosis* were to entail the destruction of all strife, then admittedly a situation would arise -- an interval of absolute peace and rest -- such as is expressly denied by several of Heraclitus' statements (p. 52)

Wheelwright finds no need to see the conflagration, which would be decisive with regard to the destruction of this world, as in itself pure and total. "Surely the cosmic fiery state would have to be somehow impure in order to allow the seeds of a future universe to emerge from it." (p. 53) Wheelwright feels this consideration weakens (2) and (3) as well. As for (4), are we to take Plato as an unimpeachable source for what Heraclitus really meant? With respect to other Platonic remarks on Heraclitus, it is rather generally agreed that Plato has in mind contemporaries of his own, like Cratylus. Argument (5) does little more than indicate that all the evidence is inconclusive, be it Stoic or otherwise. And that is just how Wheelwright would leave the issue -- unsettled. While he is alive to the arguments that can be adduced to support either side of the matter, he is convinced that this is one of many points where our knowledge can be at best conjectural and inconclusive. It is easy to subscribe to this view.

From ancient times Heraclitus has been taken as the founder of the eternal flux school of thought. All things flow, says a phrase attributed to Heraclitus. This, some have argued, is not so much an explanation of knowledge as the destruction of its very possibility. If everything is always changing, nothing is ever fixed enough to be an object of knowledge. What these two views fail to take into account is the notion of *Logos*, for beyond fire as substrate and the constant change of it and everything else, there is the *Logos* -- the orderly process whereby all change takes place. The *Logos*, it has been argued, is the true One in the doctrine of Heraclitus. Wisdom is one; Heraclitus has said, "Wisdom is one and unique; it is both willing and unwilling to be called by the name of Zeus." (Fr. 32) "Zeus" is the name we give that which governs all things in so far as we conceive it anthropomorphically, and this is all right so far as it goes. There is a law governing things in the universe, which prevents the sun from overstepping its bounds. (Fr. 94) Fire is the vehicle for expressing this divine governance: "The thunderbolt steers all things." (Fr. 64) The thunderbolt, in mythology, was the missile whereby Zeus expressed his displeasure and, by extension, his will and governance. *Logos* conveys the idea of law, intelligence, something apart from the material. Yet in Heraclitus it is inextricably bound up with fire. Thus for him, fire is a symbol. The measures of the ceaseless changes in the universe are not, however, immediately obvious to us. "An invisible harmony is better than a visible one." (Fr. 54) This harmony is the basis for wisdom; to attain to a recognition of it is the task of philosophy and its attainment sets the philosopher off from the mass of men. They are as men asleep; he alone is awake.

When men are asleep, each has his own private world; awake there is one world common to all. (Fr. 89) The waking state enables us to participate in the *Logos* which governs all and is common to all things. The soul is said to have its source in the moist (Fr. 12); while in this sense it is a thing among other things, "You could not discover the limits of the soul if you traveled every road to do so; such is the depth of its meaning." (Fr. 45) One is tempted to see in this a switch to an ethical perspective, a warning as to the difficulties involved in obeying the oracle's injunction, "Know thyself." The same may be said of some other remarks about soul. "A dry soul is wisest and best." (Fr. 118) But "Souls take pleasure in becoming moist." (Fr. 77) The cosmological priority of fire is here applied to soul and made to serve an ethical function. To be fiery and dry is best for soul, but there is a contrary tendency towards moistness, symbolized by intemperance. "A drunken man has to be led by a young boy, whom he follows stumbling, not knowing where he is going, for his soul is moist." (Fr. 117) "It is death to souls to become water, and it is death to water to become earth. On

the other hand, water comes into existence out of earth, and souls out of water." (Fr. 36) Once more there is present the cosmological perspective, the upward and downward paths. The tendency in our nature to what is harmful to us is summed up in the following fragment. "It is hard to fight against pride; whatever it wants it will buy at the cost of soul." (Fr. 85) The ethical message of Heraclitus is difficult to discern. We will be morally awake insofar as we are alive to the *Logos*; dry and fiery, insofar as we see the hidden harmony in the constant strife which is the universe. This strife is microcosmically present in ourselves; we must not be led by the masses or allow ourselves to sink into drunkenness where it is difficult to hide our ignorance. (Fr. 95) *In vino veritas* -- but this is not the truth of the *Logos*. Wisdom comes when we expect the unexpected and are stirred up in our nature to the proper proportion, for "Even the sacred barley drink separates when it is not stirred." (Fr. 125) If there is a conflict in nature we must, like the universe as a whole, impose a *Logos* on the warring opposites to achieve a harmony -- like a drink which requires constant stirring.

The very character of our contact with the thought of Heraclitus -- fragmentary, enigmatic, oracular and paradoxical -- invites prolonged speculation. But the further we go along the path of interpretation, the deeper we get into mere conjecture. But we can safely conclude by saying in Heraclitus there seems little or no distinction between statements about the universe, the constitution of our soul and the ethical demands made on the individual.

{9} Diogenes Laertius, *Lives of Eminent Philosophers* 2 vols., edited and translated by R. D. Hicks (Cambridge: Loeb Classical Library, 1950).

{10} See Diels-Kranz, *Die Fragmente der Vorsokratiker* (Berlin: Weidmann, 1954).

{11} Leon Robin, *Greek Thought and the Origins of the Scientific Spirit* (New York: Alfred A. Knopf, 1928) pp. 30ff; see also Paul Tannery, *Pour l'histoire de la science hellene* (Paris: F. Alcan, 1930); and R. Bacon, *Histoire de la science grecque* (Paris: Aubier, 1951).

{12} See John Burnet, *Early Greek Philosophy* (New York: Meridian Books, 1957), p. 45.

{13} Kathleen Freeman, *The Presocratic Philosophers* (Oxford: Clarendon Press, 1946), p. 52.

{14} See Charles H. Kahn, *Anaximander and the Origins of Greek Cosmology* (New York: Columbia University Press, 1960), p. 166.

{15} G. S. Kirk, *Heracitus, The Cosmic Fragments* (Cambridge: University Press, 1954), pp. 335-8.

{16} Philip Wheelwright, *Heracitus* (Princeton: Princeton University Press, 1959).

Chapter III

The Italians

Now we turn to philosophers who, while not uninterested in the type of problem which bothered the Ionians, differ from them in striking ways. The Pythagoreans, for example, speak of the physical world as if it were constituted of numbers. Parmenides, on the other hand, overwhelmed by what he conceives to be logical truths, denies the reality of the world we see or think we see. His arguments against motion and multiplicity provide a difficulty for natural philosophy which is not solved until Aristotle. Moreover, his distinction -- between what appears to be and what is -- is destined to have a long history in Greek philosophy and beyond.

A. The Pythagoreans

This group of philosophers takes its name from Pythagoras, a native of Samos in western Greece, but whose career was spent mainly in Italy. We speak of Pythagoreans rather than of Pythagoras, because little is known of the founder and, therefore it is difficult to assign to any individual the characteristic Pythagorean doctrines. The information we have from Aristotle about this school seldom begins with anything but the vaguest designation, e.g., "Certain Pythagoreans . . ." From an historical point of view, this presents difficulties. This is particularly true when we attempt to make use of Hellenistic testimonies, since the span of centuries, together with the anonymity of the members of the school, tends to blur the difference between very early Pythagorean teachings and later ones. For these were formulated with an eye to Plato, Aristotle and the Stoics.

1) Pythagoras of Samos

The life of Pythagoras can be told rather briefly. Born in Samos, an island in the Aegean off the Ionian coast, where he is said to have lived until his fortieth year, perhaps 532/1 B.C., he then fled the tyranny of Polycrates. He then went to Croton in southern Italy where he was well received and, according to tradition, exercised no little political influence. His pupils there were said to have numbered some three hundred. The citizens of Croton finally revolted and set fire to a house in which the elder Pythagoreans were meeting, but Pythagoras himself escaped. He went to Metapontium where he died many years later.

The school that Pythagoras formed in Croton must not be thought of as the same sort as the "school" of Miletus. The Pythagorean community was structured more like life in a religious order. The community embraced men and women. There was a common doctrine but it was not to be divulged to outsiders; indeed, we have no mention of a Pythagorean writing anything until Philolaus at the end of the fifth century before our era. Pythagoras himself wrote nothing, but the practise of the society was to attribute every doctrine to its founder. Renowned for their secrecy, the Pythagoreans and *a fortiori* Pythagoras himself early became the object of mystery and speculation. Plato and Aristotle, consequently, are not in the habit of saying that Pythagoras said such-and-such, but that he is said to have said such-and-such, or, more usually, that the Pythagoreans or some Pythagoreans say such-and-such. There seems to be no doubt, however, that Pythagoras did live.

One very early testimony is that of Xenophanes who says of Pythagoras,

Once they say that he was passing by when a puppy was being whipped, and he took pity and said: 'Stop, do not beat it; for it is the soul of a friend that I recognized when I heard it giving tongue.' (Fr.7)

Thus, Pythagoras is said to have held the doctrine of the transmigration of souls; indeed, he is said to have remembered four previous incarnations of his own! Much later, Porphyry summarized his doctrine thus: (1) he believed in the immortality of the soul; (2) that it changes into other kinds of living things; (3) that events occur in definite cycles such that, to adapt the words of one commentator, the time will come when I will once more be writing these words and you will be reading them. That is, *nihil novi sub sole*: nothing is new and unique; (4) that all living things should be regarded as akin. There is a persistent tradition, beginning with Herodotus, that such doctrines were imported into Greece from abroad. Herodotus claimed that the doctrine of the transmigration of souls was borrowed from the Egyptians; but the Egyptians seem never to have held the belief themselves.

The belief that human souls could show up in other living things is connected with certain taboos or prohibitions observed by the Pythagoreans, such as abstention from meat and an injunction against associating with butchers. The testimony on these points is conflicting, however, since Pythagoras is said to have sacrificed an ox when he discovered the Pythagorean theorem. Later writers listed rules of conduct which were said to guide the Pythagorean community -- rules such as abstention from beans, and of smoothing out the

impression left on one's bed, not wearing rings, not letting swallows nest under one's roof, etc. One that still has a peculiar force is this: "Speak not of Pythagorean matters without light."

We see, then, that the Pythagoreans were a community guided by a number of primitive and somewhat foolish rules of conduct, a secret society formed for the spiritual good of its members with a view to the survival of the soul. And this survival seems to involve reward or punishment for the deeds of one's present incarnation.

This is but half of the Pythagorean story, however; for to this mystic fervor was coupled an interest in science, particularly mathematics. Indeed, the society itself divided into two groups after the death of Pythagoras, the "Acousmatics" (hearers) and the "Mathematicians" (knowers). The former probably concentrated on the religious aspect of the society, while the latter devoted themselves to the more scientific aspect.

It is the peculiarity of the Pythagorean view of number which controls the scientific contributions of the society, and some at least of this view seems to go back to Pythagoras himself. Tradition has it that Pythagoras discovered that the chief musical intervals can be expressed in numerical ratios. If he arrived at this by measuring the length of the strings on a monochord, he would have expressed the octave as $2/1$, the fifth as $3/2$ and the fourth as $4/3$. What we have to understand about Pythagoras' attitude is that he did not think of the numerical ratios as statements about sound by way of some application, but came to hold that there was an identity between number and sound. In other words, he did not arrive at any distinction between number and what is numbered, measure and the measured. This identification of things with numbers was to become the characteristic Pythagorean doctrine. The first four numbers, moreover, were thought to contain the whole nature of number, since $1 + 2 + 3 + 4 = 10$. When one gets to 10 he simply begins counting over again. Besides the Pythagorean theorem itself, Pythagoras seemingly can be credited with the discovery of the incommensurability of the diagonal and the side of the square. There is a story that one Hippasus of Metapontium was drowned for having revealed this to outsiders. Much speculation has been spent on why he should be so punished. One explanation is that the discovery of incommensurability of the *alogon* or irrational, was such a blow to the belief that there is a proportion or harmony in all things, that the initiates were particularly enjoined against revealing what could only seem a scandal.

The figure of Pythagoras is a shadowy one, not, as with the philosophers considered earlier, because of scanty information, but almost by design. He is the founder, the master, to whom all doctrines are attributed. (The Pythagoreans were famous for introducing statements with, "He himself said so.") And who is regarded as more than human, the son of Hermes. In one legend, for example, he is described as revealing his golden thigh. Soon the historical figure is lost behind the stories and our knowledge of what he taught is reduced to a view of the kinship of all things and an interest in mathematics which, apart from some mystical interpretations on the power of numbers, seems genuinely scientific. There seems to be as well the identification of things with numbers, leading perhaps to a belief in the harmony of all things -- a belief called into question by the discovery of the incommensurability of the diagonal and side of a square.

2) Pythagorean Doctrines

A remark attributed to Pythagoras describes perhaps for the first time an important aspect of what had been begun by Thales and was carried on by subsequent thinkers.

Life, he said, is like a festival; just as some come to the festival to compete, some to ply their trade, but the best people come as spectators, so in life the slavish men go hunting for fame or gain, the philosophers for the truth. (Diogenes Laertius, VIII,8)⁽¹⁷⁾

We have here a distinction between the practical pursuits of men, the mark of which is activity and striving, and the pursuit of truth, described in terms of seeing or understanding for its own sake.

Because the philosopher wants to see, he must purge himself. The Pythagoreans held that as medicine purges the body, so does music purge the soul; and music -- proportioned sound -- is number. Number is the nature of all things. We must in this connection consider a lengthy passage from Aristotle.

Contemporaneously with these philosophers, and before them, the Pythagoreans, as they are called, devoted themselves to mathematics; they were the first to advance this study, and having been brought up in it they thought its principles were the principles of all things. Since of these principles numbers are by nature the first, and in numbers they seemed to see many resemblances to the things that exist and come into being -- more than in fire and earth and

water (such and such a modification of numbers being justice, another being soul and reason, another being opportunity -- and similarly almost all other things being numerically expressible); since, again, they saw that the attributes and the ratios of the musical scales were expressible in numbers; since, then, all other things seemed in their whole nature to be modeled after numbers, and numbers seemed so be the first things in the whole of nature, they supposed the elements of numbers to be the elements of all things, and the whole heaven to be a musical scale and a number. (*Metaphysics*, I, 5)

Here Aristotle expresses what he learned of the Pythagoreans. They were interested in mathematics, Aristotle says, and this indicates an interest in the abstract, the formal, a science which does not have for its object the sensible things around us. Now the Pythagoreans, Aristotle says, thought of numbers as the stuff out of which things are made, as the Ionians had spoken of air and water as the primal matter out of which all things are fashioned. This is a difficult transition, and Aristotle gives us a few preliminary clues as to how it should be understood. Justice is a number, as is soul, and all other things; they are different arrangements of units and are thus made up of numbers. When Aristotle says that the Pythagoreans noticed that the attributes of the musical scale were expressible in numbers, he is speaking in terms of the recognition of a distinction which was most likely not known at the outset of the Pythagorean school. There is an unavoidable tension in the Aristotelian passage between the view that number is material cause and that it is somehow formal, applied to natural things, but itself different from natural things. Because they had not adequately distinguished between material and formal causes, the Pythagoreans seem to be making the same thing do service as both kinds of cause; number is that out of which things are made, and the particular arrangement of the elements is their nature. Consequently, we are faced with a doctrine according to which there is no distinction between natural science and mathematics, according to which the study of number tells us about the natural world as natural world. For the elements of number are the elements of all things.

What is meant by the elements of number?

Evidently, then, these thinkers also consider that number is the principle both as matter for things and as forming their modifications and their permanent states, and hold that the elements of number are the even and the odd, and of these the former is unlimited, and the latter limited; and the 1 proceeds from both of these (for it is both even and odd), and number from the 1; and the whole heaven, as has been said, is numbers. (986a15-21)

In this continuation of the previously quoted passage, Aristotle recognizes that number is matter and form for the Pythagoreans. What we must understand, if we are to grasp the identification of physics and mathematics, is the notion of oddness and evenness as the "elements of number," the relation between these and the number one and the numbers proper which follow from it.

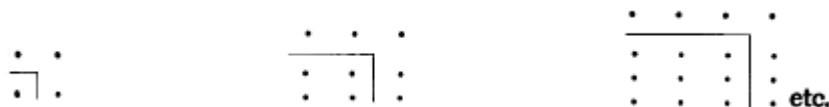
If we must set aside the distinction between mathematics and physics to get an idea of Pythagoreanism, it seems that we must also abandon any sharp distinction between arithmetic and geometry. If we find this last shift considerably easier to accept, we must not think that the Pythagoreans anticipated any later view on the relationship between geometry and arithmetic. The Pythagoreans did not have, of course, anything like a simple system of notation for numbers. Very much later, Nicomachus indicates the difference between linear, plane and solid numbers in such a way that we understand that the source of this application of geometrical adjectives to numbers is quite pictorial. Linear numbers are obtained by setting down the unit once, twice, etc. Thus, 1 is a, 2 is aa, 3 is aaa, etc. Of course, the Greeks used letters of the alphabet as shorthand for such linear numbers (e.g., iota for 10; kappa for 20), but the linear numbers are basically what they stand for. Linear numbers are of one dimension and the unit is their principle. Plane numbers are generated from linear number and have three as their root. Here is Cornford's illustration of plane numbers.⁽¹⁸⁾

			a
		a	a a
	a	a a	a a a
a	a a	a a a	a a a a
a a	a a a	a a a a	a a a a a
1+2=3	3+3=6	6+4=10	10+5=15...

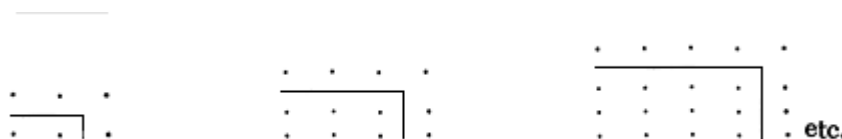
The first solid number is composed of four units and is illustrated by a tetrahedron. These remarks from later Greek mathematics give us an indication of the way in which the manner of depicting numbers leads to speaking of types of number progressing in terms of dimensions. The immediate relevance of these remarks is that they enable us to understand why the Pythagoreans spoke of odd and even numbers as square and oblong, respectively, and said that the former is finite, the latter infinite.

Further, the Pythagoreans identify the infinite with the even. For this, they say, when it is taken in and limited by the odd, provides things with the element of infinity. An indication of this is what happens with numbers. If the gnomons are placed round the one and without the one, in the one construction the figure that results is always different, in the other it is always the same. (Physics, 111,4,203a10)

If we start with the unit and enclose it with the first odd number, the figure obtained is a square; if we enclose the resultant figure with the next odd number, the figure remains the same, and the ratio of the sides is the same and the numbers obtained will be called squares, 4, 9, 16, and so on. Thus, square, odd and limit go together for the Pythagoreans. The following illustration will be of assistance here.



The oblong numbers, on the other hand, are obtained by beginning with 2 and enclosing it with the next even number. The continuation of this procedure is said to produce figures which are always different, that is, the ratio of the sides is never the same; the oblong numbers thus obtained are 6, 12, 20 etc.



Thus, oblong, even and unlimited are grouped together by the Pythagoreans.

When we grasp that numbers are conceived in terms of different configurations of units in space, we can see how the Pythagoreans could have come to believe that the elements of number are the elements of all things. The crude way of making this identification was to take pebbles and form with them a picture of an object and, by counting the pebbles used, assign the number-nature of the object.

From Aristotle's account, we know that according to the Pythagoreans numbers have magnitude. "Now the Pythagoreans also believe in one kind of number -- the mathematical; only they say it is not separate but sensible substances are formed out of it. For they construct the whole universe out of numbers -- only

not numbers consisting of abstract units; they suppose the units to have spatial magnitude . . ." (*Metaphysics*, XIII, 6, 1080b16) The illustrations we have seen indicate the identification of the arithmetical unit and the geometrical point; the point, however, while considered to be indivisible, is not without extension. Because of this, physical things could be looked on as in some way composed of such units as of their matter.

Before looking into the cosmology which followed from this view of mathematics, however, we must consider an important point made by Aristotle in his account of Pythagoreanism at the outset of his *Metaphysics*.

Other members of this same school say there are ten principles, which they arrange in two columns of cognates -- limit and unlimited, odd and even, one and many, right and left, male and female, resting and moving, straight and curved, light and darkness, good and bad, square and oblong. In this way Alcmaeon of Croton seems also to have conceived the matter, and either he got his view from them or they got it from him . . . For he says most human affairs go in pairs, meaning not definite contraries such as the Pythagoreans speak of, but any chance contrarieties, e.g., white and black, sweet and bitter, good and bad, great and small. He threw out indefinite suggestions about the other contrarieties, but the Pythagoreans declared both how many and which their contraries are. (*Metaphysics*, 1,5)

Alcmaeon of Croton, who is thought to have flourished at the beginning of the fifth century, was primarily concerned with medical matters, and one indication of his interest in contraries is to be found in his view that health is a balance of moist and dry, cold and hot, sweet and bitter, and so forth, and that sickness is the result of one contrary getting the upper hand. Certain Pythagoreans, as Aristotle says, were more systematic in pursuing the recognition of the role of opposites in the world and tried to summarize in ten oppositions the major types. We have already seen the reason for linking limit, odd and square, on the one hand, and unlimited, even and oblong, on the other. Even numbers are unlimited or infinite not, as Simplicius held, because the even number is infinitely divisible -- this is manifestly absurd -- but only in the sense that nothing prevents their being divisible into halves. The odd number is limited since, by adding one to an even number it prevents such equal division and, like three, the first odd number, has "a beginning, a middle and an end." Explanations of other contraries in the right or left column of opposites emerge when we look at Pythagorean cosmology.

If the elements of number are the elements of all things and odd and even are these elements, the number one, which is odd and even, must somehow be the source from which all things flow. Looked at from what for us would be the mathematical angle, we understand how bodies can be generated from points. The number one is represented by the point; in the order of linear numbers, 2 (..) would already be the line, where line is a cluster of at least two points. The first plane number, 3 (..) is the triangle; and 4, the first solid number, is the pyramid (..), The solid body, accordingly, is a number in the sense of a multitude of unit-points. Now, as has been mentioned earlier, just as the distinction between discrete and continuous quantity is not operative in Pythagorean mathematics, neither is there any distinction between geometrical solids and physical bodies. Since their way of depicting numbers produced plane and solid figures, bodies and even physical bodies, were taken by the Pythagoreans to be composed of units and, consequently, these bodies are numbers. Aristotle is clear on this even as he objects to it. "For not thinking of number as capable of existing separately removes many of the impossible consequences; but that bodies should be composed of numbers, and that this should be mathematical number, is impossible. For it is not true to speak of indivisible spatial magnitudes; and however there might be magnitudes of this sort, units at least have not magnitude; and how can a magnitude be composed of indivisibles? But arithmetical number, at least, consists of abstract units, while these thinkers identify number with real things; at any rate they apply their propositions to bodies as if they consisted of those numbers." (*Metaphysics*, XIII, 8, 1083b10) Once more, the Pythagoreans did not consciously maintain that physical bodies are mathematical bodies, that the physically concrete is really the conceptually abstract -- they simply failed to make the distinction. The indivisibility of the unit-point is, as Aristotle pointed out, suspect if it is to possess magnitude. But this is what the Pythagoreans maintained. They were not consciously adopting a mathematical interpretation of the universe; for them mathematics was about the physical universe. The things in the universe are numbers and they are generated from the one. Pythagorean mathematics is a cosmology.

It is strange also to attribute generation to eternal things, or rather this is one of the things that are impossible. There need be no doubt whether the Pythagoreans attribute generation to them or not; for they obviously say that when the one has been constructed, whether out of planes or of surface or of seed or of elements which they cannot express, immediately the nearest part of the unlimited began to be drawn in and limited by the limit. (*Metaphysics*, XIV, 3, 1091a12)

Now this has in common with Ionian thought the fact that it is a way of explaining how the world began; "they are describing the making of a cosmos and mean what they say in a physical sense," as Aristotle adds. Aristotle is contrasting the Pythagorean view with the Platonic one according to which there are subsistent numbers, existing apart from physical bodies. Aristotle is under no illusions about the Pythagorean view of the extent of reality. Already in his account of their doctrine at the outset of the *Metaphysics* he wrote:

They employ less ordinary principles or elements than the physical philosophers, the reason being that they took them from non-sensible things (for the objects of mathematics, except those of astronomy, are without motion); yet all their discussions and investigations are concerned with Nature. They describe the generation of the Heaven, observing what takes place in its parts, their attributes and behavior, and they use up their causes and principles upon this task, which implies that they agree with the physicists that the real is just all that is perceptible and contained in what they call 'the Heaven.'
(989b29ff)

We must keep in mind the identification made by Pythagoras himself of sounds and numerical ratios -- identification, not application -- lest we delude ourselves into thinking that the Pythagoreans have turned from the objects which concerned Ionian philosophy to other, more real entities. It is an appraisal of physical reality in both cases, not a conscious change of objects. The first stage of the cosmogonical process which Aristotle attributes to the Pythagoreans, consists of the formation of the first unit, though elsewhere he objects that the Pythagoreans are at a loss to describe the nature of his formation. (*Metaphysics*, XIII, 6, 1080b20) Subsequently, the unit draws in the unlimited and by imposing limits on it produces other units. As to the formation of the first unit, Aristotle mentions several possibilities: it could have been formed of planes or of surface, of seed or of some other elements. If composed of planes or surfaces, the first unit would be a solid. The supposition that the constituents of this first unit are seed would fit in with the location of "male" in the same column of opposites as limit. Its complement, the female, is the unlimited which it "draws in." The picture delineated becomes very much like the cosmogony of Anaximenes when we learn that the unlimited is air and that the first unit breathes it in. Air or void is drawn in and keeps things apart, for it seems likely that the first unit grows and splits and is kept apart by the void or air. The continuation of this growth results in the universe we know.

Perhaps we have here the answer to the difficulty expressed by Aristotle: "For natural bodies are manifestly endowed with weight and lightness, but an assemblage of units can neither be composed to form a body nor possess weight." (*De Caelo*, III, 1) Kirk and Raven suggest that bodies would vary in weight according as they contained more or less void. This solution, of course, presupposes that the units have weight, and it is that assumption Aristotle is questioning. Aristotle is also perplexed by the fact that the Pythagoreans seem to leave qualitative distinctions unexplained.

To judge from what they assume and maintain, they speak no more of mathematical bodies than of perceptible; hence they have said nothing whatever about fire or earth or the other bodies of this sort, I suppose because they have nothing to say which applies *peculiarly* to perceptible things. (*Metaphysics* 1,8)

Even when we recognize that the Pythagoreans did not distinguish mathematical and physical bodies, their failure to explain sense qualities is a serious gap.

The Pythagorean view of the universe represents a significant shift from the geocentric view of Ionian philosophy. Fire, not earth, is the center of things and the earth is one of the stars for which night and day is caused by its circular motion around the central fire. They are said to have invented a planet, the so-called counter-earth, to bring the number of planets to ten, the perfect number. The counter-earth follows the earth in its path around the sun, always remaining invisible to us because of the bulk of the earth. "In all this," Aristotle comments, "they are not seeking for theories and causes to account for observed facts, but rather forcing their observations and trying to accommodate them to certain theories and opinions of their own." (*De Caelo*, II, 13) It is thought that the notion of the counter-earth dates from the time of Philolaus; another astronomical theory, that of the "harmony of the spheres" is considered to be of earlier origin in the school.

From all this it is clear that the theory that the movement of the stars produces a harmony, i.e., that the sounds they make are concordant, in spite of the grace and originality with which it has been stated, is nevertheless untrue. Some thinkers suppose that the motion of bodies of that size must produce a noise, since on our earth the motion of bodies far inferior in size and in speed of movement has that effect. Also, when the sun and the moon, they say, and all the stars, so great in number and in size, are moving with so rapid a motion, how should they not produce a sound immensely great? Starting from this

argument and from the observation that their speed, as measured by their distances, are in the same ratios as musical concordances, they assert that the sound given forth by the circular movement of the stars is a harmony. (*De Coelo*, II, 9)

We do not hear this sound only because we have always heard it and are unable to contrast it with any opposed silence.

The Pythagoreans have lumped together unit and a point with magnitude, from which point the line is generated and so on to solids. Just as no differentiation is made between number and extension, so no distinction is recognized between mathematical and physical bodies, although this causes many rather obvious aspects of sensible bodies to go unexplained. The coming into being of our world is likened to the generation of the number series and the series of solids. The universe has grown from a primal unit which breathes in air or void and then splits up, imposing limits on the previously unlimited. The unit is considered to be male, the unlimited female. Earth is not at the center of the universe, but swings in a circular motion around a central fire, which motion produces day and night. In their movements, the heavenly bodies produce a wonderful music which has been singing in our ears since birth, and so is imperceptible by us.

B. Parmenides of Elea

In his dialogue, *Parmenides*, Plato gives us information which enables us to arrive at the approximate time of Parmenides' life. Plato says that Parmenides once visited Athens with his pupil, Zeno, when Parmenides was sixty-five and Zeno forty. Socrates was a young man at the time and had occasion to talk with the distinguished visitors; and that is the hook on which Plato hangs the dialogue. Since we know that Socrates was seventy when he was put to death in 399 B.C., the visit probably occurred about 451-449 B.C. With this information, we can arrive at the probable date of Parmenides' birth and say that he was in his prime about 475 B.C. This does not agree with the date assigned by Diogenes Laertius, but there is fairly general agreement that Plato can be relied on in this matter. The best argument for accepting the accuracy of Plato's information is that there would have been no need to have been so specific; and that since he was, it is more likely than not that the ages he gives are true.

Elea was a town in southern Italy, not far from Croton and Metapontium, where the Pythagoreans were influential. It is not surprising, therefore, to learn that Parmenides was a Pythagorean. It is said that Parmenides was converted to the contemplative life by Ameinias, a Pythagorean, although he is also said to have been a pupil of Xenophanes. Cornford (Plato and Parmenides, p. 28), speaks of Parmenides as "a dissident Pythagorean" and we will see the basis for this description in our analysis of Parmenides' poem. We have a good deal more to work with in the case of Parmenides than was true of any philosopher we have previously considered. We have the introductory portion of his poem which tells us in an allegorical fashion a good deal about what follows; moreover, large portions of the body of the poem have come down to us and we are able to arrange them in what seems to have been their original order. As we shall see, there is very little of the "poetic" in the Parmenidean doctrine; the prologue gives us a hint as to why he chose the form of presentation he did. As a poet, Parmenides is not held in very high esteem by knowledgeable scholars. Besides Xenophanes, there are only two philosophers who wrote in verse, Parmenides and Empedocles, who imitated Parmenides. Perhaps we can say of Parmenides what Aristotle said of his imitator: the poetry does not matter. We mention this to indicate that Parmenides presents no special instance of the problem raised in the first chapter because he expressed himself in verse. It is Simplicius, incidentally, to whom we are primarily indebted for our possession of so much of what Parmenides wrote; he introduced much of it in his commentaries on Aristotle. The prologue was preserved by Sextus Empiricus.

The fragment opens abruptly with the statement that Parmenides, the man who knows, is being borne in a car on the renowned way of the goddess through all the towns. Attended by maidens, Parmenides describes the sound made by the whirling axle and then, the daughters of the sun throw back the veils from their faces and leave the abode of night. At the entreaty of the maids, the gates of the ways of night and day are thrown open, Parmenides enters and is greeted by the goddess.

Welcome, O youth, that comest to my abode on the car that bears thee tended by immortal charioteers! It is no ill chance, but right and justice that has sent thee forth to travel on this way. Far indeed does it lie from the beaten track of men. Meet it is that thou shouldst learn all things, as well the unshaken heart of well-rounded truth, as the opinions of mortals in which is no true belief at all. Yet none the less thou shalt learn these things also -- how passing right through all things one should judge the things that seem to be.

The heightened tone of this prologue is thought to have been adopted in order to win respect for what is to follow. In fact, some think the prologue and the epic form were chosen to dress up an otherwise dull doctrine; but the prologue seems to be saying that Parmenides has in hand what must be regarded as a divine revelation. The goddess has spoken to him, even as a young man, and he is to learn all things. This is by way of conversion, and the opening sentence seems to suggest that Parmenides has sought in vain through many towns the knowledge that his heart desired, while all along, unbeknownst to him, he was being led on to the gates of night and day, and to the goddess whose revelation he would make known to us. The goddess instructs Parmenides in two ways: he is to know all things, both "well- rounded truth" and the opinions of men. The first is far from the usual thoughts of men; but it is the truth. The same cannot be said for the opinions of men. "But do thou restrain thy thought from this way of inquiry, nor let habit by its much experience force thee to cast upon this way a wandering eye or sounding ear or tongue; but learn by argument (*logos*) the much disputed proof uttered by me. There is only one way left that can be spoken of" We find in this passage, still part of the prologue apparently, an opposition between the two ways, that of opinion and of truth, expressed in terms of an opposition between the senses and argument or reason. The goddess asks that Parmenides eschew the senses and listen to the argument she will give in order that he might grasp the truth. As will appear, Parmenides is attacking the views of his predecessors, both those of the Ionian physicists and of the Pythagoreans whose cosmogony has many points of similarity with the Ionian.

The fragments which have come down to us can be divided according to the indications of the prologue into those pertaining to the way of truth and those belonging to the way of opinion. It has been a matter of much discussion as to why the poem of Parmenides should contain doctrine which he himself describes as utterly false. Why should he not confine himself to well-rounded truth and forget false opinions? Surely because the goddess revealed both; but why was this done? He is to learn all things, the false as well as the true; but above all, he is going to learn how to distinguish between the two. First we must consider the way of truth.

Come now, I will tell thee -- and do thou hearken to my saying and carry it away -- the only two ways of search that exist for thinking. The first, namely, that it is and cannot not be is the way of belief, for truth is its companion. The other namely that it is not and it must needs not be -- that, I tell thee, is a path

that none can learn of at all. For thou canst not know what is not -- that is impossible -- nor utter it; for the same thing exists for thinking and for being.

We have here the initial statement on which all else ultimately depends for Parmenides; unfortunately, it is not a very clear statement. Let us take him to be saying that, if a thing is, it is and cannot not be, since it is impossible to think of something as not being.

Parmenides is convinced that it is nonsense to speak of something as not being, since it would seem that it somehow is and then is said not to be; but he will not allow that we can think of what is not as if it were. Indeed, if something can be spoken of and thought, it is, and that is all there is to it. "It needs must be that what can be spoken and thought is; for it is possible for it to be, and it is not possible for what is nothing to be. That is what I bid thee ponder." The goddess now suggests that the false way is twofold. The truth is that it is; the false that it is not, but there is a variation in falsehood insofar as men speak as if something could both be and not be.

I hold thee back from this first way of inquiry, and from this other also, upon which mortals knowing naught wander two-faced . . . undiscerning crowds who hold that to be and not to be are the same, yet not the same, and that of all things the path is backward-turning.

Cornford suggests that the attempt to reconcile being and non-being is actually an effort to accept both reason and the senses and that, once more, Parmenides is saying that we must leave the reports of the senses behind and rely on reason alone. Xenophanes and Heraclitus expressed some doubts about sense perception; Parmenides, however, goes far beyond them, for he maintains that nothing but falsity can be gotten from the senses. The senses give rise to the notion of opposites and contraries, to the belief that one thing is such-and-such and another is not such-and-such.

Thus far the doctrine of Parmenides seems utterly abstract and unrelated to the thought of his predecessors. He has insisted, in effect, on the difference between being and non-being. Being is being and non-being is non-being, and there is an end to it. Despite the differences, we can detect a continuity between Parmenides and his predecessors. The Ionians had each spoken of some one, primal thing whose modifications and states produce a multiplicity of things -- things which are contrary to one another: this one is not that one and vice versa. Nonetheless, there is some one nature, alive and divine, which pervades all

things and which survives the ceaseless change of the many things perceived by the senses. When Xenophanes speaks of the one divine thing and the deceptiveness of the senses, he is not really going against the physical philosophers; he can be seen rather as laying even greater stress on the one which is stable and unchanging. Heraclitus finds unity in the proportionate and ordered changing of one opposite into another and back again. What is unchanging is change; and given the fact that one thing can change into another, opposites in a sense are not as other as we might think. The Pythagoreans, too, speak of the world as proceeding from a primitive unit, growing out of that unit in its multiplicity and variety. Parmenides reacted against all these views, though like his predecessors he begins with the one -- with what is. The one thing can be called being and the doctrine of Parmenides is that, when reason reflects on this one, it will seem that what philosophers had hitherto said is unacceptable to thought. The senses, of course, report multiplicity, the opposition of one thing to another and so forth; but this simply cannot be the case. Being is; non-being is not. Given the truth of that proposition, the falsity of earlier doctrines can be made manifest, even though the truth is surprising and far from the beaten path of men; for truth appeals to reason alone and can be attained only if we abandon the senses. Parmenides does not seem to be saying that anyone would explicitly maintain that nothing exists, that non-being is, but the variant of the position -- that being is somehow the same and yet not the same as non-being -- is held by many. As we shall see, it is this variant of the way of falsity which is at issue in the second part of the body of Parmenides' poem. His predecessors had all maintained in one way or another that being arose out of non-being.

What is the argument Parmenides is asked to heed?

One path only is left for us to speak of, namely, that *It is*. In this path are very many tokens that what is, is unborn and imperishable, for it is whole, immovable and without end. Nor was it ever, nor will it be; for now it is, all at once, a continuous one.

Being -- what is -- will be shown to have certain properties, namely, that it did not come to be nor will it cease to be. "For what kind of origin for it wilt thou look? In what way and from what source could it have drawn its increase? If being has come to be, we must be able to assign some source for it. "I shall not let thee say nor think that it came from what is not; for it can neither be thought nor uttered that anything is not." Can being come from nothing? This cannot be since we cannot say nor think that nothing is. We have already seen Parmenides

maintain that what is thought and what is are one, so that non-being is unthinkable and unutterable. This may seem curious, since Parmenides insists that others have thought and said just this. Cornford suggests that what Parmenides means is that false statements -- statements about non-being -- have nothing to refer to and consequently are meaningless. Moreover, Parmenides adds, if being came from nothing, what prompted it to arise when it did? There is no reason for it to come forth at one time rather than another. Being, therefore, is simply being all at once and ever, or there is simply nothing and always will be.

Our judgment thereon depends on this: 'Is it or is it not?' Surely it is adjudged, as it needs must be, that we are to set aside the one way as unthinkable and nameless (for it is no true way), and that the other path is real and true. How then can what is be going to be in the future? Or how could it come into being? If it came into being, it is not; nor is it if it is going to be in the future. Thus is becoming extinguished and passing away is not to be heard of.

Being cannot come to be because there is nothing from which it could come save nothing and it cannot come from that. Being is, right now and always the same; tenses have no meaning in speaking of being since it has neither a past nor a future. There are other properties of the one being.

Nor is it divisible, since it is all alike, and there is no more of it in one place than in another, to hinder it from holding together, nor less of it, but everything is full of what is. Wherefore it is wholly continuous; for what is, is in contact with what is.

The Pythagoreans explained the coming into being of the many by saying that the primal unit breathed in the void which then served the function of keeping the split-up units separated from one another. Thus, in this view, there are interstices in being, and the void of non-being is employed to explain a multiplicity of things. Parmenides will have none of this. Being is a plenum indistinguishable into parts as if there were more being here than there or as if something intervened between what is and what is. Being is one, homogeneous, the same throughout, containing and permitting no void or non-being. We see once more that Parmenides does not want to go beyond the one with which his predecessors can be said to begin. First there was some one nature and then by one process or another, it broke up into many things and so forth. Parmenides wants to begin and end with the one being which is wholly the same, unique, admitting no gaps or distinctions within itself, utterly homogeneous.

Not only is being ingenerable and incorruptible, it does not move.

Moreover, it is immovable in the bonds of mighty chains, without beginning and without end; since coming into being and passing away have been driven afar, and true belief has cast them away. It is the same, and it rests in the selfsame place, abiding in itself. And thus it remaineth constant in its place; for hard necessity keeps it in the bonds of the limit that holds it fast on every side. Wherefore it is not permitted to what is to be infinite; for it is in need of nothing; while, if it were infinite, it would stand in need of everything.

Being cannot be undetermined, since then it would require a determination from something else. Rather it is limited and thus complete, something which leads Parmenides to provide us with an image of the being he is speaking of. Since then it has a furthest limit, it is complete on every side, like the mass of a rounded sphere, equally poised from the center in every direction; for it cannot be greater or smaller in one place than in another. For there is nothing that could keep it from reaching out equally, nor can aught that is be more here and less there than what is, since it is all inviolable.

The well-rounded truth -- that what is, is and cannot not be -- involves a view of being as a solid sphere in which there is an equal distribution of being throughout, no gaps, no imbalance. This is the truth of the matter, whatever the senses report. There is one being, unique, immobile, absolutely unchangeable and indivisible. These properties have been deduced from the original statement by showing that any other possibility is contradictory.

Thus Parmenides has rejected the dualism which defines the Pythagorean doctrine. Indeed, it can be seen that he has chosen one column and rejected the other. The left-hand column of opposites contains limit, unity and resting or immobility -- the very contents of the way of truth. Parmenides is, in the words of Aristotle, a nonnatural philosopher, since it is not the world of change which interests him, that world being for him an illusion from which we are freed by the way of truth. The being we come to know by reason is one, timeless, unchanging, devoid of all perceptible qualities, a motionless sphere of completely homogeneous mass.

With Parmenides, philosophy reaches a crisis that is acutely felt by all who follow on him. The validity of the senses has been called into question and with them, the world of opposites, of hot and cold, smooth and soft, and all other perceptible contraries. We are driven by cold reason to a view of being which is

austere and uncompromising and which threatens the beginnings of natural science achieved by Parmenides' predecessors. We will see later the central place Aristotle accords to Parmenides and how important he feels is his own resolution of the Parmenidean dilemma which makes change and multiplicity impossible.

The methodology of the Parmenidean poem must be carefully considered. For the first time, we have a use of reasoning which moves from an initial statement to inescapable consequences. This dialectic is something which will be employed by Zeno, the pupil of Parmenides, and we must see in it the beginnings of what comes to be called logic. The basic procedure is a reduction to absurdity. Parmenides makes progress by showing that views contrary to his own involve impossible consequences; when we see this, we are prepared to accept as the truth the doctrine he would maintain: what is, is and cannot not be. What is not, is not and cannot be. There is no way in which being and non-being can be construed as in any way the same. To maintain that being is generable or corruptible, measured by time, divisible or mobile is to fly in the face of these premises. We must then accept the premises as true and abandon all hope of reconciling with them the world we see around us. Nonetheless, the poem of Parmenides has things to say about the ordinary world. "Here shall I close my trustworthy speech and thought about the truth. Henceforward learn the beliefs of mortals, giving ear to the deceptive ordering of my words."

Before turning to the way of opinion, that part of his poem in which Parmenides sets forth a doctrine about the world around us, a doctrine revealed to him by the goddess so that no other mortal will appear wiser than Parmenides, there is an important question to be asked. Is the one being of which Parmenides speaks immaterial? In favor of an affirmative answer is the sharp dichotomy Parmenides draws between what the senses grasp and what is real for mind. The latter, the only true being, is then stripped of all sensible qualities and has none of the properties we should normally associate with the corporeal or material. On this interpretation, Parmenides' example of the sphere would be just that -- an example, a simile, an aid to understanding -- but not a literal description of the one being. Perhaps it is safer to hesitate here and ask ourselves whether Parmenides can seriously be taken to have arrived at the notion of a wholly immaterial being, the only being that is. True enough, in speaking of the one being, he denies of it many properties of sensible bodies, a process which may suggest all too easily to us an incorporeal, nonspatial reality. We may even want to suggest that Parmenides was groping towards such a notion, but the fragments we have do not permit any categorical statement to

the effect that Parmenides has arrived at the recognition of immaterial reality. Aristotle was of the opinion that the Eleatics, Parmenides and Melissus, were aware of no reality beyond the corporeal although they were forced by their dialectic to the recognition of the need for something unchanging as a ground for true knowledge. (*De Caelo*, III, 1, 298b14) What they did was to question the validity of sensation, deny sensible qualities of the one being which nevertheless remained a body. In other words, the Parmenidean being retains the note of spatial extension and, if doubtless a strange one, is nevertheless a body. As with the Pythagoreans, Parmenides speaks in such a way that what he has to say could be taken to be applicable to non-physical being -- provided of course that there is such being and we have some cognitive access to it. This recognition, on our part, is no argument that the Pythagoreans themselves or Parmenides recognized there was such being. In short, Aristotle's opinion that they made no such explicit recognition seems deserving of acceptance. This is not to say that the procedure of the Pythagoreans and of Parmenides was not destined to have great influence on later attempts to achieve scientific knowledge of non-physical being. And, again, the method of Parmenides and his immediate follower, Zeno, had no little influence on the development of what came to be called logic.

In the way of opinion or of seeming, the poem of Parmenides passes from what is accessible to reason alone to what the senses report. The goddess has urged Parmenides to give ear to the deceptive ordering of her words, which may refer to the earlier remark that

language about what is not is meaningless and that, in this part of the poem, we find a doctrine concerning things which, truly speaking, are not. The reason given in the poem for this part of the revelation is that Parmenides must not appear less than anyone else. There has been much conjecture about the motive for the natural doctrine of Parmenides, and of course it will occur to one that Parmenides will enjoy a poor sort of superiority if he surpasses others in the order of falsehood. Burnet feels that Parmenides is here giving a review of popular beliefs concerning the physical world, that it is, in effect, a sketch of the Pythagorean cosmology. Kirk and Raven are unimpressed by this estimate, since they fail to find the characteristic notes of the Pythagorean doctrine. More positively, they point out that the ancients uniformly considered the cosmology to be of Parmenides' own devising. Aristotle feels that Parmenides, in the way of opinion, is attempting an explanation of the way being appears to man with his senses, something he could do without in any way changing his mind that this world involves features which are contradictory to pure reason. Whatever the

explanation of this second part of the body of his poem, it does not contain the influential portion of Parmenides doctrine; indeed, neither Melissus nor Zeno seems to exhibit the slightest interest in anything save the content of the way of truth. Moreover, much less of the way of opinion has come down to us than of the way of truth. It is mainly for this reason that we shall content ourselves with having given some slight indication of the difficulties posed by the existence of the way of opinion, and not go into the many attempts to make sense out of the few fragments of it we possess. Whatever his motives for setting it down, Parmenides' physical doctrine seems to be pretty nearly the same sort of attempt as his predecessors had made. What is utterly distinctive of the Eleatic philosopher is the dilemma he poses for anyone who would take seriously what his senses tell him of the world. All that is false, Parmenides has argued; what is, is one, immobile, indivisible, atemporal; it has neither been generated nor can it ever be destroyed. To put it most succinctly, Parmenides has called into question the existence of change and multiplicity. Change is impossible, for if something has come into being, it must first of all not have been; that is, we would have to say that something came from nothing. Multiplicity too is unacceptable. If there are two things, two beings, how do they differ? They cannot differ in being, because that is what they have in common; they cannot differ in nothing, since that is no difference. There can be no difference in what is, accordingly, and we must recognize that being is a monolithically unique body. This type of doctrine and the argumentation which sustains it is the major contribution of Parmenides to the development of Greek philosophy. We can now turn to the defence and development that doctrine received in the hands of the followers of Parmenides.

C. Zeno of Elea

We have seen that Zeno made a journey to Athens with Parmenides when the latter was sixty-five years old. At that time, Zeno was forty and, on the basis of the earlier analysis, we can place Zeno's birth at approximately 490-485 B.C. The story of his life, meager as it is, parallels the little we know of his master's. Zeno is a native of Elea, and a converted Pythagorean. In his *Parmenides*, Plato speaks of a book written by Zeno in which the pupil essays the defence of his master against those who object to his doctrine of the one, a defence that pays the attackers back in their own coin, for, as they had maintained that many absurdities follow from the position of Parmenides, so Zeno is intent to show that absurdities equally if not more great follow from adopting a view opposed to that of Parmenides. Tradition has it that Aristotle, in a lost dialogue of his, credits Zeno with being the founder of dialectic or logic. Zeno seems to have

been primarily concerned with showing that impossible contradictions issue from our acceptance of the reality of motion and of multiplicity. Let us turn immediately to the consideration of some of these arguments which have come down to us.

As we have indicated, the arguments of Zeno are directed against multiplicity and motion. What is the multiplicity against which Zeno argues? There are at least two possibilities. We may take Zeno as arguing against the possibility of there being many things in the macrocosmic world around us; or, and this is the more likely one, Zeno is arguing against the Pythagorean doctrine that things are numbers and consequently aggregates of unit-points. Thus anything is a plurality of such monads which in themselves are indivisible but have position, that is, are in space. With this in mind, we can turn to a few of the more than forty arguments Zeno is said to have devised against multiplicity.

If there is a plurality, things will be both great and small; so great as to be infinite in size, so small as to have no size at all. If what is had no magnitude, it would not even be. For if it were added to something else that is, it would make it no larger; for being no size at all, it could not, on being added, cause any increase in size. But if it is, each thing must have a certain size and bulk, and one part of it must be a certain distance from another; and the same argument holds about the part in front of it -- it too will have some size and there will be something in front of it. And it is the same thing to say this once and to go on saying it indefinitely; for no such part of it will be the last, nor will one part ever be unrelated to another. So, if there is a plurality, things must be both small and great; so small as to have no size at all, so great as to be infinite.

Zeno may here be seen as putting his finger on the difficulties which attend the attempt to identify mathematical and physical bodies. Any determinate thing will be a given number of units; thus it will be finite in arithmetical and geometrical quantity, such-and-such a number, and a body of determinate size. Zeno wants to show that if we hold that the units have magnitude, things composed of them are going to have to be infinite in size. Between any two units there must be room for another, and so on to infinity. This point is made explicitly in another fragment.

If things are a many, they must be just as many as they are, and neither more nor less. Now, if they are as many as they are, they will be finite in number. If things are a many, they will be infinite in number; for there will always be other

things between them, and others again between those. And so things are infinite in number.

In mathematical magnitude, we never run out of points; between any two points on a line, an infinity of points can be designated. When physical bodies are spoken of as composed of points having magnitude, we apply the geometrical doctrine and arrive at the need to say the physical body is infinite in size. If, on the other hand, we try to elude the difficulty by denying that the points of which physical bodies are composed have size, then we would be hard pressed to explain how something having size is made up of parts having no size. This view of things as pluralities of units, then, involves the idea that bodies are infinite in size and that they have no size at all.

On this interpretation, then, Zeno is looked on as defending the views of his master against the attacks of the Pythagoreans. The plurality which involves contradictory consequences is the plurality of units physical bodies are said to be. But Zeno was also concerned to defend his master's doctrine that being is immobile. His procedure is the same; the acceptance of motion involves one in contradictory consequences. Zeno is credited with four arguments against motion; they are preserved by Aristotle and are important enough in themselves and for subsequent philosophy to be examined in their entirety.

The impossibility of traversing a race track --

You cannot traverse an infinite number of points in a finite time. You must traverse the half of any given distance before you traverse the whole, and the half of that again before you can traverse it. This goes on to infinity, so that there is an infinite number of points in any given space, and you cannot touch an infinite number one by one in a finite time.

Once more, we have the consequences of the Pythagorean identification of the mathematical and physical drawn out. If a length is infinitely divisible and if a physical length is composed of points having magnitude, then the traversal of a finite space implies the traversal of infinite spaces. Aristotle, in first mentioning this paradox in Book Six of his *Physics* (233a21), sets it aside by pointing out that Zeno should have put the same question to time that he puts to space. The finite time in which one is held to traverse an infinite distance is itself infinite in much the same way as the finite distance is. That is, Zeno begins with some such situation as this. I run a mile in five minutes. The mile, of course, is a finite distance but it can be infinitely divided into lesser distances. Therefore, in five

minutes, a finite time, I traverse an infinite distance. But, Aristotle observes, the finite time, five minutes, is divisible to infinity in much the same way that the mile is. This is not, of course, anything like a complete answer to the dilemma, nor did Aristotle think it was. In a neglected passage in Book Eight of the *Physics* he returns to the matter, pointing out the inadequacy of the previous reply. (263a4ff.)

Achilles and the tortoise --

Achilles will never overtake the tortoise. He must first reach the place from which the tortoise started. By that time the tortoise will have got some way ahead. Achilles must then make up that, and again the tortoise will be ahead. He is always coming nearer, but he never makes up to it.

This is not essentially different from the previous argument, only more complicated, since we have to do with two moving bodies and division of space in fractions other than halves. Once more, it is directed against the view that physical magnitude has the same properties as mathematical extension.

The arrow --

The arrow in flight is at rest. For, if anything is at rest when it occupies a space equal to itself, and what is in flight at any given moment always occupies a space equal to itself, it cannot move.

Once more, the assumption is that time is composed of moments in the way that the line has been taken to be composed of points.

The moving rows --

It will be best to approach this argument by way of a diagram.

- (a) AAAA
- (b) BBBB-->
- (c) <-- CCCC

Imagine a race track on which the four bodies in (a) are stationary, and consider the bodies in rows (b) and (c) to be moving past them in opposite directions, with those of (b) occupying a position between the goal and middle point of the track and those of (a) between the midpoint and the start. With this example, Zeno wants to show that half a given time is equal to twice the same time. As

the race ends, the first body in (b) reaches the last body in (c) at the same moment as the first in (c) reaches the last in (b), but at this moment, the first in (c) has passed all the bodies in (b) whereas the first in (b) has passed only half the bodies in (a). Thus, the first in (b) has taken up only half the time as the first in (c). The movement of (c) with respect to (b) is double the movement of (c) with respect to (a), since the first body of (c) passes all the bodies of (b) and half those of (a). Thus, while each body in (b) has passed two in (a), each body in (c) has passed four in (b). Zeno can be taken to have proved that it is impossible to think of space and time as composed of indivisible units. That is, he is making the point that the sensible and the mathematical are not the same thing, that the properties of the latter cannot be attributed to the former. It is just this confusion that the Pythagoreans seem guilty of, and Zeno is justified in drawing out the implications of that confusion. Here is the way Aristotle deals with this argument.

The fourth argument is that concerning the two rows of bodies, each row being composed of an equal number of bodies of equal size, passing each other on a race-course as they proceed with equal velocity in opposite directions, the one row originally occupying the space between the goal and the middle point of the course and the other that between the middle point and the starting-post. This, he thinks, involves the conclusion that half a given time is equal to double that time. The fallacy of the reasoning lies in the assumption that a body occupies an equal time in passing with equal velocity a body that is in motion and a body of equal size that is at rest; which is false. (*Physics*, VI, 9, 239b33)

D. Melissus of Samos

Although Melissus moves us geographically back to the area where Greek philosophy began, his acceptance of the doctrine of Parmenides dictates his inclusion with the Italian school. He led the Samian fleet that defeated that of the Athenians in a battle fought in 441-440 B.C. We know little else of the life of this man; and it is dubious whether he was ever in personal contact with Parmenides, although he is said to have been his pupil. This is thought to have been based on what he wrote rather than on any knowledge of, or association with, the Eleatic philosopher.

We have already seen that Parmenides maintained that being was finite, meaning by this that it was perfect of itself and needed nothing outside itself in order to attain perfection. Melissus, many of whose fragments are simply

repetitions of Parmenides, departs from his master on this matter of the finitude of being.

Since, then, it has not come into being, and since it is, was ever, and ever shall be, it has no beginning or end, but is without limit. For, if it had come into being, it would have had a beginning (for it would have begun to come into being at some time or other) and an end (for it would have ceased to come into being at some time or other); but, if it neither began nor ended, and ever was and ever shall be, it has no beginning or end; for it is not possible for anything to be over unless it all exists. (Fr. 2)

This argument did little to earn for Melissus the esteem of Aristotle, who points out that Melissus thinks he has a right to say that if that which has come into being has a beginning, that that which has not come into being has no beginning. Despite the fallaciousness of his reasoning, we can conjecture why it was that Melissus felt he must depart from Parmenides on the matter of the finitude of being. If being were a finite sphere, one could easily imagine that the sphere was bounded by the void, which could sound very much like saying that nothing is, what makes being what it is. The fragment just quoted seems to fluctuate between talk of a beginning in time and a spatial beginning; Melissus wants to deny both of being. It is eternal in duration and infinite in extension. This infinite extension is of a very curious kind, apparently, since Melissus also denies that being can be a body. "If being is, it must be one; and being one, it must have no body. If it were to have bulk, it would have parts and be no longer one." (Fr. 9) Tannery held that Melissus had arrived at the conception of immaterial being, but that this is not unequivocally so appears from the way in which Melissus speaks of the infinity of being. It had no beginning in time; but as well it has no limits as to spatial extension. It seems one thing to say that something is not spatially limited and that the notion of spatial limitation or illimitation is inapplicable to it. Fragment 9 is surely difficult to reconcile with Fragment 2, and we must admire Melissus' grasp of what seems to have escaped Parmenides, namely, that his spherical being despite his protestation, is divisible.

If we permit ourselves a few summary remarks on the Eleatic school, it is because something of extreme importance seems almost to have been reached by its members -- though doubtless for the wrong reasons. The sharp dichotomy drawn between reason and the senses is destined to have a great impact on subsequent philosophy, together with the allied implication that the object of reason must be immutable if there is to be a foundation for true knowledge.

Whatever the senses report, change seems impossible for, if we admit it, we seem committed to the view that something comes from nothing. Better, then, to reject sense perception and accept only what makes sense to reason. This leads to extremely Pickwickian statements about being. With Parmenides we are faced with a unique, homogeneous sphere devoid of all sense qualities. Zeno defends this conception against attack by taking the opposed position and showing that it involves contradictory consequences. One might say that Zeno leaves the enemy demolished but the victory meaningless -- a recognition that has led to the description of Zeno as a sceptic for whom nothing is true, since all positions are susceptible of fatal attack. Whatever the truth of this description, Zeno is engaged solely in attacking the opposition and does not directly defend the position of Parmenides. Melissus, on the other hand, makes the Parmenidean doctrine his own, altering it when he feels that it is in need of alterations, lest it become an untenable position. Hence his great correction of the master when he says that being must be infinite. Further, being cannot be a body, since this would involve having parts. Though none of this leads to a definitely established grasp of the existence of being other than corporeal being, the language in which these men express themselves is such that what they say, as Aristotle pointed out, seems to have application beyond the physical world -- if there is such a beyond; for surely this is not evident. Doubtless it is anachronistic to speak of Parmenides as the first metaphysician but equally doubtless he has managed to present his thought in such a way that later thinkers saw a path opening before them which they were the first to trod. Another extremely important aspect of the Parmenidean school, particularly in Zeno, is its awareness of the force of the form of argumentation. An ideal is thereby set for future philosophizing and, consequently, a need gradually becomes recognized of examining for their own sake the forms of argument. It may be that when Aristotle called Zeno the father of dialectic, he meant dialectic in the sense of his own *Topics*, such that Zeno is seen to take the position granted him and to proceed from that. Nonetheless, Zeno can be called the father -- or at least the grandfather -- of logic in the wide sense, in that his rigorous procedure made men conscious of their own procedures in establishing the desired conclusion. Of more immediate impact was the doubt cast on the possibility of change and motion. If the admission of change involved the admission of something coming from nothing, then natural philosophy was in an impossible quandry. No one who wanted to continue the kind of speculation that had begun with the Ionians could conscientiously avoid the Parmenidean attack on motion and change. It is from this vantage point that we can best appreciate the efforts of Empedocles and Anaxagoras.

{17} Pythagoras is said to have coined the term "philosophy." See Kirk and Raven, p.229.

{18} See F. M. Cornford, *Plato and Parmenides* (New York: Liberal Arts Press, 1957), pp. 1-27.

Chapter IV

Empedocles of Acragas

Empedocles was said to have been in his prime in 450 B.C., to have been younger than Anaxagoras, whom we shall consider next, but to have begun his philosophical career earlier. A native of Acragas in Sicily, he was very active politically, an ardent democrat to whom many wondrous feats were attributed and duly recorded by Diogenes Laertius. It is possible to argue, as Freeman has, that these anecdotes are more likely than not based on elements in the works of Empedocles rather than on independent knowledge of what he did. The same is perhaps true of the stories that he was a miracle worker.

We possess some one hundred and fifty fragments of the writings of Empedocles, and the survivals of two poems, which together originally totaled five thousand lines, the one called *On Nature*, the other *Katharmoi* or Purifications. We are, then, in a rather better position to grasp the teaching of Empedocles than of earlier thinkers. The fragments reveal a man interested both in knowledge about the natural world and in religion. A prose work on medicine, numerous tragedies and other poems were also said to have been written by Empedocles; but the assumption is that the fragments we possess are from the two poems first mentioned and that, whatever the truth about other writings, our estimate of Empedocles must be based on his *On Nature* and *Purifications*. We will see that it has been thought difficult to reconcile the contents of these two poems. We indicated in an earlier chapter that Aristotle did not have a high opinion of Empedocles as a poet, saying that about the only thing he had in common with Homer was the fact that he wrote in verse, that he should be classified as a physiologue rather than a poet, and that his verse is a poor vehicle for expressing scientific views, since it is filled with ambiguity. (*Poetics*, 1447b17) He is one of those, Aristotle feels, who writes in verse to conceal the fact that he has nothing to say. (*Rhetoric*, 1407a31) As an example of these defects, Aristotle observes that to call the sea the sweat of the earth, as Empedocles did, makes a good metaphor but is not a scientific statement. (*Meteorology*, 357a24)

Given this judgment, one is permitted to wonder if there is not a little irony in Aristotle's calling Empedocles the father of rhetoric as Zeno is the father of dialectic. Lucretius, of course, had a different view of Empedocles and praised both his style and his doctrine. (*De rerum natura*, 714ff.) Empedocles' poem *On*

Nature is addressed to his pupil, Pausanias; *Purifications* addresses itself to the citizens of Acragas.

Empedocles is said to have been, with Zeno, a student of Parmenides, and Theophrastus says that he is an imitator of Parmenides, something which surely has in view the fact that both men wrote in verse. He is also said to have listened to Pythagoreans.

A. On Nature

Before turning to the intrinsic evidence of the influence of Parmenides on Empedocles, we shall examine the introductory portion of the poem, for that too reveals a debt to the great Eleatic thinker. Empedocles asks Pausanias to give ear to what he has to say that he might learn all that a man can learn. That it is difficult for a man to acquire wisdom is clear from the severe limitations on each of us; we are necessarily restricted to our own particular experience, which is ours largely by chance, and which is circumscribed by birth and our all too swift death. Moreover, the senses constantly break in upon and disrupt our thought.

For straitened are the powers that are spread over their bodily parts, and many are the woes that burst in on them and blunt the edge of their careful thoughts. They behold but a brief span of life that is no life, and, doomed to swift death, are borne up and fly off like smoke. Each is convinced of that alone which he had chanced upon as he is hurried every way, and idly boasts he has found the whole.

Pausanias is fortunate to have found the way here, but Empedocles promises him only what a man can learn. Empedocles then appeals to the gods to turn his own tongue from the madness of those who think they possess knowledge, and asks that he might "hear what is lawful for the children of a day. Speed me on my way from the abode of Holiness and drive my willing car." Fr. 4) The allusion to Parmenides' own divinely directed chariot ride seems clear.

The influence of Parmenides is seen in the content of the poem as well. "Fools! -- for they have no far-reaching thoughts -- who deem that what before was not comes into being, or that aught can rise from what in no way is, and it is impossible and unheard of that what is should perish; for it will always be, wherever one may keep putting it." Fr. 11,12) "And in the All there is naught empty and naught too full." (Fr. 13) We could ask for no clearer indication that

Empedocles has accepted the argument of the way of truth as utterly inescapable. Being cannot come from nothing nor can it become nothing; the all is a plenum, with no more being here than there, and the void or non-being is impossible. Now we have seen that, for Parmenides, this argument leads to the view that the physical or natural world, the world of change and becoming, is only an illusion. If Empedocles had accepted that consequence, he would hold considerably less interest for us than he does. What gives him importance in the story we are attempting to follow is the fact that he accepts the position of Parmenides, the way of truth, and yet writes a work on nature. He does not, as we shall see, discount the testimony of the senses, nor is he willing to dismiss the world around him as an illusion that simply cannot be what it so clearly is, a world of many things which come and go. The reconciliation of this unquestioned presupposition of Ionian cosmology and the way of truth of Parmenides is Empedocles' great contribution and the surest indication that Parmenides had indeed posed a problem that later thinkers did not feel they could ignore.

First of all, Empedocles comes to the defense of the validity of sense perception.

Hold not thy sight in greater credit as compared with thy hearing, nor value thy resounding ear above the clear instructions of thy tongue; and do not withhold thy confidence in any of thy other bodily parts by which there is an opening for understanding, but consider everything in the way it is clear. (Fr. 4)

The senses are to be accepted for what they are, the paths to understanding, the openings we have on the world through which reality can come to be known by us. With this departure from Parmenides, the difficulty of the task Empedocles sets himself becomes even more acute. Trust in the senses is to be coupled with the acceptance of the austere and non-natural argument of the way of truth. How is this to be accomplished?

"Hear first the four roots of all things: shining Zeus, life-bringing Hera, Aidoneus and Nestis whose tear-drops are a well-spring to mortals." (Fr. 6) Empedocles is here assigning the four roots or elements out of which all things grow and he gives them names of gods, names which were intended to convey fire, air, earth and water. The mythological flavor of Empedocles' poem is clear here, and there has been some discussion as to which divine name was to be taken for which element. Nestis is clearly water; Zeus and Hera are probably fire and air respectively, and Aidoneus earth. (See Freeman, p.181) These roots are uncreated. (Fr. 7)

And I shall tell thee another thing. There is no nature (*physis*) of any of all the things that perish, nor any cessation for them of baneful death. They are only a mingling and interchange of what has been mingled. Nature is but a name given to these things by men. Fr. 8)

The things around us that come and go are not anything other than a coming together of the four roots, a coming together that cedes to separation.

But they (hold?) that when Light and Air (chance?) to have been mingled in the fashion of a man, or in the fashion of the race of wild beasts or of plants or birds, that that is to be born, and when these things have been separated once more, they call it woeful death. I follow the custom and call it so myself. (Fr. 9)

It is just here, in Diels' ordering of the fragments, that we find Empedocles repeating the argument of the way of truth. This suggests that what we call being and becoming and perishing is not truly being, becoming and perishing; for they apply to things like men and beasts and plants which are simply the coming together of the four roots or elements. This is no true becoming and such collections have no true nature (*physis*) of their own. What truly are, are the four elements and of them there is no becoming or perishing. We can begin to see here how Empedocles is going to eat his Parmenidean pie and keep it, too. The senses tell us that there are many things in the world, things that come and go, and in our language we speak of nature, of becoming and perishing. But these are not the things which truly are and consequently we should not take the language of change too seriously, as if it applied to being in the rich sense. What brings about what we call becoming and perishing?

I shall tell thee a twofold tale. At one time it grew to be one only out of many; at another, it divided up to be many instead of one. There is a double becoming of perishable things and a double passing away. The coming together of all things brings one generation into being and destroys it; the other grows up and is scattered as things become divided. And these things never cease continually changing places, at one time all uniting in one through love, at another each borne in different directions by the repulsion of Strife. Thus, as far as it is their nature to grow into one out of many, and to become many once more when the one is parted asunder, so far they come into being and their life abides not. But, inasmuch as they never cease changing their places continually, so far they are ever immovable as they go round the circle of existence. But come, hearken to my words, for it is learning that increaseth wisdom. As I said before, when I declared the heads of my discourse, I shall tell thee a twofold tale. At one time it

grew together to be one only out of many, at another it parted asunder so as to be many instead of one; -- Fire and Water and Earth and the mighty height of Air; dread Strife too apart from these, of equal weight to each, and Love in their midst, equal in length and breadth. Her do thou contemplate with thy mind, nor sit with dazed eyes. It is she that is known as being implanted in the frame of mortals. It is she that makes them have thoughts of love and work the works of peace. They call her by the names of Joy and Aphrodite. Her has no mortal yet marked moving around among them, but do thou attend to the undeceitful ordering of my discourse. For all these are equal and alike in age, yet each has a different prerogative and its own peculiar nature, but they gain the upper hand in turn when the time comes round. And nothing comes into being besides these, nor do they pass away; for, if they had been passing away continually, they would not be now, and what could increase this All and whence could it come? How, too, could it perish, since no place is empty of these things? There are these alone; but, running through one another, they become now this, now that, and like things evermore. (Fr. 17)

This lengthy passage contains the nub of Empedocles' cosmology. We now learn that, besides fire, air, earth and water, there are two other principles, the causes of their coming together to form beings and of their separation in the destruction of those things. There seem to be opposite poles at one of which love has collected the elements together in a whole or one, at the other of which strife has driven apart into a many. There is a continuous shuttling back and forth between these two extremes in a cosmic cycle that is ever repeated. This cycle does not entail that being, the four elements, change, for they are ever the same throughout the cycle. This, we have seen, is Empedocles' way of accepting the strictures of Parmenides. Nothing really real comes to be or ceases to be but only compounds of the things that really are. Empedocles further observes that the process itself is unchanging. We have, then, overtones of Heraclitus as well as of Parmenides. What is more, it is difficult to avoid being reminded of Anaximander when we read Empedocles' description of the way in which what we call beings arise. Notice that Empedocles draws our attention to the "undeceitful ordering of his words," as opposed doubtless to the goddess' deceptive ordering of words when she speaks to Parmenides of the world of nature. The principles of Hate and Love differ in this, that Hate is outside the elements whereas Love is in them, though both are of equal weight with fire, air, earth and water. And, Love, while it is described as that which men name as in themselves, is described, as is Strife, in bodily terms; both have length and breadth. When the elements are brought together by Love to form a one, we find yet another instance of affinity with Parmenides. "But he was equal on every side

and quite without end, spherical and round, rejoicing in his circular solitude." (Fr. 28) This notion that the one brought about by the influence of Love is spherical is repeated in many other fragments. Strife is excluded from this sphere, doubtless explaining the earlier description of it as outside. Somehow or other, Strife begins to work its way into the sphere which is described in personal terms as a god.

But when Strife was grown great in the limbs of the god and sprang forth to claim his prerogatives, in the fulness of the alternate time set for them by the mighty oath . . . for all the limbs of the god in turn quaked. (Fr. 30,31)

This conjures up the image of the breaking up of the sphere by Strife and the consequent dispersal of the elements, capable, however, of lesser comings-together which produce the things of our sense experience. Before seeing a cosmogony in this disruption of the sphere, we might point out that the oath of Love and Strife seems to introduce a seventh element into Empedocles' picture of ultimate explanations. The echo of Anaximander is here, too, of course, and with it a mythological element which seems to go contrary to Empedocles' assertion that he is simply describing the way things have to be; the oath may be a metaphorical expression of this natural necessitation, but it is a factor that has been the occasion of no little discussion.

Empedocles' doctrine as to how the world we are in came to be constituted by the disruption of the sphere has not come down to us in the fragments, although ancient writers have things to say on the subject. From them we learn that air was first separated off from the sphere, then fire which hardened the air thus forming the outer sphere of our world which is half fire and causes day, the other mainly air and the cause of night. Earth was separated off and retains a central position in the cosmos because of the whirling movement of the heavens, kept in place, Aristotle suggests, in the way water stays in a swung bucket. Water is squeezed out of the earth. All this happens by chance, Empedocles insists. (Fr. 53)

Empedocles has a good deal to say about how living things arose. The following passage from Aetius indicates how strange Empedocles' doctrine was.

Empedocles held that the first generations of animals and plants were not complete, but consisted of separate limbs not joined together; the second, arising from the joining of these limbs, were like creatures in dreams; the third was the generation of whole-natured forms; and the fourth arose no longer from

the homoeomerous substances such as earth or water, but by generation, in some cases as the result of the condensation of their nourishment, in others because feminine beauty excited the sexual urge; and the various species of animals were distinguished by the quality of the mixture in them (Diels, A72; Kirk and Raven)

We have, in the fragments, some indication of these various stages. "The kindly earth received in its broad funnels two parts of gleaming Nestis out of the eight, and four of Hephaistos (fire). So arose white bones divinely fitted together by the cement of proportion." (Fr. 96)

And the earth, anchoring in the perfect harbours of Aphrodite (Cyprus), meets with these in nearly equal proportions, with Hephaistos and Water and gleaming Air -- either a little more of it, or less of them and more of it. From these did blood arise and the manifold forms of flesh. (Fr. 98)

We are asked to notice the influence of the Pythagorean notion of proportion or harmony in this description of the formation of bone, flesh and blood by the coming together of the elements. Once these have been explained, the existence of limbs seems accounted for, and Empedocles says "Solitary limbs wandered seeking for union. (Fr. 58) Or, yet more graphically, "On it (the earth) many heads sprung up without necks and arms, wandered bare and bereft of shoulders. Eyes strayed up and down in want of foreheads." (Fr. 57) These surrealistic scenes answer to the first stage mentioned by Aetius. The second stage is covered by other fragments. "But as divinity was mingled still further with divinity, these things joined together as each might chance, and many other things besides them continually arose." (Fr. 59) Empedocles tells of many handed creatures (Fr. 60) and of:

Many creatures with faces and breasts looking in different directions were born; some, offspring of oxen with faces of men, while others, again, arose as offspring of men with the heads of oxen, and creatures in whom the nature of women and men was mingled, furnished with sterile parts. (Fr. 61)

Aristotle suggests that successful compounds survived, others perished.

Wherever, then everything turned out as it would have if it were happening for a purpose, there the creatures survived, being accidentally compounded in a suitable way; but where this did not happen, the creatures perished and are perishing still, as Empedocles says of his 'man-faced ox-progeny.' (*Physics*, II, 8, 198b29)

Fragment 63 seems to describe the third stage mentioned by Aetius, and, of course, the fourth stage is covered by all the fragments in which Empedocles is speaking of the world as we know it.

Empedocles' explanation of how sensation takes place has always occasioned comment. "For it is with earth that we see earth, and water with water, by fire destroying fire; by love do we see love, and hate by grievous hate." (Fr. 109) We perceive what is outside us through the same thing within us, so that perception involves like answering to like. Moreover, Empedocles speaks of things as giving off a stream of particles which are picked up by the same element within us in sensation. "Know that effluences flow from all things that have come into being." Fr. 89; Burnet) Plutarch tells us that Empedocles saw this as an eroding process the term of which was the destruction of the thing from which the effluences flow. This theory of knowledge would seem to entail that we ought to get into the presence of the right things in order that their effluences might flow into us and transform us. Fr. 106) We saw earlier Fr. 4) that the senses provide an opening for understanding. All knowledge seems to involve the response of like to like; to know is to become what we know, in what would seem to be a physical change and, consequently, a resultant physical resemblance.

For, if, supported on thy steadfast mind, thou wilt contemplate these things with good intent and faultless care, then shalt thou have all these things in abundance throughout thy life, and thou shalt gain many others from them. For these things grow of themselves into thy heart, where is each man's true nature. But if thou strivest after things of another kind, as it is the way with men that ten thousand sorry matters blunt their careful thoughts, soon will these things desert thee when the time comes round; for they long to return once more to their own kind; for know that all things have wisdom and a share of thought. Fr. 110)

Somehow one retains possession of what he has learned by contemplation; distraction leads to the escape of what is gotten by knowledge back to like things outside of man. It is not surprising, on this explanation of thought, to learn that all things have a share of thought; this is repeated in another fragment. "Thus have all things thought by fortune's will . . ." (Fr. 103) Indeed, what we call thought is identified with the flow of blood. "(The heart), dwelling in the sea of blood that runs in opposite directions, where chiefly is what men call thought; for the blood round the heart is the thought of men." (Fr. 105) Both sensation and thought are explained in a purely materialistic fashion, then; knowledge is picking up the particles of things which flow

ceaselessly from them until they are destroyed. By the same token, distraction can cause what we have learned to seep from us. As for blood as thought, Theophrastus conjectures that this was chosen because it is in the blood that the elements are especially blended.

Empedocles had begun his poem on nature with a somber description of the evanescence of the life of man, the brief span of time allotted to him, his being buffeted hither and thither, until all too soon his life is snuffed out. While this description is by way of preparing Pausanias for the recognition that wisdom has hitherto gone undiscovered by man, Empedocles will promise to teach only what a man can learn. Fragment 111 suggests that Empedocles thought the working of marvels to be among such objects of learning, and his theory of knowledge would seem to lend itself rather easily to magical interpretations. However, the note we want to end on here, one which provides a bridge to the consideration of Empedocles' other poem, is that our Sicilian wise man has such a view of human existence that we could expect him to seek consolation somewhere to offset the chancey aspects of our existence. This is something that he seems to have done in his other poem.

B. Purifications

We have mentioned earlier that this poem is addressed to Empedocles' fellow citizens of Acragas. Approximately one-third of the fragments of Empedocles are thought to be survivals of this poem, so that we have somewhat less to go on than we had in discussing the other work. What appears to be the opening passage has its startling aspects.

Friends that inhabit the great town looking down on the yellow rock of Acragas, up by the citadel, busy in goodly works, harbors of honor for the stranger, men unskilled in meanness, all hail. I go about among you an immortal god, no mortal now, honored among all as is meet, crowned with fillets and flowery garlands. Straightway, whenever I enter with these in my train, both men and women, into the flourishing towns, is reverence done me; they go after me in countless throngs, asking of me what is the way to gain; some desiring oracles, while some, who for many a weary day have been pierced by the grievous pangs of all manner of sickness, beg to hear from me the word of healing. (Fr. 112)

Empedocles had a reputation for being arrogant, and it is not unlikely that the view was based on such passages as this. His modesty is superhuman. "But why do I harp on these things, as if it were any great matter that I should surpass

mortal, perishable men?" (Fr. 113) Empedocles goes on to explain how it is that he is immortal.

There is an oracle of Necessity, an ancient ordinance of the gods, eternal and sealed fast by broad oaths, that whenever one of the daemons, whose portion is length of days, has sinfully polluted his hands with blood, or followed strife and forsworn himself, he must wander thrice ten thousand seasons from the abodes of the blessed, being born throughout the time in all manners of mortal forms changing one toilsome path of life for another. For the mighty Air drives him into the Sea, and the Sea spews him forth on the dry Earth; Earth tosses him into the beams of the blazing sun, and he flings him back to the eddies of Air. One takes him from the other, and all reject him. One of these I now am, an exile and a wanderer from the gods, for that I put my trust in insenate strife. (Fr. 115)

An exile from the gods due to a lapse, Empedocles is rejected in turn by each of the elements -- rejections which of course involve the taking on of different and varying forms of life. "For I have been ere now a boy and a girl, a bush and a bird and a dumb fish in the sea. (Fr. 117) A god, Empedocles has fallen from a state of honor and bliss (Fr. 119) that make the earthly honors of which he is the recipient (Fr. 112) pallid things indeed, and scant consolation for what he has lost.

Empedocles' description of himself as a fallen god making retribution for a past fault by a cycle of incarnations finds its parallel in the view that all men are now in a fallen condition which calls for purification in order that they might escape from it. Indeed, Empedocles is often interpreted as uttering a fact common to all men when he speaks of himself as a fallen *daimon*, so that transmigration, the cycle of incarnations, is not his personal affliction but the common lot. In one fragment he seems to describe a golden age prior to the alliance with strife which has brought about the need for the thrice ten thousand seasons of wandering through various forms of life.

Nor had they any Ares for a god nor Kudoimos, no nor King Zeus nor Kronos nor Poseidon, but Kupris as Queen. Her did they propitiate with holy gifts, with painted figures and perfumes of cunning fragrancys, with offerings of pure myrrh and sweet-smelling frankincense, casting on the ground libations of brown honey. And the altar did not reek with pure bull's blood, but this was held in the greatest abomination among men, to eat the goodly limbs after tearing out the life. (Fr. 128)

The golden age is ended by the eating of the flesh of animals, and Empedocles seems to adopt many of the taboos we have seen to have been part of Pythagoreanism. Kupris here is Aphrodite or Love, and Empedocles seems to be describing what is called the reign of love in the poem on nature. At this time the kinship of man with all life -- another Pythagorean view -- was recognized. "For all things were tame and gentle to man, both beasts and birds, and friendly feelings were kindled everywhere." (Fr. 130) We remember that Empedocles attributes his own fall to putting his trust in strife; the god Ares, absent from the golden age, can be taken to stand for strife and, to draw a parallel between what Empedocles says in the *Purifications* and in *On Nature*. With strife, dissolution enters in and we have a descent from unity and wholeness.

Empedocles has some things to say about divinity which deserve attention; he himself does not think it a small matter what view men have of the gods. "Blessed is the man who has gained the riches of divine wisdom; wretched he who has a dim opinion of the gods in his heart." (Fr. 132) Now, God is not something which can be grasped by the senses. "It is not possible for us to set God before our eyes, or to lay hold of him with our hands, which is the broadest way of persuasion that leads into the heart of man." (Fr. 133) We have already seen how the senses are openings into man's understanding; here Empedocles suggests that there is another way to wisdom. Since man cannot attain knowledge of divinity from the things around him in any direct way, there must be some other mode of access. God is not like any of the things we encounter in the world around us.

For he is not furnished with a human head on his body, two branches do not sprout from his shoulders, he has no feet, no swift knees, nor hairy parts; but he is only a sacred and unutterable mind flashing through the whole world with rapid thoughts. (Fr. 134)

This description bears a remarkable similarity to the description of the sphere of being during the reign of love, contained in a fragment thought to belong to the poem *On Nature*. "Two branches do not spring from his back, he has no feet, no swift knees, no fruitful parts; but he was spherical and equal on every side." (Fr. 29) Such parallels make it quite likely that Empedocles is striving for a unity of thought in these two poems, the first of which speaks of the world, the second of man's relation to reality. It is man's misdeeds which disrupt the unity of things and bring him into his present sorry state. Crimes of bloodshed, not only of man against man, but the slaying and devouring of animals, cry out for punishment and man is punished. Father slays son, son father. The

consumption of animals, too, leads to impurity. "Ah, woe is me that the pitiless day of death did not destroy me ere ever I wrought evil deeds of devouring flesh with my lips." (Fr. 139) The picture of the dissolution of the sphere formed by love, when strife waxes strong in it and separates the elements bringing into being our present cosmos, is paralleled by man's fall from a state of innocence in a lost golden age, the commission of crimes which leads to the soul's migration through various forms of life. There is an escape for man from the cycle of incarnations just as our cosmos must give way once more to the reign of love. When the soul has made the rounds of forms of life, it reaches the state in which Empedocles, the immortal, finds himself.

But, at the last, they appear among mortal men as prophets, songwriters, physicians, and princes; and thence they rise up as gods exalted in honor, sharing the hearth of the other gods and the same table, free from human woes, safe from destiny, and incapable of hurt. (Fr. 146, 147)

Finally, the soul is set free from the wheel of birth and enjoys a life of bliss with the gods, where the sorrows of this life can no longer touch it.

There are, of course, great difficulties in reconciling the fragments of the two Empedoclean poems that have come down to us. In the first, the view of life and knowledge that is expressed seems to reduce them both to the material elements out of which they arise. Thought, we remember, was referred to the movement of blood about the heart. In the *Purifications* on the other hand, the soul comes to the body as to its exile; it is a fugitive from the realm of the gods and must pass through a cycle of birth, until it reaches the stage occupied by Empedocles himself, after which there is a transition to a life of bliss with the immortals once more.

Empedocles is under a number of influences and he responds to each of them. He accepts the Parmenidean way of truth, and yet finds a way to account for the validity of sensation and the reality of the world the senses report. With Parmenides he rejects the Pythagorean notion of the void or non-being, but the Pythagorean notion of harmony or proportion is employed by Empedocles; what is more, the taboos of the Pythagorean society find their counterpart in Empedocles. Fragment 141 even warns us about beans. There is certainly an echo of Anaximander in the description of the becoming of the things we sense, and the Heraclitean notion of the unchangeability of the ceaseless change of reality from one stage to its opposite is reflected in the alternating reigns of

Love and Strife. Xenophanes seems to be exerting an influence in the negations Empedocles makes about divinity.

We have in Empedocles, then, a kind of summation of what has gone before. There is the linking of the scientific and religious motifs of philosophy in his two poems, a great sensitivity to what he feels is valuable in his predecessors, but no slavish adherence to previous positions. If he learns from everyone, he seems to do so by way of assessment, rejecting some things, accepting others. For the moment, we would stress his way of continuing to account for the natural world despite the grave dilemma posed by the Parmenidean way of truth. Empedocles attempts to bypass the difficulty by beginning with a multiplicity of elements which are truly beings and consequently do not become nor perish.

When we speak of becoming and ceasing to be we are not speaking of things which truly are, but of the things of our sense experience. Since they are not truly beings, we do not violate the Parmenidean logic even when we say, as doubtless we must, that they come to be and cease to be. The further question of the relation of Empedocles' views to earlier mythological doctrines -- a point much stressed by Cornford -- and to Orphicism is not one we shall discuss now. Later, when we look back over the terrain of Presocratic philosophy, we shall try to exhibit the relevance of the questions we raised in our opening chapter. For the present, we must now turn to the attempt of Anaxagoras to meet the difficulty posed for natural philosophy by Parmenides' way of truth.

Part I: Presocratic Philosophy

Chapter V

Anaxagoras of Clazimene

With Anaxagoras we move once more back to eastern Greece whence philosophy had originally come. We have seen that Anaxagoras was older than Empedocles, but that he began his philosophical activity later than the Sicilian. This would be helpful if we could achieve more determination about Empedocles' dates and if we did not have the puzzling information to the effect that Anaxagoras came to Athens and began philosophizing at the age of twenty. It may be safe enough to think of him as flourishing about 460 B.C. We are told that Anaxagoras was forced to flee Athens and that he died at Lampsacus. We

need not here go into the speculation and discussions about when and why he left Athens. Anaxagoras is said to have studied under Anaximenes, but this remark seems prompted by certain doctrinal tenets.

Anaxagoras is said to have written only one book; this seems virtually certain, although in late antiquity many writings were attributed to him. We have a little over twenty fragments of his book and they are such that we can surmise that we have the core of his doctrine. That doctrine seems to be a quite conscious response to the challenge of Parmenides' way of truth; and Anaxagoras answers it in a way that goes beyond the efforts of Empedocles, though somewhat in the same line. The tenor of his solution is found in the first fragment.

All things were together, infinite both in number and in smallness; for the small too was infinite. And, when all things were together, none of them could be distinguished for their smallness. For air and aether prevailed over all things, being both of them infinite; for amongst all things these are the greatest both in quantity and size. (Fr. 1)

Simplicius, who has preserved most of the fragments we have of Anaxagoras, tells us that this remark was the first in Anaxagoras' book. We are immediately apprized of Anaxagoras' way of handling Parmenides. If things cannot come to be, we need not conclude with Parmenides that there is but one unchanging thing; rather let us say that in the beginning everything, the whole variety of things of our sense experience, was present in a confused whole. That is, if nothing can be said to come to be, let everything exist from the beginning, since thereby we can save what our senses tell us and not violate the Eleatic logic.

But before they were separated off, when all things were together, not even was any color distinguishable; for the mixture of all things prevented it -- of the moist and the dry, and the warm and the cold, and the light and the dark, and of much earth that was in it, and of a multitude of innumerable seeds in no way like each other. For none of the other things either is like any other. And these things being so, we must hold that all things are in the whole. (Fr. 4)

It is not mere conjecture that Anaxagoras begins as he does because of what Parmenides had taught.

The Hellenes follow a wrong usage in speaking of coming into being and passing away; for nothing comes into being or passes away, but there is mingling and separation of things that are. So they would be right to call coming into being mixture and passing away separation. (Fr. 17)

Empedocles had met the difficulty by speaking of the four roots of all things, the elements which alone are fire, air, earth and water. Anaxagoras accepts the diversity of things and says that something like gold is not to be reduced to non-gold as to its elements, but rather to particles of gold, particles which must be thought of ultimately as infinitesimally small but of the same nature as that with which we began. "How can hair come from what is not hair, or flesh from what is not flesh?" (Fr. 10)

We have here a much more radical heeding of the argument of the way of truth. To say, as Empedocles had, that what we call the coming into being of something like flesh is a new combination of elements, themselves not flesh, seems to give too much meaning to becoming. If, on the other hand, before flesh comes to be, as we would say, it has already existed as flesh, but as very small particles, then the notion of becoming is too strong to describe what is happening, and we would do better to speak of mingling and separation. If we take something like flesh and think of it as being broken into smaller particles of flesh, and each of those as being further broken down, we will never, Anaxagoras maintains, come to an end.

Nor is there a least of what is small, but there is always a smaller; for it cannot be that what is should cease to be by being cut. But there is also always something greater than what is great, and it is equal to the small in amount, and, compared with itself, each thing is both great and small. (Fr.3)

If fragments quoted earlier seem to refer to Parmenides, this one, as Kirk and Raven have argued (pp. 371-2), seems clearly to have Zeno in mind. The paradoxes of Zeno were directed against those who confused physical and mathematical magnitude. In mathematics, we can speak of infinite divisibility, but Zeno attempted to show that we encounter insurmountable difficulties if we think of physical extension in this way. Anaxagoras refuses to be intimidated and appears to insist quite consciously on the infinite divisibility of physical matter. There is no end to the process whereby flesh could be cut down into small particles of flesh; and just as that with which we begin has magnitude, so too must the infinity of infinitely small particles it contains. Thus, no matter how small the particle of flesh we imagine, it is great because it can be further subdivided into parts of the same kind. Aristotle uses the word *homoeomeriēs* to convey this notion that the parts of the whole are of the same nature as the whole. Difficult as it is, Anaxagoras is deliberately saying that something like flesh has this in common with the line that however far you divide it, you will

still be left with smaller versions of that with which you began. The assertion that things are at once great and small seems clearly an open retort to Zeno.

Thus far the view of Anaxagoras would seem to lead to an image of the world according to which things are rigorously set off from each other, since the parts of any thing are simply smaller instances of its nature. Flesh and bone, then, would seem to have nothing in common. A first corrective to this is Anaxagoras' cosmogonical remark about the way things were first of all. In Fragment I we find Anaxagoras speaking of when all things were together, at which time they were indistinguishable because of their smallness. So too in Fragment 4, quoted before, we read of the innumerable seeds present in the original mass where none of the opposites was yet obvious. More emphatically, Anaxagoras maintains that everything is in everything.

And since the portions of the great and of the small are equal in amount, for this reason, too, all things will be in everything; nor is it possible for them to be apart, but all things have a portion of everything. Since it is impossible for there to be a least thing, they cannot be separated, nor come to be by themselves; but they must be now, just as they were in the beginning, all together. And in all things many things are contained, and an equal number both in the greater and in the smaller of the things that are separated off. (Fr.6)

This would mean that we could take any object in the universe and it would contain within itself everything else. Change, consequently, would involve the separation off of some particles so that the original object would appear different afterwards than it had before what we call the change took place. Not that there could be a complete depletion of particles of a certain kind from a given physical object. There is an infinity of particles of any given type in anything both before and after what we would call a change.

Now, as Aristotle pointed out, this leads to a perplexing question. If anything contains everything, why do we call some things flesh, others bone, yet others gold, etc.? Anaxagoras' equally perplexing answer, according to Aristotle, is that what we call flesh has more portions of flesh than anything else -- although of course it has an infinity of parts of anything else. It is well to recall once more why Anaxagoras is putting forward this paradoxical doctrine. If everything is in anything, then no change, however surprising, is going to involve the coming into being of something which did not exist before. Aristotle gives a more concrete and plausible statement of this motivation. Anaxagoras accepted the rather obvious fact that a thing seems to arise from its opposite and its opposite

again from it, hot from cold, dry from wet. Now Anaxagoras, given the Parmenidean way of truth, would not want to say that before hot comes to be, hot did not exist; therefore, he is led to say that hot already existed in the cold, that it was there in parts and particles, and that when portions of cold are removed we notice the portions of hot and designate the object accordingly. Such a process could never be completed, however, as if all the particles of cold might be removed: everything is in everything. "The things that are in one world are not divided nor cut off from one another with a hatchet, neither the warm from the cold nor the cold from the warm." (Fr. 8)

The perplexing thing about the theory of Anaxagoras is that his cosmogony seems repeatable in every particular thing in the world. That is, he speaks of all things being together in the beginning, present in one mass in infinitely small particles so that the whole would not seem to have any particular nature, except perhaps that of air or aether. (Fr. 1) Then begins a process of separating off. "For air and aether are separated off from the mass that surrounds the world, and the surrounding mass is infinite in quantity." (Fr.2) This separation is accomplished by a whirling motion (Fr. 13), and after separating, things begin to mingle again.

The dense and the moist and the cold and the dark came together where the earth is now, while the rare and the warm and the dry (and the bright) went out towards the further part of the aether. (Fr. 15)

In the area of the earth, the things that we know appear. "From these as they are separated off earth is solidified; for from mists water is separated off, and from water earth. From the earth stones are solidified by the cold, and these rush outwards more than water." (Fr. 16) Thus far, the doctrine of Anaxagoras emerges as an ingenious if difficult attempt to circumvent the difficulties Parmenides had posed for those who would accept motion and change and multiplicity as requiring no explanation. Anaxagoras is more thoroughgoing in his attempt to have change without novelty, becoming without any alteration of the things that are; but the difference would seem to be one of degree rather than kind.

There is a good deal more than this in the fragments of Anaxagoras, an element of doctrine that caused Aristotle to say that he sounded like one sober man in a chorus of drunks (*Metaphysics* I 3,984b15-18). Aristotle is referring to Anaxagoras' doctrine of *Nous* or Mind. The few things said about Mind in the

fragments indicate that a great advance is being made, a positive contribution, and not simply an ingenious reaction.

If everything is in everything, there is at least one exception to this dictum.

All other things partake in a portion of everything, while *Nous* is infinite and self-ruled, and is mixed with nothing, but is alone, itself by itself. For if it were not by itself, but were mixed with anything else, it would partake in all things if it were mixed with any; for in everything there is a portion of everything, as has been said by me in what goes before, and the things mixed with it would hinder it, so that it would have power over nothing in the same way that it has now being alone by itself. For it is the thinnest of all things and the purest, and it has all knowledge about everything and the greatest strength; and *Nous* has power over all things, both

greater and smaller, that have life. And *Nous* had power over the whole revolution, so that it began to revolve from the beginning; but the revolution now extends over a larger space, and will extend over a larger still. And all the things that are mingled together and separated off and distinguished are all known by *Nous*. And *Nous* set in order all things that were to be, and all things that were and are not now and that are, and this revolution in which now revolve the stars and the sun and the moon, and the air and the aether that are separated off. And this revolution caused the separating off, and the rare is separated off from the dense, the warm from the cold, the light from the dark, and the dry from the moist. And there are many portions in many things. But no thing is altogether separated off nor distinguished from anything else except *Nous*. And all *Nous* is alike, both the greater and the smaller; while nothing else is like anything else, but each single thing is and was most manifestly those things of which it has most in it. (Fr. 12)

Empedocles of course had introduced a pair of causes beside the elements to account for the changes that occur. Anaxagoras speaks of *Nous*, of Mind, as the motive force behind the revolution, causing the separating off from the primal whole and the consequent constitution of our world. Why does this make Anaxagoras more deserving of praise than Empedocles? One way would be to suggest that when Empedocles wants to explain what he means by love he says that men have felt it as they feel the effects of Aphrodite. Now this would suggest being propelled along a given course of action, rather than choosing it with reason; and, indeed, Empedocles does not want to suggest that cosmological processes take place otherwise than by chance. Moreover, love and

strife or hate are manifestly corporeal for Empedocles, just as thought is the surging of blood about the heart. The *Nous* of Anaxagoras is described in such a way that it seems quite clear that he is striving in a wholly new way to speak of a principle which transcends the material order. *Nous* is unmixed, it does not contain portions of all things; and it is this freedom from admixture which is said to explain both its power over all things and its ability to know all things. This way of speaking of knowledge goes quite a distance beyond Empedocles' explanation, whereby knowing was like answering to like. Here it is utter difference in nature which underlies knowledge. The dichotomy between *Nous* and the material world is sharply described in the long fragment we have just quoted. *Nous* is over against the cosmos; unmixed, itself by itself. If it contained any bit of physical matter it would contain all, since everything is in everything. Thus Anaxagoras speaks of *Nous* as thinnest and purest, striving to express what is utterly unlike the other things he is speaking of. *Nous* directs the cosmogonical process and continues to govern the movements of the heavens and this means that things do not come about by chance. They are caused by mind, by intelligent direction. Later Greeks -- Socrates and Aristotle -- could not refrain from praising Anaxagoras for stating that the world involves rational direction, intelligence; if they then went on to make reservations about the use to which Anaxagoras put this new principle -- reservations we shall discuss presently -- we should not let this distract us from the magnificence of Anaxagoras' contribution to Greek philosophy. The world is no longer something that just happened, that is, a state consequent on a previous state for purely mechanical reasons; it is caused by a rational principle who initiates the process, and, thanks to his unmixed nature, knows and has power over all things.

In the *Phaedo*, Plato allows Socrates to speak of how overjoyed he was when he found that Anaxagoras had taught that all things were directed by Mind; and yet Socrates was disappointed because the principle remained so abstract and Anaxagoras had not employed it in any particular explanation. Aristotle suggests that Anaxagoras, despite the merit of his introduction of *Nous* to explain the cosmogonical process, has set *Nous* an impossible task. *Nous* is given as the cause of separating off; but this, according to Anaxagoras, is a process which can never achieve its term. Thus, Mind is engaged in an irrational enterprise. As Freeman points out (p. 269), the complaint of Socrates is not wholly justified, since *Nous* is said to be the cause of whatever was, is, or will be; the statement may be general, but it could hardly be more comprehensive. Aristotle's criticism would be a good deal harder to meet, and Anaxagoras' doctrine acquires, in the face of it, a bifurcated look. On the one hand there is

his strange and difficult response to Parmenides; on the other, the new principle, *Nous*, the directing cause of whatever goes on in the physical world.

Doubtless we must caution ourselves against attributing too much to Anaxagoras, as if his doctrine on the *Nous* emerges unequivocally from the few fragments we possess. We have seen earlier attempts -- the first notable one being that of Xenophanes -- to achieve clarity in speaking of what is beyond the things grasped by the senses. Obviously, the way which proceeds by negating of such a principle the qualities of physical things is the only way open to us. So Xenophanes denies that the divine is at all like men or other things. So, too, Parmenides would speak of a being which is devoid of all sensible qualities -- although it seems to have spatial extension. Despite the dangers of exaggeration, it seems undeniable that Anaxagoras has taken a significant step beyond earlier efforts to speak of a principle which is other than physical things. What is more, as Aristotle liked to point out, the dissimilarity of *Nous* with physical things is explained in function of such activities as knowledge and governing. If Anaxagoras is making a significant forward step in the effort to transcend the material, it is equally true that there is still a long way to go; only with Plato and Aristotle, and with significantly different underpinning, will we encounter clearcut statements of entities beyond the material. From the vantage point of their achievements, a backward glance at the contribution of the first Greek philosopher associated with Athens will enable us to make another and perhaps surer appraisal of the extent of that contribution.

The doctrines of Empedocles and Anaxagoras are not the last efforts by presocratic philosophers to escape the dilemma Parmenides had posed for natural philosophy. Extreme as their doctrines may seem as reactions to the Eleatic philosopher, there is another species of pluralism, atomism, to which we must now turn.

Chapter VI

Atomism

In this chapter, we shall discuss a doctrine first advanced by Leucippus and carried on by Democritus, although the latter cannot technically be called a Presocratic, being roughly a contemporary of Socrates. There is good reason, as we shall see, for linking the two men, something that has been done since earliest times. Leucippus is said to have been a native of Elea, but also of Miletus; there seem to be better reasons for regarding him as a Milesian, although there is little strong evidence either way. As a matter of fact, we are not very well informed about Leucippus; even in ancient times there were those who doubted that he had existed at all. Democritus tells us in his *The Little World Order* that he was forty years younger than Anaxagoras, making it likely that he was born about 460/457 B.C. Now the assumption is that Leucippus first put forward the doctrine of atomism and that Democritus accepted and elaborated it. Freeman gives 430 B.C. as the *floruit* of Leucippus and 420 B.C. as that of Democritus. Leucippus was said to have written a book entitled *The Great World Order* and a book *On Mind* of which one sentence has come down to us: "Nothing happens at random, all happens out of reason or by necessity." The sentiment of this remark hardly accords with the implications of atomism, however, and it is conjectured that the remark is not really his, but that of a follower of Anaxagoras. Democritus, on the other hand, seems to have written a good deal, certainly much more than any of his predecessors. We are told that his works were later listed in groups of four, much as Plato's were, and that the total number of his writings was fifty-two. Among his physical works were *The Little World Order*, *Cosmography*, *On the Planets*. In later lists, his works were divided into Physical, Mathematical, Musical, Technical, and Ethical; it is from these latter works that most of the surviving fragments come. These included works on Pythagoras, on the philosophical nature, on courage and imperturbability, and on the next world. There were also works grouped under the heading, Causes. We are told that many of these works may have been written after the time of Democritus, but later collectors and listers tended to attribute them all to Democritus. Our ancient sources sometimes speak of Leucippus and Democritus separately but more often together; our presentation will not lay great stress on one man or the other. We shall regard the doctrine involved as one common to Leucippus, Democritus and their followers. Although the attribution of the theory may have to remain fuzzy, the theory itself is precise and fairly easy to set forth.

Aristotle leaves no doubt as to the motivation for the doctrine attributed to Leucippus and Democritus. He tells us of it after setting forth the Eleatic rejection of nature and of other reactions to it.

But Leucippus thought he had a theory which, being consistent with sense-perception, would not do away with coming-into-being or perishing or motion or the multiplicity of things. So much he conceded to appearances, while to those who uphold the one he granted that motion is impossible without void, that the void is not-being and that no part of being is not-being. For being, in the proper sense, is an absolute plenum. But such a plenum is not one, but there is an infinite number of them, and they are invisible owing to the smallness of their bulk. They move in the void (for the void exists), and by their coming together they effect coming-into-being by their separation perishing. (*De gen.*, 1, 8, 325a23ff.)

Being in the strict sense does not come to be or perish, it simply is. But it is not one, but many, indeed, infinite in number. What truly are, are quite small particles, so small as to be invisible. If these are the things to which being properly applies, there is as well void, empty space, not-being, in which the particles move. Clusters of them form visible bodies and the things we mistakenly regard as beings in the proper sense; these things come to be and perish, to be sure, but not what is truly being, the small, invisible particles. As Aristotle remarks, this explanation saves the facts of sense perception but not at the expense of the Eleatic logic. We see that this reaction has much in common with those of Empedocles and Anaxagoras; where atomism differs is in its insistence that the void, too, must exist and that if by not-being we mean empty space, then not-being exists. Thus being and space, the plenum and the void, become the elements of explanation.

Leucippus and his associate Democritus hold that the elements are the full and the void; they call them being and not-being respectively. Being is full and solid, not-being is void and rare. Since the void exists no less than body, it follows that not-being exists no less than being. The two together are the material causes of existing things. And just as those who make the underlying substance one, generate other things by its modifications, and postulate rarefactions and condensation as the origin of such modifications, in the same way these men too say that the differences in atoms are the causes of other things. They hold that these differences are three -- shape, arrangement and position. Being, they say, differs only in 'rhythm, touching turning,' of which 'rhythm' is shape, 'touching' is arrangement, and 'turning' is position; for A differs from N in

shape, AN differs from NA in arrangement, and from H in position.
(*Metaphysics* I,4,985b4)

What deserves the name "being" are very small bodies, atoms, a word which means indivisibles, things which cannot be cut. Not only are these imperceptible, there is a countless number of them and they differ from one another in the ways indicated in the above passage, basically by size and shape. As irreducible elements of what we ordinarily call beings, the atoms cannot come from one another; each is given in its being and is not subject to change, to becoming or perishing. The void is where the atoms are not rather than where they are; each atom is full, compact, and endowed with motion whereby it can move through the void. Now, if visible, everyday bodies are compounded of atoms; the differences which strike us in such bodies are finally explicable in terms of the arrangement and shape of the component atoms. The usual opposites, hot-cold, wet-dry, etc., will be explained in this way. This reduction of the differences grasped by our senses to quantitative and local differences tends to commend the atomistic doctrine to contemporary thinkers, as if it were an anticipation of the physical theories with which we are familiar today. The atoms just happen to be the shape they individually are and they just happen to move around, so that the bodies we perceive and the world they compose seem reducible to chance.

The movement of the atoms may have been assigned to their weight, although we have conflicting testimony as to whether Democritus assigned weight to the atoms; some say that this was a later addendum of Epicurus. At any rate, that atoms had weight would seem to have been part of the theory of the Democritean school. It seems that weight was not invoked to explain particular kinds of motion -- motion in this direction as opposed to that -- but simply the random motion of the atoms in the void. In the case of the atoms, the weight would be in direct proportion to bulk or size, something not true of the bodies of our sense experience. A large mass of cotton will weigh less than a relatively small stone; this was attributed to the different portions of void in the two. A body with fewer interstices between its atoms would weigh more than another body of the same size comprising less atoms, and hence more empty space. This gives us an indication of why atoms and the void were said to be the elements of atomism. The shapes of the atoms were said to account for their clinging together when they collided, and the arrangements thus formed led to the further differences mentioned by Aristotle. If compounds result from collision, they can, of course, be destroyed in the same way, or larger compounds can be formed, the whole thing occurring by chance. The bodies which we would say

come to be, then, are such that their becoming is consequent upon the local motion of atoms. Simplicius quotes the following from a lost work of Aristotle's, *On Democritus*.

As they move they collide and become entangled in such a way as to cling in close contact to one another, but not so as to form one substance of them in reality of any kind whatever; for it is very simple-minded to suppose that two or more could ever become one. The reason he gives for atoms staying together for a while is the intertwining and mutual hold of the primary bodies; for some of them are angular, some hooked, some concave, some convex, and indeed with countless other differences; so he thinks they cling to each other and stay together until such time as some stronger necessity comes from the surrounding and shakes and scatters them apart. (Kirk and Raven, n. 581, pp. 418-9)

This passage emphasizes that nothing which is simply one results from the entanglement of the atoms -- not a further atom, surely; but just as surely the atoms are not somehow subsumed in a being of a higher order. Atoms are the only real being there is; what might appear to us to be a being of a higher order is merely the result of the collision and sticking together of the atoms.

This account of the formation of the world (or worlds) is attributed to Leucippus by Diogenes Laertius. (IX, 31-32)

He declares the All to be unlimited, as already stated; but of the All part is full and part empty, and these he calls elements. Out of them arise the worlds unlimited in number and into them they are dissolved. This is how the worlds are formed. In a given section many atoms of all manner of shapes are carried from the unlimited into the vast empty space. These collect together and form a single vortex, in which they jostle against each other and, circling round in every possible way, separate off, by like atoms joining like. And, the atoms being so numerous that they can no longer revolve in equilibrium, the light ones pass into the empty space outside, as if they were being winnowed; the remainder keep together and, becoming entangled, go on their circuit together, and form a primarily spherical system. This parts off like a shell, enclosing within it atoms of all kinds; and, as these are whirled round by virtue of the resistance of the center, the enclosing shell becomes thinner, the adjacent atoms continually combining when they touch the vortex. In this way the earth is formed by portions brought to the center coalescing. And again, even the outer shell grows larger by the influx of atoms from outside, and, as it is carried round in the vortex, adds to itself whatever atoms it touches. And of these some portions are

locked together and form a mass, at first damp and airy, but, when they have dried and revolve with the universal vortex, they afterwards take fire and form the substance of the stars.

The whirling vortex which is the first stage in the formation of a world is something we have already encountered in Anaxagoras, although then it had been begun by *Nous*. Here it just happens, given atoms and their motion; moreover, it can happen many times simultaneously, since there is no dearth of atoms and we need not think that all the atoms there are have gone into the composition of our world. The whirling motion causes like to be attracted to like. The similarity involved here would seem to be that of weight, though shape is not unimportant since the outer ring of fire-atoms were thought to be round and smooth, thus accounting for their mobility. Leucippus and Democritus were said to have maintained that each world is enclosed in a kind of skin, formed by the linking of hooked atoms, and through this more atoms were taken in after the initial stages of world-formation. Not all worlds are like our own; some have animals, some do not; some do not have sun or moon, or have ones of a much different size than ours. Presumably when a world stops growing -- by absorbing more atoms through its outer skin -- it begins to break up, its atoms returning to the common fund whence they can proceed to form another world at another time.

That the atomists meant that the world came about by chance seems clear enough. Surely it is not enough to suggest, as Freeman does, that motion, collision and formation of conglomerates is something which follows on the very nature of the elements. "The second stage was the collision of atoms, and consequent coagulation; this, the original formation of each cosmos, seems to be assigned to chance; but it was a 'chance' arising out of the essential nature of things." (p. 303) Precisely, and it is the world that comes about by chance, since the essential nature of things does not determine that their random movements, collisions and coagulations should result in just this order of things.

The first members of living species, including the human, were generated from mud or slime; simply appearing by chance. Procreation somehow becomes natural and Democritus compared the sexual act to epilepsy -- an atomic collision which is itself a kind of disease. Many biological opinions are attributed to Democritus, in the realm of embryology, generation, etc. He did not think death instantaneous, since nails and hair continue to grow; the corpse is still alive and perceptive, though heading toward complete dissolution. Democritus' views on sense perception are of interest here, although Aristotle

wrote, "Democritus and the majority of natural philosophers who discuss perception are guilty of a great absurdity; for they represent all perception as being by touch." (*De Sensu*, 442a29)

Aristotle's criticism is of a position that seems inevitable for atomism. We recall that the atoms, the only things that really are, are imperceptible because of the smallness of their size. Now, if what really is cannot be grasped by the senses, the objects of perception must be appearances, in the sense of not being wholly real. If perception is of what is not wholly real, this does not mean that it is false; what we must grasp is that the wholes we perceive are compounds of atoms, but this is something we grasp by understanding. To know about the atomic structure of macrocosmic things is knowledge and to speak of perceptible things in terms of atoms is to speak the truth. The atomistic explanation of color, tastes, hot and cold entails reducing them to atoms and their movements; thus, though we retain these words out of custom, we will not say that they really are.

Sweet exists by convention, bitter by convention, color by convention; atoms and Void (alone) exist in reality . . . We know nothing accurately in reality, but (only) as it changes according to the bodily condition, and the constitution of those things that flow upon (the body) and impinge upon it. (Fr. 9; Freeman)

If this should lead us to look with condescension on the senses, one fragment gives their reply. "Miserable Mind, you get your evidence from us, and do you try to overthrow us? The overthrow will be your downfall." (Fr. 125) Thus, perception is not dismissed as totally unreal; if we did not perceive the things we do, we could never go on to the knowledge that they are composed of atoms and the void. Our perceptions consist of an impingement of things on our senses, much as Empedocles had explained them.

They attributed sight to certain images, of the same shape as the object, which were continually streaming off from the objects of sight and impinging on the eye. This was the view of the school of Leucippus and Democritus (Alexander)

The same explanation apparently served for thought, since the soul is itself composed of atoms. "Democritus says that the spherical is the most mobile of shapes; and such is mind and fire." (*De Anima*, 1,2,405 all) These atoms are subtle, easily moved, and, if quite corporeal, at the extreme of fineness and thinness. Sensation and thought seem to involve the movement of the atoms of

percipient and thinker in response to the impingement from without. The soul is spread throughout the body, although what we call mind is a concentration of atoms in the bosom. Breathing has as its result the keeping of the soul atoms in the body; death is the escape of the soul atoms -- an escape which is gradual, not instantaneous.

Democritus does not so much account for the divine as for man's belief in the gods. Fear and awe of natural phenomena, such as under, storms, eclipses, and gratitude for unlooked for goods, tend underpin belief in the gods; but Democritus will allow for no inrporeal reality. He seems to accept the reality of visions, both those in dreams and when awake, and attaches divinity to these, ugh they are not immortal. From the point of view of his natural and psychological opinions, there is no role for the gods to play. That is why it would appear that Democritus is simply trying to account for belief of his fellows. Visions are produced by atoms present in the air; this being so, animals are as much struck by what men call the divine as are men themselves. In his ethical remarks, however, Democritus sometimes relies on this belief in gods.

We have already mentioned that the bulk of the fragments of Democritus that have come down to us is from his ethical writings. The Democritean ethics is ordered to the acquisition of happiness. "Happiness does not dwell in flocks or in gold. The soul is the dwelling place of the (good and evil) genius." (Fr. 171, Freeman) The ethical views of Democritus seem to bear little relation to his atomism; there is no mention of atoms in the ethical fragments which have come down to us. We have seen that, for Democritus, the soul pervades the body, that it is, so to say, a body within a body, since the soul atoms are corporeal; mind however is centered in the breast. This division in the soul, between what we may call its rational and irrational parts, finds its analogue in the distinction of soul from body. "It is right that men should value the soul rather than the body; for the perfection of soul corrects the inferiority of the body, but physical strength without intelligence does nothing to improve the mind." (Fr. 187, Freeman) Pleasure is to be surmounted, or at least pleasure in mortal things, since pleasure is a criterion.

The criterion of the advantageous and the disadvantageous is enjoyment and lack of enjoyment. (Fr. 188, Freeman) The best way for a man to lead his life is to have been as cheerful as possible and to have suffered as little as possible. This could happen if one did not seek one's pleasures in mortal things. (Fr. 189, Freeman)

What is called for, then, is a discrimination between pleasures, for the works of justice bring cheerfulness.

The cheerful man, who is impelled towards works that are just and lawful, rejoices by day and by night, and is strong and free from care. But the man who neglects justice, and does not do what he ought, finds all such things disagreeable when he remembers any of them, and he is afraid and torments himself. (Fr.174, Freeman)

A man should be concerned with society and with public affairs.

One must give the highest importance to affairs of the State, that it may be well run; one must not pursue quarrels contrary to right, nor acquire a power contrary to the common good. The well-run State is the greatest protection, and contains all in itself; when this is safe, all is safe; when this is destroyed, all is destroyed. (Fr. 252, Freeman)

Justice is what should be done; injustice its opposite. (Fr. 256) This is not very enlightening, of course; nor is concern with the common good urged for altruistic reasons.

To good men, it is not advantageous that they should neglect their own affairs for other things; for their private affairs suffer. But if a man neglects public affairs, he is ill spoken of, even if he steals nothing and does no wrong. And if he is negligent and does wrong, he is liable not only to be ill-spoken of but also to suffer bodily harm. To make mistakes is inevitable, but men find it hard to forgive. (Fr. 253; Freeman)

Nevertheless, it is not the censure of others we should fear, but rather our own.

One must not respect the opinion of other men more than one's own; nor must one be more ready to do wrong if no one will know than if all will know. One must respect one's own opinion most, and this must stand as the law of one's soul, preventing one from doing anything improper. (Fr. 264, Freeman)

Democritus has a lively sense of the role chance plays in human affairs.

"Courage is the beginning of action, but Fortune is the arbiter of the goal." (Fr. 269, Freeman) But only fools are shaped by the gifts of chance (Fr. 197), and more men become good through effort than by nature. (Fr. 242) An important aspect of Democritus' ethical pronouncements is his injunction that we should

limit our desires, strive for the possible. (Fr. 285, 286) The key to true pleasure is moderation. (Fr. 211)

Cheerfulness is created for men through moderation of enjoyment and harmoniousness of life. Things that are in excess or lacking are apt to change and cause great disturbance in the soul. Souls which are stirred by great divergences are neither stable nor cheerful. Therefore one must keep one's mind on what is attainable, and be content with what one has, paying little heed to things envied and admired, and not dwelling on them in one's mind. Rather must you consider the lives of those in distress, reflecting on their intense sufferings, in order that your own possessions and condition may seem great and enviable, and you may, by ceasing to desire more, cease to suffer in your soul. For he who admires those who have, and who are called happy by other mortals, and who dwells on them in his mind every hour, is constantly impelled to undertake something new and to run the risk, through his desire, of doing something irretrievable among those things which the laws prohibit. Hence one must not seek the latter, but must be content with the former, comparing one's own life with that of those in worse cases, and must consider oneself fortunate, reflecting on their sufferings, in being so much better off than they. If you keep to this way of thinking, you will live more serenely, and will expel those not-negligible curses in life, envy, jealousy and spite. (Fr. 191, Freeman)

While in no way in conflict with his natural doctrine, the ethical pronouncements of Democritus are relatively independent of atomism and seem a continuation of the Greek concern with moderation, responsibility and justice. The similarity between several of the moral dicta of Democritus and those of Heraclitus is often pointed out. If there is a more or less distinctive note, it is that the popular religion is one of fear and that moral maturity can enable one to surmount such fears. "People are fools who hate life and yet wish to live through fear of Hades." (Fr. 199, Freeman) The fact is that death is the end, and tales of an afterlife are the products of imagination.

Some men, not knowing about the dissolution of mortal nature, but acting on the knowledge of the suffering in life, afflict the period of life with anxieties and fears, inventing false tales about the period after the end of life. (Fr. 297, Freeman)

Democritean ethics promises no great solutions. Contentment, moderation, restraint -- these are the keys. When things go badly, consider that there are others worse off, but do not take pleasure in their misfortune. "Those to whom

their neighbors' misfortunes give pleasure do not understand that the blows of fate are common to all; and also they lack cause for personal joy." (Fr. 293, Freeman)

With atomism, the reaction to Parmenides' way of truth, which nonetheless accepts its logic, has gone about as far as it can go. Being can neither come to be, nor can it cease to be. Empedocles and Anaxagoras agree but see no reason why a multiplicity of beings is impossible; many beings which do not come to be, do not perish. The atomists add the void or not-being, speak of an infinity of like ultimate beings which differ only in shape and size. What all these reactions have in common is the view that the things we customarily call beings are really not such, largely because they undeniably come to be and pass away. This calls for a new look at the validity of sense perception, and while none of these reactions rejects sense perception out of hand, there is the sharpening of a distinction between appearance and reality. Anaxagoras stands out from the others because of his attempt to assign a role to *Nous* or Mind, an entity he manages to describe in such a way that he more nearly achieves success in delineating the incorporeal. Empedocles had introduced, over and above the four elements, Love and Strife, but they seem clearly corporeal principles. Atomism has no room at all for anything other than atoms and empty space; whatever regularities and apparent laws we may think to discover in our world, the world has come about by chance. Once more, these are three reactions to the austere Parmenidean pronouncement that being does not become; no direct questioning of the pronouncement is involved in these reactions, but rather all three are ways of accommodating the world we perceive to the Parmenidean laws. Only in the next period of ancient philosophy, the Golden Age of Greek philosophy, do we encounter frontal assaults on the way of truth.

A. Diogenes of Apollonia

Diogenes of Apollonia is said to have flourished in the second half of the fifth century, that is, about 440-430 B.C. He is consequently considered to be one of the last of the natural philosophers; his doctrine is often called an eclectic one and his fragments indicate his indebtedness to his predecessors. We know from Simplicius (see Kirk and Raven, n. 600, p. 428) that Diogenes wrote several works, but at the time of Simplicius only one was extant, *On Nature*. Here, as so often before, it is on Simplicius that we principally depend for our knowledge of the doctrine.

"In beginning any discourse, it seems to me that one should make one's starting point something indisputable, and one's expression simple and dignified." (Fr. 1; Burnet) In obvious conformity with this methodological remark, Diogenes writes:

My view is, to sum it all up, that all things are differentiations of the same thing, and are the same thing. And this is obvious; for, if the things which are now in this world -- earth and water and fire and air, and other things which we see existing in this world -- if any one of these things, I say, were different from any other, different, that is, by having a substance peculiar to itself; and if it were not the same thing that is often changed and differentiated, then things could not in any way mix with one another, nor could they do one another good or harm. Neither could a plant grow out of the earth, nor any animal nor anything else come into being unless things were composed in such a way as to be the same. But all these things arise from the same thing; they are differentiated and take different forms at different times, and return again to the same thing. (Fr. 2; Burnet)

Diogenes is often linked with Anaximenes because, like the earlier philosopher, Diogenes makes the common nature, air. Air is life and intelligence, which men and animals draw in by breathing; deprived of air, they necessarily die. (Fr. 4) Air is divine and pervades the universe; indeed, it has the attributes Anaxagoras assigned to *Nous*.

And my view is, that that which has intelligence is what men call air, and that all things have their course steered by it, and that it has power over all things. For this very thing I hold to be a god, and to reach everywhere, and to dispose everything, and to be in everything; and there is not anything which does not partake in it. Yet no single thing partakes in it just in the same way as another; but there are many modes both of air and of intelligence. (Fr. 5; Burnet)

It is by transformations described in terms of rarefaction and condensation that many things are produced from the divine air. The order among things so produced can only be ascribed to intelligence; this is the best of all possible worlds. (Fr. 3)

The cosmology of Diogenes is one of the last positive efforts of pre-Socratic natural philosophy and it manages to combine elements from all preceding attempts, from the earliest Ionian to the doctrines consequent upon Parmenides' way of truth. While Diogenes does little to advance natural

philosophy, his fragments indicate to us the tradition in terms of which the individual talent sought to develop. It is that tradition which not even Parmenides had been able to disrupt, which is called radically into question by the Sophists.

Chapter VII

The Sophists

If we think of a progression from Homer to Hesiod in terms of a passage from concern with human action, its sanctions and consequences, to concern with the role of the gods, not only relative to man but also with respect to the constitution of our world, the emergence of the Ionian natural philosophers can be seen as a new concern with the world itself and with man as a part of that world rather than as moral agent. To be sure, Ionian natural philosophy retains its links with the myths which form the background of the epics as well, but the beginnings of philosophy appear as a deliberate movement away from an anthropomorphic interpretation of the universe and a consequent decrease in interest in man as moral agent. There is no complete break, of course. Xenophanes and Heraclitus exhibit a deep concern with conduct; the Pythagoreans express best of all the twofold concern with a scientific explanation of the world and with man's achievement of his moral possibilities. This is present as well in Empedocles and in Democritus. We cannot say, therefore, that concern with man begins with the Sophists. It is rather the quality and causes of this concern that make the Sophists a breed apart and allow them to play an important if transitional role in the development of Greek thought.

It is not a negligible fact that the Sophists do not figure in the sketch of previous philosophy Aristotle gives at the outset of his *Metaphysics*. In that sketch Aristotle is intent on pointing out previous efforts to arrive at the various principles of explanation of the things that are. The absence of the Sophists from this historical summary would lead to the conclusion that their interests were quite different from those of others Aristotle finds important for his purposes. Now if philosophers we have considered previously were concerned with man -- with human existence, with man's place in the universe -- this was not their only concern. Is it perhaps because the Sophist was interested only in the human that he differs from his predecessors? The truth is that the Sophists can best be assessed in terms of the context which favored their flourishing.

The best analogue for the Sophists is Xenophanes. We recall that Xenophanes was a wanderer, that he recited his own verse; thus setting himself up in opposition to the declaimers of Homer, particularly in the content of his verse. He opposes the Homeric view of the gods, he ridicules the emphasis on sports, he visits city after city, seeking to make an impact -- to direct and educate his

fellows. In this he differs from the Milesians, who can be related as master and pupil only by reading back into the past later forms of philosophical education. There are, of course, many legends about the political prowess of the Milesians, but the fact of the matter is that their concern was with knowledge, with research, and not primarily with teaching, and certainly not with the changing of men's lives. Xenophanes is primarily concerned with education, with persuading his fellows; he is aware of the research of the Milesians, and makes use of its results for his own purposes, but he is not himself directly engaged in the same endeavor. If Homer and Hesiod had become the teachers of Greek youth, Xenophanes aspires to supplant them in that very role. It is in this light that we can best appreciate his criticisms of the immorality apparently set up for emulation by the epic poets. With the Pythagoreans, the blending of the efforts of the Milesians and Xenophanes results in the formation of a philosophical school, but it is restricted to members of the order -- to the initiates, who must by asceticism and purgation free themselves from the cycle of birth. The Pythagoreans were said to have been politically active in southern Italy, but their efforts ended in disaster, and their cult seems never to have achieved any political success, so as to have general educative significance. It was efficacious rather as a rule of conduct for a select few living in a community, which cut them off from the general run of mankind. The Sophists, it seems, must be looked upon primarily in terms of the educative role they strive to play with respect to the multitude; like Xenophanes, they are wanderers. They go from town to town and offer to teach what is necessary for the citizen of the city state. They do this for a fee; some of them become quite wealthy in the process. In the words of Jaeger, they literally live by their wits. This historical milieu which favored the flourishing of the Sophists was the Greek *polis*, the city state.

The Greek *polis*, from which our word "politics" comes, is an entity difficult for us to imagine. It was an autonomous unit, composed of citizens who were primarily farmers. Their houses were in town, surrounding the acropolis -- the upper city, a citadel indispensable because of the many wars -- beneath which would be the market place. The members of the polis were connected by the land from which they all came, by the gods they worshipped in common, by economic and geographic boundaries, by the political life they shared. These groupings of people were small. Plato says the ideal city should have 5,000 citizens, i.e., free males of a given age; and Aristotle feels that the citizens of a *polis* should all know one another by sight. In such a city it was possible for each citizen to have his say about common affairs, and it became a problem to prevent one or a few citizens from acquiring power over the others. The compactness and community of purpose of such a group fostered a type of

government in which everyone participated -- in which each man could speak up and be heard, in which matters of *polis* policy were objects of street corner discussion by men whose opinions could matter in a direct and efficacious way. There would be leaders, of course, and leadership in such a community required a number of skills -- those of rhetoric and jurisprudence which do not always come naturally. It was to meet this felt need for political skills, for *arete* or excellence, that the Sophists arose. They offered to teach those who came to them everything necessary for human excellence, for leadership in the *polis*.⁽¹⁹⁾ Their activity was most pronounced at the end of the fifth and the beginning of the fourth centuries B.C., that is, just before and during the life of Socrates.

The purpose of the Sophist was -- in the words Plato attributes to Protagoras in his dialogue of that name -- to educate men. This, we will see, is one thing the Sophists had in common. Theirs is a practical, a pedagogical, activity, and this they share even though we would be hard put to enumerate characteristically sophist teachings. It does not seem profitable to think of them as a philosophical school, complete with common points of doctrine; this is one reason why Aristotle would not have included them in his discussions of his predecessors. The Sophist is a professional teacher and proud of it, often overly proud; he will teach, for a fee and over a space of perhaps four years, everything a man must know in order to achieve political success. What he will teach is summed up in the word *arete*, which is often translated virtue; and this led to the socratic criticism, in the form of a question: can one man teach virtue to another? The socratic critique, presented immortally in the dialogues of Plato, has done more than anything else to make the term "sophist" one of abuse, of denigration. It was not always so.

As used by earlier writers, the term "sophistes" meant an expert, one well-versed in a particular craft; poets used it to describe the poet and the musician; Herodotus to describe seers. Plato will use it to describe the creator of the world. Notice how Cornford translates the passage. "Tell me, do you think there could be no such craftsman at all, or that there might be someone who could create all these things In one sense, though not in another?" (*Republic*, 596D) The Sophist or wise man was one who knew the principles of an art or craft; later we will see Aristotle begin from this humble notion when he wants to depict the wisdom which is First Philosophy. Moreover, Aristotle will follow early usage and speak of the seven sages as Sophists. The word early has the connotation of clever or shrewd and, we might surmise, through sarcasm comes to be a word of abuse in certain contexts. The use of the term to designate the teachers we are about to consider would not in itself involve the censure which

we rather easily associate with it; that association -- the idea that there is something reprehensible in the idea of going from town to town offering to teach wisdom for a price -- is made once and for all by Socrates and his followers. From then on, the word becomes largely unsalvageable except for purposes of criticism. By turning now to individual Sophists, we will be able to learn to what degree the judgment of Socrates was well-founded and to what degree it may have been exaggerated.

A. Protagoras of Abdera

We have it from Plato that Protagoras was a native of Abdera. He lived in the latter half of the fifth century B.C. although his precise dates are difficult to determine. Protagoras was said to have been taught by Persian Magi, though this is considered doubtful. He is also said to have studied under Democritus, an allegation which may be jointly based on their being natives of the same place and of probable influences of Democritus on the thought of Protagoras, rather than because the Sophist was actually a pupil of the great atomist. Protagoras is said to have begun his own traveling-teaching career when he was thirty and, in Plato's *Protagoras*, he is made to say that he was the first openly to declare that he taught for money. While his fees were high, it is said that Protagoras demanded payment only at the completion of the course and the disappointed student could pay only what he thought the instruction had been worth. Tradition has it that Protagoras amassed a huge fortune in his teaching career.

Protagoras visited Athens a number of times, perhaps three, and it is one of these visits that forms the setting of the Platonic dialogue, *Protagoras*. In Athens he became friends with Pericles, Gallias and Euripides; on a journey to Sicily he met Hippias. On his last visit to Athens, Protagoras was accused of impiety and all his books were burned.

It is no easy matter to determine what exactly Protagoras or any other Sophist wrote, although tradition attributes a great many titles to the Sophist from Abdera. Jaeger feels that their works did not survive because of their purpose, namely to influence the men of a given time; they were not interested in posterity or in a timeless wisdom, but wanted to persuade the hearers of the moment. The famous sentence of Protagoras, "Man is the measure of all things, of the things that are, that they are, and of the things that are not, that they are not," (Fr. 1; Freeman) was said, by Plato, to be the opening of a book entitled *On Truth*; but Sextus says it was the opening of a book entitled *Refutatory*

Arguments. Untersteiner⁽²⁰⁾ feels that *Antilogiae* is the title of one book of which the other titles indicate subdivisions. This book, he feels, (p. 10) dealt with four fundamental problems: the gods; being; laws; and arts. The work itself was divided into two books containing several sections, such as the ones entitled *On the Gods*, and *On Being*. Porphyry will claim that the latter is used extensively by Plato, who borrows its arguments. In this way, Untersteiner is able to accomodate most of the titles attributed to Protagoras, taking them to be parts of the four major sections of the *Antilogiae*. This, together with a work *On Truth*, would be the literary production of Protagoras. His oral teaching was the elaboration of themes discussed in these writing.

The work, or section, devoted to the gods is said to have begun with the following sentence. "About the gods, I am not able to know whether they exist or do not exist, nor what they are like in form, for the factors preventing knowledge are many, the obscurity of the subject, and the shortness of human life." (Fr. 4; Freeman) Is this simply a reiteration of earlier criticism, notably that of Xenophanes or Heraclitus, or is it more involved? Of Protagoras, Diogenes Laertius writes, "He was the first to maintain that in every experience there are two logoi in opposition to each other." (IX, 51) It is thought that the story to the effect that Protagoras had been instructed by the Persian Magi, considered agnostics, would have led him to see the difficulty of reconciling Greek and Oriental statements about the divine and to the further conclusion that only opinion is possible in this matter. This sentiment would seem to underlie the sentence quoted earlier, to the effect that man is the measure of all things, of those that are that they are, of those that are not, that they are not.

What does Protagoras mean by this statement? It could mean that whatever seems to me to be at any given moment is and what to me seems not to be, is not. That it may seem otherwise to me later, or to another now, indicates that no truth is possible, and that every judgment is opinion at best. The subjectivity implied may seem to apply only to sensible objects, so that a breeze will seem warm or cool to me depending on my condition, and may seem different to different individuals at the same time. There is no point in arguing whether it is *really* one or the other; we can only know how it appears to us. There is no need, however, to restrict the scope of the remark to sensible objects. The word Protagoras uses for things can mean moral judgments as well and, indeed, statements about the gods. As applied to physical things, the proposition can be taken to mean that things are when we perceive them and cease to be when we cease to perceive them. Man, then, need not be the individual man, but rather mankind. Plato apparently takes the statement to mean this, since he asks why

Protagoras did not say a pig or a baboon is the measure of all things, since they too have perception. The statement of Protagoras, then, comes to mean that things exist when they are perceived and that the perceptions of individuals are equally true.

To say that conflicting statements on the same thing are equally and simultaneously true is an assault on the principle of contradiction, and Aristotle, when he discusses that principle in the Fourth Book of his *Metaphysics*, refers to Protagoras several times. "We observe, next, that if all contradictories were true at the same time of the same thing, it is clear that all things would be one. For if anything may be affirmed or denied of everything (as those must maintain who say what Protagoras says), then the same thing would be a trireme, a wall, and a man. For, if someone should hold that a man is not a trireme, clearly he is not a trireme; then, if the contradictory be also true, he also is a trireme." (1007b18ff) If contradictory statements about the world are true, all things are one, all statements are both true and false including the statement about statements. The application of Protagoras' doctrine of equal truth to his own statement is first said to have been done by Democritus. Plato, too, made use of this device to dismiss Protagoras.

The net effect of the teaching of Protagoras would seem to be the calling into question of the possibility of fixed knowledge, and, consequently, the disparaging of the efforts of earlier philosophers. Protagoras cannot offer to give what he holds to be impossible and if he teaches at all it cannot be that he thinks some statements are more true than others but only more desirable -- in his opinion. What he can teach is the art of getting along in a society where conflicting opinions are inevitably maintained; he can teach the art of persuasion, the technique of leading others to accept one's own views. Somewhat less practically, he can teach how contradictory propositions can be shown to be equally true and equally false. Such argumentation relied heavily on word study and grammar, and on the analysis of ancient poets who were taken to be saying things of contemporary significance. In Plato's *Protagoras* there is a parody of such poetical exegesis. Protagoras wrote out rhetorical exercises which, in the opinion of Aristotle, involved not reasoning but plays on words and other eristic tricks to achieve their end.

We will close our discussion of Protagoras by posing a riddle to which we will have to return after having looked at a number of other Sophists. The doctrine of Protagoras, since Plato, has been taken to be a sceptical one, or nearly so; nonetheless, in the Platonic dialogue named for him, Protagoras is allowed to

tell a story which will justify his function and it is difficult not to feel the idealism in what he says. The story or myth speaks of how man, unlike the animals, was created naked and defenseless, and that Prometheus stole the arts of Athene and Hephaestus together with fire to compensate for man's natural lacks. These arts are not made known to all men but only to a few, and after their appearance there was still a lack of political wisdom which left men open to the attacks of animals for they had not yet grouped into cities. It was then that Zeus granted justice and reverence to men, not just to a few -- as with the arts -- but to all. The sign of this is that all men are held to profess them and are punished when they do not. Now surely, this would not be the case if men were incapable of acquiring virtue and political excellence. And, Protagoras argues, as a matter of fact, men do try to teach their sons excellence. This is what he himself can teach, and to do so is a very noble function, since it brings men to the perfection expected of them. (Protag. 320-328) The riddle of course is this: how can this high ideal of sophistic instruction be reconciled with such destructive doctrines as that of Protagoras which we have just examined. Surely it would be quite impossible if, in the *polis*, one statement were as true as another, one mode of conduct as justifiable as another. Moreover, if my opinion is no better than another's, for me to persuade him to see things my way is unwarranted despotism. This difficulty becomes more pronounced in Sophists other than Protagoras.

B. Gorgias of Leontini

Gorgias was a native of Leontini, in Sicily, and flourished in the latter part of the fifth century B.C. Gorgias came to Athens in 427 B.C. at the head of an embassy from his native city and won the Athenians to its cause over the pleas of the representative of Syracuse, Teisias, who is said to have been the author of the first textbook of rhetoric. His success was overwhelming, and he seems to have repeated it on a number of occasions, at Olympia, at Delphi about 420 B.C. where his reception was such that a statue of him in pure gold was erected, although he also is said to have paid for it himself -- an indication of his financial status. The Athenians once asked him to deliver the funeral oration for those who had died in battle, a rare honor for a non-Athenian. With these successes it is not surprising to learn that he was much in demand as a teacher of rhetoric. He knew Socrates, and Isocrates was his most notable Athenian pupil. As for his own background, Gorgias is said to have studied under Empedocles, his fellow Sicilian. Gorgias enjoyed a long life, and it is said that he lived over a hundred years.

Gorgias is said to have written a textbook of rhetoric but, if we can believe what Aristotle says in his own *Rhetoric*, these earlier efforts did not so much teach the rules of rhetoric as they gave sample speeches to be memorized. Most of the other writings of Gorgias, except that on nature, were speeches he had given -- an encomium on Helen, the funeral oration, the Olympian oration and a few others. Fragments from the orations mentioned have come down to us. It would seem that the work on nature, *On Not-being or on Nature*, is Gorgias' main claim to being numbered among the philosophers.

We want to set down in some detail the arguments of a significant section from that work which has been preserved for us in Sextus. The passage in question sets out to prove three propositions. (1) Nothing exists. (2) If anything exists, it is incomprehensible. (3) If it is comprehensible, it is incommunicable. (1) Nothing exists. The first proposition is proved by showing that neither being nor not-being can exist. Not-being cannot exist since if it existed it would have to be being, and that is clearly impossible. In showing that being cannot exist, Gorgias will show that it is not everlasting, nor created, nor both, nor is it one or many. Here is the argument which purports to show that being is not everlasting. "It cannot be everlasting; if it were, it would have no beginning, and therefore would be boundless; if it is boundless, then it has no position, for if it had position, it would be contained in something, and so it would no longer be boundless; for that which contains is greater than that which is contained, and nothing is greater than the boundless. It cannot be contained by itself, for then the thing containing and the thing contained would be the same, and Being would become two things -- both position and body -- which is absurd. Hence if Being is everlasting, it is boundless; if boundless it has not position ('is nowhere'); if without position, it does not exist." (Fr. 3; Freeman) One notes the identification of extension in time and space which gives the argument its specious cogency. Gorgias argues that being cannot be created either, since it must come either from Being or not-being, both of which are impossible. If being can be neither everlasting nor created, it cannot be both and being does not exist. In showing that being cannot be one, Gorgias points up the major flaw in the Parmenidean sphere of being. "Being cannot be One, because, if it exists, it has size, and is therefore infinitely divisible; at least it is threefold, having length, breadth and depth." (Fr. 3; Freeman) If being cannot be one, it cannot be many either, Gorgias argues, for the many is a plurality of ones and it is impossible for being to be one. Being cannot exist; not-being does not exist; a mixture of being and not-being cannot exist. Nothing exists. Q.E.D.

(2) If anything exists, it is incomprehensible.

If the concepts of the mind are not realities, reality cannot be thought; if the thing thought is white, then white is thought about; if the thing thought is non-existence, then non-existence is thought about; this is equivalent to saying that 'existence, reality, is not thought about, cannot be thought.' Many things thought about are not realities: we can conceive of a chariot running on the sea, or a winged man. Also, since things seen are the objects of sight, and things heard are the objects of hearing, and we accept as real things seen without their being heard, and vice versa; so we would have to accept things thought without their being seen or heard; but this would mean believing in things like the chariot racing on the sea. Therefore reality is not the object of thought, and cannot be comprehended by it. Pure mind, as opposed to sense perception, or even as an equally valid criterion, is a myth. (Fr. 3; Freeman)

Gorgias here embraces a radical empiricism; we accept as real what is seen but not heard and vice versa, but thought cannot be set up over against perception nor as an equal criterion of reality. Why? Because many thoughts are of non-existent things, imagined entities, which cannot be corroborated by perception. If what does not exist can be thought about, reality or existence cannot be thought, according to Gorgias. If we should point out that thought is often of existents, Gorgias would doubtless reply that we know this is the case because we perceive the thing to be. He begins by saying that if the thing thought is white, then white is thought about. But white is grasped by sight, not thought. Thought as something above perception is a myth; it has no object: being cannot be thought.

(3) If anything is comprehensible, it is incommunicable.

The things which exist are perceptibles; the objects of sight are apprehended by sight, the objects of hearing by hearing, and there is no interchange; so that these sense perceptions cannot communicate with one another. Further, that with which we communicate is speech, and speech is not the same thing as the things that exist, the perceptibles; so that we communicate not the things which exist, but only speech; just as that which is seen cannot become that which is heard, so our speech cannot be equated with that which exists, since it is outside us. Further, speech is composed from the percepts which we receive from without, that is, from perceptibles; so that it is not speech which communicates perceptibles, but perceptibles which create speech. Further, speech can never exactly represent perceptibles, since it is different from them, and perceptibles are apprehended each by the one kind of organ, speech by another. Hence, since the objects of sight cannot be presented to any other

organ but sight, and the different sense-organs cannot give their information to one another, similarly speech cannot give any information about perceptibles. (Fr. 3; Freeman)

The argument of Gorgias, then, is that nothing exists; if it did it couldn't be known; and if it could be known it couldn't be communicated. Are we to take this as a serious position? When one considers such Gorgian efforts as the defense of Helen, it seems plausible that the tripartite argument of Gorgias is but a tour de force, one more exhibition of his willingness to defend any position, however impossible it may seem. Strangely enough, the argument was seldom treated as a joke in antiquity. Isocrates, the pupil of Gorgias, seems to believe that his master seriously maintained that nothing exists. If taken seriously, the argument refers us to Parmenides and we would then see Gorgias as a bifurcated Zeno arguing, in effect, a pox on both your houses. Lycophron, a pupil of Gorgias, is referred to by Aristotle in the *Physics*.

Even the more recent of the ancient thinkers were in a potter's lot the same thing should turn out in their hands both one and many. So many, like Lycophron, were led to omit 'is', others to change the mode of expression and say 'the man has been whitened' instead of 'is white', and 'walks' instead of 'is walking', for fear that if they added the word 'is' they should be making the one to be many -- as if 'one' and 'being' were always used in one and the same sense. (1,2, 185b25)

Indeed, Aristotle is said to have written a study of Gorgias himself. That Gorgias should have been taken seriously is of course no argument that he took himself seriously; the lengthy passage we have just seen may very well have been, in the intention of the Sophist, a display of his prowess and his willingness to discourse on any subject whatsoever. An Aristotle could turn such a display to serious purpose by an analysis of the mode of, argumentation, showing how the eristic attained his objective of dazzling his audience. Once more, we shall have to ask ourselves what useful purpose someone like Gorgias thought himself to be serving. Untersteiner has an elaborate and unconvincing argument that Gorgias was bent on showing the tragic character of human thought. The reader is rather struck by a cocky ebullience and finds it difficult to see the Platonic depictions of the Sophist as unjust caricature.

C. Prodicus of Ceos

Prodicus, a native of Ceos, an Aegean island, lived at the end of the fifth century, and came frequently to Athens on official business of his island. Socrates himself is said to have paid to listen to Prodicus on one such occasion, although there seems to be a note of irony in the admission that he could only afford the one drachma course. Euripides, Isocrates, Thrasymachus and Xenophon were also said to have heard Prodicus. Prodicus is often mentioned by Plato and the references, as in Aristotle, are to Prodicus' concern with correct terminology. He is said to have written on this subject and to have produced a book entitled either *On Nature* or *On the Nature of Man*. Like the other Sophists, he also composed exercises on various themes, to illustrate his own method and for the instruction of pupils. Prodicus said that the Sophist is "on the borderline between the philosopher and the statesman." (Fr. 6; Freeman) If his own reputation as one who made a great point of terminology, of distinctions in words, etymologies and so forth can be taken to indicate what he thought his own metier was, it can be said that the philosophers did not remain uninfluenced by his attempts. If Plato allows Socrates in the dialogues to treat Prodicus quite ironically, it is nevertheless true that there is an acceptance of Prodicus' method of determining the meanings of words. Aristotle, certainly, would not be insensitive to such attempts at determining the meanings of words and the relationships between various meanings, the establishment of synonyms, etc.

Prodicus' explanation of how the gods arose is of interest. Those things which are necessary and beneficial to man are transformed into divinities. Thus, bread becomes Demeter, wine Dionysius, water Poseidon, Hephaestus, and so on. This approach to the official gods was continued by later thinkers and led to charges of atheism. Prodicus is depicted as in rivalry with Gorgias, and it seems likely that their competition for students would lead to sharpness.

D. Some Other Sophists

Thrasymachus of Chalcedon, also prominent at the end of the fifth century, was often in Athens. He is said to have written much and among his works were a Great Text Book and Subjects for Oratory. He is known primarily as a rhetorician and the fashioner, according to Theophrastus, of the middle diction, a style between the austere and the plain. We are told that he devised methods of eliciting pity and anger in audiences. Plato thought little of this method, and in the *Laws* outlaws it; Aristotle is thought to have profited from the work of

Thrasymachus in writing his own *Rhetoric*. Thrasymachus is most famous for the view of justice that he is made to propose and defend in Plato's *Republic*. According to that dialogue, Thrasymachus held that justice is nothing else than the advantage of the strong.

Hippias of Elis is mentioned in several Platonic dialogues, the *Protagoras* and the *Apology*; two are named for him and he figures in the seventh epistle of Plato. Although writings are attributed to Hippias, nothing has come down to us. Hippias appears to have been a man of universal ability as a craftsman, for there is a story that he once journeyed to Olympia entirely clothed in things he had made himself. Moreover, in mathematics, he was interested in squaring the circle. His views on the relationship between nature, virtue and law will be discussed in our summary statement on the Sophists.

Antiphon the Sophist, so called to distinguish him from others of the same name, is believed to have been a native of Athens. He wrote a book entitled *Truth* of which we possess sizeable portions; indeed, we have more of the writings of Antiphon than of any other Sophist. It seems that Antiphon accepted the Parmenidean view that all things are really one, although they appear many to the senses. He, too, was interested in squaring the circle, and is mentioned by Aristotle in this connection. The large fragments we possess of Antiphon deal with justice, indicating Antiphon's view that there is an opposition between the laws of the state and the laws of nature, a theme to which we shall come in a moment.

In conclusion, we may mention the *Dissoi logoi*, or *Twofold Arguments*, work which is thought to epitomize the method common to the Sophists. Thus, the first argument concerns the good and the bad: some say they are different, others that they are the same.

Arguments are adduced to prove now one side of the opposition, then the other. Next, arguments are given to show that the honorable and dishonorable are different; then that they are the same. The same thing is done with just and unjust, and true and false. Fifthly, some maintain that the mad and the sane, the wise and ignorant say and do the same things. This is argued pro and con. The sixth twofold argument has to do with whether knowledge and virtue can be taught; the seventh with whether offices should be awarded by lot; the eighth with whether it pertains to the same man to be politician, speaker and scientist. The manuscript concludes with a discussion of memory, its utility for knowledge and for life, with a number of rules for memorizing. This anonymous

work is taken to indicate the effect of the Sophists. One should train himself to argue either side of conflicting views; it is the implication that, as Protagoras explicitly said, the one side is as defensible as the other, that they are equally true, that Socrates, Plato and Aristotle criticize and which led to making the term "sophist" an abusive one.

E. Concluding Summary

The role of the Sophist was the practical one of training men for the life of the *polis* and if, beginning with Protagoras, there is the implication that one opinion is as good as another and that they could teach the method whereby one could make his own views most persuasive before the assembly, there is as well the view that underlying the opinions of men and the varying laws which express these opinions is a common nature which binds all men together whether they are Greek living in different city-states, or Greek and barbarian. True justice is to be had when men live in accord with their nature, not with the laws of the assemblies. This is expressed by the Athenian sophist, Antiphon.

Justice, then, is not to transgress that which is the law of the city in which one is a citizen. A man, therefore, can best conduct himself in harmony with justice if when in the company of witnesses he upholds the laws, and when alone without witnesses he upholds the edicts of nature. For the edicts of laws are imposed artificially, but those of nature are compulsory. And the edicts of the laws are arrived at by consent, not by natural growth, whereas those of nature are not a matter of consent. So, if the man who transgresses the legal code evades those who have agreed to these edicts, he avoids both disgrace and penalty; otherwise not. But if a man violates against possibility any of the laws which are implanted in nature, even if he evades all men's detection, the ill is no less, and even if all see, it is no greater. For he is not hurt on account of an opinion, but because of truth. The examination of these things is in general for this reason, that the majority of just acts according to law are prescribed contrary to nature. (Fr. 44; Freeman)

Such a view is at once constructive and destructive. It is destructive, because it inculcates a cynical attitude towards the life of the *polis*; however, the view that there are edicts of nature, of man's nature, which bind everywhere and always whatever their relation to human laws, should have led to the enunciation of what these laws were and an attempt to bring the laws men make into accord with them. This does not seem to be the direction in which the influence of the Sophists went. Rather, as we have seen in Thrasymachus, nature's law was

sometimes interpreted as meaning that the strong should rule, that men are not equal, as the structure of the *polis* implied; for in the *polis* each citizen had a voice in the government of the community even when military affairs were to be decided. Pericles, the great leader of Athens was subject, even during the trying days of the Peloponnesian war, to the judgment of his fellow citizens. The oligarchs made use of the teaching of the sophists about the law of nature to argue against Athenian democracy. As Cleon says in the debate about the fate of the citizens of Mitylene who had revolted against Athens, "This debate only confirms me in my belief that a democracy cannot rule an empire." Now Plato too will be of the opinion that men are not by nature equipped to make the decisions necessary to rule themselves or others, but he thinks the sophists' teaching is simply the formulation of the method by which the *polis* was in fact run. What the sophist did was teach a method whereby one could succeed in a popular democracy.

It is the Sophists' emphasis on method that gives them importance in the development of philosophy. Indeed, as Jaeger argues in his *Paideia* (Vol. I. pp. 313-4), they can be credited with the elaboration of what was to be called the trivium of the liberal arts: grammar, rhetoric, dialectic, since these had never been separate studies before their time. All later writers on these subjects, and particularly Plato and Aristotle, can be thought of as profiting from the attempts of the Sophists. The remnants of what can be called the doctrines of the Sophists strike us rather as exercises in a method than presentation of held beliefs. The *Dissoi logoi*, again, indicate that one can argue either side of a matter; in Protagoras we have the view stated that one opinion is as good as its opposite; but even if this be not accepted, the method involved in arguing either side is something which comes to the fore in these exercises. If the rhetoric of the Sophists tended to be sample speeches rather than principles, as Aristotle complained; if their dialectic was largely fallacious and led to the use of "sophistic" to describe an argument that was only apparently valid, they nonetheless paved the way for the fuller development of these instruments of reasoning in the hands of Plato, but principally in those of Aristotle, the founder of logic.

The paucity of fragments of the writings of the Sophists makes it difficult to give anything like an accurate assessment of their efforts; equally, nothing in the fragments we do possess gives us a reason for thinking that Plato and Aristotle were unjust in their assessment of the Sophists. If we accept their estimate there is an element of despair in the Sophists' activity, for they pride themselves on their ability to manipulate words, to sway their hearers; and yet this is done to

no justifiable purpose. What is lacking in the Sophist, even if his logic were unimpeachable, is the moral dimension of the use of dialectic. Perhaps it is not too great a simplification to say that Plato was primarily concerned with this moral lack while Aristotle was primarily interested in devising a valid logic to supplant the eristic of the Sophists.

{19} Of course, and ironically, only the wealthy could benefit from their instruction. See Burnet, p. 109.

{20} Mario Untersteiner, *The Sophists* (New York: Philosophical Library, 1954).

Part II: The Classical Period

Chapter I

Socrates

A. His Life

Socrates was born about 470 B.C. and lived for seventy years until his execution in 399. His life thus covers the time when Athens rose to political prominence and glory and then moved tragically towards the fatal Peloponnesian war. In his lifetime Aeschylus, Sophocles, Euripides and Herodotus were active; Thucydides was about Socrates' age and Aristophanes, somewhat younger than Socrates, was to give Socrates one of the forms of immortality he enjoys in the *Clouds*. As we have seen, it is not unlikely that Socrates met Parmenides and Zeno when they visited Athens and he lived at the time when some of the more important Sophists were flourishing. Protagoras was older than Socrates but nonetheless alive; Gorgias, Hippias and Prodicus were active. Socrates was an Athenian in an almost exaggerated sense of the term; tradition has it that he left the city only infrequently, mainly on military service. Plato presents him as uninterested in the countryside and longing for the city streets and the possibility of dialogue with the citizens, on the rare occasion when he was in the country. Nature had nothing to teach him, he felt, but with men and through conversation, knowledge became a possibility. But if Socrates' attachment to Athens was partly a native gregariousness, there was also a deep sense of patriotism. It was this concern for his city that can be seen to underlie the Socratic reaction to a time when the political horizon was changing rapidly, when empire was the goal and Athens its champion; this is the time when the Sophists, too, offered training for life in the city, for the future statesman. It may be said that Socrates' efforts as well as the Sophists were directed mainly at those who would one day assume political responsibility; indeed, we find that Socrates was often called a Sophist in antiquity, and we may wonder how he differs from them. The difference will be clearer to us if we recall that the sophistic position can be summarized in the denial that there is any truth apart from the convictions of men, that this is quite obvious in the matter of moral principles and that, consequently, the search for the truth is the search for a chimera and can hardly be the object of intellectual effort. The point we are making is that Socrates stands at a critical juncture of Western thought, when the previous efforts of Philosophers are made an object of ridicule by paid

teachers who call the very validity of knowledge into question. Socrates emerges not as one who furthered the destructive purpose of the Sophists but rather as one who set philosophy on a route which led to its Golden Age in the fourth century B.C. This role of Socrates is indicated in the time-honored designation by historians of all his predecessors as presocratics; by the same token, Plato and Aristotle can be called Socratics. Such designations, as well as the few timid affirmations recorded above, would indicate that we have determinate knowledge about Socrates as a person and about the content of his teaching. That we do not constitute what is called with simple eloquence the Socratic Problem.

Socrates wrote nothing. He was, however, written about a good deal. Seemingly then we are in a position with respect to Socrates similar to that we were in with respect to his predecessors. What we must do is glean from the writings about him what appear to be quotations or near paraphrases. That no such simple procedure is possible becomes clear when we consider our sources. On the face of it, they are four: the dialogues of Plato; various writings of Xenophon, particularly the *Memorabilia*; Aristophanes; and a few remarks by Aristotle. With respect to Aristotle, it must be pointed out that if our knowledge of Socrates were dependent on what Aristotle wrote, we would know extremely little about the man. The comedy of Aristophanes, although as a successful lampoon it must have borne some relation to the historical Socrates, could hardly be taken as the principal source of our knowledge. This leaves the more extensive accounts of Plato and Xenophon. Xenophon certainly knew Socrates, but there was a great difference in their ages and Xenophon was absent from Athens during the last three years of Socrates' life. Moreover, Xenophon is primarily concerned with defending Socrates against charges similar to those brought against him in the trial which led to his execution. And if Xenophon entitles one relevant work the *Memorabilia*, it soon becomes clear that he is not recording personal recollections of Socrates but borrowing those of others. Plato, who probably knew Socrates over a long period, was not an intimate of his and the style of the dialogues produces a Socrates who looks very much like a literary creation. It is unlikely, to say the least, that the extensive exchanges in the dialogues are verbatim reports of the philosophical activity of Socrates. Thus, if we think of Plato's method as artistic, as analogous to that of Aristophanes -- if, indeed, we suspect that the Socrates of the dialogues is simply a convenient vehicle for Plato's personal thought -- we end by calling into question the basic sources for any knowledge of Socrates and what he taught. The skeleton of our certitudes then becomes what a scholar has recently described.

We can only point with any real conviction to the facts that Socrates lived; that he was called Socrates; that the name of his father was Sophroniscus while that of his mother was Phaenarete; that he belonged to the *deme* Alopece; that he was an Athenian citizen: that he probably participated in some military campaigns; that he was connected with the trial of the generals in 406; that he was tried and condemned to death; and that he died in 399.^[21]

To add that he had a wife named Xanthippe and was a father hardly puts flesh on the bones of this portrait. As will have been guessed, there are also those who deny that there ever was any historical personage underlying the various Socratic legends. What seems called for now is more of that specialized detective work we call history. If the historical Socrates is not conveyed by the sources but rather obliterated, the prospect of sifting truth from the testimony of witnesses all of whom must be treated as on a level with liars is indeed a melancholy one. Against what do we test our guesses? There was a time when the Socrates of Xenophon was taken as the model and the Socrates of Plato brought into corroborate and expand the emerging picture. The assumption was Xenophon was too unimaginative and prosaic to distort his portrait, an assumption which looks foolhardy when it is pointed out that Xenophon gives us a composite portrait. Others would take Plato as a guide and use Xenophon as a check. No matter what procedure is adopted, however, the historical Socrates remains an elusive object and the only end in sight seems scholarly despair.

If it was necessary to introduce the Socratic Problem it is equally necessary to exorcise it. From the point of view of the history of philosophy it is really not an important issue at all. This can be seen by reflecting on the fact that if, *per impossible*, some intrepid scholar succeeded in unearthing the historical Socrates, that Socrates would be a distinctly modern entity; the fact of the matter is that the Socrates who has been influential in the history of philosophy is precisely the one conveyed by our sources, preeminently the personage in the Platonic dialogues. Now as we shall see when we turn to Plato, this sweeping away of the Socratic Problem can be productive of only temporary elation, for we are then faced with unearthing from the dialogues those which can be called Socratic and those which are more properly Platonic. This is not a new problem, of course; it has always been involved in attempts to discover the chronology of the dialogues. At any rate, we are now permitted to postpone the problem and to apply to Plato for a picture of the doctrine of Socrates. We shall also make appeal to the other sources to depict what may be for the Historian the Socratic legend, but what nevertheless constitutes, from the point of view of influence within the history of philosophy, the factual Socrates.

B. The Character of Socrates

As important as any doctrine or method which can be traced back to Socrates is the influence on later thinkers of his character. That character is inevitably seen against the background of the trial and death of Socrates and it is there that we must begin. In 399 B.C. Socrates was accused of speaking against the official religion, of introducing new gods and of leading youths astray. Here is Socrates' reaction in the *Euthyphro* of Plato.

What is the charge! Well, a very serious charge, which shows a good deal of character in the young man, and for which he is certainly not to be despised. He says he knows how the youth are corrupted and who are their corruptors. I fancy that he must be a wise man, and seeing that I am the reverse of a wise man, he has found me out, and is going to accuse me of corrupting his young friends. And of this our mother the state is to be the judge. Of all our political men he is the only one who seems to me to begin in the right way, with the cultivation of virtue in youth; like a good husbandman, he makes the young shoots his first care, and clears away us who are the destroyers of them. This is only the first step; he will afterwards attend to the elder branches; and if he goes on as he has begun, he will be a very great public benefactor. (2-3)

Socrates is here speaking of Meletus, his chief accuser; Meletus was joined by Anytus, a champion of Athenian democracy and Lyco. In the *Apology*, Plato gives us Socrates at the trial, replying to his accusers.

The *Apology* is, of course, a literary piece, but one in which the character of Socrates is revealed by the claim that he will not defend himself by means of a carefully written speech prepared by a professional rhetorician -- the usual courtroom procedure.

For I am more than seventy years of age and appearing now for the first time in a court of law, I am quite a stranger to the language of this place; and therefore I would have you regard me as if I were really a stranger, whom you would excuse if he spoke in his native tongue . . . (*Apology* 17)

Socrates first divides the charges he must answer into old ones that will have affected the present accusers in their youth and the actual charges. The old charges are that Socrates speculates about the heaven and earth, makes the worse appear the better cause, and teaches such things to others. Aristophanes is one who makes it appear that Socrates is concerned with natural philosophy

and Socrates is intent to show that he knows nothing of physics, though he respects those who do. This same disavowal is found in the *Phaedo*, a dialogue whose setting is the death cell of Socrates. "When I was young, Gebes, I had a prodigious desire to know that department of philosophy which is called the investigation of nature; to know the causes of things, and why a thing is and is created or destroyed appeared to me to be a lofty profession. . ." (*Phaedo*, 96) Socrates goes on to review the natural questions which agitated him and adds that mathematics too aroused his interest and wonder. Anaxagoras particularly interested Socrates because he had said that Mind is cause and director of all things, and yet Socrates was disappointed to find that Anaxagoras never seemed to invoke this cause when he dealt with particular things. Plato goes on to make Socrates a proponent of his own doctrine of Forms or Ideas, but what we are especially interested in is the turning away from natural philosophy.

The second old charge Socrates is anxious to meet is that he is a Sophist who dispenses wisdom for a fee; he denies having wisdom at all in the usual sense. Why then is he accused?

Men of Athens, this reputation of mine has come of a certain sort of wisdom which I possess. If you ask me what kind of wisdom, I reply, wisdom such as may perhaps be attained by man, for to that extent I am inclined to believe that I am wise; whereas the persons of whom I was speaking have a superhuman wisdom, which I may fail to describe, because I have it not myself; and he who says that I have it, speaks falsely, and is taking away my character. And here, O men of Athens, I must beg you not to interrupt me, even if I seem to say something extravagant. For the word which I will speak is not mine. I will refer you to a witness who is worthy of credit; that witness shall be the god of Delphi -- he will tell you about my wisdom, if I have any, and of what sort it is. You must have known Chaerephon; he was early a friend of mine, and also a friend of yours, for he shared in the recent exile of the people, and returned with you. Well, Chaerephon, as you know, was very impetuous in all his doings, and he went to Delphi and boldly asked the oracle to tell him whether -- as I was saying, I must beg you not to interrupt -- he asked the oracle to tell him whether any one was wiser than I was, and the Pythian prophetess answered, that there was no man wiser. (*Apology*, 20-21)

When Socrates heard of this reply, he did not know what to make of it; he knew that he did not possess wisdom and he knew that the god would not lie. He hit upon the plan of seeking for some one wiser than himself so that he could take this refutation to the god. Socrates sought among the politicians and what he

found were men who, though not wise, were thought by themselves and others to be wise; when Socrates pointed this out, he earned enemies. Moreover, he began to see that he himself had a slight advantage, namely, that in not being wise he knew himself not to be wise. In this he was better off than philosophers and poets as well as politicians. "This inquisition has led to my having many enemies of the worst and most dangerous kind, and has given occasion also to many calumnies. And I am called wise, for my hearers always imagine that I myself possess the wisdom which I find wanting in others: but the truth is, O men of Athens, that God only is wise; and by his answer he intends to show that the wisdom of men is worth little or nothing; he is not speaking of Socrates, he is only using my name by way of illustration, as if he said, He, O men, is the wisest, who, like Socrates, knows that his wisdom is in truth nothing." (*Apology* 23) This is the socratic ignorance which is its own kind of wisdom, not to know but to know that one does not know. Unwisdom is the not knowing which fancies itself to be knowledge.

The note of piety struck by Socrates in this reference to the oracle is sounded ever more strongly in the sequel. Socrates makes short work of Miletus, who at once accuses him of atheism and the introduction of new gods. More important is the belief expressed that his activity of philosophizing, of seeking for wisdom, is a service to the gods, one which nothing, not even the threat of death, could induce him to stop.

. . . if you say to me, Socrates, this time we will not mind Anytus, and you shall be let off, but upon one condition, that you are not to enquire and speculate in this way any more, and that if you are caught doing so again you shall die; -- if this was the condition on which you will let me go, I should reply: Men of Athens, I honor and love you; but I shall obey God rather than you, and while I have life and strength I shall never cease from the practise and teaching of philosophy, exhorting any one whom I meet and saying to him after my manner: You, my friend, -- a citizen of the great and mighty and wise city of Athens, -- are you not ashamed of heaping up the greatest amount of money and honor and reputation, and caring so little about wisdom and truth and the greatest improvement of the soul, which you never regard or heed at all? (*Apology*, 29)

Socrates describes himself as a sort of gadfly, given to the city by God to stir it to life. The sense of a divine mission is given a quite unique basis.

Some one may wonder why I go about in private giving advice and busying myself with the concerns of others, but do not venture to come forward in

public and advise the state. I will tell you why. You have heard me speak at sundry times and in divers places of an oracle or sign which comes to me, and is the divinity which Meletus ridicules in the indictment. This sign, which is a kind of voice, first began to come to me when I was a child; it always forbids but never commands me to do anything which I am going to do. This is what deters me from being a politician (*Apology*, 31)

This same *daimonion* induces him to accept the death verdict when it is passed.

Hitherto the divine faculty of which the internal oracle is the source has constantly been in the habit of opposing me even about trifles, if I was going to make a slip or error in any matter; and now as you see there has come upon me that which may be thought, and is generally believed to be, the last and worst evil. But the oracle made no sign of opposition, either when I was leaving my house this morning, or when I was on my way to the court, or while I was speaking, at anything which I was going to say; and yet I have often been stopped in the middle of a speech, but now in nothing I either said or did touching the matter in hand has the oracle opposed me. What do I take to be the explanation of this silence? I will tell you. it is an intimation that what has happened to me is a good, and that those of us who think that death is an evil are in error. For the customary sign would surely have opposed me had I been going to evil and not to good. (*Apology*, 40)

As will have been guessed, the nature of Socrates' inner voice has been the subject of a great deal of discussion, and it has been variously explained as the voice of conscience and as a type of abnormal psychic experience which, it is said, is well understood today. We shall take it, together with the reply of the Delphic oracle, as indicating Socrates' sense of mission, that he was doing the work of God. That mission has been stated by Plato in words which have become unforgettable. "I say again that daily to discourse about virtue, and of those other things about which you hear me examining myself and others, is the greatest good of man, and that the unexamined life is not worth living." (*Apology*, 38)

The person who emerges from the trial is one who is so totally convinced of the rightness of the manner in which he has spent his life, of his philosophizing, that he prefers death to ceasing to act as he always has. Not to live in the way he has is no life at all. The death of Socrates thus becomes a seal, what would be called nowadays the existential proof of the sincerity of his convictions. When we turn to Plato, we shall be assessing the arguments for the immortality of the soul

therein offered by Socrates to the small band which gathers around him as he awaits the hour when he must consume the hemlock and carry out the death sentence. It has been wisely pointed out that the true proof of immortality in that dialogue is not the arguments which are formulated, but the conduct of Socrates in the face of death. He is, as it were, living proof of the conviction that death is not the end, that we have here no lasting home, that we go through death to a better life. The *Apology* ends with these words "The hour of departure has arrived, and we go our ways -- I to die, and you to live. Which is better God only knows." But, in the *Phaedo*, where several Pythagoreans, Simmias, Gebes and Phaidondas, are listed among the close friends of Socrates, philosophy is described as the study of death, as a species of purgation whereby the soul is freed from the chains of the body and fitted for a better life elsewhere.

And what is purification but the separation of the soul from the body, as I was saying before; the habit of the soul gathering and collecting herself into herself from all sides out of the body; the dwelling in her own place alone, as in another life, so also in this, as far as she can; -- the release of the soul from the chains of the body? And the true philosophers, Simmias, are always occupied in the practise of dying, wherefore also to them least of all men is death terrible. (*Phaedo*, 67)

This sympathy with Pythagorean and Orphic attitudes is an undeniable trait of the Socrates Plato presents. Philosophy is a way of life which prepares for death beyond which the soul can enjoy a truer and better existence. The body is an impediment, a drag on the flight of the soul; the natural state, then, is a kind of sickness, and philosophy provides a remedy; through it the soul will gain health.

If there is a mystical side to this portrait, we cannot think of Socrates as an ascetic who eschewed all pleasures. The charge of pederasty is, after all, not entirely without basis, though it is difficult to concede that it is proved. We hear of Socrates as able to drink his companions under the table and begin early the next morning his interrogation of his fellow citizens as if nothing had happened. There is, however, the description of trances into which Socrates fell some of which lasted a full day.

One morning he was thinking about something which he could not resolve; he would not give it up, but continued thinking from early dawn until noon -- there he stood fixed in thought; and at noon attention was drawn to him, and the rumor ran through the wondering crowd that Socrates had been standing

and thinking about something ever since the break of day. At last, in the evening after supper, some Ionians out of curiosity (I should explain that this was not in winter but in summer), brought out their mats and slept in the open air that they might watch him and see whether he would stand all night. There he stood until the following morning; and with the return of light he offered up a prayer to the sun and went his way. (*Symposium*, 220)

We can end this discussion of the character of Socrates most fittingly by quoting the description of the death of Socrates in the *Phaedo*.

Crito made a sign to the servant, who was standing by; and he went out, and having been absent for some time, returned with the jailer carrying the cup of poison. Socrates said: You, my good friend, who are experienced in these matters, shall give me directions how I am to proceed. The man answered: You have only to walk about until your legs are heavy, and then to lie down, and the poison will act. At the same time he handed the cup to Socrates, who in the easiest and gentlest manner, without the least fear or change of color or feature, looking at the man with all his eyes, Echecrates, as his manner was, took the cup and said: What do you say about making a libation out of this cup to any god? May I, or not? The man answered: We only prepare, Socrates, just so much as we deem enough. I understand, he said: but I may and must ask the gods to prosper my journey from this to the other world -- even so -- and so be it according to my prayer. Then raising the cup to his lips, quite readily and cheerfully he drank off the poison. And hitherto most of us had been able to control our sorrow; but now when we saw him drinking, and saw too that he had finished the draught, we could no longer forbear, and in spite of myself my own tears were flowing fast; so that I covered my face and wept, not for him, but at the thought of my own calamity in having to part from such a friend. Nor was I the first; for Crito, when he found himself unable to restrain his tears, had got up, and I followed; and at that moment, Apollodorus, who had been weeping all the time, broke out in a loud and passionate cry which made cowards of us all. Socrates alone retained his calmness: What is this strange outcry? he said. I sent away the women mainly in order that they might not misbehave in this way, for I have been told that a man should die in peace. Be quiet, then, and have patience. When we heard his words we were shamed, and refrained our tears; and he walked about until, as he said, his legs began to fail, and then he lay on his back, according to the directions, and the man who gave him the poison now and then looked at his feet and legs; and after a while he pressed his foot hard, and asked him if he could feel; and he said, No; and then his leg, and so upwards and upwards, and showed us that he was cold and stiff. And he felt

them himself, and said: When the poison reaches the heart, that will be the end. He was beginning to grow cold about the groin, when he uncovered his face, for he had covered himself up, and said -- they were his last words -- he said: Crito, I owe a cock to Asclepius; will you remember to pay the debt? The debt shall be paid, said Crito; is there anything else? There was no answer to this question; but in a minute or two a movement was heard, and the attendants uncovered him; his eyes were set, and Crito closed his eyes and mouth. Such was the end, Echecrates, of our friend; concerning whom I may truly say, that of all the men of his time whom I have known, he was the wisest and justest and best. (*Phaedo*, 117-8)

C. The Doctrine of Socrates

When Aristotle considers Socrates, he pays him the same courtesy he extends to most of his predecessors and seeks a description of his teaching. We can set the stage for the subdivisions of this section by setting down a number of Aristotelian remarks on the philosophical activity of Socrates.

Socrates, however, was busying himself about ethical matters and neglecting the world of nature as a whole but seeking the universal in these ethical matters, and fixed thought for the first time on definitions; Plato accepted his teaching, but held that the problem applied not to sensible things but to entities of another kind -- for this reason, that the common definition could not be a definition of any sensible thing, as they were always changing. (*Metaphysics*, 6,987b1-5)

We will see in a moment why Aristotle describes the enquiries of Socrates as a search for definitions as well as why he thinks of this search as confined to ethical matter. That Aristotle is not simply repeating what we can read in Plato is indicated by his assertion that the theory of Forms or Ideas was not something maintained by Socrates, but an innovation of Plato; in the *Phaedo*, Plato presents Socrates as teaching the doctrine of Forms.

But when Socrates was occupying himself with the excellences of character, and in connexion with them became the first to raise the problem of universal definition (for of the physicists Democritus only touched on the subject to a small extent, and defined, after a fashion, the hot and the cold; while the Pythagoreans had before this treated of a few things, whose definitions -- e.g., those of opportunity, justice, or marriage -- they connected with numbers; but it was natural that Socrates should be seeking the essence, for he was seeking to

sylogize, and 'what a thing is' is the starting point of syllogisms; for there was as yet none of the dialectical power which enables people even without knowledge of the essence to speculate about contraries and inquire whether the same science deals with contraries; for two things may be fairly ascribed to Socrates -- inductive arguments and universal definition, both of which are concerned with the starting-point of science) : -- but Socrates did not make the universals or definitions exist apart. They, however, gave them separate existences, and this was the kind of thing they called Ideas. (*Metaphysics*, XIII, 4, 1078b17ff.)

This reiterated information that Socrates stopped short of the theory of Forms and confined his attention to ethical matters, gives us something like a criterion for distinguishing Socratic from properly Platonic doctrines in the Platonic dialogues, even when Socrates is the spokesman for Plato's own views. A characteristically Socratic doctrine, according to Aristotle, (*Eudemian Ethics*, 1, 5, 1216b6-8) is that virtue is knowledge, so that it is the same thing to know the just and be just. A corollary of this is that no one errs knowingly. Now we may ask how a man who judges rightly can behave incontinently. That he should behave so when he has knowledge, some say is impossible; for it would be strange -- so Socrates thought -- if when knowledge was in a man something else could master it and drag it about like a slave. For Socrates was entirely opposed to the view in question, holding that there is no such thing as incontinence; no one, he said, when he judges, acts against what he judges best - - people act so only by reason of ignorance. (*Nicomachean Ethics*, VII, 2, 1145b21-27)

We can summarize the testimony of Aristotle by saying that Socrates has as his distinctive characteristic the search for definitions which are the principle of syllogism; that, in fact, he indulged in inductive argumentation, confining his attention to the ethical realm, and that he thought knowledge about the object of virtue was in fact the virtue in question. We must concern ourselves, these reports suggest, with Socrates' method of procedures and his ethical doctrine.

Socratic Method. In the *Theaetetus*, Plato has Socrates describe at some length his own method of interrogation, a method called maieutics or midwifery. The comparison becomes rather elaborate, with Socrates pointing out that the midwife is one who is herself past the age of bearing and one who knows better than others who is pregnant and who is not; moreover, the midwife is able to apply potions and the like to make easier a difficult birth; finally, the midwife is the best of matchmakers.

Well, my art of midwifery is in most respects like theirs; but differs, in that I attend men and not women, and I look after their souls when they are in labor, and not after their bodies: and the triumph of my art is in thoroughly examining whether the thought which the mind of the young man brings forth is a false idol or noble and true birth. And, like the midwives, I am barren, and the reproach which is often made against me, that I ask questions of others and have not the wit to answer them myself, is very just -- the reason is, that the god compels me to be a midwife, but does not allow me to bring forth. And therefore I am not myself at all wise, nor have I anything to show which is the invention or birth of my own soul, but those who converse with me profit . . . It is quite clear that they never learned anything from me; the many fine discoveries to which they cling are of their own making. But to me and the god they owe their delivery. (*Theaetetus* 150)

The method practised by Socrates, then, purports to be a method of releasing what is already in the mind of the one being questioned; Socrates has nothing positive to teach, there is no question of transferring something from his mind into that of the object of his questions. Socrates openly professes his own ignorance, and appeals to the other for the benefit of his wisdom. The irony of Socrates appears most forcibly when one who with some condescension has agreed to enlighten Socrates finds himself revealing that he does not possess the knowledge he so confidently admitted to possessing. A less painful instance of this evokes the following description of Socrates from Meno in Plato's dialogue of that name.

O Socrates, I used to be told, before I knew you, that you were always doubting yourself and making others doubt; and now you are casting your spells over me, and I am simply getting bewitched and enchanted, and am at my wits' end. And if I may venture to make a jest upon you, you seem to me both in your appearance and in your power over others to be very like the flat torpedo fish, who torpifies those who come near him and touch him, as you have now torpified me, I think. For my soul and my tongue are really torpid, and I do not know how to answer you; and though I have been delivered of an infinite variety of speeches about virtue before now, and to many persons -- and very good ones they were, as I thought -- at this moment I cannot even say what virtue is. (*Meno*, 80)

Socrates goes on to assure Meno that he really knows what virtue is. But we have here, it would seem, Plato making use of Socrates' maieutic art and finding its possibility in the theory of anamnesis which we shall be considering later.

With Socrates, the search for definition usually ends without the discovery of the object of the quest, and the conclusion is not that the knowledge still lies unconcealed in the mind of the person being questioned; rather, the implication usually is that the one questioned would be better advised to confess his ignorance.

The Socratic method, then, consists of carefully marshalled questions directed at discovering what something is its essence, often a particular virtue, sometimes virtue itself with the inquiry usually ending without success. Thus, in the *Charmides*, Socrates is inquiring what is temperance. The *Lysis* asks, what is friendship; the *Protagoras* asks not only what is virtue but is there more than one and can virtue be learned. The *Lysis* asks what courage is but the question turns into an asking after virtue in general. And so on. Now before asking what Socrates had to say about virtue and its acquisition as such, let us by following the general development of the *Charmides* try to discern the method of Socrates.^{22}

Socrates has asked what temperance is and the first reply is that it is doing things quietly and in an orderly way; temperance is quietness. Socrates allows that it is possible that many would identify the quiet with the temperate, but he wants to know if Charmides would say that temperance falls in the class of the noble and good. Charmides agrees and Socrates asks if it is better to write quickly or quietly. Charmides replies, quickly. So too with reading, playing the lyre, and wrestling -- quickness or sharpness is better than quietness. Socrates continues to mention such activities and in every case quickness is desirable. In summary, Socrates asks and gains an affirmative answer to the question whether in all bodily actions agility and quickness is noblest and best and not quietness. But is not temperance a good? Since it is, temperance in bodily matters would seem to involve quickness and not, as Charmides had originally proposed, quietness. Socrates next moves to activities which are not bodily. Teaching, learning, remembering, understanding -- in each of these, facility and quickness of operation and not quietness is best. In this way, Socrates has done away with Charmides' first essay at a definition of temperance. Characteristically, he asks the young man to try again. This time Charmides suggests that temperance is the same as modesty. By now we suspect that Charmides is not going to do too well, but we can also sense the pull of Socrates' method; not satisfied with a simple assertion, he is going to draw out its implications and see if they leave the original remark as tenable as it sounds at first. Temperance, now, has been identified with modesty. Earlier Charmides had admitted that temperance is noble and he now agrees that the temperate are good. Socrates asks if that can

be good which does not make men good. Charmides says that it cannot, and Socrates reminds him of Homer's remark that "Modesty is not good for a needy man." If modesty can be good only sometimes, it cannot be identified with temperance which is always good. Undaunted, Charmides suggests a third definition of temperance, namely, that temperance is doing one's own business. This sounds to Socrates like a definition Charmides has gotten from someone else and, as it happens, when Socrates begins to make fun of the definition, asking if it requires the temperate man to do his own laundry, Critias steps forward to take responsibility for the definition and offers to defend it. Critias introduces a distinction between making and doing, the upshot of which is a fourth definition of temperance; it is now said to be the doing of good actions. If the shoemaker makes shoes for someone else, this making is a doing, an action, which cannot be alienated from the artisan. As Taylor⁽²³⁾ has pointed out, Socrates could have seized here on Critias' assumption that we already know what is meant by good, something which would have brought him quickly to the point made at the end of the dialogue. Nevertheless, the dialogue takes another turn. In answer to Socrates' queries, Critias admits that a man may do good unknowingly and thus be temperate unknowingly if temperance is simply the doing of good actions. Critias backs away from the notion of someone's being temperate unwittingly and is willing to withdraw everything in favor of a fifth definition: temperance is self-knowledge. Now we might expect Socrates to welcome this definition, but in fact he is most wary of it. If temperance or wisdom (*sophrosyne*) is a science, of what is it a science? Critias replies that it is the science of itself, to which Socrates objects that of other things which are called sciences there is an effect or product; of medicine health, of architecture houses and so on. Critias then introduces a distinction between sciences which have a product and those which do not, a distinction to be made much of by Aristotle; computation and geometry do not have effects although architecture does. Socrates rejoins that at least these non-practical sciences have different subject matters; computation is concerned with odd and even numbers and such numbers are not themselves the art of computation. In other words, every science differs from the object of its concern. "Now, I want to know, what is that which is not it is just in this that wisdom differs from other sciences: wisdom is the science of other sciences and of itself. This soon appears to be the reduction of wisdom to the recognition of the presence or absence of knowledge without knowledge -- of what the knowledge which may be present is about. To this Socrates objects that wisdom should not be made to consist of such ignorance, since then it is useless. To live according to knowledge is desirable only when the knowledge in question is assigned some object; if there were such a universal knowledge, surely to possess it and live according to it would

constitute happiness, but with what would such knowledge be concerned, as the art of shoemaking is concerned with shoes? Critias is gradually brought down from the notion of the knowledge of knowledge, to knowledges which have an object. He is asked successively if working with leather or brass, or the art of computation or of health can make one happy. Suddenly it dawns on him that the knowledge which can make one happy is that whereby he discerns good and evil, a remark which brings a joyful outburst from Socrates.

Monster! I said; you have been carrying me round in a circle, and all this time hiding from me the fact that the life according to knowledge is not that which makes men act rightly and be happy, not even if knowledge include all the sciences, but one science only, that of good and evil . . . But that science is not wisdom or temperance, but a science of human advantage; not a science of other sciences, or of ignorance, but of good and evil: and if this be of use, then wisdom or temperance will not be of use. (174)

By wisdom here, Socrates means any of the definitions which have been given, particularly the explanation of the fifth definition to the effect that wisdom is the science of science. Critias tries to reduce knowledge of good and evil to wisdom, but Socrates is able to prevent this by pointing out that medicine is concerned with good and evil and Critias has been rather insistent on the difference between medicine and wisdom. The upshot of the dialogue is that wisdom has eluded all attempts at definition and that they have been brought to the unfortunate conclusion that temperance or wisdom is useless.

It goes without saying that no paraphrase can convey the living movement of a Platonic dialogue; but perhaps this summary will give us something on which to pin a number of generalities concerning the Socratic method. In the first place, we can see quite clearly why Aristotle should say that Socrates was in quest of definitions. Once one is offered, Socrates sees what consequences follow from accepting the definition, a procedure which is often sufficient for rejecting the proposed formula. Consequences are shown to follow by a process of analogy; what is to be defined falls into class A, but so does something else which has such-and-such a characteristic. Does that which we are defining have this characteristic? This procedure involves assuming things not directly in question, but Socrates does not make assertions so much as he solicits assent to seemingly innocent remarks from which devastating consequences soon follow. And he is much gentler with the tentative suggestions of a young man than with the initially confident replies of one who thinks he knows. With the latter, a series of fairly vicious thrusts achieves a salutary deflation, which is then made

somewhat less unpalatable by Socrates's suggestion that the trial balloon be thought of as a common effort and not the carrier of anyone's reputation. But whoever the one questioned, Socrates cannot resist the ironic touch; someone who is led by the nose to make a certain remark is pounced upon as a sly character who has all along been restraining himself from stating what he knows.

Aristotle has also characterized the method of Socrates as inductive reasoning. Instances of what he means are present in the sweep of the *Charmides*. For example, if the art of shoemaking produces shoes, and the art of building houses, what is the product of the art which is temperance? Several times this method calls for a division among things which fall into a common classification and, as Critias points out, Socrates is ever on the lookout for the notes had in common by things which the conversation has agreed are grouped together. His method always leads Socrates to demand that one who utters a generality about a given matter show how it applies to instances; e.g., if art is such- and- such, can this be shown in the case of shoemaking? Finally, the *Charmides* is not untypical in failing to achieve its object; at the end of the dialogue, temperance remains undefined, but no participant is left unaware of the collective ignorance of what it is they have been talking about. This Socratic ignorance is qualitatively different from the scepticism of the Sophists. Kierkegaard, for example, conjectured that the reason the dialogues which seek after the definition of a virtue or of virtue itself do not reach a conclusion is that their indirect point is precisely that knowledge of the definition of virtue is a misplacement of the problem facing one who desires to be virtuous. Unfortunately, as we shall see in a moment, this suggestion is wide of the mark, although it tells us something important. The thing which separates Socrates sharply from any scepticism is the note of optimism that the failure to achieve a definition carries with it. The knowledge not presently possessed is nonetheless attainable, and nothing is more important than the continued search for it. Socrates will have nothing to do with the position that one opinion is as good as another, or as good as one's ability to sustain it in argumentation. He is not interested in triumphing over an opponent. When he is accused by Critias of desiring to refute him, he answers,

And what if I am? How can you think that I have any other motive in refuting you but what I should have in examining into myself? which motive would be just a fear of my unconsciously fancying that I knew something of which I was ignorant. And at this moment I pursue the argument chiefly for my own sake, and perhaps in some degree also for the sake of my other friends. For is not the

discovery of things as they truly are a good common to all mankind? . . . attend only to the argument, and see what will come of the refutation. (166)

The assumption always is that there is a truth founded in reality and that it makes sense for man to pursue it.

Knowledge and Virtue. In several of the Platonic dialogues we find Socrates dealing ironically with a Sophist who purports to be able to teach men virtue. His opposition to them is not prompted by the impossibility of the task, but by the fact, soon revealed by the questions he puts to the Sophist, that the meaning of virtue, what virtue is, cannot be explained by the man who would teach it to others. We have seen earlier Aristotle's reports that, for Socrates, knowledge and virtue are one -- that virtue is knowledge. Thus, to know what is just and to be just are one and the same. On the face of it, this is an extraordinary position; we are all painfully aware that it is possible to know what one ought to do and yet not do it. Socrates' retort to this would be that, if we really know, then knowledge and virtue are one. Let us look first of all at a place in Plato where Socrates is shown defending his position; afterwards, we can indicate what Plato and Aristotle made of this position.

The passage in question is to be found in the *Protagoras* (351-358) and we propose to reproduce it in full. Socrates is speaking; his interlocutor is Protagoras, the Sophist.

I said: You would admit, Protagoras, that some men live well and others ill?

He assented.

And do you think that a man lives well who lives in pain and grief?

He does not.

But if he lives pleasantly to the end of his life, will he not in that case have lived well?

He will.

Then to live pleasantly is a good, and to live unpleasantly an evil?

Yes, he said, if the pleasure be good and honorable. And do you, Protagoras, like the rest of the world, call some pleasant things evil and some painful things

good? -- for I am rather disposed to say that things are good in as far as they are pleasant, if they have no consequences of another sort, and in as far as they are painful they are bad.

I do not know, Socrates, he said, whether I can venture to assert in that unqualified manner that the pleasant is the good and the painful the evil. Having regard not only to my present answer, but also to the whole of my life, I shall be safer, if I am not mistaken, in saying that there are some pleasant things which are not good, and that there are some painful things which are good, and some which are not good, and that there are some which are neither good nor evil.

And you would call pleasant, I said, the things which participate in pleasure or create pleasure?

Certainly, he said.

Then my meaning is, that in as far as they are pleasant they are good; and my question would imply that pleasure is a good in itself.

According to your favorite mode of speech, Socrates, let us reflect about this, he said; and if the reflection is to the point, and the result proves that pleasure and good are really the same, then we will agree; but if not, then we will argue.

And would you wish to begin the enquiry? I said; or shall I begin?

You ought to take the lead, he said; for you are the author of the discussion.

May I employ an illustration? I said. Suppose someone who is enquiring into the health or some other bodily quality of another: -- he looks at his face and at the tips of his fingers, and then he says, Uncover your chest and back to me that I may have a better view; -- that is the sort of thing which I desire in this speculation. Having seen what your opinion is about good and pleasure, I am minded to say to you: Uncover your mind to me, Protagoras, and reveal your opinion about knowledge, that I may know whether you agree with the rest of the world. Now the rest of the world are of opinion that their notion is that a man may have knowledge, and yet that the knowledge which is in him may be overmastered by anger, or pleasure, or pain, or love, or perhaps by fear, -- just as if knowledge were a slave and might be dragged about anyhow. Now is that your view? or do you think that knowledge is a noble and commanding thing, which cannot be overcome, and will not allow a man, if he only knows the difference

of good and evil, to do anything which is contrary to knowledge, but that wisdom will have strength to help him?

I agree with you, Socrates, said Protagoras; and not only so, but I, above all other men, am bound to say that wisdom and knowledge are the highest of human things.

Good, I said, and true. But are you aware that the majority of the world are of another mind; and that men are commonly supposed to know the things which are best, and not to do them when they might? And most persons whom I have asked the reason of this have said that when men act contrary to knowledge they are overcome by pain, or pleasure, or some of the affections which I was just now mentioning.

Yes, Socrates, he replied; and that is not the only point about which mankind are in error.

Suppose, then, that you and I endeavor to instruct and inform them what is the nature of this affection which they call 'being overcome by pleasure,' and which they affirm to be the reason why they do not always do what is best. When we say to them: Friends, you are mistaken, and are saying what is not true, they would probably reply: Socrates and Protagoras, if this affection of the soul is not to be called 'being overcome by pleasure,' pray, what is it, and by what name would you describe it?

But why, Socrates, should we trouble ourselves about the opinion of the many, who just say anything that happens to occur to them?

I believe, I said, that they may be of use in helping us to discover how courage is related to the other parts of virtue. If you are disposed to abide by our agreement, that I should show the way in which, as I think, our recent difficulty is most likely to be cleared up, do you follow; but if not, never mind.

You are quite right, he said; and I would have you proceed as you have begun.

Well, then, I said, let me suppose that they repeat their question,

What account do you give of that which, in our way of speaking, is termed being overcome by pleasure? I should answer thus: Listen, and Protagoras and I will endeavor to show you. When men are overcome by eating and drinking and other sensual desires which are pleasant, and they, knowing them to be evil,

nevertheless indulge in them, would you not say that they were overcome by pleasure? They will not deny this. And suppose that you and I were to go on and ask them again: 'In what way do you say that they are evil, -- in that they are pleasant and give pleasure at the moment, or because they cause disease and poverty and other like evils in the future? Would they still be evil, if they had no attendant evil consequences, simply because they give the consciousness of pleasure of whatever nature? -- Would they not answer that they are not evil on account of the pleasure which is immediately given by them, but on account of the after consequences -- diseases and the like?

I believe, said Protagoras, that the world in general would answer as you do.

And in causing diseases do they not cause pain? and in causing poverty do they not cause pain; -- they would agree to that also, if I am not mistaken?

Protagoras assented.

Then I should say to them, in my name and yours: Do you think them evil for any other reason, except because they end in pain and rob us of other pleasures: -- there again they would agree?

We both of us thought that they would.

And then I should take the question from the opposite point of view, and say: 'Friends, when you speak of goods being painful, do you not mean remedial goods, such as gymnastic exercises, and military service, and the physician's use of burning, cutting, drugging and starving? Are these the things which are good but painful?' -- they would assent to me?

He agreed.

'And do you call them good because they occasion the greatest immediate suffering and pain; or because, afterwards, they bring health and improvement of the bodily condition and the salvation of states and power over others and wealth?' -- they would agree to the latter alternative, if I am not mistaken?

He assented.

'Are these things good for any other reason except that they end in pleasure, and get rid of and avert pain? Are you looking to any other standard but pleasure

and pain when you call them good?' -- they would acknowledge that they were not?

I think so, said Protagoras.

'And do you not pursue pleasure as a good, and avoid pain as an evil?'

He assented.

'Then you think that pain is an evil and pleasure is a good; and even pleasure you deem an evil, when it robs you of greater pleasure than it gives, or causes pains greater than the pleasure. If, however, you call pleasure an evil in relation to some other end or standard, you will be able to show us that standard. But you have none to show.'

I do not think that they have, said Protagoras.

'And have you not a similar way of speaking about pain? You call pain a good when it takes away greater pains than those which it has, or gives pleasures greater than the pains: then if you have some standard other than pleasure and pain to which you refer when you call actual pain a good, you can show what that is. But you cannot.

True, said Protagoras.

Suppose again, I said, that the world says to me: 'Why do you spend many words and speak in many ways on this subject?' Excuse me, friends, I should reply; but in the first place there is a difficulty in explaining the meaning of the expression 'overcome by pleasure', and the whole argument turns upon this. And even now, if you see any possible way in which evil can be explained as other than pain, or good as other than pleasure, you may still retract. Are you satisfied, then, at having a life of pleasure which is without pain? If you are, and if you are unable to show any good or evil which does not end in pleasure and pain, hear the consequences: -- If what you say is true, then the argument is absurd which affirms that a man often does evil knowingly, when he might abstain, because he is seduced and overpowered by pleasure; or again, when you say that a man knowingly refuses to do what is good because he is overcome at the moment by pleasure. And that this is ridiculous will be evident if only we give up the use of various names, such as pleasant and painful, and good and evil. As there are two things, let us call them by two names -- first, good and evil, and then pleasant and painful. Assuming this, let us go on to say that a man

does evil knowing that he does evil. But some one will ask, Why? Because he is overcome is the first answer. And by what is he overcome? the enquirer will proceed to ask. And we shall not be able to reply 'By pleasure,' for the name of pleasure has been exchanged for that of good. In our answer, then, we shall only say that he is overcome. 'By what?' he will reiterate. By the good, we shall have to reply; indeed, we shall.

Nay, but our questioner will rejoin with a laugh, if he is one of the swaggering sort, 'That is too ridiculous, that a man should do what he knows to be evil when he ought not, because he is overcome by good. Is that, he will ask, because the good was worthy or not of conquering evil?' And in answer to that we shall clearly reply, Because it was not worthy; for if it had been worthy, then he who, as we say, was overcome by pleasure, would not have been wrong. 'But how,' he will reply, 'can the good be unworthy of the evil, or the evil of the good?' Is not the real explanation that they are out of proportion to one another, either as greater and smaller, or more and fewer? This we cannot deny. And when you speak of being overcome -- 'what do you mean,' he will say, 'but that you choose the greater evil in exchange for the lesser good?' Admitted. And now substitute the names of pleasure and pain for good and evil, and say, not as before, that a man does what is evil knowingly, and because he is overcome by pleasure, which is unworthy to overcome. What measure is there of the relations of pleasure and pain other than excess and defect, which means that they become greater and smaller and more and fewer, and differ in degree? For if anyone says: 'Yes, Socrates, but immediate pleasure differs widely from future pleasure and pain' -- To that I should reply: And do they differ in anything but in pleasure and pain? There can be no other measure of them. And do you, like a skilful weigher, put into the balance the pleasures and the pains, and their nearness and distance, and weigh them, and then say which outweighs the other. If you weigh pleasures against pleasures, you of course take the more and greater; or if you weigh pains against pains, you take the fewer and the less; or if pleasures against pains, then you choose that course of action in which the painful is exceeded by the pleasant, whether the distant by the near or the near by the distant; and you avoid that course of action in which the pleasant is exceeded by the painful. Would you not admit, my friends, that this is true? I am confident that they cannot deny this.

He agreed with me.

Well then, I shall say, if you agree so far, be so good as to answer me a question: Do not the same magnitudes appear larger to your sight when near, and smaller

when at a distance? They will acknowledge that. And the same holds of thickness and number; also sounds, which are in themselves equal, are greater when near, and lesser when at a distance. They will grant that also. Now suppose happiness to consist in doing or choosing the greater, and in not doing or avoiding the less, what would be the saving principle of human life? Would not the art of measuring be the saving principle; or would the power of appearance? Is not the latter that deceiving art which makes us wander up and down and take the things at one time of which we repent at another, both in our actions and in our choice of things great and small? But the art of measurement would do away with the effect of appearances, and, showing the truth, would fain teach the soul at last to find rest in the truth, and would thus save our life. Would not mankind generally acknowledge that the art which accomplishes this result is the art of measurement?

Yes, he said, the art of measurement.

Suppose, again, the salvation of human life to depend on the choice of odd and even, and on the knowledge of when a man ought to choose the greater or less, either in reference to themselves or to each other, and whether near or at a distance; what would be the saving principle of our lives? Would not knowledge? -- a knowledge of measuring, when the question is one of excess and defect, and a knowledge of number, when the question is of odd and even? The world will assent, will they not?

Protagoras himself thought that they would.

Well then, my friends, I say to them; seeing that the salvation of human life has been found to consist in the right choice of pleasures and pains, -- in the choice of the more and the fewer, and the greater and the less, and the nearer and remoter, must not this measuring be a consideration of their excess and defect and equality in relation to each other?

This is undeniably true.

And this, as possessing measure, must undeniably also be an art and science?

They will agree, he said.

The nature of that art or science will be a matter of future consideration; but the existence of such a science furnishes a demonstrative answer to the question which you asked of me and Protagoras. At the time when you asked the

question, if you remember, both of us were agreeing that there was nothing mightier than knowledge, and that knowledge, in whatever existing, must have the advantage over pleasure and all other things; and then you said that pleasure often got the advantage even over a man who has knowledge; and we refused to allow this, and you rejoined: O Protagoras and Socrates, what is the meaning of being overcome by pleasure if not this? -- tell us what you call such a state: -- if we had immediately and at the time answered 'Ignorance,' you would have laughed at us. But now, in laughing at us, you will be laughing at yourselves: for you also admitted that men err in the choice of pleasures and pains; that is, in their choice of good and evil, from defect of knowledge; and you admitted further, that they err, not only from defect of knowledge in general, but of that particular knowledge which is called measuring. And you are also aware that the erring act which is done without knowledge is done in ignorance. This therefore is the meaning of being overcome by pleasure; -- ignorance, and that the greatest . . . Then, I said, no man voluntarily pursues evil, or that which he thinks to be evil. To prefer evil to good is not in human nature; and when a man is compelled to choose one of two evils, no one will choose the greater when he may have the less.

This particular presentation of the identification of knowledge and virtue owes a good deal to the assumption of the strict equation of good and evil and pleasure and pain; we need not take it that Socrates himself accepted this identification, although it is difficult to see that his point could be made with anything like the facility of this passage without it. Once one insists on a distinction of goods or of pleasures, say between rational and bodily goods or pleasures, the troublesome phrase "overcome by pleasure" cannot be translated out of the way as Socrates has just done. Now of course it is just such a distinction that, as we shall see, Aristotle has in mind in his discussion of the incontinent man, who, while knowing that he should do, does not do it because of the pull of the senses; with this distinction comes also a distinction in the use of the terms "knowledge" and "ignorance." A man may know the demands of justice in general but, because of bad dispositions, not recognize them in a particular instance. The great question, then, is the meaning Socrates attaches to "knowledge" when he identifies it with virtue. If he means general knowledge, then the identification is absurd, provided one does not identify pleasure and good, pain and evil. If he meant what Aristotle means by practical wisdom, then of course knowledge is virtue and is involved in the possession of virtues which are not dispositions of a cognitive faculty. Aristotle, in the passage quoted earlier, takes Socrates to be identifying virtue with that universal knowledge expressed in the definition -- an understandable interpretation when we

remember that, for Aristotle, Socrates' characteristic concern is with universals. Nevertheless, a case can be made for the view that Socrates does not intend this universal knowledge when he identifies knowledge and virtue. As we have just indicated, this would not make his position acceptable to Aristotle, but it is not quite so repellent as in the first interpretation. The examined life, which for Socrates is the only one worth living, must finally be our own, and the result of this examination would be that self-knowledge which, while ideally including awareness of what we have in common with all men, is constituted precisely by the knowledge of what is peculiar to ourselves in the particular circumstances in which we find ourselves. Moreover, the Socratic view that philosophizing is precisely a purgation of the senses, a way of escaping from the chains of the body, suggests that he was alive to those factors which, for Aristotle, are essential to the acquisition of the moral virtues.

We can conclude this brief treatment of Socrates by once more contrasting his efforts with those of the Sophists. As has been mentioned earlier and as is evident from the dialogues of Plato, Socrates does not hold every Sophist in contempt; the Protagoras, for example, exhibits quite a different attitude towards the Sophist who gives his name to the dialogue than that exhibited in the Euthydemus where the Socratic irony hardly masks his contempt for Euthydemus and Dionysiodorus. Generally speaking, however, Socrates differs from the Sophist in his concern, everywhere evident, for the improvement of his interlocutor when this is a young man desirous of such improvement. Socrates is not interested in money; he is not interested in triumph; he is not interested in getting his ideas into someone's head or in providing others with models of dialectical exercise for copying. Nothing delights him more than the awakening of independent thought on the part of his interlocutor; moreover, he is dismayed when a young man begins to imitate the Sophists (e.g., Ctesippus in the Euthydemus). From this point of view, Kierkegaard is not perhaps far wrong. Socrates wishes to inculcate moral virtue, and the habit of self-examination, but he cannot teach this as a doctrine he possesses and which needs only to be grasped intellectually by the other. His maieutic art gives birth to an impulse towards self-knowledge and that is the only way in which he can teach virtue. This gives us a somewhat Kierkegaardian Socrates, of course; but such a Socrates is surely there in the dialogues and the man who considered himself to be another Socrates was not the first to find him.

D. The Socratic Schools

We have already indicated that, if all Socrates' predecessors can meaningfully be called presocratics, Plato and Aristotle can fittingly be considered to be socratics. Before turning to these men who, with their genius, dwarf their esteemed predecessor, we must say a few words about other men and other schools spoken of in terms of the imperfect socratic or minor socratic movements. Some of the men involved were not without influence on the Stoic Philosophy and thereby deserve mention; moreover, we find efforts made to connect Socrates more or less directly with presocratic doctrines.

The School of Socrates. Under this heading we may mention Xenophon and Aeschines, two students of Socrates not associated with any of the later socratic schools. To those who receive a classical education, Xenophon is often the first author encountered, enjoying a place analogous to that of Caesar among Latin authors. And the work is, of course, the *Anabasis*, the account of the retreat of the 10,000 to the sea, under the leadership of Xenophon. The march took place about 401 or 400 B.C.; Xenophon himself was probably born around 439. Xenophon's claim to mention here rests on his authorship of the already cited *Memorabilia*, and on his *Apology*, *Oeconomicus*, *Symposium* and *Cyropaideia*. With respect to Socrates, Xenophon is primarily interested in defending his old teacher; the main attack is thought to be an *Accusation of Socrates* by Polycrates. While his style is often praised, Xenophon is never awarded high marks for philosophical penetration. His writings have retained philosophical interest largely because of the belief that they convey valuable information on the historical Socrates.

Aeschines is credited with seven dialogues and is also highly praised for their style. Diogenes Laertius (II, 62) says that Aeschines was in Sicily at the court of Dionysius at the same time as Plato and Aristippus; the former kept aloof from him while the latter befriended him. When the three were back in Athens, Aeschines did not venture to lecture in competition with the other two. The names of many other pupils of Socrates can be found in the dialogues of Plato, but very few others need arrest our attention even this fleetingly.

The Megarian School. The founder of this socratic school was Eucides of Megara who is known chiefly for the attempt to bring together Parmenidean and socratic teaching. This is thought to be exemplified in the following passage.

He held the supreme good to be really one, though called by many names, sometimes wisdom, sometimes God, and again Mind, and so forth. But all that is contradictory of the good he used to reject, declaring that it had no existence. (Diogenes Laertius, II, 106)

Eubulides of Miletus, a member of the school, is famous for the formulation of several dialectical arguments and paradoxes; he seems to have set the school in a direction which had its impact on the later Stoic logic. The paradox of the liar, to be discussed later when we turn to the Stoics, is attributed to Eubulides.

Both Eubulides and Diodorus Cronus are said to have made attacks on Aristotelian doctrines. The following is found in Sextus Empiricus. (*Adversus Physicos*, II 85-6)

And another weighty argument for the non-existence of motion is adduced by Diodorus Cronos, by means of which he establishes that not a single thing is in motion, but has been in motion. And the fact that nothing is in motion follows from his assumption of indivisibles. For the indivisible body must be contained in an indivisible place and therefore must not move either in it (for it fills it up, but a thing which is to move must have a larger place) or in the place in which it is not; for as yet it is not in this place so as to be moved therein; consequently it is not in motion. But, according to reason, it has been in motion; for that which was formerly observed in this place is now observed in another place, which would not have occurred if it had not been moved.

Aristotle tells us that members of this school denied the reality of possibility.

There are some who say, as the Megaric school does, that a thing 'can' act only when it is acting, and when it is not acting it 'cannot' act, e.g., that he who is not building cannot build, but only he who is building, when he is building; and so in all other cases. It is not hard to see the absurdities that attend this view. (*Metaphysics*, IX,3,1046b29ff.)

Another member of the Megarian school is Stilpo who taught at Athens around 320 B.C. and was the teacher of Zeno, the founder of Stoicism. He is known for his denial of predication, feeling that this implies that things which are other are not other. In ethics he taught a theory of self-dependence or autarchy which foreshadows the ethical theories of the Stoics and Epicureans.

No special remarks need be devoted to the schools of Elis and Eretria other than pointing out that Menedemus accepts Stilpo's argument against the

possibility of predication. Diogenes Laertius (II, 135) also credits him with formulating the proper answer to such questions as "Have you stopped beating your wife?" According to this account when asked "if he had left off beating his father, his answer was, 'Why, I was not beating him and have not left off.'"

The Cynics. This socratic school is of importance because, when fused with the Megarian school, it exerted great influence on Stoicism. Antisthenes, pupil of Gorgias and then Sophist in his own right before coming under the influence of Socrates, is said to be the founder of the Cynics. A great many writings were attributed to Antisthenes. What seems to have struck Antisthenes in Socrates was not the seeds of a doctrine so much as a rule of conduct, and the Cynic is set aside from his fellows primarily by his mode of life. Indeed, the term "cynic" itself may derive from the Greek word for dog and thereby indicate the nature of that mode of life. Independence from and indifference to his surroundings followed from the Cynic's conviction that only virtue is good, only sin evil and all else indifferent. Virtue is wisdom and it can be taught by means of training. Other members of this school worthy of mention are Diogenes of Sinope and Crates of Thebes.

It is customary as well to mention a Cyrenaic socratic school founded by Aristippus of Cyrene; as with the Cynics, this school was primarily concerned with moral virtue. We must now turn to the great successors of Socrates, Plato and Aristotle.

{21} A. H. Chroust, *Socrates, Man and Myth* (Notre Dame: University of Notre Dame Press, 1957), p. xii.

{22} See T. C. Tuckey, *Plato's Charmides* (Cambridge, England: University Press, 1951).

{23} see A. E. Taylor, *Plato, The Man and His World* (New York: Meridian Books, 1956).

Chapter II

Plato

A. The Man and His Work

Plato was born in 428 or 427 B.C., probably at Athens, into a family distinguished both in paternal and maternal lines, and died in 347, giving him a lifespan of approximately eighty years. It may be that "Plato" is simply a nickname and that his real name was Adstocles; whatever the truth of this, it is as Plato that he is known and recognized as standing in the very front rank of philosophers of all time. Pathetic attempts to discount his claim to greatness are not wanting, but nothing can change the fact of his awesome reputation in antiquity, and in the early middle ages when little else of his was known than the *Timaeus*, and in the West generally after the introduction of the rest of his writings. The dialogues of Plato represent one of the few indispensable sources of philosophizing, so much so that ignorance of them is tantamount to ignorance of philosophy itself. A mark of their greatness is their inexhaustibility; as instruments of philosophizing, they are at the disposal equally of the novice, the adept and the scholar. Philosophers of every view feel kinship with Plato and are at pains to show that he anticipated them. If it is difficult not to profit from a reading of Plato, it is equally true that the dialogues do not yield a comprehensive meaning easily, for reasons we shall be hinting at in a moment. For now, suffice it to say that they are the vehicles of a genius which has put its indelible stamp on what we call philosophy, the man who is the greatest pupil of Socrates and the master of Aristotle.

Plato's father was Ariston, whose ancestry can be traced to the kings of Athens; Plato's mother, Perictione, was descended from Solon. Adeimantus and Glaucon, who turn up as characters in the *Republic*, were Plato's older brothers; he had a sister, Potone, whose son Speusippus succeeded Plato as head of the Academy. Charmides and Critias, both uncles of Plato, appear as characters in dialogues. After the death of his father, Plato's mother remarried and her son, Plato's half-brother, Antiphon, appears in the *Parmenides*. Plato's aristocratic connections are sometimes invoked to explain his distaste for the Athenian democracy -- a distaste abetted, no doubt, by the execution of Socrates. We have Plato's own word for his political attitudes.

Once upon a time in my youth I cherished like many another the hope, directly I came of age, of entering a political career. It fell out, moreover, that political

events took the following course. There were many who heaped abuse on the form of government then prevailing, and a revolution occurred. In this revolution fifty-one men set themselves up as a government, eleven in the city, ten in the Piraeus (both of these groups were to administer the market and the usual civil affairs), and thirty came into power as supreme rulers of the whole state. Some of these happened to be relatives and acquaintances of mine, who accordingly invited me forthwith to join them, assuming my fitness for the task. No wonder that, young as I was, I cherished the belief that they would lead the city from an unjust life, as it were, to habits of justice and really to administer it; so that I was intensely interested to see what would come of it. Of course I saw in a short time that these men made the former government look in comparison like an age of gold. Among other things they sent an elderly man, Socrates, a friend of mine, who I should hardly be ashamed to say was the justest man of his time, in company with others, against one of the citizens to fetch him forcibly to be executed. Their purpose was to connect Socrates with their government, whether he wished or not. He refused and risked any consequences rather than become their partner in wicked deeds. When I observed all this -- and some other matters of similar importance -- I withdrew in disgust from the abuses of those days. Not long after came the fall of the thirty and of their whole system of government. Once more, less hastily this time, but surely, I was moved by the desire to take part in public life and in politics. To be sure, in those days, too, full of disturbance as they were, there were many things occurring to cause offence . . . As it chanced, however, some of those in control brought against this associate of mine, Socrates, whom I have mentioned, a most sacrilegious charge, which he least of all men deserved. They put him on trial for impiety and the people condemned and put to death the man who had refused to take part in the wicked arrest of one of their friends. (*Ep.* VII)

If these events turned Plato away from an active political career in Athens, the problem of government never ceased to occupy a primary place in his thoughts. And, as we shall presently see, Plato came to be involved in political events in Sicily which must have done little to strengthen his belief in the practicality of the ideal expressed in the seventh letter and in the *Republic*, that the only solution was that philosophers become rulers or that rulers become philosophers. Plato's distaste for democracy, then, must be seen as part of a recognition that all existing governments were bad. The fact that Plato lived at the time of the Peloponnesian war would not, of course, have induced in him a high opinion of the Athenian democracy's ability to cope with so serious a situation. Plato's lively sense of the imperfection of men and of the institutions

of his day doubtless has something to do with the combination of visionary ideals and nostalgia for the past in the dialogues. Burnet cannot be far wrong in seeing the dialogues as memorials not only to Socrates but as well to the better days of Plato's own family.^{24}

If Plato has no hesitation to introduce members of his family into the dialogues, he himself shows up rarely. From them, we learn that he was present at the trial of Socrates (*Apology*, 38) and was among those who expressed their willingness to pay the fine if that should be the court's punishment. From the *Phaedo* (59), we learn that due to illness he was not present at the death of Socrates. If the acquaintanceship of members of his family with Socrates, everywhere evidenced in the dialogues, is not pure fiction -- an unlikely possibility -- Plato must have been aware of Socrates throughout his own youth. There is no evidence that he was among the intimates of Socrates, however; indeed, the seventh letter seems to suggest that he was definitively converted to philosophy by the death of Socrates. This does not do away with the fact that he learned from Socrates, of course; other philosophical interests, in the doctrine of Cratylus, the Heracitean, reported by Aristotle (*Metaphysics*. I,6,987a32ff.), may also go back to his youth.

According to Hermodorus, an early biographer, Plato, along with other followers of Socrates, left Athens for Megara after the death of the master; there they spent some time with Eucides. Nothing is known for certain about Plato's life between the death of Socrates and twelve years later when Plato was forty. There are stories of extensive travels, to Egypt, for example, and to Cyrene, but we are only certain of a voyage to Sicily at the age of forty. This was the beginning of an extensive involvement in Sicilian politics which lasted into Plato's old age.

What were Plato's reasons for going to Sicily the first time? He probably went there to converse with Pythagoreans. At any rate, we know that he made the acquaintance of Archytas; Diogenes Laertius tells us that Plato wanted to see Mount Etna. Whatever brought him there, Plato met Dion, the brother-in-law of the tyrant of Syracuse, Dionysius, and thereby hangs the tale. Dion showed great enthusiasm for philosophy, and his contact with Plato brought about a great change in his life. Plato spent perhaps a year at Syracuse on this first visit and, when he returned to Athens, founded the Academy. Plato's first visit took place around 388/387; twenty years later, in 367, Dionysius died. He was succeeded by his son, Dionysius II, and Dion sent a request to Plato, asking him to come and influence the young king probably hoping that the change Plato

had wrought in him could be reproduced in the young tyrant. "He thought that Dionysius might perhaps become one of these [i.e., those who held virtue dearer than pleasure] through the cooperation of the gods. Moreover, if he were to become such a one, the result for him and for the rest of the Syracusans would be the attainment of a life beyond all calculation blessed. Furthermore he felt it to be absolutely necessary that I come to Syracuse as soon as possible to lend a hand in the work." (*Ep.* VII, 327c;Post) Now 60, Plato set off, accompanied it is said, by Xenocrates, a member of the Academy (Diogenes Laertius, IV, 6). When he got to Syracuse, he found that Dion was in jeopardy due to accusations that he was intent on deposing the tyrant. A few months after Plato's arrival, Dion was sent into exile, and Plato found himself in a sticky position. Dionysius *fil*s kept Plato a virtual prisoner, but finally let him go. Back at the Academy, Dion was enrolled as a pupil, but Plato continued to keep in contact with the Syracusan tyrant, not yet in despair of winning him over to philosophy. In 361, Plato responded to the request of Dionysius II that he visit Syracuse once more. The offer was made attractive in a number of ways; a trireme was sent to bring Plato, and Archedemus, a disciple of Archytas, came along to persuade Plato; moreover, Dionysius offered to accept any plan Plato might propose with respect to Dion. Not to be forgotten was the forlorn hope that the tyrant might be won over to philosophy. Thus, for the third time, with his nephew Speusippus, Plato set out for Sicily. Plato failed to accomplish either of the purposes for which he made the trip, and managed to leave Syracuse only at the intervention of Archytas. Plato seems to have continued a correspondence with Dionysius despite these set-backs. Dion had long since despaired of persuading Dionysius of anything; in 357 he returned home with an army and captured Syracuse. Dion held shaky control for three years; in 354 he was assassinated at the instigation of an Athenian companion, Calippus, who set himself up as tyrant. A year later, the party of Dion set up a son of Dionysius as tyrant and the whole sorry business came full circle. The seventh and eighth letters of Plato are addressed to these followers of Dion and partly at least they amount to an apologia by Plato of his role in the sequence of events in Syracuse.

Academy. We have already noted that Plato founded the Academy after returning to his home city from his first visit to Sicily, perhaps after twelve years of exile from Athens. It is suggested that the Academy -- it drew its name from a gymnasium outside the walls of Athens where Plato had a house and garden -- was modeled on already existing schools. That of Eucides at Megara, which Plato had known, the Pythagorean society with which he had contact on his travels, the contemporary school of Isocrates in Athens -- all of these are

thought to have influenced. Plato. The academy was organized somewhat on the lines of a religious society, with a temple and, perhaps, days of special observance. The work of research was carried on by individuals, particularly in mathematics, and it is said that Plato himself posed problems for solution, indicating that such individual efforts were part of the common task. The view of the Academy as a research institute is bolstered by the fact that Eudoxus is said to have moved his whole school of mathematics to Athens and incorporated it with the school of Plato. Now, since men of the caliber of Eudoxus, Speusippus and Xenocrates and Aristotle remained at the Academy for many years, whereas some were there for relatively short periods and then went into the world, very often into politics, it seems necessary to say that there were grades of membership. Plato, of course, was undisputed head; then there would be senior members and junior members. It appears that there were public lectures also, since we have reports that Aristotle remarked on the reaction of those who came to hear Plato's lecture on the good, a lecture, incidentally, of which Aristotle and several others were said to have published versions. There is reason to believe that the attendants at this lecture were candidates for entrance into the Academy, and in this connection, the seventh letter gives us some important indications as to Plato's method of introducing another to philosophy. After a "protreptic" discourse, whetting the candidate's appetite for philosophy, there came the grim picture of the program to be followed if one were to achieve the goal, a program consisting largely of mathematics. Here is Plato's description of his procedure with the younger Dionysius at Syracuse.

When I had arrived, I thought I ought first to put it to the proof whether Dionysius was really all on fire with philosophy or whether the frequent reports that had come to Athens to that effect amounted to nothing. Now there is an experimental method for determining the truth in such cases that, far from being vulgar, is truly appropriate to despots, especially those stuffed with second-hand opinions; which I perceived, as soon as I arrived, was very much the case with Dionysius. One must point out to such men that the whole plan is possible and explain what preliminary steps and how much hard work it will require; for the hearer, if he is genuinely devoted to philosophy and is a man of God with a natural affinity and fitness for the work, sees in the course marked out a path of enchantment, which he must at once strain every nerve to follow, or die in the attempt. Thereupon he braces himself and his guide to the task and does not relax his efforts until he either crowns them with final accomplishment or acquires the faculty of tracing his own way no longer accompanied by the pathfinder. When this conviction has taken possession of him, such a man passes his life in whatever occupations he may engage in, but through it all never

ceases to practise philosophy and such habits of daily life as will be most effective in making him an intelligent and retentive student, able to reason soberly by himself. Other practices than these he shuns to the end. As for those, however, who are not genuine converts to philosophy, but have only a superficial tinge of doctrine -- like the coat of tan that people get in the sun -- as soon as they see how many subjects there are to study, how much hard work they involve, and how indispensable it is for the project to adopt a well-ordered scheme of living, they decide that the plan is difficult if not impossible for them; and so they really do not prove capable of practising philosophy . . . This test then proves to be the surest and safest in dealing with those who are self-indulgent and incapable of continued hard work, since they throw the blame not on their guide but on their own inability to follow out in detail the course of training subsidiary to the project. (340b-341a;Post)

The test described here would have fairly general application to anyone seeking admission to the academy; moreover, if there is validity in the information we have about the Pythagorean society to the effect that in the final analysis it was appeal to the authority of the master which counted -- *ipse dixit* -- the Platonic Academy would rather be defined by its effort truly to teach, to make the doctrine a possession of the student himself; his to defend on its own basis and not because he had gotten it from the master. This could only come as the fruit of long, sustained effort in the company of others so that dispute and dialogue would be the means of achieving the goal. Personal interaction, not the reading of books, is the way to philosophy.

I certainly have composed no work in regard to it, nor shall I ever do so in the future; for there is no way of putting it in words like other studies. Acquaintance with it must come rather after a long period of attendance on instruction in the subject itself and of close companionship, when, suddenly, like a blaze kindled by a leaping spark, it is generated in the soul and at once becomes self-sustaining. (341c-d;Post)

What was this long program of study which would issue in the possession of philosophy? Scholars have always thought that we have a good indication of the practice of the Academy in the educational program set forth in the *Republic*. That program is ordered to the formation of the ruler, of course, but it is predicated on the ideal that the best ruler is the philosopher. The most striking thing about this for the higher education of the guardians scheme is the requirement that ten years, from the age of twenty to thirty, be devoted to study of mathematics. The disciplines are arithmetic, plane geometry, solid geometry,

astronomy and harmonics. We will see later, in our analysis of the *Phaedo* particularly, how the ascetic and pedagogical concerns become one in Plato; it is sufficient to point out now that mathematics are propaedeutic not only because they seem to be the only sciences recognized by Plato, but because they wean the soul from the sensible world and train its eye to see the truly real. After this decade of mathematical work, the future philosopher is to be introduced to dialectics. Plato's reason for postponing what we think of as characteristic of Socrates to this age is of great importance.

There is a danger lest they should taste the dear delight too early; for youngsters, as you may have observed, when they first get the taste in their mouths, argue for amusement, and are always contradicting and refuting others in imitation of those who refute them; like puppy-dogs, they rejoice in pulling and tearing at all who come near them . . . and when they have made many conquests and received defeats at the hands of many, they violently and speedily get into a way of not believing anything which they believed before, and hence, not only they, but philosophy and all that relates to it is apt to have a bad name with the rest of the world. (*Republic*, 539)

Plato is not interested in eristic -- in arguing solely to maintain a point; dialectic for Plato is the search for truth. We are reminded of those dialogues in which Socrates is gently prodding a Sophist to become serious. We shall see later that the precise meaning of the Platonic dialectic is not easy to discern, that it is argued that we have a change of attitude towards it in the dialogues, but the passage in question indicates that prior to the free give and take of dialectic, there must be firm grounding in mathematical subjects. Thanks to this grounding, the student comes to an appreciation of what true being and science are, and is prepared to seek such being and knowledge in a realm which transcends the mathematical. With the taste acquired in the study of mathematics, dialectic, the method of this further pursuit, is not likely to be abused and degenerate into the eristic of the Sophists. Philosophy is patently no game for Plato; it is indulged in with deep earnestness and only continuous, sustained effort can bring one to the goal. It is because philosophy has become detached from such seriousness that, according to the *Republic*, it has fallen into such disrepute. The fault lies with the men and with the community in which they live; both must be changed -- this is the whole point of the *Republic*.

If mathematics was obviously stressed in the Academy, we have some indirect evidence that concern for the natural, biological, world was also present. A fragment of a comic poet, Epicrates, describes the efforts of students at the

Academy poring over a pumpkin trying to decide what species it is. It seems likely that the anecdote is rather evidence of practice in the search for definition than of the kind of interest in the natural world we find later in Aristotle.

The Academy was at once a school, a training ground for politicians, a research institute, and a religious fraternity. There seems to be no single analogue for it in the modern world; what sets it off most definitively from what today is the general view of philosophy, is that the society Plato hoped to form was conceived of as a way of life. The perfection of the intellect was never divorced from the perfection of the man. The identification of the two movements, towards truth and goodness, must be remembered if we are to understand many remarks of Plato concerning the nature of philosophy and the steps to its acquisition. The Academy, founded to form new men and, hopefully, a new society based on the truth of things, was destined to have a long history. Although it changed its character a number of times, and even for a long period drifted into the scepticism against which Plato warned so eloquently, the Academy continued at Athens, though not in precisely the same location, until 529 A.D. when the philosophical schools were closed by Justinian. This was far from being the end of the influence of Plato, however; from the material influence evident in our use of the term "academy" and its derivatives, to the continuing role the dialogues have played as sources for philosophizing, it is evident that, for philosophers consciously and for all men unconsciously, Plato of Athens is a contemporary.

Writings. Plato's written work consists of thirteen letters, some of disputed authenticity, and the dialogues. The function of the letters is fairly obvious from their content; that of the dialogues is obscure and much disputed. That the nature and purpose of the dialogues cannot easily be decided upon becomes evident when we look once more at Plato's seventh letter (which is quite generally considered authentic). Plato there discusses the rumor that Dionysius has taken it upon himself to write about philosophy.

One statement at any rate I can make in regard to all who have written or who may write with a claim to knowledge of the subjects to which I devote myself -- no matter how they pretend to have acquired it, whether from my instruction or from others or by their own discovery. Such writers can in my opinion have no real acquaintance with the subject. I certainly have composed no work in regard to it, nor shall I do so in the future; for there is no way of putting it in words like other studies. (341 b-c; Post)

The same thought is expressed in the second letter.

Take precautions, however, lest this teaching ever be disclosed among untrained people, for in my opinion there is in general no doctrine more ridiculous in the eyes of the general public than this, nor on the other hand any more wonderful and inspiring to those naturally gifted. Often repeated and constantly attended to for many years it is at last gold with great effort freed from alloy. Let me tell you, however, the surprising thing about it. There are men, and a good many of them, too, who have intelligence and memory and the ability to judge a doctrine after examining it by every possible test, who are now old men and have been receiving instruction not less than thirty years who have just reached the point of saying that what formerly they thought most uncertain, now appears to them quite certain and evident; while what seemed most certain then, appears now uncertain. Consider these facts and take care lest you some time come to repent of having now unwisely disclosed the doctrine. It is a very great safeguard to learn by heart instead of writing. It is impossible for what is written not to be disclosed. That is the reason why I have never written anything about these things, and why there is not and will not be any written work of Plato's own. What are now called his are the work of a Socrates grown beautiful and young. Farewell and believe. (314a-c;Post)

Dionysius is urged to read the letter many times and then burn it. Thus far, we may feel that Plato's reluctance to publish is not due to the impossibility of writing his doctrine but to the fact that those who understand it imperfectly or not at all will thereby have occasion to ridicule it; to the initiate, we might feel, the written work would speak and speak the truth. However, in the seventh letter, (the same thoughts are found in the *Phaedrus*), Plato goes on to argue that words cannot be vehicles of the ultimate truth, that the stricture against writing is caution against attempting the impossible. It will be well to record here his argument for the inefficacy of words; not only does it pertain to our present subject, but it will give us a first taste of Platonic doctrine.

For everything that exists there are three classes of objects through which knowledge about it must come; the knowledge itself is a fourth; and we must put as a fifth entity the actual object of knowledge which is the true reality. We have, then, first, a name; second, a description; third, an image; and fourth, a knowledge of the object. (342 a-b;Post)

Plato exemplifies what he means by the case of circle. There is the thing itself -- the circle -- and there is the word "circle." There is the description of the circle

which is composed of nouns and other verbal expressions. Next there are the objects which are drawn and erased, made in wood and destroyed, and so forth. But when the wooden wheel is broken, nothing untoward happens to the circle as such. Finally, there is our knowledge of correct opinion concerning the circle itself. Since our understanding is in our mind where things do not have the qualities sensible bodies have, of all the things mentioned, our knowledge has most affinity with the real object in question, namely, the circle itself. These remarks can be applied to anything whatsoever, mathematical objects, the good and beautiful, artificial and natural things.

For if in the case of any of these a man does not somehow or other get hold of the first four, he will never gain a complete understanding of the fifth. Furthermore these four -- names, descriptions, bodily forms, concepts -- do as much to illustrate the particular quality of any object as they do to illustrate its essential reality because of the inadequacy of language. Hence no intelligent man will ever be so bold as to put into language those things which his reason has contemplated, especially not into a form that is unalterable -- which must be the case with what is expressed in written symbols. (342e343a;Post)

Language seems as much concerned with the particular quality, that is, with the circles we draw and make, and consequently see, touch, etc., as with the essential reality, that is, circle itself. The object of philosophy is knowledge of essential reality which cannot be conveyed by language, since language will be taken to be about sensible particulars or observable instances and the whole discussion will seem involved in contradictions. Names and verbal descriptions and particular instances are necessary if one is to achieve knowledge of essential reality, but once this knowledge is attained its vast difference from the means of achieving it will be seen and, consequently, the impossibility of expressing it in words. But the way is difficult and arduous and Plato once more insists on this.

To sum it all up in a word, natural intelligence and a good memory are equally powerless to aid the man who has not an inborn affinity with the subject. Without such endowments there is of course not the slightest possibility. Hence all who have no natural aptitude and affinity with justice and all the other noble ideals, though in the study of other matters they may be both intelligent and retentive -- all those too who have affinity but are stupid and unretentive -- such will never any of them attain to an understanding of the most complete truth in regard to moral concepts. The study of virtue and vice must be accompanied by an inquiry into what is false and true of existence in general and must be carried on by constant practice throughout a long period, as I said in the

beginning. Hardly after practicing detailed comparisons of names and definitions and visual and other sense-perceptions, after scrutinizing them in benevolent disputation by the use of question and answer without jealousy, at last in a flash understanding of each blazes up, and the mind, as it exerts all its power to the limit of human capacity, is flooded with light. (344a-b;Post)

Now all this would lead us to suspect that the dialogues that have come down to us do not and cannot represent the authentic teaching of Plato such as it could have been learned by a member of the Academy. There is, as we shall see, some ground for this interpretation. Nevertheless, the dialogues do exist, they do convey a doctrine and, as works of Plato, they can be said to convey a Platonic doctrine. That it may not be the very same sort of thing as was taught in the Academy to the initiate need not keep us from studying it with great care.

In one of the passages quoted, we heard Plato describe his writings as giving not his teaching but that of a Socrates grown young and beautiful. Now this will be seen to cover a good number of the dialogues, but certainly not all of them, since there are dialogues in which Socrates does not appear and several in which, though he appears, he is far from being the main speaker. The fact is, as Field has pointed out,^[25] that we would not be well-advised to ask ourselves, what is the purpose or function of the dialogues? Not all the dialogues proceed in the same way nor seem to address themselves to the same audience. Of many of them, particularly of what are called the early dialogues and in which Socrates is far and away the main figure, it can be said that they are addressed to the general public; their effect if not their purpose was to make known to the outsider what interested the Academy and often to induce him to enter. Difficulties arise when we find discussed in the dialogues precisely those points which Plato claimed he would never discuss in writing. Thus, in the *Republic* everything finds its focus in the doctrine of essential realities or Ideas or Forms -- the things that really are and towards which the mind strains in philosophy. Moreover, in the later dialogues the discussions become much more abstruse and difficult than in the so-called "socratic dialogues" and it does not seem likely that they were directed to a popular audience. Field suggests that one possible explanation is that Plato is addressing himself to *philosophers* outside the Academy who could be expected to profit from such advanced discussions which, nevertheless, fall short of any attempt to put into writing statements about essential realities themselves. This is not unlikely, nor is it unlikely that more difficult dialogues should have been used within the Academy as *loci* for discussion. Indeed, it is possible to multiply motives with the inspection of particular dialogues; the important thing is that we refrain from seeking some

one motive which covers all the dialogues without exception. This is important because, while we must accept the fact that there is a distinction between Plato's dialogues and what Aristotle calls his unwritten doctrine (knowledge of which we obtain from Aristotle), it is equally necessary to accept a difference in the philosophical content of the dialogues themselves. It has been maintained that Plato gave no oral teaching, that everything he taught is in the dialogues and finally, that whenever Aristotle ascribes something to Plato which is not to be found in the dialogues it must be rejected as at best a misunderstanding. Ross^[26] has drawn attention both to the implausibility of this image of a Plato completely aloof doctrinally from the members of the society he had formed and to the sufficient evidence for accepting Aristotle's remark that there were indeed things which Plato taught but did not write down. One example is the lecture on the Good given before a public audience. We need say no more of this now, since it can be discussed more fittingly when we come to those doctrines ascribed to Plato which are not clearly present in the dialogues themselves.

There is some evidence that we possess, in the dialogues that have come down to us, all that Plato wrote -- no dialogue is referred to by name in ancient authors that we do not have today. The question is, are all thirty-five of the dialogues attributed to Plato actually his? We know for certain that twenty-four dialogues are generally recognized as authentic works of Plato. These are: *Apology*, *Crito*, *Euthyphron*, *Laches*, *Protagoras*, *Charmides*, *Lyris*, *Gorgias*, *Meno*, *Euthydemus*, *Hippias Maior*, *Cratylus*, *Symposium*, *Phaedo*, *Republic*, *Phaedrus*, *Theaetetus*, *Parmenides*, *Sophist*, *Statesman*, *Philebus*, *Timaeus*, *Critias*, *Laws*. There seems to be insufficient reason to reject: *Ion*, *Hippias Minor*, *Menexenus* and *Epinomis*. The question of the chronological order in which the dialogues were written is often discussed. One way of deciding the question is from internal evidence such as the mention of historical events. The *Apology*, for example, could hardly have been written prior to 399 B.C. when Socrates was put to death. From ancient sources as well as such internal evidence, the *Laws* can be taken to be among the very last things Plato wrote. On the basis of the *Laws* as ultimate, it has been possible to subject the other dialogues to a stylometric test and guess their relative distance from the last work insofar as they are near or far from it in style. It goes without saying that any proposed chronology will amount only to a more or less educated guess.

Why is it important to decide on the order in which the dialogues may have been written? From many points of view, the question obviously has no importance. Acceptance of a chronology often leads to the quest for changes of mind, a search which usually finds its object, and we are left with several

conflicting Platos. Now while it may seem that if one Plato is good several are better, the quest for a unified doctrine is surely to be commended; however, divergence between the dialogues is not something the recognition of which is inextricably wedded to matters of chronology. We shall have to address ourselves to the question of the unity of Plato's thought since some dialogues (those which happen to be put later chronologically) seem to differ considerably from others (which, as it happens, fall towards the beginning of most chronological lists). We will want to see if arguments for the unity of the Platonic method go hand in hand with indifference to the chronological question while those who profess to find more or less irreconcilable differences base their claim on a development from a young to an older and perhaps wiser Plato. We have our own interest in chronology, of course, since, when we were discussing the Socratic Problem we grandly dismissed it in favor of the acceptance of the Socrates who has come down to us in the Platonic dialogues and disavowed all concern with the unknown and mayhap unknowable historical Socrates. Dialogues which have been located as early ones are also often called "Socratic dialogues" because in them Socrates occupies the central position. These dialogues exhibit that interest in definition in moral matters which Aristotle says was characteristic of Socrates, meaning no doubt the historical Socrates. These dialogues are of course artistic presentation, indeed in the judgment of those in a position to say, the best literary style and imagination of Plato are more uniformly present in the early dialogues. There is no point, of course, in denying that the Socrates of these dialogues is, as Plato confessed, a Socrates grown young and beautiful. This cannot simply mean that Socrates has achieved immortality by being caught and held in the net of art; there seems to be as well the admission of conscious idealization. Of course, even if, (though it is impossible), these were simply records of actual conversations, the editorial hand would alter them by selecting which ones were to be preserved for posterity and, doubtless be evident as well in omissions and ordering whereby the vagaries of daily intercourse would take on that beginning, middle and end they seldom have in real life. Plato cannot be regarded as the recording secretary of the Athenian Socratic Society; he is no Boswell. There is nevertheless some evidence that these artistic reconstructions bear an intended relation to what Socrates actually did. We need only mention the admission in the *Republic*, when Socrates urges the study of solid geometry on the part of the guardians, that this science has not yet been discovered. Burnet makes much of this, arguing that if the Socratic Dialogues were pure invention there would be no need for pointing out any anachronisms they might contain; that is, if Socrates were simply the vehicle of Plato's own thought, he could take this as sufficiently evident to those of his readers who had known Socrates and would

not have to make a point of his putting in Socrates' mouth remarks that he could not possibly have made. Dialogues which are placed earlier than the *Republic* can generally be taken to present conversations based on those Socrates actually engaged in and as hewing rather closely to what he actually did say or might have said. Plato, accordingly, either identifies himself entirely with his memory of Socrates or is quite self-effacing with respect to his own interests and allows Socrates to speak for himself. In a dialogue like the *Phaedo*, we find Socrates presenting the doctrine of Ideas without apology on Plato's part. This can mean either that Aristotle is wrong in saying that Socrates never taught this doctrine or that Plato is now using Socrates in a somewhat different way. We adopt the latter possibility and claim that there are dialogues in which Socrates is presented more or less as Plato thought him to have been, whereas in other dialogues Socrates takes on a more or less symbolic function, while things are discussed that Socrates himself never discussed. If this is tenable, our earlier dismissal of the Socratic Problem does not entail that we have no further right to distinguish between Socrates and Plato.

B. The Doctrine of Forms

The distinctive characteristic of the philosophy of Plato is the doctrine of Forms -- the "essential realities" of the seventh letter. In the first book of his *Metaphysics*, where he is offering a synopsis of his predecessors' views on causes, Aristotle gives us brief statement on this fundamental Platonic doctrine, indicating that it is the point of difference between Socrates and Plato.

After the systems we have named came the philosophy of Plato, which in most respects followed these thinkers (i.e., the Pythagoreans), but had peculiarities that distinguished it from the philosophy of the Italians. For, having in his youth first become familiar with Cratylus and with the Heraclitean doctrines -- that all sensible things are ever in a state of flux and there is no knowledge about them -- these views he held even in later years. Socrates, however, was busying himself about ethical matters and neglecting the world of nature as a whole but seeking the universal in these ethical matters, and fixed thought for the first time on definitions; Plato accepted his teaching, but held that the problem applied not to sensible things but to entities of another kind -- for this reason, that the common definition could not be a definition of any sensible thing, as they were always changing. Things of this other sort, then, he called Ideas, and sensible things, he said, were all named after these, and in virtue of a relation to these; for the many existed by participation in the Ideas that have the same name as they. Only the name 'participation' was new; for the Pythagoreans say that

things exist by 'imitation' of numbers, but Plato says they exist by participation, changing the name. (987a29-987b13)

Something of this same movement seems to show up as we pass from the "Socratic Dialogues" to subsequent ones.

Such dialogues as the *Charmides*, *Laches* and *Euthyphro* are asking about a particular moral quality -- temperance, courage, piety -- what is it? Socrates is interested in seeking an answer to such questions as "What is piety?", because he thinks that few people can answer correctly. And they do not even know they cannot answer them. Not knowing what a virtue is, they cannot possess that virtue -- since knowledge of the virtue and possession of it are one and the same thing. Socrates assumes knowledge of Greek in his interlocutor; he does not expect or try to induce puzzlement concerning the grammar of the word "Piety" for example. The word is known and certain actions to which the word is applied are known. The question then becomes, what does the word mean when it is applied to those actions? If we call several actions courageous, isn't there something in all actions that is one and the same? And can't we formulate a definition which will express this one, same thing they have in common?

And therefore, I adjure you to tell me the nature of piety and impiety, which you said that you knew so well, and of murder, and of other offences against the gods. What are they? Is not piety in every action always the same, and impiety, again -- is it not always the opposite of piety, and also the same with itself, having, as impiety, one notion which includes whatever is impious? (*Euthyphro*, 5)

There must be some one thing, piety, in terms of which all pious acts are said to be pious; so too with impiety and the rest. That is what we are seeking when we ask, what is piety? That is why Socrates is impatient with replies to the question which consists of giving an inventory, piling up examples of pious acts.

Remember that I did not ask you to give me two or three examples of piety, but to explain the general idea which makes all pious things pious. Do you not recollect that there was one idea which made the impious impious, and the pious pious? . . . Tell me what is the nature of this idea, and then I shall have a standard to which I may look, and by which I may measure actions, whether yours or those of any one else, and then I shall be able to say that such and such is pious, such another impious. (*Ibid.*, 6)

The occurrence of the word "idea" in this passage permits us to remark that, for our purposes, "Idea" and "Form" are synonyms. The Greek terms they translate both derive from the verb "to see"; their first meaning, accordingly, would be, that which is seen; soon they come to mean the visible shape -- that whereby things seen are distinguished from one another, their visible difference. Gradually the terms come to mean distinguishing mark of specific characteristic without restriction to what can be grasped by the sense of sight. It is just this distinguishing mark, grasped by thought and expressible in words, that Socrates is after in his questions about virtue. Thus, he is after what anyone is after when he asks, "Well, what is goodness?" This question can be thought of as prompted by the bewilderment consequent on noticing what diverse things are called good. It is not necessary that the posing of such a question follow on the recognition of the wider problem it involves. Recall the divisions that Plato gave in the seventh letter. There is the word, the verbal description; there are observable things to which the word is applied and which are accordingly instances or examples of what the word means. It is fairly easy to accept this breakdown; it is the next step that causes difficulty; and Plato, by taking it, seems to go beyond what Socrates' quest involved. Plato asks, must there not be, besides the word "circle" and our verbal expression of what it is and particular circles, the Circle itself? This Circle itself, the essential reality, the Form or Idea -- Plato's view of it is what takes him beyond Socrates to a distinctive doctrine of his own.

In the *Phaedo*, Socrates is made to recount his dissatisfaction with the actual practise of Anaxagoras after he had been struck by the view that Mind governs the universe and had hoped to be told why things are better as they are than otherwise. But Anaxagoras does not tell him this, giving instead the same kind of explanation of things that the other natural philosophers give. Socrates feels that they only recount those things without which the cause cannot be a cause, but leave the cause itself unstated. He turns from natural philosophy to what he calls a second-best way of proceeding -- second-best to discovering the Good which would explain why things are better as they are. The method described gives us one way of seeing into the doctrine of Forms. And because of its generally recognized importance, we reproduce it here.

Socrates proceeded: I thought that as I had failed in the contemplation of true existence, I ought to be careful that I did not lose the eye of my soul; so people may injure their bodily eye by observing and gazing on the sun during an eclipse, unless they take the precaution of only looking at the image reflected in the water, or in some similar medium. So in my own case, I was afraid that my

soul might be blinded altogether if I looked at things with my eyes or tried to apprehend them by the help of the senses. And I thought that I had better have recourse to the world of mind and seek there the truth of existence. I dare say that the simile is not perfect -- for I am very far from admitting that he who contemplates existences through the medium of thought, sees them only in images any more than he who considers them in action and operation. However, this was the method which I adopted: I first assumed some principle which I judged to be the strongest, and then I affirmed as true whatever seemed to agree with this, whether relating to the cause or anything else; and that which disagreed I regarded as untrue. But I should like to explain my meaning more clearly, as I do not think that you as yet understand me . . . There is nothing new, he said, in what I am about to tell you; but only what I have been always and everywhere repeating in the previous discussion and on previous occasions: I want to show you the nature of that cause which has occupied my thoughts. I shall have to go back to those familiar words which are in the mouth of everyone, and first of all assume that there is an absolute beauty and goodness and greatness, and the like; grant me this, and I hope to be able to show you the nature of the cause, and to prove the immortality of the soul . . . Well, he said, then I should like to know whether you agree with me in the next step; for I cannot help thinking, if there be anything beautiful other than absolute beauty should there be such, that it can be beautiful only insofar as it partakes of absolute beauty -- and I should say the same of everything . . . He proceeded: I know nothing and can understand nothing of any other of those wise causes which are alleged; and if a person says to me that the bloom of color, or form, or any such things is a source of beauty, I leave all that, which is only confusing to me, and simply and singly, and perhaps foolishly, hold and am assured in my own mind that nothing makes a thing beautiful but the presence (*parousia*) and participation (*koinonia*) of beauty in whatever way or manner obtained; for as to the manner I am uncertain, but I stoutly contend that by beauty all beautiful things become beautiful. This appears to me to be the safest answer which I can give, either to myself or to another, and to this I cling, in the persuasion that this principle will never be overthrown, and that to myself or to anyone who asks the question, I may safely reply, that by beauty beautiful things become beautiful. (99-100)

The Forms or Ideas are appealed to here through a second best way of arriving at an explanation of the things around us; the best way would be to trace in detail the working of Mind in the universe, which method would enable us to see that things are better the way they are. The second-best way explains something's becoming beautiful or big or two or just by participation in, respectively, beauty,

bigness, twoness or justice. The method of the second-best way consists in turning away from sensible things, since the object of true knowledge cannot be conveyed by sensation (*cf.* 65-66), and concentrating on propositions which are said to be images. They are images, not of the sensible world, but of the intelligible, the world of Forms. The supposition of the method of hypothesis which the passage describes is, as Hackforth puts it⁽²⁷⁾, that the world of discourse is a faithful representation of true being. The method of hypothesis consists in accepting as true as well as everything which follows from its acceptance, while rejecting whatever is incompatible with it. In the *Phaedo*, the hypothesis in question is precisely the doctrine of Forms; if there are Forms or Ideas, then the soul is immortal. We are interested now in the doctrine of Forms and not in the attempt to prove the immortality of the soul.

The situation Socrates presents here is the following. We are in a world surrounded by objects; we speak about these objects and in doing so we are, of course, applying a given word to a number of objects. If we say of an action that it is just, we can also say this of another; if we say of some stones that they are two or three, "two" and "three" can be applied to other groups as well. As Plato will say in the *Republic*, the point he is getting at is suggested whenever we have a group of things which share a common name. Now a just act is one which has the note of justice; this action is not justice itself, nor is that thing one, and so on; nor is the group of all just acts what justice is, since justice is that whereby we recognize the particular acts as just. Justice is not the word, nor is it the mental notion we form and express verbally in the definition. Plato is suggesting that we are led inexorably to recognize another type of entity, something apart from the word, concept or instance: Justice itself. Now earlier in the *Phaedo*, the question has arisen as to how we acquire knowledge of such entities as Justice itself, and so forth, with the true object of every common name. The body is denied any role in such knowledge.

Then must not true existence be revealed to her in thought, if at all? . . . And thought is best when the mind is gathered into herself and none of these things trouble her -- neither sounds nor sights nor pain nor any pleasure -- when she takes leave of the body, and has as little as possible to do with it, when she has no bodily sense or desire, but is aspiring after true being? . . . Is there or is there not an absolute justice?

-- Assuredly there is.

-- And an absolute beauty and absolute good?

-- Of course.

-- But did you ever behold any of them with your eyes?

-- Certainly not.

-- Or did you ever reach them with any other bodily sense? -- and I speak not of these alone, but of absolute greatness, and health, and strength, and of the essence or true nature of everything. Has the reality of them ever been perceived by you through the bodily organs? or rather, is not the nearest approach to the knowledge of their several natures made by him who so orders his intellectual vision as to have the most exact conception of the essence of Each thing which he considers?

-- Certainly.

-- And he attains to the purest knowledge of them who goes to each with the mind alone, not introducing or intruding in the act of thought, sight or any other sense together with reason, but with the very light of the mind in her own clearness searches into the very truth of each; he who has got rid, as far as he can, of eyes and ears and, so to speak, of the whole body, these being in his opinion distracting elements which when they infect the soul hinder her from acquiring truth and knowledge -- who, if not he, is likely to attain to the knowledge of true being? (65-66)

This withdrawal of the mind into itself is, in the *Phaedo*, described in such a way that the movement from the sensible order and the triumph over sensuality are but different ways of viewing the same process. In this fashion, there are steps by which the soul is weaned from the sensible order so that its eye can turn to the true essences of things. This true essence or being of things is not to be thought of as in particulars; it is apart from them, separated, in another realm. The world of the Forms or Ideas is another and better world than that around us. The influence of Orphic and Pythagorean religious doctrines on the Plato who wrote the *Phaedo* is often mentioned and is undeniably there; but if there is mysticism, it is a mysticism which is wedded to science. Plato is not interested in elevating what we might call "existential affinity" with real being above the cognitive grasp of it; in the seventh letter he speaks of those who have a natural affinity with moral ideals but no intellectual capacity. They are not apt students of philosophy and it is philosophy that is being discussed in the *Phaedo*.

If the senses do not enter into the knowledge of reality as constituents of such knowledge, they nevertheless have a role to play in acquiring knowledge. It is the fact that many sensible things have a common name that presents the problem which can lead to the recognition of the Forms. And yet, involved in the recognition of the problem would seem to be the very knowledge we seek. If we are able to recognize some acts as just, must we not have some sort of imperfect knowledge of justice already? A first answer to this problem can be found in the *Meno*, although it is not there explicitly connected with the Forms. In that dialogue, Socrates elicits from a slave boy, by means of carefully ordered questions, the solution to a mathematical problem although the boy has had no training whatsoever in mathematics. Socrates makes use of the incident to suggest that the boy's soul had already possessed the knowledge in question, but had simply forgotten it. What he needed was a diagram and a few questions and the knowledge comes back to him. "And if there have been always true thoughts in him, both at the time when he was and was not a man, which only need to be awakened into knowledge by putting questions to him, his soul must have always possessed this knowledge, for he always either was or was not a man?" (*Meno*, 86) Here we have the Platonism of the *Ode on the Intimations of Immortality*. Issued from the hand of God, the soul lives in the presence of the Forms, of true being; its being placed in a body is the death of that former pure existence and, due to the corrupting and obscuring effect of the body, the previous knowledge is pushed into subconsciousness. When it is elicited by questions, it is rather a matter of remembering (*anamnesis*) than of learning. This notion that the soul is imprisoned in the body and that it must purge itself from the body -- that what we call death is in fact the release and rebirth of the soul to its true life -- is everywhere present in the *Phaedo*. Moreover the notion of *anamnesis* is also involved there. When we say that two sticks are equal, we have an instance of equality. But the recognition of them as equal involves a knowledge of equality which is neither derivable from the instance before us nor from any other instance. Close inspection of any pair of equal sensible things would reveal that they are not perfectly equal. We cannot, then, have gotten the notion of perfect equality, of equality itself, from such instances. Nevertheless, the recognition of instances can prompt us to ask what is the ideal to which they approximate and, further, to ask how we acquired knowledge of such ideas.

Thus far, then, Plato seems to have divided the world into sensible things and the Forms which are the causes of sensible things. If these are the poles of the emerging Platonic universe, there is more than emptiness between. There is first of all the question as to the status of the soul. It would appear to be neither Form nor sensible thing; consequently it must be an intermediary kind of entity.

Moreover, although when it comes to exemplifying what he means by Forms, Plato speaks as often of mathematical as of moral Forms; mathematics itself will introduce a kind of multiplicity which is not that of the sensible world. If 2 and 2 are 4, the two 2's in question need not be sensible instances and, as more than one, they seem to fall short of the Form, Twoness itself. We will see later that this is precisely the problem of intermediate entities which in turn is part of the problem of the unwritten doctrine and the reliability of the testimony of Aristotle. It is well to recognize from the outset that the Platonic universe is susceptible of greater complexity; that the written doctrine of Plato itself makes that universe more complex is something that will emerge.

Republic. We find, in the middle books of the *Republic*, three connected passages which have importance for determining what Plato conceived to be the relationships between the Forms and things and, indeed, among the Forms themselves. These passages present the analogy of the sun and the Good, the "divided line," and the allegory of the cave. The Fifth Book begins with a discussion of the difference between knowledge and opinion. To know has as its object what truly is; not to know in anyway would have nothing or non-being to respond to it. To have opinion or to believe has as its object not what truly is, for opinion can be false, nor what wholly is not. Forms or Ideas are what truly are, and knowledge will be of them. Opinion or belief has as its object the instances of the Ideas which can be grasped through the senses. One who has only opinion is one who recognizes some things as beautiful, for example, but does not know what Beauty is. He who knows can discern the essence of beauty as well as the things which participate in or imitate it, and he never confuses the one with the other. The status of the objects of sense perception is somewhat anomalous, according to this view. They are in the realm of seeming, of phenomena, of what appears. To know what sensible things are, their true nature, is to know something other than the things grasped by the senses. Ignorance, opinion, knowledge, these three -- the greatest, of course, is knowledge and the passages which interest us now speak of the transition from the other mental states to knowledge.

The transition in question is once more triumph over the body as well as progress in knowledge, but the process is taken as aimed at the Good. The highest knowledge is possession of the Good.

. . . I am certain that you have heard the answer many times, and now you either do not understand me or, as I rather think, you are disposed to be troublesome, for you have often been told that the idea of good is the highest knowledge, and

that all other things become useful and advantageous only by their use of this. . . . Do you think that the possession of all other things is of any value if we do not possess the good or the knowledge of all other things if we have no knowledge of beauty and goodness? (*Republic* VI, 504-5)

The great difficulty is to acquire this knowledge, for what is the Good? Whatever it is, the guardians of the ideal commonwealth are going to have to know what it is. Not that Plato conceives the primacy of knowledge of the Good as of moment only for practical concerns; we shall see that this knowledge is primary with respect to mathematics as well. It is the comparison of the sun and the Good which begins to clarify the function Plato assigns to the Good and to knowledge of it.

In the visible order more is required than eyes to see with and things to see; there is need as well of light. The source of light is preeminently the sun, and the sun is spoken of by Plato as the child the Good has created in the visible world as a symbol of itself.

And the soul is like the eye: when resting upon that on which truth and being shine, the soul perceives and understands and is radiant with intelligence; but when turned towards the twilight of becoming and perishing, then she has opinion only, and goes blinking about, and is first of one opinion and then of another, and seems to have no intelligence. . . . Now, that which imparts truth to the known and the power of knowing to the knower is what I would have you term the idea of good, and this you will deem to be the cause of science, and of truth in so far as the latter becomes the subject of knowledge; beautiful too, as are both truth and knowledge, you will be right in esteeming this other nature as more beautiful than either; and, as in the previous instance, light and sight may be truly said to be like the sun, and yet not to be the sun, so in this other sphere science and truth may be deemed to be like the good, but not the good: the good has a place of honor yet higher. (508)

We have here the suggestion of a hierarchy among the Forms or Ideas themselves. Just as the Forms take precedence over sensible things, so in the realm of Forms, the Good takes precedence over the others: Whether or not we are to take all other Forms to be on the same level, there is no doubt as to the primacy of the Good. "And so with the objects of knowledge: these derive from the Good not only their power of being known, but their very being and reality; and Goodness is not the same thing as being, but even beyond being,

surpassing it in dignity and power." (509) We will see the implications of that passage drawn out by Neoplatonism.

We have seen a division of opinion from knowledge, of the visible from the invisible. Plato proposes to illustrate this by an example.

Now take a line which has been cut into two unequal parts, and divide each of them again in the same proportion, and suppose the two main divisions to answer, one to the visible and the other to the intelligible, and then compare the subdivisions in respect of their clearness and want of clearness, and you will find that the first section in the sphere of the visible consists of images. . .

(*Republic* VI, 509-510) Let us first set before us the illustration Plato intends.

Visible		Intelligible		
Images	Visibilia	Mathematicals	Forms	(Objects)
<i>eikasia</i> (imagining)	<i>pistis</i> (belief)	<i>dianoia</i> (thinking)	<i>noesis / episteme</i> (knowledge)	(States of Mind)

Previously the role of the sun in the visible world provided a bridge to the discussion of the role the Good plays in the intelligible world; here Plato first examines that part of the line devoted to the visible. By images Plato tells us he means "in the first place, shadows, and in the second place, reflections. . . ." what we have designated as visibilia in our diagram includes "animals which we see and everything that grows and is made." The images (shadows or reflections) relate to *visible* things as copy to reality; moreover, knowledge of a thing through its image is likened to opinion, while knowledge of the visible thing in itself is likened to science. The same relation is to be found in that part of the line devoted to the intelligible.

There are two subdivisions, in the lower of which the soul uses the figures given by the former division as images; the enquiry can only be hypothetical, and instead of going upwards to a principle descends to the other end; in the higher of the two, the soul passes out of hypotheses, and goes up to a principle which is above hypotheses, making no use of images as in the former case, but proceeding only in and through the ideas themselves.

This is a difficult passage and is recognized to be such by Plato, who goes on to clarify his meaning. Let us follow his development.

. . . you will understand me better when I have made some preliminary remarks. You are aware that students of geometry, arithmetic and the kindred sciences assume the odd and the even and the figures and three kinds of angles and the like in their several branches of science; these are their hypotheses, which they and everybody are supposed to know, and therefore they do not deign to give any account of them either to themselves or others; but they begin with them, and go on until they arrive at last, and in a consistent manner, at their conclusion. . . . And do you know also that although they make use of the visible forms and reason about them, they are thinking not of these but of the ideals which they resemble; not of the figures which they draw, but of the absolute square and the absolute diameter, and so on -- the forms which they draw or make, and which have shadows and reflections in water of their own, are converted by them into images, but they are really seeking to behold the things themselves, which can only be seen by the eye of the mind? (510)

The geometer makes use of diagrams drawn in sand; these diagrams have the status of the objects which fall into the second division and as such they can have reflections and shadows (we can see now perhaps why Plato puts things made into the class of visible things opposed to images). For the geometer, however, the diagram is only a means of turning his eye on the Form itself, e.g., triangle itself, of which the illustration is but an image and copy.

So much is fairly clear; what is not so clear is what Plato means by hypothesis. The hypotheses of the geometer seem to be the axioms which are not proved by the geometer but assumed to be true. The suggestion is that the soul must rise above these hypotheses.

And when I speak of the other division of the intelligible, you will understand me to speak of that other sort of knowledge which reason herself attains by the power of dialectic, using the hypotheses not as first principles, but only as hypotheses -- this is to say, as steps and points of departure into a world which is above hypotheses, in order that she may soar beyond them to the first principle of the whole; and clinging to this and then to that which depends on this, by successive steps she descends again without the aid of any sensible object, from ideas, through ideas, and in ideas she ends. (511)

Now it is difficult to avoid interpreting all this as coming down to the following. Among the sciences, mathematical sciences enable the soul to rise above the sensible world; they accomplish this by making use of sensible diagrams which are not the object of study, but which are instruments enabling the eye of the soul to gaze upon Forms. The mathematical concern with Forms, however, does not establish their existence so much as it assumes that there are Forms. Consequently, the mind must turn to these Forms which are assumed to exist and rise to the point where their existence is no longer a matter of assumption. This would seem to be precisely the acquisition of knowledge of the Good, knowledge which permits us to perform a contrary movement, thanks to which, by what appears to be a deduction, all of the Forms are seen to depend upon the Good.

This concern with the Forms as such, without having recourse to the sensible as mathematics must, is called Dialectic. The culmination of philosophy, it is, in a sense, philosophy. Moreover, the methodology of this passage bears obvious resemblances to that presented in the *Phaedo* (99d-102a). We recall that there Socrates had introduced a second-best method which consisted of assuming something to be true and accepting what follows and rejecting what does not follow from it. This aspect of the methodology of the *Phaedo* answers to the description of the procedure of mathematics in the *Republic*. What is more, there is a hint of the procedure of dialectic, of the passage beyond hypotheses, in the *Phaedo*.

And if anyone assails you there, you would not mind him, or answer him until you had seen whether the consequences which follow agree with one another or not, and when you are further required to give an explanation of this principle, you would go on to assume a higher principle, and a higher, until you found a resting-place in the best of the higher. (101a)

Let us turn now to the allegory of the cave. In a deep cave prisoners are chained with their backs to the entrance, facing the wall, a condition they have been in since childhood. Behind them a fire burns, but between them and the fire is a raised screen. Behind it walk men carrying images of men and animals, which they hold up in such a way that the shadows of the images are projected on the wall for the prisoners to see. It is only these shadows that the prisoners have ever seen; they cannot turn and look at the graven images. Due to the acoustics of the cave, when those carrying the images speak, the sound seems to come from the shadows on the wall, and if the prisoners talked among themselves, their words would be taken to refer to the shadows before them. ("You have

shown me a strange image, and they are strange prisoners. -- Like ourselves, I replied. . . ." 515)

Suppose now that one of these prisoners is unchained and turned towards the light and the images. His eyes would be dazzled, and he would not discern them clearly. If questioned, he would say that the images are less real than the shadows he is used to, and since his eyes would ache at the brightness of the fire, he would want to return to his wall and shadows. But suppose he were dragged back through the long entrance to the mouth of the cave, and out into the sunlight which had never penetrated to his original habitat. He would be doubly dazzled now and unable to look at any of the things he is told are real. Here; too, he would first see their shadows and images reflected in water, and only later the things themselves. Finally he would turn his gaze to the heavenly bodies and, last of all, the sun itself could be studied in its reflections.

We shall not now concern ourselves with Plato's penchant for telling a story like this one to illustrate or substitute for a difficult doctrine. What we must be concerned with is the connection between the three passages we have considered: Sun/Good; Divided Line; the Cave. In the first of these passages, Plato has distinguished between the visible and intelligible, the world of becoming and the world of being. By an analogy with sight and its need for the sun by whose light the visible is seen, the Good is spoken of as that Form or Idea in terms of which all other Forms are intelligible. In the divided line passage, a gradation of mental states was pointed out, answering to the way in which the mind must be led from its immersion in the sensible world through mathematics to dialectic which attempts to analyse Ideas into the Good and then deduce the structure of the world of Ideas from the Good. The cave parable seems to go over the steps of this ascent in story fashion, returning to the symbol of the sun -- the child of the Good in the visible world.

It is after the story of the cave that Plato outlines the various sciences which, we have already seen, were to constitute the higher studies of the guardians. These are all on the intelligible plane, of course, and it is a question of arithmetic; geometry, plane and solid; astronomy and harmonics. Finally, by means of dialectic, one turns to the Ideas themselves without any dependence on the visible order. We might ask ourselves here whether Plato leaves us with several sciences or one alone, dialectic. Cornford is of the opinion that the five disciplines mentioned are truly sciences for Plato. He observes that Plato has added solid geometry to the four traditionally recognized, and adds,

These sciences are here described and criticized with respect to their power of turning the soul's eye from the material world to objects of pure thought. They are the only disciplines recognized by Plato as sciences in the proper sense, yielding *a priori* certain knowledge of immutable and eternal objects and truths. For him there could be no "natural science" no exact knowledge of perishable and changeable things.^{28}

One wonders whether this judgment is in accord with Plato's remark on these disciplines in 534. In Cornford's own translation, "From force of habit we have several times spoken of these as branches of knowledge; but they need some other name implying something less clear than knowledge, though not so dim as the apprehension of appearances." (p. 254) Shortly thereafter Plato groups these mathematical disciplines under "intelligence concerned with true being" and we recall that mathematics earlier seemed to be drawn into dialectic itself: "... because they start from hypotheses and do not ascend to a principle, those who contemplate them appear to you not to exercise the higher reason upon them, although when a first principle is added to them they are cognizable by the higher reason." (*Republic* VI, 511d) It is as if mathematics is distinct from dialectic only in its imperfect state; at the term of the ascent, all is one science -- triangle itself, circle itself, etc., are Forms and thus objects of dialectic.

To summarize our discussion thus far: it was the search for definitions which led Plato, following Socrates, to the view that there are Forms or Ideas in which individual sensible things participate. How do we recognize the need for Ideas or Forms? "Whenever a number of individuals have a common name, we assume them to have also a corresponding idea or form." (*Republic*, X, 596) Such Ideas are the objects of knowledge as distinct from opinion. Not only are there Ideas, but they have certain relations, either of mutual exclusion (coldness is opposed to and excludes heat and vice versa -- cf. *Phaedo*, 102a-105b) or of subalternation (whatever participates in threeness also participates in oddness). In the *Republic* the point is seemingly made that there is one supreme Form or Idea in terms of which all the others have their being and their intelligibility. That is why true knowledge, which is of true being or the Ideas, must ultimately terminate in the Idea of the Good.

It should be pointed out that Plato explicitly excuses himself, in the *Republic* and elsewhere, from saying what the Idea of the Good, or indeed any other Idea, truly is, or from saying what exactly the method he calls dialectic is. We have seen that, in the seventh letter, Plato disavows ever having written about his doctrine of Ideas; surely this is why Plato's thought as it has come down to

us in the dialogues is in so many respects both teasingly attractive and maddeningly frustrating. Despite the difficulties, the dialogues we have thus far looked at present a picture of a realm of entities over, and above, separate from, the sensible things in the world around us; these separate entities are true being -- a note they have largely because of their immunity from change and apparent contradiction. They are causes of sensible things because these exist by participation in or imitation of the Forms. Obviously it is not easy to see how there can be any one type of relation of particular changeable things to the world of Forms, especially when we consider the difference between the two types of things which usually figure in attempts to describe the theory of Ideas, namely, mathematical and moral ideals. It is easier to see how a particular action can be described as striving to imitate a perfect pattern of action than to see how a figure in the sand is striving to imitate triangularity itself. Not that the Forms are restricted to the essential realities behind mathematical and moral actions. We have just quoted a passage in the *Republic* which makes it clear that any common name requires that we posit a Form -- even "bed." In the *Phaedrus*, there is a mythical account which speaks of the Ideas as dwelling above the heavens. "There abides the very being with which true knowledge is concerned: the colorless, formless, intangible essence, visible only to mind, the pilot of the soul" (247) Since these words occur in a myth, we need not take too seriously, perhaps, the location of the Ideas. Nevertheless, because Plato stresses so strongly the separation of the Ideas from sensible particulars, the question as to where these Ideas are naturally arises. Doubtless it is legitimate to point out that, since the Ideas are conceived to be utterly different from things that are in place, the question is nonsense. For all that, Plato is quite clear as to where the Ideas are not: they are not in sensible things and they are not concepts in our minds. The latter denial is made most emphatically in the *Parmenides*, a dialogue which, along with several others, can be looked upon as a product of Plato's own critical reflection on the doctrine of Ideas and the method of dialectic he had set forth with reference to the Ideas. We can point out that, already in the *Phaedrus*, which is thought to have been written prior to the dialogues we shall be considering in the next section, a slightly different conception of method is described. Socrates, looking back over a previous discussion, points out that it involved two principles: (1) a gathering of particulars under one Idea which is then defined; that is, first a process of generalization (*synagoge*); (2) the division into species. "I am myself a great lover of these processes of division and generalization; they help me to speak and to think. And if I find any man who is able to see 'a One and Many' in nature, him I follow, and 'walk in his footsteps as if he were a god.' And those who have this art, I have hitherto been in the habit of calling dialecticians; but God knows

whether the name is right or not." (266) Let us turn now to those dialogues in which Plato casts doubt on the doctrine of Ideas itself and has things to say about dialectic which seem quite different from the descriptions he has previously given.

{24} Burnet, *Greek Philosophy, Thales to Plato* (London: Macmillan, 1928), p. 208.

{25} C. C. Field, *Plato and His Contemporaries* (Oxford: Clarendon Press, 1930), pp. 5911.

{26} Sir David Ross, *Plato's Theory of Ideas*, (Oxford: Clarendon Press, 1951), Chap. IX.

{27} See R. Hackforth, *Plato's Phaedo* (Cambridge, England: University Press, 1952), p. 138.

{28} F. M. Cornford, *The Republic of Plato* (London: Oxford, 1941), p. 236.

C. The Crisis in Plato's Thought

Under this heading we intend to examine a number of dialogues, in particular four which seem intended by Plato to form a group, the *Parmenides*, *Theaetetus*, *Sophist* and *Statesman*. In them there seems to be a change of attitude on Plato's part as to the nature of the Ideas, their relations among themselves and their relevance for knowledge of the physical world. How radical this shift is and indeed whether it constitutes a shift at all, are matters of dispute among scholars. Quite apart from this controversy, there is a *prima facie* shift in attitude at the outset of the *Parmenides*; where before the doctrine of Ideas was assumed as familiar and eminently reasonable, there is now hesitation and doubt. The devil's advocate in the dialogue is Parmenides, a significant fact, and Socrates, grown young and beautiful, is in this dialogue the one to be convicted of ignorance.

The *Parmenides*, *Theaetetus*, *Sophist* and *Statesman*, often called the metaphysical dialogues, are quite obviously meant to be read together. One binding note is the meeting of Socrates with Parmenides with which the *Parmenides* begins; this is referred to in the *Theaetetus* (183E) as well as the *Sophist* (217G) in such a way that the reference seems obviously to be to the dialogue, *Parmenides*. The *Theaetetus* ends in such a way that the *Sophist* is its immediate continuation; the *Statesman* refers to the *Sophist* as to its immediate predecessor. The four dialogues form a literary whole and, as we shall see, involve a progression of thought of the utmost importance. There is some evidence that the *Theaetetus* may have been written in an interval between the opening section and the dialectical close of the *Parmenides*, but this in no way affects the order intended by Plato. We are going to try to present in as neutral and objective a way possible the content of these dialogues, with special reference to the doctrine of Ideas; afterwards we shall indicate the extremes of interpretation these dialogues have prompted with respect to the continuity of Plato's thought in the dialogues generally.

We have already discussed, in our section on *Parmenides*, the reasons for accepting the meeting of Socrates, Parmenides and Zeno as historical. This is not to say, of course, that the *Parmenides* of the dialogue represents the philosophical position which historically was his. There is, nevertheless, good reason why he should have been chosen to interrogate the young Socrates on the doctrine which, again only in the dialogues, is characteristic of him. Plato's brothers, Glaucon and Adeimantus, and his half-brother, Antiphon, figure in the opening section, since it is Antiphon's recounting of someone's recollection

of the visit of Parmenides and Zeno that forms the body of the dialogue. Zeno is said to have given one of his arguments in favor of the One of Parmenides, after which Socrates poses a question which leads into the theme of the first part of the dialogue. "What is your meaning, Zeno? Do you maintain that if being is many, it must be both like and unlike, and that this is impossible, for neither can the like be unlike, nor the unlike like -- is that your position?" (127) It is, and Socrates is right in seeing that all Zeno's arguments aim at showing the non-being of the many. Then Socrates expresses the doctrine which will be shown to involve many difficulties not alluded to in the earlier dialogues.

But tell me, Zeno, do you not further think that there is an idea of likeness in itself, and another idea of unlikeness, which is the opposite of likeness, and that in these two, you and I and all other things to which we apply the term many, participate -- things which participate in likeness become in that degree and manner like; and so far as they participate in unlikeness become in that degree unlike, or both like and unlike in the degree in which they participate in both? And may not all things partake of both opposites, and be both like and unlike, by reason of this participation? -- Where is the wonder? Now if a person could prove the absolute like to become unlike, or the absolute unlike to become like, that, in my opinion, would indeed be a wonder; but there is nothing extraordinary, Zeno, in showing that the things which only partake of likeness and unlikeness experience both. Nor, again, if a person were to show that all is one by partaking of one, and at the same time many by partaking of many, would that be very astonishing. But if he were to show me that the absolute one was many, or the absolute many one, I should be truly amazed. (129)

We have here the doctrine of Ideas, familiar from the earlier dialogues; while one Idea cannot be another, since this would involve a contradiction, it is not surprising that sensible things be both X and not-X. Socrates is both small and large, small with respect to Alcibiades, large with respect to Xanthippe. He is not taller by a head because he might also be smaller by a head; rather he is small by participating in smallness, large by participating in largeness. By means of participation, particular things can involve a veritable hodgepodge but the Ideas must be solitary and serene, utterly other one another, and so forth. Generally, the Ideas other than the Good are spoken of as if they were all on the same level; participation is of sensible things in Ideas, not of one Idea in another. It is the very thing that Socrates here says he would be surprised and amazed to be shown, that the sequence of the dialogues which now interest us seems aimed at showing. If however, as I just now suggested, some one were to abstract simple notions of like, unlike, one, many, rest, motion, and similar

ideas, and then to show that these admit of admixture and separation in themselves, I should be very much astonished. . . . I should be far more amazed if any one found in the ideas themselves which are apprehended by reason, the same puzzle and entanglement which you have shown to exist in visible objects." (129-130)

The theory of Ideas interests Parmenides and he begins to question Socrates about it. Socrates is sure there are Ideas of one and many, likeness and unlikeness; he is equally sure that there are Ideas of the just, the beautiful and the good. In other words, mathematical and moral Ideas present no problem; however, when asked if there is an Idea of man apart from us, and of fire and water, Socrates hesitates. Parmenides increases the difficulty by asking if Socrates would maintain that there are Ideas of mud, dirt, hair and other vile and paltry things. . . .

Certainly not, said Socrates; visible things like these are such as they appear to us, and I am afraid that there would be an absurdity in assuming any idea of them, although I sometimes get disturbed and begin to think that there is nothing without an idea; but then again, when I have taken up this position, I run away, because I am afraid that I may fall into a bottomless pit of nonsense and perish; and so I return to the ideas of which I was just now speaking, and occupy myself with them. (130)

Parmenides, however, reassures Socrates and tells him the time will come when he will no longer despise even the meanest of things. We remember that in the *Republic* that any common name was taken to be sufficient indication that an Idea was involved. This may be thought of as the first problem presented, namely, the extent of the world of Ideas. The second problem turns around the relation between particular things and Ideas.

Socrates maintains that there are Ideas "of which all other things partake, and from which they derive their names." (131) Parmenides wants to know the nature of this partaking. There are two possibilities: either the thing partakes of the whole of the Idea, or a part of it. The first possibility does not seem acceptable, for then something which is one will be in different places. "Because one and the same thing will exist as a whole at the same time in many separate individuals, and will therefore be in a state of separation from itself." (131) The Idea cannot be like a cloth that covers many men, since only a part of the cloth is over any individual head. This seems to leave only the possibility that individuals partake of part of the Idea. But this, too, leads to an absurdity. If a

small thing is small by partaking of only a part of smallness, then the Idea of smallness would be larger than the small thing. How, then, can all things participate in Ideas if they are unable to partake of them as wholes or as parts?

Parmenides then takes another line.

I imagine that the way in which you are led to assume one idea of each kind is as follows: You see a number of great objects, and when you look at them there seems to you to be one and the same idea (or nature) in them all; hence you conceive of greatness as one. . . . And if you go on and allow your mind in like manner to embrace in one view the idea of greatness and of great things which are not the idea, and to compare them, will not another greatness arise, which will appear to be the source of all these? . . . Then another idea of greatness now comes into view over and above absolute greatness and the individuals which partake of it; and then another, over and above all these, by virtue of which they will all be great, and so each idea instead of being one will be infinitely multiplied. (132)

This argument, that participation involves an infinite regress, since similars will always call for an Idea to explain their similarity, is more devastating than the previous criticism, and recurs in somewhat different form in Aristotle's critique of Plato. It is important to see that Plato first poses the difficulty.

Socrates attempts to avoid the infinite regress in a very interesting manner. "But may not the ideas, asked Socrates, be thoughts only, and have no proper existence except in our minds, Parmenides? For in that case each idea may still be one, and not experience this infinite multiplication." (132) But a thought must be a thought of something, and of something which is a single form or nature that the mind recognizes as one and the same in many. "Then, said Parmenides, if you say that everything else participates in the Ideas, must you not say either that everything is made up of thoughts, and that all things think; or that they are thoughts but have no thought?"

Socrates agrees that his suggestion is irrational and attempts to restate the relation of particulars to the Ideas. "In my opinion, the ideas are, as it were, patterns fixed in nature, and other things are like them, and resemblances of them -- what is meant by the participation of other things in the ideas is really assimilation to them." (132) But this is really no improvement. If things are like the Ideas, then inevitably the Ideas must be like the things which are like them; that is, Idea and participants must both participate in some further Idea of

likeness, and we are back to the infinite regress Socrates wants to avoid. What is needed, Parmenides suggests, is some other mode of participation than that of resemblance, not that there are not many other difficulties.

There are many, but the greatest of all is this: If an opponent argues that these ideas, being such as we say they ought to be, must remain unknown, no one can prove to him that he is wrong, unless he who denies their existence be a man of great ability and knowledge, and is willing to follow a long and laborious demonstration; he will remain unconvinced, and still insist that they cannot be known. (133)

The whole doctrine of Ideas becomes utterly trivial if it cannot be shown that they must exist. The Ideas, being absolute essences, cannot exist in us, for that would make them relative to us. Parmenides now suggests that the difficulties concerning participation arise from Socrates' attempt to talk about the Ideas in terms of something other than Ideas. But if Ideas are what they are

in relation to one another their essence is determined by a relation among themselves, and has nothing to do with the resemblances, or whatever they are to be termed, which are in our sphere, and from which we receive this or that name when we partake of them. And the things which are within our sphere and have the same names with them, are likewise only relative to one another, and not to the ideas which have the same names with them, but belong to themselves and not to them. (133)

One man is master, another slave, and there is nothing absolute about this: one is relative to the other. So too the Idea of mastership is relative to the Idea of slavery. The one realm would seem to have nothing to do with the other. Moreover, absolute knowledge will answer to these absolutes, to Ideas, and the kind of knowledge we have will answer to the relative things around us. In other words, the Ideas will be unknown to us, since we do not have absolute knowledge i.e., the Idea of knowledge, but simply the knowledge we have. Absolute knowledge sounds like the kind of knowledge God would have, Socrates agrees, but that leads, Parmenides observes, to the blasphemous conclusion that God would have no knowledge of us. The reason is that God, having absolute knowledge, would have knowledge of absolute things which are unrelated to the things around us or to us.

No one can think that the difficulties here put in the mouth of Parmenides could be put forward lightly by the author of the earlier dialogues; indeed, we

might think that Plato is here making a public rejection of his previous views. It soon becomes apparent, however, that these objections amount to a prelude to a new program of approach to the Ideas.

These, Socrates, said Parmenides, are a few, and only a few of the difficulties in which we are involved if ideas really are and we determined each one of them to be an absolute unity. He who hears what may be said against them will deny the very existence of them -- and even if they do exist, he will say that they must of necessity be unknown to man; and he will seem to have reason on his side, and as we were remarking just now, will be very difficult to convince; a man must be gifted with very considerable ability before he can learn that everything has a class and an absolute essence; and still more remarkable will he be who discovers all these things for himself, and having thoroughly investigated them is able to teach them to others. (135)

It will be noticed that Parmenides does not consider that the difficulties raised cancel out the theory of Ideas; the next move is not to reject Ideas, but to devise a way to defend them against such objections as have been raised. Plato undoubtedly takes all the liberty we can allow him when he makes Parmenides the defender of the Ideas, although they have in common with his One the notion of immutability and separation from the things of sense experience. The youth of the Socrates of the dialogue and the eminence of Parmenides, make it fitting that Socrates be instructed in the art which will enable him to defend the Ideas against such objections as Parmenides has been raising, namely, the art of dialectic. The rest of the dialogue consists of the exemplification of this art by discussing the theory of Parmenides that being is one. What Socrates must learn to do is to consider not only the consequences which follow from a given hypothesis but also the consequences flowing from denying the hypothesis -- "that will be still better training for you." (136) The important thing to notice is that this new conception of dialectic is introduced with a view to defending the Ideas.

And yet, Socrates, said Parmenides, if a man fixing his attention on these and the like difficulties, does away with ideas of things and will not admit that every individual thing has its own determinate idea which is always one and the same, he will have nothing on which his mind can rest; and so he will utterly destroy the power of reasoning, as you seem to me to have particularly noted. (135)

The problem of the One and the Many has been raised by the initial criticism of the Ideas in the first part of the Parmenides; if there is an absolute Idea from

which many things receive their name, what is the relation between that one and these many? It seems that the many cannot participate in it as a whole nor as part. A new explanation of participation is called for and, at the same time, an appreciation of the vagaries of "one" and "many." This is at least one of the functions performed by the dialectical training of the second part of the dialogue. The position of Parmenides is that being is one, or the One is. The dialectical exercise can be thought of as testing eight hypotheses, which is somewhat surprising since we should expect only two, namely, the One is and the One is not. What has happened is that each of these is broken into four. Thus, on the supposition that the one is, it is shown (I) that it cannot exist and that it admits of no predicates whatsoever; it is then shown (II) that if One is, it exists, can be known, spoken about, etc. Then (III) that if the One exists, the others (the many) are susceptible of contradictory predicates and (IV) that nothing can be predicated of them. On the other hand (V), if the One does not exist, each member of opposed predicates belong to it, and (VI) neither of such predicates can be said of it. Finally, (VII) if the One does not exist, the others admit of both of contrary predicates and (VIII) of neither of contrary predicates.

The conclusion of this exercise is not precisely positive in tone. Let this much be said; and further let us affirm what seems to be the truth, that, whether one is or is not, one and the others in relation to themselves and one another, all of them, in every way, are and are not, and appear to be and appear not to be.
(166)

There have been many diverse interpretations of the significance of this exercise in dialectic; a recent book groups them under five headings: Anti-Eleatic, according to which the point is the refutation of Parmenides; Neoplatonic, according to which the indirect point is that the One is above and beyond our efforts of understanding; the Hegelian, whereby we have here a foretaste of the Hegelian logic; the Logical, according to which this is an exercise in formal logic; and, finally, the Metaphysical interpretation which sees the testing of the hypotheses as incorporating positive statements about reality.^[29] The puzzling truth is that there is some merit in each of these interpretations. It is not our purpose, of course, to present these divergent views and judge them; it is sufficient to have mentioned their existence. Our own procedure will be to look for any influence of the dialectical training in the dialogues we know to have been conceived as sequels to the Parmenides.

Theaetetus. -- After he has brought forward objections to the Ideas which the youthful Socrates confesses are devastating, Parmenides adds that without the Ideas there is no anchor for knowledge. This suggests that knowledge itself must be reexamined against the background of the difficulties facing the doctrine of Ideas and, fittingly enough the *Theaetetus* concerns itself precisely with knowledge.

In the opening section of the dialogue, the sight of the wounded Theaetetus being brought back from battle reminds Euclid of Socrates' estimate of the mathematician when Theaetetus had been young and how well Socrates' predictions of his future success have been borne out. This provides an occasion for having an early conversation between Socrates and Theaetetus which had been written down read for the benefit of Terpsion, who has often desired to hear it. The dialogue proper is thus introduced as a memorial to Theaetetus fallen in battle.

The dialogue proper begins with Socrates and the mathematician, Theodorus, a Cyrenian, who introduces his student Theaetetus as worthy of the attention of Socrates. Theaetetus having been singled out as a knowledgeable boy, Socrates poses to him the question that has long bothered him and which he has never been able to solve to his satisfaction: what is knowledge? The initial exchange is reminiscent of the early dialogues and indeed this one is often compared with the *Charmides*. Theaetetus first says that knowledge is what he learns from Theodorus, sciences like geometry, as well as the art of the cobbler and other craftsmen. Socrates protests that Theaetetus is giving him much more than he asks for; he wants a single answer, a definition of knowledge, not an enumeration of men who have knowledge. Socrates indicates the kind of answer he wants by pointing out that if he asked Theaetetus what cobbling is he would want to be told that it is the art of making shoes, if he asked him what carpentering is he would want to be told it is the art of making wooden implements. In other words, the answer sought is a definition. Now can Theaetetus give that kind of answer to the question, what is knowledge? If Socrates asked what clay is, Theaetetus would not tell him that there is a clay of potters, a clay of oven-makers, and another of brick-makers.

In the first place, there would be an absurdity in assuming that he who asked the question would understand from our answer the nature of 'clay', merely because we added "of the image-makers," or of any other workers. How can a man understand the name of anything, when he does not know the nature of it?
(147)

Now this is just the nature of Theaetetus' response to the question, what is knowledge? Theaetetus, of course, understands Socrates well enough, since he has recently made a successful generalization concerned with roots in mathematics, but he says that he is unable to give Socrates the kind of answer he wants to his question.

It is at this point that Socrates gives the description of his method of questioning as a kind of midwifery, a *maieutic*, which will bring forth from Theaetetus the answer to the question if the answer is within him. We have discussed this section in Chapter One of this part when Socrates himself was our main interest. Here it raises another point. We have seen that, in the *Meno*, Plato speaks of learning as recollection (*anamnesis*); now, as Cornford points out in his commentary on our dialogue (pp. 27-8), the exchange which in the *Meno* leads up to the statement of the nature of recollection is exactly like the exchange in the present dialogue which calls forth the description of the *maieutic* method. Cornford points out that *anamnesis* cannot be appealed to in the *Theaetetus*, because it is in effect a causal explanation of knowledge and presumes that what knowledge is is already understood or accepted. Here the nature of knowledge itself is in question and the midwifery of Socrates less obviously begs the question. We must see the *Theaetetus* in the wider context of the metaphysical dialogues, those which follow on the *Parmenides*, which will explain as well the absence of any explicit mention of the Ideas in the dialogue before us. One very obvious purpose of the *Theaetetus* is to analyse in detail and then reject claims that knowledge can be furnished by the world of sense. It is well to emphasize this here, since it is the claim of many that the crisis represented by the questioning of the Ideas in the *Parmenides* indicates a new-found interest in the physical world, an interest which takes Plato very far in the direction to be pursued with such vigor by Aristotle. But let us return to the dialogue and the results of Socrates *maieutic* art.

Knowledge is sensation. -- Encouraged by Socrates, Theaetetus essays the kind of answer wanted and suggests that knowledge is sensation (*aisthesis*). There will be two other attempts at definition in the sequel, but this first one receives the most attention. The treatment of it can be divided into an exposition of the implications of the definition and then a criticism of it, relieved by a lengthy comparison of the philosopher and the lawyer. The first move on Socrates' part is to observe that Theaetetus' attempt at a definition is another way of expressing the claim of Protagoras that man is the measure of all things, of the things that are, that they are, and of things that are not, that they are not. Socrates suggests that they try to understand why a wise man has said such a

thing. The wind is not hot or cold absolutely speaking, but only in relation to us; the wind is cold to him who is cold, but not to him who is not cold. Thus, for the wind to be cold is for it to appear to be cold to him who perceives it as such. Now, on this showing it would seem that perception is only of existence and is thus unerring; there is a heavy price exacted by this infallibility.

I am about to speak of a high argument, in which all things are said to be relative; you cannot rightly call anything by any name, such as great or small, heavy or light, for the great will be small and the heavy light -- there is no single thing or quality, but out of motion and change and admixture all things are becoming relatively to one another, which becoming is by us incorrectly called being, but is really becoming, for nothing ever is, but all things are becoming. Summon all philosophers -- Protagoras, Heraclitus, Empedocles, and the rest of them, one after another, and with the exception of Parmenides they will agree with you in this. (152)

The view that knowledge is sensation, first identified with the Protagorean dictum that man is the measure, is now further identified with what is called the doctrine of Heraclitus, and indeed of all the natural philosophers, to the effect that sensible things are in constant flux, always becoming and never possessing any stable being. This doctrine that all things are in process is applied to the percipient as well as to the perceived object, in such wise that sensation becomes the intersection of two dynamic lines, an active and passive motion; thus, what we call white is just an eddy in the flux and not something independent of the percipient. This in turn leads to the view that sensation, provisionally equated with knowledge, is infallible, since things are what they seem to me to be because for them to be is to seem such-and-such to me.

Dreams, however, pose a threat to the infallibility of sensation or perception or awareness, since in my dreams I am aware of things which later I say were not as they appeared. This suggests that reality cannot be reduced to appearance. The difficulty is handled by a thorough-going relativism: dreams are real to the dreamer. In the same fashion, the wine which tastes bitter to the sick man is bitter for him, though it is sweet to the man in good health. Despite the apparent difficulties, then, things are as they seem to be as long as we are careful to let the one sensing them be the judge of what they are.

Then you were quite right in affirming that knowledge is only perception; and the meaning turns out to be the same, whether with Homer and Heraclitus, and

all that company, you say that all is motion and flux, or with the great sage Protagoras, that man is the measure of all things. . . . (160)

Socrates now turns to the attack. His first point is that, if the individual is the only judge of what he perceives, Protagoras might just as well say that the dog-faced baboon is the measure of all things as that man is, since the baboon too has sensations of which he is the only adequate judge. He then makes some initial criticisms of the identification of knowledge and sensation: we hear a foreigner speak, but we do not know what he is saying; generally we sense sounds, colors, and so forth but claim to know more than these; moreover, we could not be said to know what we remember seeing since we are not now seeing it. At this point, Socrates fabricates a defense for Protagoras. The latter admits that he teaches that what is is what appears to a man, but, that nonetheless the wise man exists, for the wise man is he who would make the evils that are and appear to a man come to seem goods which are and appear to him. Now it is just the opinion that not all opinions are of equal worth that is most widespread and if, where opinions conflict, the test should be to discover what seems to be the case to most men, then Protagoras' view is rejected because it seems false to most men. But perhaps Protagoras' statement applies only to sensible things and not to opinions about just anything. Certainly in politics Protagoras thinks that his opinion outweighs those of other men since he offered to teach others how to get along politically.

It is at this point that there is a long digression, in the course of which the philosopher and lawyer are compared. The philosopher of course cuts a pathetic figure in the law court and the sophist can make him look ridiculous. But the cleverness of the advocate is counterfeit and the goal of his striving is not worth the effort; small wonder then that the philosopher is not adept in that which makes a good lawyer: the philosopher is striving for wisdom.

But, O my friend, you cannot easily convince mankind that they should pursue virtue or avoid vice, not merely in order that a man may seem to be good, which is the reason given by the world, and in my judgment is only a repetition of an old wives' fable. Whereas, the truth is that God is never in any way unrighteous - he is perfect righteousness; and he of us who is the most righteous is most like him. Herein is seen the true cleverness of a man, and also his nothingness and want of manhood. For to know this is true wisdom and virtue, and ignorance of it is folly and vice. All other kinds of wisdom or cleverness, which seem only, such as the wisdom of politicians, or the wisdom of the arts, are coarse and vulgar. (176)

We need not detain ourselves here with the array of arguments Socrates makes use of when he returns to attack the position that all opinions are of equal worth. We do not think that the carpenter's opinion about health is worth as much as the physician's, nor vice versa when it is a question of furniture. Experts are part of everyday life. What is most important is the final rejection of the identification of knowledge and sensation. Plato makes the point that the senses do not know but we know by means of the senses; this is particularly evident with respect to our knowledge of what seeing and hearing have in common. What is at issue are "being and not-being, I likeness and unlikeness, sameness and difference, and also unity and other numbers which are applied to objects of sense." (185) The soul, in other words, views some things by herself and some things through the bodily organs, but in no case is knowledge identifiable with sensation. "Then knowledge does not consist in impressions of sense, but in reasoning about them; in that only, and not in the mere impression, truth and being can be attained." (186) The upshot is the same point made in the *Phaedo*: sensation can have no part in what we mean by knowledge.

Knowledge is true opinion. -- Theaetetus happily concedes that his first effort to define knowledge was unsuccessful and offers a second definition: Knowledge is true opinion. The word here translated opinion is *doxa*; perhaps a better translation would be judgment. The reader should bear this in mind; we shall use opinion, however, Jowett's choice, since we shall be quoting from his translation. Knowledge has been said to be reasoning about sense impressions; thus Theaetetus is led to suggest that knowledge is true opinion. He will not say opinion simply, since false opinion is possible. That observation is important since much of the present discussion is concerned with the possibility of false opinion. How can there be such a thing? Something is either known or not, and opinion accordingly must be concerned with the known or the unknown. But false opinion can be neither thinking what is known to be something else which is known and thus be ignorance of two known things, nor thinking one unknown thing to be another unknown thing; to complete the picture, false opinion cannot be explained as thinking that a known thing is an unknown thing nor vice versa. False opinion then seems impossible. Its impossibility seems to emerge with equal clarity if we shift the discussion from the sphere of knowing to that of being, and suppose that false opinion consists of thinking what is not. But isn't this like seeing something and at the same time seeing nothing; this is impossible, for to see something is to see something that is. So too, Plato suggests, to think something is to think something that is, so that false opinion cannot be explained as thinking what is not.

Perhaps then false opinion may be explained solely in terms of what is.

May we not suppose that false opinion or thought is a sort of heterodoxy: a person may make an exchange in his mind, and say that one real object is another real object. For thus he always thinks that which is, but he puts one thing in place of another, and missing the aim of his thoughts, he may be truly said to have false opinion. (189)

Theaetetus thinks this is truly false opinion defined. But his sense of security is, of course, unfounded. Can a man who knows two things ever think that the one is the other, or can he think one thing he knows is another thing he doesn't know? Despite the difficulties, a way is sought which will permit a man somehow not to know what he knows. Socrates suggests a distinction between the possession and having of knowledge, where possession would be that latent knowledge we have as the result of the impressions on a wax block each of us can be thought to be equipped with, a block blank at first but gradually filled with impressions; the having of knowledge is the actual use of it. If then we presume a distinction between sensation and knowing, the possible mixup of present sensation and knowledge or knowledge and remembered sensation seems to give us an explanation of false opinion.

The only possibility of erroneous opinion is, when knowing you and Theodorus, and having on the waxen block the impression of both of you given as by seal, but seeing you imperfectly and at a distance, I try to assign the right impression of memory to the right visual impression, and to fit this into its own print: if I succeed, recognition will take place; but if I fail and transpose them, putting the foot into the wrong shoe -- that is to say, putting the vision of either of you on to the wrong impression, or if my mind, like the sight in a mirror, which is transferred from right to left, err by reason of some similar affection, then "heterodoxy" and false opinion ensues. (193)

This entails that error and deception are confined to the things a man knows and perceives; false opinion arises not from the comparison of one perception with another, nor in thought alone, but in the union of thought and perception. Though a valiant try, this attempt at a definition of false opinion must be set aside because it does not account for error which takes place in thought alone. When someone says that seven plus five equals eleven, his error cannot be explained in the way described.

A new attempt is made to explain false opinion by likening our knowing to the catching of birds; we have a cage within where we imprison the thoughts we catch. Having them and catching them are different processes, and once they are had we can pluck them out like birds from a cage and this is using our knowledge. False opinion, then, would consist of coming forth with the wrong bird. The aviary here takes the place of the earlier image of the wax block with this difference that, whereas the wax block was the storehouse of sense impressions, the aviary houses opinions or beliefs. Falsehood seems explicable now in terms of a plucking of the wrong bird from the aviary. Now this is precisely the flaw, since the birds stand for bits of knowledge, and falsehood presupposes, on this view, knowledge of that concerning which one is mistaken. More drastically, although the discussion has gone off in search of the explanation of error, it has now turned into a begging of the original question, what is knowledge? We are explaining error in terms of the confusion of what is already known; thus, we must already be able to answer the question as to what knowledge is -- or our statements about error are vitiated. If this theory of error be accepted, knowledge would consist of plucking out a true belief; the difficulty with this, however, is that our attitude towards a true belief is indistinguishable from that we have towards a false belief. That is, if we assume, as we may, that the essence of error is to assert what is not so while believing it is so, our attitude when we make a mistake is the same as when we do not make a mistake. It is just this that leads to the third and last attempt at a definition of knowledge.

Knowledge as True Opinion plus an Account. The breakdown of the previous definition of knowledge leads naturally enough to the thought that, by adding something to true belief, we can turn it into knowledge and thus exhibit it as something quite different from false belief. This added note will be an account (*logos*). Knowledge thus is not only stating a truth but possessing the grounds for the truth. This explanation is suggested as a theory presented by philosophers and, generally, it depends on a distinction between elements and the things composed of them. Of the elements there can be no account but everything else can be explained by having recourse to the elements. Thus, by way of illustration, let the elements be letters and the things to be explained words or, better, syllables. The syllables are explained by enumerating their component letters. Notice that, if knowledge consists in giving such an account, there can be no knowledge of the elements; however, when knowledge is had it is had by enumerating that of which no knowledge can be had. We can try to escape this, and Socrates urges that we do, by making the whole something

different from the sum of its parts; however, if we move in this direction, such a whole cannot be known by enumerating parts which are not, precisely, its parts.

It is only after this preliminary criticism that Socrates suggests a search for the meaning of "account" intended in the proposed definition of knowledge.

"Account" (*logos*) is assigned three meanings. The first is speech, an unimportant meaning here, since those who have not knowledge in the sense sought can express themselves in words. A second meaning is the enumeration of the elements of the thing. Here it is not presupposed that the elements are unknowable; the point is rather that one can reduce a thing to its elements and still not have knowledge of what it is. The illustration is the child learning to write his name. When he has done it once, he has thereby set down all the elements (letters) of the thing, but this is no assurance that he has in one fell blow learned to spell it. His answer is correct, but it could be that he does not know it. A third meaning of "account" is the citing of a distinguishing mark. Although the passage may seem to suggest that Plato is here speaking of including the specific difference in the definition of a thing, Cornford has shown (pp. 161-2) that what is at issue is rather the singular thing. Since this is the case, the demand is an impossible one, since we must be able to locate the individual whose distinguishing mark we seek before we start on our quest; in other words, we must already have in hand what we think we must find. Thus, no satisfactory meaning of "account" is discovered which will enable us to add an account to a true opinion and come up with knowledge.

The dialogue ends in failure; knowledge had eluded all efforts at defining it. Nowhere in this dialogue do we find any mention of Ideas; what is crystal clear, however, is Plato's unwavering conviction that knowledge cannot be had in sensation. The *Theaetetus*, accordingly, far from evidencing a newly found predilection for sensation on Plato's part, is rather a thorough exploration of what earlier dialogues had always assumed, namely that sensation is powerless to produce knowledge, with a view towards rehabilitating the doctrine of Ideas in the face of the objections set forth in the *Parmenides*. If the net effect is negative, the dialogue is nonetheless part of a larger literary whole, and we must now turn to the other parts of that whole to see what new positive direction, if any, Plato has taken.

Sophist. The opening section of this dialogue places it on the day after the conversation in the *Theaetetus*. Once more we have Socrates, Theodorus and Theaetetus, but Socrates' interlocutors of the previous day have brought along an Eleatic Stranger, a disciple of Parmenides and Zeno, a true philosopher.

Socrates asks the Stranger how those of his persuasion would define the philosopher in order to distinguish him from the statesman and the sophist. Since the present dialogue is concerned with the sophist and is followed by another concerned with the politician, it is felt that Plato originally intended to devote a dialogue to the philosopher himself but for some reason changed his mind. This four-dialogue project may have importance with respect to the altered role of Socrates in the *Sophist*. Although his request sets the stage for the following discussion, Socrates himself withdraws quickly to the wings, leaving the center of the stage to the Eleatic Stranger. Thus while Socrates and Theaetetus are the main speakers in the *Theaetetus*, the Stranger and Theaetetus are such in the *Sophist*; in the *Statesman*, young Socrates, a contemporary of Theaetetus, replaces the latter. Cornford's argument that Socrates and the young Socrates were to carry on the discussion in the projected dialogue on the philosopher (p. 168) suggests that the eclipse of Socrates is only temporary and surely cannot be construed as a repudiation by Plato of his old teacher.

This dialogue has as its purpose the definition of the sophist but it is not too much to say that its importance lies rather in two subsidiary issues: the method whereby a definition is to be achieved, and the settling of the problem of false opinion by the granting of a kind of being to non-being. This last point represents a definitive break with Parmenides and one reason at least for Socrates' fading into the background is that both in the *Parmenides* and *Theaetetus*, Socrates has shied away from any criticism of the great Eleatic. We shall confine our comments to the various divisions made in the first part of the dialogue and the definitions of the sophist offered; the method involved in arriving at these definitions; and the problem of non-being.

First as to the method. The *Theaetetus* ends without resolution since there is no sense of account which would permit us to add an account to true opinion and achieve knowledge. The failure is due to that dialogue's attempt to speak of knowledge without having recourse to the Ideas, to restrict knowledge to the sensible order. In the *Sophist*, we are no longer interested in the individual, but in the type, the species, the intelligible reality. Account or *logos* now means definition and this in terms of what we would call genus and difference. The Stranger illustrates the method he wishes to employ in seeking the definition of the sophist by a preliminary exercise bearing on a more accessible type, the angler. What we must do is begin with a wide class which will include the angler and then by the addition of distinguishing characteristics, set him off from all other members of the general group. Thus angling is an art, acquisitive rather

than productive and by means of capture rather than trade, resulting from hunting. The illustration is well-chosen, since the Stranger will be able to use members of this original division in his quest for the sophist. Five definitions of the sophist emerge: he is a paid hunter after wealth and youth; he is a merchant in the goods of the soul; he is a retailer of the same sort of wares; he is the manufacturer of the wares he sells; finally, he is a member of the fighting class, a hero of debate professing the eristic art. There is as well a sixth division, descriptive rather of the Socratic method than that of the Sophist. It consists of purifying the soul of ignorance by the maieutic method. The Stranger says that he would hesitate to call the practitioners of this art sophists. If the sophist appears as a many-headed beast there is nonetheless one overriding characteristic of the man and that is disputation which he not only engages in but offers to teach others. The sophist is willing to dispute about all things; but, since he cannot possibly know all things, he is held in honor because he appears to know all things. What the sophist says appears to be the truth yet is not; that is, what he says is false, and the problem of the possibility of falsehood now comes to the fore.

He who says that falsehood exists has the audacity to assert the being of not-being; for this is implied in the possibility of falsehood. But, my boy, in the days when I was a boy, the great Parmenides protested against this doctrine, and to the end of his life he continued to inculcate the same lesson -- always repeating both in verse and out of verse: 'Keep your mind away from this way of enquiry, for never will you show that not-being is.' (237)

The discussion begins in a familiar fashion. Not-being can not be attributed to being; not-being is neither one nor many, it cannot be spoken of or thought about. The Stranger then expresses the objection that occurs even to the undergraduate in reading the fragments of Parmenides: what are we to make of all this talk about non-being, the burden of which is that nothing can be said of non-being? When I say that non-being is ineffable, my assertion involves attaching an is to not-being, a verb which is singular and not plural: that is, some kind of being and some kind of unity seem asserted of non-being in our very attempt to deny being and speech and unity of it. This recognition will lead the Stranger to become a parricide. "Because, in self-defense, I must test the philosophy of my father, Parmenides, and try to prove by main force that in a certain sense non-being is, and that being, on the other hand, is not." (241) Not only Parmenides is put to the test, however; the Stranger turns to an assessment of early philosophy in general.

Before turning to the opinions of the ancients, the Stranger points out that the difficulty is not simply posed by non-being.

And very likely we have been getting into the same perplexity about 'being,' and yet may fancy that when anybody utters the word, we understand him quite easily, although we do not know about not-being. But we may be equally ignorant of both. (243)

This is made clear though consideration of the teachings of the early philosophers as falling under two headings, monism and dualism. Some philosophers speak of hot and cold as principles and say that they must be two; and yet if both are, being would seem to be some third principle prior to them both. And yet if we ask what is meant by being, they can give us no answer. As for those who deny plurality, their contention that being is one involves two names, being and one, leading to the conclusion that they are either synonyms or that the real is multiple. But Parmenides cannot maintain that being and one are synonyms because he has said that being is the whole, having the fullness of a well-rounded sphere. Thus being has parts and can be one as a whole is one, that is, by participating in unity, but it cannot be unity itself. But let us try to avoid this by denying that being is a whole, then wholeness and being will differ and being will contain a defect and become in some sense not-being. And, whether being be called a whole or not a whole, we are faced with plurality. On the supposition that being is other than wholeness and unity, being cannot be said to exist nor to have come to be. The meaning of being and not-being becomes a matter of some perplexity when we consider this view, but the Stranger wants to pass to another opposition among philosophers.

There are some philosophers who will admit as real only that which can be grasped in the hand: Plato calls them giants. i.e. sons of the earth. For them being and body are one. The Stranger imagines them agreeing nevertheless that living bodies have souls, and that some souls are wise and just, and further that such things as soul, justice and wisdom are not corporeal. With this agreement, it is necessary to formulate a notion of being which can apply as well to the incorporeal. The Stranger has a suggestion.

My notion would be, that anything which possesses any sort of power to affect another, or to be affected by another, if only for a single moment, however trifling the cause and however slight the effect, has real existence; and I hold that the definition of being is simply power. (247)

The Stranger adds that we may change our minds about this definition, but that it can suffice for now as an agreement with the giants or materialists.

Opposed to the giants are the friends of the Ideas. There has been some dispute as to the identity of these friends; the most plausible interpretation seems to be that Plato has in mind earlier views of his own on the Ideas. The Stranger finds that these friends would distinguish being and becoming and continues:

And you would allow that we participate in generation with the body, and through sensation, but we participate with the soul through thought in true essence; and essence you would affirm to be always the same and immutable, whereas generation or becoming varies? (248)

They would, and the Stranger tries to get at what they mean by participation by asking their opinion of the definition of being just proposed. The Friends reject it, because to affect or to be affected seems to belong to the realm of becoming and not that of being. The Stranger counters by pointing out that "to know" is active and "to be known" passive, and we come to what has always been regarded as one of the most crucial passages of the dialogues. "And, O heavens, can we ever be made to believe that motion and life and soul and mind are not present with perfect being? Can we imagine that being is devoid of life and mind, and exists in awful unmeaningness an everlasting fixture?" (249) If only the immutable can be said to be, then mind, soul and life must be excluded from being; on the other hand, the opposed position that change is everything would equally do away with intelligence, depriving it of any anchor. The upshot of this discussion with the giants and the Friends of the Ideas is that each group must admit that reality includes both changing and unchanging things.

The Stranger now points out that these admissions do little towards clearing up the problem of not-being and being. If both motion and rest are, neither one can be identified with being since then motion would have to be rest and vice versa; let us say, then, that being is some third thing in which motion and rest participate and which in its own nature is different from both. Thus things that are are not being and being is different from the things that are. How is this possible? Are we not making one thing many and many things one? To resolve this problem, the Stranger first observes that any statement involves the same difficulty. We say of Socrates, who is one man, that he is many things: wise, white and wizened. In other words, predication always presents the problem of participation which can be thought of in three ways (1) Participation is impossible; nothing participates in anything else. This denial is taken to be a

denial of predication itself and thus self-defeating. (2) Participation can be indiscriminate; all things participate in everything else. This is impossible, however, since then motion would be rest and vice versa. (3) The remaining possibility, that participation is limited calls for a special art to determine when participation obtains.

And as classes are admitted by us in like manner to be some of them capable and others incapable of intermixture, must not he who would rightly show what kinds will unite and what will not, proceed by the help of science in the path of argument? And will he not ask if the connecting links are universal, and so capable of intermixture with all things; and again, in divisions, whether there are not other universal classes, which make them possible? . . . By Zeus, have we not lighted unwittingly upon our free and noble science, and in looking for the sophist have we not entertained the philosopher unawares? (253)

It is the knowledge of which Forms pervade a scattered multitude and which are separate and aloof which is provided by the art of dialectic, possession of which is the mark of the philosopher. It is just this art that must now be applied to the problem of being and not-being.

The discussion begins by assuming three of the most important genera, being, rest and motion. Motion is not rest and vice versa; that is, they cannot participate in one another but both participate in being. Of these three, each is the same as itself and different from the other two. Our three genera then involve two more, "same" and "other." Moreover, these, like being, are shared by both motion and rest. Thus, motion is the same as itself, yet not the same, that is, not sameness itself, since, if it were, rest in order to be the same as itself would have to be motion. So too motion is other than rest but is not otherness since, if it were, rest in order to be other than motion would have to be motion. The same could be shown with respect to motion and other, and the Stranger observes that we now have a legitimate way of saying that motion is the same and not the same, other and not other. We can see that the same argument can apply to motion and being: motion is and is not being since, while it participates in being, it is other than being.

Then not-being necessarily exists in the case of motion and of every class; for the nature of the other entering into them all, makes each of them other than being, and so non-existent; and therefore of all of them, in like manner, we may truly say that they are not; and again, inasmuch as they partake of being, that they are and are not existent. (256)

Not-being or otherness is a class like being, motion, etc., and the analysis has arrived at a way of asserting that not-being is, contrary to the doctrine of Parmenides. It seems fairly clear from the sequel that Plato fully intends that not-being is a form, a changeless reality, since it is appealed to, to explain falsehood; it is the objective complement of error in thought and speech.

If not-being has no part in the proposition, then all things must be true; but if not-being has a part then false opinion and false speech are possible, for to think or to say what is not -- is falsehood, which thus arises in the region of thought and in speech. (260)

The Stranger is now able to explain the possibility of the sophist since falsehood itself has been shown to be possible. The sophist of course can object that language and thought are not among the things which can participate in not-being. But this is shown not to be the case. Of course the not-being which underlies falsehood is just the kind of not-being which has been shown to exist, namely being other.

The *Sophist* has as its major internal problem the explanation of the possibility of false statements. Viewed in a wider context, this dialogue reveals that the problem of participation has been shifted to the realm of the Forms or Ideas and that these form a coherent system. The art of dialectic bears on the discernment of the communion or non- communion of Forms. Not-being exists as an objective reality, but is otherness in which all things participate. False statement bears on realities but consists of a statement reflecting a communion of Forms which are in fact other.

Statesman. This dialogue is connected with the preceding, as we have already seen, with the younger Socrates taking the place of Theaetetus as the interlocutor of the Eleatic Stranger. The purpose of the dialogue is achieved without the detours which make the *Sophist* important for an understanding of the doctrine of Forms and the art of dialectic which bears on the Forms. There is an aside which warns of the necessity to make all the intermediate steps in performing a division of a class in order to isolate a species like statesman (261-266), but on the whole the dialogue sticks to its subject and we need not consider it here. The Forms or Ideas continue to be looked upon as a structured universe but, as has been pointed out (Ross, pp. 118-119), the aim of dialectic seems to be much more modest than that set forth in the Republic. It is not so much a question of deriving all Forms from one supreme Form, as of dealing with those necessary to settle a particular question.

Concluding. The earlier dialogues of Plato, those which go beyond what can be taken as a heightened but not inaccurate description of the teaching of Socrates, do not explicitly present the Forms or Ideas as entities apart from and indeed far more real than the things of everyday life, particular actions, sensible particulars. In the *Phaedo* and *Republic*, however, the Forms are quite clearly better beings, other than the explanatory of particulars of our experience, and philosophy is precisely the ascent to knowledge of these beings. Indeed, in the *Republic*, a lengthy training is described the result of which is the purgation of body and mind so that the soul can ascend to the contemplation of the Forms. This is achieved in dialectic which bears primarily on the Form of the Good from which all other Forms are seen to emanate, although the notion of a hierarchical structure among the Forms is not stressed. With the *Parmenides* a highly critical attitude towards the Forms comes to the fore and a list of difficulties is compiled, difficulties which are to be taken not as a refutation of the Forms but as a program to be followed if this doctrine is to be saved from ridicule. What is the nature of the relation between Forms and particulars; how are the Forms related to one another? These may be taken to summarize the objections of the *Parmenides*, and it is rather the latter than the former question which seems answered by the *Sophist*. The *Theaetetus* we have taken as showing what had hitherto only been assumed, that sensation is not knowledge and cannot be productive of knowledge in the rich sense. If the Forms have emerged as involving in their interrelationships the participation which at one time seemed to express only the relation of Form to particular, the question of sensible particulars seems yet to be examined. In the *Republic* the things of this world were granted a shadowy kind of being, sufficient for inducing opinion but not knowledge. We shall now turn to a dialogue in which Plato is concerned with the physical world, its relation to the Forms, and the nature of our knowledge of the physical world.

D. Plato's Natural Doctrine

The dialogue to which primary appeal must be made for Plato's views on the natural world is the *Timaeus*, a dialogue written quite late in the life of its author and one which forms part of an uncompleted trilogy the purpose of which indicates that the story of the fashioning of the visible world was meant to provide an analogue of the moral life much as, in the *Republic*, the structure of the individual soul provides the model for the ideal state. The persons of the dialogue (it is not really a dialogue at all) are four: Socrates, Timaeus of Locris in Italy, Critias and Hermocrates. Of these, only Timaeus seems to be a purely fictional character. They meet the day after Socrates has outlined to the others

the plan of an ideal state, the characteristics of which he summarizes now, something he would hardly need do if the discourse in question were the *Republic*, as some have thought; moreover, the outline suggests something a good deal less extensive than the perfect commonwealth of the early dialogue. Socrates now wishes to put flesh on his skeletal outline, pleads his own disability and enlists the help of the three others. Critias provides the setting for what is to follow by recalling a story his grandfather had told him of a visit by Solon to Egypt. The Egyptian attitude towards the Greek sage is summed up in the following remark: "O Solon, Solon, you Hellenes are never anything but children, and there is not an old man among you." (22) The Athenians, it seems, are ignorant of their own past; their history has been much longer than they realize, since before the flood there was an Athens and it had the glory of defeating the forces coming from the isle of Atlantis. The story of this battle will be told by Critias in the dialogue of that name and its purpose is this: the ideal state Socrates speaks of is to be shown to be one that actually existed in antediluvian Athens. Thus, Socrates' political theory will no longer be fiction but fact. There follows the plan for a trilogy of dialogues. In the first of them, Timaeus, who is an astronomer, will discuss the nature of the universe, the generation of the world including the creation of man. In the second, Critias was to take up the story of man, and show him as the beneficiary of the education Socrates has spoken of and identify him with the citizen of that Athens described to Solon by the Egyptian priest. We may surmise with Cornford^[30] that Hermocrates was then to continue the story into the historical period, to give us in effect the material that we find in the *Laws*. Of this project, only the *Timaeus* and part of the *Critias* were finished; nevertheless, it is important that we see the intended context of the *Timaeus*; it is part of a larger discussion the import of which is clearly moral and political. The implication is that we must look to the structure of reality if we are to grasp the structure of morality. Certainly this does not diminish the importance of the *Timaeus* for gaining knowledge of Plato's views on the natural universe; it does, nonetheless, indicate the survival in the aged Plato of the interests of the author of the earlier dialogues.

Once Timaeus begins his allotted task, the dialogue becomes an unalleviated monologue, quite unlike anything else of Plato's we possess -- except perhaps the *Critias*. Despite its apparent strangeness, it is not too much to say that the *Timaeus* is the most influential writing of Plato on the subsequent history of philosophy, both in antiquity and in the middle ages. Much commented on in antiquity, it was early a focal work in the Academy, it figures often in Aristotle's discussion of Platonic doctrine, it looms large in Hellenistic philosophies and, in

the Latin translation of Chalcidius, made in the 6th century of the Christian era, it exercised tremendous influence. It is an extremely complicated and compressed piece of work and we can hope to do little else here but suggest its contents. Even before attempting that, however, we must say something about Plato's own view of the value of the teaching to be found in the *Timaeus*.

It is a commonplace of earlier dialogues that knowledge cannot be sought in the sensible world, since knowledge is of the unchanging whereas the sensible world is precisely the arena of change and, indeed, of ceaseless change. The best we can expect, with regard to particulars, is conjecture and opinion, since sensible things are not quite real. Now this would not lead us to expect a writing like the *Timaeus*. Perhaps we must accept the view that Plato has suffered some change of attitude, but we should not be misled as to the extent of the change since, as it happens, the *Timaeus* itself raises the questions which occur to one who has familiarized himself with the earlier dialogues. Indeed, *Timaeus* begins his discourse by making a distinction between being and becoming.

That which is apprehended by intelligence and reason is always in the same state; but that which is conceived by opinion with the help of sensation and without reason, is always in process of becoming and perishing and never really is. Now everything that becomes or is created must of necessity be created by some cause, for without a cause nothing can be created. The work of the creator, whenever he looks to the unchangeable and fashions the form and nature of his work after an unchangeable pattern, must necessarily be made fair and perfect; but when he looks to the created only, and uses a created pattern, it is not fair or perfect. (28)

Now the world has been created, since it is visible and tangible and has a body. The world, being sensible, is apprehended by sense and opinion. Since it is created, the world must have a cause, though it is difficult to find; this cause must have looked to a pattern, and the perfection of the created world makes it clear that the creator looked to an eternal pattern in making it.

And having been created in this way, the world has been framed in the likeness of that which is apprehended by reason and mind and is unchangeable, and must therefore of necessity, if this be admitted, be a copy of something. Now it is all important that the beginning of everything should be according to nature. And in speaking of the copy and the original we may assume that words are akin to the matter they describe; when they relate to the lasting and the permanent and intelligible, they ought to be lasting and unalterable, and, as far

as their nature allows, irrefutable and unmovable -- nothing less. But when they express only the copy or likeness and not the eternal things themselves, they need only be likely and analogous to the real words. As being is to becoming, so is truth to belief. If then, Socrates, amid the many opinions about the gods and the generation of the universe, we are not able to give notions which are altogether and in every respect exact and consistent with one another, do not be surprised. Enough, if we adduce probabilities as likely as any others; for we must remember that I who am the speaker, and you who are the judges, are only mortal men, and ought to accept the tale which is probable and enquire no further. (29)

This passage makes it quite clear that the *Timaeus* does not repudiate the early Platonic belief that there is no knowledge of the sensible world. That world is only an image of the real world, the world of Forms; the most we can expect is a likely story. The question arises as to whether one story can be more likely than another and, if so, in virtue of what its greater likelihood is gauged. There is no basis for assuming that what Plato means is that one account is closer to the way the sensible world actually is as if that world contained its own intelligibility. The source of intelligibility must be sought in those Forms which are utterly other than the sensible world and of which the sensible world is the pale copy. Thus a story will be more likely insofar as it refers the sensible world to the Forms. In the *Statesman*, Plato has pointed out that some Forms have sensible copies whereas others have none (285-6); where there are such sensible copies, these are not examined for their own sakes but for the sake of the Form of which they are copies. In the *Timaeus* Plato is not so much moving from the sensible world to the Forms as the other way round, but the Forms remain of the utmost importance; they are, after all, the only true anchor for thought, and if the sensible world is said to be perfect and good this is because it is an image of the intelligible and eternal. Insofar as it is about the sensible world, the *Timaeus* can give us only a likely story, conjecture; Plato will return to this again and again in the course of the dialogue. But the ultimate ground of the whole enterprise is the realm of Forms which are not in the sensible world (52). Any story of the universe which ignores the Forms or suggests that the physical universe has its intelligibility intrinsic to itself will be for Plato unlikely and indeed downright false.

The nature of the maker or demiurge is something that has been much discussed. There are those who maintain that Plato is here speaking of an agent indistinguishable from the creating God of *Genesis*; others claim that the demiurge or maker is simply a mythical expression of the familiar Platonic

doctrine to the effect that the sensible world mirrors the intelligible world. There are a number of positions between these extremes, but what every position must accept is that Plato is asserting by means of the likely story of the *Timaeus* that the sensible universe is a product of an intelligent power. The demiurge can be taken to be symbolic of this intelligence insofar as he looks to the eternal patterns in fashioning out of the given chaos the proportioned universe. These eternal patterns cannot be conceived as thoughts in the mind of the demiurge; rather they are realities apart from him and independent of him. In other words, he is not making the world in his own likeness, but to the likeness of the eternal patterns other than himself. As a symbol of divine power, the whole story suggests the revealed story of creation, but Christian thinkers like St. Augustine will touch up the account considerably, particularly by their interpretation of the location of the Forms or Ideas, to bring it into line with Christian faith. The very least we must take from the story of the making of the universe is that for Plato the source of order in this world comes from without, that there is an intelligence responsible for the way sensible things are, that they represent imperfectly the patterns according to which they have been fashioned.

There is a connected controversy as to whether Plato is here maintaining that the world had a beginning in time. The talk of becoming and its cause would seem to indicate that what has become beforehand was not; in a word, that it has a beginning before which it simply was not. There is no doubt that Plato speaks as if there is a beginning of becoming, and this is how Aristotle interpreted him, namely as asserting that time itself had a beginning. (Cf. *Physics*, VIII, 1, 251b17) A far more common interpretation was that originating in the early Academy of which Aristotle himself speaks.

Some of those who hold that the world, though indestructible, was yet generated, try to support their case by a parallel which is illusory. They say that in their statements about its generation they are doing what geometers do when they construct their figures, not implying that the universe really had a beginning, but for didactic reasons facilitating understanding by exhibiting the object, like the figure, as in the course of formation. (*De caelo*, I, 10)

Aristotle is no doubt right in maintaining that you cannot say that something has come to be and yet never was not -- that is, unless you are using words in a new sense which requires exposition. Plato's own point seems rather to be that the sensible world as copy must always be dependent on the eternal model. The image of the maker of the world fashioning it as a craftsman makes artifacts is less important surely than the central point.

What is the motive of the demiurge in fashioning this world? The answer that Plato gives here recalls a good many things, his dissatisfaction with those explanations which do not explain natural things in terms of finality and the good, the primacy of the Good in the *Republic*.

Let me tell you then why the creator made this world of generation. He was good, and the good can never have any jealousy of anything. And being free from jealousy, he desired that all things should be as like himself as they could be. This is in the truest sense the origin of creation and of the world, as we shall do well in believing on the testimony of wise men: God desired that all things should be good and nothing bad, so far as this was attainable. Wherefore also finding the whole visible sphere not at rest, but moving in an irregular and disorderly fashion, out of disorder he brought order, considering that this was in every way better than the other. (29-30)

The primacy of the good, of what Aristotle will call the final cause, makes this account particularly attractive to men of faith; it is not simply the imaginative translation of Jowett that suggests the biblical account of creation here. We do not wonder that the men of the middle ages will come to treat the *Timaeus* almost as they treated Scripture itself. Aristotle will make use of the notion that the gods are not jealous to indicate the fittingness of striving for the divine science; more importantly, his own ultimate explanation of reality will be anchored securely in the Good towards which the whole world strives as its end and justification.

Timaeus first describes the formation of the body of the world and then of the world soul. The world is a living creature, an animal, composed of body and soul. That on which the universe is patterned contains every intelligible thing; the universe consequently should be an animal which contains every kind of animal. The body of the world is composed of fire, air, earth and water, proportioned to one another. The world exhausts these four, they are entirely within it, and it is thereby incapable of changing as a whole. The shape of the body of the world is a globe, a perfect sphere, the figure which comprehends within itself all other figures. Timaeus then says that a soul was formed, placed in the center of the world body, from which point it diffused itself throughout the body. Actually, he cautions, he should have begun with the soul, since in order and excellence it is prior to the body: the soul is master, the body subject.

Just as the world body is composed of the four elements, so the world soul is composed of the same, the other and being. The passage in question (35)

indicates that Plato is striving to describe a kind of being intermediate between the Forms on the one hand and bodies on the other. The elements of which soul is composed are taken over from the *Sophist*, and just as in that dialogue life and motion (i.e. thinking) are said not to be excluded from reality, so here the soul is shown to have affinity with the Forms and yet to be lower than them; by the same token it is linked with the world of becoming although its acting or power is on a higher level than the *dynamis* of the sensible world. The world soul is then divided and the parts are joined to form two motions, that of the same, and that of the other. The motion of the other, at first enclosed like a band revolving horizontally by the band of the motion of the same revolving vertically and around the same center is divided into seven lesser motions whose orbits are those of Sun, Mercury, Venus, Moon, Saturn, Mars and Jupiter. Once more, Timaeus speaks of the joining of the world soul and world body.

The account of the creation of time is of particular interest; the demiurge is said to have been moved to create it in order to make the copy even more like the original.

Now the nature of the ideal being was everlasting, but to bestow this attribute in its fullness upon a creature was impossible. Wherefore he resolved to have a moving image of eternity, and when he set in order the heaven, he made the image eternal but moving according to number, while eternity itself rests in unity; and this image we call time. For there were no days and nights and months and years before the heaven was created, but when he constructed the heaven he created them also. They are all parts of time, and the past and future are created species of time, which we unconsciously but wrongly transfer to the eternal essence; for we say that he was, he is, he will be, but the truth is that 'is' alone is properly attributed to him, and that 'was' and 'will be' are only to be spoken of becoming in time . . . (37-8)

Time and the heaven are interdependent. Plato says both that they "came into being at the same instant" and that "the created heaven has been, and is, and will be, in all time." (38) The planets serve to distinguish and preserve the numbers of time; moreover, the planets are living creatures, one of the four kinds that are made. "There are four such; one of them is the heavenly race of the gods; another, the race of birds whose way is in the air; the third, the watery species; and the fourth, the pedestrian and land creatures." (40)

The demiurge delegates the task of fashioning the bodies of men and lower animals to the created gods, reserving for himself the task of furnishing the

immortal principle, the soul. The souls of men are composed of the same elements as went into the making of the world soul. Moreover, the knowledge presupposed by the doctrine of *anamnesis* is explained.

And having made it he divided the whole mixture into souls equal in number to the stars and assigned each soul to a star; and having there placed them as in a chariot, he showed them the nature of the universe, and declared to them the laws of destiny, according to which their first birth would be one and the same for all -- no one should suffer a disadvantage at his hands; they were to be sown in the instruments of time severally adapted to them, and to come forth the most religious of animals; and as human nature was of two kinds, the superior race would hereafter be called man." (41-2)

Each soul comes to its body in its first birth with equal knowledge of reality, with an equal chance of being good. But each of them must come to a body and this leads to the necessity that each should be provided with the faculty of sensation; moreover, union with the body entails the possession of emotions and being subject to the opposition of pleasure and pain. To conquer these emotions is to live righteously; to be conquered by them, to live unrighteously. "He who lived well during his appointed time was to return and dwell in his native star, and there he would have a blessed and congenial existence." (42) If he did not live well, his soul would pass into the body of a woman at its second birth, a state which could be the start of a further declension into the body of a brute. Of course there are women at the outset so that, in the first generation at least, women, though less than men, are not bad men.

After a brief discussion on the composition of the human body, Timaeus indicates that this discourse has reached a point where a new beginning is necessary. What has been recounted thus far is the work of intelligence; we must now take into account the role of necessity in the constitution of the universe. We are told that the universe has been able to come into being because reason has persuaded necessity, which is represented as a variable or errant cause. It is necessary to take into account the nature of fire, water, earth and air as they were prior to the creation of heaven; the generation of the elements is something usually left out of account, and Timaeus intends to reduce these to something yet more primary. Nevertheless, he does not intend that what he has to say should be the ultimate answer to the question, what are the elements of all things? Once more, he reminds us that he is striving for a likely story, a probable account.

From the very outset, Timaeus has spoken of the demiurge as imitating in his production the eternal patterns; it is taken for granted that being cannot be perfectly mirrored in becoming, that the eternal can have only an image in time. Necessity is introduced to explain this defect in the world of time and becoming. Accordingly, Timaeus begins once more and where earlier he had divided all things into being and becoming, he now sets forth a threefold division.

There is also a third kind which we did not distinguish at the time, conceiving that the two would be enough. But now the argument seems to require that we should set forth in words another kind, which is difficult of explanation and dimly seen. What nature are we to attribute to this new kind of being? We reply, that it is the receptacle, and in a manner the nurse of all generation. (49)

This receptacle is the arena of change, that in which things which do not have any stable nature but are alternately hot and cold, pass into their opposites and back again. "Anything which we see to be continually changing, as, for example, fire, we must not call a 'this' or 'that', but rather say that it is 'of such a nature'." (49) To call them this or that would be to imply that they have some stability, but they are in constant flux. We notice here the continuity with the attitude expressed in the *Theaetetus* and a corroboration of Aristotle's claim that Plato accepted the Heraclitean estimate of the fluidity of the sensible universe. The suggestion is that fire, for instance, is not the name of something but the designation of the state or quality of something. Timaeus illustrates this by speaking of gold which is shaped now this way now that way. If we are asked what it is, the safest answer by far will be gold, since if we cited one of its shapes which it would quickly lose, our answer would be only momentarily correct. The usual elements are spoken of as qualities or attributes of something more basic.

And the same argument applies to the universal nature which receives all bodies -- that must always be called the same; for, while receiving all things, she never departs at all from her own nature, and never in any way, or at any time, assumes a form like that of any of the things which enter into her; she is the natural recipient of all impressions, and is stirred and informed by them, and appears different from time to time by reason of them. (50)

This receiving principle is likened to a mother, the intelligible nature which is being imitated is like a father, and the process of generation is the child or

product. In order to be receptive of the copy of any Form, the receptacle must be taken to be free from any such imitation in its own nature.

Wherefore, the mother and receptacle of all created and visible and in any way sensible things, is not to be termed earth, or air, or fire, or water, or any of their compounds or any of the elements from which these are derived, but is an invisible and formless being which receives all things and in some mysterious way partakes of the intelligible, and is most incomprehensible. (51)

Thus I state my view: -- If mind and true opinion are two distinct classes, then I say that there certainly are these self-existing ideas unperceived by sense, and apprehended only by the mind; if, however, as some say, true opinion differs in no respect from mind, then everything that we perceive through the body is to be regarded as most real and certain. But we must affirm them to be distinct, for they have a distinct origin and are of a different nature; the one is implanted in us by instruction, the other by persuasion; the one is always accompanied by true reason, the other is without reason; the one cannot be overcome by persuasion, the other can; and, lastly every man may be said to share in true opinion, but mind is the attribute of the gods and of very few men. Wherefore also we must acknowledge that there is one kind of being which is always the same, uncreated and indestructible, never receiving anything into itself from without, nor itself going out to any other, but invisible and imperceptible by any sense, and of which the contemplation is granted to intelligence only. And there is another nature of the same name with it, and like to it, perceived by sense, created, always in motion, becoming in place and again vanishing out of place, which is apprehended by opinion and sense. And there is a third nature, which is space, and is eternal, and admits not of destruction and provides a home for all created things, and is apprehended with the help of sense, by a kind of spurious reason, and is hardly real; which we behold as in a dream, say of all existence that it must of necessity be in some place and occupy a space, but that what is neither in heaven nor in earth has no existence. (51-2)

This passage shows that while Plato still retains the basic bifurcation of reality into Forms and their sensible copies, he is now introducing a third thing which is real without being the copy of a Form. It is where the fleeting copies reside, their receptacle, and not another copy. The receptacle is now explicitly identified with space.

We are now invited to think of this receptacle as containing chaotically fire, earth, and the rest, prior to the persuasive ordering of the demiurge. This

activity consists of imposing form and number on them. The four elements are generated as the four regular solids which are seen as built up out of triangles. The mathematics of this generation and the chemistry subsequently based on it is far too complicated to go into here. The reader is urged to consult the concise, clear exposition in Cornford's *Plato's Cosmology*. (pp. 210 ff.)

In the sequel of the *Timaeus*, we find a discussion of meteorological matters, the mechanism of sensation, human physiology, diseases, and so forth. We can see why this dialogue is said to be the only work of Plato which can lay claim to being a kind of encyclopedia.

The *Timaeus* is in many ways an extraordinary and surprising work to issue from the pen of Plato. Nevertheless, far from undermining or repudiating the world of Forms, the distinctive doctrine of earlier dialogues, the *Timaeus* exhibits unwavering confidence in the existence of the Forms. This is manifest in a number of ways. There is first of all the repeated insistence that an account of the sensible world can be at best a likely story. The locus of the really real has not changed; it remains the world of the Forms. What is more, the attempt to give a likely story concerned with the sensible world serves to clarify the nature of the Ideas: they are outside of time, nonspatial, causes of sensible things only in the sense of models. The one stable element in the world of becoming is the receptacle in which process takes place; this receptacle which is identified with space, is grasped by the mind but it has no model in the World of Forms themselves which neither receive other things nor go out into something else. Another clarification in the *Timaeus* concerns soul, the world soul but as well the human soul. The soul enjoys an intermediate existence between Forms and the realm of becoming and the story of its creation and the manner in which its future history depends on moral behaviour, while once more expressed in the form of myth, underlines what appears to be the serious intent of Plato with respect to the soul. The soul cannot, like the body, come to be by means of motion and change. The nature of true knowledge, always the takeoff point for the assertion that Forms or Ideas exist, prevents Plato from seeing in the experience of the things of this world a basis for knowledge. The soul must already have knowledge of what is truly existent. Moreover, this knowledge is the ground of true morality. Let us agree then that the objections of the *Parmenides*, serious as they are and, indeed, unanswered as they seem to remain in the subsequent dialogues, do not dissuade Plato from his belief in the eternal Ideas of Forms. If anything, this conviction is strengthened. Doubtless, some kind of change has occurred, since we cannot imagine the author of the *Republic* undertaking the task of composing the *Timaeus* with anything but distaste. (Of

course, as we have seen, the ultimate purpose of the *Timaeus* is not unlike that of the *Republic*.) But, once more, what has not changed is the insistence that reality is not to be sought in the sensible thing but elsewhere in the realm of Forms where exists that from which the perishable things we perceive receive their name and nature. We may find this unpalatable and want to interpret this inescapable foundation stone of Platonism out of existence, but it is finally inescapable and, if Aristotle directs much of his criticism at it, we cannot say that he has no target.

We shall not go into any discussion of the myth of Atlantis as it is given in the *Critias*; as for the proposed third dialogue of this trilogy, if Cornford's conjecture is correct, our discussion of the *Laws* will give us an indication of what the sequel might have been. It is now time to take up the matter of a doctrine attributed to Plato by Aristotle which does not seem to be taught in any of the dialogues.

Mathematical Intermediates. We have already alluded to the fact that doctrines are attributed to Plato by ancient authors which do not occur in the dialogues that have come down to us. The most interesting information of this sort comes to us through Aristotle and, although it is surely the most natural thing in the world to suppose that one who was a member of the Academy for nearly twenty years while Plato was head should have heard the master say one or two things that are not mentioned in the dialogues, this possibility has been seriously questioned. Since its proposal by Cherniss this image of a mute and aloof Plato has not met with a warm reception and we shall assume the validity of the arguments against Cherniss and entertain seriously an important advance in Plato's theory mentioned by Aristotle. Of course a study of Aristotle's account of Plato's doctrine is one which requires great scope, as is evidenced by Leon Robin's *La theorie platonicienne des idees et des nombres d' apres Aristote*^[31]. We shall be interested here only in the view that mathematical entities occupy a place midway between the Forms and sensible things; Aristotle's criticism of the Forms is something we shall take up in our next chapter.

We recall that in the seventh letter, in exemplifying what he meant by a Form or essential reality, Plato chose the example of circle and there is no suggestion that there is an intermediate entity between sensible things which are circular and circularity itself. Indeed, it may safely be said that the most familiar examples of Forms in the dialogues are those of moral qualities and the mathematical aspects of sensible things. Not that the world of Forms is limited to these. In the seventh letter, Plato gives us some idea of what he conceived to

be the scope of the world of Forms by saying there are Forms of shapes and surfaces, both straight and curved, of the good, beautiful and just, of natural and artificial bodies, of fire and water and so forth, of every animal and every quality and of all active and passive states. In a word, its scope is as unlimited as that suggested in the *Republic* where every general name is said to call for a Form or Idea. This is the doctrine that has become quite familiar to us from our previous discussions; a note of unfamiliarity is added in Aristotle's summary of Plato's position.

The passage in question is found in the first book of the *Metaphysics*. Aristotle is speaking quite definitely of Plato himself.

Further, besides sensible things and Forms he says there are the objects of mathematics, which occupy an intermediate position, differing from sensible things in being eternal and unchangeable, from Forms in that there are many alike, while the Form is in each case unique. (987b14-18)

First of all, what is meant here by the objects of mathematics? If we consider that the geometer often uses two circles of the same diameter and the arithmetician in saying, "two plus two equals four" is employing "two" twice, we can see the problem the intermediates were intended to solve. According to the doctrine of Forms as we have it in the dialogues, there is only one twoness which is shared by all perishable couples. Thus "two plus two equals four" cannot be about the Form of twoness, nor does it seem to be about perceptible couples since it expresses a truth which does not perish. The plurality of "two's" then cannot belong to the realm of Forms nor to the world of becoming and an intermediate realm is called for. Aristotle's testimony is that Plato saw this and switched from a twofold to a threefold division of reality. And, while it is tempting to think that the divided line of the *Republic*, with its passage on the side of the intelligible from the hypotheses of mathematics to dialectic, is based on just this distinction between intermediates and Forms, it is extremely doubtful that the passage can bear this interpretation, although it is possible to see there the seeds of this later change.^{32}

In the continuation of the passage just quoted, Aristotle tells us of a hierarchy in the Forms of numbers, as opposed now to mathematical numbers.

Since the Forms were the causes of all other things, he thought their elements were the elements of all things. As matter, the great and the small were

principles; as essential reality, the One; for from the great and the small, by participation in the One, come the Numbers. (987b19-23)

Each number has unity and it has this by participation in the One; numbers differ from one another by addition and subtraction. We need not pursue this extremely complicated doctrine here. The point we are making is simply that the doctrine of Forms as we find it in the dialogues of Plato cannot be thought of as his final word on the subject. Whatever may be said of details of Aristotle's reports on the teachings of his master, there is far too much evidence for an unwritten doctrine of Plato that extended his views on reality for any dismissal on grounds of prejudice or misunderstanding on Aristotle's part. And, significantly enough, none of the advances recorded by Aristotle do the slightest bit towards diminishing the fact that, for Plato, there is another and better realm of things beyond the sensible particulars around us, a realm of things which is the anchor of knowledge and which subsists outside of time and space. Forms are not in sensible things nor are they concepts in our minds; they are objective realities introduced primarily to save the notion of knowledge as stable and permanent and to underwrite moral striving.

{29} W. F. Lynch, *An Approach to the Metaphysics of Plato* (Georgetown: University Publishing Co., 1959), pp. 4-5.

{30} F. M. Cornford, *Plato's Cosmology* (New York: Humanities Press, 1957), pp. 7-8.

{31} (Paris: F. Alcar, 1908).

E. Plato's View of Man

It may seem superfluous to introduce this new heading into our attempt to sketch the doctrine of Plato, since so much of what we have already said has indicated Plato's attitude towards man's place in nature, his ethical goals, the status of the human soul, the nature of the state. Nevertheless, these matters have hitherto been subsidiary to our feeble efforts to depict the central doctrine of the dialogues, the theory of Forms or Ideas. We want now to concentrate explicitly on what Plato has to say of the nature of the human soul, what ethical doctrine he enunciates, his view of the state. Needless to say it would have been possible to introduce the Forms in function of these discussions, but it is our hope that the sequel will indicate the reasons why we have chosen to proceed as we do.

The Soul. Plato's view of man is usually expressed in passages whose goal is the enunciation of what man must do, and moral obligation, in turn, is usually, or at least most significantly described in terms of the political order, of man's place in society. Nevertheless, we can glean from such passages a doctrine as to what man is, particularly what the nature of the soul is and, in the *Phaedo*, we have a discussion of the perfection of man which concentrates on the individual and makes no reference to the political context in which self-perfecting takes place. For this reason, the *Phaedo* is thought to have been written prior to the *Republic* in which the analogy between the parts of the soul and those of society becomes a major theme.

In asking what the nature of the soul is, for Plato, we shall begin our discussion with the *Phaedo* where, as we have already seen, Socrates is presented in his death cell surrounded by friends. The point of the dialogue, at least on the surface, would seem to be the formulation of a proof for the immortality of the soul. We shall see that there are reasons for qualifying this description of its purpose, but that proofs are offered is beyond doubt and by examining them briefly we can get a preliminary idea of what Plato thought the nature of the soul to be.

We have already discussed the confidence of Socrates in the face of imminent death that he is going to a better world. It is just the basis for this confidence which is sought in the arguments of the *Phaedo*, and it is sought against the background of Socrates' assertion that philosophy is a preparation for death, since philosophizing consists in the turning of the soul from the body and the realm of sense, a turning which already suggests the distinction between soul

and body. Cebes objects to the implication that the soul continues to exist after death and Socrates, though noting that in his present plight he will find it difficult to be indifferent to the outcome of the discussion, offers to seek the basis for his belief.

The first argument relies heavily on the notion that opposites are generated from one another. The just is generated from the unjust, the good from the bad, hot from cold and so forth, though the transitions are gradual and not necessarily abrupt: a hot thing cools and then is cold. If this is so and life has as its opposite death, must we not say that life comes from death as waking from sleeping? Thus, if the dead come to be from the living, it seems that the living must come to be from the dead.

To this argument is immediately linked another drawn from recollection (*anamnesis*). An allusion is made to a situation like that described in the *Meno*, where, by means of questions and a diagram, one can be shown already to know what he has not learned, at least not learned in this world. In other words, the soul once existed in another place before its being in a human form, it must have dwelt with the Forms or Ideas. "There is the same proof that these ideas must have existed before we were born, as that our souls existed before we were born; and if not the ideas, then not the souls." (76)

The reader will see that, having set out to show that the soul will survive the death of man, Socrates has twice shown that the soul existed prior to its union with the body; but has he shown that it will survive? Socrates says this is implicit in the foregoing, if we take the two proofs together. "For if the soul exists before birth, and in coming to life and being born can be born only from death and dying, must she not after death continue to exist, since she has to be born again?" (77) Socrates offers to make this explicit, and he begins by asking what it is that we think can corrupt. The compounded can corrupt, be dissolved, but the uncompounded or simple cannot. For instance, the Forms must always remain the same, unchanging, since they are identical with themselves. There are not many justices. Bodily things, on the other hand, are many and compound and always in a state of change. In an important remark Socrates asserts the affinity of our body with corruptible things and the affinity of our soul with the incorruptible Forms.

. . . the soul when using the body as an instrument of perception, that is to say, when using the sense of sight or hearing or some other sense (for the meaning of perceiving through the body is perceiving through the senses) -- were we not

saying that the soul too is then dragged by the body into the region of the changeable, and wanders and is confused . . . But when returning into herself she reflects, then she passes into the other world, the region of purity, and eternity, and immortality, and unchangeableness, which are her kindred, and with them she ever lives, when she is by herself and is not let or hindered; then she ceases from her erring ways, and being in communion with the unchanging is unchanging. And this state of the soul is called wisdom (79)

We have here the wisdom the love of which constitutes philosophy and the suggestion that there is a sense in which immortality is won by the acquisition of moral virtue, i.e. the triumph over the body. The immortality which is spoken of as deserved in this life is, so to speak, good immortality; in any case the soul will survive, but those souls which have not purged themselves of the effects of the body, will be imprisoned once more but this time in the bodies of brutes. Pleasure and pain are as nails which fasten the soul ever more surely to body; these snares can be avoided only by the study of philosophy which enables us to surmount the ignorance whose sign is vice and fasten the eye of the soul on true existence, the realm of the Forms. The passage in which Socrates describes the function of philosophy (80-84), too lengthy to be quoted and too polished to be paraphrased, has to be read in order to appreciate how, in this dialogue, the conjunction of moral excellence and the contemplation of the Forms produces an almost mystical view of philosophy.

The difference of soul from body is clearly expressed in the discussion we have reviewed and, if interaction between soul and body is admitted, the emphasis is on the deleterious aspect of the interrelationship. This distinction is underlined by Socrates' response to a doubt expressed by Simmias. Simmias cannot repress the thought that the soul may be simply the harmony or attunement of the body. Socrates responds by pointing out that one who accepts the previous arguments for the preexistence of the soul cannot make the soul a harmony, since it would be absurd to suppose a harmony could exist prior to its elements. Moreover, if soul is a harmony, the discord of vice is difficult to explain; indeed, this view would seem to make all souls good. Moreover, the fact that the soul is the ruler of the body suggests the point that the body is not in agreement with the soul nor vice versa; thus, the soul leads the elements of which she is said to be composed, opposes them and suppresses them. The soul's otherness from body is thus maintained and the theory that soul is a harmony of body rejected.

A final argument moves from the fact that whatever is three is also odd and cannot remain three and not admit of oddness to the assertion that since the

soul brings life and life cannot admit death, the soul withdraws before the approach of death. The procedure is, of course, a good deal more complicated than this and has often been criticized. For our purposes, it is one more indication that soul is taken to be other than body by Plato whether or not he is able to prove this satisfactorily. In the Tenth Book of the *Republic* the point is made that, since the soul is not destroyed by what is its own greatest evil, vice, it can hardly be destroyed by evils of the body. In the *Phaedrus*, the immortality of the soul is based on the notion of soul as self-mover.

The soul through all her being is immortal, for that which is ever in motion is immortal; but that which moves another and is moved by another, in ceasing to move ceases also to live. Only the self-moving, never leaving self, never ceases to move, and is the fountain and beginning of motion to all that moves besides. Now, the beginning is unbegotten, for that which is begotten has a beginning; but the beginning is begotten of nothing, for if it were begotten of something, then the begotten would not come from a beginning. But if unbegotten, it must also be indestructible; for if beginning were destroyed, there could be no beginning out of anything, nor anything out of a beginning; and all things must have a beginning. And therefore the self-moving is the beginning of motion; and this can neither be destroyed nor begotten, else the whole heavens and all creation would collapse and stand still, and never again have motion or birth. But if the self-moving is proved to be immortal, he who affirms that self-motion is the very idea and essence of the soul will not be put to confusion. For the body which is moved from without is soulless; but that which is moved from within has a soul, for such is the nature of the soul. But if this is true, must not the soul be the self-moving and therefore of necessity unbegotten and immortal (245-6)

This same definition of soul is found in the *Laws* as last of the ten kinds of motion there distinguished. (893 ff.) It should be said in conclusion that the proofs of immortality set forth in the *Phaedo* seem to function as much as emotive appeals as appeals to reason alone. This is true not only because of the setting but because philosophy itself consists in a movement away from immersion in the world of sense, a movement which will be begun only on the assumption that the soul is immortal. But of course, until one has made the movement, the eye of his soul is not clear enough to grasp the truth. Thus, while the purpose of the dialogue may seem to be the proof of the soul's immortality, its more subtle role is as an exhortation to philosophy. It is for this reason that, as was pointed out earlier, the true proof of the dialogue is the represented composure of Socrates in the face of death.

In the *Republic* Plato wishes to proceed from an analysis of the state to that of the individual, a procedure we will be discussing in a moment. The ideal commonwealth sketched is, of course, an ordered whole; thus, when Plato turns to the individual, it is perhaps not surprising that he begins to speak of parts of the soul. What has been said in the *Phaedo* concerning the affinity of soul with the Forms does not prepare us for this, since there the Forms are argued to be simple and uncomposed and, presumably, the soul is too. The state has been shown to consist of three groups: those who deliberate and govern, those who execute policy, and the craftsmen. Now if the state is the soul on a larger scale, we should expect this threefold division to obtain in some way in the soul.

But the question is not quite so easy when we proceed to ask whether these principles are three or one; whether, that is to say, we learn with one part of our nature, are angry with another, and with a third part desire the satisfaction of our natural appetites; or whether the whole soul comes into play in each sort of action -- to determine that is the difficulty. (436)

In order to decide the question, we must first accept the following principle, namely, that the same thing cannot act or be acted on in the same part or in relation to the same thing at the same time in contrary ways. Thus a man cannot be in motion and rest at the same time in the same respect; he can however move his hands while resting in the same place. Now much the same kind of distinction must be drawn when we consider that a man may at one and the same time desire and not desire a drink; if both desires are attributed to soul, it seems we must distinguish that part of the soul with which a man hungers and thirsts, the appetitive or irrational, from that with which he reasons, the rational part. There is as well a part with which we feel indignation and anger, a spirited or courageous part. That this is distinct from the appetitive part is clear from the fact that they are often in conflict as when we are angry because we desire something; that it is distinct from reason is shown by its presence in children and brutes who have not the use of reason. This spirited part of the soul tends to be the ally of reason in disputes with the appetitive part. Of the three parts of the soul, only the rational part is immortal (*Timaeus*, 69) and it is composed of the same mixture as the world soul. (*ibid.* 41).

It is sometimes suggested that when Plato attempts to establish the immortality of the soul all he is able to do is to indicate that spiritual substance cannot be corrupted, but that he cannot establish personal immortality, that is, that my soul as mine will survive. This objection would have force if my soul could be said to be mine thanks to dwelling in this body, and yet Plato often speaks --

although admittedly this is usually in mythical flights of fancy -- of a plurality of souls being in existence prior to their assuming a body and then returning once more, certain conditions having been fulfilled, to an existence apart from body. Thus, the earthly career of a soul could not be said to constitute it as a personal one and its persistence would be personal. Plato speaks of the soul using a body, taking over a body, ruling a body; that is, the soul is individual and substantial in its own right and could not lose these features at death. There is, however, another side to the matter. Contrary to the view expressed mythically in the *Timaeus* according to which all souls are presented as having an equal chance at least in their first birth into bodies, Plato, speaking more matter-of-factly in the *Laws* (VI, 775), indicates that the sins of parents can be visited upon their children, that intemperate parents, for instance, generate children who will inevitably stray from the right way. Hence during the whole year and all his life long, and especially when he is begetting children, he ought to take care and not intentionally do what is injurious to health, or what involves insolence and wrong; for he cannot help leaving the impression of himself on the souls and bodies of his offspring, and he begets children in every way inferior.

Such a view presupposes a closer relationship between soul and body than other remarks, but the union seems to be that of two things rather than of two principles of one thing. What seems never to recede is the view that our soul is our better being, our true being, with the body somehow alien and unnatural. Death, consequently, is not something the fear of which is justified; rather, philosophy will enable us to see that death releases the soul -- and us -- to a better life.

In the *Phaedrus* Plato describes the three parts of the soul in the following way: the rational part is a charioteer, the spirited and appetitive parts are two horses. Man's task is to bring the two steeds under the control of reason. It is because Plato asks what man is when he is seeking the answer to what man ought to do that any discussion of Plato's view of the soul must be juxtaposed to his doctrine on morality. The latter doctrine, as we have already mentioned, seems inseparable from Plato's political theories and we must determine why this is so.

Morality and Politics. The seventh letter, although written late in Plato's life, describes his outlook as a very young man. The young man he remembers is one who was vitally interested in the activity of Socrates, a Socrates who was questioning the assumptions on which his fellow Athenians based their lives, who seemed always interested in the problems of the state but who nevertheless kept himself curiously aloof from practical involvement. The letter recounts that

Socrates refused to be enlisted by the Thirty Tyrants (among whom were numbered relatives of Plato, relatives who asked the young Plato himself to join their movement) in an effort to execute a friend of the exiled democrats, and that, ironically, these same democrats executed Socrates when they had returned to power. Plato indicates how the failure of the Thirty to eradicate the evils of the city depressed him and how his depression increased when the government had changed and Socrates was condemned. The next step, we should think, would be withdrawal from politics, both as practical vocation and theoretical interest, but here as always Plato is surprising. On the practical level, he seems to have kept clear of Athenian politics, but we have seen his extended involvement in Syracusan government; moreover, the Academy became a training ground for men who wrote laws and constitutions for a number of states. On the theoretical side, as the seventh letter indicates, a question arose which intrigued Plato throughout his life; if society is corrupt because lawgivers are corrupt, does not the only hope lie in putting power in the hands of those who are not corrupt, that is, those who have studied philosophy? To break the vicious circle of social evil, there must first be some good men who would so devise a state that its citizens would be trained in virtue; vicious men in power will only perpetuate vice in themselves and in their subjects. Two of Plato's works, and they are the two longest, are devoted to this problem which is touched on in many of the other dialogues as well. The *Republic* and the *Laws* are the great sources; the first dating from what is called Plato's middle period, approximately from the time of the founding of the Academy, the second thought to be the last thing Plato wrote.

Republic. This dialogue has come down to us in the form of ten books, a division which does not divide the subject matter. The discussion breaks rather easily into the following parts: (1) introductory: consideration of certain opinions as to the nature of justice (Book I and the first third of Book II -- to 367); (2) the structure of the ideal society (to the end of Book IV); (3) how the ideal society can be achieved: the philosopher king (Books V-VII); (4) the declension of society: stages of corruption of the ideal state, (Books VIII-IX); (5) otherworldly sanctions of justice, preceded by comparison of philosophy and poetry.

The entertainment of views of justice comes about at the outset of the *Republic* in a familiar way with Socrates asking for light on the matter. He turns his attention first to Cephalus, an old man, retired from business, and asks him how life looks to him now that he stands on the threshold of the beyond. There are a few remarks on the advantages of being freed from youthful passion the echo of which we find in Cicero's *De senectute*; this quieting of the flesh provokes

retrospective thought about one's life and an uneasiness at the thought that one will be held to account for it. The advantages of ending up with a tidy fortune are summed up in terms of half of Cephalus' description of justice; one must pay his debts. The other half consists in telling the truth. While continuing to pay deference to the old man, Socrates suggests that, if justice be what Cephalus says it is, there are times when it would be manifestly wrong to be just. For example, if one has borrowed a weapon from a man who goes mad, it would surely not be right to return it to him nor to tell him the simple truth. Cephalus graciously withdraws, leaving the defense of his definition to his son, Polemarchus.

Polemarchus argues that his father's position is simply that of the poet Simonides who had defined justice as giving to each his due. That the poet is a safe guide in disputes of this sort is something which will be denied later in the *Republic*, but at this point Socrates proceeds as so often elsewhere by interpreting the poetic dictum to show that it cannot not be true. That is, the poetic utterance is treated as inspired but requiring subtle exegesis. When Socrates repeats the difficulty of the lunatic's loan, Polemarchus interprets the poet to mean that we should help our friends and harm our enemies. Of the many difficulties Socrates raises, we can select the following. If justice is doing good to friends and harm to enemies, and we can be mistaken about our friends, deeming an unjust man just, justice may require doing good to an unjust man. If however this is amended to say that we ought to do good to the just and harm to the unjust, Socrates will not allow that the function of justice is to do harm to anyone. This assertion that we ought to do good to our enemies is one of the points that lifts Plato far above the assumptions on which most men operate. It is as well what triggers off Thrasymachus, whose attitude towards justice, although it shifts in his interchange with Socrates, is an articulation of man's worst motives.

Thrasymachus (a Sophist mentioned in an earlier chapter) literally crashes his way into the dialogue; he has been represented as grumbling impatiently through the discussion between Polemarchus and Socrates. When he finally bursts out it is to take exception to Socrates' whole method; Socrates should come out with his own statement as to what justice is. The implication is that Thrasymachus himself could do this; he allows that he could, and Socrates applies for instruction. The definition Thrasymachus offers is this: justice is nothing else than the interest of the stronger. It turns out that this means that in any form of government, the ruling party makes laws for its own interest, thereby making the obeying of these laws on the part of the subjects the

protecting of the interest of the ruler. Socrates objects that since rulers are not infallible, they can sometimes make a law which they think to be in their own interest though in reality it is not; then justice will turn out to be doing that which is not in the interest of the stronger. The retort of Thrasymachus is of great importance since it will lead to the downfall of his position! He does not mean to say that the ruler deserves the name when he is doing something contrary to the art from which he is named but only when he is acting in accordance with that art. For example, the physician as physician does not make mistakes, since he is called a physician precisely insofar as he possesses the art which enables him to cure. If the man who is a physician causes harm to a patient, he does this not insofar as he possesses the art of medicine, but because of some deficiency. So too the ruler as ruler always legislates in his own best interest.

What Thrasymachus has done in this precision is to introduce the notion of an art and the connected question as to the interest of an art. Arts are devised to make up for certain defects, as the art of medicine arises from the fact that the human body is susceptible of disease. Now the interest or end of any art is precisely to supply those defects which have prompted its emergence; the interest of medicine is to eradicate illness. Thus the interest of the physician precisely as physician is to cure illness; if he is interested besides in

collecting a fee, this is not something which belongs as such to the art of medicine, since the carpenter and plumber and portrait painter can also be interested in collecting a fee, but this cannot be their interest *qua* carpenter, *qua* plumber or *qua* portrait painter. We can see what Socrates can now say of the art of governing: the ruler governs for the sake of the governed, this is his function precisely as ruler; if he is interested in self-aggrandizement, this is not insofar as he is a ruler. Thrasymachus' reply to this is not so much an argument but a stating of the facts of life for the naive Socrates. The thrust of his statement is that the unjust prosper on every level of life if only they are skillful in their injustice. In business partnerships and pickpocketing, in paying income taxes and in governing, it is the unjust man who profits. A sign of this is the success of tyrants.

But when a man besides taking away the money of the citizens has made slaves of them, then, instead of these names of reproach, he is termed happy and blessed, not only by the citizens but by all who hear of his having achieved the consummation of injustice. For mankind censure injustice, fearing that they may be the victims of it and not because they shrink from committing it. And

thus, as I have shown, Socrates, injustice, when on a sufficient scale, has more strength and freedom and mastery than justice; and, as I said at first, justice is the interest of the stronger, whereas injustice is a man's own profit and interest. (344)

Socrates returns to the notion of the interest of an art as such and will not allow that the profit one gets from performing an activity is the interest of that activity since making a profit is common to many activities.

The larger question raised by Thrasymachus to the effect that success in life amounts to acting unjustly is one that elicits the characteristically Platonic attitude. What we get, however, is not simply the counter assertion that justice is everywhere to be preferred to injustice, but an examination of the original assertion. One way this is done is by pointing out that one cannot be consistently unjust; a band of thieves is possible only if its members do not rob one another. Any united action demands justice among the members of a group and would be undermined by injustice. Thrasymachus then is prescribing a mode of conduct which is subversive of any joint action among individuals. What is more, where injustice is prescribed in this way, it cannot be productive of well-being or happiness. There is no need for us to consider the caliber of the arguments whereby Thrasymachus is made reluctantly to admit that justice is the good of the soul and injustice its defect; from this it follows that, when the soul is deficient in its proper excellence, it cannot perform its task well nor can the well-being or happiness of the agent result.

That justice is desirable in itself is not thought to have been proved, and Plato's brothers, Glaucon and Adeimantus, now urge Socrates to continue. This implies the division that Glaucon spells out: goods are either sought for themselves alone; for themselves but also for their results; and simply for the sake of something else. Socrates would put justice in the first class, but most men would disagree, placing it rather in the third. One is just because it pays, with money, honor or reputation. What Glaucon proposes to do is to adopt a position like that of Thrasymachus and see if Socrates can convince him that justice is truly a good in itself. Is not the order of justice simply a compact men have made? They will not wrong others if they themselves will not be wronged; this is the origin of law and justice which are not sought for their own sake; they are a compromise reached by those who despair of ever fully triumphing over their neighbors. Moreover, without the constraint of law, there would be no distinction between the just and unjust. If one could act with impunity, would he avoid doing what is called unjust; if I could act just as I please, would it

please me to do what is now called the right thing? Finally, if we take the just and unjust as perfect types, could the just man who is truly just but is not honored as such, but rather punished and pilloried by his fellows possibly be called happier than the perfectly unjust man who prospers and is praised? Adeimantus supplements his brother's case by arguing that justice is always commended, not for itself, but for the advantages it brings, respectability, advancement, etc. One thinks of Yeats' line to Lady Gregory: "Only God could love you for yourself alone and not your yellow hair."

Plato's brothers have presented this description as something commonly accepted and as something they want Socrates to dissuade them from accepting. They pose the central problem of the *Republic* which is to show that justice is an intrinsic good which does not require certain concomitants and effects to be seen as good and that injustice is such an evil that any concomitant or resultant advantages cannot lessen its evil.

The problem having been set, it remains to sketch the program of the subsequent discussion. On the assumption that justice is something to be found not only in the individual but in the state as well, Socrates argues that it will be easier to discern in the state and that, once discerned there, they can argue by analogy to justice in the individual. He proposes therefore that they examine the evolution of political society, and Socrates' first point is that the state comes into being out of natural needs -- as against the previous assumption that it is a kind of unnatural imposition which thwarts the individual. This can be seen by observing the dependence of men on one another for such elementary things as food, shelter and clothing. A division of labor is preferable among the arts, with an exchange of products, and Socrates is enabled to move rather swiftly from an imagined group of four or five men to a complex society comprising artisans and farmers, merchants and sailors and so forth. The sketch ends with the suggestion that, in such a situation, justice will be looked for in the economic dealings of the members of this society with one another.

It has been observed that Plato is not so much constructing an imaginary state as he is describing such a city as Athens on its fundamental level. This being so, the diet and dwellings and diversions of the citizens are of the simplest order and Glaucon would allow them a few luxuries. This entails enlarging the community, to include not only hunters and fishers, but also poets and other artists, nurses and servants, barbers, etc. etc. This seems to require expansion of territory, hence war whose origin is thus located in desire for things beyond the necessities of life. The need for war implies warriors, guardians of the state and

these must be chosen because of natural gifts. Guardians must possess courage, they must have a gentle nature and great spirit and be, like a watch dog, kind to friends and fierce with enemies. The ability to discern friend from foe comes with knowledge, so the guardian will possess a love of wisdom, of philosophy. The problem then becomes, how are we to train persons of this nature that they might become good guardians?

In describing the early training of the guardians, Plato continues to work with the existent Athens as his model, correcting where he feels correction is due. If we think of this primary education as consisting of grammar, music and gymnastic, we find that, with respect to the first two, Plato is concerned with what is read, the way it is read and the musical accompaniment. As to content, he does not want the future guardians filled with stories of the immoral exploits of heroes nor with absurd and contradictory statements about the gods. He suggests, in effect, a censoring of the poets traditionally read in the schools, and Homer is by no means exempted. The fact that the student had to give a dramatic recitation of poems, throwing himself into the story and identifying himself with the characters, increases the importance of there being acceptable heroes with which to identify himself. Finally, the musical accompaniment of the poetry has to be scrutinized to make sure that the modes employed inculcate the proper disposition in the young, harmony and the harmony of the soul. This transition indicates the role poetry was intended to play in the moral education of the young, giving them a first glimpse of that beauty the love of which enables one to transcend the order of images.

The education Plato has described is to continue to the age of twenty; at that age, a few will be selected to receive a higher training and ultimately to become rulers. That is, the state will consist of rulers, guardians who execute the wishes of the rulers, and the class of artisans. Members of each level are determined by natural aptitude, not by birth or wealth and so on. The guardians themselves will lead an ascetic life, having no private property. Given these elements of the state, Socrates can go on to inquire after the virtues of the state.

Plato assumes that there are four virtues: wisdom, courage, temperance and justice. The question now is, how are these -- particularly justice -- the virtues of the state, meaning by this, as Cornford points out (p. 119), not the virtues of some abstraction but the virtues of individuals precisely as they are citizens of the state described. Wisdom will be a kind of knowledge and can only be that which resides in the rulers, a group much smaller than any other that can be said to possess knowledge of the function it fulfills.

And so by reason of the smallest part or class, and of the knowledge which resides in this presiding and ruling part of itself, the whole state, being thus constituted according to nature, will be wise; and this, which has the only knowledge worthy to be called wisdom, has been ordained by nature to be of all classes the least. (429)

The state will be said to possess courage too because those guardians who are not as well rulers possess it. This courage is defined in terms of right opinion as to what is to be feared and what not feared, an opinion that the guardians have thanks to their early education. The temperate man is sometimes said to be the master of himself, which seems to imply that he is also in some sense the subject of himself; let us understand this to mean that there is a better and worse part of a man and that temperance consists of the mastery of the better over the worse. In the state, temperance will be a virtue, not simply of a part, but of all the citizens insofar as they willingly accept the hierarchical structure of the state. This leaves us with the need of describing what it is in which the justice of the state will consist. Socrates recalls that in describing the state on its most primitive level, it was suggested that justice had something to do with each man performing his own task or job. Can justice be something like minding one's own business? If we had to decide whether the wisdom, courage or temperance of the state, as these have been described, are more important than each citizen doing his own job, we would be faced with a difficult choice. One reason for the difficulty appears when we consider that the rulers, in judging lawsuits, will want to take care that each man is given his due, what properly belongs to him. Now if carpenters could become soldiers and soldiers rulers and one man generally usurp the function of another, we would have what might be called a community of injustice. "Seeing then, I said, that there are three distinct classes, any meddling of one with another, or the change of one into another, is the greatest harm to the state, and may be most justly termed evil-doing." (434) Justice will be the quality whereby each citizen wants to preserve the order of the state.

It is at this point that, on an analogy with the state, the soul is said to have three parts corresponding to the ruling, executive and productive classes. The genesis of virtue in the individual is linked here with the program of primary education.

And ought not the rational principle, which is wise, and has the care of the whole soul, to rule, and the passionate or spirited principle to be the subject and ally? And, as we were saying, the united influence of music and gymnastic will bring them into accord, nerving and sustaining the reason with noble words and

lessons, and moderating and soothing and civilizing the wildness of passion by harmony and rhythm? (421-2)

Wisdom and courage will then rule over the appetitive and lead to temperance. Once more now the question becomes, what is justice? We must see this question against the background of the statement of it by Glaucon and Adeimantus. Remember that they wanted a description of justice which would show that possession of it was an intrinsic good, apart from advantageous consequences.

But in reality justice was such as we were describing, being concerned however not with the outward man, but with the inward, which is the true self and concernment of man: for the just man does not permit the several elements within him to interfere with one another, or any of them to do the work of others, -- he sets in order his own inner life, and is his own master and his own law, and at peace with himself, and when he has bound together the three principles within him, which may be compared to the higher, lower and middle notes of the scale, and the intermediate intervals -- when he has bound all these together, and is no longer many, but has become one entirely temperate and perfectly adjusted nature, then he proceeds to act, if he has to act, whether in a matter of property, or in the treatment of the body, or in some affair of politics or private business; always thinking and calling that which preserves and cooperates with this harmonious condition, just and good action, and the knowledge which presides over it, wisdom, and that which at any time impairs this condition, he will call unjust action, and the opinion which presides over it ignorance. (443-4)

We have here a suggestion that justice is in some way the totality of virtue, a view that will be pursued by Aristotle. Moreover, we see the retention by Plato of the Socratic maxim that knowledge is virtue, and vice ignorance. The description of justice as a harmony in the soul, permits Plato to liken it to the health of the body; this metaphor can apply as well to the state as a whole, and a gradation of types of government be drawn in terms of a greater or lesser approximation to true health. The basic analogy of state and soul will permit Socrates to equate the best form of government, that which he has been describing and which can be called monarchy or aristocracy depending on the number of rulers, and the best form of soul, and then move through the types of government which fall short of the ideal and have as their analogues imperfect conditions of soul. This is not taken up until what we have given above as part 4

of the *Republic*; Plato discusses the status of women first and then the central doctrine of the philosopher king.

The *Republic's* attitude towards women follows on the view that nature prepares individuals for one or another role in society and, while Socrates admits that nature has devised for male and female different roles in procreation, he does not see that this in any way stands in the way of their performing the same role in other tasks. Thus women of talent can be trained as guardians and even be selected from the guardians as rulers. Socrates is willing to accept the fact that, generally speaking, women are inferior to men with respect to the best pursuits, but does not feel that this precludes the possibility that some women are better than most men even with respect to what is best. A second point is that the guardians are to have wives and children in common, so that a man will not know which children are his. Nor is the breeding of human beings to be left to chance; rather the rulers will contrive to bring together males and females who stand the best chance of producing perfect offspring. The children will be put in the care of nurses; defective children will be destroyed. In this way, it is argued, the interests of the guardians will not be distracted from their civil function by private attention to wife and family. Needless to say, having wives in common cannot be construed on the model of a harem, the limitless possibility of orgy and promiscuity. The nature and training of the guardians will ensure their virtue and temperance in matters of sex. Plato is swept so far as to see no difficulties in mixed gym classes with all participants nude. But then, returned from his flight into theoretical eugenics, Socrates admits that he is contemplating only a possible state to which existing ones can only approximate. This approximation will take place in actual states only when philosophers are kings or kings philosophers: we have reached the famous and central Platonic contention.

We need only sketch here the procedure of this famous discussion; its most important doctrines have already entered into our presentation of Plato's doctrine of Forms. The first step consists in establishing the distinction between knowledge and belief with the corresponding demand that the rulers differ from the guardians by passing from belief to knowledge. It may seem that the philosopher's preoccupation with Forms makes him unfit for the practical task of ruling, but Plato argues that this is a preoccupation which precisely rids him of impediments to right ruling. Invoking the image of the ship of state whose captain is the people with a mutinous crew, the politicians, the role of pilot is assigned to the philosopher. This story enables Socrates to argue that the apparent uselessness of the philosopher amounts to little more than mankind's

failure to make use of his wisdom. This is not to say that philosophical natures are not often corrupted in the present state of society. Rare spirits, their virtues militate against achievement of their ultimate possibilities. For example, abounding in courage, the potential philosopher will be called upon to perform tasks which prevent him from devoting himself to study. Even should he study, the actual state of education will turn him into a veritable monster: *corruptio optimi pessima*. What is more, it is now possible for people of little or no talent to devote themselves to philosophy, a fact which is not calculated to bring philosophy to a place of honor.

Turning from the actual to the possible, Socrates begins to discuss how the situation can be rectified. This calls for a return to the discussion of the four virtues, a discussion which can now be shown to have been inadequate. There is a knowledge higher than that involved in justice and the other virtues discussed:

You have often been told that the idea of good is the highest knowledge, and that all other things become useful and advantageous only by their use of this . . . Do you think that the possession of all other things is of any value if we do not possess the good, or the knowledge of all other things if we have no knowledge of beauty and goodness? (505)

It is here that the Form, Goodness is likened to the sun in the visible order; this is followed by the discussion of the divided line and then Socrates tells the parable of the cave. These connected passages lead to a description of the higher education of the rulers, rulers who have now been described as philosophers, which in turn is taken to involve contemplation of the Forms. Now we have already seen that the program of primary education was to have been pursued until the age of twenty; the years from twenty to thirty are now designated as those to be spent in the study of mathematics, of arithmetic, plane and solid geometry, astronomy and harmonics. From the age of thirty to that of thirty-five, future rulers are to be instructed in dialectic. By acquainting the young man with the realm of the Forms, it is hoped that his soul will be brought into harmony with them, that he will become attuned to genuine reality.

Just as earlier the division into members of the productive class and the guardians was treated as a simple division, to be made more complex later by a division in the latter class between those who would be selected as rulers and those who would not, so now Plato is in a position to speak of the cut-off points on the route to the term of philosophical studies. Some of those who study mathematics for ten years will be selected to study dialectic. From the age of

thirty-five to that of fifty, these men will fill posts of public service. Fifty will be an age at which some will have arrived at the vision of Goodness and their remaining days will be divided between study and service in the highest deliberative council of the state. This part is concluded by saying that the ideal described can be approximated if only there are well trained philosophers who are given the right to refashion society.

Books VIII and IX of the *Republic* present, as Nettleship observes (p. 294), the counterpart to the preceding description in which Plato has described the ascent of the human soul to its highest possible condition; now Plato will show how low the soul can fall and through what stages it may be seen to pass to its nadir. Since evil is a kind of negation, its gradation can be measured in terms of its degradation from the ideal already described. The relation between the ideal society Plato has endeavored to describe and the soul of man is given by Nettleship. "The best man would be one whose self was as nearly as possible identified with the life of the society of which he was a member, and ultimately with the laws of order of the world of which he, and the society also, were parts." (p. 299) The just man, like the just society, is such because of the organization of his parts according to their natural priority and posteriority. Where this order is lacking in a sufficient number of individuals, their disorder can come to be reflected in the state of which they are citizens. Thus, in what Plato calls the timocracy, where the spirited element takes the ascendancy and those who would be simple guardians occupy the highest rank, praise (*time*) becomes the object of action. A lower type of society, oligarchy, is the reflection on the level of government of the predominance of the acquisitive sense in the individual: oligarchy thus is plutocracy, government by those whose sole concern is wealth. The descent to democracy is to a condition where the state reflects the individual driven by all the baser appetites. The lowest type of government is the despotic or tyrannical where the tyrant balances on one end of the scale the philosopher king on the other.

The tyrant is the exact counterpart of the philosopher. The philosophic king is at one with everybody and everything about him. The tyrant -- his personality concentrated in a single dominant passion -- is absolutely alone; he is the enemy of his own better self, of the human kind, and of God. Theoretically the owner of the state, in reality he is absolutely poor. (Nettleship, p. 300)

The tyrant so described meets the specifications of the unjust man described by Thrasymachus and Plato will now deny, in answer as well to the problem posed by Glaucon and Adeimantus, that such a man can be happy. He is slave of his

passions and the licence he can allow himself cannot be confused with freedom since he is unable to perform those just actions which alone answer to the nature of man and produce his well-being. It is because his being has been made to harmonize with the pattern laid up in the heavens that the just man is happy; his happiness does not depend upon pleasure or wealth or honor. In this life, then, justice is its own reward, but at the end of the *Republic*, Plato introduces the belief in the immortality of the soul, indicating that one's condition in the afterlife is determined by the mode of existence chosen in this. It matters little then if the just man be mocked and scorned in this life -- Plato is doubtless thinking of Socrates -- his true reward awaits him beyond the grave. Despite this, Plato will not allow that it is injustice men prize and justice they condemn. Even in this life, justice has external rewards, though these are not of course constitutive of it nor the true source of the happiness it brings. In order to stress that it is not this world which confers his true reward on the just man, Plato concludes the *Republic* with the myth of Er, a story of the soul's journey after death. Similar stories are to be found in the *Gorgias*, *Phaedo* and *Phaedrus*. It is as if Plato, dissatisfied with arguments to prove immortality or recognizing the need of supplementary images, desires to give a fabulous portrayal of what lies beyond. We shall say something about this appeal to myths in our concluding section; at the same time we can take into account Plato's remarks on the relation between poetry and philosophy, another feature of the last book of the *Republic*.

Laws. While any complete account of Plato's political theory would have to take account of the *Statesman*, our brief sketch must content itself with indicating the relation between the *Republic* and the work of Plato's extreme old age, the *Laws*. The *Laws*, while it carries on the pretence of being a dialogue, is actually a long disquisition by an Athenian Stranger strongly reminiscent of Plato himself to two other old men, one from Crete, the other from Sparta. Estimates of the work vary, some holding it to be patchwork without connecting theme, others arguing that the absence of the literary flights of the earlier dialogues blinds us to the tight logical organization of the piece. The immense detail of the work make it impossible to give of it anything like the summary we attempted to give of the *Republic*; nevertheless, we must ask ourselves what relation this late work bears to the *Republic*.

The *Laws* has come down to us in twelve books and it can be divided in two, the first three books forming an introduction to the planning of a city which will approximate the ideal. That the task of the dialogue is indeed to frame a state is

not made known until the end of the third book; the remainder of the *Laws* is devoted to that task.

The introductory books leave little doubt that Plato is not writing the *Laws* to record basic differences with the views he expressed in the *Republic*. The judgment that Sparta had overemphasized the spirited element suggests the notion of oligarchy described in the earlier dialogue. The primacy of pleasure and pain and their consequent importance for moral education reveal the characteristic Platonic concern with education. "Pleasure and pain I maintain to be the first perceptions of children, and I say that they are the forms under which virtue and vice are originally present to them." (653) Education is the training of these impressions, making the young take pleasure in the good. The whole of book seven of the *Laws* concerns itself with education and what one notices is the vast detail, psychological and historical; there is no doubt that the *Republic* is a more exciting work to read, but from the point of view of content, it seems a sketch which is filled out at great length in the *Laws*. The same may be said of the descriptions of the genesis of the state in the two works. In the earlier work, Plato was content to take the fundamental commercial aspect of Athens and treat it in abstraction from the other aspects of the city; now he attempts a truly genetic description of the state, based on a cyclic view of history: civilizations advance and then are destroyed. Let us begin then with the remnants of society left after a flood: we find a few shepherds left in the hills. They have no arts or metals or means of transportation; they lead an utterly simple life, with no letters and no law, the form of authority being patriarchal. Gradually there is a movement to the foot of the mountains and the beginning of agriculture, the grouping of families, the need for a legislator. In discussing the task of the legislator, Plato insists that he must be concerned with all four cardinal virtues and not just one. That this is so, that states have in the past collapsed because of the lack of harmony described in the *Republic*, is illustrated by appeal to the Persian monarchy and the Athenian democracy. It is at this point that Cleinias the Cretan indicates that he and nine others from Cnossus have been commissioned to found a colony on the site of a destroyed town and he suggests that the Athenian indicate how one should go about framing such a state. The remaining nine books have this as their purpose.

The topography of the proposed city is first discussed and it turns out that it will fit the Athenian's specifications; he wants it sufficiently far from the sea so that it will not become engaged in exports and, we may surmise, go the way of Athens once it had become a sea power. The fact that the land is not extraordinarily good will prevent a surplus of crops and the temptation to trade.

What the Athenian wants is a self-sufficient community, fairly isolated from neighbors and the dangers of dispute, not productive enough to go into trade. There is then the matter of the selection of the colonists, followed by a discussion of the kind of ruler who will be most likely to bring about the best possible state. Plato argues that a gifted despot will be best: it is far easier to convert one man to the cause of good government than to try to persuade the multitude; moreover, the example of the ruler will be most powerful in bringing about the proper attitude in the citizens. One can hear echoes here of Plato's efforts in Syracuse. The laws themselves must be presented as leading the citizens to virtue; for this reason they must at once command and persuade, and the Athenian suggests a great preamble to the laws. The basic principles to be stressed are: respect for the gods; respect for parents; respect for self and for others. There is an order in self-respect, for one must first honor his soul and then his body. The greatest threat to lawful society is selfishness.

The size of the population is next discussed; Plato proposes 5040, meaning, it seems, that number of homes and not of people. The population is to be divided into twelve tribes; there will be thirty-seven men between the ages of fifty and seventy who will be the guardians of the constitution. The representative chamber will have three hundred and sixty members. The most important post of all is that of minister of education. (766) Before turning to the subject of education in Book Seven, Plato discusses courtship, marriage and procreation; what he is concerned with is that children be conceived with a view to the good of society, which will in turn provide apt subjects for the education he will next describe.

The discussion of education begins with the need for exercise on the part of the expectant mother and goes on to suggest frequent rocking by the nurse during the first years of the child's life. The life of the child should be happy and content, free on the one hand from softness and coddling and on the other from exposure to objects of fear. From three to six years children are to play in the village temple; at the age of six, boys and girls are to be separated for the study of music and for gymnastics. They are to be trained in the use of arms and to become ambidextrous, something useful in battle. Reading and writing are to be taught from ten to thirteen years of age after which three years are to be devoted to the study of music which includes arithmetic, geometry and astronomy. We have already suggested the relation of this book to the discussion of primary education in the *Republic*; the *Laws* can be said to contain at once both more and less than the earlier work, less because the *Republic* seems clearly presupposed, more because the *Laws* is almost tedious with detail. The

discussions of the proper melodies to inculcate virtue, of the education of women, of the value of tradition in poetry add to those of the earlier work; besides there are discussions of memorization, hunting, and the value of astronomy.

Books Eight and Nine are devoted to discussions of contests and the connected question of sexual morality; boundary disputes; commerce; conservation of resources; craftsmen; homicide and crimes against the state. So too Books Eleven and Twelve concern themselves in great detail with the various aspects of the state, the discussion ranging from the question of the marriage of fatherless daughters to funeral arrangements. Book Ten is of special interest since it contains the theology of Plato.

The Tenth is doubtless the most eloquent book of the *Laws*; it finds its place there since respect for the Gods is one of the basic presuppositions of the great preamble to the laws and yet there are those who would either reject the existence of the gods or entertain attitudes towards them which would defeat the function of belief in the schema of the laws of the state. However, although the Tenth Book has this justification for inclusion in the *Laws*, it is admittedly a great digression, though ultimately a necessary one. Its length is justified by making appeal to the leisurely procedure throughout the work; haste is not required, no one is pressing on the heels of the three old men.

There are three positions that Plato wishes to confront in Book Ten: the denial of the existence of the gods; the characterization of the gods as having no concern for the affairs of men; the claim that the gods can be bought and won over to the cause of injustice. With respect to the atheists, it will not do to allude to the order of the universe, as Gleinias suggests. The Athenian observes that there are those who would mock the attempt to prove the gods exist by appeal to the heavenly bodies, since these bodies are nothing but earth and stone in orbit. The reference seems to be to those natural philosophers who suggest that if only one can get down to the basic stuff of things, to the elements, he will see that everything else is built up from them and cannot transcend in nature the nature of the elements: that is, nothing can be more divine than fire, air, earth and water. The Athenian thus makes clear that he is not concerned with the barroom atheist, but rather with those who profess to have philosophical reasons for rejecting the gods. With these, as he observes, it is difficult to be calm.

Who can avoid hating and abhorring the men who are and have been the cause of this argument; I speak of those who will not believe the tales which they have heard as babes and sucklings from their mothers and nurses, repeated by them both in jest and earnest, like charms, who have also heard them in the sacrificial prayers . . . (887)

They have seen their parents exhibit the conviction that gods exist; they are aware that such belief is common to Greeks and barbarians -- and still they disbelieve. Is Plato here indiscriminately endorsing every popular religion? He, like the nurse he describes, would repeat these tales in jest and earnest: we have seen his impatience with tales of the gods which demean the divine; nevertheless, every religion embodies the essential truth that there is intelligence in the universe, that man is subject to a higher principle, that there are sanctions for conduct, that death is not the end. Plato seems to feel that whatever the form these convictions may take from place to place and from people to people, one would do well to respect it for what it involves. Shorey suggests that Plato has no ambition to make everyman a theologian. Disbelief is equated with youth by Plato and the Athenian, stressing the need to suppress the anger one must feel when faced with disbelief, is made to address the atheist thus:

O my son, we will say to him, you are young, and the advance of time will make you reverse many of the opinions which you now hold. Wait awhile, and do not attempt to judge at present of the highest things; and that is the highest of which you now think nothing -- to know the gods rightly and to live accordingly. (888)

But Plato does not intend to content himself with pious exhortation; he goes on now to the philosophical root of atheism.

The key tenet of the position he wants to reject is that things come about by nature, by chance or by art, and that of these nature and chance are primary, art secondary. They say that the greatest and fairest things are the work of nature and of chance, the lesser of art, which, receiving from nature the greater and primeval creations, moulds and fashions all those lesser works which are generally termed artificial. (889)

Politics is thereby relegated to the realm of art, having some connection with nature, but legislation is entirely a work of art being based on assumptions which are not true. In order to reject this Plato proposes to assert the superiority of art over nature and chance, a superiority which is in effect that of

intelligence and soul over the inanimate. The elements listed by the natural philosopher are not the first explanation of things. We shall make no attempt to trace Plato's proof in detail (891-899); it involves the same view of soul that we have seen in the *Phaedrus*, the motion which can move itself. In other words, soul is the source of those motions to which the natural philosopher appeals and his explanation is accordingly one that begins in the middle. Plato suggests that a good and an evil soul are involved in the universe.

If, my friend, we say that the whole path and movement of heaven, and of all that is therein, is by nature akin to the movement and revolution and calculation of mind and proceeds by kindred laws, then, as is plain, we must say that the best soul takes care of the world and guides it along the good path. (897)

When we consider the ordered movement of such a body as the sun, we must appeal to soul to account for this motion, a soul which may be thought to be related to the solar body in one of three ways.

Either the soul which moves the sun this way and that, resides within the circular and visible body, like the soul which carries us about every way; or the soul provides herself with an external body of fire or air, as some affirm, and violently propels body by body; or thirdly, she is without such a body, but guides the sun by some extraordinary and wonderful power. (898-9)

These souls which guide the heavenly bodies are gods and we can therefore assert that in some sense all things are full of gods.

To the second position that, though the gods exist, they have no concern for human affairs, Plato observes that it seems prompted by undeniable difficulties. "Perhaps you have seen impious men growing old and leaving their children's children in high offices, and their prosperity shakes your faith." (900) Once it is admitted that the gods see and know all and that they have all power, it seems impious to declare that they are not concerned with every singular thing and event. What the doubter must realize is that the order of the universe was not created for him, but that he is for the sake of the order of the universe. (903) What one can be sure of is that all things work for the good of the whole. This second view is said by many, for example Shorey and Taylor, to anticipate the Epicurean view. The third view, that the gods can be bribed to serve the ends of injustice, is more or less summarily dismissed as an affront to reason.

It is fitting that we have brought our discussion of Plato's views to a close with a few remarks on the theology of Book Ten of the *Laws*. The close connection between man's scientific and moral advance, present from the earliest dialogues, indicates that man's chief concern must be to inscribe in his own soul the pattern of the divine. It may be mentioned here that in the *Epinomis*, whose very title indicates its connection with the *Laws*, Plato goes on to discuss the education of those who will be members of the highest council of the state. We are not surprised to learn that the study stressed is that of number -- without knowledge of number man must remain ignorant and immoral. What Plato means is that knowledge of astronomy, of the heavenly bodies, will lead us surely to knowledge of the divine, a view which Aristotle will share. For both men, the heavenly bodies are not merely analogues of immaterial substances but, as the passage from the *Laws* suggests, the means of knowing them. It is in knowing the changeless and eternal beings, in contemplation of the gods, that the term of philosophy, the wisdom for love of which one subjects himself to the years of apprenticeship in mathematics, is reached. Plato describes this movement in passages of unsurpassed literary quality; he is, so to say, the poet of science. The dialogues do not establish the existence of the the Forms in a satisfactory way, their dialectic seldom achieves even more limited objectives; rather they present the movement of thought, exhortations to virtue, but very little of what could be called an established doctrine. We have observed that Plato has a penchant for drifting into mythical tales when very important doctrines are at issue and we get, in lieu of an argument, a likely story. It seems best to interpret the frequency of such mythical explanations against the background of Plato's own remarks on the function of his written works. He does not claim to strive for rigor there or to establish his most basic positions. This is left to the personal contact of master and pupil within the Academy. As a result, we find an unwritten doctrine attributed to Plato by students of his. In turning now to Aristotle, we will find that the bulk of his writings consists of lecture notes, precisely the doctrinal effort that, in Plato's case, has not been preserved. This is not to say that Aristotle serves simply to bolster up the written positions of Plato, but if there are fundamental differences between the two men, it will have to be remembered that we are comparing quite different types of source. Moreover, it will be seen that Aristotle emerges quite naturally from the Plato we know.

{32} Ross, *Plato's Theory of Ideas*, *op. cit.*, pp. 58-65.

Chapter III

Aristotle

A. The Man and his Work

Aristotle was born in Stagira, an Ionian colony in northern Greece, in 384 B.C. We know very little about his life, but certain bare facts seem beyond dispute. His father, Nicomachus, was a physician who died when Aristotle was still a boy; it is said that Nicomachus was both friend and physician of Amyntas II, king of Macedonia. Since his father belonged to the Asclepiad society, it is conjectured that Aristotle's interest in biology can be traced to a period when he watched his father at work. At the age of eighteen (367-6 B.C.), Aristotle came to the Academy of Plato, although perhaps the master was engaged in Syracusan business when he arrived. Aristotle remained at the Academy until Plato's death in 348, when he left for Assos where, under the patronage of the tyrant Hermeias, some students of the Academy had formed a school; indeed Hermeias himself is thought to have been a former student at the Academy. Aristotle stayed at Assos for a few years, perhaps three, married the niece of Hermeias, Pythias, and had by her a daughter of the same name. After her death, he had a son, Nicomachus, by a common-law wife. When Hermeias was executed by the Persians -- with whose help he had gained power -- because of a correspondence with Philip of Macedon, Aristotle fled to Mitylene on the isle of Lesbos. Some of Aristotle's biological investigation depends on specimens peculiar to Assos and Lesbos. In 343-2, Aristotle was invited by Philip of Macedon to educate his son Alexander, then a boy of thirteen. There has been much conjecture about this incident which is, of course, an extremely provocative one, but there seems to be little evidence that Aristotle exercised much influence on the future conqueror of the world. In fact he seems to have had little more than three years in the post; at the most he could have had six: Alexander became king at the age of nineteen. His friendship with Antipater, an extremely valuable one for Aristotle, must be traced from this period. In 335-4, Aristotle returned to Athens and began his own school, the Lyceum or Peripatos (named after a covered loggia where Aristotle lectured) under the patronage of Antipater. We are told that he lectured on difficult matters in the morning (logic and first philosophy) and gave public lectures in the afternoon on ethical and political subjects. When Alexander died in 323, Aristotle fled Athens; feeling against the Macedonian empire was high and Aristotle is said to have feared lest the Athenians should sin twice against philosophy. He went to Chalcis where in 323 he died at the age of sixty-three.

These few facts and connected conjectures are all we know of Aristotle's life. The dearth of information does not matter. The only Aristotle who can be truly meaningful for us is to be found in the writings that have come down to us; in them we find Plato's greatest pupil, the culmination of all the philosophy that had gone before him. The fact that he himself is inclined to point this out should not dissuade us from seeing that it is the case. Above all we must guard against thinking in terms of two autonomous abstractions, Platonism and Aristotelianism; to do so is to leave oneself open to "discoveries" of many points of contact between the two which in turn leads to the most breathtaking discovery of all, namely that the man who spent nearly twenty years in the Academy had a Platonist period! Aristotle himself has dwelt on his differences with Platonism, but so too, in a sense, did Plato himself. It is difficult to think of arguments more devastating to the doctrine of Forms than those set forth in the *Parmenides*; where Aristotle differs from Plato is in taking them to be conclusive enough to indicate that another direction is desirable.

Without minimizing this difference it must be said that there are countless instances where Aristotle simply takes over Platonic doctrine and builds from it, that, generally speaking, without Plato there could have been no Aristotle. It is a fact of no small importance that Aristotle looked upon himself as the true heir of Plato and had little sympathy with Speusippus and Xenocrates, the successors of Plato at the Academy. Aristotle himself had a keen sense of the way in which philosophy develops and we shall merely be employing something of that sense if we see his thought as an outgrowth of Plato's. It is only when we look on a man's philosophy as his biography that such commingling suggests the diminution of personality.

We have said that the important Aristotle for us is the author of the writings which have been handed down as his. As it happens, however, these writings present something of a problem and Aristotelian studies since the second decade of this century have been largely concerned with that problem and with the hypothesis that it can be, if not solved, made explicable enough to live with by seeing the writings as containing layers indicating different stages of their author's intellectual history. What we must imagine is an attitude somewhat like this. The Platonic dialogues present us with a thinker who, over a long period of time, undergoes a number of shifts in his basic attitude: Stenzel for example would have us see a Plato who was moving inexorably towards an interest in natural science.^[33] On the other hand, we have Aristotle whose "system" came to him whole as if, as someone has suggested, he was able to sit down and write it out in the sequence in which it appears in the Bekker edition. The *corpus*

aristotelicum contains Aristotelianism, of course, which is opposed to Platonism. There is a class of writings of Aristotle which disturb this picture, the dialogues, which were famous in antiquity, praised by Cicero and Quintilian for their style, but which exist today only in fragments. The strange thing is that several of these exhibit a doctrine which is much closer to that of the dialogues of Plato than to the "Aristotelian" writings. At one time, the reaction was to reject them as not Aristotle's at all. A much different approach was suggested by the most influential Aristotelian of our century, Werner Jaeger.⁽³⁴⁾ The dialogues suggested to him that Aristotle, as a member of the Academy, was won over to Plato's doctrine and wrote dialogues in imitation of those of the master and only gradually came to doctrines peculiarly his own. Moreover, this transition is not simply a matter of the character of the Aristotelian dialogues on the one hand and the treatises on the other; Jaeger began to see that some passages, indeed books, of Aristotle's *Metaphysics* were Platonic, some Aristotelian, and others a curious melange of both which suggested that Aristotle had tried unsuccessfully to smooth over the change in his mind with later additions. it is no overstatement to say that this approach caused a revolution in Aristotelian studies. Where the earlier view that the works of Aristotle form a well-wrought whole had dictated that apparently incompatible passages be reconciled, the tendency now became to assign such passages to different periods. One by one the works of Aristotle turned from literary wholes into patchworks of disparate and irreconcilable elements. And, even when the analysis of particular evolutions were called into question, critics were hasty to add that the overriding assertion was not thereby struck down -- as if the notion of an evolution in Aristotle's thought were somehow an *a priori* certainty and not an induction from the texts. More recently there seems to be growing doubt concerning the fruitfulness of the shelves of scholarly works Jaeger's suggestion has produced. Randall, Barker, Allen and many others feel, as Mure⁽³⁵⁾ had long before, that the feverish efforts of the philologists were perhaps somewhat irrelevant; Barker indeed went the whole distance and asserted that any genetic approach was basically subjective. While it is difficult to see any wholesale repudiation of Jaeger's approach just over the horizon, the problem which prompted that approach remains. There are the dialogues, after all, and we must say something about them here. As to the supposed evolution within the treatises, there is no point in making any generalization; we have suggested that the general theory is only as good as particular interpretations of troublesome texts. Accordingly, we shall infrequently make allusions to those interpretations in the exposition which follows. We can say now that Jaeger's opinion that Aristotle moved from a metaphysical period in the manner of Plato to empiricism not only suggests a quite different picture of the Plato Aristotle

would have known from that given by Stenzel, but seems unable to account for the detailed interest in nature exhibited during the Assos and Lesbos periods. Moreover, Theophrastus, Aristotle's successor as head of the Lyceum, is concerned with metaphysics; and this does not suggest that Aristotle had become disenchanted with such pursuits. Finally of course, all other things being equal, it can scarcely matter to us at what stage of his life Aristotle was occupied with one science or the other; if a given treatise presents a unified doctrine, that will be sufficient reason for its engaging our philosophical interest. We have drawn consolation from the fact that the genetic approach to Aristotle has not prevented the appearance of efforts to grasp Aristotle's thought as a whole in works written by men fully aware of the new approach to Aristotle and far more competent than we are to assess its validity.

Dialogues. The foregoing indicates that the writings of Aristotle must initially be divided into two main groups (we shall exclude the lists of Olympic winners and other factual data he is said to have collected, but feel constrained to mention that of the 153 constitutions of Greek and barbarian cities collected under his direction, only that of Athens is extant): first of all there are the works Aristotle referred to as *exoteric*. These works were destined for general publication and the dialogues must be placed under this heading. Secondly, there are *acroamatic* or lecture treatises, the writings for which Aristotle is principally known and which formed the basis of his instruction in the Lyceum. The *exoteric* writings have been lost and what we possess of them are fragments gleaned from ancient writings and, more recently, from Arabian authors. One will appreciate that Jaeger's theory has increased interest in these writings; his own speculation was based primarily on an edition of the fragments made by V. Rose, an editor who never believed the fragments to represent genuine works of Aristotle. Nevertheless, Rose grouped the fragments under the titles listed by Diogenes Laertius; since that time and largely under the impetus of Jaeger's work a great amount of scholarly literature has been devoted to the fragments, new editions have appeared, notably one by Ross. Quite recently, there has been doubt cast on the methods hitherto followed to collect the fragments of the lost works and it becomes clear that a good deal of work must yet be done before we can speak confidently of the relation between the fragments and the treatises.⁽³⁶⁾ What we have to say about the lost works is based on Ross's edition of the fragments.

Perhaps the three most important lost works, certainly the three on which we can form an opinion most easily, are the *Eudemus* or *On Soul*; the *Protrepticus* and *On Philosophy*. The fragments of the *Eudemus* appear to be little more than the assertion by Aristotle of views familiar to us from the Platonic dialogues.

Thus, death is the return of the soul to its home (Fr. 10); the immortality of the soul is in fact the immortality of reason and can be proved in three ways, by appeal to the doctrine of recollection (*anamnesis*), from the definition of the soul as self-moved and from the soul's likeness to God. (fr. 2) Aristotle is said to have spoken of the state of the soul prior to its descent to body as well as after its return (fr. 4) and he explains why the descent entails the forgetting of what it had hitherto known. (fr. 5) Moreover, the arguments of the *Phaedo* against the conception of the soul as a harmony are also said to have been given in the *Eudemus*. In short, testimony on the contents of the dialogue indicates that in it Aristotle held views which are either quite different from those in the *acroamatic* works or which do not enter into the treatises at all. The commentator Elias suggests that in the dialogue Aristotle contents himself with probable -- we might say, popular, literary, mythical -- arguments, reserving conclusive arguments for the treatises. Such a view is prompted by a disinclination to find a serious difference between the exoteric and *acroamatic* works; while this attitude is generally repudiated today, Elias' remark does prompt us to recognize that while it is Plato's popular works that we possess intact and his lectures which are known to us only on the testimony of others, the exact reverse is the situation with respect to Aristotle. There is no evidence, however, that the doctrine of Forms was something Plato put out only for popular consumption; Aristotle makes it clear that Plato held firmly to this doctrine. Thus, while Elias' interpretation should not be dismissed summarily, the *prima facie* evidence does not favor its accuracy. As with the other dialogues, it is extremely fruitful to ask whether the initial impression that Aristotle is simply repeating what we can find in Plato is in fact true; moreover, since the affinities are with the *Phaedo*, a dialogue presumably written well before Aristotle's entry into the Academy, we must ask with which Plato Aristotle is in agreement or disagreement.

The *Protrepticus*, an exhortation to the study of philosophy, was addressed to Themison, a prince of Cyprus, about whom we know nothing. The assumption of the work is one with which we are familiar from Plato, namely that it is in the study of philosophy that a man will achieve his proper perfection and thus happiness. The implication seems to be that there is a link between moral perfection and intellectual contemplation such that, in some sense of the phrase, knowledge is virtue. Jaeger argued (*Aristotle*, p. 81) that Aristotle had found this link in the conception of *phronesis*. Once more, while the echoes of Plato are quite distinct in the fragments of the *Protrepticus*, the doctrine of the *acroamatic* works is also foreshadowed. There is some controversy as to the extent of the Platonism exhibited in the *Protrepticus*.⁽³⁷⁾ The dialogue *On Philosophy* is

noteworthy for its rejection of Idea-Numbers and for its assertion of the Prime Mover.

More than this we can not say about Aristotle's early writings; we have already indicated the increased interest in them and it would be impossible in a work of this nature to indicate the direction or directions in which contemporary research is leading. Something however had to be said of the fragments since, if certain estimates of the stages of Aristotle's development represented by this lost dialogue or that be accepted, and if we should find close affinity between fragments of a dialogue and portions of an acroamatic work, we may find ourselves led in the direction Jaeger himself took. In other words, we may begin to despair of the possibility of finding a single, coherent doctrine in the Aristotelian treatises. Needless to say we feel no such despair. There is much to be learned from studies of Aristotle's fragments; there is infinitely more to be learned from the treatises. That the treatises present difficulties is neither news nor surprising, but the difficulties are not such as to dissuade us from sharing the optimism of those who from antiquity to the present day feel that the treatises contain a coherent and intelligible doctrine.

Treatises. The works we shall now mention were not intended for popular publication; they were rather written works which circulated among members of Aristotle's school. Their language and style and the frequent cross-references among them are sufficient indication that they were intended for the initiate. We shall mention these works under headings which will later be justified as in accordance with Aristotle's own views. First, there are logical works: *Categories*, *On Interpretation*, *Prior Analytics*, *Posterior Analytics*, *Topics* and *Sophistical Refutations*. The natural works are: *Physics*, *On the Heaven*, *On Generation and Corruption and the Meteorologica*, *On the Soul* and the *Parva Naturalia*, the latter including works on sensation, memory and reminiscence, sleep and so forth. There are also the *History of Animals*, the *Parts of Animals*, the *Motion of Animals*, the *Progression of Animals* and the *Generation of Animals*. In moral philosophy the *Eudemian Ethics* as well as the *Nicomachian Ethics* seems to be genuine work of Aristotle. There is the *Politics* as well; we can also mention here the *Rhetoric* and *Poetics*. Finally there is the *Metaphysics*.

B. The Nature and Division of Philosophy

We can attain a preliminary understanding of what philosophy is for Aristotle by asking what non-philosophy might be. This is not as negative as it may sound, because Aristotle shows what philosophy has in common with that from which

he would distinguish it. In the *Theaetetus* Plato had traced the genesis of philosophy to wonder; so too in the first book of the *Metaphysics* (982b12) Aristotle says that it is because of wonder that, now and in the beginning, men begin to philosophize, the objects of their wonder varying from obvious matters to celestial phenomena. The idea here is that wonder is a concomitant of our observing an event without understanding why it has taken place; the impetus to grasp the "why" of the event is the impetus towards philosophy. Philosophy is the flight from puzzlement and wonder. Aristotle adds that the lover of myth (*philomythos*) is in a way a philosopher (*philosophos*) because the myth is made up of wonders. (982b18-9) We might wonder just wherein the similarity alluded to is supposed to lie, asking ourselves if myth and philosophy are alike in that both attempt to dispel the ignorance which is productive of wonder, or whether Aristotle means that, whereas the philosopher attempts to dispel wonder, the mythmaker strives to produce it by creating stories that elicit our awe and amazement. In other words, we may be puzzled by the ambiguity of our word "wonder." Sometimes we wonder what the explanation of an event is, sometimes we wonder in the sense that we stand in awe of a certain happening. Now it may be argued that these are connected uses of the same term, but they are different enough to make us doubtful of the similarity Aristotle is suggesting. Fortunately, we can apply to Aristotle himself for an elucidation of the matter.

Let us first consider the term "myth" (*mythos*). It is possible to trace the history of this word in such a way that a special use of it made by Aristotle in the *Poetics* can cast light on the difficulty just posed. From signifying speech as opposed to action, the word came to signify advice, a command, and then purpose or plan. Finally it meant a story which was distinguished from a merely historical narrative. We have here a use of the term that answers to many instances of its use by Plato; moreover, it is the meaning of the term present in the *Poetics* of Aristotle on which he founds his peculiar use of it to mean what we translate as plot. (Cf. Else, *Aristotle's Poetics*, pp. 242ff.) The tragedian takes the old stories (*mythoi*) and imposes a plot (*mythos*) on them. (1451b24 ff.) The myth or plot of the play is the principle of intelligibility of the actions depicted; what is more, the plot not only explains the sequence but causes admiration and awe in the spectator. We have here the root of Aristotle's famous comparison of history and poetry according to which poetry is more philosophical and serious than history. (cf. 1451b1 ff.) Poetry is not simply a narrative of what has happened; rather it involves a kind of generalization of a type of occurrence. This entails that poetry is more explanatory than history. Now what we have done is to move from a comparison of myth and philosophy to the use of the term "myth"

in the *Poetics* of Aristotle, a movement which suggests not only a link between myth and poetry, but a similarity between poetry and philosophy in terms of universality and consequent explanatory power. That is, while we began by seeking the meaning of philosophy by asking what philosophy is not, we have actually arrived at a rough indication of what Aristotle thought philosophy is.

Aristotle derived the notion that philosophy arises from wonder from Plato. Let us now ask if he would assent to Plato's assertion that there is an ancient quarrel between philosophy and poetry. The answer to this question is going to involve an odd consequence; not only does Aristotle agree with Plato that poetry is one thing and philosophy another, he accuses Plato of being too poetic in his explanations. The way in which Aristotle speaks of the opposition between myth and philosophy is genetic in the sense that he sees philosophy arising out of a background of myth. We can see this in his criticism of Hesiod and the other poetic theologians. These men would explain immortality in terms of the consumption of nectar and ambrosia, an explanation Aristotle finds quite over his head. Aristotle concludes that it is not worthwhile to examine seriously mythical sophistries; he will only concern himself with those who speak apodictically, who use the language of proof or demonstration. (1000a18ff.) What is the sense of this opposition between mythical and demonstrative language? If we look at Aristotle's criticism of Plato's language, the meaning becomes clear. Aristotle says that to hold the Platonic view that all things are from the Forms, that they share in or participate in them, is to speak idly and in poetic metaphors (991a20). Mythical language as opposed to philosophical language thus emerges as metaphorical language. Now to speak metaphorically, for Aristotle, is to speak of one thing in terms of something else (1457b6-7), incapable of manifesting the proper causes of that which is to be explained. We might say that Aristotle is here contrasting literal and fanciful language, that philosophy for him is not the concoction of a tale but the formulation of an explanation in terms of the things to be explained. This contrast, coupled with the criticism of Plato, leads us to expect that Aristotle will have much to say about what constitutes an argument and about the nature of philosophical language. Neither expectation will be disappointed.

Philosophy is like poetry in that both attempt to dispel wonder, both attempt to explain a puzzling fact, but the two differ in the kind of language each uses. Let us now seek a more positive description of philosophy. Aristotle takes seriously the etymology of the term "philosophy": love of wisdom. For him the phrase indicates not only the efficacious desire that is a prerequisite for learning philosophy, but as well has significance in terms of his division of the sciences.

When we have identified the meaning of "wisdom" for Aristotle and gone on to see the division of philosophical sciences it implies we will be able to give an interpretation of "love" in the etymological definition which will free it from emotional overtones.

The term "wisdom" is discussed by Aristotle in a number of places, notably in the opening chapters of book one of the *Metaphysics*, and he uses it in such a way that we can see a continuity between his extended usage and Homer's use of it to signify the art of carpentry. A reading of the chapters just cited will reveal this and give one an unforgettable taste of Aristotle's mode of procedure. He begins with a sweeping generality: all men naturally want to know. There is no need to ready the example of the boy who despised geometry; Aristotle turns immediately to sensation. Everybody likes to have a look at things, even when no action is contemplated. We know immediately what Aristotle means; what he says can be verified by the shopper who tells the clerk he is "just looking" or by the person watching the sun set. However, Aristotle is beginning with the obvious in order to arrive at something quite obscure. His choice of sight is important, since it is the verb signifying this type of activity which is extended to mean intellectual activity as well; we have become familiar with this analogy in Plato and saw that it underlay the terms "Idea" and "Form." Moreover, Aristotle at the very outset of the passage is suggesting the distinction between the practical and speculative (notice once more the connection with sight in "speculative" and "theoretical") -- even when we have no further end in view, we enjoy looking. Notice as well the way in which Aristotle moves from the most elementary and indisputable type of knowledge, that of the external senses, through the collating of various memories in what he calls experience, to intellectual knowledge. A sign of the difference between knowledge and experience is that one who has the former can teach another whereas the man of experience cannot properly teach. The man of experience knows that a particular potion cures a given illness; the man with knowledge knows why it does. It is typical of Aristotle's good sense that he should point out that, where results are wanted, the man of experience is to be preferred over the man who has knowledge but no experience, e.g. the old midwife to the bright young intern. Nevertheless, it is the one who knows why who is called wise. Aristotle now moves from knowledge to wisdom. Even more deserving of the appellation "wise" is the one who knows the causes of many particular operations which he directs to an end unknown by those who have knowledge of a particular operation. His example is the master builder, the architect, what we would call today the project manager. He must direct the carpenters, bricklayers, plumbers, etc., so that a powerhouse, say, will result. Only the manager can

truly be said to know how to build a powerhouse. Wisdom then is the knowledge of the first causes, the ultimate principles, in any given area, whether in the realm of production or in the realm of things knowledge of which is sought for its own sake.

Aristotle goes on to enumerate several characteristics of the wise man and shows that they are saved preeminently by a certain kind of theoretical knowledge. The wise man is he who knows all things insofar as this is possible, who knows things which are extremely difficult to learn. This is the kind of knowledge God has and it will be an approximation of divine knowledge to know everything in its ultimate why: that is just the wisdom involved when we say that philosophy is the love of wisdom. Philosophy is the progressive conquering of wonder and ignorance which will culminate in the acquisition of this kind of wisdom.

Speculative and Practical. In the text of the *Metaphysics* just referred to, Aristotle distinguished knowledge sought for a utilitarian purpose (e.g., learning how to build a house) from that which is sought for its own sake (e.g., wanting to know what man is or what is or what an eclipse is). It may be well to examine somewhat more closely this distinction, for it is fundamental to an understanding of the doctrine of Aristotle.

It would be more accurate to say that Aristotle divides philosophy into the practical and speculative or theoretical than to say he divides knowledge this way. He gives us a threefold division of knowledge: theoretical, practical and productive. (*Metaphysics*, 1025b25) We shall be seeing more of the difference between the practical and productive later; for the moment it will suffice to indicate the difference by way of examples. Productive knowledge is exemplified by carpentry; practical knowledge by discussions of how justice can be saved in such-and-such circumstances. As for the practical and speculative, one way of distinguishing them is by their respective ends or goals. The end of theoretical knowledge is truth, that of practical knowledge action. (*Metaphysics*, 993b19; *Soul*, 433a14). The difference in end suggests a difference in the objects considered by theoretical and practical philosophy. Truth, Aristotle suggests, is of the changeless; action and the knowledge concerned with it are of the changeable and relative. In the *Metaphysics* (1025b19) Aristotle says that when the artist wants to shape a material (think of the shoemaker), or when a man makes a moral decision, the principle of what they do is within them. The artisan has an idea he wants to realize in matter; the moral agent has his own assessment of the situation in which he must act and his reason and will are

principles of what he does. The object of theoretical concern, on the other hand, is such that it has the principles of its movement within itself. Aristotle is here speaking of physics and, as we shall see, is employing the notion of nature which he will manifest in his natural doctrine in just this way by opposition to art. To express, his point somewhat differently, in practical philosophy we are concerned with things of which we are the principle; in theoretical philosophy we are dealing with things which are not products of our making or doing and our only cognitive relation to them is to know what they are. For example, we don't study human nature as something that can be produced by us, but with a view to knowing what man is.

It follows, further, that there will be a different method in these two branches of philosophy. In theoretical philosophy we will try to analyse things into their causes. In practical philosophy we will begin from causes and study how something can be brought about, whether it be a right action or an artifact.

Actually, then, there are three criteria for distinguishing speculative from practical philosophy: the end, the object and the method. Among the many difficulties which arise from this doctrine, we can entertain the following. Sometimes we study things which we could do, but our purpose is not to do them -- at least not while we are studying them. For example, an examination of just actions in an ethics class is not the same as the consideration which precedes immediately the performing of a just act. We will encounter this difficulty when we examine Aristotle's ethical doctrine, but for the moment we can suggest a distinction between our purpose and the purpose of a given kind of knowledge. In terms of our purpose, many practical considerations are in a sense theoretical. However, what is most basic in the distinction between theoretical and practical philosophy is the object. When what we are considering is something of which we can be the principal even if our purpose, and indeed even our method of studying it, is theoretical.

Division of Theoretical Philosophy. Not only did Aristotle distinguish between theoretical and practical philosophy, he also distinguished several theoretical sciences. In order to see that his division is not arbitrary nor simply the product of historical observation, it is important to see what the principles of this further division are. We have already seen that the object of theoretical knowledge exists independently of our doing or making. Let us call it the theoretical object. As such it has two characteristics, one that is due to the faculty with which we grasp it, our intellect, the other due to the determination or perfection of the faculty as it bears on the theoretical object. We shall see later Aristotle's

argument to the effect that the operation of intellection is immaterial; given this, its object too must be immaterial. Moreover, Aristotle teaches that science is concerned with what is necessary (*Posterior Analytics*, 74b5-75a17) The two characteristics of the theoretical object then are immateriality and unchangeability. That is to say, in order for something to be an object of speculative science, it must be removed from matter and motion. Insofar as there can be differences among theoretical objects with respect to these two characteristics we can speak of different theoretical sciences.

Aristotle exhibits the variation in terms of these characteristics in a manner which makes his point easily grasped. Let us consider snub and curve. In order to define snub, a material must be mentioned, since snubness is always of something, namely nose. So too, Aristotle suggests, with water, man, plant or horse: in order to grasp what these things are, we must include sensible matter in their definition. All of the objects of natural science, physics or philosophy of nature are of this kind. Of course, it is not *this* nose which is included in the definition of snub, but simply nose. In defining curve, on the other hand we do not allude to sensible matter. For example, take a straight line AB with C a point on the line between A and B, and on the same plane with the original line and points. Any line on the same plane which passes through A and B but does not pass through C is a curved line. We can thus arrive at what is meant by curve without speaking of hot or cold, smooth or rough, heavy or light, etc. Where are the lines we study in geometry? Any lines which exist in the full sense of the term are the edges of physical bodies, like the contour of the nose. But of course the mathematician is not concerned with existent line -- nor do such lines have the properties of the lines he studies, The fact that such things as lines can be defined without including sensible matter is enough for there to be a science of mathematics.

There is a third possible kind of theoretical object, one which would not only be defined without reference to sensible matter but which, unlike mathematical entities, also exists apart from matter. If indeed there are such objects, there is need for a science other than physics and mathematics. (*Metaphysics*, 1026a10) The science concerned with such objects is wisdom or First Philosophy, what has come to be called metaphysics.

The bases for this division of theoretical philosophy into three sciences are extremely difficult and we will return to them in subsequent discussions. A preliminary statement of the doctrine serves not only to familiarize us with

matters of extreme importance in the philosophy of Aristotle, but as well to give a foundation for the divisions of our own subsequent presentation.

Division of Practical Philosophy. Aristotle will also speak of different sciences in practical philosophy, but the principle of division here will not be the same as in the theoretical or speculative order. We have seen that the practical differs from the theoretical because its end is different, operation and not truth. But operations or actions are performed with a view to attaining a certain good, and insofar as we can distinguish a difference in the goods to which actions refer we can speak of different practical sciences. This mode of distinguishing is taken from Plato who spoke in the *Republic* of the virtues of the state and of the individual soul, which we saw to mean the virtues of man insofar as he is a citizen and insofar as he is a private individual. So too, for Aristotle, it is the difference between the good of the individual, his private good, and the good he shares with others as a member of the family, and the good he shares with others as a member of the political community which underlies the division of practical philosophy into ethics, economics and politics. (*Eudemian Ethics*, 1218b13)

The Order Among the Philosophical Sciences. We have accepted love of wisdom as a sufficient indication of what philosophy is for Aristotle. Wisdom, as we have seen, can mean many things, but in this context it must be taken to be First Philosophy, the theoretical science beyond physics and mathematics. In other words, philosophizing is the activity whereby we acquire various sciences with a view to acquiring the ultimate in the order of theoretical science, First Philosophy, wisdom, metaphysics. When Aristotle calls this First Philosophy, he means first in the order of importance and desirability, not the science which is or could be first learned by us. As a matter of fact, not only does Aristotle hold that we should learn this science last, he seems also to suggest the order in which the other philosophical sciences should be learned if the term of wisdom is to be reached and reached more or less easily.

There seems little doubt that Aristotle held that logic should be learned first. (*Metaphysics*, 995a12) Notice that Aristotle is not speaking historically of the order in which what he argues are different sciences may have been discovered; he is not maintaining the absurd thesis that logic was the first concern of man when he had the leisure to pursue study. He is speaking of one in the advantageous position of being helped by a teacher along the path to wisdom, a teacher who has gone this road himself. The reason that logic should be learned first is that it teaches the method to be observed if one is to acquire science, and

one cannot simultaneously learn the method and use it. Indeed, unless one first knows what the logical demands of science are, he will not possess the common methodology or procedure which must be followed if science is to be acquired. Obviously this position does not preclude the use of philosophical examples by the logician, nor exclude a familiarity on the part of the student with some such discipline as mathematics, a familiarity which will not be equivalent to scientific knowledge.

Why was no mention made of logic when we discussed the divisions of philosophy? Is logic a part of philosophy? If so, is it theoretical or practical? These are good questions; unfortunately, Aristotle neither asks nor answers them. They will form a basis for controversy between Stoics and later Peripatetics, a controversy which will be carried into the Middle Ages by way of Boethius' commentaries on Porphyry's *Introduction to Aristotle's Categories*.

Given that one has studied logic, what should he study next? Aristotle's answer seems clear -- mathematics. This science, he observes, can be learned even by the young, whereas moral philosophy, the philosophy of nature and metaphysics cannot. (*Nichomachean Ethics*, 1142a13-23) For reasons which will soon become clear, it is the philosophy of nature which should next be learned. The order to be followed in the study of nature is set forth at the outset of the *Physics*; Aristotle teaches that the study of living things, and hence psychology, is a part of the philosophy of nature. (*On the Soul*, 403a27) That moral philosophy should be learned after the philosophy of nature seems suggested by two remarks by Aristotle. The first, that one must be quite mature before engaging in ethical studies (*Nichomachean Ethics* 1095a1-11); the other, which is somewhat more conclusive, that moral philosophy presupposes psychology. (1102a14-29) A general reason for studying metaphysics last is that, in its character as wisdom, it has the function of defending the other philosophical sciences against attacks on their principles. This the particular science cannot do, since it would have to argue from the principles in question. We can see one indication of this in the remarks Aristotle makes about Parmenides in the first book of the *Physics*. The special dependence of First Philosophy on the philosophy of nature in the order of learning philosophy is clear from the fact that, if we did not have demonstrative knowledge of immaterial or separated substance there would be no grounds for supposing a theoretical object different from those of physics or mathematics. As Aristotle remarks if there were no substances other than natural ones, physics would be First Philosophy. (*Metaphysics*, 1026a28-9).

We find in Aristotle a view of philosophy which presents an ordered whole of sciences, both theoretical and practical, a fact which may make his writings seem unduly formidable. Indeed, we may tend to think of it as a "system", a body of doctrine with that perfection Aristotle demands of the good tragedy, namely, that it have a beginning, a middle and an end. This notion of system is of fairly recent origin, connoting not only finality but also personal ownership; it is much easier to verify in Hegel than in any Greek. It could be argued that Parmenides and perhaps Heraclitus were confident that they had said nearly all there was to say, but it is a monumental blunder to look on the efforts of Plato and Aristotle as systems in our latter-day sense. Of course both men saw earlier philosophy as reaching fruition or rejection in their own efforts, and each of them had a quite justified sense of accomplishment. But even if Plato or Aristotle had thought they had done all the philosophy there was to do, they would not have taken this as any indication that there would be no further need for philosophers. Neither man was writing an autobiography or contemplating his own navel; rather each was striving to understand the way things are, and this is a task passed from one generation of philosophers to the next. It would be a melancholy thought if all that was passed on were the task; solutions to problems are also transmitted, and these are the property neither of their discoverer nor of the one who learns them. They are common goods. Nowadays we deplore systems largely, it would seem, as an attempted check on the expansion of our own personality. Aristotle's system is as much program as accomplishment, but both can best be understood in terms of what he took philosophy to be about and how its efforts can be broken up in terms of various subject matters. In what follows we shall be concentrating on what Aristotle accomplished, leaving it to the good sense of the reader to see what he did not do and how what has been done since his time can profitably be thought of in terms of his delineation of the basic structure of philosophy.

C. Aristotle's Logic

The logical works of Aristotle which have come down to us are known as the *Organon*, that is, instrument or tool. The term seems to have been applied to logic first by Alexander of Aphrodisias and came to be applied to logical works generally in the sixth century of our era.⁽³⁸⁾ We have already indicated that a controversy arose on this matter insofar as some held that logic was not a part of philosophy, but only its instrument. The Stoic division of philosophy, as we shall see, was threefold: logic, physics, ethics. That controversy does not interest us now, but a difficulty we must face is this: Aristotle does not himself use the term "logic" to cover what is under discussion in those books of his we call

logical ones. If then we make statements about Aristotelian logic, we shall be referring to the contents of what we call the logical works and not to the way in which Aristotle used the Greek term *logike* from which we get our "logic."

The following works constitute the Organon: *Categories*, *On Interpretation*, *Prior Analytics*, *Posterior Analytics*, *Topics* and *Sophistical Refutations*. It became the view of at least one eminent Aristotelian of the Middle Ages that the *Rhetoric* and *Poetics* also belong to logic; we shall attempt to say something of the nature of Aristotelian logic as exhibited in the books which are undisputed candidates for membership in the Organon.

Nature and Subject Matter of Logic. Presumably something other than historical accident explains the plurality of works which make up the Organon, just as we should expect that, despite their differences, these works have something in common. We want now to see if there is some principle dividing these works; then we shall ask what all of them are concerned with and what significance there is in the traditional ordering of these works. After these points have been discussed, we can profitably turn to a brief analysis of important features of the contents of the works.

In the *Categories*, we find a distinction between complex and incomplex expressions (chap. 2, 1a16-19). Examples of the former are "the man runs" and "the man wins," of the latter, "man," "ox," "runs" and "wins." In the first chapter of *On Interpretation* we find a distinction made between thoughts which do not involve truth or falsity and those which must be either true or false. What are called simple or incomplex expressions in the *Categories* are not true or false; however, it is not just any complex expression which is either true or false, but only the enunciation or proposition. In the *Prior Analytics*, the syllogism is defined as a discourse (*logos*) in which if certain things be maintained, something else necessarily follows. (24b18-20)

What these distinctions and definitions indicate is that there is an ascending order from incomplex terms, of themselves neither true nor false, to complex expressions which are such that they must be either true or false, to discourse or reasoning which involves several propositions ordered in a rather special manner. Moreover, it seems that the *Categories* is concerned with incomplex terms, *On Interpretation* with sentences which are true or false, and the *Prior Analytics* and all other works of the Organon with discourse. The very title of the *Categories* indicates that things which are expressed incomplexly or simply

are viewed in that work from the point of view of their being possible predicates (*kategoriai*). Indeed, it is just in this light that the list of categories is introduced.

Expressions which are in no way composite signify substance, quantity, quality, relation, place, time, position, state, action or affection. To sketch my meaning roughly, examples of substance are "man" or "the horse," of quantity, such terms as "two cubits long" or "three cubits long," of quality, such attributes as "white," "grammatical." "Double," "half," "greater" fall under the category of relation; "in the market place," "in the lyceum," under that of place; "yesterday," "last year," under that of time. "Lying," "sitting," are terms indicating position; "shod," "armed," state; "to lance," "to cauterize," action; "to be lanced," "to be cauterized," affection. No one of these terms, in and by itself, involves an affirmation. It is by combination of such terms that positive or negative statements arise. For every assertion must, as is admitted, be either true or false, whereas expressions which are not in any way composite, such as "man," "white," "runs," "wins," cannot be either true or false. (1b25-2a10)

The *Categories*, then, appears to contain discussions necessary for the treatment of true and false expressions in *On Interpretation*. By the same token, as we shall see, the analysis of the syllogism into term and premiss points back to the two works just mentioned.

If we should now say that the *Categories* is concerned with simple terms, the elements of assertions, *On Interpretation* with propositions which are either true or false, and the *Analytics* and subsequent works with reasoning or discourse, would this enable us to say what these different books have in common? At the outset of *On Interpretation*, Aristotle refers to *On the Soul* when he is distinguishing between thoughts which do and thoughts which do not involve truth and falsity. The passage he has in mind seems to be this.

The thinking then of the simple objects of thought is found in those cases where falsehood is impossible: where the alternative of true or false applies, there we always find a putting together of objects of thought in a quasi-unity . . . For falsehood always involves a synthesis; for even if you assert that what is white is not white you have included non-white in a synthesis. It is possible also to call all these cases division as well as combination (430a26-430b3)

Now this reference to a work on soul and to a distinction between the mental act of conceiving, on the one hand, and that of judging, on the other, can make

it appear that what is being discussed in the books of the Organon is psychological activity.

There is a first reason why such an interpretation seems wrong. We have already seen that the discussions in *On the Soul* form part of the philosophy of nature. Apart from their lacking the principles of natural things, the matters of the Organon seem presupposed by natural philosophy as by any theoretical discipline. Aristotle, we have seen, argues that one must first be trained in the art of argumentation for it is absurd to seek knowledge and the mode of attaining it at the same time. (*Metaphysics*, 995a12-14) Secondly, there is the very manner of reference in *On Interpretation*. The distinction alluded to is said to belong "to an undertaking distinct from the present one." (16a8-9) That is, we are told as clearly as possible that the psychological doctrine is quite another matter than that of *On Interpretation*. To generalize, the logical works do not have psychological acts as their object of concern. This is not to say, of course, that these works are indifferent to distinctions made elsewhere, particularly in the study of the soul; the reference we have been discussing seems clearly to indicate that the psychological doctrine is presupposed.

This poses a problem. If logic is to be learned before any other discipline and if logic presupposes psychology, aren't we moving in a vicious circle? The answer is no. Aristotle has pointed out that the pupil must trust his master (*Sophistical Refutations*, 165b3), must accept things not directly involved in what he is being taught. One who teaches the doctrine of the Organon must know the disciplines of which logic is the instrument, and he will make use of a number of things which will be learned later by the aspiring philosopher. The latter, in being taught logic, accepts on trust matters drawn from psychology which he will learn later.

If it is clear that the works of the Organon are not concerned with psychological activity, it may appear that they are concerned with language and grammar. The constant occurrence of "incomplex expressions," "sentences," etc., seems to fortify this interpretation. It is of course a highly subtle matter to contend that logic, for Aristotle, has as its concern grammar and language. We do not have a definition of logic given by Aristotle and it is questionable that grammar, even in the ancient sense, had developed by the time of Aristotle. In the *Topics* (142b33-4) Aristotle entertains, as a definition of grammar, knowledge of writing and reading. Now this surely does not cover what is being discussed in the books of the Organon. Nevertheless, it is certainly true that these books are concerned with the nature of language. That concern, however, although it

presupposes grammar, is not itself a grammatical one. The justification of this assertion will be given later.

Thus far, we have tried to indicate that the logical works of Aristotle do not have as their subject matter either psychological activity or language understood as grammar. A third possible interpretation is that the concern of the works of the *Organon* is with things "out there." This position might take its rise from the opening sentence of the Aristotelian corpus. "Things are said to be named equivocally which have a common name but the definition signified by the name differs for each." This seems clearly to be a statement about things. So too, in the second chapter of the *Categories*, beginning a highly important division for what is to follow, Aristotle writes, "Of things themselves, some are predicable of a subject and are never present in a subject." Predicability, it would seem, is something of things themselves (*ton onton*). And, still with reference to this book, are not substance, quantity, quality, etc. designations of things "out there"? The demonstrative syllogism discussed in the *Posteriora* seems also to involve things out there in a special way. Does logic then concern itself with things? Or should we, as many today seem inclined to do, reject the *Categories* and *Posteriora*, denying that they are logical works at all? In order to do so it would seem that we must already know what Aristotelian meaning can be assigned to the term "logic," since it is of little help towards an understanding of Aristotle to be told that some works in the *Organon* contain no logic in our sense of the term.

What is logic for Aristotle and with what is it concerned? Logic is not as such concerned with things in themselves. As we have already indicated, this assertion is not a denial that the books of the *Organon* presuppose Aristotle's psychology and are unintelligible without it. Nor are we saying that the logic of Aristotle has nothing to say about things. This last remark can be understood in two ways. Aristotle's logic presupposes that, in reality, there is a distinction between substance and accident: this is not a logical doctrine although it has ramifications in logic. It is also true that logical entities, though distinct from natural entities, things-out-there, are defined and discussed with oblique reference to things out there, for the reason that they have an indirect dependence on natural things. Logic, we shall see, is concerned with natural things from the point of view of what happens to them when we know them, and this in a fashion different from psychology.

Perhaps the most economical way to illustrate the nature of logical entities is to consider a problem which will loom large in the history of medieval thought, that of universals.

The universal as universal does not exist. This is Aristotle's constant rebuke of the Platonists. What does he mean? The universal is not a substance.

"Substance, in the truest and primary and most definite sense of the word, is that which is neither predicable of a subject nor present in a subject; for instance, the individual man or horse." (*Categories*, 2a11-14) By "present in a subject," Aristotle means the mode of being of accidents such as "white," moving, "six feet tall." (1a24-5) Not only is substance in the primary sense not an accident of a subject, it is not the species "man" or "horse." Nevertheless, the species and genus can be called substance. "But in a secondary sense those things are called substance within which, as species the primary substances are included; also those which, as genera, include the species." (2a14-16) In this extended sense of the word, "man" and "animal" can be called substance. When, in the seventh book of the *Metaphysics*, Chapter Thirteen, Aristotle denies that the universal is a substance, he is denying that it is substance in the first and primary sense, that is, man in general does not exist and cannot exist in the way in which Socrates does. "Further, substance means that which is not predicable of a subject, but the universal is predicable of some subject always." (1038b15) "Man" expresses the substance of Socrates and yet Socrates is most properly an instance of substance whereas Man is not. Why is this so? "Man" expresses what Socrates is in such a way that it can be predicated of many, e.g., of Socrates, Plato, etc. For the moment we need not go into what justifies this situation. But it is this predicability of many that is meant by calling Man a universal. "But the universal is common, since that is called universal which is such as to belong to more than one thing." (1038b11-12) The universal is one and it can be predicated of many. Socrates is one and he can be predicated of nothing else. Socrates is not a universal; he is substance in the strictest sense. Man is a universal. Does this mean that predicability of many is part of the definition of Man? Certainly not, for then Man could not be predicated of Socrates. Universality is something that happens to what Socrates is as a result of our knowing that whatness. It is a relation between the nature as grasped by our mind and the individuals whose nature it is, e.g., Socrates. This relation of predicability is something following upon our knowing a thing, not part of what is known, the concept which terminates the mental act. The relation of universality, of predicability is precisely the sort of thing which concerns logic. This universality may be that of species (what is predicable of many differing in number, e.g. Man of Plato and Socrates, etc.), or genus (what is predicable of

many differing specifically; e.g., Animal of man and beast). Such relations, consequent upon our mode of knowing, are examples of the subject matter of logic.

We can see now that logic, although it is not directly concerned with things out there, is indirectly concerned with them. The mental image is an image of what exists out there, e.g. what Socrates is. This nature, as known by us, becomes the subject of logical relations, e.g., it can be called a species, a predicate in an assertion, a term in a syllogism, etc. These are quite accidental to the nature itself (not part of its definition) and accrue to it thanks to its existence in our mind. It is only because we know real things that the relations which concern the logician are there to be studied. Though only indirectly dependent on the real thing, the thing "out there," logical relations can never be severed from this indirect dependence. It is certainly not Aristotle's task, in logic, to examine what it is in things which permits their abstracted nature to be subject of a given logical relation; but logic, as he envisages it, assumes and anticipates this natural (or metaphysical) enquiry. Not just any conceived nature can become the subject of the relation of species, for example; but, again, it is assuredly not the task of logic to decide which ones can and which cannot. Aristotle has a knack for picking effective examples in his logical works, relying sometimes on common sense, often on received philosophical opinion and sometimes, finally, on his own philosophical doctrines.

Some of these points can be illustrated by alluding to a text introduced earlier as a possible corroboration for another interpretation of Aristotle's position on the subject matter of logic. The *Categories* begins with the remark that things are said to be named equivocally if they share a common name but that name has different definitions as used of each. What Aristotle has in mind is the way in which the cow in the pasture and the figure in a picture on the wall can both be called animal. These two things are equivocal: they receive a common name, but the definition corresponding to the name differs in each case. Let us say that, when speaking of the cow in the pasture we take "animal" to mean a living thing having senses. In calling an arrangement of paint on canvas an animal we would want to alter the definition and say it is an image or likeness of a living thing having senses. Now when we say things are equivocal, we do not mean that, taken by themselves they have equivocation as a property in the way both may have the same color. Their color would be known by us, but would not be attributed to them because they are known by us. But when we call things equivocals we are saying that they are named equivocally and we are certainly attributing something to them which is consequent upon our knowing and

naming them. Unknown and unnamed things simply are not equivocals. When things are said to be equivocals, or species, or genera or middle terms, etc., we are speaking of things, not as "out there," but insofar as when known by us they take on certain relations. That is why universality is not a property of any existing thing as such, why Aristotle says the universal does not exist.

This discussion of the subject matter of logic and our aside on universals will stand us in good stead when we consider Aristotle's criticism of Plato's doctrine of Forms. We can now summarize any light the foregoing may have cast on the *Categories* after which we will go on to discuss the other logical works. We have seen that the *Categories* is concerned with incomplex things, that is, things which can be so expressed that neither truth nor falsity is involved in the expression. The discussion seems thought of as preliminary to that of *On Interpretation* where expressions involving truth and falsity are composed of such incomplex things as are discussed in the *Categories*. The latter work came to be thought of as divided into three parts: antepredicaments, predicaments, postpredicaments. The first three chapters discuss things presupposed by the doctrine of categories: equivocality, univocity, denomination; the distinction of simple from complex expressions; the notion of predicability. The heart of the *Categories* is the discussion of substance (Chap. 5), quantity (Chap. 6), relation (Chap. 7), quality (Chap. 8) and action and passion (Chap. 9). What are called postpredicaments are discussed in Chapters Ten through Fifteen, the discussions of opposites and particularly of contrariety being most important.

On Interpretation. We have a preliminary idea of what this work is about from the foregoing; our procedure now will be to analyse in some little detail the first half of the work, Chapters One through Seven, and then indicate more sketchily the nature of the sequel.

At the outset of the work, Aristotle tells us that he wants to talk about the nature of the noun and verb, about negation and affirmation, and about the proposition and "speech." He will go on to indicate the order of treatment, but first he says a few things on the nature of signification. Words, generally, are signs of what is in the mind and written words are signs of spoken words. Just as written language differs from one people to another, so does the spoken, but that of which both are signs, namely what is grasped by the mind, is the same for all men. Moreover, what is in the mind is significative of things, something discussed in natural philosophy, in *On the Soul*. Mental states or concepts are the first or immediate signs, referring directly to things. Spoken or written signs, on the other hand, refer to things *through* our concepts of them. Concepts are

taken to signify things naturally as opposed to the conventional manner in which language signifies. As has already been indicated, Aristotle is not here concerned with discussing the relationship between concept and thing; he assumes that doctrine from natural philosophy. Given this view on the nature of such signs, he compares them with conventional signs, the nature of which he takes to be sufficiently manifest. We know there are different languages; hence there is no natural relation between this spoken or written word and the concept it is taken to signify. If "man" stands for what we know of certain things, "homo" and "anthropos" could do and have done just as well.

Continuing to borrow from his treatise on the soul, Aristotle notes that there is a difference between the mental state which does and that which does not involve truth or falsity. Truth or falsity is had when the mind composes or divides things which can be known apart from this synthetic act. Now if language signifies mental states, we can expect to find this same division in the spoken and written word: language sometimes expresses what is neither true nor false and sometimes what must be true or false. The noun and verb are elements of the linguistic expression of truth or falsity. "Man" and "white" are neither true nor false and do not become so until at least *is* or *is not* is added to them.

We are now in a position to explain the title of the work as well as the order of things to be discussed. An interpretation is an expression signifying what is true or false. Nouns and verbs are not interpretations, but the proximate elements of interpretations. Thus Aristotle's procedure consists in, moving from the components to the compound which is the subject of his treatise.

We are now faced with a difficulty. If the noun and verb are simply what can enter into the complex which is true or false, shouldn't they have been discussed in the *Categories*? In defence of Aristotle's procedure, it can be pointed out that to be a noun or verb is something which happens to concepts only in the proposition. These relations are not prior to the proposition but what the proposition can be formally analysed into. For this reason their treatment has been postponed to the present work.

Aristotle defines the noun as a vocal sound which signifies by convention and without reference to time, no part of which signifies alone. The definition of the verb is identical except for the replacement of "without reference to time" by "with reference to time." The first two elements of the definition, vocal sound and signifying by convention, are sufficiently clear from our previous

considerations. After discussing the last part of the definition, we will turn to the difference between the noun and verb.

What does it mean to say that no part of the noun or verb signifies separately? If we take the noun "liberty," it is clear enough that no syllable alone means anything in English. But what about "woman" and such compounds as "breakfast"? Obviously Aristotle's restriction does not apply to these, since "break" and "fast" and "man," if not "wo," are significant apart from the original nouns in which they occur. Although this is true, it does not affect the point Aristotle is making. "Man" is a word, but it is not a word insofar as it is taken to be part of "woman;" moreover, it does not signify part of what woman signifies. The same is true of the compound noun. "Break" and "fast" are both words but not precisely insofar as they are parts of "breakfast" nor does either apart signify part of the morning meal. We will see that this part of the definitions of noun and verb is introduced to set them off from that of which they are parts.

Aristotle clarifies the notion of conventional signification in his discussion of the noun. Not every sound that issues from the throat is significant conventionally. Thus a groan, though a vocal sound, is significant not of a concept but of pain. Despite the fact that groans and exclamations may differ from one language and culture to another, the difference between them and those sounds which happen to be used to signify concepts or mental states is clear. We need only consider the difference in the way the word "pain" and a groan signify pain to see what Aristotle is getting at. It might be noted parenthetically that if we want every significant activity, from groans to gestures, to be included under the heading of language, Aristotle's procedure must appear a narrow one. Precisely, since there is an overriding purpose governing what is relevant to the treatise. Aristotle is not engaged in setting down a general theory of signs. We will see more indications of narrowing in this work, but what Aristotle does is always wide enough to attain the end in view.

Aristotle now rejects what he calls the indefinite noun and the cases of the noun as irrelevant. Non-man and not-to-be-running do not signify any one thing; whatever is not a man, e.g., a tree, a horse, an angel, indeed what is nothing at all, is non-man. So too any activity other than running, all non-activity and even nothing at all are signified by not-to-be-running.

The verb is the sign of what is affirmed of another and affirmations involve time. Verbs imply a composition but do not of themselves signify a composition. In this they are like nouns, not being of themselves true or false.

Aristotle now turns to a discussion of discourse or speech (*logos*); "speech" seems an acceptable translation since we speak of parts of speech. The speech differs from the noun and verb in this that its parts signify separately, although they do not separately signify the true or false. This should be understood as meaning that the phrase or sentence includes nouns and verbs and not that any element of a phrase or sentence signifies in the way the noun and verb do.

Aristotle is not saying that "to," "in," "every," etc., etc., signify concepts. Moreover, not every speech (*logos*) will be a proposition or interpretation since not every compound of noun and verb signifies what is true or false. We are faced here with another narrowing on the part of Aristotle. As his commentators point out, he is excluding from the scope of the present work questions, pleas, commands, etc. He does not however banish these from logic if we accept the view that the *Rhetoric* and *Poetics* are parts of the *Organon*.

The interpretation or proposition is, in its simplest form, the affirmation of one thing of another. It is not simple in the way a word or definition is. We should not be misled by the fact that in reply to a question a single word may suffice to signify what is true or false. What are you doing? Reading. If "reading" signifies what is true or false here this is only because we understand a composition, e.g. I am reading. A compound proposition will have simple propositions as its components, e.g. "Socrates is white and Plato is tan," or "If you are cold, then you are ill." We might ask if "Socrates, assailed by pangs of hunger and seeing the cupboard was bare, set out, without having put on his coat, for the corner grocery where a sale was in progress" is simple or compound. For Aristotle's purposes, it is simple. The relative and adverbial clauses modify the simple conjunction of Socrates and going to the store.

We have said that the affirmation asserts one thing of another. Whatever can be affirmed can be denied and vice versa. Aristotle calls this opposition of affirmation and negation contradiction. Opposed propositions are those one of which affirms the other of which denies the same predicate of the same subject, e.g., John is white; John is not white. If John is a Negro whose family name is White, we could object that the two propositions are not opposed and we would be right, for we would be understanding them to have different predicates. The question of the opposition of propositions continues to occupy Aristotle in Chapter Seven.

There is an initial distinction between universal and singular things. We have already seen how such a division of things must be understood in a logical work;

it is clear that Aristotle is here distinguishing kinds of subjects of propositions. "Man" is an example of a universal, "Callias" of a singular subject. With respect to universal subjects, something can be predicated either universally (e.g., "Every man is white.") or not (e.g. "Some man is white"). When something is not predicated universally of a universal subject, the result is either a particular ("some man is white") or indefinite proposition ("man is white"). It is because of the nature of the universal that we cannot say, "Every man is every animal," for, since Socrates is a man, we would then have to agree that he is every animal.

Confining ourselves to propositions which have a universal subject, we can distinguish a number of oppositions and arrive at what came to be called the square of opposition. First, the opposition of contradiction. This obtains between the universal affirmative and particular negative, on the one hand, and between the universal negative and particular affirmative, on the other. Thus, "some man is white" is the contradictory of "no man is white." To contradict the universal proposition it suffices to adduce one instance in which it does not hold. Obviously it is not necessary to say that every man is white in order to oppose the claim that no man is white. The opposition between "every man is white" and "no man is white" is called contrariety. Like contradictories, contraries cannot be true simultaneously, but, unlike contradictories, it is not necessary that either contrary be true. Thus, in our example, since some men are white and some are not, neither the universal affirmative nor universal negative is true. It is clear that "some man is white" and "some man is not white" are not opposed propositions strictly speaking, since they do not satisfy the condition that one should affirm and the other deny the same predicate of the same subject. These two propositions can be simultaneously true because they have different subjects. Aristotle maintains that a proposition has only one opposite and by this he means contradictory opposite.

Aristotle's doctrine of the opposition of proposition has come in for a good deal of criticism from modern logicians, not all of it relevant. An informative and judicious discussion of this can be seen in P. F. Strawson's *Introduction to Logical Theory*.^[39]

The foregoing presentation gives a good deal of the basic doctrine as well as the flavor of *On Interpretation*. It is difficult to indicate briefly the remaining doctrine and we must content ourselves with saying that Aristotle goes on to discuss the truth and falsity of propositions having a verb in the future tense, indulging in a lengthy aside on the difficulties which attend future propositions

with a singular subject, e.g. "Socrates will arrive tomorrow." Much of this discussion goes far beyond logic and we shall be speaking of it elsewhere. Aristotle discusses a variety of problems consequent on the positions we have examined, and introduces a discussion of modal propositions, i.e. those which qualify the synthesis as possible, impossible, contingent or necessary.

Prior Analytics. -- The opening chapter of this work follows naturally enough on the matters we have just examined. The overriding concern of the *Analytics* is demonstration, the demonstrative syllogism. We should know that the division of the work into prior and posterior occurred after Aristotle's death,^{40} and the opening sentence of the *Prior Analytics* indicates the continuity of the two parts: the demonstrative syllogism is not discussed until the *Posterior Analytics* and yet it is mentioned as the subject of interest. We are told that the present treatise is concerned with demonstration, that it is for the sake of demonstrative science. We will see in a moment that the discussions of the *Prior Analytics* are a common introduction to the *Posterior Analytics* and *Topics*.

If our goal is to discuss demonstrative science, we must deal first with premiss, term and syllogism. The term, it soon emerges, is a component of the proposition insofar as the latter is a component of the syllogism which is the genus of demonstrative syllogism. Having determined the presuppositions of the discussion of the demonstrative syllogism, Aristotle will distinguish the perfect from the imperfect syllogism and then say something about the relationship between the terms.

The premiss is defined as a statement in which one thing is affirmed or denied of another and which is either universal, indefinite or particular. The division of the premiss excludes the singular proposition for reasons which will become clear. The division recalls doctrine with which we are familiar from *On Interpretation*. The division of the premiss into demonstrative and dialectical is something quite new. First of all, we are referred to contradictory opposition. (1) "Every man is risible" is the contradictory of (2) "Some man is not risible." As we have seen, one of these must be true, the other false. The demonstrative premiss is such that one who proceeds from it knows it to be true. The dialectical premiss, is understood somewhat like this: either (1) or (2) is true and if (1) is true, (2) is false and vice versa. Let us take (1) to be true and see what can be inferred. Aristotle says that the premiss can be settled on dialectically in one of two ways. By question, as: which do you think is true, (1) or (2)? Given one or the other, we proceed. Or: (1) may simply be asserted, not because it is known to be true, but because the majority (either of the learned or

simply the majority) holds it. The important thing at this point, however, is not to achieve absolute clarity in understanding this distinction, since it is not now formally under discussion; the present point is not so much how we achieve our premisses, but how we proceed from them. "But this will make no difference to the production of a syllogism in either case; for both the demonstrator and the dialectician argue syllogistically after stating that something does nor does not belong to something else." (24a26 If.) This tells us something rather important about the nature of the *Prior Analytics*. When Aristotle says that he has made clear the difference between the syllogistic, demonstrative and dialectical premisses, he is not speaking of three species of premisses. Both the demonstrative and dialectical premisses are syllogistic ones and Aristotle proposes that we teleologically forget their differences and consider only what they have in common. Thus "syllogistic premiss" is, as it were, the genus of demonstrative and dialectical premisses. Indeed, the logic of the *Prior Analytics*, came to be called formal, that of the *Posterior Analytics* and *Topics* material for just this reason. After this discussion of premisses, Aristotle says that terms are parts of premisses, i.e. the predicate and that of which it is predicated. With these matters behind him, Aristotle can now define syllogism.

"A syllogism is discourse in which, certain things being stated, something other than what is stated follows of necessity from their being so." (24b18-20)

"Discourse" (*logos*) implies here at least the complexity of the proposition; as it turns out, it involves several statements. In this discourse which is the syllogism it is the case that certain things being stated, or certain statements being made, something else, another statement, follows necessarily from their being so. Much of the sequel will concern itself with the order of terms in the original statements which necessitates that another statement should follow from them. Aristotle explains this necessity by adding (24b20) that he has in mind the necessity of the consequence being drawn or recognized; he is not saying that what the syllogism is about is always necessary.

This series of definitions with which the *Prior Analytics* begins is extremely difficult to understand. This is only to be expected; Aristotle is in effect setting forth a program to be developed in what follows and in the subsequent discussions the enigmatic pronouncements with which he begins are gradually clarified. If we look now at the final remarks of Chapter One, we will be able to give a better idea of what he means by syllogism and can then indicate what he endeavors to do in the rest of the work.

The final remark has to do with the relation of terms. If A is said of all B, it must be said of everything of which B is said. So too, if A is said of no B, it is said of nothing of which B is said. These definitions of what it means to be said of all and to be said of none enable us to flesh out the definition of syllogism. An example of syllogism is usually stated in this manner: Every A is B, Every C is A, therefore every C is B. Aristotle would not put it just that way and he would express and explain the above as follows. If B is said of all A and A is said of all C then, with necessity of consequence given the meaning of "to be said of all," B is said of all C. The definitions of "to be said of all" and "to be said of none" are the principles on which syllogism is based. Given these principles as well as the division of propositions into affirmative and negative, and into universal, indefinite and particular, Aristotle goes on to develop the logic of syllogism. It would be impossible to find a logician who does not accept the logic of syllogism, although it is often a matter of doubt whether every logician means by this what Aristotle would have meant by the phrase. In any case, no elementary logic course fails to acquaint the student with the doctrine of syllogism; there is justification, then, for giving the most summary statement of the remainder of the *Prior Analytics*, while at the same time impressing on the interested reader that no modern account of syllogism or what Aristotle taught on this matter should be taken to do away with the need for a careful study of the *Prior Analytics*. Only such a study will enable him to assess the anachronistic analyses of logicians who may have a radically different view of their discipline than Aristotle had.

The subject and predicate of the conclusion of the syllogism occur in the premisses together with another term which serves to connect them. Since in the conclusion the predicate is said of the subject, in the premisses we may find the predicate said of the third term and the third term said of the subject. This distribution of terms is the first figure of the syllogism and it enables us to see why the predicate of the conclusion is called the major term, the third term the middle, and the subject of the conclusion the minor term. These comparatives have to do with universality and the principle of "to be said of all" makes us see that the major must be said of the minor. Different moods of this figure of the syllogism can be had by varying the premisses in terms of affirmation and negation, universality and particularity, although not every combination turns out to be valid. There are two other figures of the syllogism, each of which has many moods. If the middle term is in the predicate position in both premisses or in the subject position in both we have figures different from the first. Since these figures are less obvious than the first, Aristotle is concerned to show how, by means of such devices as the conversion of terms, discussed in Chapters Two

and Three of Book One, they can be reduced to the first. It will be appreciated that we cannot have a syllogism in which the middle would be predicated of the major, and the minor of the middle; this does not serve to link them, as the definition of "to be said of all" makes clear. A good deal of Aristotle's discussion concerns syllogism involving modal propositions. After lengthy and complex discussions relating to the syllogism, Aristotle, at the end of the second book of the *Prior Analytics*, discusses arguments akin to syllogism among them induction and argument from example.

Posterior Analytics. We have already seen that demonstration involves a syllogism whose premisses must be of a definite sort, a point Aristotle made by contrasting them with dialectical premisses. Let us see how Aristotle elaborates the notion of demonstrative or scientific syllogism.

"All instruction given or received by way of argument proceeds from pre-existent knowledge." (71a1) Aristotle establishes the truth of this assertion by means of an induction. This is the case in the mathematical and other science; it is the case as well in dialectical arguments whether syllogistic or inductive; finally, it is involved in the use of enthymemes (Cf. *Prior Analytics*, II, 27) and examples. Appeal is always made to what is already manifest when new knowledge is to be acquired. It should be noted that Aristotle's statement is not that all knowledge, not even all intellectual knowledge, comes from previous knowledge. He is concerned only with intellectual knowledge got by reasoning or argumentation.

The foreknowledge referred to is of two kinds, Aristotle continues; either it is knowledge of a fact or knowledge of the meaning of a word. Sometimes both types of foreknowledge of the same thing are assumed. In order to understand this distinction, we have to keep in mind that Aristotle is here speaking of the demonstrative syllogism and the syllogism, generally, is a discourse by means of which, given the premisses, the conclusion follows. Thus, in order to arrive at knowledge of the conclusion, we must first know the premisses and the facts and meanings therein involved. Now, in the conclusion, we affirm something of something else and, according to the distribution of terms in the syllogism, the predicate and subject of the conclusion first appear in the premisses. Thus, since we know the premisses prior to the conclusion, we must have some kind of prior knowledge of the terms of the conclusion; and, since the subject and predicate are simple terms, we will have one kind of foreknowledge of them insofar as we know the propositions in which they occur are true. Of the premisses as such we know not what they are (since they are complex in a way defined things are

not), but that they are true. The example Aristotle gives of what must be known to be true is extremely general, namely that every predicate can be truly affirmed or denied of any subject. (Later he will show that such a principle, because of its generality, is never a premiss of a demonstration.) Of the predicate of the conclusion we must know beforehand what its name means, i.e. have a nominal definition of it. Aristotle gives the example of triangle which is a predicate in the demonstration whereby a triangle is constructed on a line. Of the line or unity, which are not predicated of other subjects (clearly triangle can be both a predicate and a subject of which something is proved), we must know both what the words signify and that they are. As is made clear in the second book, this is tantamount to saying that we must have a real definition of the subject. Before deciding whether or not something exists, we must know what its name means; if something in reality answers to what the name means we either have or can seek its definition, although the definition itself does not include any assertion that such a thing exists. With respect to that which in the conclusion is shown to belong to the subject, we do not know beforehand both what the word means and that it exists, since the fact that for it to be, is for it to be in the subject, is precisely what we learn in the conclusion of the demonstration strictly so called. Aristotle hints in Chapter Two (71b 16) and makes explicit later in Chapter Thirteen that "science" and consequently "scientific or demonstrative syllogism" are equivocal, but equivocal by design. (This will be discussed later.) Thus, he will first discuss the demonstrative syllogism in the most proper sense of the term and go on to discuss less rigorous demonstration.

We can say by way of conclusion to these remarks on foreknowledge that a demonstration presupposes that we first know the truth of the premisses, have a real definition of the subject and a nominal definition of the predicate of the conclusion. Aristotle makes it clear that it is not always temporal priority that is involved in such foreknowledge. Thus, one may assent simultaneously to one of the premisses and to the conclusion, but knowledge of the premiss will always be prior in the sense that it grounds the conclusion.

If there are certain things which must already be known if the conclusion is to be drawn, we can also ask if there is any way in which the conclusion itself might be said to be known before it is demonstrated. Aristotle has in mind here Plato's doctrine of recollection. Prior to its being demonstrated, the conclusion may be said to be known and not known. Absolutely speaking, it is not known, but in another way, since we must already know that from which the conclusion follows, we may be said to know the conclusion potentially. Thus, when we

come to know the conclusion thanks to demonstration, we are not learning what we already knew in the same sense: what beforehand we knew only potentially, we now know actually. In Chapter Two of the first book of the *Posterior Analytics*, Aristotle sets out to define the demonstrative syllogism; his procedure is of the utmost importance.

We suppose ourselves to possess unqualified scientific knowledge of a thing, as opposed to knowing it in the accidental way in which the sophist knows, when we think that we know the cause on which the fact depends, as the cause of that fact and of no other, and further, that the fact could not be other than it is. (71b9-12)

Aristotle states here what he thinks anyone would mean when he says he knows, really knows, that something is so. In other words, he is proceeding from a nominal definition. Whether or not a person actually has knowledge of something when he claims to have it, he thinks he knows why the thing is as it is, the cause of its being so; and, given that cause, the thing cannot be otherwise than it is. We have such knowledge, Aristotle says, only as the result of a demonstrative syllogism. What is the nature of this syllogism?

First of all, and repetitiously, it is that syllogism which enables us to have the kind of knowledge anyone thinks he has when, rightly or wrongly he says he knows. The point is to discuss what sort of syllogism can produce such knowledge. Its premisses, Aristotle asserts, must be true, primary, immediate, better known than and prior to the conclusion which is related to them as effect to cause. A syllogism can be had without such premisses, but only with premisses like these can we get the knowledge defined at the outset. Before examining these characteristics of the premisses of the demonstrative syllogism, something can here be said of the division of logic into formal and material.

There is, first of all, a way in which logic in general is formal to any matter we may reason about. Secondly, we can distinguish in the syllogism a form and matter, that is, the terms are material, their proper distribution formal. Neither of these distinctions is the one whereby we speak of formal and material logic; the first is not, because it distinguishes logic from that about which we hope to be logical; formal and material logic is presumably a distinction within logic. So too the second distinction is one which lies on the side of formal logic in the sense we are trying to determine. We said earlier that the *Prior Analytics* proceeds on a common or abstract level because it discusses syllogism apart from its division into demonstrative and dialectical. This suggests that what

makes a syllogism demonstrative, for example, is a less abstract, more material consideration. In the text we have just been considering, the characteristics of the premisses of the demonstrative syllogism indicate how material logic goes beyond formal logic of syllogism. The discussion presupposes the doctrine of the syllogism and adds to it something which goes beyond the notion of necessity of consequence. These additions do not take us out of the realm of logic; we are still discussing the method to be pursued in seeking knowledge, a method which, despite the many references to geometry in the *Posterior Analytics*, is not the method of some particular science. In that sense, material logic, like all logic, is formal with respect to the objects we might reason about.

To return to the nature of the premisses of the demonstrative syllogism; they must be true. It is possible to conclude something from false premisses, but if we want to know something in the sense defined above, we must proceed from true premisses. Moreover, the premisses must be first and immediate. "First" suggests a relation of order and "immediate" the basis for the order. Immediate propositions are those in which the connection of predicate and subject is evident without appeal to something else, to some mean which justifies the connection. Such propositions are prior to those which require a mean, of course, but we may understand "first" in a further sense. The premisses of any demonstration need not be immediate in the full sense, for they may have been demonstrated in their turn; yet this must be considered to be accidental to their role in the demonstration in which they are premisses. If referred to prior premisses, however, they will be mediate; ultimately, Aristotle is saying, the demonstration must be reducible to immediate propositions. Thus, though not every demonstration in geometry proceeds from immediate or indemonstrable propositions, all demonstrations are reducible to such propositions which are first in that order.

Moreover, the premisses must express the cause of what is expressed in the conclusion; for this reason the premisses are prior and better known than the conclusion. The priority and greater knowability is not to be equated with what is most obvious and familiar to us, but rather with that which in the nature of things is prior. It may happen that these two orders coincide, but Aristotle is here speaking of a priority of nature, as causes are always prior to their effects.

After this initial explication, Aristotle returns to the notion of immediate propositions which are such that nothing is prior to them in the way that they are prior to mediate or demonstrable propositions. The proposition is one side of a contradiction, Aristotle says, recalling here the distinction made in the *Prior*

Analytically between dialectical and demonstrative premisses. The demonstrator knows that his premisses are true and that their contradictories are false. With respect to immediate propositions, Aristotle makes a number of divisions. Some immediate propositions are such that anyone will assent to them once they are stated, so much so that disagreement with them is merely verbal and self-defeating. (Cf. *Metaphysics* IV, 3-8) These immediate propositions, known to all, are called axioms and no demonstrator need worry about their acceptance. Other propositions, though immediate in the sense of indemonstrable, are not so commonly recognized. With these it is necessary to explicate their terms in order that their indemonstrability be manifest. Some propositions, further, may be called immediate because they are indemonstrable in a given order, although they can be proved elsewhere. Aristotle has in mind the notion of a subalternated science: something may be proved of natural things by appeal to geometrical truths which are indemonstrable in natural philosophy.

Aristotle goes on to speak of theses which are either suppositions (hypotheses) or definitions. Suppositions seem to be the immediate propositions already discussed; only in the third case would we have a supposition or hypothesis in a sense close to our use of the term, but even there, propositions borrowed from another science are not considered to be of doubtful truth, simply the rules of a game. The introduction of definition here may seem strange. We have been discussing immediate propositions and it is clear enough that the definition is immediate in the sense that it cannot be proved (although, in the second book, Aristotle will present a very nuanced qualification of this assertion) that plus the fact that it is a principle of demonstration seems to explain its mention here.

What we have done thus far is to present the doctrine contained in the first two chapters of Book One of the *Posterior Analytics*; in Chapter Three Aristotle argues that circular demonstration is impossible, i.e., that we cannot first prove the conclusion from the premisses and then a premiss from the conclusion. This is another way of establishing the need for first and immediate propositions. The rest of the first book falls into two parts: the discussion of the conditions of demonstrative science (Chapters 4-23). Chapter Four is particularly important; the commensurately universal property there described is necessary if the notion of knowledge set down at the beginning is to be attained. Chapters Thirteen through Fifteen indicate that there is a less perfect kind of demonstrative syllogism than that hitherto described. The next major part comprises Chapters Twenty-four through Thirty-four where demonstrations are compared from various points of view. The second book can be divided into two parts: the first (Chapters 1-18) discusses the middle term of demonstration; the

second (Chapter 19) concerns the knowledge of the first principles from which demonstration proceeds.

Topics. Let us listen to Aristotle's description of what this work is about.

Our treatise proposes to find a line of inquiry whereby we shall be able to reason from opinions that are generally accepted about every problem propounded to us, and also shall ourselves, when standing up to an argument, avoid saying anything that will obstruct us. First, then, we must say what reasoning is, and what its varieties are, in order to grasp dialectical reasoning: for this is the object of our search in the treatise before us. (100a18)

To the kinds of syllogism mentioned at the outset of the *Prior Analytics*, Aristotle here adds the contentious, fallacious or apparent argument. This will be his concern in his *Sophistical Refutations*. To carry out the program outlined will provide us with something useful for three things: intellectual training, casual encounters, and the philosophical sciences. (cf. 101a25) This last point is developed in a way which enables us to appreciate Aristotle's procedure in his treatises.

For the study of the philosophical sciences it is useful, because the ability to raise searching difficulties on both sides of a subject will make us detect more easily the truth and error about the several points that arise. It has a further use in relation to the ultimate bases of the principles used in the several sciences. For it is impossible to discuss them all from the principles proper to the particular science in hand, seeing that the principles are the *prius* of everything else: it is through the opinions generally held on the particular points that these have to be discussed, and this task belongs properly, or most appropriately, to dialectic: for dialectic is a process of criticism wherein lies the path to the principles of all inquiries. (101a34 if.)

Noting that problems arise with respect to genus, property or accident, Aristotle sets the stage for the subsequent development. In Books Two and Three he discusses problems respecting accident; in Book Four those involving the genus; in Book Five, property; in Book Six, definition. Book Seven concerns the question of identity and definition and, in Book Eight, Aristotle discusses the use of dialectic. The *Refutations* discusses the origin of fallacies and how they may be solved.

{33} Julius Stenzel, *Plato's Method of Dialectic* (Oxford: Clarendon Press, 1940).

{34} W. Jaeger, *Aristotle*, 2nd ed. (Oxford: Clarendon Press, 1948).

{35} See Sir Ernest Barker's edition of Aristotle's *Politics* (Oxford: Clarendon Press, 1948); J. H. Randall, *Aristotle* (New York: Columbia University Press, 1960); D. J. Allen *The Philosophy of Aristotle* (Oxford: Clarendon Press, 1952); G. R. G. Mure, *Aristotle* (London: E. Benn Ltd., 1932). The following works contain invaluable surveys of the problem. Joseph Owens, *The Doctrine of Being in the Aristotelian Metaphysics* (Toronto: Pontifical Inst. of Medieval Studies, 2nd. ed., 1957); S. Gomez-Nogales, *Horizonte de la Metafisica Aristotelica* (Madrid: Facultades de Teologia y de Filosofia del Colegio Maxima S. I. de Ona, 1955).

{36} Cf. Paul Wilpert, "The Fragments of Aristotle's Lost Writings," in *ARISTOTLE AND PLATO IN THE MID-FOURTH CENTURY*, edited by I. Düring and G. E. L. Owen (Göteborg; Almqvist and Wiksell, 1960) pp. 257-264. This collection of the papers of the *Symposium Aristotelicum* held at Oxford in 1957 is of the highest importance.

{37} For one of the best analyses, see the essay by S. Mansion, "Contemplation and Action in Aristotle's *Protrepticus*," in Düring and Owen, *op. cit.*

{38} See W. D. Ross, *Aristotle* (London: Methuen & Co., 1930).

{39} (London: Methuen, 1952).

{40} W. D. Ross, *Aristotle's Prior and Posterior Analytics*, (London: Oxford Univ. Press), p.1.

D. Aristotle's Philosophy of Nature

The bulk of the writings of Aristotle is devoted to the philosophy of nature which, of course, he in no way distinguishes from the science of nature. In pursuing the study of nature, Aristotle shows himself to be uncommonly interested in everything his predecessors had to say, making use of their opinions and findings at almost every step of the exposition of his own doctrine. We should not expect, however, that Aristotle is interested in the mere enumeration of previous opinions. He quite consciously uses his predecessors as stepping stones to what he takes to be the truth of the matter. We would be well advised to look on this not as a cavalier abuse of historical truth, but as a way of taking seriously the intent of earlier thinkers. Aristotle's assumption is that whoever talked about nature was not merely providing biographical data, but was concerned with explaining the way things are. If the explanations break down for reasons that can be shown, the positions can still be used for establishing the truth of the matter. Thus, for Aristotle, not every mobile thing is alive and no divine thing is mobile in the proper sense of the term. If this is so, the use of statements of a philosopher who thought that the basic stuff of things is God-Soul-Matter is going to entail ignoring certain aspects of a statement and concentrating on those aspects relevant to a particular consideration. Aristotle is always less interested in what an author intended to say than he is in a statement's relation to what the author intended to talk about.

The first problem the science of nature faces, Aristotle seems to think arises from the fact that its possibility had been denied by Parmenides and, somewhat differently, by Plato. As Aristotle makes clear, this is really a problem that the philosophy of nature cannot solve: for natural science to answer fundamental attacks on itself is for it be in a position similar to that of the man painting the eaves of a two-story house when his partner tells him to get a good grip on his brush because he is taking away the ladder. Despite this, Aristotle feels constrained to discuss Parmenides at the very outset of the *Physics*; his reply to Plato we will reserve for the proper arena for such disputes, metaphysics. A problem more peculiar to natural philosophy has to do with its order of procedure; we will turn immediately to Aristotle's doctrine on this point.

Order of Procedure. In pursuing scientific knowledge of nature, as in any other scientific pursuit, we are after an explanation of the subject in terms of its principles, causes and elements. We observe the world around us, we wonder why it is what it is and as it is, and our wonder is dispelled when we can assign reasons. In saying this, Aristotle may be thought to be making explicit what is

implicit in the endeavors of his predecessors. He goes on to make an important distinction.

What we must not fail to recognize, he insists, is that what we first come to know about the natural world and in terms of which give our first explanations of it, must not be identified with that in nature which is most properly the cause of the phenomenon in question. What is most easily known by us is not the same as what is "most knowable by nature," the determinate cause of things. We have first a global, confused knowledge of the things around us and our first explanations are on this level. Aristotle is not arguing that there are things other than the things we first know by sensation. Rather, he is saying that what we first know about these singular things is something very general, common and confused. That is, we first know that there are mobile things, things which come into being, change ceaselessly while they are, and then pass out of being. To be changeable is something seemingly characteristic of everything in the sensible world. Thus, if we examine what is proper to them under this most universal aspect, we will be on our way towards knowledge of natural things. The next step will be to distinguish different kinds of mobile being and to study their characteristics as different, and by proceeding downwards through steps of universality, we aim to arrive at "determinate knowledge of this particular, specific mobile thing," e.g., horse.

Nothing is more important for an understanding of Aristotle than this notion of the order of procedure in the study of nature. And nothing is more commonly overlooked. In his *Physics*, Aristotle is concerned with examining the common characteristics of mobile being. What he discovers there should be true of every physical thing, but the *Physics* does not pretend to show how this kind of physical thing differs from that. For what is being sought is that which every physical thing has in common. Furthermore, it is foolish to think that Aristotle proceeds deductively from one level of universality to another, as if from the notion of "changeable being" he could infer the existence of any species of mobile being. Aristotle is the first to warn against attempting this, urging that acquaintance with nature is the only road to more determinate statements about it.

We will come back later to the way in which Aristotle's mode of procedure can be traced through his many works on nature. For the moment, let us keep in mind that the analyses of the *Physics* are intended to be true of every physical thing, but should not be confused with specific and proper knowledge of any physical thing. This work of Aristotle's is the first step in an orderly approach to

nature: the order to be followed is the first thing Aristotle treats in the *Physics* and he keeps coming back to it in his other natural writings.

The Historical Background. We must here recapitulate some of the points raised in the first part of this book concerning the doctrine of Parmenides and its aftermath. Parmenides, we recall, denied the possibility of change because it seemed to involve a passage from non-being to being, from nothing to something. Discussed on this stratospheric level, his argument appears to be irrefutable. Here is being. You say that it has come to be. But from what previous state could it come? There are two possibilities: being or non-being. But if being comes from being, there seems to be no change; where beforehand there was being, there is now being. If we should say that being came from non-being, this is to say too much. Thus, change is impossible; there is only being. For somewhat the same reasons, being must be one, unique, without parts, an utterly indistinguishable sphere; for how could two beings differ? Surely they would not differ insofar as they are being; this is what they have in common; and, should we say that they differ in non-being, this is tantamount to saying there is no difference between them, that is, they are the same. Now the obvious retort to Parmenides is oblique; we see many things each of which is. Parmenides is ready for us; sensation cannot be trusted if it appears to conflict with the logic of the foregoing arguments. The student will learn to sympathize with thinkers who followed on Parmenides by exercising his own wits to find a solution or way out of the Parmenidean dilemma. These thinkers, as we have seen, attempted to devise ways of accepting both change and the denial of change. What could not change or come to be, is what truly is. Ignoring the Parmenidean strictures against multiplicity considered apart from the problem of change, the atomists and Empedocles and Anaxagoras simply posited a multiplicity of ultimate building blocks of macrocosmic entities, which building blocks were the alphabet (elements) from which the world of appearance was spelled. What can be said to come to be is that which is a conglomeration of ultimate things, the things which really are and do not themselves come to be; it follows that what comes to be cannot truly be said to be; only what has not and cannot come to be truly is.

There is no denying that this is one way out of the difficulty posed by Parmenides; our own cultural climate may lead us to find it quite attractive, for we are accustomed to think that the things of our everyday experience are, in a mild sense at least, quite different from the way they appear to be. The solid quality of our desk, for example, is misleading if we think of the swarms of electrical charges which we believe compose the desk. Its surface then must be

thought of as anything but solid, since there are more interstices than "components." This should indicate why Democritus' plenum and void are looked upon as a crude but interesting premonition of later scientific explanatory elements. If we have dwelt a bit on this supposed affinity of atomist doctrine and modern physics, it is because our culture does set up a block to our reoccupying the position from which Aristotle surveyed these attempts and found them wanting. What exactly is Aristotle's viewpoint?

It will be well to recall here our earlier methodological remarks. Aristotle is attempting to begin at the beginning, at what for us at any rate is the beginning of knowledge of the physical world. This beginning cannot be equated with what we nowadays for a number of reasons find familiar. By this we mean that our ready acceptance of elements and electrical charges and so forth can blind us to the fact there it is certainly not such things as these that we first know. We can imagine Aristotle expressing it somewhat as follows. You tell me that what truly exist are things I do not directly encounter but components of the latter; the things I do directly encounter are said by you not truly to be. I protest that, if I am to have some acceptable notion of what it means for a thing to be, I must have recourse to the very things you claim are not, in the rich sense of the term. You are saying, in effect, that a tree or a horse or a man is not one in the sense of one being. I cannot accept this because it is of the unity of such things that I must think when I attempt to imagine the imperceptible things you assure me are really one and really are. In other words, Aristotle refuses to accept the presuppositions of earlier attempts to adjust to Parmenides. He takes seriously his initial certitudes because he sees that, by one kind of prestidigitation or another, these are assumed even when they are being called into question. Obviously, then, he is faced with the problem posed by Parmenides all over again; Aristotle will insist that a tree truly is and that it has come to be and that there are many such beings. Parmenides would reply, if the tree is a being and has come to be, it must have come either from being or non-being. Aristotle has to confront that objection head on; it is because he sees the role Parmenides' argument has played in the history of natural science, and because he feels he can meet the argument head on and answer it, that Parmenides looms rather large in the first book of the *Physics*, despite the fact that Aristotle's questioning of the basic assumption of natural science precludes an answer from the viewpoint of the contradictory assumption. Aristotle is quite clear on this point in Chapter Two, but he also feels justified in considering Parmenides.

We physicists, on the other hand, must take for granted that the things that exist by nature are, either all or some of them, in motion -- which is indeed made

plain by induction. Moreover, no man of science is bound to solve every kind of difficulty that may be raised, but only as many as are drawn falsely from the principles of the science: it is not our business to refute those that do not arise in this way: just as it is the duty of the geometer to refute the squaring of the circle by means of segments, but it is not his duty to refute Antiphon's proof. At the same time the holders of the theory of which we are speaking do incidentally raise physical questions, though Nature is not their subject: so it will perhaps be as well to spend a few words on them, especially as the inquiry is not without scientific interest. (185a12-20)

A careful reading of Chapters Two and Three of the first book of the *Physics* reveals that Aristotle's ultimate weapon is that Parmenides and Melissus use terms like "being" and "one" in such a way that they must assume a meaning for these words which can be grasped only when their referents are the very things these non-physicists wish to reject.

Dialectical Summary. In Chapters Five and Six of the first book of the *Physics*, Aristotle asks if there are any points on which his predecessors agree despite their many disagreements. Such efforts as this are often looked upon as attempts by Aristotle to show that his predecessors were groping towards his own view. Now this is perfectly true, though badly expressed. Aristotle is not putting forth a doctrine expressive of his personal way of looking at the world and his effort should not be construed as an attempt to show that Anaxagoras, for example, was trying in a lisp to attain the timbre of Aristotle's voice. Aristotle is infinitely more serious than the neutralist, unengaged historian of ideas. His viewpoint is at once obvious and profound. His predecessors looked hard at the world, the same one Aristotle and you and I are confronted by, and they said a number of things about it. Some of what they said can be understood by taking into account what others had said, but finally their statements must stand the test of comparison with the world they hoped to explain. Now it is quite clearly the latter test that most interests Aristotle. Moreover, he, like most of us, is disinclined to feel that men can look at the common universe and explain it in ways which are utterly different and utterly false. It is against this background that we must read Aristotle's efforts to discern in the cacaphony of previous natural doctrines some concordant views.

Their first agreement is in their recognition that contrariety is involved in change. This is true of the Presocratics; it is true of Parmenides (in the *Way of Opinion*) who sees the hot (fire) and cold (earth) as principles of other things. The atomists opposed the full and empty, others spoke of the rare and dense,

yet others of congregation and separation. Thus, hot comes from cold and vice versa; white comes from black and vice versa and, generally, a thing comes to be from its contrary and passes into its contrary. Changes are not capricious and their non-capriciousness is expressed by this appeal to contraries as the terms of change. This basic assumption involves another. There must be something besides the contraries, something which underlies them. To say that hot becomes cold is to say that something which is hot becomes cold, not that heat becomes its opposite.

Aristotle's conclusion is that, despite their diversities, and even when appearing to speak of only one principle, previous philosophers had all relied on two contraries and a subject of these contraries in speaking about change. For example, Anaximenes says that all things are air, but in order to explain the diversity of things, he takes air as capable of possessing two contrary states, rarefaction and condensation. The diversity of things thus comes to be looked on as forming a scale read in terms of the extreme states. None of these thinkers alluded to these assumptions but, since they all made them, it is probable that these three, two contraries and their subject, will enter into the explanation of changeable being.

Principles of Changable Being. Turning to the elaboration of the truth of the matter in Chapter Seven, Aristotle reveals the importance of his talk about order and methodology. One need expect no mention of fire, air, earth and water. What is first presented for our consideration is something we can all be reasonably expected to understand: the change involved when a man becomes a musician. With deceptive simplicity, Aristotle observes that such a change can be expressed in three ways: (1) Man becomes musical. (2) The not-musical becomes musical. (3) The not-musical man becomes a musical man.

These three expressions of the same change indicate that Aristotle is interested in the different ways in which we express that from which a change begins. Otherwise he would have set down a fourth expression of the change. The three ways of expressing the beginning of the change are "man," "not-musical" and "not-musical man." Let us call the first two simple expressions and the third composite.

Why is it that we sometimes say "X becomes Y" and at other times "From X, Y comes to be"? The grammatical distinction seems to suggest the recognition of a real difference. We would hesitate to say, "From man, musical comes to be," whereas "From not-musical, musical comes to be" feels all right. Why? Is it not

because, in the first instance, man does not cease to be when he has become musical, whereas in the second, not-musical is replaced by musical? If this is so, we can speak of things which survive the change and things which do not. Only in the first expression of the change, that is in (1), does the subject of the sentence stand for what survives the change. In (3), the composite "not-musical man," like the simple term "not-musical" of (2), does not survive the change. When we have a musical man, we no longer have a not-musical man.

In our instance of a man learning how to play the lyre, there is something which is there both before and after the change, namely the man. And, as the other two expressions of the change indicate, opposites or contraries are involved: there is a change from not possessing musical art to possessing it on the part of the man. In the first expression, consequently, although only man is mentioned, he has to be understood as not being already musical, since he would hardly be said to become what he already is.

What the analysis of this change reveals, then, is that there is a subject of the change, man, where subject is understood as what is there before and after the change occurs. Understood in the subject is the negation of that which is acquired as the result of the change, a negation opposed to the new quality, i.e. not-musical to musical. What results from the change is a composite of the subject, man, and the quality, musical. What does not survive the change is the negation of the acquired quality.

Aristotle next wants to analyse another more basic kind of change in terms of what has been clarified in the more obvious instance of a man becoming musical. In the instance already analysed, something came to be such-and-such, a man came to be musical. But what of changes where something comes to be, not such-and-such, but comes to be without qualification? For example, a tree or a man comes to be. Can we explain such changes by appeal to a subject which survives the change?

We have already insisted on the fact that Aristotle is going to take seriously the certitudes of everyday life. Before a man comes to be, he is not; before a tree comes to be, it is not. When man and tree exist, they are things in a more fundamental way than the quality musical or color or the composites musical man and green tree. A man and a tree are in themselves beings in a basic sense; they are not modifications of some basic being, nor are they accidental compounds of basic beings. In a word they are substances (Aristotle's word is *ousia*: being). Now this is something that we all already know; the fact is certain.

But how can we understand the fact? Aristotle, recall, does not want to explain it away, but to explain it. If a man and a tree come to be as the result of a change, this suggests, on an analogy with man becoming musical, a subject of the change. But, if we posit some such subject as earth or air or fire or atoms, we would be in agreement with the Presocratics and the post-Parmenidean natural philosophers. It is a costly agreement, however, for if we appeal to such subjects as these, a tree and man would be modification of it in the same way as musical is a modification of man. We know why the post-Parmenidean physicists took this route; by not claiming any substance came to be, they skirted the difficulties of the Eleatic's argument. Aristotle does not want to avoid Parmenides, however, and as a result he asserts there is a subject of unqualified change, a subject which he calls elsewhere prime matter.

If a tree's coming to be is a change, there must be a subject of change. "But that substances too, and anything else that can be said to be without qualification, come to be from some substratum, will appear on examination. For we find in every case something underlying from which proceeds that which comes to be; for instance, animals and plants from seed." (190b1-5) This instance may seem difficult to understand, but it is remarkably well chosen. Great oaks from little acorns grow here: From X, Y comes to be. Now if a tree is appreciably different from a musical man, its coming to be will have to be explained in an appreciably different way. Just as in "Man becomes musical" there is a subject, man, so too in "a tree comes to be." In the first case, the subject of the change is a substance and the result of the change is a new accidental determination of the substance. In the second case, the subject cannot be a substance, for then any new determination of it will be an accident, and we are taking seriously our conviction that such things as trees are substantially one. But what then are we to make of the example above, an example which can be expressed as "A seed becomes a tree." Notice that this is not on a par with "A man becomes musical" since, almost biblically, unless the seed die, the tree cannot be. Thus, "The seed becomes a tree" is much more like "The non-musical becomes musical" or, better, "The hot becomes cold."

The subject of unqualified becoming, Aristotle remarks, is known by a comparison or analogy.

The underlying nature . . . of the coming to be of substance . . . is an object of knowledge by analogy. For as the bronze is to the statue, the wood to the bed, or the matter and the formless before receiving form to anything which has

form, so is the underlying nature to substance, i.e., the 'this' or existent. (191a7-12)

Notice that in this statement Aristotle is appealing to a change in the realm of art as to something readily comprehensible by us; moreover, the terms he uses, "matter" and "form," seem drawn from the realm of art. The Greek equivalents of these terms, *hyle* and *morphe*, call to mind the modification of wood by imposing a new shape or form on it. The argument from analogy, then, goes beyond the example of man becoming musical, to the imposition of a shape on wood, a form on matter. When we make a bed, we impose on lumber (which is a more primitive instance of imposing an artistic shape on natural material) a new shape or form. Let us now talk of a man's becoming musical in these terms, altering their meaning somewhat as we go. Man may now be called the matter, musical the form of the product of the change. If we are to retain these same terms in speaking of the change whereby a man comes to be, their meanings must once more change; that this extension of meaning occurs is signaled by designating matter as *prime* matter and the form as

substantial form. Since prime matter is not a substance in its own right, it is said to have no substantial determination of itself, and thus can only be known by comparison with the subject of accidental change in the natural order or to the subject of an artificial change. If we speak of the subject of a substantial change, however, we need not think that what is meant is that it appears as the subject of sentences which express such changes. "The seed becomes a plant" is much more like what we are apt to say but, in saying it, we are not saying that the seed persists throughout the change. If we accept the fact that trees are substantial beings, that they come from seeds, and that a subject or matter must be involved if we are to speak of change -- we would be using "change" in a Pickwickian and mysterious sense if by it we meant that seeds disappear and plants appear on the analogy of a change of scenery -- then we must inevitably be led to an ultimate subject of substantial change, itself not a substance but a component of substance. Notice that this is to explain a fact of which we are certain, not to explain it away. The determination of matter whereby a substance is constituted is called form, but unlike musical it does not make something to be such-and-such, but makes it be absolutely and in the first instance. And, of course, matter could not be determined by a form it already possessed, but only by one it did not possess. Thus in all change, whether qualified (accidental) or unqualified (substantial) there are three principles necessary: matter or subject, form or determination and the previous privation of this determination on the part of the subject.

Parmenides Confronted. The test of a solution is its ability to withstand objections, to make things clearer than rival solutions to indicate how allied problems should be dealt with. It is to the second test that Aristotle wishes to put his solution to the problem of change, for he wants to show that Parmenides got into an unnecessary difficulty, thereby leading himself and others astray. The Parmenidean problem, once more, is that if we say that something has come to be we must show that it has come either from being or non-being and this we cannot do. It is wholly typical of Aristotle to subject to analysis the troublesome sentences, (1) "Being comes from being," and (2) "Being comes from non-being." Parmenides obviously understands them in such a way that (1) could be taken to mean something like "A trained seal becomes a trained seal" where after the supposed change we end up with what we began and ought not to talk of any change having occurred. Statement (2), on the other hand, becomes something like "From absolutely nothing at all, a trained seal came to be." Now this is one way to understand (1) and (2), but it is not the only way and it is not the way we would take them if we wanted them to agree with our certitude that change is real. Aristotle suggests there is another manner of understanding the way something comes from another, just as there are several ways to understand an activity attributed to a physician. For example, in "The physician heals," and "The physician golfs," we are speaking of the physician and what he does, but in the first sentence we are speaking of him *qua* physician, that is, just insofar as he is a physician, whereas the second activity ascribed to him is not so ascribed just because he is a physician. So too if we say hot comes from non-hot, we need not understand our remark in the way Parmenides would understand it, nor do our words imply the further assertion that hot comes from cold. Aristotle sees Parmenides' position as a result of thinking that being comes from either privation or previous determination. For example, hot (being) comes from cold (being) or from non-hot (non-being). Now, Aristotle says, hot does come from cold and non-hot, but neither of these is a principle of what has come to be, where by a principle of what has come to be he means what is a component of the result of the change. Only the subject (water) is this kind of subject-from-which. Let us listen to Aristotle himself on this solution.

We ourselves are in agreement with them in holding that nothing can be said without qualification to come from what is not. But nevertheless we maintain that a thing may 'come to be from what is not' -- that is, in a qualified sense. For a thing comes to be from the privation, which in its own nature is not-being -- this not surviving as a constituent of the result. Yet this causes surprise, and it is thought impossible that something should come to be in the way described from what is not -- in the same way we maintain that nothing comes to be from

being, and that being does not come to be except in a qualified sense. In that way, however, it does, just as animal might come to be from animal, and an animal of a certain kind from an animal of a certain kind. Thus, suppose a dog to come to be from a horse. The dog would then, it is true, come to be from an animal (as well as a certain kind) but not as *animal*, for that is already there. But if anything is to become an animal *not* in a qualified sense, it will not be from animal: and if being, not from being -- nor from not-being either, for it has been explained that by 'from not-being' we mean from not-being *qua* not-being. (191b13-26)

Aristotle by identifying Parmenides' not-being with privation and his being with the previous determination of the subject, effectively does away with the old difficulty by showing that neither privation nor the previous form is that from which the result of the change comes in an unqualified sense. The from which the result of the change comes without qualification is the subject as capable of possessing the new determination. That is, water from being hot only potentially comes to be actually hot. This is why we often read that Aristotle solves the difficulty of Parmenides by introducing the distinction between act and potency. It should be said, however, that this is only half the story and that the identification of being and not-being with two of the principles necessary for any change is the solution developed in the first book of the *Physics*.

Nature. The term "physics" is derived from the Greek term *physis* which Aristotle analyses in the *Metaphysics*, V, 4. There he points out that the term had first meant the process of being born, then the principle of that process, and then had been extended to signify the principle of any change whatsoever. It is this third sense that is operative in the *Physics* and, in the second book of that work, Aristotle undertakes to define it by comparing it with art. Things, he begins, are either natural or artificial. We say that animals, their parts, plants, fire, air, earth and water exist by nature, that is, are natural products. What we mean is made explicit by contrasting such things with works of art. A pair of shoes is not a natural product. The thing that distinguishes what exists "by art" from that which exists "by nature" is that the latter has the principle of its change within itself. Nature consequently can be defined as follows: it is the principle of motion and rest in that to which it belongs primarily, *per se* and not accidentally. By calling nature a principle, Aristotle leaves the way open to understanding it as an active or passive principle of the change, that is nature may be a power to act or a power to be acted upon. By saying that nature is a principle of motion and rest, Aristotle is alluding to the view he does not argue for here that things have a natural place in the universe: when they are in that

place they are naturally at rest. By saying that it is a principle *in* the moved thing, Aristotle is contrasting nature to art. By calling nature a first principle, Aristotle is suggesting that change can be called natural not necessarily with respect to a compound as such, but with reference to a component of it.

Now we have already seen that the components of natural compounds are, for Aristotle, matter and form. Thus we should expect that both these will save the definition of nature, and this is precisely what Aristotle goes on to show. The change of something may be described as natural either with respect to the matter of which it is composed or with respect to its form. To indicate what this means in a very general way, let us notice that death may be natural to man because of matter, and reasoning and immortality because of form. Aristotle gives several arguments to prove that form is more deserving of the appellation nature than is matter. The discussion of the second chapter of book two turns on the difference between physics and mathematics, a point we discussed earlier.

The effect of the opening considerations of the second book is to establish the meaning of "physical things" (*ta physica*) in a way that connects with the analyses of the first book. We now have a fairly clear idea of what it is of which we seek scientific knowledge, a knowledge which is had through causes.

Now that we have established these distinctions, we must proceed to consider causes, their character and number. Knowledge is the object of our inquiry, and men do not think they know a thing till they have grasped the 'why' of it (which is to grasp its primary cause). So clearly we too must do this as regards both coming to be and passing away and every kind of physical change, in order that, knowing their principles, we may try to refer to these principles each of our problems. (II, 3, 194b16-23)

This sets the stage for the analysis of causes into four types, each of which will be a principle of explanation in physics. The material cause is that out of which something is made and which remains as a component of the result. The examples are from art: bronze is the material cause of the statue. The formal cause is that which is expressed in the definition of the thing. For example, if we are asked what a statue is, we would say, not bronze, but bronze shaped in such a way. The efficient cause is the primary cause of the change; e.g. the sculptor who makes the statue. The final cause, that for the sake of which something is done, is also a cause. Why does one exercise? To be healthy. This is the end or purpose explaining why one is sweating in the gym. It can be seen that a

physical thing can be explained in terms of one or all of these four causes. Moreover, each type of cause can be designated in several ways. In terms of prior and posterior (distinguished according to universality) we may designate the efficient cause of health as, respectively, the trained man or the physician. A cause may be designated accidentally as when we say Polycitus is the cause of the statue, since it happens that the sculptor is named Polycitus. Finally, causes can be either actual or potential causes of their effects. The physicist's interest in the four causes is described thus by Aristotle.

Now, the causes being four, it is the business of the physicist to know them all, and if he refers his problems back to all of them, he will assign the 'why' in the way proper to his science -- the matter, the form, the mover, 'that for the sake of which.' The last three often coincide; for the 'what' and 'that for the sake of which' are one, while the primary source of motion is the same in species as these (for man generates man) . . . (II,7,198a21-27)

The form or essence of the generated things is that for the sake of which the process took place and the moving or agent cause is of the same species as its effect in natural generation. Aristotle has much to say of finality in nature, but before looking at that doctrine, we must say something about the discussions in Chapters Four through Six on accidental causes.

Chance and Fortune. We say that some things come about by chance, and Aristotle wants to investigate our reasons for doing this to see if we are speaking of a real cause. It is important to realize that Aristotle is concerned with chance as cause, something evident in our use of the phrase "by chance." Aristotle first notes a surprising range of opinions on chance. Some deny its reality and claim that anything ascribed to chance can be ascribed to a determinate cause. Meeting an old friend does not come about by chance but because one went to the market place and ran into him. Moreover, the early natural philosophers did not list chance as a cause, though some, like Empedocles, assign things to chance as to a cause. Indeed, some who do not attribute to chance the formation of lesser entities, say that the heavenly sphere is a result of chance. Finally, there are those who believe chance to be real but mysterious, inscrutable and beyond our ken. Since some things always come about in the same way, whereas others occur in a certain fashion only for the most part, there are some rare occurrences as well. It is this last class of events that we ascribe to chance. In arriving at an analysis of chance in the natural realm, Aristotle proceeds by analysing chance in human affairs, what we would call fortune or luck. Very briefly, his teaching is this. Something can come about by

chance only where there is an agent acting for an end. Thus, if on the way to the store I find a ten dollar bill, I would call myself lucky, the beneficiary of good fortune. In other words, I am ascribing the finding of the money to chance. What now of the first objection Aristotle recorded? Someone might say that I am guilty of fuzziness when I ascribe such an occurrence to chance; there is a determinate cause of the event. If I had not gone to the store, I would not have found the money; since I went to the store, I found the money. This objection is extremely useful, for it forces us to ask what kind of cause of my finding the money going to the store is -- that it is the cause is evident enough. But if we observe that going to the store does not usually have as its result my finding ten dollars, we are in a position to say that this is not a determinate cause of my discovery -- a determinate cause produces its effect always or for the most part (or at least, intentionally; the pole vaulter does not always or usually surpass his previous feats, but when he does it would not be advisable to congratulate him on his luck until he has laid aside his pole). The event ascribed to chance comes about rarely, outside the intention of the agent, and is good or evil for the agent. Now the regularity in nature suggests intention; freaks of nature suggest that chance is a real cause in the natural order. The universe, then, is not a concatenation of necessary occurrences for Aristotle. There are things that come about by chance, an element of caprice and unpredictability; moreover, it would make no sense to say that obviously ordered things come about by chance, since this would mean, not that there is no purpose, but that in pursuing one purpose nature accidentally brought about another. In other words, chance in nature, as Aristotle has analysed it, makes sense only against the background of nature as purposive.

Finality. Arguments against finality seem to be as old as arguments for it and Aristotle commences his discussion of nature as purposive principle by setting down objections to this view. Why can't we say that things simply come about by necessity and, rather than say that rain falls so that the crops will grow, say it rains because vapor rises and condenses and falls as rain, with the growth of the crops as an incidental effect of this necessary chain of events? So too we need not say that some teeth are for tearing and others for chewing, but that, simply given the teeth we have, it happens we find them useful for various purposes of ours. Aristotle's rejection of these objections are straightforward. Why should we ascribe to chance what comes about usually or normally? "If then, it is agreed that things are either the result of coincidence or for an end, and these cannot be the result of coincidence of chance, it follows that they must be for an end." (II, 8, 199a3-5) This view that nature acts for an end suggests the kind of necessity we may expect in our explanations of natural events.

But in things which come to be for an end, the reverse is true. If the end is to exist or does exist, that also which precedes it will exist or does exist; otherwise just as there, if the conclusion is not true, the premiss will not be true, so here the end or 'that for the sake of which' will not exist. . . If then there is to be a house, such-and-such things must be made or be there already or exist, or generally the matter relative to the end, bricks and stones if it is a house. But the end is not due to these except as the matter, nor will it come to exist because of them. (II, 9, 200a19-27)

Motion. Having defined nature as a principle of motion, Aristotle must determine what motion is if he is to have a proper understanding of the subject of physics. The discussion of motion will carry him on to allied subjects: infinity, place, void, time. We shall give a brief exposition of what Aristotle has to say concerning motion and then indicate his doctrine on the allied notions.

Motion is defined by Aristotle as the act of what exists in potency insofar as it is in potency. To understand this definition, we must make a threefold division of things into those which are wholly in act, those which are only potentially, and those which are midway between these extreme conditions. What is only potentially is not in movement; e.g., the seated man is standing only potentially. So too what is wholly in act is not in motion; e.g. the man, having stood up, is actually standing and that's that. The act of getting up out of the chair is motion and it is identified by contrast with the extreme states of potency alone and act alone. It is an imperfect act: one who is getting up is not wholly in act, i.e., not actually standing, nor wholly in potency, that is simply seated, but somewhere in between. Motion, therefore is not the potency of something existing in potency nor the act of something existing in act, but the act of something existing in potency just as such, where "just as such" indicates the relation of the imperfect act to further act. The thing existing in potency alone, e.g., the man seated, can be seen as in potency to two acts: the imperfect act which is motion and the perfect act which is the term of the motion, standing erect.

Related Notions. Since motion involves the continuum and this is said to be infinite, Aristotle goes on to define infinity. "A quantity is infinite if it is such that we can always take a part outside what has been already taken." (III,6,207a7-8) Infinity is always potential, it is that which is always further divisible. Aristotle argues against the possibility of an actual infinite. The physicist must define place as well, if only because locomotion, change of place, is the most general and common kind of motion. Place is the innermost motionless boundary of the container (IV,4,212a20-1) As to the void, Aristotle

rejects it. Time he defines as the number of motion with respect to the before and after. (IV, 11, 219b1-2)

In the fifth book of the *Physics*, Aristotle divides motion into its species, locomotion, alteration, augmentation and decrease, motions in the categories of place, quality and quantity, respectively. He then discusses the unity and opposition of motions. In the sixth book, he discusses the quantitative parts of motion.

In the last two books of the *Physics*, Aristotle discusses the Prime Mover. That such a mover exists is proved by appeal to two truths: whatever is moved is moved by another and there cannot be an infinite series of moved and moving things. Thus there must be a first unmoved mover if motion is to be explained; Aristotle proceeds here on the assumption that motion is eternal and has not had a beginning. The Prime Mover is shown to be without parts or magnitude. Having arrived at an entity which is immobile and incorporeal, Aristotle has encountered a being which does not as such fall under the scope of natural science. As we will see, it is this proof that something immaterial and immobile exists that permits Aristotle to say that "being" need not be taken to mean mobile being alone; in other words, there is a possibility for another science whose subject is being as being. That science is First Philosophy, what has come to be called Metaphysics, and in discussing it, we will examine Aristotle's notion of a Prime Mover. Treatises Consequent on the *Physics*. We saw earlier Aristotle's concern for orderly procedure in natural science. This attention to method is not confined to the materials of one work, but applies as well to the relation of the various works to one another. In this wider perspective, the *Physics* presents doctrine presupposed in the other natural works and is thus prior to all the rest. In this introductory work we find a general analysis of motion, a comparison of types of motion, and so forth. It is in terms of types of motion that the later works can be seen to be divided, and since there is an order of priority and posteriority among the species of motion, there is also an order among the works concerned with them. It has been proved in the *Physics* that local motion is the first and most common motion; thus, in *On the Heavens* both heavenly and terrestrial bodies are treated insofar as they are subject to local motion. Subsequent motions, that according to quality and that according to quantity, are thought not to be common to every natural thing: alteration, taken as ordered to generation, is confined to terrestrial things and *On Generation and Corruption* is concerned with such things from that viewpoint. The discussion of the transmutation of the elements (fire, air, earth and water) is continued in the *Meteorology*. Augmentation and decrease, looked upon as

following the taking of nourishment, lead to the discussion of living beings, a discussion commenced in *On the Soul* and carried on in *On Sense and the Object of Sense*, *On Memory and Reminiscence* and in the many works on animals. The whole sweep of Aristotelian natural doctrine thus appears as a movement from general truths which cover natural things indiscriminately to determinate, concrete statements about natural things in their specificity. We must here satisfy ourselves with a few remarks on his general treatise on the soul.

The Soul. Aristotle begins his discussion of the nature and properties of the soul by indicating the desirability and value of possessing such knowledge as well as the difficulties which face one who would ask of soul, what is it? There is no one method to be followed when one is seeking knowledge of essence as there is one method for demonstrating properties. What we must ask is what genus contains soul; whether soul is a substance, quality or something quantitative, or something else; further, we must ask if it is something potential or actual. Moreover, we must be careful to ask whether soul can be defined in general without regard to its species, or whether we must from the outset seek the definition of a determinate type of soul.

We must be careful not to ignore the question whether soul can be defined in a single unambiguous formula, as is the case with animal, or whether we must not give a separate formula for each sort of it, as we do for horse, dog, man, god (in the latter case the 'universal' animal -- and so too every other common predicate -- being treated either as nothing at all or as a later product). (I,1,402b5-9)

Again, if there are parts of the soul, should these be considered before or after soul itself? A question also arises concerning the mode of defining the passions or affections of soul, since it seems necessary to include the body in such definitions. "That is precisely why the study of the soul must fall within the science of nature, at least so far as in its affections it manifests this double character." (403a27-8) Before turning to his own answers to these and allied questions, Aristotle polls his predecessors to see what they had to say on the subject of soul, a survey that occupies the remainder of the first book.

When he turns to the task of defining the soul in the second book, Aristotle begins with a number of divisions. First, he points out that there is a threefold sense of "substance:" matter, form, and the compound of the two. Furthermore, matter is potentiality, form actuality. Finally, actuality is of two sorts, exemplified by the possession of knowledge and the use of knowledge. Now the most obvious instances of substance are natural bodies and of these some have

life, others do not. A sign that a body is living is self-nutrition or growth. Thus, living natural bodies are substances in the sense of compounds of matter and form.

But since it is also *body* of such and such a kind, viz. having life, the body cannot be soul; the body is the subject or matter, not what is attributed to it. Hence the soul must be a substance in the sense of the form of a natural body having life potentially within it. But substance is actually, and thus soul is the actuality of a body as above characterized. Now the word actuality has two senses corresponding respectively to the possession of knowledge and the actual exercise of knowledge. It is obvious that the soul is actuality in the first sense, viz. that of knowledge as possessed, for both sleeping and waking presuppose the existence of the soul, and of these waking corresponds to actual knowing, sleeping to knowledge possessed but not employed, and, in the history of the individual, knowledge comes before its employment or exercise. That is why the soul is the first grade of actuality of a natural body having life potentially within it. (II,1,412a16-28)

The soul is the substantial form of the living body; for this reason, body is described as that which has life potentially. The genesis of the living thing is a substantial generation and cannot be the addition of life to an already constituted body as if this were the addition of an accidental determination. Soul is the first actuality of living body, determining it as to *what* it is.

Aristotle goes on to say that the body which has life potentially is an organic body; a diversity of parts is required for the diversity of vital functions.

Aristotle's procedure indicates that he is in effect answering several of the questions he posed at the outset of the first book. The soul is substance in the sense of form furthermore, it is something actual, indeed the first act of living body. To the question as to whether soul in general can first be defined, Aristotle is answering in the affirmative. "We have now given an answer to the question, What is soul? -- and answer which applies to it in its full extent." (412b10) Finally, the soul is considered before its parts.

Having given a definition of the soul, Aristotle wants to explain it and to do this he returns to his distinction of natural bodies into those which are living and those which are not. How do we come to say that some bodies are living? Obviously, because they manifest life. but this is done in a variety of ways, and if any of them is present it suffices to say that a body is alive. "Living, that is, may

mean thinking or sensation or local movement and rest, or movement in the sense of nutrition, decay and growth." (413a23-5) The soul is that whereby the living thing performs any and all of these operations, but their diversity suggests parts or faculties or powers of soul and these faculties are related as prior to posterior. Thus, self-nutrition can not only be considered apart from other powers or faculties, it can also exist apart in the sense that some living things possess only this grade of life. When living things also possess the power of sensation, we say that they are animals, and among the faculties of sense, touch is the most basic, since any animal must have this at least. Thus, self-nutrition enables us to group plants and animals, possession of touch enables us to group all animals together. The power of thinking sets the human soul apart from other animal souls. There are three species of soul, then, the plant, the animal and the human, and of each of them the definition of soul given at the outset can be predicated univocally. Nevertheless, when we turn to the species of the soul, we notice that there is a certain order among them; the human soul has the capacities of the animal soul and more besides; the animal soul has the capacities of the plant soul and more besides. The most basic type of soul, consequently, is the plant soul. Aristotle likens the relations among the species of soul to those obtaining among the species of figure: the triangle is contained in the square, etc. (414b20 ff.)

The remainder of the second book is occupied with the discussion of the nutritive power and the external senses, sight, hearing, smell, taste and touch. In the third book, Aristotle speaks of internal senses. The common sense is that which accounts for sensible awareness of the differences among the objects of external sense. As St. Thomas puts it: "We know the difference between white and sweet, not only with respect to what each is, for this is done by intellect, but also with respect to a diverse immutation of sense and this can only be done by sense." (In III *De Anima*, 1.3, n. 601) Imagination is the internal sense whereby we have sensory awareness of objects no longer present to the external senses. The highpoint of *On the Soul* is reached in the discussion of that faculty with which the soul knows and thinks. Aristotle first compares intellection with sensation and then makes this important statement.

Thus that in the soul which is called mind (by mind I mean that whereby the soul thinks and judges) is, before it thinks, not actually any real thing. For this reason it cannot reasonably be regarded as blended with the body: if so, it would acquire some quality, e.g., warmth or cold, or even have an organ like the sensitive faculty: as it is, it has none. It was a good idea to call the soul 'the place

of forms' though (1) this description holds only of the intellective soul, and (2) even this is the forms only potentially, not actually. (III,4,429a22-9)

An indication of the difference between sense and intellect is found in the fact that our ability to see can be impaired by an object too bright whereas when mind concentrates on what is more knowable it is afterwards more able to think of objects less intelligible: "the reason is that while the faculty of sensation is dependent upon the body, mind is separable from it." (429b4-5) It is the realization that intellectual activity is independent of body that leads Aristotle to the assertion that the intellective soul is immortal and eternal.

And in fact mind as we have described it is what it is by virtue of becoming all things, while there is another which is what it is by virtue of making all things: this is a sort of positive state like light; for in a sense light makes potential colors into actual colors. Mind in this sense of it is separable, impassible, unmixed, since it is in its essential nature activity . . . Actual knowledge is identical with its object: in the individual, potential knowledge is in time prior to actual knowledge, but in the universe as a whole it is not prior even in time. Mind is not at one time knowing and at another not. When mind is set free from its present conditions it appears as just what it is and nothing more: this alone is immortal and eternal . . . (430a14ff.)

Aristotle speaks of two intellectual faculties: the agent, here compared to light, and the passive, that which actually becomes what it knows. Since the activity of each is separable, not involving the body, the soul of which they are faculties survives death.

This passage is extremely difficult and the interpretation we have given of it would be rather generally contested. In the Middle Ages Aristotle's doctrine of a passive and agent intellect, the separability of each, and whether they are faculties of the human soul will be points of great contention. Whether one accepts or rejects the view that Aristotle demonstrates the immortality of the soul in the third book of *On the Soul*, it is clear that neither in that work nor in any other treatise does he undertake to discuss the status of the human soul as separated from the body.

In Chapter Six of the third book, Aristotle establishes the two-fold operation of intellect he had presupposed in *On Interpretation*. Mention must be made of the famous Aristotelian remark that the soul is in a way all things, since existing things are either objects of sensation or thought. He stresses the dependence of

intellection on sensation, and teaches that intellectual activity always involves concomitant imaginative activity.

If it is accepted that Aristotle proves in *On the Soul* that the human soul does not perish at death, we find two instances in natural philosophy where thought is led to the reasoned realisation that there is an existent separable from matter and motion: the Prime Mover and the intellectual soul after death. It is against the background of these discoveries that Aristotle is able to go on to speak of a speculative science which has as its subject being as being, a phrase that suggests that his interest is no longer confined to being as mobile. Of course, it would make little sense to speak of being as being if this were tantamount to being as mobile: there would be no distinct science of it. Before turning to an examination of Aristotle's doctrine contained in the books of the *Metaphysics*, we must first consider his practical philosophy.

E. Moral and Political Philosophy

The main sources for this exposition of Aristotle's practical philosophy will be the *Nicomachean Ethics* and the *Politics*. As essays in practical philosophy, these two works will be seeking knowledge which is ordered to doing, to human action. It is always from the viewpoint of its relevance for action that we must consider the procedure of Aristotle in practical philosophy; the study of ethics and politics should not be considered something of interest for its own sake. One who would listen to discussions of what ought to be done, take copious notes and commit these to memory, without applying this knowledge to his own actions would be like one who expects to get well, not by doing what the doctor prescribes, but by listening attentively and remembering everything he is told. (cf. *Ethics* II,4,1105b13 ff.) One does not become good by philosophizing, but by performing good actions; the hope, of course, is that the considerations of practical philosophy will facilitate the choice of the correct course of action. Somewhat the same point is made in the first book of the *Ethics* (1095a5) when Aristotle observes that the young are not apt students of moral philosophy. The young in heart, whatever their age, pursue now this object, now that, as passion directs. We might object that no one needs moral philosophy more than the immature, but Aristotle will reply that, since such persons are indisposed with respect to action and the end of moral science is not knowledge but action, its study is vain and unprofitable for them. What is it that the properly disposed student has that the immature lack which enables the former to profit from moral philosophy? Moral science, Aristotle notes, in common with every discipline, must begin with those things which the student knows and of which he is a good judge. But the things with which the moral philosopher is concerned are good and just acts and the recognition of these as such requires a special disposition on the part of the student, a disposition that the science presupposes and does not confer. We will see later that Aristotle does not deny that there is a sense in which the morally immature man can learn ethics, but its true import, which is not for knowledge but for action, will be lost on such a student. It is against this background that Aristotle makes the following very important methodological remark.

Now fine and just actions, which political science [Aristotle's generic name for moral philosophy] investigates, admit of much variety and fluctuation of opinion, so that they may be thought to exist only by convention, and not by nature. And goods also give rise to a similar fluctuation because they bring harm to many people; for before now men have been undone by reason of their wealth, and others by reason of their courage. We must be content, then, in

speaking of such subjects and with such premisses to indicate the truth roughly and in outline, and in speaking about things which are only for the most part true and with premisses of the same kind to reach conclusions that are no better. (*Ethics*, I,3,109b15-23)

As practical knowledge, moral philosophy is directed to action, to singular actions, as to its term; because the circumstances in which we act and we as agents vary considerably, the generalizations of ethics and political science, both premisses and conclusions, will be unable to achieve a perfect fit with action. Nevertheless, since such knowledge, though remote and tentative, is of some value when we must decide, its pursuit is justified; we notice once more that the justification comes on the side of a disposition to make use of this knowledge. The unsatisfactory character of practical philosophy just as knowledge makes the pursuit of it for its own sake, and not for the sake of using it in action, an endeavor of little moment.

If Aristotle insists again and again that the doctrine of the *Ethics* and *Politics* is only probable, more or less likely, and so forth, we must not think that he is of the opinion that convention and custom are the only rules of action and that nature has no role to play. We saw how Plato, faced with the question as to what man ought to do, turns immediately to ask what man is. Practical norms must be anchored in knowledge of man's nature. (Cf. *Ethics*, V,7) I So too, in the first book, Aristotle will ask, what is man's proper function? The answer to this question presupposes knowledge of what man is. Before turning to that discussion, however, we must first say a word on the relation between ethics and politics.

We have already alluded parenthetically to the fact that "political science" is synonymous with moral philosophy for Aristotle. His reason for this usage is to be found in his contention that man is naturally a political animal. This statement has no more alarming purport than that man, inevitably, is born into a society: that of the family since a man must have parents; that of a community of families, since men are better enabled to survive if there is a division of labor. "Naturally," in the statement "man is naturally a political animal," obviously does not mean that states are natural products in the way trees are. Rather, man's nature suggests the state, since the individual cannot achieve human perfection easily if at all in a solitary condition. The formation of the state follows, then, on the pursuance of the goal suggested by man's nature. If man is part of various communities, these communities or wholes can possess ends which are not simply the end of the individual taken as such; but, because the

family and the state are the kinds of whole they are, their parts, individuals, can have ends or goals which are not those of the whole as such. This, as we have already seen, is the basis for the division of practical philosophy into ethics, economics and politics. Politics, since it is concerned with the common good of citizens, is preeminent in the practical order, and its direction of various activities to an end has more the nature of wisdom. For this reason, as terminal and preeminent, politics lends its name to the whole of practical philosophy.

The End of Man. Since every study, action and pursuit seems ordered to an end or good, Aristotle suggests that we inquire whether there is some end of the things we do which is desired for its own sake and for which all other things are desired. It does not seem likely that one thing could be desired for another, that for yet another, and so on infinitely; rather, there must be some chief good towards which all activities are directed. Knowledge of such an end, he feels, would clearly have practical import and, of the sciences, politics would be chiefly concerned with it. A sign of this is that it is left to political science to order business, the military and education itself to an end. That happiness is what all men seek in all their actions is a matter of widespread agreement, although the nature of happiness is not agreed upon. Most men seek happiness in pleasure, honor and wealth; some philosophers in the Form or Idea of the good. The life lived for pleasure is the first of three ways of life distinguished by Aristotle; it is not a human life, he observes, since it places human happiness in something common to men and animals. A second way of life is the political and this seems aimed at honor. Nevertheless, the man of practical wisdom seems desirous of honor because he is virtuous, and the suspicion arises that virtue and not honor is the end of the political life. A third way of life is the contemplative, but discussion of this Aristotle defers until the tenth book. Money is palpably not an end, since it is wanted for something else: The Platonic Good is dismissed as not taking sufficiently into account the variety of goods which also deserve the name.

The ultimate good is one achievable by action; obviously there are many goods which can be achieved by our action, but there may be one which is not sought because it is conducive to yet another good, but as terminal and final. Indeed, happiness seems to be such a good: everything else seems sought in order that we might be happy. Not only is happiness an ultimate or final good, it seems also to be self-sufficient; it is easier to agree that we would be satisfied with happiness alone than with any other good we seek. Given these two characteristics of happiness, we really have said next to nothing about it. To say a trifle more, we must ask what man's function is. It would be incredible if man

had no proper function when the carpenter and plumber do, as do the eye, hand, ear, etc. Man's function is not merely to live if life is something he has in common with plants; nor can sensation be his proper function, since animals too possess that. "There remains, then, an active life of the element that has a rational principle; of this, one part has such a principle in the sense of being obedient to one, the other in the sense of possessing and exercising thought." (I,7,1098a3ff.) The properly human life is a rational one, and this can mean that an activity is in accord with reason or of reason. If this is man's function to live rationally, then we have something in terms of which we can say a man performs well or ill. But to perform well is to perform virtuously and the human good, human happiness, thus appears to be an activity of soul in accord with virtue or virtues or the best of virtues. And, since "one swallow does not make a summer" (one of Aristotle's most quoted remarks), this activity must be fairly continuous if it is to constitute happiness.

Having arrived at a general designation of the ultimate human good as happiness which in turn is a life lived in accordance with virtue, Aristotle turns to other views to see if he has hit on something others would agree with. Those who divide goods into external, corporeal and goods of the soul, with the last type the best, would seem to be in agreement with the proposed description of the ultimate good. Others have located happiness in virtue, but Aristotle wants not only this corroboration of his own view; there may be a difference. "But it makes, perhaps, no small difference whether we place the chief good in possession or in use, in state of mind or in activity." (1098b32) Happiness, for Aristotle, is an activity.

Moreover, it is pleasant activity since the virtuous man will take pleasure in virtuous actions. Finally, those who say that happiness requires external goods are not wrong; the virtuous life can be lived properly only if one has a minimum of worldly goods. Nevertheless, Aristotle does not want to identify happiness with good fortune; good fortune by definition is not something one acquires by deliberate action.

The remark of Solon that no man should be called happy while he lives, prompts an interesting digression. What the sage seems to be getting at is that only the dead are beyond the reversals and vagaries of fortune, and yet the dead may not be blessed in their descendants. Aristotle does not feel that the dead can be affected by the bad fortune or vices of just any of their descendants; at the same time he thinks those of some of their descendants must have some effect on their happiness. All this is quite tentative, of course, but it reveals

Aristotle's appreciation of the strength of family ties, ties which, with the immortality of the soul, become transcendent. His more direct reply to Solon is that the sage seems to allow us to say only that a man has been happy, not that he is happy, and Aristotle wonders how something can have a past if it had no present. His final statement on how the fortunes of living descendants affect the dead is found at the end of chapter eleven. "The good or bad fortunes of friends, then, seem to have some effects on the dead, but effects of such a kind and degree as neither to make the happy unhappy nor to produce any other change of the kind." (1101b4ff.)

Aristotle's view of the effects of fortune, good and bad, on the happiness of the virtuous man steers a middle course between making happiness result as such from these and making them a matter of total indifference. A prolonged siege of bad fortune can affect happiness, though it can also make the nobility of the good man's soul shine forth and this not because he does not feel pain. Aristotle does not think pain a matter of indifference; he feels it can diminish happiness, but he hesitates to say it can stamp it out utterly. In short, Aristotle outlines a notion of happiness consonant with man's nature, a happiness which is an activity and to be possessed in this life, although it is not utterly unrelated to another life. Nevertheless, he hesitates to speak of the happiness of the separated soul except insofar as this may seem to be affected by events here below. Obviously, this reserve is quite in keeping with the purpose of practical philosophy.

Virtue. Happiness having been defined as an activity of soul in accordance with virtue, Aristotle must now turn to the discussion of the nature of virtue. Earlier, he had introduced virtue into his description of human happiness by saying that a function may be performed well or ill, and that performing it well is what we mean by virtue or excellence in that order. It is human virtue as such that now interests him and since the activities which can be performed well or ill are those of the soul, the moral philosopher must presuppose knowledge of the soul. We saw earlier that psychology is presupposed by moral philosophy and took that as suggesting the place of the latter in the proper order of learning the various philosophical sciences. Now since it is a question of accepting from elsewhere a doctrine of soul, Aristotle is willing to make use of the Platonic doctrine because he feels it is adequate enough for his purposes here. What he has in mind is the division of the soul into a rational and an irrational part. (He suggests that he would tend to discuss the related questions differently from the Platonists, but that does not matter here.) The irrational part of the soul is further subdivided into the vegetative and sensitive; the vegetative activities do

not seem to require any specifically human direction, so they are of little interest here. Digestion is not something we concern ourselves about; indeed, it takes place, and perhaps best when we are asleep. The other element of the irrational part of the soul seems to fight against reason; nevertheless, it can be brought under the control of reason and is thus rational by participation. This leads to a division of virtue, since the good activity of reason will be one kind of virtue, the good activity of the irrational part of the soul as it is brought under the suasion of reason, another kind. Let us call these intellectual and moral virtues respectively. Books Two through Five are concerned with moral virtue; Book Six with the intellectual virtues; Book Seven is concerned with continence and incontinence and we will find there Aristotle's discussion of the claim that knowledge is virtue; Books Eight and Nine concern themselves with friendship and, in Book Ten, Aristotle returns to the discussion of happiness.

Acquisition of Moral Virtue. Intellectual virtue can be gotten from a teacher, but moral virtue is the result of habituation. Moral virtues are not products of nature, but neither are they acquired quite independently of nature. "Neither by nature, then, nor contrary to nature do the virtues arise in us; rather we are adapted by nature to receive them, and are made perfect by habit."

(II,1,1103a24-5) We become just by performing just acts; we become temperate by performing acts of temperance, and so on with the other moral virtues. Aristotle observes that it is the purpose of law to make citizens perform good actions and thus to acquire the habit of virtue so that sanctions are no longer the motive. There is a cliché we hear often nowadays to the effect that you cannot legislate morality, meaning, it seems, that a law can't make people good. It can nevertheless make people perform good acts and thus, hopefully, be conducive to the acquisition of virtue. "It makes no small difference, then, whether we form habits of one kind or of another from our very youth; it makes a very great difference, or rather all the difference." (1103b24-6) Virtuous action seems to be a matter of avoiding extremes, since the virtue of temperance is destroyed both by an excess and a defect with respect, say, to food and drink. Moreover, moral excellence has to do with pleasures and pains; pleasure can induce us to do bad things and pain to refrain from good, so that the virtuous man must be well disposed with respect to pleasure and pain. Aristotle endorses Plato's view that a good deal of moral education has to do with training the young to take pleasure in and be pained by the proper objects. Virtue and vice are concerned with the same things, then, but they differ in the manner of their relation to them.

Aristotle concludes his general remarks on the acquisition of moral virtue by comparing art and virtue; one learns an art by repeated action, but the acquisition of moral virtue requires a number of things of the agent which are not necessary for art.

The agent also must be in a certain condition when he does them; in the first place he must have knowledge, secondly he must choose the acts and choose them for their own sakes, and thirdly his action must proceed from a firm and unchangeable character. These are not reckoned in as conditions of the possession of the arts, except the bare knowledge; but as a condition of the possession of the virtues knowledge has little or no weight, while the other conditions count not for a little but for everything, i.e. the very conditions which result from often doing just and temperate acts. (II,4,1105a31-b4)

Definition of Moral Virtue. Turning to an attempt to define moral virtue, Aristotle first seeks its genus. Virtue has been referred to soul and the soul comprises passions, faculties and habits and virtue must be one of these. Virtue is not passion, no more than is vice, Aristotle remarks, since we are not praised or blamed for the feelings we may have, but for what we do when we have such feelings as anger, desire and so forth. Moreover feelings are not matters of choice, whereas virtues are. The same objections would have to be brought against the supposition that virtues are faculties of the soul; in addition, we are provided with faculties by nature, but virtue we must acquire. This leaves only habit and this, Aristotle says, is the genus of moral virtue.

Given the genus of virtue, we must now seek the mark that sets it off from other habits. Now any virtue in the sense of excellent performance makes both the performer and his work good; so too human virtue will make a man good and enable him to do his proper work well. It has already been suggested that virtue is destroyed by excess or defect.

For instance, both fear and confidence and appetite and anger and pity and in general pleasure and pain may be felt both too much and too little, and in both cases not well; but to feel them at the right times, with reference to the right objects, towards the right people, with the right motive, and in the right way, is what is both intermediate and best, and this is characteristic of virtue. (II,6,1106b18ff.)

All these conditions have to be determined by reason under whose guidance we act; to fall short of any of these conditions is to fall from excellence to some

extreme, whether of defect or excess. For this reason, Aristotle will say that the virtue in a given order lies between two vices. The determination of the mean is made in concrete circumstances by a particular individual and consequently does not have a hard and fast character. Moreover, not every action admits a mean in the sense at issue: the judicious execution of a crime cannot be called virtue.

Aristotle goes on to make these remarks more concrete. The mean with respect to fear and confidence is courage; the extremes in this area are timidity and foolhardiness; with respect to money, liberality is the virtue, prodigality and miserliness the vices; with regard to honor, the virtue is proper pride, the vices vanity and obsequiousness, and so on. The two vices related in terms of excess and defect with respect to the same passions are further removed from one another than from the virtue, and the virtue is closer to one vice than to the other; e.g., courage is more closely related to foolhardiness than to timidity. One source of this is our own tendency towards one extreme and this suggests that the way to hit the mean is to strive to avoid the excess to which we are most inclined. A sign of our inclination will be the mode of action which gives us the most pleasure, and the acquisition of moral virtue will thus entail controlling pleasure.

Involuntary Acts. A person is praised or blamed for those actions he performs voluntarily; if he does something involuntarily he would be pardoned or pitied, but not praised or blamed. What is an involuntary action? Aristotle says that they are those caused by compulsion or ignorance. In this way two essential aspects of the voluntary act are removed. By compulsion or violence he means an activity the principle of which is outside the agent in such a way that the agent contributes nothing to it. He exemplifies this by a man being tossed by the wind or dragged about by others. Of some actions it is difficult to determine whether they are voluntary or involuntary; parents of the child accede to the wishes of kidnappers out of compulsion, but they act to get their child back. So too the captain who dumps his cargo in order to save his ship and crew. "Such actions, then, are mixed, but are like voluntary actions; for they are worthy of choice at the time when they are done, and the end of an action is relative to the occasion" (III,1.1110a11ff.) Speaking generally, no one wants to give his life-savings to strangers or to dump his goods into the sea, but the force of circumstances can make these actions which are chosen.

Ignorance too can be the cause of involuntary action; by ignorance here Aristotle means a lack of knowledge of the circumstances of an act. For

example, the man shoots his son thinking him an intruder. When he discovers what he has done, he is abject; only ignorance productive of a result contrary to what one wills involves the involuntary; one who collects bits of metal and learns afterward that they are extremely valuable has acted out of ignorance, but ignorance is not productive of an involuntary act since the result does not go contrary to the will of the agent. Aristotle rules out the ignorance consequent upon rage and wine as causative of the involuntary, since in such cases ignorance is a concomitant of something else to which the act must be ascribed.

Choice and Deliberation. Choice is involved in voluntary action but is not synonymous with it; choice is taken here to mean that which bears on means to an end and not on the end itself; wish or intention has the end as object. It is choice, and not action for an end, which sets the human agent off from all others. We deliberate or take counsel about the ways to achieve an end we intend, but it is not this cognitive activity alone which is choice. Aristotle makes this point by asking whether choice is identical with opinion. That opinion is involved in choice is not denied, but the question is, are the two one and the same thing? A sign of their difference is that we are not praised or blamed for what we think so much as for what we choose.

And we choose what we best know to be good, but we opine what we do not quite know; and it is not the same people that are thought to make the best choices and to have the best opinions, but some are thought to have fairly good opinions, but by reason of vice to choose what they should not (III, 2, 111a7 ff.)

Choice, Aristotle will say later, can be looked upon either as a knowing willing or a willing knowing. Deliberation and choice are important for the consideration of virtue, since the latter is concerned with the means.

Knowledge and Virtue. After the points just mentioned, Aristotle alludes to the Socratic position that no one does evil knowingly. He will turn to this position again in Book Seven after discussing in some detail the virtues of courage, temperance, liberality, magnanimity, etc. in the remainder of Book Three and in Book Four, and a discussion of justice in Book Five. Book Six is devoted to virtues he had earlier opposed to moral virtues, namely the intellectual virtues. Of particular interest in that book is the discussion of practical wisdom or prudence. We shall turn immediately to the discussion of the Socratic position and make allusions to the doctrine on prudence from that vantage point. In Book Seven, Aristotle is concerned with the continent and the incontinent man.

Virtue is a state of character thanks to which the one having it and his operation are rendered good. It is a determination to the good of action so fixed and habitual that it is accompanied by pleasure. When the virtuous man must decide, he is disposed to see his circumstances with an eye to the good and to follow with ease and pleasure the dictates of reason as to how the good can be obtained. The continent man is something less. He is not totally indisposed to the good; he can judge the circumstances in which he must act with an eye to the good and even do the right thing. However, he does not do this easily and with pleasure, but by means of a certain constraint, despite himself, so to speak. It is in this context that the problems of incontinence and the incontinent man are approached and immediately the position of Socrates looms large.. Is it possible for a man who judges rightly to behave incontinently? Socrates, we are reminded, rejected this possibility and maintained that no man can act contrary to what he judges is best. If a man does the wrong thing, this is only because of ignorance. Noting that this view plainly contradicts the observed facts, Aristotle suggests that one must ask what is the manner of ignorance to which Socrates refers. The incontinent man is one who, before he acts, does not think that he should act as he does. Perhaps then it was not really knowledge that the incontinent man had, but only opinion. Aristotle feels that this would be too great a concession, for we might sympathize with a man who acts contrary to a weak as opposed to a strong conviction, but we do not sympathize with wickedness. One thing is certain as far as Aristotle is concerned and that is that the knowledge which the incontinent man has is not that of practical wisdom or prudence. This he has earlier defined as the virtue of the practical intellect which complements the possession of moral virtue. Thus prudence is the knowledge of the one possessing moral virtues who will act on the knowledge he has. The knowledge of the incontinent man, on the other hand, allows for actions which are not in keeping with it.

Is the distinction, made by those inclined to agree with Socrates, the distinction between knowledge and opinion, relevant here? Aristotle thinks not. From the point of view of action, there seems to be little perceptible difference between those who have opinion and those who know. Striking a wry note, Aristotle says that in general men seem no less convinced of what they opine than of what they know -- "as is shown in the case of Heraclitus." What is of relevance here is the twofold way in which we use the word "know." Both the man who has knowledge and is not using it as well as the man actually using it are said to know. Thus, there would be an important difference between the man who has knowledge and uses it in acting and the man who, although he has knowledge, does not make use of it in action.

In the reasoning involved in action, there are included general judgments and singular ones; for example, "Dry food is good for every man," on the one hand, and, on the other, such judgments as "I am a man" and "This is dry food." Obviously, there is a difference between these kinds of judgment, and Aristotle observes that it would indeed be surprising if the incontinent man had both the knowledge involved in the general judgment as well as that involved in the singular ones when he acts incontinently. As a matter of fact, he claims, what the incontinent man knows is expressed in general judgments and, though he may be said to have the knowledge expressed in singular judgments, he is not using that knowledge when he acts. And, since action is concerned with the singular, the use of general knowledge by the incontinent man is compatible with not making certain singular judgments when he acts. What is important in action is seeing the singular circumstances in the light of the universal and it is just this that the incontinent man fails to do.

Since what Aristotle is doing, in effect, is pointing out that "knowledge" has many more senses than Socrates apparently suspected and that, once these various meanings are brought to light, the identification of knowledge and virtue appears unduly simplistic, resting as it does on a univocal acceptance of "to know," it is not surprising to find him going on to point out further nuances in the distinctions he has already made. Thus, the distinction between having knowledge and not using it necessitated by a variety of causes. The geometer who is asleep, gone mad or drunk is not using his knowledge and, moreover, is incapable of using it. Now this is also the case with one buffeted by passions, particularly such passions as anger and sexual appetite. Just as those mad or drunk are incapable of making use of the knowledge they have, so too the incontinent man. Nor does it matter that he can even then "use the language that flows from knowledge" -- even the mad and intoxicated can do this. Aristotle employs another important analogy, that of beginners in a science who can orally repeat what pertains to the science although they do not know it, for it has not yet become a part of themselves, an independent possession. In a moment, we will be able to say something about what it means for practical knowledge to become part of oneself.

Continuing his analysis, Aristotle next compares the reasoning of the speculative intellect with that of the practical intellect. In speculative matters, the mind need only affirm a conclusion, whereas precisely an action. For example, given "Every injustice should be avoided," and "This action is unjust," it is theoretically a simple matter to formulate and assent to the conclusion. But in practise, the judgment about the singular can involve a host of difficulties.

Consider the "practical syllogism" of a temperate man. Assume that he knows as true that every sweet ought to be tasted as well as that the tidbit on the table before him is sweet. Unless his hands are tied, the action dictated is obvious and follows smoothly. Taking a more realistic example, the just man, knowing that every injustice is to be avoided and recognizing a possible course of action as unjust, avoids it. So too, to give vice its due, the intemperate man, working in terms of the general view that every delight is to be pursued, goes in pursuit of what appears delightful to him. What, in terms of such "practical syllogisms," is the procedure of the incontinent man?

The incontinent man is distinguished from the virtuous man because he does not have his passions disciplined and under control. Nevertheless, the incontinent man knows as well as do the virtuous and continent men that delights are not to be pursued inordinately. Such a universal, however, is not decisive for action, since actions are singular. The incontinent man, whose passions are by definition uncontrolled, when faced with a concrete possibility of pleasure has present to his mind, so to speak, conflicting general judgments: that delights are not to be pursued inordinately, and, what is involved in the uncontrolled movements of his emotions, that every delight is to be caught on the wing. His passions being aroused in the presence of a pleasurable object, the universal that he and the virtuous man both know is pushed rudely aside. Then, judging somewhat in this way, "But of course this is not prohibited," a judgement prompted by his aroused passions, he dismisses the general view that prohibits seeking this pleasure and acts, in effect, under the formality of the general judgment that every delight is to be pursued.

What does all this have to do with the position of Socrates? Aristotle leads the discussion back with the surprising remark that, from his considerations, "the position Socrates sought actually seems to result." Does this mean that Aristotle agrees that knowledge is virtue and evil doing a matter of simple ignorance? We have already seen Aristotle's concern to take into account the various meanings "to know" has. Insofar as knowledge refers to a theoretical, general, knowledge about actions, knowledge is not virtue. Actions are in the singular, are concrete. What Socrates is right in suggesting is that, when a man knowingly does wrong, he is not considering the general knowledge relevant to the circumstances in which he acts, but Aristotle will insist that there is no contradiction in saying that such a man knows that general knowledge to be true. In the case of the incontinent man, this knowledge is pushed aside by passion and, when he acts, he is not using it.

It is just this distinction between judgments made in moral science and judgments made by one who possesses practical wisdom or prudence that is expressed by later distinctions between judgments "by way of knowledge" of, and judgments "by way of inclination" to, or connaturality with the good. The virtuous man is determinately disposed in his appetite with respect to the good of reason. That is why there is no impediment in his appetitive condition when it comes to judging his particular circumstances in the light of a rationally recognized norm. Well-disposed appetite is indispensable for the kind of knowledge which is efficacious for action, i.e., for practical wisdom or prudence. Lacking this disposition, that is, lacking virtue, the incontinent man does not apply the knowledge he has.

Judgments made by way of knowledge are those made in moral philosophy where it is not circumstances which are at issue, but typical circumstances. We have already seen that an efficacious study of moral philosophy presupposes a well disposed student. But whether morally well disposed or not, judgments in moral philosophy require the assignment of communicable reasons for a proposed course of action. These reasons and their validity do not depend on the subjective condition of the one giving them, a sign of this being that even the intemperate man can give useful advice on the acquisition of temperance.

Aristotle's point with reference to Socrates, then, is that knowledge is and is not efficacious for action -- it all depends on what one means by knowledge. There is on the one hand general knowledge, the sort of thing one is after in moral science, but possession of such knowledge is compatible with actions contrary to it. There is as well practical wisdom or prudence, the kind of knowledge had by one who possesses moral virtue which disposes him to judge correctly the contingent and particular circumstances and to act in conformity with that judgment.

Justice. The fifth book of the *Ethics* is concerned with the virtue of justice; we shall hint here in but the briefest way the direction the discussion takes. Noting that we can often arrive at the nature of something by examining its contrary, Aristotle observes that injustice is the unlawful and unfair or unequal. This suggests to him that justice is action in accordance with law and concern for equality. The equality or fairness is gauged in terms of goods connected with prosperity and adversity, and this suggests that we have to do with two meanings of injustice and justice. The justice which is synonymous with living in accord with the law is not restricted to those goods mentioned above, since the law sanctions the practice of every virtue insofar as a virtue brings us into

relation with our neighbor. In other words, there is a form of justice, legal justice, which is the same as virtue as such; but when we designate all virtue by the term justice we are specifying virtuous acts, not absolutely, but insofar as they bring one man into relation with others. The type of justice which is concerned with those goods involved in prosperity and adversity is a special virtue included in legal justice and divided from other virtues also included in justice in the wide sense, e.g., courage, temperance and so forth. This particular justice is divided by Aristotle into two species, one of which is concerned with the equitable distribution of goods to citizens of a state, the other of which has to do with transactions among citizens. Aristotle has much to say about how equality, the just mean, is established in each of these species of justice, and how they compare on this basis.

Friendship. It falls to moral philosophy to discuss friendship because this is either a virtue or dependent upon virtue; moreover, it is necessary for life. Indeed, at all stages of our life we are in need of others, of friends. It is not the necessity and usefulness of friends alone which commends friendship, however; we praise it as something noble in itself and feel that there is a connection between being a good friend and a good man. In describing friendship, Aristotle always has one sense of the term in mind and, if he speaks of kinds of friendship, these are not so much species of a genus, as imperfect imitations of friendship properly so-called. Generally speaking, friends are those who bear goodwill to one another in such a way that this is mutually obvious and they wish one another well for reasons of the good, of utility or pleasure. "Perfect friendship is the friendship of men who are good, and alike in virtue; for these wish well alike to each other *qua* good, and they are good in themselves." (VIII, 3, 1156b7ff.) Friendship based on utility or pleasure falls short of this.

Therefore we too ought perhaps to call such people friends, and say that there are several kinds of friendship -- firstly and in the proper sense that of good men *qua* good, and by analogy the other kinds; for it is in virtue of something good and something akin to what is found in true friendship that they are friends, since even the pleasant is good for the lovers of pleasure. (VIII, 4, 1157a30)

Aristotle's view of friendship, even of perfect friendship, is sometimes thought to be deficient because it appears to involve egocentricity. Such remarks as the following cause difficulty.

And in loving a friend men love what is good for themselves; for the good man in becoming a friend becomes a good to his friend. Each, then, both loves what

is good for himself, and makes an equal return in goodwill and in pleasantness; for friendship is said to be equality, and both of these are found most in the friendship of the good. (VIII,5,1157b33)

A minimum of reflection enables us to grasp Aristotle's point. We love that which is good and a good for ourselves. Now when a good man becomes the friend of another he becomes a good for that other and good men when they are friends love what for them is good in loving one another. What the good man loves for himself, he loves or desires for his friend; self-love accordingly is the basis of love for another, but this must not be construed in terms of utility. Even in the supernatural order, only God can be loved more than oneself (See St. Thomas, *Summa theologiae*, II-II,q.26,a.4) so that the primacy of the self in love among men can hardly be thought of as pagan.

Contemplation. In the tenth book of the *Nicomachean Ethics*, Aristotle returns to the notion of happiness, man's last end, and endeavors to say a few more things about it. He recalls his earlier remarks to the effect that happiness must consist in activity and in an activity which is sought for its own sake. Virtuous activities seem to meet this requirement and happiness should be activity in accord with the highest and best virtue. Contemplation, the activity of philosophic as opposed to practical wisdom, constitutes human happiness.

For firstly, this activity is the best (since not only is reason the best thing in us, but the objects of reason are the best of knowable objects); and, secondly, it is the most continuous, since we can contemplate truth more continuously than we can do anything. And we think happiness has pleasure mingled with it, but the activity of philosophic wisdom is admittedly the pleasantest of virtuous activities; at all events the pursuit of it is thought to offer pleasures marvellous for their purity and their enduringness, and it is to be expected that those who know will pass their time more pleasantly than those who inquire. And the self-sufficiency that is spoken of must belong most to the contemplative activity. For while a philosopher, as well as a just man or one possessing any other virtue, needs the necessities of life, when they are sufficiently equipped with things of that sort the just man needs people towards whom and with whom he shall act justly, and the temperate man, the brave man, and each of the others in the same case, but the philosopher, even when by himself, can contemplate truth, and the better the wiser he is; he can perhaps do so better if he has fellow workers, but still he is the most self-sufficient. (X,7,1177a20-bl)

He goes on to point out that it above all is loved for its own sake and depends on leisure; in a word, contemplation best saves the characteristics of happiness discerned in the first book.

Despite this, contemplation seems to be somewhat more than a human activity and a secondary type of happiness must be recognised, also rational activity, but that involved in the exercise of the moral virtues. Aristotle characteristically is not carried away by his own eulogy of contemplation to the extreme of saying that only in it can happiness be secured.

Politics. The *Ethics* ends with a discussion which provides an easy transition into the *Politics*, for Aristotle begins to point out that legislation is required if the end of man is to be achieved. We must content ourselves with the barest outline of the contents of the *Politics*. Aristotle begins with a discussion designed to show that the state is natural, that man is by nature a political animal. The state is the perfect community and aims at the perfect good; its difference from other communities is discovered by examining the parts of which it is composed, villages and households. The state is organized to provide necessities but it is also the vehicle for securing the good life. In discussing the nature of the household, Aristotle argues that there is a natural basis for slavery insofar as some men require to be ordered to the good of others. This is easily one of the most controversial doctrines in the *Politics*. Aristotle classifies constitutions in three groups: the good ones are monarchy, aristocracy and polity; their perversions are tyranny, oligarchy and democracy. He has much to say on the variations in the three types of good constitution as well as on revolutions. The *Politics* we possess is not a completed work; it ends with a discussion of education which advances no farther than a treatment of music and gymnastic.

The moral philosophy of Aristotle is most striking because of its judicious blending of the ideal with the possible; the movement from the way things are to the way they can and should be is present throughout, from the appeal to human nature to discover that function in terms of which man's peculiar excellence can be computed to research into existing constitutions as a background for the treatment of what the best constitution would be.

F. First Philosophy

The Platonic doctrine of Ideas is the assertion that there exist things separate from matter and motion. As we will see shortly, Aristotle was convinced that this doctrine was based on a confusion of our mode of knowing with the way in

which things are. This is not to say that Aristotle lacked interest in the possibility of separate or immaterial being; the work of his which has come to be called the *Metaphysics* is precisely ordered to make defensible statements about such beings. However, the separated substance that Aristotle can admit will be seen to be quite different from the Platonic Idea. Given this, an examination of Aristotle's criticism of the Ideas followed by a presentation of his ascent to separated substance in the twelfth book of the *Metaphysics* will provide us with an adequate glimpse of Aristotle's First Philosophy.

Aristotle's criticism of the Ideas occurs in a section of the *Metaphysics* where he is inquiring into the essence of material substance. Before turning to it, we must say a few words about the nature of First Philosophy, the discipline in which the criticism in question occurs. We have already seen, in our effort to describe Aristotle's division of the sciences, that philosophical wisdom, the term of philosophizing, is one of the speculative sciences. As such, its subject must be distinguished from those of other speculative sciences and there will only be a speculative science other than natural philosophy and mathematics if an immaterial substance exists.

We answer that if there is no substance other than those which are formed by nature, natural science will be the first science; but if there is an immovable substance, the science of this must be prior and must be first philosophy, and universal in this way, because it is first. And it will belong to this to consider being *qua* being -- both what it is and the attributes which belong to it *qua* being. (*Metaphysics*, VI, 1. 1026a27 ff.)

Each of the sciences considers being, but none of them studies being just as such, but rather some particular kind of being; physics considers mobile being, mathematics quantity. When we see that Aristotle holds that there will be a science beyond these only on the supposition that there is an immaterial substance, we might think that the subject of that further science is immobile being. Yet this cannot be the case, first, because First Philosophy is a general science and is about all beings just insofar as they are beings and, secondly, because there can be no science of simple or uncomposed things in the sense that these cannot be the subject of any science. (See *Metaphysics*, VII, 17) If immaterial substance is to be known, it must be known by way of material substance; this will suggest the mode of procedure we can discern in the treatises which make up the *Metaphysics*. Aristotle will first of all examine material substance, striving to discover attributes of it, not as material, but as substance. Then, if there are immaterial substances they will be known and

named from material substance. And since Aristotle will attempt to show that material substance is an effect of immaterial substance, it becomes clear that the order we follow in knowledge is just the opposite of the order in reality.

Why is it that, when he is speaking of being, Aristotle speaks almost solely of substance? Aristotle points out in many places that "being" means a number of things. (See *Metaphysics*, IV,2; V,7; VII,1, etc.) Moreover, like many other words, "being" is not a purely equivocal term, since we can discern an order in the various meanings assigned to it. Aristotle often exemplifies this by an appeal to "healthy." We say of an animal, of food, of urine and of exercise that they are healthy and, while we don't mean the same thing in each case, we don't mean utterly different things either. By "healthy" we mean most properly what possesses a certain quality; secondarily, it can mean what is productive of, significative of or restores that quality. There is then a focal meaning in the various uses of the term, something which is either simply meant or at least referred to in the above uses of the term. Much the same thing is true of "being." The term means most properly what is without qualification, substance; if substance is not meant, it is involved in any secondary meaning of the term. This is why the science which is concerned with being as being is primarily concerned with substance.

Criticism of the Ideas. One of the lengthiest criticisms of Plato's doctrine of Ideas is found in Book Seven of the *Metaphysics*; Aristotle has begun the consideration of substance and is inquiring first of all into the essence of material substance. We can distinguish three stages of his critique of Plato's views on this subject. First, he maintains that the essences of material things cannot be separate from them; secondly, that Ideas are unnecessary to explain the coming into being of material things; Ideas are neither efficient nor exemplar causes; finally, as Plato speaks of them, the Ideas cannot be substances.

Having determined that "being" means a number of things but that there is an ordered diversity in its meaning with substance primary, Aristotle commences the discussion of substance. For reasons we have suggested, he begins with the study of material substance: if any substance is obvious to us, it is this kind. Of material substances Aristotle asks whether it is possible that their essence, what they are, can be separate from them in the way in which the Ideas are said to be separate.

On the face of it, he observes, the assertion that the essence of a thing is separate from it is contrary to the common view. "Each thing is thought to be

not different from its substance, and the essence is said to be the substance of each thing." (1031a17) What a thing is, does not seem to be separate from it. Nevertheless, there are some cases where a thing and its essence differ, notably in things accidentally one, for example, "white man." The essence of whiteness, what white is, is not the same as man, for to be man and to be white are not same. Moreover, if the accidental unit, white man, were one with the essence of man, the same would be true of musical man and man. From which it would seem to follow that whiteness and musical were the same. It seems better to maintain that the essence of white is not identical with the man who is white. This is not to say, however, that the essence of whiteness is separate from white. Later Aristotle will point out the ambiguity involved in asking what is essential to what is white or musical. What is said of white or musical may be taken to refer either to the quality or the subject of the quality. In the case of man, however, what man is and man would seem to be the same and inseparable.

If the Platonists want to maintain that the essence of a thing is other than the thing, why do they not say this of the Ideas themselves? That is, why is not the essence of goodness separate from the Idea of the good, and being from the Idea of being, unity from the one, and so on? If they were separable, the good would not be goodness, nor the one unity nor being being. But if what man is, is separate from man, how can man be man? Moreover, if we have knowledge of a thing when we know what it is, and its whatness is separate from it, we are in the absurd situation of not knowing the thing we claim to know precisely when we know what it is. This is not a simple *tour de force* on Aristotle's part. Plato had posited the Ideas with the express purpose of saving the notion of science with respect to the things around us; he did this by claiming that when we know in the rich sense what we know are the Ideas in which sensible things participate. Aristotle's reaction is to say that it is odd to run to something else when we ask what the things around us are.

Before these difficulties, Aristotle feels that we must abandon the doctrine of separate essences and say that what a thing is, is inseparable from it, and that to know what a thing is is to know the thing itself and not something else. In a final flourish, he observes that once we separate an essence from that of which it is the essence, we have embarked on an infinite regress: if the essence of horse is separate from this thing we call a horse and the essence too is a thing, its essence must be separate from it and so on to infinity.

A possible objection to Aristotle's view which the reference to sophistical procedure at the end of chapter six of Book Seven prompts is the following. The

essence of man is humanity; but Socrates is not humanity; therefore the essence of Socrates and Socrates differ. The difficulty with this distinction is that "humanity" is not a proper answer to the question, "What is Socrates?" "Humanity" signifies the essential principles of man but to the exclusion of, prescinding from, the accidents which each man has. Now although accidents are not of the essence of man, nor are they formally signified by "man," nevertheless "man" does not signify in such a way that the individual accidents of Socrates are in every way excluded. "Man" signifies as a whole, "humanity" as a part. And since the essence or whatness of Socrates is expressed in his definition, and the definition can be predicated of the *definiendum*, "man" and not "humanity" expresses the essence as it is spoken of here.

If Ideas are not necessary for the existence of material substances nor for our knowledge of material substances, neither are they necessary for an explanation of the coming into being of these substances. To understand Aristotle's argument here we must recall his doctrine of the coming into being of material substance. We saw above that the result of any coming into being is a composite being. That is, a compound of matter and form. What comes to be, accordingly, is not the matter as such, for that is presupposed, nor the form as such, but the compound of the two. Thus, to begin with an artificial change, in the coming into being of a bronze circle, it is not the bronze that comes to be, nor roundness, but precisely the bronze circle. So too when Socrates comes to be. Matter does not come to be as a result of a change -- it is presupposed by the change. The form as such does not come to be, for whatever comes to be as a result of generation is composed of matter and form. What it is to be a man does not come into being when Socrates does, except accidentally. What properly comes to be is the compound of form and matter, i.e., Socrates.

Before continuing this exposition, we must advert to a difficulty which comes up here. Earlier we said that Aristotle teaches that a thing and its essence are one. If this is the case, when Socrates comes to be, his essence should come to be at the same time -- if Socrates and his essence are one. But we have just said that what man is does not properly come to be when Socrates does. It is only later in this book (Chap. 11, 1037b1 ff.) that Aristotle solves this problem, after he has considered what formally enters into the definition; we must turn now to that solution. When we said that man and his essence are one, we were careful not to say that Socrates and his essence are one in the sense of coexistence; there is of course no question of Socrates' essence existing separately in a Platonic world of Ideas. Also, when we pointed out the different ways in which "man" and "humanity" signify we laid the grounds for the distinction we must

now try to make. "Humanity," since it signifies only what is essential to human nature and prescind from individual accidents, signifies as a part and is not predicated of Socrates save obliquely. "Man" formally signifies only the essential principles of human nature, but it does not prescind in its mode of signifying from the accidents of the individual; it signifies as a whole and is predicated directly of Socrates. Nevertheless, there are many things peculiar to Socrates unexpressed in the definition of his essence or quiddity. It is because the essence signified by the definition of man is, in Socrates and every man, individualized by matter that each of them has individual traits not expressed in their definition. Material individuals cannot be defined as individuals; what can be known and defined is what is essential to them. Since this is so, Aristotle must deny any one-to-one correspondence between Socrates and his essence. What is peculiar to Socrates is not expressed in the definition of his essence. Now, it is obvious that this precision does not take Aristotle back to the Platonic position of the separated essence. The essence exists only in the individuals of the species; but in each of these individuals there is much that is not their quiddity or essence.

To return to our present concern, Aristotle is maintaining that the essence does not come into being properly speaking when Socrates comes to be. The composite which comes to be properly speaking is generated from matter and is of such and such a form. Aristotle goes on to say there is no need to postulate a separate form, that of man as such, to explain the coming into being of Socrates. In the realm of art, he explains, we see no need to assert the existence of some separate roundness to explain the coming into being of the bronze circle, nor a separate house to explain the coming into being of this house. (Aristotle reports that the Platonists did not posit Ideas of artificial things.) If there were a separate man, he could hardly be a determinate individual like Socrates and Plato. Such a separate man would be the type of determinate individuals like Socrates and Plato but not itself a determinate individual man. But when a determinate individual like Socrates comes to be, there seems always to be a cause who is also a determinate individual and the effect produced is like the generating cause. The point is that it makes sense to say that Callias generates Callias junior, but that to say that Man generates or causes Gallias junior is devoid of sense. Man is an abstraction and causes nothing. We will see a bit later that Aristotle's general criticism of the Ideas or Forms is that they are projections into reality of logical entities.

By the same token, Aristotle sees no need for Ideas as exemplar causes of the particular things that come to be in the order of nature. The form of the

composite generated is similar to that of its cause, and even in cases where the cause and effect seem to fall to different species, as when a mule is generated, there is sufficient similarity and no need for appeal to a separate hovering pattern of mulehood to explain the effect. This suggests something the importance of which will become plain when Aristotle turns to God's causality.

We turn now to the central criticism of the doctrine of Ideas. It is clear enough that there are individual men, Socrates, Plato, Callias, Alcibiades, etc. Each of them, we can say, is a man. But what of Man? The term signifies something which can be said of many, it is in some way one, and it refers to many. That is what we mean when we say that Man is universal. But if the only men who exist are individuals, none of whom can be predicated of the other (e.g., Socrates is not Gallias, nor vice versa), where do we get this Man who is said of many? Does Man too exist? We must certainly admit that what Man signifies, i.e., rational animal, exists, for this is found in Socrates, Plato, etc. What is signified by the universal, then, exists, but it exists only in individuals. If we consider Man as universal, as something one but predicable of many, then we are taking it as it is in our mind, as we consider it. What Plato has done, so runs the criticism of Aristotle, is to locate the universal outside the mind and outside the individuals of which it can be said. The Ideas, therefore, are reified concepts. Viewing them in this light, Aristotle asks; can the Ideas be substances?

There is this about substance, that it is proper to one being and cannot be in another. The universal, however, is by definition common to many. If the universal were a substance, it would have to be the substance of something, but of what? Either it is the substance of everything of which it is said, or of one of these things. It cannot be the substance of many, for many men are many substances. If the universal is said to be the substance of only one of the things of which it can be said, everything else of which the universal is said would be the one thing of which the universal is the substance. Since this seems absurd, we must conclude that the universal is the substance neither of one nor of all the things of which it is said.

Moreover, substance is by definition not in a subject. But the universal is precisely predicable of a subject. Therefore the universal is not a substance. One may object that Man is universal and signifies substance. Man is an example of what Aristotle calls in the *Categories* a second substance, not a first substance, the concrete existing thing which he is here concerned with. Indeed, the point Aristotle is making is precisely that Man does not signify a substance over and above concrete individual men.

Aristotle's judgment of the source of the Ideas is that Plato, noting that with respect to Socrates, Plato, etc. we have a notion of man that is not exclusively proper to any individual, makes of Man another substance. This notion of man, one in our mind and predicable of many, is granted the same kind of existence had by those things of which it can be said: it is outside the mind and outside particular men. Thus, a new order of substances is postulated. It is these Ideas, taken as subsistent, that Aristotle argues cannot be the quiddities of individual material things, nor can they be the causes of the coming into being of material singulars. Finally, since Plato wants his subsistent Ideas to retain the note of predicability, Aristotle argues that they cannot be substances, for what subsists is not common to many.

The consequence of this criticism is that Plato is seen to have gained access to a realm of beings apart from material things only because of a confusion. On his own part, Aristotle feels that there are legitimate ways of arguing that there are immaterial substances. It is to his own effort to arrive at such beings that we will now turn.

Aristotle's Ascent to God. Aristotle on the one hand rejects the Ideas as constituting a realm of beings other than the material ones of our acquaintance, and on the other teaches that the human soul, since it has an operation independent of matter, does not cease to exist when the human composite does. But, if souls continue to exist, they do not do so as complete substances. Nor is their separate existence a fully natural one: as substantial forms they belong with matter. Aristotle does not toy with the notion of the transmigration of souls, as Plato does in the *Phaedo*. On the other hand, he certainly presents no obstacles to the Christian theologian's arguments for the fittingness of the resurrection of the body. What we have now to consider is Aristotle's proof of the existence of a being which is not a separated substantial form, but a complete substance which exists immaterially. We find such a proof in the twelfth book of the *Metaphysics*.

Aristotle has said in the beginning of the *Metaphysics* that his inquiry is directed to things as incommensurate with the capacity of the human intellect as daylight is to the eye of an owl. We need expect no facile soaring into the beyond as to a realm more congenial to the eye of the soul. Predictably enough, in setting about to prove the existence of God, Aristotle recalls a good number of things said elsewhere about the kind of substance most knowable to us, material substance. Material substances are the products of change; they are compounds of matter and form. It is not matter that comes to be; it is not form

that comes to be; the product of generation is a composite of matter and form. What is the efficient cause of such substance? "Note, next, that each substance comes into being out of something that shares its name." (1070a5) A man is the cause of a man, a dog of a dog, etc. What a thing is, its essence, exists before it in the sense that it exists in its cause. This is not a separated universal, a reified definition, as Plato thought; it is something as concrete as the effect, e.g., the father preexists his son: the son has the same nature as his father, not numerically, but in kind. Thus form preexists the compound which is the result of generation only in the sense that the efficient cause which "shares its name" does. This sharing of name leads to the notion of a univocal cause. "Man" is said of both father and son and the term signifies the same thing in each case; that is, the term is univocal and, where cause and effect are so named we have a univocal cause. Now in the case of the human soul the form survives the corruption of the compound substance. (1070a25) "Evidently then there is no necessity on this ground at least for the existence of the Ideas. For man is begotten by man, a given man by a individual father." (1070a27)

Aristotle goes on to observe that matter, form and privation would seem to be elements common to all the categories, but he must determine in what sense they can be common. There cannot be any matter, form or privation apart from the categories, nor can the matter and form and privation of substance be identical with those of other categories. He resolves the difficulty in several ways, all of which are difficult. Suffice it to say that the principles of substance are said to be causes of the principles of accidents, a sign of which is that when substance is destroyed so too are its accidents. The principles of substance, then, are common to the categories, not in the sense that they are univocally predicable of the other categories, but in the sense that they are causes of the principles of the other categories. In a certain way matter, form and privation are universally predicable of the categories, but as such they are not causes.

Further, one must observe that some causes can be expressed in universal terms, and some cannot. The proximate principles of all things are the 'this' which is proximate in actuality, and another which is proximate in potentiality. The universal causes, then, of which we spoke do not exist. For it is the individual that is the originative principle of the individuals. (1071a17)

The cause named in a universal manner does not give another cause. This is not to say that there are no causes other than the proximate one which shares the name of its effect. Earlier, Aristotle had seen the need for something more than the intrinsic principles of a man, "something else outside, i.e. the father, and

besides these the sun and its oblique course, which are neither matter nor form nor privation of man nor of the same species with him, but moving causes." (1071a15) It is a moving cause not of the same species nor even of the same genus -- and thus not sharing its name univocally -- as material substance that Aristotle goes on to show must exist. If man is a univocal cause of man, father of son, the sun is an equivocal cause of the generation of man. And the first cause of coming into being and of being will *a fortiori* not be a cause which shares the name of its effect, material substance.

At the outset of Chapter Six of Book Twelve, Aristotle is not begging the question as to whether or not there is a substance which exists in separation from matter and motion. When he says "since there were three kinds of substance" one of which is immovable, he is actually referring to the first chapter of the same book where the views of the Platonists concerning the separate existence of the Forms and numbers are given. (1069a30) Aristotle now intends to show the necessity of maintaining that there is an "eternal unmovable substance."

He has already shown that substance is prior to accidents, for when substance ceases to exist so too do its accidents. If every substance were destructible and non-eternal, it would follow that there is nothing eternal and that everything (i.e., both substance and accidents) is corruptible. Since this is impossible, there must be some eternal substance. The puzzling element of this argument is the claim that it is impossible that there be nothing eternal. Aristotle argues for the truth of this by noting that it seems impossible that there should have been an absolute beginning of motion or that motion should ever cease entirely. He has given a proof of this in the eighth book of the *Physics*; motion is eternal. Likewise it seems impossible that time should ever have had a beginning. What was before time? And, since Aristotle teaches that time is the measure of motion with respect to before and after, the eternity of motion and time are not entirely separable questions. Moreover, if time is continuous and eternal, it must be the measure of some continuous and eternal motion. Now the only motion that can satisfy these demands is local motion and, indeed, a circular motion. But of such a continuous and eternal motion, there must be some cause. In the *Physics*, Aristotle has shown at great length that whatever is moved, is moved by another; on the basis of this, he now says that there is an eternal substance which is the cause of the first and most regular continuous and eternal motion of which time is the measure. This agent cause must have always been exercising its causality: first because of the position already taken with regard to the impossibility of motion having an absolute beginning and, secondly, because it

would seem that for this cause to begin causing would be for it to pass from potency to act. This last possibility is to be rejected because the cause of the first motion cannot itself be moved; if it were, we would have to seek a cause of it. Being always actually causing and unmoved itself, this substance has no matter and is only actuality. Thus, by basing his inquiry on the eternity of motion and time, Aristotle arrives at a substance which is immaterial and immobile.

Is there any difficulty in maintaining that such a substance is actuality alone? One might object that possibility precedes actuality and, consequently, that before this substance is actual it must have been possible. Aristotle notes that if this objection were valid, at one time there would have been nothing: but then how could there now be something? A passage from potency to act always requires an actually existing cause. Actuality, therefore, is absolutely prior to potentiality.

Again, given the eternity of motion and time, there is a cycle of generations and corruptions which involves generable and corruptible causes. Such a cycle is possible on the basis of some cause which always acts in the same way. Thus, the sun exerts its causality through eternal generations of corruptible things. The perpetuity of such a cause as the sun is in its turn reducible to that which is first; a second cause such as the sun, since it is moved, acts in virtue of a first unmoved agent. Aristotle is assuming as given "not in theory only but in fact" that there is a first cosmic movement, that of the outer sphere of the universe. "Therefore the first heaven must be eternal. There is therefore also something which moves it. And since that which is moved and moves is intermediate, there is something which moves without being moved, being eternal, substance, and actuality." (1072a22) In order to explain how the first unmoved mover moves, Aristotle stresses the notion of final causality. The end or good is not changed by being sought or loved. The end involved here is not something yet to be, but a substance which is the term or good at which a motion is directed. The first unmoved mover, then, is seen to be such because it is the final cause. Does this mean that God is not an efficient cause of the universe? It is increasingly being said that Aristotle's God is "only" the final cause and not the efficient cause of everything else. It would be difficult to come up with a more superficial criticism. First of all, the hypothesis of an eternally existing world does not preclude that the world has an efficient cause.

Secondly, if Aristotle teaches that God is the final cause of the universe *a fortiori* he must be the efficient cause of it, since the final cause is the cause of all the other causes. Moreover, nature acts for an end and if the whole of nature is

ordered to the first unmoved mover as to its end, this can only be because it is adapted to this end; that is, nature must be so fashioned as to seek the ultimate good. That this ordination of the universe towards the unmoved mover cannot be the result of chance is clear from the second book of the *Physics*. Aristotle has arrived at a principle on which "depend the heavens and the world of nature."

The proof for the existence of God involves the recognition of some divine attributes. He is immaterial, since he is actually -- there is no potency in this principle and, since potency and matter seem synonymous, it is clear that God is immaterial. He is immobile, since the first cause of all generation and change must be himself free from change. Moreover, it is fitting that we speak of the first unmoved mover as a person for, if it is intelligence which is the note of personality, God is the first of personal beings. Moreover, he is supremely happy. Aristotle shows this by noting the pleasure we find in thought, in the activity of thinking. This pleasure is had by God to a supreme degree, since he is thought itself always perfectly in possession of its object which is again itself. And, since thinking is a vital operation, God must also be said to be alive. It is important to see that the diversity of these attributes is not meant to imply any diversity or multiplicity in God. God is thought; God is life. When we name him in these ways, we are moving from perfections in the world around us and trying to say something about the limitless and simple perfection of God. Thus, by holding that God is thought, thinking itself, Aristotle is asserting God's utter independence of others. It has sometimes been concluded that God does not know anything other than himself. Aristotle seems to suggest this in Chapter Nine, and yet it is difficult to accept this interpretation as consistent with other things he teaches. We have earlier seen that the whole universe is directed to God as to its final cause and that this cannot be by chance; if not by chance then by design, and intelligent design, and if by design then whose but God's? It would seem that God must know everything in order to be the first cause of the universe. When Aristotle

seems to restrict God's knowledge to himself alone, he is concerned with the mode of such knowledge; moreover, he seems to be bothered by the problem of evil, "for there are even some things it is better not to see than to see." Such a remark can hardly apply even to the humblest of natural things; in his natural writings Aristotle rebukes those who disdain the detailed study of physical things, urging them to pursue the inquiry because they will find everywhere evidence of intelligent design: "there are gods even here."

Having established the existence of God and some of his attributes, Aristotle goes on to ask whether there are other separated substances besides the Prime Mover. He argues that there are, below God, other separated substances arranged in a hierarchy answering to the various spheres within the supposed ultimate sphere of the universe. These lesser immaterial movers are ordered to the Prime Mover in much the same way as the lesser motions are to the first. The separated substances which are as final causes of lesser spheres seem to be related, given their function, in terms of degree of power and knowledge. "That the movers are substances, then, and that one of these is first and another second according to the same order as the movements of the stars is evident." (1073b1) The immaterial substances are hierarchically arranged, ordered to the first, and this not by chance. Their nature and operation must thus depend upon the first immaterial substance which above all deserves the name God. Aristotle sums this up by quoting Homer: "The rule of many is not good; one ruler let there be."

The sketch of the doctrine of Aristotle we have attempted may serve to give some indication of the breadth of his interest, his method with respect to the opinions of his predecessors and the manner in which, while retaining Plato's interest in the immaterial and eternal, he insists that only knowledge of material substance can provide us with knowledge of such entities. If we should say that it is difficult to think of a branch of knowledge which did not attract the attention of Aristotle, we should have to add that it is largely due to Aristotle that many branches of knowledge came into being; moreover, Aristotle, as we have seen, provides us with criteria for distinguishing areas of knowledge. Coming when he did in the development of Greek philosophy, he was able to discern and describe the common method of philosophical research and demonstration; because of this he is called the father of logic. With respect to the natural world, Aristotle rescued an area of knowledge which had hung under the indictment of Parmenides and not only gave a timeless description of the general principles of natural science, but engaged in extensive research, particularly with respect to living things. Because of this he is called the father of biology. In the field of morals, Aristotle is most obviously dependent on his predecessors, on Plato above all.

Nevertheless, his contributions here are by no means negligible. With respect to the term of philosophy, wisdom or metaphysics, it is unfortunately in fashion to be so impressed by certain considerations first proposed by Jaeger that the unity of a science present in the treatises grouped under the title *Metaphysics* has been lost sight of. On this score, much can be learned from the medieval

commentators who, while they may seem to sense few difficulties in the state of the text, justify their interpretation of the *Metaphysics* as containing a unified doctrine by close textual analyses. One is struck by the fact that it is by analyses of the same kind that it is argued that the work contains contradictory attitudes. In other words, when the philologist has had his say, we are faced with the same task as has always faced the student of the text of Aristotle and we must be careful lest we bring to it the *a priori* conviction that we can expect no unified doctrine.

It is difficult to find a more pithy description of the immediate wake of Aristotle than that given by Mure.

The wisdom of Socrates was a prophecy; Plato's philosophy was a vision half seen and half communicated and his death an urgent challenge to make good his uncompleted conquest. But Aristotle's triumphant fulfilment of his inherited task was a climax from which the tide of speculation could only recede. We can trace through two and a half centuries the names of his successors at the Lyceum, but each is a lesser and a dimmer figure than the one before. In Theophrastus survived some ember of Aristotle's universal genius, and for later generations his repute remained within a still measurable distance of his master's; but Strato devoted himself almost entirely to natural science, and his brother Lyco, the fourth head of the school, seems to have been distinguished chiefly by an enthusiasm for adolescent education, and by an eloquence of speech which failed him sadly when he came to write. (*Aristotle*, p. 233)

It is surprisingly the case that Aristotle's influence in the final period of Greek philosophy is often difficult to discern; certainly his own school showed none of the tenacity for survival exhibited by the Academy (even when it bore little relation to the school Plato founded). The influence of Aristotle cannot be easily placed in one school or another; it is not too much to say that it pervades all the schools, though Plato will exercise far greater influence in the last great effort of Greek thought.

Part III: The Hellenistic Period

Chapter I

Epicureanism

Epicurus was born on the island of Samos in 341 B.C. He was an Athenian citizen since Samos was in league with Athens; consequently, in 323, Epicurus went to Athens to fulfil his two year military obligation. In 321, Epicurus rejoined his family, which had moved to Colophon. A ten-year period was devoted to study and then, in 311, when Epicurus was thirty, he set himself up as a teacher at Mytilene; the following year he transferred to Lampsacus where he taught for four years after which, in 306, he and the school that had gathered around him, moved to Athens. Epicurus spent the rest of his life in Athens, though we are told of several visits to Lampsacus; in 271 B.C. Epicurus taught in the garden of the home he bought in Athens; it is thought that he chose this location because it exempted him from a law then in effect which required approval of the Senate and the Assembly to open a school. Indeed, Theophrastus, Aristotle's successor in the Lyceum, was in exile and the Academy was under attack. Epicurus decided against public teaching.

Epicurus and the philosophy named after him have been objects of vituperation since ancient times. This was due in no small part to the cavalier way in which Epicurus dealt with other philosophies. He is said to have studied under the Platonist Pamphilus, perhaps for four years, at Samos, beginning at the age of fourteen. After his military service, he studied in Rhodes with Praxiphanes, an Aristotelian. Epicurus denied having studied under him, however, and this contributed to his reputation as an ingrate. Diogenes Laertius records some of the accusations leveled against Epicurus in antiquity, and adds,

But these people are stark mad. For our philosopher has abundance of witnesses to attest his unsurpassed goodwill to all men -- his native land, which honored him with statues in bronze; his friends, so many in number that they could hardly be counted by whole cities, and indeed all who knew him held fast as they were by the sirencharms of his doctrine . . . (Z,9)

As significant as anything is the information that of all the philosophical schools, only the Epicurean was still flourishing when Diogenes wrote in the third century of our era. DeWitt's book⁽⁴¹⁾ is a lengthy attempt to rescue Epicurus, not so much from personal denigration as from interpretations he attempts to show have little or no basis in the evidence we have.

Diogenes Laertius says that Epicurus was a prolific author, surpassing all his predecessors in the number of his writings. Diogenes himself transmits his will and three letters, one being known as the *Little Epitome*. He also tells us what the division of philosophy is according to Epicurus.

It is divided into three parts -- Canonic, Physics, Ethics. Canonic forms the introduction to the system and is contained in a single work entitled *The Canon*. The physical part includes the entire theory of nature: it is contained in the thirty-seven books *Of Nature*, and, in a summary form, in the letters. The ethical part deals with the facts of choice and aversion: this may be found in the books *On Human Life*, in the letters, and in his treatise *Of the End*. The usual arrangement, however, is to conjoin canonic with physics, and the former they call the science which deals with the standard and the first principle, or the elementary part of philosophy, while physics proper, they say, deals with becoming and perishing and with nature; ethics, on the other hand, deals with things to be sought and avoided, with human life and the end-in-chief. (X,30)

A. Canonic

This indispensable first part of philosophy concerns itself with sensations, preconceptions and feelings as standards of truth.

Every sensation Epicurus says, "is devoid of reason and incapable of memory; for neither is it self-caused nor, regarded as having an external cause, can it add anything thereto or take anything therefrom. Nor is there anything which can refute sensations or convict them of error: one sensation cannot convict another and kindred sensation, for they are equally valid; nor can one sensation refute another which is not kindred but heterogeneous, for the objects which the two senses judge are not the same; nor again can reason refute them, for reason is wholly dependent on sensation; nor can one sense refute another, since we pay equal heed to all." (X,32)

Reason, as Plato and Aristotle would speak of it, plays no role in the Epicurean explanation because Epicurus, as we shall see, simply does not recognize such

an entity. The passage just quoted indicates that he held to the infallibility of sensation, a position reminiscent, as DeWitt notes, of Aristotle's. Sight is not deceived with respect to its proper object, nor is hearing, nor any other sense. Error arises only with judgment. Sensation itself is the momentary registering of a quality -- it is this momentary character that Epicurus seems to have in mind when he says sensation has no memory. Moreover, it is dependent on the external, incapable of eliciting itself, a passive reaction, therefore. Does Epicurus want to maintain that the senses never mislead us, that the problem of wine tasting sour to the sick man and sweet to the well man is fictional? It seems plausible that he would reply that sourness and sweetness are different reports, and equally true relative to the tasters. It is not here that error lies. "He was consequently at pains to locate the source of error, and he found it in the hasty action of the automatic mind. For example the boat on which the observer is a passenger is standing still but it seems to be moving when a second boat is passing by. In such an instance the eyes are not playing the observer false; it is the hasty judgment of the automatic mind that is in error." (DeWitt, p. 137) As Diogenes puts it,

For all our notions are derived from perceptions, either by actual contact or by analogy, or resemblance, or composition, with some slight aid from reasoning. And the objects presented to madmen and to people in dreams are true, for they produce effects -- i.e. movements in the mind -- which that which is unreal never does. (X,32)

It is obvious that "true" here is synonymous with "real;" that is, dreamers really dream dreams, truly dream dreams. "Notions" at the beginning of this passage indicates something over and above the direct impression of qualities in sensation, and their derivation from sensation is emphasized. This has led to the interpretation that Epicurus allows for no knowledge which is not derived from sensation, an interpretation which can be traced in no small part to the fact that Lucretius concentrates on sensation and has nothing to say of the preconceptions and feelings as criteria. When the preconceptions are taken into account, we find a somewhat different teaching than is usually attributed to Epicurus. To conclude the statement on sensation, we can say, following DeWitt, that when Epicurus says that all sensations are true he means the proper objects of the five senses and not judgments made in terms of notions formed on the basis of such immediate sensory reports; e.g., if one tastes sweetness he tastes sweetness and no mistake; but to judge that he is tasting honey, though based on the sensation, is liable to error.

Preconceptions.

By preconception they mean a sort of apprehension or a right opinion or notion, or universal idea stored in the mind; that is, a recollection of an external object being presented, e.g., Such and such a thing is a man: for no sooner is the word 'man' uttered than we think of his shape by an act of preconception, in which the senses take the lead. Thus the object primarily denoted by every term is then plain and clear. And we should never have started an investigation, unless we had known what it was that we were in search of. For example: The object standing yonder is a horse or cow. Before making this judgment, we must at some time or other have known by preconception the shape of a horse or a cow. We should not have given anything a name, if we had not first learnt its form by way of preconception. (X,33)

This account of Diogenes is vehemently rejected by DeWitt who now calls Diogenes a "stodgy compiler" though earlier his account was said to be excellent. DeWitt would distinguish between the general and the abstract, an example of the former being "horse," of the latter justice. Now, Diogenes has made preconception or anticipation (*prolepsis*) refer to the general; DeWitt feels it must be referred rather to the abstract, and what is involved in an anticipation of experience, an inborn conception not reducible to sensation.

Perhaps to speak of preconception or anticipation as instinct would not be too misleading, since we use this word in a way which suggests what is prior to actual experience. DeWitt makes the suggestion that, as Plato appealed to preexistence and *anamnesis* to explain the learning process, Epicurus, for whom the soul neither antedates nor survives the body, opts for innate ideas. Thus, we have an idea of justice and, as well, an idea of god. These ideas are sketches and anticipations of future experience and to that degree are introduced by Epicurus as canons of thought. Needless to say, if, as many have said, Epicurus did not mean something like innate ideas by preconception, it is difficult to see why preconception would be an element of the canonic.

Feelings. "They affirm that there are two states of feeling, pleasure and pain, which arise in every animate being, and that the one is favorable and the other hostile to that being, and by their means choice and avoidance are determined." (X, 34) The elevation of pleasure and pain into criteria underlies the hedonism of Epicurus. This is not to say that the only pleasure recognized by Epicurus is that of the flesh; although for him soul was also corporeal, nevertheless, its pleasures were of a higher kind than those of the body and to be preferred. In

the first instance, of course, pleasure and pain are consequent upon sensation. And we can easily see what he meant by speaking of these feelings as criteria. Pain alerts us to the harmful, pleasure is an indication that something is conducive to our well-being; hence the role pleasure and pain are made to play in the training of the young. But pleasure is the goal as well as the beginning of the happy life, that is, happiness becomes that for which one consciously strives. From this point of view, happiness becomes a criterion according to which we judge. We shall have more to say about the feelings as criteria when we speak of the ethics of Epicurus.

B. Physics

We shall base ourselves here on the letter to Herodotus, preserved by Diogenes Laertius, in which Epicurus attempts an epitome of his physical doctrine. That doctrine, accordingly, is reduced to a few key doctrines, suitable for memorization, from which the details of the system are deducible. "To the former, then -- the main heads -- we must continually return, and must memorize them so far as to get a valid conception of the facts, as well as the means of discovering all the details exactly when once the general outlines are rightly understood and remembered. . ." (X, 36) In pursuing the elementary doctrine, we must be aware of the proper meanings of the words we use and subject what is said to the criteria set forth in the canonic. Epicurus begins with the problem of presocratic physics, with the Parmenidean difficulty.

To begin with, nothing comes into being out of what is non-existent. For in that case anything would have arisen out of anything, standing as it would in no need of its proper germs. And if that which disappears had been destroyed and become non-existent, everything would have perished that into which the things were dissolved being non-existent. Moreover, the sum total of things was always such as it is now, and such it will ever remain. For there is nothing into which it can change. For outside the sum of things there is nothing which could enter into it and bring about the change. (X,39)

We have here the acceptance of the dilemma of Parmenides: matter cannot come to be nor can it cease to be. Everything is as it has always been, since no fundamental change is possible. For a more positive account, Epicurus takes over the atomic doctrine of Democritus: ". . . the whole of being consists of bodies and space." (X, 39) The existence of bodies is clear from sensation; that of space from the fact that, were there none, bodies would have nothing in which to be and through which to move. There is nothing besides bodies and

space. The beginning of the Epicurean physics, then, brings us back to very familiar ground. Being cannot come to be; being cannot cease to be; what is are bodies and the space in which they are and move.

Again, of bodies some are composite, others the elements of which these composite bodies are made. These elements are indivisible and unchangeable, and necessarily so, if things are not all to be destroyed and pass into non-existence, but are to be strong enough to endure when the composite bodies are broken up, because they possess a solid nature and are incapable of being anywhere or anyhow dissolved. (X,41)

These elements or atoms are fundamental beings and there is an infinite number of them just as the void in which they are and move is infinite in extent. The atoms differ from one another by shape, weight and size and are always in motion. Since atoms are infinite in number and space infinite in extent, there is an infinity of worlds.

If atoms and void exhaust being, the soul like the body is composed of atoms. Bodies are constantly giving off a stream of atoms, which does not diminish them, since other atoms immediately take the place of those departing. As atoms leave the body, they retain the shape of the body; thus the air is filled with images or *eidola* of bodies and it is thanks to contact with these that vision takes place. Given this, no vision can be uncaused although, as was indicated above, this does not mean that we never make mistakes about things seen.

For the presentations which, e.g., are received in a picture or arise in dreams, or from any other form of apprehension by the mind, or by the other criteria of truth, would never have resembled what we call the real and true things, had it not been for certain actual things of the kind with which we come in contact. Error would not have occurred, if we had not experienced some other movement in ourselves, conjoined with, but distinct from, the perception of that which is presented. And from this movement, if it be not confirmed or be contradicted, falsehood results; while, if it be confirmed or not contradicted, truth results. (X,51)

Error here is seen to arise, not from sensation itself, but from a conjoined movement whose source is within us. Epicurus explicitly states that he is here concerned to defend the criteria of truth set forth in the canonic.

If sensation is explained by the flow of atoms from bodies, we must not conclude that the atoms have the qualities we perceive.

Moreover, we must hold that the atoms in fact possess none of the qualities belonging to things which come under our observation, except shape, weight and size, and the properties necessarily conjoined with shape. For every quality changes, but the atoms do not change, since, when the composite bodies are dissolved, there must needs be a permanent something, solid and indissoluble, left behind, which makes change possible. (X,54)

The soul itself is said to be composed of fine particles which are dispersed throughout the bodily frame "most nearly resembling wind with some admixture of heat, in some respects like wind, in others like heat." If there is a gradation of the atoms which accounts for the difference between body and soul, there is also a gradation in the soul atoms. Epicurus spoke of a rational and an irrational part of the soul and it is the irrational part which is dispersed throughout the body; the rational resides in the breast. Sensations are borne from the periphery of the body to the mind while emotions move from the mind in the contrary direction. "Move" here is not metaphorical, since both sensation and emotion consist of movement of atoms. The activity of mind is conceived by Epicurus as being either automatic or volitional. The automatic mind receives the data of sensation and speedily assesses and judges. Such judgments are often in error because they go beyond the sensory reports, from sweet to honey, for example. The volitional mind operates in terms of the preconceptions or anticipations and passes judgment.

C. Ethics

The philosophy of Epicurus moves towards the ethics as to its term. In the letter to Menoeceus, recorded by Diogenes Laertius, the ethical doctrines are preceded by a general statement concerning the pursuit of wisdom. There follow considerations which will aid one in the pursuit of happiness. First, Epicurus warns against accepting the usual view of the gods which makes of these vindictive entities, angered by our misdeeds, pleased by our virtues. Nor must we entertain any fear of death. Death is neither good nor evil, since these can be said only of things sensed and death is the end of sentience for us.

For life has no terrors for him who has thoroughly apprehended that there are no terrors for him in ceasing to live. Foolish therefore is the man who says that he fears death, not because it will pain when it comes, but because it pains in the prospect. Whatsoever causes no annoyance when it is present, causes only a groundless pain in the expectation. Death, therefore, the most awful of evils, is

nothing to us, seeing that, when we are, death is not come, and, when death is come, we are not. (X,125)

The wise man neither deprecates life nor fears death. Those who say that it were best for a man not be born are wrong as well as inconsistent; if they believed themselves, they should quit talking and commit suicide.

We must also reflect that of desires some are natural, others are groundless; and that of the natural some are necessary as well as natural, and some natural only. And of the necessary desires some are necessary if we are to be happy, some if the body is to be rid of uneasiness, some if we are even to live. (X,127)

For the end of all our actions is to be free from pain and fear, and, when once we have attained all this, the tempest of the soul is laid; seeing that the living creature has no need to go in search of something that is lacking, nor to look for anything else by which the good of the soul and of the body will be fulfilled. (X,128)

Pleasure is the beginning and the fulfillment of a happy life. Epicurus takes a genetic view of human life. The standard of choice and aversion of the child, of man, that is, in a natural state, is clear and obvious. In this sense feeling is a criterion of choice, not that every pleasure is pursued and every pain avoided; rather, there is a calculus, an art of perspective, whereby some pains are sustained in view of a future pleasure which outweighs them; some pleasures are ignored because they would lead to a pain incommensurate with the pleasure. "it is, however, by measuring one against another, and by looking at the conveniences and inconveniences, that all these matters must be judged." (X, 130) The mark of this hedonism is not a surplus of bodily goods and pleasures, but contentment with a minimum. Indeed, the Epicurean ideal is indifference to external goods, an effort whereby one habituates himself to simple fare, to the necessities of life. Despite the connotation "epicure" has taken on, sensual pleasure is discounted. "When we say, then, that pleasure is the end and aim, we do not mean the pleasures of the prodigal or the pleasures of sensuality, as we are understood to do by some through ignorance, prejudice, or wilful misrepresentation." (X, 131)

As the canon indicates, Epicurus wants to make nature the guide; he seeks the fundamental indication of nature and, in ethics, the drive of our own nature towards certain goals. Having rejected the Platonic and Aristotelian views according to which the soul is capable of existence apart from the body,

Epicurus must secure his ethical doctrine within the confines of birth and death. There has been no previous state, there will be no state subsequent to that in which we now find ourselves. The great good then is life. When he wants to indicate what the good, what pleasure is, Epicurus speaks of narrowly escaping death. The pleasure we then feel indicates that the radical good is to be alive. Pleasure is the fulfilment of life, but life itself is the fundamental good. Paradoxically, this placing of ultimate value in life is coupled with an exhortation to indifference towards death. Tranquility of mind is impossible when one fears death; death is quite simply the end, should not be feared; to know this is to be on the way towards peace of mind. How can nature be a guide if everything is atoms and void? In accepting the Democritean physics, Epicurus has adopted a world which is not the product of intelligence as is the world of Anaxagoras or Aristotle; it is the result of a chance collision of atoms. And yet, Epicurus introduces a doctrine of a "swerve" in the perpetual movement of the atoms, which is supposed to allow for spontaneity, freedom and consequently responsibility. Contrary to what we might think his atomism would suggest, Epicurus says the wise man will not be fatalistic.

Destiny, which some introduce as sovereign over all things, he laughs to scorn, affirming rather that some things happen of necessity, others by chance, others through his own agency. For he sees that necessity destroys responsibility and that chance or fortune is inconstant; whereas our own actions are free, and it is to them that praise and blame naturally attach. (X,133)

Better to accept the myths about the gods than the philosophical doctrine that all is determined.

The passage quoted earlier about the differences between desires contains a doctrine which led Epicurus to conclude that nature has set a limit to pleasures. If we have a natural desire for food, this does not mean that nature prompts us to eat always or in any amount. The distinction of desires is not to be taken as implying that some pleasures are better than others. Epicurus recognizes a gradation among the atoms, a difference of body and soul, but he will not allow that pleasure, any pleasure, is other than good. Pursuit and avoidance are decided on the basis of consequences, but each pleasure is a good in itself, each pain an evil in itself. The emphasis on mental pleasure, the joy of contemplation, by Plato and Aristotle is not unconnected with their conviction of the destiny of the soul after this life; Epicurus tends to reduce pleasures to a kind of sameness, although, according to Diogenes Laertius, he differs from the Cyrenaics by holding that mental pain is worse than bodily. Nevertheless, the

keynote of the Epicurean ethics is to be found in *ataraxy*, the mental state of tranquility which differs from kinetic pleasures in being a state of repose and rest. Perhaps it is this state of mind that is envisaged when Epicurus denies that virtue is its own reward, that it can somehow be justified apart from the pleasure it brings. Pleasure is the end or fulfilment of life, not virtue seen as possibly opposed to pleasure.

The role played by friendship in the Epicurean ethics deserves emphasis. The foregoing should have indicated that the ideal that Epicurus sets for man is, in effect, to make the best of a bad situation. The goal is to acquire a trouble-free state of mind, to avoid the pains of the body and mental anguish. The Epicurean wants to be let alone by society and by chance, good as well as bad. If a fortune should come his way, he ought to distribute it and win friends, thereby securing the double advantage of ridding himself from a possible source of anguish and gaining esteem and gratitude. The political order is to be avoided, since it brings little but trouble. Is this a repetition of the Platonic and Aristotelian view that the philosopher is superior to the politician? Epicurus rejects the *paideia* which for Plato and Aristotle would lead into the life of wisdom. If Epicurus has a *Canon* and a *Physics*, these are rigorously subjected to the pursuit of happiness. He goes so far as to say that if men had no fear of the gods and of celestial phenomena, there would be no need of physics. That science, then, has as its sole function to remove impediments to happiness. To conceive of mathematics and physics as goods in themselves is wrong; they are of no use in helping us live well. If the Epicurean is not interested in the honor and renown the political life can bring, he is no more interested in the praise knowledge can elicit: his only reason for doing philosophy is to attain peace of soul. It is because Epicurus did not feel that the spirit could be healed in solitude that friendship is important for him, that the society in his garden was indeed a gathering of friends. They were of mutual help to one another in understanding the teaching of the master and the discrimination present in other schools was largely absent there; women were allowed, even courtesans, and the love they bore one another is said to have remained the most attractive thing about the Epicurean way of life during the centuries it flourished. However, while friendship as well as pleasure was said to be desirable for its own sake, it is difficult not to see a selfish motivation in this mutual love. The friend, finally, would seem to be one who is eminently useful for attaining one's tranquility of spirit; if the friend used you in the same way and the advantage was mutual, this does not necessarily seem to broaden the motivation for friendship.

A final word on Epicurus' attitude towards religion. We have seen that Epicurus was intent on freeing men from fear of the gods. Nevertheless, he held that there were gods -- they were material beings living in the rare spaces between the numberless worlds, blessed beings whose happiness was in no way affected by the deeds of men. Since it is a false notion of the gods which leads to unhappiness, a true notion of what they are should be conducive to happiness. Epicurus feels that we have natural knowledge that the gods exist and that if we would stick to that, make it the criterion of statements we make about them, we must inevitably arrive at views contrary to the common ones. The gods are immortal and happy; to think of them as concerned with man or assigned the task of keeping the heavenly spheres moving is to attribute to them something inconsonant with the freedom from worry and care which is happiness. Nor are the gods in any need of our praise or sacrifice. Nevertheless, and this is an important fact, it is natural for us to honor them. This explains the way in which Epicurus himself was so devout, observing pious practises, inaugurating in his school religious feasts. Epicurus' vigorous rejection of the ordinary fears of the gods did not lead to atheism or impiety. Once more, however, there is the paradox. Since the gods do not need our praise, the justification for praising them becomes the happiness this affords us.

It is difficult to take a view of Epicureanism which does not grant a good deal to the traditional criticism of this school of philosophy. The goal is a practical one, and a severely limited one, the achievement of freedom from pain and worry. In this achievement, in *ataraxy*, happiness consists. Perhaps the best way to judge it is on its own assumptions, namely that birth is the beginning and death the end. On that basis, what would be the best human ideal? Doubtless happiness here and now, but this is not to be sought in sensual orgies nor in abstract science for its own sake. Epicurus wants to make nature the guide and he finds indications to the effect that good and bad are synonymous with pleasure and pain. But if pleasure is the fulfilment of life, there are natural bounds set to pleasure and he is issuing no call for a riotous existence. Quite the contrary. It seems to be a matter of hedging one's bets, sticking to the possible, being content with a minimum that will not be the cause of envy or care. This materialistic ethic does little to stretch the aspirations of men, but its concern for the individual at a period of history that saw the Greek political order disappearing and giving way to empires of a scale such that the old ideal of the political life became meaningless, would seem to account for its steady appeal over a period of seven centuries.

D. The History of the School

When Epicurus came to Athens he brought with him Metrodorus and Plynæus, both natives of Lampsacus who died before their master, Hermarchus of Mytilene, who succeeded to the headship of the school upon the death of Epicurus, and many others. During the first century of the school's existence its main competition came from the Platonists and Peripatetics, though its doctrines were directed against many schools, not least the Cynics and Sceptics. In the next two centuries, the main opposition came from the Stoics; there seems to be little evidence of any clash between the Epicureans and the early Stoics, Zeno, Cleanthes; the quarrel begins with Chrysippus. The reader is referred to DeWitt for a fairly detailed history of the school. Centers were set up in Antioch and Alexandria, and in the Christian era, Epicureanism finally met a victorious foe. It would not do to omit mention of Lucretius who in 54 B.C. published the *De rerum natura*, a poem intended to present Epicureanism in its fundamentals. Cicero devoted the last years of his life to an attempt to discredit the contents of that poem and emerged as a defender of Stoicism. Plutarch, much later, was to be another outstanding foe of Epicureanism. But the final assault came from Christianity, bringing to an end the school which had had the longest and greatest impact on ancient times.

{41} N. W. De Witt, *Epicurus and His Philosophy* (Minneapolis: University of Minnesota Press, 1954).

Chapter II

The Stoics

Stoicism is a philosophical school which came to prominence during the Macedonian ascendancy and continued when the Roman Empire had extended its sway over Greece. The founder of the Stoic school was Zeno of Citium on Cyprus. He came to Athens about 320 B.C. and was first associated with the Cynic, Crates. It is thought that Zeno was born about the middle of the fourth century, so he came to Athens as a young man, somewhere between the ages of twenty and thirty, shortly after the death of Aristotle. After Crates, Zeno studied with Stilpo the Megarian. He is also said to have studied under Xenocrates, head of the Platonic Academy. Diogenes Laertius says that Zeno studied altogether some twenty years before beginning to give lectures at the *stoa poikile*, the place from which his school took its name. Some twenty written works are attributed to Zeno by Diogenes Laertius, but very little of Zeno's actual teaching has come down to us; indeed, with the early Stoics, we are in much the same position we are with the Presocratics, for we must rely mainly on what later writers say they said. This should be kept in mind in assessing expositions of Stoic doctrine, including the one to follow. Zeno's successor as leader of the Stoic school was Cleanthes, a native of Assos, who has a reputation for retaining but not advancing the doctrine of his master and of being of exceptionally admirable moral character. Of the other students of Zeno, mention must be made of Persaeus who was a countryman as well as a follower of the master. Zeno himself is said to have enjoyed a long life and to have brought it to an end himself.

The successor of Cleanthes as head of the school is often called the second founder of Stoicism; he is Chrysippus, born in 280 B.C. in Cilicia. He may have been a student under Zeno; he was certainly a student of Cleanthes. Chrysippus is said to have listened to Arcesilaus and other philosophers of the Academy, and from them to have learned a critical attitude which exerted an influence on his own thought. He had a reputation for having differed widely in doctrine from Zeno and Cleanthes and is said to have been a prolific writer, author of more works than Epicurus, but, once more, very little of his actual teaching has come down to us. Chrysippus died in 206 B.C. having, we are told, brought Stoicism to a form which it continued to retain. His successor was Diogenes of Seleucia who was followed by Antipater whose successor, Panaetius of Rhodes, introduced Stoicism to the Roman world. We will consider the Stoics of the Greek world, but say a word or two about the Romans as well.

While we know that the above mentioned men, as well as a number of others, were members of the Stoic school and that many of them wrote, it is nevertheless the case that our sources for Stoic doctrine are extremely limited and indirect. For this reason, it is necessary to speak of Stoicism rather than of this Stoic or that, and to attempt to set forth the common tenets of the school. Among these common tenets is the view that philosophy is knowledge of things human and divine. Moreover, the Stoics divided philosophy into parts. "They assert that philosophic argument has three parts. One part of it concerns physics, another ethics, and the third logic. This division was first made by Zeno of Citium in his work *On Logic*." (S. V. F. 45)⁽⁴²⁾ This division was illustrated in a number of ways, according to the following testimony.

And they compare philosophy with an animal, representing logic as the bones and sinews, ethics as the flesh, and physics as the soul. Or again, with an egg. The outside is logic, next is ethics and the innermost part is physics. Or it is like an enclosed field. Its enclosing fence is logic, the fruit is ethics, and the earth and trees are physics. Or they compare philosophy with a walled city rationally governed. (38; Clark)

For some time it was the fashion to see the Stoics as primarily interested in the ethical, subordinating logic and physics to the roles they might play in the acquisition of virtue. On this view, the Stoics appear to reject the Platonic and Aristotelian doctrine that contemplation is the goal of philosophy and any contribution the Stoics may make should be sought in the ethical doctrine attributed to them. The obvious corollary is that their logic and physics do not amount to much. Recently, with a new assessment of the logic of the Stoics and, more recently, of their physics, the supposed priority of the ethical in Stoicism becomes less acceptable. It must be said that the testimonies concerning the order in which one should learn the parts of the philosophy do not seem to bear out the view that everything was subordinated to the acquisition of moral virtue.

Chrysippus thought it was necessary for the young student first to attend lectures on logic, second on ethics, and then on physics, and similarly at last to take up theology. Since he very frequently made these assertions, it will suffice to add a quotation from the fourth book on *Lives*. 'First of all it seems to me, conformably to what was so correctly stated by the ancients, that there are three types of philosophic speculation, logic, ethics and physics. Next it is necessary to arrange them by putting logic first, ethics second, and physics third. Now the final division of physics is theology, therefore also they named the teaching of this subject the initiatory rites.'(42;Clark)

This passage indicates that there were subdivisions of the main parts of philosophy, and Diogenes Laertius (VII,41) tells us that Cleanthes spoke of six parts of philosophy, Dialectic, Rhetoric, Ethics, Politics, Physics and Theology. The order of learning the parts of philosophy was based on principle.

The Stoics teach that we should begin with logic, continue with ethics, and place physics last. For first it is necessary to make the mind sure so that it will be an invincible guardian of the teachings. And dialectics serves to make the reason secure. Second we must subscribe to ethics to improve our character, for the study of ethics is without danger to one who has previously mastered logic. And finally we must proceed to physics, for it is more divine and requires more profound attention. (44;Clark)

It must be said that those who maintain that the Stoic philosophy was primarily directed to a practical end, the acquisition of moral virtue, are not insensitive to the difficulties presented to their position by such texts. Zeller observes that the relationship between ethics and physics is difficult to understand.

On the one hand, ethics appears to be the higher science, the crowning point of the system, the subject towards which the whole philosophical activity of the school was directed; for philosophy is practical knowledge and its object is to lead to virtue and happiness. On the other hand, virtue and the destiny of man consist in conformity to the laws of nature, which it is the province of science to investigate. Therefore, natural science has the higher object. It lays down the universal laws which in ethics are applied to man. To it, therefore, in the graduated scale of sciences, belongs the higher rank.^{43}

Whatever the value of Zeller's opinion that the Stoics have always a practical end in view, the texts we have cited indicate the procedure we must follow. We shall examine the doctrine of the early Stoics under the three headings of logic, ethics and physics, in that order. In the course of our examination we shall encounter facts which will enable us to assess the views that a practical end is and is not the goal the Stoics have in mind in their philosophy.

A. Logic

With regard to the ultimate aims of their philosophy, there seems to have been some difference among the earliest Stoics themselves, a difference which emerges when the value of logic is discussed. Thus Aristo, who was a pupil of Zeno, holds that the sole business of man is to pursue virtue and that logic is

justified to the extent that it furthers this end, having at best a therapeutic function. Unfortunately, in practise, logic does more harm than good. By the same token, he disparaged physics, holding with Socrates that it transcended the capacity of the human mind. (Zeller, pp. 59-61) Zeno, while he is concerned with the tricks of the dialectician and with sophisms generally, does not equate logic with its abuse. Indeed, he is said to have urged its study. "The business of a philosopher is what Zeno says: to know the elements of argument, what type each of them is, how they harmonize with each other, and what their implications are." (51;Clark) The importance the Stoics attached to logic is perhaps nowhere more evident than in their insistence, as against the Peripatetics, that logic was a true part of philosophy and not simply its instrument. Indeed, the Stoics came to be called Dialecticians and there is increasing concentration on logic during the headship of Chrysippus. This emphasis causes interpreters like Zeller some consternation. (p. 65)

Division of Logic. The Stoics divided logic into rhetoric and dialectic, but sometimes mention is made of a part which deals with definitions and a part which deals with the criterion of truth. This latter part is said to deal with the discovery of truth and to concern itself with the kinds of perceptions we have. It amounts to a theory of knowledge and was thought to have priority over the other parts of logic. Definition is concerned with the recognition of the truth, with apprehension by general notions. Rhetoric is the science of speaking well on matters set forth by plain narrative. Dialectic is the science of correctly discussing subjects by question and answer, or the science of statements, the true, the false, and those which are neither.

Criterion of Truth.

It pleased the Stoics to place first their theory of representation (*phantasia*) and sensation (*aisthesis*), because the criterion by which the truth about things is known, is generically a representation, and because the theory of assent and of comprehension and thought, the presupposition of everything else, cannot be formulated without involving representation. For representation comes first, then articulate thought puts into words what representation has conveyed. (52; Clark)

Theory of knowledge precedes any concern with the other parts of logic. Cicero has given us Zeno's graphic description of the degrees of knowledge. Extending his right hand, palm upwards, fingers extended, Zeno said, this is representation (*phantasia*); bending the fingers, he said, this is assent (*sunkatathesis*); making a

fist, he said this is comprehension (*katalepsis*); smashing the fist into the palm of the other hand, he said, that is science. The representation is an impression made on the soul, and it is either comprehensive or noncomprehensive.

The comprehensive representation, which they assert is the criterion of things, is that which is produced by a real object, resembles the object itself, and is sealed and stamped on the soul. The non-comprehensive representation either does not come from a real object, or if it does, it does not resemble the object. It is not well formed and distinct. (53; Clark)

As Sextus objects, the Stoics say that the comprehensive representation is that produced by a real object and a real object is one grasped by a comprehensive representation. Sensation is always true; representation is sometimes true, sometimes false. The following passage, reminiscent of Aristotle, indicates how the Stoics viewed the genesis of knowledge.

The Stoics say: When a man is born, the ruling part of the soul is like a sheet of paper suitable for writing. On this he writes off each single thought. -- That which comes through the senses is the first thing written down. For those who perceive something, like white, have a memory which comes from it. And when many similar memories have arisen, then we say people have experience, for experience is the manifold of similar representations. -- But of thoughts, some arise naturally in the aforementioned ways without technical skill, while others come by our teaching and conscious effort. These latter are called thoughts only (*ennoia*) but the others are also termed preconceptions. -- Now reason, because of which we are called rational, is said to have received all its preconceptions by the time a child is seven years old. And a notion (*ennoema*) is an image of the mind of a rational living being, for when the image strikes a rational soul, then it is called a notion, taking its name from the mind. -- Therefore all those which strike irrational animals are images only, but those which we or the gods have are both images, generically, and notions, specifically. (83; Clark)

If the images of animals and thoughts of men differ only thanks to their subjects, it is equally clear that knowledge is viewed as a passive reception on the part of the knower of images thrown off by things. Truth and falsity consist in an assent with respect to representations, and the comprehensive representation is one whose falsity is unthinkable; it simply has built into it its own justification. The Stoic theory of knowledge is complicated by the admission of incorporeal entities, things which do not produce images in us, but

of which we have representations. Such incorporeals are void, place and time, but most important of all the *lekton*, which is the concern of dialectics.

Categories. Our sources do not indicate to us where the Stoic doctrine of categories was situated in logic; the reports on which we rely most heavily for Stoic logic, have little to say about the categories. The Stoics speak of four categories and also of a highest notion, that of being or of the indefinite something. Being or something is called a highest genus, a designation which perhaps would separate it from Aristotle's doctrine of the way being is common to the various categories. Moreover, also in opposition to Aristotle, for whom categories are mutually exclusive, the Stoics maintained that the categories telescope, so that the first is contained or more accurately determined in the second and so on. We have no information as to how this was actually exemplified. The four categories are substrate or subject, quality, state and relation. The substrate seems to be matter with quality giving it identifiability and distinctness. We have here something like the Aristotelian matter and form spoken of as different categories. The two remaining categories, according to Zeller (p. 107), can be taken to cover whatever is taken to be non-essential, state referring to it as taken by itself, relation in its reference to other things. It is generally felt that this doctrine of categories is a physical and not a logical doctrine;⁽⁴⁴⁾ this being so, we shall let this skeletal statement suffice for now.

Dialectic is said to be concerned with signs and with things signified; thus, a first part of dialectic dealt with language, the notion of articulate sound, the letters of the alphabet, syntax, parts of speech. The *lekton* is the thing signified and is the principal subject of dialectic. To grasp what the Stoics meant by *lekton*, we must recall the triadic explanation of meaning we have already encountered in Aristotle. There is the sign, the spoken or written word which consists of articulated sound or ink arranged in a certain fashion. It stands for another corporeal thing but not directly; the external, corporeal thing for which the word stands is not the meaning of the word, but its reference or denotation. The meaning or sense of language is what we understand when we hear another speak a language we know; it is not as such the image in his mind. This meaning or *lekton* is incorporeal and that with which dialectic is concerned. The *lekton* is said to be that whose content corresponds to some rational presentation.

We have then three things: the sign or sound; what is signified, the meaning or *lekton*; and that to which the sign refers via the *lekton*, namely the external object. The *lekton* is divided into complete and incomplete *lekta*, the incomplete being the elements of a proposition, the subject alone or the predicate alone.

The complete *lekton* is a proposition, a judgment (*axioma*) which is either true or false. Complete *lekta* are simple and non-simple. Diocles, quoted by Diogenes Laertius, gives the following division of simple complete *lekta*. Negation (e.g. "It is not day"); denial (e.g., "No one is walking"); privation (e.g., "This man is unkind"); affirmation (e.g., "Dion is walking"); definite (e.g., "This man is walking"); indefinite (e.g., "Some one is walking"). Examples of non-simple or complex *lekta* are the following, hypothetical (e.g., "If it is day, it is light"); inferential (e.g., "Since it is day, it is light"); compound (e.g., "It is day and it is light"); disjunctive (e.g., "Either it is day or it is night"); causal (e.g., "Because it is day, it is light"); more and less (e.g., "It is rather day than night"). The Stoics also divided propositions into possible and impossible, necessary and non-necessary, with each of these adjectives referring to truth and falsity.

Argument. An argument consists of premisses and conclusion, as for example, "If it is day, it is light; but it is day. Therefore it is light." This would be called a full argument. A mood (*tropos*) is an outline of an argument; e.g. "If the first, then the second; but the first, therefore the second." Sometimes, for sake of brevity, a combination (*logotropos*) is used; e.g. "If Plato is alive, he breathes; but the first, therefore the second." Arguments were divided into the inconclusive and conclusive. An inconclusive argument is one the contradictory of those whose conclusion is compatible with the premisses. Conclusive arguments are either non-syllogistic or syllogistic. An example of a non-syllogistic argument is: "'It is both night and day' is false; now it is day, therefore it is not night." Or, "If Dion is a horse, he is an animal; but Dion is not a horse, therefore he is not an animal." These arguments seem to be called non-syllogistic because their major premiss is not true. The syllogistic argument is best exemplified by the five types of immediate inference.

Diocles' account states that Chrysippus taught five kinds of indemonstrable or immediate arguments. In the text these are given as arguments or as moods or with alphabetical symbols. (1) "If the first, then the second; but the first, therefore the second." This argument consists of two premisses, a conditional proposition and its protasis, from which the apodosis follows. (2) "If it is day, then it is light; but it is night, therefore it is not day." Here the premisses are a hypothetical proposition and the contradictory of the apodosis, from which the contradictory of the protasis follows. The mood of this argument would be: If the first, then the second; but not the second, therefore not the first. (3) "It is not the case that Plato is both dead and alive; but he is dead, therefore he is not alive." Here the premisses are a conjunction of a negative proposition and one of the conjoined propositions, from which the contradictory of the other

follows. (4) "Either A or B, but A, therefore not B." The premisses here are a disjunctive proposition and one of the alternatives from which the contradictory of the other alternative follows. (5) "Either it is day or it is night; but it is not night, therefore it is day." The premisses here are a disjunctive proposition and the contradictory of one of the alternatives from which the other alternative follows.

The foregoing gives something of the flavor of Stoic logic as it has come down to us by hearsay, with most of our informants unsympathetic to what they are reporting. We know that there was a goodly amount of disagreement among the Stoics on points which we have had to attribute to them as a group; moreover, the Stoics were in fairly lively debate with members of other schools on logical questions. In recent times, judgments about the value of Stoic logic have swung from one extreme to the other, with Zeller and Prantl dismissing it almost venomously as trivial and decadent and with historians like Bochenski and Mates⁽⁴⁵⁾ feeling that the Stoics have reached a level of logic which can best be appreciated from the vantage point of contemporary formal logic. The late commentators on Aristotle were quite severe with Stoic logic, although the traditional logic came to incorporate a treatise on the "hypothetical syllogism" which owes something to the Stoics. The trend of recent evaluation of Stoic logic makes any assessment of the argument depend on the large and vexed question of the relationship of contemporary formal or symbolic or mathematical logic to ancient logical works, particularly those of Aristotle. In this connection, the views of Virieux-Reymond are not without interest. She accepts the judgment that the logic of the Stoics is nominalistic.

Of essentially nominalistic inspiration, it is quite differently oriented than is Aristotelian logic. There are no genera and species in nature for nominalism . . . there are in fact only individual facts under the form of things, beings and properties. Only propositions having a singular subject can correspond to these individual facts. (p.150)⁽⁴⁶⁾

On this view, the theory of the proposition becomes the study of the links and connectives between singular propositions.

From the moment that there are no longer universal propositions, reasoning bears exclusively on individuals and groups of qualities connected by certain laws. The problem of the modes and figures of the syllogism disappears. The only task incumbent on us is to reduce all the types of conditional syllogisms to the least number possible (namely five) of elementary forms, which indicate the

connection of consequent to antecedent, these latter being indicated by ciphers to show clearly that 'it is question, not of a relation of concepts, but of an order of succession' between concrete events. (pp.150-1)

She summarizes the differences between Aristotelian and Stoic logic thus. (1) The principle of *dictum de omni et de nullo* is inoperative, since universal propositions are suppressed. (2) The connection of subject and attribute is no longer a relation of inherence or inclusion but of concomitance or sequence. (3) With constant succession replacing inference, the idea of law replaces that of essence. (4) The Aristotelian teaching that there is science only of the general is replaced by the teaching that there is science only of the necessary. That is, Stoic logic is the expression of a particular view of knowledge and of reality, one in conflict with the Aristotelian. But of course for the Aristotelian there are singular propositions and even if he should incorporate the Stoic indemonstrables, he will tend to make remarks about them that the Stoic would find unacceptable.

No one is satisfied with the information we have of Stoic logic and every attempt at assessment must be qualified with an indication that it is founded on an arguable construct of what is largely lost to us. To praise or blame the Stoics for their logic is always a risky business.

B. Physics

The physics of the Stoics proceeds in terms of a doctrine of two principles of all things.

They believe that there are two principles in the universe, the agent and the patient. The patient is unqualified reality, viz, matter, and the agent is the reason inherent in the matter, viz. God. For he is eternal and, present throughout matter, is the artificer of each thing. (300;Clark)

Zeller takes the Stoics to be accepting the notion of reality Plato sets down in the *Sophist*, namely that being is that which acts or is acted upon. The two basic principles of the universe, then, are the passive matter and the active God. God is not looked upon by the Stoic as something extrinsic to the universe; he is, in one sense of the term, an element of it, the world soul. If man is a compound of body and soul, what is compounded is not the corporeal and the incorporeal, but rather two corporeal things, since soul too is a kind of body, a body which pervades what we usually mean by body, although there is a governing part of

the soul, located in the chest or head. So too God pervades the universe, one corporeal thing pervading another, although individual Stoics would locate God in different places, in the heavens, in the center of the earth, etc. If Stoicism is a materialism it is also a pantheism.

At the very beginning of the discussion, in their assertion of two principles of all things, matter and God, of which one is the maker and one the patient, we might reasonably accuse them of saying that God is mixed with matter, extending through all of it, arranging, forming, and making the cosmos in this way. For if God is a body, as they say, being an intelligible and eternal spirit, and if matter is a body, then, in the first place there will be a body extending through a body, and second, this spirit will be either some one of the four simple bodies, which they also call elements, or a compound of them, as even they somewhere state -- for they conceive spirit to derive its reality from air and fire -- or, if it should be anything else, their divine body will be some fifth reality, asserted without demonstration or defense by those who object to a person who asserts this by proper proofs, on the basis that he is asserting a paradox. (310;Clark)

Sambursky (pp. 18-19) has pointed out the relation of the Stoic categories to this physical doctrine. Matter, the passive principle, is the category of substrate or subject; the pervading Pneuma or spirit gives matter all its qualities and is then the second category. The third category is the specific quality of the body resulting from the proportion of the two principles. The fourth category is the state with reference to an opposite term of change.

Whatever our views on the conscious relationship between the Stoics and the Presocratics, the Stoic *Physics* seems to bring back the ancient notion of the divine as that which runs like a continuum through the cosmos, which grows and its growth is the history of the cosmos. If the divine is not substrate, it is nonetheless corporeal. Zeller feels that matter and force are simply different aspects of the divine for the Stoics. The justification of this interpretation is to be found in the Stoic cosmogony.

In the beginning there is fire which turns into vapor and thence into moisture. Some of the moisture condenses to earth, some remains as water, some by evaporation becomes air from which fire is enkindled. This is taken to describe the separation of the active and passive, the soul and body of the world, with air and fire being active, water and earth being passive. The present state of the cosmos will terminate in a general conflagration, something the Stoics are

thought to have borrowed from Heraclitus, the result of which is that pure fire, the primary being is once again alone and the process can begin again. The inexorability of this process seems to have led to the Stoic notion of Fate or Destiny, which is at once a name of God or the primary being and the law governing the cosmos. Indeed, "nature," "God" and "fate" all name the same thing. One argument for fate or destiny has especial interest, because of its logical overtones.

The argument is attributed to Chrysippus and Cicero records it as follows.

Chrysippus concludes as follows: If there is a motion without a cause not every proposition, which the dialecticians call *axioma*, is either true or false; for whatever does not have efficient causes is neither true or false. However, every proposition is either true or false. Therefore, there is no motion without a cause. And if this is so, everything that happens, happens by antecedent causes. And if this is so, everything happens by fate. Therefore, whatever happens, happens by fate . . . (952;Glark)

Everything happens by necessity because every proposition must be either true or false and some propositions refer to the future, e.g., "Socrates shall die on such and such a day." Aristotle discussed this matter in *On Interpretation* and distinguished among propositions about the future those we can call future contingents. These as simple propositions (e.g., "It will rain on this date two years from now.") are neither true nor false, but a disjunction can be formed which is true (e.g., "Either it will rain on this date two years from now or it won't." -- where the negative covers the possibility that the world will be destroyed in the meantime). The Stoic view seems based on an absolute determinism, something Aristotle rejects. It is a nice question how Stoic fatalism is compatible with the view of possible propositions. Plutarch points out this difficulty.

How could there be no contradiction between the doctrine of the possible and the doctrine of fate? If indeed the possible is not that which either is true or will be true, as Diodorus postulates, but everything is possible that admits of coming true though it may never come about, then there will be many things possible among those which will not happen in accordance with unconquerable, unassailable and victorious Fate. Either the power of Fate will dwindle or, if Fate is as Chrysippus supposes it to be, that which admits of happening will often become impossible. For all that is true will necessarily be, being compelled by supreme necessity, but all that is false will be impossible, the

strongest cause preventing it from becoming true. (*De stoic. repugn.*, 1055d-e; tr. Sambursky)

As Sambursky points out, the Stoics reconciled Fate and possible propositions, not by making the course of events less determined, but by referring possibility to the imperfection of our knowledge. Things seem possible as opposed to necessary because of our ignorance of a determined world.

Monistic, materialistic, deterministic -- these adjectives best describe the cosmology of the Stoics. Other salient features are the acceptance of the supposedly Heraclitean notion of the divine fire and the notion of *ecpyrosis*, the destruction of the cosmos by a general conflagration which returns all things to fire from which the process begins again.

C. Ethics

Before turning to the Stoic doctrines on human conduct, its goal and the means to achieve the goal, we must take notice of their teaching on the nature of man. The world soul was spoken of on an analogy with the human soul and the latter as well as the former is considered to be material by the Stoics. The materiality of the soul is proved by remarking that it is affected by bodies, and that it is three-dimensional, extending through the obviously dimensional body. The soul is spoken of as fire or as breath diffused throughout the body so as to form with it one thing. The Stoics speak of seven parts of the soul, the five senses, the power of reproduction and the power of speech. Speech and reasoning are almost equated and the reasoning part of the soul is the ruling part with the others reduced to it as to their origin. Personal identity is located in the ruling part of the soul. This is not to say that the Stoics taught personal immortality. Finally all souls will be consumed in the fiery destruction of the cosmos, though some Stoics thought some souls would continue after death until this conflagration. In other words, the human soul is a part of a determined universe, a view which would seem to preclude any ethical theory. Nevertheless, the Stoics had a moral philosophy and, indeed, it was to be the most influential part of their doctrine.

How can there be freedom and responsibility in a world from whose determined course the soul is not excepted? Doubtless, we must recall the manner in which the Stoic could retain the notion of chance by an appeal to ignorance. More important is the Stoic view that the individual must see resignation to the law of the universe as the great goal and this resignation is seen as one to a higher

reason than man's. This ideal is expressed in Cleanthes' hymn to Zeus and it is one to which we must return as to the culmination of any presentation of the Stoic ethics.

Harmony with nature. The goal of conduct is happiness and this is achieved in rational activity or virtue. Every creature has a natural impulse to act in accordance with its nature, and happiness will be sought in that which is conformable to nature. This remark has a meaningful ambiguity in Stoic doctrine, since "nature" may be taken to refer to man's nature or to the cosmos; both are intended. Every creature must willy-nilly be in accord with the law of the cosmos; a rational creature has a natural impulse to become conscious of this law and to live in recognition of it. Virtue will consist in action in conformity with the recognized course of the world. Only virtue is a good for the Stoics; only vice is evil. They will not allow that riches and honor or pleasure are goods, though of a lesser order, than the good of reason. Best not to call them goods at all, but rather indifferent to the distinction of good and evil. Stoic happiness becomes, accordingly, a rather austere affair; it can consist only in the good of reason, the rational good and nothing else can increase or diminish happiness. The only pleasure that can be considered a good is that which is the concomitant of righteousness, the inner peace of the virtuous man. This ought not to be made into the object of action, however; some Stoics toyed with the idea that every pleasure was contrary to nature. If pleasure cannot be constitutive of happiness, pain cannot be destructive of it. The Stoic ideal is to rise above the gifts and blows of fortune and to place happiness where it is unassailable, the rectification of reason, bringing it into conscious conformity with the course of the world. The conformity was a conformity with law, with the law of nature and consequently the divine law; human law is at best an attempt to give expression to this law. Obedience to human law thus gains a foundation in the divine and the guidance of law is an expression of our natural call to a life of morality and virtue.

We have remarked that the Stoic ideal of virtue is an austere one. It seems to give little heed to the emotions and passions. The Stoics, however, were not silent on these matters. Emotions and passions are movements contrary to nature; they result from the rational part of the soul, indeed, but from its abuse, for they follow from precipitous judgments. Emotion, then, is not looked upon as something which may sway judgment wrongly, but as a consequent of wrong judgment. This puts the emotions in our power and makes virtue apathy, a state of being free from all emotions. In short, virtue is simply a matter of knowledge, vice of ignorance, and the Stoics had no hesitation in saying the virtue could be

taught and learned. The knowledge involved is, of course, ordered to action and cannot be taken as an end in itself.

To complete this picture of the ethical idea, the Stoics drew a hard and fast line between the virtuous and vicious and place the vast majority of men in the latter class. A man was either wholly good or wholly bad, there being such a connection between the virtues that to possess one is to possess them all and to lack one is to lack them all.

Qualifications of the Ideal. The foregoing may be seen, as Zeller would have it, as the general doctrine of morality with subsequent modifications and qualifications as the special theory; or, as Hicks would have it,^{47} the Stoics first set up a moral ideal, impossible of realization, and then went on to talk of action in terms of a striving for this ideal. In either case, we find qualifications of the doctrine just sketched, a bringing of it into line with the possible. Thus, while only the good of reason is said to be a good, the Stoics came to recognize that other than rational impulses are also part of our nature and thus intended. The objects of such impulses are then described as analogous to true goods and a scale among them spoken of. This tends to diminish the scope of objects indifferent to the dichotomy of good and evil. Now things which are conducive to virtue or tend to distract one from its pursuit are made objects of pursuit or avoidance and things of little relation to virtue or vice or none at all are called indifferent. The Stoics continue to insist on the difference between what is truly good and what is good only because it is conducive to the true good. By the same token, the Stoics come to recognize a role for the emotions or passions. While the ideal would seem to call for a complete eradication of emotions, the Stoics come to speak of the affections of the wise man and, indeed, it is difficult to read the hymn of Cleanthes without experiencing the deep-seated emotion which suffuses his statement of the need to be subject to the will of God, to the law of nature. Moreover, while the ideal permits of no intermediate stage between good and evil, the Stoics speak of a progress towards the good, the acquisition of virtue, of imperfection and perfect possession of virtue. They were led to this because of the difficulty of citing any concrete example of the virtuous man. Would Socrates be such? No, he was only tending in the right direction. Now this implies that Socrates, not being wholly good, was not wholly evil either and there is, consequently, an intermediate stage, perhaps the best any of us can make of our lives.

There is little point in emphasizing the contradictions which can be pointed up between the theory and its qualifications. Rather let us underline what was

influential in Stoicism and gave it, in the days of the Roman empire, an undisputed primacy among the philosophical schools inherited from the Greeks. The Stoic made it incumbent on every man, as a law of nature, to seek after virtue, to bring himself into conformity with the will of God which is equated with the course of the cosmos. It is this rational consciousness of and assent to the law of fate that is the peculiar demand of Stoicism. Its equivocal attitude towards the goods of this world does not obscure the fact that it created a general climate in which the good of man was identified with the rational direction of his life, a life which was not to be the plaything of the emotions or passions, but whose affective side consists of the peace which follows on the ordering of one's life in conformity with the law of nature. The generality of this demand made the individual's pursuit of virtue the pursuit of the common ideal, a matter of social consequence. Moreover, it lifted the moral ideal to a cosmopolitan level and the Stoic looked upon himself as a citizen of the world.

A seemingly paradoxical feature of Stoicism is found in its attitude towards suicide. We have already seen that legend has it that the founders of Stoicism ended their own lives. This is regarded, not as an escape, but as an expression of a man's triumph over circumstances, his indifference to pleasure. To die for one's country, to avoid being forced to do something unlawful, to avoid poverty, illness or the weakening of the mind -- all these are cited as reasons for suicide. Now we notice that these are things, which, from the standpoint of the Stoic ideal, would seem to be matters of indifference. Zeller would resolve the paradox by saying that life and death are equally indifferent to the Stoic and that in themselves they cannot constitute an act as moral or immoral. How the acceptance of suicide conforms with the Stoic goal of resignation to the course of events is a problem apparently incapable of solution.

We must also notice the way in which the Stoic explained the existence of evil. For the Stoic, the world is governed by reason, and we should therefore expect that everything happens for the best. And yet there is evil rampant in the world, both physical and moral evil. Physical evil, pain and suffering, would present little difficulty for the Stoic since these are not truly evils, but moral evil is something else again. Moral evil is real evil and, by the Stoic account, there is much more of it in the world than moral good. Why does God permit it? Is he perhaps impotent to prevent it? The Stoic explanation was one which would have a long history. God's ways are not our ways. If the world is governed by reason it is a reason a good deal more perfect than ours and if evil is permitted it is for the sake of the good. Virtue is acquired by resisting vice, and if vice did not exist, how would we know how to act? Thus, moral evil plays a role with

reference to moral good and finds its justification in this. This enables us to return to something we posed as a problem at the outset of this presentation of a sketch of the ethics of the Stoics. If ours is a deterministic universe and men are but parts of the universe, how can one man be virtuous and another vicious? There seems to be no room for responsibility and freedom if all men are fated to act as they do. Now the Stoics quite clearly want to make a man responsible for his actions and to retain a qualitative difference between good actions and bad. This difference is had by speaking of a conscious direction of actions in accord with the law of nature. As in the case of moral evil, the Stoic will have it that every element of the whole is governed by the law of the whole, that evil may be conducive to the good of the whole although it is not thereby the good of the individual guilty of it. There is, then, a responsibility to be resigned to the law of nature, freely and consciously. Whatever we may think of this as a solution, it seems to have been considered one by the Stoics.

D. The Roman Stoics

We cannot close this chapter on the Stoics without mention of the Latin authors who subscribed to Stoicism and were the instruments for giving its ethical doctrine the influence it was to have.

The first of these is L. Annaeus Seneca, born about 4 B.C. not in Rome but in Cordoba, Spain, died by his own hand at the command of the Emperor Nero in 65 A.D. He was brought to Rome in his infancy, became a successful and affluent lawyer and amateur philosopher, in the etymological and redundant application of the adjective. A disturbing fact about Seneca is that his own life does not seem to have been guided by the precepts he expressed so eloquently in his philosophical essays and letters. He has been called the favorite pagan of the Latin church, something resulting from the belief that he had carried on a correspondence with St. Paul. Seneca was the author of a number of tragedies, based on Greek themes, e.g. *Hercules Mad*, *Trojan Women*, *Medea*, *Oedipus*, etc. Among his more philosophical works is a collection of *Dialogues* which includes treatments of the happy life, tranquillity, leisure, providence and so forth. There are one hundred twenty-four *Letters to Lucilius*, seven books of *Natural Questions*. Seneca is not generally regarded as contributing to Stoicism but as disseminating it in an extremely polished style.

Epictetus (c. 50-138 A.D.) perhaps a native of Phrygia was in his youth a slave in Rome. Granted his freedom he remained in Rome, leaving in 90 A.D. when the Emperor Domitian expelled philosophers from the city. Epictetus repaired

to Nicopolis in Epirus, where he taught until his death. His teachings are preserved for us in the writings of his pupil, Arian, the *Discourses* and the *Enchiridion*. As was Seneca, Epictetus is concerned almost exclusively with ethical matters.

Marcus Aurelius, Emperor from 161 to 180 A.D., is author of the *Meditations*, which he wrote in Greek. The fact that a lawyer successful in the maze of imperial politics, a freed slave and an Emperor were all three exponents of Stoicism gives some indication of the scope of its appeal. With Marcus Aurelius there is a slight indication that the radical materialism of earlier Stoicism is being left behind. The Emperor distinguishes in man a body, soul and mind, with the latter in some sense transcending matter. It is mind or *nous* which is the spark of divinity (as Epictetus had said) in every man. Nevertheless, this *nous* will be consummated in the conflagration which ends the cosmos.

{42} Quoted from Gordon H. Clark, *Selections from Hellenistic Philosophy*, (New York: F. S. Crofts, 1940).

{43} E. Zeller, *The Stoics, Epicureans and Sceptics*, (London: Longmans, Green, & Co., 1892), p. 67.

{44} See S. Sambursky, *Physics of the Stoics* (London: L. Routledge, 1959), pp. 17-18.

{45} I. Bochenski, *Formal Logic* (Notre Dame: University of Notre Dame Press, 1960; B. Mates, *Stoic Logic* (Berkeley: University of California Press, 1953).

{46} A. Virleux-Reymond, *La Logique et l'epistemologie des Stoiciens*, Lausanne: Librairie de l'Université, 1949, p. 150.

{47} R. D. Hicks, *Stoic and Epicurean* (New York: Russell, 1961)

Chapter III

Sceptics and the New Academy

In this chapter, we want to discuss a number of Sceptics who actually antedate the Epicurean and Stoic school; our reason for postponing discussion of them will be made clear in a moment as well as the desirability of seeing in them a preliminary for a discussion of the New Academy. We will take the occasion of this chapter to indicate the subsequent history both of Scepticism and the Platonic Academy.

A. Pyrrho of Elis

Pyrrho was born about 365 B.C. at Elis and came to philosophy after an unsuccessful career as a painter. He is said to have studied philosophy under Bryson, the sophist, and Euclid of Megara; there is no doubt that he studied with Anaxarchus, a Democritean, with whom he accompanied Alexander the Great on his campaign in Asia. The variety of cultures and customs and ideas to which this journey exposed the young Pyrrho are thought to have had their influence in the philosophical position he adopted. After the death of Alexander, Pyrrho returned to Elis where about 330, at the age of thirty-five, he opened his philosophical school. He lived a long life; he died around 275 B.C. It is said that Pyrrho composed a poem in honor of Alexander, but apart from that he wrote nothing and we are dependent for an account of his doctrines on his pupil, Timon of Phlius, the so-called Sillographer whose name we have encountered in our discussion of Xenophanes, as well as on other secondary sources.

Acclaimed by later Sceptics as their founder, Pyrrho is a man whose personal doctrine is most difficult to determine, since there was always a tendency to read later formulations back into the founder. It is said that Pyrrho posed three questions as fundamental: What are things in themselves? How should we be disposed towards them? What is the result of these dispositions? The answers he proposed are somewhat bleak. Things do not differ from one another; they are equally uncertain and indiscernable. Our sensations and judgments can be productive of neither truth nor falsity. Consequently, we should trust neither sense nor reason, but strive to be without opinions, choosing neither one side nor the other of contradictories. No matter what is at issue, we should neither affirm nor deny. The result of these dispositions towards things is apathy, ataraxy, a suspension (*epoche*) of judgment, universal indifference. This attitude explains the claim that "I define nothing." One opinion is as good as another;

the ideal is to suspend judgment, say nothing, make no commitments. Pyrrho is asking us to be wary of treating either our sensations or our judgment as revelatory of reality and we may see in this both the acceptance and the surpassing of Democritus. Democritus, we remember, had cast doubt on the validity of sensation; sweet and color are only conventions, telling us nothing of what is, since what is, is exhausted by atoms and the void. Now if Democritus called sensation into question, he did so in the interest of a knowledge in no way dubitable. Pyrrho does not give reason or judgment any privileged position: a judgment is no more valuable than its negation. Indeed, there seems to be some justification in distinguishing the immediate and mediate, with judgment being mediate and dependent on sensation. The value of judgment accordingly, is dependent on that of sensation. In speaking of sensation, Pyrrho does not advocate doubt as to the reality of what seems to us. When we taste sweetness, there is no reason to doubt that we do indeed have a sensation of sweetness. But this is what seems, a phenomenon, and is no infallible index of what is. That, reality, Pyrrho suggests, is unknowable. Pyrrho has no doubt that honey tastes sweet, but he would not have the temerity to assert that it is sweet.

It is sometimes suggested that such a distinction must be referred to later Scepticism and that Pyrrho himself would subject everything to doubt: it is just as likely that I taste bitterness as sweetness. On this view, Pyrrho's tendency is a complete and utter withdrawal and he could have nothing to say even about his own sense impressions that could not be contested. In a word, he would have nothing positive at all to communicate. His message would be that we must cultivate indifference, complete apathy, a universal suspension of judgment and commitment. What then are we to make of the fact that Pyrrho started a school? A teacher may come to see that he has nothing to offer to others, but it seems unlikely that one would become a teacher to teach nothing. The reply is that Pyrrho was teaching an attitude and that, he did it as much by example as by words. The anecdotes in Diogenes Laertius are then interpreted as bearing out this point.

"Whatever the extent of Pyrrho's own position, it seems clear that it was intended as a way of producing happiness. The notion of *ataraxy* and apathy, which we have seen constitute the respective aims of the Stoic and Epicurean ethics, have their source, it would seem, in Pyrrho. Actually we can trace a connection between Pyrrho and Epicurus, since Pyrrho's pupil Nausiphanes is said to have been a teacher of Epicurus. As we shall see, the Stoic and Epicurean schools were objects of attack by later sceptics; by the same token, it seems that later Sceptics were somewhat less hardy than Pyrrho himself. Even

Pyrrho, however, is said to have possessed the great certitude that suspension of judgment and indifference is the key to happiness. Perhaps we would not trivialize his stand too greatly if we should say that, in a time of incredible political upheaval, when there was such a proliferation of philosophical doctrines, represented by warring schools, Pyrrho, who had seen tyranny at first hand as well as the variety of cultures and customs and perhaps had been struck by the impassivity of Indian holy men, chose to find happiness in a despair of philosophy, with one being considered as good as another, and total indifference to the vicissitudes of life. If one has to act, let him do so in such a way that he follows the customs of his time and place.

We have already indicated the possible influence of Democritus' critique of sensation on Pyrrho. This could be expanded, and the Eleatic doctrine and that of Heraclitus, insofar as both opposed sense and *logos* in favor of the latter, could be seen as influential. Apart from this, we may wonder about Pyrrho's affinity with the Sophists. If Pyrrho is an iconoclast, calling everything into question, he does not seem to differ in this from the sophists who, despite the attack of Socrates, Plato and Aristotle, were still around at the time of Pyrrho. The great difference would seem to be that the sophists sought to have a practical impact, going among men and seeking payment for their services. Pyrrho characteristically withdraws. He preaches resignation and, like Socrates, becomes an object of veneration. Bevan suggests,^[48] with some persuasiveness, that Pyrrho sums up the attitude of the common man before the multitude of philosophical schools and the rapidly shifting political scene. And yet, obscure as his own doctrine is, Pyrrho is hailed as their great forerunner by later sceptics, and their deference to him is qualitatively different from their attempts to find the root of their attitude in all previous philosophies.

Before turning to Timon, a word on the term "sceptic." Speaking of the later Sceptics, Diogenes Laertius writes,

All these were called Pyrrhoneans after the name of their master, but Aporetics, Sceptics, Ephectics, and even Zetetics, from their principles, if we may call them such -- Zetetics or seekers because they were seeking the truth, Sceptics or inquirers because they were always looking for a solution and never finding one, Ephectics or doubters because of the state of mind which followed their inquiry, I mean, suspense of judgment, and finally Aporetics or those in perplexity, for not only they but even the dogmatic philosophers themselves in their turn were often perplexed. (IX,70)

Like the term "sophist," "sceptic" acquired the meaning we would normally associate with it only gradually.

B. Timon of Philus

Pyrrho had a number of disciples whose names are known to us. One Eurylochos who is said to have fled from the questions of his students, plunged into the sea, and swum away; one Philo of Athens who if he was a teacher was himself his only pupil -- both anecdotes indicate how difficult it was to attain the Pyrrhonian ideal of avoiding all philosophical wrangling. Timon Phlius (c. 325 - c. 235 B.C.) is considered the more direct successor of Pyrrho. He is said to have started as a dancer, dropped that and gone to hear Stilpo at Megara; returning home, he married and went with his wife to Elis to hear Pyrrho. He went to Chalcedon to teach, went thence to Athens which became his permanent home where he died. Timon's way of following Pyrrho was unlike that of Eurylochos and Philo: Timon did not embrace poverty, liked his wine and was a contentious soul. He is known as the sillographer because of the lampoons he wrote with, philosophers as his target. Indeed, it appears that Timon was a prolific writer: epic poems, tragedies, satires, comedies and other works are attributed to him. We possess only a few fragments of his writings, drawn from *Images* and the *Lampoons* (*Silli*). The latter was, at least in part, a poem whose sub-title might have been "Timon in Hades" where he poses questions to Xenophanes concerning ancient and modern philosophers and elicits mordant and unflattering descriptions. There was as well a battle of the philosophers in which, we may suppose, they were allowed to show forth their absurdity. It also seems likely that Timon distinguished the dogmatic and anti-dogmatic philosophers, and locating the figures of the New Academy, notably Arcesilaus, among the latter, was able to point out their borrowings from Pyrrho. Certain philosophers are treated somewhat gently by Timon -- Democritus, of course, but also the Eleatics, Protagoras. It goes without saying that he held Xenophanes in esteem. Apart from his willingness to involve himself in philosophical wrangling, at least to the point of lampooning, and his rejection of the Stoic *katalepsis*, Timon serves only to indicate the as yet negative side of Scepticism. As a philosophical doctrine Scepticism undergoes change at the hands of the New Academy.

C. Arcesilaus

Arcesilaus was born at Pitane in Aeolia about 315 B.C. He studied mathematics in his native city and then came to Athens with the intention of studying

rhetoric. Taken by philosophy, he studied under Theophrastus and then Crantor. He became quite devoted to the latter and continued in the Academy after Crantor's death, listening to Polemo and Crates as well. After the death of Crates, Arcesilaus took over the headship of the Academy. He lived to the age of seventy-five. Arcesilaus was a wealthy man and his mode of life was anything but austere. He had a good number of enemies; Timon of Phlius maligned him while he was alive, but praised him after his death. Epicurus is said by Plutarch to have been jealous of Arcesilaus' fame. Arcesilaus delighted in attacking the Stoics; nevertheless Cleanthes is said to have defended him, saying that his actions made up for what was lacking in his teaching. Arcesilaus wrote nothing and we are possessed of quite scanty information about his doctrine. What information we do have indicates that Arcesilaus was most concerned with refuting the Stoic claim to certitude in knowledge.

We have seen that the Stoics held that the first task of philosophy is to provide a criterion of truth, something they found in the comprehensive representation, the representation which commanded assent. That is, this representation is so clear and precise that it cannot be confused with anything else and thus bears within itself the confirmation of the truth of its object. Such representations were compared to an open hand by Zeno, and they are the first degree of knowledge. These representations elicit from the superior part of the soul an assent which, while a response to a stimulus, comes from a willed act. The soul cannot fail to give this assent when confronted with a comprehensive representation, and its assent is the second degree of knowledge, i.e., a hand with fingers partially bent. Comprehension (*katalepsis*) is represented by a fist and science by the fist clasped by the other hand. Thus, the wise man is defined ultimately in terms of comprehensive representations: if this first degree of knowledge cannot be defended, the Stoic theory collapses in its entirety. Arcesilaus, accordingly, addresses himself precisely to the doctrine of the comprehensive representation.

In questioning the comprehensive representation, Arcesilaus first attacks the notion of assent, for it is the assent to the representation which makes it comprehensive. Now, the objection runs, the assent is said to be produced by the will and is prior to or constitutive of knowledge and not consequent upon it. True assent, however, should follow on knowledge and the Stoic assent is something precipitous and unjustified. Moreover, this doctrine of assent is incompatible with the Stoic notion of the wise man. If assent is necessary to found knowledge, it must precede it; consequently the result of assent is not

knowledge but opinion. However, the Stoics speak of the wise man who gives his assent only to the truth.

As to the notion of comprehensive representation itself, the Stoics would not want to make it the prerogative of the wise man and accordingly sometimes speak of it as between opinion and knowledge. This goes contrary to other statements which seem to say that only the wise man has such comprehensive representations. Apart from this inconsistency, Arcesilaus rejects the comprehensive representation itself as contradictory. Such a representation implies approval or assent if it is to be spoken of as always true; but judgment and approval are acts of reason, not of the senses. That is, if certain sense representations are said to be always true, truth is not something which belongs to them as sense representations, but is, as it were, superadded by reason. It seems likely that Arcesilaus also rejected the idea that there can be sensation which necessarily elicits the mental judgment that there are correlates of it in the real world, by appealing to dreams, fantasies and optical illusions. At times, such representations are equally irresistible to reason which gives assent to them. Consequently, the ideal must be the suspension (*epoche*) of judgment; the wise man despairs of ever possessing absolutely certain knowledge.

Like the earlier Sceptics, Arcesilaus' doctrine is a negative one; it goes beyond them in being dialectical, in delighting to take up the opinions of others and show they cannot command assent. Arcesilaus shies away from proposing anything like a positive doctrine, and is shrewd enough to agree that he cannot even be certain that he can be certain of nothing. But this is not the sum total of what we know of Arcesilaus. The earlier Sceptics saw the ideal of indifference and suspension of judgment as a way out of controversies which upset and distress; in a word, their negative approach was to be productive of happiness. The dialectical approach of Arcesilaus made it imperative that he answer his Stoic critics as to how action is possible if we know nothing for certain. If knowledge, both sensible and rational, is called into question, how can we perform the simplest tasks of our daily lives?

Arcesilaus, however . . . certainly seems to me to have shared the doctrines of Pyrrho, so that his Way of thought is almost identical with ours. For we do not find him making any assertion about the reality or unreality of anything, nor does he prefer any one thing to another in point of probability or improbability, but suspends judgment about all. He also says that the End is suspension -- which is accompanied, as we have said, by 'quietude.' He declares too that

suspension regarding particular objects is good, but assent regarding particulars bad. (Sextus, *Outlines of Pyrrhonism*, I, 232-3; tr. Bury)

Arcesilaus rejects probability in the sense that he denies that any representation can outweigh another; they are of equal value and one is no more worthy of assent than another. But he does admit a scale in terms of which choice is possible, and the founding note of this scale is the reasonable (*eulogon*). Thus, while rejecting any canon of truth, of knowledge as such, Arcesilaus introduces the reasonable as a canon or criterion of choice. In order to act, we do not require certain knowledge; indeed perceptions can influence the will without reason judging that they are absolutely true. The reasonable would seem to amount to a justification of action in terms of consistency but without any pretense that the statements involved in such a justification are any truer than their contradictories. The doctrine of the reasonable is thought to have been as much another attack on the Stoics as a positive doctrine of Arcesilaus; this attack won from the Stoics the concession that in acting the wise man does not rely on certain knowledge.

Before ending this discussion of Arcesilaus, something must be said of the view that his scepticism was a public stance and a device for testing students for entry into the profound reaches of Plato's philosophy.

And if one ought to credit also what is said about him, he appeared at the first glance, they say, to be a Pyrrhonean, but in reality he was a dogmatist; and because he used to test his companions by means of dubitation to see if they were fitted by nature for the reception of the Platonic dogmas, he was thought to be a dubitative philosopher, but he actually passed on to such of his companions as were naturally gifted the dogmas of Plato. (Sextus, *ibid.*, 234; Bury)

This supposition of a dogmatic teaching at the interior of the school is fairly universally rejected by modern scholars: it is put forward by those who want to denigrate Arcesilaus or to mitigate his scepticism. Sceptics who resented the Academy's intrusion into their domain countered with the claim that the scepticism of Arcesilaus was largely for external consumption and that his true interest was to pass along the dogmas of Plato; Platonists, distressed by Arcesilaus' negations, found the hypothesis pleasant for obvious reasons. But if Arcesilaus is not to be thought of as teaching the positive philosophical doctrines of Plato, whose successor after all he was, it is argued that he did preserve and pass on to others the method of dialectics and that he doubtless

used the dialogues themselves for this purpose. Since Xenocrates had extracted the "dogmas" from the dialogues, and this retail version was widely known, opponents hearing that Arcesilaus made use of Plato would think of the doctrinal précis of Xenocrates and formulate the accusation of dogmatism among the initiate of the supposedly sceptical Academy.

D. Carneades of Cyrene

Carneades was born about 219 B.C. and is said to have lived eighty-five years. A certain date in his life is 156/5 when in the company of Diogenes of Babylon and Critolaus, the embassy of philosophers, he visited Rome. On this occasion, Carneades presented the arguments of Aristotle, Plato and the Stoa on justice, and the following day refuted them all. It was a dazzling performance and is thought to be indicative of Carneades' unusual procedure. Carneades wrote nothing; he studied under the Stoic, Diogenes of Babylon, was greatly influenced by the writings of Chrysippus, but joined the Academy and became its head. He is sometimes called the third founder of the Academy, (Arcesilaus being the second).

Carneades carries on the refutation of the Stoic theory of knowledge, repeating some of the arguments of Arcesilaus and adding some of his own. These have to do with the perception of sense qualities. Carneades wants to show that sight does not perceive color; by this he means that we never see the color of the object just as such, since its color can be seen to vary depending on changes in the object and changes in the viewer's position. We are aware of these variations in our sensations of a particular object, but we can never know what its color truly is. Sense qualities, in other words, are relative to the one perceiving them and the conditions in which he finds himself. There is no comprehensive representation which would be its own guarantee of truth and a foundation for science.

Carneades launched a direct attack on the Stoic dialectic which is of no little interest. We have seen that in the Stoic logic every proposition is true or false. Carneades borrows the paradox of the liar from the Megarians to contest this claim. If you say that you are lying and it is true that you are lying have you lied or are you telling the truth? The reply of Chrysippus was that this is an insoluble paradox, an exception, which does not disturb logic as such. Carneades will not accept this as a reply: one cannot simultaneously maintain that every proposition is true or false and that there are exceptions to this rule. Carneades makes use of the same difficulty to question the view that a syllogistic (in the

Stoic usage) form guarantees the validity of an inference. The Stoic would say, "If you say that it is now day and this is true, it has to be day; but you say it is day, this is true, therefore it really is day." Carneades asks if the following is equally irreproachable. "If you say that you are lying and, in saying it are telling the truth, you are lying; but you say that you lie and are telling the truth, therefore in telling the truth you are lying." There is a relatively easy way to get rid of the difficulty posed by this paradox, for one might point out that "I am lying" and "it is daytime" are not on the same level, since the first statement must be attached to another to have any significance, i.e., it is about a proposition or propositions in the way that the second is not. Carneades, receiving no effective reply, used the difficulty to call into question the validity of logic. Nor was he at all half-hearted in his rejections. He is said to have regarded the mathematical proposition, "If equals are added to equals the result is equals" as dubious; so too "Two quantities each of which is equal to a third are equal to one another" is not necessarily true. In effect, nothing is certain, neither in the order of sense nor in that of reason. This claim is directed against every positive philosophy and not simply against the Stoics. We have the individual impression (*pathe*) and have no way of distinguishing one from the other in such a way that some are seen clearly to stand for external realities, and others not.

Carneades, then, teaches the absence of all certitude in knowledge; there is no criterion of truth, no comprehensive representation. There is on this score a definite continuity between him and Arcesilaus. A doctrine peculiar to Carneades is that of the probable (*pithanon*) which demands comparison with what Arcesilaus called the reasonable (*eulogon*). Brochard^[49] has drawn attention to the fact that there are conflicting accounts of Carneades' doctrine on the suspension of judgment (*epoche*), one stemming from Clitomachus, the pupil of Carneades, the other from Metrodorus. According to Clitomachus the *epoche* can be understood as meaning that the wise man affirms nothing, or it can mean that the wise man, while affirming nothing, prefers certain representations as being more likely. In action, of course, the wise man must choose, but this does not entail having opinions or giving assent to what is not certain. According to Metrodorus, on the other hand, Carneades did not hesitate to give assent to representations which are not certain. Thus, following Metrodorus, Carneades would appear in a midway position with respect to Arcesilaus and the Stoics. These would agree that the wise man gives his assent only to true representations and would disagree on whether there are any; Arcesilaus, feeling there are none, counseled a universal suspension. Metrodorus suggests that Carneades would let the suspension go and assent to what would be recognized

as only opinions. It is this suspension of the suspension which seems to have led Carneades to his doctrine of the probable.

Sextus Empiricus reports that the probable was possessed of degrees.

And respecting the probable impressions they make distinctions: some they regard as just simply probable, others as probable and tested, others as probable, tested, and "irreversible." For example, when a rope is lying coiled up in a dark room, to one who enters hurriedly it presents the simply "probable" appearance of being a serpent; but to the man who has looked carefully round and has investigated the conditions -- such as its immobility and its color, and each of its other peculiarities -- it appears as a rope, in accordance with an impression that is probable and tested. And the impression that is also "irreversible" or incontrovertible is of this kind. When Alcestis had died, Heracles, it is said, brought her up again from Hades and showed her to Admetus, who received an impression of Alcestis that was probable and tested; since, however, he knew that she was dead his mind recoiled from its assent and reverted to unbelief. So then the philosophers of the New Academy prefer the probable and tested impression to the simply probable, and to both of these the impression that is probable and tested and irreversible. (*Outlines*, I, 227-9; Bury)

From the point of view of the subject, not all impressions are of equal value, some have more probability than others and can thus merit our assent. More importantly, what is called the probable seems to involve a whole nest of impressions and the degrees of probability seem to be read in terms of the compatibility of various impressions. It is thought that Carneades was not attempting to judge the relationship of impressions and external objects, so much as the subjective differences among impressions. At times we cannot put an impression to the test by comparing it with others; when we do have time for this, an incompatibility may emerge, or a compatibility, in terms of which what was originally probable becomes less or more so. The notion of the probable, of giving assent to what is recognized to be merely an opinion, does not seem to have been simply a response to the exigencies of the practical life. Whether it is a question of actions to be performed or of speculative positions, Carneades is able to examine them in terms of the probable and make his choice. In this way, for example, he can assent to the proposition that nothing is certain as to something probable and thereby avoid a difficulty.

In destroying, to his own satisfaction, the Stoic theory of knowledge, Carneades had toppled the whole system; nevertheless, as we have seen, he did go on to

discuss its logic. It must also be said that he argued against the notion that the cosmos is an intelligent being, against the Stoic attributions of divinity and acceptance of the popular gods and against the notion of fate or providence.

E. Some Later Sceptics

The immediate successor of Carneades as head of the Academy was Clitomachus, followed by Philo of Larisa who was listened to in Rome by Cicero in 87 B.C. The successor of Philo, Antiochus, rejects scepticism, attempts to reconcile Platonism and Aristotelianism and, according to Cicero, is in reality the most authentic Stoic, meaning doubtless a Stoic of the stripe of Panaetius and Posidonius. With the death of Antiochus, the Academy ceases to have adherents at Athens, according to Cicero; its immediate continuation in the Greek world is to be found at Alexandria in Egypt. At the end of this chapter we will give a brief indication of the subsequent history of the Platonic Academy.

We shall not be detained by the difficulties which attend any attempt to trace the history of the Sceptic school after Timon of Phlius. We have already seen that Pyrrhonism was fairly effectively usurped by the Academy and that it flourished there until the time of Antiochus. Diogenes Laertius (IX, 116) suggests a continuity of heads of the Sceptic school, but scholars are agreed that acceptance of this chronology involves insuperable difficulties. Accordingly, we shall content ourselves with a brief mention of two of the most important later Sceptics, each of them separated by a large temporal gap from one another and from the earliest non-Academic sceptics.

1) Aenesidemus of Cnossus

Very little is known of the life of Aenesidemus; a good deal is known of his teachings. A native of Cnossus, on Crete, he is thought to have been alive in the first century before Christ. He taught at Alexandria in Egypt. He was the author of *Pyrrhonian Discourses*, *Against Wisdom*, *On Inquiry* and perhaps several other works. We possess information of the content of the *Discourses*. Aenesidemus is intent on showing that scepticism and the Academy must not be confused; the Platonists are essentially dogmatists, he feels, whereas the sceptic is never certain that something is true or not true. He always and everywhere suspends judgment.

Aenesidemus argued against the possibility of truth, causality and proof, and is famous for his doctrine of the ten tropes, or modes by which suspension of judgment can be brought about. As to truth, he argues that it cannot be sensible, cannot be intelligible and cannot be both. (Cf. Sextus, *Ad. Log.* II, 40-47) His arguments against causality consist in showing that body cannot cause body, nor the incorporeal the incorporeal, nor can body cause the incorporeal or vice versa. (Sextus, *Ad. Phys.* I, 218-226) Various arguments against the possibility of proof which are set forth by Sextus Empiricus are taken, as by De Vogel^[50] to have Aenesidemus as their source. The arguments against proof attempt to show that the propositions from which something is shown are themselves in need of proof and that, ultimately, the validity of proof itself is in need of proof. Nor will the sceptic accept the admission that not everything stands in need of proof nor can be proved.

But, say they, one ought not to ask for proof of everything, but accept some things by assumption, since the argument will not be able to go forward unless it be granted that there is something which is of itself trustworthy. But we shall reply, firstly, that there is no necessity for their dogmatic argumentations to go forward, fictitious as they are. And, further, to what conclusion will they proceed? for as apparent things merely establish the fact that they appear, and are not capable also of showing that they subsist, let us assume also that the premisses of the proof appear, and the conclusion likewise. But even so the matter in question will not be deduced, nor will the truth be introduced, so long as we abide by our bare assertion and our own affection. And the attempt to establish that apparent things not merely appear but also subsist is the act of men who are not satisfied with what is necessary for practical purposes but are eager also to assume hastily what is possible. (Sextus, *Ad. Log.*, II, 367-8)

But, if assumption lies at the beginning, is what is assumed trustworthy because it is assumed, and if not, why say that the true is such by assumption?

The ten modes attributed to Aenesidemus have the same object as any sceptic device, to show that affirmation is as ungrounded as negation in the same matter. This leads to the suspension of judgment. Thus, appearances are opposed to objects of thought, or appearances to appearances, thoughts to thoughts. A tower appears round from a distance, square close up; the order of the heavens induces to a belief in providence, the sufferings of the good calls providence in question. Let us see how the ten modes of Aenesidemus operate.

(1) The first mode is based on the variety among animals, e.g., as to sense organs, some being more keen sighted than others, some having a more acute sense of smell. Thus, it would seem that when confronted with the same objects, they have different impressions. On what basis would preference be given to one impression over another?

(2) The second mode has to do with differences among men, on the assumption that, faced with the difficulties of the first mode, one retorted that the impressions of men are to be taken as normative. But are men so much alike? Some men sweat in the shade and shiver in the sun; some go a long time without water, others not. (3) The third mode is based on the differences between the senses: to sight, an apple is red; to taste, sweet, and so forth. The eye finds dimensions in a painting which touch does not; some things are pleasant to taste and not to smell, and vice versa.

(4) The fourth mode is based on differences of condition: things appear differently to us depending on whether we are well or ill, sleeping or awake, young or old, happy or sad, etc.

(5) The fifth mode relies on the different customs of men with respect to what is beautiful or ugly, good or bad, true or false. A Persian father may marry his daughter while this shocks the Greek.

(6) The sixth mode takes its rise from the fact that a color differs in moonlight, sunlight and lamplight; that a stone in water can be moved by one man, while two are needed to lift it on land.

(7) The seventh mode proceeds by observing that in different positions and from different distances things appear differently (e.g. the sun at rising, at noon, at sundown), and concludes that we never know them as they are in themselves.

(8) The eighth mode notes how the properties of things are said to vary in quantity and quality: thus what is hot or cold is not absolute, but relative to us and our condition, as is the amount of wine which is healthful.

(9) The ninth mode points out that things are called rare for subjective reasons. For some, earthquakes are rare occurrences, for others usual.

(10) The tenth mode has to do with relatives of all kinds, and suggests that, since we can know them only with reference to something else, we cannot know them in themselves.

Sextus Empiricus devotes much time to these modes (*Outlines*, I 36-163), which he lists in a slightly different order than that of Diogenes Laertius whose order we have followed. Sextus feels these tropes are based on differences in the judging subject, on variances in the object, or on both, and he groups the tropes accordingly.

2) *Sextus Empiricus*

Sextus Empiricus is thought to have lived in the second century of our era; he was head of the Sceptic school, a physician (hence the Empiricus, denoting a particular approach to that art). It is difficult to say where he lived; he exhibits knowledge of Rome, Athens and Alexandria. For our purposes, his main interest is the fact that he is our primary source for the Sceptic school in general. As it happens, his summary of sceptic attacks on the Stoics makes him a source for Stoic doctrine as well. We shall confine our comment here to the purely bibliographical level. This is justified in any case, since Sextus is primarily a compiler of the teachings of the school he represents and we have been relying on him heavily for that teaching.

The surviving works of Sextus can be thought of as two in number, *Outlines of Pyrrhonism* and *Adversus Mathematicos*. The latter is often subdivided, as in the Loeb Classical Library edition by Bury, something which makes references hard to verify by the beginner when equivalences are not given. In that edition, volume one contains the *Outlines*; volume two contains two books "Against the Logicians" which are books VII and VIII *Adversus Mathematicos*, respectively, the first six books being found in volume four. Volume three contains two books "Against the Physicists" (IX and X of *Adversus Mathematicos*) and a book "Against the Ethicists" (XI).

F. The Subsequent History of the Academy

With Antiochus, the identification of the Academy with Scepticism ceased and his eclectic efforts to show the fundamental agreement of Stoicism, Aristotelianism and Platonism was carried on at Alexandria, whence Antiochus had gone to Athens to succeed Philo as head of the school. The Athenian Academy seems to have broken up after the death of Antiochus, but his influence remained with what is called the eclectic Academy in Alexandria under the leadership of Eudorus (c. 25 B.C.) Eudorus is thought possibly to have written a commentary on the *Timaeus* and on the *Metaphysics* of Aristotle, as well as a work against the *Categories* of the latter. He is also said to have

written a work on the division of philosophy. Eudorus was responsible for a new edition of the dialogues of Plato, arranged in tetralogies, as well as a collected edition of Democritus.

Perhaps the most famous of later Platonists is Plutarch of Chaeroneia (45 A.D. - c. 125 A.D.) in Boetia. He studied in Athens and possibly in Alexandria, returned to his home town to what appears to have been a leisurely life, although he gave lectures. Plutarch went to Rome in 90 A.D. on public business and lectured in Rome; the lectures were later published as the *Moralia*. Once more at Chaeroneia, he wrote his *Parallel Lives of Illustrious Greeks and Romans*. Plutarch was a deeply religious man and was a priest of Apollo at Delphi. His *Lives* have exerted great influence, not least because they are a source for many of Shakespeare's plays. Plutarch is regarded as a forerunner of Neoplatonism. As against the Stoics who identified God with the world, Plutarch teaches God's transcendence. Only God truly is, since he is unchangeable; so too only God is truly one. God is the highest principle and is goodness; to account for evil in the world, Plutarch introduces a principle other than God. Moreover, he introduces various daimons or spirits as intermediaries between God and man. In his ethics, Plutarch gives love of one's fellow men, philanthropy, as the sum of the virtues.

In speaking of the Academy of the second century of our era, we are no longer speaking of a group of men at Athens; some Academicians were at Athens, of course, e.g. Atticus, but of equal if not greater importance are such men as Albinus and Theon of Smyrna and Gaius; at Alexandria there was, perhaps, Celsus, who made a written attack on Christianity, and Ammonius Saccas, a Christian and teacher of Plotinus. Maximus of Tyre was teaching philosophy at Rome during this period. Apuleius of Madaura expounded the Platonic philosophy in a number of works written in Latin. Though some of these men are doubtless of some importance, that importance is eclipsed by Neoplatonism.

{48} Bevan, *Stoics and Sceptics* (New York: Barnes & Noble, 1959).

{49} Victor Brochard, *Les Sceptiques Grecs* (Paris: Imprimerie Nationale, 1932), pp. 133-4.

{50} C. J. DeVogel, *Greek Philosophy*, Vol. III, (Leiden: Brill, 1959), pp. 221ff.

Chapter IV

Neoplatonism

In an earlier chapter we have seen that in the later Platonic Academy attempts were made to reconcile the Plato's thought with that of other schools, particularly the Peripatetic and the Stoic. It is this tendency above all others which is characteristic of what is called Neoplatonism, a movement of thought which can be taken to achieve its zenith with Plotinus. Before turning to Plotinus himself, we must mention a number of significant events in the first centuries of our era.

A. Revival of Pythagoreanism.

We have ample evidence that a revival of Pythagoreanism took place, perhaps in the first century B.C. Diogenes Laertius records a brief account of Alexander Polyhistor (VIII, 25-35) which begins as follows. "The principle of all things is the monad or unit; arising from this monad the undefined dyad or two serves as a material substratum to the monad, which is cause; from the monad and the undefined dyad spring numbers; from numbers, points; from points, lines; from lines, plane figures; from plane figures, solid figures; from solid figures, sensible bodies, the elements of which are four, fire, water, earth and air; these elements interchange and turn into one another completely, and combine to produce a universe animate, intelligent, spherical, with the earth at its center, the earth itself too being spherical and inhabited round about." What is of present interest in Neopythagoreanism is this notion of all things flowing from the One which prefigures the theories of emanation we shall find in other late schools.

Moreover, coupled with an exaggerated reverence for Pythagoras and a revival of the religious practises of the early school, is the effort to apply

Pythagoreanism to other philosophical schools. Thus, we find Pythagorean interpretations of Plato. Between the One and the world, there are daemons whose function it is to govern the world. Nicomachus of Gerasa who lived in the middle of the second century of our era and who wrote an *Introduction to Arithmetic* can be counted among the Neopythagoreans. He is noteworthy for his insistence on knowledge of mathematics as a prerequisite for wisdom and for locating numbers in the mind of God. The order and harmony of the universe indicate that it is patterned on numbers and their proportions which accordingly must exist in the mind of the fashioner of the world. (Cf. DeVogel, III, 2388b)

The Hermetic writings (*Corpus Hermeticum*) were probably collected around 300 A.D. These are attributed to Hermes Trismegistus, a name apparently derived from an attempt to identify the Greek god Hermes and the Egyptian god Tat. The section of the *Corpus* known as Poimander seems to represent a type of gnosticism, that is, an amalgam of Christianity and paganism, although the Hermetic writings contain much less Christian reference than popular gnostic writings. The production of the world is explained in the following fashion. A part of God breaks off and becomes chaotic nature on which a Logos also proceeding from God descends, separating the elements from one another. The Father God is present in the cosmos in the form of innumerable powers which seem to be akin to the Platonic Ideas which are located apparently in the divine mind. The original God and the Logos fashion a third god, a demiurge who is also an intellect and the fashioner of the heavenly bodies which govern the sensible world. The first god fashions man equal to himself, while other living things are made by the second and third gods. When man comes to see his origin, he faces the task of returning to it: the route back is by way of purgation as well as by an illumination from God. In the Hermetic writings there is a tension between reunion with God by means of knowledge of a scientific type and by means of a mystical vision. Of interest is the notion of intermediaries between God and the sensible world and the ambiguity involved in treating the sensible world as evil in itself but also as an image of God. Man is a compound of light and matter and his task is to free himself from the clutches of the body and matter and return to his source: this ascent becomes a matter of celestial geography, a movement from sphere to sphere and then beyond.

Numenius of Apameia in Syria, who lived in the second half of the second century of our era, also speaks of three gods, the father, the maker and the made. The first god is transcendent and has nothing to do with the formation of the cosmos; the second is the demiurge, the cause of becoming who forms the world. The third god is the formed world. The first god is absolutely one and is identified with Plato's God as well as with Aristotle's self-thinking thought. If the first god is transcendent and indifferent to the cosmos, the demiurge is not; it is by his thinking of man that man is kept alive. There is a dispute as to whether the third god is to be taken to be the sensible world itself or the conception of the world in the mind of the demiurge. Matter itself is conceived to be positively evil by Numenius; consequently, the soul's union with body is an evil and death a liberation which permits the soul to rejoin its principle.

Mention must be made here of the Jewish philosophers of antiquity, particularly those at Alexandria, although with them a factor is introduced which sets them definitely off from all other thinkers we have considered, a factor which gives them far greater affinity with the men to be considered in the next volume of this series. This can best be exemplified by Philo Judaeus (c. 25 B.C. - 40 A.D.), a man who accepted as revealed truth the Jewish Scriptures and was at the same time drawn to the philosophy of the Greeks. This led him to interpret the Scriptures in terms of Greek philosophy, an effort actually begun, according to Wolfson,^[51] when the Pentateuch was translated into Greek around 260 B.C. The effort of Philo carries with it difficulties which we shall not undertake to discuss here, preferring to postpone the matter until our consideration of the confrontation of philosophy on the part of the Christians. At that time, we shall endeavor to sketch the main lines of the effort of Philo Judaeus. For the present, we must turn to Plotinus.

B. Plotinus

Plotinus was born in Egypt (we are not sure of the city) in 203/4 A.D. He studied at Alexandria, spending many years as the pupil of Ammonias Saccas who is said to have been the founder of Neoplatonism. When he was forty, Plotinus came to Rome where he was a very successful teacher, numbering senators among his students and acquiring the favor of the Emperor himself. When Plotinus was sixty, Porphyry became his student and we are indebted to the latter not only for a life of Plotinus but also for the present form of the writings. Porphyry arranged the writings of his master into six groups each containing nine tractates; it is because each book contains nine tractates that the work is known as the *Enneads*. Each tractate is divided into chapters but there is no uniform number of these. References to Plotinus involve the citation of the *Ennead*, the tractate, the chapter; e.g., I, ii, 1. Porphyry tells us that Plotinus was a most effective teacher. Plotinus, as might be expected from his doctrine, was a deeply spiritual man, ascetic in his habits, who according to Porphyry, achieved ecstatic union with God a number of times during the six years Porphyry was his pupil. There are no references to Christianity in Plotinus, though he does criticize the Gnostics. We are told that Plotinus had an unrealized hope of founding a city to be

called Platopolis in which a perfect society could flourish. In the *Enneads*, Plotinus sets forth the structure of the world and our place in that structure, the manner in which all things proceed from the One and return once more to it.

There is multiplicity and unity, consequently, and Plotinus speaks of the unity in a striking fashion.

The elements in their totality, as they stand produced, may be thought of as one spheric figure; this cannot be the piecemeal product of many makers each working from some one point on some other portion. There must be one cause; and this must operate as an entire, not by part executing part; otherwise we are brought back to a plurality of makers. (VI,v,9)

The many things which are in the universe must refer back to one principle and this one principle contains everything within itself. The procession of all things from the One is a necessary procession; each level of reality proceeds necessarily from its superior as the first proceeds necessarily from the one. If we call this procession creation, a creation in time is excluded from the Plotinian universe; any priority and posteriority is based on nature rather than temporal sequence. In the beginning was everything, but everything is so structured that there is a first from which all things proceed, if only mediately. Thus, if Plotinus speaks of production and procession, it is an ontological dependence only that he wants to speak of; time has nothing to do with it and the processions are not results of acts of will. This is a sketch of the Plotinian universe:

The One is all things and no one of them; the source of all things is not all things; all things are its possession -- running back, so to speak, to it -- or, more correctly, not yet so, they will be. But a universe from an unbroken unity, in which there appears no diversity, nor even duality? It is precisely because that is nothing within the One that all things are from it: in order that Being may be brought about, the source must be no Being but Being's generator, in what is to be thought of as the primal act of generation. Seeking nothing, possessing nothing, lacking nothing, the One is perfect and, in our metaphor, has overflowed, and its exuberance has produced the new: this product has turned again to its begetter and been filled and has because its contemplator and so an Intellectual-Principle (*Nous*). That station towards the one (the fact that something exists in presence of the One) establishes Being; that vision directed upon the One to the end of vision, it is simultaneously Intellectual-Principle and Being; and, attaining resemblance in virtue of this vision, it repeats the act of the One in pouring forth a vast power. This second outflow is a Form or Idea representing the Divine Intellect as the Divine Intellect represented its own prior, the One. This active power sprung from essence (from the Intellectual-Principle considered as Being) is Soul. Soul arises as the idea and act of the motionless Intellectual-Principle -- which itself sprang from its own motionless

prior -- but the soul's operation is not similarly motionless; its image is generated from its movement. It takes fullness by looking to its source; but it generates its image by adopting another, a downward, movement. This image of the Soul is Sense and Nature, the vegetal principle. Nothing however is completely severed from its prior. (V,ii,1)

We have here the three hypostases of Plotinus: the One, *Nous* and Soul. Elsewhere Plotinus speaks of them as light, sun and moon respectively. (V, vi, 4) *Nous* has received the light into its very essence, but above *Nous* must be that which gives the light. Soul has an essentially borrowed light. Plotinus argues for the exhaustiveness of these, arguing that there can be neither more nor less. (Cf. II, ix, 1). We must examine each of the hypostases in turn, Soul, *Nous*, One. We adopt this order because the philosophy of Plotinus is not merely the depiction of the hierarchy of the universe, but an exhortation to man to mount to the first principle.

Therefore, first let each become godlike and each beautiful who cares to see God and Beauty. So, mounting, the Soul will come first to the Intellectual-Principle and survey all the beautiful Ideas in the Supreme and will avow that this is Beauty, that the Ideas are Beauty . . . What is beyond the Intellectual-Principle we affirm to be the nature of Good radiating Beauty before it. (I,vi,9)

The Term at which we must arrive we may take as agreed: we have established elsewhere, by many considerations, that our journey is to the Good, the Primal-Principle; and indeed the very reasoning which discovered the Term was itself something like an initiation. (I,iii,1)

Soul. Soul occupies a middle position between *Nous* and the corporeal world; it is the reflection of the former and the organizer of the latter. As we have just seen, Plotinus is urging us to ascend to the first principle, something which would seem to involve a turning away from bodies; nevertheless, soul has a function relative to bodies. This introduces a kind of tension into the Plotinian conception of soul, although there is no doubt as to which region the soul itself belongs. "In that allocation we were distinguishing things as they fall under the intellectual or the sensible, and we placed the soul in the former class." (IV, ii, 1) It is because of its reference to body that the soul can be said to be divisible.

The nature at once divisible and indivisible, which we affirm to be the soul has not the unity of an extended thing: it does not consist of separate sections; its

divisibility lies in its presence at every point of the recipient, but it is indivisible as dwelling entire in the total and entire in any part. (IV,ii,1)

Plotinus is not in agreement with Aristotle's teaching that the soul is the form or entelechy of the body. His arguments against this are not all on the same level. On the one hand, he says that if soul were so wedded to body as form to matter, sleep would be impossible, sleep being conceived as the soul's withdrawal upwards. On the other hand, Plotinus holds that the soul serves as a principle of organization for many bodies, though successively; consequently, it does not come to be with the coming into being of the living thing.

The substantial existence of the soul, then, does not depend upon serving as form to anything: it is an essence which does not come into being by finding a seat in body; it exists before it becomes the soul of some particular, for example, of a living being, whose body would by this doctrine be the author of its soul. (IV, vii, 8E)

The soul is essence or substance (*ousia*), the permanent, whereas body and generally the realm of the corporeal is process and change; the corporeal comes and goes and is not so much being as it participates in being, a capacity to participate in what authentically is, namely, the soul. The soul itself has a superior and inferior part, the former looking to *Nous*, the latter to matter which is formed in the image of the soul just as soul is an imitation proceeding from its superior, *Nous*.

If soul acts as a genus or species, the various (particular) souls must act as species. Their activities will be twofold: the activity upward is intellect; that which looks downwards constitutes the specifically different powers; the lowest activity of soul is in its contact with matter to which it brings form. (VI,ii,22)

Thus far what we have said would indicate that Plotinus wishes to speak of the human soul somewhat after the manner of Aristotle, though with disagreement as to the notion of soul as form of the living body. But soul, for Plotinus, is one of the three hypostases, and in his hierarchical universe "soul" has four meanings. There is first of all a transcendent soul, proximate to *Nous*, and, secondly, there is the soul of the visible world which, in its totality, is a living creature. The human soul parallels this division, there being a superior and inferior human soul. These notions bring us close to Plato, of course, but Plotinus is careful to point out that he does not subscribe to the alleged purport of the *Philebus* to the effect that particular souls are simply parts of the

universal soul; the *Timaeus* makes it clear that other souls are of the same nature as the universal soul but distinct from it. (IV,iii,7) The suggestion is made that we form part of the cosmic order thanks to the inferior part of our soul, but transcend that order because of the superior part of the human soul which is destined for union with the first principle. Nevertheless, the plurality of souls is not a function of their union with bodies; the plurality pertains to the intelligible order itself.

The distinctive character of soul is not thought but ordering and governing. It is this which distinguishes soul from *Nous* and explains its progression from *Nous*.

There is the *Nous* which remains among the intellectual beings, living the purely intellectual life; and this, knowing no impulse or appetite, is for ever stationary in that realm. But immediately following upon it, there is that which has acquired appetite, and, by this accrument, has already taken a great step outward; it has the desire of elaborating order on the model of what it has seen in the *Nous*: pregnant by those beings, and in pain to the birth, it is eager to make, to create. (IV,viii,13)

This passage will become clearer when we have examined the nature of *Nous*; for the moment, we can point out the kinship between the Plotinian soul and what Aristotle called the practical intellect. Soul for Plotinus looks outward, fashions matter to the image of what has been contemplated in *Nous*, the realm of what Plato called the Ideas or forms. Thus, the cosmic order is due to a world soul; organic bodies are due to particular souls. Soul, for Plotinus, is the source of providence.

We can see that Plotinus speaks of the downward movement of soul, its emanation from *Nous* and movement towards matter as organizing principle, as the natural function of the soul. And yet the soul is called to transcendence, to retracing the stages of procession and to reunion with the first principle. Thus, while perfectly natural, the soul's involvement with the corporeal constitutes a danger. What the human soul must do is to seek safety in a return to the universal soul; in union with the universal soul it can exercise governance without care or trouble. The body is not necessarily the prison of the soul, but it can become so if the soul should so concentrate on the particulars of this world as to become forgetful of its origin and destiny. But even in this fallen state there is hope.

But in spite of all it has, for ever, something transcendent: by a conversion towards the intellective act, it is loosed from the shackles and soars -- when only it makes its memories the starting point of a new vision of essential being. Souls that take this way have place in both spheres, living of necessity the life there and the life here by turns, the upper life reigning in those able to consort more continuously with the divine Intellect, the lower dominant where character or circumstances are less favorable. (IV,viii,5)

The soul by its very nature is divine, and evil must be looked upon as an accretion to the soul due to its commerce with the corporeal. Goodness is of the essence of the soul, evil is accidental and not constitutive. For Plotinus it is hardly necessary to argue that the soul is immortal; it is not constituted by its union with body, with the realm where mortality has meaning. It preexists this state and of itself is divine and eternal. The ontological status of the soul enables Plotinus to put the oracular dictum "Know thyself" to a use which epitomizes the ambiguity of his doctrine, a doctrine in which the religious and the speculative are inextricably commingled, where union with the One and knowledge of the structure of the universe are but two aspects of the same effort. For it is by reflecting on itself that the soul will discover the order of the universe and begin its ascent to the Father. The soul must recall that the world that lies before the eyes of the body has as its author another soul, that every wonderful thing in the cosmos has as its source soul: this recognition draws the soul towards knowledge of that universal soul.

That great soul must stand pictured before another soul, one not mean, a soul that has become worthy to look, emancipate from the lure, from all that binds its fellows in bewitchment, holding itself in quietude. Let not merely the enveloping body be at peace, body's turmoil stilled, but all that lies around, earth at peace, and sea at peace, and air and the very heavens. (V,i,2)

By contemplating the heavens and their orderly movements, we will become conscious of the living soul behind that order and harmony and the human soul will become conscious of its likeness to the engendering father of the cosmos. Thus, the awe induced by this ascent to the universal soul comes to be directed on one's own soul, which is of the same nature as the universal soul. Once ascent is made to the universal soul, the stage is set for the next step in the upward movement, for the universal soul, awesome and great as it is, is the sign of something greater beyond it, the *Nous* of which it is the image. As speech is an image of the reason within the soul, so is soul the utterance of *Nous*, indeed

it is the activity of *Nous* as the procession from it. Having arrived thus at *Nous*, let us leave our preliminary sketch of Soul and turn to the higher hypostasis.

Nous. We have just seen Plotinus speak of the ascent of the human soul to *Nous* via the universal soul which is a procession from *Nous* and its image. Earlier we have seen Plotinus employ a metaphor of light according to which the Soul would be a moon, illuminated not by its own light but by that of the sun, the *Nous*. The *Nous*, if it is light in a more essential way than the Soul, has nevertheless received its light from a higher source. This indicates that *Nous* is at a midpoint between the other two hypostases and will lead us onward to the summit. This must be mentioned since, just as in speaking of Soul we had to make reference to *Nous*, any discussion of *Nous* requires reference to both other hypostases. A less metaphorical way of speaking of *Nous* is to call it beauty and thereby less than the good of which it is the image. The good is the One, that which is beyond and the primal principle, whereas *Nous* is its articulation into Ideas; the reference to the *Republic* is clear and intended. *Nous* is divine, is god though not the highest god, is the divine intellect in which resides the multiplicity of ideas and archetypes of the sensible world.

That archetypal world is the true Golden Age, age of Kronos, who is the *Nous* as being the offspring or exuberance of God. For here is contained all that is immortal: nothing here but is Divine Mind; all is God; this is the place of every soul. Here is rest unbroken; for how can that seek change, in which all is well; what need that reach to, which holds all within itself; what increase can that desire, which stands utterly achieved? All its content, thus, is perfect, that itself may be perfect throughout, as holding nothing that is less than the divine, nothing that is less than intellective. Its knowing is not by search but by possession, its blessedness inherent, not acquired; for all belongs to it eternally and it holds the authentic Eternity imitated by Time which, circling round the Soul, makes toward the new thing and passes by the old. Soul deals with thing after thing -- now Socrates; now a horse: always some one entity from among beings -- but the *Nous* is all and therefore its entire content is simultaneously present in that identity: this is pure being in eternal actuality; nowhere is there any future, for every then is a now; nor is there any past, for nothing there has ever ceased to be . . . (V,i,4)

Nous is identical with its objects; knower and known are one. This may seem to imply that to be is to be thought since *Nous* has been said to be being and now is said to be identical with what is thought. Nevertheless, Plotinus expressly rejects the view that to be is to be thought; he does this by asserting that *Nous*

does not produce its objects; it contains them but, as with Plato, Ideas are not taken to be concepts.

If the *Nous* were envisaged as preceding Being, it would at once become a principle whose expression, its intellectual act, achieves and engenders the Beings: but, since we are compelled to think of existence as preceding that which knows it, we can but think that the Beings are the actual content of the knowing principle and that the very act, the intellection, is inherent to the Beings, as fire stands equipped from the beginning with fire-act; in this conception, the Beings contain the *Nous* as one and the same with themselves, as their own activity. Thus, Being is itself an activity: there is one activity, then, in both or, rather, both are one thing. (V,iv,8)

Plotinus attributes our difficulties on this score to the fact that we necessarily separate things which are one in our thinking of them. In somewhat the same way, the *Nous* as emanation from the One and as its image, introduces multiplicity in its very imitation, the variety of Ideas. Nevertheless, as compared to Soul and to the visible world, *Nous* must appear as highly unified and this is explained precisely by the fact that it is an imitation of the One itself. Looked at from below, however, in *Nous* the spatial discreteness and temporal succession of the visible world are done away with and there each thing is everything and everything is an "each," although each thing is the whole in a somewhat different way. Brehier^[52] notes the similarity between such descriptions in Plotinus and the monadology of Leibnitz.

The *Nous* emerges as most ambiguous. It is the Platonic world of Ideas, the ground, law and guarantor of the things in the visible world; it is also that which contemplates these Ideas. But if *Nous* has as its function to contemplate its own content, to be a perfect union of thought and its object, it also is directed beyond and above itself.

Nous, thus, has two powers, first that of grasping intellectually its own content, the second that of advancing and receiving whereby to know its transcendent; at first it sees, later by that seeing it takes possession of *Nous*, becoming one only thing with that: the first seeing is that of Intellect knowing, the second that of Intellect loving; stripped of its wisdom in the intoxication of the nectar, it comes to love . . . (VI,vii,35)

Thus once more we see the way open to a further ascent.

It may be well to recall at this point the double aspect of the doctrine of the *Enneads*. We find discussed in an essentially interdependent fashion the structure of the world and the journey of the human soul back to the primary source of itself and the universe. *Nous* has proceeded from the One, which involves multiplicity; thus *Nous* is other than the One because it is an intelligible universe being the archetypes of the visible world and the ground for our moral and aesthetic judgments. Soul in turn proceeds from *Nous* and, unlike *Nous*, is ordered to govern not to contemplate. The human soul is a being in itself, possessing being prior to and thus apart from its union with a body; nevertheless it is the organizing principle of the body and this is natural to it and not evil in itself. Contact with the corporeal and the particular constitutes a danger, however, and the soul must withdraw within itself and, by knowing itself, know all that is. It must first arise to its higher sibling, the universal Soul which will inevitably lead it on to that of which the universal soul is but the image, namely *Nous*. Here too we seem to have the objective counterpart of a function of the human soul itself, intellect. Our intellect must learn to contemplate what *Nous* itself contemplates, the Ideas; in this way our minds become one with the divine mind although *Nous* is not the first but only the second god. As we have just seen, such contemplation leads to a further upward movement to that which is superior to *Nous* itself, the One or the Good.

The One. We must now attempt to sketch the most difficult aspect of the teaching of Plotinus, the summit and absolute source of being. Now, simply to call the One the source of being indicates the difficulties which face us; we have seen that Plotinus identifies *Nous* with being, for to be is to stand off from the One. The One, accordingly, would have to be said not to be. Plotinus does not flinch from this consequence. The One is beyond being as it is beyond knowledge; it is unknowable and ineffable. However, as will appear, Plotinus tends to speak of the One in an affirmative as well as a negative manner.

The One, then, is not *Nous* but something higher still: *Nous* is still a being but that First is no being but prior to all being: it cannot be a being, for a being has what we may call the form of its reality but the One is without form, even intellectual form. Generative of all, the One is none of all; neither thing nor quantity nor quality nor intellect nor soul; not in motion, nor at rest, not in place, not in time: it is the self-defined, unique in form or, better, formless, existing before form was, or movement or rest, all of which are attachments of being and make being the manifold it is. (VI,ix,3)

To speak of the One as a cause is not to say that something happens to it, but rather that something happens to other things. For the One to be a cause is not for it to change or to lose anything but for other things to gain by coming into being. But is not Plotinus caught up here in inextricable difficulties? To speak of the One seems to imply asserting that there is a One, that is, that something is One, an assertion which seemingly involves a multiplicity; what is One and the unity whereby it is so? Plotinus is aware of these difficulties. One is not a predicate of the One, nor is the One a numerical unity.

That awesome Prior, the One, is not a being, for so its unity would be vested in something else: strictly, no name is apt to it, but since name it we must there is a certain rough fitness in designating it as unity with the understanding that it is not the unity of some other thing. Thus it eludes our knowledge, so that the nearer approach to it is through its offspring. Being: we know it as cause of existence to *Nous*, as fount of all that is best, as the efficacy which, self-perduring and undiminishing, generates all beings and is not to be counted among these its derivatives, to all of which it must be prior. (VI,ix,5)

The best way to know the One is to know what we can know and thereby transcend through love and unknowing to the One. Plotinus is prepared as well to be faced with the paradox of his statements about the One which is said to be ineffable. To say the One is ineffable is thereby to make it effable, is it not? His reply is that speech about the One is not really about the One as if this first principle were grasped and being conveyed in language; his writings and talk urge us towards the One, they are an appeal to vision, a pointing of the path: "our teaching is of the road and the travelling; the seeing must be the very act of one that has made this choice." (VI,ix,4) Plotinus indicates the way by urging self-contemplation, that turning of the soul inward upon itself by which turning it will see that it is to this that its whole history points, that it is its very nature to turn toward its own center. The One is not the center of the Soul, but the center of the Soul is analogous to the One and provides a kind of bridge for the necessary transcendence towards the One.

The negative designation of the One can perhaps be best understood if we see the hypostases as objectifications of the spiritual life. The sensible world is to provide us with an occasion to rise to contemplation of the organizing principle of the cosmos, the universal soul; this in turn leads us onwards and upwards to *Nous*, which, as the realm of Ideas, is the objective correlative of our intellectual life, the life in which knower and known become one. In this scale, the One functions as the yonder or beyond what we can know: it is the ground ultimately

of intelligibility, unknowable in itself but drawing us towards it to a union which transcends the cognitive. From the point of view of the universe, this emphasis of Plotinus puts the One outside the universe of being although it is the cause of that universe. Any cognitive efforts to speak of the One must be in terms of finding analogies in what we do know, saying the One is like this or that, e.g. the center of a circle, while cautioning about the inability of our language to convey what transcends the capacity of our understanding. There is, however, another side of the doctrine of the One, a more positive side.^{53}

Porphyry, in his life of Plotinus, tells us that the whole of the *Metaphysics* of Aristotle is to be found compressed in the *Enneads*. The significance of that remark is nowhere more evident than in certain statements about the One. First of all, there is the utter simplicity of the One.

In us the individual, viewed as body, is far from reality; by soul which especially constitutes being we are in reality, are in some degree real. This is a compound state, a mingling of Reality and Difference, not, therefore reality in the strictest sense, not reality pure But in That which is wholly what it is -- self-existing reality, without distinction between the total thing and its essence -- the being is a unit and sovran over itself; neither the being nor the essence is to be referred to any extern. (VI,viii,12)

Plotinus goes on to say that the One which is identical with its essence (*ousia*) is equally one with its actuality (*energeia*). We have here the other side of the coin: when the One is said not to be, the being denied of it is an imperfect being, one which involves duality and otherness and thus dependence on something beyond; but if the One is said not to be in an imperfect way, it is a most perfect being, is in the most perfect sense of the term. Plotinus is not of course contradicting himself here, since this notion of perfect being is formed on an analogy with the imperfect being we can know, the being which is known to be non-self-sufficient and dependent on something further which can thereby be indirectly described. The One is not-being in the sense that it cannot be like the beings we know; if it were, we would be involved in an infinite regress. However, to know what the One cannot be is to be able to formulate a statement descriptive of what it is, (a necessarily imperfect statement, because made by reference to something else) of the perfection of its being. Now there is a momentum to this approach which carries Plotinus on to admit activities of the One, but activities which are one with its substance. Thus, the One has will in the sense that it is will; (VI,viii, 13) it is good in the sense that it is goodness. (V,v,13) The One is not concerned at all with the things which emanate from it,

a concern which Plotinus seems to feel would argue against its self-sufficiency and transcendence. Much the same thing can be said of *Nous*, of course, since governance and providence are assigned to the third hypostasis, Soul. And, to correct the possible import of a previous quotation, we must point out that Plotinus will say that the One is actuality (*energeia*) without being (*ousia*). (VI,viii,20) It is often pointed out that Plotinus has a tendency sometimes to speak of the One in the same way he speaks of *Nous* while at other times he sharply distinguishes between them. Various passages are extremely difficult to reconcile with one another and a certain impatience is justified. As a consistent theoretical doctrine, the *Enneads* leave much to be desired. But even to formulate this criticism is to indicate that one has lost touch with what is doubtless the most important aspect of Plotinus' upward journey. He does not promise us speculative accuracy; his many attempts to speak meaningfully of the One must always give way before what is the essential way to achieve contact with the first principle. This is not had by knowledge, but rather by presence (*parousia*).

The main part of the difficulty is that awareness of this Principle comes neither by knowing nor by the Intellection that discovers the Intelligible Beings but by a presence overpassing all knowledge. In knowing, soul or mind abandons its unity; it cannot remain a simplex: knowing is taking account of things; that accounting is multiple; the mind, thus plunging into number and multiplicity, departs from unity. Our way then takes us beyond knowing; there may be no wandering from unity; knowing and knowable must all be left aside; every object of thought, even the highest, we must pass by, for all that is good is later than This and derives from This as from the sun all the light of the day. (VI,ix,4)

The term of the teaching of Plotinus is the fulfillment of the spiritual life, the perfection of the individual; it does not seem too much to say that every theoretical statement is ultimately subordinated to enticing the soul upwards, beyond the theoretical to communion with the One, to that presence to the One in which our happiness consists. This term of the ascent of the spiritual life is not something remote and far off; the yonder or beyond is actually nearby and our distance from it is in function of our distance from our true self.

Thus the Supreme as containing no otherness is ever present with us; we with it when we put otherness away. It is not that the Supreme reaches out to us seeking our communion: we reach towards the Supreme; it is we that become present. We are always before it: but we do not always look . . . (VI,ix,8)

The World. The three hypostases, One, *Nous* and Soul give us the structure of the intelligible universe in which *Nous* proceeds necessarily from the One as its image, introducing a multiplicity expressed in the Platonic Ideas, a multiplicity which is nevertheless unified with respect to the next hypostasis, Soul, which emanates necessarily from *Nous* as its image. With the third hypostasis we have not yet reached the sensible world: this world proceeds from Soul, according to Plotinus, and it is to that aspect of his doctrine that we must now turn.

Something besides a unity there must be or all would be indiscernibly buried, shapeless within that unbroken whole: none of the real beings would exist if that unity remained at halt within itself: the plurality of these beings, offspring of the unity, could not exist without their own nexts taking the outward path; these are the beings holding the rank of souls. In the same way the outgoing process could not end with the souls, their issue stifled: every kind must produce its next; it must unfold from some concentrated central principle as from a seed, and so advance to its term in the varied forms of sense (IV,viii,6)

The universal soul, in its contemplation of *Nous* is filled with its object and overflows in an image. Soul as creative involves a secondary phase of the Soul, accordingly, and its production is something lower than itself. This creation by the Soul is not to be looked upon as a fall on its part, since it can create only if it ascends.

We assert its creative act to be a proof not of decline but rather of its steadfast hold. Its decline could consist only in its forgetting the Divine: but if it forgot, how could it create? Whence does it create but from the things it knew in the Divine? If it creates from the memory of that vision, it never fell. (II,ix,4)

Plotinus asserts that the production of the sensible world does not take place in time. Creation is an eternal process, coeval with the hypostases themselves. By the same token, the world is imperishable, being held together always by Soul.

And is it conceivable that the Soul, valid to sustain for a certain space of time, could not so sustain for ever? This would be to assume that it holds things together by violence; that there is a "natural course" at variance with what actually exists in the nature of the universe and in these exquisitely ordered beings; and that there is some power able to storm the established system and destroy its ordered coherence, some kingdom or dominion that may shatter the order founded by the Soul. (II,i,4)

We have seen that Soul produces the sensible world as a result of its primary activity which is contemplation; the lowest part of the soul responsible for this production is what Plotinus means by nature (*physis*). In keeping with his doctrine on the emanation of the hypostases, Plotinus does not intend that nature be a conscious production. "Nature, thus, does not know, it merely produces: what it holds it passes automatically to the next; and this transmission to the corporeal and material constitutes its making power." (IV,iv,13)

In the fourth tractate of the second *Ennead*, Plotinus discusses the notion of matter and decides that he must admit two kinds, sensible matter and intelligible matter. Thus, in the intelligible order where there are many ideal forms, these differ from one another precisely by the differences of their forms; nevertheless, such difference involves a similarity as well and Plotinus assigns this function to matter. This intelligible matter is the correlate of matter in the sensible world, but whereas the former has real being and life the latter does not, it is inert and lifeless. Opposed to the logos flowing from the creating soul or nature, which is a participation in light, matter is darkness and opacity. The argument leading to a recognition of matter is Aristotelian.

An additional proof that bodies must have some substratum different from themselves is found in the changing of the basic constituents into one another. Notice that the destruction of the elements passing over is not complete -- if it were we would have a Principle of Being wrecked in Non-Being -- nor does an engendered thing pass from utter non-being into Being: what happens is that a new form takes the place of an old. There is then a stable element, that which puts off one form to receive the form of the incoming entity. (II,iv,6)

This matter is utterly devoid of determination; moreover, the advent of form which is productive of body does not affect matter itself, and since form is the image and vehicle of the good, matter is other than good and unreachable by it. In a sense, then, matter is evil; its evil, however, is something negative so that evil is not set up as a positive being.

Man. Plato had maintained that man is his soul; Aristotle that man is a compound of soul and body. Plotinus agrees with Plato, but finds some justification for the Aristotelian view as well.

The soul of that order, the soul that has entered into matter of that order, is man by having, apart from body, a certain disposition; within body it shapes all

to its own fashion, producing another form of man, man reduced to what body admits, just as an artist may make a reduced image of that again. (VI,vii,5)

There is a way of considering man apart from body, then, "man yonder" or "man beyond" who differs from man here, man in the body, in this, that the latter is characterized by discursive thinking while man yonder, residing in *Nous* as ideal, has the characteristic of its abode, namely intuitive thought. (VI,vii,9) Man in the sense of soul has pre-existed man in the body and once in the body it is his divine origin which is the keynote of his goodness: as long as he remains in contact with it, strives towards it, man is good; evil will be the turning away from that origin, concern with things below. Looking at man below, in the body, it is his capacity for returning whence he came, his higher soul, which is his true self. Bodily passions are not states of the soul which use the body as their instrument; the soul itself is impassible. This produces a difficulty with respect to sense perception.

The faculty of perception in the soul cannot act by the immediate grasping of sensible objects, but only by the discerning of impressions printed upon the animate by sensation: these impressions are already intelligibles while the outer sensation is a mere phantom of the other which is nearer to authentic existence as being an impassive reading of ideal forms. (I,i,7)

The authentic self remains independent of its involvement in body and its knowledge is not to be reduced to states of the body. True knowledge will consist in transcending the body and its passions and here the homogeneity of human soul and universal soul, human intellect and divine intellect or *Nous* will be the explanation of knowledge. Evil is possible because of the lower side of the soul.

When we have done evil it is because we have been worsted by our baser side -- for a man is many -- by desire or rage or some evil image: the misnamed reasoning that takes up with the false, in reality fancy, has not stayed for the judgment of the reasoning principle: we have acted at the call of the less worthy, just as in matters of the sense-sphere we sometimes see falsely because we credit the lower perception, that of the couplement (of soul and body) without applying the tests of the reasoning faculty. (I,i,9)

The curious thing about this ascription of evil to the lower, less authentic part, is that the soul itself, what is really man, remains guiltless. It is nonetheless true that man sins and must make recompense for it.

By the soul subject to sin we indicate a groupment, we include that other, that phase of the soul which knows all the states and passions: the soul in this sense is compound, all-inclusive: it falls under the conditions of the entire living experience: this compound it is that sins; it is this, and not the other, that pays penalty. (I,i,12)

Soul as never subjected to body -- even the lower part of the soul being other than and above matter -- retains a freedom from the causes which govern the bodies of the sensible universe. The soul is not changed by its surroundings; rather the noble soul will change its surroundings or, where this is impossible, soul can retain its innocence. (III,i,8) Plotinus can accept the view that the course of the sensible world is necessitated by the sidereal movements and at the same time insist on the freedom of man. Indeed, Plotinus has a tendency to identify moral evil and involuntariness.

We admit, then, a necessity in all that is brought about by this compromise between evil and accidental circumstance: what room was there for anything else than the thing that is? Given all the causes, all must happen beyond aye or nay -- that is, all the external and whatever may be due to the sidereal circuit -- therefore when the soul has been modified by outer forces and acts under that pressure so that what it does is no more than an unreflecting acceptance of stimulus, neither the act nor the state can be described as voluntary: so, too, when even from within itself, it falls at times below its best and ignores the true and highest laws of action. (III,i,9)

Virtue, for Plotinus, is that state of soul which is devotion to its like; just as evil results from frequenting things unlike and below it. Virtue, therefore, while it does not consist in purgation or catharsis, is consequent upon it. There will be a gradation of virtue insofar as the hierarchy within the soul reflects the objective hierarchy of hypostases; this upward path is trod by the use of dialectic. The whole of philosophy is ordered to getting the soul to retrace the path to its ultimate origin, the first principle, the ineffable One, which is attained in a spiritual union which is beyond doctrine and communicable thought. Virtue will describe way-stations on this route, necessary states of the soul as it turns inward, away from the distractions of otherness, and finds within the correlate of the transcendent and its own destiny as union with the One, the supreme God.

Summary. If the Plotinian doctrine begins with God, the One, this is because that is the term of human striving. From the One proceeds the realm of Ideas,

Nous, and from these proceeds the governing and organizing principle of the material world. The sensible world itself is the ultimate product of emanation and although Plotinus speaks of matter as evil, he refuses the Gnostic claim that this world is itself evil. For one thing, the sensible world is more than matter; it has logos in it, form, an image even though remote of the good. The sensible world is good and beautiful, not so much in itself, as because it participates in goodness and beauty. This makes it a sign of the beyond, a token of where we must go. For man, to become aware of the goodness of the world is to be impelled to transcend it towards its governing principle and this in turn impels towards the Ideas, *Nous*, the realm of law and ideal forms. Here knowledge is intuitive, one becomes like what he knows, but in this knowledge becomes aware of the beyond, of that which is greater than the ideas, of a possession which is more noble than intellectual knowledge. Love now becomes the guideline and the union which is its term cannot be communicated as a philosophical doctrine. Plotinus can only hint and urge; he becomes thereby a spiritual director, his sayings a catalyst which may bring about union. It is impossible to separate Plotinus' theoretical descriptions of the intelligible universe from his overriding concern with the spiritual life; the doctrine of hypostases, the story of the formation of the sensible world, all function in getting man to see his destiny. In achieving his aims, Plotinus is at pains to take into account the doctrine of earlier philosophers. It is not difficult to see that the principal goal of his teaching singles out an important aspect of Plato and subjects everything else to it; at the same time, he incorporates much of Aristotle, particularly from the *Metaphysics* and the *De Anima*. His admission of providence indicates a deference to the Stoics, but for Plotinus providence cannot mean the conscious concern of the higher for the lower; rather it is a way of recognizing the necessary impact of the higher on the lower. So Plotinus can admit the causality of the sensible world and at the same time argue for man's transcendence of that world, thus preserving human freedom. It may be said that Plotinus represents the culmination of Greek philosophy precisely by showing the inadequacy of philosophy just as such, as an intellectual effort alone. Philosophy at its best makes us aware of something utterly beyond the sensible world, beyond the best efforts of our intellect. In Plotinus this calls for an effort on man's part to go through philosophy to this recognition and then by means of love and ecstasy to go beyond philosophy to union with that primal principle of all things. But, if this is to go beyond philosophy as intellectual exercise, it is to rejoin the notion that philosophy is a way of life, man's way of achieving his perfection. It is noteworthy that Plotinus' recognition of the primacy of the spiritual life does not lead to a repudiation of the intellectual, or to the degenerate attempt to achieve ecstasy by a return to the primitive: revival

of the primitive, the sophisticated attempt to rid oneself of sophistication, is always less innocent than the primitive itself. For Plotinus the route to ecstasy and union with the One leads through philosophy as an indispensable element. This serves as a brake on the willed irrational and the *Enneads* present one of the noblest natural attempts to cope with the demands of man as spiritual being.

C. After Plotinus

Of the pupils of Plotinus, Amelius may be mentioned, but Porphyry of Tyre (born 232/3 A.D.) is far and away the most important. We have already seen that he was with Plotinus in Rome for six years and was responsible for arranging the writings of his master in the form of the *Enneads*. Porphyry is credited with having written a great number of works, notable among them being his *Isagoge*, an introduction to the *Categories* of Aristotle. This work discusses the five universals or predicables, knowledge of which is presupposed to an understanding of the Aristotelian logical work. Porphyry excuses himself from undertaking a resolution of the controversy between Plato and Aristotle on the ontological status of universals, a modest withdrawal from controversy which was destined to provide an occasion to take up just that controversy not only for Boethius when he commented on the *Isagoge* but far into the Middle Ages, long after contact with the literary context of the dispute had been lost.

Since it is necessary, Chrysaorius, both to the doctrine of Aristotle's *Categories*, to know what genus, difference, species, property and accident are and also to the assignments of definitions, in short, since the investigation of these is useful for those things which belong to division and demonstration, I will endeavor by a summary briefly to discuss for you, as in the form of introduction, what in this subject has been delivered by the ancients, abstaining, indeed, from more profound questions, yet directing attention in a fitting manner, to such as are more simple. For instance, I shall omit to speak about genera and species, as to whether they subsist (in the nature of things) or in mere conceptions only; whether also if subsistent, they are bodies or incorporeal, and whether they are separate from or in is most profound and requires another more extensive investigation. (Porphyry, *Isagoge*, chap.1)

Porphyry goes on to discuss five universals, genus, species, property, difference and accident, their respective natures and the interrelationships between them. Porphyry is also said to have written two commentaries on the *Categories* themselves, a significant fact since Plotinus had rejected the Aristotelian

categories in favor of those to be found in the Sophist of Plato. He is also said to have written commentaries on Plato, e.g., on the *Timaeus*. Fifteen books written against the Christians have been lost, although a few fragments are extant. There has been some discussion of the possibility that Porphyry was an apostate Christian, but it seems to be a moot point.

Aside from the immediate school of Plotinus, it is customary to speak of later Neoplatonists in terms of schools associated with particular geographical areas. Thus Iamblichus who died about 330 A.D. is the most important member of what is called the Syrian school. He is looked upon as one who discounted the Plotinian doctrine that the way to ecstasy was through theory and sought it rather by means of theurgy, occultism and magic. Iamblichus describes himself as a Pythagorean and considers mathematics as a preparation for knowledge of the gods, a doctrine that casts some doubt on the belief that he emphasized theurgy. Certain mathematical works of Iamblichus have come down to us, a work *On General Mathematical Science* and an *Introduction to the Arithmetic of Nicomachus*; a third mathematical work, often attributed to him, is now thought not to be his. Iamblichus considered the doctrine of Pythagoras to be a gift of the gods and thought divine grace necessary to comprehend it; he advocated the use of the mathematical method in philosophy. In reference to Plotinus, Iamblichus tends to elaborate the doctrine of emanation by positing intermediate stages between the three Plotinian hypostases.

The School of Pergamon, an offshoot of that of Iamblichus, is noteworthy for its attempt to bring back polytheism because it influenced the Emperor Julian who had been brought up a Christian but during a brief reign was a fierce opponent of Christianity and defender of the traditional polytheism.

The School of Athens produced several commentaries on Aristotle, e.g., a commentary on the *De Anima* by Plutarch of Athens and on the *Metaphysics* by Syrianus: both men died towards the middle of the fifth century of our era. Proclus, born in Constantinople in 410, studied with Olympiodorus at Alexandria and then came to Athens where he studied under Plutarch and Syrianus, succeeding the latter to the headship of the school. A number of his writings have come down to us, commentaries on *Alcibiades I*, *Parmenides*, *Republic* and the *Cratylus* among them. As well, we have his works on Plato's theology, on providence and evil and his famous *Elements of Theology*. There is also an *Elements of Physics*. The *Elements of Theology* were destined to have a great impact on the Arabians and in the Latin West, particularly because much of it appeared in the work known as the *Liber de causis* until St. Thomas Aquinas

pointed out its origin in Proclus. The literary form of both *Elements* of Proclus consists in setting forth a proposition and following it with a proof. The one on physics is based largely on Aristotle; it is the *Elements of Theology* which give us a characteristically Neoplatonic doctrine.

If we concentrate on the *Elements of Theology*, it must be pointed out that the form, and to a great degree the contents of this work, present a difficulty of no little moment when compared to Proclus' commentaries on Plato and his work on the theology of Plato. Proclus, in common with Iamblichus and Syrianus, under whom Proclus studied, opts strongly for the theurgy side of the dichotomy *theoria* / *theourgia*. Iamblichus differs from Plotinus in maintaining that union with the One is not attained through knowledge and speculation, but rather through a ritualistic magic, by being taking possession of by the divine. The Chaldean Oracles thereby become a sacred book to be interpreted and even the Platonic dialogues are studied for symbolic intent and hints of magic formulae. We must remember that Proclus believed in mermaids and dragons, automotive statues, the man in the moon and a whole raft of astral gods. Against this background the severe literary form of the *Elements* is almost a shock. If it does not represent a good deal of material original with its author, it is nonetheless a sustained attempt at a rigorous and austere formulation of the processions of Neoplatonism as it had been developed in the wake of Plotinus.

The *Elements of Theology* consists of 211 propositions each of which is followed by what purports to be a proof of it. The procedure is not one of citing authorities, then, (though the sources of the propositions and the proofs can be found), but an argued presentation of the great synthesis of reality as it moves out from its ultimate principle according to the familiar Plotinian triad: One, Process, Return. Dodds^[54] suggests that the *Elements* contains two main sections.

The first of these (props. 1 to 112) introduces successively the general metaphysical antitheses with which Neoplatonism operated -- unity and plurality, cause and consequent, the unmoved, the self-moved and the passively mobile, transcendence and immanence, declension and continuity, procession and reversion, *causa sui* and *causatum*, eternity and time, substance and reflection, whole and part, active and passive potency, limit and infinitude, being, life and cognition. The remaining part (props. 113-211) expounds in the light of these antitheses the relations obtaining within each of the three great orders of spiritual substance, gods or henads, intelligences, and souls; and the relations connecting each of these orders with the lower grades of reality.

If we consider the opening propositions of part one, we shall be able to say something about the doctrine contained in part two of the *Elements*.

At the very outset, Proclus wants to establish the absolute priority of the One. His first proposition is: "Every manifold participates unity in some way." The proof is by dichotomy and *reductio*. Let us take the opposite possibility, a manifold which in no way participates unity; it will thus not be one whole nor can its parts be ones. For the part must be either one or not one; and if not one then either many or nothing; if the part is nothing, the whole is nothing; if many, we are embarking on an infinite regress. And, since nothing can be made up of an infinity of infinities we must accept the original proposition that the manifold in some way participates unity. Proclus goes on to argue that what participates unity is not unity itself; therefore it must be something else besides the unity it participates. Having shown that the manifold must be logically posterior to the One, Proclus asserts (prop. 6) "Every manifold is composed either of unified groups or of henads or units." We have already noticed that Iamblichus had felt constrained to introduce another One between the Ineffable One and *Nous*. It is characteristic of later Neoplatonism that it tends to multiply the strata of the intelligible universe beyond the three hypostases of Plotinus. One great motivation for this was the desire to achieve a gradual shading off into the sensible world and thereby mask a difficulty of which Plotinus himself had been aware: how do you get multiplicity from unity unless the one is already in some sense many? Proclus' notion of henads or units as what first proceeds from the One amounts to a population increase in the area of Iamblichus' intermediate one. Moreover, Proclus has no hesitation in identifying these henads with the gods of Greek mythology.

This proliferation continues in the realm of *Nous*, in which sphere of Being, Life and Thought are first distinguished and then each sphere subdivided; the realm of Soul becomes quite densely populated. As for the sensible world, it is formed and looked after by Soul but, as with Plotinus, it is as if matter is there awaiting this formation and is not taken to proceed from the higher order. Of course, the processions described are not the history of the universe in any chronological sense: Proclus, like Plotinus, is describing logical priority and posteriority. From that point of view, the notion of matter, formless and chaotic, "awaiting" determination and governance is not a problem for Neoplatonism. What remains the problem is the initial emergence of multiplicity from the One, and later Neoplatonism seems in effect to be attempting to obscure this difficulty by filling in the interstices of the intelligible universe of Plotinus with more and more grades -- as if one could construct a line from points.

Proclus was succeeded as head of the school of Athens by his student Marinus who wrote a life of Proclus. The last of the heads of this school was Damascius. We have a work of his dealing with difficulties and their solutions concerning the first principles occasioned by the Parmenides of Plato. Damascius insists on the utter transcendence of the first principle: it is beyond our language and our understanding. All the talk about processions is but a groping way to speak of what is, after all, quite beyond our grasp. A student of Damascius was Simplicius, the author of several very important commentaries on Aristotle, e.g., on the *Categories*, *Physics*, *De Coelo* and *De Anima*. The reader will remember, from Part One above, the importance of Simplicius' commentary on the *Physics* from the point of view of Presocratic fragments. In 529 A.D. the Emperor Justinian closed the school at Athens; Simplicius and Damascius, upon the invitation of the Persian king, went to Persia around 531. They rereturned a year or two later but Athens' long history as the capitol city of philosophy had come to an end.

The Alexandrian school is particularly important, not only for its commentaries on Plato and Aristotle, but also because we find there a sustained proximity with Christianity and the evolution of something like a *modus vivendi* between philosophy and religion; indeed, we see at Alexandria the entry of a number of philosophers into the Church. The commentators who may be mentioned are Ammonius, Johannes Philoponus and Olympiodorus. Since these men bring us into the sixth century of our era and some of them make an attempt to reconcile philosophy and the Christian Revelation, discussion of them must be postponed till volume two of this series. When one considers that Johannes Philoponus, a Christian living after Augustine, in his commentaries on Aristotle is anticipating some of the debates of the thirteenth century, it will be appreciated that it is difficult to place him among the ancient philosophers.

{51} H. A. Wolfson, *Philo*, vol. I (Cambridge, Mass.: Harvard University Press, 1948), p. 94.

{52} E. Brehier, *The Philosophy of Plotinus* (Chicago: University of Chicago Press, 1958)

{53} See A. H. Armstrong, *The Architecture of the Intelligible Universe in the Philosophy of Plotinus* (Cambridge, England: University Press, 1940).

{54} E. R. Dodds, *Proclus, The Elements of Theology* (Oxford: Clarendon Press, 1933).

Bibliography

Since the point of this reading list is to cite books the student might consult, the books are mentioned in terms of their present availability and the dates are often the dates of reprints whether in hard covers or soft.

READINGS FOR PART ONE

A. SOURCES

1. Charles M. Bakewell, *Source Book in Ancient Philosophy*, (New York: Charles Scribner's Sons, reprinted 1963).
2. Kathleen Freeman, *Ancilla to the Presocratic Philosophers*, (Oxford; Basil Blackwell, 1948).
3. R. D. Hicks, editor and translator, *Diogenes Laertius*, 2 vols., (Cambridge: Harvard University Press, 1950).
4. C. S. Kirk and J. E. Raven, *The Presocratic Philosophers*, (Cambridge: University Press, 1957).

B. GENERAL STUDIES

1. John Burnet, *Early Greek Philosophy*, (New York: Meridian Books, 1957).
2. F. M. Cornford, *From Religion to Philosophy: A Study in the Origins of Western Speculation*, (New York: Harper Torchbooks, 1957)
3. Kathleen Freeman, *The Presocratic Philosophers, A Companion to Diels*, (Oxford: Basil Blackwell, 1946).
4. W. K. C. Guthrie, *A History of Greek Philosophy, Volume I, The Earlier Presocratics and the Pythagoreans*, (Cambridge: The University Press, 1962). This is the first of a projected five volume history by Guthrie and is a veritable monument abuilding.
5. Leon Robin, *Greek Thought*, translated by M. R. Dobie, (London, 1928).

C. PARTICULAR STUDIES

1. Cyril Bailey, *The Greek Atomists and Epicurus*, (Oxford: The Clarendon Press, 1928).
2. F. M. Cleve, *The Philosophy of Anaxagoras*, (New York: King's Crown Press, 1949).
3. F. M. Cornford, *Plato and Parmenides*, (New York: The Liberal Arts Press, 1957).
4. Charles H. Kahn, *Anaximander and the Origins of Greek Cosmology*, (New York: Columbia University Press, 1960).
5. G. S. Kirk, *Heraclitus: The Cosmic Fragments*, (Cambridge: The University Press, 1954).
6. J. E. Raven, *Pythagoreans and Eleatics*, (Cambridge: The University Press, 1948).
7. Mario Untersteiner, *The Sophists*, translated by Kathleen Freeman, (Oxford: Basil Blackwell, 1954).
8. Philip Wheelwright, *Heraclitus*, (Princeton: Princeton University Press, 1959).

READINGS FOR PART TWO

A. SOURCES AND STUDIES: PLATO

1. *The Dialogues of Plato*, translated into English by Benjamin Jowett, 4 (New York: Random House, 1937); *The Collected Dialogues of Plato Including the Letters*, edited by Edith Hamilton and Huntington Cairns, Bollingen Series LXXI, (New York: Pantheon Books, 1961).
2. H. F. Cherniss, *The Riddle of the Early Academy*, (Berkeley: University of California Press, 1945).
3. F. M. Cornford, *Plato's Theory of Knowledge*, (New York: The Liberal Arts Press, 1957).
4. F. M. Cornford, *Plato's Cosmology*, (New York: The Liberal Arts Press, 1957).

5. F. M. Cornford, *Plato and Parmenides*, (New York: The Liberal Arts Press, 1957).
6. Paul Friedlander, *Plato, Volume I, An Introduction*, translated by Hans Meyerhoff, Bollingen Series LIX, (New York: Pantheon Books, 1958).
7. John Gould, *Development of Plato's Ethics*, Cambridge: The University Press, 1955).
8. G. M. Grube, *Plato's Thought*, (Boston: Beacon Press, 1958).
9. R. Hackforth, *Plato's Phaedo*, (New York: The Liberal Arts Press, 1952).
10. Eric A. Havelock, *Preface to Plato*, Volume One of a projected History of the Greek Mind, (Cambridge, Massachusetts: Belknap Press of Harvard University Press, 1963).
11. R. L. Nettleship, *Lectures on the Republic of Plato*, (London: Macmillan, 1937).
12. Richard Robinson, *Plato's Earlier Dialectic*, (Ithaca, N.Y.: Cornell University Press, 1941).
13. Paul Shorey, *What Plato Said*, (Chicago: University of Chicago Press, 1933).
14. Julius Stenzel, *Plato's Method of Dialectic*, translated by D. J. Allan, (Oxford: The Clarendon Press, 1940).
15. A. E. Taylor, *Plato, The Man and His Work*, (New York: Meridian Books, 1956).

B. SOURCES AND STUDIES: ARISTOTLE

1. *The Works of Aristotle Translated into English*, 12 Volumes, edited by W. D. Ross, (Oxford: The Clarendon Press, 1908-1952); this translation is utilized in *The Basic Works of Aristotle*, edited and with an introduction by Richard McKeon, (New York: Random House, 1941).
2. D. J. Allan, *The Philosophy of Aristotle*, (London: Oxford University Press: 1952).

3. Ernest Barker, *The Politics of Aristotle*, Translated with an Introduction, Notes and Appendixes, (New York: Oxford University Press, 1958).
4. S. H. Butcher, *Aristotle's Theory of Poetry and Fine Art*, (New York: Dover Publications, 1951).
5. L. M. De Rijk, *The Place of the Categories of Being in Aristotle's Philosophy*, (Assen: Vangorcum, 1952).
6. I. During, *Protrepticus: An Attempt at Reconstruction*, (Goteborg: Almqvist & Wiksell, 1961).
7. I. During and G. E. L. Owen, *Aristotle and Plato in the Mid- Fourth Century*, Papers of the Symposium Aristotelicum held at Oxford in August, 1957, (Goteborg: Almqvist & Wiksell, 1960).
8. Leo Elders, *Aristotle's Theory of the One, A Commentary on Book X of the Metaphysics*, (Assen: Van Gorcum, 1961).
9. Werner Jaeger, *Aristotle, Fundamentals of the History of His Development*, Second Edition, translated by Richard Robinson, (New York: Oxford Paperbacks, 1962).
10. H. H. Joachim, *The Nicomachean Ethics: A Commentary by the late H. H. Joachim*, edited by D. A. Rees, (Oxford: Oxford University Press, 1951).
11. J. Lukasiewicz, *Aristotle's Syllogistic from the Standpoint of Modern Formal Logic*, (Oxford: Oxford University Press, 1951).
12. G. R. G. Mure, *Aristotle*, (London: E. Benn, 1932).
13. Joseph Owens, *The Doctrine of Being in the Aristotelian Metaphysics*, (Toronto: Pontifical Institute of Mediaeval Studies, 1957).
14. William David Ross, *Aristotle*, (New York: Meridian Books, 1959).

READINGS FOR PART THREE

1. A. H. Armstrong, *The Architecture of the Intelligible Universe in the Philosophy of Plotinus*, (Cambridge: The University Press, 1940).

2. Cyril Bailey, *The Greek Atomists and Epicurus*, (Oxford: The Clarendon Press, 1928).
3. E. Bevan, *Stoics and Sceptics*, (New York: Barnes and Noble, 1959).
4. I. Bochenski, *Formal Logic*, translated by I. Thomas, (Notre Dame: The University of Notre Dame Press, 1960).
5. E. Brehier, *The Philosophy of Plotinus*, (Chicago: The University of Chicago Press, 1958).
6. Gordon H. Clark, *Selections from Hellenistic Philosophy*, (New York: F. S. Crofts, 1940).
7. N. W. DeWitt, *Epicurus and His Philosophy* (Minneapolis: University of Minnesota Press, 1954).
8. C. J. DeVogel, *Greek Philosophy*, Volume III, (Leiden: Brill, 1959).
9. E. R. Dodds, *Proclus The Elements of Theology*, (Oxford: The Clarendon Press, 1933).
10. A. J. Festugiere, *Epicurus and His Gods*, translated by C. W. Chilton, (Cambridge: Harvard University Press, 1956).
11. H. D. Hicks, *Stoic and Epicurean*, (New York: Russell, 1961).
12. W. H. Inge, *The Philosophy of Plotinus*, 2 volumes, (London: Longmans, Green, 1929).
13. B. Mates, *Stoic Logic*, (Berkeley: University of California Press, 1953).
14. S. Sambursky, *Physics of the Stoics*, (London: L. Routledge, 1959).
15. H. E. Witt, *Albinus and the History of Middle Platonism*, (Cambridge: The University Press, 1937).
18. E. Zeller, *The Stoics, Epicureans and Sceptics*, (London: Longmans, Green & Company, 1892).

Index to Volume I

- Anytus: accuser of Socrates, 112, 114
- Aphrodite: 7, 69, 74
- Apollodorus: 117
- Apology (Plato): 112-15, 145
- Apology (Xenophon): 131
- Apuleius of Madaura: 338
- Aquinas, St. Thomas: 267
- Arabic authors: writings from, 219
- Arcesilaus: 308, 327-31; birth, 328
- Archedemus: 137
- Archytas: 136, 137
- Ares: 7, 74
- Argument: 314-6; contentious, fallacious or apparent, 249; definition, 314
- Arian: 323; Discourses, 323; Enchiridion, 323
- Aristippus of Cyrene: 131, 133
- Aristo: 311
- Aristocles: 134
- Ariston: 134
- Aristophanes: 9, 109; *Clouds*, 109; lampoon of Socrates, 110
- Aristotelian doctrine: 309
- Aristotelianism: 217; opposed to Platonism, 218
- Aristotle: 6; account of Pythagorean doctrines, 43-50; accounts of Thales' doctrines, 18-20; acromatic works, 220, 221; appraisal of Empedocles, 65-6; appraisal of Melissus of Samos, 63; birth, 216; *Categories*, 222, 2317, 337; change in mind, 218; criticism of Plato's language, 223-4; *De Caelo*: 49, 50, 57; death is return of soul to home, 220; dialogues, 218-21; division of practical philosophy, 228-30; division of theoretical philosophy, 227-8; doctrine of opposition of proposition, 241; early writings, 221; *Eudemian Ethics*, 118, 222, 228; Eudemus, 220; evolution in thought, 218; exoteric works, 219; First Philosophy, 228-30, 284-96; Generation of Animals, 222; History of Animals, 222; importance of order and methodology, 255; influence on Alexander the Great, 216-17; interest in everything said by predecessors, 250; interest in nature, 219; judgment of Xenophanes, 30-1; logical works, 230-49; lost works, fragments of, 220; man and his work, 216-22; *Metaphysics*, 5-6, 16, 19, 218, 222, 224-6, 230, 233, 235, 260, 269, 337; Meteorologica, 65, 222, 225-6; moral

and political philosophy, 269-84; Motion of Animals, 222; nature and division of philosophy, 222-30; Nicomachian ethics, 222, 229, 269-84; on Anaxagoras' doctrine of *Nous* or Mind, 81; *On Democritus*, 87; *On Generation and Corruption*, 222; *On Interpretation*, 222, 231, 237-44, 318; *On Philosophy*, 220, 221; *On the Heaven*, 222; *On the Soul*, 220, 222, 229, 265-9; order among philosophical sciences, 228-30; order of procedure in natural science, 250-1; *Organon*, 230-1; *Organon*, plurality of works in, 231; *Parts of Animals*, 222; *Parva Naturalia*, 222; philosophy of nature, 31, 149, 250-69; *Physics*, 21, 222, 250, 253-4, 260, 264-5; Platonist period 217; Plato's greatest pupil, 217; *Poetics*, 6, 65, 222, 223, 231; *Politics*, 222, 269-84; *Posterior Analytics*, 222, 227, 231, 244-9; *Prior Analytics*, 222, 231, 241-4, 247-8; *Progression of Animals*, 222; *Protrepticus*: 220-2; *Rhetoric*, 65, 222, 231; sciences of nature, 250; science of nature, order of procedure, 250-1; Sophistical Refutations, 222, 231; speculative and practical philosophy, 225-7; stages of development, 221; statement on doctrine of Forms, 147; student in Academy, 138, 216; study of Gorgias, 104; "system" came to him whole, 218; *Topics*, 64, 222, 231, 249; treatises, 218, 221-2; true heir of Plato, 217; view of philosophy, 230; writings, 218

- Arithmetic: 44-5, 158
- *Arrow: The*, 61
- Artisans: 204
- Asclepiad society:
- Assertions: elements of, 232
- Assumptions: basic and contradictory, in natural science, 253-4
- Astronomy: 158, 214; Thales' knowledge of, 15-7
- Atheists: 212, 213
- Athene: 7, 101 Academy: See Academy; democracy: 210; Socratic Society: 146; stranger: 209
- Athenians: 181
- Athens: champion of empire, 109; sea power, 210
- Atlantis: isle of, 181; myth of, 190
- Atomism: 84-108; and modern history, 252-3; doctrine of Leucippus and Democritus, 84-5; motivation for doctrine of, 85-6; reaction of Empedocles and Anaxagoras to, 85
- Atomists: belief that world came about by chance, 88
- Atoms: definition of, 86; imperceptible, 88; movement of, 86-7; only things that really are, 87, 88; structure of, 89; weight of, 86
- Augustine, St.: 184

B

- Babylon: wisdom of, 16
- Barbarians: myths and rituals, 12
- Beauty: 153
- Becoming: and perishing, 68, 76; beginning of, 184; unqualified, 257
- Being: 178; and becoming, 187; and not-being: 170, 174-9; does not move, 55; finitude of, 55, 62-3; ingenerable and incorruptible, 55; is finite, 55, 62-3; is one, 165; monolithically unique body, 58; non-physical, 57; properties of, 54 ff; source of, 54
- Beings: atoms, as irreducible elements of, 86
- Belief: See Knowledge
- Blood: as thought, 72
- Bodies: compounded of atoms, 86; generated from points, 47; living, 267; living natural, 266; mathematical and physical, identification of, 60; physical, and geometrical solids, 47; physical, composed of points having magnitude, 60
- Body: definition of, 266; organic, 266; solid, 47; triumph over, 153; world, 185-6
- Boethius: commentaries on Porphyry's *Introduction to Aristotle's Categories*, 229
- Boundless, the: 34; doctrine of Anaximander: 21-4
- Briseis: 8
- Bryson: 324

C

- Caesar: 131; *Anabasis*, 131
- Caldean Oracles: 358-9
- Calipus: 137
- Callias: 98
- *Cannon, The* (Epicurus): 298
- Canonic: 298-300
- Carneades of Cyrene, 331-3; birth, 331
- Carpenter: 167
- Categories: 312-4
- *Categories* (Aristotle): 222, 231-7, 337
- *Causes* (Democritus): 84; four types of, 261
- Cave: allegory of, 153, 157-8
- Cebes: 115, 193
- Celsus: 338
- Cephalus: 199
- Chaerephon: 113
- Chalcidius: 182

- Chance: 2, 91, 261-2; analysis of, 262-3
- Change: analysis of, 256; Anaxagoras' Paradoxical doctrine of, 79-81; ceaseless, 182; in realm of art, 257-8; principle of, 26, 251
- Changeable being: principles of, 255-8
- Chaos: 9
- Charmides: 120-1, 134
- *Charmides* (Plato): 120-3, 145, 147
- Children: 211
- Choice and deliberation: 277
- Christ: 16
- Christianity: 358; history of, 307
- Christians: 357
- Church: entry of philosophers into, 361
- Chrysippus: 307, 308, 331; death, 309
- Cicero: 307; *De senectute*, 199
- Circle: 148, 191; knowledge of, 142-3, 148
- City: planning of, 209; population of, 211
- City-state: See Greek *polis*
- Cleanthes: 307, 308
- Cleinias: 212
- Cleon: 107
- Clitomachus: 332-34
- *Clouds* (Aristophanes): 109
- Cobbler: 167
- Colophon: 27
- Commerce: 211-2
- Conjecture: 182
- Conservation of resources: 212
- Contemplation: 283-4
- Contests: 211
- Continuum: 264
- Contradiction: principle of, 100
- Contrariety: involved in change, 254-5
- Corpus Hermeticum: 340
- Corruption: 22
- Cosmic cycle: 69
- Cosmogony: 9, 12
- *Cosmography* (Democritus): 84
- Cosmology: of Diogenes of Apollonia, 93-4; of Empedocles, 69; Pythagorean, 46-7, 58; (Plato): 189

- Cosmos: 20
- Courage: 204, 205; definition: 204
- Craftsmen: 212
- Crantor: 328
- Crates of Thebes: 133, 328
- Cratylus: 37, 136
- Cratylus: (Plato): 145
- Created world: perfection of, 182
- Creation: 70-1, 181
- Creation of time: 186
- Creator: 182; See also Demiurge
- Crimes against state: 212
- Critias: 121-3, 134, 181
- *Critias* (Plato): 145
- Crito: 117
- *Crito* (Plato): 145
- Critolaus: 331
- Croesus, King: 17
- Croton: 40-1
- Cthonic deities: 12
- Cthonic religion: 12; snake god, 28
- Cycle of incarnations: 73-5
- Cycle of the Great Year: 36
- Cynics: 132-3, 307; doctrines of, 132-3; mode of life, 133
- Cyrenaic socratic school: 133
- *Cyropaideia* (Xenophon):

D

- Daemons: 20
- Damascius: 360
- Day: 9
- *De Anima* (Democritus), 89
- *De Caelo* (Aristotle): 49, 50, 57
- *De Rerum Natura* (Lucretius): 66
- *De Senectute* (Cicero): 199
- Death: neither good nor evil: 303; not instantaneous, 88; releases the soul, 198; return of soul to home, 220
- Deception: 172
- Definition: definition of, 311

- Definitions: search for (Plato), 158-9
- Delphic oracle: 114, 115
- Demiurge (maker), 183-4; motive in fashioning world, 184-5; See also Creator
- Democracy: 208; cannot rule empire, 107; Plato's distaste for, 135
- Democritean ethics: 90-2; independent of atomism, 91
- Democritus: 84; accounts for man's belief in gods, 89-90; Causes, 84; *Cosmography*, 84; *De Amina*, 89; doctrine of atomism, 84-5; *Little World Order, The*, 84; on role of chance in human affairs, 91; *On the Planets*, 84; plenum and void, 252; works, 84
- Demonstration: 244; circular, 248-9
- Demonstrative science: 241-2; syllogism: 241
- Desires: differences between, 303, 305
- Despotism or tyrannical government: 208
- Destruction: 22
- Dialectic: 66, 159, 165, 179, 310, 311; definition, 156, 311, 313; founded by Zeno of Elea, 59, 66; Platonic, 140; See also Logic
- Dialecticians, 160
- Dialogues of Aeschines: 131
- Dialogues of Aristotle: 218-21
- Dialogues of Plato: 111, 134, 136, 141, 144-6; authentic, 145; chronology, 145; metaphysical, 160; purpose or function of, 144; Socratic: 111, 144, 146, 147
- Dianoia (thinking): 155
- Diodes: 313-4
- Diodorus Cronus: 132
- Diogenes Laertius: 15, 136, 220; air as primary stuff, 25; Empedocles works and feats recorded by, 65; on perception, 299-300
- Diogenes of Apollonia: 92-4; cosmology of, 93-94; fl. about, 440-430 B.C., 92; eclectic doctrine, 93; makes common nature, air, 93; natural philosophers, 92; *On Nature*, 93
- Diogenes of Babylon: 331
- Diogenes of Selencis: 309
- Diogenes of Sinope: 133
- Dion: 136-7
- Dionysius: 131, 136-7
- Dionysius II: 136-7, 142; introduction to philosophy, 138-9
- Disbelief: 213
- Disciplines: five, 158; mathematical, 158
- Discourse (logos): 242-3

- Discourse or speech (logos), 239
- Discourses (Arian): 323
- Diseases: 189
- *Dissoi logoi*, or *Twofold Arguments*: anonymous work, 106
- Divided Line: 153, 157
- Divine: reconciliation of Greek and Oriental statements about, 99
- Divine being: Ionian doctrine of, 53; Xenophanes view of, 53
- Divinity: 20; Empedocles' description of, 74; should not be localized, 28
- Doctrine of *anamnesis*: 186
- Doctrine of Forms: 118, 144, 147-60, 179, 192, 206, 217, 220; point of difference between Socrates and Plato, 147; second-best way to discover Good, 149-51
- Doctrine of Forms or Ideas: 118, 144, 192
- Doctrine of Ideas: 159-66, 160-80
- Doctrine of the One and the Many: 161-6
- *Doxa* (opinion): 171
- Dreams: 169
- Dualism: 176; of being and non-being, 56; of Pythagorean doctrine, 56

E

- Earth: 10, 34, 36, 70; (Aidoneus): 67; formation, 26; (Gaia): 8; gods: 12; origin of heavenly bodies, 26; shape, 26
- East-of-Egypt: wisdom of, 16
- Echecrates: 116
- Eclectic Academy in Alexandria: 337
- Eclipse of sun: Thales' prediction of, 16, 17
- Ecpyrosis: 36-7; arguments for and against, 36-7
- Education: 203, 210, 211
- Egypt: 16
- Egyptian mathematics: 16-7
- Egyptians: animal gods, 28
- Eikasia (imaginary): 155
- Elea: 27, 51
- Eleatic logic: 78
- Eleatic school of philosophy: 27, 30; summary remarks, 63-4
- Eleatic stranger: 174, 179
- Eleatics: 57
- Elements: 5, 34, 67-8, 70, 78; as qualities or attributes, 188; of number, 44, 46, 47; principle of change among, 26

- *Elements* (Euclid): 16
- *Elements of Physics* (Proclus): 358
- *Elements of Theology* (Proclus): 358
- Elias: 220-1
- Emotions of men: 10
- Empedocles of Acragas: 51, 64-76; cosmology of, 69; description of divinity, 74; doctrine of formation of world, 70; explanation of love, 81-2; father of rhetoric, 66; fl. 450 B.C., 65; influence of Parmenides on, 66-8; interested in natural world and religion, 65; on creation of living things, 70-1; *On Nature*, 65-72; on roots or elements of living things, 67, 70; on way of truth, 68; Purifications, 65, 72-6; reputation for arrogance, 73; writings, 65-6
- *Enchiridion* (Arian): 323
- End of world: 36-7
- Enemies: do good to, 200
- Enneads, the (Plotinus): 341-42
- Epicrates: 140
- *Enneads, The* (Plotinus): 341-2
- Epictetus: 323
- Epicurean physics: 301-3
- Epicurean school: 297-8; history, 307
- Epicureanism: 297-307; object of vituperation, 297; traditional criticism of, 306
- Epicurus: 307; accepts atomic doctrine of Democritus, 301-2; accusations leveled against, 297-8; Athenian citizen, 297; birth, 297; *Canon, The*, 298; defends criteria of truth in canonic, 302; division of philosophy by, 298; epitomy of physical doctrine, 301; infallibility of sensation, 298-9; *Little Epitome*, 298; makes nature the guide, 304; military service, 297; *Of the End*, 298; *On Human Life*, 298; *Of Nature*, 298; prolific author, 298; teaching career, 297; says wise man is not fatalistic, 304
- *Epinomis* (Plato), 145
- Equivocal things: 236
- Erebus: 10
- Eristic: of Sophists, 140
- Eros: 10
- Error: 172
- Essence: of generated things, 261
- Essential realities: 143, 147, 149, 191
- Eternal flux school of thought, 37-8
- Ethics: 303-7, 310, 318-22
- Ethics: Democritean, 90-2

- Ethiopians: 28
- Eubulides of Miletus: 131-2
- Euclid of Megara: 16, 167, 324; *Elements*, 16
- Eucides of Megara: 131
- *Eudemian Ethics* (Aristotle), 118, 222,
- 228
- Eudemus: 16
- *Eudemus* (Aristotle): 220
- Eudorus: 337; commentary on Aristotle's *Metaphysics*, 337; commentary on the *Timaeus*, 337; school of mathematics, 138
- Euripides: 98, 109
- Eurylochus: 327
- Eurystratus of Miletus: 25
- *Euthydemus* (Plato): 130, 145
- *Euthyphro* (Plato), 112, 147, 148
- *Euthyphron* (Plato), 145
- Evil: 129
- Evils besetting man: 10-1
- "Existential affinity": 151
- Explanation of sensation, 71-2
- Expressions: complex and incomplex, 231-2; true and false, 232

F

- Falsity or truth: 237-8
- Fatalism: 304
- Falsehood: 52ff; way of, as depicted by Parmenides, 52ff
- Father: 346
- Feelings: 300
- Finality: 263
- Finite space: traversal of, 60-1
- Finitude of being: 55, 62-3
- Fire: 23, 34, 35, 36, 38, 187; (Hephaistos): 70
- First philosopher (Thales of Miletus): 3, 4, 15, 18; (Aristotle): 228-30, 284-96; (Plato), 284-96
- *Floruit*: 84
- Foreknowledge: 244-6
- Form: 224, 260, 261, 266; of good: 180; substantial, 258
- Formation of world: 181; account of, attributed to Leucippus, 87; Anaxagoras' theory of, 80; Empedocles' doctrine of, 70; first stage, whirling

vortex, 87-8; Pythagorean doctrine of world, 48ff; second stage, collision of atoms, 88; story of, 180-1

- Forms: 147-59, 179, 180; doctrine of: See Doctrine of Forms; moral, 152; suggestion of hierarchy, 154; mathematical, 152
- Forms of numbers: hierarchy in, 192
- Forms or Ideas: 180; doctrine of: See Doctrine of Forms or Ideas; structural universe, 179
- Fortune: 262-3
- Friends of ideas: identity of, 177
- Friendship: 282-3; role in Epicurean ethics, 305-6

G

- Gaia (earth): 8
- Gaius: 338
- Galen: 18
- Genera: 175, 178
- Generation: 22
- *Generation of Animals* (Aristotle): 222
- Generation of world: See Formation of world
- *Genesis*: 183
- Genesis (source): 7
- Genus: 249
- Geometry: 167; plane, 158; solid, 158; Thales' knowledge of, 16-7
- Giants (sons of earth): 176-7
- Glaucon: 134, 161, 202, 205, 208
- Gnomon: 25
- Gnosticism: 340
- Gnostics: 341
- God: 29, 337; Judaeo-Christian, 30; unity of, 29-30
- God-Soul-Matter: basic stuff of things, 250
- Gods: 4-5, 7-14, 20; bought and won, 212; denial of existence of, 212; discussion about, 28-9; earth, 12; no concern for human affairs, 212, 214; See also Olympian gods
- Gold: 78, 188
- Golden Age of Greek philosophy: 92, 110, 346; ended, 74
- Good: 129, 144, 149; analogy to sun: 153; primacy of, 185; role in intelligible world, 155
- Goodness: likened to sun, 207

- Gorgias of Leontini: 101-4, 109; embraces radical empiricism, 103; fl. in 5th century B.C., 101; *On Not-being or on Nature*, 102; studies under Empedocles, 101; teacher of rhetoric, 101; textbook of rhetoric, 101; three propositions in *On Not-being or On Nature*, 102-4
- *Gorgias* (Plato): 145
- Government: forms of, 208; Plato's interest in, 135
- *Great World Order, The* (Leucippus): 84
- Great Year: 36
- Greece: 16
- Greek polis: 101, 106; description of, 96-7; milieu for Sophists, 96
- Greek primitive religion: 12-4
- Greeks: 3
- Guardians: 204, 207, 208; early training of, 203; higher studies of, 156; lead ascetic life, 204

H

- Hades: 7, 92
- Harmonics: 158
- "Harmony of the spheres" (Philolaus): 49
- Harmony with nature: 319-20
- Harrison, Jane: *Prolegomena to the Study of Greek Religion*, 13
- Hate: 69
- Hearing: 34, 267
- Heavens: 23
- Hector: 6
- Hegel, Georg Wilhelm Friedrich: 230
- Hegelian interpretation of dialectic in *Parmenides*: 166
- Helenese: 181
- Hephaestus: 101
- Hephaistos (fire), 70
- Hera: 7, 67
- Heraclitus of Ephesus: 4, 15, 32-9; arresting style, 33; belief in Ecpyrosis, debatable, 36-7; concern with change, 35, 53; concern with conduct, 95; critical of multitude, 32; doubts about sense perception, 53; ethical message of, 39; (fire): 34; founder of eternal flux school of thought, 378; obscure utterances, 33; *On Nature*, 33; theory of end of world, 36; view of natural world, 36; wealth of quotations throughout ancient literature, 32
- Hermarchus of Mytilene: 307
- Hermes: 43

- Hermes Trismegistus: 340
- Hermeias: 216
- Hermetic writings (*Corpus Hermeticum*): 340
- Hermocrates: 181
- Hermodorus: 136
- Herodotus: 6, 109; concerning transmigration of souls, 41
- Hesiod: 4-14, 223; Influence on Greeks, 6; *Theogony*, 8-14; *Works and Days*, 11-12
- Hippasus of Metapontium: 42
- *Hippias Maior* (Plato): 145
- *Hippias Minor* (Plato): 145
- Hippias of Elis: 98, 105, 109; interested in squaring the circle, 105; Sophist, 105
- *History of Animals* (Aristotle): 222
- Homer, 4-13; *Iliad*, 6, 8; influence on Greeks, 6; *Odyssey*, 7, 8
- Homeric deities: 28
- Homeric epics: 7
- Homicide: 212
- Human body: composition of, 187
- Human reproduction: 28
- Human soul: 190; ascent to highest condition, 208; degradation of, 208
- Hybris (pride): 7
- Hypotheses: of geometer: 156; definition, 156
- Hypothesis: method of, 150

I

- Iamblichus: 357; *Introduction to the Arithmetic of Nicomachus*, 358; *On General Mathematical Science*, 358
- Idea: 148, 224; definition of, 148; synonymous with Form, 148; See also Form
- Idea-number: 221
- Idea of Good: 159-60
- Idea of knowledge: 164
- Ideal: qualifications, 320-2
- Ideal state: modelled after individual soul, 181; plan of, 181; political theory, 181
- Ideas: absolute essences, 164; as dwelling above the heavens, 159-60; criticism of, 286; doctrine of: See

- Doctrine of Ideas; of One and Many: 162, 165; of which other things partake, 162-63
- Ignorance: 153, 160
- *Iliad* (Homer): 6, 8
- Image makers: 167
- Images: 155
- *Images*: (Timon): 327
- Imagination: 267
- Imagining: 155
- Immortality: 20; as reward for moral virtue, 194; of soul, 193-7
- Incarnation: 42
- Incontinence: 129
- Infinity: 25; definition, 264
- Instruction: proceeds from pre-existent knowledge, 244
- Intellect: passive and agent, 268
- Intellectual faculties: 268
- Intelligible nature: likened to father, 188
- Interpretation: definition, 238; or proposition: 239-40
- Interstices and "components": 252
- *Introduction to Arithmetic* (Nicomachus): 340
- *Introduction to the Arithmetic of Nicomachus* (Iamblichus): 358
- Involuntary acts: 276-7
- *Ion* (Plato): 145
- Ionian natural philosophy: 95
- Ionians: 14-40
- Iris: 29
- Isagoge: 357
- Isocrates: 103-4
- Italian philosophers: 40
- Italian school of philosophy: 27

J

- Jewish philosophers of antiquity: 341
- Joy: 69
- Judaeo-Christian God: 30
- Judgment: 171
- Julian, Emperor: 358
- Jupiter: 186

- Justice: 106-7, 204, 205, 281-2; definition, 200; in the state, 202-3; is intrinsic good, 202; Plato's views of, 199; to be preferred to injustice, 201
- Justinian, Emperor: 360-1

K

- Knowing: is active, 177
- Knowledge: 33, 34, 37-8, 119-23, 155; absolute, 164; and opinion, 153; and virtue, 277-8 1; as true opinion plus an account, 173-79; definition, 166-8; division from opinion, 154; fixed, possibility of, 100; is sensation, 168-71; is sensation (*aisthesis*): 168; is true opinion, 171-3; of essential reality, 142-3; of natural world, 251; threefold division of, 225; validity of, 110; and belief: distinction between, 206; and sensation: identification of, 170; and virtue: identification of, 124-30; of Good: highest possession, 153
- Kronos: 7, 10, 11, 74
- Kupris, Queen: 74

L

- *Laches* (Plato), 145, 147
- *Lampoons* (Timon): 327
- Lampsacus: 110, 131
- Land, 36
- Laws, human, 107; (Plato): 20, 145, 209-15
- Lawyer: compared with philosopher, 170
- Learning: 34, 168
- Letters (Plato): 141-2
- *Letters to Lucilius* (Seneca), 323
- Legislation: 213
- Leibnitz, Gottfried Wilhelm: 348
- Leucippus: 84; doctrine of atomism, 84-5; *Great World Order, The*, 84; *On Mind*, 84
- Life: 20, 177
- Life of man: evanescence of, 72
- Life of polis: cynical attitude toward, 107
- Light: 154
- Little Bear: 17
- *Little Epitome* (Epicurus): 298
- *Little World Order, The* (Democritus): 84
- *Lives* (Plutarch): 337

- Living bodies: have souls, 177
- Living species: first members of, 88
- Living things: creation of, 70-1; possess power of sensation, 267
- Logic: 57, 310-6; Aristotle's, 230-49; division of, 311; formal and material, 247; nature and subject matter of, 231-7; to be learned first, 228-9
- Logical entities: 234
- Logical interpretation of dialectic in
- Parmenides: 166
- Locomotion: 264
- Logos: 32-4, 38-9; (account): 173-4; (account): assigned three meanings, 173; (speech), 239
- Love: 9, 10, 69, 74; Empedocles' explanation of, 81-2
- Lucretius: concentrates on sensation, 299; *De Rerum Natura*, 66, 307
- Lyceum: 216, 297
- Lyco: accuser of Socrates, 112
- Lycophron: 104
- Lydians: 15
- Lysis: 120
- *Lysis* (Plato): 145

M

- Macedonian ascendancy: 308
- *Meditations* (Marcus Aurelius): 323
- Magnitude: mathematical, 60; of numbers, 46, 50
- Maieutic: method of questioning, 167-8, 175
- Maker (demiurge): 183, 184
- Man: 100, 186, 353-56; creation of, 181; end of, 271-3; five ages of, 11; "is measure of all things." 98, 99-100; origin of, 11; Plato's view of, 192-215
- Marcus Aurelius, Emperor: *Meditations*, 323
- Marinus: 360
- Mathematical intermediates, doctrine of (Plato): 190-3
- Mathematical numbers: 192
- Mathematical sciences: 156
- Mathematics: 155, 159
- Mathematics: 214; and natural science, 44; and physics, 44; Egyptian, 16-7; firm grounding, 140; Pythagoreans' interest in, 43, 44; stressed in Academy, 140; to be learned second, 229
- Matter: 260, 266; prime, 257
- Maximus of Tyre: 338

- Medes: 15
- Meeting: of Socrates, Parmenides and Zeno, 161
- Megarian school: 131-2
- Meletus: chief accuser of Socrates, 112, 114
- Melissus of Samos: 53, 58, 62-4; fl. 441-440 B.C., 62; divisibility of spherical being, 63; fallaciousness of reasoning, 63; Italian philosopher, 62; on finitude of being, 62-3
- *Memorabilia* (Xenophon): 110, 131
- Memory: 143
- Men: races of, 11-2
- Menedemus: 132
- Menexenus (Plato): 145
- Meno: description of Plato, 119; (Plato): 119, 145
- Mental image: 235-6
- Mental states: 153, 155
- Mercury: 186
- Metaphysical dialogues of Plato: 160; content of, 161
- Metaphysical interpretation of dialectic in Parmenides: 166
- *Metaphysics* (Aristotle): 5-6, 16, 19, 218, 222, 224-6, 230, 233, 235, 260, 269, 337; (Aristotle): 65, 222, 225-6
- Meteorological matters: 189
- Method of hypothesis: 150
- Metrodorus: 307
- Milesian philosophers: 34
- Miletus: 15; "school," 20
- Mind (*Nous*): 81-3, 92, 343, 347-52; all things are directed by, 82; governs universe, 149; is cause, 113; withdrawal into itself: 151; working, in universe, 150; See also *Nous*
- Mist: 23
- Modesty: 121
- Mitylene: 107
- Monism: 176
- Moon: 186
- *Moralia* (Plutarch): 337
- Moral actions, 159
- Morality, 198-215; structure of, 181
- Motion: 59, 177, 263-4; is eternal, 264; quantitative parts of, 264; related notions, 264; unity and oppositions of, 264; Zeno's arguments against, 60; *of Animals* (Aristotle), 222
- Mount Etna: 136

- Mount Olympus: 11
- Multiplicity: 53, 55, 58, 59
- Muses: 8
- Musical intervals: expressed in numerical ratios, 42
- Musical scale: attributes expressible in numbers, 44
- Mysticism: 151
- Myth: 3, 5; (mythos): definition of, 222-3; (*philomythos*): 222
- Mythos (myth): definition of, 222-3
- Myths: of barbarians, 12

N

- Natural doctrine: of Plato, 180-92
- Natural intelligence: 143
- Natural philosophy: 31, 149
- Natural Questions (Seneca): 323
- Natural science: See Science of nature
- Natural world: 36, 65, 140; knowledge of, 251; Plato's views on, 180
- Nature: 20, 260-3; boundless, doctrine of Anaximander, 21-4; common, binding all men together, 106; definition, 260; first known written account of, 21; harmony with, 319-20; philosophy of, 250-69
- Nausiphanes: 326
- "Nautical Star Guide, The:" 17
- Neoplatonic interpretation of dialectic in Parmenides, 166
- Neoplatonism: 337, 339-61; after Plotinus, 356-61
- Neopythagoreanism: 339-40
- Neopythagoreans: 340
- New Academy: 324-38; See also Academy
- *Nicomachian Ethics* (Aristotle), 222, 229, 269-84
- Nicomachus, son of Aristotle: 216
- Nicomachus, father of Aristotle: 216
- Nicomachus of Gerasa: 45, 339; *Introduction to Arithmetic*, 339-40
- Night: 9
- Nile: 16
- Nile delta: 19
- Noesis/episteme (knowledge): 155; (water): 67,70
- Non-physicists: 254
- Non-philosophy: 3-4, 222
- Non-physical being: 57
- Nouns: 238; definition, 238; indefinite, 239

- *Nous*: Anaxagoras' doctrine on, 81-3; exception to dictum that everything is in everything, 81; given as cause of separating off, 82; infinite and self-ruled, 81; motive force behind revolution, 81; (mind): 81-3, 92, 343, 347-52
- Number: 214; elements of, 44, 46, 47; whole nature of, 42
- Numbers: as primary stuff, 44; even, 47; Forms, 192; have magnitude, 50; have unity, 192; identified with sound, 42; identified with things, 42, 43; linear, 45; mathematical, 46, 192; mathematical, 192; oblong, even and unlimited, 46; odd, 47; plane, 45; solid, 45
- Numenius of Apameia: 340
- Numerical ratios: identification with
- sounds, 48; musical intervals expressed in, 42
- Nutritive power: 267
- Nymphs: 10

O

- Objects: 150, 155
- Objects of mathematics: 191-92;
- Objects: three classes of, 142
- Ocean: 19
- *Ode of the Intimations of Immortality*: 152
- *Odyssey*, Homer: 7
- Oeconomicus (Xenophon): 131
- *Of the End* (Epicurus): 298
- Okeanos: 7
- Oligarchy: 208
- Olympian gods: 7-13; genealogy, 32; sequence of generation, 9, 28
- Olympian religion: 12
- Olympians: See Olympian gods
- Olympiodorus: 358, 361
- Olympus: 7
- *On being* (Protagoras): 99
- *On Democritus* (Aristotle): 87
- *On General Mathematical Science* (Iamblichus): 358
- *On Generation and Corruption* (Aristotle): 222
- *On Human Life* (Epicurus): 298
- *On Inquiry* (Aenesidemus): 334
- *On Interpretation* (Aristotle): 222, 231, 237-44, 318
- *On Logic*: Zeno of Citium, 309
- *On Mind* (Lucippus): 84

- *On Nature* (Anaximander): 21; (Diogenes): 93; (Empedocles): 65-72; (Epicurus): 298; (Heracletus): 33; (Prodicus): 104; (Xenophanes): 27
- *On Not-being or on Nature* (Gorgias): 102
- *On Philosophy* (Aristotle): 220, 221
- *On the Heaven* (Aristotle): 222
- *On the Nature of Man* (Prodicus): 104
- *On the Gods* (Protagoras): 99
- *On the Planets* (Democritus): 84
- *On the Soul* (Aristotle): 220, 222, 229, 265-9
- *On truth* (Protagoras): 99
- One, the: 59, 161, 165, 343, 348-52; and the Many: problem of, 161-6
- One being: properties of, 54ff
- Opinion: 153, 182; and knowledge, 153; division from knowledge, 154; (*doxa*): 171; false, 171-4; false, definition, 174-5; true, 171-9; way of, as depicted by Parmenides, 52, 57-8
- Opposition of proposition: doctrine of, 241
- Order and methodology: importance of, 255
- Organon (Aristotle): 230-1; (Aristotle): plurality of works in, 231
- Orphic religious doctrines, 151
- Osphicism: 76
- Ouranos (Sky): 7-8
- *Outlines of Pyrrhonism* (Sextus Empiricus): 336

P

- Pain: 129
- Panaetius of Rhodes: 309
- Pamphilus: 297
- Paradox of the liar: 131-32
- *Parallel Lives of Illustrious Greeks and Romans* (Plutarch): 337
- Parmenidean dilemma: 56, 83, 252-3
- Parmenidean poem: 51ff; methodology, 56; prologue, 51ff
- Parmenidean sphere of being, 56, 57, 102
- Parmenidean way of truth, 52ff, 68, 75-6, 92
- Parmenides of Elea: 161-6, 174; attempt to reconcile being and nonbeing, 53-5; confronted, 258-60; denies possibility of change, 252; denies science of nature, 250; doctrine about world around, 57, 252; doctrine, as reflected in poem, 51ff; doctrine of, 252; doctrine utterly arbitrary, 53; fl. about 475 B.C., 50; inquiry into falsity and truth, 52ff; interested in Theory of Ideas, 162; interrogates young Socrates, 161; natural doctrine of, 58; on finitude of being,

55, 62-3; pupil of Xenophanes, 51; Pythagorean, 51; rejects dualism of Pythagorean doctrines, 56; source of being, 54; way of opinion, 52, 57-8, 254; way of truth, 52ff, 68, 75-6, 92

- *Parmenides* (Plato): 50, 59, 134, 145, 160
- Participation: 177; involves infinite regress: 163-5; limited, 178
- Particulars: 180, 182
- Patroclus: 6
- *Parts of Animals* (Aristotle): 222
- *Parva Naturalia* (Aristotle): 222
- Pausanias: 66
- Peleus: 8
- Peloponnesian war: 135
- Pentateuch, The: 341
- Perception: Protagoras statement on, 99-100
- Pericles: 98
- Perictione: 134
- Persaeus: 308
- Persian Magi: considered agnostics, 99
- Peripatetics: 307
- Persian Magi: Protagoras taught by, 98-99
- Persian monarchy: 210
- Persuasism: art of, 100
- *Phaedo* (Plato): 82, 113, 115-8, 145, 148-51, 220
- *Phaedrus* (Plato): 145
- Phaenasete: 111
- *Philebus* (Plato): 145
- Philip of Macedon: 216
- Philiponus, Johannes: 361
- Philo Judaeus: 341
- Philo of Athens: 327
- Philo of Larisa: 334
- Philolaus: 41; "harmony of the spheres," 49
- *Philomythos* (myth): 222
- Philosopher: compared with lawyer, 170; definition, 174; (*Philosophus*): 222
- Philosophers: as rulers, 206-7
- Philosophical sciences: order among, 228-30
- Philosophy: and non-philosophy, 222; arises from wonder, 223; definition of, 224; First, 228-30, 284-96; indispensable first part, 298-9; love of wisdom, 224; meant to purify religion, 4; moral and political, 269-84; natural: See Philosophy of nature; nature and division of, 222-3Q.

- Philosophy of nature: 31, 149, 250-69; bulk of Aristotle's writings devoted to, 250; historical background, 251-4
- Philosophy: ordered whole of sciences, 230; origins of, 3ff; practical, division of, 228; preparation for death, 193; quarrel with poetry, 4-6; six parts of, 310; speculative and practical, 225-7; Stoic division of, 231; study of death, 115; theoretical, division of, 227-8
- *Phronesis*: 221
- Phrynacus: 307
- Physical things: definition, 260-1
- Physical universe: 183
- Physical world: 58
- Physics: 260, 301-3, 310, 316-8; 260; and mathematics, 44; of Stoics: 316-8; (Aristotle): 21, 222, 250, 253-4, 260, 264-5; (Simplicius)
- Physiology: human, 189
- Piety: 147-8
- Pistis (belief): 155
- Place: definition, 264
- Planets: 10, 186
- Plants: 267
- Plato: Academy founded by, 137; *Apology*, 112-15, 145; aristocratic connections, 134; authentic dialogues: 145; birth, 134; *Charmides*, 120-3, 145, 147; claim to greatness, 134; *Cosmology*, 189; *Cratylus*, 145; crisis in thought, 160-80; *Critias*, 145; *Crito*, 145; death, 134; denies science of nature, 250; dialogues: See Dialogues of Plato; distaste for democracy, 135; Doctrine of Forms or Ideas, 118, 144, 147-60; doctrine of mathematical intermediates, 190-2; *Epinomis*, 145; *Euthydemus*, 130, 145; *Euthyphro*, 112, 147, 148; *Euthyphron*, 145; extensive travels, 136; family, 134, 136; First Philosophy, 284-96; follower of Socrates, 136; Gorgias, 145; head of Academy, 138; *Hippias Maior*, 145; *Hippias Minor*, 145; influence of Orphic and Pythagorean religious doctrines on, 151; involved in Syracusan government, 198; *Ion*, 145; knowledge is sensation, 168-71; *Laches*, 145, 147; *Laws*, 20, 145, 209-15; letters, 141-2; *Lysis*, 145; man and his work, 134-47; mathematical intermediates, doctrines of, 190-3; members of family in dialogues, 136; *Menexenus*, 145; *Meno*, 119, 145; method of introducing students to philosophy, 138-9; natural doctrine: 180-92; nature of soul, 193-8; on morality and politics, 198-215; no oral teaching given by, 144; *Parmenides*, 50, 59, 134, 145, 160; perhaps nickname, 134; *Phaedo*: 82, 113, 115-8, 145, 148-51, 220; *Phaedrus*, 145; *Philebus*, 145; political attitudes, 135-6; present at Socrates' trial, 136; *Protagoras*, 98, 124-9, 145; reasons for going to Sicily, 136-7; remark on unity of god, 30; *Republic*, 4, 134-5, 139-40, 144-6, 150, 153-60, 180, 199-209; search for definitions, 158-9; search for

truth, 140; Socratic dialogues: See Dialogues of Plato: Socratic; *Sophist*, 145, 160, 174-80; *Statesman*, 145, 160, 179-80, 183; *Symposium*, 145; *Theaetetus*, 119, 145, 160, 166-74; *Timaeus*, 134, 145, 180-90, 337; unwritten doctrine, 144; view of man, 192-215; views on natural world, 180; voyage to Sicily, 136; writings, 141-7

- Platonic Academy: See Academy
- Platonic dialectic: 140
- Platonic doctrine: 142, 143, 309; fundamental: See Doctrine of Forms
- Platonic method: unity of, 145
- Platonism: 217, 221
- Platonists: 307
- Platopolis: 341-2
- Pleasure: 129; and pain, primacy of, 210; fulfillment of happy life, 303-4
- Plotinus: 3, 338, 341-56; birth, 341; *Enneads*, *The*, 341-3; hope of founding Platopolis, 341-2
- Plurality: 176
- Plutarch of Athens: commentary on
- Aristotle's *De Anima*, 358
- Plutarch of Chaeronea: 71, 337; *Lives*, 337; *Moralia*, 337; *Parallel Lives of Illustrious Greeks and Romans*, 337
- Plutocracy: 208
- Plynacus: 307
- Poetics (Aristotle): 6, 65, 222, 223, 231
- Poetry: 3-6; and philosophy, relation between, 209; quarrel with philosophy, 4-6
- Poets: theological, 6-12
- Points: bodies generated from, 47; geometrical, 46
- Po emarchus: 199, 200
- Polemo: 328
- Polis: See Greek *polis*
- Politics: 198-215, 310; (Aristotle): 222, 269-84
- Polyclitus: 261
- Polycrates: 40; *Accusation of Socrates*, *The*, 131
- Polytheism: 29
- Pontus: 9
- Population of city: 211
- Porphyry of Tyre: birth, 356; student of Plotinus, 341; summary of Pythagoras doctrine, 41
- Poseidon: 7, 74
- Posidonius: 334

- Possibility: reality of, 132
- *Posterior Analytics* (Aristotle): 222, 227, 231, 244-9
- Potone: 134
- Praxiphanes: 297
- Preconceptions: 299-300
- Predication: possibility of, 132
- Prediction: 178
- Premiss: definition, 242; demonstrative, 242; dialectical, 242; division into demonstration and dialectical, 242
- Premises: 242
- Presocratic philosophy: 76
- Presocratics: 110, 130, 254, 256, 308
- Priam: 6
- Pride (*hybris*): 7
- Priests in Babylonia: 16
- Prime mover: 221, 264, 269
- Prime Mover: immobile and incorporeal, 264
- *Prior Analytics* (Aristotle): 222, 231, 241-4, 247-8
- Proclus: 16; birth, 358; commentaries, 358; *Elements*, 358; *Elements of Physics*, 358; *Elements of Theology*, 358 works on Plato's theology, 358
- Procreation: 88
- Prodicus of Ceos: 104-5, 109; concern with correct: terminology, 104-5; fl. end of 5th century, 104; on how gods arose, 105; *On Nature*, 104; *On the Nature of Man*, 104; Sophist, 104
- Productive class: 207
- *Progression of Animals* (Aristotle): 222
- *Prolegomena to the Study of Greek Religion*: 13
- Prometheus: 98-101, 109, 124-30
- Properties of being: 54ff
- Property: 249
- Proposition or interpretation: 239-40
- Propositions: compound, 240; having universal subject, 240-1; immediate, 248; simple, 240; truth or falsity of, 241
- Protagoras of Abdera: 98-101, 109, 124-30; accused of impiety, 98; *Antilogiae*, 99; fl. in 5th century B.C., 98; *On being*, 99; *On the Gods*, 99; *On Truth*, 99; pupil of great atomist, 98; *Refutatory Arguments*, 99; sceptical doctrine of, 100-1; Sophist, 124; statement on perception, 99-100; statement that "Man is measure of all things," 98, 99-100; taught by Persian Magi, 98-99; writings, 98-9
- *Protagoras* (Plato): 98, 124-9, 145

- Protrepticus (Aristotle): 220-2
- Psychology: 233
- Public service: 208
- Punishment for misdeeds: 75
- Purifications (Empedocles): 65
- Pyramids: measured by Thales, 16
- Pyrrho of Elis: 324-7; birth, 324; founder of sceptics, 324
- *Pyrrhonian Discourses* (Aenesidemus): 334
- Pythagoras of Samos: 40-3 belief in identity between number and sound, 42; birth, 40; incommensurability of diagonal and side of square, 42, 43; son of Hermes, 43; theories of number, 42, 44-7; pupils, 40-1
- Pythagorean communities: rules of conduct, 42
- Pythagorean community: 41
- Pythagorean cosmogony, 52; 46-7, 58 Pythagorean doctrines: 43-50; concerning pursuit of truth, 43; identification of things with numbers, 42; dualism of being and non-being, 56; magnitude of numbers, 46, 50; on formation of world, 48ff; role of opposites in world, 47; things are numbers and aggregates of unit-points, 59; view of extent of reality, 48; view of universe, 49, 50
- Pythagoreans: 40-50; abstention from meat, 41; influential in southern Italy, 50; interest in mathematics, 43-50; renowned for secrecy, 41
- Pythagorean mathematics: as cosmology, 48
- Pythagorean religious doctrine, 151
- Pythagorean society: 42, 137, 139; scientific contributions, 42
- Pythagorean theorem: 41, 42, 43
- Pythagoreanism: revival, 339-41
- Pythias, daughter of Aristotle: 216; wife of Aristotle: 216

R

- Race track: impossibility of traversing, 60, 61-2
- Real world: 183
- Reality: structure of, 181
- Reason: as depicted by Parmenides, 52ff
- Reasoning or discourse: 232
- Receptacle: arena of change, 188; identified with space, 189-90; likened to mother, 188
- Recollection (anamnesis): 168, 194, 220
- Reflections: 155
- *Refutatory Arguments* (Protagoras): 99

- Relativity: 35
- Religion: 3; Greek primitive, 12-14; Cthonic, 12; earth, 12; Epicurus' attitude toward, 306; Olympian, 12
- Remembering (anamnesis): 152
- *Republic* (Plato): 4, 134-5, 139-40, 144-6, 150, 153-60, 180, 199-209
- Rest: 177-8
- Rhea: 7
- Rhetoric: 66, 310, 311
- *Rhetoric* (Aristotle): 65, 222, 231
- Rituals: of barbarians, 12
- Roman Empire: sway over Greece, 308
- Roman Stoics: 322-3
- Roots: See Elements
- Rulers: 204; legislate in own interest, 200

S

- Samos: 40
- Saturn: 186
- Sceptics: 307, 324-38
- School of Athens: 358
- "School" of Croton: 41
- School of Elis: 132
- School of Eratria: 132
- School of Isocrates: 137-8
- School of Miletus: 20-1
- School of Pergamon: 358
- Science of nature: 250; and mathematics, 44; basic and contradictory assumptions, 253-4; dialectical summary, 254-5; historical background, 251-4; order of procedure, 250-1
- Scripture: 185
- Sea: 36
- Seasons: 10
- Second-best method of discovering true existence, 149-50, 156
- Self-knowledge, 130
- Self-nutrition: 266, 267
- Seneca, L. Annaeus: birth, 323; death, 323; Dialogues, 323; Letters to Lucilius, 323; Natural Questions, 323; tragedies, 323
- Sensation: 7 1-2; (aisthesis): 168; infallibility of, 298-9; Lucretius concentrates on, 299; mechanisms of, 189; powerless to produce knowledge, 174

- Sense and intellect: difference between, 268
- Sense perception: Democritus' views on, 88; Parmenides' view of, 52ff; validity of, 56
- Senses: external, 267; internal, 267
- Sensible universe: fluidity of, 188
- Sensuality: 151
- Sensible things: 151-2
- Sensible world: 183; arena of change, 180; cause, 182
- School of Plato: 138
- School of Socrates: 131
- Separating off: 81; Anaxagoras' theory of, 80; Anaximander's theory of, 23, 25; *Nous* given as cause of, 82
- Seven sages: called Sophists, 98
- Sextus Empiricus: 51, 132, 312, 333, 336-7; *Adversus Mathematicos*, 336; *Outlines of Pyrrhonism*, 336
- Sexual morality: 211
- Shadows, 155
- Sicily: Plato's visits to, 136-7
- Sight: 267
- Simmias: 195
- Simonides: 199
- Simplicius: 51; commentary on Aristotle's *Physics*, 21; *Physics*, 25
- Sins, of parents: visited upon children, 197
- Sky: 10; (Ouranos): 7-8
- Smell: 267
- Snake god of the Cthonic religion: 28
- Society: corrupt, 198; entrusted to philosophers, 198
- Socrates: 5, 50, 109-33, 143, 181; accepts death verdict, 114, 115; accused of atheism, 114; accusers of, 112; attachment to Athens, 109; belief in immortality, 193; birth, 109; called Sophist, 109; character, 112-7; charged with pederasty, 116; charges against, 112ff; concern with universals, 130; death, 50, 109, 111, 112, 115-7, 136; death, description of, 116-7; doctrine, 117-30; doctrine of Forms, 118; ethical doctrine, 118; factual, 112; his life, 109; historical, 145-6; ideal state: See Ideal state; inner voice, 115; interested in problems of state, 198; irony of, 119; lampoon by Aristophanes, 110; maieutic art, 119, 120; meaning attached to knowledge, 124-30; method of questioning, 167; midwifery (maieutic), 167-8, 175; of Platonic dialogues, 110-1; of Xenophon, 111; philosophical activity, 117; Pythagorean and Orphic attitudes, sympathy with, 116; reply to his accusers, 112; search for definitions, 118; Sophist, teaching for

fee, 113; trances, 116; trial, 112; view of philosophizing, 130; wisdom, 119; wrote nothing, 110; writings about, 110

- Socratic dialogues: See Dialogues of Plato: Socratic
- Socratic legends: 111
- Socratic method: 119, 120-1
- Socratic Problem: 110-2, 145
- Socratic schools: 130-3
- Solon: 134, 181
- Sophist: definitions, 174-5; (Plato): 145, 160, 174-80
- "Sophistes": earlier definition of, 97
- *Sophistical Refutations* (Aristotle): 222, 231
- Sophocles: 109
- Sophroniscus: 111
- Sophrosyne (wisdom), 121
- Sophists: 37, 94-108; assessment, by Plato and Aristotle, 108; concern with man, 95; connotation of word, 97-8; dialectic, largely fallacious, 108; doctrines, as exercises in method, 107; education of men by, 97; flourish in Greek polls, 96; live by their wits: 96; paid teachers, 97-8, 110; training men for life in *polis*, 106, 109; wanderers, 96; term of abuse or denigration, 97
- Soul: 177, 193-8, 343-8; basic type of, 267; composed of atoms: 89; creation of, 190; definition, 196, 220, 265-7; determinate type of, 265; exists before birth, 194; first birth of, 187; genus containing, 265; Heraclitus' remarks about, 38-9; human, 267, 268; immortality of, 193-7; imprisoned in body, 182; Intellective, immortal and eternal, 268; nature and properties of, 265; nature of, 193-8; parts of, 265-6; parts or faculties of, 267; pervades universe, 19-20; rational part immortal, 197; setting free of, 75; structure of: mode[for ideal state, 180-1; substantial form of living body, 266; three parts of, 196, 198; threefold division of, 196; world, 185
- Souls: animal, 267; guide heavenly bodies, 214; transmigration of, 41, 73
- Sound: identified with number, 42
- Source (genesis): 7
- Species: 175
- Speech (*logos*): 239
- Speusippus: 137, 138, 217; head of Academy, 134, 138
- Sphere: description of, 70; of being: 56, 57, 74-75, 102; well rounded: 176
- Spheres: See also "Harmony of the spheres," 49
- Spherical being: divisibility of, 63
- Stars: 10
- State: 90
- States of mind: See Mental states

- *Statesman* (Plato), 145, 160, 179-80, 183
- Stilpo: the Megarian, 131, 308, 327
- Stoa: 331
- *Stoa poikile*: 308
- Stoic division of philosophy: 231
- Stoic doctrine of categories: 312-3
- Stoic *katalepsis*, 327, 328
- Stoicism: 132, 307, 308
- Stoics: 36, 307-23
- Stoics: Roman, 322-3; theory of end of world, 36
- Strife: 35, 37, 69
- Structure of morality: 181
- Structure of reality: 181
- Styx: 19
- Substance: 234-5, 256-7; threefold sense of, 266
- Sun: 38, 186; ordered movement of, 213-4; role in visible world, 155; source of light, 154
- Sun / Good: 157
- Survival: 71
- Syllogism: demonstrative, definition, 246; demonstrative, nature of premisses, 247-8
- Syllogisms: definition, 242; demonstrative, 241-2, 245; demonstrative or scientific, 244; perfect and imperfect, relationship between, 242; subject and predicate of, 243-4
- Symmias: 115
- *Symposium* (Plato): 145; (Xenophon): 116, 131
- Syracusan Tyrants: 136-7
- Syrranus; 358
- Syrian school: 357
- Syrianus: commentary on Aristotle's *Metaphysics*

T

- Tartarus: 9
- Taste: 267
- Teachers: fees, 98; travelling, Sophists as, 98
- Teisias: author of first textbook of rhetoric, 101
- Temperance: 120-1, 204, 205; definitions, 121; is a science, 121
- Terms: relation of, 243
- Tethys: 19

- Thales of Miletus: 3-4, 15-20; anecdote by Plato concerning, 17-8; Aristotle's view of, 18-20; doctrines of, 18-20; first philosopher, 3, 4, 15, 18, 19; fl. sixth century B.C., 16; knowledge of astronomy, 15-7; Ionian, 27; knowledge of Geometry, 16-7; measurement of pyramids, 16; mythical figure, 20; predecessors called non-philosophers, 3; prediction of eclipse of sun, 15-7; one of seven sages of ancient world, 15; reputation and importance, 17; (water): 34
- Theadorus: 167, 174
- Theaetetus, 174; (Plato): 119, 145, 160, 166-74
- Themis: 10
- *Theogony*: Hesiod: 28
- Theological poets: 6-12
- Theology: 214, 310
- Theon of Smyrna: 338
- Theophrastus: 21, 66, 328
- Theses: either suppositions (hypotheses) or definitions, 248
- Things: universal and singular,
- distinction between, 240
- Thinking: 155
- Thirty Tyrants: 198
- Thought: identified with flow of blood, 72; true existence revealed in, 151
- Thracians: 28
- Thrasyarchus of Chalcedon: 105, 200; Great Text book, 105; Sophist, 105; Subjects for Oratory, 105; view of justice, 105
- Thucydides: 109
- Thunderbolt: 38
- *Timaeus* (Plato), 134, 145, 180-90, 337; most influential writing, 182; treated as Scripture, 185
- Timaeus of Locris: 181; astronomer, 181; discusses nature of universe, 181
- Time and heaven: interdependent, 186
- Time: beginning of: 184; definition, 264
- Timocracy: 208
- "Timon and Hades:" 327
- Timon of Phlius: 324, 326-8; Images, 327; prolific writer, 327; Lampoons (*Silli*), 327
- Titans:
- *Topics* (Aristotle): 64, 222, 231, 249
- Touch: 267
- Transmigration of souls:
- Treatises (Aristotle): 218, 221-2; consequent on Aristotle's *Physics*, 264-5
- Trilogy of Platonic Dialogues: depicting forming of world, 180-1

- Troy: fall of:
- True existence: revealed in thought, 151; second-best way of discovering, 149-50, 156
- Truth (Antiphon the Sophist), 105; criterion of, 311-2; pursuit of, 43; way of: as depicted by Parmenides, 52ff, 68, 75-6, 92; way of: Empedocles on, 68; Parmenidean, 52ff, 68, 75-6, 92; or falsity: 237-8
- Typhaeus:
- Tyrants: success of, 201; Syracusan, 137; Thirty, 198

U

- Units: arithmetical, 45, 46; configurations in space, 46; have spatial magnitude, 46
- Unity of God: 29-30
- Unity of opposites: 37
- Universals: 236-7
- Universe: 38; Anaximanders picture of, 21-4; anthropomorphic interpretation, 95; nature of, 181; Pythagorean view of, 49, 50; working of mind in, 150

V

- Venus: 186
- Verbs: definition, 238
- Virtue: 119-20, 273-4, 355; and knowledge, 277-81; and knowledge, identification of, 124-30; moral, acquisition of, 274-5; moral, definition of, 275-6; taught by Sophists, 97; four, 204
- Visibilia: 155
- Void: 264

W

- War: 35; need for, 203
- Water: 34, 70, 187; (*nestis*): 67; primacy of, 18
- *Way of Opinion* (Parmenides), 254
- Wind: encloses whole world, 27
- Winds: 10
- Wisdom: 34-5, 38, 204, 205; and knowledge, 121-2; definition, 121-2, 224; is a science, 121; love of, 228; (*sophrosyne*): 121
- Wise man: characteristics of, 225

- Women: 206; education of, 211
- *Works and Days* (Hesiod): 11-12
- World: 35; beginning in time: 184; coming into being, 9; common, to all awake, 38; consumed by fire, 367; divided into sensible things and Forms, 152; end of, 36-7; formation of: See Formation of world; of Homer, 12; private, of men asleep, 38; See also Created world; Natural world; Physical world; Sensible world; World of Forms, etc.
- World body: 185-6
- World of Forms: 183, 189-91
- World soul: 185, 190

X

- Xanthippe: 111, 162
- Xenocrates: 137, 138, 217, 308
- Xenophanes of Colophon: 4, 15, 29-32; affirmative remarks about god: 29; analogue for Sophists, 96; anecdote concerning Pythagoras, 41; birth, 27; complaints concerning Olympian gods, 28-9; concern with conduct, 95; divinity, major concern of, 29; contribution to philosophical theology, 31; divinity should not be localized, 28; doubts about sense perception, 53; (earth): 34; founder of Italian or Eleatic school of philosophy, 27; criticism of poets, 32; Ionian, 27; natural philosophy, 31; *On Nature*, 27; poet, 27; remarks on unity of god, 29-30; view of divine, 53
- Xenophon: *Apology*, 131; birth, 131; concerned with defending Socrates, 110; *Cyropaideia*, 131; *Memorabilia*, 110, 131; *Oeconomicus*, 131; *Symposium*, 131 382

Z

- Zeno of Citium: 307, 308, 328; birth, 308; founder of Stoicism, 132; *On Logic*, 309
- Zeno of Elea: 50, 58-62, 174; arguments in favor of the One, 161; birth, 59; founder of dialectic or logic, 59, 66; four arguments against motion, 60-2; Pythagorean, 59; forty arguments against multiplicity, 59; traversal of finite space, 60-1
- Zeus: 7-13, 38, 67, 74, 101
- Zeus Meilichos: 13

Volume II

Part I: The Age of Augustine

Chapter I

Faith and Philosophy

In a witty inaugural lecture C. S. Lewis spoke of the difficulties that attend dividing history into periods.⁽¹⁾ With less wit than embarrassment we call attention to the fact that the period of philosophy whose history we hope to sketch in the present volume requires that we begin by saying something of a number of men who lived before Plotinus, whose philosophy provided the final discussion of our Volume One. We do have, as it happens, a reason for excluding those men before and including them now, a reason we alluded to when we made the briefest of mentions of Philo Judaeus. (Cf. Vol. I, p. 341.) That reason is this: all of the men who are included in this volume were heavily influenced in their philosophical thinking by revealed religion. Indeed, so decisive is this influence that it is only gradually that anything like an independent charter is reissued to philosophy, and, after its issuance, it is seemingly the rare thinker who pursues philosophy in any independent fashion. The men we shall be considering are believers, religious men, who claim to have an access to truths got not by strenuous intellectual effort but by the gratuitous gift of God. In short, their faith gives them answers at the outset to many of the questions that the pagan philosopher posed. It may well be asked how such men can be considered figures in the history of philosophy, and indeed for centuries the great stretch of time from Plotinus to Descartes was regarded as an exclusively theological period of no interest to the historian of philosophy, at least in any substantive sense. This assessment has not completely disappeared today.

Prior to addressing ourselves to the problem just foreshadowed -- the possible coexistence of faith and rational thought -- in the period that interests us, it may be useful to allay our fears in a generic way by recalling an aspect of ancient thought we were concerned to underline in our first volume. Ancient philosophy did not spring full-formed from the brow of Thales nor were its problems got by a free and unfettered gaze at the natural world. From its inception ancient philosophy was theological in orientation -- the very term suggests the pursuit of a wisdom which consists in knowledge of the divine -- and this orientation

can be looked upon as the bequest of mythical thought which both antedated and to some degree was concomitant with the origin of philosophy. Mythical thought had become, as it were, institutionalized in literature as well as in the official religion, and ancient philosophers can be regarded as pursuing their task with an eye on these institutionalized attitudes. It would be difficult to say how tongue in cheek is Plato's statement that the poet is a vehicle of something like a divine revelation, or how unserious is Aristotle's notion that mythical tales are allegories of profound philosophical thought which alone survive in fallow periods. Both men, though Aristotle to a far far less degree, were prone to treat the poetic statement as an allegory of a straightforward literal truth.

Both men, too, and in this they were tributaries of earlier efforts, saw their philosophies as a replacement of popular religion, almost we might say as a better kind of religion: a way of life, a total commitment to the ultimate acquisition of knowledge of the divine. Differences there were between these two giants of ancient thought and certain it is that ancient philosophy is not perfectly homogeneous, but it is nonetheless a safe generalization that ancient philosophy did not proceed, either in fact or in desire, in complete isolation from ancient religion. There was a quarrel between them, to be sure, but it was a quarrel between a dreamt-of norm and a degenerate instance.

Viewing ancient philosophy from this angle, we can see a slight similarity between the pagan philosopher and the man of faith who began to meditate on the content of his beliefs. However, a fundamental difference can be introduced. The Christian thinker did not regard his faith as something in need of a firm rational footing, as if what he had accepted out of trust in the word of God must finally be deduced by him from the evidence of things seen. This is a generalization and therefore a simplification; one of the major motifs of the study before us is contained in that assertion, and there will be many variations on it. What will emerge in the golden period of medieval thought is the hard-won conviction that it is faith which measures natural reason and that it is eminently reasonable that this be so. For the Christian it is philosophy which must first be justified, not the faith, and if this period opens with men called Apologists, men who defend the faith it is the addressee of the apologia, not its writer, who is thought to be in need of what is said. It is almost as if the difficulties are thought to reside only in the mind of him who has not the faith.

But of course since the recipient of faith is a man, a rational creature, it is as right as it is inevitable that he will meditate on revealed doctrines, apply natural reason to them, and that this effort will have intramural ramifications and

benefits. For one thing, it is necessary that believers retain a clear and accurate knowledge of what has been proposed for their faith. The orthodox expression of the content of revelation is something which, from the beginning, is attained against the background of the heterodox, the heretical understanding. It is in this sense that there is and must be a development of Christian doctrine, a gradual clarification in the light of hitherto unthought-of difficulties and interpretations of what the true sense of Scripture is. This true sense is not had merely by pointing at biblical passages; it is the interpretation of the passages that is at issue, and the orthodox interpretation, as much as the heterodox, will consist in bringing to bear on the documents of revelation an apparatus of interpretation which is not itself revealed. From the beginnings of the Christian period there is fairly widespread agreement that this apparatus is something which can be provided by philosophy.

We do not mean to suggest that there is at the outset a clear understanding of philosophy as an autonomous and legitimate activity. Far from it. Tertullian (c.160 - c.240), one of the first Christian writers to present his thought in Latin, had little but contempt for pagan philosophy. For him philosophy was the locus of error, Christianity the summation of truth, and what has truth to learn from error? It may have been this contempt for natural reason that led Tertullian to hyperbolic excess in the claim that the truths of Christianity are absurd.

A far more widespread attitude was that expressed by Eusebius (c.265 - c.339) in the title of a book, *Praeparatio evangelica*. Ancient wisdom, pagan philosophy, should be regarded as struggling toward the truth which has been revealed whole and entire, once and for all, by Christ. Christianity is the true philosophy, the telos toward which antiquity tended. St. Justin Martyr (c.100 - 164) was an early exponent of this view; St. Clement of Alexandria (c. 150 - c.219) was another; St. Gregory of Nyssa (c.335 - c.395), who was to have such influence on John Scotus Erigena, yet another. Men who felt as they did can be expected to show a sympathetic interest in the writings of the philosophers and, generally speaking, they do exhibit such an interest. By the same token, of course, such an attitude implies that philosophy, in the Greek sense, is an historical moment that has been surpassed, since whatever there is of good in ancient philosophy is contained in an eminent and perfect fashion in Christianity.

Before describing further the thought of men who saw some positive good in ancient philosophy, it might be well to indicate how they can be grouped together. First, there are the Greek Apologists, the most important of whom are Justin Martyr, already mentioned, St. Irenaeus (born c.126), and Hippolytus

(died c.236). Second, note must be made of Clement of Alexandria and Origen (c.185 - 254) of the Catechetical School of Alexandria. Of the Latin Apologists, Tertullian, Arnobius (c.260 - c.327), and Lactantius (c.250 - c.325) are the most important. Other men of importance in what we may call the pre-Augustinian period are St. Athanasius (died 373), St. Gregory of Nazianzus (died 390), St. John Chrysostom (died 406), St. Basil (died 379), and his brother, St. Gregory of Nyssa.

Justin Martyr's own route to Christianity is presented by him in such a way that it exhibits the historical process writ small. As a pagan he went to philosophers in the expectation that they would speak to him of God, and though he was dissatisfied with the Stoic, the Peripatetic, and the Pythagorean he encountered, his needs were met when he came under the tutelage of a Platonist. Here at last he had the sense of being introduced to immaterial things, and in his efforts to contemplate the Ideas he half expected to see God. He then describes an encounter with a Christian who casts doubt on salient features of Plato's thought: his views on the nature of creation, the soul, and its immortality. The man speaks with such assurance that Justin asks him where he has learned so much, and he is directed to the Scriptures. Upon reading them, his soul was set aflame, and he concludes that he has found the safe and profitable philosophy. Besides seeing Christianity as the true object of the philosophical quest, Justin points out similarities between statements of Scripture and the theories of Plato. Justin felt that the reason for such similarities was that the Greeks had borrowed ideas from the Jews. His suspicion that Greek philosophy had been influenced by the Old Testament was shared by Clement of Alexandria, as it would be by St. Augustine.

Clement, however, held that the pagan thinkers were influenced by the divine Logos in somewhat the same way that Moses and the Jewish prophets had been influenced. Pagan philosophy, like the Old Law, was a preparation for Christianity. Clement is one of the first to insist that philosophy may also provide an instrument for understanding the faith. With the aid of philosophy the truths of faith can be approached in an effort to understand them. The result is a negative rather than a positive knowledge, Clement feels, thus opening a question which will be asked again and again by later Christian thinkers. What is the import of the various names attributed to God? Can the things of this world provide us with an access to what God is? Clement's answers here are cautious, as most subsequent answers will be, and it is possible to see him anticipating the negative theology which is developed by Gregory of Nyssa and later by the Pseudo-Dionysius and which, mediated to some degree

by Scotus Erigena, is continued in the twelfth and thirteenth centuries and beyond.

Origen exhibits some of the dangers for the faith that can follow attempts to reconcile it with philosophy, for he reads into Scripture the Neoplatonic view that creation is a necessary process, an emanation from the Divine Monad that could not not have taken place. Origen also maintained that men are in effect fallen angels, their souls being imprisoned in bodies because of some sin prior to birth. Furthermore, perhaps influenced by the Neoplatonic doctrine of return, which complements emanation, he held that ultimately all creatures would be reconciled with God, thus denying the eternity of punishment for the wicked.

Gregory of Nyssa allows the rightness of bringing philosophical conceptions to bear on revealed truths but insists that an interpretation, to be valid, must be consonant with Scripture. There is, indeed, considerable optimism on the part of Gregory as to the reach of reason, for he seems to suggest that it is possible to establish the Trinity of Persons in God on the basis of natural reason alone. There will be later attempts along the same lines, attempts which betray an unorthodox view and which tend to blur the difference between the realms of faith and reason.

Generally speaking, those of the early Fathers who look with favor on pagan philosophy, particularly that of Plato, see it as a way station to Christianity, which is the true philosophy. Moreover, when they find sympathetic doctrines in pagan thinkers, they are inclined to treat these as borrowings from the Old Testament. Finally, the utility of philosophy as an instrument for interpreting the Scriptures and clarifying the nature of belief is stressed. It is this use of philosophy to explicate and defend the faith which constitutes theology according to a definition which will emerge; thus, the question arises whether philosophy is considered an autonomous pursuit by the Christian. This is a difficulty which crops up repeatedly in the period whose history we are attempting to sketch. It has often been said that during the Middle Ages it is the theologians who do such philosophizing as is done and that by and large they do so as an adjunct to developing their theology. Philosophical contributions there may be, it will be said, but they are made *ad hoc*, with a view to their theological utility. Consequently, a man's original philosophy, as well as what he borrows, is to be found scattered through his theological writings, and it becomes difficult to determine what organizational principles we can use to

construct a system of these fragments should they be extracted from their theological context.

There is some justice to this observation, but the outlook is not as bleak as it implies. We will find many philosophical works in the period before us, and we will often find more than a hint as to the structure of the philosophical system to which fragmentary contributions are made in theological writings. Moreover, there will be many commentaries on ancient philosophical works which are their own kind of contribution to philosophy. There is, in short, a great deal of autonomous philosophy in the medieval period.

Nowadays it is particularly necessary to insist on this. From many quarters come statements which, if true, would call for an ironic reversal of recent assessments of the medieval period. As has been mentioned, until fairly recent times it was fashionable to dismiss the Middle Ages as a period when only theology was done and no philosophy. Much careful scholarship has made clear that there were any number of philosophies maintained in the Middle Ages. This variety told against the view that medieval world outlooks were simply explications of what was believed, since if the matter were that simple, we would expect but one philosophy, not several. The Middle Ages thus slowly gained recognition as a period when much vigorous philosophizing took place. Of late, however, some men whose work had much to do with this recognition have been asserting that medieval philosophy cannot be considered autonomous, that not only did it flourish in a theological context but it is inseparable from that context. If there was philosophy in medieval times, this position would have it, it was a Christian philosophy. Presumably, a major note of Christian philosophy is that one must be a Christian to accept its arguments. If this is the implication, the only conclusion must be that this is not what is meant by philosophy -- least of all in the golden period of medieval thought.

Generalities are difficult on the threshold of our task, but the tone of the preceding paragraph will indicate our lack of sympathy with the latter-day notion of Christian philosophy. If that phrase accurately described the philosophical contribution of the Middle Ages, we would see little point in writing the present book. Our conviction is that the Middle Ages saw a genuine flourishing of philosophical thought. There are peaks and valleys, of course; social and political upheavals rendered any unbroken development impossible -- but that is true of any period in the history of philosophy. What will particularly interest us in this, as in the other volumes of this series, are the giants of the period. As we move toward the thirteenth century, we will discern an evolving

clarity as to the relation between philosophy and theology and the limits of the two. Quite unabashedly we will find the highest peak on the medieval terrain in the thirteenth century, particularly in the thought of Thomas Aquinas. In our treatment of his doctrine we will attempt to underline the fact that his is a philosophical as well as a theological achievement, that in his thought we find the clearest and most lasting answer to the puzzles we have seen emerging so far in the present chapter. There is no need to discount the Christian faith of medieval thinkers, or to deny its encompassing influence on whatever they did, to maintain that throughout the period there is a striving toward the position which reaches its full clarity in Aquinas: that philosophy is independent and autonomous. A kind of *praeparatio thomistica*, if you will. If we take Thomas as the telos of this development, we are better able to appraise his predecessors, just as his predecessors give the clue to the comprehensive and synthetic nature of his philosophizing.

Much could be gained from a close and thorough study of the early Christian writers we have mentioned in this chapter. However, given the nature of our objective, we turn now to the thought of Augustine, who is beyond contest the greatest thinker of the early Church.

{1} C. S. Lewis, *De descriptione temporum* (Cambridge, 1955).

Chapter II

Saint Augustine

A. The Man and His Work

When Augustine died in 430, the Vandals were laying siege to Hippo, his episcopal city; the Roman Empire, overextended and moribund, was soon to be a thing of the past; the Western world stood at the edge of its Dark Ages. If the empire is taken as symbolic of past pagan splendor, the dying Augustine reciting the penitential psalms represents a major effort to juxtapose the Christian revelation and the wisdom of the ancients, an effort which would be renewed after the Dark Ages and would culminate in the thirteenth century in such men as Aquinas, Bonaventure, and Albert. Augustine has a lasting appeal because his own life is a dramatic representation of the triumph of grace over nature. In his *Confessions* Augustine has described his struggle against the flesh, a struggle which forms the background for his intellectual development.

Augustine was born in 354 in Tagaste, Numidia, to Patricius, a pagan who was to die baptized, and Monica, already a Christian. Since infant baptism was not the custom, Augustine was simply enrolled as a catechumen, but his mother endeavored to instill in him a reverence and love of Christ which, as he attests, was indelible. Augustine had a Christian education and once even asked to be baptized when he fell ill, but he got well and baptism was put off. But if his mother was teaching him the tenets of Christian truth, his official education was quite another matter. Augustine does not paint a flattering picture of himself as a student, describing himself as giddy, lazy, and a hater of Greek. He studied grammar in his native city and then went to Madaura, where, in his early teens, his moral life went into decline. Despite his attachment to the flesh, Augustine did well at school, and his father decided to send him to Carthage. Since he could not immediately take on the expense, he brought his son home for a year of leisure before he continued his studies at Carthage. Augustine looked back on this year of idleness as a disastrous one. In 370 he went to Carthage where he was to study rhetoric. The pagan atmosphere of the city completed Augustine's downfall, and he seemed forever beyond the influence of Christian doctrine. In 372, Augustine's son Adeodatus was born of a woman with whom Augustine lived until his thirty-third year. A turning point in his life came in 373 when, at the age of nineteen, he read the *Hortensius*, a dialogue of Cicero, which exhorts to the love of immortal wisdom. He writes: "That book transformed my feelings, turned my prayers to you, Lord, changed my hopes

and desires. Suddenly I despised every vain hope and desired with an unbelievable fervor of heart the immortality of wisdom and I began then to rise and return to thee." (*Conf.*, III, iv, 7)

Augustine became a teacher of rhetoric in 373, first in Tagaste and the next year in Carthage, where he taught until 383. The change that the reading of Cicero brought about in him led him to embrace, not Christianity, but Manicheism. Augustine himself felt that he became a Manichean out of pride. The Manichean doctrine purported to be based on reason alone and did not demand that one first believe. This appeal to his intellectual pride was enhanced by the contradictions the Manicheans professed to find in the Scriptures. Perhaps the greatest attraction of the Manichean doctrine lay in the way it accounted for evil, lifting the burden of guilt from the sinner. Augustine was well disposed to accept the exoneration: "For before then it had seemed to me that it is not we who sin but some unknown nature within us and it soothed my pride to be guiltless and, having done something evil, not to have to confess I did it in order that you might heal my soul which sinned against thee; I loved to excuse myself and accuse that unknown something in me that was not I." (*Conf.*, V, x, 18) Augustine was a Manichean through 383. During the time he belonged to the sect he was a listener as opposed to one of the elect, but he devoted himself to the study of the doctrine with great gusto. When he encountered difficulties, he was assured that they could be resolved by a Manichean bishop, Faustus. After nine years Augustine withdrew from the sect. He was prompted by a number of factors, among which was that Faustus had been quite unable to answer his intellectual difficulties with Manichean doctrine; he found the Manichean bishop to be little more than a popular orator.

At the age of twenty-nine Augustine went to Rome to open a school of rhetoric. He hoped that he would attract more promising students than he had at Carthage, and, in a sense, he did. Whenever the fees came due, however, his clientele disappeared. Disgusted, Augustine applied for and received a position as teacher of rhetoric at Milan.

Having freed himself from the bonds of Manicheism, Augustine at first devoted himself to the study of Academic philosophy, but this led him only to doubt; he continued to associate with Manicheans for a time, but then drifted away from them. Having met St. Ambrose Augustine attended his sermons, became once more a catechumen, and pondered over arguments to refute the Manicheans. Fervor came into his life once more when he read some Platonic writings,

probably translations of Plotinus made by Marius Victorinus. If the Academics had led him to despair of the possibility of finding truth, his present reading rekindled in his breast the hope he had first felt upon reading the *Hortensius*. Filled with a passion for philosophy, Augustine desired nothing but to devote his life to the quest for truth. He thought of a common life with friends of like mind, a community ordered to the pursuit of truth. But, alas, he had not yet conquered his flesh. He sent away the mistress of his youthful years, the mother of Adeodatus, and on the urging of his mother was contemplating marriage. In the meanwhile he took on another mistress.

The attraction of Platonism served to lead Augustine to a reading of Scripture, and he began to struggle against his passions. When he was told the story of the conversion of Victorinus to Christianity, Augustine yearned to be baptized; the story of St. Anthony of the Desert made Augustine see his own carnal enslavement and to long to be freed from it. A struggle at once intellectual and moral raged within him. All was resolved when he found himself in his garden, with the Scriptures beside him. From over the wall a child's voice repeated insistently, "Take and read, take and read." It occurred to Augustine that the phrase belonged to no child's game, that the voice was addressing him. He picked up the Scriptures and read from the Epistle to the Romans 13:13: "not in revelry and drunkenness, not in debauchery and wantonness, not in strife and jealousy, but put on the Lord Jesus Christ, and as for the flesh, take no thought for its lusts." With one blow all Augustine's incertitude was swept away. It was 386; he was thirty-three years old.

Shortly after his conversion Augustine took the occasion of an illness and a vacation to resign his post at Milan; he retired with his mother and son and a few friends to the country home of one Verecundus located at Cassiciacum. There Augustine prepared himself for baptism while he and his friends engaged in philosophical disputes which were taken down and preserved. We shall turn to those dialogues in a moment. On Holy Saturday of 387, St. Ambrose baptized Augustine. We can imagine the joy her son's baptism gave Monica; it was truly the fulfillment of her lifetime wish. When they were returning to Africa shortly afterwards, she died en route at Ostia.

At Tagaste, Augustine set up what amounted to a monastic community, striving to realize that ideal which had presented itself to him shortly before his conversion. Augustine enjoyed this solitary life for a few years until he was ordained a priest by popular petition in 391. This caused Augustine to move to Hippo, but there he once more set up a monastery. His life was devoted to

preaching and to writing against the enemies of the faith. He wrote polemical works against Manichean doctrines and against the Donatist heresy, beginning the literary activity which would continue throughout his long life. Augustine was consecrated coadjutor bishop of Hippo in 396 and succeeded the following year. He remained as bishop of this obscure diocese for the rest of his life, profoundly influencing the history of the Church in Africa, and finally that of the whole Church, becoming one of her most authoritative doctors. Augustine's inclination toward a monastic existence did little to prevent his ceaseless activity in the cause of truth. His conviction that there is a changeless truth made him an indefatigable adversary of anyone who would call that truth into question, pervert or dilute it in any way. Manicheans, Donatists, Pelagians -- Augustine dealt with each in turn, but always with an eye to bringing the person in error into the truth. Augustine was seventy-six when he died on August 28, 430.

Writings. It is convenient to group Augustine's writings according to the major phases of his life. Augustine published one prose work prior to his conversion, *De pulchro et apto*. The writings dating from Augustine's stay at Cassiciacum (386-387) are *Contra academicos*, *De beata vita*, *De ordine*, *Soliloquia*, *De immortalitate animae*, *De musica*. The period from his baptism to his ordination (387-391) includes among others *De quantitate animae*, *De libero arbitrio*, *De magistro*, *De vera religione*. As a priest, Augustine wrote, among others, the following works: *De utilitate credendi*, *De duobus animabus contra Manichaeos*, *De fide et symbolo*. Only the very earliest works of Augustine could be called purely philosophical efforts, for as his life becomes that of priest and then of bishop, his interests become almost exclusively theological, homiletic, etc. We shall shortly say something about the possibility of distinguishing faith and philosophy in Augustine; for the moment we must cite, from the period of his episcopacy, the following works as pertinent to the history of philosophy: *Confessions* (400), *De doctrina christiana* (397-426), *De trinitate* (400-416), *De civitate dei* (413-426), *Retractationes* (427).

B. Philosophy and the Arts

One way of approaching Augustine's views on the nature of philosophy is to examine his teaching on the arts which are propaedeutic to philosophy. This approach has chronological justification, since at Cassiciacum Augustine and his companions occupied themselves with the liberal arts. As he says in the *Retractationes* (I, 6): "At the same time, when I was preparing for baptism at Milan, I tried to write books on the arts (*disciplinae*) by interrogating those who were with me and who had no distaste for such pursuits, since they wished to

arrive at the incorporeal through the corporeal by means of determinate stages. But of these I was able to finish only a work on grammar, which afterwards disappeared from my bookcase, six volumes on music, getting to that part called rhythm. But those six books were written after my baptism and return to Africa; I had only begun them at Milan. Of the other five arts begun there in much the same way, namely on dialectic, on rhetoric, on geometry, on arithmetic, on philosophy, only the beginnings remained, which indeed we have lost but I think others have them." With one notable exception, what Augustine has given here as the arts or disciplines are what came to be called the liberal arts. Before examining the role these arts play in the doctrine of Augustine, it will be wise to recall the remote and proximate background of the notions involved.

The remote background is to be found in Plato and Aristotle. Both men stress the need for an orderly approach to the inner sanctum of philosophy. It seems to be only in Roman times that these arts begin to approach the limited number and codification which became so familiar in the scholastic period.⁽¹⁾ Varro (B.C. 116-27), a contemporary of Cicero, was the author of the lost work *Libri novum disciplinarum*, in which, together with the latter seven liberal arts, were listed medicine and architecture. Seneca (B.C. 8 - A.D. 65), in his *Epistle to Lucilius* (*Epist. Moral*, Lib. XIII, Ep. 3, 3-15) mentions five arts: grammar, music, geometry, arithmetic, and astronomy -- in that order. Quintilian (A.D. 35-96), a highly influential author in the Gaul of imperial times, his *Institutes of Oratory* forming the programme of studies in the provincial schools (cf. M. Roger, *L'enseignement des lettres classiques d'Ausone a Alcuin*, pp. 7-18), mentions many of the liberal arts but does not seem to have settled on seven as their number. The work which seems to have fixed the number of the liberal arts is that of Martianus Capella entitled *De nuptiis philologiae et mercurii*; this is thought to have been written in Carthage between 410 and 439 A.D., which would put its composition in the very lifetime of Augustine. It is assumed that Capella's work is inspired by the lost work of Varro, although Capella explicitly excludes architecture and medicine from the list of liberal arts.⁽²⁾ Capella comes up with exactly seven liberal arts which are ordered thus: grammar, rhetoric, dialectic, geometry, arithmetic, astronomy, and music. These arts, we are told,⁽³⁾ formed the basis of the curriculum in the imperial rhetorical schools -- such as that at Milan where Augustine taught.

The passage from the *Retractationes* already quoted expresses Augustine's intention, as he prepared for baptism at Cassiciacum, to write on each of the liberal arts, presumably in dialogue form. The passage certainly assumes that there are seven such arts, but it is noteworthy that "philosophy" takes the place

usually occupied by astronomy. It has been plausibly suggested that this is a quite conscious substitution by Augustine prompted by his abhorrence of what we would nowadays call astrology. We can surmise that Augustine's whole career prior to his conversion would have put him into daily contact with the various arts. Indeed, we read in the *Confessions* (IV,xvi,30) that in his youth Augustine had read "all the books of the so-called liberal arts."

Following the lead of Marrou (*S. Augustin et la fin de la culture antique*, pp. 187-193), we can find a fairly uniform doctrine on the arts in various statements of Augustine. In the second book of the *De ordine*, St. Augustine describes reason as discovering progressively grammar (nn.36-37), dialectic (n.38), rhetoric (n.39), music (n.40-41), and geometry (n.42). Earlier in the same work mention is made of arithmetic and astronomy as well as of the relation of the arts to philosophy. "Now in music, in geometry, in the movements of the stars, in the fixed ratios of numbers, order reigns in such manner that if one desires to see its source and its very shrine, so to speak, he either finds it in these, or he is unerringly led to it through them. Indeed such learning, if one uses it with moderation -- and in this matter, nothing is to be feared more than excess -- rears for philosophy a soldier or even a captain so competent that he sallies forth wherever he wishes and leads many others as well, and reaches that ultimate goal, beyond which he desired nothing else (n.14, trans. R.P. Russell, O.S.A.) The acquisition of these arts is difficult, Augustine admits, but without them it is impossible to go on to philosophy. These arts comprise a twofold science, the science of reasoning and that of numbers; armed with this knowledge, one can turn to philosophy, "to which a twofold inquiry belongs, one having to do with the soul, the other with God." (n.47) These are the two great concerns of philosophy, to know ourselves and our origin, and the study of the liberal arts paves the way for the fruitful asking of those questions.

In the *De quantitate animae* Augustine is speaking of the seven degrees of the soul's perfection, and this prompts him to mention the liberal arts. The soul, in the first degree, vivifies the body; in the second, it makes use of the senses; thirdly, the degree proper to man, the soul is possessed of arts and sciences; then, by purgation, purity, and conversion to God, the soul finally comes into possession of the Supreme Good. It is the third degree that interests us now. "Rise now to the third plane of the soul's power and think of memory, which is proper now to man, not in the way of a habit of things usual, but by way of reverting to notes and signs of innumerable things treasured and retained: -- so many arts of skilled workers, the tilling of the soil, the building of cities, the manifold marvels of varied constructions and their achievement: the invention

of so many signs in letters, in words, in gesture, in the sound of such things, in paintings and things moulded (or carved) . -- Note the languages of so many peoples, the manifold teachings, some new, some renewed. -- Note the great number of books and such like documents for the safeguarding of memory, and all this provision for posterity. -- Note the order of duties and powers and honors and dignities, in family life, in the state, in peace and in war; in the administration of things profane and things sacred. -- Note the power of reasoning and of thinking out reasons. -- Note the flowing streams of eloquence, the varieties of poetry; the thousands of means of imitation for purposes of play and of jest, the art of music, accuracy of measurements, the science of numbers, the conjecturing of things of the past and the future from the present. Great are these things and distinctively human. But yet this abounding property common to (rational) souls is shared in degrees by the learned and the unlearned, by the good and the bad." (Trans. F.E. Tourscher, O.S.A.) Augustine indicates at the close of this lengthy enumeration that he is not confining himself to the arts possessed by the learned; thus we find mechanical and fine arts side by side on his list. Also listed are grammar, reasoning (dialectic), eloquence (rhetoric), arithmetic, music, geometry, and astrology. So too in the *Confessions* Augustine mentions the liberal arts. "Whatever was written either on rhetoric, or logic, geometry, music, and arithmetic, by myself without much difficulty or any instructor, I understood" (IV,16) We find Augustine reflecting what appears to have been the common attitude of his time with respect to the liberal arts; of course, his enumerations of the arts have importance for medieval thought since his authority dictates that an interest be shown in the arts he mentions. More important for our present purpose is the attitude Augustine takes toward the liberal arts as conducive, not simply to the wisdom of the philosophers, but to Christian wisdom itself.

C. Philosophy and Beatitude

In his *De civitate dei* (XIX,1) Augustine writes, "Quandoquidem nulla est homini causa philosophandi nisi ut beatus sit." It is man's desire for happiness which explains his philosophizing. Thus, even while arguing for the role played by the liberal arts, Augustine does so by showing that they prepare the mind for the two great questions of philosophy which have to do with the nature of God and the nature of the soul. The ability to answer these questions gives one knowledge of the self, knowledge of whence he has come and whither he is going. The importance of these two questions, the fact that they sum up man's desire for truth, is clear in the following famous passage from the *Soliloquies* (I,ii,7) where Augustine and Reason are conversing. "A. My prayer is finished.

R. What then do you wish to know? A. All those things I have prayed to know. R. Sum them up briefly. A. I want to know God and the soul. R. Nothing more? A. Absolutely nothing." Concern with God and the soul is what sets philosophy off from the liberal disciplines or arts; unlike them, apparently, philosophy is concerned with the intelligible order. It is because Plato stressed the existence of the intelligible order and its distinction from the sensible order that he was able to devise the perfect philosophy.

It was the failure to achieve this intelligible order which led to the skepticism of later Academic philosophy. Just as Augustine himself had been rescued from skepticism by the reading of Plato and Plotinus, so he felt each man must overcome the temptation to skepticism. Indeed, he sees the achievement of Plato precisely against this background:

Plato, the wisest and most learned man of his time, spoke in such a manner that whatever he said took on importance and he spoke of such things that, no matter how they were treated, they could not become trivial. This Plato, after the death of his beloved master Socrates, learned, we are told, much more from the Pythagoreans. Pythagoras, dissatisfied with Greek philosophy, at the time either quiescent or too obscure, was persuaded by the arguments of Pherecydes, a Syrian, to believe in the immortality of the soul. Plato listened, moreover, to a great many wise men in the course of extensive travels. He thus added to what he already possessed of socratic charm and subtlety in moral matters, the knowledge of things human and divine diligently learned from the men just mentioned. He crowned these elements with a discipline capable of organizing and judging them, namely, dialectic, which is, he thought, wisdom itself, or at least that without which wisdom is impossible, and he composed thereby the perfect philosophy. Leaving that aside for now, it is sufficient for my present purpose that Plato thought there were two worlds, one intelligible, another manifest to us by sight and touch. The former is the principle of pure and serene truth in the soul which knows itself, whereas the latter can engender opinion in the minds of the foolish but not science.^[4]

One's ability to grasp the existence of the intelligible world was dependent on one's moral condition; a person given over to sensuality would not be able to come to knowledge of intelligible truth. Skepticism and moral turpitude are rather closely linked.

Augustine's praise of Plato's achievement as the perfect philosophy would seem to indicate that he sees no difficulty in relating faith and reason. Philosophy,

Augustine has said, is concerned with God and soul. Let us look at Augustine's appraisal of some philosophical statements about God. In the *De civitate dei* (VI,5) he accepts from Varro a threefold division of theologies. There is a mythical theology fabricated by the poets suitable for the theater; there is a natural theology taught by the philosophers; there is a civil theology which is for the people as citizens. Augustine has difficulty seeing the difference between mythical and civil theologies (VI,6) and goes into a lengthy criticism of the pagan deities. It is not until the eighth book that he returns to the question of natural theology. He notes that "philosophy" means love of wisdom. "But if God is wisdom, through whom all things are made, as the divine authority and truth have shown, the true philosopher is one who loves God. But since the reality of which this is the name is not to be found in all those who glory in the name (for surely not anyone who is called a philosopher is a lover of true wisdom), we should select from all those whose opinions and writings can be known by us those who have treated this question not unworthily." (VIII,1)

If we look to what philosophers have had to say on the matter of God, we find some who surpass Varro's notion of natural theology. He had defined natural theology as concern with the world and its soul; some philosophers, however, speak of a God above nature, cause not only of the sensible world but of souls as well, even of human souls which are beatified by participation in the divine light. "There is no one who has even a slender knowledge of these things who does not know of the Platonic philosophers who derive their name from Plato. Concerning this Plato, then, I will briefly state such things as I deem necessary to the present question, mentioning beforehand those who preceded him in time in this kind of writing." (VIII,1) Augustine praises the Platonists for recognizing that God is incorporeal, immutable, surpassing every soul, cause of all else, life, understanding, and beatitude. (VIII,6)

If God is beatitude and men philosophize only that they might become happy, is it possible that men can attain happiness by means of philosophy? But Christians have been warned about philosophy (Cf. VIII,10). St. Paul has said that we must be wary lest we be led astray by philosophy: "Take care not to let anyone cheat you with his philosophizings, with empty phantasies drawn from human tradition, from worldly principles." (Col. 2:8) We need not think that every philosopher falls under the censure of St. Paul, for the Apostle has also written: "The knowledge of God is clear to their minds; God himself has made it clear to them; from the foundations of the world men have caught sight of his invisible nature, his eternal power and his divineness, as they are known through his creatures." (Romans 1:19-20) The same Paul, speaking to the Athenians

those difficult words "in whom we live and move and have our being," added "as some of your own have said." (Acts 17:28) Philosophers are to be feared only in their errors. It is a sad fact that even when they have come to know the existence of God, philosophers have not adored and thanked him as they ought; thus, their wisdom has been turned to folly. (Romans 1:21-23) Nevertheless, there is no reason to be suspicious of philosophers who teach truth and thereby agree with us. This is why Augustine singles out the Platonists for praise: "This, therefore, is the cause why we prefer these to all the others, because, whilst other philosophers have worn out their minds and powers in seeking the causes of things, and endeavoring to discover the right mode of learning and living, these, by knowing God, have found where resides the cause by which the universe has been constituted, and the light by which truth is to be discovered, and the fountain at which felicity is to be drunk." (*De civ. dei*, VIII,10)

Augustine's defense of philosophy is not reluctant; the role that Platonism, or, more accurately, Neoplatonism, played in his own conversion led him to effuse and, as he later remarked, exaggerated praise of Plato and his followers. "Many centuries and much discussion were required in order that a perfectly true philosophy be achieved, but I believe it has been done. For this is not a philosophy of the world, which our mysteries properly condemn, but of another and intelligible world to which the subtlety of reason could never have led souls blinded by the manifold darkenesses of error and weighted down by bodily sordidness, if the most high God, animated by mercy for his people, had not bent down and subjected the authority of the divine reason to a human body, so that souls, stirred up not only by his precepts but also by his deeds, might without the disputes of the schools turn within to themselves and find the kingdom." (*Contra academicos*, III,xix,42) This easy transition on Augustine's part from the efforts of the pagan philosophers to the Incarnation and Christian revelation has caused interpreters to multiply opinions as to Augustine's own position at Cassiciacum (was he converted to Christianity or to Neoplatonism?) and as to his doctrine on the relation between faith and reason.

D. Criticism of Platonism

While it is possible, particularly in his early writings, to find praise of Plato and Platonism flowing from the pen of Augustine, from the outset there are also criticisms of Plato. At the end of his career, surveying his various works and commenting on them in the *Retractationes*, the aging bishop regrets the unqualified character of his earlier praise. He expressed regret that he had spoken of learning as remembering and the soul's ascent after death as a

returning, since this seems to involve acceptance of the Platonic view that the soul antedates its imprisonment in the body and that death is the soul's return to its natural habitat. Significantly too, he regrets the emphasis he had put on the liberal arts as propaedeutic to philosophy, since the simple faithful are capable of attaining wisdom without them.

Augustine believes that the Platonists were aware of the difference between God the Father and the Word; in the *Confessions* (VII,ix, 13-14) he comments that he found in Neoplatonic texts the equivalent of the prologue to the Gospel of John in which we read, "In the beginning was the Word and the Word was with God and the Word was God." He found nothing in these texts, however, concerning the Word become flesh. As he brings these philosophic gropings toward revealed truth into perspective we sense that he thought that Neoplatonism paved the way for acceptance of revelation: it achieves knowledge of a realm above the sensible. While Augustine seems at times to suggest that the metaphysical success of pagan philosophy almost attains the supernatural order, this is not Augustine's true opinion. Although he once interpreted the remark "My kingdom is not of this world" as a reference to the world of Ideas, he later thought better of it. (Cf. Portalie, p.98.) He feels that the Platonists, though they knew the Father and Son, were ignorant of the Holy Spirit and thus of the doctrine of the Trinity. (*De civ. dei*, X,23) Portalie in his excellent guide to Augustine provides us with a list of items from Neoplatonism which were either accepted or rejected by Augustine. There are besides, as we have already indicated, a number of alterations in viewpoint on Augustine's part. Perhaps it is enough to indicate that Augustine, even while he was virtually overwhelmed by the reading of Plotinus, not only for the beauty he found in the style and thought but also for the crucial role this reading played in lifting him out of a stultifying skepticism and restoring the passion for truth earlier instilled by the *Hortensius* of Cicero, was nevertheless a critical reader. Portalie has put the matter succinctly: "The *Confessions* describe the enthusiasm enkindled in him by Platonic writings; this enthusiasm, a source of magnificent and repeated eulogies, was to die a slow death in the heart of Augustine." (p.95)

The same author has pointed out that Augustine borrowed the Neoplatonic conception of philosophy and never seriously questioned it. This can be seen quite clearly in the connection Augustine makes between moral rectitude and a grasp of the truth. The goal of philosophy, as we saw above, is happiness; not a dispassionate correctness of judgment, but the fulfillment of all our deepest aspirations, those of will as well as intellect. Thus, the beauty of order "will be seen by him who lives well, prays well, studies well." (*De ordine*, II,xix,51) One

who wants to see the truth must take pains lest the eye of his soul be clouded by sensual attachments, by vanity and pride; with these beams removed, one may come to contemplate the truth and experience that *gaudium de veritate* which is the end and purpose of human life. Now the truth in which he must rejoice is none other than God. "The happy life consists in rejoicing in the truth, that is, rejoicing in you who are the truth, my God, my light, my salvation. All want this happy life, this life which alone is happy each one wants, everyone wants to rejoice in the truth." (*Conf.*, X,xxiii,33)

To love truth is to love God; is philosophy then religion, is there a transition from philosophy to faith? At the end of the *De beata vita* to be wise and to be happy are identified, and true wisdom is identified with the Son of God, who is truth as well. In the *De ordine* (II,v,16) we find the following comparison of philosophy and faith. "There is a twofold path we can follow when the obscurity of things bothers us; reason or, in any case, authority. Philosophy promises us reason but it frees scarcely a few; nonetheless it leads them not only not to condemn the Christian mysteries but to understand them as they ought to be understood. No other task falls to true philosophy, to authentic philosophy, if I may so put it, than to teach that there is a supreme principle of all things, itself without a principle, and how great an intelligence dwells therein and that all flows from it, without any diminution, for our salvation. This principle is the one God, omnipotent and tripotent, Father, Son, and Holy Spirit, of which the venerable mysteries teach, the sincere and unwavering profession of which frees people, without confusion (as some assert) or humiliation (as many claim) In this passage Augustine sees philosophy as a fitting preparation for Christian faith, disposing the mind for it and equipping it to deal with mysteries as they deserve. Thus, in the *Contra academicos* (III,xx,43) he writes that truth can be approached by way of reason or authority. "Now in the matter of authority I have chosen Christ for my leader, from whose direction I will never deviate. As regards the matters which are to be investigated by close reasoning, I am such that I impatiently desire to grasp the truth not only through faith but also through understanding, and I am confident that there will be found in the Platonists nothing repugnant to our faith."

Despite fluctuating attitudes toward Platonism and some of its doctrines, Augustine teaches with increasing clarity a distinction between what is believed and what is understood: not everything that is believed by the Christian has been or could be known by the non-Christian philosopher, and one who believes may nevertheless concern himself with the arguments of philosophers as at least dispositional toward a more lively faith. Both philosophy and faith

aim at the same goal, beatitude; if the latter is more efficacious, it is because it can direct man toward an incarnate wisdom unknown to philosophy. The two questions of philosophy remain the great questions of faith, however, and we want to turn now to what Augustine had to say of God and man, for this provides at least an outline of what can be called the philosophy of St. Augustine.

E. What Is Man?

When Augustine says that philosophy has two concerns, the soul and God, he does not mean that these are distinct and separable problems; indeed, we will find it quite impossible to concentrate on Augustine's doctrine of man to the exclusion of what he has to say about God, and vice versa. It is convenient to begin our discussion of what he had to say about man by examining what he had to say about the distinctive character of Platonic philosophy, the Ideas or Forms. In the forty-sixth of the *83 Diverse Questions* St. Augustine discusses the Ideas in a way which was to be decisive for scholastic philosophy. The question turns on four points: the word "Idea," its definition, the location of the Ideas, how we can know the Ideas.

1. *Augustine and Plato's Ideas.* Augustine says that although Plato was the first to use the term "Idea" in the sense that now interests him, the Ideas existed before Plato and were known by men. It is inconceivable that philosophers did not know them since "unless these be known no one can be wise." The journey of Plato which took him to Southern Italy and Sicily and to the Pythagorean communities in those places makes it at least probable, Augustine feels, that these philosophers knew of the Ideas, though they might have had another name for them. If we are interested only in transliteration, the Latin terms "*species*" and "*forma*" are equivalents of the Greek "idea." To call them "*rationes*" (reasons, notions) is to give the Latin term for the Greek "logoi," but, for all that, "*ratio*" expresses what is often meant by "idea." This is clear from the definition of Ideas: "The Ideas are the chief forms or the stable and unchangeable notions of things which have not themselves been formed and thus are eternal and unalterable; they are contained in the divine intelligence."⁽⁶⁾ We notice immediately some characteristics of the Platonic Ideas: they have not come to be and will not cease to be and consequently are necessary, incapable of being other than they are. However, the remark that the Ideas are in the divine intelligence goes beyond Plato. It makes little difference whether or not Augustine was the first to identify the Platonic Ideas and God's creative knowledge.⁽⁷⁾ We are told that this identification was first made by Philo Judaeus

(*Opif. mundi* 94) and was a commonplace among the Christian Alexandrines (see, for example, Clement, *Strom.*, VII,2). Nevertheless, it is an identification of great importance, and it was largely thanks to Augustine that it became a commonplace in the Middle Ages; clearly, when by Ideas a writer means God's creative knowledge, it becomes almost equivocal to characterize his remarks as Platonist.

How can the Ideas be known by us? They are knowable only by the rational soul because of its inner or intelligible eye. (Cf. *Soliloq.*, I,i,3.) The rational soul is not by its nature alone equipped to know the Ideas, however; it must be prepared for this vision by holiness and purity whereby its inner eye is made healthy, clear, serene. Before saying more on the fitness of the rational soul to see the Ideas, St. Augustine points out why the man of religion, even when he himself does not have the vision of the Ideas; cannot deny their necessity. He above all will know that God has created and gives being to all things. Thus he must admit that God had a notion of what he created and that the *ratio* of man is not that of horse. These different notions are precisely the Ideas of creatures, and they can exist only in the divine mind. Since whatever is contained in the divine mind is eternal and unchangeable, the Ideas must be so. Such precisely are the Platonic Ideas, St. Augustine concludes, and it is by participation in such Ideas that all other things exist.

Of all created things the rational soul, at least when it is pure, is closest to God. To the degree that the soul is united to God by charity it can contemplate the Ideas, and in this contemplation consists that beatitude which all men seek. When it possesses charity, the rational soul is illumined by an intelligible light which renders the vision of the Ideas possible. We are all free to select our own name for them, be it "Ideas," "Forms," "Species," or "Rationes." Few indeed are able to grasp them as they are.

Needless to say, this doctrine of St. Augustine is a very difficult one to interpret. He seems to be speaking of a terrestrial vision, a knowledge of the Ideas which man can attain in *via*. By linking this knowledge of the Ideas essentially to the theological virtue of charity and by speaking of illumination, Augustine seemingly prevents us from viewing his remarks as a philosophical doctrine. The Augustinian doctrine of illumination is as vexing a problem for historical interpreters as it is influential among philosophers and theologians. When Augustine speaks of illumination, he is not necessarily concerned with explaining some privileged kind of knowledge; rather, his concern is to defend the validity of knowledge as such. This is made quite clear in the *De magistro*.

How do we come to know what we did not know before? We can answer quite easily, it would seem; we come to know things by means of experience. Experience seems to connote sensation, and it is just that which would make Augustine hesitate. The knowledge he is interested in defending is unchanging knowledge, and the examples which come immediately to mind are mathematical. If we have certain knowledge of numbers, can we attribute this to experience, to the influence of sensible things? "In no wise; for even if I perceived numbers by the bodily senses, I was not able by these same senses to perceive the laws of the division and addition of numbers. For it is by the light of the mind [*luce mentis*] that I correct anyone who gives me the wrong result of adding or subtracting. Moreover, I know nothing of how long sensibly perceived things like the heaven, this earth, and the other bodies therein will endure, but seven and three are ten, not only now but always, nor was it ever true in the past that seven and three were not ten nor will seven and three sometime in the future not be ten. Such then is the incorruptible truth of number which, as I have said, is common to me and anyone else who reason." (*De lib. arb.*, II, viii, 21) What distinguishes intellect from sense is that intellect grasps truth and the senses do not; if there are truths about numbers, and there are, they are not grasped by the senses. Knowledge of numbers is not drawn from sense perception, something easily seen when we consider that every number involves the one. A true notion of unity cannot be formed from perceptions of corporeal things.

These remarks may seem to owe their tone to the peculiar character of mathematical entities, yet Augustine speaks in much the same way of our knowledge of other things: "If therefore it is certain that we wish to be happy, it is also certain that we desire wisdom, for no one is happy without the highest good which is known and possessed in that truth we call wisdom. Thus, just as before we are happy, the notion of happiness is impressed on our minds (thanks to which we firmly and without hesitation know and say that we desire to be happy), so too before we are wise, we have impressed on our minds the notion of wisdom thanks to which each of us, asked if he wants to be wise, replies without a shadow of a doubt that he does." (*Ibid.*, II, ix, 26). The suggestion here that if we did not already know what wisdom is we could not seek it, is reminiscent of the Platonic doctrine of *anamnesis* which had particular application to knowledge of mathematical and moral ideals. Learning was equated with recalling for Plato; if one already knew what he is said to learn, the problem of the genesis of knowledge is conveniently postponed; he can say that the soul came to the body with knowledge impressed upon it and has forgotten what it knew because of the drag and weight of the body.

In the *De quantitate animae* (XX,34) Augustine underlines a disagreement with Evodius: "You raise there a great problem, so great indeed that I know of none greater. With respect to it our opinions are quite opposed, for it seems to you that the soul brings no knowledge with it, to me that it brings with it every art. Nor is that which is called learning anything other than recalling and remembering." The same position is maintained more elaborately in the *Soliloquies* (II,xix,35). Despite the obvious reliance on Plato, Augustine does not accept from reminiscence the transition to the assertion that the soul existed before its union with the body, a transition made, we remember, in the *Phaedo*. In the *Retractationes*, it should be stressed, Augustine came to regret his youthful choice of words. He wishes that he had not spoken of the soul's "returning" to heaven because there are those "who think that human souls have fallen or been ejected from heaven and been placed in bodies as punishment for their sins." (I,i,3)⁽⁸⁾

Augustine's reflections in the *Retractationes* on the *Soliloquies* and *De quantitate animae* bring the doctrine of illumination to the fore. Augustine is unhappy that he had suggested that students of the liberal arts were simply recalling knowledge which had fallen into oblivion. "I disapprove of that statement. Is it not more credible that those who are ignorant of certain disciplines and yet reply correctly when well interrogated do so because there is present in them, to the degree that they lay hold of it, the light of eternal reason [*lumen rationis aeternae*] in which they see immutable truths?" (I,iv,4) The remark that what is called learning is nothing but recalling and remembering "should not be understood as an approbation of the opinion that the soul at some time lived either here below in another body or elsewhere in a body or outside of a body, such that it would have learned in another life the answers to questions it has not studied here below." (I,viii,2) The idea of a light within the human soul replaces the idea of remembered knowledge. In the *De trinitate* (XII,xv,24) Augustine compares the mind with the eye and notes that just as the eye grasps things because a light is proportioned to it and its objects, the mind grasps incorporeal things with which it has affinity in a light of the same order as itself and its objects. We would not say that the eye, because it can distinguish black from white without being taught, must have known colors before receiving life in a body; no more should we say that the soul's grasp of truths previously unknown argues that it existed prior to its creation with the body. In each case it is the appropriate light which enables eye or mind to attain its objects.

2. *Augustine on the Teacher*. In the *De magistro*, a dialogue between Augustine and his son Adeodatus, there is a fairly extensive discussion of illumination. The

dialogue begins with the fairly innocent question "What are we trying to do when we speak?" The answer, that we are trying to teach or be taught, leads to a discussion of what is happening when we learn something. Augustine prefixes this discussion of the learning process with a long section on words as signs. Although sensible things are incapable of bringing about thought, we make use of words, which are sensible things, presumably to express our thoughts. What effect do words have on the one hearing or seeing them? We might expect Augustine to brush aside the difficulty posed by language as quickly as possible; his actual procedure is quite the opposite, revealing, we may suppose, his former professional interests as a master of rhetoric. The greater part of the dialogue is concerned with words as signs and then with various other kinds of signs.

The *De magistro* contains fourteen chapters and can be conveniently divided into two parts at n. 33 of chapter ten; at that point the dialogue form is dropped and the apparent implications of the preceding discussion are dismissed. The dialogue begins by asking for the purpose of speech. We speak to teach, it is decided, for even when we ask questions we are teaching what we want to know. This teaching takes place by reminding. In speaking "we do nothing but remind, since memory in which words inhere causes by revolving them to come to mind the things of which the words are signs." (1,2) A sign is that which signifies something, and words are signs. What then of "nothing"? Augustine suggests (11,3) that this word signifies an affection of the mind, a point to which they can return. First, he draws his son's attention to the fact that when he asks him what one word ("de") means he answers with another word ("ex"), that is, a sign is explained by appeal to another sign. Is it possible to convey something without the use of a sign? Well, one can point, use gestures; in fact by means of pantomime one can dispense with words entirely. This does not dispense with signs, however, for gestures too are signs. Yet if someone should ask me what walking is, I could rise and show him the thing itself rather than a sign of it. Augustine allows this possibility, as long as we are not asked what walking is while we are walking, for then if we would speed up, the velocity and not the activity might be understood. Talking does not seem to be demonstrable in this way; nevertheless, persistence can make clear that it is the activity of speaking itself which we are trying to exhibit. At the beginning of chapter four Augustine divides things which can be shown without signs from signs which can be shown through other signs. The second class is discussed first.

Signs can be shown by means of signs. Words are signs either of other signs or of things. "Stone" is an example of the latter, "gesture" and "letter" of the former.

Romulus is signified by "Romulus," "Romulus" by "noun," and "noun" by "word." Things which are not signs can be called signifiabiles. Some signs signify themselves as well as other signs, for example, "word" signifies itself and other things, as does "noun." Although "word" signifies noun and vice versa, there is a restriction on their reciprocity, since every noun is a word but not every word is a noun. Each signifies itself as well as other things and each is an instance of the other, but they do not appear to have the same extension. There is, however, a way of understanding "noun" and "verb" which gives them the same extension. If "noun" (*nomen*) is thought of as imposed from knowing (*noscendo*) and "word" (*verbum*) from reverberation (*verberando*), then any vocal sound is a word insofar as it is audible and a noun insofar as it makes something known. The dialogue has now to discover signs which are fully reciprocal, that is, signs which signify one another, which signify themselves as well as other parts of speech, and which differ only in sound. This has not yet been achieved with "noun" and "word" since they are imposed from different things, as we have seen. Perfect reciprocity of signification is thought to be achieved with "noun" and "vocale."

This summary conveys most inadequately the lively development of the dialogue, which contains, as every effective dialogue must, an artful balance of play and earnestness.^[9] Chapter eight extracts the earnest by warning in effect that since signs sometimes signify other signs and sometimes things, what are to be called signifiabiles, it is important to be aware of this and in disputations honor only questions which bear on things. The reason for this is that signs are for signifiabiles, and that which is for something else should not be preferred to its end. But is this always true? Is it not better to know "filth" than filth? This objection enables Augustine to be more explicit about the triadic character of signification: there is the thing, the sign, and cognition, and the last consequently is the end or purpose of the sign and is always to be preferred to it.

Thus in the middle of chapter ten Augustine can say that they have settled a number of things: they have seen that some things can be shown without signs, they have asked whether some signs should be preferred to the things they signify, and they have concluded that cognition of things is always superior to signs. Indeed, the meager reward of their effort can be summarized as follows. "It is established therefore that nothing can be taught without signs and that the knowledge itself ought to be dearer to us than the signs by which we know, although not everything that can be signified is better than its sign." (X,31) At this point a major shift in the direction of the dialogue occurs. Augustine asks whether what they think they have established is beyond doubt. First of all, he

attacks the contention that nothing can be taught without signs. The activities which had earlier been thought to manifest themselves when one asks for the meaning of a word have been argued to be signs of that meaning. Nevertheless, we do learn from observation. If I watch a man fish or hunt, observe his equipment, his movements, and so forth, I can come to knowledge of the arts involved. Some things then, indeed a veritable infinity of things, can be learned without signs. Now the fundamental question is posed: Can anything in fact be learned by means of signs? "If we should consider the matter more diligently, perhaps you will discover nothing which is learned through its sign." (X,33) Augustine will attempt to show this.

His point can be reduced to this. If I do not already know what the sign signifies, I cannot be apprised of that thing by the sign alone; but if an understanding of the sign presupposes knowledge of the signifiable, the signifiable cannot be conveyed by the sign. Words do not exhibit things; they can only remind, direct us to things. We learn, not from the signs, but from the truth within. "Concerning all things that we understand, we do not consult the one speaking without but the truth presiding within the mind itself, admonished perhaps by the words to do so. He who is consulted teaches: Christ it is who is said to dwell in the interior man, that is, the changeless power of God and the sempiternal wisdom, whom indeed every rational soul consults and he reveals himself to each according to his capacity due to his good will or bad. (XI,38) Thus, he vindicates the verse: *Magister vester unus est Christus*.

When speech is concerned with present sensible things, we learn from these things, not from the words; if speech is concerned with absent sensible things, we learn by consulting our memory for images of them. Finally, when speech is concerned with things perceived by the mind itself, these things are seen, if at all, in the interior light of truth by which the interior man is illumined. In no case, then, do the words themselves teach. If someone learns from me, he does so, not from the words I speak, but from the things which the words recall. Questions and discussion can be useful to direct the mind to things, and from the inner light it can learn. By learning within, the listener becomes a judge of the spoken word. Finally, Augustine observes that words are not always signs of what is in the speaker's mind; indeed, even when they are, the purpose of teaching is to exhibit, not what the teacher thinks, but the way things are. That words are an indispensable instrument of teaching Augustine does not wish to deny; his purpose has been to show that they are but instruments to remind, that they do not exhibit that of which they are signs, and that, consequently, the human teacher is not the principal cause of learning.

Augustine's interpretation of the nature of the Ideas and his doctrine of illumination exhibit his dependence on Plato as well as his originality. That his views on the guarantee of knowledge and the learning process owe much to Plato is obvious; nevertheless, we should stress that his identification of the Ideas with the creative ideas of God is an important adjustment of the Platonic doctrine. Moreover, it is productive of a number of problems. We may say that the Ideas played a double role for Plato, what can be called an ontological and an epistemological role. It is through participation in the Ideas that sensible things exist in the deficient way they do exist, and the Ideas give a fixed object of knowledge. This latter role led Plato to what may have been only a myth of the preexistence of the soul; prior to its incarceration in the body the soul was acquainted with things themselves, with the Ideas. The soul's incarnation induces forgetfulness, and learning is the recollection of what is already known, with sensible things playing a role similar to that assigned to words in the *De magistro*: they neither produce knowledge nor are they the objects of knowledge, but they can point the mind to its true object.

By identifying the Ideas with God's knowledge Augustine forces us to ask if we learn by contemplating the divine Ideas. He retains the Platonic vocabulary, speaking of learning as recalling. However, as he warns later (*Retract.*, I,viii,2), this "should not be understood as an approbation of the opinion that the soul at some time lived either here below in another body or elsewhere in a body or outside of a body, such that it would have learned in another life the answers to questions it has not studied here below." The doctrine of an interior light is intended to replace the appeal to a previous existence and a view of learning as remembering what was once known. In the *De trinitate* (XII,xv,24) Augustine compares the mind with the eye and notes that just as the eye grasps things because a light is proportioned to it and to its objects, the mind grasps incorporeal things with which it has affinity in a light of the same order as itself and its object. We would not say that the eye, because it can distinguish black from white without being taught, must have known this distinction before being created with the body; no more should we say that the soul's grasp of truths previously unknown argues that it existed prior to its creation with the body. In each case it is the appropriate light which enables eye or mind to attain its objects. The participation in the divine light which enables man to come to knowledge of immutable truths is seen to be necessary since the mutable things perceived by the senses cannot be the cause of immutable truths in the mind. Is the divine illumination a miraculous intervention in the natural order? It seems perfectly clear that Augustine invokes the inner light to explain the natural activity of the mind. It is something at the disposal, so to speak, of Christian

and pagan alike. As we shall see later, St. Thomas Aquinas in his own *De magistro* (*Q.D. de veritate*, q.11) will interpret the light of which Augustine speaks in terms of Aristotle's doctrine of the agent intellect, to which Aristotle was led for reasons analogous to those of Augustine. It has been objected that this is a misleading identification because Aristotle recognizes that our intellectual knowledge depends upon an abstraction from sense images, whereas Augustine explicitly denies that what is sensibly perceived can cause knowledge of eternal truths. Perhaps this difference is not as great as it seems. Although Augustine's light cannot be simply identified with what Aristotle and St. Thomas call the agent intellect, Augustine, nevertheless, arrived at his doctrine for much the same reason that prompted Aristotle to recognize the need for an agent intellect. In response to a difficulty raised a moment ago, it can be said with some certainty that Augustine does not wish to attribute our knowledge of truth to a knowledge of things in the divine creative Ideas. Augustine denies that our mind is capable, in the natural course of things, of seeing God directly. Some men have such knowledge due to a rare and mystical privilege. The divine illumination, on the other hand, is an abiding and natural phenomenon common to all men. On at least one occasion (*Contra Faust.*, XX,7) Augustine explicitly denies the identity of the light in which we grasp the truth and the light which is the Divine Word. Speaking of the former, he says, "this light is not that light which is God."

Portalie, who does not think too highly of Augustine's doctrine of illumination (p.114), summarizes it as follows. "In our opinion, Augustine's doctrine is the theory of divine illumination of our understanding, so much in favor in the Middle Ages which borrowed it from him. It can be formulated this way: Our soul cannot attain to intellectual truth without a mysterious influence of God which does not consist in the objective manifestation of God to us, but in the effective production of a kind of image in our soul of those truths which determine our knowledge. In Scholastic language, the role of producing the impressed species which the Aristotelians attribute to the agent intellect is assigned to God in this system. He it is, the teacher, who speaks to the soul in the sense that He imprints that representation of the eternal truths which is the cause of our knowledge. The ideas are not innate as in the angels, but successively produced in the soul which knows them in itself." (pp.112-113)

3. *Faith and Understanding.* Earlier we touched on Augustine's apparently conflicting doctrine on the relationship between faith and understanding, authority and reason. It is important for us to grasp Augustine's thought on this

matter in order to grasp better his teaching on the nature of faith and to see its implications for the possibility of philosophical truths apart from revelation.

Augustine teaches that there is a twofold impetus to learn: authority and reason (*Contra academicos*, III,xx,43). The question then arises as to the relation between these: Are they simultaneous or does one precede the other, and, if so, which takes precedence? "Likewise with regard to the acquiring of knowledge, we are of necessity led in a twofold manner: by authority and reason. In point of time, authority is first; but in the order of reality, reason is prior. What takes precedence in operation is one thing; what is more highly prized as an object of desire is something else. Consequently, although the authority of upright men seems to be the safer guide for the uninstructed multitude, yet reason is better adapted for the educated. And furthermore, since no one becomes learned except by ceasing to be unlearned, and since no unlearned person knows in what quality he ought to present himself to instructors or by what manner of life he may become docile, it happens that for those who seek to learn great and hidden truths, authority alone opens the door." (*On Order*, II,9,26; trans. Russell [Chicago, Franciscan Herald Press]) What has been accepted and lived on authority can come to be understood. Faith in authority precedes understanding, and since it is reasonable that we proceed in this way, we can say that reason precedes the faith that precedes understanding. "God forbid that He should hate in us that faculty by which He made us superior to all other living beings. Therefore, we must refuse so to believe as not to receive or seek a reason for our belief, since we could not believe at all if we did not have rational souls. So, then, in some points that bear on the doctrine of salvation, which we are not yet able to grasp by reason -- but we shall be able to sometime -- let faith precede reason, and let the heart be cleansed by faith so as to receive and bear the great light of reason; this is indeed reasonable. Therefore the Prophet said with reason: 'If you will not believe, you will not understand' [Isa. 7:9]; thereby he undoubtedly made a distinction between these two things and advised us to believe first so as to be able to understand whatever we believe. It is, then, a reasonable requirement that faith precede reason, for, if this requirement is not reasonable, then it is contrary to reason, which God forbid. But, if it is reasonable that faith precede a certain great reason which cannot yet be grasped, there is no doubt that, however slight the reason which proves this, it does precede faith." (*Letter 120; The Fathers of the Church: Saint Augustine*, Volume 10, Letters 83-130; trans. Sr. Wilfrid Parsons, S.N.D. [N.Y., 1953], p.302) Prior to belief, reasons for credibility must be given, and these reasons bear, not on the content of the statements in question, but on the authority of the one uttering them. It will be noticed that the process Augustine outlines is

applicable not only to revealed truth but also to philosophy. Thus, he will distinguish between human and divine authority; human authority, however, is often deceiving. Despite this parallel, Augustine's interest is the faith which bears on what God has revealed to us. Faith is *cum assensione cogitare* (*On the Predestination of the Saints*, 2,5), to think with assent, and although we know what we see and may believe what we do not see, the reward of faith is to see what we believe. (*Sermon* 43,1)

These few words indicate that Augustine's position on the relations between faith and understanding, while subtle and nuanced, has an obvious meaning and truth. The reasonableness of the assent of faith must be established before that assent is given; once given, the mind can go on to examine the believed truths with an eye to understanding them. Faith opens the door to greater understanding, even of things which in principle can be understood without divine faith.

It is possible to say that Augustine holds that divine faith is necessary even with respect to truths which the philosophers have been able to foreshadow. Thus, we have seen that Augustine held that Plato had arrived at striking truths in the absence of all supernatural aid. For the most part, however, men need the help of the Christian faith to overcome the mental darkness and proclivity to vice consequent upon sin. We have seen the stress he puts on the need for rectitude of life if one is to understand: the virtuous life removes the impediments to understanding, clears the eye of the soul so that it might contemplate truth. The Christian religion enables us to prepare for understanding. "Since the blindness of our minds is so great, by reason of the gluttonous excess of our sins, and the love of the flesh, that even those monstrous ideas [of some pagan philosophers] could make learned men waste their time discussing them, will you, Dioscorus, or anyone gifted with an alert mind, doubt that there was any better way to seek the welfare of the human race than that Truth Itself should have ineffably and miraculously become man and, playing His part on our earth by teaching right principles and performing divine actions, should persuade us to believe, for our own advantage, what could not yet be understood by human wisdom?" (Letter 118; *ed. cit.*, pp.291-292) In the same letter he indicates that Plato and Plotinus had arrived at truth, but since the faith was lacking to them, many of their followers fell into error. If we believe by divine faith things which philosophers were able to know, it seems to follow that the man of faith can search for cogent reasons and come to understanding. This does not mean that whatever is believed can be understood in the way in which the philosophers understand, however. Yet the possibility of philosophical proofs is not prejudiced by the role

Augustine assigns to faith. What is most striking in his view, perhaps, is the implication that the Christian is in a privileged position with respect to philosophy itself.

4. *Immortality of the Soul*. To complete this sketch of Augustine's teaching with respect to the first great question of philosophy, What is man?, let us turn to his proof of the immortality of the human soul.

Man is composed of body and soul; neither alone is man. (*De moribus eccl.*, 1,4,6) The view that the soul is in the body as a result of previous sins is vigorously repudiated. However, if the natural state of the human soul is to be in a body, the soul itself is not a body, but spirit. The soul, therefore, is far more perfect than body and is, indeed, similar to God himself: *vicina est substantiae Dei*. (*En. in Ps.*, 145,4) It is the mind's ability to grasp truth which enables Augustine to maintain the preeminence of soul over body as well as the incorruptibility of the soul itself.

Augustine's discussion in *On the Immortality of the Soul* is not always persuasive; indeed, often his reasoning is sophistical. Perhaps the very fact that he piles argument upon argument indicates that he was far from satisfied with some that he sets forth. If we were to extract the common thread from all the arguments there given, it would run as follows. The mind is able to contemplate truth, and in this activity it turns away from body, for body can contribute nothing to this activity. In its operation the mind is capable of grasping truth, of knowing the unchangeable. Consider, for example, our knowledge of mathematical truths; of these we do not say that they were or will be but rather that they are, immutably, eternally. For the soul to know such immutable truths demands an affinity between it and its object. The subject of immutable knowledge must itself be immutable. This is not to say that knowledge of the truth is constitutive of the immutability of the soul, for the soul does not perish when it makes false judgments. The capacity of the soul is exhibited in its actual grasp of truth, its substantial affinity with what is immutable and eternal. Given this, the soul cannot perish; it must be as lasting as the truth which is its object.

Although Augustine is unwavering in his view of the destiny of the soul, the same cannot be said of his view of its origin. We have already seen that early in his career as an author he adopted Platonic modes of speech which suggest that the soul antedates its union with the body. While he later repudiated this, he had great difficulty in explaining the origin of the souls of the descendants of Adam, since any theory had to allow for the transmission of original sin. This is a

problem which can best be treated when we speak of Augustine's view of creation and his doctrine of the *rationes seminales*.

F. God

Man is made for God and, as a rational creature, he is made to know God and to love him: "you have made us for yourself and our heart is restless until it rests in you." (*Conf.*, 1,1,1) Given man's destiny, Augustine believes that only a few men can deny God's existence: "There can be found only a few of such impiety that these words of Scripture would be verified of them, 'The fool has said in his heart, there is no God.' This madness is restricted to a few." (*Sermon* 69,2,3) Even before the spread of the Christian faith most men knew of God; we can see in this a sign of the divine power. God cannot be wholly hidden to the man who uses his reason. Except for a few depraved individuals, the world called men to a recognition of God. (*On the Gospel of John*, 106,4) God is closer to us than is the world he made: "He who made us is closer to us than the many things which have been made. 'In him we live and move and have our being' (*Acts* 17:28); from which it follows that it costs us more labor to discover them than it does to find him by whom they were made (*Literal Com. on Genesis*, V,16,34) This is a hint of the distinctively Augustinian approach to a proof of God's existence; however, texts in great number can be cited which speak of the world as directing our mind immediately to its maker. The following is typical: "Behold the heaven and earth: they cry out that they have been made, for they are changed and altered. Whatever has not been made, yet is, contains nothing within itself which before was not, which would be to be changed or altered. So they cry out that they have not made themselves: Therefore we are because we have been made, nor were we before we were that we might have come to be by ourselves. The words of the speakers is evidence itself. You then, Lord, have made them, and you are beautiful since they are beautiful, you are good since they are good, you are since they are." (*Conf.*, XI,4)

In connection with the idea that the things of this world call us to knowledge and love of God, Augustine makes frequent use of a famous passage of St. Paul: "For the wrath of God is revealed from heaven against all ungodliness and wickedness of those men who in wickedness hold back the truth of God, seeing that what may be known of God is manifest to them. For God has manifested it to them. For since the creation of the world his invisible attributes are clearly seen -- his everlasting power and also his divinity -- being understood through the things that are made." (*Romans* 1:18-20) Augustine explains that an inspection of this world, of the heaven and earth and the creatures in both,

forces the mind to recognize that God exists. "It is this that noble philosophers have sought and from the art have known the Artificer." (*Sermon* 142,2,2) If men have known God and then not gone on to honor him, their wisdom is turned into folly; they make idols and fall into the most bestial vices. If we then say they are ignorant of God, this ignorance is consequent upon vice and is, as the Apostle says, inexcusable.

Augustine teaches that it is relatively easy for men to come to a knowledge of God from the world around them. This knowledge can become distorted and be an indictment if men take pride in their wisdom or fall into other vices. Augustine is very sensitive to the errors into which philosophers have fallen concerning the nature of God. He even attributes the materialism of Democritus to viciousness. This leads one to conclude that the recognition of God, which Augustine feels is widespread, is compatible with a good deal of error. Only a few philosophers, notably Plato and Plotinus, have arrived at a proper conception of God. In the absence of an authority to inform the multitude of what they had learned they made their doctrine a secret, a matter for the initiate. Christianity remedies this complex situation, for now the existence of God as well as his attributes are made known to all men on the authority of God himself, the Truth Incarnate.

Since the Christian accepts on divine authority many things which can be understood, he can go on to seek understanding. If the man of faith sets forth an argument to show that God exists, he will not have to proceed from what he believes or demand that the conclusion be accepted on faith. It is because Augustine attempts proofs of the immortality of the soul and of the existence of God which are of this nature that we can speak meaningfully of the philosophy of Augustine, despite the fact that it seems impossible to maintain that he taught a separation between philosophy and theology with anything like the clarity of an Aquinas. The note of Augustinian philosophizing is struck in the following remark: "Although I hold these things with unwavering faith, since I do not yet grasp them with knowledge, let us so inquire as if all these things were uncertain." (*On Free Choice*, II,ii,5) What will make them certain is evidence, not an appeal to the faith that has not wavered during the inquiry.

Augustine is clearly guided in his philosophizing by his faith; thus, if we define philosophy in terms of not knowing how the argument will turn out, Augustine will not be a philosopher. If we define philosophy in terms of the quality of the evidence adduced to support a proposition, however, evidence which can be grasped whether or not one has faith (Augustine would insist that good moral

dispositions are supposed, although these are not constitutive of assent), then we can expect to find philosophy in Augustine. Much will be said later on whether one can understand and believe the same truth. Augustine does not seem to have been bothered by that question. Given the fact that one who believes God exists can seek to prove this fact, as can one who does not have divine faith, it seems to follow clearly that there is no formal difference in their mode of argumentation -- if both are unsuccessful. Whether or not Augustine succeeded in discovering proofs of the immortality of the soul and the existence of God, it seems clear that he was seeking proofs which did not require divine faith for their acceptance. The fact that philosophical proofs are advanced by a man who has the gift of faith does not add anything intrinsic to those proofs. These comments must suffice for now; the problem of a Christian philosophy is one to which we shall return later in terms of the efforts of the thirteenth century.

While the passages we have cited thus far indicate that Augustine feels we can be led directly from the world around us to God, this is not his most characteristic approach to the proof of God's existence. He usually maintains that one must retreat from the world to oneself and from thence to God. The role the mind plays in the ascent to God becomes central: "Go not abroad but enter into yourself: truth dwells in the inner man; and if you should find your nature mutable, transcend yourself." (*On True Religion*, 39,72) The Augustinian approach to God receives one of its most developed expressions in *On Free Choice* (II, chaps. 3-18). His argument moves in steps to the assertion that God exists.

This argument exhibits how intimately the two great issues of philosophy, God and the soul, are intertwined in the thought of Augustine. The first point established is my certitude of my own existence. St. Augustine notes that the fact of my existence is indubitable since I would have to exist to doubt it or to be deceived in regard to it. (11,3) He says in the *De trinitate* (XV,12,21): "He who is not can certainly not be deceived; therefore, if I am deceived, I am." It is customary to suggest a parallel between these remarks and Descartes' *cogito ergo sum*. The resemblance is at best superficial. Augustine is not saying that things other than my existence are dubitable, for he adds immediately that other things are just as certain as the fact that I exist. Our awareness of our soul, of the principle of life in ourselves, is derived from observation of corporeal movements. (*En. in Ps.*,73,25) Augustine's principle was formulated to overcome the skepticism of others; it does not reflect even a methodical doubt of his own.^{10}

It is indubitable that I exist; equally evident are the facts that I am living and that I understand. This indicates three levels of being: some things simply exist; others exist and live; yet others exist, live, and understand. Man falls under the final heading and is more perfect than things on the first two levels. In man there are bodily senses: seeing, hearing, smelling, tasting, and touching, each with its proper object. As well as these there is an inner sense. When we see we do not see seeing, so that if we are sensibly aware that we are seeing, this must be accomplished by another and, Augustine argues, interior sense. This inner sense is said to be common to beasts and men. In man, over and above the exterior and interior senses, there is reason. A sign of its presence is the fact that man seeks to define seeing and the interior sense, something which neither of these senses would themselves attempt. Augustine sums up: "These things have been shown: by the sense of the body corporeal things are sensed; this sense cannot be sensed by itself; however by means of an inner sense corporeal things are sensed through sense as well as the sense of the body itself. By reason all these things as well as itself are made known and brought under knowledge (*De trin.*, 11,4,10)

This hierarchy is now elaborated. The object of the senses is something which is: sense itself is an instance of living being. Moreover, the inner sense is more perfect than the outer senses, and reason more perfect than sense. Now, if we can prove that there is something more perfect than our reason, something eternal and unchangeable, that will be God.

Augustine turns once more to the senses and notes this difference among them. Although several men cannot simultaneously touch the same portion of a body or eat the same food, several can hear the same sound and see the same color simultaneously. (11,7) In much the same fashion one truth is common to many minds. For example, many minds can possess a common truth about numbers; each man does not have his own private mathematics. (11,8) Is there one wisdom for all men? "If there is a highest good common to all, so too the truth whereby it is discerned and held, that is, wisdom, is one and common to all." (11,9) Truth is what is more perfect than our mind or reason, and if there is something more excellent than truth, that will be God. Or, if there is nothing more excellent than truth, God will be truth. (11,15)

We have already seen Augustine cite numbers to illustrate a truth which is common to many minds. This is not a casual allusion. The role of number in Augustine's proof is difficult to overestimate. We are asked to move from the number that we encounter in the sensible world to the eternal realm of number:

"Every changeable thing you see can only be grasped by the senses or considered by the mind because it has received from number a certain perfection without which it would fall back into nothingness. If this is so, doubt not that for these changeable things not to cease to be but to continue with measured movements and a variety distinct from their perfection to travel the grooves of time, as it were, requires an unchangeable and eternal perfection which is not limited and extended in space or prolonged and diversified in time. By it all these things are capable of receiving their perfection and fulfill it while realizing, each according to its own species, the numbers of place and time."
(11,16)

The argument reaches its crescendo in the following passage: "But if you can find creatures other than those which exist without life, those which exist and have life but not understanding, and those which have existence, life, and understanding, then you might dare affirm that there is some good which does not come from God. These three types can be designated by two names: body and life. The name 'life' applies properly either to those beings having only life without intelligence, like the animals, or to those having intelligence, like men. But these two, namely body and life, insofar as they pertain to creatures (for the creator too has life and that is life supreme), these two creatures, then, body and life, being perfectible, as we have seen above, and such that they would fall into nothingness if they should completely lose their perfection, sufficiently indicate that they derive their existence from that which exists ever the same. That is why every good, be it ever so great or ever so small, can come only from God. For what in creation is greater than the life of understanding and what less than the body? And no matter what their deficiency whereby they tend to nothingness, it is no less true that some form belongs to them such that in a certain manner they are. That which is from being, however little, is from the perfection which knows no deficiency and does not allow the changes of things which corrupt and are perfected to exceed the laws of number. Therefore, whatever worthy of praise be found in the natural world, whether it be deemed worthy of greater praise or less, ought to be referred to the ineffable and incomparable praise of the creator." (11,17,46)

Insofar as the rational soul recognizes truth as something more perfect than itself, for it is the measure of the soul, it has come to a recognition of God. (*On True Religion*, 30,56) The recognition of truth, of God, is the recognition of something immutable and eternal, and the good of the soul is seen to consist in being joined to God.

Augustine's view that recognition of God's existence is had by the majority of men has been seen not to exclude a good deal of error concerning the nature of God. This error, Augustine feels, can be largely explained in terms of moral turpitude. But of what quality is the knowledge of God that is had when knowledge is at its best, as with the philosophers whether Christian or not? This is a question we must pose, particularly since the supposed parallel between Augustine and Descartes on the certitude of our own existence may seem to suggest a further parallel with respect to our knowledge of God. Descartes suggests that the divine nature is known in much the same way as the nature of the triangle is known. Such a remark would be an abomination to Augustine.

Although nothing in this life is to be preferred to knowledge of God, our knowledge of and talk about God is quite imperfect: "God is known more truly than he is spoken of and he is more truly than he is known." (*De trin.*, VII,4,7) It is not remarkable that when God is spoken of, we do not understand. If we could understand, it would not be God who is the object of our knowledge. When we speak of God, it is more pious to confess our ignorance than boldly to claim knowledge. Nevertheless, while comprehension of God is quite impossible, our happiness consists in whatever knowledge of him we can attain. (*Sermon* 117,3,5) We must come to realize that what we can know is what God is not rather than what he is. (*Letter* 120,3,13; *On John's Gospel*, 23,9) "We understand God, if we can, to the degree that we can, as good without quality, great without quantity, creating without needing to, present but not located, containing all things without 'having' (*habitu*), wholly everywhere but not contained, sempiternal without time, making mutable things without himself changing, altered by nothing. Whoever thinks thus of God, though he cannot yet discover in every way what he is, is piously cautious to think of him as far as possible in terms of what he is not." (*De trin.*, V,1,2) God said to Moses "Ego sum qui sum" (I am who am); his name is "Qui est" ("He who is"). Being is God's proper name because God can not change; there is no past nor future for God, since the dimensions of time are revealed by change. God is wholly unchangeable and immutable: he is. (*Sermon* 6,3,4) The perfection of God is such that any perfections we encounter in creatures are one and simple in God, a fact which makes our language inevitably inept to express even our imperfect knowledge of God. Even the most common or universal terms tend to express a perfection distinct from others and

so fall short of expressing God; thus, to call God a substance connotes a perfection differing from accidents, from other perfections, and seems prejudicial to the divine simplicity. (*De trin.*, VII,5,10) It is better then to call

God essence or being and to realize that he does not have the perfections attributed to him but is each of them, for example, God is wisdom.

The name God attributes to himself, being, indicates that in God there is no distinction between the divine nature and the various perfections we are constrained to affirm of God. The attributes Augustine stresses are the divine simplicity, immutability, omnipresence, eternity, and providence. Augustine insists that God's foreknowledge of our free acts does not lessen their freedom, since what God foresees is precisely our free choice. (*On Free Choice*, 111,3,8)

While we will not enter here into St. Augustine's remarkable and influential doctrine of the Trinity of Persons in the divine nature, something must be said of his procedure in discussing this mystery. We have seen that Augustine holds that man can come to knowledge of God even apart from revelation; indeed, the one who has the gift of faith can seek understanding of what he believes, can regard as doubtful things in which he has unwavering belief, in order to learn reasons for them. That God exists is a truth which can be known by philosophical reasons as well as by faith. Philosophical knowledge does not lead to comprehension of the divine nature; the term here is an understanding that our knowledge of God is its own kind of ignorance. The characteristically Augustinian approach to God proceeds via man. Now if the divine nature always retreats before our efforts to understand it, it surely follows that we cannot understand how there can be three persons in one divine nature. If we seek in creatures things to proportion this mystery to our minds, analogies to the Trinity, what we find will not be conclusive in the way arguments for the existence of God and for attributes of the divine nature may be. True, Augustine sometimes speaks as if philosophers had arrived at a recognition of the Trinity, but finally he denies that even the Platonists grasped this. The man to whom this truth has been revealed by God himself will seek created analogies of this truth, but any certitude concerning it will always be a result of faith. Typically, Augustine seeks analogies of the Trinity in man, who has been made in the image of God. Thus, he finds in man these trinities: mind, knowledge, love; memory of self, understanding, will; memory of God, understanding, love. (See Gilson, *The Christian Philosophy of Saint Augustine* [New York, 1960], pp. 219 ff.) Concern with the Blessed Trinity thus leads Augustine to an extensive analysis of the human soul.

Our suggestion is that Augustine's procedure lays the basis for the later distinction between the *praeambula fidei* and truths which are of faith alone. The role of reason is appreciably different with respect to each of these. Although

the distinction is not explicitly made by Augustine, he appears to have honored it in practice.

G. Creation

Augustine holds that an inspection of the world reveals at once that it has been made; things virtually cry out that they have been made, pointing beyond themselves to their maker. Not only have things been made by God, they have been made from nothing. They were made from no subject matter which was not also made by God; nothing apart from God is but what has been made by God. "Some have tried to argue that God the Father is not omnipotent; not because they have dared to say this, but they are convinced by their traditions to feel and believe this. For they say that there is a nature which the omnipotent God has not created and of which he fabricated this world . thus they deny that God is omnipotent, for they do not believe the world could be made unless in the making of it he made use of some other nature already given which he had not made. . . . In this way they understand that the fabricator of this world is not omnipotent, since they hold that he could not make the world save by using as matter some nature unmade by him." (*On Faith and the Creed*, 2,2) If God is omnipotent, he can create from nothing; if his creative action presupposes matter, God is not omnipotent.

That God creates from nothing is an index of his omnipotence. We know from Scripture that the world had a beginning. What is the relation of creation and time? "For if eternity and time are rightly distinguished by this, that time does not exist without some movement and transition, while in eternity there is no change, who does not see that there could have been no time had not some creature been made, which by some motion could give birth to change -- the various parts of motion and change, as they cannot be simultaneous, succeed one another -- and, thus, in these shorter or longer intervals of duration time would begin? Since then, God, in whose eternity is no change at all, is the creator and ordainer of time, I do not see how he can be said to have created the world after spaces of time had elapsed, unless it be said that prior to the world there was some creature by whose movement time could pass. And if the sacred and infallible Scriptures say that in the beginning God created the heavens and the earth, in order that it may be understood that he had made nothing previously -- for if he had made anything before the rest, this thing would rather be said to have been made 'in the beginning' -- then assuredly the world was made, not in time, but simultaneously with time. For that which is made in time is made both after and before some time -- after that which is

past, before that which is future. But none could then be past, for there was no creature by whose movements its duration could be measured." (*Civ. dei*, XI,6) Augustine says that the six days of creation cannot be understood as days in the ordinary sense, for then there would have been three days without the heavens according to which we measure our days. Creation is instantaneous. However, Augustine does not hold that a fully organized world came into being at once. He puts the notion of instantaneous and simultaneous creation together with the fact of the gradual appearance of things as a result of change and the coming into being of human souls which do not simply arise out of preexisting matter. "Contrary to most of his contemporaries, however, he does not presume that the instantaneous act of the Creator produced an organized universe such as we see today. He distinguishes between creation properly so called and the formation or development of the world. This second action is due, at least in great part, to forces placed by the Creator in the depths of nature which have gradually and progressively passed through the various phases to which the Mosaic account gives an approximation." (Portalie, p. 137)

God, in producing the world, has produced not only a certain number of things but things which are causes of other things to appear in the course of time; the things which will come to be only in time are present in their causes at the very outset and, thus, do not escape the creative causality of God. Augustine speaks of the primitive elements, of seminal reasons (*rationes seminales*): "All things were created by God in the beginning in a kind of blending of the elements, but they cannot develop and appear until the favorable circumstances are realized." (*De trin.*, 111,9,16) Just as there is invisibly present in the seed everything which will later appear fully developed in the tree, so the world at the beginning of time contained in seed everything which would one day appear, including what has not yet appeared.

Augustine does not hesitate to apply his interpretation to Adam and Eve, who were not made in the very beginning of time. He must make certain adjustments in his theory, however, since he will not allow that the human soul was precontained in a causal principle. A special intervention of God is required for the formation of man. Augustine is guided in his remarks by the account of creation in Scripture. Man's body is formed from the slime of the earth, but his soul does not come into being in this way. It must be created in the same way as the primitive elements, out of nothing. This is true not only of the souls of the first parents but of every soul. Augustine is careful not to suggest, however, that all human souls are created at the outset and exist prior to their union with a body.

H. *The City of God*

In 410 A.D. Rome was sacked by Alaric the Goth, himself a Christian. Historians assure us that this was not the worst invasion of Rome, but the effect of this fall on the times of Augustine cannot be underestimated. There was a great exodus from Rome and all of Italy, and refugees appeared in North Africa and in Jerusalem filled with tales of horror. What explained the fall of Rome, the seat of what had been so proud and farflung an empire? A conviction spread that the conversion of Rome to Christianity may have been responsible, that disloyalty to the old gods under whose aegis the city had been built and the empire spread accounted for the present ignominious fall.

The suspicion that Christianity had brought political and military disaster was not confined to pagans. Rickaby suggests that the Christianity of the converts within the empire could still have amounted to little more than a patina covering a good deal of latent paganism. This conjecture would explain the vigor with which Augustine undertook to refute the argument when he began composing *The City of God* in 412. This work was to occupy him sporadically over the next fifteen years. The polemic tone dictated by its immediate occasion and the fairly negative purpose with which he began were gradually replaced; thus, the work took on an uneven, frequently erratic tone but retained its fundamentally unified purpose. Augustine wanted not only to show the inadequacies of the pagan religion but to emphasize the perfection of Christianity. The book that had been occasioned by the fall of Rome became the tale of two cities, the city of man and the city of God. In the course of describing the origins, goals, and ends of these two cities Augustine brought to bear such immense erudition and indefatigable zeal that the result is one of the great classics of all times; *The City of God* is, perhaps, the most influential work of the great bishop of Hippo.

This is Augustine's own account, in the *Retractationes*, of the writing of *The City of God*:

Meanwhile Rome was overthrown by a raid of Goths, led by King Alaric, a most destructive invasion. The polytheistic worshippers of false gods, whom we commonly call pagans, endeavored to bring this overthrow home to the Christian religion, and began to blaspheme the true God with unusual sharpness and bitterness. This set me on fire with zeal for the house of God, and I commenced to write the books *Of the City of God* against their blasphemies or errors. This work occupied me for a number of years, owing to numerous

interruptions of businesses that would not brook delay and had a prior claim on me. At last this large work *Of the City of God* was brought to a conclusion in twenty-two books. The first five of them are a refutation of their position who maintain that the worship of many gods, according to the custom of paganism, is essential to the prosperity of human society, and that the prohibition of it is the source and origin of calamities such as the fall of Rome. The next five books are against those who, while allowing that such calamities are never wanting, and never will be wanting, to the page of mortal history, and are now great, now small, under varying conditions of place, time, and person, yet argue that polytheistic worship, and sacrifice to many gods, is profitable for the life that follows after death. These first ten books, then, are a refutation of these two vain opinions adverse to the Christian religion. But not to expose ourselves to the reproach of merely having refuted the other side, establishing our own position is the object of the second part of this work, which comprises twelve books; though, to be sure, in the former ten, where needful, we vindicate our own, and in the latter twelve we confute the opposite party. Of the twelve following books, four contain the origin of the two cities, the one of God, the other of this world. The next four contain the course of their history; the third and last four their several due ends. Thus the whole twenty books, though written of two cities, yet take their title from the better of the two, and are entitled by preference *Of the City of God*.

This succinct sketch of the work does not indicate its patchwork character, but it provides us with a generally accurate and convenient division for the following presentation.

Refutation of Paganism. As Augustine indicates, a twofold defense of pagan polytheism has been put forward. On the one hand, it is maintained that the worship of many gods is more profitable in this world; on the other, that it is more profitable for the next. St. Augustine attempts to dispose of the first contention by indicating that the plight of Rome at the hands of Alaric might have been far worse if it had not been for the Christian influence, thanks to which at least some mercy was shown the conquered. Although the women of Rome were objects of the conqueror's triumph, Augustine is concerned to dismiss the pagan belief that in such an extreme situation a woman should prefer death to dishonor. An unconsenting suffering of rape is preferable to the great sin of suicide, since if consent is not given, true chastity is not destroyed. Anticipating what will be conceded by the second defense of paganism, Augustine observes that in this life misfortune comes to both the just and the unjust and that even the true religion is no guarantee against external evils. We

have here, after all, no lasting city, though another city whose destiny is eternal is intertwined with the earthly city.

This first allusion to the great opposition which is the theme of the work and the source of its title indicates that Augustine is not distinguishing between time and eternity, this life and the next. He has in mind the extension beyond this life of the option men make here below where the city of God is becoming a living reality in time, destined to continue in eternity. Men become citizens of the earthly city by preferring self to God; they become citizens of the eternal city of God by preferring God to themselves according to the true religion established by Christ. The populations of these two cities are fluid in time; not all those who are now members of the city of God will remain in that camp, and many who presently ally themselves with the earthly city will, with time and God's grace, become citizens of the eternal city. Thus, the import of the dichotomy goes beyond the political. Augustine's view of history is a Christian one; beneath the visible and evident political dispositions he is able to discern the more meaningful politics constituted by the priorities recognized by men in their minds and hearts. The city of God is not, however, something hidden and secret. Its expression is the Church. The state is not, as such, opposed to the city of God. The earthly city is not a political reality but the congress of those whose lives are governed by self-love to the detriment of God.

Augustine describes in some detail the public spectacles of indecency which had been engaged in as worship of the pagan gods and indicates that the secret rites were even less worthy of men, let alone of gods. In a vein reminiscent of the Greek philosophers, Augustine says that such gods are not fit models of imitation for men. Indeed, the Roman heroes have been worthier models than the Roman gods. The vast number of pagan deities and the conflicting and confusing roles assigned them is discussed at some length. Under paganism Rome knew much adversity and injustice; moreover, under Christianity the city and empire enjoyed much success. The main point of these first five books is that good and bad fortune befall both the just and the unjust according to God's providence.

Providence is not to be confused with fate, however. Augustine is interested to show that human freedom is not jeopardized by God's causality. God has foreknowledge of the evil men will do, but that does not diminish their responsibility for it, and the same must be said of the good men do. Augustine does suggest that the Romans were rewarded with temporal goods for the

natural virtues they practiced, but he sees this as a poor substitute for the eternal felicity which awaits the elect.

When he has finished his reply to those who would argue that worship of the pagan gods insures temporal success, Augustine turns to a variant of the argument. Some agree that good and bad fortune in this life come equally to pagan and Christian, but they maintain that we shall be better off in the next life if we worship the pagan gods in this. Augustine's reply to them is twofold. First, he shows that the pagan theology can scarcely pass as a spiritual religion. Second, he turns to the philosophers who have attempted to transform the popular religion into something more exalted. His principal concern here is Neoplatonism.

In books eight, nine, and ten of *The City of God* Augustine gives a sketch of ancient philosophy to which we have already had occasion to refer. A matter which looms large in these books is Augustine's interpretation of the airy spirits or daimons in religious Neoplatonism. These daimons occupy a middle region between the gods and men. Augustine interprets this teaching as a crude attempt to assign an intermediary between the human and the divine. A need for an intermediate is seen both from the point of view of man, who is so much less than the gods that he is sensible of his inability to approach the gods directly, and from the point of view of the gods, to whom it would seem unfitting that they should concern themselves directly with men. We are already acquainted with Augustine's praise of Plato, a good deal of which is to be found in these books. Though he praises Plato, Augustine feels constrained to reprimand the Platonists, especially Porphyry. From this critique emerges a deep appreciation of the fundamental inadequacy of any human attempt to bridge the gap between man and God. The attempts of the Neoplatonists, while partially commendable, seem a mere parody of the Christian revelation. Man is, indeed, in need of a mediator, but that mediator is Christ, and it was necessary for God to humble himself and lift man up if there was to be any intimate converse between creature and creator.

This is the negative or critical part of *The City of God*. The pagan religion has been shown to be no guarantee of good fortune in this life and wholly inadequate as a commencement of eternal life. Christianity cannot promise an absence of misfortune in this life, but by providing knowledge of the end that awaits us and the grace to achieve it, it enables us to assess both temporal goods and evils as of little moment when compared with permanent citizenship in the eternal city.

The Two Cities. "Two loves therefore have given origin to these two cities, self-love in contempt of God unto the earthly, love of God is contempt of one's self to the heavenly. The first seeks the glory of men, and the latter desires God only as the testimony of the conscience, the greatest glory. That glories in itself, and this in God." (XIV,28) With book eleven Augustine begins the discussion of the origin, progress, and ends of the two cities. First, the question arises as to how we can know God; this leads to a discussion of revelation and the canonical books of Scripture. After that, though not in an altogether orderly fashion, Augustine discusses the nature of God and the Trinity of Persons in God. He then turns to the doctrine of creation, speaking of the work of the six days. He writes of the fall of some angels and the consequent division of them in terms of light and darkness. The fall of the angels is portentous for the subsequent fall of man and the constitution of the city which is the opposite of the heavenly city. The creation of man and man's fall involve lengthy treatments of the nature and possibility of original sin, of man's state prior to it, and the consequences for the race of that first sin. With the advent of sin, two contrary courses open up for the human race: men divide themselves into the sons of flesh and the sons of promise, symbolized by Cain and Abel. Augustine sees a parallel in the fact that Cain, the murderer of his brother, founded the first earthly city, just as the founder of Rome killed his brother. Political society is seen by Augustine as a result of sin; he traces private property to the same root. Through book eighteen he provides a narrative of the history of the human race, which is derived largely from the Old Testament. The goal of part of mankind is the heavenly city and bliss with God, while the other part of mankind elects to find its lot with the fallen angels.

This is the main line that Augustine follows in the second part, the last twelve books, of *The City of God*. We shall discuss some points in detail, starting with book nineteen, which is an extended development of Augustine's view of order and is sometimes said to contain Augustine's notion of morality.

The controlling question is: In what does human happiness consist? Augustine accepts without question the Greek eudaimonistic interpretation of human action. In their moral life, in their choices and decisions, men aim for felicity or happiness. The philosophers have said much on this question. Augustine appeals to Varro, a favorite source of his in *The City of God*. (Unfortunately, Varro's *Antiquities*, a work of forty-one books, has been lost.) Augustine is clearly impressed by Varro's manner of asserting that there are 288 distinct views on the primary good held by philosophers, which can, nevertheless, be reduced to three. Either man's elementary desires are sought for the sake of virtue, virtue is

sought for the sake of man's elementary desires, or each is sought for its own sake. Varro holds that human happiness consists in both bodily pleasures and the practice of virtue; thus, elementary desires are pursued for their own sake, although virtue is the best good of man. Human happiness, as described by Varro, is a well-rounded thing: health of body and soul, and a harmonious family life in the wider context of an ordered and peaceful society.

Augustine agrees that this is a most attractive statement of human happiness, but he adds that it is little like reality. Bodily health is at best imperfect, and even the most exemplary men seem to have but a tenuous hold on virtue. A man's wife and children are too often unfaithful, and in society at large, injustice seems rampant. There is never an end to lawsuits, which often cause the innocent to suffer. Worst of all is war, which seems endless. The Stoic may judge such evils to be of little or no account, but we know he is wrong. The absence of these evils is a very real good -- that is the strength of Varro's description of happiness. However, even if Varro's ideal could be reached, it would still not assuage the deepest desires of man.

The harmonious life that we accept as the ideal cannot be perfectly achieved in this life. Consequently, happiness must be redefined in terms of the degree of harmony possible to man in an exceedingly imperfect situation. God has given us a desire for human happiness, and it is unlikely that this desire is given only to be frustrated. In pursuing the peace and harmony of the good life we are, at least implicitly, longing for the true peace of the eternal city.

Peace is the key word in Augustine's account of what men finally seek. Even the evil man seeks it, though his goal may be but the parody of peace as found, for instance, in the domination of others. (Chap. 11) "The body's peace therefore is an orderly disposal of the parts thereof; the reasonable soul's, a true harmony between knowledge and performance; that of body and soul alike, a temperate and undiseased habit of nature in the whole creature. The peace of mortal man with immortal God is an orderly obedience unto his eternal law performed in faith. Peace of man and man is a mutual concord; peace of a family an orderly rule and subjection amongst the parts thereof; peace of a city an orderly command and obedience amongst the citizens; peace of God's city a most orderly coherence in God and fruition of God; the peace of all things is the tranquillity of order." (Chap. 13)

The concept of peace and harmony is the thread that must run through the whole of society. If we are to have a total view of the peace of society, our view

must be theological. Again, we have here no lasting city; the ultimate purpose is achieved, if at all, only in an inchoative fashion in this life. We are destined for eternity, and only in the fullness of time will peace, order, and harmony establish themselves in a definitive way. The citizens of the city of God are one people here below in a far more perfect fashion than men can be citizens of a nation or empire. What constitutes one people is their union in pursuit of a common object of love. This community can transcend national boundaries and differences in language.

The last three books of *The City of God* deal with judgment, hell, and heaven. Hell and heaven are the respective terms of the earthly and heavenly cities; the goal of history is beyond history; in this life man is a pilgrim. The distinction between the two cities is not one between the political order here below and a heavenly city somewhere yonder; nor is it a distinction between two kinds of political organization here below. Charlemagne loved to have *The City of God* read to him, and it is thought to have played a great role in the elaboration of the concept of a Christian Empire, but this is an adaptation of Augustine rather than his own teaching. According to Augustine, what distinguishes the inhabitants of these two cities is their response to a basic moral choice. Does an individual serve himself to the detriment of God or God to the detriment of self? The earthly city consists of all those who make the first choice; the city of God claims all those who make the second choice. Membership in the city of God is not identical with membership in the Catholic Church. Many Catholics, nominal Catholics, as we should say, have actually made the first choice, and many of those currently outside the Church have or will make the choice that gives them membership in the city of God. Thus, here below in time the situation is fluid. With the end of time, at the final judgment, man's ultimate choice is ratified by God. One who has chosen the earthly city has chosen hell; one who has chosen to serve God rather than self has chosen heaven. Thus, what Augustine means, while it is not something covert or secret, cannot be translated into simple political terminology. He has any number of significant asides on the relation between Church and state, but that is not the real burden of his book and that is not the significance of the distinction between the earthly city and the city of God.

I. Conclusion

Augustine's influence on subsequent ages is due entirely to the force of his thought. While he lived, he was bishop of what has been called a third-rate city, and he had little or no impact on the course of practical affairs. In another see,

in another post, he would have been attended as a matter of course. Surprisingly, as Bishop of Hippo, men turned to him constantly for the resolution of theoretical and principally theological difficulties, even though he did not seek their notice. However, the influence he had on the thought of his own times is as nothing compared with the undiminishing influence he has had through the centuries, even to our own day. The fact that he is today held in almost equal esteem by Catholics and Protestants suggests the hope that he may yet have his greatest role to play in the current movement toward Christian reunion.

Together with Boethius, who lived about a century after him and professed the hope that his doctrine would be identical with the great bishop's, Augustine was destined to be the vehicle whereby some knowledge of classical antiquity was transmitted to the men of the Dark and Early Middle Ages, when most direct contact with the early sources had been lost. This was a role that Boethius deliberately assumed, but in the case of Augustine it is merely one of the significant, if adventitious, effects of his prodigious scholarly efforts. In the High Middle Ages Augustinianism was the traditional approach to theology, and if his prominence seems temporarily eclipsed by the problems and opportunities consequent upon the introduction of the works of Aristotle into the West at the end of the twelfth century, this eclipse is, if not merely apparent, certainly temporary. Aquinas, the greatest of the thirteenth-century synthesizers of the old and new, is actually proceeding in the spirit of Augustine and doubtless would have been surprised to have what he was doing assessed as an alternative to Augustinianism. For, while it does not achieve the clarity in Augustine's thought that one might wish, the thirteenth-century distinction between philosophy and theology, as well as the conception of the nature of speculative theology, owes a great deal to the efforts of Augustine.

There can be little doubt that what is called the philosophy of Augustine is principally Platonic in inspiration, although some Aristotelian elements are apparent. The philosophy of Aquinas, on the other hand, is principally Aristotelian. Whether the Thomistic philosophical synthesis is devoid of Platonism or whether Platonism is one of its principal components is a point we shall examine later. What cannot be questioned is the massive impact of the thought of Augustine on Aquinas. Indeed, it may be said that anything like an understanding of Aquinas depends on a previous understanding of Augustine. Thus, these two chief Christian Doctors must be regarded as complementary, rather than opposed, inspirations in the continuing Christian task of bringing to bear on truths of faith whatever of validity can be found in natural thought.

Bibliographical Note

Augustine's works can be found in Migne PL, 32-46, but better editions of many of his works exist. For English translations one can go to M. Dods, *The Works of Aurelius Augustinus*, 15 vols. (Edinburgh, 1871- 1876). Individual works of Augustine have been put into English by so many hands and under so many imprints it would be impossible to mention anything like a representative sampling here. J. J. O'Meara has made a list of available translations in his version of H. I. Marrou, *Saint Augustine* (London, 1958). The excellent introduction to Augustine's thought written by Portalie for *Dictionnaire de théologie catholique* has been brought out in English by Henry Regnery: *A Guide to the Thought of Saint Augustine* (Chicago, 1960). Of profound importance, of course, is E. Gilson, *The Christian Philosophy of Saint Augustine* (New York, 1960). For recent work on Augustine see *Augustinus magister: Communications et actes du congrès international augustinien* (Paris, 1954). For the nonspecialist the *Confessions*, *The City of God*, the philosophical dialogues, are of first importance.

{1} Ozanam, *La civilization chrétienne*, p. 389, would trace the seven liberal arts to Philo Judaeus, *De congressu*. Cited in Rashdall, I, p. 34, n. 2.

{2} ". . .sed quoniam his mortalium rerum cura terrenorumque sollertia est nec cum aethere quicquam habent superisque confine, non incongrue, si fastidio respuuntur." Eyssenhardt's edition, p. 333.

{3} See Margaret Deanesly, "Medieval Schools," *Cambridge Medieval History*, vol. 5, chap. 22, Pp. 765-779.

{4} *Contra academicos*, III, xvii, 37; cf. *De civitate dei*, VIII, 4.

{5} In chapter 2 of book VIII Augustine discusses the theories that Plato's theology resulted from hearing Jeremias when Plato was traveling in Egypt, or perhaps from reading his book. Augustine points out that he himself had embraced this theory in some of his works (cf. *De doctrina christiana*, II, xxviii, 43), but he feels it certain that Plato lived at least a hundred years after Jeremias and thus could not have listened to him. Nor could he have read his book, since it had not yet been translated into Greek. Nevertheless, Augustine argues, Plato could have learned of the Jewish faith through interpreters, and this would explain the similarities between *Genesis* and the *Timaeus*. In chapter 12,

however, Augustine concedes that Plato could have come to his views about God without any contact with revealed truth, and he cites Romans 1:20 again. Augustine is reluctant to say that knowledge of God is easily attained except through revelation; indeed, his whole attitude here exhibits his conviction that it is a matter of the utmost difficulty to reason to the existence of God.

{6} See too *Retractationes*, III, 2: "Plato called the intelligible world the sempiternal and changeless plan according to which God made the world. If anyone should deny it he must by way of consequence maintain that God did irrationally what he did, since if there was in him no plan of making while he created or before he created, he would not know what he was doing. If there is such a plan, and there is, Plato seems to have called it the intelligible world."

{7} Cf. *Oeuvres de S. Augustin*, 1er Series, Opuscules, vol. 10, Melanges Doctrinaux (ed. Bardy, Beckaert, Boutet), p. 726.

{8} In *De civitate dei* (XI, 23) Augustine takes issue with Origen, who maintained that souls are consigned to bodies as punishment for sins committed in a pure state.

{9} In view of the fact that Wittgenstein begins his *Philosophical Investigations* by quoting a passage from the *Confessions*, which he then proceeds to criticize as expressing a whole theory of language, it is important to notice that Augustine has many extensive treatments of language and is highly sensitive to its many subtleties. Of course, Augustine does not come within hailing distance of the linguistic philosophy of Wittgenstein. {10} See Michael Mason, *The Centre of Hilarity* (London: Sheed & Ward, 1959), p. 88.

Chapter III

Denis the Areopagite

Our only certitude regarding this author concerns who he was not. For long centuries he was believed to have been Denis or Dionysius the Areopagite, a convert of St. Paul, and the *Corpus Areopagiticum* received the attention and respect commensurate with that belief. The works were translated into Latin by John Scotus Erigena in the ninth century and were commented on by him and many other outstanding medievals, among them, Hugh of St. Victor, Albert the Great, Thomas Aquinas, and Denis the Carthusian. Internal evidence suggests that the works of Dionysius could not have been written much before the end of the fifth century. By placing his *floruit* in the year 500 we are being intentionally conservative.

The works of the Pseudo-Dionysius are the following: *De coelestia hierarchia* (*On the Celestial Hierarchy*), *De ecclesiastica hierarchia* (*On the Ecclesiastical Hierarchy*), *De divinis nominibus* (*On the Divine Names*), and *De mystica theologia* (*On Mystical Theology*). There are also ten letters.

Dionysius is a theologian; the whole burden of his works might be described as the exposition of what man can know of God and how, knowing him, he can name God. He is interested in proceeding, not according to the words of human wisdom, but in terms of Scripture. (*Div. nom.*, 1) In search of knowledge of God in terms of what Scripture has said, however, he will also appeal to the efforts of philosophers. The most striking point about Dionysius is his insistence that the object of his concern is wholly beyond the ability of man to comprehend. The language Scripture uses to speak of God cannot express with any degree of adequacy what he is; *a fortiori* the attempts of men to speak of God must fail. His thought on this subject represents a division of theology which was to have a profound influence.

Dionysius says that to see and know God is to be accomplished through not seeing and not knowing him, for not to see God is truly to see him, not to know him is truly to know him, for we can adequately praise what is above all being by removing from him everything which pertains to existent things. In other words, our ignorance of God is something which must be achieved, for we will best know what he is not by attempting to work up to him through the grades of being. (*Myst. theol.*, 2) First, there must be an affirmative theology (*theologia kataphatika*) in which we argue that God is a unique nature, that he is a Trinity

of Persons. In the *Divine Names* Dionysius attempts to show what words can be applied to the divine nature, for example, Good, Light, Love, Being. Besides these names of intelligibles, we must discuss those words which are transferred from creatures to God in what may be called symbolic theology, that is, the many metaphorical names of God. Dionysius asks his reader to consider how names for God become more numerous as we move into metaphorical language. Negative theology (*theologia apophasika*) begins on the level of symbolic theology and ascends upwards, denying as it goes, until it becomes clear that God is ineffable, uncomprehended by our names taken singly or together.

While Dionysius' mystical works present the negative theology just described and the other works are all seemingly part of affirmative theology, these are not wholly distinguishable theological activities. The name of anything that is can be transferred to God as to its cause; this is simply a symbolic way of speaking. However, when God is named by means of "intelligibles," such as one, good, and so forth, he is indeed named from a created perfection, but there must be an accompanying denial understood: God is intelligent, and he is thereby named from what we know as intelligence, created and therefore limited intelligence, but the limitation must be denied of God. We end then with the assertion that God is superintelligent, that is, intelligent wholly above our ability to understand. That he escapes our ken is even more clear when we consider that he is superlife and supergood as well, and that in him these are but one perfection. The twofold theology thus implies a threefold procedure in naming God: affirmation, denial, and then the affirmation of a perfection which wholly exceeds our experience and ability to name.

The defect in our language and knowledge of God is explained with reference to us; on the side of God there is, of course, no defect. He is imperfectly named because he surpasses in perfection our ability to understand. The supreme Monad, he is the source of all the perfections we find scattered and distinct in creation; creation refers us back to him as the source of what we know only as limited and separate. The emanation of all things from God as their source and the return of all things to him as to their end is but one Neoplatonic note struck by Dionysius. His preference for the word One as the name of God, his utilization of the metaphor of light, with creatures as so many rays springing from a source too strong for our intellectual eye, the view that creatures are images -- all these reveal the influence of Plato, Plotinus, and Proclus. There is a processus or emanation of creatures from God (*Div. nom.*, 5), and God, while one, indeed superunity, is thereby multiplied in his effects. We will find this

extremely delicate concept in John Scotus Erigena as well, with the latter arguing that in this sense God can be called created. In phrases which will echo in Erigena, Dionysius speaks of God as "all in all" and of the divine Ideas as "predestinations." There is the distinct reminder of a stratified world, with the Ideas emanating from God as primordial caused causes and other things from them, as if existence-in-itself exists between God and the things that are. Moreover, the voluntariness of creation is somewhat diminished by Dionysius, and one detects a Neoplatonic suggestion that the levels of creation proceed from God in some necessary way.

Perhaps these few remarks will suffice to indicate the power as well as the obscurity of the thought of Dionysius. By far the most influential aspect of these writings is their doctrine on the unnameability of God, and thinkers of all persuasions will make an effort to adjust their thought to this claim. Those who find in Dionysius grounds for steering between the extremes of denying that we can know anything about God and claiming that God is a proportioned object of our mind would seem to be most faithful to him. That our knowledge of God is, compared to its object, no knowledge at all, in the sense that we cannot comprehend him, does not mean that creation provides no indirect way to meaningful language about its cause. In Cusa's phrase, our ignorance of God is a learned one, and Dionysius would hardly deny that we are better off after the efforts of affirmative and negative theology than we were before. It is a matter of some importance to note that not even Scripture, which is God's revelation to man, transcends the human mode of naming, which is to apply to God names of perfections best known to us in creatures.

Bibliographical Note

For the works of Pseudo-Dionysius see Maurice de Gandillac, *Oeuvres complètes du Pseudo-Denys l'Areopagite* (Paris, 1943); J. Parker, *The Works of Dionysius the Areopagite* (London, 1897). See too A. B. Sharpe, *Mysticism, Its True Nature and Value* (London, 1910); E. C. Bolt, *Dionysius the Areopagite: On the Divine Names and The Mystical Theology* (New York, 1951); René Roques, *L'univers Dionysien* (Lille, 1954); and Denys Rutledge, *Cosmic Theology* (London, 1964).

Chapter IV

Boethius

A. The Man and His Work

Anicius Manlius Severinus Boethius (c.480-524), "the last of the Romans and the first of the Scholastics," in the famous phrase, was born in Rome of a politically prominent family. His father had been a consul, he himself became one in 510, and his two sons achieved the same distinction in 522. Boethius married a woman named Rusticana, the daughter of Symmachus; as will appear, Boethius held his father-in-law in more than ordinary esteem. Boethius was a consul under Theodoric the Ostrogoth and came to an untimely end when he was accused of conspiring with Justin, Emperor of the East, against Theodoric. There were theological undertones to his fate since Theodoric subscribed to the Arian heresy, while Boethius, like Justin, was a Catholic. Boethius protested his innocence, but he was cast into prison and executed without a trial in 524.

Although he was a statesman, Boethius produced a surprisingly large and influential body of work in philosophy. His major task was to translate Plato and Aristotle into Latin and, their teachings having been made available, to show the fundamental agreement of the two philosophers. While Boethius did not, so far as we know, even approach this awesome goal, what has come down to us indicates that he conceived his role to be considerably more than that of a middle man. His surviving translations are of logical works of Aristotle. We can conveniently divide his total production into philosophical and theological works.

Philosophical Works. Boethius translated the following logical works of Aristotle: *Categories* and *On Interpretation*. While translations of the *Prior* and *Posterior Analytics*, the *Topics*, and the *Sophistical Refutations* are included in editions of Boethius' work, scholars are now inclined to cast doubt on their authenticity. Boethius also translated the *Isagoge*, an introduction to the *Categories* of Aristotle, written by the Neoplatonist Porphyry. Besides translating, Boethius wrote a number of excellent commentaries: two on Porphyry's *Isagoge*, one on Aristotle's *Categories* (a second was projected), two on *On Interpretation*. He is also credited with a commentary on the *Topics* of Cicero. The following independent logical works are included in editions of his work: *Introduction to Categorical Syllogisms*, *On Categorical Syllogisms* (2 books), *On the Hypothetical*

Syllogism (2 books), *On Division*, *On Definition*, *On Topical Differences*, *On Rhetorical Connexion*, *The Distinction of Rhetorical Loci*. Besides these logical works, a work on arithmetic and another on music are attributed to Boethius. Finally, there is the great *Consolation of Philosophy*.

Theological Works. The theological writings of Boethius comprise works on the Trinity, on the union of the divine and human nature in Christ, and on the participation of goodness. We shall mention their titles later.

Our discussion of Boethius will center on two points: the relation between faith and reason and the problem of universals. Not only are these central concerns of his own effort, but they contain factors which were highly influential in the Middle Ages.

B. Faith and Reason

The problem of the relationship between faith and reason acquires curiously personal overtones in Boethius. We have mentioned that Boethius set as the great task of his lifetime the translation of the works of Plato and Aristotle. This task is of such magnitude that we may doubt that Boethius could have seen it through to completion even if he had not devoted much of his time to statesmanship and, as a result, come to an untimely end. As Boethius languished in prison, aware of the end that awaited him, he, like Socrates in a similar position, first devoted himself to the writing of verse. After a time, however, he turned to the composition of the work which ever since has constituted his claim to widespread fame, the *Consolation of Philosophy*. While the *Consolation* must be classified as a philosophical work, the fact that this can be done is, given the circumstances of its composition, somewhat of a mystery. The difficulty was well stated by Samuel Johnson, quoted of course by Boswell: "Speaking of Boethius, who was the favorite writer of the middle ages, he said it was very surprising, that upon such a subject, and in such a situation, he should be *magis philosophus quam Christianus*." That Boethius, on his own insistence the victim of gross injustice, should have attempted to reconcile himself to his condemnation and approaching execution by appeal to philosophical truths alone, and indeed to the example of philosophers alone, is quite surprising. We should expect that the innocent victim par excellence would have provided him consolation and example, yet no mention is made of Christ, no explicit quotation from Scripture is to be found in the *Consolation*.⁽¹⁾

What is the explanation of this strange situation? Does philosophy in the *Consolation* stand for a wisdom which would embrace both sacred and profane knowledge? We shall see that this is not the case. Was Boethius perhaps not a Catholic at all, and the theological tractates are incorrectly ascribed to him? We have the statement of Boethius' contemporary Cassiodorus that these tractates are from the hand of Boethius. Any solution to this puzzle can be at best conjectural. H. M. Barrett, in *Boethius, Some Aspects of His Times and Work* (Cambridge, 1940), gives a good sampling of proposed solutions and offers one of her own. Hers appears to be no more cogent than those she sets aside. She argues that Boethius had devoted his life to translating Plato and Aristotle into Latin and that this, by his own word, constituted his overriding interest. (In the *De syllogismo hypothetico*, PL, 64, 831A, he refers to his titanic effort as *summum vitae solamen*, the greatest consolation of his life.) It is not surprising, therefore, the argument continues, that in his extremity it would be to philosophy, to Plato and Aristotle, that Boethius would turn. Without any intention of offering a solution of our own, we might note that we are far from convinced by Barrett's. Whatever the explanation of this enigma, however, its very existence underlines the fact that a distinction between reasoning which depends upon faith and reasoning without such dependence is unquestionably present in the work of Boethius.

The *Consolation* is so purely philosophical that at one time scholars doubted that it could have been written by a Christian in the circumstances in which the text and tradition say it was written. Some of his other works, however, are clearly attempts by a believer to make intelligible in the light of truths taken from philosophy central objects of Christian faith. Thus, there is *prima facie* evidence that Boethius recognized a distinction between what is held by reason and what is held by faith. Moreover, the theological tractates provide overt statements about the relationship between these two areas. We intend to examine in turn the tractates and the *Consolation* in order to express as explicitly as possible the views of Boethius on the relationship between faith and reason.

The Theological Tractates. The theological tractates are five in number and seem generally to meet the description of Cassiodorus: "He wrote a book on the Holy Trinity, certain dogmatic treatises [*capita*] and a book in refutation of Nestorius." The *Quomodo trinitas unus deus et non tres dii* (or, more simply, *On the Trinity*); its apparent sequel, *Utrum pater et filius et spiritus sanctus de divinitate substantialiter praedicentur* (Are Father, Son, and Holy Spirit Predicated Substantially or Essentially of the Divine Nature?) and the *Contra Eutychen et Nestorium* (or, more simply, *On the Two Natures*) are mentioned by name; there

is no doubt that the *How Substances Are Good Insofar As They Are* (called the *De hebdomadibus*) is by Boethius. There is still doubt as to the authenticity of *On the Catholic Faith*. Our brief discussion will rely only on the four tractates of uncontested authenticity.

What is Boethius attempting to do in these tractates? Their subject matters are, first, the doctrine of the Trinity, second, a discussion of the Incarnate Word, and, finally, a treatment of the proposition that whatever is is good precisely insofar as it is. St. Thomas Aquinas, in the prologue to his exposition of the *On the Trinity* of Boethius, attributes an order to these tractates not unlike the order of his own *Summa theologiae*. First of all, Boethius is concerned with the one nature of God and the three Divine Persons: this is accomplished in *On the Trinity*. In *Utrum pater* Boethius "treats of the mode of predication we use in the distinction of the Persons and the unity of the essence. Secondly, in the *De hebdomadibus* St. Thomas sees Boethius treating of "the procession of created goods from the good God." The third division of the tractates has to do with the reparation of creatures through Christ. The faith taught by Christ is presented in *On the Catholic Faith*, and the way in which the human and the divine nature are united in the person of Christ is discussed in the work directed against Eutychus and Nestorius. Whether Boethius intended this order is irrelevant to our ability to see that the tractates do so arrange themselves. It is important to notice, moreover, that St. Thomas regards these tractates as theological.

How does Boethius go about the discussion of the tenets of the Christian faith? In the dedication of *On the Trinity* to his father-in-law, Symmachus, Boethius says, "You must however examine whether the seeds sown in my mind by St. Augustine's writings have borne fruit." The reference of course is to Augustine's work on the Trinity, but St. Thomas sees a methodological import in this reference. "There are two ways to discuss the Trinity, as Augustine says in *De trinitate*, 1,2, namely by appeal to authorities or through argumentations [*rationes*], both of which Augustine used, as he himself pointed out. Some of the holy Fathers, like Ambrose and Hilary, pursued the one only, namely appeal to authorities; Boethius chooses to proceed according to the other manner, namely argumentations, presupposing what has been set forth by others by means of authority." This is not to say that Boethius does not accept the fact of the Trinity on the authority of faith as something taught by Scripture, interpreted by the Church, and expounded by tradition and the Fathers. In the first chapter the point is clearly made that it is a matter of Christian belief that the Father is God, the Son is God, and the Holy Spirit is God, but that there are not thereby three gods, but one only. Boethius does not proceed by showing that this

doctrine is contained in Scripture and has been taught by the Church or by collecting what others have said about this belief. Instead, he wants to show the intelligibility of this accepted belief by appeal to argumentation. From what then will he argue? "So I purposely use brevity and wrap up the ideas I draw from the profound inquiries of philosophy in new and unaccustomed words which speak only to you and to myself (*Proemium*) Boethius appeals to philosophical truth to explain the unity of the divine nature and the Trinity of Persons. He does not intend these arguments to lead to the conclusion that there must be a Trinity of Divine Persons: this is ever assumed as a belief. Nor will his arguments eliminate the necessity of belief in the Trinity. At the end of the tractate he writes: "If with God's help I have furnished some support in argument to an article which stands by itself on the firm foundation of faith, I shall render joyous praise for the finished work to him from whom the invitation comes. But if human nature has failed to reach beyond its limits, whatever is lost through my infirmity must be made good by my intention." Boethius' method amounts to an effort to speak in a manner intelligible to one trained in philosophy of those things which every Christian firmly believes. The article of faith is not held more firmly because of the arguments given, yet Boethius sees the attempt to "conjoin" faith and reason as something incumbent on himself and others. "If I am right and speak in accordance with the faith, I pray you to confirm me," he writes to the deacon John at the end of the second tractate. "But if you are in any point of another opinion, examine carefully what I have said, and if possible, join faith and reason [*et fidem si potent rationemque coniunge*]."

The *rationes* of Boethius in these tractates are undertaken with a view toward supporting belief; in this they differ from the efforts of philosophers. The points he considers would not even be discussed apart from divine faith, and the arguments adduced do not so ground the truths in question that faith becomes unnecessary to hold them as certainly true. When Boethius remarks that he is borrowing from the inquiries of philosophers, we must not understand him to mean that his task consists simply of the application of ready-made philosophical views. Many of the philosophical points he makes appear to be original contributions. An indication of the philosophy which enters into the tractates may be had by examining somewhat closely *On the Trinity*.

In the first chapter of the tractate Boethius states that it is a matter of Christian faith that the Father is God, the Son is God, and the Holy Spirit is God and that nevertheless there are not three gods but one only. This dogma is simply asserted as a proposition of Christian belief; it is no part of Boethius' task to

establish that it is contained in Scripture. Given this revealed truth, which is accepted on the authority of God, a man who is trained in philosophy will reflect on it in such a way that he will bring it into juxtaposition with naturally known truths. The term of such reflection will not be a knowledge of the Trinity of Persons which is independent of faith. Faith in the doctrine is the starting point of the tractate, and, at its end, it is by faith alone that one accepts the Trinity as a truth.

Many of the naturally known truths which Boethius brings to bear on the doctrine of the Trinity have an Aristotelian origin. For example, in the first chapter, having noted that Catholics maintain that the unity of three Persons in the Trinity involves an absence of difference, Boethius undertakes an analysis of three kinds of difference whose immediate source is probably Porphyry but which derive ultimately from Aristotle. The denial of difference in the Persons of the Trinity is ambiguous until we have examined the kinds of difference and seen that none of them is applicable to the Divine Persons. Things differ generically, specifically, and numerically; similarly, things are generically, specifically, or numerically the same. Since sameness and difference are correlatives, Boethius can proceed by analyzing these types of sameness. Things are generically the same which share a common form which admits of further formal differentiation. For example, a man and a horse are generically the same with respect to animality. Things are specifically the same which share a common form which is not susceptible of further formal differentiation. For example, Cato and Cicero share the common form humanity. Things are numerically the same which differ only in name. For example, Tully and Cicero are but one person. Individuals of the same species differ because of their accidents.

Before applying these distinctions to the dogma of the Trinity, Boethius begins his second chapter by recalling Aristotle's division of theoretical philosophy into physics, mathematics, and theology. We will return to this subject later. All we need note now is the characterization of divine things, the objects of theology, as things which are free of matter and motion. Therefore, in treating them we must relinquish any appeal to the imagination. Material things are compounds of matter and form which owe their being principally to their form. That which is not pure form is not identical with its essence (a man is not humanity), but that which is form alone is identical with its essence. God, being pure form, is his own essence, and specific and generic differences cannot apply to God. In composed things we must trace their possession of accidents, not to their form, but to their matter or substratum. Thus, while it may be true to say that a man

is white, it is not humanity that is white. Therefore, to be white is accidental to man and inheres in him because of the subject of the form and not because of the form itself. God, since he is pure form and without subject or substratum, will not be the subject of any accidents. But numerical difference has been said to arise from accidents. Therefore, there can be no numerical difference in God.

God is completely one because no difference or plurality of the admitted kinds is applicable to him. Nevertheless, Boethius observes in chapter three, when we say the Father is God, the Son is God, and the Holy Spirit is God, we use the term "God" three times. Since three is a number, this seems to predicate numerical difference of God, whose nature is supposed not to permit numerical diversity. In response to this difficulty Boethius distinguishes two kinds of number. They are exemplified by the abstract and concrete terms "unity" and "one." A thing is one; unity is that whereby the oneness of the thing is signified. So too with "duality" and "two." Now, in speaking of one and the same thing we may say of it that it is one coat, one garment, and one vestment. This verbal repetition does not multiply the thing we are talking about. Neither does the repetition of "God" in the statement that the Father is God, the Son is God, and the Holy Spirit is God mean that we are enumerating three Gods.

The point Boethius has tried to make is that the Father, Son, and Holy Spirit must be the same God because none of the modes of difference is applicable to them. Nevertheless, the Father is not the Son, nor the Son the Father, nor is either or both the Holy Spirit. Belief in the unity of the divine nature does not, therefore, exclude the difference of Persons, and where there is difference there is number. But the only source of numerical difference mentioned so far is that which follows on the possession of accidents, and God cannot have any accidents since there is no subject or substratum of the divine form. Boethius will return to this difficulty, but first he wants to discuss the manner in which predicates are applied to God.

In chapter four Boethius has recourse to the Aristotelian doctrine of categories, the ten categories which can be universally predicated of things. As predicated, some of the categories are substantial predicates, namely, substance, quantity, and quality, while the rest are accidental predicates. Boethius states that none of these categories can mean the same thing as predicated of God and creature. Thus, while "God" predicated of God would seem to denote a substance, Boethius suggests that we think of it as a supersubstantial predicate. Likewise, when we say that God is just or great, these predicates must be taken to signify supersubstantial quality and quantity since we do not mean to suggest any

composition of the divine substance or any accidental attribute. God is justice; God is greatness. Boethius goes on to discuss the rest of the categories with a view to denying that any of them has application to God. He tentatively concludes that substance is the only category that applies to God, although this must not be taken to mean that he is a subject. That is, again, the term "substance" does not mean the same thing as predicated of God and creature.

In running through the categories in chapter five Boethius omits any discussion of relation; he turns to this category in his sixth chapter, indicating that this has been his goal all along. Relative terms, it may be said, do not alter the substance to which they are applied. For example, a man is called a master because of his relation to a servant. If the servant dies or leaves his employ, the man ceases to be a master, but this does not alter his substance in any way. From this observation Boethius wants to conclude that the category of relation does not increase, decrease, or in any way alter the substance to which it is applied, and on this basis he can say that if Father, Son, and Holy Spirit relate to the divine nature as predicates of relation, they will not introduce any difference into the divine nature itself, although they indicate a difference between the Persons in that nature.

Boethius' general conclusion is that the category of substance preserves the unity of the divine nature and the category of relation differentiates the Persons without introducing difference into the divine nature as such.

This glance at *On the Trinity* gives an indication of the way in which Boethius employs philosophy in meditating on the truths of faith. We have stressed his use of philosophical doctrines already at hand. Boethius made any number of philosophical contributions himself; however, his definitions may have the greatest influence, especially those he gave of eternity and person. In the third chapter of his work on Nestorius and Eutychus he defines person as *naturae rationabilis individua substantia* (an individual substance of a rational nature). With that definition in hand he was able to refute the two heresies. The tractates generally, along with Augustine's works, figure in all subsequent theological discussion on the Trinity and Incarnation.

The theological tractates of Boethius reveal a use of reason and a reliance on philosophy in discussions of doctrines of faith which justify calling Boethius the first Scholastic. Let us turn now to the work which, as we have indicated, is almost disturbingly restricted to the philosophical level.

The Consolation of Philosophy. This work is divided into five books, in each of which a prose section alternates with a verse section. This literary form can be traced back through Martianus Capella (who wrote a work on the liberal arts in this form) to Varro and on to a Greek origin in the Menippean Satire. (See Barrett, p. 76.) Quite apart from its content, on which we shall concentrate, the *Consolation* enjoyed an almost unparalleled fame during the Middle Ages as a work of art. The meters of its verse are varied and the result highly esteemed; the style of its prose passages is a thing of beauty. One is reminded of the *Phaedo* of Plato, but with this overwhelming difference. Socrates did not compose his immortal epitaph; Plato did -- and in retrospect. The *Consolation*, on the other hand, must have been composed by the victim in his cell. This increases the enigma of Boethius. That a man, particularly a man of Boethius' talent and background, should have the thoughts expressed in the *Consolation* is understandable enough; that he might write them down does not unduly strain the imagination; but that he should cast them into the exacting literary form he did is a severe test of our credulity. Nevertheless, there seem to be no grounds for skepticism about the facts.

The central question to which the *Consolation* addresses itself is this: What rational explanation can be found for the fact that the innocent suffer while the wicked not only go unpunished but prosper? This seemingly irrational state of affairs must be examined to see if it is not, after all, reasonable and tolerable.

In the opening poem of book one Boethius laments his outcast state. In the prose section following he describes the entry into his cell of a woman, tall, majestic, her eyes flashing and her manner authoritative. She is Dame Philosophy and she grandly dismisses the poetical muses who have been attempting to give solace to a man brought up by Eleatic and Academic studies. The muses can only increase his sorrow and self-pity. "'But it is rather time,' saith she, 'to apply remedies than to make complaints.'" (I, pr. 2) She reminds Boethius that he should know this, since he has spent much time under her tutelage. Boethius' spirits begin to rise slightly when he is reminded that Philosophy did not abandon Socrates, Anaxagoras, and Zeno in their hour of need, and no more will she abandon him. Encouraged, Boethius responds with a lengthy account of the evils that have befallen him despite his many contributions to the public weal and asks Dame Philosophy why the sovereign harmony which is apparent in the cosmos is so conspicuously and sadly absent from the affairs of men (pr. 4). Dame Philosophy is distressed to find that Boethius has sunk so low, and she undertakes a gradual process of consolation.

The therapy begins with a number of questions which will enable her to ascertain the present condition of Boethius. Boethius is asked if he would say that the world is merely the arena of chance and caprice or that it is ordered and directed; he replies that it is governed by reason. The world is the handiwork of God who has fashioned it and now directs and governs it. What then is man? Boethius knows that he is a rational animal, but that is the extent of his answer. Philosophy remarks that he is in worse straits than she had thought. Confused about the end of things, Boethius has become so forgetful of himself that he thinks the prosperity of the wicked a good and the misfortune of the virtuous an evil. "But thanks be to the author of thy health, that nature hath not altogether forsaken thee. We have the greatest nourisher of thy health, the true opinion of the government of the world, in that thou believest that it is not subject to the events of chance, but to divine reason. Wherefore, fear nothing; out of this little sparkle will be enkindled thy vital heat." (pr. 6) Nonetheless, given the depths of his depression, the first remedies will not be the strongest.

In book two Philosophy uses the "sweetness of Rhetoric's persuasions" to prepare Boethius for more solid consolation. First, they must examine the nature of fortune or luck, a natural topic since Boethius considers his present plight to be a misfortune and professes surprise at what has befallen him. Philosophy assures him that fortune has not changed but with consistent inconsistency now takes away without cause what was bestowed without cause. Whether good or bad, fortune is beyond man's control and comes to him from outside. Boethius' difficulty is that he does not see that his prior state, when he was the recipient of the goods of fortune, was just as irrational as his present unfortunate condition. In these restless times Boethius should have been impressed by the inconstancy of luck and learned thereby to seek happiness within, in an arena where his own efforts can play an essential role. "If blessedness be the chiefest good of nature endowed with reason, and that is not the chiefest good which may by any means be taken away, because that which cannot be taken away is better, it is manifest that the instability of fortune cannot aspire to the obtaining of blessedness." (pr. 4) Fortune is more profitable to man when she takes away what has been given because then a man must ask what true happiness is.

Again and again in the sequel he returns to the idea that happiness does not simply happen to a man. The third book makes the point in great detail. Happiness cannot be a matter of riches or honor or worldly power. Nor can carnal pleasure of whatever sort make a man happy. The true good, that in which human happiness lies, cannot be found in terrestrial things. Indeed, when

we seek the marks of the good we find that they must all be found in one substance and that this substance must exist outside the material world. God is the sovereign good, and he is also true human happiness. All beings aspire to rejoin their source; since all things have the same source, God is the universal or common end of everything in the universe. Boethius is urged to turn his eyes from earth to heaven if he would find consolation in his darkest hour.

This sunny view becomes clouded as book four begins. The idea of a benevolent God who is the source of the universe and who continues to direct each thing in it seems to be contradicted by the existence of evil. Dame Philosophy must be able to solve the problem of evil, or what has been said up to now is as nothing. She bends her best efforts to the task. If God is the benign governor of the universe, it would seem to follow that the good are never without reward and that the evil never go unpunished. To see that this is actually the case, we must acquire a perspective which will reveal the prosperity of the wicked as only apparent and the suffering of the virtuous as something less than unhappiness. Dame Philosophy urges Boethius to the heights where he may gain the proper perspective. Boethius is dubious but willing. Philosophy argues that it can be shown that if the virtuous are strong, the bad must be weak. He is strong who is able to attain the end he seeks, and the end sought by all men is nothing else than true happiness. But who can attain this good if not the virtuous, and who fail to attain it if not the vicious? Therefore, good men attain the object of their desires and evil men do not. The change of perspective Philosophy is trying to induce follows on the judgments made in the second and third books. The judgment that happiness cannot be constituted by honor, fame, riches, bodily pleasures, and so forth must be stringently applied; one must see that though wicked men enjoy any or all of these things, they are not thereby happy. The wicked want happiness yet are powerless to attain it since they are committed to pseudo-goods. There is an echo of Plato and Aristotle in this section. Boethius realizes that the wicked are not and cannot be happy. How silly then to envy them. What they require is our pity.

That Boethius is able to acquiesce to all these conclusions is a sign to Dame Philosophy that his sanity is returning. She urges him to recognize that whatever happens happens because God wills it, and, consequently, everything is ultimately ordered to the good. Both good fortune and bad fortune play an edifying role if we have the eye to see it. In a profound sense there is no misfortune for the virtuous who, similarly, do not view good fortune as a true good.

The final book of the *Consolation* takes up the question of the compatibility of providence and human freedom. If God directs all things, if his providence encompasses everything in the universe, it must direct the acts of men as well. But are not human acts precisely those which cannot be directed from without but have their source within man? We seem forced to say that free human acts either escape the providence of God or, being included in it, are not what they appear, namely, free. Dame Philosophy will try to show the compatibility of providence and free will by beginning with a discussion of chance events. Aristotle's definition of the chance event is accepted. Aristotle had taught that when a determined cause, called such because it is ordered to producing a determinate effect, brings about as well or instead an unintended result, that result is said merely to happen, to be a chance effect. If it is referred back to the cause, the cause is not a determinate explanation of it. If I dig for water and strike oil, the discovery of oil is the result of my digging for water, but it is unintended and accidental to my intention. Such accidental events may be unintended and unforeseen by me, but this does not prevent their being foreseen and intended by God. In somewhat the same way, Dame Philosophy suggests, we can find a compatibility between our undeniable certitude that we are free agents and the fact that our free acts come within the scope of divine providence.

As the *Consolation* reaches its term, Boethius is a changed man. At the outset he was a sobbing, self-pitying, broken man who was convinced that everything had turned against him, that the world, which had hitherto been a fairly reasonable place, had become suddenly and inexplicably absurd. Dame Philosophy has led him gradually from the view that external events and what other men can confer constitute happiness. Good luck is as absurd, finally, as bad luck. Happiness is not thrust upon us; it is something we must earn. We learn from considering this world that our happiness consists in something beyond this world. A reversal of fortune can be a stroke of good luck if we take its occasion to reassess the nature of luck and reflect that the world is a whole whose order demands a governor. Our sense of values must alter when we contemplate God's governance of the world. The wicked are not happy; the unlucky virtuous man is not less virtuous, less truly happy. We can come to see that in this world all things work together for good, though it is not our part to grasp this truth in detail. Thus, Boethius, unjustly accused and condemned to death, draws consolation from these philosophical considerations and is able to face death with equanimity.

As befits philosophy, there is no discussion in the *Consolation* of the punishment of the souls of the wicked after death (IV, pr. 4). The immortality of the soul is said to be demonstrable (II, pr. 4). Let us conclude by examining the way in which the *Consolation* treats God, its theology, to determine if it is an example of a theology different from that exhibited in the tractates.

The most striking thing about the *Consolation*, when compared with the tractates, is the absence of any concern with the Trinity. God is often referred to as Father in the *Consolation*, but the word seems to function as the name of a nature, not of a person; moreover, it is Plato who suggests the appellation. What attitude is expressed in the *Consolation* with respect to the attainment of philosophical knowledge of God's existence? Some have suggested that Boethius has no intention of offering a proof for the existence of God since his existence is assumed from the very beginning of the work. It is true that God's existence is taken for granted from the very outset, but Boethius also argues to that fact on several occasions in the *Consolation*. In prose ten, book three, a proof is found which has been likened to the later proof of St. Anselm.

In prose twelve of the same book another argument is presented. Boethius had said in *Quomodo substantiae*, with respect to the First Good, that his "being is admitted by the universal consensus of learned and unlearned opinion and can be deduced [*cognosci potest*] from the religious beliefs of savage races." In the *Consolation* he gives a learned basis for the assertion that God exists: "This world could never have been compacted of so many divers and contrary parts unless there were one that doth unite these so different things; and this disagreeing diversity of natures being united would separate and divide this concord unless there were one that holdeth together what he united. Neither would the course of nature continue so certain, nor would the different parts hold so well-ordered motions in due places, times, causality, spaces, and qualities unless there were one who, himself remaining quiet, disposeth and ordereth this variety of motions. This, whatsoever it be, by which things created continue and are moved, I call God, a name which all men use." (III, pr. 12)

It seems legitimate to conclude that Boethius recognizes in the *Consolation* that God's existence can be known from reason alone. Although he was a Christian, the *Consolation* seems a conscious attempt to remain on the level of natural reason, unaided by faith, in order to show that a rational preparation for faith is possible. There is a God who governs all things, and it is in him that perfect happiness is to be found. Christian faith teaches us far more of God than

philosophy can and elevates us to the level of friendship with God. Nevertheless, one can find the beginnings of consolation in philosophy.

C. Division of Philosophy

Having seen Boethius' de facto recognition of the autonomy of philosophical reasoning, let us turn now to his remarks on the nature and division of philosophy. While these remarks are fairly schematic and derivative, they are important because they were the vehicles whereby the Aristotelian division of philosophy was made known to later thinkers to whom the treatises of Aristotle containing the doctrine which makes the division meaningful were unknown. This fact led to some rather curious commentaries on the texts of Boethius which we want now to examine. However, because of the influence of Boethius the way had been more or less paved for the Aristotelian corpus as it became known at the end of the twelfth century.

In his first commentary on Porphyry, Boethius must ask what philosophy is and what its main divisions are to explain the role the *Isagoge* was intended to perform: "First of all we must ask what philosophy itself is. For philosophy is the love, pursuit of, and, in a certain way, friendship with wisdom." (PL, 64,10D) This love of wisdom is described as an illumination of the intelligence by pure wisdom itself and is, therefore, the study of divinity. Truth in speculation is caused by this illumination as well as by rectitude of action: "For philosophy is a genus having two species, one which is called theoretical, the other practical, that is, speculative and active." (11A) Each of the species of philosophy is further subdivided into three parts. In the second chapter of his *De trinitate* Boethius had written:

There are three parts of speculative philosophy. Natural philosophy considers things in motion which are not abstract; it considers the forms of bodies together with their matter since such forms cannot be actually separated. These bodies are in motion (for example, earth is borne downward, fire upward) and a form conjoined to matter is in motion. Mathematics considers inabstract things without motion, for it speculates on the forms of bodies without the matter and therefore without motion. These forms, since they are in matter, cannot be separated from it. Theology is concerned with abstract things separable from motion since the substance of God lacks both matter and motion.

Thus, in this text Boethius seems to be giving a fairly straightforward statement of the Aristotelian position according to which the division of the speculative

sciences does not argue for three distinct realms of entities. However, the approach of the commentary on Porphyry links the three theoretical sciences to three types of things: "There will be just as many species of speculative science as there are things worthy of speculation." (PL, 64, 11B) He names these types of things intellectibles, intelligibles, and naturals. Intellectibles are defined as things which always subsist one and the same in their proper divinity and are grasped, not by the senses, but by intellect alone. Examples are God and the soul. Intelligibles are causes of sublunary things, and soul is mentioned here too because, due to its contact with body, it degenerates from the state of being an intellectible and becomes an intelligible. Beatitude will consist in turning toward intellectibles. A third branch of theoretical science is concerned with bodies and their properties and can be called physiology. It is noteworthy that Boethius, while he associates intellectibles with theology and bodies with physics, does not align intelligibles with mathematics.

The passage in the commentary on Porphyry suggests a Neoplatonic declension toward matter, and we seem faced with a real hierarchy. This impression is strengthened by a passage in *On Arithmetic*, one quoted, incidentally, by Scotus Erigena (PL, 122,498C). Here we read that qualities, quantities, forms, magnitudes, places, times, and such are, in their proper nature, incorporeal, immutable substances; they are changed, however, by their participation in body. (PL, 63,1079D - 1081A)

Boethius has presented the Aristotelian division of theoretical philosophy in the *De trinitate* in terms of abstraction or nonabstraction from matter in being and in thought. Elsewhere, however, he speaks of a hierarchy of entities in terms of degeneration from true being, a falling off into matter, which is redolent of Neoplatonism. Which of these positions Boethius himself held has been the object of lengthy discussion. We will be able to propose an answer against the background of Boethius' treatment of the problem of universals.

D. The Status of Universals

Pascal once mused that the whole history of the world would have been different if Cleopatra's nose had been a bit longer. It is far less remote to say that much of the philosophy of the Early Middle Ages would have been utterly different if it had not been for a brief remark of Porphyry in his *Isagoge*, that is, introduction, to the *Categories* of Aristotle. In this work Porphyry proposes to discuss the notions prerequisite to an understanding of Aristotle's work on the ten genera of being. Porphyry mentions the five predicables: genus, species,

difference, property, and accident. Before getting down to them, however, he sets aside the problem posed by two widely different opinions regarding the status of the predicables, the opinions of Plato and Aristotle: "For the present I shall not discuss the question whether genera and species really exist or are bare notions only; and if they exist, whether they are corporeal or incorporeal beings; whether they are separate from sensible things or exist in them and in relation to them. Such matters are of the highest difficulty and demand a higher kind of inquiry." What could be more challenging to a reader than to be told that there is a profound and difficult problem, namely, such and such, which will not be treated in the present work? Boethius rose to the bait twice in his commentaries on Porphyry, and, because of the influence of Boethius, the problem was transmitted to the Christian schools, where many were to follow his example and propose solutions to the problem Porphyry considered too difficult to discuss in an introductory work.

The problem of universals, as it is stated by Porphyry, comprises three questions: Are genera and species subsistent entities, and, if so, are they separate from the things of sense experience or is the universal somehow present in sensible singulars? What explains Porphyry's reluctance (and distinguishes Boethius' treatment from most others until the end of the twelfth century) is the recognition that the quarrel to which he alludes is as much or more a metaphysical than a logical one. Boethius was acquainted with the works of Plato and Aristotle, but for centuries during which the problem of universals was discussed all the Aristotle known to the disputants was a few logical works translated by Boethius. Of Plato, all that was directly known was the *Timaeus* in the translation of Chalcidius. (Of course, much "Platonism" was known.) While the various theories on the status of universals, which grew ever more complex, were presented in a time when the historical background in Greek thought was but dimly perceived, they cannot be viewed as a mere waste of time. The problem involved logic, psychology, and metaphysics; moreover, its association with the divine Ideas and creation makes proposed solutions important.

Boethius' first commentary on the *Isagoge* opens as a dialogue, but there is less and less concession to that literary form as the commentary proceeds; the second commentary is a straightforward one by previous design. We shall concern ourselves with the second commentary. (PL, 64,82A - 86A) The discussion is organized as follows: having noted Porphyry's reluctance to treat the problem of universals, Boethius first indicates the triple question involved. Next, he undertakes the solution of the three difficulties, first by noting the ambiguity of the question and then by presenting his solution. In following his

division we shall make some mention of Boethius' first commentary and rely as well on other writings of his. Finally, because of his closing statement, we will seek elsewhere indications of disagreement with the Aristotelian solution Boethius here sets forth.

The Questions. In dismissing the problem of universals Porphyry has indicated that it involves three questions. In his first commentary Boethius is content with a clarification of these three questions; in the second, this clarification is prefatory to a solution. Three activities of the mind (*animus*) are mentioned. Mind conceives with the understanding or intellect (*intellectus*), describes to itself what has been so conceived with the reason (*ratio*), or depicts for itself by empty imagination (*imaginatio*) what is not. To which of these activities of mind should genera and species be ascribed? Are they due to true understanding or to the empty play of imagination? In this fashion Boethius sets up the first Porphyrian problem: Do genera and species exist or are they bare notions only, that is, are they had by true understanding or made by mendacious imagination? If we decide that they are objects of true understanding, it remains to determine the nature of genus. Whatever is is either corporeal or incorporeal: if genera exist, they must fall under one of these headings. And this is the second question.

The third question, arising on the assumption that genera exist and are incorporeal, is this: Do genera subsist only in bodies or in themselves? There are, Boethius points out, two kinds of incorporeal things, namely, those which subsist separately from bodies -- for example, God, mind (*mens*), and soul (*anima*) -- and those which cannot exist separately -- for example, line, surface, particular qualities. The latter are incorporeal in the sense that they are not tridimensionally extended in space.

The Solution. If these are the three questions to be answered, there remain certain ambiguities which must be dispelled before a solution can be proposed. By ambiguity Boethius here means dichotomy or antinomy, for he examines the apparent impossibility of either the existence or truth of genera and species. Genera and species either subsist and exist, or they are products of understanding (*intellectus*) and thought (*cogitatio*) alone. Arguments are adduced to show that genera and species cannot exist and that they cannot be true notions.

To show that it is impossible for genera and species to exist, Boethius argues that if genus, for example, is common, it cannot be one, and if it is one, it

cannot be common. Whatever is common cannot be one. But the genus is in many species, and wholly not partially in each of them. Therefore, the genus cannot be one. But if it is not one, it simply cannot exist, for whatever is, is one. Moreover, if the genus is not numerically one, but multiple, we shall always have to seek its genus, and we would thereby be involved in an infinite regress.

If, to avoid this, we say that the genus is numerically one, we compound the difficulty, for how then could it be common? Boethius enumerates three modes of community: (1) If a single thing is common, it is common by parts and not as a whole. Thus, a common dish at the table is common to all the diners in that each will receive part and not in that each will receive the whole dish. (2) Or it is common successively; for example, several men may share the same automobile, each having the use of the whole car, but at different times. (3) Or a thing can be simultaneously and totally common, as a film is common to everyone seated in the theatre -- but of course it is not substantially common to them. None of these ways in which something numerically one is common to many can explain the community of genus, for the latter must be wholly, simultaneously, and substantially common to individuals. Such a mode of community seems impossible. The genus cannot be one because it is common, and its community prevents our ever arriving at a supreme genus; if taken to be one, the genus cannot be common. Either way, then, it seems that the genus cannot be said to exist.

Turning now to the other side of the original dichotomy, Boethius examines the possibility that genera and species do not exist but are merely products of thought. This too involves an ambiguity or dichotomy. Whatever is in a concept (*intellectus*) refers to a subject thing and either reflects the way the subject itself is constituted or the way in which it is not constituted. If genera and species are *intellectus* of the subject as it exists, they cannot be simply in the mind but are truly in things as well. In other words, they would exist, and we are thus led back to the previous consideration. The alternative, then, is to say that the *intellectus* of the genus is not taken from the thing as it exists, that it is a vain idea. This cannot be the solution, for it consists in understanding the thing otherwise than as it exists.

The upshot of these analyses is that genus and species neither exist nor, when thought, are true ideas, conclusions which, as Boethius points out, are calculated to disturb one about to investigate the predicables. If he cannot solve these problems, whose difficulty Boethius has just heightened remarkably, he will be in the position of examining what may neither exist nor be true. The

following schema summarizes Boethius' presentation of the "ambiguities" which attend the Porphyrian problem:

- Genera and Species
 - either exist and subsist, which is impossible, since they are
 - either one and thus not common
 - or many and involve an infinite regress
 - or they are formed by thought alone, which is impossible, for they are
 - either ideas of things as they are and then themselves exist
 - or ideas of things as they are not and thus are false.

Boethius leads us out of the dilemma by denying the exhaustiveness of the division. Relying on Alexander but using primary Aristotelian doctrine, Boethius argues that not every idea which is not of a subject as it exists is false. The truth of this is established by noting the difference between the mind's act of understanding and its act of composition. Only the latter can properly be said to involve true or false opinion. Boethius' example is the composition of man and horse in the notion of centaur. (Of course, false opinion is had only in the assertion that centaurs really exist.) Mental acts of division and abstraction are productive of ideas not constituted as the thing is, but such ideas are not thereby false. Thus, the mind can consider line apart from sensible bodies, although the line could not actually subsist in this way. This example is a familiar one in Aristotle. (See *Physics*, II, 2.) The line, then, is an incorporeal thing which the mind can separate and distinguish from the confused thing given to the senses. Thus genera and species are found either in incorporeal or in corporeal things; in the latter case the mind abstracts "the nature of incorporeals from bodies, and beholds it alone and pure as the form itself is in itself." (85A)

Genera and species are gathered from the individuals in which they are, not by a mental composition, but by abstractions and divisions. Genera and species are in the individuals, that is, and become universal insofar as they are thought: "Species must be seen to be nothing other than the thought collected from the substantial likeness of individuals unlike in number, and genus the thought collected from the similarity of species." (85C) In things this similarity is sensible; in universals it is intelligible. Thus, genera and species subsist in individuals: what becomes universal when it is thought subsists only in sensibles. We have here the solution of Porphyry's problem. Boethius has indicated in which sense genus and species subsist (in sensibles, not as

universals), that although incorporeal in themselves, they are found in sensible bodies, and that they are not false, though they do not reflect things as they exist.

The solution proposed by Boethius is intended to be an Aristotelian one. From this point of view the likening of line and man on the basis of incorporeality seems to pose a great difficulty. In his first commentary, while discussing the first question, Boethius observed that man's mind understands things present to sense through sensible qualities and that concepts formed from these prepare a way toward understanding incorporeal things; thus, when I see singular men, I also know that I see them and that they are men. The species man, we are told, should not be called corporeal because it is grasped by the mind and not by the senses. "Incorporeal things are those which can be grasped by none of the senses, but what they are is made known solely by the consideration of the mind." Nevertheless, in pursuing the question whether the genus is corporeal or incorporeal, Boethius begins to speak of the corporeal genus. Substance, he notes, is a genus, and its species are corporeal and incorporeal. Since the genus is not identical with that which divides it into species, that is, the differences substance is neither corporeal nor incorporeal *qua* substance. "But some species are corporeal, others incorporeal. For if you place man under substance, you would introduce a corporeal species; if God, an incorporeal one."

The apparent contradiction involved in saying that genera and species are incorporeal and that some species are corporeal, when resolved, will resolve as well the difficulty inherent in likening line and man on the basis of incorporeality. Boethius himself asks how the incorporeal can be called corporeal. When one says the genus is incorporeal, he explains, the genus is not being considered insofar as it represents some nature, but insofar as it is a genus. Therefore, when substance is the genus, we do not consider it insofar as it is substance, but insofar as it has species under it. This surely distinguishes being predicable of many from the corporeal nature to which this relation attaches; the relation of predicability is not itself corporeal nor is the nature as it actually takes on this relation, that is, in the mind. This distinction should allay the reader's fear that Boethius, by likening line and man on the basis of incorporeality, means to suggest that Aristotle taught their definition would exhibit an equal freedom from sensible matter. What line and man have in common is that each involves considering apart from sensible things what cannot exist apart. As species, that is, given their condition in the mind and the relation of predicability attributed to them in that state, they can both be called

incorporeal. Nevertheless, the nature reflected by the intellectus will in one case be incorporeal (insensible) and in the other corporeal.

F. Plato or Aristotle?

By saying that he has presented an Aristotelian solution to the problem of universals, not because he agrees with it, but because the *Isagoge* is an introduction to an Aristotelian work (86A), Boethius leaves the impression that he himself may prefer Plato's position on the matter. And Plato's position, according to Boethius, is that "genera and species and the rest not only are understood as universals but also are and subsist without bodies." (86A) To settle this question, we are referred to texts in the *Consolation* and in *De trinitate*.

In the fifth book of the *Consolation* Boethius is concerned in a particular way with the relationship between God's providence and man's free will. Already in the third poem of this book a Platonic note has been struck, for it invokes the preexistence of the soul and knowledge as remembering.^[2] Indeed, earlier, having written in a poem (III, xi), "If the muse of Plato does not mislead, whatever we learn is a science forgotten that we but recall to memory," he goes on to say in prose twelve: "'But I passionately ascribe to the view of Plato,' I cried, 'for this is the second time you have recalled what my spirit had forgotten, first due to its contact with the body, then when I was crushed under the weight of woe.'"

Such remarks form the basis for judgments that Boethius is at heart a Platonist. Prose four of book five of the *Consolation* is most frequently cited as indicating that Boethius personally favored the Platonic solution to the problem of universals. Boethius is speaking of divine foreknowledge and our free acts. He points out that we ourselves foresee things which do not come about by necessity. For example, we watch an artisan at work and know that soon he will do such and such, although he is not compelled to do so. "There you have facts known in advance the realization of which is free. For, if present knowledge does not impose any character of necessity on events, foreknowledge of the future does not render future facts necessary." But is it not wrong to think one has certain knowledge of what will not come about necessarily? "If facts whose realization is uncertain are foreseen as certain, we are faced with the obscurity of conjecture and not the truth of science; for you believe that to think something to be other than it is is to fall short of the integrity of science. The cause of this error is that all one knows is thought to be known from the very nature and essence of the object, which is false. In fact every known object is

grasped not in terms of its own essence but in terms of the capacity of the knower."

He goes on to illustrate the different ways in which sense, imagination, reason, and intelligence know man: "The senses pronounce on the form constituted in a particular subject matter, whereas imagination judges the form without the matter. Reason goes beyond this and, by a universal examination, determines the species which is in the singulars. The eye of intelligence is at a yet higher level; it perceives, by the unique penetration of its proper activity, the simple form itself." Now in this cognitive hierarchy the upper stages comprise and go beyond the lower: "Reason, once it distinguishes the universal, no longer has need of sense or imagination to understand the objects of sense and imagination. Reason it is that gives the definition as its proper work: man is a two-footed animal endowed with reason. Once the general notion is had, no one is unaware that it is an object pertaining to sense and imagination, but reason examines it without the aid of sense and imagination." The point of this passage is that the existing man does not, as such, explain the different ways he is known by sense, imagination, and reason.

The passage just quoted, moreover, throws light on a point we discussed earlier and seems to argue for an abstractive view of knowledge while at the same time cautioning against taking knowledge as a mere passive reflection of reality. The next poem (v. 4) stresses this point, taking issue with the Stoics. Knowledge requires that the knower be agent as well as patient. "Here is a power far more efficacious than that which receives the imprint of matter." There must be a prior passion of our living body if knowledge is to take place, a passion which incites the first motions of the mind.⁽³⁾ Aroused by impinging colors or noises, the mind forms species intrinsic to itself which can then be applied to exterior things. The use of the participle *excitans* could seem to suggest something innate and dormant in the mind. This impression is strengthened by the next prose section (v. 5). Boethius (more accurately, Dame Philosophy) argues that if our mind has its own inner forms, although it requires the prior passion of the body, so much the more independent of body will be those minds which are not in bodies. The description of the coming into being of inner forms from a quiescent state suggests a Platonic view of human intellection.⁽⁴⁾ This and not the previous prose section could be cited as exhibiting a Platonic rather than an Aristotelian bent in Boethius.

This same prose section indicates that intelligence is not a human faculty. Reason is proper to man, and reason is concerned with the universal. Once

more we are reminded that reason comprises in itself the objects of sense and imagination. Then follows this passage, important for the problem of universals:

What would happen if sense and imagination would resist reason and deny the universal reason sees? What pertains to sense and imagination cannot have the status of universality; therefore, either the judgment of reason is true and nothing sensible exists, or since it knows that the majority of its notions depend on sense and imagination, it is the work of reason which is vain when it considers what is sensible and imaginable as universal. If reason reply that it considers the data of sense and imagination from a universal point of view but that these faculties cannot pretend to a universal knowledge since they cannot transcend corporeal forms, if it says that in knowledge it is necessary to prefer the most sure and advanced judgment -- given such a debate, would not we who enjoy both reason and sensing incline to the cause of reason?

It will be noticed that Boethius, while insisting on the *sui generis* activity of reason, always allows for the necessary precedence of sensation and imagination. Taken as such, this permits either the Platonic or Aristotelian theories, but in the *Consolation* abstraction does not loom as large as the view that forms, quiescent in mind, are awakened when the mind considers the data of sensation.

Turning now to the *De trinitate*, let us recall first that, in his *proemium* to the tractate, Boethius asks Symmachus to seek in the work the fruit of the seed sown in his mind by the doctrine of Augustine. In the second chapter, having distinguished the three kinds of theoretical sciences, Boethius goes on to distinguish God, who is pure form, from all other beings which are not pure forms but images. Nevertheless, everything is because of its form. "*Omne namque esse ex forma est.*" A statue is a statue because of its shape or form, not because it is bronze; bronze is bronze, not because of the earth which is its matter, but because of its form. Earth is not earth because of prime matter but due to the forms of weight and dryness. "Nothing is said to be because of its matter but because of its proper form." The divine substance is form without matter, one, its own essence:

Other things are not what they are, for each of them has its being from those things of which it is made, that is, from its parts; it is this and that, a compound of parts, but neither this nor that alone, as earthly man is made up of soul and body, he is soul and body, and neither soul nor body alone; therefore, he is not

identical to what he is. What is not this and that, but only this, truly is what it is and is best and most because dependent on nothing.

F. K. Rand, in his edition of Boethius, tells us that this passage shows that Boethius is definitely committed to Plato's position regarding universals. It is difficult to accept this without qualification. Does Boethius, by speaking of "earthly man," mean to imply that there is another man not composed of body and soul? A man subsisting separately from the singular men of our experience? Boethius does point out that humanity can appear to have properties which are really accidents of the man whose form humanity is and not those of humanity as such. Other forms, those which are without matter, "cannot be subjected to or be in matter, for they would then be images not forms. From these forms outside of matter those forms come which are in matter and body." Does this mean that the form, humanity, subsists separately from singular men? In a sense, yes; indeed, forms in matter are properly speaking not forms but images. "For the others which are in bodies we abusively call forms; in fact they are images. They are assimilated to those forms which are not constituted in matter." What the things of this world image, surely, are the divine Ideas. We have here, it would seem, the fruit of Augustine's seeds of doctrine a Platonism, perhaps, but again a highly modified one.

F. Conclusion

Boethius, even more than Augustine, is a bridge between the world of classical philosophy and the medieval world to come. Many centuries will intervene before we will encounter another figure in whose mind a thorough knowledge of Greek philosophy combines with theological interests and talents. It is a cause for lamentation that Boethius had hardly the time to begin the massive task of translation he had set himself, although we can only speculate on what the results of a complete knowledge of Aristotle and Plato would have meant in the immediately following centuries. Perhaps it is better to be grateful that Boethius did manage to translate some works of Aristotle, for, in periods when men had at least a fleeting leisure for such pursuits, these works provided a basis for speculation and generally interesting discussion. Moreover, something of Greek philosophy is passed on in the independent works of Boethius, and even when the context of those fragmentary retentions is unknown, some intellectual benefit was derived from attempting to grasp their meaning. In sum, the writings of Boethius may be said to be a reminder of a soon-to-be-lost philosophical greatness and the promise of a theological flowering to come many centuries later. Before that later renaissance could come, there were many

centuries during which the best that men of the West could do was to strive to preserve what had been handed down to them. Infrequently, but sometimes, a man arises who surmounts the restrictions of his time, but it will not be until the twelfth century that we encounter thinkers who approximate the stature of Boethius.

Bibliographical Note

The works of Boethius can be found in Migne PL, 63-64. See too *Boethius: The Theological Tractates . . . The Consolation of Philosophy*, trans. H. F. Stewart and E. K. Rand (Loeb Classical Library) (New York, 1918). H. M. Barrett, *Boethius, Some Aspects of His Time and Work* (Cambridge, 1940); H. R. Patch, *The Tradition of Boethius: A Study of His Importance in Medieval Culture* (New York, 1935). P. Courcelle, *Les lettres grecques en occident* (Paris, 1948), pp. 257-312; E. K. Rand, *Founders of the Middle Ages* (Cambridge, Mass., 1928); J. Collins, "Problems and Progress in the Reassessment of Boethius," *The Modern Schoolman* (1945), pp. 1-23. From among recent and important work on Boethius we can mention the following. P. Courcelle, "Étude critique sur les commentaires de Consolation de Boece, IX - XV siècles," *Archives d'Histoire Doctrinale et Littéraire de Moyen Age*, XII (1939); Gangolf Schrimpf, *Die Axiomenschrift des Boethius (De hebdomadibus) als Philosophisches Lehrbuch des Mittelalters* (Leiden, 1966); Siegfried Neumann, *Gegenstand und Methode* (Münster, 1965); Karl Durr, *The Propositional Logic of Boethius* (Amsterdam, 1951).

{1} Gilson, in *History of Christian Philosophy in the Middle Ages*, p. 102, finds quotation from Scripture (Wisdom 8:1), in book three, prose twelve.

{2} "Now beclouded by body, it (the soul) has not wholly forgotten its pristine state but keeps the memory of the whole, though it has lost the detail. He who seeks troth finds himself therefore in an intermediary state: he knows not and yet he is not wholly ignorant; he consults the whole of which he has retained the memory, by recalling what he saw above, so that it might be able to add what has been forgotten to what has been retained."

{3} "*Praecedat tamen excitans/ Ac vires animi movens/ Vivo in corpore passio.*" (11,30-33)

{4} "If in the perception of objects the organs of sense are struck by exterior impressions and the activity of spiritual energy is preceded by a physical sensation which provokes the action of intelligence and awakes in it the inner forms sleeping there, if, I say, in the perception of objects the mind is not informed by sensation but judged by its proper power, the data of sense, so much the more will beings free from all physical influence be independent of the external world in their judgments. . . ." (v. pr. 5)

Chapter V

Cassiodorus, Isadore, Bede

A contemporary of Boethius, as we have already noted, Cassiodorus Senator (c.480 - c.570) is sometimes thought to have been a student of Boethius as well. Like Boethius he was engaged in political affairs under the Goths, in the tradition of his family, but unlike Boethius he managed to survive his service. Various reasons are given for this, and it is not uncommon to accuse Cassiodorus of obsequiousness and opportunism, a charge which finds some foundation in his flattering appraisal of the Goths in his historical works. The importance of Cassiodorus for our purposes resides in the fact that he was the founder of a monastery at Vivarium, his family estate in Southern Italy, where the finest library in the West was collected. While he himself seems to have never become a monk, Cassiodorus was the patron of the monastery and lived in its neighborhood. For the monks he wrote a book called the *Institutiones*, the first part of which dealt with Scripture, the second with the liberal arts. In urging the monks to intellectual pursuits Cassiodorus was instrumental in making the monastery the repository of ancient culture during the ages when contact with the past might quite easily have been wholly lost. Indeed, the *Institutiones* of Cassiodorus begins a tradition of summarizing and epitomizing ancient wisdom. Of this work Cassiodorus said that he would not there command his own doctrine but that of the ancients. This heritage must be praised and taught, for it would be impious to shrug off what the ancients did by way of praise of God.

The second part of the *Institutiones*, which deals with the liberal arts, was particularly influential, often being copied separately. Although Cassiodorus, true to his promise, gives us very little in it that cannot be found in earlier writers, he passed on the divisions of philosophy, both the Stoic and the Aristotelian, the division of the liberal arts into the trivium and quadrivium, and had something to say under each heading which was of increasing interest when the original sources were lost from view.

Cassiodorus is insistent that the number of the liberal arts is seven, going so far as to adduce scriptural passages to support it.⁽¹⁾ If there are seven liberal arts, what is meant by "liberal" and what by "art"? When he says that he will first speak of grammar since it is the source and basis of liberal letters, Cassiodorus pauses to discuss the meaning of "liber." In Latin this term can mean either book or free, and Cassiodorus is concerned to explain this equivocation. Book is

signified by "liber" because in early times writing was done on bark freed from trees. Thus, "liberal" in the phrase "liberal arts" refers to the fact that books are involved in their pursuit. Cassiodorus thus does not attach the same significance to the term in this context as did the Greeks.

With respect to the etymology of "art" Cassiodorus suggests that the word has come from the fact that art binds and limits (*artet*) us with its rules, or it may come from the Greek term for excellence or skill (*arete*). From this passage, then, one might conclude that liberal arts are those skills or rules gathered in books. Whatever the case, there is a most interesting problem raised if not solved by Cassiodorus, namely, what is the relation between art and science? Are they the same or different; can we speak interchangeably of seven liberal arts and seven liberal sciences, or are some of the seven arts and some sciences? The question is raised first with regard to logic, "which some prefer to call a discipline and others an art, saying that when someone discourses in apodictic or true disputations it ought to be called a discipline, and when it is something likely and of opinion, it takes on the name art. Thus it has either name depending on the quality of its argumentation." (II.e, n. 17) He notes that Augustine speaks of grammar and rhetoric as disciplines (that is, sciences) as Varro had, and that Capella entitled his work (which Cassiodorus did not have an opportunity to see) *On the Seven Disciplines*. Discipline indicates that it can be learned, and something will be called such insofar as it attains to unchangeable things by the rule of truth. The difference between art and science, in short, is that science involves necessity while art does not. Insofar as some arguments are certain and some probable, logic can, on this basis, be sometimes called science, sometimes art. Cassiodorus returns to this point later, referring to Plato and Aristotle: "Between art and science Plato and Aristotle, esteemed masters of secular literature, intended this difference, namely, that art is concerned with the relations of contingent things, which can be otherwise than as they are, whereas discipline is concerned with things which cannot be otherwise." (II, 3, n. 20)

Whether this settles much is extremely doubtful. Given that art is concerned with the contingent and science with the necessary, the question remains whether we can call geometry, for example, a liberal "art." It would certainly not be said to concern itself with the contingent. With respect to logic itself, for which Cassiodorus elaborates the distinction, he can be said to have confused the logic of probable argumentation and a probable argument.

On the basis of this one sounding in search of a personal contribution, Cassiodorus does not reveal himself to have been an astute thinker. However, his claim to fame lies rather in his patronage of the monastery at Vivarium, his concern that the monks there devote themselves to both divine and liberal letters, and his pointing the way to the encyclopediac type of epitome which performed so useful a function throughout the Early Middle Ages. Some attention has also been paid to Cassiodorus' attempts in his *De anima* to prove the immortality of the soul. He shows that the soul cannot be material because it can know spiritual being and must therefore have affinity with such an object. This spiritual soul is diffused throughout the body, but everywhere distinct from it. Cassiodorus is thought to be trying in this work to reconcile conflicting traditions according to which the soul is on the one hand a substance in its own right and on the other the form of the body. This difficult reconciliation is not achieved by Cassiodorus and indeed must await the advent of St. Thomas in the thirteenth century. In the final analysis, then, Cassiodorus deserves mention as patron, compiler, and preserver of ancient culture and not as an independent thinker of any magnitude.

In continuity with Cassiodorus, we may mention here the efforts of Isadore of Seville (died 636) and the Venerable Bede (673-735), Anglo-Saxon monk of the monastery of Yarrow. Isadore's work on *Etymologies* covered in twenty books a vast range of subjects and has been called the first encyclopedia. The first three books of the work are devoted to the liberal arts, and Isadore's dependence on Cassiodorus is immediately apparent. Indeed, his general method is to reproduce his sources verbatim. We find here that art is concerned with the contingent and science with the necessary.⁽²⁾ Nevertheless, Isadore may seem to be straddling the distinction when he says, "Disciplinae liberalium artium septem sunt" (there are seven sciences of the liberal arts). (I, 2) It is interesting to watch Isadore collate the liberal arts with the divisions of philosophy. (See *Differentiae*, PL, 83, 93-94.) He has been discussing the difference between eloquence and wisdom (col. 93, n. 148) and goes on (n. 149) to point out that the ancients identify wisdom and philosophy, which is the science of things human and divine. Moreover, they held that there were three parts of philosophy: physics, logic, and ethics. Natural philosophy is ordered to the contemplation of the natures of things, logic determines the true from the false, ethics is ordered to correct living, its theory and practice. "This three-fold genus of philosophy is divided thus by the wise of this world. They say that to physics pertain the seven disciplines, of which the first is arithmetic, the second geometry, the third music, the fourth astronomy, the fifth astrology,⁽³⁾ the sixth mechanics, the seventh medicine." (Col. 94, n. 150) The seven disciplines here

listed are, of course, not the traditional liberal arts. One wonders if the distinction of eloquence from wisdom does not relegate the trivium to the former and demand an expansion of the quadrivium to attain the number seven. Under the heading of ethics, Isadore discusses the four cardinal virtues.

Isadore's sources in this discussion are Cassiodorus and the Augustine of book eight of *The City of God*, but the attempt to fit the seven liberal arts into the threefold division of philosophy, derived from the Stoics and ultimately perhaps from Plato, appears to be original with him. The attempt raises a good many questions. Do the divisions of the arts assigned to a part of philosophy produce subdivisions of that part of philosophy? For example, if the quadrivium belongs to physics, are there sciences of nature which are not mathematical? Isadore adds that not all the arts he refers to physics are suitable for a monk. We may close this brief mention of Isadore by noting that Isadore finds the threefold division of philosophy verified in Scripture: physics may be found in Genesis and Ecclesiastes, ethics in Proverbs, and logic in the Canticle of Canticles and the Gospels.

The Venerable Bede, like Isadore in Spain, was lucky enough to be living away from the turmoil on the Continent, and he is the beneficiary of a continuous tradition of learning in England. Bede is perhaps best known for his *Ecclesiastical History of the English People*, but he also wrote on the liberal arts, composing works on orthography, prosody, and figures of speech. His *De naturarum rerum*, an encyclopedia after the manner of Isadore, is an ambitious compilation. He wrote as well on time and on the computation of the date of Easter.

The works of Bede were to have great influence both at home and on the Continent, the last in large part thanks to Alcuin. Through Bede, Isadore, and Cassiodorus, as well as independently, Augustine and Boethius emerge as the great authorities in the liberal arts.

Bibliographical Note

The works of Cassiodorus can be found in Migne, PL, 69-70. *An Introduction to Divine and Human Readings*, trans. L. W. Jones (New York, 1946). See P. Courcelle, *Les lettres grecques en occident* (Paris, 1943). The works of Isadore of Seville can be found in Migne, PL, 81-84. *Etymologiae*, edited in 2 volumes by M. Lindsay (Oxford, 1911). On Bede see H. M. Gillett, *Saint Bede the Venerable* (London, 1935).

{1} *Institutiones*, II, praef., n. 2 (ed. Mynors, p. 89) points out that Scripture makes it clear that there are seven arts. Do we not read in the Psalms that David praised God seven times a day and that Wisdom has built herself a house, erected on seven pillars? So too, in Exodus God tells Moses to make seven lights to illuminate his way. The utility of each art for reading Scripture is stressed in the preface to the first book, and we sense the influence of Augustine's *De doctrina christiana*.

{2} Unfortunately, in his *Differentiae* Isidore does not contrast science and art.

{3} "Astronomy is the law of the stars. Astrology defines the changes of the heavens, their signs, powers, the rise and fall of stars." (Col. 94, n. 152)

Part II: The Carolingian Renaissance

Chapter I

Alcuin and Rhabanus Maurus

A. Charlemagne and the Schools

Already in the time of Boethius, it is fair to say, the lights of learning were out or going out across the European continent -- a fact that indicates the urgency as well as the poignancy of Boethius' plan to put into Latin the writings of the two greatest philosophers of antiquity, Plato and Aristotle. His failure to complete even a significant portion of that task is understandable but portentous. The age called for a holding operation, and this commences with the plan of Cassiodorus to have the monks of Vivarium devote a good part of their time to the copying of books, a way of preserving the cultural heritage which was to become particularly important. Isadore of Seville and the Venerable Bede were not original thinkers; they were primarily concerned with transmitting in summary form the lore that had come down to them. The period known as the Dark Ages, those centuries when learning in any formal or institutional sense was all but unknown, may be considered to extend to the ninth century, when Charlemagne made a concerted and momentarily successful effort to reestablish the schools.

During the Dark Ages there were, of course, isolated instances of learned men; Gregory of Tours (539-594), for instance, who wrote a *History of the Franks*. Gregory chronicled the sad plight of the Church in a disruptive and violent age and lamented the limits of his own intellectual formation. An individual priest teaching a gifted youngster could hardly be expected to turn the tide of the times, even if the times were conducive to learning; what was needed was the establishment of schools, of formal education, a systematic and sustained effort to roll back the barbarism brought on by successive waves of invaders. The motives for this increasing concern for education were at once ecclesiastical and political, and the greatest beneficiaries of education were the present and future clergy. However, the move to reestablish the schools was extremely important, and its consequences justify talk of a Carolingian Renaissance. As will become apparent, the curriculum Charlemagne instituted was hardly more than

elementary, and the level of instruction, particularly at the beginning, remained low; yet, considered against its historical background, Charlemagne's reestablishment of schools marked a dramatic forward step, without which the later and gradual rise in the quality and quantity of instruction would scarcely have been possible.

The chief mentor and instrument of Charlemagne's plan was Alcuin, but it should not be thought that the Emperor's interest in learning began with his contact with Alcuin. Prior to the great Briton's arrival on the scene a number of Italian masters who were brought back by Charlemagne laid much of the groundwork for later efforts. The first of these was Peter of Pisa, who was an old man when Charlemagne induced him to come to his court to teach grammar. Peter was also a poet, as was Paul the Deacon, another Italian, a monk of Monte Cassino. Paul the Deacon was an historian of some accomplishment, the author of a *History of the Lombards* and a *Roman History*. He wrote a history of the bishop of Metz which traces the origins of the Carolingian dynasty, and a homiliary, a book of lessons for the Divine Office which also served as a book of sermons. A third Italian, Paulinus, a grammarian, was at the court at the same time as Alcuin.

Alcuin was to speak of the palace school that he directed at Aachen as not only equal to that of ancient Athens but, because of its Christianity, the superior of even the cultural milieu that produced Socrates, Plato, and Aristotle. He was doubtless in a sanguine mood when he penned those lines; the historical facts render the parallel ridiculous. Indeed, we have to wait until that later Renaissance which has come to usurp the very name before we encounter similarly inflated self-estimates. In the so-called capitulary of 787 we find a description of what Charlemagne set out to accomplish. This document, probably written by Alcuin, gives a clear picture of the modesty of their aims. The capitulary addresses the bishops and abbots as follows:

Be it known to your devotion, pleasing to God, that in conjunction with our faithful we have judged it to be of utility that in the bishoprics and monasteries committed by Christ's favor to our charge care should be taken that there shall be not only a regular manner of life and one conformable to holy religion but also the study of letters, each to teach and learn them according to his ability and the divine assistance. For even as due observance of the rule of the house tends to good morals, so zeal on the part of the teacher and the taught imparts order and grace to sentences; and those who seek to please God by living aright should also not neglect to please him by right speaking. It is written "by thine

own words shall thou be justified or condemned," and although right doing be preferable to right speaking, yet must the knowledge of what is right precede right action. Everyone, therefore, should strive to understand what it is that he would fain accomplish, and this right understanding will be the sooner gained according as the utterances of the tongue are free from error. And if false speaking is to be shunned by all men, especially should it be shunned by those who have elected to be the servants of truth. During past years we have often received letters from different monasteries informing us that at their sacred services the brethren offered up prayers on our behalf, and we have observed that the thoughts contained in these letters, though in themselves most just, were expressed in uncouth language, and while pious devotion dictated the sentiments, the unlettered tongue was unable to express them aright. Hence there has arisen in our mind the fear lest if the skill to write rightly were thus lacking, so too would the power of rightly comprehending the Sacred Scriptures be far less than were fitting, and we all know that though verbal errors be dangerous, errors of the understanding are yet more so. We exhort you, therefore, not only not to neglect the study of letters but to apply yourselves thereto with perseverance and with that humility which is well pleasing to God, so that you may be able to penetrate with greater ease and certainty the mysteries of the Holy Scriptures. For as these contain images, tropes, and similar figures, it is impossible to doubt that the reader will arrive far more readily at the spiritual sense according as he is the better instructed in learning. Let there, therefore, be chosen for this work men who are both able and willing to learn, and also desirous of instructing others, and let them apply themselves to the work with a zeal equaling the earnestness with which we recommend it to them. . . .

The capitulary obviously aims at the very rudiments of learning. Subsequent instructions are somewhat more specific regarding the content of the schooling envisaged. Psalms, musical notation, chant, computation of the seasons of the liturgical year, and grammar were to be taught. Parish priests were later enjoined to set up schools for the children and to teach without payment, although they were allowed to accept small gifts from grateful parents. At the same time, teachers were cautioned to make certain that they had corrected copies of the books used.

We will return to the effects of Charlemagne's exhortations; we want now to indicate something of the background which produced Alcuin, who was induced to leave his native England by Charlemagne and who, more than anyone else, was the spirit behind the letter of such capitularies.

The Barbarian invasion of the British Isles did not extend to Ireland, where learning continued to flourish when it had been all but extinguished elsewhere. The Irish monks were missionaries, moreover, and it was through their efforts that the learning retained in Ireland was brought to Scotland and Northern England. This is not to say that England was totally devoid of remnants of past splendor. In the seventh century, with the appointment to the archbishopric of Canterbury of Theodore of Tarsus, learning experienced a forward surge in England. The twin monasteries of Wearmouth and Jarrow, founded by Benedict Biscop (628-690), soon became a repository of books, and it was there that one of Benedict's pupils, the Venerable Bede, acquired the learning that enabled him to write his great compilations and thesauri. Bede's friend, Egbert, became archbishop of York in 732 and founded the cathedral school there, amassing a great library for it. Aelbert was his *scholasticus*, or schoolmaster, and it was there that Alcuin studied and later taught, becoming in time the *scholasticus*. Thus, when he was asked by Charlemagne to become master of the palace school at Aachen in 782, he brought to it a training in divine and secular learning perhaps as great as could be had at that time.

B. Alcuin (735-804)

Little is known for certain of Alcuin's origins, although he is thought to have been born of noble Northumbrian parents. He was a young boy when he entered the cathedral school at York where he was to become master in 767. For fifteen years he devoted himself to this school, putting considerable emphasis on the expansion of the library. He made several trips to the Continent to seek copies of books. In his poem "On the Saints of the Church of York" he describes the life at his school and indicates the contents of its library. The curriculum consisted of liberal studies and Scripture, the same general plan that was to be followed in the palace school. Alcuin met Charlemagne in Parma while he was returning from a trip to Rome, and the following year he accepted the invitation to Aachen.

We have commented that the liberal arts formed the basis of instruction both at York and later at the palace school. In earlier chapters we have indicated the traditional content of the liberal arts and the work of Martianus Capella, which had set down the doctrine in an allegorical fashion. It is a matter of some interest to see how Alcuin speaks of these arts and how he relates them to philosophy.

Among Alcuin's pedagogical writings is a dialogue entitled *On Dialectic*,⁽¹⁾ in which he is being questioned by Charlemagne. Before turning to the subject of the dialogue, the king asks about more general matters, and when he asks "What is philosophy?" Alcuin replies with the words of Isadore (*Etym.*, VIII, 6), who in turn had borrowed them from Cassiodorus (*Inst.*, III, 3, n. 5), who is expressing yet earlier views: "Philosophy is an inquiry into natures, knowledge of things human and divine insofar as this is possible for man." (PL, 101, col. 952) Moreover, it is rightness of life concerned with living well, meditation on death, and contempt for the world, "which is especially fitting in Christians who have with discipline conquered secular ambition and live in imitation of a future life."

Alcuin goes on to say that philosophy is made up of science and opinion and proceeds to define each. Asked what the parts of philosophy are, he replies that they are three: physics, ethics, and logic. At this point he attaches the discussion to the liberal arts. There are, he notes, four parts of physics: arithmetic, geometry, music, and astronomy. Logic, on the other hand, has two parts: dialectic and rhetoric.⁽²⁾ Finally, he reduced the quadrivium to physics and the trivium to logic. Since philosophy is also divided into inspective and actual, that is, theoretical and practical, ethics would presumably fall within the practical part of philosophy. Elsewhere, in *On Grammar* (PL, 101, 853), Alcuin calls the liberal arts *septem gradus philosophiae*, the seven stages on the way to wisdom; they are the seven pillars which support wisdom, and one will acquire science only if he is lifted up by the seven arts. But if the liberal arts are considered a necessary preparation for the reading of Scripture, the Scriptures themselves are thought to be divisible according to the threefold division of philosophy. Thus, Genesis and Ecclesiastes are concerned with nature, Proverbs as well as other books with morals, and (believe it or not) the Canticle of Canticles and the Gospels with logic. All this is quite derivative, of course, and it seems that Alcuin had only the haziest notion of the relation of the liberal arts to the divisions of philosophy with which his sources acquaint him.

What books were used to convey these various arts? To learn grammar, the students used texts by Priscian and Donatus and studied reading and composition in Latin prose and verse. Cicero and Quintilian were read for rhetoric, and logic, or dialectic, was studied by using Porphyry's *Isagoge* and Aristotle's *Categories* and *On Interpretation*, together with the commentaries on them by Boethius. Bede's *Liber de temporibus* and *Liber de ratione temporum*, which dealt with the liturgical cycle, were studied after the rudiments of arithmetic were acquired. Some Euclid was studied for geometry, and Pliny and

Bede were the sources for astronomy. Boethius and Bede provided the texts for music. Despite the scope indicated by the curriculum and booklist, not all of the arts were studied with equal thoroughness. Actually, the emphasis was placed on grammar and rhetoric, with not only the quadrivium but also dialectic treated lightly. Later, when a shift from rhetoric to dialectic occurs, a shift of no little significance for the development of scholastic theology, there will be impassioned resistance to the change. Alcuin's dialogue on rhetoric, which is basically an adaptation of Cicero, relates the art of preaching but conveys as well something of the scope rhetoric had in antiquity.^{3}

In 796 Alcuin was made abbot of St. Martin of Tours and, what was unusual at the time, took up residence there. He devoted himself to strengthening the monastery school and collecting books. There are grounds for believing that the palace school was now divided, with an Irishman named Clement undertaking the instruction of the young at the palace while Alcuin gave theological instruction at Tours. As previously at York and Aachen, students came from far and near, and Alcuin's influence spread through them when they left to set up their own schools and/or to become prominent churchmen. Rhabanus Maurus studied under Alcuin at Tours, and later the Abbey of Fulda, to which he returned, was to exercise a tremendous influence. Fredegisus was Alcuin's successor at Tours. Other important men of the time may be mentioned here, notably Theodolphus of Orleans, a Spaniard by birth, and the author of the *Gloria, Laus* which is sung on Palm Sunday. There was also one Dungal the Recluse, another Irishman, to whom Charlemagne was to write concerning Fredegisus' strange little work, *De nihilo et tenebris*.

Before discussing other figures, however, we must attempt a summary statement on Alcuin. While no original contributions to philosophy were made by him, Alcuin's pedagogical work helped to remove from eclipse some of those disciplines without which philosophy in the classical sense is not even a possibility. It would be wrong to adopt a condescending attitude toward Alcuin because of the derivative character of his writing on the arts. While his own understanding of the ultimate sources of what he passes on seems in many cases to be severely limited, his own efforts were deliberate attempts to proportion to the recently awakened interest of his contemporaries the content of works summarizing a lost tradition. Through his teaching Alcuin played a great part in feeding the spark of curiosity in his students, acquainting them with the achievements of an all-but-forgotten time and thereby preparing remotely for the resurgence which was to begin several centuries later. A second Athens the court of Charlemagne assuredly was not, and there is something at once

delightful and sad in the report that the men gathered there were wont to appropriate the names of ancients; Alcuin was called Horace, Charlemagne David, others Homer, and so on. But this palatine parody was unintentional, and what we should see in the picture this report induces is a sincere delight in learning, an openness to pagan and secular learning, always in conjunction with the Christian vocation. How easily the effort might not have been made, and if not If we cannot discern in history the cunning of Reason, we can at least appreciate the contingent character of important efforts.

We cannot leave Alcuin without mentioning his theological endeavors. He was an exegete of power, and his commentary on John's Gospel is said to betray the salutary influence of that great man the Venerable Bede. Alcuin's works on the Trinity and the procession of the Holy Ghost and his views on the Adoptionist heresy have won praise for their sureness and force. Finally, he was a poet, and if not the best, nevertheless interesting and good.

C. Fredegisus of Tours

We have already mentioned that Fredegisus succeeded Alcuin as abbot of St. Martin of Tours. He wrote a letter to the scholars at the palace school entitled *De nihilo et tenebris* (*On Nothing and Darkness*; PL, 105, 751-756), which is curious but of some interest because it raises questions concerning the signification of terms, questions which have their importance for the dispute about universals which was later to engage the attention of many.

Few words suggest the problems attached to meaning more clearly than "nothing," as Augustine suggested in his dialogue *On the Teacher*. What do we mean by "nothing"? What is signified by the term? If we say that "nothing" means nothing, we begin to appreciate the difficulties that attracted Fredegisus.

Is nothing something or, indeed, nothing? If we say it is nothing, we seem to get into the position of saying that there is something which is not. In other words, in order to affirm that nothing is not, it seems necessary to suggest that somehow it is. Fredegisus suggests that we admit that nothing is indeed something. He will endeavor to show that is the case both by argument and by an appeal to authority. The argument moves from the assertion that every finite noun signifies something to the inevitable conclusion that the finite noun "nothing" signifies something. As soon as a finite noun is uttered, we understand at once what it means. The noun "man," we are told, designates the "universality of men placed outside any difference." So too "rock" and "wood"

are said to "include their generality." In the same way, "nothing" refers to what it signifies; it means something, And since every signification is of something which is, "nothing" signifies an existent thing.

Fredegisus then appeals to Scripture to bolster his point. God, we read, created the world from nothing. Consequently, nothing must be one of the first and principal creatures. Since Fredegisus also reads in Scripture that darkness lay over the face of the deep, we are prepared for his defense of the reality, indeed, the corporeality, of darkness. His argument is quite grammatical. Whatever functions as the subject of an affirmative proposition is, according to Fredegisus, asserted to exist. "Darkness" can function as the subject of an affirmative sentence. Therefore, darkness is asserted to exist.

This rather crude theoretical flight was rebutted by Agobard of Lyon. Agobard is the author of *Contra objectiones Fredegisi* (PL, 104, 159-174), in which the Archbishop takes the Abbot to task for a number of theological errors. Fredegisus' thought has detained us only because he anticipates disputes to come. Quite apart from the example of "nothing," the little work suggests the problems associated with the recognition that such common nouns as "man" involve a universality whose source and locus are not easy to determine.

D. Rhabanus Maurus (784-856)

Rhabanus Maurus, called the Teacher of Germany (*Praeceptor Germaniae*), entered the monastery at Fulda when quite young. After studying under Alcuin at Tours, he returned to his own monastery, where he was put in charge of the monastic school. The zeal with which Rhabanus performed his task was apparently unshared by his abbot, Ratgar; the latter felt that monks were more profitably employed in building than in study. The monastic school was shut down for a time, and, it is said, Rhabanus' notebooks were confiscated by the Abbot. The setback was temporary, however, and eventually Rhabanus himself was elected abbot. In his new capacity he not only put the monastic school on a firm footing but also completed the building program started by his predecessor. Rhabanus became archbishop of Mainz in 847. He was a voluminous writer -- five volumes are devoted to his works in the collection of Migne. There are many commentaries on Scripture, an encyclopedia, and the *De clericorum institutione* (*On the Formation of the Clergy*). This last work, despite its immediately clerical goal, became a model of German education and won for Rhabanus the title mentioned above.

Before considering the *De clericorum institutione*, let us pause for a moment before Rabanus' encyclopedic work, *De universo*. The title could be translated *On Everything*, and the twenty-two books of the work justify the title. The work begins with a discussion of the Triune God and ends with a discussion of garden tools and bridles and reins. In between, Rhabanus has treated the important figures of the New and Old Testaments, discussed the matter of the canonical books of Scripture, and spoken of man's body, the ages of man, procreation, family relations, and death. He treats of beasts, serpents, worms, fish, birds, and bees; in successive books he takes earth, time, water, and world as leading ideas and scoops into the discussion whatever can conceivably be attached to those ideas; languages, rocks, weights and measures, agriculture, the military -- everything is brought into play. The procedure is noteworthy. Rhabanus will appeal to Scripture as to a source book of biology. In book fifteen, when he gives a list of philosophers, he quotes verbatim from Isadore. (*Etym.*, VIII, 6) One pages through this massive work with fascination and disbelief, trying to imagine what lay behind the industry that is almost palpable even on the yellowing pages of Migne with their crowded, cracked type and intimidating double columns. There is a drive toward unity certainly, a zestful desire to dominate knowledge and to turn it to religious advantage. Perhaps it is not fanciful to catch a different tone here, or at least a sharpening of the tone one hears in Cassiodorus. This encyclopedist looks backward still, but there is that naive optimism of the Carolingian Age which makes the *De universo* seem less like twenty-two sandbags against a siege than a summary of the basis from which one may proceed.

The *De clericorum institutione* is, as has been mentioned, a manual outlining what the monk should know. It is a kind of seminary curriculum, we might say, and its first two books are almost exclusively concerned with the religious life; the third sketches the profane knowledge which can also be of use to the religious. The first book deals with ecclesiastical orders, with vestments, and with sacraments. It emphasizes baptism, the Eucharist, and the Mass according to the Roman rite. The second book deals with the Divine Office, or canonical hours, and goes on to discuss fasting, confession and penance, lessons and chant. it ends with a discussion of the Catholic faith with reference to various heresies. Of the third book Rhabanus in his preface says, "it teaches how all the things written in the sacred books are to be investigated and learned as well as whatever in profane studies and arts is useful to a churchman." From chapters eighteen through twenty-five (PL, 107, 395-403) Rhabanus devotes himself to the liberal arts. Grammar, rhetoric, dialectic, mathematics (arithmetic), geometry, music, and astronomy -- Rhabanus devotes a chapter to each. The

influence of the Augustine of *De doctrina christiana* is evident in this third book of the *De clericorum institutione* as are traces of Cassiodorus, Isadore, and Bede. The great justification for studying the liberal arts remains religious and utilitarian. One well-versed in these arts is better equipped to understand Scripture. This is the purpose and ideal that was contained in the capitulary quoted earlier, of course, and it would be surprising indeed if Rhabanus would have thought otherwise, particularly in a work aimed as his was at the formation of monks.

Rhabanus Maurus figured in the Eucharistic controversy which began after the appearance of Paschasius Radbertus' work *De corpore et sanguine Christi* (PL, 120, 1255-1350). Paschasius insisted on the identity of the Sacrament of the Altar with the Body of Christ that had been born of Mary and been crucified. Rhabanus, in a difficult statement, speaks of the reception of the sacrament as uniting us in faith with Christ, so that we form with him one body. Gottschalk, in *Dicta cujusdam sapientis*, flails Paschasius, whom he makes to mean that Christ on the altar suffers again and dies again. Yet Gottschalk does not in any way deny that the body and blood of Christ are an objective reality on the altar. Ratramnus of Corbie, in his own *De corpore et sanguine domini* (PL, 121, 125-170), continues the criticism of Paschasius, who took the occasion of a commentary on Matthew for a reply (FL, 120, 890-899). The controversy is of interest because it exhibits the need for a precise language if theological debate is to be effective; moreover it presages the later debate between Berengar and Lanfranc when the nature and status of reason in settling such matters will be the real topic of discussion.

Another theological controversy of the time centered on the question of predestination and involved Gottschalk, Rhabanus Maurus, Hinemar of Rheims, John Scotus Erigena, and many others. Far more bitter and involved than the Eucharistic controversy had been, it is yet another instance of theological debate which had not yet found its method and vocabulary.

Candidus of Fulda is known to us through an opusculum entitled *Dicta Candidi de imagine dei*, which Hauréau printed in his *Histoire de la philosophie scolastique* (vol. 1 [Paris, 1872], pp. 134-137). It proceeds in fairly catechetical fashion through twelve dicta, relying heavily on Augustine. The twelfth is entitled *Quo argumento colligendum sit deum esse?* (*From What Argument Can It Be Inferred That God Exists?*). Here is Candidus' reply: "The totality of things can be divided into three kinds: what is, what lives, what understands; and these, as they differ in power, differ as well in goodness. For example, as the beast which lives can do

more than the stone which does not, so man who both lives and understands can do more than the beast who lives but does not understand. Moreover, in the same way, just as that which is and lives is better than that which is alone and does not live, so what lives and understands is better than that which lives and does not understand. The least among things with respect to power and goodness, then, is that which is alone and is not alive; in the middle range falls what is and lives; the highest is that which is, lives, and understands. Therefore, as this argument shows, the most perfect among things is that which has understanding, namely, man who understands, and he attempts to understand his understanding and to examine the power of understanding itself. He asks if he who because of understanding is better and more powerful than other things is omnipotent, that is, capable of doing whatever he wishes. Now if he finds, as indeed he would, that he cannot do whatever he wills . . . he knows that there is one superior to and better than himself possessing the power which permits man to remain in the bodily realm so long as he wishes and, when he wishes, causes him to leave it. No one can doubt that this omnipotent one who dominates those who live and understand is God." We recognize here the influence of Augustine, of course, but the repetition of the proof is important. Gilson tells us that it is the first dialectically developed proof we come across in the modern part of the Middle Ages. (*History of Christian Philosophy in the Middle Ages*, p. 608, n. 4)

E. The Carolingian Heritage

Under the impetus of imperial decrees two kinds of schools came into being in northern Europe. First, there were the monastic schools, which had a twofold purpose. Primarily they were intended for the instruction of oblates (literally, the "offered," the children offered to the religious life by their parents) and young boys who lived in the monastery; the monastery also provided schooling for young men who did not live in the monastery, although this second purpose was the first to be dropped in difficult times. Secondly, there were cathedral schools, set up by the bishop and presided over, as we saw to be the case at York, by a schoolmaster, a *magister scholarum* or *scholasticus*. On rare occasions this was the bishop himself. Of these two main types of schools the more permanent was the monastic. Not every bishop had a school, but it was a rare monastery which did not have at least a school for its oblates. We have seen that Alcuin himself came to be situated at the monastery at Tours, and Rhabanus Maurus at that of Fulda. From the latter the influence spread to Reichenau, where Walafrid Strabo lived. Rhabanus' influence was also felt in France, where Lupus Servatus was abbot of Ferrières. Schools were also set up at Rheims, Auxerre,

Laon, and Chartres, some of which would eventually provide an education for the most illustrious men of the Early Middle Ages. Schools came into being in the Lowlands and, to the south, in Northern Italy. Thus did the leaven of the palace school spread throughout the empire, renewing what already existed but principally causing centers of learning to be inaugurated. The invasions from the north prevented a continuous development, and the great beginning was checked, receding for the most part back to the monastic schools during the period known as the Benedictine centuries. Despite this gloomy end to the Carolingian revival there are many figures of interest to us as the darkness closes again. The most important by any standards is John Scotus Erigena, to whom we now turn.

{1} Besides the *De dialectica* there are two dialogues on grammar, one on orthography, another on rhetoric and the virtues, and an astronomical work. See Migne's *Patrologiae latinae cursus completus* (PL), 101.

{2} "In his quippe generibus tribus philosophiae etiam eloquia divina consistunt. -- C. Quomodo? -- A. Nam aut de natura disputare solent, in Genese et in Ecclesiaste; aut de moribus, ut in Proverbiis et in omnibus sparsim libris; aut de logica, pro qua nostri theologiam sibi vindicant, ut in Cant. Cant. et in sancto Evangelio. -- C. Theologia quid sit? -- A. Theologia est, quod latine inspectiva dicitur, qua supergressi visibilia de divinis et coelestibus aliquid mente solum contemplamur. Nam et in his quoque partes philosophia vera dividitur, idest in inspectivam et actualem." (col. 952) The text fairly echoes with echoes, of course, and however faintly we can catch Boethian strains.

{3} See Wilbur Samuel Howell, *The Rhetoric of Alcuin and Charlemagne: A Translation, with an Introduction, the Latin Text, and Notes* (Princeton, 1941).

Chapter II

John Scotus Erigena

A. His Life and Works

We know very little of the life of John Scotus Erigena. As his name redundantly suggests, he was Irish; the date of his birth is approximately 810. It seems fairly certain that he was educated in his homeland before coming to France, where he became head of the palace school under Charles the Bald. We have already seen the salutary influence on Continental schooling that Alcuin had when he came earlier to the court of Charlemagne. But if Alcuin was a luminary, John Scotus Erigena was a good deal more. Indeed, there is no one like him in the ninth century, and historians quite properly marvel that a man of Scotus Erigena's intellectual range and daring should appear when he did. Nor is his brilliance merely a comparative thing, as if he were "fair as a star when only one is shining in the sky." His work is an authentic landmark in the Early Middle Ages, great not only in its immediate historical context but in the broader sweep of time which includes the twelfth century.

Scotus Erigena knew Greek well, a rare accomplishment and one which he put to good purpose. He translated into Latin the *Celestial Hierarchy*, *Ecclesiastical Hierarchy*, *Mystical Theology*, and *Divine Names* of Denis the Areopagite as well as his ten letters; he also wrote commentaries on the *Celestial Hierarchy*. Other works include translations of the *De hominis opificio* of Gregory of Nyssa and the *Ambiguities* of Maximus the Confessor and a commentary on the work of Martianus Capella. His translations from the Greek made the basic tenets of Neoplatonism known in the West. The thought of Scotus Erigena himself reveals the strong influence of Denis and Maximus the Confessor, particularly in his masterpiece, the work that assures Scotus Erigena a place as one of the great original thinkers of the Early Middle Ages, his *On the Division of Nature*. Also among his writings are *On Predestination*, in which he disputed the position of Gottschalk, only to have his own position condemned by two councils, and fragments of a commentary on St. John's Gospel.

The major characteristic of Scotus Erigena's original work is the attempt to combine Christian revelation and Neoplatonic elements in a speculative synthesis. The result is a panoramic view of the whole of being or nature which cannot fail to impress the modern reader with its philosophical daring. We can imagine how Scotus Erigena's contemporaries must have reacted to a work of

such strangeness and comprehension. His influence is difficult to trace, but it is thought to be visible in the School of Chartres and elsewhere, notably in Hugh of St. Victor. This is not to say that the work of Scotus Erigena was ever accepted as a whole; rather, certain elements of his system were taken over and introduced into other, more familiar contexts. Indeed, it was the fate of his *On the Division of Nature* to be condemned by the Council of Paris in 1210. The council ordered that all copies of the book be burned. The command was apparently not obeyed with alacrity, for Pope Honorius III in a letter of January 23, 1225, to the archbishops of France ordered that copies of the book -- complete or incomplete -- be sought out and sent to Rome to be solemnly burned.

Scotus Erigena is thought to have died around 877, perhaps after returning to his native land. Many legends surround the story of his life, among them a story that he was attacked and killed by his students with their pens. Whether or not he died by the pen, he has managed to survive in his writings, to the content of which we will now turn.

B. Faith and Philosophy

We recall that Alcuin, having accepted a threefold division of philosophy, applied a similar division to Scripture: Genesis and Ecclesiastes treat of nature, Proverbs and similar books of morals, the Canticle of Canticles and the Gospels of logic. Moreover, Alcuin and Rhabanus Maurus considered the liberal arts to be preparatory to the study of Scripture. Scotus Erigena, although he was a thinker of far greater sophistication than his predecessors in the palace school, seems to hold to the same identity of faith and reason. There is, for example, his famous identification of true religion and true philosophy: "For what else is it to treat of true philosophy than to set forth the rules of true religion by which God, the chief and highest cause of all things, is at once humbly served and rationally investigated? Conclude, then, that true philosophy is true religion and, conversely, that true religion is true philosophy." (*On Predestination*, chap. 1; PL, 122, 357-358)

Sacred Scripture contains the whole of the liberal arts (*Exposition of Celestial Hierarchy*; PL, 122, 140); in fact, it contains everything philosophy is thought to contain: "Divine Scripture is like an intelligible world composed of four parts as its elements. The earth which is found in the middle in the manner of a center is history around which, like water, flows the sea of the moral sense: this the Greeks call *ethike*. Beyond history and ethics, which are as it were the inferior

parts of this world, extends the air of natural science, called *physike* by the Greeks. Beyond and above all these is found the subtle and ardent fire of the empyrean heaven, that is, the highest contemplation of the divine nature called *theologike* by the Greeks. Beyond that no intelligence can go." (*Homilies on John*; PL, 122,291) Given all this, we are not surprised to read "*Nemo intret in celum nisi per philosophiam*" (No one enters heaven save through philosophy). (*Notes on Martianus*, 38,11)

Despite this identification of faith and philosophy Scotus Erigena was for a long time considered one for whom reason is the measure of faith. He has said that "true authority cannot contradict true reason nor can true reason contradict true authority." (*Div. Nat.*, I,66,511) Such contradiction is impossible because both stem from the same source, the divine wisdom. Such an opinion does not suggest rationalism, surely, but there are times when Erigena reduces authority to reason: "But reason never proceeds from authority, for every authority which is not approved by reason is seen to be inferior. Therefore, authority proceeds from true reason." (*Div. Nat.*, 1,69,513) True reason, on the other hand, stands by itself. Now if "authority" were meant here to stand for faith and Scripture, Erigena would be saying that Scripture can be acceptable only if it can be measured by our reason. This would indeed be rationalism, but it is difficult to see how such a position could be reconciled with the quotations we have given earlier. That the suggested understanding of "authority" in the present passage is unacceptable is clear from Erigena's admonition that the authority of Sacred Scripture is to be followed in all things. (*Div. Nat.*, 1,64,509)

What authority is it which must be subjected to reason? Cappuyns has argued that for Erigena Scripture is simply given and its authority is never to be questioned. When Erigena compares reason and authority, he has in mind two methods of interpreting Scripture: rational argumentation or appeals to the Fathers. Erigena holds that the authority of the Fathers must commend itself to reason if it is to be accepted. Where such authority is true, it cannot disagree with true reason, since both proceed from a common source. Reason and authority are complementary, and it is necessary to use both to arrive at pure knowledge, that is, of course, pure knowledge of Scripture. (*Div. Nat.*, 1,56,499) Erigena is, therefore, arguing for the use of reason as well as of the Fathers in the interpretation of Scripture. And, although the authority of the Fathers must be tested by reason, Scripture itself is an authority which must never be subjected to the doubt of reason.

Does Erigena think that man can come to knowledge of God apart from Scripture? Consider the following passage:

I would not say that this world surpasses the intellectual capacity of our rational nature since it was for this it was made. Not only does divine authority not forbid it, it counsels us to seek knowledge of both visible and invisible things. The Apostle says that it is through that which has been made that the terrestrial creature comes to knowledge of the invisible things of God. This is not something small, then, but something great and most useful, namely, that the knowledge of sensible things is ordered to the understanding of intelligible things. For just as one proceeds from sense to understanding, so by way of the creature one goes to God. We ought not then like irrational creatures only consider the surface of visible things but seek to comprehend what is perceived by our bodily senses. The eagle sees more clearly the form of the sun; so the wise man sees more clearly its position and movement in space and time. Are we to think that if man had not sinned and by falling become like unto the beasts he would then have ignored what is proper to him, namely, the world which he should govern justly according to the laws of nature? Another angel would have been required to praise God in sensible creatures. Man did not lose completely the dignity of his nature after sin. He still has a rational appetite which seeks to know things and does not want to be mistaken, although it often but not always is. If at the moment of transfiguration Christ's two vestments appeared as white as snow, namely, the letter of Divine Scripture and the form of visible things, why should we be obliged so carefully to attach ourselves to one of these vestments and merit to find him who wears it, and prevented from considering the other, namely, the visible creature? I do not see clearly for what reasons this could be maintained. Abraham, for example, knew God not by the letter of Scripture, which did not yet exist, but by the movement of the stars. Or did he perhaps, in the manner of the animals, consider only the forms of the stars, unable to comprehend their natures? I would not have the temerity to say that of this great and wise theologian. And if someone thinks we are wrong for employing philosophical arguments, let him consider the people of God fleeing Egypt, admonished by divine counsel to gather spoils and irreprehensibly use them. Much more those who take up the wisdom of the world ought to be accused not of wandering among visible creatures but of not having sought sufficiently in these creatures their author, for then they wil' have found the creator by means of the creature, something, we read, that Plato alone has been able to accomplish. (*Div. Nat.*, III, 23, 689)

This is a very tantalizing passage. Abraham, the father of faith, can hardly be considered to have been in the same position as a Plato. By referring to Plato, who has found God by means of the study of creatures, Erigena seems to be recognizing a distinction between philosophy and the knowledge of faith. The passage can also be construed as a defense of the use of reason, that is, rational argumentation, in the interpreting of Scripture.

If we consider reason and authority, with the latter comprising both Scripture and the Fathers' interpretation of it, we can say that reason comes before the authority of the Fathers -- we do not blindly accept their views -- but that in the study of Scripture faith must precede reason. Erigena uses the example of John and Peter running to Christ's tomb on the first Easter morning. John arrives before Peter, but he waits and allows Peter, the symbol of faith, to go in before him. John is the symbol of understanding. "For thus, since it is written, 'Unless you believe, you shall not understand,' faith necessarily precedes and goes first into the monument of Sacred Scripture, and reason, taking second place, follows along behind, its entrance being prepared by faith." (*Homilies on John*, 284-285)

But what precisely is the value of rational argumentation in relation to Scripture? At the end of an argument in astronomy, Erigena writes, "Such are the philosophical arguments concerning the spaces of the universe. If someone should find them superfluous because they are neither transmitted nor confirmed by Scripture, he should not thereby blame us. For he can no more be assured that they are false than we are able to affirm that they are true." (*Div. Nat.*, 11,34,723) If we can generalize on this, we would say that Erigena grants only a borrowed cogency to rational argumentation. If also explicitly taught by Scripture, the conclusions of an argument are true; if the contrary of the conclusion is taught by Scripture, the argument is invalid. If Scripture says nothing one way or the other, the conclusion is neither true nor false. True reason is such due to its conformity with Scripture; there seems to be no way for reason to arrive at a body of doctrine independently of Scripture, and a philosophy other than that already contained in Scripture is not possible. It would be difficult to say whether this means that the pagan philosophies can be judged true only by the test of revealed truth. Thus, the reference to Plato in the earlier quotation does not have any clear meaning. What is quite clear, however, is that Erigena himself is uninterested in any philosophy other than that revealed in Scripture. When his arguments conclude to something not contained in Scripture, he considers them neither true nor false. This brings us inexorably back to the identification of true reason and true religion: "I greet

nothing more gladly than an argument bolstered by the firmest authority." (*Div. Nat.*, 1,64,509)

C. The Division of Nature

As has been mentioned, the single most important work of John Scotus Erigena, the one to which he owes his claim to our particular attention, is the *De divisione naturae*. It is a long work, comprising five books, in the literary form of a dialogue between master and pupil. However, it has nothing like the give-and-take between the participants in a Platonic dialogue. The master is just that: he pronounces, asserts, states his views. The pupil, while not a simple foil -- he is the vehicle of much of what Scotus Erigena wants to say -- is not the occasion for dialectical progression.

The Meaning of "Nature." This term is employed by Erigena to mean everything that is and everything that is not. This may seem to be a curious definition, but Erigena presents five different understandings of the opposition of being and nonbeing which make his usage understandable: (1) In the first place, if by "being" one understands only what can be grasped by the senses, then whatever is immaterial will be nonbeing. Erigena goes further, however, thus bringing us face to face with one of the main difficulties in *On the Division of Nature*. Whatever escapes reason and intelligence will also be called nonbeing; as examples, Erigena gives the essences of things. He reasons here that God surpasses the reach of both reason and understanding; God is the essence of all things; therefore, the essences of all things escape reason and understanding. Only God truly is, Erigena continues, to quote Denis the Areopagite (*esse omnium est superesse divinitas*) and to cite Gregory of Nyssa: "Just as God as he is in himself is beyond the comprehension of any created intellect, so too in the deepest recesses of the creature made by him the essence considered as existing in him is incomprehensible." (1,3,443) The difficulty here is that by speaking of God's eminent being as the being of all things, as Denis had before him, Erigena seems to become involved in pantheism. This point will come up again in the sequel.

(2) A second way to understand the being/nonbeing dichotomy is drawn from the fact that creatures are hierarchically ordered, that a given creature is more perfect than another and less perfect than yet another on the scale of reality. Thus, the affirmation of one thing, say an inferior thing, is the negation of a superior thing. That is, to be a man is not to be an angel and vice versa.

(3) A third way in which what is can be distinguished from what is not is by confining existence to the material order. That is, we may restrict the range of the term "being" to those things which have achieved their own perfection and are independent of the causes that brought them into being. Those things which are not yet, which have not yet been perfectly formed, will then be instances of nonbeing.

(4) A fourth and more philosophical usage is that whereby only those things which do not come to be, which are not spatial and temporal, are called beings. Changeable, spatiotemporal things are then instances of nonbeing.

(5) A fifth and final way of making this distinction pertains to human nature alone. To be in the state of grace is for a man to be, whereas to be in a sinful condition is for a man not to be.

Nature for Erigena, as the foregoing indicates, is the totality of reality. The initially strange statement that nature includes both being and nonbeing can now be seen as a necessary remark if both God and creature are to be brought within the scope of a single term. To complete these preliminary but necessary remarks about the title of the work, we should understand that by "division" Erigena means a separation or emanation which has as its counterpart a resolution or return. From this we can conclude that the title of the work is not intended to convey simply a distinction of the various meanings of "nature" or a list of the various things which fall under the scope of the term. What Erigena suggests in the title is the characteristic Neoplatonic doctrine that there is a One, a first principle, from which all things emanate in such a way that a hierarchical scale is created by the graded falling away from this first principle. At the term of emanation the route is retraced by the process of return. That this is indeed the implication of the title becomes clear when we consider the fourfold division of nature that Erigena proposes, a division which provides the basic structure of the work.

Nature, which includes whatever is and whatever is not, is divided thus: first, there is the nature which creates and is not created; second, the nature which is created and creates; third, the nature which is created but does not create; finally, the nature which neither creates nor is created. When we see that these refer, respectively, to God as efficient cause, the divine Ideas, external creatures, and God as final cause, it becomes clear that in the system of *On the Division of Nature* Erigena is attempting a panoramic description of the way things have

taken their origin from God, how this is accomplished, what such things are, and how creatures necessarily return to their source.

Nature Which Creates and Is Not Created. This phrase pertains truly to God alone, for he alone is *anarchos*, without any cause. (1,11,451) Himself without cause, God is the beginning, middle, and end of all other things: the efficient, sustaining, and final cause of all things. By making this identification Erigena would seem to have made the essential point. Once more, the question he chooses to raise is surprising: Cannot God be said to be created in some sense of the term? Erigena asks after the etymology of the Greek term for God, "theos," and suggests that it comes either from the Greek word for seeing or from the word meaning to run. The latter possibility makes at least metaphorical sense if we think of God as running through or permeating all creatures. "God is said to run, therefore, not because he literally runs outside himself, he remains always and immutably in himself, but because he makes all things run from being nonexistent to being existent." (1,12,453) It is necessary to point out now that God is not created in the sense of being dependent on anything other than himself. However, insofar as in making things he, in a certain fashion, comes to be in them, it is possible to say that God is created in his effects.

This suggestion becomes the occasion for raising the broader question concerning the possibility of talking about God. Erigena adverts to previous remarks of his own and to the nature of the theologian's task in stating that since it seems clear that assertions about God are based only on what we can know of him in his effects, no statement about God can be expressive of what God is like in himself. From the essences of things we can conclude that God is, from the marvelous order among creatures we conclude that He is wise, and from their activity that God is life. Erigena attaches these attributes to the various Persons of the Trinity, but his point, once more, is the broad one that none of our names can be applied to God in such a way as to be expressive of what he is. His source here is Denis the Areopagite.

Erigena says that we must either refrain from saying anything at all about God or speak of him with great caution in terms of the twofold division of theology made by Denis, affirmative and negative. Affirmative theology takes names from creatures and applies them to God on the assumption that what is found in the effect must also be found in some fashion in the cause. Affirmative theology will say of God that he is truth, goodness, being, light, justice, sun, star, spirit, water, lion, and innumerable other things. Erigena says that such predicates, a list of

which could be derived from Scripture alone, all involve metaphor. By metaphor he means simply transference from creatures to God. His general assumption is that our language is fashioned to signify the things we know first, and, of course, what we know first are finite things; thus, our names are the names of creatures. Any use of them to speak of God must involve transference, or metaphor. If affirmative theology comes up with a vast number of terms which can be predicated of God, negative theology will deny the same predicates of God. It is considered nevertheless, as complementary to affirmative theology, for the negations serve to remind us that our terms cannot be applied to God in the same way that they are applied to things that exist. He is beyond our ken, incomprehensible, accessible only indirectly and imperfectly by way of his effects.

Nothing can be coeternal with God, Erigena observes, for this would be prejudicial to the divine unity and absolute transcendence. This observation leads Erigena to introduce the third moment in any attempt to talk about God. First, it would appear, we affirm predicates of God because they express what is found in his effects. Second, noticing that there is always something in the meaning of these terms which is not appropriate to God -- if only because all our terms are appropriate to creatures -- we deny these same predicates of God. Third, we can prefix these terms to suggest that what the term signifies is found in God in a fashion which surpasses our understanding. Thus, an illustration of these stages would be: "God is truth," then "God is not truth," and, finally, "God is supereminent truth." Without such additions, Erigena says, such names are metaphorical; with them they are, as it were, proper names of God.

Erigena continues by raising the objection that it does not seem right to say that God is ineffable and then go on to discuss how we can speak of him. Moreover, he claims, the distinction between affirmative and negative theology seems to get us into the position of making contradictory statements about God, for example, that he is truth and that he is not truth. "This appears to be a contradiction, but if we consider the matter closely this is seen not to be the case. For one who says 'he is truth' does not affirm that the divine substance is properly truth but that such a term can be transferred by way of metaphor from creature to creator; considered with respect to their proper signification, such terms simply do not attain the divine essence. On the other hand, to say 'he is not truth,' knowing clearly that the divine nature is incomprehensible and ineffable, is to say, not that he does not exist, but that he cannot properly be called or be truth." (1,14,461) To which of the two kinds of theology, negative or affirmative, belong the statements that God is more than truth, is

supergoodness, and so on? Erigena replies that such statements encompass the two theologies, for they have both affirmative and negative overtones. God is goodness, but his goodness is of a much more eminent kind, utterly unlike created goodness.

In an attempt to determine what predicates can be attributed to God, Erigena appeals to the Aristotelian categories. These categories are taken to be the most general predicates applicable to finite or creaturely being and thus are examined in terms of possible transference to God as cause of the things to which the categories properly apply. Augustine is quoted to the effect that the categories lose their power when we attempt to speak of God, but Erigena gets rid of his objection by appealing to the general assumption that whatever can be properly predicated of creatures can be transferred metaphorically to their creator. However, Erigena is swift to agree that none of the categories, not even that of relation, can be attributed to God properly. God transcends the limited mode of being which is involved in the signification of any and all of the categorical names. The conclusion is the familiar one: the categories do not in any way call into question the general truth that creaturely names cannot provide us with knowledge of what God is in himself. God is transcendent, ineffable, incomprehensible. This is Erigena's point from first to last, and if we rightly hear the echo of Denis in this section, we are also hearing what will remain the orthodox view. The human mind, in this life, whether it be considered in its own nature or as elevated by grace, cannot know God as he is in himself.

In the first book of *On the Division of Nature*, however, Erigena is not content with a general statement concerning the inadequacy of the categories to give us knowledge of God; he proceeds to take up the ten categories one by one. This thoroughness lands him in a difficulty he might have avoided had he settled for the universal statement. One of the Aristotelian categories is action, the Latin term for which is also the term for making. Of course, God makes all things, being the Creator of all things. Must we say however that God does not properly make things because the category of action pertains to him only metaphorically? Erigena is not faced with a serious difficulty. He points out that making in the categorical sense involves motion and that motion cannot be found in God. It is his further statement concerning the nature of God's making that is troublesome. When we read that God makes all things, says Erigena, we should take this to mean that God is in all things, that he is in fact the essence of all things. "He alone truly is in himself, and everything which is truly said to be in the things that are is him alone, since none of the things that are truly is in itself." (1,72,518) Once more we encounter one of the most difficult aspects of

the doctrine of Erigena. Such statements as this have led interpreters to find pantheism in his writings. In the context of the foregoing quotation it should be pointed out Erigena says that things other than God are and are what they are by participation in God.

But we do not want to dwell on the putative pantheism of Erigena. The first book of *On the Division of Nature* concludes with a reiteration of some of the points we have stressed: the transference of names of creatures to God and the need for both affirmative and negative theology. This will suffice for Erigena's doctrine on the nature which creates and is not created.

Nature Which Is Created and Creates. This phrase signifies what Erigena calls the primordial causes. These are the predestinations or patterns of external creation which are formed in the divine Word; as formed, they are created. As the patterns or ideas of external creatures, they can be called causes.

The Neoplatonic influence on Erigena is particularly clear in his discussion of this division of nature. When he speaks of primordial causes, he has in mind such ideas as Wisdom itself, Goodness itself, and so on. For the Neoplatonist, we may generalize, such entities were considered subsistent and apart from the first principle. Erigena, in orthodox fashion, locates these patterns or ideas in the Second Person of the Trinity. While the Son is coeternal with the Father, Erigena maintains, however, that the primordial causes or ideas are not quite coeternal. As creatures, they are theophanies, that is, manifestations or appearances of God. The notion of theophany as the chief characteristic of the creature should be referred to the earlier contention that God can be said in some sense to be created. He comes to be in his manifestations, or theophanies. We will return to this notion when we discuss the charge of pantheism which has been made against Erigena.

Erigena is set definitively apart from the Neoplatonism which is exercising at least an indirect influence on him by his insistence that God creates freely. The Neoplatonic tendency was to assert that God could not not create, that things emanate from him necessarily, independently of his will. Erigena indicates his opposition to this view by the very language he uses in speaking of the primordial causes. They are the wishes of God, the predestinations of God, who is a free cause. Of course, we are reminded here that it would be less inaccurate to speak of God as a supercausal principle.

The second division of nature deals with created creating causes. The primordial causes occupy a station midway between God and the creature proper; they are intermediaries. External creatures exist by way of participation; that in which they participate are the primordial causes. Here Erigena is quite close to the Neoplatonic view that the lower creature is referred to the first principle not directly but by way of an intermediary hierarchical order. This goes a long way -- some would say too-far -- toward preserving an ontological distance between God and external creation.

The primordial causes or Ideas are in the Word; while they are many, the Word is one. In a fashion that will become common, Erigena suggests that the multiplicity of primordial causes should be read in the direction of external effects. He feels that by so saying he is calling into question neither the oneness of the Word nor the simplicity of the divine nature.

He makes a further point about the primordial causes as patterns of external creatures. Creatures exist in a more perfect fashion in the primordial causes than in matter. Erigena considers existence apart from God a diminished sort of being. Such a remark is a recognition of the need to return to the source which is the other side of the created coin.

Nature Which Is Created and Does Not Create. The universe to the Neoplatonic eye and to a certain extent to Erigena's eye is a declension from the incomprehensible and ineffable unity of God, a declension which begins with the Ideas or primordial causes and then in a graded falling away from completeness and simplicity, which implies increasing complexity, arrives finally at material individuals. This concept of intermediates involves for Erigena a ceaseless flirtation with the reification of the Porphyrian tree, as if more universal terms named a higher and more perfect type of being. This third division of nature, that which is created and does not create, is the whole of external creation; in this realm man occupies a privileged position.

Erigena wants to maintain that the meaning of the statement in Genesis that man has been created in the image of God is that all things have been created in man. Man is not merely an element in the cosmos; in a sense the reverse is true, for man is a microcosmos, a world writ small. We may think that this position of Erigena's would lead to the conclusion that man is the only creature in the cosmos, that other things have whatever existence they have in man, but Erigena does not opt for this kind of idealism. Nevertheless, there is a kind of parallel here to his earlier contention that the better being of creatures is the

existence they have in the primordial causes, their being as known. Similarly, with respect to external creation, the better being of things other than man is had in man's knowledge of them. This will have dramatically important consequences in Erigena's theory on the return of things to God. The Ideas exist in man insofar as he is united to the Word. As a consequence of sin, man is unaware of the presence of the Ideas in himself, although they are innate to him. Furthermore, knowledge of the Ideas could not be derived from material things. The most important aspect of this teaching of Erigena's is the conviction that the substance of things, their real being, consists in their being known. This is preeminently the case with the primordial causes, which are the true essences of things, but it is also true with respect to man's cognitive relation to material creatures.

Nature Which Is Neither Created Nor Creates. This phrase refers to God, not as the source of creatures, but as that to which all creatures must ultimately return. At this point the profound import of Erigena's insistence that man is a microcosm is revealed. Because all things have been created in man, it is through man that they will be returned to God until that final stage is reached when, in the words of the Apostle, God will be all in all. The Incarnation is introduced here; Christ's reparation of our nature makes possible the return of man which is described as a deification. Here we must dispose of the charge of pantheism. Erigena insists that the individual soul does not lose its individuality when it has returned to God. Moreover, while Erigena employs in surprising ways the Pauline statement that God will be all in all, his firm view on the transcendence of God and the vast difference between him and creatures is as clear as anyone could wish in his treatment of the failure of our names to express what God is. It is this very incomprehensibility of God, on which Erigena insists in talking about the reach of our language, that leads him to speak of creatures as manifestations of God, or theophanies. Although he is unknowable in himself, God can be known in his effects. Furthermore, when God is said to be the essence or being of creatures, Erigena does not seem to intend an identification in being of God and creature; rather it is the dependence of creatures on God which he wants to emphasize.

The return of being to God through human nature is accomplished in five stages according to Erigena. First, at death there is a dissolution of the material body into the four elements. Second, at the resurrection the soul reclaims once more its body, gathering it from the elements so that, third, the body is changed to spirit. Fourth, there is a return of the spirit, and the whole of human nature which has become spirit, to the primal causes. Finally, there is the passage of

the spiritualized nature, together with its causes, into God, *quando nihil erit nisi solus deus* (when there will be nothing save God alone).

In *On the Division of Nature* Erigena gives a view of reality as rhythmic movement, the emanation of creatures from the One, a cascading away from the source which is productive of a hierarchy, with an ultimate overcoming of this diversity in the return of everything to the source via man, creation's lieutenant. A satisfying picture, perhaps, but dissatisfying as well; it is a blend of nature and grace, and the assertions of otherness seem to clash in the final apotheosis when creation apparently dissolves into God. Erigena's departure from orthodoxy was not merely imagined, for no matter how genial an interpretation we attempt, there are too many passages which do not lend themselves to irenic treatment. Nonetheless, Erigena's influence on later men was significant, though in a somewhat underground fashion. Eric and Remigius of Auxerre exhibit that influence, as does Berengar. Anselm of Laon; and, more importantly, Gilbert of Poitiers and Abelard take from the thought of Erigena; indeed, the Victorine school as a whole can be said to come under the influence of Scotus Erigena. Thus, Erigena was not an isolated and insulated ninth-century phenomenon; he cannot be read from the lists as an aberration and excursus from the mainstream of medieval thought. However hidden, he is in that mainstream, one influence among others, but always one to be reckoned with.

Bibliographical Note

The works of Scotus Erigena may be found in PL, 122. See Henry Bett, *Johannes Scotus Erigena* (Cambridge, 1925); M. Cappuyns, *Jean Scot Erigène* (Paris, 1933); J. Huber, *Johannes Scotus Erigena: Een Beitrag zur Geschichte der Philosophie und Theologie* (Munich, 1861); M. Del Pra, *Scoto Erigeno ed il neoplatonismo medievale* (Milan, 1941).

CHAPTER III

Other Ninth and Tenth Century Figures

What had been begun in the Carolingian Renaissance was never fully extinguished during the subsequent difficult centuries, and it is a fairly widespread opinion nowadays that the Twelfth-Century Renaissance, to which we shall turn in the next part of this volume, had its roots in the Carolingian. The present chapter will attempt to touch briefly on selected figures who insured that continuity.

A. Heiric of Auxerre (c.835 - c.887)

Heiric studied at Fulda, not under Rhabanus Maurus, but under one of his students; afterwards he repaired to Ferriere, where he studied under Servatus Lupus, whose humanism had a lasting effect on Heiric. He wrote home to his abbot in verse, extolling Servatus Lupus and the recreation to be had from profane studies. Heiric also wrote a life of Saint Germanus in verse which Manitius does not hesitate to call his masterpiece. When he returned to his own monastery to teach, he made its school famous. Charles the Bald is said to have sent his son Lothar to study under Heiric. Perhaps Heiric's most famous student was Remigius of Auxerre. Only fragments of the writings of Heiric are preserved, and very little has been edited. Nevertheless, it is possible to gain some small appreciation of Heiric and thereby to understand the magnitude of his reputation in his own and later times.

In a marginal note to his poem on the life of Saint Germanus, Heiric penned what was thought to be a remarkable anticipation of the Cartesian *cogito*.⁽¹⁾ "In every rational intellectual nature these three are seen always and inseparably to obtain: essence, power, and act (*ousia, dynamis, energeia*). By way of example, no nature whether rational or intellectual can ignore that it itself exists, though it may be ignorant of what it is. When I say therefore 'I understand myself to be,' does not the verb 'I understand' signify three things inseparable from one another? For I show myself to be and to be capable and to understand myself to be. For I could not understand if I were not, not understand if I lacked the capacity to understand; nor is that power at rest in me, but it bursts forth in the activity of understanding." Hauréau, having quoted the passage (vol. 1, p. 182), makes short work of its claim to originality by showing that it was borrowed almost verbatim from Scotus Erigena (*De divisione naturae*, I, 50), who in turn got it from Augustine. Well, we have already seen the relevant passage in

Augustine, but Hauréau seems to be rather insensitive to the liveliness of minds which would seize on this provocative Augustinian suggestion.

Heiric wrote glosses on Boethius' translation of Aristotle's *On Interpretation* and on the *Categoriae decem*, which was wrongly attributed to Augustine. Heiric is aware that Aristotle was the author of the *Categories* and is also aware that the work before him is not a translation of the Greek work. However, he considers it a free version in Latin of the Greek work and suggests that it be considered an exposition rather than a translation. This caution would, of course, have been suggested by a close reading of the work itself. In setting out to gloss the work, Heiric is delayed by a verse of Alcuin's which was placed as prologue to it and which serves him as an occasion to say something of the word nature. What he has to say indicates the strong influence on him of Scotus Erigena, for he provides us with a contracted version of the meanings with which the *De divisione naturae* begins. The influence of Erigena is also apparent when Heiric glosses the remark stating the permanent character of substance. This permanence must be ascribed to simple substances, such as the four elements; bodies composed of the four elements are not permanent but can be resolved into their elements. However, with respect to the problem of universals Heiric shows himself more independent. The categories other than substance are general or common modes of being and have whatever being they have thanks to the subjects which enjoy those modes of being. Of course, they may be said to enjoy some kind of being in the mind of God, but they have no real existence apart from their subjects. In short, Heiric is not disposed to increase the created population by listing alongside particular substances their common attributes as if the latter too enjoyed some independent mode of existence. Furthermore, when he comments on the statement that whatever can be said of animal can be said of man Heiric raises the following difficulty. "Genus" can be predicated of animal, but we would not thereby wish to say that man is a genus. He resolves the difficulty by saying that "genus" is not predicated of animal as to its reality or substance (*secundum rem, id est substantiam*). "Genus" does not express part of what animal is; it does not enter into its definition anymore than species enters into the definition of man. These predicates advert to the predicability of what is defined as "animate sensitive substance," for example, in the case of animal. Without forcing what he says, we can conclude that for Heiric "genus" and "species" are not names of real things. Hauréau concludes from this, surprisingly it would seem, that Heiric is not only a nominalist but a naive one. Let us pursue the matter.

There are three things, Heiric writes, which are involved in any speech or disputation: things, concepts (*intellectus*), and words. Since words may be either spoken or written, we may say that there are four things thus involved. Written words signify spoken words which in turn signify the concepts whereby the mind grasps things. Of these four, two are natural, namely, things and concepts, and two, spoken and written language, are conventional (*secundum positionem hominum*). Disputes arise, then, from three sources: from what is, from what is perceived, from what is said. Now, if we can attribute to Heiric, as Hauréau feels we can, the definition of genus as knowledge gathered from the similarity of its parts (*genus est cogitatio collecta ex singularum similitudine partium*), the most we can say is that we have insufficient evidence for saying what Heiric's views are. One would want to know whether Heiric distinguished between "animal" as expressive of something real in such entities as Socrates, Lothar, and Fido, and "animal" as predicable of many specifically different things. That is, did he feel that "animate sensitive substance" is a concept of something real, whereas "predicable of many specifically different things" is not a concept of anything out-there? What we do know of Heiric suggests that he was drawing attention to the different status of such words as "animal" and "man" on the one hand, and "genus" and "species" on the other. Once this different status is recognized, the perplexities that can arise from considering the statement "animal is a genus" can be handled. Heiric clearly does not want to say that "Socrates" and "man" and "animal" name three distinct individual substances; they are three names which can be applied to one single thing. That reluctance separates him from the blatant realist, to be sure, but it does not of itself make Heiric a nominalist.

B. Remigius of Auxerre (c.841 - c.908)

Remigius was a monk of Auxerre, where he had the good fortune to study under Heiric, whom he succeeded as master of the school in 876. About 883, together with his fellow student, Huebald, he was called to Rheims, where Archbishop Fulco wanted to restore the cathedral school. Remigius' task was to instruct young clerics in the liberal arts, and it is said that Fulco himself became his student. Under the direction of Huebald and Remigius the school flourished, but it appears that Remigius left Rheims, perhaps after the death of Fulco in 900, to go to Paris, where once more his fame as a teacher caused the school to flourish. The school must have been a monastic one, and it is the first school in Paris of which there is any record. Rashdall conjectures that it was the monastic school of Saint-Germain-des-Pres. Among his Parisian students mention must be made of Odo of Cluny. A vast number of works are attributed to Remigius, and his fortune among editors has been a good deal happier than

Heiric's. First, there are a number of commentaries on Scripture: on Genesis, Psalms, the Canticle of Canticles, the Epistles of Paul, the Gospel of Matthew, and the Apocalypse. He also wrote homilies, a work on the celebration of the Mass, commentaries on Boethius, Donatus, and Priscian, and many other works.

Remigius' commentaries on Boethius convey to us the flavor of his teaching. Two things strike one about Remigius as commentator: first, his dependence on others, especially, in at least one notable instance, on Scotus Erigena; second, the almost complete lack of speculative originality on his part. Let us confine ourselves to Remigius' commentary on the ninth poem of the third book of Boethius' *Consolation of Philosophy*. This poem, the "O qui perpetua," provides something like a sketch of Plato's *Timaeus*. Together with Chalcidius' commentary on the *Timaeus*, Macrobius' commentary on Cicero's *Dream of Scipio*, and the so-called Hermetic writings, this poem of Boethius is one of the sources of the Platonism of the Middle Ages.

As we have seen, Boethius is an enigmatic figure; it is a matter for amazement that the same man could write the theological tractates and the *Consolation of Philosophy*. Moreover, the Platonism of Boethius is a matter of interest since, while it is Aristotle he translated and on whom he commented, it is by no means clear that Boethius accepts without qualification key Aristotelian doctrines. For example, the division of speculative science in chapter two of the *De trinitate* seems at first blush simply Aristotelian, but when we read it more closely, when we compare it with remarks Boethius makes in a commentary on Porphyry, the initial interpretation seems questionable. A more important aspect of Boethius' Platonism is revealed in the "O qui perpetua." Is the doctrine of this poem compatible with Christian faith? There are many who maintain that it is not, that the pagan and Platonic view presented there is quite opposed to what Christians believe about the relationship between God and the world.

Erigena, in his commentary on this poem, has little difficulty in seeing its compatibility with Christianity. Remigius seems to have borrowed liberally from the commentary of Erigena; however, as H. Silvestre has argued, Remigius' version is in many ways inferior to that of Erigena. Like Erigena, Remigius reads Boethius in the light of Christian faith, but to move from Erigena's to Remigius' commentary is like moving from the clear to the smudged. Both men, it must be said, are less concerned to clarify the intention of Boethius than to take off from the poem to develop more or less related ideas. In this they are in

striking contrast to Bovo of Corvey. Bovo, whose commentary on the "O qui perpetua" may have been intended as an answer and antidote to Erigena's, is noteworthy for two things. First, he is convinced that the content of Boethius' poem is Platonic and that it is contrary to Christian doctrine. Second, Bovo's commentary is a good deal more faithful to the text of Boethius; he provides us with a great quantity of historical material so that we can grasp the meaning of the poem. Once we see what it means, Bovo feels, it will be quite clear to us that no easy adjustment can be made of this Platonic doctrine and what Christians believe about God and the world, creation and time.

Consider what Erigena and, consequently, Remigius do with the following verses (13-17):

Thou in consenting parts fitly disposed hast
The all-moving soul in midst of threefold nature placed,
Which, cut in several parts that run a different race,
Into itself returns, and circling doth embrace
The highest mind, and heaven with like proportion drives.

This allusion to the world-soul is said to be susceptible to two interpretations. Philosophers take it to be the sun; it can also be understood in terms of the human soul. The first interpretation is reported at some length, but the second is said to be better, by Erigena, and more prudent, by Remigius. We are then given a highly imaginative but quite ungrounded dissertation on the human soul as divisible into irascible, concupiscible, and rational parts. And so forth. This has little or nothing to do with what Boethius has written and less of course with Plato, on whom Boethius is depending. It is instructive to compare Erigena and Remigius, on the one hand, with each other, and, on the other, with Bovo of Corvey. How odd that Bovo, who is convinced the text is dangerous and incompatible with Christianity, should give us a closer and more accurate reading of it, while those who would assimilate it to Christian teaching seem only slightly detained by the text before them. And yet, if one is going to use a text as an occasion for speaking of things only tenuously connected with or grounded on it, how much better to do this on one's own, as Erigena did, than simply to borrow, as Remigius did.

When we turn to Remigius' commentaries on the theological tractates of Boethius, we find him staying so close to the text that what he has to say about it seldom goes beyond suggesting synonyms, making the most obvious kind of statement, or quoting Scripture and the Fathers. One cannot fail to be

impressed with Remigius' learning; at the same time he strikes us as one whose learning is not an instrument for independent thought.

C. Gerbert of Aurillac (c.940-1003)

Gerbert, who was to end his life as Pope Sylvester II, was one of the most famous teachers of his time, a tireless collector of books, and an intimate of the great of his day. Having entered the monastery at Aurillac at an early age, he was taken to Spain by a visiting noble on the recommendation of his abbot in order that he might receive what instruction could be had there. It is certain that he studied at Barcelona, but the story that he studied under Arabian masters at Cordova and Seville is mere legend. Nevertheless, he seems to have been acquainted, indirectly at least, with Arabian science, particularly astronomy and mathematics. From Spain he went to Rome, where the pope recommended him to the Emperor Otto I, who sent him to Rheims. As a teacher in the cathedral school there, Gerbert continued to seek far and wide for books to broaden his knowledge. It was while he taught at Rheims that he took part in the dispute with Otiric to which we shall return. In 983, Otto II appointed him abbot of Bobbio. This was a rich abbey, possessing lands throughout Italy, but the wealth was illusory since it required an army to collect. Upon the death of Otto II, Gerbert resigned and returned to Rheims. There he once more taught, became deeply involved in secular and ecclesiastical political affairs, and, after the deposition of Archbishop Arnulph, a natural son of King Lothar, in 991, Gerbert was elected archbishop of Rheims. In 995 he was temporarily suspended from his episcopal office, and subsequently Arnulph's deposition and Gerbert's election were declared invalid. Gerbert then repaired to the court of Otto III, where he became the teacher of the youthful Emperor. He was named archbishop of Ravenna in 998, and in 999 was elected pope.

Richer, the biographer of Gerbert, recounts a public dispute between Gerbert and Otiric which had to do with the division of philosophy. Picavet develops the hints of Richer in such a way that Otiric appears intimidated by the fame of Gerbert, a fame which had spread from Rheims into Saxony. Otiric, older than Gerbert, had reason to expect that his years of teaching would be crowned by the award of a bishopric, and Gerbert's fame might have seemed a threat to this ambition. So he planted a spy in Gerbert's class and was supplied with a schema purporting to give his supposed rival's views on the parts of philosophy. Considering Gerbert's views in error, Otiric hastened to take the matter to the Emperor as evidence of Gerbert's incompetence. The upshot was that Otiric

and Gerbert were summoned to settle the matter in a debate before the imperial court.

When we try to get at what the dispute was all about, we seem to find that it involves Otiric's acceptance of a division of philosophy which was known to the West through Augustine and which is ultimately the Stoic division of philosophy. According to this division philosophy has as its parts physics, ethics, and logic. Gerbert, on the other hand, accepted the Aristotelian division of philosophy as made known by Boethius. He puts his position thus: "Philosophy is a genus whose species are the practical and theoretical; I assign the dispersive, the distributive, and the civil as species of the practical. Under theoretical, on the other hand, it is not surprising that we should place physics (natural science), mathematics (the science of intelligibles), and divinity (the science of intellectibles)." (PL, 138, 107C) Apparently what bothered Otiric was that physics, which for him was one of the three genera of philosophy, should be presented as a species. The source of the dispute, again, would seem to be two different notions of how philosophy is divided. In the report of the disputation that Richer gives, there is an indication that Gerbert is suggesting the basic compatibility of the two divisions, but this is not developed. What does come out quite clearly is Gerbert's assumption that the division handed on by Boethius is the most complete and nuanced schema of philosophy.

Aside from its further importance for Gerbert's own thought, which we will develop in a moment, the dispute with Otiric foreshadows a difficulty which seems never to be faced head-on during the Carolingian period and its more or less immediate wake, but which occupies men considerably more during the twelfth century. We have seen in Alcuin, for example, a stress on the importance of the seven liberal arts for describing the nature of philosophy; furthermore, he will allude to the threefold division of philosophy passed on by Augustine. A third factor is the Aristotelian division of philosophy transmitted through Boethius. What is the reconciliation, if any, between these various traditions? Although Gerbert's dispute with Otiric seems to have swung around certain aspects of this problem, it is hard to see that Gerbert proposed even a partially definitive solution.

A point that arose in the dispute with Otiric was further developed by Gerbert in his *De rationali et ratione uti*, which could be translated as *On Being Rational and Reasoning*. (PL, 139, 159-168) The topic under discussion in this little work can be summed up in the following question: How can "to use reason" or "reasoning" be predicated of "rational" if every predicate is wider than its

subject? Some have suggested that "to use reason" is broader than "rational" because the former signifies a capacity together with its use, while the latter signifies the capacity alone. Gerbert himself resolves the difficulty by distinguishing between substantial and accidental predicates. On that basis, just as one can say "man sits" because it is true to say 'Socrates sits," so one can predicate actual reasoning of what is rational because some rational being is reasoning. Predicates like "sits" and "using reason" are not part of the definition of the subject of which they are predicated, and since the rule that the predicates must be broader than or at least equal to its subject in predicable scope refers to the hierarchy of substantial predicates, the difficulty as stated is not a real one.

What is of interest in this opusculum is not so much the difficulty it sets out to resolve as (1) the wide acquaintance with Aristotelian thought it evidences and (2) the fact that it has been used as an occasion to assert that Gerbert was a realist with respect to the status of universal terms. As for the knowledge of Aristotle, this far exceeds what one would expect from acquaintance with the *Categories*, *On Interpretation*, and the Boethian and Porphyrian adjuncts to these works. Gerbert observes that "potency" is equivocal when we take it as common to an act which is always actualized and an act which is temporarily consequent upon a capacity. So too he distinguishes between simple things whose actuality is such that they can never not be and things which, so long as they are, manifest a given activity (for example, fire is always hot; water is always wet) but which can not be, and things which are and may or may not perform an act which they are capable of performing. "To use reason" is an activity of the last kind. Gerbert had a penchant for schemata (he is said to have enjoyed using the abacus and other mathematical machines), and he provides us with a summary outline of the ontology just sketched.

Hauréau views the *De rationali et ratione uti* as a resolute but premature attempt to reconcile Aristotle and Plato, but he observes that Gerbert is far better acquainted with Aristotle than he is with Plato. Where does Gerbert stand with respect to the opposition between Plato and Aristotle on universals? "Do not we find firmly stated, in the passages of this treatise we have quoted, the thesis of universals *ante rem*, separated from the divine intelligence? He says it; he believes in eternally substantial intelligibles, in forms of forms, permanent acts, which are located in the vaguely described circumscribed space through which man's reason passes when it attempts to elevate itself to God. We must then definitively place this odd interpreter of the *Categories* in the ranks of the declared realists." (Hauréau, vol. 1, pp. 218-219) The passage Hauréau has in

mind constitutes chapter eleven of the treatise, and he takes it to mean that "rational," or what the term signifies, exists eternally and necessarily in the sempiternal form of man, which exists elsewhere than in individuals like Socrates and Plato. Gerbert could be taken to mean this by one who reads the passage independently of what has gone before it if he omits, as Hauréau does in citing the passage, a rather important portion of it. Prior to this passage, Gerbert had distinguished between eternal and necessary entities on the one hand, and contingent entities on the other. The latter have some activities without which they are never found and others which they sometimes exercise and sometimes do not. "To use reason" is the second kind of activity. Now, in chapter eleven Gerbert uses the terms Boethius had used, namely, "intellectibles," "intelligibles," and "naturals," which in Boethius were the respective objects of divine science, mathematics, and physics. Gerbert begins with intelligibles and says that rational can be a specific difference of sempiternal and necessary entities. Surely he can be taken to refer to objects higher than man which are also rational. Rationality, however, which is always actuated in sempiternal and necessary things (they are always actually reasoning), alters when it enters into the corporeal order as a capacity which is sometimes actuated and sometimes not. At this point Gerbert mentions intellectibles. All the things which are genera, species, and differences are in (or as) intellectibles the forms of things. Imagine now that Gerbert is here referring to the divine Ideas, the creative patterns or archetypes of creatures. He then suggests another meaning for intelligible: this may be the status of something as understood by man. There follows this remark: "Rational therefore is considered in one way in the sempiternal species of man, whether in intellectibles or intelligibles, and in another way in natural things. There forms or acts are sempiternal, here a power which may be actualized." The sempiternal form of man may be understood either as a divine Idea (intellectible) or as the mental concept (intelligible) or better the object or content of the concept, whereas the form as it is found in individuals is spoken of by Gerbert as natural. On this interpretation there is no need or clear warrant for making the assertion Hauréau has made, and it is noteworthy that he omits the passage in which Gerbert speaks of intelligibles in terms of mental concepts (*passiones animae*).

The *De rationali et ratione uti* remains an obscure and difficult work, and the difficulty is compounded by its employment of the Boethian triad: intellectibles, intelligibles, and naturals. There is some plausibility in Hauréau's interpretation of it; we hope there is at least equal plausibility in the interpretation we have suggested. Perhaps the safest summary remark on it is that it is deliciously obscure as to Gerbert's views on the status of universals.

It is hardly surprising that Gerbert entered into the Eucharistic dispute we mentioned earlier. In his *De corpore et sanguine domini* (FL, 139, 179-188) Gerbert sides with Paschasius, but at the same time he attempts to show that the position of Paschasius is not as different from that of Ratramnus of Corbie as these men, and others, had thought.

Our impression of Gerbert is that of a man of immense erudition with an indefatigable desire for new sources of knowledge, a builder of libraries, an inspiring teacher, an able dialectician. At the same time he is a political animal both in the secular and ecclesiastical worlds, worlds which were not far separated in his day. His career has some aspects of a roller-coaster ride, but when it ends with Gerbert in the papacy, he exhibits his magnanimity by certifying his old rival's right to the archbishopric of Rheims. In a bleak period Gerbert was an undeniable source of light; even if much of it was reflected light, he nonetheless forms an important link in the chain binding the Carolingian Renaissance with that of the twelfth century.

Bibliographical Note

Barthélemy Hauréau, *Histoire de la philosophie scholastique*, Part One (Paris, 1872).

{1} For a discussion of this doctrine of Descartes see volume three of this series (pp. 168-174).

Part III: The Twelfth Century

CHAPTER I

Introduction

What links the twelfth century with Carolingian times is the survival of the monastic and cathedral schools which had been the objects of imperial concern. The cathedral school of Chartres is one of the most important centers of learning and inquiry, more so than the cathedral school of Paris. In Paris the monastery of St. Victor is the locus of continuing intellectual liveliness, with William of Champeaux and Hugh of St. Victor as outstanding instances of the type of men who taught there. Abelard is a moveable feast, teaching at Laon, Paris, and elsewhere, but, wherever, it is he who enhances the school rather than vice versa. There is a split in the monastic influence in the twelfth century. When we consider St. Victor and Cluny, the influence is a positive and fairly conservative one; when we consider Bernard of Clairvaux and the monastic reform with which he is associated, the monastery appears as an alternative to the learning of the schools.

The Eucharistic controversy of the eleventh century with its opposition between dialectician and nondialectician carries over into the twelfth. Berengar of Tours appeared to elevate reason above authority in discussing matters of faith; Roscelin, in discussing the Trinity from a logical point of view, arrived at tritheism. The question then arose as to whether heresy was a necessary product of applying logic to objects of faith or was simply an indication that a legitimate endeavor had gone astray. In the twelfth century men who are in most senses opponents grope toward a proper understanding of the relation between faith and reason. St. Anselm of Canterbury, who in the context of the century seems the least polemical of men, sums up what will be the shared attitude in a phrase: *fides quaerens intellectum*. The believer is a creature endowed with reason, and it is fitting and natural that he should meditate on what he believes in an effort to grasp its meaning. There is much room for diversity within the sense of the phrase. Is the meditation on what is believed to be understood as the spiritual life, a meditation on Scripture with the aid of the Fathers in order to incorporate its message into one's own life? Or is this meditation something more abstract, making an appeal to logic and philosophy generally? These two attitudes agree that faith is not a result of natural reasoning; it is that from which one begins, what is firmly held before, during and after the meditation.

Bernard of Clairvaux represents the view that pagan philosophy not only has nothing to contribute to the Christian's effort, but is a temptation to pride and vanity. In varying ways, Hugh of St. Victor, Anselm of Canterbury, the men of Chartres, and, of course, Abelard will see philosophy as something of positive importance. Its importance is one more or less controlled by its relevance for understanding the faith. There are a number of logical writings which can be counted as purely philosophical, but by and large the writings of the men we have mentioned are theological in character. Actually, such a judgment cannot be made in terms of any clear-cut distinction between philosophy and theology operative in the twelfth century. That distinction, the distinction between knowledge of God attainable by natural reason, philosophical theology, and knowledge of God gained by faith, does not become truly effective before men of the West are confronted with the documents exhibiting philosophical theology as it was developed by the Greeks.

Much of the importance of Chartres lies in its Platonism, a Platonism revealed in the interest shown in the *Timaeus*. That dialogue, surely one of the most difficult of Plato's writings, conveys a picture of the universe that many of the teachers at Chartres tried to put into relation with the creation story of Genesis. As we examine their efforts, we can get some notion of the awakening that will follow the influx of Aristotle, his Neoplatonic commentators, Plotinus and Proclus, and the philosophy of Islamic thinkers. It is not easy to trace the introduction into the West of Islamic thought.

The points of contact are Southern Italy and Sicily, on the one hand, and Spain, on the other. Already with Gerbert there is the possibility of contact; Islamic medical writings are translated into Latin very early in Italy. It is held that we can see an acquaintance with Avicenna's *Fons vitae* in Gilbert of Poitiers' commentary on the *De trinitate* of Boethius. Peter the Venerable will be instrumental in having the Koran translated into Latin. But it is at Toledo that the work of translation is first systematically undertaken, and later at the court of Frederick II. Gundissalinus, who was connected with the translating effort in Spain, also tried to bring the new sources into contact with the traditional ones in the West, and in that he is truly a harbinger of the work of the thirteenth century.

From the middle of the twelfth century onward we are faced with the emerging situation that will define the thirteenth. The universities come into being at the end of the twelfth century, having their antecedents in the cathedral schools whose masters, at Paris, gain autonomy from the chancellor and form a guild

which is self-governing. The new entities are not recognized or granted charters until the thirteenth century, but in many cases, notably that of Paris, they are already there to be recognized. The university, with its division into various faculties, the faculty of art and that of theology particularly, provides the scene for the effort to absorb the new sources which come from antiquity through Islam to the Latin West.

If the relation between faith and reason is the fundamental motif of medieval thought, the context within which the relation is discussed shifts and varies, so that although we seem to see the same questions asked over and over, the sense of the questions alters as new data are brought to bear on their discussion. The important variable for our purposes is the amount of weight that is attached to natural reason: Of what is unaided reason capable? The answer to that question is in large part controlled by the amount of Greek philosophy that is known. That is why there is such a decline in the quality of the discussion from Augustine and Boethius to Alcuin and Rhabanus Maurus; that is why Scotus Erigena looms so large in the Carolingian period -- his knowledge of Greek enables him to bring into play Pseudo-Dionysius and Gregory of Nyssa. What this means, of course, is further variations on a basically Platonic or Neoplatonic theme, and since this is the tenor of thought emanating from Augustine as well, it is possible to speak of the tradition in the West as a Platonic one. The employment of the *Timaeus* at Chartres in the twelfth century, while it introduces novelties, does not really disturb that tradition. The increase in knowledge of Aristotle's logical writings relates to the ongoing tradition, although it alters the emphasis in instruction in the trivium. A far more disturbing alteration of the discussion of faith and reason is due to the introduction into the West of Islamic and medieval Jewish attempts to reflect on objects of faith in the light of the philosophy of Aristotle. We will see in the next part that Islamic versions of Aristotle are in fact Neoplatonic, but together with these interpretations came what was being interpreted, the *Metaphysics* of Aristotle as well as his writings on physical nature. After that point, things would never be the same again: the relation of faith and reason would be discussed in terms of philosophy and theology understood in quite new ways.

The twelfth century, then, is a complex one. It seems a continuation of the Carolingian effort -- and it is -- yet the quality of discussion and the caliber of the men involved is so much higher that it seems discontinuous with what had gone before. But much more importantly, from roughly the middle of the century onward new factors begin to be introduced into the West, a whole new statement of the problem of faith and reason. Because these factors are not

widely and fully known until the thirteenth century, the men of the twelfth suffer by comparison with those of the thirteenth. For the moment, however, we want to look at them in their own terms. When we do so, we find an impressive group of thinkers.

Chapter II

Saint Anselm of Canterbury

A. The Man and His Work

Saint Anselm was born near Aosta in 1033. His education commenced under the tutelage of the local Benedictines. When his mother died, Anselm knew a period of grief and sadness and, after three years of wandering, came to the monastery at Bee, drawn there by the reputation of Lanfranc. He became a monk of Bec in 1060 and, when Lanfranc went to Caen in 1063, succeeded him as prior of the abbey. He was a teacher in the monastery and became abbot in 1078. After fifteen years in this post he was summoned to England in 1093 to become the archbishop of Canterbury. His years at Canterbury were filled with controversy, and it was in that post that death overtook him in 1109. A rather extensive biography by his pupil Eadmer has come down to us.

This skeletal outline of the life of Anselm seems to present us with a busy ecclesiastic. Despite this impression, it is generally held that Anselm was a reluctant administrator and that he had no real relish for the many controversies into which he was drawn. He seems to have been prompted by a sense of obligation rather than by any deep inclination of his own nature. His essential self, it would seem, was inclined to withdraw into study and contemplation. Eadmer suggests that Anselm was so intent on the life of a teacher that he considered leaving Bec because Lanfranc already occupied the teaching post there. Later Anselm was to chastise himself for this worldly ambition, which he felt to be incompatible with the cloistered vocation that was his. Nonetheless, that ambition symbolizes his deep-seated desire for study, for teaching, for the calm of contemplation. Anselm's dislike for administration and active posts was based on his conviction that he had no real competence for leadership. Twice he asked the pope to relieve him of the see of Canterbury. He sought to return to the peace and tranquillity of the cloister, to prayer, meditation, and the teaching that awaited him there. Although he was a reluctant archbishop, his troubles in the post seem not to have been due to any incompetence of his. He was nonetheless twice exiled from his see, something that caused him no little anguish, but perhaps he derived a kind of ambiguous pleasure from those absences, for during those periods he recaptured in some measure the life he truly desired. But even in his active periods as archbishop he was as much theologian as spiritual administrator, composing some of the works on which his fame was to repose.

Of the writings of Anselm the following are the most important for our purposes. First, the *Monologion*, written for the monks at Bec, completed in 1076. Second, the *Proslogion*, written around 1077- 1078, with the replies to his objector, Gaunilon, coming in subsequent years. Third, between 1080 and 1085, three works: *De grammatico*, *De veritate*, and the *De libertate arbitrii*. Fourth, the *De casu diaboli*, written perhaps between 1085 and 1090. Fifth, begun in 1092 and completed in 1094, the *Epistola de incarnatione verbi*, more frequently referred to as the *De fide trinitatis*. Sixth, the famous *Cur deus homo*, which reached its completion in 1098. Finally, the *De conceptu virginali et de originali peccato*, written between 1099 and 1100. There are other works, notably prayers and meditations, as well as official letters. Those we have mentioned are easily the most important, some obviously more important than others for an assessment of Anselm the philosopher.

Just as the sketch of his life can mislead us into thinking that in Anselm we are confronted principally with a Church leader, so this seemingly meager list of writings could cause us to think that we will not find Anselm to be a significant thinker. He is a major figure nonetheless. His teaching represents one of the highest points reached by what may be referred to as the Augustinian tradition. It has often been suggested that Anselm has suffered unfairly from the tendency of students to hurry past him in order to arrive at the giants of the thirteenth century. But Anselm is a man of the eleventh century, and it is in its terms that he must be viewed. Thus regarded, he looms above the men of his own time. If we must say, as we must, that the men of the thirteenth century knew much more than Anselm, we may add that Anselm was one of the sources of their knowledge.

B. Faith and Reason

The list of his writings makes it immediately evident that Anselm's major contributions must be classified as theological. This is not to say that he had no philosophical contributions to make, of course, and with respect to the major methodological question of the Middle Ages, the relative status of philosophy and theology, reason and faith, Anselm has much to say that is of abiding importance.

Anselm is a thinker who has submerged himself in the writings of Augustine. One scholar feels that we would be struck by the Augustinian influence on Anselm even if he did not stud his works with overt references to his great predecessor. If we were to seek a motto for the total effort of Anselm, we could

do no better than to select the original title of the *Proslogion*, a phrase which Anselm felt was the best expression of the spirit of Augustine: *fides quaerens intellectum*, faith seeking understanding. Anselm, like Augustine before him, is a believer; he accepts on faith and without the slightest wavering or doubt whatever God has revealed. Yet, since he is a man, a rational animal, he must meditate and reflect on what has been proposed for his belief. Out of such study and meditation, understanding issues.

The very simplicity of this motto conceals the difficulty of grasping its meaning. Is faith merely the starting point, a transient condition, which is to give way when understanding has been achieved? Or is faith as present at the end of the effort as it is at the beginning? In his preface to the *Monologion* Anselm says that he is seeking to base truths, not on Scripture, but on arguments and the necessity of reason (*rationis necessitas*). Anselm will also say that Scripture is the source of every problem he discusses. His method, however, is so to consider what Scripture has taught that his considerations will not derive their persuasive force from the authority of Scripture. This makes it clear that faith, the acceptance of Scripture as true, is the starting point.

Given faith, one can concern himself dialectically with what he believes. This is why, after the Apostles, the holy Fathers and Doctors have said so much about the content of faith. Their writings are ordered not only to confuting the foolish and correcting the hardness of heart of those who do not have the faith but also to nourishing those whose hearts are already cleansed by faith and who can take delight in reasoning about their beliefs. That we ourselves may undertake to reason about our faith is clear from the fact that the Fathers and Doctors have certainly not exhausted the matter. Far from it. Mortals could spend an infinite time on revealed truths without exhausting their content. The scriptural basis for his position is the same as Augustine's: "Unless you believe, you shall not understand." (Is. 7:9) This text is seen by Anselm as a clear invitation to reason about our beliefs, and he goes on to suggest that such reasoning can bring us to a point midway between blind faith and the perfect vision of the next life. (See the dedicatory letter to Pope Urban II prefacing *De incarnatione verbi*.) Faith provides the conclusion, Anselm holds, and one seeks reasons for that conclusion. Chiding others, he remarks that no Catholic should entertain the possibility that what the Church believes and confesses is untrue; rather, holding tenaciously to the faith, humbly loving and living according to its truth, he can seek reasons why it is so. If understanding be achieved, one should thank God; if understanding is not forthcoming, one must nevertheless submit his reason to the incomprehensible truth. It is a vast mistake to attempt to reverse

the order given in the scriptural passage quoted above, as if reason unaided by faith could bring us to a firm adherence to revealed truth.

Nor is it enough, Anselm continues, to be confirmed in the faith (*fide stabilitus*) in order to undertake reasoning about revealed truths safely and profitably. One must also possess wisdom and moral maturity lest by sophism and levity he be led astray even to the point of embracing falsehood. That is the difference between those who commendably and continuously approach Holy Writ and those "dialecticians of today, indeed those heretical dialecticians." (PL, 158, 265)

In chapter six of the *De fide trinitatis* there is a passage in which Anselm describes what he had tried to do in the *Monologion* and *Proslogion*. Having said that many of the Fathers, especially Augustine, have given irrefutable arguments that there is but one God though the Persons be three, he continues: "If anyone would deign to read two short works of mine, namely, the *Monologion* and *Proslogion*, which were written precisely to show that what we hold by faith concerning the divine nature and Persons, apart from the Incarnation, can be proved by necessary arguments [*necessariis rationibus*] and without the authority of Scripture -- if, I say, one should read them, I think he will find there nothing that he can disprove nor would wish to reject." That is one of the strongest statements -- though it is by no means isolated or unique -- of Anselm's doctrine of *fides quaerens intellectum*. An obvious understanding of his claim would be that while faith is necessary to come into acquaintance with the fact of the Trinity, once one has developed necessary arguments he would accept the Trinity on the basis of those arguments and not because it has been revealed. But is that what Anselm wishes us to find in his remarks? Some of the passages mentioned above would suggest that this is not his meaning.

There can be little doubt that Anselm wishes to surpass faith in some sense and to arrive at what he calls reason or understanding. Nevertheless, he seems to want this understanding to be supported by faith. Furthermore, the understanding he seeks assumes a number of different forms. Sometimes the understanding at which he aims is of the fact of the revealed truth and not what that truth is, as if he had comprehended it. In the *Proslogion*, for example, having given a proof for God's existence, Anselm, addressing God, says that now even if he chose not to believe that God exists, he would still know that he exists. But the argument he has given does not enable him to penetrate to an understanding of the God about whose existence he has no doubt. At other times, Anselm notes, our arguments consist merely in the presentation of

analogies to and approximations of the truth that we firmly believe. "Often too we see an object only imperfectly as to what it is, only by way of image and semblance, as when we see someone's face in a mirror." (*Monologion*, chap. 65) In such cases we cannot understand the thing in terms of its essential properties. Thus, in attempting to know God we can never attain to what is proper to him but can only approach him by way of the similarities we find in other things.

The "necessary arguments" that Anselm mentions quite often have as their purpose to exhibit the coherence of the objects of faith. Thus, in *Cur deus homo* he will try to give reasons for the Incarnation, will try to show that it was necessary for God to become man. The arguments are sought by Anselm against the background of his own firm faith in the Incarnation. He seeks them because those without the faith deride this belief, and many of the faithful wonder in their hearts about the grounds and reasonableness of it. Such arguments, then, will silence the infidel and reassure the faithful concerning the reasonableness of the objects of faith.

What in sum is Anselm's view on the relation between faith and reason? Not only does faith happen to precede reason in the case of the Christian but faith must always precede reasoning about the highest matters. However, unless faith is conjoined with rectitude of life, the effort to understand what is believed will have disastrous consequences. In reflecting on the content of his faith one becomes aware of the reasonableness of what God has done to effect our salvation. The way God has chosen, one becomes sure, is the best way. In collating the various objects of his faith he will see their interconnections, the compatibility of these various truths. The expression of the recognition, the attempt to show the reasonableness of faith -- it is this that Anselm has in mind when he speaks of "necessary arguments." He does not use the phrase loosely. In his writings he is striving for the greatest possible rigor. Moreover, he is aware when he is presenting only an analogy or semblance.

Anselm's arguments are addressed to the infidel, not with the idea that they may lead him into faith, but rather to silence his objections. If such an objector acquired faith, he might then return to Anselm's arguments and see them in a new and more positive light. The term of argumentation, of the search for understanding, is such that one realizes he has not exhausted the object of faith, has not comprehended it. Anselm's remark, after having offered a proof for the existence of God, that he would now have to affirm it even if he did not have faith, may be interpreted in several ways. First, it may refer to that truth alone

and not be a generalization about every effort to understand what is believed. Second, if we should want to think of the remark as applying as well to Anselm's "proofs" of the Trinity and the Incarnation, we would have to stress what he stressed, namely, that he in no way comprehends the truths of whose factual existence he feels certain.

One check to the interpretation that Anselm felt reasoning goes beyond faith is found in his insistence that faith is always the guide of the search for understanding. Anselm does not seem to hold, with Erigena, that we can conclude truly only to what has been revealed, but he will say that when we think we have a good argument which concludes to something contradictory to the faith, we can be sure by that fact alone that our argument is faulty. "We accept everything which is clearly demonstrated and that Holy Writ does not contradict, for since it is not opposed to the truth, it does not favor any falsehood, and from the fact that it does not deny any affirmations of reason, it sustains them by its authority. But if Scripture were evidently repugnant to our senses, no matter how irrefutable our reason may seem, we must believe to be sure of truth." (*De concord. grat.*, 6) St. Anselm's *fides quaerens intellectum* does not elevate reason into an absolute criterion of truth. To have done that would have been to engage in that philosophy against which St. Paul warns us lest it lead us astray.

Anselm's position on faith and reason is complex and not in every way clear. Nevertheless, it contains a good many precisions which will be operative in later, more definitive resolutions of the question. In the light of his views, can we say that Anselm was a philosopher? If the question means Did Anselm consider himself a philosopher? the answer would likely be negative. Given his principal purpose, to show the reasonableness of what is believed, we must call him a theologian. This does not, of course, preclude the possibility that much philosophy will be found in his writings -- that is, arguments which do not bear on the object of faith as such and whose cogency is independent of faith, antecedent or concomitant.

C. The Proof of God's Existence

It is not only a convenience historians avail themselves of, or invent, that explains our tendency to identify a thinker with one point or item of his doctrine, however extensive that doctrine may be. The historian considers the chronological progression of thinkers, and this consideration brings to light what in a given doctrine has been most influential on later thought. Whether or

not what has been most influential in a doctrine is the key to that doctrine itself is another question, of course, although its otherness is not always recognized by historians. At any rate, the single most influential item in Anselm's works is the so-called ontological argument for the existence of God. In his own lifetime it quickly became a source of controversy, and in later ages it is almost possible to classify philosophers in terms of their response to it. It has had its champions, and there are champions of it today; it has never been without its critics, and there are critics of it today. Its historical importance, gauged in terms of its influence, is accordingly beyond dispute. Moreover, it is perfectly clear that Anselm himself regarded it as a most important achievement of his thought. This is not to say that it provides us with a key which will unlock every door of Anselmian doctrine, but it is certain that we are not faced with a position which, while of little importance to the man who first held it, came to loom large in later estimates of his accomplishments.

In concentrating on the ontological argument (Anselm never called it that), we would not want to convey the impression that it represents Anselm's only attempt to prove that God exists. There are a number of proofs offered in the *Monologion*, but there is nothing particularly novel or original in them or in Anselm's presentation of them. The proof of the *Proslogion*, which came to be called the ontological argument, is both novel and original, and we will go into some detail in our presentation of it.

In his preface to the *Proslogion*, which, as we have seen, was first entitled *Fides quaerens intellectum*, or faith seeking understanding, Anselm recalls that in the *Monologion* there was a great concatenation of arguments which lead to knowledge that God truly exists. This complexity bothered him when he looked back on it, and the thought grew in his mind that it would be desirable to have a single, self-sufficient proof of this truth. This thoughtful wish seemed doomed to frustration, however; Anselm sought in vain over a period of time for that single clinching proof, and though often he had the feeling it hovered just out of reach, he was unable to formulate it. Yet he could not set aside the hope. However he tried to turn his mind to other things, he found himself importuned anew by that drive for simplicity and cogency and self-sufficiency. And then, as is the way with thought, with inspiration both good and bad, one day he had it whole: the proof of which he had despaired simply came. Out of the charity that motivated his intellectual life, Anselm wanted to convey this proof to others and thus communicate to them the joy he had felt in discovering it. He presents the proof in the role of one seeking to elevate his mind to contemplation of God, of one seeking to understand what he believes. This explains the style of the

Proslogion, where we find Anselm communing with his God, addressing him as the object of love and faith, the Being toward which Anselm's whole being tends. The first chapter is an exhortation and prayer in which Anselm approaches the God of his faith. He wishes some degree of understanding of the truths he believes since he believes in order that he might understand, and unless he believed, he would not understand.

What believed truth is it that Anselm would understand? That God is as he believes him to be and that God is that which he believes him to be. How can the God of belief be described? He is that being than which nothing greater can be conceived. Can anyone who knows that this is what the word "God" means possibly think that God does not exist? Perhaps, but as the psalmist has sung (14:1), it is the fool who says in his heart there is no God. But even the fool, hearing God described as that than which nothing greater can be conceived, understands what he hears, and what he understands is in his understanding, even if he does not understand God to exist. What Anselm is getting at is the difference between two modes of existence: existence in the mind and existence outside the mind. He illustrates the distinction by reference to the painter who, before he executes something on canvas, has in mind what he will paint. Idea precedes execution in this case; existence in the mind precedes existence outside the mind. Furthermore, this example shows that something can exist in the mind prior to, and thus without, its being instanced outside the mind. We may surmise that Anselm would also agree that, at least with respect to human minds, existence out-there can be independent of, or unaccompanied by, mental existence. Once the painter has executed his idea, the subject may be said to exist both in the mind and on canvas.

Anselm now returns to the fool, for whom God enjoys at least mental existence since he knows that God is said to be that than which nothing greater can be conceived, but who would deny that the idea is exemplified or instanced outside his mind. Anselm's argument attempts to show that the fool is indeed a fool if he thinks his denial is reasonable. That than which nothing greater can be conceived cannot, Anselm maintains, exist only in the mind. Why? Because if that than which nothing greater can be thought existed only in the mind, it would not be that than which nothing greater can be thought; for if it exists only in the mind, it can be conceived to exist in reality as well, which is more. "Consequently, if that than which nothing greater can be thought is in the mind alone, that than which nothing greater can be thought is something than which something greater can be thought. It is beyond doubt, consequently, that there

exists something than which nothing greater can be thought, and it exists both in the mind and in reality." (Chapter 2)

This is Anselm's first statement of the proof. No one can deny its simplicity, and few have failed to be at least momentarily attracted by it. The word "God" means something, involves an idea, such that whoever gets that idea lodged in his mind cannot, except at the risk of contradicting himself, deny that there is an entity, something outside the mind, which responds to or instances the idea. In short, the argument as stated relies on the validity of a passage from the conceptual to the real order, from the grasping of a definition or description to the assertion that there exists outside the mind something which this description describes. One can appreciate the elation of Anselm at having come up with so succinct an argument. The term "God" means, to put it in a less indeterminately comparative way, the summation of all perfection. Surely then, our notion of God must include existence outside the mind, since not to exist outside the mind would be to lack a basic perfection. Say then that God is the greatest existent being. Is it not at the least odd to suggest that the greatest existent thing does not exist? It is that oddity that struck Anselm. So there you are, Anselm would say. To know that by the term "God" is meant the greatest possible existent is to know that it makes no sense to deny that such an entity exists. Only a fool would do so, and his denial must be considered merely verbal. One can say that two and two are five, but one cannot really mean it if he knows what he is saying.

The objection to Anselm's argument that comes fairly quickly to mind is one that can be found already in the work of his contemporary Gaunilo, who wrote a reply to the opening of the *Proslogion* which he entitled *On Behalf of the Fool*. In a number of ways Gaunilo points out the truth that it is indeed possible to think of the greatest existent thing, to entertain the notion of something which lacks no perfection, without thereby being committed to the judgment that such a thing exists. We will try to convey the apparent purpose and content of Gaunilo's reply without great concern for putting the matter in his exact words.

Both the believer and the unbeliever can agree on this: the term "God" means the greatest existent thing, the most perfect existent. In Anselm's terminology, then, they both can be said to agree that God exists in their understanding. Now, it should be noticed that "to exist in the understanding" is no part of what either means by "God," although this is obscured by the phrase Anselm uses to express the meaning of the term "God," namely, that than which nothing greater can be thought or conceived. Surely he does not mean by this

description the limit of our abilities to think of objects. So we have the believer and unbeliever established on a common ground; they both know that when men speak of God they are speaking of the greatest existent. Now to say either that there is nothing in reality responding to this idea or that there is something in reality responding to it is to go beyond a grasp of what the term "God" means. Only in this going beyond would the unbeliever claim that God is only an idea, and when he says this, he should not be taken to mean that other men, particularly believers, mean to speak of some mental activity of theirs when they use the term "God." By the same token, when the believer says that God exists, he is not claiming that an idea of his exists outside his mind as well as in it, but that there is something in reality which responds to the content of the idea he has when he uses the term "God."

The objections of Gaunilo enable us to see an ambiguity latent in Anselm's presentation of this thought. The unbeliever understands that the believer means the greatest existent thing when he uses the term "God." External existence, consequently, is built right into the concept in the way most of us would think merely imaginary existence is built into the concepts of elves and unicorns. In short, Anselm means by the term "God" the greatest existent you can think of, but the "you can think of" is only the usual concomitant of attending to any object and not part of what the object is or is presumed to be. Now Gaunilo has trouble in grasping Anselm's insistence that simply by allowing that he is thinking of the greatest existent he is committed to asserting that there is such a thing. For him "Does the greatest possible existent exist?" is still a fair question. That is, is there something which is all perfect and good and on which everything else depends for its being? Gaunilo cannot allow that that question is answered as soon as one understands that by the term "God" men mean an all-perfect and good being on which everything else depends for its being. In summary, Gaunilo is expressing his misgiving about the view that a mental act whereby we understand the meaning of the word "God" necessitates the further mental act whereby we affirm that God exists.

The objection of Gaunilo may be thought of as more or less the usual reaction to the argumentation of Anselm. So forceful and obvious has the objection seemed that many have been content with the curtest dismissal of the ontological argument. For example, St. Thomas Aquinas, after having pointed out that it is by no means obvious that just anybody would take the term "God" to mean what Anselm wishes it to mean, since after all there have been men who thought of trees as divine, proceeds on the assumption that the desired meaning of the term can be presupposed. "Once it is granted that everyone

would understand the term 'God' to mean what has been mentioned, namely, that than which nothing greater can be thought, it does not from that fact follow that everyone would understand that what is signified by the name exists in the external world [*esse in rerum natura*] rather than in the mind alone. It cannot be argued that it exists in reality unless it is granted that there is given in reality that than which nothing greater can be thought, something which would not be granted by those who maintain that God does not exist." (*Summa theologiae*, Ia, 2, 1, 2m)

Recently there has been a growing chorus of voices suggesting that such a dismissal of the ontological argument is cavalier because it takes Anselm's weaker presentation of his argument as the definitive one. In other words, it is suggested that, despite his avowal that he had hit on one simple proof, Anselm, perhaps in a way of which he himself was insufficiently aware, actually stated his proof in two ways and that, however weak and vulnerable his first statement of it may be, the second is a different kettle of fish entirely. That second statement has been called, by Professor Charles Hartshorne, the "modal proof" for the existence of God. He maintains that if it were presented without any allusions to the history of the Anselmian argument, it would meet with a far more favorable reception than is actually the case.

The second statement of the proof is made in terms of possibility, impossibility, and, by implication, necessity. The merit of this alternative statement is that it brings out what is so easily overlooked in the first, namely, that by the term "God" one means a being for whom it is impossible not to exist. In short, God is a necessary being, and his existence cannot be confused with the mere factual givenness of anything else, of any creature, since presumably of any creature it can be said that, however true that it now exists, it is possible for it not to exist and possible for it not to have existed. Now those who have difficulty seeing that existence without qualification can function as a predicate (by which they mean a further descriptive note of an entity) do not have the same difficulty in seeing that necessary existence, the impossibility not to exist, is significantly descriptive. Consequently, those passages in which Anselm makes it clear that God is such that for him to exist is not some merely factual matter, something that happens to be the case, but that God is such that it is impossible for him not to exist -- these passages are considered to contain an alternative presentation of his argument that is not open to the criticisms we mentioned earlier. Thus, it is the kind of existence that is God's that functions in the proof, not mere existence. When this is considered, things are not so bad with the proof as may have been thought. Peter exists. Let us take this as a true

statement signifying that there is in the external world an individual man and his name is Peter. While Peter exists, it is impossible for Peter not to exist, that is, only in virtue of some Pickwickian sense could it be true that Peter exists and does not exist. Yet it does not require any exhaustive acquaintance with Peter to realize that he could very easily not have been and that, however true that he now exists, he can in the future cease to be. His existence, on this basis, may be described as possible or contingent. Accordingly, there are things, and by far the vast majority of things, of which we can say that it is possible for them to be or not to be. But God is not one of those things. He is a being such that it is impossible for him not to exist; he is a necessary being; he necessarily exists. That, it has been suggested, is the full meaning of Anselm's phrase "that than which nothing greater can be thought": that which is thought of as necessarily existing.

One must agree that this is a far more nuanced way of putting the matter than we find at the end of chapter two of the *Proslogion*. But does it follow that we are faced with an ineluctable need to agree that once we grasp the significance of God being defined as a necessary being, we must affirm that God exists? Is it not possible to retort that now we have definitions of necessary and contingent being, but we still do not know if the definition of necessary being applies to anything? My own view is that concentration on the modal statement of the proof changes nothing at all with respect to the central move Anselm wants to make, namely, from the conceptual to the real order. That movement remains suspect, and the valuable precision we have just sketched does nothing to validate the desired move. In saying this, I think I am expressing what underlies Aquinas' admittedly peremptory dismissal of Anselm, namely, that it is only by examining that region of being populated by entities of which it is true to say that their existence is contingent and by coming to knowledge of their constituents that one will find grounds for claiming that an ultimate cause of them must be present. Thus, what provides the nexus for assenting to the proposition that there exists something which is the first cause of all we survey is precisely our knowledge of what we survey, and not concentration on the descriptions we may have ready at hand for that cause should it come to be learned that it does indeed exist.

While a foreshadow of the ontological argument has been discovered in Augustine (*De moribus Manichaeorum*, II, xi, 24; PL, 32), the proof itself is fittingly ascribed to Anselm. We have already mentioned that in subsequent ages this proof has had its champions and its opponents. Descartes offers a variant of the proof; Spinoza and Leibniz thought some version of the ontological proof

valid. In addition to the opponents we have already mentioned, it should be stated that Kant and Schopenhauer were convinced that the proof is invalid. Kant's criticism, which has been perhaps the most influential in modern times, is the more serious because he maintains that other attempts to prove the existence of God participate in the flaw he finds in the ontological argument and thus, together with it, must be consigned to the wastebasket of history. In our own times there has been a remarkable renewal of interest in the argument, an interest which is so intense that a strident note enters both the refutations and defenses of it. We can be certain that the discussion will continue so long as men philosophize and, in philosophizing, recognize that it is such ultimate Questions as that concerned with the existence of God which must occupy us. If the treatment of such questions makes us aware of both the grandeur and debility of the human mind, the persistent role of the ontological argument in the discussion amply attests to the importance and influence of Anselm of Canterbury.

D. Anselm and Dialectics

Anselm lived at a time when the quarrel between the dialecticians and antidialecticians was raging, and it was doubtless inevitable that he would be drawn into it. There is some reluctance in Anselm's entry into the fray, and it is certain that his language was a good deal more moderate than that of other disputants. This has led to the following judgment: "Thus Anselm's interest lay in a field above the controversies of logic; his thoughts did not readily move within that formal circle. He joined of necessity in debates to which one cannot believe that he devoted his best faculties." (R. L. Poole, *Illustrations of the History of Medieval Thought and Learning* [New York: Dover Publications, 1960], p. 92). Anselm, in short, was not only a reluctant logician; he was a poor one. To counter this unfortunate attitude, we want to consider two things: first, Anselm's treatment of the errors of Roscelin with respect to the Trinity; second, Anselm's little work *De grammatico*.

Refutation of Roscelin. The position of Roscelin concerning the doctrine of the Trinity is as follows. The Father, Son, and Holy Ghost must be three things and not merely one; if this were not so, if they were but one thing, then we could not say that only the Son became man; rather the one thing which is the Father, Son, and Holy Ghost was united with human nature. Since our faith forbids us to accept this consequence, we must agree that the three Persons are not one thing, but three, and that if usage permitted it, we could say there are three

Gods. The three Persons are three things in the same way as there may be three angels or three souls.

In presenting this position Roscelin invoked the authority of Lanfranc and Anselm, and, as De Vorges has shown (pp. 74-75), there is some basis for Roscelin's appeal to Anselm in the latter's preface to the *Monologion*. There Anselm notes that the Greek phrase "*mia ousia, treis hypostaseis*" can be rendered in Latin as "*una essentia, tres substantiae*." In short, a transliteration of the Greek into the Latin suggests that the persons of the Trinity can be referred to as three substances in one essence. Now this is quite misleading, since the traditional rendering of the Greek term "hypostasis" had been "persona," while "substantia" had quite another function in Latin. That this is indeed ambiguous had been pointed out to Anselm, but he was not convinced of the possible danger until Roscelin put the translation to such alarming use. Roscelin took it as warrant for claiming that the three Persons are three substances in exactly the same way as three men or three angels are three substances.

Before launching his refutation, Anselm expatiates on the proper approach to an analysis of truths of faith, a discussion we drew on in speaking Anselm's views on the relation between faith and reason. First, Anselm forestalls the misunderstanding that he is out to establish the truth of the Trinity of Persons in God. This is something he accepts on faith, a truth which cannot be grounded on pure reason. Nevertheless, although this truth exceeds the comprehension of reason and because it seems to be repugnant to reason, it is important to show that this repugnance is only apparent. Second, he warns against temerity in undertaking such a discussion. The Christian ought not to undertake to show that any truth believed and confessed by the Church is impossible; rather, holding any such truth to be indubitable, loving that truth and living in humble accord with it, he may rationally seek to understand the fact. If he succeeds, let him give thanks to God; if he does not succeed, his head should be lowered, not in preparation for a defiant charge, but in venerating submission. Third, he observes that one who presumes to combat a truth confessed by the universal Church cannot be considered a Catholic; further, one who, without faith, undertakes to dispute about believed truths, simply cannot be dealt with as if he had the faith. We have already seen Anselm's insistence that faith is a prerequisite for doing theology; without faith one simply does not have the appropriate experience of what is up for discussion. "For he who does not believe does not experience; and he who is not an expert [*qui expertus non fuerit*] will not know." (*De incarn. verb.*, 1) This suggests his approach to Roscelin, who Anselm bluntly says is not a Catholic. If he were of

good faith, it would be a simple matter to show him on the authority of Scripture that there is one God and three divine Persons. Lacking this simple approach, being unable to avail himself of it, Anselm proposes to show Roscelin's error in a rational manner (*ratione*), which is here opposed to showing it by appeal to authority.

There are dialecticians nowadays, Anselm begins, indeed heretical dialecticians, who maintain that universal substances are nothing other than vocal sounds (*flatus vocis*), who are unable to distinguish between a body and its color, who see no difference between a man's soul and the knowledge he has. It is such men as these who presume to discuss spiritual questions, men for whom reason is unable to rise above bodily imaginings. How, Anselm rhetorically asks, how can men who are unable to understand that many men are specifically one man grasp how it is that in the exalted and hidden nature of God there are several Persons, each of whom is God, and yet that there is but one God? A mind so dim that it cannot distinguish a horse from its color cannot be expected to be able to distinguish the one God and his several relations. He who identifies man and individual man can only think of man as person. How then can he understand the assumption of human nature by the Word of God? Christ is not a union of two persons, but the union of a divine Person with human nature. But how could a nominalist grasp that?

With respect to trinitarian doctrine as such Anselm's reply can be briefly stated. When it is said that Father, Son, and Holy Ghost are three things, what is meant by thing? If thing refers to Persons which are diverse relations, there is no difficulty in the phrase, but if it refers to the divine substance, to what the Persons possess in common, then the statement is heretical. Anselm's remarks on the analogues to which Roscelin appeals are of interest from the point of view of opposition to nominalism. Roscelin argues that the Persons are three as three angels or three souls are. In what way are three members of the same species one for Anselm? The question is directed at the possible realism of Anselm. A realistic answer to the question would maintain that there is some one thing which is referred to by the common name. Is there, over and above individual men, a human nature which is referred to by "man"? Given his attitude toward Augustine, as well as the general tradition, we would expect that Anselm will accept the doctrine that in the Divine Word are to be found the exemplars of whatever is. In fact, Anselm holds this. (*Monologion*, 10) Furthermore, with respect to individual men he will maintain that however much they may be alike with respect to human nature, they differ from one another because of the collection of accidents peculiar to each. (*De proc. spirit.*

sancti, 28) Thus, we have already seen Anselm asking rhetorically how those who cannot understand how many individual men are one man in species can expect to understand the Trinity. But in what sense are all men one man? "Specie," Anselm says: specifically. But what does that mean? The fact is that it is difficult to come up with a clean-cut answer when we ask how Anselm stands on the question of universals.

De grammatico. The remark of Poole that we quoted earlier concerning Anselm's general disinterest in dialectics and his obvious incompetence when he overcomes his disinterest and indulges in it expresses a widespread estimate which is currently being questioned. The negative assessment of Anselm's talent as a logician was in large part based on a little dialogue, *De grammatico*, but the recent edition of that text with a commentary by Desmond P. Henry provides grounds for believing that it is Anselm's critics who may come out badly.

In the first place, the *De grammatico* is solid proof that Anselm was interested in logic apart from current debates. In the preface to his *De veritate* Anselm suggests that the *De grammatico* could be useful for introducing people to the study of dialectics. If, contrary to received opinion, that dialogue presents us with an Anselm not only adept at logic but original and exciting when he turns his mind to it, a reappraisal is obviously called for. Furthermore, the dialogue may cast some light on Anselm's position with respect to the problem of universals.

The topic under discussion in the *De grammatico* is the meaning of denominative terms, and the title is taken from the common example of such terms, "grammatical" or "literate." The dialogue revolves around the difficulties which ensue when one fails to distinguish between the qualities signified by such terms and the bearers of these qualities. "White," for example, signifies whiteness and is applied to such things as cloth, skin, clouds, and so on. Since so many different things can possess the quality, it would be a mistake to identify the meaning of "white" with any of its possible bearers, for then it might seem to follow that we must identify cloth and skin, for example. If we think that whatever can possess the quality is a substance and notice that when we use the concrete quality-word (as "white"), we do so to speak of substances (for example, of skin, cloth, clouds, and so forth), then we may seem forced to accept both (1) "white is a substance" and (2) "white is a quality." The difficulty with (1) is that we think of any substance without thinking of it as white, and the difficulty with (2) again is that what is white is always a substance. Anselm suggests two kinds of meaning to dissolve these difficulties. First, there is

precise meaning. In this kind of meaning "white" only signifies "what possesses whiteness." Second, he speaks of oblique meaning. In this sense the vehicle or bearer of the quality is meant by the denominative term. Anselm's point is that no determinate type of hearer is included in the precise meaning of a quality-word or denominative term. When a particular bearer of the quality is referred to and thus meant in a given context, it is only the context and not the precise meaning of such a denominative as "white" which enables us to see what is referred to. Anselm gives the following example. We are standing with someone and looking at two horses, a black one and a white one. He says, "Hit it." We look confused, and he adds, "Hit the horse." We ask which one, and he replies, "The white." It is not the meaning of "white" (precisely it means only what possesses whiteness) but the context which enables us to know that it is the white horse which is meant.

So far so good. Substances are named or denominated from qualities which are not part of what they are, not part of their essence or nature. A man may be and be called short, fat, learned, and so forth. "Short" and "fat" do not have human nature in their meanings and cannot, in the sense of precise signification, be said to signify or mean man. In certain contexts they are used to speak of man; we can then say that man is obliquely signified or referred to by them, but this does not commit us to the view that whatever is short is man and vice versa. Now Anselm wants to equate "grammaticus," or "literate," with "short." We may find it difficult to agree with him in this. If, as is sometimes held, "literate" is a proper accident of man, then man must enter into the definition of "literate." Anselm denies this. He explicitly says that "literate" is just like "white" and "short" and the like. One way he employs to show this is by comparing the relation between genus and species, on the one hand, and the denominative term and the denominated, on the other. He observes that while it would be silly to say of man that he is animal man, it is not silly to say that he is literate man. This is because man is not part of the definition of "literate." But, of course, with respect to the former example, we could say that man, or a man, is a human animal.

There may be restrictions on the applicability of the point Anselm makes in the dialogue. What comes through clearly is the point that a denominative word signifies chiefly the denominating form and not anything which happens to possess that form. Such a term as "white" may be taken to mean "whatever possesses whiteness." If it were taken to mean, in the strong sense of "mean," the bearer of the quality, at least one of two absurdities would follow. Either there are different bearers of the quality, which we will then be committed to

identifying, or, given there is but one bearer, we will find ourselves involved in infinite repetitions. To exemplify the first undesirable consequence, given that snow is white and swans are white, if these bearers are involved in the meaning of "white," or indeed if only one of them is, we would seemingly have to say that to be a swan and to be snow are the same. If there should be but one bearer of the quality and it be understood to be part of the meaning of the denominative, or quality, word, then "snow is white" can be analyzed into "snow is white snow" and that into "snow is white snow snow," and so on. Our earlier qualms about Anselm's generalization may be reexpressed now in terms of a distinction between qualities which just happen to have a single bearer and a quality which could not have more than one bearer. If "literate" be an example of the second type, then its analysis would have to proceed differently than the *De grammatico* suggests.

It is not our intention to enter into a formal discussion of the logical doctrine of Anselm's little dialogue. Our principal historical point is that this dialogue exhibits, in a manner which cannot be gainsaid, Anselm's interest in dialectics for its own sake. Thus, not only did he employ dialectics in his other works but he was interested in the study of dialectics itself. Furthermore, and this is the point of Henry's study, he does so with an expertise and fruitfulness which ought to be appreciated. In commending this reassessment, Henry employs devices of recent logic and experiences none of the misgivings we have shown in our brief exposition of the subject matter of the *De grammatico*.

In his work on free will Anselm is concerned to analyze a definition of Augustine's according to which free will is a power to do good and evil, a definition which would seem to preclude our speaking of God and the angels as free. In his work on truth Anselm distinguishes many meanings of "true" and extracts from them the core meaning of rectitude or correctness. He is thereby able to compare and distinguish the meanings involved in speaking of God as truth, of judgments and statements as true, of willing as correct or true. Both works repay close study and exhibit a fine mind at work.

The thought of St. Anselm by and large proceeds within a context provided by faith, but if his is a believing intelligence, his writings give us the fruit of an activity which is not simply a reiterated act of faith. He wanted to understand what he believed, and this ideal, as we saw at some length above, is not a simple or uniform one. Furthermore, with respect to the controversy between the dialecticians and the antidialecticians, the placement of Anselm is not a black-and-white matter. He was understandably harsh with those he felt were trying

to subject matters of belief to the canons of natural reason in a crude and distasteful manner, but his writings exhibit, deliberately and consciously, the bringing to bear of a questioning intelligence on matters of faith. Finally, Anselm wrote a logical work which, though it was for a long time dismissed as unimportant and inept, has recently undergone a significant reappraisal.

It is not the task of the historian to predict the influence Anselm may have on future philosophy, but it can be asserted that it could be a broader and consequently different influence than he has exercised up to the present. Looking backward, it is safe to say that the single most important Anselmian doctrine is the proof of God's existence attempted in the *Proslogion*. We can be certain that Anselm's ontological argument will continue to be discussed. For the Christian, Anselm can be a model of the intellectual life; his was an intellect captivated by faith but not, for all that, indisposed to range as far and wide as possible. His writings convey, not so much by an argument to this effect as by their pervading spirit, that no rational truth could be inimical to or incompatible with what God has chosen to reveal to man. That conviction and his efforts to exhibit its grounds in particular matters are indication enough that obscurantism and narrowness are not necessary concomitants of religious faith.

Bibliographical Note

In English there is *Saint Anselm's Basic Writings*, translated by S. N. Deane with an introduction by Charles Hartshorne (LaSalle, Illinois: Open Court Publishing Co., 1962). *Obras completas de San Anselmo* (Madrid: Biblioteca de Autores Cristianos, vol. 1, 1952; vol. 2, 1953) gives the Latin text together with a Spanish translation. Two works of Charles Hartshorne are important: *Anselm's Discovery: A Re-examination of the Ontological Proof for God's Existence* (Open Court, 1964) and *The Logic of Perfection and Other Essays* (Open Court, 1962). Also, Karl Barth, *Anselm: Fides Quaerens Intellectum* (New York: Meridian Books, 1962); Alvin Plantinga, ed., *The Ontological Argument* (New York: Doubleday Anchor, 1965); Desmond P. Henry, *The De Grammatico of St. Anselm* (Notre Dame: University of Notre Dame Press, 1964). An old but still interesting book is that of Domet de Vorges, *Saint Anselme* (Paris: Alcan, 1901). A recent edition of *Cur deus homo* has appeared in the series *Sources chrétiennes*: Rena Roques, *Pour quoi Dieu s'est fait homme?* (Paris: Editions du Cerf, 1963). Roques presents the Latin text, gives a French translation of it, and has written an interesting introduction. Also on the *Cur deus homo* is John McIntyre, *St. Anselm and His Critics* (London: Oliver and Boyd, 1954). For the influence of Augustine on Anselm see Armando Ciecchetti, *L'agostinismo nel pensiero di Anselmo d'Aosta*

(Rome, 1951); C. B. Phelan, *The Wisdom of Saint Anselm*, Wimmer Lecture (Latrobe, Pa., 1960); and *The Life of St. Anselm . . .* by Eadmer, edited with introduction, notes, and translation by R. W. Southern (London: Nelson and Sons, 1962); also by Southern, *Saint Anselm and his Biographer* (Cambridge, 1963). A fairly recent addition is Desmond P. Henry, *The Logic of Saint Anselm* (Oxford, 1967).

Chapter III

Peter Abelard

A. The Man and His Work

His name should be spelled "Abailard," but it is as "Abelard" that he is known, Peter Abelard, and just as he was wont to distinguish between *vox* and *res*, word and reality, we must take into account the difference between the myth or reputation of Abelard and what the man really was. The tradition of misspelling his name can be taken as almost symptomatic. Abelard has been for a long time a personality, an interesting, even tragic, character; there is a temptation, which few resist, to take sides first and then view the controversies in which he was involved from the vantage point of the *parti pris*. Was he the victim of William of Champeaux, of Anselm of Laon, of St. Bernard of Clairvaux, of the uncle of Heloise? Or was he the victim of his own pride and vanity, of the *hubris* which seemed to characterize him until his last year? To such questions we should perhaps respond with the title of one of Abelard's works: *Sic et non*, yes and no. He was an exceedingly complex character, at once congenial and abrasive, and no event of his life seems free of a fundamental ambiguity. Heloise and Abelard have been called the first modern couple -- I believe this is intended as a compliment -- and perhaps they were; perhaps that explains the ambivalence which marks not only their doomed affair but other events of his life as well. There is no label that has been attached to Abelard that cannot be questioned or at least qualified. He may not in this differ from others -- the convenience of labels seems inseparable from their inconvenience -- but here as elsewhere what may be true of many seems particularly true of Abelard. There is an element of exaggeration in the man, no matter how we view him. Always controversial, seldom dull, he seems never to have run out of surprises for his contemporaries. One is tempted to say that his ultimate trick was to end his life in so edifying a way that he elicited the unstinting praise of Peter the Venerable, and one wants to think that St. Bernard of Clairvaux, if not William of St. Thierry, must finally have come to admire his enemy.

Peter Abelard was born in Palais, or Le Pallet, in Brittany in 1079. The stock from which he came was said to produce men good for the clerical life and not much else. Peter was early interested in things of the mind, and it may have been in 1094, at the age of fifteen, that Peter studied under Roscelin. There is reason to believe that he studied under Thierry of Chartres as well, and this too may have occurred while he was still a boy. We are not certain when exactly he

first went to Paris. During this first stay he studied under William of Champeaux, at which time a characteristic of his manifested itself in a dramatic way. He began to quarrel with his teacher, to take exception to him, and, by his own account, to get the better of William. The upshot was that Abelard set up his own school, first at Melun, soon after at Corbeil in order to be closer to Paris; for his school, begun around 1104, was intended to rival that of William. Sometime before 1106 Abelard fell ill and returned to Brittany, where he remained for several years.

In 1108 Abelard returned to Paris and to the classroom of William of Champeaux. William was now teaching at St. Victor in Paris, having become a monk. Abelard attended William's lectures on rhetoric, and the old quarrel began anew. Abelard forced William to change his view on the status of universals and, thus triumphant, once more set up his own school, this time just outside Paris at Mont Ste. Genevieve. He continued to teach and to cause consternation among William's loyal students until his mother summoned him home. His father had joined a religious order and his mother intended to do so, and she seems to have wanted him home before she took the step.

Abelard returned from home around 1113, but now he had an entirely different ambition. He had decided at the age of thirty-four to study theology, and for this purpose he went to Laon, where Anselm and his brother Ralph taught. Their reputation was high, and it seemed a good choice, but almost immediately upon arriving at Laon, Abelard began to voice his criticism of Anselm. Taunted by the other students, he offered to comment on the Book of Ezekiel to show them how theology should be taught. They laughed when he sat down to the Bible. And yet, Abelard assures us, he dazzled his putative peers. They came to chortle; they stayed to take notes; they urged him to continue. Anselm was not to be counted among those elated at this outcome and became, in his turn, critical of Abelard.

Predictably, Abelard's next move was to set up his own school of theology. Actually he was offered a chair at the cathedral in Paris, students from Laon followed him, and his career was on the ascendant. Then, as eventually it does to most men, love came to Abelard. He was no callow youth; he was mature in years, he had devoted his life to study and teaching, and his academic and ecclesiastical future looked bright indeed. But Heloise, when he met her, seemed brighter still and certainly preferable. She was the niece of Fulbert, a Canon of Notre Dame, a girl of much talent and some education. Abelard suggested to Fulbert that he, Abelard, move into the house where he could

direct the education of Heloise. Abelard's teaching was the first casualty, he tells us. He no longer prepared; he taught only what he had taught before; he wrote poetry. Heloise became pregnant, and Abelard took her off to Brittany, where in the house of his sister their son Astralabe was born. Fulbert, who had been flattered by Abelard's interest in his niece, was infuriated by this turn of events. Abelard wanted to marry Heloise, but she refused. Her reasons came down to this, that a married Abelard could not achieve the heights beckoning to an unmarried Abelard. She was not suggesting a clandestine relationship; she did not propose to be his mistress or his wife; rather, she wanted the affair to end. Heloise emerges as a genuinely selfless young lady, while Abelard in his *Historia calamitatum* confesses that his own attitude was essentially selfish. Nonetheless, he refused to accept the self-effacing offer of Heloise, insisted they marry, but agreed that it should be kept secret. The marriage seems to have taken place in Paris, to which they had returned, having left little Astralabe in Brittany with his aunt. Fulbert would have nothing to do with a secret marriage, however, and he bruited about that the nuptials had taken place. Heloise and Abelard, for a multitude of reasons, were incensed by this, and Abelard took Heloise to a nunnery at Argenteuil, the abbess of which he knew and where Heloise had been raised. Infuriated by this, Fulbert in company with friends burst into Abelard's room and emasculated him. The uncle's rage is surely curious in its intensity -- and of course its effect on Abelard was decisive and permanent.

In the wake of his maiming, Abelard repaired to the Abbey of St. Denys near Paris, where, he tells us, he reflected on the justice of the punishment that had been inflicted upon him. He made his profession as a monk at St. Denys around 1118, devoted himself to study and prayer -- and became critical of the house. It is generally agreed that this time his criticism had an unequivocal target. Bernard of Clairvaux was also critical of the mode of life at St. Denys. Old students sought out Abelard at the monastery, and he resumed teaching; it was at this time that he wrote his first theological work, in response to student requests and in criticism of Anselm of Laon. In 1121 he was summoned to a council at Soissons, where he expected to engage in public debate with Bernard of Clairvaux but where, to his surprise, he found a tribunal already convinced of his guilt. The charge was Sabellianism, but Abelard insists he was not found guilty of heresy. It was the fact that he had no license to teach theology that seems to have been his undoing, and in the event his book was burnt. As punishment he had to recite the Creed publicly and was entrusted to the abbot of St. Medard. Eventually he was freed by the papal legate and sent back to St. Denys. There he wore out his welcome by assuring the brethren that their St. Denys could not possibly have been Denys the Areopagite. One night he slipped

away to Champagne and, once there, petitioned his abbot for permission to lead a monastic life elsewhere than at St. Denys. This was refused, but by the time Abelard got back to Paris there was a new abbot, permission was granted, and Abelard built an oratory at Quincey dedicated to the Paraclete. Once more students sought him out and Abelard resumed teaching, but he seems to have been somewhat nervous about doing so, perhaps mindful that he was again teaching theology without papal authority. In 1125 the monks at St. Cildas invited him to come as their abbot, and he agreed. It is conjectured that sometime during his stay at the Paraclete Abelard was ordained a priest. The monastery of which he became abbot was a literal nightmare. The monks kept concubines, and the place was impoverished. There was reason to suspect that the monks had both imagined that Abelard was a lenient religious and expected that students, with their fees, would follow him to St. Gildas. Abelard at this time gave the Paraclete to Heloise and her nuns. His attempts to reform his monastery put Abelard's very life in danger, and in 1131 he requested a papal investigation of the place. He himself left St. Gildas, in either 1131 or 1132, intending to go to Paris. It is here that the *Historia calamitatum* ends, and it has been conjectured that Abelard wanted the book to precede him to Paris and pave his way.

Our next firm word about Abelard comes from John of Salisbury, who studied under Abelard at Mont Ste. Genevieve in 1136. Abelard seems to have taught until the convening of the Council of Sens in 1140. William of St. Thierry had written to Bernard of Clairvaux concerning Abelard's teaching, to receive encouragement, and Abelard once more was headed for trouble. There is reason to believe that Abelard and Bernard met to discuss the former's teaching, but Bernard was unsatisfied and Abelard was charged. Abelard appealed immediately from the council to the pope, but the council was upheld. Abelard was condemned and excommunicated, and his works were burnt at St. Peter's in Rome. Abelard set out for Rome to see if he could not reverse the judgment. He never got there. En route, he stopped off at Cluny, where Peter the Venerable was abbot. The abbot persuaded Abelard to make his peace with Bernard, and this was done. Abelard settled at Cluny, where his humility and devotion were a source of edification to the monks and to Peter the Venerable himself. Abelard died on April 21, 1142.

Putting the *Historia calamitatum*, his poetry and letters to one side, the writings of Abelard fall into two main groups: logical and theological. Reliable editions of Abelard's logical writings are of fairly recent date, all within the present century, some within the decade. They fall into four groups: the so-called

Introductiones parvulorum (1114), which are glosses of a fairly close type on Porphyry, Aristotle, and Boethius; the *Logica ingredientibus* (1120), containing glosses of increasing originality on Porphyry and Aristotle; the *Logica nostrorum petitioni* (1124), a very elaborate gloss on Porphyry's *Isagoge*; finally, the *Dialectica*, which is thought to have achieved its final form while Abelard was at Cluny. The dating of these works is, of course, conjectural and controverted. In 1958 two further works were attributed to Abelard. One of the most fruitful periods for historians of logic is the twelfth century, and for this reason it has been attracting so much attention that we may expect that our knowledge of Abelard's own logical work, and the context in which it was done, is bound to increase.

Among Abelard's theological writings are *De unitate et trinitate divina* (about 1120), *Sic et non* (1122-23), *Theologia Christiana*, *Theologia* (1124-1136), *Expositio ad Romanos*, *Scito teipsum* (this is Abelard's ethics), and *The Dialogue Between a Philosopher, a Jew, and a Christian*.

Since it was chronologically his first interest, we will begin our consideration of Abelard's doctrine with his logic and then go on to his theological work. Finally, we will have something to say about the ethical doctrine contained in his *Know Thyself*.

B. Abelard's Logic

The Nature of Logic. In his *Dialectica* Abelard tells us that there are seven works which are in common use among the Latins when logic is engaged in. They are the *Isagoge* of Porphyry, the *Categories* and *On Interpretation* of Aristotle, and four works of Boethius -- the *Book on Divisions*, the *Topics*, *Categorical Syllogisms*, and *Hypothetical Syllogisms*. Actually the influence of Boethius is very apparent in Abelard's logical works. Even in commenting on Porphyry and Aristotle he follows Boethius closely. It is a matter of curiosity whether Abelard knew any of Aristotle's logical works other than the *Categories* and *On Interpretation*. Fairly general agreement can be obtained that Abelard knew the *Sophistical Refutations* and that he had seen at least some of the *Prior Analytics*. Through Boethius he, of course, had some indirect knowledge of the complete *Organon*.

Abelard simply takes over Boethius' solution of the controversy between the Stoic and Peripatetic schools (which Boethius, in turn, probably took from Ammonius). Should logic be regarded as a part of philosophy or only its instrument? The Stoics, who subdivided philosophy into physics, ethics, and

logic, felt that logic had as much reason to be regarded as an autonomous part of philosophy as physics and ethics. It had an end of its own which was irreducible to those of physics and ethics. The Peripatetics, on the other hand, insisted on the instrumentality of logic and maintained that its goal was simply to aid us in achieving the goals of speculative and practical philosophy. Boethius, following Ammonius, wanted it both ways. He invoked the analogy of the hand, which is at once a part of the body and its instrument. More often than not, Abelard uses "dialectic" as synonymous with "logic." He is of course aware of the narrow use of the term "dialectic" when it refers to merely probable arguments; when he comments on the *Topics* he follows Boethius in likening the dialectician in the narrow sense to the rhetor or orator. In its broad sense, when dialectic is logic, it is a science. These two meanings of the term, the broad and the narrow, reflect Stoic and Aristotelian usage, respectively. When Abelard discusses the nature of logic, he appeals to the Stoic tripartite division of philosophy. Physics, or speculative philosophy, is concerned with the nature and causes of things; moral philosophy, or ethics, gives norms for the conduct of life. What does logic do? It treats of the way to construct arguments (*de ratione argumentorum compenenda*). It may be defined as *ratio disserendi*, that is, the science of discourse. Its task is to establish the truth or falsity of discourse. Abelard accepts from Boethius the notion that logic comprises both the art of discovering arguments and the art of confirming them, of judging their truth or falsity according to certain rules. These are constitutive parts of logic and not subdivisions of it, he says. What makes an argument true? Two things: the disposition of terms and the nature of things. If the goal of logic is the construction of true or scientific discourse, it is possible to see the task of logic subdivide into a study of names, propositions, the discovery of arguments, and, finally, their confirmation. Abelard goes to some trouble to distinguish logic from metaphysics, from psychology, from grammar and rhetoric, and from the mere ability to formulate arguments without knowing what it is that makes an argument valid or invalid.

The logic of Abelard, whether in the various glosses or in the independent work *Dialectica*, takes its scope and direction from the authoritative logical works then available in Latin. This is not to say that Abelard was not an independent and interesting logician. For a lengthy analysis of Abelard's *Dialectica*, the reader is referred to W. and M. Kneale, *The Development of Logic*, pp. 202-224. To give some flavor of Abelard the logician, we will devote ourselves here to an analysis of one of his glosses on Porphyry, that of the *Logica ingredientibus*.

The text being glossed is the famous one in which Porphyry states the problem of universals. Abelard lists the three questions raised by Porphyry and adds three of his own: (1) What is the common cause of our imposing universal names? (2) How do we understand universal names in which no particular thing seems to be conceived? (3) Would the name rose continue to have a meaning if all roses were destroyed? Promising to resolve these questions, Abelard notes that he will discuss the problem of universals only from the point of view of genus and species, leaving the other three predicables aside.

A definition of universal is needed at the outset. Abelard invokes the definition given by Aristotle in *On Interpretation*: a universal is that which is naturally apt to be predicated of many. As for the particular, Porphyry's definition is taken to be accurate enough: the particular is that which is predicated only of one. Not only words but things too are called universals, Abelard says. What he has in mind is Aristotle's remark, "Since of things, some are universals and others are singulars, I call that universal" So too, Porphyry has located genus and species in the nature of things. From all this Abelard concludes that things themselves are contained in the universal name.

How can the universal definition be applied to a thing? It would seem that no one thing, or no collection of things, is predicated of many things taken one by one. Yet that seems to be the characteristic of the universal. How is one thing, or a collection of things, called universal? Abelard proposes to examine all the available opinions on the matter.

(1) *First Opinion*. Some have tried to resolve the difficulty by saying that things which differ from one another in form nonetheless have essentially the same substance. This is the material essence of the individuals in which it is; moreover, it is one in itself and diverse only through the forms of its inferiors. Were these forms removed, there would be absolutely no difference between the things, for their diversity is due simply to forms: the matter is in essence absolutely the same. Thus, the same substance is made to be Plato by these accidents and Socrates by those.

Abelard thinks Porphyry would agree to this solution since he had written, "By participation in the species many men are one, but in particulars the one and common is many." Boethius too would seemingly agree, for he maintained that the same universal is at the same time entirely present in the different things of which it constitutes the substance materially; though universal in itself, it is individual thanks to advening forms, without which it subsists naturally in itself.

Apart from such forms it by no means exists actually; in actuality it is always individual, although by nature it is universal. According to Boethius, Abelard concludes, individuals subsist, whereas universals are understood.

We need not be terribly concerned with the degree of accuracy with which Abelard ascribes positions to his predecessors; our present interest is in his reaction to the position as he has formulated it. He objects to it by saying that it is contrary to nature. Consider this one nature which is said to be essentially the same beneath diverse forms. Where is it? It is in individuals and individuals are many, and that entails that some one thing which is affected by certain forms be another thing which is affected by other forms. For example, animal is a genus, a species of universal. All right. Animal is essentially the same thing as it takes on the form of rationality and as it takes on the form of irrationality. But this is tantamount to saying that the irrational animal is the rational animal.

One might reply to Abelard by saying that irrational animal and rational animal can be identified to the extent that they are animal, but not insofar as they are rational and irrational. Abelard is ready. If substance is said to be the same and different only because of different qualities, we are merely postponing the problem. Quality too is a genus, and this would seem to entail that all qualities are the same. Finally, Abelard says, if difference is always something other than substance, how can we possibly talk about a plurality of substances? The import of that question is clear. If there are not many substances, there will not be many individuals for a universal to be common to.

(2) *Second Opinion.* Another opinion, one Abelard feels is close to the truth, would have it as follows. Individual things do not differ from one another because of forms; rather they are discrete personally in their essences. That which is in one is in no way to be found in another, whether it be matter or form. Even were all their forms removed, things would not subsist less discrete in their essences. Their personal differentness, that thanks to which this one is not that one, does not come from forms. It is the diversity itself of essence, just as the forms themselves are diverse one from another in themselves. If we do not say this, the diversity of forms would have to proceed *ad infinitum*, appeal always being made to further forms to explain the difference between these forms. Well, if forms can just simply differ from one another, why cannot individuals?

Abelard reacts to this opinion by wondering how those who hold it, hold namely that things are utterly different from one another, can admit universals

at all. They do so by a distinction. True enough, they hold, things are not essentially the same, but they can be said to be indifferently so. Thus, individual men, different from one another in themselves, as individuals, are the same in man, that is, they do not differ (are indifferent) with respect to humanity. The universal is grounded on this indifference.

There is a subdivision of this second opinion. (a) Some hold that the universal is simply a collection of many individuals, for example, all men taken together are the species. Abelard thinks that Boethius would be in agreement with this. He quotes Boethius as follows, "Species must be considered to be nothing other than the thought collected from the substantial likeness of individuals, and genus from the likeness of species." That collected likeness, Abelard suggests, amounts to a collecting of many. (b) Others hold that the species is not only men brought together but the individuals also insofar as they are men. When it is said that what Socrates is is predicated of many, this has to be understood figuratively, that is, many are the same as he. This means that there will be as many species and genera as there are individuals, but because of likenesses of nature those who hold this position would assign a smaller number of universals than there are individuals.

In replying or reacting to (a) Abelard asks how the whole collection of men together can be called the species if the species is predicated of each of them? The species is not predicated partially of an individual; what it expresses must be wholly in each one of them. Furthermore, why would not small groups of men constitute a species, with the result that the species, man, would contain a great number of species? And what happens to the species if one member of the collection is removed?

As for (b), Abelard asks how we are to distinguish the universal from the particular in terms of "predicated of many" if Socrates like man can be said of many things? That is, if what Socrates is as man is said of many because they are the same as he, why cannot other men be called Socrates for the same reason? To the possible retort that to say Socrates agrees with Plato "in man" means that he does not differ from him in man, Abelard replies that we could just as easily say that Socrates does not differ from Plato in stone.

Abelard feels that the discussion has brought us to a point where it seems clear that things cannot be called universals, whether things be taken singly or collectively. The only alternative is to ascribe universality to words alone. The grammarian distinguishes appellative from proper nouns, Abelard observes, and

the logician has a similar distinction to make between universal and particular words. By a universal word is meant one that can be predicated of many, for example, the term "man" can be conjoined with the particular names of men because of the nature of the subject things on which it is imposed. Particular words, a proper noun like "Socrates," are predicable only of one thing unless it be used equivocally, in which case it is no longer one word but many. Abelard does not mean to equate the grammatical and logical distinctions. He points out that a construction satisfies the grammarian if it makes sense, even though it does not show the status of a thing. Thus, "Man is a stone" is good grammar; it clearly indicates a meaning, but it does not truly demonstrate the status of man. The universal is never just the appellative, for the appellative includes oblique cases which are of little interest to the logician, who is primarily concerned with the proposition.

Having defined words as universal and particular, we must inquire into the properties of universal words. The thing about universal words is that they seem to stand for no one thing and to constitute no clear meaning of anything. For it is clear that a universal word does not apply to something, insofar as it differs from something else to which the same universal word is applicable. Thus, it might seem that universals do not derive their meaning from things. "Man," for example, does not stand for Socrates or any other individual or for the collection of individuals. We cannot infer from the proposition "A man sits in the house" that Socrates or any other particular man is sitting there. "Man," then, seems to signify no one thing, or even nothing. Where does it get its meaning? Abelard suggests the following. Universal words signify different things by naming them. Take the word "man." It names individual things for a common reason, namely, that they are men, and that is why it is called a universal. It also forms a certain conception which is common, not proper, and which pertains to the individuals in which it conceives the common likeness. Abelard considers this sufficient preparation for answering the three questions he has added to those of Porphyry. First, what is the common cause of the imposition of the universal word? Abelard says that individual men are discrete in essence as well as in accidents, but are united insofar as they are men. He does not mean to say that they are united in man, since nothing is man except a discrete thing. They are united in being man. To be man is not the same as man or indeed as anything. "Not to be in a subject," one of the characteristics of substance, is not itself something; the same can be said of "not to undergo contrariety" and "not to be subject to more or less." Yet these phrases express what Aristotle says is true of substance. What Abelard is trying to avoid is the position according to which individuals are said to be the same because of some

other individual thing. Socrates and Plato are alike in being man; horse and ass are alike in not being man. Consequently, for different things to agree is for the individuals to be the same or not to be the same, as to be white or not to be white, to be man or not to be man. Abelard is not desirous of avoiding the issue by appeal to negative phrases. When we say of two things that they agree in the status of man, we are saying that they are alike in being men, that in this they do not differ in the least. Abelard adds that we can make that assertion without any appeal to essence. We call it the status itself of man to be man and that is not some further thing. To be a man -- that is the common cause of the imposition of the universal term "man" on individuals. It is in being man that the individuals agree with one another. What is Abelard saying here? Is he denying that Socrates and Plato have the same nature? Is he denying that they agree in essence? Or is he simply saying that it is not human nature as it exists in Socrates or as it exists in Plato that is signified by "man"? And what does he mean by status? We shall return to these questions.

Abelard's second question had to do with the understanding of universal words. He begins by distinguishing understanding and sense. The former does not make use of a corporeal organ, and it bears on the likeness of things constructed by the mind for itself. Thus, if a tower we saw is destroyed, we can no longer see it, but we still have the mental likeness formed of it. And, as seeing does not constitute the tower, neither is understanding the form or likeness. Abelard goes on to disagree with Aristotle, who, Abelard feels, equated an operation of the soul with the form by saying that the *passiones animae* are likenesses of things. Abelard prefers to call the image the likeness of the thing. Nonetheless, he concedes that understanding too can be called a likeness since it conceives what is properly called the likeness of the thing. Actually there is no disagreement with Aristotle here; the phrase "passions of the soul" which occurs in *On Interpretation* does not mean, as Abelard thinks, mental acts, but concepts. Abelard maintains that universal words are common and confused images of many things. Universality is achieved at the expense of distinctness. "Man" stands vaguely for this man, that man, and so on, and does not evoke a sharp image in the way "Socrates" does. Abelard would seem to have answered his third question as well: although the conception of rose would depend upon existent roses, once the mental image is formed, it can be retained and "rose" will preserve its meaning even if all roses should cease to exist.

Abelard's conclusion is that universal words signify the common form which is present to the mind, although he notes that the most forceful explanation of universals is that they are caused by a common conception formed in accord

with the nature of things. Abelard feels that solution is closed to him, and he is left with the view that the universal word signifies a fuzzy, indistinct image which is formed by the mind.

These preliminaries done, Abelard turns to Porphyry's questions. He feels he can say, first of all, that universal words name existent things, but only in the way explained above. They owe their universality to the operation of our mind; the universality is not something existent in the sense of extramental. Are universals corporeal or incorporeal? This question, like the foregoing one, is ambiguous, Abelard observes. Some universal words may signify incorporeal substances, others corporeal substances. In the latter case, does the name common to many corporeal things signify something incorporeal? *Sic et non*, Abelard replies. It signifies individual corporeal things in a common fashion which presupposes an incorporeal image in our mind. The third question, whether universals exist in corporeal things, is answered by noting that the universal concept or image does not exist in corporeal singulars, although it represents what is in them. As for the dispute between Plato and Aristotle on this matter, Abelard has a swift reconciliation. Aristotle correctly maintained that what universals signify exists actually only in singulars; Plato just as correctly maintained that there is nothing to prevent their existing apart.

What, in sum, is Abelard maintaining with respect to universals? Few scholars discern an absolutely clear-cut doctrine emerging from Abelard's several discussions of the problem. Perhaps the two outstanding difficulties with Abelard's position are (1) the view that the universal word signifies something vague and (2) that this vague something is the status rather than the essence of individuals. As for the first, are not we able to have a very distinct notion of what "house" or rose means without at the same time asserting that those meanings are snapshots of individual houses or roses? When Abelard discusses Boethius' treatment of universals, it becomes clear that Abelard sees no way in which things named by the same universal word can be said to have a common essence. And yet, what can he mean by his status theory if not something pertaining to the substance -- that is, the essence -- of individual men? He has emphatically excluded the view that this man is this one thanks to his accidents; by the same token, it would seem, he cannot maintain that 'many' signifies a similarity among individual men based on their accidents. What is left save to say that they are essentially similar? At this point, Abelard hits on the notion that human nature is one and predicable of many due to our mental image; what the name stands for is the common conception, the result of understanding. But again we must ask, what does the common conception

stand for, of what is it a conception? Abelard, it is true, admits that something other than the common conception causes the imposition of the universal word, but he is far from clear as to what this something is -- except that it is the status of the individuals.

C. Faith and Reason

In the controversy between dialecticians and antidialecticians Abelard must of course be counted among the dialecticians. It would indeed have been curious if one who had devoted himself to logic as long and profoundly as Abelard had did not, when he turned to theology, seek some relation between his new interest and his previous one. He was convinced of the utility of logic for theology because he wanted, not to reduce faith to the level of reason, but rather to defend and understand the faith with a most powerful weapon. Abelard's difficulties with ecclesiastical authorities should not lead us to think that he questioned authority or that he had an inadequate sense of the harmful effects of heresy. Indeed, much of his theological work was prompted by a desire to refute heresy. He wrote that he had no desire to be a philosopher if that entailed turning away from St. Paul; indeed, if it meant separation from Christ, he would not care to be Aristotle himself. But he had little patience with those who warned against the study of dialectic. In the prologue to the fourth tractate of his *Dialectica* Abelard gives a strong defense of logic in relation to faith. He asks why he should be forbidden to read authors by men who apparently read them assiduously themselves. More seriously, he observes that logic offers strong weapons against the sophisms of heretics. He adds that many are foes of logic because they have not the talent to understand it.

Besides its defensive role, logic has a more positive part to play with respect to faith. Not only is its study useful in order that the believer may dispute well with those who attack the faith but logic also has a constructive role to play, insofar as the believer strives for an understanding of his faith. If Abelard finds it fairly easy to argue in favor of a conjunction of reason and authority when it is a matter of defending the faith, he is somewhat less clear on the constructive understanding of faith. Before considering this further, we must look at what Abelard had to say on the nature of faith.

Against the view that faith can be explained solely in terms of a voluntary assent to what is not understood, Abelard defines faith first in terms of intellectual assent: *id quod mente firmiter tenemus*. We must, he insists, know the meaning of what we believe. In writing to his son he says that faith comes not from force

but from reason (*ratione*). To sustain this point, he will cite the persuasive efforts of apostolic and patristic writers. On another occasion Abelard defined faith as an *existimatio* of things unseen, and this drew fire from critics who felt that he was maintaining that belief is merely an opinion. How, they asked, can this square with the certainty of belief? Abelard may seem to have argued himself into a strange position here, holding both that we must understand what we believe and that faith is opinion. He makes a number of distinctions which make it clear that he is consistent with himself and is maintaining a position that is far from dangerous.

As others had before him, Abelard distinguishes three modes of faith which can be expressed by three phrases: *credere Deum*, *credere Deo*, *credere in Deum*. It is not easy to find English equivalents for these nuanced expressions. The first (*credere Deum*) covers acceptance of the existence of God, and this as a kind of minimal assent. The second (*credere Deo*) involves trust in God's words and promises. The third (*credere in Deum*) involves loving and cherishing God. As Sikes points out, the progression Abelard has in mind here is not unlike the distinction Aquinas will employ between *fides informata* and *fides formata*, the latter a faith informed by charity. Abelard is willing to say that faith in its minimal sense can be possessed by one who does not love God, one who is in a state of sin. This indicates that Abelard does not think of faith as some abstract, merely mental assent to what God has revealed. Beyond revealed truth is the revealing Truth, and we must convert ourselves to him by means of love. Thus, Abelard can speak of the *primordia fidei*, the beginnings of faith, which one has when one accepts the truth of Christian doctrine. But the term of faith is not a dialectical exercise on the contents of faith; it is a loving union with God.

As for the meaning of *existimatio*, we must take into account a distinction Abelard makes when he says that we must understand what we believe. The understanding referred to is contrasted with comprehension; Abelard, no more than any other medieval theologian, is not suggesting that we can comprehend what God has revealed. Since the understanding of faith falls short of comprehension, it is not altogether surprising that Abelard speaks of it as an *existimatio*. Later writers like Aquinas will say of faith that it is less than science and more than opinion. Since Abelard is not suggesting that what we believe is probable and is insisting that it is less than comprehension, his *existimatio* could be taken to foreshadow the view of Aquinas.

Abelard is thought to be a very important figure in the history of theology, particularly from the point of view of method. De Chellinek writes, "The

prologue as well as the content of the *Sic et non* had an enduring influence on the theological movement, even on the canonical." (*Le mouvement théologique du XIIe siècle*, p. 164) The method of *Sic et non* is to set before the reader conflicting authoritative texts on a variety of points, a device calculated to stimulate the student to effect a resolution of seeming contradictions. Many of the passages brought into seeming contradiction are from Scripture, and in the prologue to the work Abelard has things to say about the language of Scripture. We must, he points out, take into account that copyists' errors may present us with difficulties. Beyond that, his view of the inspiration of Scripture is that it consists of an indwelling of the Holy Ghost in the sacred writer, an indwelling which does not amount to the dictation of what is to be written. Rather, thanks to the indwelling of the Holy Ghost, the sacred writer expresses in his own words and in his own way what he has learned. Scripture is extremely important for Abelard as the vehicle of what is to be believed, but he sees a need for interpretation and critique of it. And, after all, Abelard is a Catholic, and will insist on the role of tradition and of the Church in defining the content of faith.

We have already said that Abelard must be numbered among the dialecticians. Actually, his position may seem a little ambiguous on this controversy. In the *Theologia Christiana* we read: "Is God subjected to the rules established by philosophy? No, he breaks them and shows their vanity by every miracle. Is not the healing of a blind man in contradiction to that rule of Aristotle, 'For neither will one made blind see again'? And does not the divine maternity of Mary contradict this other rule: 'If she gives birth, she has been with a man'?" (PL, 178, 1245C-D) Furthermore, "It should suffice for human reason to know that human intelligence cannot comprehend him who so far surpasses all things and completely exceeds the powers of human discussion and comprehension." (1124B) It would be easy to multiply such texts in which Abelard sounds as negative as any antidialectician. However, in the *Introductio* a different view is taken. Commenting on a remark of St. Gregory to the effect that faith loses its merit if grounded on reason, Abelard replies to those who would use this remark to counsel against thinking on what is believed: "If the interpretation given were true, St. Gregory would be in opposition to himself and to all the holy Doctors who have recommended the use of reason to establish and defend the faith. Moreover, St. Gregory opposes with arguments those who doubted the fact of the resurrection, although precisely on this question he had said 'Faith loses its merit if reason gives an argument for it.' Does not his procedure go contrary to the opinion attributed to him, namely, that one ought not to reason concerning the faith? No more has he said that it loses all its merit when it was engendered by rational argumentation rather than by divine authority

and when one believes not because God has said it but because reason has been convinced." Those who condemn reasoning about faith are seeking an excuse for their own ignorance. "I think no one can ignore the fact that it is rational study rather than sanctity which has caused progress in those instructed in divine science."

The ambivalence here disappears when we consider the purpose of the two works. The *Theologia Christiana* was written to counter the abuses of Roscelin; in the *Introductio* Abelard is countering the abuses of the antidialecticians. Moreover, in the former work Abelard speaks of the utility of the arts, especially dialectic, for reading Scripture, and in the latter insists on the incomprehensibility of mysteries. There does not seem to be any strong basis for arguing that Abelard's thought developed on this point. Rather, we must consider his main target of the moment and the balance he strives to maintain.

For Abelard God is the source of all knowledge of faith. There is, however, much ambiguity in his position on the Trinity. Is the Trinity a mystery which can be attained only on the basis of faith or is it accessible to natural reason? Abelard says at least that the Trinity of Persons has been revealed by God through the Jewish prophets and through Gentile philosophers. He says explicitly that Plato came closest to the Christian faith in this matter. It is true that he makes this claim on the basis of what he has read in the Fathers. What is more important, however, is Abelard's insistence that if pagan philosophers possessed knowledge of the Trinity, this was because God had revealed it to them. But what does he mean by revelation? In commenting on the Epistle to the Romans 1:20, a passage where Paul says that the invisible things of God were available to the Romans through the visible things of this world, Abelard seems to mean by revelation that God's nature and the Trinity of Persons can be known by reflecting on God's effects. Scholars are at odds on the significance of such assertions. Some see here an explicit contradiction with Abelard's teaching elsewhere that a mystery cannot be comprehended; others try to distinguish between factual awareness of the mystery and comprehension of it. The defender of Abelard's orthodoxy on this point has his work cut out for him.

D. Abelard's Ethics

Abelard was more impressed by the pagan philosophers in the area of morals than in divinity, however, and it is in their ethical concern that he sees the center of their philosophical effort. Indeed, the pagan philosophers were far closer to Christianity than were the Jews, because Jewish law is largely an

external matter, whereas pagan philosophers saw the importance of interior justice, of chastity, of contempt for this world. This assessment of pagan moral philosophy betrays the salient feature of Abelard's own moral teaching.

In his *Ethics*, the subtitle of which is *Know Thyself*, Abelard insists on the priority of the interior in morals. It is our inner intention, our consent, that makes an external act good or bad and not vice versa. The measure of morality thus firmly located in the interior act, in intention and consent, Abelard had opened himself to the charge of subjectivism. And indeed among the propositions condemned at the Council of Sens was the contention that those who crucified Christ did not sin because they acted in ignorance, and we cannot ascribe guilt to those who know not what they do. Another stated that a man is not made good or bad by his acts, presumably his external actions.

Is Abelard's ethical position subjectivistic? He appears to be maintaining, not that our interior intention constitutes what is good or bad, but rather that one must personally recognize what is good and bad. The emphasis, nevertheless, is on the inner man. Purity of heart is what must be striven for, since, once it is had, perception of the true end is possible. Abelard will distinguish between vice and sin. Vice is a tendency to perform bad actions, but sin consists in consenting to the tendency and acting in accord with it. What this distinction enables Abelard to do is to separate himself from the position that sin resides in the will. This is not so, he maintains; it is possible to sin while having a good will or disposition and not to sin while having a bad will. The nub of the matter lies in consent and intention.

Abelard's ethical doctrine, with its strengths and weaknesses, has always been regarded as remarkable and as a further sign of his genius. It would not be long before ethical discussions would be carried on within the framework of the *Nicomachean Ethics*, something which will enable them to advance rapidly beyond the tentative steps Abelard took. And yet, when it is considered that Abelard's work on ethics was composed in the almost complete absence of a model, we must marvel at his accomplishment. This is not to say, of course, that Abelard did not draw on previous authors. The point is rather that, surprising as it may seem, there is a case that can be made for the contention that Abelard was more original in ethics than in logic. His logical writings consist either of glosses or commentaries on the writings of others or, in his *Dialectica*, a more or less independent presentation of the contents of the works which had been previously commented on and largely in the same order as the commentaries. The *Dialectica* is, so to speak, a commentary without the text. The *Ethics* is

Abelard himself, from beginning to end: the form is his, the problems and their order are his. If, as is charged, he emphasized the subjective in such a way that he seems to cut it adrift from adequate criteria of good subjectivity and bad subjectivity (a charge which is surely an exaggeration), it would have to be added that such an emphasis is always a salutary balance to the tendency to an excessive exteriorization of the criteria for good and bad action.

Bibliographical Note

A critical edition of the *Historia calamitatum* by J. T. Muckle in *Mediaeval Studies*, 12 (1950), pp. 175-211. The *Logica ingredientibus* and *Logica nostrorum petitioni* in *Philosophische Schriften*, ed. Geyer (Munster, 1919-1933); *Introductiones parvulorum in Scripti Filosofici*, ed. Mario Dal Pra (Rome, 1954); *Dialectica*, ed. De Rijk (Assen, 1956); *Abaelardiana inedita*, ed. Minio Paluello (Rome, 1958). See M. T. Beonio-Brocchieri Fumagalli, *La logica di Abelardo* (Florence, 1964); W. and M. Kneale, *The Development of Logic*; Boehner, *Medieval Logic* (Chicago, 1952); E. A. Moody, *Truth and Consequence in Medieval Logic* (Amsterdam, 1953); Otto Bird, "The Logical Interest of the Topics as Seen in Abelard," *The Modern Schoolman*, 37 (1959), pp. 53 -- 57; Meyrick Carre, *Realists and Nominalists* (Oxford, 1946). Abelard's theological works can be found in Migne's *Patrologia latina*, vol. 178. Abelard, *Christian Theology*, trans. J. Ramsay McCallum (Oxford, 1948). J. G. Sikes' *Peter Abailard*, first published in 1932, reissued in 1965, is still the best general work. Maurice de Gandillac's introduction to his edition of *Oeuvres choisies d'Abelard* (Paris, 1945) is brief, delightful, and informative. A must is É. Gilson, *Heloise and Abelard* (Chicago, 1951).

Chapter IV

The School of Chartres

A. From Fulbert to Bernard

The fame and influence of the cathedral school of Chartres during the twelfth century is beyond dispute. The writings of the men whom we shall consider in this chapter would be sufficient argument for the importance of the school, but we have as well the unstinted praise of John of Salisbury, himself a notable figure, who records the merits of the men under whom he studied at Chartres. There is, moreover, the opposition to Gilbert of Poitiers and William of Conches by St. Bernard of Clairvaux and William of St. Thierry, an opposition whose vigor witnesses to the importance of the target. Furthermore, there is the rivalry between Chartres and St. Victor at Paris to indicate that in the twelfth century the school of Chartres was widely recognized as a force to be reckoned with. Before speaking of the character of the twelfth century, something must be said of the first real fame of the school during a period straddling the millennium.

Fulbert (c.960-1028), who was bishop of Chartres from 1006 until his death, is generally recognized as the man who put Chartres on the map of medieval education. He was not, of course, the founder of the school. Fulbert studied under Gerbert at Rheims, came to Chartres about 990, and for about ten years was an assistant in the cathedral school. It is unlikely that Fulbert continued to teach after becoming bishop, but under his patronage the school achieved great fame. Its purpose was to prepare young men for the clerical life, and there was no ideal of a general secular culture. The course of studies was based on the liberal arts, and Fulbert himself seems to have known some medicine. The quality and scope of instruction at Chartres under Fulbert have probably been greatly exaggerated, as have the accomplishments of Fulbert himself. Chartres was not the only center of learning during the time of Fulbert, and the schools at Liège and Cologne were undoubtedly more advanced in mathematics than was Chartres. Qualifications in the usual estimate must be made accordingly, but when adjustments are made for the excessive praise of his contemporaries at Chartres, the fact still remains that Fulbert presided over a definite strengthening of the cathedral school. It should be noted that Berengar of Tours, who was to provoke a lively theological controversy, studied at Chartres under Fulbert, although it is doubtful that Fulbert himself was then in the classroom.

A new flowering of the school took place under Bernard of Chartres (died before 1130), of whose teaching we know through John of Salisbury, although John himself had studied, not under Bernard, but rather under two of his pupils, William of Conches and Bernard Bishop. John gives us a description of Bernard's method of teaching grammar. There are, he writes, four things which are of chief importance in the pursuit of philosophy and the exercise of virtue. They are reading, doctrine, meditation, and good works. The first three lead to knowledge, and from knowledge good works flow; by the same token, the cultivation of virtue naturally precedes the quest for knowledge. Grammar is the foundation for and presupposition of all else and must therefore be learned first. Thus, reading (*lectio*) is the first step in the study of philosophy. In what does this reading consist? John suggests a distinction between prereading and reading, the former being the task of the teacher in the classroom, the latter solitary reading. Now what the grammaticus does in the prereading is this: he breaks the text into parts of speech, explains the metrics when it is verse, points out barbarisms and other breaches of the rules of language, explains tropes and figures of speech. A grammaticus like Bernard apparently employed the prereading as an occasion to discourse about all the arts.⁽¹⁾ John tells us that he would assess the arguments of the text (logic), comment on its eloquence and persuasiveness (rhetoric), and, when the text permitted it, expatiate on the quadrivium of mathematics and on physics and ethics. John assures us that this is the desirable way of prereading the *auctores*, the authors who came to function as authorities.

When he mentions the doctrine of the *Timaeus* of Plato, according to which the coming to be of the things of this world involves Ideas and matter, John of Salisbury calls Bernard the best Platonist of his time. He quotes some verse of Bernard in which a distinction is made between what is not and what truly is. What truly is comprises God, the Ideas, and matter.⁽²⁾ Of these three, God alone is unqualifiedly eternal, since Bernard is reluctant to speak of matter and Ideas as coeternal with him. John quotes a few lines from Bernard's exposition of Porphyry which cast some light on this. "There are two kinds of effect of the divine mind, one which he creates from a subject matter or which is created along with it, another which he makes of himself and contains in himself, requiring no outside aid. The heavens indeed he made in his intellect from the beginning, and to form them there he needed neither matter nor extrinsic form." (*Metal.*, IV, 35) The Ideas appear to be the patterns of external divine creativity, but as Ideas they are described as *velut quidam effectus*: as certain effects. John returns to the Platonism of Bernard in another text. "He posited Ideas, emulating Plato and imitating Bernard of Chartres, and said that apart

from them there is no genus or species. An Idea, in the definition of Seneca, is an eternal exemplar of those things which come to be by nature. And since universals are not subject to corruption nor alterable by movements . . . they are truly called universals." (*Metal.*, II, 17) A common noun, then, names an unchanging reality, an Idea contained in God, though an effect of God and not quite coeternal with him; as for the sensible things around us, John agrees with Plato that they "await no naming due to their instability."⁽³⁾

The Platonism John of Salisbury attributes to Bernard is a common characteristic of the school in the twelfth century, and its source is, aside from the information that could be gleaned from the Fathers (principally Augustine) and Boethius, the *Timaeus* as translated and commented on by Chalcidius. Another source of Chartrian Platonism was Macrobius' commentary on Cicero's *Dream of Scipio*. Moreover, there is evidence that the so-called Hermetic writings exercised an influence on the school. The Plato of the *Timaeus* is of course a philosopher seeking to explain the cosmos. During this time Plato, as natural philosopher, is often contrasted with Aristotle, the dialectician. When the Chartrian thinkers employ Plato, it is to aid in understanding the content of their faith: the *Timaeus* is considered to be an explication of Genesis. In short, we must not expect to find in the twelfth-century school of Chartres anything like a clear distinction between philosophy and theology. The problem here, as with Anselm of Canterbury, is rather one of applying reason to faith in order to occupy a middle ground between the simple acceptance of what God has revealed and the full knowledge of truth. Full knowledge is not something that can be attained in this life. The pertinent dyad, then, is faith and reason. As a school, the men of Chartres are convinced that they have an obligation not only to believe but to understand, to the degree that this is possible, the contents of their belief. This approximation to an understanding is gained by appeal to such works as the *Timaeus*. In this effort they quite often offended the sensibilities of others who felt they were compromising the clear intent of revelation and ridiculing the faith of the simple. Bernard of Clairvaux and William of St. Thierry are as shocked by some Chartrians as they are by Abelard himself -- and often with good reason. Bernard and William feel that the way to explicate Scripture is to have recourse to the Fathers, not to pagan philosophers.

We will see some particular points of dispute later in this chapter and in the next; from a distance of centuries, and with the intermediary of much development in theology, the modern reader finds himself drawn sometimes to the side of the antidialecticians, sometimes to that of the dialecticians. There

were excesses on both sides, to be sure; perhaps the greatest temptation to the historian is to look with lofty condescension on the whole dispute. That attitude is not a serious possibility for one who senses the utter seriousness of what is at issue in the clash of the dialecticians and antidialecticians. Perhaps the best attitude here is suggested in a remark attributed to Bernard of Chartres by John of Salisbury. "We are like dwarfs sitting on the shoulders of giants; we see more things and more distant things than did they, not because our sight is keener nor because we are taller than they, but because they lift us up and add their giant stature to our height." (*Metal.*, III, 4) T. S. Eliot put the same thought more succinctly in replying to those who say we ought not read the old authors because we know so much more than they did: "Yes," Eliot said, "And they are what we know."

By stressing the efforts at cosmology at Chartres we do not mean to suggest that the schema of the seven liberal arts no longer provided the basic pattern of education. It did. But what differentiates the twelfth century from earlier ones, and what justifies calling it a renaissance, is the fact that the various arts were no longer considered to be summed up in encyclopedias or collections of statements by ancient authors. Each of the arts now achieves new vigor thanks to the introduction of fundamental works dealing with each of them. Pagan authors hitherto unavailable were read avidly, and with the increase of such material for the study of each of the arts there was a natural tendency toward specialization. The ideal of a cycle of education, a panoramic view of things to be gained by moving through each of the arts and arriving finally at a reading of Scripture, became jeopardized. From quite different viewpoints both William of Conches at Chartres and Hugh of St. Victor in Paris would speak out against the tendency to specialize, against the demand for a "quickie" course. When we realize that John of Salisbury devoted twelve years to study, moving from master to master, from school to school, we get a picture of what was thought to be necessary for an adequate education. (Of course, there were not as yet set courses of study in the manner of the universities to come at the end of this century.) Thus, the dialecticians had enemies other than the antidialecticians; these others are the adversary John of Salisbury dubs with the name of an opponent of Virgil, Cornificius. The Cornificians wanted to be propelled through their studies in three, perhaps even two years; they wanted the emphasis put on the practical and useful, on what it takes for a man to get ahead in the world. The controversy was not merely one of educational theory. William of Conches actually had to give up teaching under the onslaught of Cornifician demands.

The men we shall now discuss are of great, if unequal, importance in the effort, which intensifies in the twelfth century, to conjoin faith and reason, in the phrase of Boethius. The old structure of the seven liberal arts as a preparation for biblical studies is retained, but it begins to be altered somewhat insofar as the Stoic division of philosophy into dialectics, physics, and ethics, and the Aristotelian division according to theoretical and practical sciences, takes on a growing meaning with the advent of more substantive ancient philosophical works. But no ultimate clarity with respect to a division between philosophy and theology is reached by the masters of Chartres.

Bibliographical Note

On the school of Chartres the following are fundamental: A. Clerval, *Les écoles de Chartres au moyen âge du Ve siècle au XVIe siècle* (Paris, 1895); R. L. Poole, "The Masters of the School of Chartres in John of Salisbury's Time," *English Historical Review*, 35 (1920), pp. 321-342; C. Paré, A. Brunet, P. Tremblay, *La renaissance du XIIe siècle: Les écoles et l'enseignement* (Ottawa, 1933); F. M. Powicke and A. B. Emden, Rashdall's *Universities of Europe in the Middle Ages* (Oxford, 1936), vol. 1, chap. 2; J. M. Parent, *La doctrine de la création dans l'école de Chartres* (Ottawa, 1938); Tullio Gregory, *Anima mundi: La filosofia di Guglielmo di Conches e la scuola di Chartres* (Florence, 1955); C. H. Haskins, *The Renaissance of the Twelfth Century* (New York: Meridian Books, reprint, 1957); M. D. Chenu, "Les Platonismes du XIIe siècle," in *La théologie au douzième siècle*, Études de Philosophie Médiévale, XLV (Paris, 1957); William Harris Stahl, *Macrobius' Commentary on the Dream of Scipio* (New York, 1952); Raymond Klibansky, *The Continuity of the Platonic Tradition During the Middle Ages* (London, 1939). On Fulbert see Loren C. MacKinney, *Bishop Fulbert and Education at the School of Chartres* (Notre Dame, 1957).

B. Gilbert of Poitiers (1076-1154)

Gilbert, a native of Poitiers, studied first at Chartres and then at Laon under Anselm. He started his teaching career in his native city but returned to Chartres as a teacher, becoming chancellor of the school in 1126. He held this post until 1138, and seems to have taught at Paris as well (John of Salisbury is our authority for that). In 1142 he was named bishop of Poitiers. In 1147 and 1148 his views on the Trinity were called into question, and he publicly retracted some of his statements.^[4] These difficulties did not affect his reputation in his own day or his influence on men of the thirteenth century. Gilbert is often cited by the French and Latin versions of his name, which are,

respectively, Gilbert de la Porée and Gilbertus Porretanus (or Gilbertus Pictaviensis). The works of Gilbert which are of unquestioned authenticity are his commentaries on the theological tractates of Boethius. The *Liber de sex principiis* was attributed to Gilbert, but most scholars express deep doubt that it is his.

Because Gilbert commented on the *De trinitate* of Boethius, we need only turn to his remarks on the three types of speculative science mentioned in chapter two of that work to find Gilbert's views on the scope and divisions of philosophy. He begins by observing that speculative sciences are opposed to practical science. In a speculative science we ask whether something is, what it is, what its properties are, and what its causes are (*intuemur an sint, et quid sint, et qualia sint, et cur sint singula creata*). (PL, 64, 1265C; Häring, p. 46) An active or practical science is ordered to operation, says Gilbert, who cites medicine and magic as examples. Having given these definitions, Gilbert sets aside practical sciences and says he will be interested only in the speculative. The first division of speculative science which he introduces is the familiar tripartite division into physics, ethics, and logic, and it is clear that for Gilbert moral science and logic are speculative sciences. He puts these two to one side now and, retaining only physics, says that what Boethius is doing in the text is giving us a subdivision of physics, or natural science. Physics is thus a generic name. one of whose species is also called physics, or natural science; the other two species of course are mathematics and theology. Scotus Erigena, at this point in his commentary, had linked the quadrivium with mathematics, but Gilbert makes no effort to connect the divisions of philosophy with the liberal arts. What is the principle of division whereby we arrive at physics, mathematics, and theology? "He describes these through motion, separation, and their contraries, placing a twofold difference in the definition of each." (1265C) Gilbert indicates that the threefold division of speculative science given by Boethius is not a reference to three kinds of existing things. "It is not only as they are, but indeed sometimes otherwise than as they exist, that some things are often truly conceived. That is why the mind's speculation is divided and denominated either on the basis of the things inspected or on the manner of inspection." (1267A) When Gilbert turns to Boethius' remark that natural science is concerned with things in motion which are inabstract or inseparable, he proceeds to explicate this with reference to matter, because natural science considers forms together with their matter, and goes on to give a list of meanings of the term "matter." Moreover, he follows this up with a discussion of several meanings of the term "form."

In the first place, "matter" means that origin of all things that Plato calls necessity, receptacle, womb, mother, and the locus of all generation; his students call it *hyle*, that is, building material (*silva*), while Plato himself called it *prime matter*. Second, the four elements -- fire, air, earth, and water -- are called matter. Third, specifically different bodies -- like bronze, wax, and stone -- are called matter. Fourth, general and special subsistencies may be called matter. Now, this fourth type would seem to be peculiar to Gilbert, at least with respect to the term he uses; what he is referring to here are the common predicates which are genera and species and out of which, as out of something material, particular things may be thought to be constituted. Particular things exist owing to these subsistencies, but the subsistencies may be said to be owing to the existence of that which is constituted out of them. We will have to return to this.

"Form," too, has many meanings. First of all, it means the essence of God, the artificer due to whom whatever is something and whatever is a being is. "Nam essentia Dei, quo opifex est quicquid est aliquid, et quicquid est esse, unde illud aliquid est, et omne quod sic inest ei quod est aliquid, ut ei quod est esse adsit, prima forma dicitur." (1266B) Second, it refers to the forms of the four elements, which are as Ideas or exemplars to those unions of concrete form and prime matter which result in the four elements as they are named matter. Such forms Gilbert calls *substantiae sincereae*. Third, that whereby subsistent things are something, namely, subsistencies, is called form. For example, corporeality is the subsistency thanks to which body is body. Finally, the fourth species of quality, namely, the shape or figure of bodies, is called form.

Of those things called matter there is one kind which is unformed and simple, namely, prime matter; there is another kind which is complex, for example, body. Only the first two meanings of "form," God and the Ideas, or exemplars of the four elements, signify *substantiae sincereae*. In order to understand how Gilbert can speak of the four elements as true or pure (*sincerae*) substances, we must distinguish between the four elements and those imitations of them perceived by the senses. The pure forms, or Ideas, dwell in a region apart. (1266D) What we perceive possesses, not such a pure form, but rather an engendered form, a *forma nativa*. The *forma sincera* is naturally separate from matter, and it is only its image, the form of this composite, which is in sensible objects. The *forma nativa* is a participation in the pure form and therefore has its origin from it. The *forma nativa* which gives being to sensible body is not truly a form.

Before looking into what this means for the status of universals -- and we will find Gilbert drawing the consequences for us -- we must first see how he employs all this to explicate what Boethius had said of the distinction between the three speculative sciences.

Matter taken simply is not formed; pure forms are not in matter. Where matter and form are conjoined in sensible things, there is motion. It is formed matter which we first know, since it falls under the senses, but in knowing composites reason can abstract the forms from their matter, constructing in the process a concept of matter and a concept of form. The form thus abstracted is freed from motion and thus imitates things which can exist separately from motion and from matter. Primary matter and the primary form which is the substance, or *ousia*, of the creator, and the Ideas of sensible things, require neither forms nor matter in order to be and thus lack motion. (1266D) The form that is abstract thanks to an operation of our minds is not the *forma sincera*. It is because forms thus abstracted are considered otherwise than as they exist (*aliter quam sint*) that concern with them belongs not to physics but to mathematics. Gilbert is quite explicit that mathematics is concerned with native forms, but he considers them in a manner other than that in which they exist. He suggests a dependence of physics on mathematics in that the latter deals with corporeality and width, knowledge of which is presupposed by a physical concern with body and wide things. Having accounted for two speculative sciences by saying that physics deals with native forms along with their proper matter, while mathematics deals with native forms abstractly, Gilbert goes on to speak of theology. Theology goes beyond native forms to deal with true and pure forms (*formae sinceræ*). By intellectual intuition the mind, in theology, looks to God, to the exemplar Ideas and to simple or primary matter. In theology, in other words, the mind attains to what is simple, without matter, immobile, and eternal.

We have already seen Gilbert make reference to abstraction. The *forma nativa*, he holds, cannot exist apart from its proper matter; however, it can be considered apart by our mind (*ratione*). What is thus abstracted by the mind must, it would seem, be distinguished both from individual substances and from the Ideas. We have already alluded to the curious terminology Gilbert employs when he distinguishes particular existents and universals. Individual things are subsistents and substances, for they "stand under" accidents; for example, this body is the "support" of this color. Besides substances there are subsistencies.⁽⁵⁾ That this distinction is important for determining the status of universal is clear from the fact that Gilbert calls universal subsistencies. "Therefore genera and

species, that is, general and special subsistencies, only subsist and are not truly substances [*non substant vere*], for accidents inhere neither in genera nor in species. That which is requires accidents in order to be, but genera and species have no need of accidents in order to be. It is individual things which truly subsist, for individuals no more than genera and species require accidents in order to be. That this is true of individuals supposes that they are already informed by the proper and specific differences whereby they subsist. However, they do not only subsist; individuals are substances as well since they confer being on accidents; while they are subject to these accidents, they are, in the reasonable order of creation, their causes and principles." (1375C)

Does this mean that genera and species exist apart from individuals? To say that they are not substances is simply to deny of genera that they are supports of accidents. There seems to be every reason to say that Gilbert had no intention of giving separate existence to genera and species. He speaks of universals as what our mind collects (*colligit*) from particulars.¹⁶¹ Universality is something which seems to be the sense of John of Salisbury's remark: "He attributes universality to native forms. . . . A native form is an example of an original form and it is not something in the divine mind but inheres in created things. This is what the Greeks called "eidos" (form) and is to the Idea as example to exemplar. It is sensible in the sensible thing but insensible as conceived by the mind; singular in singular things but common to all." (*Metal.*, II, 17) Thus, it is amply clear that Gilbert does not identify universals and the divine Ideas, but it is seemingly by appeal to those Ideas that he justifies the applicability to individuals of the universal which has been collected from them, for the individuals are similar owing to imitation of the same Idea.

The verb and derivative noun "colligere" and "collectio" that Gilbert uses when he talks about universals are rather difficult to interpret. Does he mean that the mind gathers together the similarities to be found in the individuals and ends by forming an abstract concept common to all the individuals? Or is he identifying the universal with the collection or class of all similar individuals? De Wulf seems to adopt the second alternative. "The genus and species are the sum total of the beings in which those similar realities (subsistencies) are found, belonging in proper to each of them." His basis for this interpretation is the text quoted earlier. It is probable, I think, that this is what Gilbert intends; if it be what he intends, if Gilbert holds that the species is a class, then "man," for example, would stand for the class of all men. On this interpretation, to say "Socrates is a man" would have to be unpacked in the following manner:

Socrates belongs to the class of those objects called "man." Of course, this is not to say that "man" signifies "to be a member of the human class."

Gilbert himself approaches it as follows. Wishing to contrast the way in which "God" is predicated of Father, Son, and Holy Ghost to the way in which "man" is predicated of three individuals, say Plato, Cicero, and Aristotle, he says that in the case of the Persons of the Trinity, although what is predicated is predicated of numerically diverse Persons, there is a repetition not merely of the predicate but of the *res* signified by the predicate. This is not the case when Plato, Aristotle, and Cicero have "man" predicated of them. The word is repeated, of course; *rem tamen predicatum non repetunt* (the same *res* or reality is not repeated): *sed quamvis conformes, tamen diversas: imo quia conformes, ergo numero diversas a se invicem natures de numero a se diversis affirmant, et haec trium de tribus praedicatorum necessaria differentia non patitur hanc adunationem, ut dicatur, Plato et Cicero et Aristoteles, sunt unus singulariter homo* (but though similar, yet diverse; indeed, because similar, therefore natures numerically diverse from one another, and this necessary difference between the predicates of the three prevents the unity which would lead to saying Plato, Cicero, and Aristotle are one single man). (1262B) Gilbert seems to want to read these affirmative propositions thus: Plato is this man, Cicero is this (other) man, and Aristotle is this (yet other) man. Thus, the *res* signified by the apparently common predicate, "man," is different in the three affirmations. Oddly enough, this makes Gilbert sound like a nominalist, and yet he is traditionally classified as a realist. Like most of us, he seems to have been a bit of both.

Perhaps his extremely nuanced views can be summarized as follows. Consider the statement "Socrates is a man. The predicate of that sentence can be regarded in at least three ways by Gilbert: (1) it refers to this singular instance of human nature which is Socrates, (2) it refers to the divine creative Idea which is more real and out-there than Socrates himself, (3) it involves an *intellectus*, or concept, that the mind has formed against the background of experiencing that Socrates is like Cicero, Aristotle, and so on. Now, if we ask if this third thing, this concept, answers just as such to something out-there, independent, real, but neither the divine Idea nor this singular human being or that, we are led inexorably to Gilbert's notion of subsistency. Do subsistencies exist? Does human nature exist elsewhere than in individuals, where it is associated with collections of accidents which are signs of, if not causes of, that nature's individuation? There is no simple answer to this question in Gilbert of Poitiers. Subsistencies exist in individuals that are also substances. Gilbert *seems* to say that that is the only way subsistencies can exist. He wants to avoid saying that

my concept of such a subsistency as human nature commits me to the view that there is some numerically one *res* existing in, say, Socrates, Cicero, and Plato. Many men are specifically but not numerically one. They are specifically one because they are *conformes*. Is not the concept the expression and recognition of that conformity, and are not the objective bases and guarantees of the concept singular men and the divine Ideas? If this suggests only that the utmost caution must be exercised in applying labels like "realist" or "nominalist" to Gilbert of Poitiers, my purpose will have been attained.

Bibliographical Note

Gilbert's commentaries on Boethius can be found in PL, 64, accompanying Migne's presentation of the Boethian texts. More recent edition: Nikolaus Haring, "The Commentaries of Gilbert, Bishop of Poitiers, on the Two Boethian *Opuscula sacra* on the Holy Trinity," in *Studies and Texts, Nine Medieval Thinkers*, ed. J. Reginald O'Donnell, C.S.B. (Toronto, 1955), pp. 23-98. The *De sex principiis* is found in PL, 188, 1257-1270. Secondary literature: A. Forest, "Le réalisme de Gilbert de la Porée dans le commentaire du *De hebdomadibus*," *Revue Néoscholastique de Louvain*, 36 (1934), pp. 101 -- 110; N. Haring, "The Case of Gilbert de la Porée, Bishop of Poitiers," *Medieval Studies*, 13 (1951), pp. 1-40. For the influence of Gilbert see J. DeGhellinek, *Le mouvement théologique du XIIe siècle* (Paris, 1948), pp. 175-180.

C. William of Conches (c.1080 - c.1184)

William, a native of Conches in Normandy, studied under Bernard of Chartres and stayed on at the cathedral school as a teacher of grammar. He speaks of having taught for twenty years and more, and his teaching was at last interrupted by the Cornifician controversy. Did he resume his teaching career? Tullio Gregory conjectures that he did not. The Cornificians were routed we know, but William had been charged with heresy by William of St. Thierry, and it is not impossible that, soured by this, he retired to his native Normandy, where he wrote his *Dragmaticon* under the protection of Geoffrey the Fair, Count of Anjou and a Plantagenet. An early work of William's, which he calls simply *Philosophia*, is printed as *De philosophia mundi* among the works of Venerable Bede, and it is to be found as well among those of Honorius of Autun. The *Dragmaticon*, a more mature work, takes into account the objections that had been made to the earlier systematic work; indeed, William formally retracts a number of positions he had held as a younger man. We have as well some glosses on Boethius' *Consolation of Philosophy* as well as on the *Timaeus* of

Plato. The *Moralium dogma philosophorum* (*Teachings of the Moral Philosophers*) has been attributed to William, but it is quite doubtful that this anthology is actually his. In the glosses on Boethius William announces his intention to comment on Macrobius and Martianus Capella, but these glosses have not been found, if indeed he wrote them. For what comfort we may want to derive from it, books which are announced as forthcoming only to appear tardily or not at all are not a twentieth-century achievement.

Division of Philosophy. In his glosses on Boethius, William provides us with a schema of the sciences which tells us a good deal about his own predilections. There are two kinds of science, he begins, wisdom and eloquence. Wisdom is true and certain knowledge of things; eloquence is the science of expressing in ornate words and sentences what is known. William likes to quote Cicero on the relative value of these two. In the *De inventione* Cicero warns that eloquence without wisdom is dangerous, whereas wisdom without eloquence, while it can accomplish something, can accomplish much more with it. Consequently, both eloquence and wisdom are important, but wisdom is preeminent. Philosophy and wisdom are identical (*sapientia vero et philosophia idem sunt*). Eloquence, therefore, is an aid to and a requirement for philosophy, but not actually a part of it. The term "eloquence" is here taken to cover the arts of the trivium, but wisdom is not equated with the arts of the quadrivium. When he turns to wisdom, William introduces the Aristotelian distinction between theoretical and practical sciences; the former are pursued in contemplation by the leisured (*otiosi*), while the latter are the concern of the busy (*negotiosi*). The practical sciences are economics, politics, and ethics; the theoretical sciences are physics, mathematics, and theology. The arts of the quadrivium show up as subdivisions of mathematics. William takes a certain pedagogical pleasure in translating the divisions mentioned to diagram form.

- Knowledge
 - Eloquence
 - Grammar
 - Rhetoric
 - Dialectic
- Wisdom
 - Theoretical
 - Theology
 - Mathematics
 - Arithmetic
 - Music

- Geometry
- Astronomy
- Physics
- Practical
 - Ethics
 - Economics
 - Politics

A further division of music is given in the text of the glosses on Boethius.

Given now that philosophy comprises all these various sciences, can one begin just anywhere? "This is the order of learning," William writes at the end of *Philosophia*. "Because all teaching employs eloquence, we should first be instructed in eloquence. But there are three parts of it: to write correctly and correctly to pronounce what has been written; to prove what needs proving, which is taught in dialectics; to adorn words and sentences, and this rhetoric teaches. Therefore, we should be initiated in grammar, then be taught dialectics, and afterward rhetoric. Armed with these, we should proceed to the study of philosophy. The order to be followed here is such that we should first be instructed in the quadrivium, and, in it, first in arithmetic, secondly in music, thirdly in geometry, finally in astronomy, and thence in Holy Writ so that we might, from knowledge of creatures, come to knowledge of the creator." In the glosses on Boethius the order of learning is expressed somewhat differently by William. Speaking of the sciences which fall under wisdom, he says that one should first study the practical sciences and after that turn to contemplation. First, we contemplate corporeal things in our study of mathematics and physics, and then we move on to the incorporeal in theology.

Our Knowledge of God. In the preface to his *Philosophia* William says that he will begin with the first creation of things and continue the discussion until he reaches man, of whom he will have much to say. Philosophy is concerned with two sorts of thing, the invisible and incorporeal, on the one hand, and, on the other, the visible and corporeal. We begin with the first, and our discussion will bear on the creator, the world soul, angels, and human souls. The first concern of all will be God. Immediately we encounter difficulties. When we seek knowledge, William observes, there are eleven questions we can ask. Of the object at issue we must first ask if it exists; if this is answered in the affirmative, there remain ten further questions based on the Aristotelian categories: What is it? Of what kind? and so forth. But none of these questions seems to be pertinent when we are seeking knowledge of God. We must conclude that

whatever knowledge we have of him will be both imperfect and indirect, and William suggests that there are two kinds of argument that can be devised to provide knowledge of God, one based on the creation of the world, the other on its daily course. The argument from creation is as follows. The world is made up of contrary elements -- hot, cold, wet, and dry -- and their compounding is due either to the operation of nature, or to chance, or to some artificer. But nature avoids the contrary and seeks the similar, so the conjunction of contrary elements cannot be ascribed to nature. Nor can chance be the cause, since, in the first place, if chance could cause the world, it is surprising it does not produce simpler effects like houses. William's more serious opposition to chance as the cause of the world is based on the explanation of chance which Boethius gives in the *Consolation of Philosophy*. According to that view of it, chance is an unlooked-for result of the crossing of two lines of causality; thus, if chance is the cause of the world, there are causes prior to the first cause of everything. But only the creator antedates the world, William says, so chance is out and the cause of the world must be some artificer. Could it be man or an angel? No, for man appears in a world already made, and angels are made simultaneously with the world. Consequently, God alone created the world.

When he is commenting on the *Timaeus*, William has no difficulty in interpreting the demiurge there described as God the creator; nor does he have any difficulty with the rather clear implication of the text that the demiurge finds a material chaos ready at hand, which he then fashions after the patterns of the Ideas into sensible things. For William, as for his contemporaries, the *Timaeus* is a creation story and, as the product of a pagan philosopher, a remarkable corroboration of what is revealed in Genesis. Whatever comes to be requires a cause; the world has come to be and its cause is the creator. But there are four kinds of cause: formal, efficient, final, and material. William proposes that we divide the causes into two classes. On the one side we have as efficient cause the divine essence, as formal cause the divine essence, as final cause the divine goodness. On the other side we have the four elements as material cause. The efficient, formal, and final causes are one with God, and there is no principle of his existence; we can say of these three causes of the world that they are eternal and uncaused, where by eternal we mean, not unending survival through time, but being free from time's tenses utterly. The eternal has no past and no future, and we can speak of it as always in the now or present. The material cause of the world, like everything fashioned from the elements, has a principle of its being. Matter, then, is a caused cause. This approach to creation through the *Timaeus* ends with the dyad creator and created. God depends on nothing outside himself in his act of creative causality.⁽⁷⁾ The Ideas to which the

demiurge looked as to entities independent of himself are now equated with the divine wisdom. The archetypal patterns of created things are explained by appeal to Augustine's interpretation of Plato's Ideas. Whoever sets out to make something works up in his mind beforehand what he would effect. The archetypal patterns of creatures, the Ideas, are one with the wisdom of God. So too, matter, or chaos, is not something which awaits the divine causality as if it could exist apart from that causality. Everything other than God is an effect of God. Others in interpreting Plato here had spoken of chaos as the first effect of God out of which order gradually emerged. William of Conches emphatically rejects that view; he feels it is heretical and prejudicial to the divine goodness. Men like Hugh of St. Victor thought that God's gradual imposition of order would reveal the divine goodness rather than call it into question. The opposing views bear on Genesis as much as on the *Timaeus*, of course; the scriptural account speaks of God laboring for six days in creating the world. William thinks we ought not to think of six literal days here, whereas Hugh resists the view that the hexameron has merely figurative import. (See J. Taylor, p. 227, n.3.)

Given that the world has been created by God and that nothing other than God (save evil) escapes the divine causality, are we to say that the world has always been or that it had a beginning in time? If time measures the alterations of material things, time and material things come into being together, and we can say that there was no time when the world was not. This does not amount to the assertion that the world is eternal, however, if eternity is the prerogative of a being fully in possession of its perfection and thus beyond time. The second proof of God's existence that William offers is drawn from the daily disposition of the world. Beginning with the observation that the things of this world are wisely disposed -- that is what "world" means -- he points out that this presupposes a wisdom responsible for it. There are three possible candidates: human, angelic, or divine wisdom. It can hardly be human wisdom; nor can it be the wisdom of some angel, since angels too are wisely ordered and what wisdom would be responsible for that? There remains only the divine wisdom. "This is the formal cause of the world, because according to it he forms the world by creation. Just as an artisan when he wishes to make something first conjures it up in his mind and then, having found the right material, works in accord with his conception, so the creator, before he creates anything, has it in his mind and then accomplishes it in an effect. It is this that Plato calls the archetypal world because it contains whatever is in the world; 'archetype,' that is, originative form, for 'archos' is first, and 'typos' form or figure." (*In Tim.*, cited by Parent, p. 50)

The World Soul. The demiurge in Plato's *Timaeus* is said to make but one world because he fashions the world after the model or Idea of living creature. The Idea of living creature contains within itself the Ideas of the many and various things found in the world. If the model for the world is the Idea of living creature, then the world as a totality can be spoken of as a living thing, a cosmic animal, and there will be a world soul.

What does William of Conches make of this notion of the world soul? There are, he notes, various possible interpretations. "According to some, the world soul is the Holy Ghost, for, as we have said, it is owing to the divine will and goodness, which the Holy Ghost is, that all the living things of this world live. Others say that the world soul is the natural force (*vigor*) which God has put in things whereby some only live, some both live and sense, some live, sense, and understand (*discernunt*). For there is nothing which lives or senses or understands in which such a natural force is not found. Yet others say the world soul is some incorporeal substance which exists as a whole in every body, although, because of the dullness (*tarditatem*) of some bodies, it does not effect the same thing in all. . . . Thus in man there would be both his own soul and the world soul, from which one might conclude that man has two souls. We think this conclusion is false, however; the world soul is not a soul anymore than the head of the world is a head. Plato speaks of it as being excogitated from the indivisible divine substance, composed of the same and the different: if one wants to know what that means, let him consult other works of ours."

(*Philosophia*, I)

Now, William of Conches' own interpretation is (not without qualification) the first one given in his list, but the problem of the world soul leads us inevitably to his statements on the Persons of the Trinity, statements which called forth objections from such critics as William of St. Thierry. Speaking generally, we must say that what attracted William in Plato's talk of the world soul, what perhaps has an inevitable attraction for the Christian if we can gauge this by the many responses to it before and after William, is that it seems to express God's presence in the world. St. Paul is reported in Acts of the Apostles (17:23-30) to have likened God to the *deus ignotus* worshipped by the pagans. He goes on to say that God is he in whom we live and move and have our being, and he quotes a pagan poet: "Ipsius enim et genus sumus." Knox translates this, "For indeed we are his children." God's children, his kind -- this sense of man's kinship with God, of the world's kinship with its creator, of God's presence in his effects may be thought of as the essence of religion; it is surely a salient note of the Christian attitude. Just as St. Paul found in pagan thought suggestions of the

true faith, so such interpreters of Plato as William of Conches will look for secular approximations of the Christian mysteries. It is in this light that we must approach his remarks about the world soul. The sestet of Gerard Manley Hopkins' "God's Grandeur" expresses the same sense.

And for all this, nature is never spent;
There lives the dearest freshness deep down things;
And though the last lights off the black West went
Oh, morning, at the brown brink eastward, springs --
Because the Holy Ghost over the bent
World broods with warm breast and with ah! bright wings.

In his glosses on Boethius, William suggests an interpretation of the world soul which blends two of the items on his list of possible interpretations. "The world soul is the natural force whereby some things have it in them to be moved, some to grow, some to sense, some to understand. But it is asked what force is. It seems to me that that natural force is the Holy Ghost, that is, the divine and benign harmony, which is that whereby all things have being, movement, growth, sense, life, and intelligence." This soul, which is the divine love, the diffusiveness of the divine goodness, grants existence to both corporeal and spiritual things. In explicating Plato's statement that the world soul is composed of the same and the different, William says that it is one and undivided in itself, but can be thought of as multiple in its effects. (*In Tim.*, ed. Parent, p. 170) Thus, the world soul is a philosopher's way of expressing the creative causality of God, and William does not feel that it in any way jeopardizes the distinction between creator and created, that what Plato said of the composition of the world soul in any way prejudices the divine simplicity and divisibility. The phrase is interpreted, not as symbolic in intent, but as naming the ultimate cause of the physical world. It can also draw our attention to the imitation of God by his effects, so that the natures of things, *vigor insita rebus*, in all their diversity, point toward the one simple cause of them all.

Faith and Reason. The effort of William of Conches to bring reason to bear on faith (*conjunge rationem et fidem*) was, if we can judge by the defenses of what he is doing which stud his *Philosophia* and other early works, an object of constant criticism. He asks, somewhat plaintively, how what he says can be construed to be contrary to Scripture if he is attempting to explain the manner in which that was done which Scripture tells us was done. More sharply, he writes of his critics, "Because they do not know the forces of nature, desiring that all men should be companions of their ignorance, they will not permit others to engage

in research and want us to believe like countryfolk, asking no reason; thus would the prophecy be fulfilled: the priest shall be as the people. We say a reason must be sought in all matters, and then if failure ensues, we must entrust the matter to the Holy Ghost and to faith, as Divine Writ says." (*Philosophia*, PL, 172, 1002E) William does not feel intimidated by the reminder that God regards the wisdom of this world as foolishness. "The wisdom of the world is foolishness with God: not that God thinks the wisdom of this world is foolishness, but because it is foolishness in comparison with his wisdom; it does not follow on that account that it is foolishness." (*Philosophia*, I, 19)

There is a discernible difference in William of Conches after the attack on him by William of St. Thierry. The latter wrote a letter to St. Bernard of Clairvaux which has come down to us under the title *De erroribus Gullielmi de Conchis* (*On the Errors of William of Conches*). (PL, 180, 333ff.) In his letter William of St. Thierry objects to William of Conches' statements on the Trinity, and he takes violent exception to the master of Chartres' theory that the body of the first man need not be thought of as directly created by God (as the soul is): it can be thought of as immediately the effect of the stars and spirits, which are, of course, the effects of God. A further charge has to do with William of Conches' view that the biblical description of the creation of Eve from a rib of Adam should not be understood literally.

This attack had a great impact on William of Conches. He had written what he had written in all sincerity; he had no desire to be or to be considered a heretic. One is a heretic, not simply by writing error, he observes, but by defending it when it is pointed out. The *Dragmaticon* emerges as an attempt to go over the same ground as the *Philosophia* in such a way that he would not unduly offend the sensibilities of his fellow believers. "There is another book of ours on the same subject," he writes there, "one entitled *Philosophy*, composed in our youth, and it is, being the product of one imperfect, itself imperfect. In it truths were mixed with falsehoods, and many necessary things were not touched on. Our plan is to set down what was true in it, to condemn what was false, to add what had been overlooked." He goes on to list specific errors of the earlier work and to retract them; the list follows closely the accusations of William of St. Thierry. Moreover, he adds, any errors he does not now mention and retract but which may later be found ought to be brought to his attention and he will be prompt to root them out.

It would be easy to see here an obsequious and spineless capitulation to antidialecticians whose views William of Conches did not actually share. But

there is something more, I think, and something quite edifying. Scholars have pointed out that the *Dragmaticon* continues to exhibit William's search for an understanding of what he believes. He has not dropped that ideal, nor is he simply masking it in a shrewd way. Rather it seems that he came to see the underlying justification of the charges that had been made against him, namely, that his earlier interpretations were too freewheeling, that what he had said could indeed endanger the faith more than it explicated it. William of Conches had no desire to do that. The *Dragmaticon* differs from *Philosophia*, not in substance, not in method, but in style; the youthful zip and vigor, the taunting tone, the suggestion that every invitation to caution indicates obscurantism -- these are absent from the later work. In the *Dragmaticon* we find a remark that, perhaps as much as any other, gathers together the elements of this controversy and focuses on the essential. "I am a Christian," William writes, "not a Platonist."

Bibliographical Note

As was mentioned in the text, William's *Philosophia* is found in two places in Migue, each time attributed to a different author: PL, 90, 1127-1178, and as well as at PL, 172, 39-102. The *Dragmaticon philosophiae*, edited by Grataroli (Aregentorati, 1567); J. Holmberg, *Moralium dogma philosophorum* (Upsala, 1929). Glosses on the *Timaeus* and on the *Consolation of Philosophy* in J. M. Parent, *La doctrine de la création* (Ottawa, 1938). Of the secondary literature we may mention P. Duhem, *Le Systèm du monde*, t. 3, pp. 90-112; H. Flatten, *Die Philosophie des Wilhelm von Conches*; Tullio Gregory, *Anima mundi: La filosofia di Guglielmo di Conches e la scuola di Chartres*; Eugenio Garin, *Studi sul Platonismo medievale* (Florence, 1958). The last two are particularly important.

D. Thierry of Chartres (died before 1155)

Thierry, or Theodoric, was the brother of Bernard of Chartres and, like him, served as chancellor of the cathedral school of Chartres. Thierry is a mysterious figure on several counts: we know next to nothing about his life, and it is a matter of some difficulty to identify his writings. As for hard biographical data, apart from certainty that he taught at Chartres, we know that he was present both at Abelard's trial at Soissons in 1121 and at Gilbert of Poitier's trial at Rheims in 1148 and that sometime between those two dates he taught briefly in Paris. He is said to have retired to a Cistercian monastery and to have died a monk.

Because he was not in the habit of signing his works, perhaps motivated by humility, a great deal of scholarly detective work has gone into identifying Thierry's writings. We know that he wrote an *Heptateuchon*, a work on the seven liberal arts; this has not yet been edited for the modern reader. He wrote a work on creation, the work of the six days recounted in Genesis, a critical edition of which is now available to us. Thierry's commentaries on Boethius have most recently become available owing to the labors of Nikolaus Haring; if Haring's arguments hold, we actually have three different commentaries by Thierry on the *De trinitate* of Boethius. John of Salisbury has nice things to say of Thierry, who seems to have enjoyed the reputation of being a good teacher, particularly of logic.

Account of Creation. In his commentary on the biblical account of creation Thierry proceeds in a manner similar to that of William of Conches. There is a preliminary reference to the opening sentence of Genesis, but, rather than continuing with an exposition of the text, Thierry turns immediately to a physical account of the origination of things in which he employs whatever science was available to him. Only after doing this does he turn to the text itself, and the impression is given that the scriptural account can be seen as verifying the earlier physical doctrine.

There are four causes of the world. God is the efficient cause; the divine wisdom is the formal cause; the divine benignity is the final cause. The material cause of the world is the four elements. The things of this world, being changeable and perishable, must have an efficient cause; their order and arrangement show that they are effects of wisdom; and since creatures cannot be thought of as filling any deficiency in the creator, his motive in creating must be an overflow of his own goodness, the desire to let others participate in his fullness. Thierry attaches this interpretation to the opening sentence of Genesis by saying that there we are told of God as efficient cause and of the material cause. Wherever we read that "God said" we can take it that reference is made to God as formal cause; the remarks that God found what he had made good tell us of God as final cause.

First, God created matter. Heaven, being extremely light, did not proceed in its movement in a straight line but began to revolve, and one of its revolutions can be taken to represent one day. In the rotation which constituted the first day, fire assumes the highest location and illuminates air, which is just below it; this activity has the further effect of warming water and earth. Thus, matter and light are first created, and the heating of the water causes a vapor to be drawn

up into the air; this is the origin of the clouds and, by way of consequence, of rain and snow. The drawing-up of water causes islands to emerge and then greater areas of earth. In subsequent rotations living things and stars are quite naturally brought into being by the natural activities of the elements. Thierry holds that the stars are formed from the water rising from below because of the heating effect of fire through the mediation of air. The visibility of the stars must be accounted for by their ability to refract light. Only water and earth have the necessary density to refract light; only water can be thought of as achieving the necessary elevation, and this came about by the process already mentioned.

Thierry's physical account of the coming into being of the world is thus an appeal to the natural activities of the elements created by God. A rotation begins immediately, thanks to the nature of the elements, and six such rotations are sufficient to account for the furniture of the cosmos. Every possible natural mode of becoming is employed during those six rotations; that is the meaning of the scriptural statement that after six days God rested. He employs no new method of generation after the first six rotations of the universe, for during this time seminal causes (*causae seminales*) are so embedded in the elements that all subsequent natural history is, in a sense, present from the beginning in natural causes.

Within the world, fire has a special role to play and may be thought of as the efficient cause and artificer of all other things. Earth is the material on which it works. Thus, fire is the active element, and earth the passive element.

The foregoing speculation, although it involves references to Genesis, is obviously presented as natural, physical knowledge of the origin of things. The doctrine involved is not presented as a discovery of Thierry so much as a summation of what philosophers have been able to learn on the subject. Having stated the findings of physical or natural philosophy, Thierry then turns to an explication of the text of Genesis itself. What he does, in effect, is to attempt to show both that the biblical narrative bears out what physical philosophy teaches and that the text can be illuminated by the philosophical doctrine. Thus, when he reads that the Spirit of the Lord moved over the water, Thierry observes that this has been taken to be a reference to the element, air, which can be likened to the divine Spirit because of its spiritual qualities. His own view is that it is the world soul which is being referred to, since Plato's world soul is precisely what Christians call the Holy Ghost. Thierry identifies the Holy Ghost with the power of God, something for which both Abelard and William of Conches were severely criticized.

Having turned to the text of Genesis, Thierry must say something of God, since it is God to whom all this creative activity must be referred. It is the quadrivium, the mathematical arts, which leads to knowledge of the creator. Thierry's conviction that mathematics is the key to knowledge of God is clear in his employment of otherness or duality (*alteritas*) and unity (*unitas*). All multiplicity or otherness takes its rise from the number two, and one naturally precedes two. Thus, prior to all multiplicity and otherness is the one; moreover, we can say that the number one precedes all change, since change is consequent on otherness or multiplicity. To be changeable is to be capable of turning one way or the other, consequently to be multiple. Now if every creature is subject to change and if being in its totality comprises both the eternal and the created, the eternal must escape multiplicity and otherness. The eternal, which is the One, must precede all creatures. The upshot is that we can identify the One, the divine, and the eternal. The One is the cause of being in all creatures, their *forma essendi*, since for them to be is to derive their being from the divine or eternal. It is this pervasiveness of the divine causality which is meant when it is said that God is everywhere; it is the dependence of all else on the eternal and divine One which is meant when it is said that every being that exists exists because it is one.

To say that God is the *forma essendi* of creatures, to say that God is the One at the root of the duality or otherness any creature is, is to run the risk of being severely misunderstood, and Thierry knew it. He asked not to be understood to mean that God is some kind of intrinsic form of the creature; what he is insisting upon is that apart from the divine causality there is nothing. Creatures, he says, exist neither in God nor apart from him. In short, Thierry attempts to forestall the pantheistic interpretation of his remarks. The vocabulary of his doctrine of participation has one expected and one unexpected result. Apart from the One, which is eternal and divine, there are also created units: things which are and are called ones. They are one and deserve the appellation owing to their participation in the One; a sign of the difference between created and eternal unity is that in the former case we can speak of a plurality of ones. But just as what partakes in the divine unity can be called a one, so too can it be called divine or a god. This is somewhat surprising, and it does not require a limber imagination to guess that misunderstandings of it will be plentiful. But these observations permit Thierry to stress the utter unity of God and to state the inappropriateness of speaking of any plurality or number in God. What consequences will that assertion have for the Trinity?

Thierry speaks first of square and oblong numbers; the former are obtained by the multiplication of a number by itself, for example, two times two, three times three, which generates tetragons, cubes, circles, and so on. The multiplication of a number by a different number generates oblong numbers. But what result is obtained when one is multiplied by itself? Obviously the result is simply one. The one considered as begetter and the one considered as begotten, then, are one and the same nature. This kind of multiplication (the generation of the Son by the Father) fittingly precedes all subsequent kinds of multiplication which refer to creatures. In speaking of the Trinity, then, Thierry arrives, in the manner sketched, at the One and the Equal One; these are spoken of as Persons because nothing can generate its own self. Since the generation of the Son precedes that of creatures, the Son is equally the cause of the existence of creatures; furthermore, as generated from the One, the Equal One is the image and splendor of the One. In the Equal One, then, are the patterns of all other things that can imperfectly reflect the One, and the Equal One is therefore called the divine wisdom. The little treatise we are relying on here promises to explain the third Person of the Trinity as the link (*connexio*) between the One and the Equal One, but at this point the manuscript ends.

Thierry's procedure in speaking of the physical origins of things prior to considering revelation is somewhat more risky when it is employed in speaking of the Trinity. The hope that, quite apart from revelation, men can arrive at knowledge of the natural origin of things may be easy enough to accept, even when we notice the crudity of the science Thierry uses; but it is quite another matter to agree that the kind of analysis he performs on unity and otherness secures us, just as such, anywhere near knowledge of divinity and of the divine Persons. It has been observed that Thierry concentrates on what Augustine would call a trinity of things, in this case, of numbers. This is opposed

to the more traditional and Augustinian manner of approaching the mystery of the Trinity via an analysis of intellection. Thierry seems to be proceeding in the direction of a mathematical proof of the Trinity.

Häring's conjecture that Thierry could not go on with his analysis because he had denied relations in God is interesting but not conclusive. Thierry, in his effort to distinguish the One from all multiples or creatures, had denied of God all consequences of otherness in things: among these consequences are form, weight, measure, place, time, and relation. To exclude relation from God, Häring thinks, cuts Thierry off from the traditional approach and dooms his own. But surely we can expect that Thierry could have overcome this,

particularly since he has already employed the relation of equality between the Father and Son. Moreover, the exclusion of *forma* does not prevent talk of God as *forma essendi*. Häring's essential point, however, namely, that Thierry is off on a different and risky direction and is shoring up difficulties for himself, is beyond contest. Finally, Thierry's procedure in his trinitarian doctrine has been the cause of speculation about the possibility that a Latin translation of Plato's *Parmenides* was available to him; it is certain that indirectly, by way of references something of that dialogue as well as of the doctrines of Pythagoras was known. However he would have handled it, Thierry's difficulty is not unlike that facing the Pythagorean doctrine: how to derive from a consideration of mathematical entities nonmathematical properties.

For whatever significance it may have, it may be pointed out that Thierry does not pursue this mathematical interpretation of the Trinity in the three works of his which deal with Boethius' *De trinitate*. Indeed, in his lectures on that Boethian opusculum, which their editor, N. Häring, calls the *Quae sit* version, the only allusion we have to a mathematical treatment comes in reply to a question. There are three ways of speaking of the Trinity, we read: theologically, mathematically, and ethically. Augustine is cited as one who speaks mathematically, and we are reminded that he maintained that unity is in the Father, equality in the Son, and the connection of unity and equality in the Holy Ghost. What follows is reminiscent of the One and the Equal One. As for the Holy Ghost, Thierry says that unity desires equality and equality unity, and that this desire or love is their connection.

Man and Philosophy. In commenting on Boethius, Thierry must face the division of speculative science set down in chapter two of the *De trinitate*. His remarks on the passage tend to be a description of man as much as anything.

Thierry's attempt to locate the *De trinitate* itself has interesting overtones. Boethius' opusculum belongs to speculative philosophy, Thierry says, and to precisely that part of speculative philosophy which is called theology. There are, he continues, three parts of philosophy: the ethical, the speculative, and the rational. The speculative is subdivided into theological, mathematical, and physical. Now Thierry speaks of this division as of a declension. Theology takes its start from a consideration of the most high God and the Trinity and then descends to angelic spirits and souls, concerning itself with incorporeal things which are outside bodies (*de incorporeis quae sunt extra corpora*). The start of mathematics is a concern with numbers, whence it descends to proportions and magnitudes and is generally, concerned with incorporeal things which are in

bodies (*circa corpora*). Physics is concerned with bodies themselves and takes its start from the four elements.

Answering this declension of the objects of a science, and the hierarchy among the speculative sciences consequent upon their range of objects, is an ascension described by man because of the multiplicity of his powers of knowing. Thierry says that we must know the powers of the soul and their modes in order that all things may be compared with them, that we might know how things can be grasped and by what knowing powers of our soul they are grasped. He mentions five powers of the soul: sense, imagination, reason, intelligence, and intelligibility. Sense is that power of the soul which is comprehensive of bodies, as when we see colors, touch, taste, and so on. Imagination is comprehensive of forms and of images, which are corrupted by their involvement in matter, though they are imagined without matter. Reason is a power of the soul which in its agility moves itself and abstracts from many things of the same general or special nature that very thing they partake in, a form which is immattered and subject to mutability, for example, when I abstract from all men the nature in which they agree (*conveniunt*), I consider it as participated by them, somewhat separated from mutability by mind. Intelligence (*intelligentia*, properly called *disciplina*) is a power of the soul which considers the single qualities and properties of forms, or the forms themselves as they truly are, in such a way, however, that the single terms (*terminos*) are not removed from them, for example, when I attend to "humanity" or "circle" in its true being. Thus, I see that neither is varied by the flux of matter, and I find the nature it cannot have in a subject matter: as that all the lines from the center to the circumference of the circle are equal or, in humanity, that every monstrosity is repelled by its nature. Intelligibility (*intelligibilitas*) is the power of soul which removes from forms all limits whereby they were distinct from one another, contemplating only *esse atque entiam*, rejecting all plurality and seeing only the union of all things, for example, if we ignore the limits of circle and humanity, their difference, only being remains. This is what all things have: being is the simple simplicity of all things.

The very definitions of these powers of the soul indicates the order Thierry sees among them. Sense leads to imagination and that to reason, which bears on the universal; a higher truth beckons to intelligence, and then when the soul extends itself to the simple unity of all things, it becomes intelligibility, which is of God alone and had by few men.

The soul is made for the totality of things, and the totality of things is such that it exists in four manners. God is all things without being any of them singly; if he were any one of them, he would not be the totality, Thierry says. All things are made by God, and He Who Is is prior to them all and in some way the totality of them, for they were first in him and whatever is in God is God and is eternal. God's being, being being, is independent of all dependence: God is He Who Is. God is Absolute Necessity, the form of forms, eternity, unity. God is not, of course, an immattered form; things other than God are form and more. Possibility, that is, is included in all things, Absolute Possibility. Absolute Possibility is descriptive of primordial matter and, Thierry insists, is created by God. Thierry now has set up two poles, God and matter, Absolute Necessity and Absolute Possibility, and these are modes of the totality of things. Between these two poles he will locate two other modes of the totality of things, what he calls Determined Necessity and Determined Possibility. The former describes the realm of Ideas, the world soul; the latter, Determined Possibility, is the result of the fusion of Idea and matter, that is, the things of this world. Thierry can now speak of the three speculative sciences in terms of these modes of the totality of things. Physics, he says, considers both kinds of possibility; mathematics considers determinate necessity; theology considers Absolute Necessity.

Much more could be said of the ideas Thierry has brought into play here; there is much to be gained by comparing the treatments of these ideas in the different commentaries Thierry wrote on the *De trinitate* of Boethius. Perhaps enough has been said to indicate in an introductory fashion the flavor of Thierry's thought.

Bibliographical Note

The edited writings of Thierry are the following: B. Hauréau published an edition of *De sex dierum operibus* in *Notices et extraits de quelques manuscrits latins de la Bibliothèque Nationale*, vol. 22, pp. 170-186. Thanks to the efforts of Nikolaus Häring our knowledge of Thierry has taken a quantum jump in recent years: "The Creation and Creator of the World According to Thierry of Chartres and Clarenbaldus of Arras," AHDL (1955), pp. 137-216; "A Commentary on Boethius' *De trinitate* by Thierry of Chartres (Anonymus Berolinensis)," AHDL (1956), pp. 257-325; "The Lectures of Thierry of Chartres on Boethius' *De trinitate*," AHDL (1958), pp. 113-226; "Two Commentaries on Boethius (*De trinitate* and *De hebdomadibus*) by Thierry of Chartres," AHDL (1960), pp. 65-136. The Eptateuchon, still unedited, discussed by A. Clerval, *Les écoles de Chartres* (Paris, 1895); its prologue has been edited by E. Jeaneau in *Medieval*

Studies, 16 (1954), pp. 174 if. Of the secondary literature, mention may be made of P. Duhem, *Le système du monde*, vol. 3, pp. 184-193; J. M. Parent, *La doctrine de la création dan l'école de Chartes* (Ottawa, 1938); Pare, Brunet, Tremblay, *La renaissance du XIIe siècle* (Ottawa, 1933) and, of course, Häring's introductions to the editions mentioned above.

E. Clarenbald of Arras (died c.1160)

The connection of Clarenbald with the school of Chartres lies both in that he studied there under Thierry of Chartres and in that he was a critic of Gilbert of Poitiers. He was an opponent of Abelard as well and a friend of St. Bernard of Clairvaux. Besides having been a student of Thierry, Clarenbald studied under Hugh of St. Victor. Clarenbald is known to us through his commentaries on the *De trinitate* and *De hebdomadibus* of Boethius as well as through a work appended to one of Thierry's and called *Liber de codem secundum* (*Another Book on the Same Subject*). This last work was just recently identified as Clarenbald's by Nikolaus Häring and published under the title *Clarenbaldi tractatulus*.

Account of Creation. Clarenbald refers to the teachers under whom he studied with a deference whose sincerity cannot be questioned; in the *Tractatulus*, which he appends to Thierry's account of creation, he promises no more than to collate the thoughts of others and to show that their doctrines are actually corroborated by Scripture. A modest task, we might expect, and certainly not likely to lead to an original book. Indeed, when we leaf through it, our eye is struck by passages reminiscent of William of Conches, of Thierry of course, and of others. Were we to be satisfied with this superficial estimate, we would be doing both Clarenbald and ourselves an injustice. Even what he takes from others has a way of altering in his hand and often of taking on a precision and clarity it did not have in its source.

Clarenbald's *Tractatulus* begins with a reference to Genesis, goes on to relate it to the other books of the Pentateuch, speaks of the various senses of Scripture, and promises to proceed in terms of the literal sense. But it is not really a commentary on Scripture. The comparison of the books of the Pentateuch to Roman law is apparently original with Clarenbald, although of course the notion of senses of Scripture is not. When these preliminary matters have been treated, Clarenbald turns to the opening line of Genesis and observes that the book can only gain in intelligibility if we discuss the creation of things. For created things speak to us of their creator. Clarenbald then gives a faithful version of William of Conches' first argument for the existence of God.

Ignorance of creation can lead to heretical views concerning the nature of God, Clarenbald continues, and he makes reference to the heresies discussed in Boethius' *De duabus naturis*.

Clarenbald speaks of three inchoative principles: primordial matter, seminal reasons (*rationes seminales*), and the beginning of time. These three inchoative principles have the Son of God as their creator. Relying on Augustine, Clarenbald speaks of God as forming all things in his Word and then as forming them in an unformed way in matter and seminally in seminal reasons. In the succession of time God operates actually and reparatively. In these four ways, he adds, the totality of things exists. We are reminded of Thierry. Indeed, Clarenbald employs the same quartet: Absolute Necessity, the Necessity of Concatenation (Determinate Necessity), Absolute Possibility, and Determined Possibility. The influence of Thierry is also evident in Clarenbald's use of what he calls the Pythagorean doctrine, but with the addendum of the number ten as the perfect number, since ten is the sum of the first four numbers. Clarenbald identifies Absolute Necessity as One; Absolute Possibility as Two, since matter is the source of otherness and otherness is reducible to duality; The Necessity of Concatenation with Three, since three is the first number to be connected by a middle term; Determinate Possibility with Four, since matter is first actualized by the forms of the elements -- fire, air, earth, and water. Clarenbald's discussion of the meaning of the word "day" presents a variation on Thierry's account and a rejection of Augustine's speculation that it may refer to angelic knowledge.

For Parent the *Tractatulus*, not yet established as the work of Clarenbald, serves as yet another illustration of the spirit of the school of Chartres. Häring, who made the identification, agrees with Parent's estimate and puts the point stylistically: what the *Tractatulus* shares with the typical product of the Chartres of the day, and what sets it off from contemporary writings emanating from elsewhere, is the niggardly appeal to the Fathers and the prominence of quotations from the doctrines of the philosophers. This has as a general effect the seeming attempt to make Scripture agree with philosophy rather than the reverse; therein lay the so-called rationalism of Chartres, a tendency which, if Clarenbald himself displays it in his *Tractatulus*, he is suspicious and critical of in others. By his ties to his friends and his professors he was on both sides of the dialectician/antidialectician controversy of his day; in a sense, by his very existence he provides hope that the opposite tendencies of these factions would ultimately be reconciled.

Being and Goodness. In his commentary on the *De trinitate* of Boethius, Clarenbald again exhibits the influence of his mentors, and once more it is Thierry who is perhaps most prominent, although he may be thought to share this honor with Gilbert of Poitiers. Given Clarenbald's opposition to the latter, the second influence is interesting; it is the opposition that seems to come to the fore, however, thereby obscuring Gilbert's positive influence on Clarenbald. Gilbert's teaching on the Trinity involved, as we have seen, the question of individuation. In a difficult doctrine Gilbert had sought to maintain that not only can we speak of a universal humanity but we must also speak of a humanity proper to Socrates, another proper to Plato, and so forth. Clarenbald finds this nonsense. What individuates is not part of the shared nature itself but is derived from accidents; therefore, there is one and the same humanity whereby individual men are men. Here as elsewhere we must be careful in employing the term "realism" to describe what Clarenbald is doing. He does not seem to be clear on the locus of that identical nature, and this very lack of clarity prevents unqualified ascriptions of an apriori definition of realism to him.

While the commentary on the *De trinitate* deserves and repays an attentive reading, we shall turn immediately to Clarenbald's commentary on the *De hebdomadibus*, one of his works which has not hitherto received much attention. This opusculum of Boethius asks, we remember, whether everything that is is good. The point of the question is this: How can things be good just insofar as they are unless they are substantially good, that is, good in their very substance? Posing the question in this way seems to force a denial, since only God is good in his very substance. But the reply that creatures are good only accidentally is not without its difficulties. Boethius will suggest as a satisfactory answer, which avoids the apparent options, that creatures are good by participation, by a participation which differs from that whereby they partake of accidents. In the opusculum Boethius says he is striving for mathematical rigor and, first, lays down axioms from which he hopes to deduce the desired result. Let us see what Clarenbald makes of this Boethian effort.

Clarenbald sees Boethius employing at the outset an *accessus*, or approach which, by stressing the obscurity of the question, renders the reader attentive. Furthermore, he renders the reader docile and benevolent in the appropriate rhetorical fashion. Now, what in the question is referred to by "the things that are"? Things may be said to be in three ways: in the divine mind, in matter, in existence. Only in the final way can they be said to exist absolutely, and it is on things thus existing that the question bears. Clarenbald then goes on to

distinguish between things as existent and as understood; the passage is obscure, but it appears to be an effort to distinguish the logical or conceptual order from the real order rather than, as Häring suggests, an effort to distinguish substance from accidents. If our interpretation is correct, our earlier caveat about speaking of Clarenbald's realism is strengthened. Clarenbald interprets the hebdoniads of the title to refer to common mental conceptions, that is, axioms. How does Clarenbald now explicate the question Boethius sets out to answer?

The good of substances does not seem to be substantial goodness because good is not predicated of them as genus, species, difference, or definition. In this, "good" is like "being"; when we have a substantial predicate we know in virtue of it, at least in part, what the thing of which it is predicated is, but "being" does not give us this kind of knowledge of that of which it is predicated. If, further, we understand by substantial goodness that whose essence is goodness, the phrase can apply to God alone. How then can created substances be and be called good?

"Diversum est esse et id quod est" (being and that which is are diverse). Clarenbald takes this Boethian dictum to refer to the distinction between God and creatures. God is being, the *forma essendi*; creatures have being by partaking in the being God is. What is meant by partaking or participating? It is used here to signify the difference between God and creature; God does not partake of anything, whether prior to himself (there is nothing prior to God) or posterior (for this would indicate dependence on something which, being posterior to God, depends on him). "*Ipsum esse nondum est*" (being itself is not yet). This enigmatic remark of Boethius means that God who is being is not that which has being; he does not partake of being. The mark of the creature is found in participation or partaking. "*Quod est, partici pare aliquo potest.*" That which is, that is, created substance, can partake of something which is not constitutive of its nature, of accidents, that is. Boethius' doctrine of participation enabled him to distinguish between *what is* and *what is such and such*, with the former referring to substantial and the latter to accidental being. Clarenbald prefers to interpret *to be such and such* (*esse aliquid*) as covering both substantial and accidental determinations; prior to both modes of being there is participation in the *forma essendi*, thanks to which the thing is or exists. In short, Clarenbald argues that existential participation is prior to any essential or accidental participation. Thus, he can interpret Boethius' statement that in every composite its being is one thing and what it is is another as referring

respectively to participation in the *farina essendi* and to participation in a determinate form.

Now to the question itself. What do we mean when we say that whatever is is good? Whatever is tends toward the good, but such a tendency is toward what is similar to that which has the tendency; therefore, whatever is, is good. Is that which is good good substantially or by way of participation? We can of course guess that the answer will be that they are good by way of participation, but before he gives that answer, Clarenbald carefully distinguishes between participation in the various substantial predicates which constitute the Porphyrian tree and participation in accidents which are not constitutive of substance. The expected answer, moreover, is a nuanced one. That which is is by participation in being; that which is is good by participation in goodness. But it is by participation in being that created substances are substances, and we can say that these substances are good. The doctrine of participation, therefore, leads to the conclusion that created substances are substantially good, but this assertion cannot be understood as it would be in the case of God.

These few remarks may suggest something of the doctrine of Clarenbald. His reading of Boethius' *De hebdomodibus* makes it abundantly clear that, as Häring's introductory remarks imply, the view that prior to Aquinas no one had undertaken to speak of the existence of things is simply without historical foundation.

Bibliographical Note

The writings of Clarenbald of Arras can be found in the following editions: W. Jansen, *Der Kommentar des Clarenbaldus von Arras zu Boethius De Trinitate* (Breslau, 1929); Nikolaus M. Häring, "A Commentary on Boethius' *De hebdomodibus* by Clarenbaldus of Arras," in *Nine Medieval Thinkers*, edited by J. Reginald O'Donnell, C.S.B. (Toronto, 1955), pp. 1-21; N. Häring, "The Creation and Creator of the World According to Thierry of Chartres and Clarenbaldus of Arras," *AHDL* (1955), pp. 137-216; edition of *Tractatus*, pp. 200-216. The secondary literature is not extensive, but mention can be made of the following: M. de Wulf, "Clarembaud d'Arras," in *Melanges Louis Arnould* (Poitiers, 1934), pp. 22-27; R. L. Poole, *Illustrations of the History of Medieval Thought and Learning* (London, 1920), pp. 320-322; J. M. Parent, *La doctrine de la création* (Ottawa, 1938); N. Häring, "A Hitherto Unknown Commentary on Boethius' *De hebdomodibus* Written by Clarenbaldus of Arras," *Medieval Studies*,

15 (1953), pp. 212-221. Finally, mention must be made of Häring's most recent contribution, *Life and Works of Clarembald of Arras* (Toronto, 1965).

F. John of Salisbury (1110-1180)

The connection of John of Salisbury with the school of Chartres is a multiple one: he studied there as a young man, he provides us with a sketch of the teachings of its masters, and he ended his life as bishop of Chartres. He also studied under teachers elsewhere, for example, Abelard; indeed, he seems to have been acquainted with most of the prominent thinkers of the time. But John was no mere academic. After his studies he returned to England, where he lived in Canterbury and was associated with, among others, Thomas à Becket. When he fell out of favor with Henry II, John returned to the Continent and eventually was elected bishop of Chartres. His importance for medieval history in general is undeniable; here we are interested in what further light he can throw on the school of Chartres in the twelfth century.

In chapter seventeen of book two of his *Metalogicon* John gives a sketch of current views on the status of universals. His tone is one of gentle irony, his manner offhand; the general impression given is of tolerant condescension. The endless dispute is, John opines, largely verbal, the oppositions being not as clear-cut as proponents of the various positions believe. John suggests that with a little application of common sense the disputants could be shown to be in basic agreement. He chides the masters of the day for putting an impossible burden on beginners in philosophy by their tendency to launch immediately into the vexed and sophisticated questions connected with the problem of universals. When he himself decides to enter the dispute, John notes that he will thereby be liable to the same kind of picayune criticism that other contributors have invited when they commit their thoughts to writing. But enter it he does, and with the clear conviction that he can settle the matter definitively by pressing what he bills as the Aristotelian solution as against the Platonism he finds rampant with few exceptions among the current views on the status of universals.

In chapter twenty of the second book of the *Metalogicon* John of Salisbury argues that Aristotle's teaching on the status of genera and species is supported by reason, the facts, and much that has been written on the subject. The fact is, John writes, that genera and species do not exist, as Aristotle had said. How melancholy then to contemplate the array of opinions which have multiplied on the mode of existence proper to genera and species. Genera and species lack

substance and, therefore, cannot be identified with *voces*, *sermones*, sensible things, ideas, native forms, or collections. Such identifications go contrary to the simple statement of Aristotle that universals do not exist, and, according to John, all those who made these identifications profess to be followers of Aristotle. However, although those genera and species do not enjoy any substantial existence, we need not fear that in attending to them our mind is empty. Recalling Aristotle's distinction between what can be called simple apprehension, the simple attending to what is thought, and affirmations and denials which follow on composing or dividing what has been simply understood, John of Salisbury says that in both kinds of mental acts we sometimes consider things as they are and sometimes otherwise than as they are. We can consider line or surface without considering the body to which it attaches, and when we do this, we need not be taken to affirm that line or surface exists apart from any such body. The mind just considers the form without considering the matter. In much the same way, John suggests, the mind can consider man as this form does not exist, because no individual man is being considered in the process. There is simply no point in asking what in nature corresponds as such to man considered as a species, since for man to be considered as a species follows on the abstractive character of our thinking whereby we draw away, as it were, from the natural world. What happens in the formation of a species is that reason, considering the mutual substantial resemblances of a given range of individual things, formulates the resemblance in a general concept. Thus, species are mental representations of actual things in the natural world.

There is a good deal more to John of Salisbury's exposition, but this may suffice to indicate that his calm, common-sense approach to the matter does introduce some much-needed light. One may contest whether the Aristotelian position emerges in all its clarity, but surely the elements of a realist solution are present in John's lengthy chapter twenty. Furthermore, one sees the basis for his claim that his contemporaries are really not as far apart as they think. By the same token, it must be said, however, that many of the positions John criticizes are more alive to real difficulties in the problem than is John himself. One comes away from reading this section of the *Metalogicon* impressed by what John has to say concerning universals, of course, but rather more impressed by the mood he conveys that the problem of universals has been discussed beyond the point of fruitfulness. In a word, John seems to suggest a weariness with the dispute and the hope that dispute will pass to other and more rewarding and certainly less picked-over topics.

Bibliographical Note

We are indebted to C. Webb for editions of the two most important works of John of Salisbury: *Polycraticus*, 2 volumes (Oxford, 1909) and *Metalogicon* (Oxford, 1929). The latter has been translated into English by D. McCarry, *The Metalogicon* (Berkeley, 1955). See as well Hans Liebeschuetz, *Mediaeval Humanism in the Life and Writings of John of Salisbury* (London, 1950).

{1} "Sequebatur hunc morem Bernardus Carnotensis, exundantissimus modernis temporibus fons litterarum in Gallia, et in auctorum lectione quid simplex esset et ad imaginem regule positum ostendebat; figuras gramatice, colores rhetoricos, cavillationes sophismatum, et qua parte sui propositae lectionis articulus respiciebat ad alias disciplinas, proponebat in medio; ita tamen ut non in singulis universa doceret, sed pro capacitate audientium dispensaret eis in tempore doctrine mensuram." (*Ioannis Saresberiensis Episcopi Carnotensis Metalogicon*, ed. Webb [Oxford, 1929], p. 55.)

{2} The verses of Bernard which John quoted are

Non dico esse quod est, gemina quod parte coactum

Materiae formam continet implicitam:

Sed dico esse quod est, una quod constat earum:

Hoc vocat Idem illud Acheus et hylen.

(*Metal.*, IV; PL, 199, 938)

{3} "Sed appellatione verbi substantivi non satis digna sunt, (quae cum tempore transeunt, ut nunquam in eodem statu permaneant, sed, ut fumus, evanescent: 'fugiunt enim,' ut idem ait in *Timaeo*, 'nec exspectant appellationem.'" (*Metal.*, IV, 35)

{4} See Ganfredus' letter (PL, 185, 587-596) and *Libellus eiusdem contra Gilliberti Porretani Pictaviensis episcopi* (PL, 185, 596-617). This author was St. Bernard's secretary and later became abbot of Clairvaux.

{5} "Subsistit enim illud, et quadam ratione est per se, quod non indiget accidentibus ut esse possit; imo accidentia, eo quod hac ratione subsistere et per se esse dicitur, adeo indigent, quod nisi illa adsint, nulli inesse possunt." (*In de duab, Nat.*; PL, 64, 1375)

{6} "Genus vera nihil aliud putandum est, nisi subsistentiarum secundum totam earum proprietatem ex rebus secundum species suas differentibus similitudine comparata collectio (*In de trin.*,)

{7} ". . . haec tria scilicet *existens* id est archetipum mundum, *locum* id est primordiale materiam, *generationem* id est sensilem materiam, ante *exornationem sensilis mundi*, non dixit ante creationem quia etsi ante creationem fuit archetipus mundus, non tamen materia nec generatio potuit ante esse, sed dicit ante exornationem" (*In de trin.*, ed. Parent, p. 174, 28-31)

Chapter V

Monastic Thought

If Abelard and the school of Chartres are indications of things to come, heralding as they do the age of the university, we must not think that the monastic centers were in decline. Abelard's experiences as monk and as abbot were not unique, but the twelfth century saw a great resurgence and reformation of the monastery. The Monastery of St. Victor in Paris was part and parcel of the intellectual life out of which the University of Paris would grow. However, at the very time when feudalism was breaking down and giving way to the rise of cities and communes, there was a flight to monasteries with the founding of hundreds of new monasteries and the sound of voices warning against some of the newer dialectical tendencies. In this chapter we want to look briefly at some men associated with this remarkable resurgence of the monastic ideal, men who were not simply criers in the wilderness but who made their presence known in the cities and, indeed, throughout Christendom.

A. Hugh of St. Victor (1096-1141)

Hugh, already a canon regular of St. Augustine, came to the monastery of St. Victor in Paris in 1115, and it was there that he lived out his life. He was elected head of the school of St. Victor in 1133. Among his works are the *Didascalicon*, an introduction to the arts; a work on grammar; a work on the sacraments of the Christian faith; commentaries on Scripture and on Denis the Areopagite. His mystical writings include a work on contemplation and its kinds as well as a work on the vanity of this world.

The *Didascalicon* presents a survey of all the areas of knowledge and attempts to show that they are parts of a whole that is necessary for a man if he would achieve his natural perfection and his heavenly destiny. The work was written for students who came to the school of St. Victor, and its purpose was to provide them with a synoptic view of the object of their study. With the shift of the schools to urban centers there had come about both a specialization and secularization of knowledge, and Hugh, in the *Didascalicon*, may be regarded as combating such tendencies. Knowledge is a whole, and it must be understood both with reference to man's fall in Adam and to the ultimate calling of mankind. Professor Jerome Taylor, in a magisterial introduction to his translation of the *Didascalicon*, shows how Hugh's insistence on the need for learning, in its totality and with reference to both temporal and eternal life,

contrasts with a number of other tendencies. Various cathedral schools were becoming centers of specialization in law or medicine or the poetic arts; many influential authors advocated a more or less literary humanism; there was the Platonism of Chartres, on the one hand, and, on the other, the emphasis on dialectic by Abelard and others; finally, there was the retreat from secular learning -- indeed, an impassioned opposition to it -- in many monastic centers. By depicting the map of learning the *Didascalicon* provides a way to avoid both exaggerating and narrowing tendencies by retaining an ultimately religious telos in study.

The definition of philosophy which is the guiding principle of the *Didascalicon* is taken from tradition. "*Philosophia est disciplina omnium rerum humanarum atque divinarum rationes plene investigans.*" Philosophy is a thorough investigation into the nature of all things, both human and divine. Hugh takes this definition quite seriously and includes the mechanical arts within the scope of philosophy; on the other side, the study of Sacred Scripture is also a component of philosophy. This novelty conveys the flavor of Hugh's synopsis. He considers another definition of philosophy, this one taken from Boethius, according to which philosophy is the love, pursuit of, and friendship with wisdom. Boethius goes on, it would seem, to distinguish the knowledge that would be included in this definition from the arts of making. Hugh insists that such an exclusion is not intended. He adds that something can be included within philosophy in the sense that knowledge of it is included, even though its use or practice is excluded. For example, knowledge of agriculture is necessary to the philosopher, but the actual tilling of ground is the work of the farmer. Furthermore, artifacts may not be natural objects, but since they imitate nature, knowledge of them falls within the scope of philosophy.

Philosophy is divided into four basic kinds of science which include all others. First, there is the theoretical part of philosophy, which speculates on truth; second, there is practical philosophy, which considers moral discipline; third, there is the mechanical, which governs the action of this life and repairs part of the damage due to original sin; finally there is logic, the science of correct speech and disputation. Hugh proceeds to subdivide each of these.

The division of the theoretical part of philosophy is taken from Boethius. There is theology (*theologia, intellectibilis, divinalis*), mathematics (*mathematica, intelligibilis, doctrinalis*), and physics (*physica, physiologia, naturalis*). The theoretical sciences, far more than logic and the practical and mechanical

sciences, deserve the name of wisdom because they contemplate the truth of things.

There is a threefold division of the practical as well. Actually Hugh gives a number of alternative expressions of this division, perhaps to achieve symmetry with the data on the theoretical sciences which he took from the *De trinitate* of Boethius. The division may be said to be a division into the solitary, the private, and the public; into ethics, economics, and politics; or into moral, dispensative, and civil. The various options are combined in the manner suggested by the parentheses in the foregoing paragraph on the division of speculative or theoretical science.

Hugh gives a list of seven mechanical arts which is deliberately parallel to the traditional seven liberal arts. The mechanical arts are spinning, armor-making, navigation, agriculture, hunting, medicine, and the theatrical arts.

Logic, the fourth part of philosophy, is first divided into two parts: grammar and the art of discourse. The latter is subdivided into probable and sophistic, with rhetoric and dialectic falling under probable discourse. These are divisive or subjective parts of logic; the integral or constitutive parts of logic are discovery and judgment. Hugh raises the question whether discovery and judgment could be divisive as well as integral parts of logic and, in giving a negative reply, enunciates a general principle. Any science which is an art or discipline can be said to be a part or subdivision of philosophy, but not every instance of cognition is an art or discipline. In order to be a subdivision of philosophy, in order, that is, to be considered an art or discipline, an instance of cognition must have its own end and be complete in itself. Discovery and judgment do not satisfy these criteria: neither is complete in itself. Thus, they are elements of discourse and not special parts of philosophy. This discussion is reminiscent, of course, of Boethius' discussion, with which Hugh would have been familiar, of Ammonius' resolution of the dispute between Stoics and Peripatetics on the question of whether logic is a part of philosophy or merely its instrument. The devices used to solve that far broader question are applied by Hugh to the narrower question just mentioned.

Has Hugh accounted for the traditional liberal arts? He speaks of the quadrivium when he discusses mathematics, and notes that the four arts of the quadrivium are the divisions of mathematics. Moreover, he compares the seven liberal arts with the seven mechanical arts he lists. Three of the mechanical arts pertain to the extrinsic cloaking of nature (weaving, armament, and,

presumably, theater), while four are concerned with sustaining inner nature (navigation, agriculture, hunting, and medicine). So too with the liberal arts. Three are extrinsic, being concerned with speech, while four are concerned with thought conceived within. The mechanical arts are concerned with repairing the damage done to man's bodily nature by original sin, whereas the liberal arts are concerned with repairing the damage done to reasoning and its expression in speech. Hugh returns to the liberal arts when he has enumerated the various parts of philosophy, noting that of all the sciences listed, the ancients singled out certain ones for special attention because of their peculiar utility. One who was well-versed in these was well-disposed to acquire the others. These then are the rudiments as well as instruments whereby the soul is prepared for the full knowledge of philosophical truth. That is why they are called the trivium and quadrivium, respectively, being three and four ways whereby the soul is introduced into the secrets of wisdom. Thus, no one is thought to deserve the title "master" unless he is proficient in the knowledge of these seven. But men have lost sight of the appropriate way to concern themselves with these arts; that is why, while they spend much time on them, they come away with little wisdom.

With respect to the terms "art" and "science," Hugh recounts earlier efforts to explain their different meanings and adds something of his own. If philosophy is, as Isadore writes, the art of arts and science of sciences (*ars artium et disciplina disciplinarum*), we can say that an art can be called a science since art consists of precepts and rules. Hugh's own explanation is this. An art can be said to be anything which has a subject matter and is explicated by an operation, like architecture, whereas a discipline or science consists of speculation explicated through reason alone, like logic. Thus he succeeds in distinguishing mechanical art and science, but does not illuminate why logic is called a liberal "art." Later, Hugh distinguishes mechanical and liberal arts, but not as arts. Mechanical arts are those which alter the form of nature. The liberal arts are so called either because they require a liberated soul or because in antiquity free men and not slaves engaged in them.

Hugh's original breakdown of philosophy into four parts is not intended to replace the traditional emphasis on the liberal arts, as his eulogy of these arts adequately shows. The liberal arts are ways to, the mode of entry to, the other parts of philosophy. Indeed, he writes, in the seven liberal arts we find the foundation of all learning. These above all must be acquired, since without them no one can explain or defend any other philosophical discipline.

It is difficult in this rather bloodless résumé to convey the impact of Hugh's *Didascalicon*, which, besides the careful divisions we have recounted, devotes a great deal of time to the moral virtues required for the intellectual life. Despite the fact that Hugh relies throughout his work on the doctrine of his predecessors, bringing to bear the whole testimony of the tradition, there is something peculiarly his own in every part of his book. Of particular interest is his insistence on the broadest possible scope for philosophy, which does not lead him to depreciate the importance of the traditional liberal arts. Those arts are fundamental and propaedeutic to the other parts of philosophy. Why then does he list logic last? When he is setting down the four parts of philosophy, Hugh is not attending to the pedagogical order; in that order, as has been made clear, logic, or rather the liberal arts, would occupy pride of place.

In order to understand the broadening of philosophy that Hugh has effected, we must realize that for him the term "philosophy," the love of wisdom, has as its ultimate telos Wisdom in the sense of the Second Person of the Trinity. The learning Hugh is commending in the *Didascalicon* is part and parcel of the Christian vocation; he is recommending to the neophytes to whom the *Didascalicon* is addressed that they set out with their supernatural destiny firmly in mind and that they continue to assess and understand the pursuit of any science in the light of their calling to union with God. What philosophy seeks to do, the whole point of Christianity, is to restore man and to remedy the effects of sin. That is the basic reason for including the mechanical arts within the scope of philosophy; this is simply to show their importance for achieving our goal as Christians. Man must make his way in this world, he must heal the wound sin has opened between man and nature, and this is the task of the mechanical arts Hugh mentions. That task is, of course, a subservient one. All human tasks must work together for the attainment of man's ultimate good.

This orientation of Hugh's treatment of philosophy must be kept in mind when we compare him with his contemporaries. He is not irenic, as are certain Chartrians, regarding the compatibility of the *Timaeus* and Genesis. Hugh's mentor is Augustine; he will bring everything to the measure of the truth that has come down from above. Hugh has no interest in, indeed he is fundamentally suspicious of, any effort to update revelation by accommodating it to what is currently regarded as the last word of science. With respect to the dialecticians he would seem to be very dubious of efforts which seem to lose sight of the whole theological enterprise. Dispute for its own sake is a perversion, and any interpretation of Scripture which seems to explain it away or needlessly obscure it, or, perhaps worse, to treat it as if its function were to

provide grist for dialectical mills, is repugnant to Hugh. For all that, he is no obscurantist. The various arts and sciences which men have discovered are viewed by Hugh as part of the divine economy of salvation. They are not to be condemned because of the abuses to which they are subject; rather, they are to be taken over by the Christian as his rightful possessions and put to the purpose for which they are intended. The attitude of the *Didascalicon*, if we may risk yet another generality, would seem to be a balanced one. Hugh counters the excesses of those who are overwhelmed by pagan knowledge to a point where their adherence to it jeopardizes their faith; at the same time, he seems to be providing a corrective to excessive repudiations of the pagan sources of philosophy. To both extremes Hugh issues one fundamental reminder. Philosophy is the way to wisdom, and we know that Wisdom is the Second Person of the Trinity. The salutary consequence of this reminder is that the Christian cannot regard his interest in and study of pagan documents as a recess from or an alternative to his ultimate vocation. Hugh is a mystic, not in the sense that he depreciates secular learning, but in the sense that he insists on the ultimate ordering of every human effort to man's restoration in Christ.

Bibliographical Note

The works of Hugh can be found in Migne's *Patrologia latina*, 175-177. The *Didascalicon* is in a critical edition by Brother C. H. Buttimer (Washington, 1939). Jerome Taylor's *The Didascalicon of Hugh of St. Victor* (New York, 1961) is a translation of the Buttimer text; Taylor's introduction and notes make his volume a valuable sourcebook for the twelfth century. See Roger Baron, *Science et sagesse chez Hugues de Saint-Victor* (Paris, 1957). Baron has edited Hugh's *Epitome dindimi in philosophiam* (*Traditio*, XI, 1955, pp. 91-148) and his *Practica geometriae* (*Osiris*, XII, 1956, pp. 176-224). There is an English translation of *On the Sacraments of the Christian Faith* (Cambridge, Mass., 1951) by Roy J. Deferrari. For further secondary literature consult the works of Taylor and Baron.

B. Other Victorines

Although Hugh of St. Victor is far and away the most important figure of this Parisian monastery in the twelfth century, there were, of course, other teachers associated with the school of St. Victor in this century. Richard of St. Victor, a Scot by birth, came to Paris around 1139. In 1162 he became master of theology at St. Victor. He died in 1173. His writings, which are to be found in Migne (PL, 196), include a *De trinitate*, the *Benjamin minor*, and *Benjamin*

maior. The last two works deal expressly with the contemplative life and, as such, with man's ultimate concern. Like Hugh before him, Richard sees both reason and faith as necessary if we are to arrive at contemplation, at the *gaudium de veritate* in Augustine's phrase. Reason has a natural ordination to contemplation, and this natural ordination is aided and enhanced by grace. Man's effort is viewed as a drive toward understanding, toward vision. In speaking of the need to go beyond authority to seeing, Richard is not denying the limits of reason nor is he suggesting that faith is something which can be surpassed in this life. Like Hugh, Richard both recognizes a distinction between nature and grace, reason and faith, and insists on a continuity between them in our drive toward contemplation, a drive which has to be sustained by love or charity. There is, consequently, a subordination of all knowledge to the experimental or loving, mystical knowledge of Wisdom, but this subordination is not a suppression. The Victorine impulse is to make all knowledge a component of man's effort to arrive at his true goal. Contemplation, of course, is not something man can achieve by his own power. In his works on contemplation Richard dwells on the various degrees or stages of the interior life whereby the soul is brought to spiritual perfection. In his work on the Trinity Richard proposes to proceed by reason alone, and he offers a number of proofs of the existence of God. There is also an effort to show by reason that there are three Persons in God.

Godfrey of St. Victor was born around 1130, entered the monastery of St. Victor shortly after the midpoint of the century, and died in 1194. His works can be found in Migne, PL, 196; the *Microcosmos* has been edited critically by Philippe Delhaye (Lille, 1951). In the *Fons Philosophiae* (*The Font of Philosophy*), having recorded the vagaries of his contemporaries, particularly on the nature of universals, Godfrey suggests that the best sources of philosophizing are ancient: Plato, Aristotle, Martianus Capella, and Macrobius. Godfrey seems to have had to combat obscurantist tendencies within his own monastery, but he himself remained faithful to the Victorine ideal as it had been set down by Hugh.

C. Bernard of Clairvaux (1090-1153)

Bernard, one of the dominant figures of the twelfth century, was born in Burgundy in 1090. At the age of twenty-two he entered the monastery of Cîteaux, which had been founded by a group of monks intent on adhering to the letter of the Rule of St. Benedict. Bernard, a nobleman, turned away from worldly possibilities of power and pleasure. His birth would have assured him of power, his looks of the latter. According to the Roman Breviary, in a second

nocturne lesson for his feast, Bernard as a youth was so handsome the ladies lost their heads over him, but he never reciprocated this emotional decapitation. (*Bernardus, Fontanis in Burgundia honesto loco natus, adolescens propter egregiam formam vehementer sollicitatus a mulieribus, numquam de sententia colendae castitatis, dimoveri potuit.*) Scott Fitzgerald once compared himself with Hemingway by saying that Ernest speaks with the authority of success, I with the authority of failure. We could adopt the phrase, make it refer to the flesh, and have Abelard play Fitzgerald to Bernard's Hemingway. Characteristically, Bernard did not seek the solitude of the monastery alone. He brought with him thirty-two other nobles whom he had convinced to leave the world. At the time of their arrival the reform of Cîteaux seemed doomed, the house dying out. Its fortunes changed dramatically with the arrival of Bernard. At twenty-five, Bernard became abbot of the Cistercian monastery of Clairvaux, and it was in this post that he became a leading spokesman for the monastic ideal, a leader of the Cistercian reform, and an influence as well on the abbeys of Cluny. If Bernard entered the monastery to leave the world, for much of his life he was nonetheless drawn into the disputes of the outside world, both ecclesiastical and secular. He preached the Second Crusade; he was consulted by kings and popes; he intervened in the disputes of the schools. He was, by any account, a fantastic man and one whose stature cannot be explained on a purely natural level. Bernard of Clairvaux was a saint. He died in 1153 and was canonized in 1174. He is known as the Mellifluous Doctor, as much for what he said as for the way he said it.

The writings of Bernard are for the most part sermons and letters, but there are also a number of treatises, written at the request of others. These exhibit his principal interest, which, of course, he did not see as a narrow or exclusive one. He wrote on the degrees of humility, on loving God, on conversion, on meditation, and on the errors of Abelard.

While it is risky to attempt a general definition, there may be some point in trying to say what is meant when Bernard is classified as a mystic. Bernard does not differ from Abelard in seeing that man's ultimate end is a supernatural one; the two do not differ because Bernard held that everything must be subordinated to man's religious calling. Where perhaps the difference lies, what leads us to call Bernard a mystical thinker but not Abelard, is the organic unity Bernard saw between the life of prayer, the spiritual life, and the intellectual life. For a thinker like Abelard there is a connection between studying the logical works of Aristotle and being a Christian, but it is an adventitious, almost extrinsic, connection. For Bernard it is not so. Everything the Christian does

must be intimately and essentially ordered to his final end. This need not lead, and in Bernard seldom led, to pietistic excursions away from the topic at hand. But one is struck in Bernard by the living unity of everything he did and wrote, its subordination to his drive for spiritual perfection. We see this in all his activities, whether he is counseling popes, chiding kings, criticizing other religious, refuting Abelard, preaching, or building monasteries. It is all one; everything must be subjected to a single criterion if it is to be justified and considered important. Man is made to know and love God. It is that simple. The loving knowledge of God in contemplation, the experiential knowledge of God, is the central thing -- not merely abstract arguments, not dialectical finesse, but loving union with the source of truth who is Truth, the source of knowledge who is Wisdom. Bernard, who had been granted that mystical union with God, could not take seriously the suggestion that a bloodless and neutral logic must preside over our talk of that infinite reality.

Now this indicates that there is a vantage point from which all human activity can be assessed. Bernard has much to say about the route that takes us to that vantage point. Let us consider what he has to say about the triad opinion, faith, and intelligence. Human knowledge bears first of all on created things, the things of this world. The visible world is a book in which divine truth can be read, but the script is smudged, the knowledge thus gained imperfect. Beyond such knowledge or opinion is faith. Faith marks an advance because of its certitude -- Bernard is therefore extremely critical of what he takes to be the import of Abelard's description of faith as *existimatio* -- but faith is a dark knowledge: the truth is hidden for it behind a veil. Beyond opinion and faith there is intelligence or understanding. Here not only is truth had, but knowledge that it is the truth. Here there is a similarity between knower and known. Understanding is had, if it is had, because of the presence of the Word in the soul. Understanding is beyond images, a gift; it is the purity and perfection of love where one is concerned only with the good of the other. How far this love is beyond love as we first know it! There is, first in time, a carnal love, selfish love. The direction in which we must go is from self-love to love of God. The perfection of love is the perfection of knowledge because love unites us with Wisdom itself. The progression here is a progression in freedom as well. Bernard will distinguish between various kinds of freedom. There is a natural freedom, one that belongs to us essentially because we are men. But there are two kinds of freedom which are added to us, which are not ours because of our nature. These are the freedoms of grace and glory.

Bibliographical Note

Thomas Merton's *The Last of the Fathers* (New York, 1954) is a brief and interesting introduction to St. Bernard; Watkin Williams' *Saint Bernard of Clairvaux* (Manchester, 1935) is useful if one can survive the style and format. Étienne Gilson's *The Mystical Theology of Saint Bernard* (New York, 1940) is the eminent medieval scholar at his best.

D. Other Figures

1. *Peter the Venerable* (1092-1147). We have encountered Peter the Venerable in our discussion of Abelard. The Abbot of Cluny gave Abelard asylum in the last year of the latter's life and was instrumental in effecting a reconciliation between Abelard and Bernard of Clairvaux. Peter the Venerable was a defender of the Cluniac interpretation of the Rule of St. Benedict and an advocate of an adaptation of the monastic ideal to the changing times in opposition to St. Bernard's call for the strict and literal Cistercian interpretation.

Peter visited Toledo, where he became acquainted with the translations being made there and was instrumental in getting the Koran translated into Latin. Lest this be seen as indicating sympathy on his part, we must add that he then wrote a refutation of the Islamic religion and tried to interest St. Bernard in doing the same. Peter also wrote against the Jews and various heretics. An important figure in the history of monasticism as well as in the history of spirituality, Peter the Venerable is of interest for us insofar as his conception of the monastic life did not preclude the kind of scholarly work which had long been associated with the monastery schools. His works can be found in Migne, PL, 189.

2. *William of St. Thierry*. William was born in Liège around 1080, studied at Reims or perhaps Laon, where he might have come into contact with Abelard. He became a monk at St. Thierry in 1113 and was elected abbot in 1119. In 1135 he resigned and became a Cistercian. He died in 1148.

William had little more than disdain for secular learning, both in itself and in its application to the faith. A man is called to love God, and this is not aided by study of Ovid or dialectics. The school of divine love is the cloister. In his various writings William attempts to set down the itinerary of man's will. His works, which are found in Migne, PL, 180 and 184, include *Epistola ad fratres de*

monte dei, speculum fidei, Aenigma fidei, De contemplando deo, De natura et dignitate amoris, De natura corporis et animae.

In William the stress is on love rather than knowledge, but ultimately there is a knowledge which issues from love. Speaking of man's nature, of body and soul, William distinguishes the life of the body, the life of the soul, and the life of spirit. This distinction between kinds of life provides him with the structure of the spiritual life. That life consists of three stages or moments. First, man finds himself bound by the senses and passions; he is as it were outside himself, and if he responds to the promptings of spirit, he does so with a sense of being constrained or forced. Second, there is the life of virtue. Virtue is a voluntary consent to the good. The contrast here is between the voluntary and the constrained; the spiritual life is a movement toward greater and greater freedom. The good may be known by natural knowledge and it may be desired, but there is not yet the fullness of love. Monastic asceticism is the school of charity which turns desire to love. Third is the spiritual life properly so called, which is marked by spontaneity and freedom. The perfect are prompted and led by the Holy Ghost. Such perfect inwardness cannot be learned from the masters of the schools; it comes only from complete docility to the movement of the Holy Ghost. The spiritual life is a condition of union with God: *cum fit homo unus cum Deo*. This is a unity of grace, not of nature; it means to will what God wills, so that there is no longer any difference between our will and God's. William will speak of the progression of the spiritual life in various ways, but always with the emphasis on will and love. Sometimes the progression is expressed as *voluntas, amor, caritas, sapientia* (will, love, charity, wisdom); sometimes as *amor, dilectio, caritas, unitas, spiritus* (love, affection, charity, unity, spirit). But more often than not, William stresses that love is the vehicle of knowledge or, better, of wisdom. *Amor crescit in caritatem, caritas in sapientiam* (love grows into charity and charity into wisdom). The knowledge given man by the Holy Ghost, not the knowledge of the schools, is what life is all about. We do not know God by disputation, by dialectics, by endless wrangling. Charity is the eye with which we see God (*ipsa caritas est oculus quo videtur Deus*).

One can see here the difference between a mystic like William of St. Thierry and one like Hugh of St. Victor. For the latter there is no need to choose between secular learning and the interior life. All things work together for good; secular learning responds to something real in man, something which remains in him as Christian, and he can turn it into an instrument for arriving at his supernatural goal. William, on the other hand, convinced of the vanity of this world, is more struck by the way in which secular learning can be an impediment to the one

thing needful, and he warns against it. What we are called to, what will perfect us, is not something we can achieve by our own efforts; it is not something within the grasp of the naturally talented but withheld from the unlearned and simple. William, we may be sure, would not understand the charge that his position is an obscurantist one. He would no doubt reply that to devote oneself to spiritual perfection, to be responsive to the promptings of grace and the Holy Ghost, to live the life of charity -- that is to come into possession of the fullness of wisdom. What could be lacking in one who has the fullness of wisdom? If the cautiously inclusive attitude of Hugh of St. Victor seems preferable, we must nonetheless keep in mind that what William was confronted by was not dialectics in the abstract but singular dialecticians, men like Abelard. Abelard may not in the long run and in his writings have been so distant from the emphasis William made, but on the hoof, so to speak, Abelard must have appeared a dangerous and disruptive force, an almost demonic presence. The remedy, at least as far as William of St. Thierry was concerned, was to eschew what Abelard engaged in, retire from the world, and let God work his marvels in the soul.

3. *Isaac of Stella*. An Englishman by birth, Isaac became a Cistercian and, in 1167, was elected abbot of the abbey of Stella near Poitiers. He was an unusual Cistercian in that he employed dialectics in his writings effectively and unapologetically. But the dialectics is at the service of a constant theme of St. Bernard of Clairvaux: the vanity of the world, the nothingness of creatures. Isaac makes this point by engaging in the discussion over the status of universals. He begins by distinguishing substance and accident. The being of accidents is to inhere in substance; accidents enjoy no autonomous existence. But how is it with substance? Well, we must distinguish between first or primary substance, for example, this man, and second substance, for example, Man. The individual man would not exist if there were not Man, or human nature, and human nature exists only if there are individual men. Thus, not only accidents are imperfect beings but also substance, whether considered as universal or singular. From this Isaac draws the surprising conclusion that creatures are nothing, certainly nothing in themselves, since creatures are either substances or accidents and these have been shown to have at best a precarious hold on existence. God alone exists of himself: God is both autonomous in existence and immutable. Thus, God is distinguished from accidents and from both first and second substance. God's existence is discoverable by reflecting on the "nothingness" of creatures; their being, because it is so precarious that it deserves to be called nothing, demands the being God is. Can we speak of God? Isaac distinguishes levels of theology: divine theology consists of negations,

claiming that we can affirm nothing literally of God; symbolic theology is metaphorical and speaks of God as a lion and so forth. Between these two is another kind of theology which speaks of God neither literally, for that can produce only negations, nor metaphorically. God is said to be wise and just, not metaphorically, as he may be called a lion, and not literally either, since to say God is just is not to say the same thing as to say that a man is just.

In speaking of the soul and its faculties, Isaac, like Alcher of Clairvaux, is interested in relating the powers of the soul to the stages of the spiritual life. Through sensation the soul is in touch with the corporeal world; through its highest faculty, intelligence, the soul attains to the Holy Ghost and then, thanks to the influence of the Holy Ghost within it, the soul comes to knowledge of the Word and then of the Father. Isaac defines soul as *similitudo omnium*, the likeness of all things; the plurality of faculties of the soul is taken to be an image of the Trinity.

The works of Isaac of Stella are to be found in Migne, PL, 194.

4. Alcher of Clairvaux. Alcher is noteworthy for a work on the soul, the *De spiritu et anima* (PL, 40, 779-832). Aquinas had a low opinion of it and dismissed the suggestion that it was a work of Augustine. "This book, *Concerning Spirit and Soul*," he wrote, "is not by Augustine; it is said to have been written by some Cistercian. As for its contents, they are not worth bothering about." (*Q.D. de anima*, a. 12, ad 1) The work is a compilation of texts taken from Augustine, Boethius, Cassiodorus, Alcuin, Hugh of St. Victor, and Bernard of Clairvaux. The definition of soul (*animus*) Alcher gives became famous: the soul is a substance which participates in reason and is so fashioned as to rule the body (*animus est substantia quaedam rationis particeps, regendo corpori accomodata*). But the *De spiritu et anima* is not simply concerned with the nature of soul and its faculties. It goes on to discuss the spiritual life. The route of perfection is Augustinian. The soul must turn upon itself if it would go to God, for the soul is the image of God.

5. *Alan of Lille*. Poet, theologian, apologist, philosopher, Alan of Lille (Alanus de Insulis) was born about 1128. He is noteworthy for his contributions to theological method, which indicate a profound influence of Boethius. In his *De hebdomadibus* Boethius proposed to proceed by first setting down a set of propositions or maxims and then subjecting them to analysis in such a way that he seems to be elaborating an axiomatic system. As there are echoes of the Proclus of the *Elements of Theology* in this work of Boethius, so there are echoes

of Boethius in Alan's *Rules of Sacred Theology*. What Alan thought he was doing is clear; his method is an application to theology of something common to the other sciences. Each science proceeds from maxims or axioms: in rhetoric these are commonplaces (*loci communes*); and in dialectic, ethics, geometry, and music there are analogous common principles. Theology must also begin from rules or axioms, although these are very obscure and subtle and may be called paradoxes or enigmas. Given the character of the starting points of theology, they should not be given over to discussion by the uninstructed or those whose thoughts are completely bound to sensed objects. Many of the rules Alan set down, pithy axiomatic statements, became the common currency of theological discussion, though, of course, not all of them were original with him. Thus, he takes from Boethius the identity of essence and existence in God: *omne simplex esse suum et id quod est unum habet*. A Neoplatonic influence is apparent in the very first maxim Alan sets down: *Monas est qua quaelibet res est una* (the Monad is that whereby anything is one). The influence of Pseudo-Dionysius is apparent in another. Only negations can be truly and properly predicated of God since by them we remove from God what cannot inhere in him (*negationes vero de Deo dictae et verae et propriae sunt, secundum quas removetur a Deo quod ei per inhaerentiam non convenit*). Alan's work is also influenced by the so-called Hermetic writings.

Taking his cue from Boethius as well as from Chalcidius, Alan develops a remarkable doctrine on the nature of matter. This aspect of his teaching links him with the school of Chartres. In discussing the various meanings of the word "nature," Alan singles out a meaning according to which nature is an intermediary between God and the world, something reminiscent of Erigena's *natura quae creatur et creat*.

Alan's apologetic work not only is directed against the Albigensians but also takes into account the Jewish and Islamic religions. His apologetic effort is guided by a very intense feeling for the unity of Christianity, and his trump card against heretics is that they threaten that unity; as for non-Christians, their failure to take sufficiently into account the unity of mankind's religious experience is taken as a mark against them.

The poetic work of Alan includes the *Anticlaudianus*, in which he argues for the unity of nature and virtue, and *The Complaint of Nature*. The latter, a mixture of prose and poetry, also has as its theme the relation between nature and virtue. His poetry has earned Alan the title of Christian humanist. He resigned his

chair of theology and retired to the monastery at Citeaux, where he died in 1202.

Bibliographical Note

The works of Alan can be found in Migne, PL, 120. These include *Ars predicatoria*, *De fide catholica contra haereticos sui temporis praesertim Albigenses*, *Regula de sacra theologia*, *Anticlaudianus*, *De planctu naturae*. There are English translations of the *Anticlaudianus*, by W. H. Cornog (Philadelphia, 1935), and of the *De planctu naturae*, by D. M. Moffat (New York, 1908).

Chapter VI

Dominicus Gundissalinus

Dominicus Gundissalinus was a member of the Toledo school of translators of Islamic and Judaic writings which was established by the archbishop of that city, Raymond (1126-1151). Others of the school were John of Spain, Gerard of Cremona, and, later, Michael Scot and Herman the German. The writings of Gundissalinus are now placed in the second half of the twelfth century, probably under Archbishop John (1151-1166). Prior to the establishment of the Toledo school there had been translations made (for example, by Adelhard of Bath), but such efforts were sporadic and unorganized. Spain was the logical place for such work, for there intimate contact between Latin Christian culture and Judaism and Islam was a fact of life. Converts from these faiths were a major source of the works which came to be translated into Latin.

Gundissalinus is thought to have been a convert from Judaism. The ancient texts which were thus introduced into the West had been filtered through a number of languages before finding their way into Latin. The Nestorian school at Edessa (431-489) translated many works from Greek into Syriac. It is interesting that Cassiodorus mentions both Alexandrian and Syrian scholarly efforts. (PL, 70, 1105) The object of these efforts was not Aristotle alone but also the works of the Alexandrian commentators. Such work is thought to have continued from the fifth to the eighth century. In the eighth century Syrian scholars were summoned to the courts of the Caliphs of Bagdad. One of these scholars, Henin Ben Isaac, translated works from Syriac into Arabic. In the eleventh and twelfth centuries this heritage became available to Jews and Christians. The last step, into Latin, was in itself a somewhat complicated one. At Toledo, for example, an Arabic or Jewish text was first translated into the vernacular, Spanish, and it was that version that someone might put into Latin. What was thus translated was not simply a text of Aristotle, say, but such a text together with an Arabian commentary on it. Thus at the same time that Aristotle and his Alexandrian commentators were introduced to the Latin West, Alkindi, Alfarabi, Algazel, Avicenna, and Averroes came to be known. This fact was to have not a little influence on Aristotle's fate in European universities.

Unlike most other translators, Gundissalinus wrote independent philosophical works. Besides his work on the divisions of philosophy (*De divisione philosophiae*), at which we will take a sustained look, he wrote on the creation of the world, the immortality of the soul, and unity.

De divisione philosophiae. Although this work exhibits a great deal of community with the tradition on the relationship between the arts and philosophy, a tradition to which we have been alluding in what has gone before, Gundissalinus strikes a note that is definitely new, a note which anticipates the sort of approach to the nature of philosophy which in the thirteenth century will be taken by Thomas Aquinas. In devoting a modest amount of space to Gundissalinus' map of philosophy, we leave it to the reader to compare what the Spaniard has to say with what has been said earlier about other twelfth-century views on the division of philosophy.

The *De divisione philosophiae* begins with a fairly familiar lament: *Felix prior aetas*: Alas, for the good old days. The phrase, as it happens, is lifted from Boethius' *Consolation of Philosophy*. Although once avidly pursued, philosophy is nowadays fallen into oblivion, for men are too concerned with worldly matters. To help rectify this situation, Gundissalinus proposes to write a kind of summary of wisdom in which he will do three things: show (1) what wisdom is, (2) what its parts are, and (3) the usefulness of each part.⁽¹⁾

Since everyone prefers some things to other things, and things are preferred either with reference to flesh or to spirit, we must examine what is sought by flesh and what is sought by spirit. The goods sought by the flesh are of three kinds. Some are necessary, for they sustain us, and they are either provided by nature (for example, food and drink) or by art (for example, medicine). Others are such that they are pleasant (for example, fine clothes, well-prepared food, sex). Finally, some things are sought by flesh out of curiosity (*curiositas*), for example, superfluous possessions and riches. Those who seek such things are corrupt and abominable.

The concerns of spirit are also threefold. Some are harmful, such as moral vice; others are vain, such as worldly honor and magic; finally, some are useful, such as virtues and worthwhile sciences. It is in the latter that human perfection consists, since human perfection cannot be had in virtue without knowledge or in knowledge without virtue.

Sciences are of two kinds, human and divine. Divine science is that which is revealed to man by God, for example, the Old and New Testaments. The sign of such a science is that it is introduced by "The Lord God spoke . . ." and "Jesus said to his disciples"

Human science is that which is attained by human discourse, for example, all the arts which are called liberal. Some human sciences pertain to eloquence, others to wisdom. Grammar, poetry, rhetoric, and law belong to eloquence, for they enable one to speak correctly and ornately. Those belong to wisdom which enlighten the soul with respect to the knowledge of truth or elevate it to the level of the good. Now all of these sciences belong to philosophy; there is no science which is not part of philosophy. Here Gundissalinus sets himself a fourfold task: he will determine what philosophy is, what its intention or end is, what its parts are, and what each part is concerned with.

A. What Is Philosophy?

This discussion is divided into two parts, in the first of which Gundissalinus gives us definitions of philosophy. The second part establishes the intention of philosophy and assigns its parts. With respect to the definition of philosophy. Gundissalinus suggests some definitions drawn from what is proper to it and others taken from the effect of philosophy. As a matter of fact, six definitions of philosophy are given, four arising from what is proper to philosophy and two from its effect.

The first four definitions are (1) philosophy is the assimilation of man to the works of the creator insofar as humanly possible, (2) philosophy is the study of death, (3) philosophy is the knowledge of things human and divine conjoined with the effort to live well, (4) philosophy is the art of arts and the discipline of disciplines. Baur gives Isaac's book of definitions as the immediate source of the first two definitions and Isadore as the source of the next two. He also indicates (p. 169ff.), however, the ancient sources for these definitions, citing *Theaetetus* 176AB as the source of the first two.

As found in Plato, these seem to reflect a Pythagorean influence, a mystical direction of thought in which philosophy is ordered to religion, speculation to intuition and ecstasy. For Gundissalinus, however, the first definition has a straightforward, scientific meaning. Philosophy is the assimilation of man to the works of God in the sense that it is the perception of the truth of things, the truth of knowledge, and the truth of operation. But to know the truth of things is to know them in their causes. Gundissalinus then enumerates the four species of cause taught by Aristotle, dividing each species into spiritual and corporeal. The second definition, that which sees philosophy as solicitude for death, is interpreted as meaning the mortification of base desires, a prerequisite for the pursuit and acquisition of truth.

Baur's research into the sources of the third and fourth definitions is particularly interesting for us since, as has already been said, they are taken from Isadore. Where did Isadore get them? Baur sees these definitions as Stoic in origin, citing a fragment of the Pseudo-Plutarch in the *Placita philosophorum*. The fourth definition is thought to be derived from Aristotle. (*Meta.*, 1,2)

The definition of philosophy from its effect is "Man's complete knowledge of himself." The relation of this definition to the dictum of the Delphic Oracle is noted by Baur, a dictum whose ethical import is clear. It can be seen to suggest that introspection is a source of knowledge of the macrocosm. We can see in this the option of Neoplatonism and Augustinianism: "*Noli foras ire, in te redi: in interiore homine habitat veritas, et si tuam naturam mutabilem inveneris, transcede et te ipsum*" [Go not about, retire within: truth dwells in the inner man, and should you find your own truth mutable, go on beyond yourself]. (*De vera religione*, chap. 39, n. 72) Gundissalinus' interpretation of this definition could hardly be less mystical. In man substance and accident are found, and not only that but both spiritual and corporeal substance and accident. Now since whatever is is either substance or accident, spiritual or corporeal, man is a sort of compendium of being; for him to know himself will be in a way to know whatever is.

The sixth definition given of philosophy is etymological: philosophy is the love of wisdom; the philosopher is one who seeks wisdom. Wisdom itself is definable in two ways: first from its proper nature, second from its effect. "Wisdom is the true knowledge of first and sempiternal things." These first things are described in terms of emanation, somewhat redolent of the *Fons vitae* of Avicenna. The first genus is created immediately by God, and from it come other genera. Individuals and species receive their names and definitions from the genera, and, thus, owing to the genera each this is what it is and has what truth it has. Truth is that which is. Thus, we can say that wisdom is true knowledge of the first and sempiternal things. Finally, wisdom is the intellectual comprehension of what is true and false in every area.

B. The Division of Philosophy

Philosophy is an attempt to understand all things insofar as this is humanly possible. A first division of things is that into those which result from our willing (for example, laws, constitutions, wars, rites, and so on) and those which do not. Only God in no way comes to be; every creature comes to be, whether before time (angels and matter), with time (celestial bodies and earthly

elements), or in time (everything else). Those which come to be in time either will never have an end (for example, soul) or will have an end. Of those things which will have an end, some are due to nature, others to art.

1. *Theoretical and Practical Philosophy*. Since whatever is is due either to our willing or to God or nature, philosophy is first divided into two parts. The first, having to do with human affairs, is *practical* philosophy, which seeks to know what we ought to do; the second, *theoretical* philosophy, having to do with everything other than human works, seeks to learn what ought to be known. Gundissalinus goes on to make several distinctions calculated to clarify this initial division of philosophy. Theoretical philosophy is in the intellect, consisting only in the mind's knowledge; practical philosophy is in doing (*in effect*) and consists in the execution of a work. Philosophy is sought for the perfection of the soul, and this is achieved by science and operation. Operation pertains to the sensible part of the soul, speculation to the rational part. The rational part of the soul is divided by the concern with divine things not elements of our work and with human things. The end of the speculative is in knowledge, the end of the practical in knowledge of what ought to be done. The principles of this division are, in the first place, objects (divine and human things), mode (knowing and knowing for doing), and the parts of soul involved (rational and sensible). But, lest one think that practical philosophy is action, Gundissalinus makes clear that, as philosophy, it is rational knowledge of what ought to be done.

2. *Divisions of Theoretical Philosophy*. In assigning the parts of theoretical philosophy our author gives two accounts of how the division is made and then compares them with the doctrine of Boethius, *De trinitate*, chapter two. We will set down in schematic form the divisions given in the text.

First Division

Theoretical knowledge has as its object whatever does not result from our willing. But such things are

- 1. either such that motion cannot belong (*accidit*) to them (God, angels),
- 2. or such that motion can belong (*accidit*) to them,
 - (a) some of which can exist without motion (e.g., one, cause)
 - (b) while others cannot,
 - (i) though some can be understood without proper motion (e.g., square)

- (ii) while others cannot be so understood (e.g., humanity).

With respect to the things which would fall under 2a, our author notes that they can be considered in two ways, either according to proper matter and motion or without them. Examples of the first mode would be the consideration that fire is one, the elements are four, hot and cold are causes, the soul is a principle. These things -- namely one, four, cause, and principle -- can exist apart from matter. The second mode pertains to 2b (i), and refers to the consideration of mathematical apart from proper matter and motion. What this seems to be saying is this: mathematical exist in material and mobile things but can be considered without including proper matter and motion. So too such things as cause, principle, and unity are found in material and mobile things, but, unlike mathematical, they are also sometimes found existing apart from matter and motion. Thus, although matter and motion are not accidental to material and mobile causes, matter and motion *accidunt* to cause as such.

Second Division

Whatever is understood

- 1. either exists altogether apart from matter and motion,
 - (a) some such that it is impossible for them to exist in matter and motion, such as God and the angels,
 - (b) others such that it is not necessary that they be in matter and motion, such as unity and cause,
- 2. or exists in matter and motion,
 - (a) although some can be understood without matter and motion, such as figure, square, circularity, curve, etc.,
 - (b) while others cannot be understood without matter and motion, such as man, vegetable, animal, etc.

That each of these divisions is in agreement with the doctrine of Boethius is next shown. The two divisions come down to saying that speculation is concerned either (1) with what is not separate from its matter either in existing or in the intellect, or (2) with what is separated from matter in the intellect but not as it exists, or (3) with what is separate from matter in existing and in the intellect. The science concerned with the first kind of things is called physics or natural science; the science concerned with the second, mathematics; that concerned with the third, first science or first philosophy or metaphysics. It is precisely these sciences and such objects which Boethius describes when he says

that physics is *in abstracta* and with motion; mathematics, abstract and with motion; theology, abstract and without motion. What is more, this division of speculative philosophy is the one given by Aristotle.

3. *Division of Practical Philosophy.* Our author introduces this division by noting that future happiness requires not only science of what should be understood but also knowledge of what is good. Thus, practical philosophy too is necessary. And, as it happens, practical philosophy too is divided into three parts.

One part of it is the science that has to do with intercourse with all men, something which requires knowledge of grammar, poetry, rhetoric, and secular law. These provide for that science of ruling states and of knowing the rights of citizens which is called political science. Second, there is a science concerned with the household and one's own family. By means of it knowledge is had of the relations of man with wife, children, and servants and of all domestic matters. This science, usually called economics (from the Greek *oikia*, home), Gundissalinus calls family government.

A third science is that by which a man knows how to regulate himself. This is ethics or moral science. Since a man lives either alone or with others and, if with others, either with his family or with his co-citizens, the division of practical philosophy is seen to be adequate.

4. *Logic and the Schema of Philosophy.* The six sciences already enumerated contain whatever can and should be known. Because this is so and because they are precisely the parts of philosophy, the intention of philosophy is said to be the understanding of all things insofar as this is humanly possible. And since philosophy has as its effect the perfection of the soul, it has been pointed out that the end of practical philosophy is the love of the good, that, of speculative philosophy the knowledge of the truth.

Truth, however, is either known or unknown. Examples of known truths are that two is more than one and that the whole is greater than its part. Unknown truths, such as that the world began and that angels are composed of matter and form(!) require demonstration. What is unknown comes to be known through something else previously known. Logic is the science which teaches how to bring about this transformation. For this reason logic is naturally prior to every theoretical science and is necessary to each. However, since truth is expressed in propositions and these are composed of terms, grammar, whose concern is the composition of terms, must precede logic.

Gundissalinus holds that every science is either a part or an instrument of philosophy. Examples of parts would be mathematics and physics; of an instrument, grammar. Grammar is only an instrument, for although it is necessary in order to teach philosophy, it is not necessary in order to know it. But since philosophy inquires into the dispositions of its subject, logic is not only an instrument but also a part of philosophy.

Baur (p. 193) draws up the following schema to represent the doctrine of Gundissalinus' *De divisione philosophiae*.

- I. Propaedeutic Sciences (Sciences of Eloquence)
 - 1. Scientia litteralis: grammar
 - 2. Scientiae civiles: poetics and rhetoric

- II. Logic

Logic is situated midway between the sciences of eloquence and the sciences of wisdom. However, two of the sciences of eloquence enter into the parts of logic Gundissalinus sets down: *Categories*, *Perihemeneias*, *Analytica Priora*, *Analytica Posteriora*, *Topica*, *Sophistica*, *Rhetorica*, *Poetica*. These "parts" are simply names of Aristotelian works.

- III. Properly Philosophical Sciences (Sciences of Wisdom)
 - 1. Theoretical
 - (a) physics
 - (b) mathematics
 - (c) metaphysics
 - 2. Practical
 - (a) politics
 - (b) economics
 - (c) ethics

It is clear that the liberal arts do not as such constitute the main concern of Gundissalinus in his division of philosophy. He mentions them but once, and then seemingly suggests that any human science can be called a liberal art. The mechanical arts come up for discussion, briefly, when economics is considered. His understanding of them would seem to be that these arts transform natural

matter in order to make objects useful for man. Gundissalinus suggests a division of them according to the natural matter transformed, according to whether their matter is inanimate or (formerly) animate body. When medicine is distinguished from the liberal arts, we are not faced with anything new, for Isadore (IV, 13, 1-5) makes the same distinction. We need not look in the *De divisione philosophiae* for any precisions on the meaning of the phrase "liberal art" or for any distinction of art from science. Gundissalinus is content to accept Cassiodorus on the definitions of art and science, indicating that these are simply different names for the same thing. Indeed, for Gundissalinus metaphysics is an art and the metaphysician an artifex. What the schema just set down indicates (if it be supplemented by the division of mathematics into its parts⁽²¹⁾) is that the trivium and quadrivium have been wholly subsumed under the more important division of philosophy as a whole. It is this division and the use he makes of it that sets the work of Gundissalinus off from all other views discussed earlier, even those of Hugh of St. Victor. The exact nature of Gundissalinus' difference from the others we shall now endeavor to make plain.

The reader will have noted the similarity of the division of Gundissalinus and that of Aristotle discussed in volume one (McInerny, *A History of Western Philosophy: From the Beginnings of Philosophy to Plotinus* [Notre Dame, 1963], pp. 222ff.). Should this be a surprise? As Gundissalinus himself indicates, the division of speculative philosophy that he sets down is that of Aristotle, but it is as well the division Boethius gives in his *De trinitate*, chapter two. Moreover, this division, doubtless thanks to Boethius, is present in many of the books which influenced the tradition of the liberal arts up to and into the twelfth century. In Isadore, for example, we find this division of speculative philosophy. Nevertheless, there is a difference. The division of philosophy which is most operative in the tradition we have been examining is that which divided philosophy into logic, physics, and ethics. This is the division set down by St. Augustine, (Cf. *De civ. dei*, VIII, 4; II, 7; XI, 25.) Moreover, it is the division favored by Cicero. This division is clearly the one most influential on Cassiodorus, Isadore, Alcuin, Rhabanus Maurus, Scotus Erigena, Gilbert of la Porrée, and John of Salisbury. Baur sees the work of Cassiodorus as primarily an introduction to the study of Sacred Scripture, having this in common with Augustine's *De doctrina christiana* before him and, after him, with Rhabanus Maurus' *De clericorum institutione*. For this reason, secular sciences were shrunk to the seven liberal arts, and physical and metaphysical speculation was set aside. "Theology, in the sense of the theology of Christian revelation, takes the place of metaphysics as the queen of the sciences." (p. 353) The justice of Baur's remark will be clear if one considers the manner in which Rhabanus Maurus,

for example, discusses the notion of wisdom as it enters into the definition of philosophy. The wisdom involved is precisely that revealed by Christ in the Scriptures. If the liberal arts are useful for the Christian, if they are the pillars on which wisdom is raised, this is simply, according to Augustine, Cassiodorus, Alcuin, Rhabanus Maurus, and so forth, because they are useful in reading the Scriptures.

The new note struck by Gundissalinus, a note possible only because of the influx of the Arabian Aristotelianism, is that there is a wisdom distinct from what has been revealed, a metaphysics to which philosophical sciences are ordered. It was impossible for earlier thinkers to so interpret the third member of Boethius' division in the *De trinitate*, chapter two. Theology was the knowledge of God handed down in the Scriptures; philosophy was a melange of propaedeutic arts and revealed wisdom. The materials with which Gundissalinus is dealing are precisely those which in the thirteenth century will pose the problem of a relationship between philosophy and theology. This problem is not formally posed prior to the introduction of the corpus aristotelicum into the Latin West. There is, of course, the problem of faith and reason, but that is not the same problem as that of the relation between philosophy and theology.

It may not be immediately evident that a difference exists between Gundissalinus and Hugh of St. Victor. That such a difference does exist is clear from the fact that the Victorine school is usually considered to be a mystical one. What does this mean? As we have seen, the purpose of philosophy, the goal of philosophy, is a wisdom which will rectify the nature of man which has been disintegrated by sin. "Omnium autem humanarum actionum seu studiorum, quae sapientia moderatur, finis et intentio ad hoc spectare debet, ut vel naturae nostrae reparetur integritas, vel defectuum quibus praesens subiacet vita temperetur necessitas" [The end or intention sought in any human action or pursuit, guided by wisdom, is either that the integrity of nature might be restored or that the harshness stemming from the flaws to which our present life is subjected be tempered]. (*Didase.*, I, 5, p. 12, 3-6) The wisdom with which Hugh is concerned is not the speculative science which is metaphysics, anymore than it would appear to be that theological science whose beginnings had long been had. The goal of philosophy is one of union with God, a condition of man which is no more attained in the sciences than it is attained in the mechanical arts or in philosophical knowledge of them, but via them. We have tried to indicate how Gundissalinus, on the other hand, can so interpret definitions of philosophy whose origins are mystical or ethical that they have a straightforward

scientific meaning. In this Gundissalinus is the precursor in a special way of the directions taken by philosophical thought in the thirteenth century.

In his study of unity, in his work on the soul, Gundissalinus, while paying deference to such writers as Boethius, draws much of his inspiration from Arab thinkers. Thus, as translator and independent thinker, Gundissalinus provides for us, writ small as it were, the problem which faces the West with the influx of the Aristotelian corpus together with Arab commentaries. Like the last generation of Greek commentators on Aristotle, men contemporary with Boethius, there is a good deal of Neoplatonism among the Arab commentators on Aristotle. This creates difficulties not only for a true understanding of the text of Aristotle but also with Christian orthodoxy.

Bibliographical Note

Dominicus Gundissalinus: De divisione philosophiae, edited by Ludwig Baur, *Beitrage*, vol. 4, II/III (1903); *De processione mundi*, edited by G. Buelow, *Beitrage*, vol. 24, III (1925); *De unitate*, edited by P. Correns, *Beitrage*, vol. 1, I (1891). See J. T. Muckle, "The Treatise *De anima* of Dominicus Gundissalinus," *Mediaeval Studies*, 2 (1940), pp. 23-103; A. H. Chroust, "The Definitions of Philosophy in the *De Divisione Philosophiae* of D. Gundissalinus," *The New Scholasticism*, 25 (1951), pp. 253-281.

{1} A general admission of indebtedness to the remarkable work of Ludwig Baur must be made here. Baur not only edited the text and wrote a brilliant analysis of it from the point of view of its sources but also traced the history of Einleitungslitteratur from antiquity to the Middle Ages. Written in 1903, these historical studies of Baur remain today indispensable for research into questions concerning introductions to philosophy, divisions of philosophy, the Platonic and Aristotelian currents in Scholasticism, and so on. Exact reference to Baur is in the Bibliographical Note at the end of this chapter. {2} On page 32 of the text Gundissalinus gives the division of mathematics to be found in Cassiodorus: mathematics is concerned either with magnitude or multitude. Magnitudes are either immobile and the concern of geometry, or mobile and the concern of astrology; multitude is either considered in itself, as by arithmetic, or with reference to something else, as in music.

The Thirteenth Century

Chapter I

The Background

What Freud said of the life of an individual can be applied to history at large: in retrospect it takes on an inevitability and natural progression that it does not possess when one is at the beginning or in midstream. When we consider the thirteenth century as the point of arrival of what had gone before, it is possible so to arrange the data that the rise of universities, the full development of Scholastic theology as against philosophy, and all the rest seem to flow almost effortlessly from their antecedents as if any other outcome would be unthinkable. The contrary view leans rather heavily on the much overworked term "renaissance." We are confronted first with the Carolingian Renaissance, next with the twelfth-century renaissance, then with the renaissance of the thirteenth century, and finally with the Renaissance with the resultant picture of discontinuous bootstrap efforts which bear little positive relation to one another, though each points back in various ways to classical times. It would be difficult to decide definitively for either view, particularly if we advert to the original analogue, the life of the individual. Our lives may seem at once a continuum of deeds culminating in what we now are and a discontinuous succession of turning points at each of which we refashioned ourselves. Neither view alone would be sufficient; each has an interpretative value.

In the present chapter we want to look at the thirteenth century both as the *telos* toward which earlier efforts in the philosophy of the Christian West tended and as something surprising, unforeseeable, and quite *sui generis*. The first viewpoint is valuable in discussing the rise of the universities, which can be regarded as evolving out of previous modes of instruction; the second seems called for when we consider the impact on the West of Islamic philosophy, which was the vehicle whereby the integral Aristotle first came into view. Islamic philosophy and its influence on the West force us to see the need for both of the viewpoints we have mentioned. On the one hand, the Arabian Aristotelians represent a threat to the Augustinian tradition which was dominant in theology and, on the other hand, their Aristotelianism must be viewed with relation to the Aristotle already known and influential in the West. Furthermore, the Neoplatonism of the Islamic philosophers provides a common note with that operative in Augustine and Boethius, and yet because Islamic philosophy brings with it closer contact

with Neoplatonic sources, there is an element of strangeness and difference. In short, Islamic philosophy and its influence on the West demand that we see the ambiguity of the thirteenth century with respect to what had gone before. There is both continuity and disruption, a modification of an ongoing effort and quite fundamental changes in the conception of the nature of that effort. Finally, we will make some general remarks about the sources of the philosophizing of the thirteenth century.

A. The Universities

The preceding chapters have acquainted us with the palace school begun by Alcuin as well as with the fact that Alcuin was already associated with a cathedral school when the invitation from Charlemagne came to him. During the Carolingian Renaissance, as we have seen, great emphasis was put on the establishment of cathedral and monastic schools, and during the twelfth century the men we have considered were associated with one or the other of the latter types of school. At Paris there were schools on Mont Ste. Genevieve, at the monastery of St. Victor, and at the cathedral; it was from the last, the cathedral school, that the University of Paris evolved. The thirteenth century saw the rise of a great many universities, those of Salerno, Bologna, Paris, Montpellier, and Oxford. We shall study this phenomenon in terms of the University of Paris if only because so many of the men to be considered in the following chapters were associated with that university.

The cathedral school of Paris first came into real prominence with William of Champeaux, and the city's importance on the educational map was further enhanced by Abelard's tenures there. It is generally recognized that the University of Paris did not exist at the time of Abelard. The first statute of the university dates from 1215, though this seems to be a confirmation of something already established however inchoatively. But what is it we are talking about when we talk of a university?

The model of the university was the medieval guild; the university is a society of masters and scholars. Sometimes the guild was made up of the students, as in the south; sometimes, as was the case at Paris, the guild comprised the masters. In the latter case students can be regarded as apprentices who are candidates for full membership in the guild, that is, to the society of masters. It is thought that the masters formed a corporation because of a struggle with the chancellor of the cathedral school at Paris. With the recognition of the autonomy of the university, or society of masters, control of the granting of licenses to teach

passed from the chancellor to the rector of the university, who was elected by his peers; at Paris the rector of the faculty of arts was also the rector of the university.

There were four faculties at Paris -- arts, law, medicine, and theology -- with the faculty of arts serving as preparation for the others and thus as the undergraduate college, so to speak. The principal purpose of the university was to train future masters who, after prescribed courses of studies and the successful passing of examinations, were granted degrees. The degree arose quite naturally out of the license to teach. However, not all those who received a degree became teaching masters at the university, thus the distinction between the *magistri regentes* and *magistri non regentes*. The striking thing about the medieval university as it came to be constituted was its autonomy, its freedom from pressure of both an ecclesiastical and a political sort. The University of Paris was from the outset an international university; indeed, besides the division into faculties there was a division of the masters into nations. Of course, since the masters were members of the clergy, both secular and regular, freedom from religious pressure often amounted to little more than freedom from the local bishop. Moreover, since the masters were believers, the constraints of faith on their work, if "constraint" is the right term, could scarcely be considered as emanating from an external source. It is safe to assume that no master wanted or intended to teach anything contrary to the received doctrines of the Church; often it was judged that he nonetheless was so teaching, and condemnation was certainly not unknown. Academic freedom in its most responsible sense was surely present in the medieval university; a master was answerable to his peers, and free and open debate, public occasions when he would defend his views against all comers, both students and fellow masters, were frequent. For sheer hurly-burly of debate and disputation there has probably been nothing to equal the medieval university.

As has been pointed out, the chief purpose of the society of masters was to train others to become masters in their subject. The student entering the faculty of arts was thirteen or fourteen years old, and he embarked on a course of studies which continued for something over four years (even more at universities other than Paris). The curriculum of the arts faculty can conveniently be thought of in terms of the trivium and quadrivium, and the basic mode of instruction was the *lectio*, which was a lecture, not in the modern sense, but in the older sense of a reading. Stated books were read and commented on: in grammar, Priscian; in logic, Porphyry and subsequently the entire *Organon* of Aristotle. Some of the *Nicomachean Ethics* was also read; in the quadrivium no particular books were

prescribed in the statutes of 1215, but the *Metaphysics* of Aristotle as well as his natural writings, newly introduced in the West, are excluded from consideration and may not be read. This prohibition was later lifted -- certainly it came not to be heeded -- and with the passage of time other books were prescribed for the arts course.

To finish the arts course was to obtain a license to teach in that faculty and to pursue studies in one of the others. The hours of instruction in theology, for example, were such that a master from the faculty of arts could do his teaching and then attend lectures in theology. As a student of theology one followed lectures on Scripture for four years, after which two years were spent attending lectures on the *Sentences* of Peter Lombard. When one had finished this course and had attained the age of twenty-six, he received the baccalaureate and himself lectured on Scripture for two years and subsequently on the *Sentences*. The doctorate of theology could then be awarded if one had achieved the age of thirty-four and fulfilled other requirements such as holding public disputations.

Besides the *lectiones* there were two kinds of disputation or inquiry, the *Quaestio Disputata* and the *Quaestio Quodlibetalis*. The former could be a fairly regular classroom feature. The procedure was as follows. A thesis was proposed, objections to it were entertained, and finally a resolution was given and the objections resolved. In the classroom the *baccalarius* might make the first attempt at replying to the difficulties proposed, to be followed by the more magisterial resolution of the master. Disputed questions swinging around a common theme could be entertained in the course of a year and be productive of the sort of thing we have in Aquinas' *Disputed Question on Truth*, which is a series of *quaestiones*. The written form of such classroom disputations was sometimes the report of a student, sometimes the composition of the master himself. The quodlibetal questions were just that, on anything at all, and they were entertained at specific times during Advent and Lent when the participants or interlocutors could be other masters. These seem to have been very arduous affairs for the master who undertook them; they were certainly occasions when he would have to prove his mettle or suffer a diminution of prestige. It was not incumbent on a master to subject himself to this ordeal, however; the master was also free in setting the number of disputed questions he would handle.

The style of the *quaestiones* gradually made inroads on the *lectiones*, so that commentaries on the *Sentences*, for example, quickly became a suite of questions. The style of the *Summa theologiae* of Aquinas (but not of his *Summa*

contra gentes) reflects that of public disputations, though this was from the outset a written work. The style of the schools, the Scholastic method, exhibits, even on the printed page, the flavor of inquiry, disputation, and dialectic that animated the medieval university. A *quaestio* of the *Summa theologiae*, for example, is first articulated into a number of subsidiary questions or articles. An "article" of the *Summa* begins with a question and is followed by an answer which is the thesis for what follows. Immediately after the statement of the thesis a number of reasons for not accepting it are given; these are terminated by the *sed contra*. There follows the *respondeo*, or sustained answer, to the question, after which each objection to the initial thesis is taken up in turn. Debate is easily controlled in writing, of course, but when we consider that this literary style reflects the debate of the classroom or open disputation, we get some inkling of what the medieval university was like.

Commenting on set texts in the lectures was an effort to expound what an admitted authority had to say on a given subject; indeed, the very term "authority" suggests, in Latin as well as English, reference to an author. The principal concern of the reader or lecturer would be to expound what the author had to say. But we need not think of this as slavish adherence to the text. What animated the effort was the search for truth, and the exposition must be seen in terms of this larger quest. We are of course speaking of the ideal, and we can surmise that in the medieval, as in modern universities, the very good teacher was a rare entity and that a mode of instruction which, in the hands of a talented teacher, might soar would, in lesser hands, bore. The clue to this mode of instruction was inquiry, questioning, disputation which took their rise from received authors (whether directly or indirectly) as well as from the difficulties the subject matter suggested to master and student. The dangers inherent in the system are clear: authorities might block the way to inquiry; debate can become overly stylized; the mere repetition and manipulation of available material can replace serious and independent research; and so forth. When the system became rigid and an impediment, "Scholasticism" became a pejorative term. This should not lead us to forget that in its heyday it simply covered the method of the medieval schools, a method which was open and lively, disputatious and dialectical, striving for an ideal blend of respect for tradition and openness to novelty. Scholasticism, intimately linked with the medieval university, is, when all is said and done, that out of which modern university instruction arose.

B. Translations

In the previous part, in discussing Dominic Gundisallinus, mention was made of the translation into Latin of the works of Greek and Islamic authors. Toledo in Spain was one of the centers of this effort. In that city Muslim, Jew, and Christian were in contact with one another, and under the patronage of Archbishop Raymond (1126-1151) the task of the translators was given impetus. Among those engaged in this work in the twelfth century, besides Gundisallinus, were John of Spain, Gerard of Cremona, Michael the Scot, and Herman the German. Already in the twelfth century efforts at commentary and assimilation are apparent, and, once more, Gundisallinus is a major example.

Naples was another scene of translation work; the Emperor Frederick II (1197-1250) invited Islamic and Jewish philosophers to his court. The Emperor also founded the University of Naples, where Aquinas was to attend the faculty of arts and where Peter the Irishman commented on Aristotle and Porphyry. Michael the Scot came to Naples and with a team of translators rendered Averroes into Latin about 1230. The papal court was also the locus of translating, notably by William of Moerbeke; during his sojourn in the papal court Aquinas urged William on. Thus, translations into Latin were being made from the original Greek as well as through the medium of Arabic.

Almost the entire Aristotelian corpus was available in the West when the thirteenth century began, but the versions of the *Metaphysics* and *Nicomachean Ethics* were partial ones. Of Plato, part of the *Timaeus* was translated; the *Phaedo* and *Meno* were translated into Latin about the middle of the twelfth century. The Neoplatonism which was part of the patrimony of the West was augmented by translations of Neoplatonist commentaries of Aristotle, the *Liber de causis*, and the so-called *Theology of Aristotle*, derivative from Proclus and Plotinus, respectively. In the thirteenth century the spate of translations increased, and largely through the efforts of William of Moerbeke the complete Aristotle together with the Greek commentaries on him were turned into Latin. William also translated a number of works of Proclus as well as his commentaries on the *Timaeus* and *Parmenides*. The result of his labor was an Aristotle who had been freed from the interpretation of the Islamic commentators.

C. Islamic Philosophy

Now that we have some notion of the academic setting in which the men we are soon to consider lived their lives, we must say something about the impact of

the Islamic philosophers on the thought of the thirteenth century. It is only under this aspect that we propose to say a few things about a number of thinkers, for the most part Arabs, who lived prior to the thirteenth century but who exercised a considerable influence on the masters of the universities. Our knowledge of these men is in a considerable state of flux, and it increases almost daily. For this and other reasons the following sketch is attempted with more than the usual trepidation.

We are already aware of the fragmentary way in which Greek thought came into the Latin West. Of Plato little was known directly, apart from the *Timaeus*; for a long time Aristotle was represented only by portions of his *Organon*, then by all of it as well as by the first three books of the *Nicomachean Ethics*. Meanwhile, Greek thought was traveling a circuitous route that would eventually bring it into contact with the Christian West in Spain, a route through Syria and Persia and Arabia. As it traveled this route, Greek thought underwent translation from one language into the next with all the dangers that are involved with respect to fidelity to the original Greek. Furthermore, there was not simply transmission but interpretation, and the thinkers of Islam, like their Christian counterparts, were bent on establishing a harmony between pagan philosophy and their religious beliefs. When Aristotle finally came into the West, he came together with the writings of his Arabian interpreters. This had consequences of an interesting kind.

Al-Kindi (c.801-873). The first Muslim philosopher was al-Kindi. He is said to have written 270 works, but most of them are lost, and it is probable that sections of works have been counted as whole works. His writings, as they are described, are encyclopedic in scope, ranging from logic through medicine and science to theology. Some of al-Kindi's works were translated into Latin by Gerard of Cremona, and, until recently, he was known only through these Latin translations. He revised the Arabic version of Plotinus' *Enneads*, a work he thought to be one of Aristotle's.

It was owing to al-Kindi that philosophy became part of Islamic culture; he became known as the "philosopher of the Arabs," and his task as he saw it was to reconcile the wisdom of the Koran with Greek philosophy. This will be the continuing task of al-Farabi, ibn-Sina (Avicenna), and ibn-Rushd (Averroes). Philosophy, al-Kindi observed, depends upon reason, religion upon revelation; logic is the method of the former, faith of the latter. Al-Kindi's view of philosophy is quite comprehensive; it embraces the whole of human science. The divisions of it that he offers are Aristotelian, distinguishing speculative and

practical philosophy and subdividing the former into physics, mathematics, and divine science, the latter into ethics, economics, and politics. The fact that divine science, or theology, is a part of speculative philosophy provided al-Kindi with one of his reasons for the compatibility of philosophy and religion, though this reason led to an ambiguity. He also suggests a common source, ultimately, of the prophet's revelation and philosophical truth and goes on to speak of religion as the ultimate ordination of philosophizing.

Al-Kindi's use of the term "theologian" varies. Sometimes he uses it to describe those who opposed the study of philosophy and argues against them in a manner reminiscent of Aristotle's *Protrepticus*. Either the study of philosophy is necessary or it is not. If it is necessary, it should be pursued; if it is said to be unnecessary, one must show why this is the case, and to do this he must engage in philosophy. Thus, willy-nilly, philosophy is necessary. Further, although he sometimes seems to identify Aristotle's metaphysics and divine science without qualification, al-Kindi makes the following contrast between the divine science of the Koran and that of the philosophers. That of the Koran is strictly a divine science, while that of the philosophers is finally a human science. The knowledge of the prophet is immediate and inspired, whereas that of the philosophers is reached by way of logic and demonstration. Confronted with Aristotle's view that the world is eternal, al-Kindi will deny this because of his faith. Only God is eternal; everything else is created and finite. The denial of infinitude of anything other than God is found in the *De quinque essentiis*, a work which holds that matter, form, space, movement, and time attach to every physical body. Holding that any body must be finite, al-Kindi argues that the sum of finite magnitudes must be itself finite. In his *De intellectu* al-Kindi argues that man has four intellects: the agent intellect, the passive intellect, the latter as actuated, and the use of knowledge already had. We can take it that he is distinguishing four senses of "intellect."

Al-Farabi (c.870-c.950). Al-Farabi was a Turk by birth and of the Islamic faith. He came to the study of philosophy late in life, perhaps at fifty years of age, and half of his writings deal with logic and consist of commentaries on the works of Aristotle's *Organon*. One of the striking things about al-Farabi's conception of philosophy is that he holds that the various philosophical schools teach, not many philosophies, but different aspects of the one philosophy. He shares the Neoplatonic hope, expressed by Porphyry, that the doctrines of Plato and Aristotle can be reconciled and shown to be complementary. The fact that Porphyry, Plotinus, and Proclus, together with Neoplatonic commentaries on Aristotle by Alexander of Aphrodisias, Ammonius, and Themistius, had been

translated into Arabic doubtless gave fuel to this hope of synthesizing the great philosophers of antiquity.

In al-Farabi we find a picture of the universe which is quite clearly Neoplatonic, one relying on a doctrine of emanation and insisting on a hierarchy such that God acts on lesser orders only through the medium of intervening orders. The picture of the universe is contained in al-Farabi's theory of the ten intelligences. First, there is God, the One, who in thinking of himself produces a first intelligence which emanates from him. God is necessary, but the first intelligence is possible in itself, though necessary with respect to another, that is, to God. When the first intelligence thinks about God, this is productive of another intelligence, and the chain of emanations continues, reaching the tenth intelligence, called the "agent intellect," which directs the sublunary world. As with Neoplatonism, *esse est percipi*, in the sense that to be thought is to be created; the first nine intelligences hierarchically ordered are productive of the souls of the nine celestial spheres of the astronomy of Ptolemy. Prime matter issues in some way from the tenth intelligence, and prime matter underlies the four elements out of which all physical things are ultimately made; the forms of bodies also emanate from the tenth intelligence, and it is here that room is found for Aristotle's teaching on the hylomorphic composition of physical bodies. Al-Farabi's writings on the intellect were translated into Latin and are influential in the West; his interpretation of Aristotle's agent intellect, an interpretation reflecting the influence of the school of Alexandria, will have an impact on the University of Paris in the thirteenth century. The various senses of intellect we have seen distinguished by al-Kindi have their counterpart in al-Farabi, but there is the further note of illumination from superior intelligences, a kind of infused knowledge, which enabled al-Farabi to make a rather smooth transition from philosophy to religion. The counterpart of the Neoplatonic emanation is the theory of return to the One, and al-Farabi, like Plotinus, speaks of this return as it is effected by the human intellect in religious and even mystical terms.

Avicenna (980-1037). Perhaps the greatest of the Islamic philosophers, ibn-Sina, or Avicenna, was known by the men of the thirteenth century chiefly through his *Sufficientiae*, whose parts are devoted to the principal divisions of philosophy -- logic, physics, mathematics, psychology, and metaphysics. Avicenna's vision of the world is essentially that we find in al-Farabi, and his procedure in treating of God is reminiscent of the ontological argument, as Fazlur Rahman has pointed out. God is a necessary being and cannot not exist; from him, considered as premise, creation emanates as if it were a conclusion. In knowing himself, God

effects the first intelligence, which is not a necessary being considered in itself but only possible. Any being other than God is not necessary of itself, in its nature, but receives its necessity from God. It is here that Avicenna develops a thought of al-Farabi, who, taking up a distinction Aristotle had made in the *Posterior Analytics* between knowing what a thing is and knowing that it is, had maintained in creatures a difference between essence and existence. Essence here stands for nature, which is possibility, which does not include existence. If a nature exists, this must be explained by something other than itself. In short, existence is accidental to essence or nature. By accident Avicenna did not mean what would be meant if red were said to be an accident of a thing, for the thing might continue to be while ceasing to be red. His point is simply that if existence is not part of what a thing is, part of its essence, when it exists existence befalls it; it happens to exist. Existence seems to identify the created nature's dependence on God and would be, if Rahman is right, a relational notion. God is existence, is necessary existence. Avicenna intends to say, not that in God essence and existence are the same, but that God has no essence or nature. This denial can doubtless be explained in terms of the Neoplatonic notion that nature or essence is a limitation or restriction on existence.

The difference between essence and existence in creatures provides Avicenna with the great ontological difference between creatures and God. Like al-Farabi, Avicenna interpreted a remark of Aristotle's in the *Metaphysics* to mean that God is wholly aloof from the world, neither knowing things other than himself nor caring about them, and perhaps not the cause of other things either. No doubt inspired by his religious beliefs here, Avicenna wants both to insist on the ontological difference between God and creature and to put God into contact with the world. This contact or relation introduces the problem of the one and the many, and the doctrine of emanation commends itself to those who feel that by placing God at the top of a hierarchy, the first of whose constitutive members he accounts for directly, God's immediate influence can be kept to a minimum, while his mediated influence has total scope. Just as with al-Farabi's theory of the ten intelligences, Avicenna's theory of emanations is productive of angels and of celestial spheres, to the tenth intelligence, the agent intellect, whose name is the angel Gabriel.

The agent intellect is the giver of forms (*dator formarum*), responsible for forms not only in the sense of the substantial forms of physical bodies but also in the sense of man's mental concepts. Our glance at al-Farabi has already acquainted us with this projection of a faculty of the human soul into the status of a separate entity, an angel. As for man himself, Avicenna denies that the human

soul is the form of the body. Rather the union of soul and body is the union of two substances. This doctrine is based on Avicenna's reflections on the difference between mental and corporeal activities, which he sees to be heterogeneous and which he then concludes cannot pertain to one and the same substance. However, if soul and body are two substances, their link is something so intimate that the soul retains after its separation from the body in death a relation to the matter which entered into its body. For this and other reasons Avicenna will deny that souls coalesce into one in their separated state; *a fortiori* he rejects the Neoplatonic conception that the ultimate goal of the return to God which complements emanation will be the fusion of the soul with God. Unlike al-Farabi, who had made immortality an achievement of good men, Avicenna maintains that every human soul is immortal.

A matter that elicited criticism within Islam was Avicenna's inability to account for creation in time. The emanation from God of creatures cannot be understood as something willed by God in the sense that this is a process which takes place but might not have taken place. When Avicenna speaks of God willing the emanation of creatures, he means little more than that God consents to its necessity. Rahman's image of premise and consequence is helpful here: creatures are thought of as emanating from God with a kind of logical necessity such that their not emanating would be unthinkable and contradictory.

In speaking of God's "nature" and attributes Avicenna will use terms like "will" and "knowledge" and "power," but it is his opinion that all such terms are either negative or relational and finally coincide with existence, which is what God is. He will speak of all things preexisting in God as Ideas or forms, but this is not taken to be a denial of his basic claim that God knows only himself; it is in knowing himself that God knows whatever emanates from him. Avicenna wants to say that God knows individuals as individuals, not merely types or universals, but his way of maintaining this left him open to the criticism of al-Ghazzali. It must be said that Avicenna's teaching on universal nature is a difficult one in itself and has been the topic of much comment. Avicenna can speak of a threefold existence of a nature: in God's knowledge, in the created mind, and as it exists. The first two differ for reasons already hinted at, but further because human knowledge, indeed created knowledge, is discussed in terms of an illumination from above. In the case of man the proximate illumination comes from the agent intellect, that is, the angel Gabriel. As for the difference between the nature as known (by man) and as it exists in nature, Avicenna will say it is universal in the former, singular in the latter, and he goes on to say that neither mode of existence pertains to the nature in itself. This is the *natura absoluta*

considerata that we encounter in Aquinas' *De ente et essentia*. Some present-day critics of this doctrine have seen in it a claim that a nature exists absolutely apart from the various kinds of existence it can enjoy, which would of course be an absurd claim. But it is not what Avicenna is trying to say. We must no doubt return to his position that in creatures essence and existence are distinct; when we do this, the present point is little more than a corollary. When we consider a nature, a whatness, we need make no reference to its status as being thought by us nor to the accidents which attend it in an individual, and even when such references are made, we are adverting to what is accidental to, not a part of, the nature in question. A consideration of manness, for example, need not advert to accidents which accrue to that nature insofar as we think of it (for example, universality) nor of accidents which accrue to it because of Socrates who is a man (for example, being bald). Aquinas quite rightly sees this Avicennian doctrine as part and parcel of his distinction between essence and existence; it is highly surprising, therefore, to find Thomists criticizing Avicenna on this point with no apparent awareness that their criticism, if valid, would undermine their confidence in what Aquinas has had to say about *esse* and *essentia*.

Averroes (1126-1198). Averroes was born in Cordova and, unlike Avicenna whose works had a more independent cast, expressed himself most influentially through commentaries on Aristotle. For the medievals of the West he became simply the Commentator, and his direct association with the writings of Aristotle make his influence more palpable than that of Avicenna and a good deal more controversial. There are three sorts of commentary Averroes wrote on a given work of Aristotle, for example, the *Metaphysics*, but they seem to differ largely in terms of quantity and detail.

The closeness to the text of Aristotle that his role as commentator demanded of Averroes led him to separate himself from the more Neoplatonic views of al-Farabi and Avicenna. Thus, he will deny the theory of emanation, though he retains the notion of a hierarchy of intelligences. Furthermore, he will accept as true Aristotle's doctrine that the world is eternal. His treatment of the human soul is also markedly different from that of Avicenna. For Averroes the human soul is the substantial form of the body, and as the form of a body it has whatever existence it has as a bodily form. This was taken to mean that the soul of Socrates does not survive the death of Socrates in any meaningful way.

An attitude characteristic of Averroes' procedure, fidelity to the text of Aristotle despite the apparent conflict between his understanding of it and his religious faith, got Averroes into trouble in Islam. Al-Ghazzali was highly critical of him

(as well as of philosophers generally), and Averroes attempted to reply to this criticism in *Destructio destructionis*. What emerges from his attempts to explain the relation between philosophy and faith becomes definitive of Latin Averroism as well, the so-called two-truth theory. By this Averroes seems to mean that the statement of truths in the Koran is not as exact and accurate as might be, and this is only fitting since the Koran addresses itself to all, not merely to the learned. For a clear and distinct statement of a truth we must turn to philosophy. Philosophy thus becomes the measure of faith, and revealed statements are considered not to be in straight conflict with philosophical ones (they are, again, couched in a different, more symbolic language), but to be inadequate as they stand. As adequately expressed, revealed truths come to say something apparently different from what they are taken to mean in their original habitat, and it is not surprising to learn that Averroes was sent into exile and his books proscribed and even burnt.

★ ★ ★

There were other Islamic philosophers whose works became known to the medievals and were influential on their thinking. What is generally important about all these men, for our purposes, is that they modified the appearance that Greek thought had for the men of the medieval universities. Among the Arabs there was an unfortunate confusion of Plotinus and Aristotle; a portion of the *Enneads* came to be known as *The Theology of Aristotle*, and the *Liber de causis*, which consists of borrowings from Proclus, was not associated with its true author. Al-Farabi and Avicenna gave a version of Greek philosophy which appeared to be a restatement of what Aristotle had taught, and Averroes as the commentator on Aristotle was taken to be unpacking the text and revealing what Aristotle had really taught. Sometimes this is what he was doing; on other occasions his and other Islamic versions of Aristotle's doctrine were wide of the mark. In many cases the Aristotle they presented to the Christian West was a teacher whose tenets were in sharp contrast to revealed truths. Thus, the first reaction is one of caution. Aristotle's writings were proscribed at Paris in 1210, but later a commission was set up to study and evaluate the Aristotelian corpus. This was in 1231, and from that time the earlier prohibition seems largely to have been ignored. Along with the translations which had been made in Spain newer translations, made directly from the Greek, were becoming available. Resistance continued to this influx of a strange and different Aristotle, a thinker whose range, like that of the Islamic philosophers, was significantly greater than that of anyone in the West. Aristotle was thought to be a threat to the great tradition of Western theology, to be inimical to the faith, to be wrong on

significant points. The assessment of Aristotle was surely in large part an assessment of the Aristotle interpreted by the Arabs, particularly by Averroes, but even the unadorned text of Aristotle presented massive difficulties for the Christian thinker. Fortunately there were some, most notably Aquinas, who held themselves to the task of getting at what Aristotle really meant and assessing the result of that inquiry in terms of both natural and supernatural criteria of judgment.

Besides Islamic philosophy, brief mention must be made of the influence of Jewish thought on the thirteenth century. Avicenna, mistakenly thought to be an Arab, lived from 1021 to about 1070; his work *The Origin of Life (Fons vitae)* is often cited. Moses Maimonides (1135-1204) was born in Spain and died in Egypt. His *Guide for the Perplexed* is an attempt to make use of the philosophy of Aristotle to interpret Scripture. Rabbi Moses, as Aquinas refers to him, was well acquainted with Islamic philosophy, and we may surmise that his effort to effect a concordance of philosophy and faith was influenced by their similar effort. Whatever the principles that guided his interpretation, Maimonides comes up with rational defenses of items of religious faith which his Islamic counterparts tended to call into question on the basis of philosophy. For example, since it is clearly revealed that the world has not always existed and the philosophical arguments for its eternity are inconclusive, Maimonides concluded that we must accept the position of Scripture. Maimonides stands ready to abandon an item of belief if it can be disproved by philosophy, but this led to no wholesale housecleaning of religious tenets.

★ ★ ★

Intellectual culture reached a crest in Islam long before it did in the Christian West. The medicine, the mathematics, the science of the Arabs surpassed what was known in the West; more importantly for our immediate purposes, there was in Islam a long tradition of study of the Greek philosophers, a study which led to assimilation, interpretation, appropriation. The finest fruits of two intellectual cultures, the Greek and Islamic, entered Europe with what can only be described as suddenness and at roughly the moment when the universities were assuming the shape that would define them for centuries. The diet was extremely rich against the background of the Western tradition, and it is not surprising that caution was exercised by some, while adulation of an uncritical kind was displayed by others, and that only gradually the medievals became equipped to assimilate and appropriate in their turn.

Bibliographical Note

On the rise of the universities *Rashdall's Medieval Universities*, brought up to date in three volumes by F. M. Powicke and A. B. Emden (Oxford, 1936), is still a must. Also L. J. Pactow, *The Arts Course at Medieval Universities* (Champaign, 1910). Charles Homer Haskins' *The Renaissance of the Twelfth Century* (reprint, New York, 1957) is still useful, particularly chapter twelve.

On the Arabs see *A History of Muslim Philosophy*, edited in two volumes by M. M. Sharif (Wiesbaden, 1963); A. M. Goichon, *La Philosophie d'Avicenne* (Paris, 1951); F. Rahman, *Prophecy in Islam* (London, 1958); DeLacey O'Leary, *Arabic Thought and Its Place in History* (London, 1922); É. Gilson, *Being and Some Philosophers* (Toronto, 1952); R. Walzer, *Greek Into Arabic* (Harvard, 1962); G. F. Hourani, *Averroes* (London, 1961).

Chapter II

The Beginnings

Before turning to progressively more important responses to the influences just mentioned, we want in the present chapter to give some sense of the initial efforts to cope with and profit from the new influx of philosophical literature. The men we shall presently mention -- William of Auvergne, Alexander of Hales, and Robert Grosseteste -- though subsidiary figures in the broad sketch we are trying to give, are, when considered for themselves alone, a good deal more interesting than the following may suggest.

A. William of Auvergne (c.1180-1249)

William was a master of theology at the University of Paris, who in 1228 became bishop of Paris. Since William was a member of the commission appointed by Pope Gregory IX to study and correct the writings of Aristotle, his works take on a special interest. William writes at a time when the attitude toward Aristotle is at least wary, and what William has to say gives us a fair indication of an early thirteenth-century response to the inundation of new philosophical literature.

The first thing that must be said about William's treatment of Aristotle is that he seems imperfectly aware of the line of demarcation between Aristotle's doctrine and the doctrines of the Islamic thinkers. Often criticisms are made of Aristotle and his followers when the point at issue is one that Aristotle emphatically did not hold. For example, William attributes to Aristotle the view that from the one God only one effect can immediately proceed and that that first creature, the first intelligence, goes on to create its own effect, and so on to the constitution of the ten intelligences, with the tenth the creator of matter, corporeal forms, and human souls. We are familiar with this Neoplatonic emanation theory in Islamic thinkers, but it is possible that William attributes it to Aristotle as well because Gerard of Cremona, the translator of the *Liber de causis*, had called that work a work of Aristotle's. Furthermore, William often cites specific works of Aristotle and accurately identifies Aristotelian doctrines. Consequently, if some of William's criticisms of Aristotle are misconceived, others are not.

The mode of William's critique is interesting. Whenever he encounters a philosophical doctrine contrary to Christian belief, he will label it an error. But he does not leave the matter there. He will go on and try to show by argument

that the position is false or ill-founded. An error that particularly incenses William is the contention that there is but one human soul, that the present diversity of souls is merely a function of matter, and that when death comes, the distinction between souls disappears. He says that this error should be countered not only by proofs and arguments but also with steel and torments! This problem raises the question of the principle of individuation. William's discussion of the matter is critical of the doctrine of Boethius, found in the *De trinitate*, according to which it is the "variety of accidents" which individuates members of the same species. William transfers the question to the angels and goes on to criticize the identification of the nine intelligences and the choirs of the angels, since in each choir there is a plurality of angels. It is difficult to know whether William is suggesting a plurality of angels of the same species or not.

Let us consider William's treatment of the eternity of the world, a question he takes up in his *De universo*, the second part of the first principal part, chapters one to eleven, especially chapter eight.

Some have tried to suggest extenuating reasons for Aristotle's claim, William observes, but it is quite clear that Aristotle held the world to be eternal, that it did not begin to be, and that he held the same to be true of motion; Avicenna followed him in this and added reasons and arguments to sustain the claim. The first reason the philosophers give is as follows: either the creator precedes the world or he does not; if he does not, there would be neither creator nor world, since the creator is the creator of the world and the world is the effect of the creator. We might want to say that they come to be simultaneously (*incoepit esse cum mundo*), but this will not do since the first cause cannot come to be.

William's resolution of the difficulty is to free the relation of eternity to time of the temporal sense of before. Eternity is not just endless time; its priority to time is a priority of nature. To say that God is eternal and before the world is not to say that he is older than the world in the usual sense of older which is temporal. To ask what is before time sounds as funny as it would to ask what is beyond the world. Time like space is intramundane; therefore, to speak of God as before the world is not to speak of a time before time.

William criticizes the view, associated with Avicenna, to the effect that all creatures are composed of matter and form, even angels. In order to sustain that position, it was necessary to speak of a different kind of matter in angels, a spiritual matter. For William, *natura* means what it meant for Aristotle: it is a principle of generable and corruptible things. Thus, he holds that angels are not

composed of matter and form. Matter of course is the cause of the contingency of material things, that because of which they can cease to be. Would not a creature which was not composed of matter and form, a creature who was pure form, be a necessary being?

The Islamic view that creation is necessary was one that William of Auvergne strongly contested. Despite this opposition, William took over from Islam the distinction between essence and existence, finding in this the fundamental ontological difference between creator and creature. In God there is no distinction between essence and existence: what God is is necessary existence. The existence of any creature is other than what it is and thus relates to its essence or nature as an accident. The composition of essence and existence calls for a cause and thus is the basis for the contingency and dependence on God of every creature.

It is in the first eight chapters of his *De trinitate* that William develops his doctrine on essence and existence. First, he distinguishes between essential being and participated being. He draws an analogy between the predication of "good" and of "being": some thing may be essentially good, be goodness, while other things participate in goodness. These different modes, being and having, are explained by Boethius in his *De hebdomadibus*, William observes, and he goes on to construct a general rule. Whatever is predicated is predicated either essentially or accidentally, that is, the predicate expresses what the thing is or something other than what the thing is. Such a predicate as being cannot be an accidental predicate of everything of which it is said; it must be predicated essentially of something. "Ens igitur de unoquoque aut substantia aut participatione dicitur. Dicitur autem de quodam substantialiter, de quodam participatione dicetur: et quoniam non potest dici de unoquoque secundum participationem, necesse est ut de aliquo dicatur secundum essentiam." William underpins this move by saying that we must either accept it or get involved in an infinite regress. Let A be said to be good by participation, that is, by having good; call the good it has B, and then if B is good by having good, call that good C, and so on to infinity. To avoid this we must say that something is good in the sense of being goodness, not having it, and the same must be said of being.

The Latin infinitive "esse" that we have been translating as "existence" has two senses, according to William. It can mean essence or nature: the residue left when all accidents are removed, the substance, what the defining notion rather than the specific name expresses. In another sense "esse" means something other than essence: it is not part of the essence or nature of any creature. It is,

however, the essence of what is said to be essentially. Given these two senses of "esse," we can say that in God they are one, in creatures they differ. Only that being whose essence is existence is truly called being, and being is its proper name. God is the only being of whom it can be said that it is impossible to think of him without thinking of him as existing. Such a being is uncaused; every being which is such that existence is other than its essence must be a caused being.

William goes on to draw the corollaries of this doctrine. The being in whom essence and existence are identical is simple, unique, and so forth. With William of Auvergne's discussion of essence and existence we have a first contribution to what will be a continuing discussion of his century and beyond. The influence of Islamic thinkers is palpable here, but the doctrine is also referred to Boethius. Islamic thinkers, as we have seen, trace the distinction back to Aristotle. We stress its importance for a further reason as well, for it reveals that William of Auvergne was not merely a critic of the new philosophy, but a judicious borrower from it. His obvious acquaintance with the new sources of philosophy was bound, we should think, to leave its mark on him; and so it did.

The Aristotelianism that William never ceased to oppose can be conveniently summed up in the point suggested by Pierre Duhem. (1) It contends that the whole of creation is necessary and exists from all eternity; that it necessarily issues from the first cause. (2) It holds that God creates directly but one creature, which, though created, in turn creates; of these further creatures God is only a mediate cause. (William's major objection to this is that it suggests that God can do more things in conjunction with creatures than he could do on his own.) (3) Human souls are not really distinct from one another save in their bodily existence and thus must coalesce into one after death. In treating these points William does not merely label them heresies; he seeks arguments to refute them.

As for the philosophical teachings he accepts, it is difficult to say what larger whole they become parts of. With William we are not yet at the point where the structure of Aristotelian philosophy is accepted as fundamental, with the errors it may contain refuted in terms of its own principles. Perhaps we must conclude that William remains very much an *ad hoc* philosopher; his *De universo* is reminiscent of earlier writings bearing that same title which were more or less random collections of everything under the sun. Nevertheless, in many ways William is the herald of an emerging style both in theology and philosophy, a style that developed with astonishing rapidity.

Bibliographical Note

The 1674 edition of William of Auvergne's works was reproduced in 1963: *Guiliemi Alverni opera omnia*, 2 vols. (Frankfurt am Main: Minerva). See too, Pierre Duhem, *Le système du monde*, vol. 3, pp. 249-260; É. Gilson, "La notion d'existence chez Guillaume d'Auvergne," *Archives d'Histoire Doctrinale et Littéraire du Moyen Age*, 21 (1946), pp. 55-91.

B. Alexander of Hales (c.1185-1245)

Alexander was an Englishman who came to Paris, where he was a master of arts prior to 1210 and went on to become a master of theology probably around 1220. He went home for several years (1229-1231), where he was named archdeacon of Coventry, but he reclaimed his chair in theology at the University of Paris in 1232. He joined the Franciscan Order in 1236.

Alexander was the first master of theology to employ the *Sentences* of Peter Lombard as the text for his courses. His commentary on it, discovered in 1945 and recently published (1951- 1957), was written prior to his entrance into the Franciscans. Several volumes of *Disputed Questions*, published in 1960, contain work done by Alexander before becoming a Franciscan. He continued to teach in the university as well as in the Franciscan house of studies after he had entered religion; indeed, it is held that he taught until the time of his death. His *Summa theologica* is in many ways a Franciscan effort, since it is generally agreed that the *Summa* consists of a compilation from Alexander with contributions by other hands, all of them Alexander's confreres in religion. The recognition that the *Summa* is not in every sense a personal work does not lessen its interest or importance.

Alexander's explanation of the *Sentences* clarifies the text principally by appeals to Scripture and St. Augustine, but he cites Aristotle both in his logical writings and in the *Physics*, *On the Soul*, and *Metaphysics*. References to other philosophical writings are infrequent, and Van Steenberghen sees in Alexander's commentary on the *Sentences* the first tentative effort at speculative theology in the presence of Aristotelianism. So too, the *Disputed Questions* thus far published exhibit a modest interest in the new influx of philosophical writings. The *Summa* is something else again. Its attitude toward the new literature is open but critical, though questions have been raised as to the extent of the acquaintance with the new literature that the *Summa* exhibits. The general attitude of the *Summa* has been compared with that of William of Auvergne, but

there are many substantive differences of judgment about particular points of doctrine. Let us mention some salient points of the teaching of the *Summa* attributed to Alexander of Hales.

We find a discussion of the nature of the human soul and human intelligence. Aristotle is said to have distinguished a material, a potential, and an active intelligence. Though it is usually held that the material intelligence is the potential intelligence for Aristotle, the *Summa* takes them to be distinct and identifies the former with the sensitive soul. The role of the active intelligence is abstraction; it illumines and actuates the potential intelligence. As for the status of the active intelligence -- is it a part of the human soul or separate from it? -- the *Summa* suggests that the reason for maintaining that it is separate is that there are intelligible forms nobler than those attained by means of abstraction. In order for the mind to grasp such divine forms, it seems necessary that it be aided by something other than and apart from it. The agent intellect is assigned this role. In discussing this position, the *Summa* distinguishes between the form and matter of the soul, The agent intellect relates to the form of the human soul whereby it is spirit; the possible intellect relates to the matter of the soul, and the use of the term "matter" here is suggested by the fact that the soul is in potency to knowable things. Thus, the *Summa* would deny that the agent intellect is something apart from soul. The reason for suggesting that it is, alluded to above, is contested by saying that the agent intellect is said to be in act, not in the sense that it actually knows all forms, but in the sense that it receives from the first agent an illuminating power which relates to forms. The agent intellect is thus a participated light. However, if the agent intellect is not considered to be a power separate from the human soul, both the possible and agent intellects are held to be separable in the sense that they can continue in existence apart from the body.

A point of difference between the *Summa* and William of Auvergne lies precisely in this talk of the matter and form of the soul. Indeed, the *Summa* holds to a universal hylomorphism: every creature is composed of matter and form; hylomorphic composition is the mark of the created. Boethius' distinction between *quod est* and *quo* is invoked in this connection, and Albert the Great, who denied hylomorphic composition of the soul, is explicitly contested. The matter which enters into the composition of spiritual creatures is, of course, a spiritual matter and not to be confused with the component of physical things. Thus, there is not one and the same matter present in all creatures; if there were, transmutation between spiritual and physical things would be possible, and it is not. The term "matter," that is, covers any and every potentiality, and it

must be recalled that the motive for universal hylomorphism is to retain a distance between creator and created. Speaking of this initially surprising claim, Aquinas will say that while it is misleading to say that angels, for example, are composed of matter and form, once one understands the intention of those who say this, it is possible to agree with them without admiring their vocabulary; but, in the final analysis, *sapientis non est curare de nominibus*.

Mention should also be made here of John of Ia Rochelle, a disciple and contemporary of Alexander of Hales, who is probably the one most responsible for the *Summa* attributed to Alexander, although he wrote other things as well and indeed was a master of theology at the same time as Alexander. The Augustinian doctrine of illumination, employed in the *Summa*, was a favored doctrine of John's and was to receive even greater development at the hands of Bonaventure.

C. Robert Grosseteste (1175-1253)

William of Auvergne, Alexander of Hales, and John of Ia Rochelle give us a rather good indication of the initial reaction to the influx of new philosophical writings at the University of Paris. Meanwhile, at Oxford the example of Robert Grosseteste is an indication of a quite different response to the new literature. Robert, who was later to become bishop of Lincoln, was well acquainted with the works of Aristotle. Roger Bacon, whose admiration for Grosseteste knew no bounds and whose contempt for such Parisian masters as Alexander of Hales was equally unrestrained, liked to portray Grosseteste as the easy equal of the likes of Aristotle and indeed as one who opposed the Greek philosopher on all important points.

The thing that strikes the reader of the philosophical writings of Grosseteste, edited in 1912 by Ludwig Baur, is the preponderance of mathematical and scientific topics. It is easy to feel that here is independence and originality of a sort unknown in William of Auvergne and Alexander of Hales. Further consideration leads, however, to the judgment that, despite the mathematics, Grosseteste is actually representative of a conservative mentality, that in him Augustinianism lives on in a less adulterated form than in his continental contemporaries. It is customary, convenient, and fitting that the flavor of Grosseteste's work be exhibited by his contribution to Augustine's theory of illumination.

Among the philosophical writings of Grosseteste is one entitled *De luce seu de incohabatione formarum* (*On Light and the Beginning of Forms*). The following amounts to a rough translation of the beginning of that essay. I think, Grosseteste writes, that the first bodily form, what some call corporeity, is light, for light of its very nature (*per se*) diffuses itself in all directions such that, given a point of light, a sphere of light of whatever size is immediately generated unless something opaque (*umbrosus*) impedes. Matter's extension in three dimensions follows necessarily on corporeity, but matter itself is a simple substance lacking dimensions. So too, form is a simple substance also lacking dimensions, and it cannot account for the dimensions matter comes to have. To account for the extension of matter, Grosseteste says, I nominate light. Extension in all directions is a *per se* property of light; it diffuses and multiplies itself everywhere. Whatever performs the task of introducing dimensions into the compound of form and matter must therefore be either light or something that does this just insofar as it participates in light. Corporeity, bodily extension, is either light or a participation in light: something which acts through the power of light. Grosseteste's own opinion is simply put. Light is the most noble form of bodies and is that in bodies which makes them most akin to separate substances.

If light is employed to explain the extension of bodies, it is also used to explain the constitution of the universe. We mentioned earlier that the diffusion of light can be checked by the interposition of an obstacle; Grosseteste also holds that any given point of light has an intrinsic limitation on the extent of its diffusion. As for the constitution of the cosmos, then, he can begin with a single body which may be thought of as light and matter, a compound of form and matter: its diffusion to the extent of its intrinsic power will produce a sphere which is finite and whose limit is the heaven. Then, by thinking of that outer limit of light reflecting on the center from which it radiated, Grosseteste speaks of the generation of the celestial bodies. The picture that results is quite geocentric. The degree or intensity of light provides Grosseteste with a scale on which he can compute the ontological status of entities, so that the universe for him is a hierarchy of lights or a hierarchy based on degrees of participation in light.

Thus far Grosseteste's use of light to explain the cosmos may seem only the inspiration of one who had been impressed by the application of mathematics to natural phenomena, like the distribution of light from a source and like the rainbow. Bacon was to laud Grosseteste for having views about the natural world which derived not simply from what he read but from his own careful observations. Historians of science dispute the importance of the contribution

Grosseteste made to the emergence of scientific method as we know it. At any rate, beyond his attempt to interpret the physical world by means of light as his basic concept, Grosseteste's theory must be seen as a continuation of the Augustinian doctrine of illumination. St. James spoke of God as the Father of lights and St. John of Christ as the light of the world, and it may not be too much to say that what Augustine had developed from such scriptural remarks as these is as important for the development of Grosseteste's universe of light as anything of an observational nature.

In speaking of the relation of creatures to God two of Grosseteste's philosophical essays are of particular importance. One asks if God is the single form of all things (*De unica forma omnium*), and the other deals with the emanation of creatures from God (*De ordine emanandi causatorum a deo*). In discussing the first point Grosseteste employs Augustine's reinterpretation of the Platonic Ideas and is careful to deny that God is the form of all things in the sense of their constitutive or inherent form. The second point deals with the need to distinguish the difference between the procession of the Son from the Father and the emanation of creatures from God. In the course of the essay he distinguishes between the measure of God's duration (eternity) and that of creatures (time) and removes some of the confusion that surrounds the claim that a creature might be eternal. The text Grosseteste seems to be commenting on here is taken from proposition two of the *Liber de causis*. Only the Son is coeternal with the Father; angels and soul are measured by something other than time, which is the measure of the duration of corporeal things. When God is said to exist before every creature, the adverb must not be understood as temporal, since God is not measured by time. (Cum dico creator est quando non fuit creatura, illud quando significat aeternitatem . . . et est sensus: creatorem esse in aeternitate, in qua non est vel fuit creatura. . . .) The truth of things, Grosseteste maintains in *De veritate pro positionis*, consists in a conformity with the creative Idea of God. To know the truth, consequently, involves ultimately knowing that conformity. This is the twist Grosseteste, in the familiar Augustinian manner, will give the dictum that truth resides in a conformity of thing and mind. Grosseteste wants no more than Augustine to hold that when we know the truth, we are attending both to things and to the divine pattern, the Word of God, and seeing the conformity between the two. Rather, he suggests that our mind is a participation in the light that is the Word and that as a participation in light our mind is capable of knowing the truth.

Bibliographical Note

See S. H. Thomson, *The Writings of Robert Grosseteste, Bishop of Lincoln* (Cambridge, 1940); L. Baur, *Die philosophischen Werke des Robert Grosseteste* (Münster, 1912) and *Die Philosophie des Robert Grosseteste* (Münster, 1917). A. C. Crombie, *Robert Grosseteste and the Origins of Experimental Science* (Oxford, 1953) and D. A. Callus, ed., *Robert Grosseteste, Scholar and Bishop* (Oxford, 1955).

Chapter III

Albert the Great

A. The Man and His Work

Albert was born in Lauingen, in Bavaria, in 1206 and died in Cologne in 1280. His long life, superimposed on three-quarters of the thirteenth century, makes him a particularly interesting figure since he was very much a part of the intellectual developments of his century, of his order, and of his country. Thomas Aquinas studied under Albert, and although it is no easy matter to compare the doctrine of master and pupil on many points, it is safe to say that Albert's indefatigable energy and the scope of his interests inevitably had their impact on Aquinas. It is a tempting thought that the influence might in some cases have gone in the opposite direction, though there is as yet no scholarly agreement on the extent of such influence. In some ways the interests of Albert were broader than those of Aquinas, a fact that does not seem explicable merely in terms of Albert's longer life. There is little or nothing in Aquinas that echoes Albert's concern with what he called experimental knowledge. Aquinas will insist that our knowledge of nature arises only out of experimental contact with it, but we do not find any of the detailed natural descriptions in Aquinas that we find in Albert; nor do we find in Aquinas, as we do in Albert, such judgments passed on classical texts as "I have tested this," "I have not tested this," "This does not accord with experience.

Albert began his university studies at Bologna and Padua, but he seems to have spent scarcely more than a year in the Italian schools. In 1223 he joined the Dominican Order and was sent to the convent at Cologne to make his novitiate and pursue his studies. One of Roger Bacon's complaints against Albert was that the latter had not pursued the study of philosophy in the university. With the exception of his brief sojourn in Italy as a boy, this charge is accurate. From 1228 to 1240 Albert taught theology in various Dominican convents in Germany; in 1240 he was sent to the University of Paris. After two years of study he occupied one of the two Dominican chairs (1242-1248). He was then sent to Cologne to set up the Dominican *Studium generale*; he was Dominican provincial of Germany from 1254 to 1257; he returned to teaching at Cologne for three years and became bishop of Ratisbon in 1260, remaining in the post for two years. After resigning his see, Albert devoted the last eighteen years of his life to teaching, preaching, research, and writing.

It is convenient to consider the writings of Albert from the point of view of chronology, and the chronological periods distinguished by Van Steenberghen and the location of writings within these periods serve our purposes ideally. A first period, what Van Steenberghen calls the first theological period, extends from 1228 to 1248 and comprises Albert's first teaching in Germany and the Parisian sojourn (1240-1248). Apart from many of the biblical commentaries, three important works may be assigned to this period. First, there is the *Tractatus de natura boni* (*Treatise on the Good*). Second, there is the *Summa de creaturis* (*Summa on Creatures*), perhaps written in the first half of Albert's stay in Paris and containing five parts: (1) On the Four Coevals, (2) On Man, (3) On Good and the Virtues, (4) On the Sacraments, and (5) On the Resurrection. Finally, there is Albert's commentary on the *Sentences* of Peter Lombard.

A second period, extending from 1248 to 1254, is called by Van Steenberghen the mystical, or Dionysian, period. In this period, of course, are located Albert's commentaries on the entire corpus of Pseudo-Dionysius. Furthermore, Albert's first commentary on Aristotle's *Nicomachean Ethics* was written during this time span.

Albert's so-called Aristotelian, or philosophical, period is located between the years 1254 and 1270. To this period are assigned his paraphrases of Boethius as well as of a vast number of works of Aristotle. It may be noted here that Albert is quite insistent on the fact that his own views do not appear in these commentaries or paraphrases; his task, he claims, is simply to set forth the tenets of the Peripatetic philosophy. His description of the philosophy he is relating is significant, for what we find in these writings of Albert is not simply a restatement of the works of Aristotle, or Boethius, which give them their titles. Albert draws on the various commentators on these works and is thought to show a marked preference for the interpretations of al-Ghazzali and Avicenna. To show a preference is, of course, to make a judgment, and to make a judgment is to reveal one's criteria as a judge; thus, a good deal of Albert inevitably gets into these works. In his exposition, or paraphrase, of the *Liber de causis* Albert is particularly drawn by the views of al-Ghazzali.

As Duhem has pointed out, we find Albert addressing himself in these writings to issues which must vex the Christian thinker. For instance, in his exposition of the *Metaphysics* (bk. XI, treat. 3, chap. 7) Albert brings up the matter of the relationship between philosophy and theology. He describes the two disciplines in a manner consonant with his treatment in his *Summa theologiae*, as we shall see below, but adds here that we cannot discuss theological questions in

philosophy. Moreover, Albert suggests that in philosophy we merely follow the argument, not concerning ourselves with what religion teaches. This suggestion of the autonomy and neutrality of philosophy vis-à-vis faith becomes quite explicit when Albert asks what results when philosophy arrives at positions contrary to faith. Albert uses the specific example of Peripatetic philosophy's contention that from one only one proceeds and that not all things are direct effects of God, but only one thing. Well, he will say, theology contradicts the "one-from-one" principle, and, besides, the philosophy he is setting forth is not his own; for his personal opinions we will have to consult his theological writings. Despite this disclaimer, however, Albert attempts to adjudicate such divergences from faith within the context of his philosophical writings. In presenting the doctrine of book eight of Aristotle's *Physics* and its arguments that motion could not have begun absolutely in some past time, Albert interjects that he holds that everything has been simultaneously created by God and that, anyway, one well acquainted with Aristotle will know that Aristotle has nowhere proved the eternity of the world, has nowhere shown that the beginning of time and of motion are one with the beginning of the heavenly movements. Albert will even review natural or philosophical arguments which attempt to show that the world and motion and time had a beginning, but, like Maimonides, he is not much impressed with these, holding that none is completely cogent.

We will, as we have already indicated, return to Albert's views on the relation between faith and reason, theology and philosophy, but it seemed appropriate to say something here about Albert's curious insistence that what he is doing in his philosophical writings does not engage his personal thought and amounts to nothing more than a neutral relation of the contents of an ambiguous aggregate dubbed Peripatetic philosophy.

The fourth and final period of Albert's career is called by Van Steenberghen the second theological period. It extends from 1270 to 1280. The major work of this period is Albert's *Summa theologiae*, which was composed after that of Aquinas and which, like that of Aquinas, was not completed. It shows little, if any, influence on Albert of his most important student.

Albert was a prodigious writer, and the length of his life and active career makes his collected writings an imposing, even intimidating, edifice. It will be appreciated that in the present sketch we have referred to his writings largely by way of class and type rather than by individual title. To do the latter would amount to giving a very long list indeed.

B. Faith and Reason

The relationship between faith and reason is most profitably discussed, as far as the bulk of thirteenth-century authors is concerned, in terms of the nature of theology, its relation to philosophy, and so forth. We have already said that Albert indicates something of his views on this matter in writings that are billed as ignoring the relation involved, and that even when some glimmer of his own thought is seen, he directs us for his definitive stand to his overtly theological writings. In following this advice we now turn to Albert's *Summa theologiae*, which, as the foregoing makes clear, is a late work and should provide us with Albert's mature and developed thoughts on the matter in question. Obviously, a full understanding of Albert's doctrine would involve comparing the doctrine of the *Summa theologiae* with earlier explicit as well as oblique treatments, but the narrower procedure we shall follow, while not aiming at such a complete statement, nevertheless sketches what would have to be a significant component of the wider treatment.

At the very outset of the *Summa* Albert asks if theology is a science and, if so, what kind of science it is, what its subject matter is, and so forth. He has little doubt that theology is a science; indeed, beyond being a science it is also a wisdom. He gives the following definition of it: *theologia scientia est, ea quae sunt ad fidem generandum, nutriendam, roborandum considerans* (theology is a science that considers whatever pertains to generating, nourishing, and strengthening faith). Referring to Paul's Epistle to Titus (1:1), he emphasizes that theology is concerned with what is knowable, though not with every knowable thing, but only with the knowable as it inclines to piety. Piety is defined as the cult of God which is perfected by faith, hope, and charity, prayers and sacrifices. (*Summa theologiae*, first part, treat. 1, quest. 2, sol.) Theology, in short, is knowledge of those things which pertain to salvation, and for this reason it is concerned with those things from which faith is generated and by which it is nourished and strengthened in us with respect to our assent to the first truth. What unifies theology, for Albert, is its end, namely, salvation. But what precisely is the subject of theology? Albert remarks that the subject of a science can be understood in several ways: thus, God is the subject of metaphysics in the sense that knowledge of God is what is principally sought in that science; being is the subject of metaphysics in the sense of that whose properties and causes are sought. So too, the subject of theology can be variously designated. God is its subject since knowledge of God is principally sought and intended by the theologian; theology's subject in the sense of that whose properties are sought is Christ and the Church, or the Incarnate Word and all the sacraments with

which he perfects the Church. That is to say, the subject of theology is the work of reparation. (*Ibid.*, treat. 1, a. 3) The comparison of theology and metaphysics in this passage leads us to ask in what they differ, and Albert states the difference succinctly. "Ad secundum dicendum quod prima philosophia est de Deo secundum quod substat proprietatibus entis primi secundum quod ens primum est. Ista autem de Deo est secundum quod substat attributis quae per fidem attribuntur." (*Ibid.*, q. 4) In metaphysics God is known in terms of being and its properties because he is the first being; theology considers God in terms of attributes known to be his through faith. The difference in mode of access to a common concern of theology and metaphysics is accompanied by another difference, namely, that theology is more certain than metaphysics. "Certior est scientia quae magis primis innititur, quam quae secundis, et sic deinceps. Theologia autem innititur primae veritati incircumscriptae et increatae et aeternae: aliae vero scientiae veritatibus creatis, et ideo non primis, nec immutabilibus, nec aeternis: quia omne creatum, ut dicit Damascenus, vertibile sive mutabile est. Theologia ergo certior omnibus est." (*Ibid.*, q. 5) If we range the sciences and assess their certitude in terms of the ontological hierarchy of their objects, then theology, which is concerned with and founded on what is first, immutable, and eternal, is the most certain science. We will come back to this comparison and claim later.

The foregoing suggests that theology is knowledge of God which, as knowledge, is in some way comparable to other sciences, the philosophical sciences, but with the difference that God is studied in theology in terms of those attributes of his which are revealed to faith. Moreover, the term of the study is not knowledge as such but salvation. In the *Summa theologiae* Albert spends a great deal of time discussing the knowability of God, and the contrasts and similarities already mentioned are further elaborated. Let us consider the way in which Albert handles the question Can God be known on a purely natural basis? He replies by saying that God can be known in many ways: positively, by knowing that he is, what he is, and so on, or privatively, by knowing what he is not, how he is not, and so forth. "Dicimus igitur quod ex solis naturalibus potest cognosci quia Deus est positivo intellectu: quid autem, non potest cognosci, nisi infinite. Dico autem *infinite*: quia si cognoscatur, quod substantia est incorporea, determinari non potest quid finite genere, vel specie, vel differentia, vel numero illa substantia sit. Et remanet intellectus infinitus, qui constituitur ex negatione finientium ad nos ex constitutione infiniti. Dicimus enim, quod cum dicitur *substantia* Deus, non est substantia quae nobis innotescit finite genere, vel specie, vel differentia, vel numero: sed est substantia infinite eminens super omnem substantiam. (*Ibid.*, treat. 3, q. 14, memb. 1) On

a natural basis, then, we can have positive knowledge that God is, but we can know what God is only "infinitely," that is, in a manner that leaves undetermined what precisely God is. Albert uses the example of our saying that God is an incorporeal substance; to know this about God is not to know him as a finite substance may be known, a finite substance whose genus, species, difference, and so forth are known. Rather, when we know God is substance, our knowledge is of a substance more perfect than any we can determinately know, and we fashion our notion of the divine substance by negating the characteristics of finite substance.

To know God privatively is to know that he is not a body and is not measured by any corporeal measures, that is, God is not measured by place and time and so forth. The kind of natural knowledge Albert is discussing here is, he suggests, that indicated by Paul in his Epistle to the Romans (1:20), where we read that the invisible things of God can be known by knowledge of what God has made. Later (*ibid.*, q. 15, memb. 1) Albert expatiates on the discursive knowledge suggested by the Pauline text. "Dicendum, quod in praesenti vita cognitio Dei sine medio esse non potest: quod medium effectus Dei est in natura, vel gratia, in quo Deus monstratur. Talis enim cognitio per medium ad viam pertinet, et *cognitio viae* vocatur. . . Notandum tamen est, quod medium est duplex: ex parte visibilis, et ex parte videntis. *Ex parte visibilis* formaliter et effective medium est, quod ut actus visibilium, invisibilia potentia actu facit esse visibilia. *Ex parte videntis* medium duplex: commune scilicet, et speciale. Commune est illud, quod sub uno vel duplici situ formam visibilis ad visum est deferens. Specialis est, quo utitur visus ad excellens, sicut ad solem in rota videndum. . . ." In this life all knowledge of God is through knowledge of something else, through some mean or medium, whether a natural mean or grace as the mean. The natural mean can be considered either from the side of the known or the knower. What are the means whereby God can be naturally known? Albert invokes the Augustinian notions of vestige and image. The vestige is an imperfect similitude of creature to God, the image a less imperfect similitude. Albert's development of the way in which God is diversely known through his vestiges and images in creatures, besides recalling Augustine, is reminiscent of Bonaventure rather than of Aquinas, though the latter too employs this Augustinian distinction. As for knowledge of God through his image, we find that man or the human soul is treated as what gives us the best access to God. Thus, the vestige of God is found in all creatures, his image in some, and both these means of knowing God are distinguished from the way we know God through faith. "Dicendum quod fides medium est in cognitione viae, sive sit fides informis sive formata. . . . Et cum supra multiplex medium est distinctum, fides est medium dicens in

scientiam crediti, et coadjuvans credentem ad intelligendum: per quod medium quaeritur et invenitur intellectus crediti." (*Ibid.*, treat. 3, q. 15)

In comparing the knowledge of God gained through natural means and that had through faith, Albert returns to the comparison of philosophical and theological knowledge in terms of certitude, this time distinguishing kinds of certitude. "Certitudo multiplex est. Est enim certitudo simpliciter et certitudo quoad nos: et certitudo quoad nos duplex, scilicet certitudo inclinantis ad actum, et certitudo rationis quasi arguentis. Et quaelibet istarum certitudinum ducit ad alterum minus certum. Certitudine ergo simpliciter nihil est adeo certum sicut Deus et divina. . . . Hoc modo certissima cognitionum est cognitio divinorum facie ad faciem, et sub illa cognitio per fidem, infima vero cognitio per naturalem rationem. . . . Certitudo autem quae est quoad nos, ex notioribus est quoad nos, secundum quod animales sumus enutriti sensibus. . . . Et hoc modo nihil prohibet cognitionem per naturales rationes esse certissimam, et post hoc cognitionem fidei, et minime certam quam quae est facie ad faciem." (Treat. 3, q. 15, memb. 3, a. 2) What Albert now compares is the knowledge of vision in the next life when God is known face to face and the knowledge through faith and the knowledge of God through natural reason in this life. Certitude is said to be of two kinds, absolute and relative (to us). Albert does not here say what constitutes certitude in the absolute sense, but we can get a glimmer by recalling the earlier passage where it is a function of our mode of access to an object as well as the perfection of the object. Since in all cases being compared here it is God who is known, it must be the way he is known that produces the variation in certitude. Knowledge of vision would seem to be God's unmediated presence to the blessed; the other two kinds of knowledge are through a medium, and faith is more certain than reasoned knowledge of God in that the former is explicitly based on the authority of the first truth. In terms of the mode of knowledge most in tune with our nature, that intimately related to sense perception, the scale of certitudes is exactly reversed. It must be said that Albert's handling of this issue does not have the clarity of Bonaventure's or Aquinas' treatment.

When Albert turns to the precise manner in which God's existence can be naturally known, he sets down his own *quinque viae*, but these five ways of proving God's existence are quite distinct from the more famous five found in Aquinas. Indeed, it is only after Albert lists five proofs taken from Ambrose, Augustine, and Peter Lombard that he adds two others, one from Aristotle, the other from Boethius. The proof taken from Aristotle is that of the prime mover. "His viis ego addo duas. Una quae sumitur ex octavo Physicorum, in cuius

principio probatur, quod motor primus non potest esse motus ab aliquo. Deinde probatur, quod movens motum nec movere, ne moveri habet nisi per influentiam a primo per omnia media moventia et mota usque ad ultimum quod est motum tantum. Propter quod si cessaret motus in primo secundum quod est actus moventis, cessaret in omnibus mediis in quibus est actus moventis et mobilis, et cessaret in ultimo in quo est actus mobilis tantum. Destruatur ergo consequens: quia videmus, quod non cessat in mediis, nec in ultimo. Ad sensum enim patet esse multa mota; et multa esse moventia et mota: ergo necesse est esse unum primum movens, in quo non cessat motus, secundum quod est actus moventis et non mobilis." (*Ibid.*, treat. 3, q. 18) This is a rather bland summary of the proof of the light of the controversies Albert alluded to in his philosophical commentaries. No mention whatsoever of real or apparent conflicts with truths of faith; no mention, for that matter, of the eternity of the world or of the "one-from-one" principle Albert had attributed to the philosophers. The proof, as he presents it, relies on a distinction between what is only moved and what is moved and moves, and the question becomes, Is there something which only moves and is not moved? A moved mover cannot move save under the influence of the first mover, something true of the whole range of such movers between the unmoved mover and what is moved alone and does not move something else. Thus, if the activity of the first mover ceased, all activity subsequent to it would cease. This of course does nothing toward proving that there is a first mover. In fact, it is difficult to say that Albert, even indirectly, has set forth the premises of the Aristotelian proof. One is shocked by the apparent indifference of Albert in this regard, and it is tempting to suggest that we are faced here with the effort of an aging man, that, ironically, the promissory notes issued in those impressive philosophical works are not too persuasively redeemed in the later theological works.

C. Conclusion

Although we found it necessary to conclude the preceding section with a criticism of the manner in which Albert presents an important and much discussed Aristotelian doctrine, and this in a work where we have reason to expect a nuanced study, we must end by saying that the massive effort Albert undertook in endeavoring to present in a narrative fashion the Peripatetic philosophy in its full scope did as much as any other single thing to make respectable the study of Aristotle. That Albert himself tended to favor the lead taken by al-Ghazzali and Avicenna in interpreting Aristotle has its significance; insofar as the philosophers of Islam were heavily influenced by certain tenets of Neoplatonism, we should expect to find the same influence in Albert. Many

students of Albert's writings have insisted that there is much Neoplatonism there.

Bibliographical Note

The collected works (*Opera omnia*) have long been available in the Borgnet edition in thirty-eight volumes (Paris, 1890-1899), but a new and better edition is currently in preparation, several volumes of which have already appeared (Munster, 1951-). See as well James A. Weisheipl, OP., "The Problemata Determinata XLIII Ascribed to Albertus Magnus (1271)," *Mediaeval Studies*, 22 (1960), pp. 303-354.

Chapter IV

Roger Bacon

A. His Life and Work

Roger Bacon was born in England about 1219. He may have studied the arts at Oxford and then gone on to Paris to teach, or he may have begun his studies at Paris; at any rate, he began his teaching career at Paris, in the faculty of arts, and prided himself on the frequency with which he commented on the works of Aristotle. This is the first phase of his career, and the writings of Bacon representative of this phase, while of interest, do not prepare us for what he was to become. It is often said, perhaps unjustly, that they are indistinguishable from the philosophical efforts of the typical lecturer at Paris at the time. About the year 1247 Roger returned to England, where at Oxford he came under the influence of Robert Grosseteste. Bacon was a man of violent likes and dislikes, and for Grosseteste he conceived an almost unbounded admiration. His own work took a dramatic turn. For ten years Bacon devoted himself to scientific studies, though his conception of the scope of such studies must temper the judgment that in Bacon we have a forerunner of modern science. Alchemy and astrology fascinated Bacon, and his interest in mathematics, pure and applied, carried him back and forth across the border between magic and science. He was heavily influenced by a pseudo-Aristotelian work, *The Secret of Secrets*, and was charged with necromancy. The picture Bacon sketches of the ten years of study he undertook in the wake of his contact with Grosseteste may seem overdrawn, but his whole career was one of such intensity and indefatigable energy that it must be accepted. In that picture Bacon comports himself in the shifting role of mad scientist, dedicated scholar, and a university master who is progressively less patient with the usual academic fare. The disputes of the schools seemed to him airy and ungrounded; he was shocked by the spectacle of men without training in philosophy occupying chairs of theology. (Bacon himself must for a time have been a student in the faculty of theology.) In approximately 1257 Bacon joined the Franciscan Order.

As a Franciscan, Bacon seems to have stopped teaching; furthermore, because of a rule of his order, he could not write for external publication without obtaining permission. Another decade went by and reached its culmination when Bacon contacted the future Pope Clement IV, seeking patronage for a work he wanted to compose in which a reform of university education would be set forth. The sequel to this contact has its bizarre moments. Guy de Foulques,

the cardinal Bacon contacted, was elected pope. He seems to have had the impression that the work Bacon mentioned was completed and needed only to be copied before being sent to him. He requested that Bacon send it on, enjoining him to do so secretly. The Pope is sometimes described as Bacon's patron and benefactor, even as being enthusiastic about the writings Bacon eventually sent him. This is all conjecture. The most that can be said is that the Pope accepted the offer of Bacon's book and that he received what Bacon sent him. The claim that the Pope was on the verge of introducing Bacon's proposed reforms when death cut him down is pure fable. Were one to permit his imagination a bit of leash here, it would be just as easy to imagine the Pope chuckling over the inflated offer of the friar, concealing his mirth as he writes that he will accept the book, and being alternately overwhelmed by the four huge parcels that contained the *Opus majus* and miffed when Bacon tried to dun him for expenses. The fact is, we simply do not know what the Pope's attitude in this matter was. We know he received the *Opus majus* since it is still in the Vatican library.

Bacon was upset when he learned from the Pope's letter that the impression had been created that he had already written a work he had simply proposed to write. Indeed, there is some basis for thinking that what Bacon had in mind was something like an encyclopedia by several hands rather than a personal work. Nonetheless, he took the papal letter as a mandate enjoining him to secret composition, and he set to work on what became the *Opus majus*. This was followed shortly by the *Opus minus* and the *Opus tertium*. The Pope is thought to have received these works in 1267; in 1268 he died. Bacon turned to other works then, but he was now an object of suspicion in his own order. His teachings were condemned by the minister general, and Bacon was put in prison by the Franciscans. We do not know how long he was in prison, only that he was out before 1292, when he wrote his last work, a *Compendium of Theological Studies*. his death may be placed in that same year.

Roger Bacon is a polyvalent figure -- dedicated, irascible, caustic, vain, credulous, and critical. With few exceptions he despised his contemporaries, and he voiced his views in untempered language. His jeremiads become tedious, his promissory notes seem unredeemable, his self-importance is comic. And yet, and yet . . . Bacon himself would not have been content to bring home half a loaf, but what he had to say about university education in his time had its merit. He saw the danger in a theology unanchored in philosophy and science; he knew the book of the world had not yet been reduced to books. The final irony

is that it was likely his abrasive personality more than his ideas that denied him the hearing he craved.

B. The *Opus majus*

The writing of the *Opus majus* occupied a very small portion of Roger Bacon's scholarly life. Yet it represents a period when he was at the height of his powers, had done a good deal of the research which set him off from other masters of arts, and was quite unrestrained with respect to the scope of his vision and ambition. This work provides us with a convenient source to give something of the spirit and character of Bacon's thought. As we have seen, it was followed by other writings, writings whose importance is undeniable; nonetheless, the *Opus majus* is vast and representative of the mature Roger Bacon.

It should be said at once that Bacon conceived the work to be a program rather than an accomplishment; it points beyond itself and seeks to summarize not so much what has been done as what must be done. The work is divided into seven parts and is written with an urgency, directness, and forcefulness that make it quite personal. One would not go far wrong in describing it as a voluminous letter to the Pope. This is not to say that it lacks some of the common features of the scholarly style of the day, but even when familiar stylistic notes are present, their familiarity is dimmed, and intentionally so, by Bacon's conviction that the times called, not for encyclopedias, but for a vast concatenation of scholarly efforts. It would be both an anachronism and a disservice to Bacon to say that his work reads something like a prolonged and prolix appeal to a foundation for funds to support research -- but the parallel does suggest itself.

Part one of the *Opus majus* discusses the four general causes of human ignorance. These causes are subjection to unworthy authority, the influence of habit, popular prejudice, and false conceit of our own wisdom. It is noteworthy that Bacon is concerned here with moral faults rather than with what might be called intrinsic causes of human error. His choice is of course dictated by the fact that he is launching a general critique of his milieu. Thus, though Bacon prided himself on his own sustained attention to the writings of Aristotle during his tenure as master in the faculty of arts at Paris, he points out that while Aristotle was undoubtedly one of the wisest of men, he is not without his defects. Thus, while deference to Aristotle may bring one to the truth, it may lead one into error. Indeed, Bacon holds that it is only rarely that authority, habit, and popular prejudice have positive effects in the search for truth. But the most prominent target of Bacon's criticism, and the one from which he

considered himself to have suffered the most, is popular prejudice. He goes on to make a number of very useful remarks having to do with the attitude of the student, who must question authorities and enter into discussion with them, for a later generation can often detect flaws in a great man which were concealed from his contemporaries. Though his ire mounts most noticeably when discussing the deleterious effects of popular prejudice, Bacon holds that false conceit of one's own wisdom is the most injurious factor in the pursuit of truth. In the first place, knowledge must always be of less scope than religious faith; second, the sum total of what is known and of what has been revealed is as nothing when compared with what can be known. A boy of today may know more than the wisest men of yesteryear, Bacon observes, and thus how stupid to be puffed up if one is abreast of the present status of knowledge. Addressing the Pope directly, Bacon says that his point is not that anything of substance now being taught in the universities should be proscribed; rather, he is directing attention to the vast areas of inquiry which are presently ignored.

The first part of the *Opus majus* is thus quite moralistic in tone; Bacon chides, laments, urges, prescribes, pleads. Nor should we think that Bacon's concern with the morals of the intellectual is a rhetorical device. He was of the opinion that moral philosophy is the aim and goal of speculative philosophy. The moral philosophy he advocates is continuous with pagan ethics but goes far beyond it because of the influence of Christianity. This poses the problem of Bacon's conception of the relationship between philosophy and theology, a problem to which he devotes part two of the *Opus majus*.

Bacon sees the search for truth as divided into three avenues: Scripture, canon law, and philosophy. All truth is contained in Scripture, but to elicit it we need canon law and philosophy. Wisdom, however, is one. Canon law is an articulation of what is contained in Scripture, and so is philosophy. Bacon is not suggesting that all the scholar need do is pore over the Scriptures in order to arrive at the truth. His point is rather this: no truth can be incompatible with Scripture because wherever truth is found it belongs to Christ. It is at this point that Bacon, although he makes reference to Augustine, takes a stand on one of the vexed points of Aristotelian interpretation. The agent intellect, Bacon says, is not a part of our nature, not a faculty of the human soul. Avicenna is right in this interpretation of Aristotle: the agent intellect is outside us, something to whose influence we are susceptible, something divine. And since all human knowledge requires the influence of this separated agent intellect, knowledge is something divine. The effective source of all knowledge in God indicates the ultimate goal of knowledge, which, again, is God, God as final cause. Bacon

accepts an Augustinian suggestion, also made by Abelard, that the giants of pagan philosophy were recipients of a revelation from God. What then of the distinction between philosophy and theology? Philosophy and theology are parts of one whole. The whole purpose and function of philosophy is to lead us to the threshold of divine truth; across that threshold would seem to be Scripture. Bacon guards against the view that philosophy is something to be gotten through hastily in order to arrive at revelation in Scripture. The task of philosophy is one that never ends, for what do we indeed know? The Christian must not simply borrow bits and pieces from the philosophers; he must engage in philosophy, in the ceaseless pursuit of truth, with a constant eye for its relevance to what God has revealed in Scripture. Bacon's conception of philosophy and theology, while obscure, seems clearly distinct from that held by Aquinas, for example. The notion of a theology fashioned on the model of Aristotelian science is absent from Bacon's writings. What we find is the much earlier notion that in some vague and *ad hoc* way all human knowledge serves to illustrate the truths God has revealed in Scripture. The appeal to illumination, the interpretation of the status of the agent intellect, serves to blur the distinction between faith and knowledge, although Bacon frequently mentions such a distinction. What Bacon seems most concerned with is that human knowledge, philosophy, be open with respect to revealed truth and see it as its complement. The very limitations of philosophical knowledge enable the philosopher to devise an argument to the effect that God must have revealed to man truths which are of the greatest importance.

It was said earlier that Bacon does not refer to the truth contained in Scripture in such a way that he is condemning the need for philosophical research. That is true, but it must be added that Bacon does tend toward the view that the totality of truth was known by the Patriarchs and has been lost because of the moral defects of men. The Scriptures then come to seem repositories of esoteric knowledge which must be elicited by the appropriate means. Since philosophy is a principal means of eliciting this truth, Bacon can at one and the same time hold that in a hidden fashion everything is contained in Scripture and that we must bend our best efforts to discover knowledge.

Although it is difficult to establish an order among Bacon's enthusiasms, since he went all out for anything he favored, his recommendation of the study of languages is impassioned and is the subject of part three of the *Opus majus*. He expatiates on the difficulties of accurate translation and urges the study of Greek and Hebrew. The obvious advantages of knowledge of these languages for grasping the meaning of Scripture are dwelled on, but Bacon points as well to

the advantages for ecclesiastical diplomacy and for preaching the Gospel to all nations to be gained from the study of languages.

Parts four, five, and six constitute the bulk of the *Opus majus* and treat, respectively, mathematics, optics, and experimental science. Bacon considers mathematics to be the key to all the other sciences; consequently, having said something of mathematics itself, he will show its importance not only for other human knowledge but to divine knowledge and to the governance of the Church. Bacon makes a teasing remark on the affinity of logic to mathematics, and speaks of the greater cogency of mathematical demonstrations and the tendency we show, in other domains, of selecting examples from mathematics to illustrate our points. The discussions which ensue are, from the outset, devoted to applied mathematics, its use in astronomy, in understanding the propagation of light, and so forth. It becomes clear that in part four it is precisely the utility of mathematics in other sciences that Bacon is out to show. Its application to sacred subjects is of particular interest.

Bacon reverts to his point that knowledge of nature is needed if we are to unpack the message of Scripture. Distinguishing between the literal and spiritual meanings of Scripture, Bacon says that we need to grasp the former to get at the latter and that to grasp the literal meaning of Scripture mathematical knowledge is necessary. He illustrates this by appealing to astronomy; this enables us to see the relative insignificance of the earth in the universe. Geography enables us to determine the exact location of the places mentioned in Scripture. The chronology of Scripture can be established by appeal to astronomy. Furthermore, what we know about the rainbow is particularly fruitful for understanding the literal and then the spiritual meaning of Scriptural passages. It is in this section of the *Opus majus* that Bacon proceeds with the most gusto. When he begins to discuss the terrestrial effects of celestial bodies, Bacon does not go immediately to astrology. Rather he goes on at great length to relate character traits, and even religious profession, to regional and climatic conditions. Only afterward does he insist that the stars exercise an influence on the affairs of men, an influence that can be understood and thus can become a powerful force in human foresight and governing.

Part five, which deals with optics, is thought to be the section that best illustrates Bacon's own work. He begins with the physiology of eyesight, the eye, and the brain, and goes on to discuss the conditions of seeing: light, distance, position, size. There are other conditions as well, and, after considering them, Bacon relates what he has said to Aristotelian psychology. He goes on to discuss

direct vision, reflected vision, and refraction. Typically, after a quite lengthy treatise, whose scope and content may surprise one whose opinion of medieval science is dictated by myth rather than history, Bacon typically discusses the spiritual significance of optics. Here he considers, for example, the meaning of the prayer in which we ask God to guard us as the apple of his eye.

Bacon's discussion of experimental science in part six must be correctly understood. Although he begins with the observation that there are two ways of acquiring knowledge, reasoning and experience, he goes on to divide experience into sense experience (our own or that of trustworthy witnesses) and internal experience. Internal experience has as its object spiritual things and is aided by grace; a by-product of internal experience is often knowledge of earthly things. Bacon lists seven grades of spiritual experience and, in one of those asides that make the *Opus majus* the singular work it is, tells the Pope that the young man who has carried the book to him is a good example of the intellectual benefits to be derived from a spotless life. Although he distinguishes these various meanings of "experience," the experimental science Bacon wishes to discuss is that which tests tentative judgments about natural things. Bacon discourses on various experimental apparatuses and goes on to picture the inventions that may be expected if studies are turned in the direction he advocates.

The culminating discussion of the *Opus majus* is to be found in part seven, which is devoted to moral philosophy. We have already mentioned that for Bacon knowledge is ordered to virtue, an opinion that dictated the structure of his work. The study of our practical conduct, that due to which we are adjudged good or bad, deals with the final purpose of all human wisdom. The conclusions of the other sciences are the starting points of ethics. That is to say, in moral philosophy we try to set forth the practical implications of all other knowledge.

The task of moral philosophy is threefold, dealing with duties to God, duties to our neighbor, and duties to ourselves. A recognition of the nature of the universe and its dependence on God is the basis for maxims having to do with worship of and reverence for God. Civic morality commences with reflections on the propagation of the species and moves quite naturally into matters of the state, the functions of the citizens, reward, punishment, and law. It is in the third part of moral philosophy that Bacon explicitly joins the discussions of Aristotle and that he treats of virtue and vice in general and then of the special virtues. His discussion of the moral virtues leans heavily on the fact that until our sensual desires are reined, the mind is not free for its pursuit of the truth. Bacon, who seems to have been a most irascible man, dwells on the topic of

anger: its sources and remedies and, predictably, the way it impedes the intellectual life. His discussion of the proper attitude toward death and the way to peace of mind exhibits the influence of Stoicism on Bacon. The section on moral philosophy concludes with a discussion of the sacraments, placing special emphasis on the Mass and on the Eucharist.

C. Conclusion

The life of Roger Bacon almost spans the thirteenth century. Perhaps it can be said that he was typical of that century precisely in not being typical of it. The man to whom he would seem to be closest in spirit, Albert the Great, was a man Bacon professed to despise; his distance from the spirit of Aquinas, and even from that of Bonaventure, does not require emphasis. It seems necessary to say that Roger Bacon was at one and the same time a very traditional figure and a daring innovator. That kind of phrase, lacking as it does all sharpness and precision, could be applied to almost any figure in whatever time who excites our interest. It does, nonetheless, have a somewhat illuminating application to Roger Bacon.

Bacon's conception of the task of the Christian thinker was in fundamental continuity with St. Augustine. All truth, wherever it might be found, must be seized by the believer as rightfully his. All knowledge illumines and is illumined by what God has revealed in Scripture. Like Augustine, Bacon does not seem to have drawn a sharp distinction between philosophy and theology. This is due to the manner in which he compares knowledge and faith. Theology is taken to be the content of Scripture, but at other times it seems to consist of the application to revealed truth of human knowledge. The influence of Aristotle is seen in men like Aquinas in the conception of theology as a science which derives its principles from Scripture and by reasoning relates what is believed to what is known by natural powers. The model of theology, in short, is demonstrative science in the Aristotelian sense. It is not here that the influence of Aristotle on Roger Bacon is evident; rather it is in his passion to know the natural world.

The student of Aquinas finds that Thomas knew the natural writings of Aristotle and that he wrote commentaries on many of them, but he will search in vain for any contribution Thomas has made to our knowledge of the world around us. This is one of Bacon's complaints. Let us, he urges, carry on the work that Aristotle had begun. Let us study the natural world. Aristotle is not the last word here. Islamic thinkers who came into contact with Greek thought assimilated it and tried to continue it with their own work in mathematics and

astronomy. Bacon's fear that the universities of his day would become too bookish, too tied down to authors and authorities, was not unfounded. No doubt he underestimated the value of the speculative work his contemporaries were doing, but he did so because he was so impressed by the promise of other studies, studies more experimental, studies that could correct and prolong what earlier thinkers had discovered. If his own vision of philosophy involves much credulity and fancy, it must nonetheless be said that Roger Bacon, though certainly not alone, was insisting on the quest for a certain kind of knowledge that could never be attained by the developing Scholastic method. There is no need to exaggerate his achievements, no need to deny his unlovableness, to recognize the importance of Roger Bacon for the history of human thought.

Bibliographical Note

The 'Opus Majus' of Roger Bacon, edited by John Henry Bridges in two volumes (Oxford, 1897); *Fr. Rogeri Bacon opera quaedam hactenus inedita*, ed. J. S. Brewer (London, 1859); *Opera hactenus inedita Fratris Rogeri Baconis*, edited by F. M. Delorme and A. G. Little in sixteen volumes (Oxford, 1905-1940). See as well the volume edited by A. G. Little, *Roger Bacon: Essays Contributed by Various Authors on the Occasion of the Commemoration of the Seventh Centenary of His Birth* (Oxford, 1914); Theodore Crowley, *Roger Bacon: The Problem of the Soul in His Philosophical Commentaries* (Dublin, 1950); and E. Westacott, *Roger Bacon in Life and Legend* (New York, 1953).

Chapter V

Saint Bonaventure

A. The Man and His Work

The saint we know as Bonaventure was born John Fidenza about 1217 near Viterbo. It is thought that he studied the liberal arts in Paris from 1236 to 1242. In 1243 he entered the Franciscan Order, and it may be that he had begun studying theology before becoming a Franciscan; at any rate, he studied theology until 1248 under Alexander of Hales and others. Bachelor of Scripture in 1248, Bachelor of the Sentences in 1250, he received the *licentia docendi* in 1253 and was master of the Franciscan school in Paris from 1253 to 1257. Because of the opposition of the secular masters, Bonaventure was not admitted as a master of the faculty of theology in the university until 1257, being admitted at the same time as Thomas Aquinas. Some months before, Bonaventure had been elected master general of the Franciscan Order -- which effectively ended his university career, although he did lecture at the convent in Paris on several occasions. Bonaventure became a cardinal in 1273 and died in Lyon in 1274 while attending the ecumenical council which was held in that city. He was fifty-seven.

Bonaventure's commentary on the *Sentences* of Peter Lombard and disputed questions on the knowledge of Christ, the Trinity, and evangelical perfection, together with the *Breviloquium* (which Van Steenberghen has called a kind of summa which résumés substantially Bonaventure's commentary on the *Sentences*), the *De reductione artium ad theologiam*, and the *Itinerarium mentis in deum* are among his most important works. There are also commentaries on Scripture, of course, and the publication of his lectures in the Paris convent after he became the head of his order: *De decem praeceptis* (1267), *De donis spiritus sancti* (1268), and the *In Hexaemeron* (1273).

Despite their proximity in time and place, it is doubtful that Bonaventure and Aquinas were close friends. Although Aquinas was successful in avoiding the active life and Bonaventure was not, the two men seem complementary, each representative of the spiritual and intellectual life of his order, and both looming above the other giants of the thirteenth century.

B. The Nature of Philosophy

"Philosophical knowledge is a preparation for other sciences and he who wishes to stop there falls into error." (*De donis*, col. IV, 12)

It is an extremely difficult matter to determine what for Bonaventure the nature of philosophy is. The point has been debated by Gilson and Van Steenberghen, but we shall address the question without explicit reference to the views of these leading medievalists. Can a philosophy be isolated from the theological writings of Bonaventure and, if so, how much autonomy does it have? Further, what kind of philosophy would such a constructed system be? To say that it would be Augustinian presupposes that St. Augustine's conception of the nature of philosophy is clear and easily grasped. Before we can possibly adjudicate the differences of opinion between the scholars mentioned, we must of course turn to Bonaventure himself. When we do this, we find that while it is difficult to maintain that any one of Bonaventure's works is specifically philosophical, in many of them he does say things about philosophy.

Let us begin with the division of knowledge into four kinds, a division Bonaventure makes in his commentary on the Sentences. First of all, knowledge may be purely speculative and founded on principles of reason: this is the science of human philosophy. Second, there is a knowledge which resides in the intellect insofar as it is inclined by appetite: when founded on principles of faith, such knowledge is the science of Sacred Scripture. Third there is a science or knowledge which resides in intellect inclined by appetite toward operation: such knowledge is founded on the principles of natural law. Finally, there is a kind of knowledge which is in the intellect considered both as inclined and inclining, that is, inclined by faith and inclining to good works. This knowledge is founded on principles of faith and finds its source in the gift of grace. Such knowledge is called a gift of the holy Ghost. (*III Sent.*, d. 35, q. 2, c.)

We find in this passage a distinction of philosophy (the first and third members) from theology (the second and final members of the division), and in both philosophy and theology there is a distinction between the speculative and the practical. Philosophy is based on principles of reason (*principia rationis*), theology on principles of faith (*principia fidei*). Elsewhere Bonaventure compares philosophy and theology in the following fashion. Sacred doctrine, or theology, is principally concerned with the First Principle, that is, with God as one and three; nevertheless, theology is concerned with other things as well. "The reason for this truth is that since sacred doctrine, or theology, is a science

giving knowledge of the First Principle sufficient for our present state, insofar as it is necessary for salvation, and since God is not only the principle and effective exemplar in creation but also the restoring cause in redemption and the perfective cause in reparation, this science treats not only of God the creator but also of creation and the creature. . . . Thus, it alone is perfect science because it begins at the beginning, with the First Principle, and proceeds to the term, which is the eternal reward; it begins with the highest, the most high God, creator of all, and descends to the least, which is the punishment of hell. That alone is perfect wisdom which begins with the highest cause, that is, with the principle of caused things -- which is where philosophical knowledge ends." (*Breviloquium*, p. 1, cap. 1, 2-3) This passage clarifies the grounds for the distinction between philosophy and theology. The former, being based on what is naturally known to us, must begin with creatures and ascend to knowledge of God as to its term. Theology, since it is based on faith, begins with God and considers everything else in the light of revealed truth.

In discussing the subject of theology in his commentary on the *Sentences* Bonaventure distinguishes a variety of meanings for the phrase "subject of a science." In effect, he says, this phrase may mean either (1) that to which all else in the science is referred as to its radical principle, or (2) that to which everything in the science is referred as to an integral whole, or (3) that to which everything is referred as to its universal whole. Thus, the subject of grammar, following these three possibilities, is either the alphabet, or perfect and correct speech, or articulated sound capable of signifying something as itself or in another. In geometry the three subjects are, respectively, point, body, and immobile continuous quantity. As for theology, the subject to which it reduces everything as to its cause is God; the subject to which everything is reduced as to an integral whole is Christ, who unites in himself human and divine, created and uncreated nature. "The subject to which all things are reduced as to a universal whole can be named in two ways, by a disjunction, and then it is reality and sign, where sign means sacrament; or it can be named by one word, the credible, insofar as the credible takes on the note of intelligibility by having reason brought to bear on it [*prout tamen credibile transit in rationem intelligibilis et hoc per adductionem rationis*]; properly speaking, then, the credible is the subject of this book." (*I Sent.*, proemium, q. 1, c.)

Theology and philosophy are different ways of considering things, and things are named differently because of the different light in which they appear to the theologian and to the philosopher. We will be seeing more of Bonaventure's use of the metaphor of light; for the moment consider the following remark

concerning an innate and infused light. "The innate light is the natural light of the judging faculty or reason; superinfused light is the light of faith." (*De donis*, col. IV, 2) The first is impressed by the creator on the rational creature: it is the possible and agent intellects. In this context Bonaventure speaks of philosophical knowledge, theological knowledge, the knowledge which is the gift of the Holy Ghost, and the knowledge of the blessed in heaven. "Philosophical knowledge is nothing other than certain knowledge of the truth as what can be investigated [*ut scrutabilis*]. Theological knowledge is loving knowledge of truth as credible. The gift of knowledge is holy knowledge of the truth as lovable [*ut diligibilis*]. The knowledge of glory is sempiternal knowledge of truth as desirable [*ut desiderabilis*]." (*Ibid.*, 5)

To sum up our findings thus far: (1) philosophy is based on principles of reason, theology on principles of faith. (2) The former sees things under an inborn light, something belonging to the nature of the rational creature; the latter is dependent on an infused light, the gift of faith. (3) The subject of theology is the credible; the subject of philosophy is the naturally knowable. (4) Philosophy begins with creatures and arrives at knowledge of God as its term; theology begins with God and considers everything else in the light of what God has revealed to us about himself.

Speaking of the philosophical sciences, Bonaventure says: "All these sciences have certain and infallible rules which are as lights and rays descending from the eternal law into our mind." (*Itinerarium*, cap. 3, n. 7) Given this participation, it is possible for us to be led to contemplation of the eternal light. Things are the object of philosophy insofar as they can be investigated in the light of principles naturally known; they are objects of theology insofar as they are credible in the light of faith. From these considerations there seems to emerge a picture of philosophy as an autonomous science having its own light, principles, and certitude. What is more, philosophy is more certain than theology. "That concerning which no doubt is possible is known more certainly than that about which we can doubt; but what is known by scientific knowledge [*scientiæ cognitione*] is so known that it cannot be doubted, as is obvious." (*III Sent.*, d. 23, a. 1, a. 4, a. 2) When something is seen in the light of those principles naturally inserted in the mind, absolute certainty is attained. (*De reductione*, n.4) "Someone can know something so certainly through science that he can in no way doubt nor disbelieve it nor in any way contradict it in his heart, as is clear in knowledge of axioms [*dignitatum*] and first principles." (*III Sent.*, d. 23, a. 1, q. 4, c.) Thus, not only are philosophy and theology distinct but philosophy is more certain than the faith on which theology is founded.

This greater certitude of natural reason has to be correctly understood, however. No science is more certain than that which the blessed enjoy in heaven. "In another way, science means knowledge had in this life and is of two kinds: either it concerns things which are objects of faith or it concerns other knowable objects. If it is concerned with objects of faith, then absolutely speaking faith is more certain than knowledge. Hence, if some philosopher knows a given article [of faith] by reason, for example, that God is the creator or that God is the rewarder, he can in no wise know it more certainly through his science than the true believer through his faith. If however we are speaking of knowable objects other than those of faith, then in some ways faith is more certain than science and in other ways science is more certain than faith. For there is the certitude of seeing [*speculationis*] and the certitude of adherence [*adhaesionis*]: The first pertains to intellect, the second to the affections."

"In terms of the certitude of adherence the certitude of faith is greater than that of the habit of science, since the true faith causes the believer to adhere more firmly to what is believed than science causes the knower to adhere to any known thing. If, however, we speak of the certitude of seeing, which pertains to intellect and bare truth [*nudam veritatem*], then it can be granted that the certitude of any science is greater than that of faith insofar as someone can know a thing so certainly in a science that he can in no way doubt or disbelieve or gainsay it in his heart, as is clear in the case of first principles." (*III Sent.*, d. 23, a. 1, q. 4, c.)

We have here a very nuanced position, since a distinction between objects and modes of certitude is implied. When the objects of philosophy and theology are different, then of course the certitude with which they attain truths about them will differ. But Bonaventure seems to begin with an instance of the same object simultaneously known and believed. This is particularly intriguing, and we must pursue it since it will lead us to call into question our earlier tentative conclusion that philosophy, for Bonaventure, is a quite autonomous enterprise.

C. Simultaneity of Knowledge and Belief

Is it possible for the same truth to be known certainly with reference to the principles of reason and to be believed on divine faith? Bonaventure, in speaking of knowledge of God as creator and belief of this same truth, could, of course, be referring to different men, for example, to the pagan pre-Christian philosopher and the simple faithful of the Christian era. Such a "simultaneity" of knowledge and belief would present no problem. In the area of secular

knowledge the teacher may know what the student for a time only believes. But it would seem odd to say that the same man simultaneously knows and believes the same truth, particularly if the following Bonaventurian distinction is accepted. "We must say that what is true is an object of faith differently from the way in which it is an object of knowledge: the object of knowledge, I say, is the seen truth [*verum visum*], whereas the object of faith is truth not as seen but as salutary. (*III Sent.*, d. 23, a. 1, q. 1, ad 2) On this basis, to say that the same thing can be simultaneously known and believed is like saying that the same thing can be simultaneously seen and not seen. In short, such simultaneity seems contradictory. Nevertheless, Bonaventure is quite commonly represented as maintaining such a simultaneity. Thus, we must examine the relevant passages to see if he does indeed do this and, if so, in precisely what manner.

St. Bonaventure asks whether faith bears on the same objects as does scientific knowledge, and he approaches the question with a fine feeling for useful distinctions. (See *III Sent.*, d. 24, a. 2, q. 3, c.) He first points out the difference between knowledge which is of open comprehension" and that which results from reasoning (*duplex est cognitio, scilicet apertae comprehensionis et inanuductione ratiocinationis*). If we speak of that open comprehensive knowledge whereby God is known in heaven, then faith is not compatible with it such that the same thing could be simultaneously known and believed, for such knowledge absolutely excludes any darkness (*aenigma*). With respect to this knowledge it is the view of the saints and the common opinion of masters of theology that here the same thing cannot be simultaneously known and believed.

With respect to the knowledge which results from reasoning there is, Bonaventure notes, a division of opinion. Some hold that it is incompatible with faith, since with such knowledge the intellect assents because of an argument, necessarily, and to a thing inferior to itself. Faith, on the other hand, is an assent to the first truth for its own sake and voluntarily, which elevates reason above itself. Thus, science and faith mutually exclude one another.

Others are of the opinion that with respect to one and the same thing it is possible to have science as the result of reasoning and faith. St. Augustine and Richard of St. Victor are invoked as authorities, and the upshot is this: "Hence, someone believing God to be one and the creator of everything, if he begins to know the same thing by necessary arguments, does not thereby cease to have faith; nor, if he first knew, would the advent of faith destroy his knowledge, as is clear from experience." The reason such science can be had simultaneously with faith concerning the same thing, neither destroying the other, is that science,

which results from reasoning, although it gives some certitude and evidence about divine things, is not in every way clear as long as we are in this life. For though one be able by necessary reasons to prove that God is and that God is one, to discern the divine being itself, the very unity of God and the way in which that unity does not exclude a plurality of persons, is impossible until one is cleansed by the justice of faith. The illumination and certitude of such science are not so great that, science being had, the illumination of faith becomes superfluous. On the contrary, given that science, faith is all the more necessary. A sign of this, Bonaventure feels, is in the fact that some philosophers, while they knew many truths concerning God, committed many errors because they lacked faith. In short, Bonaventure holds that knowledge and belief are simultaneously possible concerning the same object.

Now, as will be seen, it is the view of St. Thomas that one cannot at the same time know and believe the same truth. It is often said that this is a point of open conflict between him and Bonaventure. But is this really the case? Is Bonaventure claiming that it is precisely the same thing that is known and believed? Let us consider the example he gives. A contemporary of Christ looks at him and sees a man; his divinity is hidden to every sense and is an object of faith alone. Bonaventure says that this makes it perfectly clear that doubt and certitude concerning the same thing are simultaneously possible. I can be certain that this is a man: I see that he is. But that he is God -- that I cannot see. "Therefore, it must be conceded that faith and vision can bear on one and the same thing, although not in the same respect [*quamvis non secundum idem*]." (*III Sent.*, d. 24, a. 2, q. 1, c.) It is that final phrase which is all important. Bonaventure is not saying that the same thing can be simultaneously known and believed where "the same thing" would mean same in every respect. The propositions "Christ is a man" and "Christ is God" are about the same thing, but they do not say the same thing about it. So too, the philosopher who proves that God is one knows that about God; what he does not know and cannot know in this life by natural reason is that the unity of God admits of a plurality of persons. In other words, faith and knowledge do not bear unequivocally on the same thing, since the thing is an object of faith insofar as it is not seen and an object of knowledge insofar as it is seen. Thus, while the same thing may be a *credibile* and a *scibile*, to be a *credibile* and to be a *scibile* are not the same. One cannot simultaneously know and believe the proposition "God is one" unless the predicate is made to bear two meanings. For example, (1) there is not a plurality of gods, and (2) the one divine nature admits of a plurality of persons. In conclusion, the position of Bonaventure, while complex, seems to be the same as that maintained by Thomas Aquinas: the same man cannot at the same

time both know and believe the same truth. However, if the philosophical as opposed to the theological understanding of a proposition admits of certitude, it is not thereby wholly autonomous for Bonaventure. For the great Franciscan philosophical truth all by itself is dangerous; indeed, it is its own kind of error.

D. Is Philosophy Autonomous?

There are things which to a certain degree and in a certain respect are evident to sense and in another fashion and respect are hidden; owing to this complexity there can be vision or knowledge, in part, and credulity or faith, in part. Knowledge will bear on what is evident, faith on what is hidden. With respect to divine things Bonaventure grants that philosophers can know with certitude, because of necessary arguments, some truths concerning God. For example, that God exists and is one. Bonaventure says that such philosophical proofs cannot be resisted. (*In Ioannem*, proemium, n. 10)

Does this mean that philosophy can enjoy a life of its own quite apart from faith? The following somewhat lengthy passage gives a first approach to Bonaventure's thought on this matter. "To the objection that faith concerns what is above reason and science what is below, it must be said that just as nothing prevents one and the same thing from being both evident and hidden, so nothing prevents the same thing from being above and below reason according to different modes of knowing and thus being both known and believed. For though 'the sempiternal power and divinity' (Romans 1:19) can be known either through acquired or even innate science, yet, as compared with the plurality of persons or with the humbleness of our humanity which God assumed, it is wholly above reason and science. For should someone base himself on the judgment of reason and science, he would never believe it to be possible that the highest unity could admit a plurality of persons or that the highest majesty could be united with our humility or that the highest power should from not acting come to act without any change in itself, or other similar things which seem to go contrary to the most common concepts of the mind according to philosophy. Thus it is that science attains precious little in the way of knowledge of divine things unless it is based on faith, because in one and the same thing what is most obvious to faith is most hidden to science. This is clear in the highest and most noble questions, the truth of which is hidden from philosophers, for example, concerning the creation of the world, concerning the power and wisdom of God -- matters that were hidden from philosophers but are now manifest to simple Christians. Because of this Paul writes, 'God has made foolish the wisdom of this world,' since any faithless wisdom concerning

God in this life is stupidity rather than true science. For it will drag the inquirer into error if he is not directed and aided by the illumination of faith; it is not destroyed by faith, consequently, but rather perfected." (*III Sent.*, d. 24, a. 2, q. 3, ad 4)

While maintaining that science and faith can coexist, Bonaventure makes it clear that philosophy is not sufficient to itself; it needs the aid and light of faith lest it be turned into foolishness. It is difficult to express Bonaventure's thought accurately here. He has shown that science and faith, philosophy and theology, are distinct, that from one point of view science is more certain than faith, although from another point of view faith is more certain than science. If these can coexist, can they exist separately? Bonaventure wants to deny that they can. First of all, theology without philosophy does not seem to be possible. We remember from earlier considerations how Bonaventure described the subject of theology. That subject is not the credible as credible -- this is the object of simple faith, and faith is presupposed by theology. Theology is concerned with the credible, with the believed, to the degree that it can take on the note of understandability. In order for this to come about, philosophy is presupposed. "Since the teachings of philosophers are often useful for understanding truth and refuting error, there is nothing to be feared from the study of it, particularly since many questions concerning the faith cannot be resolved without it." (*De tribus quaestionibus*, n. 12) "Philosophy is concerned with things as they are in nature, or in the soul, according to knowledge naturally inserted or even acquired, but theology as a science founded on faith and revealed by the Holy Ghost deals with things pertaining to grace and glory and even to Eternal Wisdom." (*Breviloquium*, prologue) But these are not wholly separable pursuits. First of all, the theologian must employ philosophy. The simple faithful accept revealed truths owing to an infused and gratuitous light. So does the theologian, but, having accepted what God has revealed, he reflects on these truths, bringing to bear on them the findings of the philosophers. This results in an organization of articles of faith according to a pattern not followed by Scripture itself. The theologian defends the truths of faith against his own and others' doubt, and philosophy is an apt instrument for this task.

The credible, in short, can be looked upon in three ways. "For the credible insofar as it has in itself the note of First Truth to which faith assents for its own sake and above all else, pertains to the habit of faith; insofar as the note of authority is added to that of truth it pertains to the teaching of Holy Writ, of whose authority Augustine said that it is greater than any insight of human genius; but insofar as to the notes of truth and authority the note of being

susceptible of proof [*probabilitas*] is added to the credible, it pertains to the consideration of the present book." (*I Sent.*, proemium, q. 1, ad 5) It is out of love for what is believed that man naturally seeks arguments on behalf of what has been revealed. (*Ibid.*, q. 2, ad 6) Thus, theology cannot exist without philosophy.

This would seem to indicate that philosophy must be able to enjoy a separate existence. And yet, the very fact of theology seems to call philosophy into question. "Beyond philosophical science God gave us theological science, which is a pious knowledge of credible truth, because the eternal light which is God is a light inaccessible to us as long as we are mortals and have the eyes of owls." (*De donis*, IV, 13) The reference is to Aristotle, who had pointed out the weakness of our minds with respect to divine things. On this very point, however, Bonaventure deals somewhat harshly with Aristotle. Of what good is it to recognize the weakness of the human mind and be unable to understand the cause of it, to recognize the illness and be unable to provide the remedy? "This, then, is the medicine: the grace of the Holy Ghost. This aid and this grace philosophy cannot attain." (*In Hexaemeron*, VII, 11) The weakness of the human mind is an effect of original sin. This weakness is present in the theoretical as well as the practical intellect, and Bonaventure will point out the errors into which philosophers were led, errors they might have avoided if they had had the grace of faith, which is the cure for original sin.

In his lectures on the work of the six days (*Collationes in Hexaemeron*) Bonaventure gives us a veritable catalogue of the errors of philosophers. The philosophical doctrine on virtue is unsatisfactory: no philosophical doctrine can provide the means for healing our wayward affections. (IV, 12) Philosophical moral doctrine fails to recognize man's true end, which is supernatural; consequently, it is mistaken about the sufficiency of the merit for acts done and cannot cure the weakness of our faculties. "Only faith can divide light from this darkness." (IV, 13) Bonaventure's charge here is not that philosophy failed to do what it could and should do, but that philosophy is radically inadequate in matters of morality. "We must then go on to the light of faith, which the philosophers did not have, for they knew only by the natural light. The most perfect virtues, however, are known by faith and lead on to the end." (III, 32)

As for Aristotle, Bonaventure summarizes his defects in the following manner. Asking how philosophers fell into darkness, he answers, "For this reason, that while all recognized a first cause, the principle and end of all things, they disagreed about the in-between. For some denied that the exemplars of all

things were in God, the chief being Aristotle, who, at the beginning and the end of his *Metaphysics* and in many other places, rejects the Ideas of Plato. Hence, he says that God knows only himself and has no need of knowledge of other things and moves as what is desired and loved. From which it follows that he knows nothing of particular things. Aristotle execrates the Ideas in his *Ethics* as well, denying that the highest good can be an Idea. But his arguments are worthless and are disproved by the Commentator. Now from this error another follows, namely, that God has neither foreknowledge nor providence since he has not the notions of things in himself whereby to know them. Moreover, they say there is no truth concerning the future except in necessary events, the contingent having no truth. From which it follows that all things come about either by chance or by fatal necessity; and since it is impossible that everything come about by chance, the Arabs held for the necessity of fate, saying that the substances which move celestial bodies are necessary causes of all events." (*In Hexaemeron*, VI, 2-3) Thus, a rejection of exemplars, of the Platonic Ideas, leads inexorably to a rejection of providence and thus to a fatalistic view of the happenings in the world. (Bonaventure betrays no acquaintance here with the discussion of fate and providence in the fifth book of *The Consolation of Philosophy* or of Boethius' handling of chapter nine of Aristotle's *On Interpretation*.) The opinion that the world is eternal, that there is but one intellect for all men, and that, consequently, there is neither punishment nor reward for deeds done in this life -- these too follow from the rejection of the Ideas. The unicity of the intellect was maintained to avoid having to affirm an actual infinity of human souls, which would seem to follow if the world and time had no beginning. And if there is but one intellect, only it survives the demise of particular men, and no personal immortality is possible, nor, of course, personal reward or punishment. (*Ibid.*, 4; see VII, 2)

It is a melancholy picture that Bonaventure paints of the philosophy of Aristotle. Since the whole sorry story has been made to hang on the rejection of the Ideas, we would expect to see Plato treated somewhat more kindly. And so he is. "Other enlightened philosophers posited the Ideas, and they were worshippers of the one God, for they placed all good in God as the best good." (VII, 3) Plato, Plotinus, and Cicero are cited and praised in this regard. Nevertheless, Bonaventure feels constrained to go on to enumerate the deficiencies of these men, which were due to their not having the faith.

Bonaventure does not seem to be saying that the philosophers just happened to commit errors. Rather, his point is that a man who does not have grace and faith will inevitably make philosophical mistakes; he has not the remedy with

which to cure the disorder and weakness of his natural powers, and as long as there is not faith the malady lingers on. Thus, while Bonaventure holds that theology needs philosophy, he wants also to maintain that philosophy has need of faith if it is to achieve its own ends. We might wish to object to this position in the following manner. Granted that philosophers made mistakes (though I, for one, cannot accept the Bonaventurian estimate of Aristotle), the history of philosophy is not simply a catalogue of errors; indeed, the errors there are stand out precisely because of the background of truth. Now Bonaventure himself is willing to concede that philosophers, even Aristotle, recognized the existence of God; moreover, Plato and others are commended because they recognized the Ideas. Immediately after listing the three great errors of Aristotle, Bonaventure adds, "But some, seeing that Aristotle was so good in other matters and had said so much that was true, could not believe that in these instances he was not speaking the truth." (*In Hexaemeron*, VI, 5)

Can we not then accept the truths philosophers offer and reject their errors? This is just what theology must do. "Hence, making use of philosophical knowledge and taking from the nature of things what it needs to construct a mirror in which divine things are reflected, it erects a ladder the foot of which rests on earth and whose head reaches heaven." (*Breviloquium*, prologue) Theologians must take from philosophers in the way the sons of Israel took from the Egyptians, following the counsel of Augustine. (*De tribus quaestionibus*, n. 12) What criteria enable us to recognize the errors of philosophers? Not a more adequate philosophy. Bonaventure says that the simple faithful, knowing that in the beginning God created heaven and earth, see the falsity of the claim that the world is eternal. It is in the light of faith then that philosophical errors are recognized as such -- and presumably philosophical truths as well. But if a philosophical truth is the conclusion of an argument, to accept that truth because it is also revealed is not to accept it as philosophically established. An examination of the proofs and a philosophical assessment of them is a different undertaking altogether, and it seems difficult to hold that Bonaventure urges us to refrain from such activity. He has said that philosophical arguments are irresistible, and he will recognize a good number of them as valid. But these arguments demand a fairly wide context. In that sense Bonaventure recognizes the existence of philosophy. But philosophy is completed by faith and the theology based upon it. Moreover, philosophy is best carried on under the extrinsic control of faith, which warns against blind alleys and guides us toward the truth. But since faith provides no proofs, it does not constitute philosophy. Perhaps one could say that for Bonaventure faith has an important role to play in philosophizing, though not in philosophy as such. The openness of

philosophy to a truth above reason would save us from thinking that the philosophical conclusion "God is one" is opposed to the truth of faith "the one God is three persons."

Perhaps something can now be said of the controversy mentioned earlier. First of all, no one maintains that Bonaventure ever devoted himself to a specifically philosophical work. The question is, Can we find a philosophical doctrine in what he has written? Our answer must be in the affirmative. Bonaventure's whole view of the nature of theology indicates that it must make use of philosophy. But did he himself contribute to philosophy or only borrow from it? I think we must say that for the most part he only borrowed, though he did make contributions.

It is often said that Bonaventure was primarily a theologian. While true enough, this remark could be misleading. We feel prompted to observe that most theologians are primarily theologians, particularly when they are doing theology. The remark is made to establish the point that insofar as Bonaventure dabbled in philosophy, he did so with a view to the ends of theology. But of course theology was no more an end in itself for Bonaventure the man than was philosophy. Theological knowledge was to lead to mystical union and that to the beatific vision. The perspective in which something may be seen, the subjective reason for doing it, need not alter what is seen or done. In making use of philosophy the scholastic theologian may or may not find ready at hand the philosophical doctrine he requires. If none exists and he elaborates one and then goes on to employ it to explicate or defend truths of faith, the historian of philosophy will be able to examine the doctrine elaborated in legitimate isolation from the theological context and from the theological purpose it is made to serve. Some medieval masters of theology, and Bonaventure is in this class, do whatever philosophy they do in the context of works which are formally theological; others not only do this but also engage in philosophy within the limits and dictates of natural reason itself. Thomas Aquinas falls into this second class. The great difference between these two classes, as represented by Bonaventure and Aquinas, is the following. When we find philosophical doctrines elaborated in theological works of Aquinas, we know what for him is the wider philosophical whole to which they are contributions. The same is not true of Bonaventure. Consequently, it must always be a work of some daring and imagination to try to make a coherent whole of Bonaventure's philosophical doctrines, which are scattered piecemeal through his theological works. To maintain that any whole constructed from such pieces depends upon the acceptance of truths of faith would be to vitiate the whole enterprise. It would

be better not to call such a systematic whole a philosophy at all if it presupposes religious faith for its acceptance.

What is for the moment clear is that philosophy represents for Bonaventure a given level of knowledge which is best interpreted with an eye to the hierarchy in which it fits. Most of his remarks about philosophical doctrines amount to more or less symbolic interpretations: they are seen as prefiguring what lies beyond philosophy. Bonaventure, in this characteristic stance, is clearly looking on philosophy from the vantage point of faith, and this is a perspective which presupposes philosophy as given and betrays no interest in contributing to it. From this sapiential point of view philosophy is a sign of what lies beyond it and at the same time is seen as a useful, if not necessary, rung on the ladder to heaven.

E. The Division of Philosophy

Scattered throughout the works of Bonaventure are the divisions of philosophy into various disciplines that indicate their sources in Augustine, Hugh of St. Victor, and, of course, Aristotle. Bonaventure sets down the threefold division of philosophy into rational, natural, and moral in *The Reduction of the Arts to Theology* and argues that it is an adequate one since there are but three kinds of truth -- that of speech, that of things, and that of morals. Another argument is more typically Bonaventurian. "Again, just as in God we can consider the note of efficient cause, of formal or exemplar cause, and of final cause, since he is the cause of subsisting, the means of understanding, and the order of living, so in philosophical illumination, since it illumines either for knowing the causes of being, and then it is physics, or for knowing the means of understanding, and then it is logic, or for knowing the order of living, and then it is moral or practical philosophy." (n. 4) Bonaventure goes on to subdivide each of these, assigning grammar, logic, and rhetoric as the parts of rational philosophy. "Further, since our intellect has to be directed in judging according to formal notions [*rationes formales*] and these can be considered in three ways -- with reference to matter, and then they are called formal notions; with reference to soul, and then they are called concepts; or in relation to divine wisdom, and then they are called Ideas -- so natural philosophy is divided into three parts. Physics, accordingly, considers the generation and corruption of things according to natural forces and seminal reasons; mathematics considers the intelligible notions of abstractable forms; metaphysics is knowledge of all beings, which it reduces to one first principle from which they come according to the Ideas, or it reduces them to God as principle, goal, and exemplar. There

is, however, a great divergence among metaphysicians on the subject of Ideas." Finally, Bonaventure subdivides moral philosophy into monastics, economics, and politics. The following schema summarizes his views:

- Philosophy
 - moral
 - monastics
 - economics
 - politics
 - natural
 - physics
 - mathematics
 - metaphysics
 - rational
 - rhetoric
 - logic
 - grammar

While Bonaventure gives these divisions, with which we are well acquainted from previous chapters, what characterizes much of his

work, and indeed the *Reduction* itself from which we have just been quoting, is the way in which philosophy occupies only a few of the rungs on that ladder which reaches from earth to heaven. We shall try now to give a sketch of *The Reduction of the Arts to Theology*; having done that, we shall discuss a famous text in which Bonaventure argues that all science and wisdom are summed up in Christ, who is the mean (*medium*) of every science. We will then go on to the allied question of the way in which whatever is known is known in the eternal notions or divine Ideas.

The Reduction of the Arts to Theology is an elaborate interpretation of a remark by St. James in the first chapter of his Epistle: "Every best gift and every perfect gift is from above, coming down from the Father of lights." From this basic image of God as a sun from which flow rays of light Bonaventure distinguishes a variety of participated lights. He speaks of four: an *external* light, or the light of mechanical art; a *lower* light, or the light of sense knowledge; an *inner* light which is the light of philosophical knowledge; a *higher* light which is the light of grace and of Sacred Scripture. Bonaventure's discussion of the mechanical arts simply recalls the relevant part of Hugh of St. Victor's *Didascalicon*. Given the threefold division of philosophy already mentioned, he arrives at six ways of

looking at the light emanating from God: mechanical arts, sense knowledge, rational philosophy, natural philosophy, moral philosophy, and Sacred Scripture. These make way for a seventh, the light of glory. (n.6)

Since these six lights have their origin in one source, they are all ordered to knowledge of Sacred Scripture: they are contained in it, perfected by it, and through it ordered to eternal illumination. (n.7) Bonaventure's task now becomes one of showing how the other illuminations of knowledge are to be led back to Sacred Scripture. He then goes on to show how sense knowledge can be distinguished in terms of a *medium*, the exercise of knowledge, and the delight concomitant with it. The means (*medium*) suggests to Bonaventure the Divine Word; the exercise of sensation gives a pattern for human life, since each sense is directed to its proper object and shrinks from what could harm it. The delight which accompanies sensation is a sign of the soul's union with God. "Behold the way in which divine wisdom is contained hidden in sense knowledge." (n.10) This may suffice to indicate how each of the levels of light leads Bonaventure inexorably back to Sacred Scripture and theology. "And thus it is obvious how the manifold wisdom of God, lucidly revealed in Sacred Scripture, is bidden in all knowledge and in nature itself. It is also clear how all kinds of knowledge serve theology, which takes examples and terms belonging to every branch of science. So too, it is clear how wide is that illumined path and how within each thing sensed or known God himself lies hidden." (n.26)

The procedure of the Reduction is not wholly unlike that of *The Mind's Journey to God*, the purpose of which is "rather the stirring of the affections than intellectual erudition." (Prologue, n.5) Doubtless such an approach lends itself to parody; moreover, the literal intent of philosophical doctrine becomes swiftly a matter of indifference as one seeks signs of what is believed. Perhaps the most difficult example of Bonaventure's attempts to take a scriptural text as programmatic for a summary assessment of human knowledge is to be found in his lectures on the six days of creation.

The first conference opens with the following passage from Ecclesiasticus (15:5): "In the middle of the Church he will open his mouth and the Lord will fill him with the spirit of wisdom and understanding and will clothe him with the mantle of glory." The *medium ecclesiae* of the text is identified with Christ, who is *mediator Dei et hominum*, the mediator between God and man (I Timothy 2:5). Bonaventure then sets forth his intention. "Our plan then is to show that in Christ are hidden all the treasures of the wisdom and knowledge of God and that he is the means (*medium*) of every science. There are seven kinds of mean:

of essence, of nature, of distance, of doctrine, of modesty, of justice, and of harmony. The consideration of the first falls to the metaphysician, of the second to the physicist, of the third to the mathematician, of the fourth to the logician, of the fifth to the moralist, of the sixth to the politician or lawyer, of the seventh to the theologian. The first mean is primary because of its eternal origin; the second weighty because of its efficacious diffusion; the third profound because of its central position; the fourth clear because of its rational manifestation; the fifth important for moral choice; the sixth important in judicial compensation; the seventh pacifying by its universal conciliation. Christ is the first mean in his eternal generation, the second in his Incarnation, the third in his Passion, the fourth in his Resurrection, the fifth in his Ascension, the sixth in the future judgment, the seventh in eternal reward and punishment." (*In Hexaemeron*, I, 11) From this we construct the following table:

	Medium	Science	Christ
1.	essence	metaphysics	eternal generation
2.	nature	physics	Incarnation
3.	distance	mathematics	Passion
4.	doctrine	logic	Resurrection
5.	modesty	morals	Ascension
6.	justice	politics	judgment
7.	harmony	theology	heaven / hell

Most of these juxtapositions are, to say the least, initially surprising. At any rate, this is Bonaventure's plan. To get some glimmering of how he makes good on it, let us see what he has to say about metaphysics. Metaphysics, he observes, although it rises from a consideration of the principles of created and particular substance to the universal and uncreated and to that being (*ad illud esse*) as it has the note of beginning, means, and ultimate end, does not grasp it as Father, Son, and Holy Ghost. It is because philosophy ultimately arrives at God, who is the efficient, exemplar, and final cause of all else, that Bonaventure sees new significance in the tripartite division of philosophy. The metaphysician, insofar

as he sees God as the first efficient cause of all things, is like the natural philosopher who considers the origin of things; and when the metaphysician considers God as final cause, he is like the moral philosopher who refers all to the highest good. "But when he considers that being as exemplifying all things, then he is a true metaphysician." (n.13) The defining role of metaphysics is concern with the eternal exemplars, or Ideas. Christ as the Word of God is the locus of the divine Ideas, and with him all things have their origin; hence, knowledge will achieve its perfection in Christ and metaphysics in its reduction of things to the Ideas. "For the beginning of knowledge is the same as the beginning of being. For if, as Aristotle says, the knowable (*scibile*) as such is eternal, it must be that nothing is known save through immutable, unchangeable, unlimited truth."

Just as Christ lies at the center of metaphysics since the peculiar concern of metaphysics is the divine Ideas, so Bonaventure would have us see that Christ is implied by or prefigured in the principal concern of each of the sciences. Physics is concerned with two worlds, the macrocosm and the microcosm, man; and the centers of these worlds are, respectively, the sun and the heart, both of which are signs of Christ in his Incarnation. Mathematics is said to be chiefly concerned with the measurement of the world and with the movements of the heavenly bodies. In his crucifixion on earth Christ stands at the center of the world; moreover, his passion is the measure of the Christian life. In logic we are concerned with the exterior manifestation of the truth, and this is a sign of the Resurrection, which is a proof of Christ's divinity. Ethics of course is concerned with virtue, and virtue lies in the mean between extremes. The Ascension corresponds with ethics insofar as the Christian is supposed to rise from virtue to virtue; moreover, rectified reason determines the mean of virtue, and faith is such a rectification of reason. The jurist or politician who must pass judgment is a sign of Christ in the last judgment. The theologian is chiefly concerned with the return of all things to God and thus with Christ as the means of eternal beatification.

One can only marvel at Bonaventure's ingenuity, though at the same time he may be baffled by its results. We have dwelt on this passage in order to give an indication of the way in which Bonaventure, while he sets down traditional divisions of philosophy, goes on in what we may well take to be his characteristic manner or style to reduce all intellectual pursuits to theology. Once more, in such efforts a minimum of time is spent in sketching aspects of philosophical doctrine, and the tendency is to hurry on to an interpretation of the whole endeavor as a sign of some role of the Incarnate Word in the

supernatural order. Generally speaking, Bonaventure's attitude toward philosophy would seem to be one which assumes philosophy as given, as already there awaiting the kind of symbolic interpretation at which he is so adept. The goal again is spiritual edification rather than intellectual enlightenment. As we shall see, it is a matter of knowing things not simply in themselves but as vestiges and images of the divine.

F. The Divine Ideas

St. Bonaventure has summed up the "whole of our metaphysics: it deals with emanation, exemplarity, consummation; that is, to be illumined by the spiritual rays and be led back to the highest is to be a true metaphysician." (*In Hexaemeron*, I, 17) We have seen that these various aspects come down to a consideration of the beginning, exemplar, and end of all things and that since in some fashion the metaphysician shares his interest in the origin and goal with the physicist and moralist, the defining metaphysical concern will be with God as exemplar cause. The proper way to approach the doctrine of Ideas is to ask after Bonaventure's theory of knowledge.

For Bonaventure knowledge is of three kinds: sense knowledge, scientific knowledge, and sapiential knowledge. It is the difference between the last two which interests us now; the fact that he admits both is testimony to Bonaventure's desire to keep what he considered best both in Augustine and in Aristotle. For Bonaventure the human intellect at the moment of creation is a blank slate, a pure possibility as far as knowledge goes. (*II Sent.*, d. 3, p. 2, a. 2, q.1) Experience is the beginning of science. "It is true beyond doubt that, as the Philosopher says, knowledge is generated in us by way of the senses, memory, and experience, from which we derive the universal which is the principle of art and science." (*Sermo, Christus magister*, 18) However, Bonaventure, while accepting the Aristotelian doctrine on the abstractive character of our intellectual knowledge, makes some notable additions to that doctrine. Is it the case that all of our intellectual knowledge comes from sense experience? "The reply must be in the negative. For we must hold that the soul knows God and itself and the things in itself without any help from the external senses. Hence, if the Philosopher sometimes says that 'nothing is in the intellect which was not first in the senses' and that 'all knowledge takes its rise from sense,' this should be understood as referring to those things which are in the soul owing to an abstracted likeness." (*II Sent.*, d. 39, a. 1, q. 2, c.) God and soul -- these fundamental, and Augustinian, concerns are excepted from the scope of

Aristotle's doctrine of abstraction. Let us first see what Bonaventure has to say about abstractive knowledge, *cognitio scientialis*.

St. Bonaventure will here make use of the Aristotelian analysis which led to the distinction between an agent and passive intellect; this distinction permits us to speak of understanding both as an activity and as a kind of receiving of an impression. The agent intellect itself is a kind of light, Aristotle had said, which plays upon the images garnered from sense experience. The agent intellect then is the innate light because of which we can attain knowledge of what things are. We do not want to emphasize just now the divergence between Bonaventure and other Scholastics on the interpretation and use of Aristotelian doctrine. What we do want to stress is that Bonaventure allows for a theory of knowledge which it is easy to think is diametrically opposed to that which comes from Plato and Augustine. Bonaventure sees the merit in the teaching that intellectual knowledge of sensible things is abstractive. Intellectual knowledge is not simply a passive reception of objects: it is an activity and there is an agent intellect which may be compared to a light and which is an indispensable element in the doctrine of abstraction. However, while he accepts the necessity of abstraction, Bonaventure will also argue against its sufficiency. He will argue that intellection requires as well a kind of illumination which is independent of abstraction. It is to that difficult Bonaventurian tenet that we must now turn.

The *locus classicus* of Bonaventure's doctrine on the necessity of illumination as the complement to and foundation of abstractive knowledge is the *Disputed Question on the Knowledge of Christ*, question 4. The thesis Bonaventure would defend is this: "For certain knowledge it is necessary that the intellect, even here below, in some way grasps the eternal notion as normative and efficient cause, not by itself and in its own clarity, to be sure, but along with the created proper notion and as known in a glass darkly." We must follow in some detail Bonaventure's defense of this thesis.

The claim that whatever is certainly known is known in the light of the eternal Ideas (*rationes aeternae*) is susceptible of three interpretations. First, one might take it to mean that in certain knowledge the evidence of eternal light is the sole and whole cause of knowing. This does not commend itself, however, since it comes down to saying that all knowledge is knowledge of things in the Word, and then there is no difference between terrestrial knowledge and the beatific vision. Moreover, there would be no difference between knowing something in the Word and knowing it in itself (*in proprio genere*), and no difference between scientific and sapiential knowledge, between natural knowledge and that of

grace, between rational knowledge and that of revelation. Since all these consequences amount to false identifications, the interpretation must be rejected. Augustine has observed that skepticism is the final result of this opinion. The Academicians, maintaining that nothing could be known with certainty save in the intelligible, archetypal world, and recognizing that that world is hidden from us, had to conclude that we have no certain knowledge, that all is opinion and open to doubt.

A second way of interpreting the claim is this. In certain knowledge an influence of the eternal Idea is necessarily involved, but not in such a way that the knower attains the eternal notion itself except in its influence or effect. This view is inadequate, however, and this according to Augustine, who expressly asserts and clearly argues that the mind has to be regulated in certain knowledge by eternal and changeless rules, not as by a possession of its own, but by things above it in the eternal truth. Thus, to say that our mind in knowing with certitude does not go beyond the effect or influence of uncreated light is to say that Augustine was deceived, since his remarks cannot be interpreted in this way. Bonaventure takes this to be an absurd accusation to make against so great a Father and Doctor, who is the most reliable expositor of Sacred Scripture.

We may add that either this "influence" of light is the general causality of God with respect to all creatures or it is special, like God's causality in grace. If the former, then God need be named the giver of wisdom with no more propriety than he is called the fructifier of the earth, nor should we ascribe wisdom to him anymore than honey; if the latter, in the fashion of the special effect of grace, then all knowledge is infused and none acquired or innate. But all this is absurd.

But there is a third way of understanding the claim, one which is a mean between the two unsatisfactory interpretations. Certain knowledge necessarily requires the eternal Idea as normative and efficient cause, not alone and in its proper brilliance, but together with the created notion, that is, it is "known in part" to the degree that this is presently possible. This is what Augustine tells us. "Let the impious one reflect in order that he may be converted to the Lord as to that light whereby even as he turns from it he is touched. So it is that even the impious recognize eternity and rightly grasp and rightly praise many things concerning the morals of man." (*De trinitate*, XIV, 15) And he adds that they accomplish this through rules "written in the book of that light called truth." In order that our mind in its certain knowledge might in some wise attain those rules and changeless Ideas, there must necessarily be both nobility of knowledge and worthiness of the knower.

By nobility of knowledge Bonaventure means that certain knowledge requires immutability on the part of the knowable and infallibility on the part of the knower (*ex parte scibilis immutabilitas et infallibilitas ex parte scientis*). Created truth is not absolutely, but potentially, immutable; similarly the light of the creature is not wholly infallible in virtue of itself -- both are created and come forth from nonbeing to being. Both truth and the light which enables us to see it are daughters of time. If then the fullness of knowledge necessitates recourse to a truth in every way immutable and stable and to a light wholly infallible, there must needs be in such knowledge recourse to the supreme art as to light and truth: to the light as to that which gives infallibility to the knower, to the truth as to that which gives immutability to the known. Since then things enjoy a threefold existence -- in the mind, in themselves, and in the eternal art -- the truth of things which follows on the second mode of being is not sufficient for the soul's certain knowledge, nor is the existence of these things in the mind sufficient. In some way things must be attained as they are in the eternal art.

Certain knowledge also requires worthiness (*dignitas*) on the part of the knower. The rational spirit has a superior and inferior part, and just as the inferior part has need of the superior for a full deliberative judgment as to what should be done, so too with respect to a complete judgment in speculative matters. The superior part of the soul is that owing to which it is an image of God; it adheres to the eternal rules, and it is through them that it defines and judges with certitude: this belongs to it insofar as it is an image of God.

The creature can be compared to God as a vestige, image, or likeness. The creature is a vestige insofar as it is referred to God as to its principle, an image insofar as it is referred to God as to an object, a likeness insofar as it is referred to God as to an infused gift. Every creature since it is from God is a vestige; every creature which knows God is an image; only that creature is a likeness in whom God dwells. Following on this threefold reference to God, there is a threefold gradation of divine cooperation.

God cooperates in the effect of the creature as vestige by way of the creative principle and in the work of the creature as likeness, a work meritorious and pleasing to God, by way of an infused gift. God cooperates in the effect of the creature as image in the manner of an effecting notion (*per modum rationis moventis*), and such is the work of certain knowledge which is not from inferior reason alone but involves the superior reason. Since then certain knowledge belongs to the rational spirit insofar as it is an image of God, in such knowledge it attains the eternal Ideas. But because in this life it is not fully like God, it does

not attain them clearly, fully, and distinctly; rather, to the degree that it is more or less like God, to that degree it attains them, but always only in a certain fashion, since it cannot rid itself of the status of image. Hence, in the state of innocence, when the soul is an image without the deformity of guilt, it nevertheless did not have the full likeness with God which is glory; thus, it attained the Ideas in part but not in darkness. In the state of fallen nature the soul both lacks "deiformity," likeness to God, and has deformity, and so attains them in part and darkly (*in aenigmate*). In the state of glory the soul will lack deformity and have full likeness to God and will then attain the Ideas fully and perspicuously.

Again, because the soul is not an image in its entirety (but only owing to the *ratio superior*), it attains the eternal notions along with the similitude of things abstracted from phantasms, and these similitudes are proper and distinct means (*rationes*) of knowing. Without these concepts the light of eternal reason is insufficient for knowledge, at least in this life, unless perhaps through some special revelation this state is transcended. This happens in states of rapture and in revelations to the Prophets.

C. The Nature of Illumination

The key text we have just examined provides us with a hook on which to hang all subsequent discussion of St. Bonaventure. We have seen that for Bonaventure true metaphysics is occupied with three things: creation or emanation, exemplarity, and consummation or return. Of these three the exemplars, or divine Ideas, are finally the most important since, as we shall see, it is via these that things emanate from God and because of these that in their different ways creatures return once more to God. This metaphysical program will strike us as Neoplatonic, and as a philosophical program that is essentially what it is. The theologian, of course, could have no alternative plan: creator, creatures, return. That is, as we shall see, the plan of Aquinas' *Summa theologiae*. Yet Bonaventure is no Neoplatonist, or if he is, he is something else besides. In the first place, while in some sense the Ideas are the beginning of knowledge as well as of being, they are also a culmination and conclusion from a philosophical point of view. The text we have just considered exhibits a typical deference to Augustine, and the Augustinian influence gives a Platonic and Neoplatonic tone to what Bonaventure has to say. True and certain knowledge entails infallibility on the part of the knower and immutability on the part of what is known. If I am certain that "X is Y", there can be no possibility that "X is not Y" will be true. The truth, in short, is immutable.

Whence comes this immutability, particularly when I consider that the value of X is something sensible and changeable? Here is the familiar source of idealism, whether it leads to the positing of another realm of entities, Xness and Yness, or whether it posits innate rules of thought according to which "X is Y" can never be overturned. Notice that Bonaventure, following Augustine, has dismissed the latter possibility: the eternal rules are not the innate grooves of the

created mind. Why? Precisely because it is created. As created, the human soul is *vertibilis in non-esse*. Being contingent, it cannot of itself account for the necessary. Changeable things are in themselves changeable and not necessary; moreover, their existence in the mind cannot as such confer unchangeability and necessity on them. If we take a proposition which expresses an eternal truth, "the truth signified by it can be signified either as it is in matter or as it is in the soul or as it is in the divine art or taken to be certain in all these ways at once." (*Ibid.*, ad 23-6) Now, if we say that things are true as spoken, such statements are signs of mental states. What we must do is see the soul as occupying the middle ground between things and the divine Ideas. Owing to inferior reason (*ratio inferior*), the soul is referred to things in themselves by way of abstractive forms or concepts which are accordingly *rationes creatae*; because of superior reason (*ratio superior*), which makes it an image of God, the soul refers to the divine Ideas which are *rationes increatae*. True and certain knowledge is had by bringing the *rationes aeternae* to bear on the *rationes creatae*. We have then a blend of abstractive and illuminative knowledge: the former without the latter is mere contingency; the latter without the former is empty. Illumination, then, an intellectual participation in the divine Ideas, is the perfection and *sine qua non* of abstractive knowledge; scientific knowledge must be anchored in and guaranteed by sapiential knowledge.

As to the nature of illumination, we are first told what it is not. Neither is it another name for the general cooperation of God in the operations of creatures nor is it something as special as grace. Illumination is not something supernatural. It lies somewhere in between these two possibilities. More positively, it is said to be God's cooperation with the activity of the creature as image; a creature is an image insofar as it can know God, and thus illumination is the divine cooperation with the activity of the intellectual creature. God as exemplar cause is the guarantee of the certitude and immutability of knowledge in the strict sense. No matter how fluid and evanescent the created thing which is known, it is a vestige of the creator, an exemplification of a divine Idea. As such it can be a factor in abstractive knowledge, in scientific knowledge. But scientific knowledge is not yet sufficiently grounded. Bonaventure accepts

Aristotle's definition and discussion of *scientia*, or *episteme*, and asks what is the guarantee of its immutability. It cannot be the things known, the things out-there in the visible world, for they are mutable and changeable; it cannot be a constituent conferred by our mind because we too are creatures, our minds are *vertibiles in non-esse*. Beyond abstraction and science, then, it is necessary for recourse to be had to the art behind the things and our souls, the art which operates through eternal notions and Ideas. By reference to these science is anchored and justified. Without this reference to Ideas there is no metaphysics. Aristotle, consequently, is a philosopher of nature, a scientist, but he is not a metaphysician or *sapiens*.

This is not to say that scientific knowledge requires explicit reference to the Ideas. Augustine has said that the eternal light reaches men even when they are turned from it. The scientist -- think of Aristotle -- achieves certain knowledge and thus implicitly at least is aware of the Ideas -- that is why his knowledge is certain: the Ideas are operative in it. But he is not, for all that, wise. "The wise man [*sapiens*] attains these Ideas in one way, the scientist [*sciens*] in another. The latter attains them as causes [*ut moventes*], the former as that in which he rests [*ut quietantes*], and to this wisdom no one attains 'unless he first be cleansed by the justice of faith' (John 1:9)." (*Ibid.*, ad 2) Alas, this passage complicates matters once more. In his defense of his thesis Bonaventure had made no mention of Ideas *ut quietantes*, but only *ut moventes*. Presumably the latter was sufficient to show the way in which abstraction is perfected by illumination, science by wisdom, and there was no question of a supernatural gift. Now, wisdom is associated with a knowledge which terminates in the Ideas as opposed to a knowledge in which the Ideas are operative but perhaps not explicitly alluded to. Is Bonaventure distinguishing now the wisdom which is a gift of the Holy Ghost from theology and *a fortiori* from philosophical wisdom, or is he distinguishing the supernatural from the natural order? If unequivocally the latter, then philosophical wisdom, or metaphysics, is impossible and Plato is no better off than Aristotle. Perhaps it is not a trivialization of Bonaventure's position to say that the Ideas are operative in scientific knowledge insofar as God is the cause of the known and cooperates with the activity of the knower. He need not be taken to mean that there are other objects of knowledge to which we must turn in order to have scientific knowledge, but that we can have scientific knowledge of the things we do because those things are what they are and we are what we are. The divine truth is known to us, not in itself, but rather insofar as it is revealed generally in the truths we know: "The eternal Idea not only causes us to know but is known, not specially in itself, but generally together with the truth of principles; thus, it does not follow that it is known to

us in itself, but to the degree that it shines forth generally in principles." (ad 16) Scientific truths, then, are referred to the Ideas, the *rationes creatae* to the *rationes aeternae*, not as objects to further objects. Rather, in reflecting on the demands of truth and knowledge the mind is led to know God and the way in which God creates.

The danger here of course is that eternal truths appear to be, *salva reverentia*, whimsical or capricious choices of God. Bonaventure's point is more subtle and Augustinian. Self-evident principles are signs of God, who is eternal truth. In short, and we shall return to this, Bonaventure is here treading the Augustinian way of truth as a proof for the existence of God. The following summary statement of his position by Bonaventure perhaps makes it as clear as it can be made. "To the objection that if we know in these notions or Ideas

[*rationibus*], every knower is a wise man, let it be said that it does not follow, because to attain these Ideas does not make one wise unless he rests in them and knows that he attains them, which is proper to wisdom. For such Ideas are attained in the concepts of scientists as instruments [*ab intellectibus scientium ut ductivae*], but in the concepts of the wise as terms and resting points [*ab intellectibus sapientium ut reductivae quietativae*]. And since they are few who so attain them, the wise are few, though the knowing are many. Few indeed are those who know that they attain such Ideas and, what is more, few wish to believe there are such Ideas since it appears so difficult for an intellect which has not been elevated to the contemplation of divine things and thus to have God present and near, although Paul says in Acts 17:27 that 'He is not far from each of us.'" (ad 19)

Few indeed maintain that Bonaventure's doctrine on illumination is an easy one to grasp. We can only hope that we have managed to remove some of the initial ambiguity from it. We must go on now to discuss a number of allied doctrines: What is the status of proofs for the existence of God? What has Bonaventure to say about creation? What is Bonaventure's doctrine on universal hylomorphism? What is the nature of the human soul and can its immortality be proved? In attempting to answer these questions we shall have more to say on the distinction of reason into a superior and inferior part and on the division of created perfections into vestiges, images, and likenesses. We shall want to ask what is the relevance of that triad to the question, How can God be named by us?

H. Proofs of God's Existence

Bonaventure's doctrine of illumination is such that God is involved in all certain knowledge, not as object, but as the regulating and motivating cause. The theory according to which intellectual knowledge is abstractive is good as far as it goes, Bonaventure feels, but it is ultimately insufficient to explain the certitude of knowledge. For that, appeal must be made to the Ideas. Now, if Bonaventure were saying that we know all things in God, it would be fair to call him an "ontologist," but, as it happens, he insists that in this life we always know God in or through something else. God, the divine Ideas, are the ultimate guarantee of knowledge, but we cannot have direct and comprehensive knowledge of God in this life. Thus, he counsels us to interpret carefully any authoritative statements which seem to say that we can know God in this life; they must be taken to mean, not that we can know God in his essence, but rather that we can know him in some inner effect. (*II Sent.*, d. 23, a. 2, q. 3) The universe is a scale we must ascend in order to arrive at knowledge of God; therefore, the universe is that through which we know God. We must begin with God's vestige in the corporeal and temporal. Although he more often than not will exempt ecstatic and mystical knowledge from the scope of this claim, Bonaventure says that our knowledge of God is always an achievement, the term of rational discourse by which we move from effects to God as their cause. If, however, knowledge must begin with what is sensed, we cannot make the object of intellection coterminous with what has been abstracted from sense experience. Once more, abstraction is a necessary, but not a sufficient, condition of intellectual knowledge. Bonaventure will insist that our knowledge of God is not abstractive; no more is our knowledge of ourselves. Not unlike Kant, he will say that while all knowledge begins with experience, not all knowledge is derived from experience. Our knowledge of sensible things may be the occasion for our self-awareness, but knowledge of self is not derived from what is sensed. In order to stress this, Bonaventure will speak of an intuition of the self as opposed to abstractive knowledge, which pertains to the corporeal and temporal. In the same connection he will speak of an intuition or cointuition of God.

Knowledge of self and knowledge of God are linked when Bonaventure wants to oppose them to abstractive knowledge. "It is necessary to say that the soul knows God and itself and its own activities without any aid from the external senses." (*II Sent.*, d. 39, a. 1, q. 2) The route that Bonaventure is taking here is certainly Augustinian. In our ascent to knowledge of God the sensible world has its role to play, not so much because one moves from corporeal things directly to their incorporeal cause, but rather because corporeal things, as vestiges of

God, lead us within ourselves, to the image of God we are. What is the next step? If there is an intuition of the self and of mental activities which is occasioned by experience of corporeal things, is there also a direct intuition of God? The difficulties here are precisely the difficulties we encountered in trying to understand Bonaventure's remarks on the divine Ideas. On the one hand, he seems to be saying that we know God by direct intuition; on the other, he vehemently denies that God is directly accessible to the human mind in this life. What he seems to be saying is that since the human soul is the image of God, reflection on it permits us to arrive at knowledge of God by moving from the image to the original. This is a discursive knowledge, a movement from effect to cause, but Bonaventure insists that it is knowledge of God. The whole purpose of the universe is to lead men to knowledge of God, and Bonaventure's teaching will not allow that this purpose is systematically frustrated. But since our discursive knowledge of God must betray its origins, our knowledge is always imperfect, and Bonaventure will again remind us of the distinction between attaining God cognitively and comprehending God. The latter is impossible. Bonaventure employs the distinction between affirmative and negative knowledge to indicate how we attempt to surmount the imperfections of what has permitted us to come to knowledge of God.

Bonaventure is often represented as accepting the ontological argument for God's existence, and there are grounds for this contention. However, here as elsewhere the precise position of Bonaventure is a nuanced one, and we must be careful in ascribing to him an unqualified acceptance of the Anselmian proof. The text which is most pertinent to this inquiry is the *Disputed Question on the Mystery of the Trinity*, question one, article one. Bonaventure is asking whether "God exists" is an indubitable truth. He begins by giving twenty-nine reasons why the truth that God exists is indubitable, and among those twenty-nine reasons are several borrowed from Saint Anselm. When we turn to the *Respondeo* of the article, however, we find Bonaventure remarking that a truth can be indubitable in itself and nonetheless dubitable by us. That is the kind of indubitable truth "God exists" is taken to be. Insofar as one does not correctly apprehend the meaning of the term "God," he can doubt that God exists. Nevertheless, Bonaventure concludes as follows: "But that God exists cannot only not be doubted, its contradictory cannot even be thought by a mind which fully comprehends the meaning of the term 'God,' namely, a being than which nothing greater can be thought." Bonaventure thinks it highly unlikely that anyone would not know the meaning of the term "God," however, and if the ontological argument is less an argument proving that God exists than the denial of the need for such proofs, one must ask what Bonaventure takes the

status of attempted proofs to be. Having recalled that "God exists" is dubitable only from a defect on the part of our mind, a defect which calls forth proofs, he adds, "Hence, reasonings of that kind are intellectual exercises rather than arguments giving evidence and manifesting a proved truth." (*Ibid.*, ad 12m) So-called proofs do not so much prove as remove impediments to our seeing that "God exists" is indisputably true.

If the whole of creation bespeaks its cause, nevertheless because the universe is graded and hierarchical, things reveal their creator in various ways.

Bonaventure, we have seen, distinguishes between vestige and image.

Sometimes he speaks of shadow, vestige, and image. Earlier we proceeded as if the difference between vestige and image were simply the difference between the corporeal and spiritual. This is not exactly true, since vestigial traces of God are found in spiritual creatures as well. "With respect to the difference between vestige and image some assign the following: the vestige is in sensible things, the image in spiritual. But this will not do because the vestige is found in spiritual things as well, for unity, truth, goodness, in which the vestige consists, are quite universal and intelligible conditions. Others say that something is called a vestige because it is a partial representation, while an image gives the whole. But this position will not do either, because since God is simple, he cannot be represented according to a part. And since he is infinite, he cannot be represented totally by any creature or indeed by the whole world. So we should recognize that when creatures lead to knowledge of God as shadows, vestiges, and images, the difference among these three, as their names suggest, is taken from mode of representing. For a shadow is that which represents in a remote and confused manner, a vestige in a remote but distinct manner, and an image more closely and distinctly. . . . Creatures are called shadows with respect to properties which refer them to God in any genus of causality, but according to an indeterminate notion of the cause. The vestige is that whose property refers it to God under the aspect of a threefold cause: efficient, formal, and final, like one, true, and good. A creature is called an image because of conditions which refer to God not only as cause but as object -- properties like memory, understanding, and will. From this we can arrive at other differences taken from the cognitive destination of these three. For the creature as shadow leads to knowledge of the common as common; as vestige, to knowledge of the common as appropriated; as image, to knowledge of the proper as proper." (*I Sent.*, d. 3, q. 2, conel.) From creatures which are only shadows we can have only the most remote conception of what God is, whereas vestiges enable us to know attributes common to the three Persons of the Trinity, though these attributes of nature can be appropriated to one Person rather than another, for example,

wisdom to the Son. The rational creature alone is an image of God and is an *imago trinitatis*.

I. Creation and Universal Hylomorphism

The principle that Bonaventure invokes to discuss creation is that the good is diffusive of itself (*bonum est diffusivum sui*), what might be called the principle of the generosity of the good. Since he invokes this same principle in speaking of the procession of Persons in the Trinity, however, some distinctions are clearly called for. The procession of Persons is eternal, whereas Bonaventure is a staunch opponent of the claim that the world is eternal. The distinction brought forward is that between production within the Godhead and production without. The procession of Persons is within the Godhead and is a necessary one, whereas the production of the world is without and is not necessary. The reason for the latter claim is that production without bears on that which can be and not be, that is, on the contingent. The created world depends for its existence on God's free will. "The reason why this causality is attributed to the will is the following: the reason for causing both from the point of view of efficient and from that of final causality is goodness, for the good is said to be diffusive of itself and the good is that for the sake of which all things are. The efficient cause becomes actually such because of the end. . . . Therefore, it is will that unites the effective with the end." (*I Sent.*, d. 45, a. 2, q. 1) It is according to the divine reason and power that creation takes place, but not automatically or necessarily -- free will is required. God has eternally within himself the patterns of creatures, but he creates them freely and in time. Creatures come to be in time and from nothing owing to the power and free will of God.

That the world was created freely by God, in time and from nothing, is for Bonaventure a truth accessible to human reason. However, though this is true in principle, in fact men have recognized this only under the influence of Scripture. "It must be said that this is the truth: the world was produced in being, not simply as a whole but also with respect to its intrinsic principles, which do not come from anything else but from nothing. This truth, though it is now open and clear to every believer, was hidden from the wisdom of philosophers, who on this matter long wandered on errant ways. (*II Sent.*, d. 1, p. 1, a. 1, q. 1, concl.) Since the most eminent philosophers have erred on this matter, Scripture has come to our aid and made the point clear. Once the truth of creation in time and from nothing is clarified by Scripture, it is easy, Bonaventure feels, to see that the opposed view is untenable. Bonaventure held that to maintain that the world is eternal not only contradicts Scripture but

involves within itself a contradiction. He gives a number of reasons for this, among which are the following. If the world had always existed, the sun would have described its revolutions an infinity of times; but the sun is still revolving, and this entails further additions to an already infinite number, which is a contradiction. Furthermore, if the world as we know it is eternal, there would be an infinite number of souls of the departed, and an infinite number is a contradictory notion. By an infinite number Bonaventure understands an actual, not a possible, infinity; an actual infinite number is one to which no further additions could be made.

Bonaventure's assertion that the notion of an eternal world is self-contradictory is directed against Aristotle. Nevertheless, Bonaventure makes use of an Aristotelian conception when he describes finite beings, namely, the notion that the creature is composed of matter and form. Aristotle had argued that an entity which comes to be as the result of a change is composed of form and matter. There are other beings, separate substances, devoid of matter. For such a thinker as Aquinas there are beings other than God and still quite immaterial. Bonaventure held to a universal hylomorphism, claiming that every finite being has matter as a principle of its limitation. "The principle of any limitation is matter or something material" (*principium omnis limitationis est materia vel aliquid materiale*). (*Q.D. de myst. trin.*, q. 4, a. 1) Only divine being, which is pure act, lacks materiality and thus is without limitation. It is clear that for Bonaventure matter is the name of the principle of limitation whereby finite being is precisely finite and not infinite; it is the source of possibility and potentiality. That is why he can speak of matter in the angels and in rational souls. "Matter considered in itself is neither spiritual nor corporeal; therefore, the capacity following on the essence of matter relates indifferently to spiritual and corporeal forms." (*II Sent.*, d. 3, p. 1, a. 1, q. 2, ad 3) On the basis of this Bonaventure will speak of spiritual matter which is simple and unextended. All this sounds odd, of course, but the strangeness recedes somewhat when we recall that what Bonaventure was seeking to emphasize is the difference between finite and infinite being. Since finite being is limited and some principle or source of its limitation must be recognized, Bonaventure chose the term "matter" to designate the principle of limitation of finite being. Souls and angels, while not corporeal, are finite beings and thus, in Bonaventure's odd locution, must contain a material component. He is not suggesting that everything other than God is material in the sense of corporeal and extended.

The human soul is immortal or incorruptible, something that could hardly be said of it if it were material or a composition in the usual sense, for matter, in its

usual sense, is the principle of change or of the corruptibility of the composite. Matter itself is incorruptible, and that provides Bonaventure with one of his arguments for the immortality of the soul. The order of the universe involves both prime matter and an ultimate form. The rational soul is the ultimate form, and if prime matter is incorruptible, then, Bonaventure argues, the ultimate form too must be incorruptible. This is the first of twelve arguments given by Bonaventure for the immortality of the rational soul. (See *II Sent.*, d. 19, a. 1, q. 1.) On other occasions Bonaventure will found the soul's immortality on its ordination to beatitude. "For the soul is the image of God because it has a capacity for God and can participate in his being, and thus it is made for beatitude and is apt for beatitude, which, I say, can only belong to an immortal substance; it is necessary therefore that the soul be immortal." (*II Sent.*, d. 26, q. 4, ad 1) In connection with discussions of the immortality of the soul Bonaventure will stress that his doctrine of universal hylomorphism is a metaphysical and not a physical doctrine. From a physical point of view the soul is wholly simple, since a composition of parts makes a body, whereas the soul is a spirit since it is simple and without extension. Metaphysical composition, such as of act and potency, is, Bonaventure notes, admitted to obtain in all creatures, even angels, and angels are certainly immortal. That is why he concludes that the soul's metaphysical composition of form and matter is not a root of corruptibility.

J. Conclusion

In form and intent the work of St. Bonaventure is always the work of a theologian; he writes as one for whom the only angle of vision and the proximate criterion of truth is the Christian faith. This fact influences his importance for the history of philosophy; when coupled with his style, it makes Bonaventure perhaps the least accessible of the major figures of the thirteenth century. This is true, not because he is a theologian, but because philosophy interests him largely as a *praeparatio evangelica*, as something to be interpreted as a foreshadow of or deviation from what God has revealed. We find in his writings something like a charter for philosophy, but not for anything like a fully autonomous philosophy. Despite this, however, Bonaventure himself seems little interested in engaging in philosophical work. In a way that is not true of Aquinas or Albert or Scotus, Bonaventure does not survive well the transition from his time to ours. It is difficult to imagine a contemporary philosopher, Christian or not, citing a passage from Bonaventure to make a specifically philosophical point. One must know philosophers in order to read Bonaventure, but the study of Bonaventure is seldom helpful for understanding philosophers

and their characteristic problems. Bonaventure as a theologian is something else again, of course, as is Bonaventure the edifying author. It is in those areas, rather than in philosophy proper, that his continuing importance must be sought.

Bibliographical Note

Bonaventure, *Opera omnia*, 10 vols. (Quaracchi, 1882-1902). English translations: C. Boas, *The Mind's Road to God* (New York, 1953); the same in Latin and English, with introduction by P. Boehner (St. Bonaventure, 1956); Sister E. Healy, *Saint Bonaventure's De reductione artium ad theologiam: A Commentary with an Introduction and Translation* (St. Bonaventure, 1939); St. Anthony Guild Press is publishing elegant English translations of Bonaventure's opuscula in a series which began in 1960. See Vollert, Kendzierski, and Byrne, *St. Thomas Aquinas, Siger of Brabant, St. Bonaventure: On the Eternity of the World* (Milwaukee, 1964). See Étienne Gilson, *The Philosophy of St. Bonaventure* (London, 1938) and F. van Steenberghen, *La philosophie au xiiième siècle* (Louvain, 1966); P. Robert, *Hylomorphisme et devenir chez S. Bonaventure* (Montreal, 1936); Efrem Bettoni, *Saint Bonaventure* (Notre Dame, 1964).

Chapter VI

Saint Thomas Aquinas

A. The Man and His Work

Almost twenty years ago a professor of philosophy remarked that there are more Thomists in the world than any other kind of philosopher, a remark that lost its surprise when one reflected that it was prompted by identifying every Catholic philosopher -- if not every Catholic -- as a Thomist. Things have changed in the meantime, for better or worse, but the remark retains some interest if only because it reveals how Thomas Aquinas has sometimes been swallowed by Thomism, that in his case we seem to be dealing less with an individual thinker than with an institution. In a broken rhythm but with general constancy since his death, men have come forward as intermediaries between Thomas and his reader. Commentaries on Aquinas, monographs devoted to particular points of his teaching, a bewildering barrage of journal articles, popularizations of his doctrine, even popularizations of the popularizations -- all this has sometimes had the effect of putting Thomas himself further and further away from the possibility of direct contact. Dominicans, members of Thomas' own order, have always been in the vanguard of these efforts to explicate, expand, and apply the teachings of Aquinas, but they have always been joined by other religious, by secular priests, and, more recently, by laymen, Catholic and non-Catholic.

The remarkable attention that has been paid the thought of Aquinas, while it relates to the essential quality of his work, cannot be understood on that basis alone. The favor shown Thomas by the Catholic Church over the centuries, the unique deference paid him in the Leonine revival at the end of the nineteenth century, added the weight of the ordinary magisterium of the Church to the attraction of the intrinsic qualities of Thomas' teaching and to his previous historical impact to insure for Aquinas an attention to his writings on the part of Catholics which goes far beyond that owed and paid to other important thinkers. This antecedent deference to the thought of Thomas has had disadvantages as well as advantages, and we are currently in a time when the mood of deference to Aquinas seems almost wholly absent from some Catholic philosophers. Just as earlier there were a few who equated repetition of what Thomas had written (a repetition that seemed most comfortable when it was of the original Latin) with understanding and philosophical argument, so now a few seem to take hostility to Aquinas as a sure sign of philosophical seriousness.

One childish attitude is scarcely preferable to another; the importance of Aquinas cannot be decided by hoisting a moist finger to catch the winds of current fashion. Our task here must be to get through the Thomistic tradition -- so much of which, taken in moderation, is undoubtedly an aid -- to the teachings of Thomas himself. When we do this, when we catch something of the flavor and style of his procedure and make soundings in the vast expanse of his teachings, we begin to see why he has been singled out for the attention he has received, why that vast tradition of scholarship and commentary arose, why he is one of a handful of truly major and perennial thinkers. Only then can we hope to speak intelligently of the role Thomas may play today -- and that he has a contemporary role to play is the continuing and insistent conviction of the teaching Church. But, of course, it will not be our task here to enter into any lengthy discussion of the timeliness of Thomas.

Thomas was born in 1225 in Roccasecca near Naples, and Aquinas is the family name taken from Aquino, where the feudal family into which Thomas was born ruled. Thomas' early education was at the Benedictine monastery of Monte Cassino. From 1239 to 1243 he studied the arts at the University of Naples, which had been founded by Emperor Frederick II in 1224. When Thomas entered the Order of Preachers in 1244, his decision met with resistance from his family; taken into custody by his brothers, he was held prisoner for several months, but they set him free in 1245. His first years as a Dominican are obscured for us, but we know that Thomas studied under Albert the Great at Cologne from 1248 to 1252. In the latter year he was sent to Paris, where he was a student of theology until 1257. He was a bachelor of Scripture from 1252 to 1253 and a bachelor of the Sentences from 1253 to 1256 in the Dominican convent in Paris. At the end of this period he was admitted as a master of theology and granted a license to teach on the faculty of theology of the University of Paris. His inaugural lecture was delivered in the summer of 1256.

It must be pointed out that these years of study at Paris were far from serene, for it was just at this time that the efforts of the secular clergy to keep the religious, particularly the mendicant friars, from faculty positions reached a peak of what can only be called frenzy. Thomas and Bonaventure were granted their degrees at the same time, but their admission to chairs on the faculty of theology was delayed. In October of 1256 Pope Alexander intervened to demand that they be received into the academic community. This demand was not complied with until August, 1257, by which time Bonaventure had been elected master general of the Franciscans. But for Thomas this marked the start

of a professorial career which, one way or the other, defined his life until he died.

From 1256 to 1259 Thomas held one of the Dominican chairs of theology at the University of Paris. During this period he wrote his *Commentary on the Sentences*, or rather completed his comments on the work of Peter Lombard begun in 1253. To this period also belong his commentary on the Gospel of St. Matthew, the *Disputed Questions on Truth*, the expositions of the *De trinitate* and *De hebdomadibus* of Boethius, the opuscula *On Being and Essence* and *On the Principles of Nature*, and several quodlibetal questions. These few years represent the first stage of Thomas' career as a teacher and are called the first Parisian period.

The first Italian period begins perhaps in 1260 and extends to 1268. During this period Thomas taught first at Orvieto, where Pope Urban IV was in residence; from 1265 to 1267 he taught in Rome, at the convent of St. Sabine, and then perhaps at Viterbo, where the papal court had gone. During the first Italian period Thomas wrote the *Summa contra gentiles* as well as the first part of the *Summa theologiae*, the masterpiece which was still to be incomplete at the time of his death. He also wrote at this time the *Disputed Questions On the Power of God* and *On Spiritual Creatures* and commented on Pseudo-Dionysius' *On the Divine Names*. The commentaries on Isaiah and Jeremiah seem also to belong to this period. Of great importance for our purposes are commentaries on works of Aristotle, some of which, notably those on the *Metaphysics* and *On the Soul*, were begun during this period.

The second Parisian period took place between 1269 and 1272, when, on the orders of his superiors, Thomas reclaimed his chair on the faculty of theology at the University of Paris. In many ways this could be called the Aristotelian period of Thomas' career since during it he completed his commentaries on the *Metaphysics* and *On the Soul* and commented as well on the *Physics*, *Nicomachean Ethics*, *Meteorology*, *On Interpretation*, and the *Posterior Analytics*. At the same time he commented on the *Liber de causis* (pointing out that it amounted to a selection from Proclus' *Elements of Theology*). He also commented on Job, St. John's Gospel, and the Epistles of St. Paul. He continued working on the *Summa theologiae* during this period and engaged in many *Disputed Questions*, those *On the Soul*, *On Evil*, *On the Virtues*, and *On the Union of the Word*, as well as in many quodlibetal questions. It is to this period that the opusculum *On the Unity of the Intellect against the Parisian Averroists* belongs.

The second Italian period begins in 1272 and ends with the death of Aquinas in 1274. Thomas taught at the University of Naples, where he was sent in 1272 to found a Dominican House of Studies. Besides organizing the curriculum and teaching, Thomas wrote more commentaries on Aristotle, those *On the Heavens*, *On Generation and Corruption*, and on the *Politics*. Moreover, he worked on part three of the *Summa theologiae*. In 1274, on orders of the pope, Thomas set out for the ecumenical council to be held at Lyons. He never made it. Falling ill on the way, he was taken into the Cistercian abbey at Fossanova, located between Naples and Rome. It was there, when he was not yet fifty, that he died on March 7, 1274.

To summarize a life in this way gives us everything but the man who lived it, and of the man Thomas it must be remembered that he was priest and Dominican, teacher and mystic, scholar and saint. In Thomas we find a blend of the natural and supernatural virtues, the moral and intellectual virtues, and truly, insofar as man can say of man, Thomas exhibited in his own life the ideal of Christian perfection of which he wrote with authority and in a style that is almost never unctuous. There is a story that as a child he asked, "What is God?" He followed the trail of that question throughout his life, not as the statement of a curious intellectual puzzle, but in quest of the ultimate meaning of life and of the universe. With respect to that controlling question of his life he was intent to run out the string of reason as far as it could go, and by doing so he came to hold that in the end the most we can do is know what God is not; we cannot know what or who he is with clarity in this life. This final position is not a sign of resignation or lethargy; it represents a learned ignorance which refers not simply to the limits of pure reason but also to the darkness and obscurity of the spiritual life. Intellectual and saint, Thomas speaks audibly both to the sophisticated academic and to the lyric mystic. St. John of the Cross professed to see in the treatise on the contemplative life in the *Summa theologiae* the map, insofar as it can be mapped, of the profundities of the spiritual life. Only the traveler who has returned can make us maps, and Thomas' credentials, unsettling for a certain view of the intellectual life, are incorrigibly dual: his holiness and his austere arguments.

It is not Thomas the saint who is our interest here, but Thomas the thinker, and however inseparable these were in his person, they can be considered apart. The writings of Aquinas which have come down to us are intimidating in their bulk and number; editions of his complete works fill shelves. Our sketch has indicated that they fall into classes. The first great division of his writings could be between those which comment on, explicate, or expose the writings of others

on the one hand and independent works on the other. Among the latter we might include those which grew out of academic debates, the *Disputed Questions* and the *Quodlibetal Questions*, but more particularly various opuscula and, of course, the two great summaries of Christian theology, the *Summa contra gentiles* and the *Summa theologiae*.

Another way of grouping his writings, one more important for our purposes, would be into philosophical writings and theological writings. The former, of course, are what seem to interest us most, but what would they be? Such opuscula as *On Being and Essence*, *On the Principles of Nature*, and *On the Unity of the Intellect* certainly -- these are, on Thomas' own criteria, philosophical works, and, while dependent on his predecessors, they are original in their conception and development. Further, there is the great mass of his commentaries on Aristotle. There has been a dispute among students of Thomas during the present century concerning the status of those commentaries, a dispute which often swings around varying ideas of what Thomas was doing when he commented on Aristotle. Some have simply used the commentaries (despite the deplorable condition of the text of crucial ones like that on the *Metaphysics*) as unqualified repositories of the teaching of Thomas; others have suggested that in his Aristotelian commentaries Thomas is trying to make as clear as he can (and some would say he succeeds to a fault) what Aristotle is saying, without giving his own personal views on the matters in question.

One cannot, of course, in a few lines hope to settle a dispute in which men of acumen and sincerity have disagreed deeply, but some working conceptions for our present purposes seem both possible of formulation and necessary to our task. It seems unreasonable to ask that we ascertain the motives of Thomas the commentator, which could in no way be verified in the evidence we have. We must look at the commentaries, consequently, and see what we find there. We find, as some have insisted, a Thomas who takes great pains to discover precisely what Aristotle is teaching, what his questions are, what the order and development of the text are, and what is the structure of his arguments. Say then that we find Thomas the commentator showing us in great detail that such and such is the problem, so-and-so is the solution, and these are the reasons for accepting the solution. Are we then to ask whether he agrees or disagrees? This suggestion seems to me to be an invitation to distraction. Our attention must be on the question, solution, and arguments; the point is, Do we agree or disagree? If the discourse as Thomas presents it is cogent, to ask if he finds it so is irrelevant. Why should we doubt that he does? What I am suggesting, with

respect to those who would downgrade the commentaries, is simply that a cogent argument, a reasoned position, in a commentary of Aquinas must quite naturally be assumed to be the one with which Thomas agrees.

Now this seems a terribly simple dissolution of a long-standing problem when we had said earlier that it would be difficult and indeed inappropriate to try to settle the matter in a few lines. I want now to develop the solution suggested in such a way that both sides in the dispute are given their due. In commenting on Aristotle, Thomas often takes into account other efforts to explicate the text; when he rejects those efforts, as he often does, it is because they make nonsense of Aristotle or negate the order and development of his thought. Now this sort of thing, which is so far from being rare that it is almost the mark of Thomas' Aristotelian commentaries, suggests that Thomas was deeply sympathetic with the philosophy of Aristotle, that he held Aristotle in an esteem that is quite unique in his relations to previous thinkers, and that this explains both the volume of his commentaries on Aristotle and the fact that he urged his Dominican friend, William of Moerbeke, to provide him with more accurate Latin translations of Aristotle. In short, we can employ Thomas' commentaries on Aristotle with the assurance that in them Thomas is striving for an accurate reading of Aristotle and he is doing so because, by and large, he agrees with the results of such a reading.

Having stated the matter in this way, we must hasten to add that no one doubts for a minute the impact of Aristotle on Aquinas. Indeed, save for infrequent exaggerated statements, no one would doubt for a minute what has been said about our proper response to cogent arguments in commentaries which are clearly regarded to be cogent by Thomas when he formulates them. That is not the point, we would be told; of course Aquinas agrees with Aristotle as far as he goes, but Aquinas often goes far beyond a text of Aristotle in commenting on it. One sees how the dispute now alters in character. At first it seemed to be a disagreement between those who insisted that there was only Aristotelianism in Thomas' commentaries and those who would find Thomism in those commentaries. Now it is those in the first group who insist that the commentaries are Thomistic and beyond mere Aristotelianism. This apparent switch points, however, to a most important truth.

No one could possibly doubt that Thomas is an Aristotelian; what those who object to an almost exclusive reliance on the Aristotelian commentaries have in mind is the fact that we cannot equate Aristotelianism and Thomism. This means at least the following: that there were other sources of the philosophizing

of Thomas than the texts of Aristotle. Some insist on the influence of Christian faith, an influence which led him to philosophical positions, philosophically arrived at, which, if not incompatible with Aristotelianism, nevertheless represent a deepened version, perhaps even a transformation, of it. Surely this is not *a priori* an implausible suggestion. Others will call our attention to philosophical influences other than Aristotle which leave their mark on Thomas' teaching. Consider for example Aristotle's opposition to Plato. Does Aquinas too reject Plato? It would seem so, on the basis of the commentaries, but a number of recent works -- notably those of Fabro, Geiger, and Henle -- have pointed out the influence of Platonism on Thomas. A crucial consideration here is that of participation, which Aristotle rejects as a useless metaphor but which permeates the doctrine of Aquinas.

Our simple solution of the dispute thus gives much to both sides. We seem to find here as we often find elsewhere that when intelligent men disagree, they may on a more profound level be in agreement. However, to agree that not every philosophical teaching of Aquinas is already in Aristotle or derived from Aristotle is not of course to agree about the nature of the more commodious entity, Thomistic philosophy. One can accept, one must accept, the reminder that there is much Platonism in Aquinas, but that is not *eo ipso* to accept certain descriptions of the resultant Thomistic synthesis. It is the assumption of the present presentation that Aquinas was so fundamentally an Aristotelian that he accepted Platonic and Neoplatonic suggestions on an Aristotelian basis. That assumption is borne out by Thomas' procedure in commenting on the *Liber de causis* and on Pseudo-Dionysius.

When we distinguish between the philosophical and the theological writings of Aquinas, we are not suggesting that a history such as this one must restrict itself to the former. For reasons that will be made clear in the following section, formally theological works of Thomas contain much that is crucial for understanding his philosophy. A presentation of the philosophy of Aquinas must rely heavily on the *Summa theologiae*. Thomas' exposition of Boethius' *De trinitate* gives us one of the most lucid presentations of the nature of metaphysics and its relations to other philosophical sciences. For the reader who has come to this chapter from the preceding ones that fact will not be terribly surprising, of course, but let us now turn to what Aquinas has to say on one of our recurrent themes, the relation of faith and reason as well as on one of the allied themes, the relation of theology and philosophy.

B. Philosophy and Theology

The very first question Aquinas asks in his *Summa theologiae* is, Why do we need any doctrine or inquiry beyond philosophy? To pose such a question suggests, of course, that philosophy is or was a going concern and that later and by way of addition a new study was introduced, call it "theology." Thus, it is theology and not philosophy which must be justified. Thomas provides a number of reasons why theology seems superfluous. In the first place, if a man ought not try to know what is beyond his intellectual capacities and whatever is within those capacities falls to the concern of philosophy, philosophy is certainly sufficient for man. Further, teaching is concerned with something, with being, but no type of being seems excluded from philosophical consideration, certainly not divine being since Aristotle's name for metaphysics is theology.

Having made things look difficult indeed for the work he is undertaking, Thomas begins to move toward the contrary position by first quoting from St. Paul's Second Epistle to Timothy (3:16), where Scripture is said to be useful for teaching, arguing, correction, and so forth. It becomes clear that what Aquinas is thus opposing to philosophy, which includes a theology or teaching about God, is Scripture, in which God tells man of himself. "Divinely inspired Scripture does not belong with the philosophical disciplines, which are discovered by human reason. We see that the difference lies in the source of a doctrine: it is discovered by human reason or it is revealed by God. It is this opposition which Thomas stresses in the body of the article. "I reply that we must say that it was necessary for human salvation that there be a doctrine according to divine revelation in addition to the philosophical disciplines which are investigated by human reason. For, in the first place, man as man is meant for God, who is a goal surpassing reason's comprehension. . . . But the goal must be foreknown by men, who are to direct their intentions and actions to it. That is why it was necessary for man's salvation that things which exceed human reason be made known to him by divine revelation." (*ST*, I, 1, 1, c.) The assumption here is that God as man's goal is beyond the ken of man. And yet, had not Thomas conceded earlier that philosophy arrives at valid knowledge of God? He had, and this will of course lead to ambiguity in the use of the term "theology." There is the theology of the philosophers, and there is the theology contained in and based on Scripture.

Aquinas now says something quite important about the theology of the philosophers. "It was necessary that man be instructed by divine revelation even concerning those things which human reason can know about God. The truth

concerning God as discovered by reason comes to only a few men after a protracted period of study and with the admixture of much error -- and yet on knowledge of such truth man's whole salvation depends, for that lies in God. Thus, it was necessary, if salvation was to come to men more fittingly and certainly, that men be instructed about divine things by divine revelation." (*Ibid.*) That last remark may seem confusing, as if Thomas were suggesting that through reason men could, though with difficulty and over a great span of time, arrive at truths concerning God which God has revealed to man. That suggestion is not wholly unintended, as we shall see, but precision is required.

For the moment Thomas is after another point. We remember that one of the arguments for the superfluity of theology pointed out that philosophy studies God. Notice what Thomas now says of that. He first gives an example of how different sciences can prove the same truth, as the physicist and astronomer might prove that the earth is round by quite different reasons. If that is true, "nothing prevents that the same things be treated philosophically insofar as they are knowable in the light of natural reason and by another science insofar as they are known in the light of divine revelation. Thus, the theology which pertains to Sacred Scripture is generically different from the theology which is a part of philosophy." Without for the moment going into the fact that there are truths about God revealed in Scripture that would be inaccessible to human inquiry no matter how much time and effort were expended, Thomas concedes that philosophy arrives at certain truths about God which can also be gleaned from Scripture. With respect to such truths the argument for revelation bears on the difficulty of arriving at them philosophically, on the few men who have succeeded in doing so, and on the errors that mar their success. Yet even here Thomas draws our attention to the different ways in which the same truth may be held to be such, either on the basis of natural reason or on the basis of divine revelation.

This seems curious because two types of rational activity are apparently being compared, and then one is described as a rational activity, namely, doing philosophy. What are we to call the other, the acceptance of a truth as a truth because it is revealed by God? Thomas will call it "believing" the truth. For example, to hold, to affirm, to say that God is one, that there is but one God, because God says so, is to believe that truth. There is a mental attitude here vis-à-vis an object which finds form in a linguistic expression, "God is one." In speaking of faith as a condition or state of mind Thomas will want to contrast it with other mental attitudes, with other states of mind. Thus, he will compare "believing that A is B" with "knowing that A is B" and "intuiting that A is B" and

"thinking that A is B." The symbols stand for subjects and predicates, of course, and perhaps we would normally say that we know or believe or think, not propositions, but what propositions are about. Nonetheless, we can say that propositions or sentences express what we know or think or believe -- that, at any rate, is the way Thomas understands the matter, though in speaking of faith he will insist that we believe someone and something. What is the relation between the subject and predicate when we intuit that A is B? (I am using intuition here for Thomas' *intellectus*.) We intuit that A is B when the connection between the two is immediate, when no link other than that of subject and predicate is required to see that the proposition is true. A frequent example of such a truth in Aquinas is "the whole is greater than any of its parts." In order to see that this is true we need know only what a whole is and what a part is. (Notice that I did not say that we need know only what "whole" and "part" mean, since this could suggest that Thomas thought it is simply a matter of convention and that he shares the suppositions of many current discussions of "analyticity.") Once we know what a whole is and what a part is, the truth is immediately seen. To know that A is B, on the other hand, is to know a statement is true whose predicate and subject are connected or mediated by some third thing. That third thing is, of course, a middle term, and the connection of knowing with the conclusion of a demonstrative syllogism is apparent; so too, intuition bears on the principles of such syllogisms. To think that A is B, finally, is to assert a connection between subject and predicate on other than conclusive grounds. That is, intuition and knowledge are certain, whereas thinking, or opinion, is not. Enough has been said, perhaps, to introduce Thomas' notion of "believing that A is B."

Is believing that A is B like thinking that A is B? It is like it in that, as believed, no cogent reasons for the truth are known; believing is unlike opinion or thinking in that believing is unwavering and, in that sense, certain. Thus, faith is like knowledge and intuition in that it is certain and unwavering, but it is unlike both in that it does not involve the same sort of clarity. Further, what is known or intuited is, in principle, within the reach of any normal man if he pays attention and follows the argument. But faith, the acceptance of what God reveals, is not just the natural employment of a natural capacity. There is something surprising about faith, an intrusion into human affairs of something outside the normal course of events. That something more is, for Thomas, grace, the power of God, and it reaches the mind through the will. "To believe belongs to intellect insofar as it is moved by the will to assent." (*ST*, IIaIIae, 2, 2, c.) What the mind assents to in faith is not seen, that is, to continue the earlier analysis, we do not see the connection between A and B. Faith involves an

intellectual assent to what is believed, but intellectual assent is of course involved in intuiting, in knowing, even in opinion. "There is another way in which intellect assents to something, not because it is sufficiently moved by its proper object, but rather by a choice voluntarily inclining it to one side [of a contradiction] rather than to the other. If this is done with doubt or fear that the opposite might be true, there is opinion; if however it is done with certitude and without fear, it is faith." (*ST*, IIaIIae, 1, 4, c.) The proper object of intellect is what is seen; when the mind assents to something it does not see, whether mediately or immediately, it may do so because it is prompted by desire. Thomas suggests that such motivation is common to opinion and faith, but the great difference is that the believer is without doubt. Or is he?

Like Bonaventure, Thomas distinguished the certitude of adherence from the certitude of comprehension. Knowledge or intuition would possess both kinds of certitude; faith has only certitude of adherence and has it, Thomas argues, to a greater degree than do intuition or knowledge. It is because the assent of faith is prompted by the will, moved by grace, that while there is no wavering with respect to adherence to what God has revealed as true, there is a kind of movement, of mental discomfort, on the part of the believer with respect to believed truths. His assent, while mental, is not prompted by what is proper to intellectual assent as such. Thus, the believer reflects in some unease on the truths he has accepted on the authority of God, and this reflection or meditation may, when it is of a certain sort, give rise to what Thomas means by theology. Thomas makes this point by comparing believing with intuiting and knowing. Intuiting involves assent without prior cogitation, while the assent of the knower to what he knows follows on cogitation. "The knower has both cogitation and assent, but a cogitation causing assent and an assent which terminates cogitation." Belief or faith involves both cogitation and assent but, as it were, on an equal footing (*quasi ex aequo*). "For the assent is caused, not by cogitation, but by the will. In this way the intellect is not determined to one [side of a contradiction] as it is when it is led to its proper term, which is the vision of something intelligible; that is why its movement is not at rest, but cogitation and inquiry remain concerning those things which are believed, though the believer most firmly assents to them." (*Q.D. de ver.*, 14, 1, c.) The mind of the believer is thus portrayed as restless since it has given its assent under the influence of will and not because of the evidence of what is assented to. The mind of the believer has been rendered captive, its assent prompted by something extrinsic to intellection as such. "Thence too it is that in the believer there can arise an impulse toward the contrary of what is most firmly held, something that does not happen in intuiting and knowing." (*Ibid.*) Having noted

that certitude involves both firmness of adherence and evidence, Thomas can speak of the certitude of faith in various ways. With respect to firmness of adherence, "faith is more certain than any intuition or knowledge, because the First Truth, which causes the assent of faith, is a stronger cause than the light of reason which causes the assent of intuition or knowledge. It [certitude] also implies evidence concerning that to which assent is given; in this sense, faith has no certitude, though intuition and knowledge do." (*Ibid.*, ad 7)

To know what Thomas meant by believing and knowing is to possess the prerequisites for understanding his distinction between theology and philosophy. Philosophy aims at knowledge which is discourse terminating in an assent prompted by the evidence of what the mind is attending to. The starting points of such discourse are truths knowable by everyone. This is why, when Thomas compares teaching and discovery, he will insist that teaching must imitate the route we would go if we were finding out for ourselves, at least in the sense that it must start from what may be presumed to be already known. What the teacher proposes must be shown to follow from what is already known by the pupil, and it is from that connection that it derives its force and commands assent -- not from the authority of the teacher. "If someone should propose to another something unconnected with self-evident principles or whose connection with such principles is not shown, he does not cause knowledge in him, but perhaps opinion or trust." (*De ver.*, 11, 1, c.) The point is that each man has in principle the capacity for knowledge; there are certain truths that no one could fail to know. Such truths are the object of what we have been translating by intuition (*intellectus*). What is known is connected with such self-evident truths, which are premises or guidelines for reasoning. If another presents to us a statement that A is B, he must give us some grounds for assenting to the connection between the terms when this is not self-evident. What mediates between A and B in knowledge is not someone's say-so, but the evidence of what is being talked about. Thus, knowledge and, consequently, philosophy are portrayed as what any man in principle can come to have owing to his natural powers.

To believe through revelation that A is B is precisely to accept the connection on someone's say-so, namely, God's, and the motive force is the will drawn by the promise that such assent will lead to man's saving good. The will is a cause here under the influence of grace, a special intrusion of God's causality. To believe is not to be a theologian in Thomas' understanding of theology, although one cannot be a theologian unless he believes.

What does theology add to belief? Theology is the science of Sacred Scripture, that is, it is a discourse bearing on the truths revealed by God in Scripture. Revealed truth, the articles of faith, are the principles of theology, and in discussing the theologian's attitude toward them Thomas will invoke the practice within certain philosophical disciplines of accepting the principles and attempting to prove things other than the principles of the discipline. Metaphysics, unlike the other sciences, disputes with those who would deny its principles. Theology, like philosophical sciences in general, accepts its principles (the philosophical sciences accept theirs because of their evidence; theology accepts its principles on faith), and, like the metaphysician, the theologian disputes with those who would deny the principles of theology, that is, who would deny revealed truth. (*ST*, Ia, 1, 8, c.) The mark of theology, as Thomas conceives of it, is the use that it makes of what men naturally know, that is, of philosophy, in its discourse about what God has revealed. We remember Thomas saying that while faith involves the firmest assent, cogitation, a kind of discursive wavering with respect to what is assented to, remains. Theology may be regarded as addressing itself to this cogitation or, perhaps better, as being an instance of it insofar as the theologian tries to bring into relation with one another what is believed and what is known. Predictably, Thomas shows concern with the procedure or methodology of the theologian, asking whether it is licit to employ philosophical reasoning in theology. (*De trin.*, q.2,a.3) Against this practice he arrays a barrage of quotations from Scripture and the Fathers, and fashions arguments against it by appealing to the methodological rules of the philosophers. Having done this, he shows that St. Paul himself used philosophical doctrines in his Epistles, cites the practice of the Fathers, and so on. With that dialectical background he develops his own position.

He begins by remarking that grace does not destroy nature but rather perfects it and that, thus, the light of faith, which is infused in us by grace, should not be thought to destroy the light of natural reason, which is also God-given. Of course, natural reason is inadequate with respect to the object of faith; nonetheless, it is impossible for the truths God has revealed to conflict with those known by the reason God has given us. Rather, naturally known truths are similitudes of a sort to what has been revealed. Thomas makes the point stronger. Natural truths are preambles to revealed truths. This is an extremely important teaching of his. We saw earlier that he holds that some things which in principle can be known by man have been revealed by God. Now, if some of the things God has revealed can be known in the strong sense of known, that is, by natural reason, this is a sign that other revealed truths, which are beyond our understanding, are also intelligible in themselves. There is thus a kind of bridge

between natural knowledge and what can only be believed; it is this bridge that is meant by the phrase *praeambula fidei*.

Thomas distinguishes three ways in which philosophy can be used in theology. It can be used for demonstrating those things which are preambles to faith, that is, to prove by natural reason that God exists, that he is one, and other things concerning God and man which can be proved in philosophy and which faith implicitly or explicitly holds. Second, it can be used to make known what is believed by appealing to philosophical doctrines, as Augustine finds many similitudes to the Trinity in philosophical doctrines. Third, philosophy is useful to the theologian to resist what is said against the faith by showing the attacks to be false or at least inconclusive. The assumption here is that anything contrary to faith is false and that since it is false, it can be shown to be such on philosophical grounds; if it is only probable, that too can be manifested by philosophical reasoning.

There are, of course, dangers involved in the theologian's use of philosophy. Thomas mentions two of them. The theologian might employ philosophical teachings which are contrary to faith, which are corruptions of natural reasoning since they are false. Thomas mentions Origen as guilty of this. Second, he might try to submit revealed truth to natural reason as to an absolute measure. For example, he might want to believe only what can be proved by philosophical reasoning. The order should be the reverse, Thomas says. *Philosophia sit ad metas fidei redigenda* (philosophy should be submitted to the measure of faith). (*De trin.*, q. 2, a. 3, c.)

Thomas sees no problem whatsoever with respect to the use of dialectic in reasoning about revealed truths. Except in the case of the *praeambula fidei*, truths which have been revealed but can be known by natural reason, there is no possibility of proving revealed truths. Insofar as dialectic is construed as simply a method of reasoning, the theologian can use it to prove, for example, that, given two revealed truths, a further truth can be derived from them. It is in this, as it happens, that he sees the possibility of an explication in time of the content of faith. (*ST*, IIaIIae, q. 1, a. 7) But beyond the employment of the method of philosophy in reasoning about the contents of faith the theologian, according to Thomas' conception of him, will often develop philosophical points in order to cast some poor light on the mysteries of faith. Thus, the notions of person, nature, relation, and so forth are clarified to a remarkable degree in the course of the theologian's deliberation on the mysteries of the Trinity and the Incarnation. There is a striking amount of such philosophical clarification in the

Summa theologiae, as, for further example, in the treatise on man and in the moral parts.

Because of this, interpreters ask if Thomas is then to be thought of as doing philosophy or theology. It is not facetious to reply that he is doing both. His overall purpose in a theological work is of course theological, that is, to achieve such understanding as is possible concerning the objects of faith. But his very conception of how to do this indicates that much philosophy must be brought into play. With respect to philosophical arguments and clarifications in theological works of Aquinas it must be said that insofar as they are philosophical, their worth and acceptance does not depend upon the acceptance of faith. That is why Thomists, in the course of philosophical writing, will often refer to passages which occur in the theological writings of St. Thomas. When they do this, they are not asking their philosophical reader to accept the overriding assumptions of theology, that is, the truths revealed by God. The passages referred to are found in a theological context, but if they are philosophical, acceptance of them demands appeal only to principles knowable by every man on a natural basis. Thus, in presenting a sketch of the philosophy of Thomas, as we are attempting to do here, we can legitimately cite passages from theological works precisely because these theological works are filled with philosophical passages.

A caution must be made nonetheless. Often in a theological work of Aquinas we can read for pages without encountering significant appeals to truths accepted on faith. Nevertheless, any such section, being a section of a theological work, draws its order and plan from the work of which it is a part. The order and procedure of the *Summa theologiae* is not, in its main lines, philosophical but precisely theological. Reasons can be adduced from Thomas to show that the order of the *Summa* is not and could not be the order of philosophical reasoning. Let one reason suffice. The *Summa* takes up at the very outset the existence of God, but this is a question which philosophy can fruitfully ask only in its culminating part. In the case of Thomas, however, we encounter far less trouble than we do in the case of a theologian like Bonaventure in determining what is the philosophical whole into which the philosophical passages which occur in theological works fit. Thomas provides us with a highly developed notion of what the term "philosophy" covers and clues as to what part of philosophy a random consideration would belong. Let us turn now to the notion of the nature and parts of philosophy.

C. The Division of Philosophy

Thomas' conception of what philosophy is and of how it is divided is basically Aristotelian, but he strives to incorporate into this conception the other major traditions. Thus, he will on occasion make use of the Stoic division of philosophy, which was used by Augustine, into natural science, ethics and logic. "Because reason's consideration is perfected by habit, there are diverse sciences following on the diverse orders which reason properly considers. For it pertains to natural philosophy to consider the order of things which human reason considers but does not cause, insofar as metaphysics is included under natural philosophy. The order that reason introduces into its own act of consideration pertains to rational philosophy, which considers the order of the parts of discourse to one another and the order of principles to one another and to conclusions. The order of voluntary actions pertains to the consideration of moral philosophy." (*In I Ethic.*, lect. 1, n. 2) Further, he will incorporate into his conception of philosophy the tradition of the liberal arts. This can be seen if we first consider his notion, developed from hints in Aristotle, of the proper order of learning the philosophical sciences. For the term "philosophy" covers a variety of disciplines.

When he comments on Aristotle's discussion of wisdom at the outset of the *Metaphysics*, he, like Aristotle, is impressed by the etymology of the term "philosophy": the love of or quest for wisdom. Wisdom is the knowledge of all things in their ultimate causes, and philosophy, accordingly, is seen as a drive toward knowledge of the highest and best reality, that is, knowledge of the divine. Any intellectual inquiry is philosophical to the degree that it is necessary for or useful to acquiring knowledge of God. Hence, the various philosophical disciplines are ranged with respect to the culminating consideration of philosophy, and this is part of what underlies Thomas' notion of the order of learning. "Thus it is that the chief intention of the philosophers was that they might come to knowledge of the first causes by means of everything they considered in reality, and thus they placed the science of first causes last and assigned the consideration of it to the final period of life. Beginning with logic, which treats the mode of sciences, they proceeded to mathematics, something which even the young can grasp; they then went on to natural philosophy, which, because of experience, requires time; fourth came moral philosophy, since youth are not good students of it. Finally they came to divine science, which considers the first causes of beings." (*In librum de causis, proemium*) The tradition of the liberal arts had divided the seven arts into two groups. The arts of the trivium (grammar, rhetoric, and dialectic) Thomas reduces in the above

list to logic; the arts of the quadrivium (arithmetic, geometry, astronomy, and music) he reduces to mathematics, and thus he absorbs into his Aristotelian conception the arts preparatory to wisdom. (*De trin.*, q. 5, a. 1, ad 3) He actually cites Hugh of St. Victor in this regard, but in doing so he alters the view of Hugh himself, since the wisdom Hugh had in mind as the goal of these arts as ways (*viae*) was not metaphysics in the Aristotelian sense.

In what way are the various disciplines which are ranged in the order of learning distinguished from one another? Thomas accepts from Aristotle a first division of philosophy into speculative and practical. "It must be said that the theoretical or speculative intellect is properly distinguished from the operative or practical in that the speculative has for its end the truth which it considers, the practical orders the truth considered to operation as to its end." (*De trin.*, q. 5, a. 1)

Thomas does not mean that we have two intellects; he is drawing attention to the two uses we make of our mind. (*ST*, Ia, q. 79, a. 11) Thomas introduces three criteria which must be taken into account when we speak of knowledge as speculative or practical, one of which, the end of the knowledge, has already been mentioned. The other two are the object and the method of knowing.

In the speculative use of our mind we have in view no end beyond the perfection of the act of knowing itself, and perfection is truth, to be in conformity with the way things are. When our thinking is aimed at the perfection of an activity other than thinking, say the perfection of choice, then it is called practical. We can see that these different uses of our mind can be dictated by the nature of its objects. Thomas will point out that there are some things that we cannot do or make, and thus our only cognitive attitude toward them is speculative. His examples are God and natural objects. Insofar as our purpose in knowing is the perfection of the act of knowing or some other activity, our method of knowing the object will differ. Thomas' point here can perhaps be captured by saying that the method of practical knowing is expressed in something resembling recipes. That is, if you want to build a house, first do this, then that and that, and, voila, there is your house. Speculative knowledge of an object does not reduce it to the steps whereby we might bring it into existence, but proceeds by a resolution into its defining principles. Much more could be said of all this, of course. We might point out that insofar as there are various criteria of speculative and practical knowledge, knowledge can be to a greater or lesser degree speculative or practical insofar as it saves one, two, or all of the criteria of the one kind of knowledge. This is a point to which we will return when we consider Thomas on moral philosophy.

Division of Speculative Philosophy. Thomas calls the objects of practical and speculative philosophy, respectively, the operable and the speculable. It is by considering the notes of the latter that he finds grounds for distinguishing various speculative sciences. There are two proper characteristics of the speculable object, Thomas argues, and these are drawn from the nature of the intellect and the demands of science, which is the quality of intellect as it bears on the speculable. The intellect, Thomas says, and we will look later at his reasons for this assertion, is an immaterial faculty; consequently, if anything is to be an object of intellect, it must be in some way immaterial. Science, knowledge in the strong sense, bears on what is necessary. To know that the sum of the internal angles of a plane triangle is equal to two right angles is to know what cannot be otherwise and is thus necessary. But what cannot be otherwise is immobile or unchangeable. All this is shorthand for matters which are not self-evident and are not taken to be such by Thomas. Nevertheless, on these assumptions he is able to conclude that immateriality and unchangeability are essential characteristics of the speculable, the object of speculative philosophy. "Therefore, it is according to the order of removal from matter and motion that the speculative sciences are distinguished." (*De trin.*, q. 5, a. 1) This removal from matter and motion is first indiscriminately described by Thomas as a separation or abstraction, terms which later acquire meanings owing to which they are opposed.

If separation or abstraction from matter and motion is essential to the objects of speculative thinking, insofar as there are different types or degrees of such abstraction we will have formal differences among speculative sciences, since the difference will be read in terms of what is essential to the speculable as such. We will seek this difference in definitions, for reasons which become clearer when we consider Thomas' doctrine on the paradigm of scientific reasoning. Now, we do find different modes of defining with respect to removal from matter and motion. "There are some speculables which need matter in order to be, since they cannot exist except in matter, but these are further distinguished, since some depend on matter both to be and to be understood, like those in whose definitions sensible matter is put and which are thus unintelligible without sensible matter; for example, it is necessary to put flesh and bones in the definition of man. Physics, or natural science, is concerned with things of this kind. Others indeed depend on matter in order to be but not to be understood since sensible matter is not put in their definitions, for instance, line and number; mathematics is concerned with these. Further, there are speculables which do not depend on matter in order to be because they can be without matter either because they never are in matter, like God and angels, or

in some cases are in matter and in others not, like substance, quality, being, potency, act, one and many, and the like, with all of which theology is concerned, that is, divine science, the chief object of which is God, a science also called metaphysics, that is, beyond physics, because for us, who must proceed from sensible things to that which is not, it is studied after physics." (*De trin.*, q. 5, a. 1)

This is a very difficult doctrine, but perhaps something can be seen of the precision with which Thomas handles what might seem to be acceptable merely as a *de facto* division of intellectual labor into natural science, mathematics, and metaphysics. If he is to admit that there are different speculative sciences, he must seek the difference in what is essential to the object of speculative philosophy. By citing the essential characters of the speculable, Thomas is able to give a statement of Aristotle's division of speculative philosophy which is a good deal clearer than that of Aristotle.

Division of Practical Philosophy. The principle of the division of moral philosophy is drawn from the fact that practical thinking is concerned with the perfection of an activity other than thinking. Thomas, in commenting on the discussion of prudence in book six of Aristotle's *Nicomachean Ethics*, makes a distinction, called for by the text, between political and ethical prudence (practical wisdom). (*In VI Ethic.*, lect. 7, n. 1196) These are substantially, that is, generically, the same, he argues, in that both involve right reason with respect to what ought to be done concerning human goods and evils, but they differ specifically. What we are calling ethical prudence is the concern of a man with his own good, whereas political prudence is a concern with the goods and evils of the whole civic community. Besides these two kinds of prudence there is economic prudence, which is concerned with the good of more than one and of less than the whole civic community, that is, the good of the family. These three are intellectual virtues and, as such, presuppose a right disposition of the will, but the distinction enables Thomas to proceed to distinctions of practical, or moral, philosophy. "It should be noticed that, as has been pointed out, prudence is not of reason alone, but depends on appetite. What we have been speaking of here are species of prudence insofar as they do not consist of knowledge alone, but depend on an appetitive condition. Insofar as they are in reason alone, they are called practical sciences, namely, ethics, economics, and politics." (*Ibid.*, n. 1200)

To make this less obscure, let us recall what was said earlier about the various criteria of speculative and practical thinking. The first and minimal criterion of

practical thinking is that the mind be concerned with something we can do or make. Moral philosophy for Thomas is concerned with man's rational choices. The task of moral philosophy, he writes, "is to consider human operations insofar as they are related to one another and to the end." (*In I Ethic.*, lect. 1, n. 2) But a human action may be considered in various ways; it may be approached in much the same way as we think of objects whose existence is not dependent on any choice of ours. Thomas' distinction of types of prudence or practical wisdom is an example; he is dividing a genus into species. A more practical method would be to know how to perform, what must be done, in order to achieve a given practical goal. That is why "normative discourse" rather than distinctions, definitions, and so forth, would be thought of as particularly ethical. Thomas would say that to know an operable thing as to how it can be done is a knowledge more practical than that whereby we know an operable in the same way we know speculable objects. What happens to knowledge when the third criterion of practical knowing is saved, namely, intention? Completely practical knowing, for Thomas, is exemplified in acting, and since actions are singular, we must say that completely practical knowledge is singular. Therefore, what Thomas in commenting on the *Ethics* gave as the distinction between types of prudence and types of practical philosophy, namely, presence or absence of a certain appetitive condition, can now be clarified by saying that practical science is at a level of generality in a way in which prudence is not. Practical sciences (the division of moral philosophy) are general judgments of man's good and of the way it can be attained, and insofar as the good is the good of the individual, of the family, or of the whole civic community, the judgments differ accordingly and so too do the sciences.

Like Aristotle, who is his mentor here, Thomas grants philosophy a charter so broad that it includes every natural intellectual pursuit insofar as it is necessary for or conducive to the attainment of knowledge of God. The point of moral philosophy is not cognitive perfection as such, but virtuous action. But moral virtue is conceived by Thomas as dispositional with respect to our seeking the perfection of our mind as such. That is why moral philosophy is philosophical.

Let us go on to give brief sketches of basic areas of Thomas' philosophical doctrine.

D. Logic

What did Thomas conceive logic to be, and with what is it concerned? In the preceding section we quoted a passage in which Thomas made use of the

division of philosophy into natural science, logic, and ethics. That same passage contains a very brief statement of the object of logic, namely, the order introduced into reason's very act of considering objects. We must now try to understand that remark, and we begin by citing another. "As Aristotle says at the beginning of the *Metaphysics*, the human race lives by art and reason, a remark in which the Philosopher seems to hit on something proper to man whereby he differs from the other animals. For the other animals seem to be led by instinct in their actions, whereas man is directed in his by the judgment of reason. Thus it is that the various arts serve to perfect human acts so that they take place easily and in an orderly fashion, for art seems to be nothing else than a determinate ordination of reason whereby human acts arrive at their appropriate ends by determined means. But reason can not only direct the acts of inferior parts; it is even directive of its own act. It is proper to the intellective part that it reflects on itself, for intellect understands itself, and similarly reason can reason about its own activity. Now, if, as a result of reasoning about manual activity, the building art is discovered, an art which enables man to perform acts of a certain kind easily and in an orderly fashion, by the same token an art seems necessary which is directive of the act of reason itself, through which art man might proceed in reasoning in an orderly fashion, easily and without error. This art is logic, or rational science." (In *I Post. Analytic.*, *proemium*, n. 1)

The assumptions of this passage are several. First, reasoning is taken to have a goal, namely, truth; and, second, it is not so determined to that goal that the possibility of error is excluded. Reflection on the reasoning process will permit us to devise an art which will direct reasoning more surely to its goal. Well, we might say, if an art, then artifacts. What are the products of this art? Notice that there is a necessary duality implied by the notion of reason reflecting on its own act, reasoning on reasoning -- for in the first, or basic, type of reasoning we are presumably concerned with known objects other than reasoning itself. The things we first understand, what reasoning first intends, leads to talk of *prima intellecta* or *primae intentiones*. On the assumption that unless we are thinking of or reasoning about something there would be no activity to reflect on, what is involved in reflection comes to be called *secunda intellecta* or *secundae intentiones*.

In the passage from the *Commentary on the Ethics* Thomas spoke of this reflective reasoning as constituting an order. What is the logical order? Thus far we have a few clues. Logic is not reasoning about just anything, but reflective reason, reasoning about reasoning. This makes logic sound like an introspective psychology, and logic is not psychology for Thomas. For one thing, Thomas will

make use of a distinction between real being (*ens reale*) and rational being (*ens rationis*), and psychological activities are instances of real being.

Perhaps the best way to achieve clarity here is to compare a list of sentences: (1) man is rational, (2) man is a species, (3) man is white. The subject of each sentence is the same, so clearly it is the predicates that interest us. Consider first the difference between the predicate of (1) and the predicate of (3). In a word, the predicate of (3) is said to be accidental, because it is not predicated of everything of which man is predicated, and even when it can be truly predicated of a man, it does not tell us *what* he is, or something of what he is, as does rational. Let us call rational an essential predicate -- it expresses the very nature of that of which it is predicated. To get at the difference between (1) and (2), consider the following discourse. Man is rational, and Socrates is a man. We feel no hesitation in formulating a further sentence: Socrates is rational. But if we should say, "Man is a species and Socrates is a man," we would hesitate to go on to say, "Socrates is a species." "Man is a species," we would want to say, is a lot more like "Man is a noun" than it is like "Man is rational." Take "Man is a noun." This tells us something about man, not in terms of what it might stand for in the world, but in terms of grammatical relations. What are we saying of man when we say, "Man is a noun" or "Man is a species"? Well, again, we are not attributing something to human nature, mentioning it in terms of an intrinsic component, as when we say that man is rational. Are we then predicating something accidental of it? Surely it is accidental to human nature that the linguistic expression for it is in grammar a noun. But if it is accidental to what "man" signifies that "man" should be a noun, just as it is accidental to human nature that the English word "man" is a three-letter word and its Latin equivalent a fourletter word, such accidental predicates are not like the predicate of (3) above. "Man is white" involves an accidental predicate, but the sentence is true because it happens that in the real, extramental world some things that are men are white. That "man is a noun depends on the intrusion of man into the world and results from a characteristic activity of his, the formation of grammars. With this as background, let us approach (2) above. If "noun" is a grammatical term, "species" is a logical term. Other examples of logical terms are "predicate," "syllogism," "proposition," "middle term," and so forth. But let us stay with "species." What does "man is a species" tell us; what does "species" mean? For Thomas "species," like other logical words, signifies a relation a nature takes on as known by us. Something is a species if it is predicable of many numerically distinct things, as "man" is predicable of Socrates and Plato and so forth. We are back at the problem of universals, since species is a type of universal, one of Porphyry's predicables. To be predicable

pertains to a nature like human nature accidentally; it is not an accident of a nature because of its presence in individuals as is the case with whiteness and man.

The logical order, as Thomas sees it, can now be defined as the relations which obtain among things as they are known and named by us. Furthermore, the logical order is intimately tied to our abstractive way of knowing. Like Aristotle before him, Thomas is struck by the fragmentation reality undergoes in our knowing process, a fragmentation which calls for the kind of ordering and binding together provided by logical relations. For example, we might first know of something simply that it is something-there, a being, then that it is a substance, then that it is living, then that it has senses, and finally that it is rational. If we stop there and collapse these steps, we would have the meaning of the term "man." But we can also label the preceding steps: "substance," "living substance," "animal"; and we might call them genera, as in the sentences "substance is a genus" or "animal is a genus." Now, what is expressed by "substance," what it means, is such that it can be predicated of objects in the world (the same would be true of "animal"), but to be a genus, in the sense of being predicable of many specifically different things, is true of the nature only as it exists in the mind. Furthermore, it is only in the mind that substance exists apart from further determinations like "living" and "nonliving." Thus, in this case certainly it is our abstractive mode of knowing, the fact that we move through progressively less vague "fixes" on things to determinate knowledge, that is productive of the "things" related by logical relations. This is something we must keep in mind when we consider the question of whether the categories (literally, predicates) of being are logical or real, but that is another consideration.

The objects of logic are the relations which accrue to things as they are known by us, relations which are accidental to the nature known. The divisions of logic, for Thomas, are precisely what he takes to be Aristotle's divisions. The bases for the division are the various acts of reason, since these acts are what logic is said to order and direct. Thomas speaks first of an understanding of incomplex things which expresses itself in definitions. Obviously, if something can be defined, it cannot be wholly incomplex or simple. Let us then start with rational discourse, with the syllogism. If C is predicated of everything of which B is predicated, then C is predicated of everything of which A is predicated. Such discourse can be seen as composed of such symbolically expressed propositions as "Every B is C" and "Every A is C." Apart from and prior to considering the relations among propositions in such discourse, we can consider the relations

involved in affirming or denying one thing of another just as such. But the things which enter into affirmations and denials must first be known as to what they are, that is, must be defined. Thus, the parts of logic are, in a sense, the parts of rational discourse. "There are three acts of reason of which the first two belong to reason insofar as it is intellect [*intellectus*]. For one act of intellect is the understanding of indivisible or incomplex things, insofar as it conceives what a thing is. To this operation of reason is ordered the doctrine Aristotle treats in the *Categories*. The second operation of intellect is composition or division, where the true or false first obtains. The doctrine Aristotle treats in *On Interpretation* serves this act of reason. The third act of reason is one proper to reason as such, namely, discourse from one thing to another so that from what is known knowledge is gained of what was unknown. The rest of the books of logic serve this act." (In *I Post. Analytic.*, *proemium*, n. 4)

The act of reasoning is sometimes necessary, sometimes probable, and sometimes fallacious. The first, necessary reasoning, is scientific and is called by Thomas judicative logic "because judgment is had with the certitude of science." Judgment is said to have its certitude owing to a resolution or analysis into principles, so this part of logic is also called analysis or analytics. "The certitude of judgment, however, which comes from resolution is either from the form of the syllogism alone, and the *Prior Analytics* is ordered to this, or is also from the matter, because it involves propositions which are selfevident and necessary, and with this the *Posterior Analytics* is concerned, which deals with the demonstrative syllogism." (*Ibid.*) The second part of the logic of reasoning is called inventive logic or the logic of discovery. Now what is discovered must be judged, and the term may be science if the judgment is with certitude; if not, if the resultant knowledge is only probable, then it is the concern of what Aristotle calls dialectics and is dealt with in the *Topics*. One can see Thomas the commentator at work here, incidentally; he is surely trying to say something accurate about the contents of and relations between the works of Aristotle's *Organon*, but what comes first, and in his own name, are the statements about the subject matter. He goes on to link the *Rhetoric* and *Sophistical Refutations*, and even the *Poetics*, with the logic of discourse. One is tempted to associate this treatment of the *Poetics* with Thomas' reduction of the arts of the trivium to logic. The only completed commentary on a logical work of Aristotle is that on the *Posterior Analytics*, but Thomas began one on *On Interpretation* and carried it forward a good distance. Among his collected writings are a number of opuscula whose authenticity has been questioned, one on modal propositions, another on demonstration, one on the square of opposition, and another on fallacies. In the commentaries on Aristotle's logical works Thomas exhibits his

usual acuity, and it is easy to wish he had commented on the whole *Organon*. But with logic, as with many subjects, we find illuminating remarks scattered throughout Thomas' works. A notable example is his teaching, which must be pieced together from many sources, on systematic ambiguity, or analogy.

Logic has, of course, come a long way since Thomas, come so far indeed that it is questionable whether the term logic must not be taken to be ambiguous as applied to what Thomas meant by it and to what logicians do today. Historians of logic, writing from the standpoint of twentieth-century logic, are seldom detained by Thomas' contributions to the subject, but quite often such historians are indifferent to what logic may have meant in earlier times and seek only foreshadows of what logic has come to mean today. This procedure, while it achieves results of value, is finally perhaps as historically suspect as a "Thomistic" critique of the *Principia Mathematica* would be. To see how contemporary thought has "gone wrong" from a thirteenth-century vantage point is as dull as seeing how medieval logic "fell short" from a twentieth-century vantage point. We have yet to see a comparison of "logics" which does justice to historical periods taken on their own terms. When such a history of logic is written, Thomas may well occupy a prominent place in it, not as an innovator, but as a lucid exponent of the view that logic is not concerned with the most abstract language, with symbols or variables whose values are the things of this world (or nothing), but with rational relations accruing to things as the result of our knowing them. That is, that logic is incorrigibly human, all too human, and that its purpose is the quite human one of assuring the correctness of discursive reasoning about objects other than logical entities.

E. Natural Philosophy

One of the most shocking things about the Thomist in the eyes of his colleagues must surely be the attention he pays to what Thomas had to say about our knowledge of the natural world. The source of the shock is not simply that Thomas lived in the thirteenth century, which is prehistoric enough as far as natural science goes, but more basically still that in natural philosophy Thomas is spiritually in the same place as Aristotle in the fourth century B.C. It would be quite easy to list tenets of Thomas in natural science which could seem quaint at best and weird at worst; even if we should think that such is the stuff of which history is made, at least the history of science, we might question the advisability of devoting time to it in a volume as restricted in length as this one. Much better, it might be thought, to pass over in generous silence this part of the philosophy of Thomas and push on into his metaphysics. But that is

precisely the problem. For Thomas there is little point in pushing on into metaphysics unless we have gained some purchase on the physical; in a word, if his physics is totally undermined, his metaphysics is *a fortiori* undermined. This is why Thomists pay so much attention to such writings of Thomas as his commentaries on Aristotle's natural works and his own *On the Principles of Nature*. It is true that these writings are so different from what we nowadays call natural science that the tendency often is to call the doctrines contained in them metaphysical. Nevertheless, for Thomas they amount to natural science, and that is how we shall consider them. To put what we shall try to say in a proper perspective, consider the following question. Is knowledge of the natural world possible, knowledge which in a significant sense is scientific, which does not employ the methodology of current natural science? If such knowledge is possible, there is no reason why it could not have been had prior to the development of scientific methodology in more recent senses of the phrase; it could have been had in the thirteenth century, even in the fourth century B.C., and need not be thought of as a competitor with or substitute for what we now call natural science.

In this section we shall consider three topics, the first of which is Thomas' statement of the hylomorphic composition of natural, or physical, things. The next two topics bear on what Thomas takes to be the presuppositions of metaphysics, the proof of the separability or immortality of the human soul, and the proof for the existence of an unmoved mover. These proofs are the reasoned ground for the conviction that "physical being" and "real being" are not synonymous and that, consequently, the science of being as being is distinct from natural science.

Hylomorphism. In *On the Principles of Nature* Thomas sets down this doctrine in a swift, staccato way. Some things can be, some things already are; the first are said to be in potency, the second in act. There are two kinds of actual being, however, substantial and accidental; it is one thing to be a man and another thing for a man to be white, and something can be in potency to either kind of being; what is in potency can in either case be called matter, though that which is in potency to substantial being might be called the matter out of which (*ex qua*) something comes actually to be, whereas that which is in potency to accidental being might be called the matter in which (*in qua*) something comes to be. "Again, properly speaking, that which is in potency to substantial being is called prime matter, but that which is in potency to accidental being is called the subject, for the subject gives being or existence to the accident, since the accident has no being save in its subject; hence, it is said that accidents are in a

subject, but substantial form is not said to be in a subject. Matter differs from the subject in this: that the subject is not something which exists because something advenes to it; rather it is autonomously (*per se*) and has complete being; for example, a man does not come to be (a man) thanks to whiteness. Matter, on the other hand, has being from that which advenes to it, since of itself it is incomplete, indeed has no being. . . . Hence, absolutely speaking, form gives being to matter, but the accident does not give being to the subject, but the subject to the accident (*De princ. nat.*, chap. 1) Having introduced two kinds of composition, that of prime matter and substantial form and that of subject and accident, Thomas goes on to speak of the coming into being of these two kinds of composites as, respectively, substantial and accidental becoming.

Accidental becoming is exemplified by a man's becoming pale. The acquisition of this quality does not make a man be a man, but a pale man. Man is the subject of the change, and prior to the acquisition must have been capable of possessing the quality, in potency to it, though at the time not in possession of it and thus deprived of it. A man moves from not being pale to being pale. Despite the restrictiveness of his earlier definitions, Thomas allows that the subject can be called matter. "Therefore, there are three principles of nature, namely, matter, form, and privation, of which one, form, is that for the sake of which the generation takes place. The other two are that from which the generation takes place. Hence, matter and privation are the same in subject but are different in conception (*ratione*), for bronze and unshaped are the same thing before the advent of form, but from one point of view the thing is called bronze and from another unshaped. Hence privation is said to be a principle not *per se* but *per accidens*, because it resides in the matter (*Ibid.*, chap. 2)

Substantial becoming, the coming to be, not of pale man, but of man, is similarly analyzed. If "man" names something one and autonomous, something substantial, the form that makes a man to be is not like the quality which presupposes a substantial subject. Humanity is not something that advenes to an already existing thing to make an accidental compound like white man. The subject of a substantial change is called prime matter precisely to distinguish it from the subject of an accidental change; prime matter is not a substance as is the subject of an accidental change. For much the same reason the form involved in the substantial change is called substantial form: the being it constitutes by advening to prime matter is a substance. Whatever comes to be as the result of a change is a compound of matter and form. This is what the term "hylomorphism" conveys, of course, fashioned as it is from the Greek terms for matter and form. Matter and form are thus two ways of accounting for a

physical thing, two causes, or principles, of its being. Besides these intrinsic causes there must be an efficient, or moving, cause which effects the composition of matter and form. Like Aristotle, Thomas also speaks of a final cause, that for the sake of which the change takes place and which in that sense terminates it. The form or the product of the change is the final cause of the change, but the final cause of the change is not of course the final cause or goal, that for the sake of which the product of the change exists. That is, a man may be the final cause of a substantial generation, but man's goal or final cause is not simply substantial existence.

The Unmoved Mover. We will present this proof in the statement Thomas gives it in his *Summa contra gentiles*, book one, chapter thirteen, a statement which is fuller than that found in the *Summa theologiae* and closer to the proof as it is developed by Aristotle in his *Physics*.

"Everything that is moved is moved by another." The fact of motion is evident to the senses, and the nature of motion demands that what is in motion is moved by another. But the mover is either itself moved or it is not. If the latter, then we have an unmoved mover (which can be taken to be a description of God); if the former, either what moves the mover is moved by another or it is not. Now either we must posit an infinite series of moved movers or we arrive at an unmoved mover. But an infinite series of moved movers is impossible, so there must be an unmoved mover.

As Thomas points out, there are two assumptions here that must be proved, and they are precisely the premises of the proof: "Every moved thing is moved by another" and "An infinite series of movers and things moved is impossible." He selects from Aristotle several proofs of the first premise.

First, if something moves itself, it must have within itself the principle of its motion, for otherwise it would manifestly be moved by another. Further, it must be what is first moved, not some part of it, as an animal being moved by its feet, for then it is moved not by itself but by its part, and, indeed, one part by another. And it is necessary that it be divisible and have parts, since whatever is moved is divisible, as is proved in the *Physics*, book six, chapter four. Given all this, the following argument can be devised. "That which is posited as moved by itself is itself first moved [*primo motum*], and thus the repose of one of its parts entails the repose of the whole. For if when one part comes to rest, another part should remain in motion, then the whole itself is not what is principally moved [*primo motum*], but its part which is moved while other parts are at rest. Nothing

which comes to rest when another thing comes to rest is moved by itself, for whose repose follows on the repose of another must be such that its motion is a consequence of another's motion, and thus it is not moved by itself. Therefore, that which was posited as being moved by itself is not moved by itself.

Therefore, it is necessary that whatever is moved is moved by another." The nub of the argument, according to Thomas, is this: "If something moves itself first and as such and not by reason of parts, it is necessary that its being moved is independent of anything else; however, for a divisible thing to be moved, as for it to exist, depends upon its parts and that is why it cannot move itself first and as such [*primo et per se*]."

Another proof of the first premise is this. "Nothing is simultaneously in act and in potency in the same respect. But whatever is in motion is, just as such, in potency, since motion is 'the act of that which is in potency just as such.' But whatever moves is as such in act, since nothing acts except insofar as it is in act. Therefore, nothing can be, with respect to the same motion, mover and moved. Thus, nothing moves itself."

Thomas offers several proofs in support of the second premise of the argument which concludes to the existence of a mover which is itself unmoved by another. That there cannot be an infinite series of subordinated moved movers is a good deal more difficult to prove, and the arguments are too technical and demand too much subsidiary commentary to go into here. In examining these arguments, as in examining those brought forward in support of the first premise, one is struck by the dependence on the *Physics* of Aristotle, and when one reflects that the proof of the unmoved mover comes at the end of the eight books of that work and depends on nearly everything that has come before, it is not surprising that Thomas, in giving a résumé of the argument, must presume so much. That presupposed doctrine is the source as well as the corollaries of the proof. If whatever is moved is moved by another and if this series cannot proceed to infinity, so that there must be a mover not itself moved, the nature of this unmoved mover can be approached by denying of it characteristics of things which are moved. Suffice it to say for now that matter is a component of what is moved and that thus the unmoved mover must be immaterial. The point is that in the course of doing natural science Aristotle and Thomas following him feel they must admit the existence of something immaterial. Thus, "being" is no longer synonymous with "material being," and the need for a science beyond the natural sciences is seen.

The Immortality of the Human Soul. There is another instance within natural science where one comes to see the existence of something immaterial, this time in biology. It should be said that considering natural science generically, we can say that its subject is mobile being. The predicable scope of such a phrase is, of course, great, and Thomas accepted Aristotle's view that on this level of generality it is possible to formulate proofs which would conclude to properties commensurately universal with the subject. That is, as a first step in natural science we can arrive at some scientific knowledge of what must pertain to any mobile thing, whatever differences among kinds of mobile being must later be taken into account. The *Physics* of Aristotle is precisely an attempt at a general science of nature, and its doctrine is thought to transcend the differences between living and nonliving natural things. One would not be content with such general knowledge of the natural world, of course, and in *On the Soul* Aristotle commences the study of living being. What distinguishes the living from the nonliving is precisely the former's possession of soul. What is meant by "soul"? The soul is that owing to which we live, move, sense, and understand. This definition, which Thomas takes from Aristotle, indicates that the soul is denominated from a variety of vital operations of which we have a privileged experience in ourselves. To wish, to fear, to love, to think, to see, and so forth are activities of our own whose existence we are not likely to doubt. If we perform these activities, we must of course be capable of performing them, and the actual performance is not equatable with the capacity since sometimes we perform them and sometimes not. This is the origin of talk about potencies or faculties of the soul: we have various capacities for vital activities like seeing, hearing, wishing, knowing, and so forth. What is the relation of these capacities to the soul? Is the soul identical with them, a class term signifying them all, or distinct from them? Thomas regards the soul as distinct from these capacities or faculties and as related to them as substance to accidents. One reason he gives for this is that if the soul were identical with a capacity to perform a vital act and if there are several such capacities (and there obviously are in man), then since two things identical with a third are identical to one another, the several capacities would actually be one. But surely it would be odd to identify our capacity to see with our capacity to will or to hear or to think. One can see that the soul is something of an inferred entity and that the procedure is from activities to faculties and from faculties to their subject, the soul.

Since this analysis is considered to be part of natural science and the hylomorphic analysis of natural things occurs at the very outset of natural science at its most general level, we are not surprised to find soul spoken of as a form. It is a kind of substantial form, in other words, and the living thing is

thought of as a compound of soul and body. Hence, a further definition of soul as the first actuality of an organic body having life in potency. Now, as was clear above, substantial form and prime matter are not so much substances as they are principles or components of substance. Neither matter nor form is thought of as capable of existence apart from a compound. For Aristotle and for Thomas too the question as to the continuance in existence of the human soul after death comes down to asking if the substantial form which is the human soul survives the dissolution of the human being, this compound of body and soul.

How could this question be answered in the affirmative? First, let us point out that Thomas speaks of kinds of soul insofar as souls are denominated from the characteristic or highest activity of the living thing of which the soul is the principle in the sense of substantial form. So we find Thomas speaking of the vegetative soul, of the animal, or sensitive, soul, and of the rational soul. Man is thought to have the capacities for the vital operations found also in lower things like plants and beasts, but beyond those to have the power of reason, and his soul is denominated from his distinguishing and defining activity. The various vital operations seem to involve the body essentially, since seeing, hearing, smelling, fearing, hoping, imagining, and so forth intrinsically involve corporeal aspects. But is the same true of thinking? Here is the crux of the matter for the question of the immortality of the soul as it is discussed by Aquinas. If a living man performs an activity which does not intrinsically and essentially involve his body, we would seem to have some basis for saying that the soul which is the subject of that activity is not dependent for its continued existence on the body.

On many occasions Thomas attempts to show that the human soul is incorruptible because it is capable of a kind of knowing which reveals that it is wholly immaterial. Question 75 of the first part of the *Summa theologiae* and the *Disputed Question on the Soul* might be particularly cited. An indication of his procedure can be had from the following sketch. Thomas will use the hylomorphic model to speak of cognition. (*De ver.*, q. 2, a. 2) Just as in things it is their forms which make them actual and what they are, so to know things can be described as coming into possession of the forms of things, of what they are. Thus, Thomas will define knowing as having the form of another as other. Now to have the form of a physical thing in knowledge is a different kind of possession of that form than is exemplified in a concrete physical thing. When the form or nature of rose is united with matter in the genesis of a rose, the result is a singular rose, this one and not that one. In short, the form is individuated as received in matter. However, when we know what a rose is, when, in Thomas' terminology, the form of rose is received in the mind, the

result is not another rose but an intentional form which enables us to know the material rose. Thus, the mode of existence of the form in the mind is an immaterial one. This is the source of the claim that the human soul is in a sense all things (*anima est quodammodo omnia*) since it can know all things. A physical thing can possess but one substantial form, but the mind can know many forms. "It should be said that the principle of intellectual operation which we call man's soul is an incorporeal and subsistent principle. For it is manifest that man owing to intellect can know the natures of all bodies. That which can know other things cannot be those things in its own nature [*oportet ut nihil eorum habeat in sua natura*] because that which is in it naturally would impede knowledge of other things as we see that the tongue of someone ill which is infected by a choleric and bitter humor cannot perceive what is sweet but everything seems bitter to it. If therefore the intellectual principle had in itself the nature of some body, it could not know all bodies, for every body has some determinate nature. Therefore, it is impossible that the intellectual principle be a body." (*ST*, Ia, q. 75, a. 2) can

Once more, we face a most difficult matter and a doctrine which be assessed only when all its presuppositions are examined, but this outline may convey something of the flavor of Thomas' procedure. As we have said several times before, the upshot of these two proofs within natural science is that one sees that "being" must be predicated of things which are not material, that the science of being as being is different from the science of natural, or material, being.

F. Metaphysics

Let us begin this section by taking a fairly close look at the preface Thomas wrote to his *Commentary on the Metaphysics of Aristotle*. This preface sketches the terrain of metaphysics and suggests a number of points we can develop in order to convey the nature of Thomas metaphysical doctrine.

Thomas begins the preface by remarking that whenever many things are organized into one whole there must be something which directs and orders the many. He illustrates the principle by noting that man is one thing composed of several "parts," namely, body and soul, and that while it is the role of soul to command, it is that of body to obey. All arts and sciences, he goes on to say, are ordered to one thing, namely, to the perfection and happiness of man, but it is necessary that one of them be directive of all the others and, that science will be called wisdom because it is the role of the wise to order: *sapientis est ordinare*.

We can get some inkling of what this directive science would be and what its subject matter is by pursuing the analogy and asking what makes a man fit to rule others. Well, Thomas says, choosing between brain and brawn, would not we say that men of vigorous intellect are more fit to rule others than are men of great bodily strength but weak minds? Could not we say, then, that the science which is most intellectual is naturally fit to be regulative of others? But what would we mean by the "most intellectual science?" Thomas suggests that it would be the science concerned with the most intelligible objects and adds that "the most intelligible" can be understood in three ways.

First, that which grounds certitude of understanding is what is meant by intelligible. Since to have certitude is to know the cause of what is known, a science which is concerned with first causes can meaningfully be said to deal with the most intelligible things and to be directive of all other sciences which deal with lesser and thus less intelligible causes.

Second, the "most intelligible" can also be explicated by comparing intellection and sense perception. "For, since perception is cognition of particulars, by that very fact it seems to differ from understanding which grasps universals. Hence, that science is most intellectual which concerns itself with the most universal principles -- these are being and what follows on being like one and many, potency and act, which ought not be left wholly uninvestigated since without knowledge of these knowledge of what is proper to a given genus or species cannot be had." Very abstract notions like being, one, act and potency, do not fall to the consideration of any particular science; indeed, since knowledge of them is needed to undertake the study of any determinate type of being, one would have to say that if the study of them falls to one particular science, it falls to every particular science. Better that in all their scope and generality they be treated in one common science which is thereby most intellectual and directive of the others.

Finally, if we consider the nature of intellectual knowledge, which involves abstraction from matter, we can say that the most intelligible things are those most free of matter. What is most free of or separate from matter will not be what is free of individuating characteristics alone, as man is free from the peculiarities found in Socrates, Plato, and so on, nor what is free from all sensible matter in conception alone, like mathematical objects, but rather most free are existent immaterial things, like God and the angels. The science concerned with immaterial things seems most intellectual and, accordingly, directive of the others.

We recall that Thomas started by saying that he was looking for the one science that would be directive of all the others and that this would be the science concerned with the most intelligible objects. Since he has introduced three criteria for understanding "most intelligible," he must go on to show that it is one and the same science that is referred to no matter which criterion of "most intelligible" is used. We are talking, he says, of one science, not three. "For the aforementioned separate substances are the universal and first causes of being [*essendi*]. It belongs to the same science to consider the proper causes of a genus and the genus itself, as the natural scientist considers the principles of natural body; so it belongs to the same science to consider separate substances, universal being [*ens commune*], which is the genus of which the foregoing substances are the common and universal causes." If the science considers the three things mentioned, it does not consider all of them as its subject; the subject of the science is being (*ens commune*). In a science we seek to know the causes and properties of the subject, but the causes of the subject of a science are not the subject of the science. However, though the subject of this science is being in general, it is said to bear on what is separate from matter both in conception and in existence, since this is taken to mean not only what never exists in matter, like God, but also what is sometimes material, sometimes immaterial, like being.

Three names for the science follow from these considerations. It is called "theology" insofar as it is concerned with immaterial existents, the chief of whom is God; it is called "metaphysics" because it comes after physics, which studies a type of being, while metaphysics is concerned with being as being. Finally, it is sometimes called "First Philosophy" because it is concerned with primary realities, first causes.

Thomas has gotten a tremendous amount into this short preface, has in fact taken stands on a number of controverted and difficult questions about the nature of metaphysics and its relation to other sciences. In the sequel we want to unpack this preface a bit and speak of the relation of metaphysics to the other sciences, and of the way in which it is both a general science and a theology. This will lead to a discussion of analogy as an explanation of talk about God.

Abstraction and Separation. When we discussed the division of speculative philosophy earlier, we made use of a text to which we must now return, a text from Thomas' exposition of Boethius' work on the Trinity. In distinguishing types of speculative science Thomas appealed to the nature of the speculable, which he characterized as immaterial, and argued that insofar as speculable

objects are more or less separated from matter, the sciences which deal with them will differ formally. The various degrees of immateriality are revealed in definitions. This is important since the model of scientific knowledge is a demonstrative syllogism whose middle term is the definition of the subject of the conclusion that links it with its predicate, a property. The order of removal (*ordo remotiois*) from matter is called by Thomas, in article one of Question Five of the *Expositio*, a separation (*separatio*), and here the term covers indiscriminately the kind of freedom from matter exhibited in the objects of natural science, mathematics, and metaphysics. In article three the term "*separatio*" acquires a narrower meaning which restricts its application to metaphysical abstraction, a fact which has occasioned much discussion.

There are four articles in question five of the *Expositio*. After the distinction of the different kinds of speculative science in article one the remaining articles take up, in order, natural science, mathematics, and metaphysics. Article three, therefore, is concerned with mathematics, but recent discussion has turned on the remarks on the nature of metaphysics to be found there. The guiding question of the article is, Does mathematics consider without matter and motion things which exist in matter? Let us turn immediately to the body of the article. Aquinas begins by saying that there are two kinds of abstraction which follow on two kinds of intellectual activity: simple apprehension, and composition and division. The first is that whereby we grasp what things are and is expressed in definitions. Notice that he says these are two kinds of abstraction.

The first kind of mental activity, he continues, looks to the very nature of a thing according to which the thing understood has what rank in reality it has (*aliquem gradum in entibus obtinet*), whether it is something complete like a whole or incomplete like a part or accident. The second kind of mental activity mentioned looks to the very being of the thing (*respicit ipsum esse rei*) which in compound things results from the conjunction of its components or principles and in simple things is a concomitant of nature.

Truth consists in the mind's conformity with reality; consequently, we cannot truly abstract one thing from another by means of the second type of intellectual activity when they are united in reality. The reason for this is that abstraction would here be expressed in a negative judgment: A is not B. "By this type of activity the mind can truly abstract only those things which are separate in reality, as when we say, 'Man is not an ass.'" Throughout this discussion of

composition and division Thomas uses "abstrahere" to signify the mental act of negative judgment and "separatia" to signify otherness in reality.

The first type of mental activity, the apprehension of the nature of a thing, is relatively freer from reality, so to speak, insofar as something can be understood and defined without reference to things with which it exists. This is not total freedom, of course. The part as part cannot be understood without reference to its whole or the accident without reference to its subject or the parent without reference to children. But of two things which exist together, "if the one does not depend on the other with respect to what constitutes its nature, it can be abstracted by the mind from the other and understood without it." Some parts can thus be understood without their wholes, as letters can be considered apart from syllables, though not vice versa, and accident apart from a determinate subject, like whiteness.

Up to this point, again, Thomas uses "abstrahere," "to abstract," both for negative judgments, the denial that one thing is another or with another, and for conceiving or considering which would be expressed in a definition. He now introduces a new term, "distinguishing" ("*distinguere*"), and speaks of distinguishing one thing from another in such a way that the phrase comprehends the two kinds of abstraction mentioned. This permits him to give abstracting and separating narrower meanings according to which they are opposed to one another as types of distinguishing. "*Sic ergo intellectus distinguit unum ab altero aliter et aliter secundum diversas operationes.*" The narrow meaning of "separation" confines it to the distinguishing proper to the second type of mental activity, that expressed in a negative judgment. Here one thing is distinguished from another when it is understood not to be with the other (*quia secundum operationem qua componit et dividit, distinguit unum ab alio per hoc, quod intelligit unum alii non inesse*). Conceptualization, the understanding expressed in a definition, may be called abstracting in a narrower sense, namely, when one thing is understood, without another, though the two are together in reality (*sed tunc tantum quando ea quorum unum sine altero intelligitur sunt simul secundum rem*). In this narrow sense an animal is not considered abstractly when it is considered apart from stone, since they are not one in reality; examples of abstraction in the narrow sense would be considering a form apart from matter and considering a whole without its parts.

The thing that has interested scholars here is the fact that Thomas goes on to speak of natural science and mathematics in terms of these two kinds of abstraction in the narrow sense, applying the consideration of form without

matter to mathematics and of a whole without its parts to natural science. This would seem to leave separation in the proper, or narrow, sense to metaphysics, and the conclusion to be drawn is that metaphysical thinking involves a negative judgment, an assertion that in reality something is separate from something, is independent in existence from something else. We know from the foregoing what central negative judgment provides the charter for metaphysics. It is precisely insofar as we can judge that some being is separate or independent from matter in existence that we can say that a science of being as being is possible, a science which will be distinct from natural science, which is concerned with a particular kind of being, mobile or physical being. (And of course we know natural science is a particular science, that is, a science concerned with a particular kind of being, just insofar as we know there is another kind of being.) This science will differ from mathematics, which, though it defines its object without sensible matter, does not assert that it so exists out-there.

Being as Being. The subject of metaphysics is being as being; metaphysics inquires into what pertains to being, not insofar as it is mobile and material, but precisely insofar as it is being. It is concerned with separate being, with whatever can be considered apart from all matter and asserted to enjoy existence in separation from all matter. Now, all this sounds extremely enigmatic, particularly when we try to put together various statements of Aquinas. When he says that there is a science of being as being he is talking of wisdom, the culminating philosophical consideration, that which is appropriately placed last in the order of learning the sciences since it would be folly to expect wisdom until one had studied for a long time. Yet Thomas will also say that being is the first thing we know (*ens est quod primum cadit in intellectu*), an observation that suggests what anyone would expect, namely, that to know of something that it is, that it is a being, is to know as little of it as is possible. But does not the description of metaphysics as wisdom suggest that to know being as being is the most profound and desirable knowledge possible?

The difficulty we are trying to elaborate can be put in another way. Is metaphysics a general science of being, an ontology, or is it rather a science of a particular kind of being, immaterial being, and thus a particular science, a theology? Remember that in the preface to his *Commentary on the Metaphysics* Thomas had said that metaphysics is concerned both with things which never exist in matter (like God and the angels) as well as with things which sometimes exist in matter and sometimes do not (like being, substance, act and potency). Does this solve our problem? It would seem not, since if substance is defined

without any matter, the definition would be appropriate only to separate substances, and once more metaphysics would seem to be a special science, not a general science of all being.

As it happens, Aristotle raised just this question in book six of his *Metaphysics*. Let us consider Thomas' formulation of the problem and its solution. "Someone might wonder if First Philosophy is universal, concerned with being generally, or if it considers some determinate genus, some one nature (which does not seem to be the case) Unlike mathematical science, which deals with a determinate kind of things, "First Philosophy is universally common to all things." (*In VI Metaph.*, lect. 1, n. 1169) Metaphysics then would seem to be about everything insofar as everything has something in common with everything else, namely, being. But would not such scope entail meager and impoverished knowledge? Here is Thomas' formulation of the solution. "If there were no substance other than those which exist in nature [*secundum naturam*] with which physics is concerned, physics would be the first science. But if there is an immobile substance, this kind will be prior to natural substance, and the philosophy considering it will be First Philosophy. And because it is first, it will be universal and it will fall to it to consider being as being. . . . The science of the first being and of common being is the same." (*Ibid.*, n. 1170)

The solution seems to retain the difficulty. This science is concerned with the first being, presumably God, and therefore must be concerned with common being. Common being is not a synonym for first being here; it seems to stand for being insofar as it is predicable of all that is, both immaterial and sensible things. Earlier Thomas had written, "Notice however that although things which are separate both in definition and in existence from matter and motion pertain to the consideration of First Philosophy, not only such things do, but sensible things as well, insofar as they are beings [*in quantum sunt entia*]." (*Ibid.*, n. 1165)

Being considered universally (*ens commune*), being as being (*ens in quantum ens*) -- these signify the subject of metaphysics, and the subject of the science is that about which we want to discover attributes or properties which belong to it because of what the subject is. What does Thomas mean by "being," what is the "*ratio entis*"? He answers this question in a number of ways: what is, what has existence (*quod est, habens esse, id quod habet esse*). Such a definition does not include sensible matter, but neither does it exclude sensible matter in the sense of prescinding from it. Thus, with this meaning "being" can be predicated of Socrates or a rose, but it surely would not tell us a great deal about them. Far better to know of Socrates that he is a man, of a rose that it is a plant -- better in

the sense of more informative. If that is true, and surely it is, what kind of an advance is metaphysics supposed to be? To answer that question we must draw back a bit and ask ourselves what for Thomas is the ultimate and crowning concern of philosophy. Philosophy drives toward knowledge of the divine, toward knowledge of God, and this is preeminently the concern of metaphysics. Metaphysics is not undertaken to give us more adequate and appropriate knowledge of physical things (if there were no immaterial substance, physics would be first philosophy); it is undertaken to give us less inadequate knowledge of the divine. The formulation "being as being" therefore should not be regarded as a more profound approach to physical or sensible beings. To know sensible beings insofar as they are beings (*in quantum sunt entia*) is simply less informative than to know them as sensible (*in quantum sunt sensibilia*). The whole point of formulating definitions of "being" and "one" and "act" which do not include sensible matter is to provide us with a less inadequate language with which to talk about God. It would seem that for Aquinas it is not even the principal business of metaphysics to prove the existence of God, since for him that is one of the presuppositions of metaphysics. Rather, for Aquinas metaphysics would seem to be a prolonged reflection on what we know of sensible being, a purification of concepts formed in knowing sensible beings so that they become means of describing less inadequately the immaterial or divine. This interpretation of the metaphysics of Aquinas may be novel, but I feel it accurately reflects both what he says about metaphysics and what he does as a metaphysician. In reply to the earlier question, we can say that for Thomas metaphysics is an ontology in order to be the only kind of theology it can be. God, simple substance, cannot be the subject of a science, Thomas argues, so metaphysics cannot be a theology in the sense that God is its subject matter. Its subject matter is being as being, that is, conceptions which do not involve sensible matter and thus are inadequately informative of sensible things but which, because of this absence of matter, provide a less inadequate bridge to talk about God.

The Analogy of Being. It is a commonplace that for Thomas being is analogous, but, before discussing his teaching on this point, it must be clear to us what analogy is for him. What are we saying when we say that being is analogous? Ultimately what we are aiming at is the fact that some beings are substances and some are accidents, that some being is finite and one infinite, but while all this is what analogy is applied to in this instance, that is not what "analogy" means. In order to get at the type of word "analogy" is, we might consider another sentence, "Being is a genus." Thomas agrees with Aristotle that that sentence is false, but we have already seen the type of predicate "genus" is, we

know what it means to say that "genus" is a logical term. Well, "Being is analogous" is the affirmation Thomas offers when he decides that "Being is a genus" is false. "Analogy" must be a logical term too, and if we imagine three statements each of whose predicates is a logical term -- "being is analogous," "being is univocal," and "being is purely equivocal" -- Thomas will say that only the first is true.

As a logical term, "analogy" signifies the relations among several meanings of a given word; analogy is a kind of signification, and it is usually exemplified by "healthy." Consider the following list: (1) Fido is healthy, (2) urine is healthy, (3) food is healthy. Although the same term occurs as predicate in each of these sentences, it does not seem to have the same meaning in all of them as "man" does in "Socrates is a man" and "Plato is a man." Nor does it seem to have entirely unrelated meanings as "top" does in "he spins the top" and "he opens the top." That is, the meanings of "healthy" in our list, while different, seem related. "Healthy," to use Thomas' language, is imposed to signify from health, and we might formulate a common meaning for the various uses in (1), (2), and (3) above by saying that "healthy" means "related in some way to health" or "referring to health in some way." This would be what Thomas means by the common notion (*ratio communis*) of an analogous name, but unlike the common notion of a univocal term (the example of "man" above) it does not apply equally to the things of which it is predicated. By applying equally Thomas means that when I say Socrates is a man I make no reference to anything else called a man, something else that might be thought to have prior right to the name. The common notion of the term Thomas calls analogous is unequally common to many things in this sense, that it applies to one thing primarily and to others secondarily. That is, beyond the *ratio communis* of "healthy" (referring in some way to health), we can formulate a proper notion (*ratio propria*) which expresses a determinate reference to health, say, "subject of health," which is the principal meaning of the term and is the meaning it has in (1). In (2) it would mean "sign of health," and in (3) "preservative of health." However, if for an analogous term there is a common notion and also a number of determinate notions or meanings, these determinate meanings are fashioned in such a way that one of them is controlling or privileged, the focal meaning of "healthy." In our list the focal meaning (*ratio propria*) or primary analogate is "subject of health." Why does Thomas say this? How does he know one meaning is more basic than the others? He arrives at this by observing that in explicating the meaning of "healthy" in (2) he must make reference to the meaning it has in (1). Thus, its meaning in (2) is "the sign of health in the subject of health," and its meaning in (3) is "preservative of health in the subject of health." Its meaning in

(1) makes no reference to its meanings in (2) or (3), and we can safely conclude that the meaning "healthy" has in (1), "subject of health," is the primary and controlling meaning.

"Healthy" is an example of the analogical community of a term, just as "animal" would be an example of genus. The doctrine of analogical signification is no more tied down to its examples than is any other logical doctrine -- and no less so. As a logical relation, analogy is a second intention and thus is a relation obtaining among real things (or other logical entities) as they are known by us. "Healthy" is one instance of an analogous term, "being" is another. To say of such terms that they are analogous is to say something of the way they are predicated of a variety of things, but just as "genus" does not say something about animal as it exists in reality, apart from our knowing animal nature, neither does "analogy" refer just as such to things as they exist, but as they are known and named by us.

Before going on to "being," we might formulate the technical language Thomas uses in discussing analogous signification. What the term is imposed to signify, health in the case of "healthy," is called the thing signified, or *res significata*; the various ways of signifying it, the *modi significandi*, make up the determinate meanings or rationes, of the term, one of which is primary (*per prius*), the others secondary (*per posterius*). What is called the common notion (*ratio communis*) is quite indeterminate and might be thought of as involving the thing signified and a place-marker for determinate modes of signifying it, something like "health," where the blank can be filled by "subject of," "sign of," and so forth, though, again, one mode of signifying will be controlling and enter into the secondary modes of signifying the *res significata*. Let us watch Thomas apply all this to being.

The common notion of being is "that which exists," so that existence (*esse*) is the *res significata*, and "that which" (or "having" in "having existence") may be regarded as a place-marker for determinate modes of being. That there are different ways of being may be recognized by constructing a list in the way we did with "healthy": (1) George is a man, (2) George is tall, (3) George is tan. This list does not look like the earlier one since we seem to have three different predicates, "is a man," "is tall," "is tan." Nevertheless, these predicates express different modes of being, different ways of existing -- the substantial, quantitative, and qualitative, respectively -- and we could say that our list suggests another: (a) substance is, (b) quantity is, (c) quality is, which suggests a further list: (i) substance is being, (ii) quantity is being, (iii) quality is being.

"Being" now emerges as the common predicate, and, as in the case of our list of sentences where "healthy" was the common predicate, Thomas holds that "being" cannot mean exactly the same thing in (i), (ii), and (iii). What the term is first predicated of, the primary analogate, is substance: the mode of signifying *esse* involved in the predicate of (i) is the *ratio propria entis*, the controlling signification: "That which exists autonomously, not in another" (*id cui debet esse in se et non in alio*). The other ways of signifying *esse* involve reference to the substantial mode of being and thus are secondary meanings of the term. The analogy of "being," therefore, tells us of the way the term "being" is common to many things according to an ordered variety of meanings. The ways of signifying *esse* express, of course, various ways of being, *modi essendi*; the various meanings of "being" express various modes of existence. Thus, though the relation of the meaning of "quantity" to the meaning of "substance" is logical, the dependence of accident on substance is real and ontological. That is why, mistakenly, the analogy of being is often understood as a direct statement about the way things are. The coincidence here between the primary meaning of "being" and what primarily is, substance, is, from the point of view of analogical signification, just that, a coincidence; the principal meaning of a term is often ontologically secondary since priority and posteriority among the meanings of a term reflect the process of our knowing and not directly the ontological hierarchy.

Being and Essence. In the foregoing we spoke of the community of being in such a way that we seem unable to account for talk of logical entities or beings, nor do we seem able to account for what Thomas called "being in the sense of [being] true" (*ens verum*). For example, "There is no one in the room." "There is" means that it is true to say that no one is in the room. This is a secondary sense of "being," as is also the case when logical relations are called beings; the primary sense of being for Thomas is real being (*ens reale*), being out-there. That real being has many senses is what we were trying to show in our talk of the analogous predication of being with respect to substance and accidents. We might also say that being is analogously common to real being, being as true and logical being on the basis of a list like (1) John is a substance, (2) "the President is not here" is true, (3) analogy is a second intention. The meaning of "being" that could be formulated on the basis of (2) and (3) would make reference to that which could be formulated on the basis of (1), and the reverse would not be the case.

Of real being Thomas will say that it posits something in reality (*aliquid in re ponit*), so we might call it positive being. That owing to which it "posits" is its essence: only real being is said to have essence. Thus, as Bobik has shown, the

title of Thomas' opusculum *On Being and Essence* suggests just this transition from being as comprehending more than real being to real being which alone possesses essence. And, since real being is analogous and substance is the primary kind of real being, essence will be found par excellence in substance. Essence is that through which and in which a thing has being; we can see the connection between essence and the modes of being (*modi essendi*) expressed in the various meanings of real being. The essence or nature of a substance is that which makes it what it is and is the measure of its actuality or *esse*.

This brings us to Thomas' teaching on the relation between the essence and existence (*esse*) of a substance. This is often presented as a novelty of Thomistic metaphysics, but it should be pointed out that Thomas himself exhibits no sense of being an innovator when he holds that essence and *esse* must be really different. He attributes the distinction to Plato, Aristotle, and of course Boethius. It is in the *De hebdomadibus* of Boethius that Thomas finds what he takes to be a capsule statement of the real distinction: *diversum est esse et id quod est* (to be and what is differ). In Aristotle a phrase which conveys the point is found in the second book of *On the Soul*: *vivere est esse viventibus* (for living things to be is to live). Thomas approaches the matter by saying that the essence or nature of a physical substance is composed of matter and form; neither of these alone is the nature of the thing. For a thing of such a nature to exist is for there to be a conjunction of its essential components or principles (*ipsum esse resultat ex congregatione principiorum rei in compositis*). (*De trin.*, q. 5, a. 1) In a living thing essence is composed of body and soul, and this conjunction makes the thing live. To live is of course a determinate kind of existence following on an essence of a given type. When Thomas speaks of a distinction between essence and *esse*, he does not mean simply that there is a difference between a possible man and an actual man; he does not mean simply that there is a difference between the abstract nature (for example, humanity) and a concrete instance of it (for example, this man). What he intends is this: in an actually existing substance we cannot identify its essence or nature and the actuality or existence which is a consequence of the essence and measured by it. The essence of a thing relates to its *esse* as potency to act. Thus, Thomas will say that *esse* is the actuality of all other acts, even of forms. The form is act with respect to matter as potency, but for the act which is form actually to be in matter is an act other than the act the form is. This absolutely fundamental actuality is what Thomas means by *esse*, and it can be equated neither with form nor with essence in material substances.

The nonidentity of essence and *esse* does not obtain only in physical substance however; Thomas holds that there are immaterial essences other than God, and as other than God their existence is dependent on God as cause. In such substances essence is identified with form, and the form is regarded as having *esse*, sharing in it, participating in it in such a way that their essence is other than their existential actuality. In order to pursue this we must first turn to what Thomas has to say about the names of God.

God and Language. We have said that the whole thrust of philosophy and *a fortiori* of metaphysics is, for Thomas, toward knowledge of God. how such knowledge is possible and how it can be expressed in language are two sides of the same coin, and for purposes of this sketch we will concentrate on the linguistic side.

In question thirteen of the first part of the *Summa theologiae*, Thomas discusses the divine names, the meaning of terms predicated of God. Some such terms pose relatively little difficulty, for example, what Thomas calls negative names. Thus, when we say of God that he is immaterial or immobile, we may be thought simply to be denying of God certain characteristics of material creatures. Problems of a more pressing sort arise in the case of affirmative names. Let us take "wise" as our example. It should be said in the first place that Thomas approaches the question of talk about God by assuming that the problem arises because a term is predicated both of God and creature, that is, that we are confronted by "Socrates is wise" and "God is wise" and ask ourselves if the predicate has the same meaning in the two uses. Thomas will say that it does not have exactly the same meaning, but neither does it bear utterly unrelated meanings, that is, neither univocity or equivocity seems to handle the case. Well, we know what remains for Thomas: he will say that "wise" is predicated analogously of God and creature. Let us try to explicate this example in terms of what we have already learned about analogical signification.

What would the common meaning (*ratio communis*) of "wise" be? Surely something like "having wisdom." Thus, wisdom is what the term "wise" chiefly signifies, its *res significata*. How is wisdom signified when Socrates is said to be wise? The *modus significandi* here would be somewhat elaborate: to say that Socrates is wise is to say that he is a substance possessing a quality of cognition such that he assesses everything in the light of what is truly first and important. That cannot be the way wisdom is signified when we say that God is wise, if only because in God, who is simple, there is no distinction between substance and accident. To be wise is an accidental attribute of Socrates, but if God is wise

and wisdom is no accident in him, we might want to say that God is wisdom. This is quite a different way of "having wisdom" than is the case with Socrates, and the term "wise" must be construed to convey this difference as it is affirmed of God.

Thomas invokes the procedure of Pseudo-Dionysius here and suggests that there are various "moments" in our analysis of the meaning of "wise" in "God is wise." First, there is the affirmation (*via affirmationis*), but we then go on to deny of God (*via negationis*) the way of being wise that is expressed in the meaning of the term as affirmed of Socrates, for example. Thus, as is generally the case with analogous predicates, there is the same *res significata* but different *modi significandi*. As to how the *res significata*, wisdom, is found in God, we do not know. We say that it is different from the way it is found in creatures for the reasons given, and this is all quite negative. Finally, we can say that the perfection exists in God in an eminent way (*via eminentiae*). Nothing in what the term "wise" means (its *res significata*) prevents our attributing wisdom to God, but we cannot have anything like determinate knowledge of the way (*modus*) this perfection is found in God. This is why Thomas will say finally that we know what God is not rather than what he is. This is not a charter for calling God anything whatsoever, of course, since it is a consequence of analysis and reflection rather than a refusal to undertake them.

Participation. Let us turn now to the question of "being" as predicated of God and creatures so that we may rejoin our earlier discussion of the real difference between essence and *esse* in simple substances other than God. We have already seen how the term "being" is common to substance and accidents; the question now is, How is it common to God and creature, to infinite and finite being? The sentences to compare, accordingly, are "Socrates exists" and "God exists," or "Socrates is a being" and "God is a being." The *res significata* of the term is *esse*, and in the case of physical substance *esse* is an actuality consequent upon the conjunction of its essential principles, matter and form. This mode of being cannot obtain in God, and his mode of existing is approached by denying of him the creaturely mode of existing. God is thus thought of as existing in an eminent way, to be existence. This kind of talk leads to a distinction between essential being and participated being.

The common notion of being -- having *esse* -- has to be so strained when we call God being that it becomes "is *esse*." Creatures, on the other hand, have *esse* and in various ways; they partake of *esse*. To participate or partake means, etymologically, to take a part of, to share in, to possess in a diminished manner.

To be something essentially, as opposed to by participation, means to be it wholly, completely, and in an unrestricted fashion. When creatures are called beings by participation, when they are said to participate in *esse*, the following is what is meant. *esse* means actuality, but no creature is actuality *tout court*: any creature is this kind of thing or that, and its nature is consequently the measure of its actuality. From this point of view, essence as we have discussed it emerges as a limitation on the actuality *esse* is, and *esse* is considered abstractly as actuality without qualification.

Now, of course, the essence of a given thing is not a limitation of its *esse*, since it is precisely the measure of the kind of *esse* appropriate to it. We must proceed with delicacy here since it is precisely at this point, it seems to me, that some champions of the Platonism of Aquinas have gone astray. The creature's *esse* is either *esse substantiale* or *esse accidentale*, substantial existence or accidental existence. *esse substantiale* is simply a general and abstract phrase which covers to be alive or to live, which, in turn, is generic with respect to the *esse* or ultimate actuality of man, beast, and plant. From the point of view of richness of information it is far more exact to say of a living thing that it lives than that it exists (*vivere est esse viventibus*); in short, more and more determinate designations move us in the direction of greater and greater determinate perfection. There is no doubt that this is true of creatures, but when we attempt to talk about God, we seem to reverse the procedure and put a premium on vagueness.

This can be seen when we consider Thomas' discussion of "being" as the most appropriate, or least inappropriate, name of God. When we say of God that he is a being, as opposed to wise, merciful, and so forth, we seem to be saying the least possible about him. But to say the least possible means here that we are making no reference to determinate creaturely modes of existence, modes which restrict and limit *esse* considered abstractly as actuality or perfection. It is this very freedom from determinate creaturely modes of being which makes "being" the least inappropriate name of God. And since God does not partake of *esse*, does not have actuality in some restricted mode, we can speak of God as subsistent existence (*ipsum esse subsistens*).

The Platonic, or Neoplatonic, aspect of this approach is evident when we see Thomas speaking of a generic expression of *esse* as if it contained in an eminent manner the specific types of *esse* below it; as if "to live" were not a vaguer expression of the type of *esse* appropriate to men or beasts or plants, but a richer concept, containing eminently the subtypes. On that assumption we can press

on and think of "to be" (*ipsum esse*) as containing telescoped within itself all determinate types of *esse substantiale* and indeed of *esse accidentale*. *esse* then becomes a kind of dialectical limit at which various kinds of actuality are considered to meet in an eminent way -- what Cusa will call a *coincidentia oppositorum*. In Fabro's phrase, *esse* has then become *esse ut actus*, the fullness of actuality, as opposed to *esse in actu*, minimal or brute being-there, mere factual givenness. *Esse ut actus*, a dialectical construct, provides us with the least inadequate name of God, for when we say he is existence, we are saying that he is total perfection and actuality and no more -- that is, without the diminution and restriction which in creatures is read from their determinate natures or essences. Anything other than God has only as much actuality and perfection as its essence permits. In short, everything other than God partakes of *esse*, has from the point of view of total perfection only a partial perfection, its own limited one. That is how Thomas establishes the difference between essence and *esse* in simple substances other than God.

★ ★ ★

When the metaphysics of Aquinas is regarded as a lengthy meditation on what man can know of God, which is what essentially it is, something can be seen of what Maritain has called *la grandeur et la misère de la métaphysique*. In metaphysics man is straining against the limits of his knowing powers, so much so that Aristotle spoke of it as something inhuman, in the sense of superhuman. For Thomas the proportioned object of the human mind is the essence of sensible things, and it is what man knows of the material world which must always provide the lens through which he attempts to see beyond the material world. The elaboration of the subject of metaphysics, being as being, is an effort to formulate concepts which will be less obscure lenses, but their obscurity remains dual: when we consider the physical world through such concepts, we see it more vaguely than we do when we look at it through the more appropriate concepts of natural science; when we use them to gain some purchase on the divine, we are brought to the melancholy realization that all our concepts, all our names, are defective with respect to their mode of representation (*quantum ad modum significandi, omne nomen cum defectu est*). (*I Contra Gentiles*, ch. 30). And yet a little knowledge of the divine, no matter how defective and distorting, is infinitely preferable to much clear and certain knowledge of lesser things.

G. Moral Philosophy

In speaking of the division of philosophy into speculative and practical, we pointed out that for Thomas there are three criteria to which attention must be paid in assessing whether an instance of knowing is speculative or practical, namely, object, method, and end. The object of practical knowledge is called generically the operable, something we can do or make. The types of operable object call attention to the distinction between man as moral agent and man as artisan. The process of making something, of art, is one whose perfection is to be found in a product beyond the process producing it. Thus, Thomas will say that art aims at the perfection of the artifact and not at the perfection of the artisan as man. Of course, a man who makes good shoes would be called a good shoemaker, but one can be a good shoemaker without being a good man in the moral sense. Doing or the do-able (*agibile*) -- it is with this that our choices and decisions have to do, and the perfection of our choices is the perfection of us as choosers. The standard of perfection here will not be the demands of an artifact. Thomas will emphasize the difference between art and prudence, or practical wisdom, by saying that we can choose the end of art, that is, to make artifacts and to make this one as opposed to that, but in an important sense the end of practical wisdom chooses us, imposes itself upon us. Of course, to act as an artisan, to make things, is something so natural to man that we must say that it would be impossible for man not to be an artisan in the sense this term has for Thomas. His point would seem to be that beyond the englobing necessity, the direction of such activity, the end it seeks to produce, is quite arbitrary and up to us. It is not like this with moral decisions, as we shall see.

Earlier we offered a description of practical knowing according to which its perfection involves the perfection of something other than mere knowing. The perfection of moral knowledge lies in its direction of voluntary acts, of choices. What perfects is a good, and moral philosophy begins for Thomas with the asking of the question, What is the good for man, what is his perfection? Since the good or perfection is looked upon as relative to a process, that which is sought in an act of becoming, the question could be stated, What is man's ultimate end? On the philosophical level Thomas is here a faithful student of Aristotle. For purposes of moral philosophy man is something that comes to be, something striving for its good and doing this in a conscious way. Unlike other cosmic entities, man, though fashioned for a given purpose, is not directed toward it in an unconscious and willy-nilly way. Rather, it is the mark of man that in reflecting on himself in his voluntary activity he asks what is the purpose of such activity, in what will its perfection consist? For a man to act or to do is

for him to know what he is doing. The question "Why are you doing that?" might never be addressed to a being less than man, but it is always a good question to put to him. The implication of the question is that man is consciously directing himself to certain ends or goals. This can be taken to be a given of moral philosophy: we do make choices, we do pursue goals. The question of moral philosophy is, How can we do this well?

One could give a first statement of the human good in terms of what has already been said. The human good is to do well man's characteristic activity. Since this characteristic activity is reasoning, performing it well is man's good, and virtue is the term which designates the perfection of an activity. Thus, Thomas, like Aristotle, will say that virtuous rational activity is the human good. That is, as Aristotle observed, little more than a platitude, but it does involve a discrimination among possible answers to the question, What is the human good? Man has not only reasoning but also a desire for pleasure, an impulse to avoid physical harm and pain. He is also the seat of any number of acts of sensation of various kinds. Yet more basically, he grows, takes nourishment, moves from place to place. Man on this basis comes to be regarded as a kind of epitomization of processes which are found in lesser beings as well, for it is not man alone who moves and grows, who senses, who seeks pleasure and avoids pain. But man, to do these humanly, must do them rationally insofar as such activities are amenable to rational control. To seek pleasure in a human way is to subject the objects which give physical pleasure to a goal beyond themselves and so to assess them; to seek pleasure pell-mell and irrationally is possible for man, but could a man who did this be considered a good man? Not if the mark of man is to use his reason, for then the specifically human good must attach to what is peculiar and characteristic of man.

"Reasoning" must be distinguished, however. There is the process of reasoning itself, with its appropriate objects and perfection; there is also reasoning which bears on activities other than reasoning and seeks to perfect them. The latter is practical reasoning, and it is when man's appetite is responsive to such rational direction that we have the perfection of rational activity which is called moral virtue. A life lived according to reason -- that is the human good -- and this covers a multitude of virtues: the human pursuit of sensual pleasure, a human avoidance of physical pain. That is, when instinctive processes become permeated with rationality, they are more fully human. A life lived for pleasure is not a human life because the objects of pursuit which cause physical pleasure are not peculiar to human appetite and because human appetite, in a broad sense, encompasses other and higher objects than these.

To talk of the end or goal of human or rational choice may seem to refer to consequences of choice, to some quietus, some state achieved when choice is done. The human good, as Thomas sees it, is not beyond action, but in action; it is the style or formality or quality of our choices. To be a man is to choose and decide and live, and to do these things well is man's goal or end, and it will be had, if it is had, in acting and choosing and deciding, not after these are done. That is why Thomas will agree with Aristotle that man's happiness is an activity, not a state or capacity.

Man's moral goal is fixed Thomas holds; he has no choice concerning what will, in the nature of things, perfect him and be his good. Since he is a rational agent, his perfection can only be the perfection of rational activity. Rational activity can be either pure reason or practical reason, and it is the latter that is the concern of moral philosophy. The perfection of practical rational activity, again, is what is meant by moral virtue. To say that man's good is fixed by his very nature means that a man cannot be perfected as man by the pell-mell pursuit of pleasure, for example. Such an activity is not commensurate with human nature.

Thomas calls judgments or precepts concerning what we must do which are anchored in our nature, and thus sure and inflexible, natural law precepts. He uses the plural. (*ST*, IaIIae, q. 94, a. 2) Thus far, we have seen one such precept, which may be stated normatively as: Act virtuously. Thomas will sometimes state this overriding precept of human moral activity in the following way: Do good and avoid evil. But of course one must understand that in terms of the human good, and when one does, "act virtuously" is synonymous with "do good" as this is addressed to man. Are there any other natural law precepts? We can get a plurality of precepts which will have the fixity of the generic one just mentioned by appealing to the tradition of the cardinal virtues. Temperance is the name given to the rational control of the appetite for objects which cause physical pleasure. Fortitude, or bravery, is the name given the rational control of the impulse to avoid objects which cause physical pain. Justice and prudence are the other two cardinal virtues, and, without going into the nature of these, we can suggest that "Be temperate," "Be brave," "Be just," and "Be prudent" are precepts which always bind a man. Their plurality is gained by articulating the regions of human activity in which rational direction is required. There will never be a time when temperance, bravery, justice, and prudence will not be the guiding ideals of human choice. That is what is meant by saying these precepts are fixed, unchanging, and so forth.

Of course, the great difficulty in human action is not in settling on the major guidelines of choice. The difficulties begin when we ask, "But how should I be temperate in such and such circumstances?" And so too with bravery, justice, and prudence. If we think of these cardinal virtues as the articulation of man's good or end, the further questions can be said to deal with means of incorporating these ideals of human conduct in our lives in the particular decisions we must make.

The precepts of natural law are judgments in the practical order analogous to self-evident judgments in the speculative order. A mark of the self-evident judgment is that to contradict it lands one in absurdity, and with respect to the most basic such judgment, the so-called principle of contradiction, the denial of it requires one to employ it. Something like this may be said about the first principle of the practical order, "Act rationally." If one questioned this, he might be thought of as asking a question as to why he should ask questions. For surely one who wants a justification of the principle is already exhibiting in his conduct his acceptance of the rule that he ought to act rationally. The articulation of the human good into the cardinal virtues is such that something of the same kind of imperviousness to contradiction attends such a precept as "Act temperately." If this principle lays on us the obligation to seek pleasure in a manner befitting the kind of agent we are, it would be difficult to gainsay it without calling into question the kind of agent man is and thus the kind of agent the questioner reveals himself to be.

At a level of great universality, then, Thomas feels that there are inflexible guidelines for human choice; the target at least is clear. But how is one to be temperate in such and such circumstances? As soon as we move into the area of enunciating means of realizing our end or goal, our judgments become corrigible, usually as opposed to always true. The elaboration of less general moral judgments not only depends more and more on experience, our own and others, but reflects increasingly the evanescence of circumstances, the particular historical epoch in which such judgments are made. Thomas will insist on the corrigibility and probability of general judgments less than the most common ones of natural law, but at the same time he will argue that this is no reason against the formulation of more particular moral judgments.

There are thus two levels of generality, what we might call the level of principles and the level of rules, with the former certain and inflexible and the latter true, even if true only for the most part or usually. The moral rule may be thought of as a statement of the means whereby we can usually achieve our end. The

justification of moral rules, therefore, is to be found in the principles, in the end. But since the rules are true only for the most part, they admit of exceptions. What would justify an exception to a rule if the principle justifies the rule? It seems that we must say that the principle justifies the exception as well as the original rule, since if we should judge that in some cases it would be wrong to abide by a rule having to do with temperance, say, we might so judge because acting in accord with the rule will thwart the end of temperance or of some other articulation of our overriding good. And, of course, the recognition of exceptions to a rule can give rise to the formulation of another and lesser rule.

Since both principles and rules are in the practical order, they are not sought for their own sake but point beyond cognition to the perfection of our choices. Thomas follows the lead of Aristotle in saying that beyond the common principles, naturally and easily knowable, and beyond moral rules which might be taken to be the domain of moral philosophy and of human law, there is a third level of moral knowledge, completely practical knowledge, where we are cognitively engaged in applying principle or rule to a concrete set of circumstances. By calling this area the realm of prudence, completely practical knowledge, we are suggesting that it saves all three of the criteria of practical knowledge mentioned earlier. With Aristotle, Thomas calls the discourse of prudence a practical syllogism.

The practical syllogism, the discourse of practical reasoning in the concrete direction of choices, is analyzed as follows. The major premise is a generality, a principle or rule; the minor premise is an assessment of our present singular circumstance in the light of the principle or rule relevant to it. The conclusion, ideally, is a choice in accord with this assessment. As soon as we move away from the major premise, we move away from generality and into an area where the state of our appetites becomes influential. The principle or rule expresses an object of appetition, a good, at a level of generality, and we can assent to such judgments in a way that does not engage us fundamentally where we live. Practical discourse, the practical syllogism, when it moves into the area of the concrete and singular, will reveal the actual condition of our appetites, what for us is really considered good. Now what for us is really considered good, what we have a bent toward due to past choices, may be other than and in conflict with the good expressed in the major premise. Imagine that the major premise has to do with temperance. We find ourselves in a situation where at the back of our mind we are aware of the obliging character of temperance with reference to our choices; with that awareness we regard the circumstances in which we find ourselves. In doing so we might repress or dismiss the moral principle because

of the attractiveness of an object which promises pleasure and go on to pursue that object. This would reveal that the good expressed in the principle or rule is not effectively our good; that our choices are really governed by a rule we might hesitate to express as a generality, namely, "Pursue pleasure mindlessly."

A man acting on such a basis would be incontinent, in Thomas' use of the term; in moments of repose, as a student of ethics, say, he might assent to a moral principle or rule, but when the chips are down, the good he seeks is not that expressed in the moral principle. In order to move from the principle to an assessment of our circumstances in the light of it and to a choice in accord with that assessment, we must effectively love the good expressed in the principle. That is why Thomas will say that the truth of the ultimate practical judgment is to be read, not as a conformity with the way things are, but as a conformity with rectified appetite. In short, only the virtuous man will easily and without pain make the transition in practical discourse from principles to choice. One can see why Thomas places such importance on moral training and upbringing; it makes a great deal of difference what objects of aspiration are placed before us in our early years. Mature reflection on human action always takes place against a background of formation and education, of ideals which have become familiar because of habituation. It is not the case that the ideal can be recognized to be such only by those who strive to incorporate it into their lives, but Thomas will insist that when it comes to how the ideal can be realized here and now, our ability to know this is a function of our past moral history.

We might summarize Thomas' view of moral knowledge, then, by seeing three levels of it. First is the level of the most common precepts, those which enunciate the ideal of rational love or loving reason: man's perfection and its articulation into the cardinal virtues. This is the domain of natural law without qualification, for as long as man is what he is, his perfection is the perfection of the kind of agent he is. On a second level would be the formation of precepts or rules which elaborate at a level of generality how the ideal can usually be achieved. All such rules are corrigible, of course, and they increasingly reflect experience and thus the changing situations in which man strives for the ideal. Finally, there is the concrete level, the singular choice in which knowledge must be proportionate to these changing, concrete circumstances and where knowledge is inevitably influenced by the condition of the agent's appetite.

H. Thomas and His Time

The foregoing presentation of various aspects of Thomas' doctrine has made little effort to relate what he taught to the currents of his day, though of course the reader will have connected elements of Thomas' teaching with controversies mentioned in earlier chapters. Since the fame of Aquinas reposes in large part on the fact that he brought together in a new whole the various strains and traditions which met in the thirteenth century, it is only right that we say some few things about his general attitude as it is exemplified in a number of key controversies.

If the men of the late twelfth and thirteenth centuries had been confronted only with accurate Latin translations of the *Metaphysics*, *Physics*, and *Nicomachean Ethics* of Aristotle, unaccompanied by commentaries, they would certainly have had their work cut out for them; we can surmise that there would have been a good dose of defensiveness in their response to such a powerful statement of the nature of the world, since in many particulars it seems contrary to what the Christian believes to be true. But of course the matter was not at all that simple. The Aristotle who came to be known came together with Islamic interpretations which confused Aristotelian and Neoplatonic doctrines. We have seen that in Islam portions of Plotinus and of Proclus were considered to be the work of Aristotle. Furthermore, the Greek commentaries on Aristotle, with their Neoplatonic bent, came into Latin directly as well as through the medium of the influence they had had on the thinkers of Islam. That meant that the whole Neoplatonic apparatus of emanation and the mediated causality of the first cause were added to the real problems in the text of Aristotle, those associated with the scope of the divine knowledge, the eternity of the world, and the survival of the individual soul after death. Before this onslaught it is not surprising to find that the first reaction was one of caution. Aristotle's works were proscribed at the University of Paris. Balancing this, however, was the commission set up by Gregory IX to study and interpret the teaching of Aristotle. The invitation to seek in Aristotle what truth might be there was clear, and masters responded to it to the detriment of the ban.

One of the members of the papal commission, William of Auvergne, indicates one mode of response to the new literature. He is quite confused as to what is authentically Aristotelian doctrine and what is not, but he has a double measure with which to confront the new teachings. First, there is faith; if the Aristotelians teach something contrary to faith, that is *prima facie* indication that their teaching is false. William does not leave the matter there, however; he goes

on to try to show the falsity of such teachings by arguments. What exercises its influence on him as he does this is the Augustinian tradition which had been dominant, but, unsurprisingly, William also takes over from Aristotle and Islamic thinkers whatever he takes to be sound. The whole into which such borrowings are assumed is somewhat difficult to describe, and to call it traditional Augustinianism is to label it rather than analyze it.

Another attitude, far less ironic, is represented by Bonaventure; yet another by Albert and, more brazenly, the Latin Averroists. We have seen the curious neutrality Albert claims when he is exposing the tenets of Peripatetic philosophy; more curious still is the silence he attempts with respect to the theological verdict on the philosophy he is narrating. His model here seems to be Moses Maimonides, and the term of the attitude could be either that theology must be measured by philosophy or, perhaps, that the truth of philosophy and that of theology may conflict and contradict in a way that is ultimately acceptable.

With Aquinas we have an Aristotelian, a man who accepts the fundamental validity of the philosophy of Aristotle and will employ it as the context into which other contributions must be fitted. This attitude is possible for him because he was able to distinguish between Aristotle and Neoplatonism, between Aristotle and his interpreters, in a way that had not been done to the same extent before. This is not to say that he rejects Neoplatonism, whether that of Proclus or of the Greek and Islamic commentators, out of hand. But there is a new clarity as to what is what. For example, in the *proemium* to his exposition of the *Liber de causis* Thomas matter-of-factly states the origin of the work in Proclus; apparently this is the first time the identification was made, and Thomas was able to make it because William of Moerbeke had provided him with a Latin translation of Proclus' *Elements of Theology*. When he approaches the text of Aristotle, Thomas seems possessed of the certitude that, accurately understood, it can withstand the criticisms that have been directed at it. Let us examine a few instances of this.

In the twelfth book of the *Metaphysics* Aristotle had said that God knows only himself, and this had been taken to mean that God knows nothing other than himself, which would seem a plausible enough interpretation. From that would follow a denial of providence and so forth, and Aristotle emerges as teaching things inimical to belief. Thomas comes at the passage in question with little indication that it has been the subject of controversy. Aristotle, he explains, identifies the First Cause with his act of understanding and says that the

nobility of that act of understanding can be read in terms of its object. The object of the act of understanding (identified with the First Cause) must be itself, since if it were something else, that object would be more noble than the First Cause. But the First Cause is most noble, and so forth, so it is necessary that it understand itself and that in it understanding and what is understood be the same. "It should be considered however that the Philosopher intends to show that God understands, not something else, but rather himself insofar as what is understood is the perfection of the one understanding and of his act of understanding. It is quite clear that nothing else can be understood by God in this sense that it would be the perfection of his intellect. Nor does it follow that other things are unknown by him, since in understanding himself he understands all other things." (*In XII Metaphys.*, lect. 11, n. 2614) The point is put more succinctly in Thomas' discussion of proposition thirteen of the *Liber de causis*: "Since according to the opinion of Aristotle, which in this matter is more in accord with Catholic doctrine [more than the opinion of Proclus, that is], we posit, not many forms above intellect, but one alone which is the First Cause, we must say that just as it is its existence [*ipsa est ipsum esse*] so it is one with its life and its intellect. Hence, Aristotle in the twelfth book of the *Metaphysics* proves that he [God] understands only himself, not that knowledge of other things is lacking to him, but because his intellect is not informed by any intelligible species other than himself. What Thomas is doing in such a case is, not bending Aristotle to Catholic doctrine, but insisting that Aristotle correctly understood is in accord with the faith. So too, for Thomas, Aristotle's God is the creative cause of the universe. "From this manifestly appears the falsity of their opinion who hold that Aristotle thought that God is the cause, not of the substance of the heaven, but only of its motion." (*In VI Metaphys.*, lect. 1, n. 1164) It has become fashionable to say that Aristotle's God is only the ultimate final cause of the world; that "only" would have mystified Thomas, for whom the final cause is, as it was for Aristotle, the *causa causarum*, the cause of the other causes. Although he never developed the argument, since he was not confronted with this curious notion of "only the ultimate final cause," it would be a simple matter for Thomas to prove that if God is the ultimate final cause of the world, he is *a fortiori* its first efficient cause.

There is considerable confusion in recent discussions of Aristotle and creation. Sometimes it seems to be suggested that Aristotle's world cannot be a created one because it is eternal. Thomas is quite sure that it was Aristotle's firm opinion that the world is eternal, but he insists, as others had, that the arguments he brings forward for this claim are at best probable. Would the eternity of the world preclude its being a created world? This was a matter

Thomas investigated, notably in an *opusculum* entitled *On the Eternity of the World Against Murmurers*. For those who take this sort of thing to be a mark of humanness, we might observe that in this *opusculum* Aquinas treats his unnamed adversaries with sarcasm, saying how marvelous it is that men like Augustine and the best philosophers had not seen the contradiction involved in an eternally created world. Those who see a contradiction in the notion, he adds icily, must alone be men, and wisdom arrived in the world with them.

In the *opusculum On the Unity of the Intellect Against the Averroists* Thomas goes to great lengths to show the inaccuracy of the Averroistic interpretation of Aristotle's doctrine on the faculties of the soul. Perhaps nowhere else is it as clear that what is at stake is what Aristotle taught and that accuracy of interpretation goes hand in hand with acceptance of the result. In this *opusculum*, as in his work *On Separate Substances*, Thomas exhibits his knowledge of Islamic, Jewish, and Greek interpretations of Aristotle.

We mentioned earlier in this chapter how unwise it is to identify the philosophy of Aquinas and the philosophy of Aristotle when this leads to a failure to recognize the influence of other philosophers on Thomas. It would be far unwise to suggest that Thomas' thought is in principle Neoplatonic. In its principles Thomas' philosophy is Aristotelian, and, as we have observed, whatever else enters into his philosophy is subjected to an Aristotelian test, is brought into a fundamentally Aristotelian context. One may cheer or lament this, but he may not deny it or ignore it.

Thomas' contemporaries found it difficult to ignore what he had accomplished, and we shall see in the next chapter something of the history of his immediate influence and reactions to it. Insofar as Thomas the philosopher is a model for the twentieth-century thinker, it may be well to distinguish two aspects of the model. On the one hand are substantive doctrines to be understood and assessed; on the other hand there is what can be called the spirit of Thomas. That spirit applied nowadays to the thought of Thomas himself would doubtless see it as a component of a larger whole, and would see the Thomist as a philosopher for whom Thomas functions as Aristotle functioned for Thomas.

Bibliographical Note

Aquinas, *Opera omnia* in 25 vols. (Parma, 1862-1870); the Leonine edition (Rome, 1882-1948) is not yet complete, but 16 vols. have been published. English translations: J. Rowan, *Commentary on the Metaphysics*, 2 vols. (Chicago,

1961); Pegis, Anderson, Bourke, and O'Neil, *On the Truth of the Catholic Faith* (*Summa contra Gentiles*), 5 vols. (New York, 1954-1956); A. Pegis, ed., *Basic Writings of Saint Thomas Aquinas*, 2 vols. (New York, 1948); Mulligan, Schmidt, and McClynn, *Truth*, 3 vols. (Chicago, 1952-1954); J. Bobik, *Aquinas On Being and Essence, A Translation and Interpretation* (Notre Dame, 1965); a new translation of the *Summa theologiae* under the auspices of the English Dominicans began to appear in 1964 (New York: McGraw-Hill); Foster and Humphries, *Commentary on the De anima* (London, 1951); Blackwell, Spath, and Thirlkel, *Commentary on Aristotle's Physics* (New Haven, 1963); A. Maurer, *The Division and Method of the Sciences* (Toronto, 1953). Studies on Aquinas are incredibly numerous; the reader is referred to the following bibliographies: V. Bourke, *Thomistic Bibliography* (St. Louis, 1945); P. Mandonnet and J. Destrez, *Bibliographie Thomiste* (Paris, 1960); P. Wyser, *Thomas von Aquin* in the series *Bibliographische Einführungen in das Studium der Philosophie* (Bern, 1950). The *Repertoire bibliographique de Louvain* (supplement to *Revue philosophique de Louvain*) is invaluable for philosophical works generally and can be used to bring the above bibliographies up to date. A recent work on the continuing role of Aquinas is R. McInerny, *Thomism in an Age of Renewal* (New York, 1966; reprint: Notre Dame, 1968).

Chapter VII

Conclusion

In the preceding chapters of this part we have concentrated on a number of thinkers, selected for their generally recognized importance, and discussed what they taught with only glancing references to the milieu in which they carried on their activity. That milieu -- the university, and particularly the University of Paris -- was one in which many divergent currents flowed, in which the kind of activity we have regarded in isolation comes into confrontation and conflict with opposed teachings, attitudes, aspirations. We have made reference to the dispute that raged concerning the academic status of Franciscans and Dominicans at Paris, a dispute which had its import for the two most eminent men of the century, Bonaventure and Aquinas. Similarly, we have referred to the caution with which the new writings invading the West were met, the proscriptions that were laid down, and so forth. The men we have concentrated on have represented, by and large, the effort to learn from the new and to assimilate it by putting it into relation with what had been had before. Yet efforts at assimilation varied insofar as one or the other middleman between the Western Scholastics and Aristotle was given prominence. We have suggested that one of the signal accomplishments of Aquinas is to be found in his distinguishing between the doctrines of Aristotle and those of his interpreters in Islam, something that required a painstaking and direct reading of Aristotle in less and less defective texts. No doubt it was this effort that enabled Aquinas to state with the clarity he did the distinction between philosophy and theology. We must now say a few words of the context in which he did this.

We have been considering the universities of the thirteenth century, taking Paris as the great model, as places where the faculty of arts functioned as a stepping stone into the faculty of theology (or those of medicine and law). This suggests that the faculty of arts, and philosophy, which was its principal concern, had nothing terminal to it, that philosophy had its destiny in theology. It should be noted, moreover, that most of the men who have been the objects of our attention in the chapters of this part were theologians, wrote as theologians, saw philosophy from the vantage point of the principles of theology, and assessed its status accordingly. Yet it was at Paris that a conflict arose with regard to the relationship of philosophy and theology which was, in many of its aspects, a conflict between the faculty of arts and the faculty of theology.

It will be remembered that if there was anything that contributed to the determination of the relationship between philosophy and theology on the part of an Aquinas, it was the striking fact of the philosophical writings of Aristotle. Here, for the first time in centuries, the Christian intellectual found himself face to face with an elaborate and nuanced theory of reality in its various aspects, a theory reached in utter independence of the influence of faith. What greater proof could be required of the autonomy and viability of philosophy? Hand in hand with the appreciation of the autonomy of philosophy there went a sensitivity to those things Aristotle had taught which went contrary to Christian revelation. We have seen that, in large part due to the interpretations of Arabian commentators, it came to be a commonplace that Aristotle had taught that the world is eternal, that the survivability of the individual soul is a doubtful matter, and so forth. The typical reaction of Aquinas was to ask, first, whether this was what Aristotle actually taught. Quite often he reached the conclusion either that he had not taught what he was claimed to have taught or that he taught it in such a way that his doctrine was not in conflict with Christian belief. Others, like Bonaventure, lacking interest in the accuracy of the historical ascription of positions to Aristotle, applied the criteria of revealed truth to assess the alleged doctrines as false. Now while there is certainly a difference in approach between Aquinas and Bonaventure in this matter, the difference does not lie in the fact that the one thought revealed truth was a useful criterion and the other did not. Aquinas, as much as Bonaventure, may be taken to be guided by his faith in assuming that the position that holds firmly that the world has always existed must be false, or at least not provable. What is particularly interesting in the case of Aquinas is that he takes this initial judgment to be, not a foreclosing of argumentation, but an invitation to philosophizing. Let us, he suggests, look at the arguments; let us consult the texts. Animating his whole approach is the assumption that wherever there is a contradiction in terms the truth cannot lie on both sides of the contradiction. Now it was something like the latter position, the so-called two-truths theory, that came to be held by some masters of the faculty of arts in the middle of the century.

The two men most important for this controversy are Siger of Brabant and Boetius of Dacia. The movement associated with their names is commonly called "Latin Averroism." They were the objects of various polemical opuscula by figures already treated, for example, in the *De imitate intellectus* of Aquinas. Now, once more it must be emphasized that what Aquinas set out to do, and what he accomplished, in that little work was to save Aristotle, not to bury him - to show that he had not taught what was being attributed to him concerning the faculties of the human soul. The controversy in question, then, may not be

viewed as arising from the efforts of theologians to condemn philosophy, to restrict it, to destroy its autonomy, and so forth. Rather, it appears to be an effort of theologians to save philosophy from the philosophers -- just as nowadays it sometimes seems that it will fall to Christian philosophers to save theology from theologians who misread the import of current philosophical trends.

When we look at the writings of Siger of Brabant, we find a repetition of interpretations of Aristotle with which we are familiar from the Arabian commentators. Thus, God's causality is restricted to a first effect, a primary intelligence, to whose activity the next level of reality must be ascribed and so on. So too, Siger teaches the "eternity of the world. Since these positions call into question, respectively, the universality of God's causal efficacy and revealed truth, it becomes a matter of some interest to inquire how Siger squared his philosophical tenets with his Christian belief. Siger seems to have adopted an ambiguous stand on this matter. On the one hand, he suggests that he is merely examining the teachings of the (pagan) philosophers; on the other hand, he implies that these tenets are unavoidable conclusions of reason. Boetius of Dacia held views that are both more openly abrasive and unequivocal in their implications. As if from an excess of professional pride, Boetius held that the pursuit of philosophy is the highest human pursuit, that only philosophers are wise, and that there is absolutely no restriction on philosophical activity. More substantively, Boetius is said to have held that creation is impossible, even though faith requires us to believe that it is possible. With what one sometimes suspects must have been perverse delight, Boetius went on to list a number of other articles of Christian faith which are philosophical absurdities, though he seems never to have urged that men cease and desist believing them.

Latin Averroism, then, would seem to be grounded on what must seem a psychological impossibility, since it asked believers to accept logical contradictions. Not only did this call into question the reasonableness of faith -- something insisted on from the beginning of theological study -- it also characterized the philosophizing of the Christian in a way that would require him to be schizophrenic. The remedy called for was one that assigned their proper notes to faith and to philosophizing, and it was this remedy that was offered by theologians -- as well as by masters of arts not in sympathy with Siger and Boetius, and these nonsympathizers, we might note, were the majority in the faculty of arts at Paris.

We began by saying that this dispute became a dispute between the faculty of arts and the faculty of theology and went on to say that it must not be viewed as prompted by an animus against philosophy on the part of theologians. This last claim, while true in the terms we made it, must now be qualified. In 1277 the bishop of Paris, at the behest of the pope, condemned a list of 219 propositions. In presenting this list, the bishop, Stephen Tempier, made quite clear who the targets of his condemnation were: masters of the faculty of arts who taught things contrary to faith and who, when accused of heresy, took refuge in the claim that there is a distinction between the truth of philosophy and the truth of faith. One of the great ironies of the Condemnation of 1277 is that several tenets of Thomas Aquinas were among the propositions condemned. Further, if animosity between philosophers and theologians was not at the source of the dispute, it was certainly one of the consequences of the condemnation. Theologians became increasingly suspect of the activities of philosophers, and there was subsequently a tendency for the theologian to pursue his proper effort in growing independence from philosophical speculation. The reverse side of this coin, of course, was the tendency of philosophers who were also believers to ignore the relevance of their faith to their philosophizing.

If doctrines have a history, it is not linear, and if the history of philosophy is interesting, this is not because of movements which carry men along but rather because of individual philosophers. Men may think in a context, but the men who interest us as philosophers are less products of their times than producers of the spirit of their own and later times. To consider the movement into the thirteenth century of the complete Aristotelian corpus and of the Arabian commentators on Aristotle, as well as of Neoplatonic doctrines of an earlier time, is not to consider something that is independent of individual thinkers; on the contrary, such movements are ideal continua whose points are individual minds. All this is preface to our unwillingness now to discuss the fate and destiny of the movements of the thirteenth century. Finally, what is of philosophical interest in the thirteenth century are the writings of men like Bacon and Bonaventure, Albert and Aquinas, writings which are to a surprising degree accessible without paying great attention to the "spirit of the thirteenth century" or other abstractions taken as the putative antecedents of the efforts of such men. One of pitfalls of the historian is to interpret individuals in the light of their times, forgetful that those times are largely defined by us in terms of the great individuals who inhabited them. It would be easier to sustain the thesis that Aquinas thought against the grain of his age than that he is the perfect mirror of it, easier but perhaps no more fruitful. The suggestion that learning is a matter of entering into conversation with the great men of all times via their

writings, while it may be marred by simplifying or overlooking the real difficulties and impediments that may obscure those writings to a later mentality, is, finally, the only defensible attitude toward the great minds of the past, of the thirteenth or any other century. We may compare the twelfth and thirteenth centuries by saying that in the former the schools were where the great teachers were, whereas in the latter the great teachers were where the schools were; however, in either case it is the great teachers who interest us -- and their greatness consists in large part in the way they transcend their times.

The Fourteenth Century

Chapter I

Introduction

This volume of our history ends by juxtaposing chapters on two thinkers who represent, respectively, what came to be called the old way (*via antiqua*) and the modern way (*via moderna*) of doing philosophy. Duns Scotus, for all his differences from the great thinkers of the thirteenth century, is nonetheless in basic continuity with them. William of Ockham, on the other hand, represents a rupture with the old and the beginning of something new, a modern way whose fundamental characteristic is nominalism. Scotus is, however, something of a bridge between the new and the old, not because Ockham was a Scotist, but rather because Scotus, though often criticized by William, provided the later Franciscan with some of the formulas for his own thought. As will become clear, however, these formulas tend to take on a quite different meaning in the hands of Ockham.

When we view the thirteenth century as a whole, we see it as a time when Christian thinkers, confronted with an amazing array of new sources which threatened the very foundations of traditional thought, won through in a variety of ways to a new and larger synthesis. But great intellectual achievements cannot be simply passed on to others; guidance they may offer, but each must reenact the achievement in his own mind if it is to be his in any significant way. We recall Roger Bacon's criticism of the thirteenth-century attitude toward the writings of Aristotle. The attitude he lamented soon came to characterize those who grouped themselves around the doctrines of the great thinkers of the thirteenth century. Thomism, for example, became a possession of the Dominican Order, and adherence to its contents a matter of orthodoxy. Josef Pieper has pointed to the striking fact that Aquinas seems to have had no immediate notable disciples; subsequently he was to have legions of them, but the question can be raised whether their attitude toward his writings was in all cases a matter of personal philosophizing or theologizing. And yet it is in such a dissident Dominican as Durandus that one senses the seeds of the *via moderna*, the foreshadowing of nominalism. Since we have concentrated on the major thinkers, such transitional discussions as the present one have been taken as occasion for generalizations to which the reader may respond as he will. What I would like to do now is to indulge myself in a significant generalization about

what happens to philosophy as we move from the thirteenth to the fourteenth century, a generalization which points to a paradox.

It is often noted that the great thinkers of the thirteenth century are theologians by profession, and of course the same must be said of those of the fourteenth century. What strikes me as the great achievement of the thirteenth century, and I have Aquinas particularly in mind, is the establishment of a clear distinction between faith and reason in the terms required by the influx of Aristotle. That is, the distinction clearly becomes one between theology and philosophy. It is no doubt the case that Aquinas was a theologian, but this did not prevent him from granting an autonomy and scope to philosophy which followed on philosophical positions with respect to the nature of the world and the capacity of man's intellect. In Aquinas' vision of philosophy there is a complementarity between the mind and being. The world is intelligible and man can understand it, and the great divisions of philosophy that Aristotle had recognized indicate the range and possibilities of philosophical thought. What has impressed most historians about Aquinas is the fact that while his faith provided him with the context of his thinking, that context required him to admit that belief is not a prerequisite for understanding the world. Rather, even for the believer philosophical thought is a prerequisite for that "understanding" of revelation that is theology. The general point I wish to make about the thirteenth century is that, contrary to what might have been expected and contrary to the judgment of some historians, in that era faith, rather than restricting the range of natural reason, came to function as a kind of motivation for asserting man's natural capacity to understand himself and the world in which he finds himself. It is one of the ironies of the history of philosophy that whenever philosophy sets out to separate itself from a religious context, its scope, its vision of its range, becomes drastically limited. More to the present point -- and this is the paradox to which I referred earlier -- in the *via moderna* of the fourteenth century there is a devaluation of philosophy which is a function of the devaluation of reality that nominalism requires.

Already in Scotus we find the beginnings of skepticism as to the natural capacities of human reason. To be sure, his doubts are expressed when he is discussing the need for revelation and the importance of religious faith. Nevertheless, Scotus seems to see as a danger what Aquinas clearly regarded as a strength of philosophy as it was developed by Aristotle, namely, the assumptions that reality is intelligible, that it delivers itself up to man according to the canons of science in the Aristotelian sense, that there are necessary truths which express the real order. For Ockham the world is a world of individuals

that happen to have been created by God; a completely contingent world is incapable of delivering up objects for necessary judgments. If having science, really knowing, is knowing what cannot be otherwise, then it is difficult to see how we can have knowledge, in the required sense, of a world which could have been otherwise, could indeed not have been at all. The only evidential knowledge for Ockham will be immediate judgments of present fact. To judge that Socrates is seated when we see him sitting there carries its evidence on its face, and of course Socrates cannot both be seated and not seated. But anything like a universal necessary empirical judgment is excluded by Ockham. The consequences of such doubts about philosophy have their ramification for theology, which ceases to be a science for Ockham; he ends in a fideistic stance where nothing counts for or against what the Christian believes.

The *via moderna* does not involve any turning away from Christian faith, but the reasonableness of faith and the intelligibility of what is believed are called into question at the same time that philosophy as it was understood in the thirteenth century is rejected. At the present time, it seems fair to say, English language philosophers feel more affinity with such men as Ockham than with the men of the thirteenth century. To no small degree this interest is due to the judgment that the terminist logic which Ockham espouses foreshadows some recent developments in logic, but it is also true that commentators on Ockham tend to find similarities to Hume and to the logical atomism of the early twentieth century. That philosophers of the twentieth century should feel this compatibility with the *via moderna* of the fourteenth century may tell us something quite important about both. What is at issue here is not simply logic but the philosophy of logic, and under the latter must be grouped various epistemological and ontological views or attitudes. Far more than notation is involved when such judgments as "Every man is animal" become expressed by "(x) Fx. D . Cx." Ultimately, for many, such a symbolic formulation comes to express a vision of the world and of human knowledge, a vision according to which everything is itself and nothing more, where the similarity of things is reduced to their otherness, and the paradigm of knowledge is sought in such judgments as "Socrates is seated now." Such a vision is an advance only if it adequately represents reality. If we are now turning away from this desert horizon by noticing that such a vision is an inadequate account of how we speak, we have still a long way to go, and the direction, of course, is not backward to the thirteenth century, but ahead to new expressions of the intelligibility of being and of man as *capax Dei* because he is first *capax entis*.

Chapter II

John Duns Scotus

A. The Man and His Work

John Duns, the Scot, was born about 1266 and entered the Franciscan convent at Dumfries in 1277. Extremely little is known of his early life; there is controversy about his birthplace, where he joined the Franciscans, and so forth. It is held that he lectured on the *Sentences* of Peter Lombard both at Cambridge and at Oxford, at the latter university about 1300. He taught at Paris from 1302 to 1303, was exiled because of the part he took in a dispute between king and pope, but was back in Paris in 1304. In 1307 he was sent to the Franciscan house of studies in Cologne, where he died the following year, 1308. Meager facts, no doubt, and facts about a brief life, but Duns Scotus, called the Subtle Doctor, is a most important figure despite the brevity of his career.

The writings of Scotus are of difficult access for a number of reasons, although we now have reason to hope that reliable texts of all his works will be forthcoming. The great difficulty concerns Scotus' commentary on the *Sentences*. He commented on this work at Cambridge, Oxford, and Paris. The so-called *Opus Oxoniense* grows out of his Oxford lectures, but was continued and worked over in Paris; also called the *Ordinatio*, it seems to have been an effort to incorporate in one commentary what was best from Scotus' various expositions of the *Sentences*. A critical edition of the *Ordinatio* began to appear in 1950. Besides his commentaries on the *Sentences*, Scotus produced some *Quaestiones quodlibetales* and *Quaestiones subtilissimae in metaphysicam*; the *Collationes*, conferences given at Paris and Oxford; and logical works based on Porphyry's *Isagoge* and Aristotle's *Categories*, *On Interpretation*, and *Sophistical Refutations*. A little work on the First Cause, *Tractatus de primo principio*, is of particular importance, not least because it seems to be a late work. The so-called *Theoremata* is of disputed authenticity; those who favor its authenticity feel that Scotus blocked it out, perhaps dictated it. Like the work on the *First Principle*, it is valuable for determining Scotus' thinking on the reach of reason with respect to knowledge of God. The writings of Scotus strike us as inchoative rather than complete, as elements of a developed view rather than that development. But such a judgment refers to them as a totality. Particular treatments, individual discussions, exhibit sureness, profundity, and, yes, subtlety.

B. Being and God

To begin our discussion of Scotus with a treatment of what he had to say about the nature of being and of our concept of it, and of the relation of that to our knowledge of God, is admittedly to begin with a much controverted and contradicted portion of his doctrine. However, no one can overlook the centrality of these questions for gaining some purchase on Scotus' characteristic teachings, and so, with half apologies, this is where we shall begin.

The medievals had learned from Aristotle that there is a science whose subject is being as being, a science which, unlike those called special or particular, is concerned, not with this kind of being or that, but rather with the characters of being prior to its particularization into kinds. For Aristotle this meant that while natural science concerns itself with mobile or changeable being and mathematics with quantitative being, a further science, what came to be called metaphysics, is concerned with being as being. Being as the subject of metaphysics, then, is being as prior to particular kinds of being. Now this position can of course mean several things, but what it did not mean for Aristotle is that insofar as "being" is a more general term than "mobile being" there must be some reality that answers to the more general term "being," just insofar as it is a general term. We can study what an animal is without introducing into our explanation what pertains to this animal or that, but this does not suggest that there is an animal which is no kind of animal. In short, the recognition of levels of generality in our conceptions is not tantamount to the claim that there are levels of reality which respond just as such to those levels of generality.

Scotus would introduce a qualification here, his famous formal distinction, and we will return to that, but even in the terms we have used in presenting the context of what he has to say about being, he would not be in complete agreement with the foregoing. The whole point of the science of being as being, he would want to say, is that there is some being which is not confined to the types or kinds or categories of being -- a being which is not, accordingly, finite. Scotus ultimate interest in being is in infinite being, and what he has to say about being as transcendent must always be understood with reference to his basic theological, or God-centered, interest. What is the relation between finite and infinite being, between God and man? How do our concepts, and the names which express those concepts, include both God and man?

Transcendentals. We must first see something of what Scotus had to say about the relation between categories and transcendental attributes of being. The categories of being are what Aristotle thought they were, types or kinds of being of which the clues are to be found in an analysis of predication. Some general predicates of individuals express what they are, others how much, yet others how, and so forth. The enumeration of such types of predicates results in an enumeration of the fundamental or basic kinds of being: substance and the various accidents -- quantity, quality, place, time, and so forth. One category of being is different from another; that which falls into the category of substance is different from that which falls into the category of quality. The categories represent diversity and fragmentation in reality, and the question arises as to the meaning of a term that can be predicated of all of these diverse kinds, of a term like "being." Substance can be said to be, as can quantities, qualities, and so forth, and if we make of the verb a common noun, "being," our question emerges. Does "being" express something other than what the categorical terms express? Unlike such terms as "substance," quantity, and so forth, the term "being" is said to be a transcendental term, which means at least this: what it expresses is not confined to one of the categories. Scotus accepted a long tradition according to which there are other terms which function as does "being," other transcendental terms, that is, terms which are said to be convertible properties of being. That difficult phrase may be translated as follows: another term is, like "being," a transcendental term if it can be predicated of whatever "being" can be predicated. Since "being" can be predicated of each of the categories, any other terms portraying the same characteristic were called transcendental terms. Among such terms the tradition recognized "one," "true," and "good." Anything that can be said to be can also be said to be, in some sense, one, true, and good. As has been mentioned, Scotus accepted this notion of transcendental, but he did not accept its apparent restriction to transcendentals of this type. In a most important passage he gives a list of various kinds of transcendentals (*Opus ox.*, lib. 1, dist. 8, q. 3, nn. 18-19).

The passage occurs in the context of his asking what kind of predicates can be predicated formally of God. What does the question mean? Formal predication here refers to a real correspondent to the content of a concept whose name is predicated. Thus, from the very outset Scotus' concern with transcendental concepts is a concern with concepts whose content indicates or means or signifies something in the real, that is, extramental order. This of course does not distinguish transcendental concepts from other, categorical concepts. The concept expressed by the term "substance" expresses something in the real

order, and so too with the concept expressed by "quantity." What then is distinctive of transcendental concepts? Here is Scotus' description of the nature of a transcendental: "*Ita transcendens quodcumque nullum habet genus sub quo contineatur*" (that is transcendent which is not contained under some one genus). This definition certainly applies to the more traditional notion of transcendental properties of being such as those which are convertible with it, namely, one, true, and good. But it soon becomes quite clear that Scotus has no intention of confining transcendentality to that traditional list. Those who would identify being transcendental with the items on that list would consider that, like being, that which is transcendental is common to everything, that it can be predicated of whatever is. But Scotus wants to speak of transcendental concepts which are not thus common or universal. It is the contextual question that provides a clue to what he has in mind. What Scotus wants to say is that all notions (and the terms expressing them) which are common to God and creature, or which are proper to God alone, are transcendental concepts. Thus, such a term as "wise" (or "wisdom") is not predicable of everything of which "being" is predicated, but it saves the notion of transcendentality because it is not confined to any one genus. Wisdom is found in God and in man, and while the latter is categorizable, God is not. In short, for Scotus, as for Aquinas, the categories of being enumerated by Aristotle are categories or divisions of finite being. God, therefore, cannot be in a category. Any name which is proper to God, or is common to God and creature, cannot express something which is included under some one category, is confined to a single category. Scotus can now amend his description of the transcendental: it is whatever rises above all genera and transcends all categories.

Following Father Wolter, we can here summarize Scotus' views in the following manner. For Scotus being is the first of the transcendentals. There are certain convertible or coextensive attributes of being which are also transcendentals, namely, one, true, and good. Furthermore, there are disjunctive attributes of being which are also transcendental as disjunctive. For example, finite-or-infinite and substance-or-accident. We can say that being is finite or infinite, substance or accident, and when we say this, we are thinking of being and saying something about it prior to its division into categories. Moreover, and here we touch on something of special interest to Scotus, pure perfections are transcendental. What he has in mind here are the divine attributes, concepts which are proper to God or common to God and creature, as the example of wisdom above. What can be predicated of God is a transcendental, that is, above the categories, precisely because God cannot be contained in the categories of

finite being. This is true both of such attributes as omniscience (predicable of God alone) and of wisdom (predicable of God and of some creatures).

The Univocity of Being. It can be said that little if anything of the foregoing goes contrary to what may be found, at least implicitly, in such a predecessor of Scotus as Aquinas. We come now to what turned out to be one of the most controversial claims in Scotus, namely, that "being" is predicated univocally of substance and accident and, indeed, of God and creature. The controversy on this doctrine has been long and intense, and it has most frequently been conducted as a conflict between religious orders. Because of the desire to triumph over an opponent, injustice has been done on both sides; when one looks into some of the literature on this controversy, it is very difficult to discover just precisely what the point of difference is supposed to be. Here we will first give a brief summary of the position according to which "being" is analogically, and not univocally, common to substance and accident or to God and creature. Then we will look at precisely what Scotus meant when he maintained the opposite. Most importantly, we shall ask if there is a real or only a verbal opposition.

In our chapter on Aquinas we went to some lengths to portray his conception of what an analogous term is, and we want now only to recall salient aspects of that presentation. A term is univocally common to many if it is predicated of them with exactly the same meaning. A term is equivocally common to many if it is predicated of them in such a way that it has a totally different and unrelated meaning in each occurrence. An analogous term is one which is neither univocal nor equivocal in the sense defined. For Aquinas a term is analogously common to many if it has the same *res significata*, and that reality is signified in different ways or modes in each occurrence. The great example, we recall, was "healthy." Health is the *res significata*, but when an animal, its coat, and its exercise are called healthy, that *res* is being signified in various related ways: the subject of health, the sign of health, the preserver of health. For Aquinas "being" is an analogous term as common to substance and accident. The reality signified is existence, *esse*, but the way in which it is signified varies insofar as substance is called being or quantity, quality, and so forth are called being. Substance is that which exists *per se*; the accidents are various modes of inherent being, of inherence or modification of substance. The parallel with the example of healthy is clear, the verdict expected: "being" is an analogous and not a univocal or equivocal term.

Scotus maintains that "being" is univocal. Now one thing is perfectly clear. If Scotus means by univocal what Aquinas means by it, then there is a contradiction. However, and obviously, if the two men have different understandings of what "univocal" means, the opposition between them is verbal and may cover an agreement when the verbal discrepancies are cleared up.

What now does Scotus mean by a univocal term? "Lest there be

any quibble about the term 'univocation,' by a univocal concept I mean one that is one in such a way that its unity suffices for contradiction when it is affirmed and denied of the same thing. It is sufficient as well for it to be a syllogistic middle term, such that the extremes united in the middle are one in such a way that the conclusion follows without the fallacy of equivocation." (*Oxon.*, 1, d. 3, q. 2, n. 5) While not unrelated to what Aquinas meant by a univocal term, these two stipulations of the univocal term that Scotus gives would not be for Aquinas peculiar to what he, Aquinas, means by a univocal term, but would be characteristics common to univocal and analogous terms. What Aquinas meant by a univocal term, again, is one which is common to several things according to exactly the same meaning. Can one conclude that Scotus has, by making quite clear what he meant by univocal, made it equally clear that there is no dispute between Aquinas and himself on this matter?

One is tempted to think so. It has often been pointed out that the adversaries Scotus had in mind when he insisted on the univocity of "being" as common to substance and accident, or on the univocity of terms common to God and creature, did not include Thomas Aquinas. Furthermore, it must be said that the way in which some opponents of Scotus on this matter define the unity of the analogous term is unintelligible in itself and not the position of Aquinas. Thus, many Thomists have held that the analogous term has many meanings which are similar to one another, but not in the sense that there is a common component of the various meanings, but rather in some obscure sense of similarity of structure. As a matter of fact, many attempted explanations of the unity of the various meanings of an analogous term are a postponement of an explanation rather than an explanation. For example, in discussing what is going on when "being" is predicated of God and creatures, many Thomists have said that what this means is that as existence is related to the created essence so is existence related to the divine essence, where the similarity is imperfect and where no term in the one proportion bears the same meaning as it has in its occurrence in the second proportion. But surely this comes down to saying that

there is no intelligible community of meaning between "being" said of God and "being" said of a creature. On this understanding the analogous term, as Scotus suggests, neither permits contradiction nor escapes the fallacy of equivocation.

It is necessary to admit the force of Scotus' opposition to the unity of the analogous term thus "explained." At the same time it must be stressed that the explanation Scotus opposes is not to be found in Aquinas. If there is a *ratio communis* of an analogous term, as there most assuredly is for Aquinas, this is so because of what Aquinas called the *res significata*. The common notion of being as predicable of substance and accidents is, for Aquinas, *habens esse* or *id quod habet esse*. The key thing signified by being, the *res significata*, is *esse*, existence, actuality. For Aquinas, when a thing is said to be, it is not said to be actuality but to be actual; what is is a mode of being, a way of being actual. Substance is actual in one way and accident in another, and when an accident is said to be, we must understand the actuality of substance. The kind of actuality substance has is the proper notion of being, and that proper notion is referred to when anything else is said to be actual. Thus, while it is necessary to insist that substantial existence is one thing and accidental existence another, the second kind of existence cannot be understood without reference to the first.

The great flaw in the explanation of the relationship between the two meanings of being, accidental and substantial, that Scotus rejects is that it imagines these as two unrelated kinds of being which in some mysterious way are related. There is nothing mysterious about the way they are related for Thomas Aquinas. Accidental being is a secondary mode of being which, in order to be explained, must be referred to substantial being. For Aquinas, then, what insures the unity of the meanings of "being" is, not *esse* considered abstractly, but the proper mode of existing which is substance. When "being" occurs as a middle term in a syllogism, consequently, it must be understood either as meaning substantial being directly or by way of reference. Undeniably this complicates discussions of contradiction when "being" occurs as the predicate in two propositions where the subject of one may be a substance and that of the other an accident. The point for Aquinas, however, is that substantial and accidental being are not distinct and autonomous modes; rather one is primary and the other secondary, and the secondary must always make reference to the primary.

Is this sufficient to bring Scotus and Aquinas into agreement? Unfortunately not. What Scotus wants to maintain is that "being" has a meaning which is quite independent of substantial and accidental modes of being. He wants being to

have a common notion or meaning such that there is no difference whatsoever in what is meant when one says that Socrates is a being and that red exists. Scotus does not deny, of course, that the substantial mode of existence is one thing and the accidental mode of existence another, but he feels that we can prescind from or ignore this further difference and understand "being" as meaning some utterly one and simple thing as said of substance and accident. One might feel that it was not necessary for him to make this claim in order to insure the possibility of contradiction and the avoidance of the fallacy of equivocation; nevertheless, Scotus clearly thought it was necessary and there is, consequently, a divergence between him and Aquinas on this point. This is true, despite the undeniable historical fact that Aquinas was not the opponent Scotus had in mind. Scotus' critique of such men as Henry of Ghent is well taken, and the Thomist who has a correct understanding of the behavior of analogous terms can accept that critique; furthermore, he can accept as necessary the stipulations Scotus lays down for "being" as common to substance and accident and for terms common to God and creature, namely, that they must have sufficient unity of meaning to permit significant contradiction, on the one hand, and to avoid the fallacy of equivocation, on the other.

What he must nonetheless recognize as a great difference between Scotus and Aquinas is the former's insistence that being has a common meaning which makes no reference to, say, substantial being and accidental being, and which is accordingly absolutely one and the same. The long prominence in Thomistic circles of discussion of "analogy of proper proportionality" should not be permitted to obscure this fundamental difference. "Analogy of proper proportionality" involves much confusion, but it is historically inaccurate to suggest that Aquinas and Scotus are as one, once this confusion is recognized and "analogy of proper proportionality" is put in proper perspective. Scotus wants "being" to be univocally common to substance and accident in such a way that "univocal" involves, besides the two characteristics he mentions, precisely the meaning it has for Aquinas. That is, Scotus wants "being" to be common to substance and accident in such a way that it has exactly the same meaning as predicated of both. This Aquinas emphatically denies. Scotus and Aquinas are unalterably opposed on the matter of the univocity of "being."

The Scotist's Reply. Father Allan Wolter has addressed himself to this question, and it may be well to examine his critique of analogous terms, a critique launched in behalf of Scotus and in an effort to clarify what Scotus had to say. Father Wolter asks us to consider the following syllogism: Whatever is divine is God. But the Mosaic law is divine; Therefore, the Mosaic law is God. This

sylllogism involves a fallacy of equivocation since the middle term is analogous and does not mean the same thing in its two occurrences. The Mosaic law can be said to be divine, not because it is God, but because it issues from God. Furthermore, and consequently, we can say of the Mosaic law both that it is divine and that it is not divine. If "good" is an analogous term, we can say of God that he is good and that he is not good. And so forth. All this is perfectly true; equally, it is true that it is this sort of thing that bothers Scotus. What is interesting here is that it indicates to us the nature of Scotus' moves. One might be inclined to say that since "divine" has diverse meanings, we are going to have to be careful in understanding sentences in which it occurs. And so too with "good," "wise," and so forth. It is this inclination that led to the development of the notion of analogous meaning. Scotus, on the other hand, bridles at this ongoing diversity of meaning in the terms of our language. He does not like it. He wants to insist that underneath this diversity there is a unity that is absolute and simple and unalloyed. That, for example, when we say that substance is and that an accident is, although on one level we must recognize that our verb means different things, on a more profound level it means exactly the same thing. Let us not ask how this would work out with the syllogism originally offered for our consideration, since even Scotus himself would be hard pressed to find a meaning for "divine" that enabled him to escape the fallacy of equivocation that syllogism involves. Let us not detain ourselves any further with the way in which "being" is common to substance and accident. I think it is fair to say that anyone who thinks he is making an identical claim when he asserts that men exist and that colors exist is confused about men and colors rather than in possession of a more profound grasp. What really motivates Scotus all along is the question of how our language can function when we talk about God. It is terms common to God and creature that are Scotus' true interest.

The Divine Names. The way in which "being" (and other names) is common to God and creature differs from the way it is common to substance and accident. In the latter case we want to say that one limited or finite mode of being is prior to others and must be referred to in understanding the secondary modes. This, of course, is the basis for claiming that the science of being as being is principally concerned with substance, the primary mode of finite being. The categorical modes of being are participated modes. Substance is not existence, but a mode of existing. When we turn to predicates common to God and creature, the tendency is to say that the created mode of perfection is participated and the divine mode is unparticipated. That is, Socrates is wise, but God is wisdom. We do not mean exactly the same thing when we say Socrates is

wise and God is wise. Wisdom is an accident in Socrates; it is something he comes to possess and, alas, might come to lose. To say that God is wise is to speak of what he is. Now "wise" is a word of our language, and it acquires the uses it has with reference to the things we know and are likely to speak about. If we assume that it signifies something we first encounter in creatures, when we say that God is wise we are speaking of him, denominating him, from his creatures. But we would also want to say that in this case what we first noticed and spoke of in creatures is an effect in them of God, a sharing or imitation in something that God is, not in the sense that he shares in it, but in the sense that he is it. Thus, we might want to say that, taking all this into account, the creature is denominated from God.

Scotus will object to this as the final word on the matter, since it clearly invokes the notion of analogy. Such a term as "wise" has different meanings as said of God and creature. What he wants is to insist on some core meaning of the term owing to which, given that meaning, the term means the exact same thing as said of creatures and God. In short, he wants the term to be univocal, and not only in his sense but in the sense univocity has for someone like Aquinas.

Scotus is confronted with something far more difficult here, since what univocity is being called upon to bridge is the difference between finite and infinite being. Let there be no mistake about it. Scotus is not inadequately aware of the radical difference between God and creature. He does not want to say that being in the sense of unparticipated being is univocally common to God and creature, or that being in the sense of participated being is common to God and creature. What he wants is a meaning of "being" which prescind from finitude and infinitude, a meaning owing to which we can understand "being" as univocally common to God and creature. In his search for that meaning he asks a most important question: Whence comes our knowledge of God?

The explanation of analogy that Scotus rejects maintained that "being" has two meanings, one of which was proper to God and the other of which was proper to creatures. Analogy was then invoked to speak of the relationship between these two proper meanings. But, Scotus asks, where do we get the meaning of "being" which is proper to God? Do we just have it? Is it given to us directly by God by way of some kind of illumination? The Aristotelian teaching on the origin of intellectual knowledge had it that all our concepts are formed against the background of our knowledge of sensible things. But sensible things are all of them finite, and thus it would seem that all our concepts have the limitation of finitude. How then can the terms signifying such concepts be predicated of

infinite being? Scotus is asking, in effect, how we can abstract knowledge of the infinite from knowledge of the finite. The effect here would then be so incommensurate with its cause that the claim seems at the least shaky.

Scotus maintains that we have no concept, gained as a result of abstraction, that expresses what is proper to infinite being. Any concepts expressive of something proper to God (for example, Pure Act, First Cause, Infinite Being) are the results of reasoning and not products of abstraction. Now if concepts expressive of what is proper to God are arrived at in the conclusion of an argument, there must be in the premises of the argument a concept functioning as middle term which bridges the gap between the finite and infinite. Thus, if we should reason to the view that God is infinite being, we are in the conclusion restricting or modifying the concept of being which figured in the premises, and in the premises our concept of being must be such that it is open to this modification. Thus, what is proper to God is infinite being, not being; likewise, what is proper to the creature is finite being, not being. Being, prescinding from finite and infinite, is common to them both. In short, the being proper to creatures and the being proper to God involve a common sort of being which is proper to neither.

At this point we must take into consideration a passage in Scotus that has often puzzled students. It is a passage in which Scotus denies that "being" is predicated univocally of all things. "As to the question, I grant that 'being' is not predicated univocally of all things. Neither is it predicated equivocally, for something is said to be predicated equivocally when those things of which it is predicated are not attributed to one another. For when they are attributed, then it is predicated analogically. But because it ('being') does not have one concept corresponding to it, it signifies all things essentially according to their proper perfection and simply equivocally according to the logician. But because those things which are signified are essentially attributed to one another ('being' is predicated) analogously, according to the metaphysicians." (*Metaph.*, 4, q. 1, n. 12) This passage had led to the gleeful claim that everything we have been concerned with earlier, while it is undeniably to be found in Scotus, is not his only, and perhaps not his final, view on the matter. Father Wolter does not find that the passage presents any particular difficulty. For him, all Scotus is admitting is that when "being" occurs in two propositions and in one of them bears the meaning of substantial being and in the other the meaning of accidental being, or when it occurs in two propositions and in one bears the meaning of finite being and in the other the meaning of infinite being, then the term "being" is not univocal but analogous. But, he goes on to say, this is true

because what is under consideration are two proper meanings of "being," proper meanings which imply and/or contain a common meaning, and when it is this common meaning we have in mind, then "being" is univocally common.

It seems to me that Father Wolter's interpretation is valuable. Considered all by itself, it may seem excessively subtle, but after all it is the teaching of the *Doctor Subtilis* that is being examined, and Father Wolter's subtle explanation has the great merit of retaining a unified Scotistic doctrine (and we must remember that the writings of Scotus were not composed over a great span of time). It is of course a further question whether one can accept the claim that being, owing to a common as opposed to proper meaning, is univocally common to substance and accident, or to God and creature.

C. Faith and Reason

Like the majority of medieval thinkers, Scotus was conscious of the limits of human reason and of the necessity of revelation. As a theologian he considered the teachings of the philosophers from the vantage point of his faith and, not surprisingly, found them inadequate on a number of crucial points. The question has been raised whether Scotus recognized anything like an independent or autonomous philosophy. Our earlier considerations will have prepared us to recognize this as an extremely tricky question. For the man of faith, needless to say, what the philosophers have had to say about man, his nature and destiny, and about God will seem insufficient. Insofar as the insufficiency and inadequacy are assessed from the viewpoint of faith no judgment of philosophical inadequacy is being passed. It could hardly be construed a fault on Aristotle's part if he failed to speak of man's supernatural destiny, of the Incarnation and the Trinity, and so forth. Philosophy is never enough for the Christian; for him what men can come to know by their natural powers must be supplemented by what God has chosen to reveal. Many who point this out go on to suggest that for many medievals, and Scotus is said to be one of them, philosophy as philosophy requires the influence of faith in order to achieve its own goals. This claim too is ambiguous, however, since it may mean simply that the Christian mentality should provide an impetus to do philosophy and that the faith of the believer gives extrinsic guidelines to philosophy. Sometimes the suggestion seems to be that for such a thinker as Scotus truths of faith are regulative within philosophy. That this suggestion is less than well-founded in the case of Scotus is best seen by consulting his work on the First Principle, *Tractatus de primo principio*. That this is a philosophical work by a believer is manifest from the beginning when Scotus invokes the aid of God in

his effort to show by reason alone that God is what he has revealed himself to be in Exodus, namely, being. The whole direction of the work is such that it is clear that the arguments formulated are taken to be conclusive for establishing what is attributed to God, conclusive in such a way that none of them makes any appeal to faith or revelation. Scotus' attitude as a believer provides the personal setting for this effort, but the whole point of the work is that one who follows the arguments must, on the basis of the arguments, assent to the conclusions reached. In the same way, many of Scotus' judgments of the inadequacy of philosophical positions are such that he wants to show that inadequacy on philosophical grounds. That his personal motivation, that the starting point of his own suspicion of inadequacy, may have been his Christian faith is neither here nor there in assessing the validity of such attempts to show philosophical inadequacy. The relations between belief and reasoning are various and nuanced in the most straightforward of medieval thinkers; we should expect that they will be yet more subtle in the case of Scotus, as indeed they are. On the face of it, almost *a priori*, it is unlikely that Scotus would have held to a simplistic position according to which philosophizing is theologizing. A few soundings in his works suffice to convince that here as elsewhere he is incorrigibly complicated and nuanced, and this in response to the phenomenon in question.

Bibliographical Note

The Vives Edition of the *Opera omnia* in twenty-six volumes will eventually give way to critical editions. Father Balic and the Scotist Commission began to publish a critical edition in 1950 which is projected to something like half a hundred volumes. Muller has published a critical edition of the work on the First Principle: *Ioannis Duns Scoti Tractatus De Primo Principio* (Freibourg, 1941). A. Wolter, *Duns Scotus: Philosophical Writings* (Edinburgh, London, 1962). Wolter's article on Duns Scotus in *The Encyclopedia of Philosophy* (ed. Paul Edwards) is recommended, as is his classic, *Transcendentals and Their Function in the Metaphysics of Duns Scotus* (St. Bonaventure, N.Y., 1946). Cf. Roy Effler, O.F.M., *John Duns Scotus and the Principle "Omne quod movetur ab alio movetur"* (St. Bonaventure, 1962). Volume three of *Studies in Philosophy and the History of Philosophy, John Duns Scotus, 1265-1965*, edited by John K. Ryan and Bernadine Bonansea (Washington, 1965) is an important collection of pieces on Scotus. The proceedings of the Scotus Congress, held at Oxford, Edinburgh, and Duns in September, 1966, soon to be published, will be of great usefulness.

Chapter III

William of Ockham

A. The Man and His Work

William was born in Ockham, near London, perhaps in the year 1285, but certainly between 1280-1290. No definite information about his early life is had except that he joined the Franciscan Order while quite young and began his studies at Oxford, where he received his master of theology degree before 1320. The story that he studied under Scotus at Oxford seems just that, a story. This is not to say that Ockham was not heavily influenced by the writings of Scotus. Ockham's career at Oxford was cut short; at the completion of his lectures on the *Sentences* he was not awarded a teaching chair at the university. Indeed, the chancellor of the university accused Ockham of teaching dangerous and heretical doctrines. This accusation was made at the papal court in Avignon in 1323, and William was summoned there in 1324 to answer the charges. He spent four years at Avignon while a commission of theologians considered his case. During these years William continued to write, both treatises and commentaries, and he may have entertained quodlibetal questions at Avignon. The assumption is that he spent these years at the Franciscan convent there. It was during this period that William was caught up in the debate within the order concerning poverty. The dispute soon became one between the master general, with whom William sided, and the pope, John XXII. When the dissenting Franciscans fled Avignon to avoid papal censure, they sought the protection of the German emperor, Louis the Bavarian. It should be said that Ockham's opposition to the Pope was based on his judgment that the current Pope was in conflict with earlier papal pronouncements on the matter in question. Unfortunately, the appeal to the Emperor made the matter political as well as theological. In 1328 the Pope convoked a general chapter of the order and demanded that the Franciscans elect a new general. It was at this point that William fled Avignon in company with the general and other leaders of the opposition to the Pope. The upshot was that William and the others were excommunicated both by the Pope and their fellow Franciscans. Ockham settled in Munich and, after John XXII's death in 1334, continued his opposition to the Avignon popes. In 1347 Louis of Bavaria died, and Ockham sought reconciliation with the pope and his order. A formula of submission has survived, but we do not know the outcome of this effort at reconciliation. Ockham died in 1349 and was buried in the Franciscan church in Munich. He is thought to have been a victim of the Black Plague.

Ockham's difficulty with the papacy led to a series of writings on political themes; our interest is in the philosophical and theological writings, all of which were perhaps written during his Oxford and Avignon periods. His theological works include, of course, a commentary on the *Sentences*, quodlibetal questions, writings on the Eucharist and predestination. He commented on Aristotle's *Physics* and composed a number of allied works. Of particular interest are his writings on logic, which consist of commentaries on Aristotle and also various independent logical treatises.

B. Knowledge

A fair notion of Ockham's importance can be had by considering his rejection of previous medieval solutions to the problem of universals. What he has to say here has ramifications throughout his doctrine, ramifications which set him definitively aside from his great predecessors and put philosophy and theology on a new track, the *via moderna*.

The problem of universals, as we have seen, requires for its solution the introduction of logical, epistemological, and ontological considerations. As the problem was transmitted to the Middle Ages by Porphyry, the options for solution were the Platonic and Aristotelian. At first blush the choice may seem to be between a theory which admits more than individuals and one that does not; rather, it should be said that the difference resides in Plato's admission of a class of individuals beyond sensible, material individuals, namely, the Ideas, and that Aristotle's criticism of Plato comes down to a rejection of the imposition on *any* extramental individuals of the mental relations established by our act of knowing. If we confine ourselves to the Aristotelian viewpoint, one according to which the world is a world of individuals (to be is to be one), the further question arises as to what it is in individuals that permits us to form universal concepts and that in turn can lead to asking after the grounds of real similarities among existing individuals. There seem to be certain checks and controls exercised by individuals over our tendency to generalize. In one sense, it seems possible to group individuals in any way that pleases us, whatever their "real" similarities or dissimilarities (for example everything in this room), but, on the other hand, with respect to certain kinds of questions that we address to reality, our generalizing or grouping seems constrained by the way things are. The range of questions relevant to the Aristotelian conception of demonstrative science and the Porphyrian tree exercised great influence over medieval discussions of the problem of universals, a fact that accounts both for the

undeniable precision of many of the solutions proposed as well as for the suspicion that a number of allied and difficult issues are being overlooked.

The problem of universals was most often discussed in terms of common nouns like "man" which occur as predicate in such a list of statements as the following: (1) Socrates is man, (2) Plato is man, (3) Aristotle is man, and so forth. If the predicate in these sentences does not refer to some individual other than those denoted by the proper names which function as subjects, it would follow that what "man" means is found in Socrates, Plato, Aristotle, and so forth. That is, we employ "Socrates" and "man" to talk about the same thing, and yet when we use "man" to talk about that thing we are not thinking of it in the same way as when we employ "Socrates." That is, sentence (1) above is not a roundabout way of saying, "Socrates is Socrates." If it were, of course, sentence (2) could be taken to mean that Plato is Socrates. As this was thought about, the tendency was to say that such common nouns express, not what is peculiar to Socrates and Plato, but what is common to them. But, obviously, "man" does not mean "what is common to Socrates and Plato," and this aspect of community (another term for universality) came to be located in the mind of the knower. "To be said of many or to be said of many numerically different things," which are less and more specific ways of explicating universality, were thought to be what happens to a nature, say human nature, as it is known by us. This led Aquinas and Scotus, though with notable differences, to the notion of a common nature, or rather to a nature not yet modified by universality *or by individuality*. Human nature is individuated in Socrates, Plato, and so forth, and it is universalized, that is, taken to be some one thing that is predicable of many individuals. Once more, "to be predicable of many individuals" is no part of the meaning of "man" it is not a constitutive of human nature, and as an accident of that nature universality was seen as something conferred by the human mind. Individuation, on the other hand, while equally accidental to human nature, for Aquinas is conferred not by mind but by matter with quantitative designations.

If universality as such was considered to be not part of the furniture of the extramental world, this is not to say that the natures to which it is attached in our knowing are not really different. The difference between a man and a horse, for example, was not considered to be the result of our mode of knowing; rather the difference is there awaiting our recognition. Another sort of difficulty arose in terms of an ascent of the Porphyrian tree, one that led to a difference between Scotus and Thomists, if not precisely between Scotus and Aquinas. Once we notice that the individual referred to by "Socrates" and man in our sentence (1) above can also be referred to by a number of other predicates, like

"animal," "living thing," and "substance," predicates which are arranged hierarchically on the Porphyrian tree, the question arises as to what in the individual permits this variety of cognitive "fixes" on it, fixes which are explained in terms of greater and lesser universality. Are they grounded on real slices in the thing, so to speak, on formal differences, or do they rather attest to our mode of knowing, which proceeds from confused to less confused knowledge of things?

Perhaps these few words will be sufficient to recall the context of the problem. The solution that has been dubbed "moderate realism," Aquinas' solution to the problem of universals, would hold, from a logical point of view, that "predicability of many" is a logical relation consequent on our mode of knowing, from an epistemological point of view, that the concept of human nature, to which universality attaches, answers, with respect to its content, to a nature really found in various individuals, but that our concept prescind from the marks of individuation and thus is not commensurate with individuals as individuals. Finally, from a metaphysical point of view the source of such a nature really found in many individuals would be sought in God, and reference would be made to the divine Ideas since the various natures in the created world are different expressions of God's awareness of his own imitability. What now of Ockham?

We cannot avoid the fact that Ockham's position on this problem is described as nominalist, but we must emphasize that the use of such tags does not provide us with a clear and distinct notion of nominalism which can then be blithely applied to Ockham. It is far wiser, having noted that Ockham is called a nominalist, to go on to examine what he had to say, with the idea that the results of such an inquiry will provide us with at least one sense of "nominalism." There is little doubt that Ockham wishes to discard the epistemological and ontological dimensions of earlier medieval discussions of the status of universals. In a fashion that is, or at least has become, characteristic of his countrymen in philosophy, Ockham exhibits an empiricist bent that leads him to want to ground whatever he says on individuals and his experience of them. Here we may mention another cliché of treatments of Ockham, namely, that famous razor. The formulation that is so much cited, *entia non sunt multiplicanda praeter necessitatem*, does not, as it happens, occur anywhere in Ockham's writings, but that is really of little importance. His spirit is summed up in that dictum: let us never introduce more than is required for an explanation. Parsimony in the apparatus of explanation is a mark of Ockham fully evident in his discussion of universals.

In his criticism of the position just sketched Ockham goes right for the jugular, that of a human nature really present in many individuals. Why should we speak of that nature as in some way awaiting individuation as it awaits the universality conferred by the mind? The world is a world of individuals through and through; the nature is not something distinct, in reality, from its individuating characteristics. If the nature predicated of Socrates does not include those individuating characteristics without which Socrates is not, well then we are saying of Socrates that he is what he is not. The *natura absolute considerata* of which Aquinas speaks, the nature regarded without reference to universality or particularity, such as human nature in itself, is, if not a fiction for Ockham, certainly nothing he considers deserving of serious attention. The crux of the matter for him seems to be that he can proceed very well without making any reference to a *natura absolute considerata* (it is this that commentators have in mind when they speak of the "common nature"). Universality and individuation can be handled as purely logical problems; they attach to language and consequently to thought, and the problem of universals becomes the logical problem of the behavior of common and proper names. The logical explanation should make no reference to unindividuated natures which are somehow in individuals (there are no such natures); consequently, the logical explanation need not bother itself with reference to the mind's ability to abstract such a nature from its individuating conditions. There is, nonetheless, an epistemological problem: How does our mind form universal notions against the background of its experience with a world of individuals?

With Ockham it is not so much the terminology of explanation that changes as the sense of the terms used. Thus Ockham will speak of abstractive knowledge, which he contrasts with intuitive knowledge, and we must achieve an understanding of this distinction. By intuitive cognition Ockham means a judgment whereby it is evident that the object exists or does not exist. He does not mean by intuitive knowledge sense perception, since the knowledge involved is intellectual, but the object of intuition is present existent fact. This description must be modified in two ways. First, it is taken to include what might be called negative facts, for example, that something does not exist or that it does not have such and such a quality. Second, Ockham admits the logical possibility of an intuition to which no existent fact corresponds, a possibility allowed to accommodate the divine omnipotence, since Ockham holds that God could cause us to have an intuition of a nonexistent object. This possibility is taken to have been realized in cases of prophecy, when God causes men to have intuitive knowledge of facts which do not yet exist. A further

modification of the meaning of intuition for Ockham must be introduced when we find that he includes mental states among the objects of intuitive cognition.

Abstractive knowledge is described by way of contrasting it with intuitive cognition. A first description of abstractive cognition, accordingly, would be that it is such that it provides no evident knowledge that its object exists or does not exist. Further, abstractive knowledge is said to follow on intuitive cognition. If I have known the color of a particular rose, I can later think of that rose and its color even without the presence of the fact. Such cognition does not permit me to say that the rose exists, however, since it may have ceased to do so. Abstract knowledge, for Ockham, is the pale reflection of intuitive cognition.

It is the concept's loss of sharpness and particularity in abstract cognition that provides Ockham with his understanding of the nature of universality. The remembered and abstractly thought-of rose, in losing the particular notes of the once-seen rose, involves a concept answering to the term "rose" that is vague enough to apply to several roses. Notice that this explanation requires no distinction in the particular rose between its thisness and its rosehood; that transition is explained solely in terms of types of human thought and in no wise prejudices a world of individuals through and through. Abstract knowledge expressed in judgments involving universal concepts makes no assertion with respect to factual existence; only when linked with a judgment of intuitive cognition is such a claim involved. Ockham's treatment of supposition corresponds to this duality.

Has Ockham really bypassed the solution to the problem of universals dubbed moderate realism? This question is bound to impose itself when we find that when Ockham asks why in abstractive cognition we form a concept of rose that is not tied down to a particular rose as known in intuitive cognition and which can be applied to many and indeed all roses, he replies that the concept fastens on to the similarity between actually existing individual roses. Now this similarity is real, and that was the fundamental motivation for the introduction of the so-called common nature, that is, the *natura absolute considerata*. Ockham will reply that he is not appealing to a common nature, but of course, as Aquinas' presents the solution of moderate realism, the nature is not actually common or universal in individuals; rather, the real similarity among individuals is the foundation in reality of the one nature expressed in the concept to which attaches the relation of predicability to many. Ockham's demur here may seem to us to be anything but persuasive, and yet it would be quite mistaken to conclude that Ockham's nominalism turns out to be identical with Aquinas'

moderate realism. "Every man is risibile" has interest and importance for Ockham only because it is a roundabout way of saying that Socrates is risible, Plato is risible, Aristotle is risible, and so on. For a thinker like Aquinas the universal judgment would not be formed if we did not have experience of some men, but the universal judgment is not merely a shorthand expression for an open series of singular and contingent judgments. Ockham may finally have confessed that universals constitute an occult problem, but rather than construe this confession of difficulty with a wavering toward moderate realism, we must see that it goes hand in hand with an attitude toward knowledge, toward contingency and necessity, which tends in the direction of a radical empiricism of individual fact and away from an empirically grounded thought whose *telos* is universal and necessary judgments which involve a qualitative move beyond judgments of particular existent fact. It is not without reason that commentators have expressed Ockham's view of the universal judgment as an open sentence, a propositional function, according to which "Every man is risible" becomes "For all x, if x is a man, x is risible," a translation that makes no ontological commitment to the effect that there are values for x, and which suggests that if there are such values, the meaning of the universal statement is the sum total of relevant judgments of individual fact.

C. Logic

The new or terminist logic of the fourteenth century, of which Ockham is an adherent and not the founder, does not differ from earlier medieval logic in being divorced from metaphysical and epistemological considerations. Indeed, as we shall see, many of the moves that Ockham makes in logical theory are dictated by his metaphysical and epistemological views.

Ockham gives three senses of "term" and concludes that its precise meaning is such that it is a term which can be either the subject or the predicate of a proposition. "Term" requires a broader sense when we wish to distinguish between categorematic and syncategorematic terms. Of the former, Ockham says that they have a definite and fixed meaning, and his examples are "man," "animal," and "whiteness." The latter, syncategorematic terms, are exemplified by "every," "none," "some," "only," and so forth. These are said not to have any definite and fixed meaning, nor do they signify things distinct from those signified by categorematic terms. "Every" attached to "man" makes the term man stand for all men; when attached to "stone" it stands for all stones.

In distinguishing between absolute and connotative terms Ockham wants to draw a difference between those which signify primarily and have no secondary signification and those on the other hand that have a primary and secondary signification. An example of the latter, of a connotative term, is "white." If we ask what "white" means, we may be told that it means "something that has whiteness," and thus it means both the quality and its bearer. Where Ockham's ontological penchants become clear is in his discussion of absolute terms, terms that signify substances. He does not want to say that "man" signifies the nature humanity and its bearer, since this conjures up for him the whole business of the common nature. This leads him to the view that "man" signifies all men and "animal" signifies cows, horses, asses, men, and so on. Away with animality and humanity as *id quod nomen significat*, and we are left ultimately only with individuals. Ockham will insist on the importance of supposition, a doctrine that may be seen to foreshadow contemporary distinctions between meaning and reference, but on crucial occasions, and influenced by his nominalism, Ockham comes very close to identifying meaning and reference. It seems difficult to see that he is doing anything else when he says that "animal" means what it can stand for. He actually defines supposition as the use of a term in a proposition for some thing or things, and normally for the things it signifies. And although he employs the standard distinction between material supposition (for example, "Man is a noun"), simple supposition (for example, "Man is a species"), and personal supposition (for example, "Every man is an animal"), when he speaks of personal supposition, which is the chief kind, he defines it as obtaining when a term stands for what it signifies and is used in its significative function. His doctrine of meaning becomes his doctrine of supposition or reference; logically, what absolute terms mean are individual things because, metaphysically, the only things that are are individual substances and individual qualities. We may no longer find such logical atomism quaint, but it must be insisted that Ockham represents a significant break with the philosophy of logic of the thirteenth century and that, in his case, his logic goes hand in hand with his metaphysics. It is not for us to ask now whether the logic Ockham clearly foreshadows is accidentally or essentially linked to an impoverished metaphysics.

D. Metaphysics

The criticism of Ockham is not that he has no metaphysics but that he has an excessively diminished one. The basis for this criticism can be seen by asking how Ockham deals with the questions of classical metaphysics, but in this

sketch we will confine ourselves to what he has to say about the subject of metaphysics.

That subject, of course, is being as being. Ockham's general approach to metaphysics is by way of criticism of the views of his predecessors, views he conceives to be fraught with confusions he is able to avoid. There is no surprise nor indeed novelty in Ockham's reminder that being as being is not some individual entity apart from other beings, but what is distinctive of him is his contention that being is univocal in such a way that it applies in a wholly undifferentiated way to whatever is and, of course, whatever is is an individual. There is nothing whatsoever that mediates the maximum universality of the term "being" and this or that individual to which it can be applied. By the same token, it stands for accidents in a direct and unmediated way. This is what Ockham means by saying that it is predicated *in quid*. Given our earlier discussions, we can assume that for Ockham the meaning of "being" is all actual existents, and no doubt this is at the basis of his rejection of the distinction between essence and existence. Ockham's simplistic interpretation of the views of others goes hand in hand with his contemptuous disdain. It is difficult to take too seriously one who reminds others that existence must not be understood as a qualification of what does not exist, for he thereby seems to suggest that "essence" means nothing and "existence" existence and that his predecessors had held that being (*ens*) is a nothing which exists. His treatment of the distinction between actuality and potentiality is equally profound; a possible being, he seems to think, was taken to be an existent nonexistent. His prescription for avoiding such undocumented paradoxes is that we learn that actuality and possibility are modalities of statements and not of things. This, while possessing the allure of a partial truth, since the characteristics of modal propositions are not precisely the characteristics of things, far from providing an alternative answer to a question asked earlier, fails even to raise the question. The principle of parsimony is here running amok. To say that Socrates can pitch a curve is not the same as to say that it is possible that "Socrates is pitching a curve" is true. Ockham may seem to be saying that the latter is not the former, but what he is really saying is that the former need be taken to mean exactly the same thing as the latter. There remains little doubt as to who is confusing modalities of statements and characteristics of things, Ockham or his unnamed adversaries.

Once the univocity of being is understood in terms of a universe of things wholly undifferentiated in terms of their being, a veritable flatland of reality where the only solid truth is that an individual thing is itself, Ockham hurries

on to show that this enables us to settle the question of God's knowability. There is no common feature in reality that the term "being" (or, in a more limited range, any other general term) is thought to pick out. Now, since this does not impede ordinary linguistic usage, being only an inadequate account of it, since we can go right on saying of Socrates and of Plato and so forth that they are men while denying that there is any foundation in reality other than their individuality, that is, their otherness, for so speaking, what is to prevent us from saying that God can be said to be, even though there is absolutely no common feature between God and anything else? In this Ockhamian wonderland everything is just itself and no other, whether we be speaking of creatures or of creatures and God. Lest our tone seem to be too strident here, we should say that it is one thing to discuss the behavior of common nouns without raising the allied epistemological and ontological questions -- this could result from mere boredom or lack of intellectual curiosity. It is quite another thing to go into a discussion of those epistemological and ontological questions and to say little more than that common nouns are common because they are common and then to lament the confusions of one's predecessors. Ockham does the second and quite different thing. And it is important to be very clear about the metaphysical results of his thought and, once we are clear about them, to lament them. A world where each thing is only itself and no other, where its very otherness is said to be its similarity with others -- well, this is a confused and paradoxical terrain. If we excuse ourselves from any consideration of Ockham's discussions of proofs of God's existence and proofs that God has certain attributes, we do so because his apparently affirmative results are such only when terms bear the peculiar Pickwickian sense according to which Ockham can predicate the same term of two entities while denying that he is doing anything other than pointing to two utterly different things. It is only when we forget Ockham's nominalism -- as he on occasion was prone to do himself -- that the results of his inquiries amount to something other than the melancholy one just stated.

Bibliographical Note

A very useful anthology of Ockham's writings together with a helpful introduction is *Ockham: Philosophical Writings*, edited and translated by P. Boehner (New York, 1957); Boehner also edited the *Summa totius logicae* (St. Bonaventure, 1951-1954). See E. A. Moody's edition of *Expositio in librum Porphyrii de praedicabilibus* (St. Bonaventure, 1965). See too P. Boehner, *Collected Articles on Ockham* (St. Bonaventure, 1956) and E. A. Moody, *The Logic of William of Ockham* (New York, 1935).

Index

A

- Abbey of St. Denys, 142, 143
- Abbey of St. Gildas, 143
- Abelard, Peter, 106, 119, 140-156, 160, 175, 182, 186, 189, 190, 196, 198, 199, 216, 249
- Abraham, 97
- Abstraction, 64, 65, 69, 73, 163, 164, 188, 208, 272, 273, 280, 302, 303, 308, 319-321, 371, 372
- Accidents, 57, 58, 73, 185, 223
- Adam, 40
- Adelhard of Bath, 203
- Adeodatus, 9, 24
- Adoptionist heresy, 89
- Aelbert, 86
- Agent intellect, 28, 29, 222, 224, 234, 249, 273
- Agobard of Lyon, 90
- Alan of Lille, 201, 202
- Alaric, 40, 41, 42
- Albert the Great, 9, 49, 234, 238-245 253, 285, 288, 338, 346
- Albigensianism, 202
- Alchemy, 246
- Aicher of Clairvaux, 200, 201
- Alcuin, 79, 83-93, 94, 95, 108, 113. 121, 201, 211
- Alexander of Aphrodisias, 69, 222
- Alexander of Hales, 229, 233, 234, 235
- Alexander, Pope, 288
- Alfarabi, 204, 222-223 226
- Algazel, 204, 239, 245
- Alkindi, 204, 221-222
- Ambiguity, 67, 68. *See also* Analogy
- Ambrose, 10, 55, 244
- Ammonius, 144, 191, 222
- Analogy, 310, 324-326, 329, 358, 359, 360, 361, 362, 363
argument from, 125
- Anamnesis, 23
- Anarchos, 100

- Anaxagoras, 60
- Anderson, J. F., 341
- Angels, 44, 230, 284, 285
- *anima est quodammodo omnia*, 319-317
- Anselm of Canterbury, 63, 119, 120, 122-139, 159
- Anselm of Laon, 106, 140, 141, 142, 162
- Apologists, 4
- Appellatives, 148
- Apprehension, simple, 187, 320
- Aquinas, Thomas, 8, 9, 28, 47, 48, 49, 55, 78, 131, 133, 152, 153, 186, 200, 204, 218, 220, 225, 234, 238, 240, 244, 250, 253, 255, 261, 267, 276, 285, 287-342, 343, 344, 345, 346, 358, 369, 370, 372
- Arabian thought. *See* Islamic thought
- Arianism, 52
- Aristotle, 4, 13, 28, 47, 52, 53, 54, 57, 65, 66, 69, 70, 74, 77, 80, 84, 110, 113, 114, 120, 143, 144, 145, 151, 159, 187, 195, 196, 203, 204, 206, 215, 217, 220, 222, 226, 227, 229, 235, 239, 245, 248, 249, 252, 253, 264, 265, 266, 271, 274, 283, 289, 290, 291, 292, 305, 315, 332, 336, 337, 338, 339, 340, 341, 343, 344, 352, 354
- Arithmetic of Boethius, 65
- Arnobius, 4
- Arnulph, Archbishop, 112
- Article, 218
- Arts, liberal, 12, 13-15, 16, 76, 77, 78, 87, 91, 95, 109, 157, 158, 160, 162, 175, 191, 192, 193, 210, 211, 217, 255, 268, 301, 302
- Astralabe, 142
- Astrology, 246
- Astronomy, 251, 253
- Athanasius, 5
- Auctores, 158
- Augustine, 5, 8, 9-48, 55, 74, 77, 79, 89, 92, 108, 112, 113, 121, 123, 124, 133, 136, 159, 178, 183, 193, 194, 201, 206, 211, 215, 233, 249, 253, 256, 260, 268, 271, 276, 301, 338
- Authority, 29, 33, 55, 96, 97, 98, 194, 219, 248, 249
- Averroes 204, 220, 225-226,
- Avicbron, 120, 204, 206, 227, 230
- Avicenna, 204, 223-225, 226, 239, 249
- Avignon, 367, 368

B

- Baccalarius, 218
- Bacon, Roger, 235, 246-254, 346, 351
- Balic, C., 366
- Baron, R., 194
- Barrett, H. M., 54, 59, 74
- Barth, K., 139
- Basil, 5
- Baur, L., 204 n, 205, 206, 211, 212, 235, 237
- Becket, Thomas, 187
- Becoming, accidental and substantial, 312
- Bede, Venerable, 78, 79, 83, 85, 87, 89, 168
- Being, as being, 321-324, 355
 - being and being known, 105
 - being, finite and infinite, 355, 357, 362, 363
 - being, hierarchy of, 104
 - being and nonbeing, 99
 - being, univocity of, 357-362. *See also* Essence, Esse, Existence, Categories
- Believing a truth, 295, 296, 297, 298
- Benedict, St., 195, 198
- Benedict Biscop, 85
- Bernard Bishop, 158
- Bernard of Chartres, 158, 159, 160, 167, 175
- Bernard of Clairvaux, 119, 120, 140, 142, 143, 157, 160, 162, 173, 195-197, 198, 201
- Berengar of Tours, 119, 158
- Bett, H., 106
- Bettoni, E., 286
- Bird, O., 156
- Blackwell, B., 341
- Blessedness, 61
- Boas, C., 285
- Bobik, J., 327, 341
- Boehner, P., 156, 376
- Boethius, 47, 52-75, 76, 79, 83, 87, 108, 110-111, 113, 121, 143, 144, 145, 147, 159, 161, 162, 163, 168, 169, 170, 173, 175, 179, 181, 183, 184, 186, 190, 191, 201, 204, 207, 208, 211, 215, 230, 232, 234, 239, 244, 265
- Boetius of Dacia, 344, 345
- Bologna, University of, 216, 238

- Bonansea, B., 366
- Bonaventure, 9, 235 244 253 255-286, 288, 296, 301, 338, 343, 344, 346
- Borgnet, 245
- Boswell, 53
- Bourke, V., 341
- Bovo of Corvey, 111
- *Breviloquium*, 255, 257, 263, 266
- Brewer, J. S., 254
- Bridges, J. H., 254
- Brunet, H., 161, 182
- Buelow, C., 212
- Buttimer, C. H., 194
- Byrne, P., 285

C

- Callus, D. A., 237
- Cambridge, University of, 354
- Candidus of Fulda, 92
- Canon law, 249
- *Capax dei*, 353
- Capella, Martianus, 13, 59, 77, 86, 94, 96, 168, 195
- Capitulary of Charlemagne, 84-85
- Cappoyns, M., 106
- Cardinal virtues, 334, 337
- Carolingian Renaissance, 84, 85, 86, 107
- Carre, M., 156
- Cassiodorus, 54, 76-80, 86, 91, 201, 203, 210, 211
- Categories, 58, 102, 356
- *Categories* of Aristotle, 52, 58, 65, 87, 108, 114, 309, 354
- Catechetical School of Alexandria, 4
- Cathedral schools, 84-86, 87, 120, 216
- *Causae seminales*, 176
- Cause, 62, 100, 170, 171, 176, 207, 268, 302, 303
- Certitude, 243, 244, 258 259, 262, 274, 275, 296, 297
- Chalcidius, 66, 110, 159, 201
- Chance, 60, 61, 62, 63, 265
- Changeable things, 36
- Charlemagne, 46, 83, 84

- Charles the Bald, 94, 107
- Chartres, School of, 120, 157-188, 193. 202
- Chenu, M.-D., 161
- Christ the Teacher, 27
- Christian Philosophy, 33
- Christianity, 33 and passim
- Chroust, A. M., 212
- Cicchetti, A., 139
- Cicero, 10, 13, 110, 168, 211
- Citeaux, Abbey of, 195, 196
- Clarenbald of Arras, 182-186
- Clement of Alexandria, 4, 5, 6, 21
- Clement IV, 247
- Clement the Irishman, 88
- Clerval, A., 161, 182
- Cluny, Monastery of, 119, 143. 144. 196, 198
- *Collationes* of Scotus, 354
collectio, 165
- Collins, J., 74
- Community, kinds of, 67
- Concept, 68, 69, 109, 150, 166. *See also* Intellectus, ratio
- Condemnation of 1277, 346
- *Confessions* of Augustine, 9, 10, 12, 14, 15, 19, 20, 33
- Cosmology, 160
- *Consolation of Philosophy*, 53, 54, 59-64, 70, 71, 73, 265
- Contemplation, 194, 195, 196, 290
- Contingent things, 77, 78, 114, 277
- Contingent and necessary existence, 132 ff., 372, 373
- *Contra academicos*, 12, 18, 20, 29
- Cornificians, 161, 167
- Cornog, W. H., 202
- Correns, P., 212
- Council of Lyons, 290
- Council of Paris, 95
- Council of Sens, 143, 155
- Council of Soissons, 142, 175
- Courcelle, P., 74, 75, 80
- *Credible / scibile*, 257, 261, 263
- Crowley, T., 254
- Cusa, Nicholas of, 51

D

- Dame Philosophy, 60, 61, 62, 72
- *De beata vita*, 12, 20
- *De civitate dei*, 12, 15, 16-18, 19, 40-47
- *De decem preceptis*, 255
- *De doctrina christiana*, 12, 77
- *De duabus naturis*, 54
- *De ente et essentia*, 225
- de Gandillac, 51
- *De hebdomadibus*, 54, 55, 289, 327
- *De immortalitate animae*, 12, 31-2
- *De intellectu* (Alkindi), 222
- *De libero arbitria*, 12, 23, 34
- *De luce seu de incoatione formarum*, 235
- *De magistro*, 12, 22, 24-29, 89
- *De musica*, 12
- *De ordine*, 12, 14, 19, 20, 30
- *De ordine emanandi*, 236
- *De pulchro et opto*, 12
- *De quantitate animae*, 12, 14, 23, 24
- *De quinque essentiis*, 222
- *De reductione artium ad theologium*, 255, 258, 268, 269 ff.
- De Rijk, 156
- *De syllogismo hypothetico*, 53, 54
- *De tribus quaestionibus*, 263, 266
- *De trinitate* (Augustine), 12, 15, 24, 28, 35, 40
- *De trinitate* (Boethius), 54, 64, 65, 70, 73, 230, 289, 293, 299, 300, 302, 304
- *De unica forma omnium*, 236
- *De universo*, 230, 232
- *De vera religione*, 12, 34
- *De veritate pro positionis*, 237
- De Vorges, D., 134, 139
- De Wolf, M., 165, 186
- Deane, S. N., 139
- Deanesley, M., 13
- Deferrari, R. J., 194
- Definition, 303, 304

- Delhay, P., 195
- Delorme, F. M., 254
- Del Pra, M., 106, 156
- Democritus, 33
- Demonstrative science, 253, 277, 303, 304
- Demiurge, 170, 171
- Denis the Areopagite, 6, 49-51, 94, 99, 101, 102, 121, 142, 189, 201, 239, 289, 293, 329
- Denis the Carthusian, 49
- Denomination, 137
- Denominatives, 136 ff.
- Descartes, R., 3, 35, 107 n, 133
- Destrez, J., 341
- *Destructio destructionis*, 226
- Dialectic, 87, 309. *See also* Logic
- Dialectician vs Antidialectician, 119, 133-136, 151, 160, 173-174, 183
- Difference, numerical, 57
- Discovery, 191, 298
- Dioscorus, 31
- *Diversae quaestiones* 83, 21
- *Diversum est esse et id quod est*, 185, 327. *See also* *De hebdomadibus*
- Divine monad, 6, 201
- Divine nature, 38. *See also* God, knowledge of
- Dods, M., 48
- Donatist heresy, 11, 12
- Donatus, 87
- Doubt, 35, 296
- Duhem, P., 175, 182, 232, 233, 239
- Dungel the Recluse, 88
- Duns Scotus. *See* Scotus, Duns
- Durandus, 351
- Durr, K., 75

E

- Eadmer, 122
- Edessa, Nestorian School of, 203
- Edwards, P., 366
- Effect, 62
- Effler, B., 366

- Eghert, 86
- Eleatic, 60
- Elements, four, 163, 183, 206, 207
- *Elements of Theology*, 289, 339
- Eliot, T. S., 160
- Eloquence, 168, 205
- Emanation, 50, 103, 222, 223, 224, 226, 236, 272, 276, 338
- Emden, A. B., 161, 228
- *Enneads*, 221, 226
- *Ens commune*, 319, 323
- *Ens reale / ens rationis*, 307, 327
- *Ens verum*, 327
- Eric of Auxerre. *See* Heiric
- Erigena. *See* Scotus Erigena
- Errors of Philosophers, 264, 265
- *Esse*, 73, 99, 131, 223, 231, 320, 326, 358, 360
- *Esse Subsistense*, 331
- Essence, 73, 134, 151, 223
and existence, 223, 224, 225, 231, 232, 326-328
- Eternal City, 42 ff.
- Eternal law, 46
- Eternal truths, 28, 29, 32, 36, 37, 278
- Eternity, 39
- Eternity of world, 171, 222, 230, 240, 245, 282, 283, 340, 344, 345
- Ethics, 79, 155-6, 191, 249, 252, 304, 305, 332-337
- *Etymologies* of Isadore, 78, 80
- Eucharistic controversy, 91, 92, 115, 119
- Euclid, 87
- Eusebius, 4
- Evil, 45, 60, 171
- Evodius, 23
- Existence, 68, 69, 73, 104, 108, 129, 185, 200, 223
kinds of, 129, 130, 131, 225
as predicate, 132
- *Existimatio*, 152, 153, 197
- Experience, spiritual, 252
- Experimental knowledge, 196, 198
- Experimental science, 238, 250, 251, 252, 253
- Eyssenhardt, 13

F

- Fabro, C., 293
- Faculty of Arts, 217, 345
- Faculty of Theology, 344, 345
- Faith, 34, 56, 197, 256, 295, 296, 297, 298
and philosophy, 3-8, 20, 29-31, 33, 53-63, 95-98, 119, 121, 123-127,
151-155, 173, 174, 226, 227, 229, 239, 240, 243-244, 256-267, 345,
365, 366
- Fallacy of equivocation, 359, 360, 361
- Falsity, 68, 69
- Fate, 43, 265
- Faustus, 10
- *Fides quaerens intellectum*, 119, 124, 127, 128
- First cause, 302, 318, 339, 340
- Flatten, H., 175
- *Flatus vocis*, 135 248, 343, 344, 216, 217, 343,
- *Fons vitae*, 227
- Foreknowledge, 71
- Forest, A., 167
- Form, 57, 64, 71, 73, 146, 163, 164, 223, 235
- Form, substantial and accidental, 312, 315, 316
- *Forma essendi*, 177, 179, 185
- *Forma nativa*, 164
- Formal distinction, 355
- Fortune, *See* chance
- Fossanova, 290
- Foster, 341
- Fredegisus, 88, 89-90
- Frederick II, 120, 220
- Freedom, 38, 62, 70, 71, 197, 198
- Freud, 215
- Fulbert of Chartres, 157
- Fulbert of Notre Dame, 141, 142
- Fulco, 109, 110
- Fulda, 88, 90, 107
- Fumagalli, 156
- Future facts, 71, 265

G

- Ganfredus, 162 n
- Garin, E., 175
- Gaunilon, 123, 130, 131
- Geiger, L. M., 293
- Genesis, 159, 170, 177, 182, 193
- Geoffrey the Fair, 167
- Gerard of Cremona, 203, 219, 221, 229
- Gerbert of Aurillac, 112-116, 157
- Germanus, St., 107
- Gilbert of Poitiers, 106, 120, 157, 162-167, 175, 182, 184, 211
- Gillett, H. M., 80
- Gilson, É., 38, 48, 53 n, 93, 156, 197, 228, 233, 256, 286
- God, man's knowledge of, 32-39, 49, 96-97, 127-133, 169 ff., 241 ff., 260, 261, 279 ff., 294
and metaphysics, 323
naming, 50, 51, 58, 100, 101, 102, 105, 225, 328-329, 331, 356, 362-364
- Godfrey of St. Victor, 195
- Goichon, A. M., 228
- Gottschalk, 91, 92, 94
- Government of world, 60, 61
- Grammar, 87, 145, 148, 158, 307
- Grataroli, 175
- Gregory, Tullio, 161, 167, 175
- Gregory the Great, 154
- Gregory of Tours, 83
- Gregory IX, 338
- Grosseteste. *See* Robert Grosseteste
- *Guide for Perplexed*, 227
- Guilds, 216
- Gundissalinus, 120, 203-212, 219, 220

H

- Häring, N., 162, 167, 175, 179, 181, 182, 184, 186
- Happiness, 15-17, 45, 61, 62, 63
- Hartshorne, C., 131, 139
- Haskins, C. H., 161, 228
- Hauréau, 92, 108, 114, 115, 116, 181
- Healy, Sr. E., 285

- Heiric of Auxerre, 106, 107 ff.
- Hell, 46
- Heloise, 140, 141, 142
- Henin Ben Isaac, 203
- Henle, 293
- Henry, D., 136, 138, 139
- Henry of Ghent, 361
- Henry II, 187
- Herman the German, 203, 219
- Hermetic writings, 110, 159, 201
- Hilary, 55
- Hincmar of Rheims, 92
- Hippolytus, 4
- Holmberg, J., 175
- Holy Ghost, 172, 173, 177, 199
- Honorius of Autun, 168
- Honorius III, 95
- Hopkins, G. M., 172
- Hortensius, 10, 11, 19
- Hourani, G. F., 228
- Howell, W. 5., 88
- Huber, J., 106
- Huchald, 109
- Hugh of St. Victor, 49, 119, 120, 160, 171, 182, 189-194, 195, 199, 201, 210, 211, 212, 268, 302
- Hume, B., 353
- Humphries, 341
- Hylomorphism, 311-313, 315
universal, 234, 279, 282-285

I

- Ideas, the, 21, 22, 27, 28, 51, 100, 103, 104, 158, 159, 163, 164, 166, 170, 236, 265, 268, 271, 272 ff., 279
- Illumination, 22, 24, 28, 29, 235, 236, 272 ff., 276 ff., 363
- Image of God, 104, 243
- Imagination, 67, 71, 72
- Immortality of soul, 31-32. *See also* Soul
- *In Hexaemeron*, 255, 264, 266, 270 ff.
- Incarnation, 59, 105, 126, 300

- Incorporeal, 69, 70
- Injustice, 60
- Intellect, 23, 29, 67
- Intellectibles, 50, 65. *See also De trinitate* of Boethius
- *Intellectus*, 68, 70, 109
- Intelligence, 71, 72
- Intelligence, Tenth, 222
- Intelligibles, 50, 65
- Irenaeus, 4
- Isaac of Stella, 199, 200
- Isadore of Seville, 78, 79, 83, 86, 91, 192, 205, 206, 210, 211
- *Isagoge*, 52, 64, 65, 66, 87. *See also* Porphyry
- Isaiah, 289
- Islamic thought, 120, 121, 202, 203, 211, 215, 216, 219, 220-228, 229, 232, 253, 339, 343

J

- Jansen, W., 186
- Jeaneau, E., 182
- Jeremiah, 289
- John Chrysostom, 5
- John of the Cross, 290
- John of la Rochelle, 234-235
- John of Salisbury, 143, 157, 158, 160, 162, 165, 175, 186-188, 211
- John of Spain, 203, 219
- John XXII, 367, 368
- Johnson, Sam, 53
- Jones, L. W., 80
- Judgment, 72
- Justin, Emperor, 52
- Justin Martyr, 4, 6

K

- Kant, 133, 280
- Kendzierski, L., 285
- Klibansky, R., 161
- Kneale, 145, 156

- Knowledge, 27, 34, 76
active, passive, 72
and faith, simultaneity of, 259-262
- Knox, R. A., 172
- Koran, 120, 198, 221, 222, 226

L

- Lactantius, 4
- Lanfranc, 122, 134
- Language, 24, 109
study of, 250
- Latin Averroism, 338, 344, 345
- Learning, 23, 24-29
- *Lectio*, 158, 217, 218
- Leibniz, 133
- Leonine revival, 287
- Lewis, C. 5., 3
- *Liber de causis*, 220, 226, 229, 237. 239, 289, 293, 302, 339
- Liberal arts. *See* Arts, liberal
- *Licentia docendi*, 255
- Liebeschuetz, H., 188
- Light, metaphysics of, 235, 236, 258. 268 ff.
- Lindsay, M., 80
- Little, A. C., 254
- Logic, 66, 77, 78, 79, 87, 119, 133, 136-139, 144-151, 1S2, 190, 193.
209, 251, 306-310, 337 ff.
- Logical atomism, 353
- Logical terms, 307, 308
- Logos, divine, 5
- Lothar, 107, 112
- Louis the Bavarian, 367, 368
- Luck. *See* Chance
- *Lumen rationis*, 24
- Lupus Servatus, 93, 107

M

- Mackinney, L. C., 161
- Macrobius, 110, 159, 168, 195

- *Magistri*, 217
- Man, fall of, 44, 105
image of God, 104, 243, 282
microcosm, 104, 105, 181
- Manicheism, 10, 12
- Manitius, 107
- Marius Victorinus, 11
- Marrou, H. I., 14, 48
- Martianus Capella, 13, 59, 77, 86, 94, 96, 168, 195
- Mason, M., 35 n
- Mathematics, 23, 36, 64, 164, 168, 177, 178, 208, 210 n, 235, 246, 250, 251, 253, 304, 321
- Matter, 64, 65, 71, 73, 158, 163, 164, 170, 176, 201, 207, 208, 222, 230, 231, 234, 235, 245, 284, 303, 311, 316
- Maurer, A., 341
- Maurus, Rhabanus. *See* Rhabanus Maurus
- Maximus the Confessor, 94
- McCallum, J. B., 156
- McGarry, D., 188
- McGlynn, 341
- McIntyre, J., 139
- Medium, 269, 270
- Memory, 71
- Menippean satire, 59
- Meno, 220
- Mental activity, 67, 69, 187, 307, 371
- Merton, T., 197
- Metaphor, 50, 101, 102, 200
- Metaphysics, 66, 145, 211, 212, 241, 242, 271, 272, 278, 293, 294, 298, 302, 304, 311, 317-332, 355, 374-376
- *Metaphysics* of Aristotle, 121, 217, 220, 224, 226, 233, 239, 264, 289, 291, 301, 306, 322, 337, 339
- Michael Scot, 203, 219, 220
- Middle term, 359, 360, 363
- Minio-Paluello, 156
- Mobile being, 314, 315, 355
- Moderate realism, 370, 372, 373
- *Modi essendi*, 326, 327, 360
- *Modi significandi*, 325, 326, 329
- Moffat, D. M., 202

- Monad, supreme, 50, 201
- Monastic Schools, 78, 83, 84, 85, 189-202
- Mont Ste-Genevieve, 143, 216
- Monte Cassino, 288
- Montpellier, University of, 216
- Moody, E. A., 156, 376
- Moral philosophy. *See* Ethics
- Moses, 5, 37, 77 n
- Moses Maimonides, 227, 240, 338
- Motion, 64, 207, 208, 313
- Muckle, J. T., 156, 212
- Muller, 366
- Mulligan, 341
- Mynors, 77 n
- Mythical thought, 4

N

- Naples, University of, 220, 288, 290
- *Natura absolute considerata*, 225, 371, 372
- *Natura* in Scotus Erigena, 98 ff., 108
- Natural law, 334 ff.
- Natural reason, 64, 101, 121, 299
- Necessary things, 114, 132
- Necessity, 71, 76, 77, 132, 181, 223, 372, 373
- Negative theology, 50. *See also* Theology
- Neoplatonism, 6, 18, 43, 44, 50, 51, 65, 94, 100, 102, 201, 215, 245, 338
- Nestorian School of Edessa, 203
- *Nichomachean Ethics*, 217, 220, 239, 264, 301, 304, 338
- *Nomen*, 25
- Nominalism, 351, 352, 370
- Numbers. *See* Mathematics

O

- Obedience, 46
- Ockham. *See* William Ockham
- Ockham's razor, 370
- Odo of Cluny, 110
- O'Donnell, J. B., 167, 186

- Old Testament, 5, 6, 7, 44. *See* Scripture
- O'Leary, Delacey, 228
- O'Meara, J. J., 48
- Omnipotence, 39
- *On Being and Essence*, 289, 291
- *On the Eternity of the World*, 340
- *On Generation and Corruption*, 290
- *On the Heavens*, 290
- *On interpretation*, 52, 114, 265, 289, 309, 354
- *On the Principles of Nature*, 289, 291, 311, 312
- *On Separate Substances*, 340
- *On the Soul*, 233, 289, 327
- *On the Unity of Intellect*, 289, 291, 340, 344
- One, the, 100, 177, 222
- One, numerically, 67
- O'Neil, C., 341
- Ontological argument, 127 ff., 223, 281
- Ontological difference, 224
- Ontologism, 279
- Ontology / theology, 322 ff.
- Opinion, 69
- Opinion, knowledge and belief, 295, 296
- Optics, 250, 251
- *Opus majus*, 247, 248, 252
- *Opus minus*, 247
- *Opus oxoniense*, 354
- *Opus tertium*, 247 "O qui perpetua," 110 ff.
- Order of learning, 302
- Ordinatio, 354
- Organon, 217, 220, 222, 309, 310
- Origen, 4, 6, 24, 300
- Original sin, 44, 105, 192
- Orvieto, 289
- Otiric, 112, 113
- Otto I, 112
- Otto II, 112
- Oxford, University of, 216, 146 354 367
- Ozanam, 13

P

- Padua, University of, 238
- Pactow, L. J., 228
- Paganism, 42, 44
- Pantheism, 99, 105, 178
- Paraclete, Oratory of, 143
- Pare, C., 161, 182
- Parent, J. M., 161, 170 n, 171, 173, 175, 182, 183, 186
- Paris, University of, 216, 229, 233, 238, 246, 255, 338, 343, 345
- Parker, J., 51
- Pormenides, 179, 220
- Parson, W., 30
- Participation, 103, 104, 146, 178, 184, 185, 186, 231, 237, 258, 269, 329-331, 362, 363
- Pascal, B., 65
- Paschasius, 91, 115
- Patch, H. B., 74
- Paul, St., 17, 18, 49, 172, 241, 262, 288, 294, 299
- Paul the Deacon, 84
- Paulinus, 84
- Peace, 45, 46
- Pegis, A., 341
- Pelagianism, 12
- Peter the Irishman, 220
- Peter Lombard, 218, 233, 244, 255
- Peter of Pisa, 84
- Peter the Venerable, 120, 140, 153, 197-198
- *Phaedo*, 24, 59, 220
- Phelan, G. B., 139
- Pherecydes, 16
- Philo Judaeus, 3, 13, 21
- Philosophy, ancient, 3-4
 - division of, 57, 64, 65, 79, 87, 112-113, 144, 161, 162, 168-169, 179-180, 199-194, 204 ff., 221, 223, 267-272, 301-306
 - practical, 64, 191, 208-209
 - and faith, 3-8, 29-31, 53-63, 95-98, 119
 - of nature, 310-317
 - and theology, 249, 250, 256, 257, 258, 259, 262, 263, 264, 265, 267, 293-301
- Physics, 64, 79, 162, 168, 368
- *Physics* of Aristotle, 233, 240, 244, 313, 338

- Pieper, J., 351
- Plantinga, A., 139
- Plato, 4, 5, 7, 11, 13, 16, 17, 18, 20, 21, 23, 27, 31, 33, 44, 50, 52, 53, 54, 59, 63, 66, 70, 71, 74, 77, 80, 84, 97, 111, 154, 159, 170, 171, 173, 222, 265, 293
- Platonism, 11, 18-20, 27, 47
- Pleasure, 61
- Pliny, 87
- Plotinus, 3, 11, 16, 31, 33, 50, 120, 220, 221, 222, 226
- *Poetics*, 309
- *Politics*, 290
- Poole, R. L., 133, 136, 161, 186
- Porphyry, 44, 52, 57, 64, 65, 66, 110, 114, 143, 144, 145, 146, 149, 159, 217, 220, 222, 368
- Portalie, 19, 29, 40, 48
- Possible / impossible, 132
- Possibility / necessity, 181, 183
- *Posterior Analytics*, 289, 306, 309, 310
- Potency and act, 311, 314
- Powicke, F. M., 161, 228
- Practical syllogism, 336
- *Praeambula fidei*, 38, 299, 300
- *Praeparatio evangelica*, 4, 285
- Predicability, 70, 108, 109, 146, 308, 372
- Predicables, 65, 145
- Predicate, 113, 147, 166, 185, 230, 261, 295, 296, 307, 308, 309, 356, 369
- Premises, 298, 364
- Prime matter, 222, 284, 312, 315
- Prime Mover, 245, 313, 314
- *Primordia fidei*, 152
- *Principia Mathematica*, 310
- Principles, moral, 334, 335
- Principles, self-evident, 278, 279, 295, 296, 298, 335, 352
- Priscian, 87, 217
- Proclus, 50, 120, 201, 220, 222, 226, 289, 338, 339
- Prohibition of Aristotle, 217
- Proper proportionality, 361
- Property, 65
- Proportions, similarity of, 359

- Proposition, 295, 296, 309
- Protrepticus, 221
- Providence, 43, 60, 61, 62, 70, 264
- Prudence, 304, 305, 335, 336
- Pseudo-Dionysius. *See* Denys the Areopagite
- Pseudo-Plutarch, 206
- Psychology, 307
- Ptolemy, 222
- Pythagoreans, 16, 21, 179

Q

- Quadrivium, 76, 87, 158, 162, 168, 169, 177, 191, 210, 217, 302
- *Quaestio disputata*, 218, 233, 290
- *Quaestio disputata de scientia Christi*, 273 ff.
- *Quaestio disputata de veritate*, 289
- *Quaestio quodlibitalis*, 218, 289, 290, 354
- *Quaestiones subtilissimae in Metaphysicam*, 354
- Quality of a word, 137
- *Quinque viae*, 244
- *Quod est / quo est*, 234

R

- Rahman, F., 223, 224, 228
- Ralph of Laon, 141
- Rand, E. K., 73, 74
- Rashdall, 13, 110, 161, 228
- *Ratio* (reason), 67, 71, 72, 73
- *Ratio communis*, 324, 325, 329, 359
- *Ratio disserendi*, 145
- *Ratio entis*, 323, 360
- *Ratio propria*, 325, 326
- *Rationes aeternae*, 273, 274
- *Rationes seminales*, 32, 40
- Ratramnus of Corbie, 92, 115
- Raymond, Archbishop, 203, 219
- Reason, 35
- Reasoning, 306, 308, 309, 310, 333
- Relation, 58, 59, 102

- Relation, logical, 308, 310
- Remigius of Auxerre, 106, 107, 109-111
- Reminiscence, 24, 70, 71
- Renaissance, 215
- *Res significata*, 166, 325, 326, 329, 358, 359, 360
- *Respondeo*, 219
- *Retractationes*, 12, 13, 18, 21, 24, 41-42
- Revelation, 38, 44, 193, 294, 295, 298, 344
- Rhabanus Maurus, 88-93, 95, 107, 121, 211
- Rhetoric, 60, 87, 145
- *Rhetoric* of Aristotle, 309
- Richard of St. Victor, 194-195, 260
- Richer, 112
- Rickaby, 41
- Robert, P., 286
- Robert Grosseteste, 229, 235-237, 246
- Roger, M., 13
- Rolt, E. C., 51
- Romans 1:19-20, 17, 33, 154, 243, 262
- Roques, R., 51, 139
- Roscelin, 119, 134 ff., 141, 154
- Rowan, J., 341
- Rule, moral, 335, 336
- Russell, R. P., 14, 30
- Rusticana, 52
- Rutledge, B., 51
- Ryan, J. K., 366

S

- Sabellianism, 142
- Saint-Germain-des-pres, 110
- St. Sabine, Convent of, 289
- St. Victor, Monastery of, 11.9, 141, 157, 189, 216
- Salerno, University of, 216
- Sameness, essential, 147
- Schmidt, R., 341
- Scholastic, 59, 89, 219
- Scholasticism, 219, 253
- Schopenhauer, 133

- Schrimpf, G., 75
- Science, demonstrative, 253, 277, 303, 304
- Science, subject of, 257
- Science and art, 77
- Scotus, Duos, 285, 351, 352, 354-366, 369, 370
- Scotus Erigena, John, 4, 5, 49, 51, 65, 92, 93, 94-106, 108, 110, 111, 121, 162, 211
- Scripture, 5, 10, 39, 40, 44, 49, 51, 55, 56, 79, 90, 95, 119, 124, 135, 153, 173, 182, 190, 193, 202, 211, 227, 249, 250, 269, 274, 294, 295, 299
- *Secret of Secrets*, 246
- Secular vs religious, 255, 343
- *Sed contra*, 219
- Seneca, 13, 159
- Sense, 22, 23, 35, 36, 71, 72
- *Sentences* of Peter Lombard, 218, 233, 239, 255, 354, 367, 368
- *Separatio*, 64, 73, 164, 319-321
- Sharif, M. M., 228
- Sharpe, A. B., 51
- Siger of Brabant, 344, 345
- Signification, 26, 89, 102, 149
- Signs, 24, 25, 26
- Sikes, J. G., 156
- Silvestre, H., 110
- Socrates, 16, 53, 60, 84
- *Soliloquies*, 12, 15, 22, 23, 24
- *Sophistical Refutations*, 52, 309, 354
- Soul, 65, 70, 201, 207, 224, 226, 232. 233, 284
creation of, 40
immortality of, 31-32, 284 ff., 314 ff.
powers of, 180-181, 200, 233-234. 249, 315, 345
- Southern, R. W., 139
- Spath, 341
- Species, 65, 67, 68, 69, 70, 109, 145, 146, 147, 165, 187, 307, 308
- Speculative / practical, 302, 303. 305, 332
- Speech, 24, 25
- Spinoza, 133
- Stahl, W. H., 161
- Status, 151
- Stephen Tempier, 346

- Stewart, H. F., 74
- Stoic, 45, 72, 76, 112, 144, 145, 161, 191, 252, 301
- Strabo, Walafrid, 93
- Subject / predicate, 295, 296
- Subsistence, 68, 163, 165, 166, 167
- Substance, 57, 58, 59, 70, 108, 134, 137, 147, 164, 185, 200
- *Substantiae sinceræ*, 163
- Substratum, 58
- Suffering, 60
- *Sufficientiae* of Avicenna, 223
- *Summa de creaturis*, 239
- *Summa contra gentes*, 289, 291
- *Summa theologiae* of Albert, 239, 240, 241, 242
- *Summa theologiae* of Aquinas, 55, 218, 289, 291, 293, 300, 301, 328
- *Summa theologica* of Alexander of Hales, 233, 235
- Supposition, 374
- Syllogism, 309, 359, 360
- Sylvester II. *See* Gerbert of Aurillac
- Symbolic theology, 50. *See also* Theology
- Symmachus, 52, 55, 73
- Syncategorematic terms, 373

T

- Taylor, J., 171, 190, 194
- Teaching, 24-29, 298
- Terminist logic, 353, 373
- Terms, logical, 307, 308, 324
- Terrestrial things, 61
- Tertullian, 4
- Thales, 3
- *Theaetetus*, 205
- Themistius, 222
- Theodolphus, 88
- Theodore of Tarsus, 85
- Theodoric, 52
- Theologian, 7, 8, 100
- Theology, 7, 16, 47, 57, 64, 87, 100, 162, 164, 201, 241, 249, 250, 253, 267, 268, 269, 293-301
affirmative, 50-51, 101, 103

- negative, 50-51, 101, 328
- philosophical, 294, 295
- divisions of, 50, 101, 102, 200, 328, 329
- Theology of Aristotle, 220, 226
- Theophany, 103, 104, 105
- *Theoremata*, 354
- Theoretical philosophy, 64. *See also* Philosophy, division of
- Theos, 100
- Thierry of Chartres, 141, 175-182
- Thirlkel, 341
- Thomism, 287, 288, 292, 300, 351
- Thomson, S. H., 237
- *Timaeus*, 17, 65, 110, 120, 121, 158, 159, 160, 168, 170, 171, 193, 220
- Toledo, 120, 198, 203, 219
- *Topics* of Aristotle, 52, 309
- *Topics* of Cicero, 52
- Tourscher, F. E., 15
- *Tractatus de natura boni*, 239
- *Tractatus de prima principio*, 354, 365
- Transcendentals, 356, 357
- Tremblay, P., 161, 182
- Trinity, 38, 44, 55-59, 103, 119, 125, 126, 134ff., 154, 162, 166, 172, 174, 178, 179, 184, 195, 237, 262, 266, 282, 300
- Trinities in man, 38
- Tritheism, 119, 134
- Trivium, 76, 121, 168, 191, 210, 217, 302
- Truth, 28 31, 37, 68, 69, 237, 276, 306
- *Truth, Disputed Questions on*, 289, 297, 298
- Truth, Two, 226, 344, 346

U

- Universal nature, 225
- Universals, problem of, 65-70, 71, 72, 89, 108, 114, 145-151, 165ff., 187 ff., 195, 200, 225, 368, 369, 370, 371, 372
- Universities, 120, 121, 215, 216-219, 247, 253
- Univocal term, 358, 359, 360, 361, 362, 363, 364
- Univocality of being, 357-362, 363, 364, 375
- Unmoved Mover. *See* Prime Mover
- Urban II, 124

- Urban IV, 289

V

- Vandals, 9
- Van Steenberghen, F., 239, 240. 255. 256, 285
- Varro, 13, 16, 17, 45, 59. 77
- Verbum, 25
- *Vertibilis in non-esse*, 277
- Vestige / image, 281, 282
- *Via moderna*, 351, 352, 353, 368
- Victorine school, 106, 189 ff.
- Virgil, 161
- Virtue, 61, 304, 305
- Virtue and happiness, 61, 63
- Vivarium, 76, 78
- *Vivere est esse viventibus*, 327
- Viterbo, 289
- Vollert, 285
- Vox, 187

W

- Walzer, R., 228
- Wearmouth, 85
- Webb, C., 188
- Weisheipl, J., 245
- Westacott, E., 254
- William of Auvergne, 229-233, 235, 338
- William of Champeaux, 119, 140, 141, 216
- William of Conches, 157, 158, 160, 167-175, 182, 183
- William of Hirschau, 168
- William of Morebeke, 220, 292, 339
- William of Ockham, 351, 352, 367-376
- William of St. Thierry, 140, 143, 157, 160, 167, 172, 173, 174, 198-199
- Williams, W., 197
- Wittgenstein, L., 26
- Wolter, A., 357, 361, 364, 366
- Word, the, 104, 273
- Words, 24, 25

- World, 171, 176
eternity of, 171, 222, 230, 240, 245, 282, 283, 340, 344, 345
- World soul, 171, 172, 173, 176
- Wyser, P., 341

Y

- Yarrow, 78, 85

Z

- Zeno, 60